

Math Notes

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1 Miscellaneous

1.1 Averages

Aside from arithmetic and geometric averages, the last Pythagorean average is the harmonic average which is often used for rates.

If a vehicle drives for 1 hour at $x = 20\text{km/h}$ and another hour at $y = 40\text{km/h}$, then the average speed is the arithmetic average. This is because the denominator of speed is time and a simple addition of fractions of common denominator.

$$\text{Average speed} = \frac{d}{t} = \frac{xt + yt}{2t} = \frac{x + y}{2}$$

If a vehicle drives for 1 km at $x = 20\text{km/h}$ and another 1km at $y = 40\text{km/h}$, then the average speed is the harmonic average.

$$\text{Average speed} = \frac{d}{t} = \frac{2d}{\frac{d}{x} + \frac{d}{y}} = \frac{2}{\frac{1}{x} + \frac{1}{y}}$$

Since the harmonic mean of a list of numbers tends strongly toward the least elements of the list, it tends (compared to the arithmetic mean) to mitigate the

impact of large outliers and aggravate the impact of small ones.

$$\begin{aligned}
 (\text{Average speed})^{-1} &= \frac{t}{d} \\
 &= \frac{v_1^{-1}d_1 + \dots + v_n^{-1}d_n}{d_t} \\
 &= v_1^{-1}\frac{d_1}{d_t} + \dots + v_n^{-1}\frac{d_n}{d_t}
 \end{aligned}$$

Another perspective.

$$\begin{aligned}
 \text{time} &= \frac{\text{distance}}{\text{velocity}} \\
 \text{time} \cdot \text{velocity} &= \text{distance}
 \end{aligned}$$

The velocity is just the sum of the velocities, e.g. $60\text{km/h} + 20\text{km/h} = 80\text{km/h}$ is like a car is driving at 60km/h and receives a boost of 20km/h. We can do this because the denominator is one hour.

If one person is much faster than the other, then the slower person does not make much difference. At the edges, when one person takes “zero time”, the two people take zero time when working together, but that, while sensible in one way, is just math being strange in another. Note the assumption of perfect additive efficiency and no synergy.

Electrical resistors in parallel. Resistance is like a barrier to speed, slowing down current. It might be thought of as a rate,

$$\begin{aligned}
 \frac{1}{R_t} &= \frac{1}{R_1} + \frac{1}{R_2} \\
 R_t &= \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}
 \end{aligned}$$

The current travels a unit of one (over the parallel resistance circuit) being slowed down by resistance R_1, R_2 for a total of R_t .

The harmonic mean is one of the three Pythagorean means. For all positive data sets containing at least one pair of nonequal values, the harmonic mean is always the least of the three means, while the arithmetic mean is always the greatest of the three and the geometric mean is always in between.

Root-mean square. Use when negatives are involved. For instance, to find the average velocity of particles in a container, the individual velocities cannot be added as they would sum to 0 or close to 0. So instead average their squares and take the square root. This is the standard deviation.

1.2 Functions

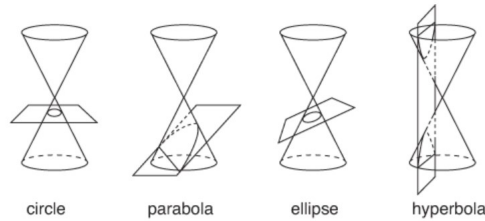
Functions can be split into even (symmetric) and odd (rotationally symmetric) functions.

$$f_{\text{even}}(x) = \frac{f(x) + f(-x)}{2}$$
$$f_{\text{odd}}(x) = \frac{f(x) - f(-x)}{2}$$

A Taylor expansion is a sequential series of odd and even functions.

1.3 Conic sections

Circles and ellipses, parabolas, and hyperbolas. The conic section actually also includes straight lines and the vertex (degenerate conic sections). The conic structure is like swiveling a tilted line around the vertex. It can also be intuitively understood as space-time, with the singularity as the event. So the conic is like the lowest energy/information configuration. The conic is also an equation in two variables with the highest degree being a square term.



The one equation for a conic is

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

where conic sections are various cross-sections. There is no explicit z-axis. The conic sections are not intersections of a plane with the conic equation. It's just that conic sections can be assembled into the conic equation.

Circle: When x and y are both squared and the coefficients on them are the same - including the sign. F is the radius.

Ellipse: When x and y are both squared and the coefficients are positive but different. When a is larger than b, the equation of a circle tilts upward on the x-axis, elongating the x-axis. So it enhances the x-values. However, the other side of the x-axis will shrink, so the ellipse center has to move from the origin and/or the ellipse is drawn by smooth changes in altitude along the conic. The

case is reversed when b is greater than a . Note the equation of the ellipse is actually:

$$\frac{x^2}{b} + \frac{y^2}{a} = \frac{1}{ab}$$

where a and b become like compression or expansion factors. The ellipse has two foci, and the proof for how this is visualized in the conic is complicated - it relies on two balls each tangent to the conic section.

Parabola: When either x or y is squared - not both. F is part of the vertical translation. Basically an infinitely elongated ellipse because if a or b is 0, then the ellipse equation has either x^2 or y^2 being infinitely stretched. The parabola only exists at the angle parallel to the cone.

Hyperbola: When x and y are both squared, and exactly one of the coefficients is negative and exactly one of the coefficients is positive. Rotating towards the vertical from the parabola. The hyperbola will have a shifted center but it will always be symmetric. The hyperbola is like a 'negative' circle or a circle turned inside-out, so at least one of the coefficients of the squared terms is negative. A hyperbola is different from a quaternion circle inside-out on a plane; a hyperbola is equivalent to just a circle inside-out which is just a plane, a plane intersecting with the conic.

As the ellipse turns into a parabola, the two branches race to infinity very rapidly. But when the plane is tilted as a hyperbola, the branches move less slowly and slower still as the plane intersects the conic vertically. As the plane of intersection moves past the vertical, the asymptotes get narrower. So the asymptotes are widest when the intersection plane is vertical. So if the intersection plane is off the vertical, both branches of the hyperbola have narrower asymptotes because they move slower.

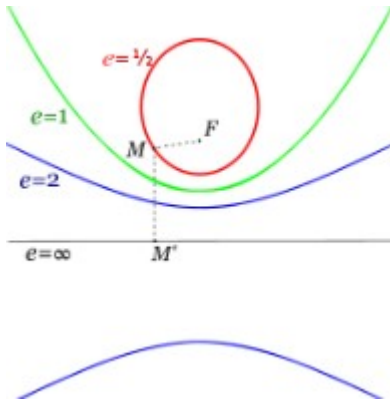
Hyperbola applications. A hyperbola is also $xy = k$ or something fixed like the number of chocolates as k and x being the number of people and y the number of chocolates per person. A hyperbola is also the shape of perfect elasticity as revenue remains fixed. This is also why the axes are asymptotes, if k is fixed, then it cannot ever hit the horizontal and vertical axes. A catenary is also a hyperbola.

Note that xy is an axis rotation term so that b is a rotational factor.

F is altitude, the z -axis. A rising F is the top conic and declining F is the bottom conic. As F rises or declines, the x and y that trace the conic will need to change, expanding (rising F) and shrinking (declining F) for positive F and vice versa.

Conic sections are also like a circle that is stretched infinitely, to an ellipse, parabola, and then hyperbola. This is the same movement when rotating the

intersecting plane through the conic.



Conic sections all have two foci because they are all fundamentally an ellipse. An ellipse and hyperbola obviously have two foci each. A parabola has a focus and directrix - though the directrix is just the focus at infinity. A circle has the two circle in the same point, the center.

1.4 Exponents

Exponents are about scaling the number line by the exponent base value, e.g. Richter scale has a base-10. The exponent index is the x-value, the independent variable.

1.4.1 Fractional exponents

$2^{1.5}$ is unintuitive. However,

$$2^{1.5}2^{1.5} = 2^3 = 2 \cdot 2 \cdot 2$$

$$2^{1.5} = 2 \cdot 2^{0.5}$$

And $2^{0.5}$ is about what multiplied by itself (ie twice, which is the inverse of the exponent 0.5) gives 2. So, fractional exponents are about understanding roots (exponents between 0 and 1).

1.4.2 Negative exponents

2^{-1} is unintuitive. However,

$$2^{-1}2 = 2^{-1}2^1 = 2^0 = 1$$

Therefore

$$2^{-1} = \frac{1}{2}$$

Division is the reciprocal of multiplication. Using a base of 2, the negative side of the exponential scale is a reciprocal.

1.4.3 Imaginary exponents

2^i is unintuitive. However,

$$2^i 2^i = 2^{-1}$$

The imaginary exponent rotates the base-2 around the fulcrum of the exponential number line, which is 1. Rotating it twice leads back to the real number line at the 'opposite' side of 2 of the fulcrum, which is $\frac{1}{2}$.

1.4.4 Tetrations

The hyperoperators are addition $a + n = a + (1 + 1 + \dots + 1)$, multiplication $a \times n = a + a + \dots + a$, exponentiation $a^n = a \times a \times \dots \times a$, tetration, pentation, etc. Addition is one-dimensional, multiplication is two-dimensional, exponents are multi-dimensional, tetrations are multi-multi-dimensional, etc.

Tetration (or hyper-4) is an operation based on iterated, or repeated, exponentiation. Tetration is denoted by $\uparrow\uparrow$ or the left-exponent ${}^x b$.

$${}^3 a = a^{a^a}$$

It can be extended to real and complex numbers. Inverses are the super-root and super-log. Note as a matter of notation that

$$3^{5^7} = 3^{(5^7)} \neq (3^5)^7$$

Some numbers tetrated infinitely, ${}^\infty a$, can converge or diverge depending on how big a is. For instance, ${}^\infty \sqrt{2} = 2$. The largest value of convergence is e .

Tetration calculus. Example with differentiation.

$$\frac{d}{dx} {}^2 x = \frac{d}{dx} (e^{\ln x})^x = \frac{d}{dx} (e^{x \ln x}) = (e^{x \ln x}) \frac{d}{dx} (x \ln x)$$

1.4.5 Pentation

$$\begin{aligned} a \uparrow\uparrow\uparrow 3 &= a \uparrow^3 3 = {}^a {}^a a \\ 3 \uparrow\uparrow\uparrow 3 &= 3 \uparrow\uparrow (3 \uparrow 3) \end{aligned}$$

1.5 Factorials

1.5.1 Gamma function

Euler's approximation to the factorial began with

$$x! = \int_0^1 (-\log s)^x ds$$

but substituting $s = e^{-t}$,

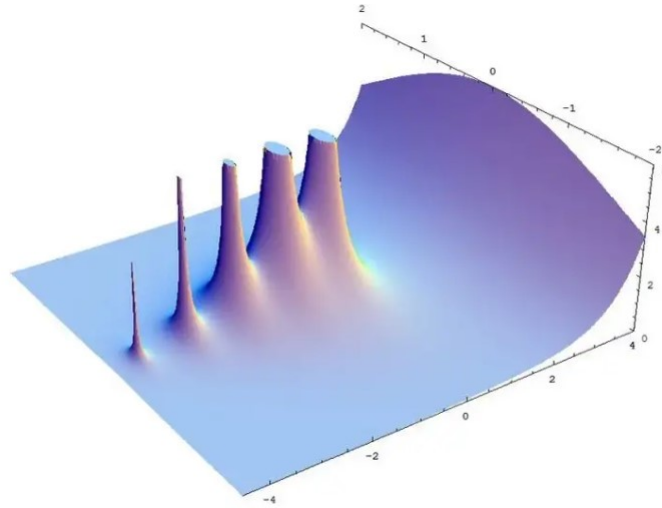
$$x! = \int_0^\infty t^n e^{-t} dt$$

and with a minor adjustment,

$$\Gamma(x) = (x-1)! = \int_0^\infty t^{x-1} e^{-t} dt$$

works for all numbers (even complex numbers) except non-positive integers, though it doesn't work well below $0! = 1$. Note that $0! = \Gamma(1) = (1-1)! = 0! = 1$. The variable t is a dummy variable and is therefore like a simplex for the calculation (of a roughly bell-shaped curve). It is calculated by repeated integration by parts, which is like a factorial. The repetition is also like a Taylor series, as the integration by parts involves repeated differentiation. The repeated integration by parts involves factorials. The calculation can be simplified to the factorial by taking the limits from 0 to infinity.

The gamma function is basically the area under the curve of the integral. So it maps whole numbers to areas and an analytic continuation is then possible.

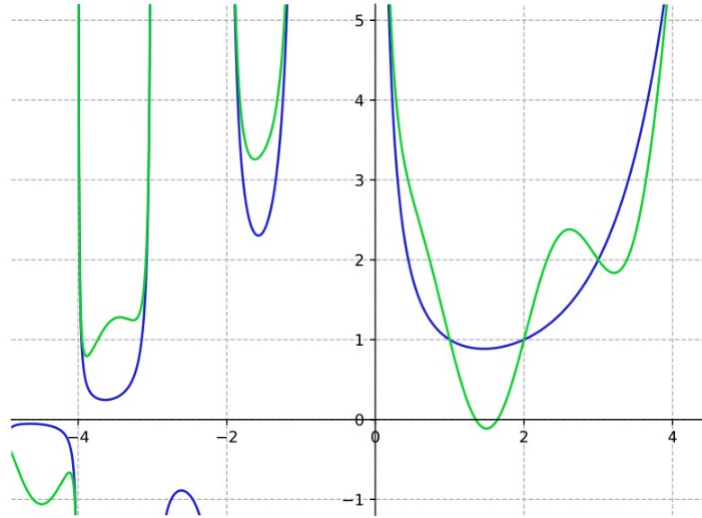


The gamma function does not do well below $x = 1$ which corresponds to $0! = 1$, and it was not meant to be used there, which is beyond a infinite value in Euler's original log formulation. Non-positive integer factorials are undefined:

$$0! = 1 = 0 \cdot (0-1)!(0-2)!... = 0 \cdot (0-1)!$$

$$\frac{1}{0} = (-1)! = \infty$$

The gamma function is not the only approximation to the factorial. Other functions, such as with sin, can also be used, though it is less natural because it is less smooth (blue is gamma, green is sin):



However, under certain conditions ($f(1) = 1$, $f(x+1) = xf(x)$, smooth, log convex (i.e. second derivative is positive)), the gamma function is unique.

Factorials (including the gamma function of course) can be approximated at high numbers by Stirling's approximation:

$$\Gamma(x+1) \sim \sqrt{2\pi x} \left(\frac{x}{e}\right)^x$$

1.5.2 Gamma function intuition

Factorial values rise fast, like exponential functions - but even faster. Use logs to rescale them, mitigating this effect.

$$\ln(n) \sim \ln(n+1) \sim \ln(n+2)$$

Then,

$$\ln(n) \sim \ln(n+k)$$

Note these recursive expressions:

$$\ln(n!) = \sum_{k=1}^n \ln(k)$$

$$\ln((x+n)!) = \ln(x!) + \sum_{k=1}^n \ln(x+k)$$

If x is very big,

$$\ln((x+n)!) \sim \ln(x!) + \sum_{k=1}^n \ln(x)$$

$$\ln((x+n)!) \sim \ln(x!) + n \ln(x)$$

Then,

$$\ln(x!) + \sum_{k=1}^n \ln(x+k) \sim \sum_{k=1}^N \ln(k) + n \ln(x)$$

$$\ln(x!) \sim \sum_{k=1}^n \ln(k) - \sum_{k=1}^n \ln(x+k) + n \ln(x)$$

$$\ln(x!) \sim \sum_{k=1}^n \ln\left(\frac{k}{x+k}\right) + n \ln(x)$$

$$\ln(x!) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \ln\left(\frac{k}{x+k}\right) + n \ln(x)$$

$$x! = \lim_{n \rightarrow \infty} \exp\left(\sum_{k=1}^N \ln\left(\frac{k}{x+k}\right) + n \ln(x)\right)$$

$$x! = N^x \lim_{n \rightarrow \infty} \exp\left(\sum_{k=1}^N \ln\left(\frac{k}{x+k}\right)\right)$$

$$x! = N^x \lim_{n \rightarrow \infty} \Pi\left(\frac{k}{x+k}\right)$$

This also shows the intuition for Euler's number in the gamma function. Another intuition for Euler's number in the gamma function is that it makes the integral of the gamma function recursive using integration by parts.

1.5.3 Beta function

The Beta function is:

$$B(z_1, z_2) = \int_0^1 t^{z_1-1} (1-t)^{z_2-1} dt$$

where the real parts of $z_1, z_2 > 0$.

$$B(z_1, z_2) = \frac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1 + z_2)}$$

which is similar to the binomial coefficient formula:

$$B(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}$$

1.5.4 Fractional factorials

The gamma function interpolates factorials but might lack intuition.

Natural number factorials like $3!$ are intuitive. It can be viewed as the permutations of 3 cards. For decimal factorials,

$$3.5! = 3.5 \cdot 2.5 \cdot 1.5 \cdot 0.5!$$

So, the intuition of factorials between 0 and 1 (like $0.5!$) is the intuition of decimal factorials. The lowest point is at around 0.46... (might be an irrational number). Below $x < 1$, the number of permutations should be below 1 as there is less than one card to shuffle - you cannot even have one way to shuffle it. At the same time, there cannot be 0 ways to shuffle it because there's something there. It reaches the lowest point at 0.46... and above this number, it resembles more like the integer 1, which has one way of shuffling. Below the minimum point, the fractional card resembles more 0, which also has a concrete interpretation of one shuffle or permutation. The point should be below 0.5 because factorials are asymmetric, as they accelerate as the number rises.

1.5.5 Negative factorials

Must use the gamma function here too.

At -1 factorial, or $\Gamma(0)$, the number of permutations approaches infinity because it is non-sensical to shuffle -1 objects. Mathematically,

$$(-1)! = (-1) \cdot (-2) \cdot (-3) \cdot (-4) \dots$$

which approaches either infinity or negative infinity. So, there are infinite permutations or this is undefined. This is also the same for all non-positive integers: $(-1)!, (-2)!, (-3)! \dots$

However, there are finite factorial values between these negative integers (i.e. negative decimals) because it mimics the shape between 0 and a high number factorial, which sags with minimum around 0.46.... The gamma function is like an interpolated parabola between 0 and a high number factorial, and this shape is also found between negative integers. So, negative decimal factorials are interpolations between two non-sensical shuffling situations. The negative integers are the 'known' non-sensical shufflings, and the negative decimals are the 'unknown' non-sensical shufflings.

1.5.6 Imaginary factorials

This is just the factorial extended out to the complex plane, but only the real number changes, so the factorial runs 'parallel' to the real plane or number line. See the 3D image above. It does not approach infinity because of the moderating effect of the imaginary part.

Intuition. Imaginary numbers are about real numbers with a 'half' or 'potential' quality to them, which stabilises the factorial result.

1.6 Dimensions

Fractional dimensions are fractals. Negative dimensions are already part of the number line (negative numbers). Can also think of them as reversing the signs of all coordinates. Complex dimensions are just ordinary dimensions with their complex planes.

1.7 Random matrix universality

Random matrices are stochastic matrices which can characterize bus time arrivals, energy of atomic nuclei, chicken retinas, where the constituent elements are not completely independent of each other. The matrices have eigenvalues that appear in a pattern on a line with probabilities that take on a consistent probability distribution. The consecutive eigenvalues are never too close to each other or too far apart from each other. This distribution of eigenvalues is also seen with gases and Riemann zeta function zeroes, indicating a possible connection to prime numbers.

1.8 Fractional calculus

1.8.1 Fractional derivative

For exponents, the regular derivative is

$$\frac{d}{dx}x^n = \frac{n!}{(n-k)!}x^{n-k}$$

and since

$$\begin{aligned} n! &= \Gamma(n+1) \\ (n-k)! &= \Gamma(n-k+1) \end{aligned}$$

where $0 < k < 1$. Then,

$$\frac{d}{dx}x^n = \frac{\Gamma(n+1)}{\Gamma(n-k+1)}x^{n-k}$$

Examples.

$$\begin{aligned} \frac{d^{1/2}}{dx^{1/2}}x &= \frac{\Gamma(2)}{\Gamma(\frac{3}{2})}x^{\frac{1}{2}} = \frac{2}{\pi}x^{\frac{1}{2}} \\ \frac{d}{dx}x &= \frac{d^{1/2}}{dx^{1/2}}\left(\frac{d^{1/2}}{dx^{1/2}}x\right) = \frac{d^{1/2}}{dx^{1/2}}\left(\frac{2}{\pi}x^{\frac{1}{2}}\right) = 1 \\ \lim_{k \rightarrow \infty} \frac{d^{1/k}}{dx^{1/k}}x &= \frac{\Gamma(2)}{\Gamma(2 - \frac{1}{k})}x^{1 - \frac{1}{k}} = x \end{aligned}$$

For $y = e^{kx}$,

$$D^1 = ke^{kx}$$

$$D^2 = k^2 e^{kx}$$

$$D^{1/2} = k^{1/2} e^{kx}$$

$$D^a = k^a e^{kx}$$

Check:

$$D^{1/2} D^{1/2} e^{kx} = D^{1/2} k^{1/2} k^{1/2} e^{kx} = ke^{kx}$$

1.8.2 Imaginary derivative

Recall

$$D^a = k^a e^{kx}$$

Then

$$\begin{aligned} D^i &= 2^i e^{2x} \\ &= (e^{\ln 2})^i e^{2x} \\ &= (e^{i \ln 2}) e^{2x} \\ &= \underbrace{(\cos(\ln 2) + i \sin(\ln 2))}_{\text{complex}} e^{2x} \end{aligned}$$

by Euler's identity. As another example,

$$\begin{aligned} D^i e^{ix} &= i^i e^{ix} \\ &= (e^{\pi/2i})^i e^{ix} \\ &= \underbrace{e^{-\pi/2}}_{\text{real}} e^{ix} \end{aligned}$$

1.8.3 Cosine and sine

Use this from above

$$D^i e^{ix} = e^{-\pi/2} e^{ix}$$

on

$$\begin{aligned} \cos(x) &= \frac{e^{ix} + e^{-ix}}{2} \\ \sin(x) &= \frac{e^{ix} - e^{-ix}}{2i} \end{aligned}$$

to get

$$\begin{aligned} D^\alpha \cos(x) &= \cos\left(x + \frac{\pi}{2}\alpha\right) \\ D^\alpha \sin(x) &= \sin\left(x + \frac{\pi}{2}\alpha\right) \end{aligned}$$

1.8.4 Hyperbolic cosine and sine

Cosh looks like a parabola but it's exponential:

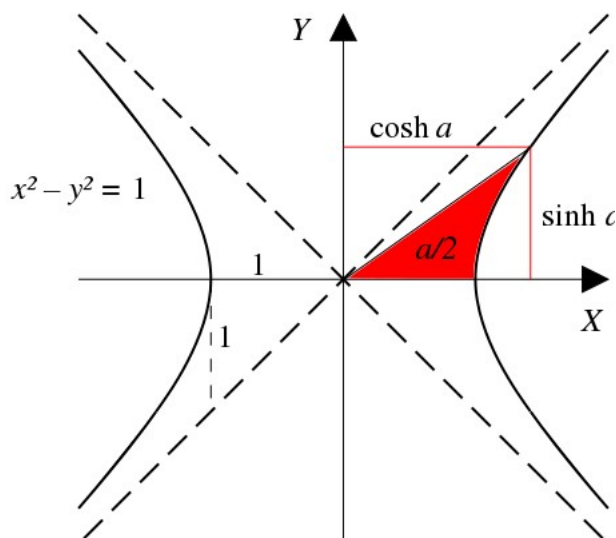
$$y = \cosh(x) = \frac{e^x + e^{-x}}{2}$$

Sinh is rotationally symmetric:

$$y = \sinh(x) = \frac{e^x - e^{-x}}{2}$$

The derivative of cosh is sinh and vice versa (no sign changes). This is intuitive.

Just as $\cos^2(\theta) + \sin^2(\theta) = 1$ in a circle using Pythagoras, $\cosh^2(\theta) - \sinh^2(\theta) = 1$ in the hyperbola, though θ is actually *twice* the area in red below:



This is analogous to circular trig functions, where θ in a sector of a unit circle has area $\frac{\theta}{2}$. A unit circle sector of angle θ is equal to $\frac{\theta}{2}$ because $\frac{\theta}{2\pi} \cdot \pi = \frac{\theta}{2}$. Using the angle in the hyperbola does not mean much as the angle just approaches the asymptote. It is better to look at the area in red above. The connection between the circle and hyperbola trig functions exists because the hyperbola is a circle that is ripped apart.

The cosh is the even part of the function and sinh is the odd part of the function. So,

$$e^x = \cosh(x) + \sinh(x)$$

this is the hyperbolic analog of Euler's identity. Both Euler's identity and its hyperbolic analog break up changes denoted by e^x, e^{ix} , which are the trig functions (including hyperbolic trig functions). In turn, the trig functions are related to triangles on the circle and hyperbola, which are themselves related to Euler's number which denotes change.

1.8.5 Non-locality

Fractional derivatives are non-local, while integer derivatives are local. So, combining fractional derivatives (e.g. a half-derivative of a half-derivative is a first derivative) might be reflecting the harmonic nature of integers.

The fractional derivative is defined as a weighted average of the function values over the entire domain, rather than just at the point of interest.

The weightings used in the fractional derivative depend on the order of the derivative and can be thought of as a form of smoothing or filtering that captures the long-range behavior of the function. This means that the fractional derivative takes into account the behavior of the function at all points in the domain, rather than just in the immediate vicinity of the point of interest.

As a result, the fractional derivative is able to capture features of a function that are not visible at the local level, such as the overall curvature or shape of the function. This makes it a powerful tool for analyzing and modeling complex phenomena in a wide range of fields.

1.9 Cosine and sine

Since cosine fluctuates between 1 and -1 with a changing independent variable x , the below is intuitive:

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

where e^{ix} is the CCW rotation and e^{-ix} is the CW rotation. We take the average of these because we need to cancel out the imaginary components of these functions. We want to isolate just the horizontal (real) component. While the real component can just be written as $\cos(x) = e^{ix}$, it contains an imaginary component whereas the cosine is purely real function. The above cosine representation also shows how Euler's number, i , rotations, and cosine are interrelated.

For the sine function:

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

It is a difference (unlike the cosine which is a sum) because the result is the vertical component of the rotation e^{ix} . Further, simply using e^{ix} leaves a horizontal component by-product. To remove this by-product, take the difference

and divide by 2. Divide by 2i because the vertical component is the imaginary component which is converted to a real value.

Putting these together,

$$e^{ix} = \cos(x) + i\sin(x)$$

so that cos is the 'even' part of the function and sin is the 'odd' part. Euler's number more generally then refers to circles and hyperbolas which are shapes with 'changing' or accelerating tendencies.

1.10 Euler's number e

Euler's number in e^x is the derivative of itself. It therefore has a fixed property, and it may be the fixed point, like the fixed point in Brouwer's fixed point theorem.

1.11 Fourier transforms

Uses Euler's identity, which for a real-valued (i.e. not complex-valued) wave, has a symmetrical frequency spectrum because, for

$$A(x) = a + bi, A(-x) = a - bi$$

Then

$$A(x)e^{ix} + A(-x)e^{-ix} \\ (a + bi)(\cos x + i\sin x) + (a - bi)(\cos x - i\sin x) = 2a\cos x$$

1.12 Exponential distribution

The exponential distribution is a continuous probability distribution that assumes events occur at a constant rate λ over time. The PDF is

$$f(x) = \lambda e^{-\lambda x}$$

which can be derived by differentiating the CDF

$$F(x) = 1 - e^{-\lambda x}$$

The intuition of the PDF is that the probability of failure or occurrence (over an interval of $f(x)$) declines as time x increases from the base value of λ (when $x = 0 \rightarrow f(x) = \lambda$). The PDF is exponentially declining because the probability value depends on the time elapsed. The PDF is more intuitive when considering $\lambda = 1$, then 'scaling up' or 'scaling down' $f(x)$ with a different λ . It is exponential in the first place because the discrete counterpart to the exponential distribution, the geometric distribution, is 'exponential':

$$p(x) = (1 - p)^k p$$

where p is the success.

1.13 Weibull distribution

The PDF is

$$f(x) = \left(\frac{k}{\lambda}\right) \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}$$

which can be derived by differentiating the CDF

$$F(x) = 1 - e^{-(x/\lambda)^k}$$

The PDF intuition is similar to that of the exponential distribution. The k is the shape parameter and is analogous to the exponential distribution's rate parameter. If $k > 1$ then the failure rate is increasing, $k = 1$ is a constant failure rate, $k < 1$ is a decreasing failure rate. The scale parameter λ indicates the scale of the failure rate. The base value at the start $x = 0$ is $\frac{k}{\lambda}$, which is the scaled failure rate. The $\frac{x}{\lambda}$ is the scaled time to failure.

1.14 Clustering

Hierarchical methods. In agglomerative hierarchical algorithms, we start by defining each data point as a cluster. Then, the two closest clusters are combined into a new cluster. In each subsequent step, two existing clusters are merged into a single cluster. In divisive hierarchical algorithms, we start by putting all data points into a single cluster. Then we divide this cluster into two clusters. At each subsequent step, we divide an existing cluster into two clusters.

Non-hierarchical methods. In a non-hierarchical method, the data are initially partitioned into a set of K clusters. This may be a random partition or a partition based on a first "good" guess at seed points which form the initial centers of the clusters. Then data points are iteratively moved into different clusters until there is no sensible reassignment possible. The initial number of clusters (K) may be specified by the user or by the software algorithm. The most commonly used non-hierarchical method is MacQueen's K-means method.

Model based methods. A model based method uses a mixture model to specify the density function of the x -variables. In a mixture model, a population is modeled as a mixture of different subpopulations, each with the same general form for its probability density function and possibly different values for parameters, such as the mean vector. For instance, the model may be a mixture of multivariate normal distributions. In cluster analysis, the algorithm provides a partition of the dataset that maximizes the likelihood function as defined by the mixture model.

1.15 Bias-variance tradeoff

Assume

$$y = f(x) + \epsilon$$

Mean-squared error MSE of an estimator is:

$$\begin{aligned}\text{MSE}(\hat{\beta}) &= E(\hat{\beta} - \beta)^2 \\ &= \text{Var}(\hat{\beta} - \beta) + (E(\hat{\beta} - \beta))^2 \\ &= \text{Var}(\hat{\beta}) + \text{Bias}^2(\hat{\beta})\end{aligned}$$

because $E(X^2) = \text{Var}(X) + (E(X))^2$.

The expected prediction error EPE:

$$\begin{aligned}\text{EPE}(x) &= E[(y - \hat{f}(x))^2] \\ &= E[(y - f(x) + f(x) - \hat{f}(x))^2] \\ &= E[(y - f(x))^2] + E[(f(x) - \hat{f}(x))^2] + E[2(y - f(x))(f(x) - \hat{f}(x))] \\ &= \text{Var}(\epsilon) + \text{Bias}^2[\hat{f}(x)] + \text{Var}[\hat{f}(x)] + \text{Var}(\epsilon)\end{aligned}$$

1.16 Regularization

Adds a penalty for a larger β . Lasso cost function:

$$\sum (y_i - \bar{y})^2 + \sum |\beta|$$

Ridge regression cost function:

$$\sum (y_i - \bar{y})^2 + \sum \beta^2$$

In Lasso regression, the penalty term is the L1 norm of the coefficients, while in Ridge regression, it is the L2 norm. The difference between these two methods lies in the shape of the penalty term and how it affects the coefficients.

The L1 penalty term used in Lasso regression has the effect of driving some of the coefficients to zero, effectively performing feature selection. This is because the penalty term creates a "sparsity" constraint on the coefficients, forcing some of them to be exactly zero, while the remaining coefficients are shrunk towards zero.

On the other hand, the L2 penalty term used in Ridge regression shrinks all the coefficients towards zero, but none of them are set to exactly zero. This results in a more "even" shrinkage of all the coefficients, regardless of their initial values.

The intuition behind why the regularization penalty only allows shrinkage of beta values and not an increase in their magnitude is that large coefficients can lead to overfitting by allowing the model to fit noise in the data. By constraining the magnitude of the coefficients (i.e. introducing bias), regularization prevents

this overfitting by making the model simpler and more generalizable (i.e. reduces variance). We do not want to increase the betas because that would be overfitting - we prefer underfitting to overfitting.

1.17 Bayesian thinking

Bayesian thinking is intuitive, updating a prior with likelihoods to get posteriors. However, the prior is subjective. To get an 'objective' prior, assign uniform priors, which is the same as a frequentist approach. Frequentism is a special case of Bayesian probability.

2 Time series

2.1 White noise

There are several types of white noise. The most basic is a discrete-time, discrete-step (+1 or -1), equiprobability (50-50 up or down) white noise. It is 'white' because the steps are uncorrelated with each other, and if the steps are independent of each other (a stronger condition), it is independent white noise. Gaussian white noise has step sizes that vary with the normal distribution.

White noise can be made more complex by making the time steps continuous, varying the step sizes, and varying the probabilities. The most common white noise is

$$\begin{aligned} E(e_t) &= 0 \\ E(e_t^2) &= \sigma^2 \end{aligned}$$

(if σ^2 is constant and any distribution is constant, then it's identically distributed uncorrelated white noise) and the errors are uncorrelated through time:

$$E(e_t e_{t+h}) = 0, h \neq 0$$

But the above includes a potential relationship between the errors. For instance, the errors can still be dependent even though they're uncorrelated:

$$e_t = \begin{cases} \text{uniform}(-1, 1) & i \neq 3N \\ e_{t-1} e_{t-2} & i = 3N \end{cases}$$

where N is an integer. Essentially, the first two time steps are Rademacher-distributed. Then, $t, t+1$ will each be independent but the third $t+2$ will be dependent on the first two. The errors are uncorrelated but dependent.

To prove independence, need to show that for every t ,

$$f_{X_{t_1}, X_{t_2} \dots}(x_1, x_2 \dots) = \prod_t f_{X_{t_n}}(x_n)$$

This is even stricter than pair-wise independence, and it's a common pitfall for people who think uncorrelated + Gaussian = independence.

Independent white noise is stronger than uncorrelated white noise: e_t and e_{t+h} , $h \neq 0$ are independent.

If the *independent* white noise has a Gaussian distribution, i.e.

$$e_t \sim N(0, \sigma^2)$$

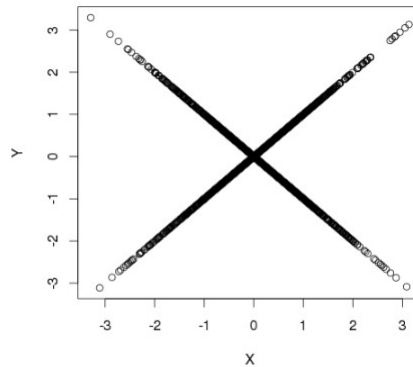
then it is a *Gaussian white noise process*. It is *not* enough to have uncorrelated white noise plus the white noise being Gaussian. Independence requires the white noise be uncorrelated and *jointly* (bivariate or multivariate) Gaussian.

In most time series analysis, assume iid Gaussian white noise:

$$e_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

2.1.1 Aside: Uncorrelated and Gaussian

Suppose $X \sim N(0, 1)$. Let W be a Rademacher distribution with $W = 1$ or $W = -1$, each with probability $1/2$, and assume W is independent of X . Let $Y = WX$. Then X and Y are uncorrelated, they both have the same normal distribution, *but they are not independent*.



However, if the variables were jointly (i.e. bivariate) Gaussian, then the graph would show independence, i.e. a symmetric circular cloud of dots. The joint Gaussian would 'shape' the relationship between the variables as represented in the graph so that they're independent.

Note two random variables X and Y are bivariate normal if $aX + bY$ has a normal distribution for all $a, b \in R$. This is due to the probabilities of X and

Y being intuitively 'coordinated' as a pair (X, Y) in a bivariate Gaussian distribution so that sum $aX + bY$ will always yield a Gaussian itself.

This is why to prove independence of white noise above, it's necessary to involve the *joint* distribution.

The typical (constant variance uncorrelated) white noise is usually enough. Even if it were not independent and/or not identically distributed, it probably won't significantly interfere with analysis.

When white noise is summed, it forms a random walk. If the steps are determined by a normal distribution, it is a Gaussian random walk.

2.1.2 Differencing white noise

The first difference of an iid white noise process is *not* iid white noise. For instance, if

$$y_t = e_t$$

where $e_t \sim \text{iid w.n.}$,

$$y_t - y_{t-1} = e_t - e_{t-1} = w_t$$

and w_t will not be iid w.n. because

$$\text{Cov}(w_t w_{t-1}) = \text{Cov}((e_t - e_{t-1})(e_{t-1} - e_{t-2})) \neq 0$$

because of the dependence of e_{t-1} .

However, differencing unrelated (i.e. completely independent of each other at all times) iid white noise likely yields iid white noise, even if the distributions are different. For instance,

$$y_t = e_t$$

$$z_t = f_t$$

Then $y_t - z_t = e_t - f_t = g_t \sim \text{iid w.n.}$. The variables must be iid white noise, and merely uncorrelated white noise will not work.

2.2 Ljung-Box test

To determine autocorrelation, can look at the ACF. Can also use tests.

The Ljung-Box test tests whether a group of autocorrelations of a time series are different from zero. Instead of testing randomness at each distinct lag, it tests the "overall" randomness based on a number of lags, and is therefore a portmanteau test.

H_0 : The data are independently distributed (i.e. the correlations in the population from which the sample is taken are 0, so that any observed correlations

in the data result from randomness of the sampling process). H_a : The data are not independently distributed; they exhibit serial correlation.

The test statistic is the sum of accumulated sample autocorrelations with some adjustments to make it fit a chi-squared distribution:

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$$

where n is the sample size, $\hat{\rho}$ is the sample autocorrelation at lag k , and h is the number of lags being tested. Under H_0 the statistic Q asymptotically follows a chi-squared distribution.

The Box-Pierce test is a simplified version of the Ljung-Box test.

The test is used on regression residuals to determine whether they're autocorrelated or not.

2.3 Causality and invertibility

AR(p) processes can be converted to an infinite MA(q) process in a Koyck transformation (by manual algebraic expansion or using operator algebra with the lag operator), showing causality. The value of the times series y is then a function of the finite series of the present and past errors. If the coefficient is less than 1, then the value of the time series y is finite.

Using a Koyck-like but opposite transformation for MA(q) processes (again by either manual algebraic expansion or using the lag operator), we can express the error in terms of the present and past time series values y . Again, if the coefficients are less than 1, then the value of the error is finite.

Example of the non-uniqueness of connection between values of θ_1 and $1/\theta_1$ in MA(1) Model. In the MA(1) model, for any value of θ_1 , the reciprocal $1/\theta_1$ gives the same value for:

$$\rho = \frac{\theta_1}{1 + \theta_1^2}$$

As an example, use $\theta_1 = 0.5$, and then use $1/\theta_1 = 1/(0.5) = 2$. You'll get the same ρ in both instances.

The example is problematic because if $\theta_1 = 2$, the Koyck transformation equation will have the time series value y become infinite. The equations are then unusable and unrealistic. So, to rule out the $\theta_1 = 2$ and such consequences, restrict the MA(q) equation such that $\theta_1 < 1$. In practice, will estimate the ρ from the data, and will be left with two possibilities - so rule out any coefficient greater than 1. Then, the MA(q) model can be made algebraically equivalent to an AR(p) - this is invertibility.

A finite AR process can be represented as an infinite MA process, and a finite MA process can be represented as an infinite AR process.

AR and MA processes are FUNDAMENTALLY different. They are not just mathematically related processes. The x_t in an AR system is a sum of infinite error terms. The e_t in an MA system is the sum of infinite time series terms and so is like a 'residual' that balances the past iterations x_{t-h} with x_t .

2.4 ACF and PACF

The ACF has no conditioning. It is the 'real-life' correlation. If a time series variable x_{t-3} is changed, then this might change later iterations like x_{t-1} (i.e. multicollinearity) which in turn can influence x_t to create a correlation between x_t, x_{t-h} :

$$ACF = \frac{cov(x_t, x_{t-h})}{var(x_t)var(x_{t-h})}$$

The PACF has conditioning of terms between x_t and x_{t-h} , i.e. events that happen after the variable in question moves. The PACF is a 'mathematical' result that ignores ramifications of changing a variable.

$$PACF = \frac{cov(x_t, x_{t-2}|x_{t-1})}{var(x_t|x_{t-1})var(x_{t-2}|x_{t-1})}$$

The ACF and PACF are usually testing the correlation and partial correlations within or between time series values (e.g. within x_t or between x_t, y_t) in ARIMA. They can also be applied to errors, e.g. to look for autocorrelation.

In AR(p) models, PACF cuts off after lag p because a PACF shows partial correlations, corresponding to AR process coefficients. Even if the AR(p) model equation is expanded, the PACF will cut off because e.g. x_{t-5} is conditioned on x_{t-2} which already takes into account the future movement of x_{t-5} . However, the AR(p) model's ACF will not cut off because it can be expanded - e.g.:

$$\begin{aligned} x_t &= \mu + \theta x_{t-1} \\ &= \mu + \theta(\mu + \theta x_{t-2}) \\ &= \mu + \theta\mu + \theta^2 x_{t-2} \\ &= \mu + \theta_\mu + \theta^2(\mu + \theta x_{t-3}) \end{aligned}$$

So, there is a correlation between x_t, x_{t-3} in the ACF. If the x_{t-3} moves, it will affect earlier iterations like x_{t-1} which in sum affect x_t .

In MA(q) models, the ACF cuts off because that's where the correlations stop. For e.g. as x_t, x_{t-1} in an MA(1) model only has the e_t in common, they can

only be related in that one way and the ACF cuts off after that. Further, there's no transformation that can change this.

$$x_t = e_t + \theta e_{t-1}$$

$$x_{t-1} = e_{t-1} + \theta e_{t-2}$$

$$x_{t-2} = e_{t-2} + \theta e_{t-3}$$

Substitute the second and third equations above into the first:

$$x_t = e_t + \theta^3 e_{t-3} - \theta^2 x_{t-2} + -\theta x_{t-1}$$

If x_{t-2} is increased by 1, it must come from changing one or both e_{t-2} and e_{t-3} . Suppose e_{t-2} is raised by 1. But that increases x_{t-1} by θ . Then, there's no change and therefore no correlation:

$$x_t = 0 + \theta^3(0) - \theta^2(1) - \theta(\theta) = 0$$

Now suppose e_{t-3} is raised by 1. Then this raises x_{t-2} by θ and when these values are plugged into the equation, the change in x_t is 0. In MA(q) models, the PACF does not cut off because, in the above example, if x_{t-2} is raised by 1 by raising e_{t-2} , this will also raise x_{t-1} by θ . This means to condition on x_{t-1} , e_{t-1} must decline by θ . But this raises x_t by θ^2 .

The above shows that AR and MA processes are FUNDAMENTALLY different. They are not just mathematically related processes. The AR process is an infinite impulse response - an error infinitely far in the past affects the present value. The MA process is a finite impulse response - it can also be viewed as an infinite impulse response where the coefficients after order q are 0.

ARMA models (including both AR and MA terms) have ACFs and PACFs that both tail off to 0. The orders will not be particularly obvious. Basically have to guess that one or two terms of each type may be needed and then test the model.

If the ACF and PACF do not tail off, but instead have values that stay close to 1 over many lags (like a random walk), the series is non-stationary and differencing will be needed.

If the series autocorrelations are non-significant, then the series is random.

If first differences were necessary and all the differenced autocorrelations are non-significant, then the original series is a random walk. The data are dependent and are not identically distributed; in fact both the mean and variance are increasing through time.

When you request a graph of the ACF values, "significance" limits are shown by R and by Minitab. In general, the limits for the autocorrelation are placed

at standard errors of . The formula used for standard error depends upon the situation.

Within the ACF of residuals as part of the ARIMA routine, the standard errors are determined assuming the residuals are white noise. The approximate formula for any lag is that s.e. $r_k = 1/\sqrt{n}$.

For the ACF of raw data (the ACF command), the standard error at a lag k is found as if the right model was an MA($k-1$). This allows the possible interpretation that if all autocorrelations past a certain lag are within the limits, the model might be an MA of order defined by the last significant autocorrelation.

Note the 'Box-Jenkins' methodology is the ARIMA model, where the AR indicates the stationarity (controlled in part by the integration order) and the stochastic portion MA.

2.5 Selecting a model among multiple choices

Possibly choose the model with the fewest parameters. Examine standard errors of forecast values. Pick the model with the generally lowest standard errors for predictions of the future.

Compare models with regard to statistics such as the MSE (the estimate of the variance of the wt), AIC, AICc, and SIC (also called BIC). Lower values of these statistics are desirable.

One reason that two models may seem to give about the same results is that, with the certain coefficient values, two different models can sometimes be nearly equivalent when they are each converted to an infinite order MA model.

2.6 Lag operator

Lag operators are like function operators and as such can be split from their arguments. The formula with the lag operator

$$\frac{1}{1 - aL} = 1 + aL + a^2L^2 + a^3L^3 + \dots$$

can be seen as its factored equivalent:

$$(1 - aL)(1 + aL + a^2L^2 + a^3L^3 + \dots) = 1$$

or just recoding the r from the infinite geometric series as aL in the above formula. After all, both r and a (numbers with absolute values less than 1) are operators - it's just that they operate differently on the succeeding number than the lag operator.

The above formula per se also makes intuitive sense, as $(1 - aL)$ is just rescaling by a and lagging. The $1 + aL + a^2L^2 + a^3L^3 + \dots$ is just rescaling by a and lagging too. So whatever the value used for x_t , the left and right sides are equal in the above formula.

2.7 Differencing

Differencing a random walk process gives white noise. Differencing white noise once gives an MA(1) process (i.e. has two white noise terms). Differencing white noise twice gives an MA(2) process.

2.8 Integration order

If there is a linear trend, difference once. If there is a quadratic trend, difference twice.

Overdifferencing can unnecessarily add model complexity and instability (as you will introduce dependence and therefore unnecessarily INCREASE variance). So differencing removes trends and therefore removes 'trending variance' but increases 'stationary' variance.

2.9 ARIMA

2.9.1 Lower-order ARIMA models

ARIMA(1,0,0) is AR(1).

ARIMA(0,0,1) is MA(1).

ARIMA(0,1,0) is just a random walk:

$$y_t - y_{t-1} = e_t$$

which is just:

$$y_t = y_{t-1} + e_t$$

ARIMA(1,1,0) is

$$y_t - y_{t-1} = a + b(y_{t-1} - y_{t-2}) + e_t$$

It uses first differences as the y-variable and the x-variable, but the error e_t is not differenced. If e_t were differenced, ARIMA(1,1,0) would just be a sum of AR(1) processes, which is just a sum of AR(1)s without the constant. For instance, summing AR(1) below:

$$\begin{aligned} y_t &= a + by_{t-1} + e_t \\ y_{t-1} &= a + by_{t-2} + e_{t-1} \end{aligned}$$

would lead to:

$$y_t - y_{t-1} = b(y_{t-1} - y_{t-2}) + (e_t - e_{t-1})$$

To get a different result, the e_{t-1} is eliminated. It is not the case that $e_t - e_{t-1}$ can be made into a new white noise variable like w_t because even if $e_t \sim \text{iid w.n.}$, the first differences of such variables are *not* white noise (see above).

ARIMA(0,1,1) is single exponential smoothing (SES).

$$y_t - y_{t-1} = c + de_{t-1} + e_t$$

While the y-variable is differenced, the errors are level. If errors were differenced, it would just be MA(1) without the constant. For instance, summing:

$$\begin{aligned} y_t &= c + de_{t-1} + e_t \\ y_{t-1} &= c + ee_{t-2} + e_{t-1} \end{aligned}$$

leads to:

$$y_t - y_{t-1} = d(e_{t-1} - e_{t-2}) + (e_t - e_{t-1})$$

To get a different result, the e_{t-2} is again eliminated and a constant is added. Then,

$$\begin{aligned} y_t - y_{t-1} &= c + de_{t-1} + (e_t - e_{t-1}) \\ &= c + (d - 1)e_{t-1} + e_t \\ &= c + fe_{t-1} + e_t \end{aligned}$$

ARIMA(0,2,2) is double exponential smoothing.

2.9.2 Long-memory ARIMA: Integration

This type of model may possibly be used when the ACF of the series tapers slowly to 0. The usual solution in this situation is to explore the first differences of the series. Often, data for which a first difference is successful will typically have a first lag autocorrelation quite close to 1. This is because these equations have a lot of recent memory, as the y_{t-1} term will be relatively prominent compared to ARMA models. An ARIMA(1,1,0) is

$$\begin{aligned} y_t - y_{t-1} &= a + b(y_{t-1} - y_{t-2}) + e_t \\ y_t &= a + y_{t-1} + b(y_{t-1} - y_{t-2}) + e_t \end{aligned}$$

An ARIMA(0,1,1) is

$$\begin{aligned} y_t - y_{t-1} &= c + de_{t-1} + e_t \\ y_t &= c + y_{t-1} + de_{t-1} + e_t \end{aligned}$$

2.9.3 Fractional differencing and ARFIMA

A persistent pattern of non-zero correlations (long memory) might however begin with a first lag correlation that is not close to 1. In these cases, 'fractional

'differencing' may be useful. A simple model that utilizes fractional differencing is

$$(1 - B)^d x_t = w_t$$

where d is a value such that $|d| < .5$ and w_t is the usual white noise term. Use a software package to estimate d .

A low value of d often leads to a long memory ACF because it's similar to compared x_t with something like $x_{t-0.5}$. As x_t is affected more by $x_{t-0.5}$ than by x_{t-1} because the former is 'closer in time' than the latter, a fractionally differenced model can more easily lead to a long memory ACF.

The model can be expanded to include AR and MA terms as well as the fractional difference. These models are called ARFIMA models. To identify an ARFIMA model, we first use the simple fractional difference model above and then explore the ACF and PACF of the residuals from this model.

2.9.4 ARIMA forecasting

To find the forecast value, simply propagate the model with 0 error or residual. For this model:

$$y_{t+1} = \phi_0 + \phi_1 y_t + e_t$$

To find the forecast variance, start with one time step ahead forecast:

$$\begin{aligned} E(\hat{y}_{t+1} - y_{t+1})^2 &= E(\phi_0 + \phi_1 y_t + e_{t+1} - (\phi_0 + \phi_1 y_t))^2 \\ &= E(e_{t+1}^2) = \sigma_e^2 \end{aligned}$$

For the two step ahead forecast:

$$\begin{aligned} E(\hat{y}_{t+2} - y_{t+2})^2 &= E(\phi_0 + \phi_1 y_{t+1} + e_{t+2} - (\phi_0 + \phi_0 \phi_1 + \phi_1^2 y_{t+1}))^2 \\ &= E(\phi_0 + \phi_1(\phi_0 + \phi_1 y_t + e_{t+1}) + e_{t+2} - (\phi_0 + \phi_1 \phi_0 + \phi_1^2 y_t))^2 \\ &= E(\phi_1 e_{t+1} + e_{t+2})^2 \\ &= E\phi_1^2 e_{t+1}^2 + E(\phi_1 e_{t+2} e_{t+2} + E e_{t+2}^2) \\ &= \phi_1^2 \sigma_e^2 + \sigma_e^2 \\ &= \sigma_e^2(1 + \phi_1^2) \end{aligned}$$

So the h-step ahead forecast variance is

$$\sigma_e^2(1 + \phi_1^2 + \phi_1^4 + \dots + \phi_1^{2(h-1)})$$

2.10 SARIMA

The seasonal ARIMA model incorporates both non-seasonal and seasonal factors in a multiplicative model. One shorthand notation for the model is $ARIMA(p, d, q) \times (P, D, Q)_S$, with p = non-seasonal AR order, d = non-seasonal differencing, q =

non-seasonal MA order, P = seasonal AR order, D = seasonal differencing, Q = seasonal MA order, and S = time span of repeating seasonal pattern. Without differencing operations, the model could be written more formally as

$$\Phi(B^S)\phi(B)(x_t - \mu) = \Theta(B^S)\theta(B)e_t$$

2.11 Decomposition

The following two structures are considered for basic decomposition models: (1) Additive: = Trend + Seasonal + Random, (2) Multiplicative: = Trend * Seasonal * Random. Additive decompositions useful when trends are relatively constant over time. Multiplicative trends useful when trends grow or shrink over time. Use moving averages or LOWESS/LOESS to smooth and therefore find the trend.

2.12 Smoothing and filtering

Whereas in the simple moving average the past observations are weighted equally, exponential functions are used to assign exponentially decreasing weights over time.

2.12.1 Moving averages

Moving averages. Can have centered or non-centered moving averages, and these can be used to smooth out seasonality. For even time spans (e.g. quarterly data), add two staggered time-spans to get an odd time span.

For non-seasonal series, you aren't bound to smooth over any particular span. For smoothing you should experiment with moving averages of different spans. Those spans of time could be relatively short. The objective is to knock off the rough edges to see what trend or pattern might be there.

2.12.2 Single exponential smoothing

Exponential smoothing is smarter than moving averages because the weight of the past instance can be adjusted. Further, exponential smoothing accounts for ALL past data, whereas moving averages only keep track of k points.

The basic forecasting equation for single exponential smoothing is often given as

$$\hat{x}_{t+1} = \alpha x_t + (1 - \alpha)\hat{x}_t = \hat{x}_t + \alpha(x_t - \hat{x}_t)$$

We forecast the value of x at time +1 to be a weighted combination of the observed value at time and the forecasted value. Although the method is called a smoothing method, it's principally used for short run forecasting.

The value of α is called the smoothing constant. For whatever reason, $\alpha = 0.2$

is a popular default choice of programs. This puts a weight of .2 on the most recent observation and a weight of $1.2 = .8$ on the most recent forecast. There is no formally correct procedure for choosing α . Sometimes the statistician's judgment is used to choose an appropriate factor. Alternatively, a statistical technique may be used to optimize the value of α . For example, the method of least squares might be used to determine the value of α for which the sum of the quantities below is minimized:

$$(x_t - x_{t+1})^2$$

SES is equivalent to ARIMA(0,1,1). Consider an ARIMA(0,1,1) with mean = 0 for the first differences, $x_t - x_{t-1}$.

The model is $x_t - x_{t-1} = w_t + \theta_1 w_{t-1}$.

Equivalently, $x_t = x_{t-1} + w_t + \theta_1 w_{t-1}$.

To forecast at time $t + 1$, we consider $x_{t+1} = x_t + w_{t+1} + \theta_1 w_t$.

Because $w_{t+1} = x_{t+1} - \hat{x}_{t+1}$,

$$\begin{aligned}\hat{x}_{t+1} &= x_t + \theta_1 w_t \\ &= x_t + \theta_1 (x_t - \hat{x}_t) \\ &= (1 + \theta_1)x_t - \hat{x}_t\end{aligned}$$

Then let $\alpha = (1 + \theta_1)$ then $-\theta_1 = 1 - \alpha$.

SES is 'exponential' because

$$\begin{aligned}\hat{x}_{t+1} &= \alpha x_t + (1 - \alpha)\hat{x}_t \\ &= \alpha x_t + (1 - \alpha)(\alpha x_{t-1} + (1 - \alpha)\hat{x}_{t-1}) \\ &= \alpha x_t + \alpha(1 - \alpha)x_{t-1} + \alpha(1 - \alpha)^2\hat{x}_{t-1}\end{aligned}$$

etc. Intuitively, single exponential smoothing is ARIMA(0,1,1) because it dampens MA innovations.

2.12.3 Double exponential smoothing

Double exponential smoothing applies single exponential smoothing twice, once to the original data and then to the resulting single exponential smoothing data. It accounts for linear trends that single exponential smoothing does not consider.

$$\begin{aligned}s_0 &= x_0 \\ b_0 &= x_1 - x_0 \\ s_t &= \alpha x_t + (1 - \alpha)(s_{t-1} + b_{t-1})\end{aligned}$$

$$b_t = \beta(s_t - s_{t-1}) + (1 - \beta)b_{t-1}$$

This is equivalent to ARIMA(0,2,2) because it takes a difference in trend and value (so two differences) which are related to MA innovations.

2.12.4 Triple exponential smoothing

This smoothing method considers trend and seasonality. It is related to SARIMA. It can come in additive and/or multiplicative forms.

2.13 Spectral density and the periodogram

The periodogram is a sample estimate of a population function called the spectral density, which is a frequency domain characterization of a population stationary time series. The spectral density is a frequency domain representation of a time series that is directly related to the autocovariance time domain representation. In essence the spectral density and the autocovariance function contain the same information, but express it in different ways.

Note. The autocovariance is the numerator of the autocorrelation. The autocorrelation is the autocovariance divided by the variance.

Suppose that $\gamma(h)$ is the autocovariance function of a stationary process and that $f(w)$ is the spectral density for the same process. In the notation of the previous sentence, h = time lag and w = frequency. Then the autocovariance and spectral density are Fourier transform pairs:

$$\gamma(h) = \int_{-1/2}^{1/2} e^{2\pi i w h} f(w) d(w)$$

$$f(w) = \sum_{h=-\infty}^{h=\infty} \gamma(h) e^{-2\pi i w h}$$

The spectral density is defined for both negative and positive frequencies. However, due to symmetry of the function and its repeating pattern for frequencies outside the range -1/2 to +1/2, we only need to be concerned with frequencies between 0 and +1/2.

The “total” integrated spectral density equals the variance of the series. Thus the spectral density within a particular interval of frequencies can be viewed as the amount of the variance explained by those frequencies.

The raw periodogram is a rough sample estimate of the population spectral density. The estimate is “rough”, in part, because we only use the discrete fundamental harmonic frequencies for the periodogram whereas the spectral density is defined over a continuum of frequencies.

One possible improvement to the periodogram estimate of the spectral density is to smooth it using centered moving averages. This is a non-parametric approach. An alternative approach to smoothing the periodogram is a parametric estimation approach based on the fact that any stationary time series can be approximated by an AR model of some order (although it might be a high order). In this approach a suitable AR model is found, and then the spectral density is estimated as the spectral density for that estimated AR model.

2.13.1 Periodogram intuition

Any time series can be expressed as a combination of cosine and sine waves with differing periods (how long it takes to complete a full cycle) and amplitudes (maximum/minimum value during the cycle). A periodogram is used to identify the dominant periods (or frequencies) of a time series.

The periodogram is the square of the Fourier transform of the time series signal or the Fourier transform of the autocorrelation of the time series signal.

Why square in the first place? Squaring the amplitude = variance of the time series (and in the physics context = power). Moreover, the Fourier transform decomposes a process into its constituent frequencies. Since we are decomposing the autocorrelation (which is the variance-scaled autocovariance), the clu density indicates how much variance contributed to a process at each frequency.

As the periodogram uses the Fourier transform, there is a real and complex component to the frequencies. To get just a real value, square both the sin and cos terms to get the square of the modulus of the wave - which is the power.

The periodogram is the Fourier transform of the autocorrelation of the time series signal, and the autocorrelation indicates the periodicity of the signal, which 'squares' the wave to get power.

Power (=energy) is like variance of the wave (it is actually the force times distance back and forth). The autocorrelation of a time series is like a covariance of power (like the force of one wave on the distance of the covarying wave).

If the raw time series signal is squared, then it may not be possible to extract the signal periodicity. For instance, negative values will become positive, so that a perfect sine wave's periodicity will become halved. Instead, we can take the Fourier transform first and then get the power by summing the squares of the amplitudes of the sin and cos components. Or, we can take the signal's autocorrelation because that contains information about the signal's periodicity, and then take the Fourier transform of that. The autocorrelation works because it gives a squared periodicity.

Sometimes, you don't have any good reason to believe that the time series you

are looking at can be decomposed (nicely) into sines and cosines. In fact, the thing you're looking at may be a realization from some stochastic experiment, in which case you know that nice representations of that realization may be hard to come by. In these cases, we can treat the time series as a realization from some stochastic process. The periodogram, which is a squared value (square of the Fourier transform or Fourier transform of the autocorrelation), is like the variance (=power) of the stochastic process.

A periodogram is similar to the Fourier Transform, but is optimized for unevenly time-sampled data, and for different shapes in periodic signals. The autocorrelation 'centers' the signal and makes it periodic.

2.13.2 Power is amplitude squared

Use a simple harmonic oscillator. A wave is a collection of simple harmonic oscillators. The energy of a SHO equals the potential energy at maximum displacement. The PE at maximum displacement is the work done by an external force in pushing the oscillator out from the equilibrium position to maximum displacement. The external force acts against the oscillator's internal force which obeys Hooke's Law $F = -kx$. Integrate this. The proportionality constant k is everywhere so just remove it. From the trough to the peak, the change in amplitude centered at 0 is $2A$, so the energy is $1/2A^2 + 1/2A^2 = A^2$.

2.14 Bivariate time series regressions

In univariate time series such as ARIMA, we want the 'pure' time series with only exogenous (iid) errors and perhaps some dependency on past values. In bivariate or multivariate time series, one univariate time series is regressed on the other univariate time series.

2.15 Integration orders

There are two different integration orders. Univariate time series integration orders (e.g. in ARIMA) are just differences between successive time periods. Bivariate or multivariate time series integration orders (e.g. regressing a univariate ARIMA time series y_t on another univariate ARIMA time series x_t).

2.15.1 Autoregressive errors

If y_t is regressed on x_t , then ideally there is only randomness in e_t . But, if both y_t, x_t are AR univariate time series, then this will cause e_t to be autocorrelated with e_{t-1} because the differences

$$e_t = y_t - ax_t$$

$$e_{t-1} = y_{t-1} - ax_{t-1}$$

should be related to each other. That is, the relationship between x_t, y_t will be affected by the relationship of x_{t-1}, y_{t-1} . So, e_t, e_{t-1} will also be correlated.

The following equations can be combined

$$\begin{aligned} e_t &= \theta e_{t-1} + w_t \\ e_{t-1} &= \theta e_{t-2} + w_{t-1} \\ y_t &= ax_t + \theta e_{t-1} + w_t \\ y_{t-1} &= ax_{t-1} + \theta e_{t-2} + w_{t-1} \\ y_t &= ax_t + \theta(y_{t-1} - ax_{t-1}) + w_t \end{aligned}$$

to get

$$\begin{aligned} y_t &= ax_t + \theta y_{t-1} - a\theta x_{t-1} + w_t \\ (1 - \theta b)y_t &= a(1 - \theta B)x_t + w_t \end{aligned}$$

2.15.2 Cochrane-Orcutt procedure

If the error term e_t in

$$y_t = a + bx_t + e_t$$

is found to have autocorrelation, e.g. through the Durbin-Watson statistic, then standard statistical inference as normally applied to regressions is invalid because standard errors are estimated with bias. To avoid this problem, the residuals must be modeled. If the process generating the residuals is found to be a stationary first-order autoregressive structure,

$$e_t = \rho e_{t-1} + w_t$$

where $w_t \sim W.N.$ then the Cochrane–Orcutt procedure can be used to transform the model by taking a quasi-difference:

$$y_t - \rho y_{t-1} = a(1 - \rho) + b(x_t - \rho x_{t-1}) + w_t$$

The Cochrane-Orcutt transformation disregards the first observation of a time series, causing a loss of efficiency that can be substantial in small samples. A superior transformation is by Prais-Winsten, which retains the first observation with a weight of $\sqrt{(1 - \rho)^2}$.

If ρ is not known, then it is estimated by first regressing the untransformed model and obtaining the residuals, leading to an estimate of ρ and making the transformed regression sketched above feasible. (Note that one data point, the first, is lost in this regression.) This procedure of autoregressing estimated residuals can be done once and the resulting value of ρ can be used in the transformed regression, or the residuals of the residuals autoregression can themselves be autoregressed in consecutive steps until no substantial change in the estimated value of ρ is observed.

2.15.3 ARIMA errors

Estimating the Coefficients of the Adjusted Regression Model with Maximum Likelihood.

Specify a regression model with errors that have an ARIMA structure. An iterative procedure to determine adjusted regression coefficient estimates and their standard errors. Remember, the purpose is to adjust "ordinary" regression estimates for the fact that the residuals have an ARIMA structure.

Steps. Use OLS on the bivariate or multivariate time series regression. If the residuals do have an ARIMA structure, use maximum likelihood to simultaneously estimate the regression model using ARIMA estimation for the residuals. Examine the ARIMA structure (if any) of the sample residuals from the model in step 3. If white noise is present, then the model is complete. If not, continue to adjust the ARIMA model for the errors until the residuals are white noise.

2.16 Cross-correlation function

Relates y_t with x_t or a lag of x_t . Can use ACF functions and scatterplots to determine the lag(s).

In full transfer-function models, the regression includes lags of y_t and more lags of x_t . These are more complicated. Use pre-whitening to determine simultaneously the x_t lags and y_t lags. Steps: (1) Determine a time series model for the x-variable and store the residuals from this model. (2) Filter the y-variable series using the x-variable model (using the estimated coefficients from step 1). In this step we find differences between observed y-values and "estimated" y-values based on the x-variable model. (3) Examine the CCF between the residuals from Step 1 and the filtered y-values from Step 2. This CCF can be used to identify the possible terms for a lagged regression. Can use a reference table to determine the lag pattern, or can think through and determine the lag by looking at the ACF.

2.17 Intervention analysis

Intervention analysis in time series refers to the analysis of how the mean level of a series changes after an intervention (e.g. change in policy at time t), when it is assumed that the same ARIMA structure for the series holds both before and after the intervention.

The ARIMA model is

$$x_t - \mu = \frac{\Theta(B)}{\Phi(B)} w_t$$

where w_t is white noise, $\Theta(B)$ is the MA polynomial and $\Phi(B)$ is the AR polynomial. With z_t being the amount of change attributable to the invention,

$$x_t - \mu = z_t + \frac{\Theta(B)}{\Phi(B)}w_t$$

First find ARIMA prior to intervention point. Then fit the ARIMA past the intervention point and compare with actual values. Then try to determine, perhaps empirically, the intervention. There are many patterns, including (1) permanent average shift, (2) temporary average shift, (3) gradual increase, and (4) spike and gradual decline.

2.18 Repeated measures and longitudinal data

The term repeated measures refers to experimental designs (or observational studies) in which each experimental unit (or subject) is measured at several points in time. The term longitudinal data is also used for this type of data. One approach to analyzing these data is the ANOVA model

$$Y = Treatment + Animal(Treatment) + Time + Treatment \times Time + Error$$

where the time can be linear, quadratic, or other non-linear form.

Correlations among observations throughout time of a sample being measured. Can assume compound symmetry, i.e. the correlation is equal for all time gaps between observations. For instance, the correlation between data at times 1 and 2 is the same as the correlation between data at times 1 and 3, and is also the same as the correlation between times 1 and 4. This is called the compound symmetry assumption.

$$\begin{pmatrix} 1 & \rho & \rho \\ \rho & 1 & \rho \\ \rho & \rho & 1 \end{pmatrix}$$

Use GLS to adjust for different correlations among observations throughout time. Among them are the following. (1) Compound symmetry with the equal variances at the different time points (the assumption above). (2) A completely unstructured set of correlations, meaning that the correlations are different for the various time gaps and have no ARIMA structure. This also allows for either equal or unequal variances at the different time points. (3) An AR(p) structure, either with or without unequal variances. (4) A general ARIMA structure.

Use for e.g. AIC and BIC to determine optimal regression above.

2.19 ARCH and GARCH

An ARCH (autoregressive conditionally heteroskedasticity) model is a model for the variance of a time series. ARCH models are used to describe a changing, possibly volatile variance. Although an ARCH model could possibly be used

to describe a gradually increasing variance over time, most often it is used in situations in which there may be short periods of increased variation. (Gradually increasing variance connected to a gradually increasing mean level might be better handled by transforming the variable.)

$$Var(y_t|y_{t-1}) = \sigma_t^2 = a + by_{t-1}^2$$

$$y_t = \sigma_t \epsilon_t$$

An ARCH model has a drawback in some cases. If it suddenly rises, then in the next time period it will likely be near the mean again. It has no 'memory' or 'inertia'. But many processes have some inertia. To address this, use a GARCH (generalized autoregressive conditionally heteroskedasticity) model which includes the previous time period's σ_{t-1}^2 . A GARCH(1,1) model for instance is

$$Var(y_t|y_{t-1}) = \sigma_t^2 = a + by_{t-1}^2 + c\sigma_{t-1}^2$$

$$y_t = \sigma_t \epsilon_t$$

ARCH is related to GARCH in the same way that AR is related to ARMA. The AR by itself may not have enough inertia, so using ARMA gives the extra e_{t-1} . ARCH may not have enough inertia, so add the σ_{t-1}^2 as GARCH. Why not just add y_{t-2} which appears to be effectively the same thing? This is possible too (in the same way AR(1) can be made AR(2), this will result in ARCH(1) becoming ARCH(2)) but because σ_{t-1}^2 gives more flexibility with the y-intercept a . Moreover, this makes the current variance depend on the variance of the past value, not just the value from two periods ago.

To identify ARCH and GARCH models, look at the time series plot and/or the ACF/PACF of the y_t^2 . If y_t appears to be white noise and appears to be AR(1), then an ARCH(1) model for the variance is suggested. If the PACF of the y_t^2 suggests AR(m), then ARCH(m) may work. GARCH models may be suggested by an ARMA type look to the ACF and PACF of y_t^2 .

2.19.1 ARIMA-GARCH

Combines ARIMA model with GARCH errors. Example with ARMA(p,q)-GARCH(s,r).

$$y_t = a + \sum b_i y_{t-i} + \sum c_i e_{t-i} + e_t$$

$$e_t = \sqrt{\sigma_t} z_t$$

$$\sigma_t^2 = d + \sum e_i e_{t-i}^2 + \sum f_i \sigma_{t-i}^2$$

2.20 Vector models

2.20.1 VAR

VAR (vector autoregression) models generalize univariate AR models to the multivariate context. Each variable is a linear function of past lags of itself

and other variables. Each univariate time series must be stationary or the VAR equations must include one or more trend components (e.g. $+ft$, where t is time) - the approaches will yield very similar results.

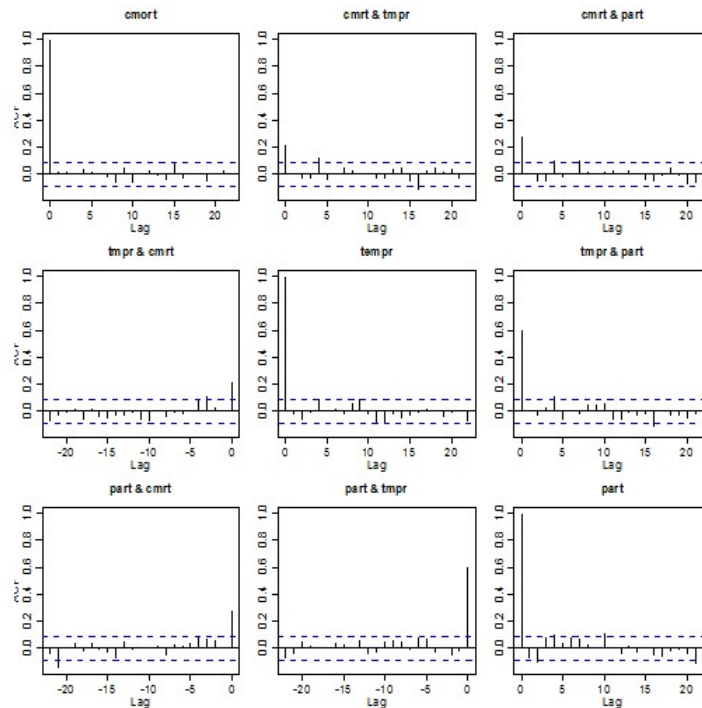
Example of 3-variable VAR(1) (here and following examples assume no trend and variables are stationary):

$$\begin{aligned}x_t &= a_1 + b_1x_{t-1} + c_1y_{t-1} + d_1z_{t-1} + e_{t,1} \\y_t &= a_2 + b_2x_{t-1} + c_2y_{t-1} + d_2z_{t-1} + e_{t,2} \\z_t &= a_3 + b_3x_{t-1} + c_3y_{t-1} + d_3z_{t-1} + e_{t,3}\end{aligned}$$

which is

$$\vec{x}_t = \vec{a} + A\vec{x}_{t-1} + \vec{e}_t$$

Residuals - this applies to VAR(1), VAR(2), VAR(p), etc. Upon fitting the VAR model, check the ACF of the residuals for any univariate autocorrelation (this is different from the ACF of the variables!). Check the covariance matrix of the residuals. The matrix diagonal will have the diagonals, while the cross terms will indicate any cross-correlations (i.e. vector auto-correlations). Can use a cross-correlation matrix plot of residuals to do this visually too:



Ideally, these would also resemble white noise at every lag, including lag 0, however there are remaining cross-correlations, especially between temperature and pollution. This model does not adequately capture the complete association between these variables in time.

In the covariance matrix of residuals, the residual variances are down the diagonal and could possibly be used to compare this model to higher order VARs (by looking at information criterion values at different VAR lag orders). The determinant of this matrix is used in the calculation of an information criterion (e.g. AIC, BIC) statistic that can be used to compare the fit of the model to the fit of other models. An information criterion is better than R^2 which increases even if near-useless independent variables are added. An information criterion adds a penalty for more independent variables. Anyways, the information criterion will include the residual sum of squares or alternatively the maximum likelihood estimate (or the fit) of the model plus the penalty for the additional variables.

2.20.2 VMA

VMA (vector moving average) models generalize univariate MA models to the multivariate context. Each variable is a linear function of past errors. For instance, 3-variable VMA(1):

$$\begin{aligned}x_t &= a_1 + e_{t,1} + b_1 e_{t-1,1} + c_1 e_{t-1,2} + d_1 e_{t-1,3} \\y_t &= a_2 + e_{t,2} + b_2 e_{t-1,1} + c_2 e_{t-1,2} + d_2 e_{t-1,3} \\z_t &= a_3 + e_{t,3} + b_3 e_{t-1,1} + c_3 e_{t-1,2} + d_3 e_{t-1,3}\end{aligned}$$

which is

$$\vec{x}_t = \vec{a} + B\vec{e}_{t-1} + \vec{e}_t$$

2.20.3 VARMA

VAR and VMA are subsets of more general VARMA models. For instance, 3-variable VARMA:

$$\begin{aligned}x_t &= a_1 + b_1 x_{t-1} + c_1 y_{t-1} + d_1 z_{t-1} + e_{t,1} \\y_t &= a_2 + b_2 x_{t-1} + c_2 y_{t-1} + d_2 z_{t-1} + e_{t,2} \\z_t &= a_3 + b_3 x_{t-1} + c_3 y_{t-1} + d_3 z_{t-1} + e_{t,3}\end{aligned}$$

which is

$$\vec{x}_t = \vec{a} + A\vec{x}_{t-1} + B\vec{e}_{t-1} + \vec{e}_t$$

2.20.4 VARIMA

This is just VARMA with an integration order.

2.20.5 VARX, VMAX, VARMAX, and VARIMAX

More generally yet, the above are special cases of VARX, VMAX, VARMAX, and VARIMAX models that include extra exogenous predictors. A VAR(1,1) model example:

$$\vec{x}_t = \vec{a} + A\vec{x}_{t-1} + B\vec{y}_{t-1} + \vec{e}_t$$

where y_t is an exogenous variable to x_t .

2.20.6 VARIMA-GARCH

This merely combines the most general model of VARIMA with GARCH.

2.21 Regime-switching models (RSM)

In RSM modeling, different models apply to different intervals of values of some key variable(s).

2.21.1 Threshold models

Threshold models are used in several different areas of statistics, not just time series. The general idea is that a process may behave differently when the values of a variable exceed a certain threshold. That is, a different model may apply when values are greater than a threshold than when they are below the threshold.

Threshold autoregressive models (TAR) for univariate time series. In a TAR model, AR models are estimated separately in two or more intervals of values as defined by the dependent variable. These AR models may or may not be of the same order. For convenience, it's often assumed that they are of the same order.

The difficulties are determining the need for a TAR model, the threshold value to use, and the order(s) of the AR models. One feature for data that a TAR model may work is that the rates of increase and/or decrease may differ when the values are above some level than when the values are below that level.

The estimation of the threshold level is more or less subjective. Many analysts explore several different threshold levels in an attempt to provide a good fit to the data (as measured by MSE values and general features of the residuals). The order(s) of the AR model(s) can also be a trial and error expedition, especially when the inherent model for the data may not be an AR. Generally analysts start with what they think may be a higher order than necessary and then reduce the order as necessary.

3 Foundations

3.1 Formal logic

Truth tables, statements, negation ($\neg$), conjunction ($\wedge$), disjunction ($\vee$), tautology (always true, e.g. $\neg(A \vee B)$), logical equivalence ($\Leftrightarrow$), conditional statements ($\rightarrow$), biconditional (iff, $\leftrightarrow$), contrapositive ($A \rightarrow B \Leftrightarrow \neg B \rightarrow \neg A$).

Deduction: modus ponens, syllogism, contradiction.

3.2 Principles

Pigeonhole principle. If n objects are placed in m containers and $n > m$, then at least one container must contain more than one object.

Inclusion-exclusion principle.

$$A \cup B = A + B - A \cap B$$

This principle is used in many applications, e.g. to show the set of real numbers is infinitely larger than the set of integers, to show that at least two people in a city have to have the same number of hairs on their head, etc.

3.3 Limits of Mathematics

3.3.1 Completeness

Mathematical axioms aren't ever complete or will not know whether it's complete. There will always be true statements that can't be proved within the system.

Godel's incompleteness theorem. "This statement cannot be proved from axioms". If it were false, then it can be proved, which is a contradiction (and therefore inconsistent). So it must be true, so the axioms must be incomplete. It must be added as another axiom for more complete math (i.e. so it is adding to the size of outside the system of math). But there are infinitely many statements outside of an axiomatic system.

"This statement that $1+1=2$ cannot be proved from axioms" is false so there is no completeness or consistency issue.

3.3.2 Consistency

Godel's second incompleteness theorem: "This statement that math is consistent cannot be proved from axioms". If false, then it can be proved so leads to inconsistency. If true, can't be proven so incomplete. No set of axioms can prove its own consistency.

Russell's paradox shows naive set theory is inconsistent. It was addressed by axiomatic set theory - so using logic from outside the system and changing the system that was naive set theory.

In **Russell's paradox**, if a set A contains all sets that don't contain themselves and it doesn't contain itself then the set must contain itself because A doesn't contain itself. But if the set now contains a copy of itself B, then A is no longer a set that contains only sets that don't contain themselves because it contains B.

3.3.3 Decidability

Halting problem. Turing's proof by contradiction. Assume program H takes input and correctly outputs whether it halts or loops forever. Add a program I that takes H's output and reverses it. Assume $H+I=J$ is one unit. Feed J into itself. If J will halt, H will output halt but I will reverse the output action and loop forever. If J will not halt, the opposite occurs. So machine can't decide whether it will halt or not so contradiction, so H cannot exist. Shows logic is undecidable. Conway's game of life is another example - only way to decide whether it stops or keeps going is to run it.

3.3.4 Uncomputability of complexity

Berry's paradox, describe an undescribably large number: 'the smallest positive integer that cannot be described in less than fifteen words'. Occam's razor: simplest rule is usually the best one out of competing explanations. Related to ML: which one should the machine choose among competing explanations? How to codify this? So need to compare complexity. Use information to compare complexity. The most fundamental unit of information is the bit, and information resolves uncertainty. Everything theoretically can be described as bit strings. So now can compare them with Kolmogorov complexity, the absolute shortest bit string, $C(x) \geq |x|$ the length of the string or $C(x) = \min_p(\text{length}(p) | U(p) = x)$ (U=universal Turing machine, i.e. computer with unlimited memory). (Randomness=incompressibility). If $C(x) = |x|$ then Kolmogorov random. Related to data compression. But how to find or know shortest string? Go from shortest to longer bit strings and if it (1) interprets and makes sense, and (2) describes the input data. This is the Kolmogorov complexity. But Berry's paradox means a 140-bit program can find the first program that requires more than 150 bits to describe, which is a contradiction. So the Kolmogorov complexity is uncomputable.

3.3.5 Chaos theory

There are limits to predictability because of the impossible accuracy required in some cases for modelling. At least for practical purposes, this may contradict the existence of a Laplace's demon or supercomputer that can predict all.

3.4 Limits of physics

Heisenberg uncertainty. There is a fundamental limit to the accuracy with which the values for certain pairs of physical quantities of a particle, such as position and momentum, can be predicted from initial conditions.

Such variable pairs are known as complementary variables or canonically conjugate variables; and, depending on interpretation, the uncertainty principle limits to what extent such conjugate properties maintain their approximate meaning, as the mathematical framework of quantum physics does not support the notion of simultaneously well-defined conjugate properties expressed by a single value. The uncertainty principle implies that it is in general not possible to predict the value of a quantity with arbitrary certainty, even if all initial conditions are specified.

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}$$

Historically, the uncertainty principle has been confused with a related effect in physics, called the **observer effect**, which notes that measurements of certain systems cannot be made without affecting the system, that is, without changing something in a system. Heisenberg utilized such an observer effect at the quantum level (see below) as a physical "explanation" of quantum uncertainty. It has since become clearer, however, that the uncertainty principle is inherent in the properties of all wave-like systems, and that it arises in quantum mechanics simply due to the matter wave nature of all quantum objects. Thus, the uncertainty principle actually states a fundamental property of quantum systems and is not a statement about the observational success of current technology. It must be emphasized that measurement does not mean only a process in which a physicist-observer takes part, but rather any interaction between classical and quantum objects regardless of any observer.

Other physics concepts related to knowledge limits include the **speed of light and causality**, including how much of the universe is observable, and possibly finite **information entropy** in the universe.

4 Set theory

4.1 Cardinality

Number of elements in a set. Two sets A and B have the same cardinality if there exists a bijection $f : A \rightarrow B$. A is finite if $A = \emptyset$ or A has the same cardinality as $(1, 2, \dots, n)$ for some $n \in \mathbb{Z}^+$. A is infinite if it's not finite.

A cardinal number is a symbol assigned to a set that indicates cardinality. $n(A), \text{card}(A), |A|$. $\aleph_0$ sets are countably infinite. $n(\mathbb{N}) = n(\mathbb{Z}^+)$.

4.2 Countability

Countable: 1-1 correspondence with $\mathcal{Z}^+$. So arranges set elements in a sequence. Positive integers, integers (0,-1,1,-2,2...), pairs of positive integers (group all ordered pairs with same 'distance' from origin, going further in these 'radii'), rational numbers are countable.

Uncountable: not countable. Reals $[0,1]$, reals, planes, etc. To prove reals are uncountable, can use Cantor's diagonalization argument.

The Cantor set, which removes numbers from $[0,1]$, has length zero (as the length is $(2/3)^n$ with n iterations), is also uncountable - it has the same size as $\mathcal{R}$. Note: the Cantor set 'strip' dimensions are shrinking but the number of these 'strips' are growing (it's also a fractal and self-replicating). Note: the Cantor function (Devil's staircase) is a representation of the Cantor set but vertically rises by 1 despite the length being 0.

Unknown whether there are intermediate infinities between naturals and reals. Continuum hypothesis asserts not, and this is independent of ZFC.

4.3 Relations

4.3.1 Preorders

Order relation satisfying reflexivity and transitivity. Weaker and more general than a total order. Some elements may not be related by an order, so some chains might not be linked.

4.3.2 Orders

Partial order. A relation $\leq$ on a set that is (i) reflexive, (ii) antisymmetric ($a \leq b$ and $b \leq a$ implies $a = b$), and (iii) transitive. Forms parallel chains where at least some elements may not be comparable but all chains linked at some point. A partially ordered set (poset) is a set with a partial order.

Total order. A specific partial order. Partial order + (iv) comparable (trichotomy law): $\forall a, b \ a \leq b$ or $b \leq a$. All elements are comparable so chains form a lattice. A totally ordered or linearly ordered set (loset) is a set with a total order. Well-ordered. A totally ordered set is well-ordered iff every non-empty subset has a smallest element. Every finite totally ordered set is well ordered. $\mathcal{Z}$ has no smallest element so it not well-ordered.

Well-ordering principle. Every nonempty set of nonnegative integers has a smallest element. So the set all positive integers are well-ordered.

4.3.3 Equivalence relations

Partitions induce equivalence relations. Partition elements form equivalence classes. Equivalence relations generalize equalities ($=$ is the identity equivalence relation). Equivalence relations require (i) reflexivity ($a \sim a$), (ii) symmetry ($a \sim b$ iff $b \sim a$), and (iii) transitivity. E.g. modular arithmetic.

4.3.4 Directed sets

(i) Preorder + (ii) every 2 elements have an upper bound.

4.4 Naive set theory

Cantor's naive set theory too broad and simple. Leads to Russell's paradox due to unrestricted comprehension, occurs when there exists a set of all sets that don't contain themselves: $S = (x | x \notin x)$ then $S \in S$ and $S \notin S$. Since S doesn't include sets that don't contain themselves, S itself should be in the set to fulfill the rule - but if it were included, it would break the rule! There are other paradoxes of naive set theory.

4.5 Zermelo-Fraenkel set theory

Axioms 1-8 are ZF, 1-9 are ZFC:

- Extensionality: If two sets have the same elements, they're the same. So the set is uniquely determined by its elements.
- Unordered pair (Pairing): For any a and b there exists a set a,b that contains exactly a and b.
- Subsets (Separation, Restricted comprehension, Specification, Specialization): If ϕ is a property (with parameter p), then for any X and p there exists a set $Y = (u \in X : \phi(u, p))$ that contains all those u in X that have the property ϕ . Prevents Russell's paradox. Subsets mean $(x \in A | p(x))$ are allowed but $(x | p(x))$ are disallowed as they lead to $(x | p(x) : x \notin x)$ which causes Russell's paradox. Note $S = (x \in A | x \notin x)$ has no contradiction.
- Union: For any X there exists a set $Y = \bigcup X$, the union of all elements of X.
- Power set: Every set has a power set.
- Infinity: There exists an infinite set.
- Replacement: If F is a function, then for any X there exists a set $Y = F(X) = (F(x) : x \in X)$.
- Foundation (Regularity or well-founded): No set contains itself. Every non-empty set x contains a member y such that x and y are disjoint sets.

- Choice (Well-ordering): Every family of nonempty sets has a choice function.

4.5.1 Axiom of choice

For any (even infinite) collection of pairs of shoes, one can pick out the left shoe from each pair to obtain an appropriate selection; this makes it possible to directly define a choice function. For an infinite collection of pairs of socks (assumed to have no distinguishing features), there is no obvious way to make a function that selects one sock from each pair, without invoking the axiom of choice.

A choice function (also called selector or selection) is a function f , defined on a collection X of nonempty subsets, such that for every subset A in X , $f(A)$ is an element of A .

Definition. For any set X of nonempty subsets, there exists a choice function f that maps each subset to an element of the subset.

$$\forall X \left(\emptyset \notin X \Rightarrow \exists f: X \rightarrow \bigcup X \quad \forall A \in X (f(A) \in A) \right)$$

Equivalent definition. A product of sets is inhabited if each set is inhabited.

Axiom of choice (or something similar) leads to the Banach-Tarski paradox.

Well-ordering theorem (different from the well-ordering principle). Every set can be well-ordered. There is a total order on the set such that every nonempty subset has a least element. Example: use absolute values on real numbers. The order may not be the intuitive or obvious order - the ordering may have to be 'forced'.

The well-ordering theorem is equivalent to the axiom of choice. Well-ordered subsets each have smallest values, so there is at least a choice function mapping subsets to their smallest values.

Zorn's lemma. A partially ordered set where every chain has an upper bound contains a maximal element. Equivalent to the well-ordering theorem and therefore the axiom of choice.

Tychonoff theorem (from topology). The product of compact sets is compact.

The nature of the individual nonempty sets in the collection may make it possible to avoid the axiom of choice even for certain infinite collections. For example, suppose that each member of the collection X is a nonempty subset of the natural numbers. Every such subset has a smallest element, so to specify our choice function we can simply say that it maps each set to the least element of that

set. This gives us a definite choice of an element from each set, and makes it unnecessary to apply the axiom of choice.

The difficulty appears when there is no natural choice of elements from each set. If we cannot make explicit choices, how do we know that our set exists? For example, suppose that X is the set of all non-empty subsets of the real numbers. First we might try to proceed as if X were finite. If we try to choose an element from each set, then, because X is infinite, our choice procedure will never come to an end, and consequently, we shall never be able to produce a choice function for all of X . Next we might try specifying the least element from each set. But some subsets of the real numbers do not have least elements. For example, the open interval $(0,1)$ does not have a least element: if x is in $(0,1)$, then so is $x/2$, and $x/2$ is always strictly smaller than x . So this attempt also fails.

The reason that we are able to choose least elements from subsets of the natural numbers is the fact that the natural numbers are well-ordered: every nonempty subset of the natural numbers has a unique least element under the natural ordering. One might say, "Even though the usual ordering of the real numbers does not work, it may be possible to find a different ordering of the real numbers which is a well-ordering. Then our choice function can choose the least element of every set under our unusual ordering." The problem then becomes that of constructing a well-ordering, which turns out to require the axiom of choice for its existence; every set can be well-ordered if and only if the axiom of choice holds.

Perhaps the axiom of choice may be unnecessary for countably infinite cases, as these are theoretically finite - and therefore 'fit' in the present universe. This number is beyond the finite universe (countable = theoretically finite) so there is not enough 'time' to specify all of them.

4.6 Banach-Tarski paradox

Given a solid ball in 3-dimensional space, there exists a decomposition of the ball into a finite number of disjoint subsets, which can then be put back together in a different way to yield two identical copies of the original ball.

More than just stretching a line, which increases an uncountable set to another uncountable set. The Banach-Tarski paradox when repeated over and over, leads to a countable set becoming an uncountable set.

Banach-Tarski paradox. Spheres have uncountable points. Create countable set of rotations comprising every combination of U, D, R, L like LRUUD. Pick starting point on sphere and apply countable rotations. Pick a starting point not covered by that countable union and repeat. Repeat for uncountable starting points. This is equivalent to applying the axiom of choice. Isolate sphere

with last move e.g. U. Then rotate down D which cancels all the $DU = \emptyset$, which makes it nearly equivalent to the D, R, and L spheres. So the one U sphere has become nearly three spheres, creating something from nothing.

Earlier versions: Vitali set and Hausdorff paradox.

Two ways to resolve: (1) reject axiom of choice and/or (2) leads to non-measurable sets.

5 Category Theory

5.1 Categories

Category consists of objects and sets of arrows (or sets of morphisms $hom_c(x, y)$). Categories can comprise other categories.

If $f : A \rightarrow B$ and $g : B \rightarrow C$, then $g \circ f = C$ then commutes. Every object A has an identity arrow: $1_A : A \rightarrow A$. Law of composition: associativity ($f \circ (g \circ h) = (f \circ g) \circ h$ and unity ($f \circ 1_A = 1_B \circ f$)).

Duality. For $f : A \rightarrow B$ in C . In C^{op} , $f^* : A^* \leftarrow B^*$. The composition directions are all reversed. Get any theories proved in C^{op} free if they're proved in C .

5.2 Functors

Expresses relationships between categories.

(Covariant) functor $F : C \rightarrow D$. Image of categories. If $f : A \rightarrow B$ and $g : B \rightarrow C$, then $g \circ f = C$. Then $F(f) : F(A) \rightarrow F(B)$ and $F(g) : F(B) \rightarrow F(C)$, then $F(g \circ f) = F(g) \circ F(f) = F(C)$.

Contravariant functor $F : C \rightarrow D$ is just $F : C^{op} \rightarrow D$. Instead of $F(g \circ f) = F(g) \circ F(f)$, for contravariant functors it is $F(g \circ f) = F(f) \circ F(g)$.

Example. Powerset functor $P : sets \rightarrow sets$, $P(A) = (X | X \subseteq A)$. So $f : element \rightarrow element$ but P maps entire sets.

Example. In λ -calculus, functions are covariant in their output types: if $B \leq C$ then $(A \rightarrow B) \leq (A \rightarrow C)$. Duality: functions are contravariant in their input types: if $B \leq C$ then $(B \rightarrow A) \geq (C \rightarrow A)$ (if the larger+nested sets are preserved, then this is a subset of the nested set being preserved).

Functors can be isomorphic. Moreover, functors preserve isomorphisms.

5.3 Natural transformations

The natural transformation η is a morphism between functors. $F : C \rightarrow D$ and $G : C \rightarrow D$ is a natural transformation $F \Rightarrow G$.

For $f : x \rightarrow y$, $F(x) \xrightarrow{\eta_x} G(x)$ and $G(f) \circ \eta_x = \eta_y \circ F(f)$. In the latter equation, the upper right side = lower left side in the diagram below.

$$\begin{array}{ccc} F(x) & \xrightarrow{\eta_x} & G(x) \\ F(f) \downarrow & & \downarrow G(f) \\ F(y) & \xrightarrow{\eta_y} & G(y) \end{array}$$

Special cases. (1) F and G are constant. If $F, G : C \rightarrow D$ and both F and G are constant functors then $F : C \rightarrow d$ and every morphism to id_d and $G : C \rightarrow d'$ and every morphism to $id_{d'}$ where $d, d' \in D$. Then a natural transformation from F to G is simply a morphism $d \xrightarrow{\eta} d'$. (2) F is constant and G is any functor. Then η is a cone over G . (3) F is any functor and G is constant. Then η is a cone under G . (4) Natural isomorphism. Each η_x is an isomorphism. Then:

$$\begin{array}{ccc} F(x) \xrightarrow{\eta_x} G(x) & & F(x) \longrightarrow G(x) \\ F(f) \downarrow & \downarrow G(f) & \downarrow \\ F(y) \xrightarrow{\eta_y} G(y) & \eta_y \text{ is invertible} & F(y) \longleftarrow G(y) \\ \eta_y \circ F(f) = G(f) \circ \eta_x & & F(y) = \eta_y^{-1} \circ G(f) \circ \eta_x \end{array}$$

5.4 Representable functors

Representable functors input arbitrary categories and output set categories. They are natural transformations. $F : C \rightarrow Set$ is representable if $x, c \in C$ s.t. F is naturally isomorphic to $hom_C(c, x)$, i.e. $F(x) \cong hom_C(c, x)$. This is for covariant functors. For contravariant functors, $F(x) \cong hom_C(x, c)$.

From representable functors, it follows elements are maps and therefore objects are functors. Let A be any set and $*$ is a set with one element. So a function $*$ to A has exactly one element in its image a or range of $*$ to A for $a \in A$. This suggests that a map $*$ to A is just a choice of element in A . So the elements of A are bijective with functions $*$ to A and $A \cong hom_{Set}(*, A)$. Further, the isomorphism $A \rightarrow hom_{Set}(*, A)$ which sends $a \in A$ to $a_f : * \rightarrow A$ where $a_f(*) = a$ is natural. So:

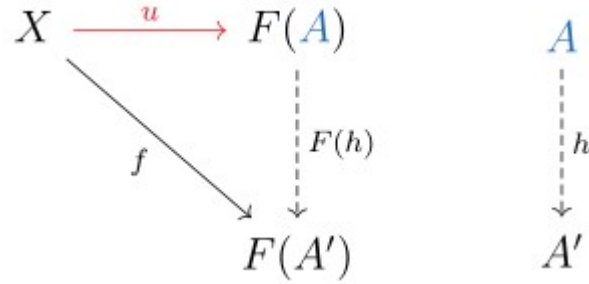
$$\begin{array}{ccc} A & \longrightarrow & Hom(*, A) \\ f \downarrow & & \downarrow f_* \\ B & \longrightarrow & Hom(*, B) \end{array}$$

5.5 Universal properties

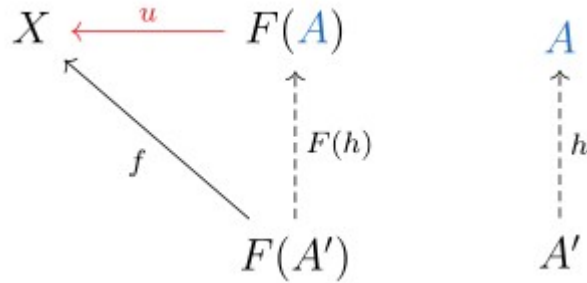
Forgets the intrinsic details of objects, and instead looks at arrows between the objects. So abstracts away from intrinsic details.

An object A in a category is called initial if for every object B there is a unique morphism $A \rightarrow B$. An object Z is called final if for every object B there is a unique morphism $B \rightarrow Z$. If an object satisfies either of these properties, it is called universal. If an object satisfies both, it is called a zero object.

Initial object:



Final object:

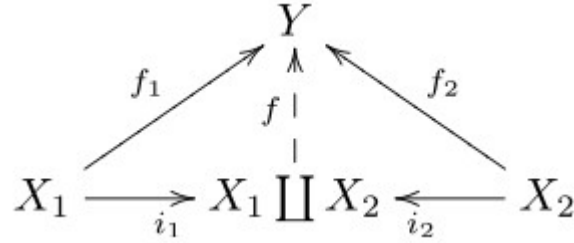


Objects that are both initial and final are zero objects. Example. In the category of groups, the trivial group (e) is both initial and final, because group homomorphisms must preserve the identity element. Hence the trivial group is a zero object.

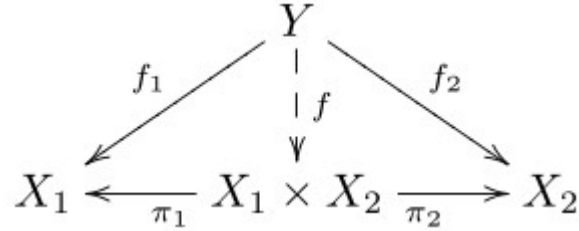
Universal objects are always unique up to isomorphism. So if there is a theorem proved for one member of a universal object, it applies to all others of that universal object. If A, A' are both initial in a category, then $A \cong A'$ are isomorphic. If Z, Z' are both final, then $Z \cong Z'$. Proof. Recall that a morphism $f : A \rightarrow B$ is an isomorphism if it has a two sided inverse, a $g : B \rightarrow A$ so that $gf = 1_A$ and $fg = 1_B$ are the identities. If A, A' are two initial objects there are unique morphisms $f : A \rightarrow A'$ and $g : A' \rightarrow A$. Moreover, these compose to be

morphisms $gf : A \rightarrow A$. But since A is initial, the only morphism $A \rightarrow A$ is the identity. The situation for $fg : A' \rightarrow A'$ is analogous, and so these morphisms are actually inverses of each other, and A, A' are isomorphic. The proof for final objects is identical.

Examples. There are many universal objects comprised from initial and final objects. Quotients are initial objects. Products and coproducts. The product of sets is universal with respect to the property of having two morphisms to its factors. $X \times Y$ is a final object as there are unique final morphisms.



The coproduct $X \amalg Y$ is an initial object.



5.6 Yoneda lemma

Yoneda perspective. Mathematical objects are completely defined (up to isomorphism) by their relationships (morphisms) to other objects by what the object looks like from other viewpoints (all $hom(x, y)$ and $hom(y, x)$).

So there are two points. (1) Everything we need to know about X is encoded in $hom(-, X)$. So $X \mapsto hom(-, X)$. (2) X and Y are isomorphic iff their represented functors $hom(-, X)$ and $hom(-, Y)$ are isomorphic.

A presheaf on a category C is a functor $F : C^{op} \rightarrow Set$. The category $Set^{C^{op}}$ contains objects which are functors F and whose morphisms are natural transformations. Therefore, there is a functor $\mathcal{Y} : C \rightarrow Set^{C^{op}}$ that sends X to $hom(-, X)$ and a morphism $f : X \rightarrow Y$ to the natural transformation $f_* : hom(-, X) \rightarrow hom(-, Y)$.

This happens if as $\mathcal{Y}$ sends X to the presheaf category, it preserves the relationships X has with the other objects in C . So the following $X, Y \in C$ defined

by $f \rightarrow f_*$ must be bijective: $\text{hom}(X, Y) \rightarrow \text{Nat}(\text{hom}(-, X), \text{hom}(-, Y))$ where 'Nat' is the natural transformation.

Injectivity is clear: if $f, g : X \rightarrow Y$ are distinct morphisms, then their push-forwards f_* and g_* are distinct (if $f \neq g$ then $f_* \neq g_*$). For surjectivity, given any natural transformation $\eta : \text{hom}(-, X) \rightarrow \text{hom}(-, Y)$, there is a morphism $f : X \rightarrow Y$ s.t. $\eta = f_*$. Then $\mathcal{Y}$ is fully (surjective) faithful (injective) and embeds $\mathcal{C}$ into $\text{Set}^{C^{op}}$ (the Yoneda embedding).

Yoneda lemma. For any functor $F : C^{op} \rightarrow \text{Set}$ and any object $X \in C$, natural transformations $\text{hom}(-, X) \rightarrow F$ are bijective with elements in the set $F(X)$. So, $\text{Nat}(\text{hom}(-, X), F) \cong F(X)$.

$$\begin{array}{ccc}
 \text{Hom}(A, B) & \xrightarrow{\text{Hom}(h, B)} & \text{Hom}(A', B) \\
 \text{Hom}(A, f) \downarrow & & \downarrow \text{Hom}(A', f) \\
 \text{Hom}(A, B') & \xrightarrow{\text{Hom}(h, B')} & \text{Hom}(A', B')
 \end{array}$$

The above diagram must commute because $\text{Hom}(A', B)$ and $\text{Hom}(A, B')$ must end up at $\text{Hom}(A', B')$. Then, every morphism $h : A' \rightarrow A$ gives rise to a natural transformation $\text{Hom}(h, -) : \text{Hom}(A, -) \rightarrow \text{Hom}(A', -)$ because $\text{Hom}(h, B')$ is uniquely determined by elements in $\text{Hom}(A, B)$. and every morphism $f : B \rightarrow B'$ gives rise to a natural transformation $\text{Hom}(-, f) : \text{Hom}(-, B) \rightarrow \text{Hom}(-, B')$ using a similar argument. Because of the way the hom-functors work, all natural transformations must be of these forms. So the hom-functors are fully faithful.

5.7 Category examples

Products are like disaggregations or projections. Examples are Cartesian products (sets), direct products (groups), and product topologies (topology). A product of X_1 and X_2 is an object X , typically denoted $X_1 \times X_2$, equipped with a pair of morphisms $\pi_1 : X \rightarrow X_1$, $\pi_2 : X \rightarrow X_2$ satisfying the following universal property. For every object Y and every pair of morphisms $f_1 : Y \rightarrow X_1$, $f_2 : Y \rightarrow X_2$, there exists a unique morphism $f : Y \rightarrow X_1 \times X_2$ which is assembled into a commutative diagram.

Coproducts are dual to products. Examples are disjoint unions (sets) and direct sums (vector spaces).

Quotients are another example of categories.

5.8 Limits and colimits

Not the same as limits from analysis and topology.

Limits. Limits are like subdivisions or subsets. The 'shallowest' natural transformation cone OVER a functor. A limit of a diagram $F : D \rightarrow C$ in a category C is an object $\lim F$ of C equipped with morphisms to the objects $F(d)$ for all $d \in D$, such that everything in sight commutes. Moreover, the limit $\lim F$ is the universal object with this property, i.e. the "most optimized solution" to the problem of finding such an object.

Limit examples. Terminal objects (one-point set in set theory and topology, (e) in group theory, 0-vector in vector spaces, 0 in $\mathcal{N}$), products (Cartesian products in sets, topology, and vector spaces, gcd in $\mathcal{N}$), inverse limits (subsets, subset topologies, subgroups, gcd, intersections), pullbacks (subsets, subspaces, subgroups for products $A \times B$ with $fa = gb$; they're often subobjects of Cartesian products and can be intersections, gcds, and others) inverse limits (smallest units; intersections in sets and classical limits), equalizers (subsets, subspaces, subgroups with where $fa = ga$; they're often solutions to equations).

Colimits. Coproducts are aggregations so glue together. The 'shallowest' natural transformation cone UNDER a functor. Examples include quotients, coproducts, unions.

5.9 Isomorphisms and equivalences

Category isomorphism. Strict. Isomorphism exists if there exist strict functors $F : C \rightarrow D$ and $G : D \rightarrow C$ s.t. $FG = id_D$ and $GF = id_C$. So (F, G) is an isomorphism between C and D . G is the strict inverse of F , and vice versa. Too strict and not useful in many contexts, e.g. ordered pairs $(a, b) \neq (b, a)$.

Category equivalence. Weaker than isomorphism but can be useful. Replace equalities with isomorphisms. $FG \cong id_D$ and $GF \cong id_C$. Example: homotopy equivalences from topology.

Category adjunction. Even weaker comparison is an adjunction, but the categories may not be equivalent anymore.

5.10 Adjunction

Generalization of inverse. Replace isomorphisms from equivalences with regular morphisms which are natural transformations. Unit $\eta : FG \rightarrow id_D$ and co-unit $\epsilon : GF \rightarrow id_C$. F and G are adjoint functors. F and G are bijective.

Example for pre-ordered sets. If C is a pre-ordered set $(0, 1, 2)$ and D is (p, q, r) , then $L \dashv R$ if $L(a) \leq x$ and $a \leq R(x)$. They will preserve the pre-orders, but

$RL \neq LG$.

Another example. $Grp(Free(S), G) \cong Sets(S, Forget(G))$.

5.11 Further category theory

Higher category theory generalizes categories to k-categories (infinity category). Enriched category theory generalizes categories to monoids.

5.12 Homological algebra

Higher-dimensional linear algebra. Chain complex of vector spaces (homomorphic to algebraic topology chain complexes) with $\partial^2 = 0$ (geometric consistency). Maps between chain complexes.

6 Abstract Algebra

6.1 Group theory

Group: set of elements, has an operation, closed under operation, identity element, each element has an inverse (use inverse operation on element to get to identity element), associative property (grouping doesn't matter). Modular arithmetic, triangle operations (rotate, flip, etc.), integers. Simplest definition to solve a basic algebraic equation. Can have subgroups.

Integers, even integers, integer multiples of 7 under addition are groups. Integers under multiplication isn't a group. Odd integers under addition aren't a group.

Cayley tables are like multiplication tables. Use to find/build groups. (1) Each first row and column have all the elements because of the operation with the identity element. (2) Each row and column must have the identity element (because each element has an inverse). (3) Diagonally symmetrical if abelian. (4) No row with duplicate elements. Proof. If an element b of G repeats twice in a row or column, the equation $x * a = b$ has to have two solutions. If the solutions be x_1 and x_2 , then $x_1 * a = x_2 * a = b$. Operate both sides on the right by inverse of a and then $x_1 = x_2$. This means an element b cannot repeat in a column. Intuition: if there's a duplicate, it means one of header elements is a duplicate.

6.1.1 Lagrange's theorem

A subset that is also a group. Every group has at least two subgroups: (i) G and (ii) e (identity element).

Lagrange's theorem. $H \leq G \Rightarrow |H|$ divides $|G|$. So there are limits on subgroup cardinalities. Example. If $|G| = 323$ then $|H| = 1, 17, 19, 323$ but since the whole group $|H| = 323$ and the identity group $|H| = 1$, then there could be two other orders possible: 17 and 19.

Proof of Lagrange's theorem. G is a finite group with $|G| = n$. $|e| = 1$ divides n , and $|G| = n$ divides n . For $H < G$, pick $g_1 \notin H$. Then $g_1H = (g_1h \forall h \in H)$ (the left coset) with $H \cap g_1H = \emptyset$ - prove by contradiction: assume there is an element in H and g_1H . Then:

$$\begin{aligned} g_1h_i &= h_j \\ (g_1h_i)h_i^{-1} &= h_jh_i^{-1} \\ g_1(h_ih_i^{-1}) &= h_jh_i^{-1} \\ g_1e &= h_jh_i^{-1} \\ g_1 &= h_jh_i^{-1} \end{aligned}$$

Since $h_jh_i^{-1}$ must be $\in H$ as they're elements of H , it must be that $g_1 \in H$. Contradiction, so the sets don't overlap. Choose g_2 not in H or g_1H . Repeat until no element not in a coset. So non-overlapping left cosets. Each coset is the same size. Proof. Pick a left coset gH and suppose it has duplicate elements made from different elements. Then:

$$\begin{aligned} gh_1 &= gh_2 \\ g^{-1}(gh_1) &= g^{-1}(gh_2) \\ h_1 &= h_2 \end{aligned}$$

Contradiction as the elements are actually the same. So each coset has the same order $|H|$, and let it be equal to d . Note: all the cosets except H are only sets as they exclude the identity element. Let the number of cosets $= k$. Index of H in G $|G : H| = k$ so $d \cdot k = n$ so d/n and $|H|$ divides $|G|$. Can use the right cosets for proof (for abelian groups, same as left cosets).

Cauchy's theorem, which works for prime number divisors only, is a corollary. This generalizes to the Sylow theorems which help characterize the groups based on their primes.

6.1.2 Quotient groups and normal subgroups

Group $\mathcal{Z}$. Quotient group (e.g. $\mathcal{Z}/5\mathcal{Z}$) comprises congruence classes: subgroup (e.g. $5\mathcal{Z}$) and cosets (e.g. $1 + 5\mathcal{Z}, 2 + 5\mathcal{Z}, 3 + 5\mathcal{Z}, 4 + 5\mathcal{Z}$). The subgroup and cosets can each be seen as elements. The quotient group isn't the same as the group. Only normal subgroups can be used to make quotient groups.

N is a normal subgroup ($N \trianglelefteq G$, an invariant subgroup) if it is invariant under

conjugation: $g^{-1}Ng$. The cosets are a 'factor group' G/N (identity element is N and the inverse is $x^{-1}N$). Proof. Let 2 cosets be xN, yN . Since $e \in N$, $x = xe \in xN$ and $y = ye \in yN$. For cosets to be a group, $xy \in (xN)(yN)$ or in other words, $= xyN$. Pick xn_1 from xN and yn_2 from yN .

$$(xn_1)(yn_2) = xyn_3$$

$$x^{-1}(xn_1)(yn_2) = x^{-1}yn_3$$

$$y^{-1}n_1yn_2 = y^{-1}yn_3$$

$$y^{-1}n_1yn_2n_2^{-1} = n_3n_2^{-1}$$

$$y^{-1}n_1y = n_3n_2^{-1} \in N$$

The left term is the conjugate. More generally, by showing the other direction:

$$y^{-1}Ny = N$$

So N has an identity: eN since $(eN)(gN) = egN = gN$. Inverse: $(gN)^{-1} = g^{-1}N$ since $(g^{-1}N)(gN) = g^{-1}gN = N$. So if N is a subgroup and the cosets form a subgroup, then $y^{-1}Ny = N$ for any y . Other direction proof:

$$(xn_1)(yn_2) = x(yy^{-1})n_1yn_2 = xyn_3n_2 = xyn_4 \in xyN$$

So $(xN)(yN) = xyN$ and the cosets form a group.

Normal subgroup is the identity element in the factor group. Every group has at least two normal subgroups: the trivial group and the entire group. But if these are the only two groups, then G is a simple group.

Intuition of a normal subgroup. Group of rotations of cube has a subgroup of rotations around vertical axis. This subgroup $A1$ has 4 elements: rotate the cube for 0, 90, 180 or 270 degrees. There is another subgroup $A2$: rotation around one of horizontal axes. Subgroups $A1$ and $A2$ are obviously different but look alike. $A1$ and $A2$ are conjugated. g is the change to another view, $g^{-1}Ng$. A normal subgroup is very "symmetric". No matter the perspective of the whole group G the subgroup N remains at place.

6.1.3 Group homomorphisms

Define groups $(G, *)$ and $(H, \circ)$. Define $f : G \rightarrow H$ with maps $x \mapsto f(x)$ and $y \mapsto f(y)$. Then if $x * y \mapsto f(x * y)$ s.t. $f(x * y) = f(x) \circ f(y)$, f is a group-homomorphism or a homomorphism. If f is bijective, it is isomomorphic (can be determined by examining Cayley tables). Some transformations like $(\mathbb{Z}, +) \rightarrow (\mathbb{Z}/2\mathbb{Z}, +)$ are homomorphic in one direction only (there is a loss of information).

Example 1. $f : \mathbb{Z} \rightarrow \mathbb{Z}$ under $+$ with $f(x) = 2x$. $f(x + y) = f(x) + f(y)$

is $2(x + y) = 2x + 2y$ so f is homomorphic.

Example 2. $f : x \mapsto e^x$ with G under $+$ and H under $\times$. Then $f(x + y) = f(x) \times f(y)$ with $e^{x+y} = e^x e^y$ so f is homomorphic.

Example 3. Complex numbers in polar form.

Example 4. All equalities unless periodic, etc.

Homomorphisms send identities to identities and send inverses to inverses. Proof that homomorphisms map identities to identities. Suppose $f : G \rightarrow H$ is homomorphic so $f(x * y) = f(x) \circ f(y)$. Define 1_G and 1_H as the identities. Then:

$$x * 1_G = x$$

$$f(x * 1_G) = f(x)$$

$$f(x) \circ f(1_G) = f(x)$$

$$y^{-1}y \circ f(1_G) = y^{-1}y$$

$$f(1_G) = 1_H$$

Proof that homomorphisms map inverses to inverses.

$$x * x^{-1} = 1_G$$

$$f(x * x^{-1}) = f(1_G)$$

$$f(x) \circ f(x^{-1}) = 1_H$$

$$y^{-1}y \circ f(x^{-1}) = y^{-1}1_H$$

$$f(x^{-1}) = y^{-1}$$

6.1.4 Kernel of a homomorphism

$\ker(f) = \{x \in G \mid f(x) = 1_H\}$. If kernel contains only 1_G then it's bijective. But if f is not injective, the kernel will contain more than 1 element.

6.1.5 Automorphism

Endomorphism: a group that permutes a set. If it is further isomorphic, it is an automorphism.

Cayley's theorem. If G is a group of order n , then G is isomorphic to S_n . Proof. Let S be the set of all elements of G . Consider the action of G on S : $G \times S \rightarrow S$ $a \cdot b = ab$. This is a homomorphism and further, an isomorphism.

6.1.6 Stabilizer

Group action on fixed points. Determines the frame of reference.

Orbit-stabilizer theorem: the order of the orbit times the order of the stabilizer group = group order.

6.1.7 Cycles and group orbits

Cycle notation. Can write symmetric group permutations in cyclic notation. Example. $(1, 2, 3, 4, 5) \rightarrow (3, 5, 4, 1, 2)$ can be written as $(1, 3, 4)(2, 5)$, the cycle decomposition of a 3-cycle and a 2-cycle. Leave out 1-cycles. Order of cycles doesn't matter. If cycles are multiplied, they will commute if elements are independent but may not commute otherwise.

These subcycles are group orbits resulting from group actions.

Orbit decomposition. A group is decomposable into X/G .

6.1.8 Burnside lemma

Can help when counting objects with symmetry taken into account. It gives a formula to count objects, where two objects that are related by a symmetry (rotation or reflection, for example) are not to be counted as distinct.

Objects $\times$ symmetries = sum of symmetries. This is the same as Orbits $\times$ group actions = group elements.

6.1.9 Direct products

$G_1 \times G_2$. Operations applied component-wise, e.g. if G_1 under $+$ and G_2 under $\times$. $(a, b)(x, y) = (a + x, by)$ with $e = (e_1, e_2) = (0, 1)$. The group order is $|G_1||G_2| = |G|$, so can be finite or infinite. If even one group is abelian, the direct product will be non-abelian because of component-wise operations.

6.1.10 Specific important groups

Cyclic group. A group that can be generated by a single element X (the group generator). Cyclic groups are abelian. Every infinite cyclic group is isomorphic to $\mathbb{Z} = \langle 1 \rangle, +$. Every finite cyclic group is isomorphic to quotient group $\mathbb{Z}/n\mathbb{Z}, +$. These are the only cyclic groups. Fundamental theorem of finitely-generated abelian groups. Any abelian group that is finitely-generated can be decomposed into a finite number of cyclic groups.

Symmetric group. The group whose elements are all the bijections from the set to itself, and whose group operation is the composition of functions. Symmetric group (permutation group) is S_n , a group of permutations on a set with

n elements, so has $n!$ elements. $|S_n| = n!$ which is a group order (as opposed to an element order).

Alternating group. Group of all even permutations. Like a 'mirror image' group. Created using a transposition. Note: A_n is generated by 3-cycles because $(ijk) = (ij)(jk) \in A_n$ and disjoint 2-cycles are also $\in A_n$ to begin with. Moreover, since A_n contain the 2-cycles, they contain all the 3-cycles too. For A_n with $n \geq 5$, all the 3-cycles are conjugate because there are $n \geq 1$ transpositions (2-cycles) that commute. for A_4 , this doesn't happen.

Solvable group. Group that forms subnormal series $e = N_0 \trianglelefteq N_1 \trianglelefteq N_2 \dots \trianglelefteq N_k = G$ where each quotient N_i/N_{i-1} is abelian. Any abelian group is trivially solvable because $e \trianglelefteq G$ and G/e is abelian. Example. D_4 (square) is solvable because it forms multiple subnormal series: $(e) \trianglelefteq (f) \trianglelefteq (r^2, f) \trianglelefteq D_4$, $(e) \trianglelefteq (r) \trianglelefteq D_4$, $(e) \trianglelefteq (r^2) \trianglelefteq D_4$, each of which forms abelian quotients.

Note: For $n \geq 5$, A_n (alternating group) is (1) simple and (2) non-abelian. So the subnormal series is: $e \trianglelefteq A_n \trianglelefteq G$ and A_n/e is not abelian because A_n is simple and non-abelian.

Dihedral group. Finite group of symmetries of a regular polygon. n rotations + n flips = $2n$. Non-abelian: $rf = fr$. Denotation: D_n with n being the number of sides, or D_{2n} with $2n$ being the number of symmetries.

Klein-4 group. Smallest non-cyclic group. Abelian group. $V = (h, v, r, e)$ where e is identity. Binary operation between two non-identity elements yields the third. $hv = r$, $hr = v$, $rv = h$, $h^2 = r^2 = v^2 = e$. Only subgroups are $\langle h \rangle = (e, h)$, $\langle r \rangle = (e, r)$, $\langle v \rangle = (e, v)$, and is a subgroup of each $\langle e \rangle = (e)$. All are cyclic. So every proper subgroup is cyclic. Subfield lattice would be V at top, $\langle h \rangle, \langle r \rangle, \langle v \rangle$ in middle, and $\langle e \rangle$ at bottom. Isomorphic to $Gal(\mathbb{Q}(\sqrt{2}, i))$. Is basically the identity, a horizontal flip, vertical flip, and 180-degree rotation.

Primes are the building blocks of integers. Analogously, simple groups are the building blocks of groups. G is a simple group if its only normal subgroups are the trivial group (just the identity element) and the entire group.

Simple groups are the building blocks of finite groups. Proof. Let N_1 be a 'maximal' (i.e. the largest subgroup) and 'proper' (not equal to the entire group G). Repeat to get the normal series $1 \triangleleft \dots \triangleleft N_2 \triangleleft N_1 \triangleleft G$. The longest one possible is the composition series. Jordan-Holder theorem: the composition series of G are all isomorphic.

4 sets of finite simple groups (proof is 10,000-100,000 pages).

(1) $\mathbb{Z}/p\mathbb{Z}$ where p is prime. Since p is prime, only divisors are 1 and p . So only

subgroups are 0, $\mathbb{Z}/p\mathbb{Z}$. So the group is simple. Since there are infinite p , there are infinite simple groups in the class. All abelian.

(2) A_n (alternating group) when $n \geq 5$. All non-abelian.

(3) Lie groups. Group + manifold.

(4) Sporadic groups. Don't fit into (1)-(3). 20 in the 'Happy family' and 6 in the 'Pariah group'.

Matrices. Categorize groups. Abelian, infinite: integers under addition. Non-abelian, finite: S_n with $n > 2$. Abelian, finite: integers mod n . Non-abelian, infinite: matrices. General linear group: invertible square matrices. Special linear group: invertible square matrices with determinant = 1 as a subgroup.

6.2 Ring theory

Group with extra operation. Closed under $+$ and $\times$. Addition is commutative and has an inverse. Multiplication may not be commutative (if it is, then it's a commutative ring), may not have an inverse (division), and may not have an identity. Multiplication obeys associative laws. Multiplication and addition are linked by distributive laws.

Examples. $\mathbb{Z}$, many matrices, polynomials, even integers (no multiplicative identity but multiplicative commutativity). Even integer 2×2 matrices - no multiplicative identity, no multiplicative commutativity. $\mathbb{Z}/n\mathbb{Z}$ is a finite ring, where $\mathbb{Z}$ is a quotient ring (similar to quotient group) and $/n\mathbb{Z}$ is an ideal (analogous to normal subgroup). Note: if n is a prime, $\mathbb{Z}/n\mathbb{Z}$ is a field.

6.2.1 Units in a ring

Rings may not have multiplicative inverses or identities. But some elements may have them, forming a group - a unit of a ring. Example. -1 and 1. Example. $/12\mathbb{Z}$ with 1, 5, 7, 11 which are closed. Happens because each element has an inverse, and the inverse of $x \cdot y$ is $y^{-1} \cdot x^{-1}$.

6.2.2 Integral domains

Zero divisors. Example $\mathbb{Z}/12\mathbb{Z}$, solving a quadratic equation, factoring, and setting solutions to 0 won't give all solutions. Insufficient as a technique. However, can use factoring and setting to 0 when there are no zero divisors. Example. $\mathbb{Z}/11\mathbb{Z}$ as 11 is a prime.

Integral domains. Commutative ring with multiplicative identity and no zero divisors. (If no commutativity, then just 'domain' not integral domain). Has 'cancellation property' for solving equations.

6.2.3 Ideals

Ideals are to rings as normal subgroups are to groups. Requirements: (1) $I \leq R$ (subgroup), (2) $r \in R$ and $x \in I$, $rx, xr \in I$. If only xr the left ideal and if only rx then right ideal. $I \trianglelefteq R$ which is also closed under multiplication.

6.3 Field theory

Group with two operations that are commutative under both operations, and operations are connected through distributive laws. Real polynomials $R[x]$ is a ring but not a field because division is a problem. To include this, need field of rational functions $R(x)$. Z is a ring but not a field.

Multiplicative inverses - the hardest requirement.

Infinite field: Q . Finite field: Z/nZ where n is a prime. These are the only fields, the 'prime fields'. The characteristic of a field ($0=Q$, $n=Z/nZ$) says which prime field it extends.

C, R are extensions of Q .

6.3.1 Field extensions

$Q(\sqrt{2})$ is a field extension of Q so that elements can be written as $a + b\sqrt{2}$. Other field extension examples: $Q(\alpha)$ where α is the solution to a polynomial. Note: $R(i) = C$. Can field-extend with variables, example: $C(x, y, z)$.

6.4 Vector spaces

Commutative group under addition (vectors) with a field (scalars). Scalars and vectors both obey distributive and associative laws. Polynomials and continuous functions can form vector spaces.

A vector space's bases are fields so the vector space is isomorphic to F^n .

6.4.1 Modules

Generalization of vector spaces. Instead of a scalar field, it is a scalar ring (so have left modules if scalar is multiplied by module element, instead of element times scalar which is a right module). The left module is just called the module. Can have submodules.

Example. Scalar ring can be a 3x3 matrix.

Unlike vector spaces, modules don't have to be isomorphic to F^n . The bases can be different, example: $Z \times Zp/Z, nZ$.

6.5 Galois theory

Splitting field: smallest field extension to split a polynomial. Example 1. $\mathcal{Q}(\sqrt{2})$ is the splitting field of $f(x) = x^2 - 2$.

Basis of field extension. Example 2. $\mathcal{Q}(\sqrt{2})(i)$. So $(1, \sqrt{2}, i, i\sqrt{2})$ is the basis of the field (like bases of vector spaces): $a + b\sqrt{2} + ci + di\sqrt{2}$ or $\text{span} = (1, \sqrt{2}, i, i\sqrt{2})$.

Field automorphism. An automorphism is a homomorphism, and since this is a field, the field automorphism requires:

$$\phi(a + b) = \phi(a) + \phi(b)$$

$$\phi(ab) = \phi(a)\phi(b)$$

It is a permutation of the field extension elements. However, the ground (base) field is invariant. Proof. If $\phi(q) = q$, $q \neq 0$ because $\phi(2) = \phi(1 + 1) = \phi(1) + \phi(1) = 0 + 0 = 0$ so not bijective (which an automorphism $\in$ homomorphism requires). $\phi(1) = 1$ because

$$q = \phi(1) = \phi(1 \cdot 1) = \phi(1)\phi(1) = q^2$$

$$q = \phi(1) = \phi(1 \cdot 1 \cdot 1) = \phi(1)\phi(1)\phi(1) = q^3$$

So $q = 1$. For $\phi(2) = r$, $r = \phi(2) = \phi(1) + \phi(1) = 1 + 1 = 2$ and etc. The extensions form orbits. Example. $\mathcal{Q}(\sqrt{2}, i)$ leads to four automorphisms: (1) identity $e(i) = i, e(\sqrt{2}) = \sqrt{2}$, (2) horizontal flip $h(\sqrt{2}) = -\sqrt{2}, h(i) = i$, (3) vertical flip $v(\sqrt{2}) = \sqrt{2}, v(i) = -i$, (4) 180 degree rotation $r(\sqrt{2}) = -\sqrt{2}, r(i) = -i$

Galois group. Group of field automorphisms: $\text{Gal}(f(x)) \cong S_n$. So $\text{Aut}(\mathcal{Q}(\sqrt{2}, i)) = \text{Gal}(\mathcal{Q}(\sqrt{2}, i))$. Can be organized into a lattice. Another example. $\text{Gal}(x^2 + 1) = \text{Gal}(\mathcal{Q}(i)) \cong C_2$. C_2 is the cyclic group (just a transposition).

Fundamental Theory of Galois. The Galois group lattice is isomorphic (but inverted) of the corresponding S_n (symmetrical group).

The e (identity element) is invariant, as is the ground field. So the lattices are inverted.

6.6 Lie Theory

Lie groups provide a way to express the concept of a continuous family of symmetries for geometric objects. Lie algebras are the algebras accompanying Lie groups.

6.6.1 Lie groups

An 'infinitesimal' group. $(G, \cdot, \mathcal{A})$ on an atlas $\mathcal{A}$. General linear group is the largest Lie group: $GL(n, \mathcal{R})$, which is a group of all $n \times n$ matrices (square) that are invertible.

Orthogonal ($R'R = 1$) groups. $O(1)$ is just a point and not interesting. $O(2)$ is a subset of the general linear group. $1 = \det(1) = \det(R'R) = \det(R')\det(R) = \det(R)\det(R)$ so $\det(R) = 1$ which is $SO(2)$. This is the group along S_1 (1D manifold: unit circle) where angles are just added. So $SO(2) \cong S_1$. $SO(3) \cong \mathcal{RP}^3$ (RP: real projective space) which is a solid ball and has 3 degrees of rotation. $\frac{N^2-N}{2}$ degrees for $SO(N)$.

$SO(3)/SO(2) = S_2$ symmetric space. $SO(N+1)/SO(N) = S_N$.

6.6.2 Lie algebra

Encodes continuous symmetry. Algebra that defines Lie groups. The Lie algebra L is a vector space over a field with the Lie bracket or commutator $[\cdot, \cdot] : L \times L \rightarrow L$ which satisfies (i) bilinearity (like distribution), (ii) alternativity (indicates symmetry), and (iii) the Jacobi identity (order of operations).

7 Point-Set Topology

7.1 Topological spaces

A topological space is (X, T) . Any set can be equipped with a topology. The collection of subsets T is a topology of set X if:

- $\emptyset, X \in T$ (note: not necessary if generalize from other two axioms)
- Any finite intersection of open subsets is $\in T$ (can get $\emptyset$ as intersection)
- Arbitrary unions are $\in T$ (can get entire set X)

7.2 Topological bases

A base for a topology X is a collection β of subsets (called basic open sets, which are the ϵ balls in $\mathcal{R}^n$) s.t. given any $B_1, B_2 \in \beta$ we have $B_1 \cap B_2$ is a union of basic open sets. Since a topology can be defined using β , such a topology fulfills the two criteria (unions and intersections) of topologies. Can be used to build or check a topology instead of using the definition of topological spaces.

7.3 Countable bases

First-countable space. Each point has a countable neighborhood basis. So each point has an $\aleph_0$ -limited neighborhood.

Second-countable space: the space has countable bases. Note: Euclidean spaces are second-countable. Although the usual base of open balls is uncountable, one can restrict to the collection of all open balls with rational radii and whose centers have rational coordinates.

7.4 Nets and filters

Nets and filters are generalizations of sequences with potentially uncountable indices.

Sequences are limited. Example: Let (X, τ) be a topological space. Then the space is Hausdorff iff every net in X converges to at most one point. Another example: Compactness and sequential compactness are identical. Can prove forward direction with sequences but not backwards. However, can be reversed if (i) the topological space is first-countable (which limits neighborhoods to $\aleph_0$ cardinality), and/or (ii) sequences are replaced with nets (or filters).

Nets. Maps a directed set to a topological space. Can have subnets, analogous to subsequences. Directed sets can have uncountable elements with multiple chains that better detect limits. Note: sequences are also a kind of net, as they map the natural numbers (which are specific instances of directed sets) to topological spaces.

Net convergence. A net converges to a point in topological space if after a certain index it is in the neighborhood of that point.

Filters. Definition. Let X be a set. A nonempty collection $F \subseteq P(X)$ is a filter on X if (i) $\emptyset \notin F$, (ii) F is closed upwards: if $A \in F$ and $A \subseteq B$, then $B \in F$, and (iii) F is closed under finite intersections: if $A, B \in F$, then $A \cap B \in F$. A filter F on a set X is an ultrafilter if it is not properly contained in any other filter on X . Can have subfilters, again analogous to subsequences.

Filters are equivalent to directed set domains. Nets are equivalent to derived filters. Nets and filters have accumulation points, which are used to define equivalent notions of convergence more general than sequential convergence.

Compactness. Let (X, τ) be a topological space. Then the following are equivalent:

- X is compact.
- Every net in X has a convergent subnet.
- Every net in X has an accumulation point.
- Every filter on X has an accumulation point.

- Every filter on X can be extended to a convergent filter.
- Every ultrafilter on X converges.

7.5 Compact sets

Let X be a topological space. K is a compact set if it is non-empty and for any union of open sets containing K , there is a finite subcover that contains K .

Being boundedness is not a topological property. Example. $y = e^x$ from $(-\infty, 0)$ (unbounded) to $(0, 1)$ (bounded). However, being bounded AND closed (=compact) is a topological property. For a bicontinuous function (=homeomorphism) $f : X \rightarrow Y$, if X is compact, then $f(X)$ is also compact. Corollary. If a function is continuous, $f : X \rightarrow \mathcal{R}$, then if X is compact, then $f(X)$ has a min and max.

Tychonoff theorem. The product of compact spaces is compact. Can prove with nets or filters. Equivalent to the axiom of choice.

7.6 Paracompactness

A paracompact space is a T_2 -space where every open cover has a locally finite open refinement. Space doesn't have to be closed - e.g. Euclidean space, $\mathcal{R}$, every metric space. A refinement X of a cover Y is a cover such that every element $x \in X$ is a subset of an element $y \in Y$. A locally finite space is where every point has a neighborhood that meets only finitely many elements of any cover. Every compact space is paracompact.

Paracompactness is similar to the compactness property, but generalized for slightly "bigger" spaces.

7.7 Connectedness

Disconnectedness: (i) X is disjoint union of non-empty open sets V_1 and V_2 (which implies both are open and closed sets since they're complements of each other), (ii) disjoint union of non-empty closed sets, and (iii) X has a proper non-empty subset which is open and closed. A set that is not disconnected is connected. Disconnectedness and connectedness are topological properties.

Path connectedness is a stronger, more restrictive concept than connectedness. Can have two or more path components, but they can still be connected (but not path connected, obviously) - e.g. Swine curve.

Locally path connected. Path connected inside open subsets.

Connected + locally path connected = Path connected.

7.8 Types of topologies

Types of topologies:

- Discrete topology: all open subsets.
- Trivial topology: $T_d = (\emptyset, X)$.
- The usual topology on $\mathcal{R}$ is the collection of open sets O : $T = \{ \text{all sets } O \text{ s.t. } \forall x \in O, \exists \epsilon > 0, (x - \epsilon, x + \epsilon) \subset O \}$. E.g. $(1, 2)$ is open for usual topology while $[1, 2)$ is closed.
- $f : X \rightarrow Y_i$ can make f continuous with discrete topology. But want coarsest topology (=initial topology), as (x_n) will more easily converge if there are fewer open sets. For this, build the minimum topology: start with $\emptyset$ and X , add intersection and unions.

Use open instead of closed sets because (i) less work, (ii) can get neighborhoods around points without including other points but encompassing more 'space' beyond the point in question so generalizes better (note: topologies can also be defined using closed sets), and (iii) convention.

If $T_1 \subset T_2$, then T_1 is coarser/weaker/smaller and T_2 is finer/stronger/larger.

7.9 Separation axiom and T_k spaces

Separation axiom. For every pair of distinct points, there are open neighborhoods where the intersection is empty. This is a topological property - i.e. given $X \simeq Y$, both have the property.

T_0 (Kolmogorov): topologically distinguishable points. For any two points, there is an open set that contains one of the points but not the other. Example. A has a neighborhood that excludes B and C, B has a neighborhood that excludes C, but neighborhood of A contained in neighborhood of B.

T_1 (Frechet): topologically distinguishable (Kolmogorov) and symmetrical (points are separated). Neighborhoods are separate for each point but can intersect. For any two points, there are open sets that include one but not the other. Like Hausdorff space but the sets may not be disjoint.

T_2 or separated space. A topological space is Hausdorff if, given 2 distinct points in X , can find 2 open disjoint sets, each containing one point. In T_2 , L is unique for sequences.

T_3 (Regular Hausdorff): same as Hausdorff, except one of the points is a closed set.

T_4 (Normal Hausdorff): same as Hausdorff, except both points are closed sets.

$T_{2.5}$ (Urysohn): same as Hausdorff except with closed neighborhoods.

Non-metrizable topological spaces can be useful for looking at non-invertibility, showing how to merge points to convert non-metrizable to metrizable spaces, and distinguishing from metrizable spaces.

7.10 Subspace topology

$f : X \rightarrow Y$ where X is a set and Y is a topological space. Then, $X = (f^{-1}(v)|V \subseteq Y \text{ is open})$. Can prove f is continuous: $V_1 \cap \dots \cap V_n \subseteq Y$ is open, then $f^{-1}(v_1) \cap \dots \cap f^{-1}(v_n) = f^{-1}(v_1 \cap \dots \cap v_n)$ is open. X is a subspace topology of Y .

7.11 Quotient topology

For $f : X \rightarrow Y$ being surjective with X being a topological space while Y being just a set, Y is a topological space if $Y = (V \subseteq Y | f^{-1}(V) \subseteq X \in (X, \tau))$. It will be an open mapping. Then f is the quotient map, and Y is a quotient topology. Note: the quotient map is always continuous.

7.12 Product topology

Let B_1 be a basis of $X = (X, \tau_X)$ and B_2 be a basis of $Y = (Y, \tau_Y)$. Then the basis for product topology $X \times Y$ is $(U \times V | U \in B_1, V \in B_2)$. The product topology is the Cartesian product. Sometimes also called the Tychonoff or weak topology.

7.13 Manifolds

A topological space which is locally Euclidean. In more detail: paracompact Hausdorff space that is locally homeomorphic to Euclidean space.

7.14 Continuity

Continuity is a concept for FUNCTIONS and OPERATORS. Completeness is a concept for SPACES. However, functions and operators can be elements of spaces, e.g. vector spaces.

Continuity doesn't require completeness, e.g. can have continuous functions on $\mathbb{Q} \rightarrow \mathbb{Q}$ but $\mathbb{Q}$ isn't complete.

Let f be a function defined from topological space X to topological space Y , then f is said to be continuous at a point $x \in X$ if for every neighborhood V of $f(x)$, there exists a neighborhood U of x , such that $f(U) \subseteq V$.

Stronger topologies have more open sets so it is easier to be continuous. The discrete topology is always continuous.

7.15 Homeomorphism

Topological isomorphism. $f : X \rightarrow Y$ is homeomorphic (bicontinuous) if it's bijective and the inverse is continuous. $X \simeq Y$. Examples. A bicontinuous function. Spinning a circle around an axis and taking a quotient space of a square or rectangle's sides as equivalence classes.

7.16 Homotopy

A continuous transformation from one function to another. A homotopy between two functions f and g from a space X to a space Y is a continuous map H from $X \times [0, 1] \rightarrow Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$, where denotes set pairing. A homotopy is an equivalence relation. Another way of saying this is that a homotopy is a path in the mapping space $\text{Map}(X, Y)$ from the first function to the second. Homotopy is more general than homeomorphism.

Homotopy equivalence. A map $f : X \rightarrow Y$ is a homotopy equivalence if there exists a map $g : Y \rightarrow X$ such that $F \circ G$ is homotopic (continuously deformable) to id_Y and $G \circ F$ is homotopic to id_X . G is a homotopy inverse for F (F does not need to be invertible). A continuous map $f : X \rightarrow Y$ between two topological spaces X, Y , is said to be null homotopic if it is homotopic to a constant map g . A homeomorphism is a homotopy equivalence but the reverse isn't necessarily true as a contraction to a point is homotopy equivalent but not homeomorphic (as it lacks bijection). Another example showing the difference: the S^2 and a Möbius strip are both 2D manifolds and homotopy equivalent but not homeomorphic.

8 Algebraic Topology

If $C \subseteq \mathcal{R}^n$ is a convex, compact and non-empty interior, then $C \cong D^n$ and $\partial C \cong S^{n-1}$.

8.1 Retraction

Retraction. Fix $A \subseteq X$ as a topological subspace. A retract of X to A is a continuous map $r : X \rightarrow A$ s.t. r restricted to A is the identity on A : $r|_A = id_A$. Example. A cylinder $S^1 \times I$ retracted to circle $S^1 \times (0)$.

Theorem. There is no retraction from $D^2 \rightarrow S^1$. Proof by contradiction. Assuming such a retraction, $S^1 \hookrightarrow D^2 \xrightarrow{r} S^1$ and $S^1 \rightarrow S^1$ by id_{S^1} . Call this whole topological space A . Apply functor π_1 to A which takes the topological space to groups: $\pi_1(S^1) \hookrightarrow \pi_1(D^2) \xrightarrow{\pi_1(r)} \pi_1(S^1)$ and $\pi_1(S^1) \rightarrow \pi_1(S^1)$

by $\pi_1(id_{S^1}) = id_{\pi_1(S^1)}$. Then, $\pi_1(S^1) \cong \mathcal{Z}$ and $\pi_1(D^2) = 0$ (additive) or $\pi_1(D^2) = 1$ (multiplicative). Then it must be that the map is $\mathcal{Z} \rightarrow 0 \rightarrow \mathcal{Z}$ with $\mathcal{Z} \rightarrow \mathcal{Z}$ through id , this is impossible.

8.2 Topological holes

A hole in a mathematical object is a topological structure which prevents the object from being continuously shrunk to a point. A hole is like a distortion or limitation of space so provides information. Want to detect and characterize this information.

Euler characteristic $V - E + F = 2$ for convex polyhedra (note: vertex is 0D, edge is 1D, and face is 2D), and different for other shapes (e.g. $V - E + F = 0$ for a torus and $V - E + F = 1$ for a disk). So it could depend on the shape AND holes. There's a sense that shapes are characterized at least in part by holes.

Riemann counted holes based on how many times it could be cut while remaining in one piece. So a straw has one hole as it can be cut once, and a torus has two holes as it can be cut twice.

Poincare generalizes Riemann's idea to higher dimensions with the number of holes being Betti numbers. This is homology. Noether then applied algebra to homologies.

There are many ways to measure holes in a space. Some holes are picked up by homotopy groups that are not detected by homology groups, and some holes are detected by homology groups that are not picked up by homotopy groups. (For example, in the torus, homotopy groups "miss" the two-dimensional hole that is given by the torus itself, but the second homology group detects it.) In addition, homology groups don't detect the varying hole structures of the complement of knots in three-space, but the first homotopy group (the fundamental group) does.

8.3 Homotopic invariants and fundamental groups

Sphere $\pi_1(S^2) = 0$ and torus $\pi_2(\mathcal{T}^2) = \mathcal{Z}^2$. S^1 is not homeomorphic (as it's not bijective) to I Homotopic paths. Null-homotopic paths do nothing.

$\pi(M)$ is the fundamental group of a surface. It's based (starting and ending) at one point x_0 on the surface. So it's a null-homotopic loop, homotopic to the constant loop. Since multiplication doesn't depend on head-tail of homotopic paths, can multiply any two such objects. It is an equivalence class and algebraic object.

For a disk or polygon, $\pi(M) = \pi(disk) = \pi(polygon)$. It collapses to a point.

S^1 . $t \in [0, 1]$. $\alpha_0(t)$ is the constant loop. $\alpha_1(t)$ is the single loop, $\alpha_2(t) = 2t$ that loops at double speed, etc. Reverse direction is inverse. Multiplication is $(\alpha_n) * (\alpha_m) = (\alpha_{n+m})$ (brackets indicating homotopically equivalent classes). Group $\pi(S^1)$ behaves as $\mathbb{Z}$ under addition.

Theorem. $\pi(S^1)$ is a group: (1) there is an identity element $(e) = \alpha_0$, (2) there is an inverse α_{-n} of α_n , and (3) associativity (indices are additive and therefore associative). Group can also be commutative. Note closure is already present as any two loops result simply results in another loop.

Torus. Unrolling the torus surface into a rectangle, the edges are a and b are the loop 'generators', and $aba^{-1}b^{-1}$ is null-homotopic! So $aba^{-1}b^{-1} = e$ and therefore $ab = eab$ and therefore $ab = ba$ so commutative! So $a^n b^m$ and therefore the fundamental group is $\mathbb{Z} \times \mathbb{Z} = \mathbb{Z}^2$.

Functoriality. If $f : X \rightarrow Y$ is a continuous map, $x_0 \in X$, $y_0 \in Y$, and $f(x_0) = y_0$, then every loop in X with base point x_0 can be composed with f to yield a loop in Y with base point y_0 . This operation is compatible with the homotopy equivalence relation and with composition of loops. There is a resulting group homomorphism called the induced homomorphism.

8.4 Simplices

Generalizations of triangles to construct spaces. Fundamental higher-dimensional simplices are comprised of lower-dimensional simplices.

Barycentric coordinates for a point inside a triangle. 1D: $v_1 + \lambda(v_2 - v_1)$ where $\lambda \in [0, 1]$. 2D: $\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3$ where $\lambda_i \in [0, 1]$ and $\lambda_1 + \lambda_2 + \lambda_3 = 1$. So it's a linear combination. 3D: Tetrahedron. n -simplex $\Delta_n = \sum \lambda_i v_i$ where $\lambda_i \in [0, 1]$ and $\sum \lambda_i = 1$.

Standard form of Barycentric coordinates: use standard basis vectors. 1D: $x_1 + x_2 = 1$, 2D: $x_1 + x_2 + x_3 = 1$, n D: $\sum x_i = 1$.

Simplicial complexes. Build up spaces with simplices. So combine different n D simplices to build up a space.

8.5 Homology

Fundamental group π_1 is good for low-dimensional information as it's just about loops. Generalize to higher dimensions (homotopy groups): $\pi_n(S_k)$ where $n \geq 2$. π_2 is a 'loop of loops', or the void of a hollow sphere. Same as a square sheet wrapping around and being sealed. Homotopy groups are hard to compute and counter-intuitive with $n > k$. So want alternative to homotopy groups: homology.

Homology, $H_n(x)$. Commutative group alternative to homotopy. n also measures the number of holes but differently from homotopy.

Examples. With directed graph x-y-z connected as a triangle with a-b-c edges, the order doesn't matter and so it's commutative: can represent as $a + b + c = b + c + a = c + a + b$. This is a cycle or a closed loop. If go backward on a directed edge, it's negative. Can represent directed graphs as algebra.

Can have free abelian groups of linear combinations of vertices C_0 (0D chains) and free abelian groups of linear combinations of edges C_1 (1D chains). C_2 are 2D planes with orientation (front and back) and are '2D chains' (a '2-cell').

Each edge has a boundary $\partial(\text{edge}) = \text{final}_{\text{vertex}} - \text{initial}_{\text{vertex}}$. This algebraically links edges and vertices. So $\partial : C_1 \rightarrow C_0$ is a group homomorphism. The basis would be the vertices. Cycles are when $\partial = 0$ (a loop). So can solve a linear system of equations all equal to 0. In linear algebra, this is solving for the nullspace or kernel $Ax = 0$. Solving this, get a linear combination, which is the basis of cycles (loops). So the kernels are cycles. If there are two cycles, it's isomorphic to $H_1(x) = \mathcal{Z} + \mathcal{Z} = \mathcal{Z}^2$, the planar graph with 2 loops with each loop being a basis.

Homology of a planar graph. The number of edges in a spanning tree is $v - 1$ and the total number of edges is e , the number of non-spanning edges is $e - (v - 1)$ which each forms a cycle. So $H_1(x) = \mathcal{Z}^{e-v+1}$.

Homology of a non-planar, 3D graph. $H_1 = \frac{\mathcal{Z}_1}{B_1}$ where $\mathcal{Z}$ is 1D cycles and B is 1D boundaries. So $H_1 = \mathcal{Z}^2 / \mathcal{Z}$ for example, which is a 2-dimensional cycle (a graph that is '3D') and generates H_2 , which captures a 2D hole as it cycles around it. Can fill the 2-cell which is hollow with a 3-cell (a solid ball) which is 0.

Homology = kernel divided by image. 0D homology are connected components. 1D homology are number of 1D holes. Homology groups correspond to vector spaces. 2D holes are voids (e.g. inside of hollow sphere). Sphere. 1 0D hole, or connected component (so vector space rank 1). 0 1D holes, as any loops can be shrunk to a point. 1 2D hole. Torus. 1 0D hole, or connected component. 2 1D holes as it has the inside of the donut and the hollow inside. This includes linear combination of these two 1D holes to that can form a Hopf link from the torus. 1 2D hole.

Homologies are quotient groups. The kernel is a cycle. The boundary of edge is final minus initial. $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$. C_2 for instance is the 2D chain. Theorem: $\partial^2 = 0$. Moreover, ∂_i is the image of C_i (the group of boundaries), and ∂_{i-1} is the kernel of C_{i-1} (the group of cycles). So image $\subseteq$ kernel. So the homology $H_i = \frac{\text{ker}}{\text{image}}$ is a quotient group.

H_0 , connected components. If there are two connected components with 2 vertices and 1 node each, then ker/img is 4 vertices divided by 2 edges = $(\mathcal{Z} + \mathcal{Z} + \mathcal{Z} + \mathcal{Z})/(\mathcal{Z} + \mathcal{Z})$.

Homologies with simplices: circle S^1 is homeomorphic to a triangle simplex. C_0 is $\mathcal{Z} + \mathcal{Z} + \mathcal{Z}$ because it's just the vertices. C_1 is $\mathcal{Z} + \mathcal{Z} + \mathcal{Z}$ because it's just the edges. $C_2 = 0$ as it's 2D. $H_0(S^1) = (ker)/(img) = (\mathcal{Z} + \mathcal{Z} + \mathcal{Z})/(\mathcal{Z} + \mathcal{Z}) = \mathcal{Z}$ because img is just a repetition of $x = y = z$. H_0 is number of connected components. Also, $H_1(S^1) = (ker)/(img)$ at higher-D so $= \mathcal{Z}$ as $ker = \mathcal{Z}$ and $img = 0$ (i.e. the whole simplex).

Homologies with simplices: disk D^2 . Same as S^2 but with C_2 as it is a 2D solid surface with an orientation in one way (typically consistent with the directed edges). $H_0(S^2) = H_0(D^2)$. But $H_1 = \mathcal{Z}/\mathcal{Z} = 0$. $H_2 = 0$ because $C_3 = 0$. Sphere S^2 . Homeomorphic to tetrahedron simplex. $H_1 = 0$ since boundaries equal the cycles. $H_2(S^2) = 0$.

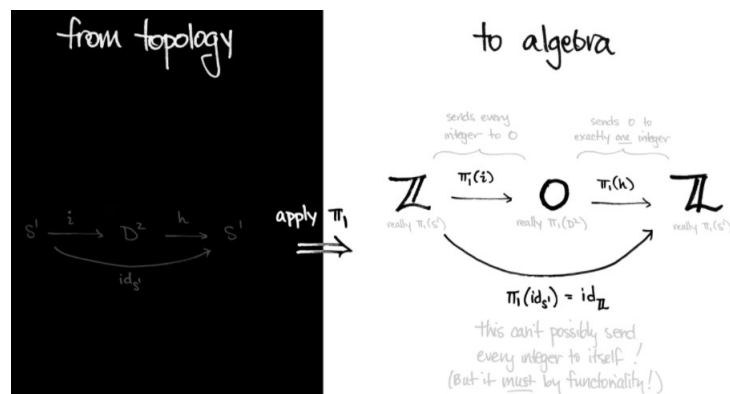
Simpler method of computing homologies. Eilenberg approach is simpler, uses combinations of simplices. Example with S^2 . Think of sphere as two flat triangles blown up. Then use their vertices, edges, and faces.

8.6 Brouwer's fixed point theorem

Any continuous function of a disk in Euclidean space $\mathcal{R}^n$ has a fixed point. More generally, it works for compact sets and is homeomorphic to a convex set. The function only needs to be injective (doesn't have to be bijective, surjective).

Proof for 1D case. Consequence of the intermediate value theorem with box $[0,1]$ for x and y and the function must cross the 45-degree line ($f(x) = x$ so need $f(x) - x = 0 = g(x)$ where $g \in [0,1]$ as $f(a) - a \geq 0$ and $f(b) - b \leq 0$). If it's not continuous, then function can 'jump' over the 45-degree line. If it's not bounded (so not compact) then the function can just run parallel to the 45-degree line. If it's not closed but still bounded then the function may not touch the 45-degree line at the corner.

Proof for 2D case by contradiction. Assume there is a continuous function $g : D^2 \rightarrow D^2$ s.t. $g(x) \neq x$. Let $h(x) \in S^1$ be a retraction of a straight line from $g(x)$ through x to the boundary S^1 . Let $i : S^1 \rightarrow D^2$ be the inclusion of the circle in the disk so $i(x) = x \forall x \in S^1$. Then $h(i(x)) = x \forall x \in S^1$ so $h \circ i = id_{S^1}$. Apply the functor $\pi : Top \rightarrow Grp$ to get $\pi(h) \circ \pi(i) = id_{\mathcal{Z}}$. Then $S^1 \xrightarrow{h} D^2 \xrightarrow{g} S^1$ which is id_{S^1} is turned into a different category which is a contradiction:



So cannot map the set of all integers to zero and then reverse the process. So cannot map the boundary of a disc to the disc and then reverse the process. So $g(x) \neq x$ doesn't exist and there is a fixed point.

Alternatively, can show that Sperner's lemma implies Brouwer's fixed point theorem. For each point P in a simplex, let $f(P)$ be its image. Consider the vector $v = f(P) - P$, which points towards some corners of the simplex and away from some. Color P with the color of the vertex it points most away from (choosing arbitrarily in the case of a tie).

This satisfies the Sperner coloring constraints: any point on an edge will point most away from one of the two vertices defining that edge, so it will be colored with one of those two colors. Thus, by Sperner's lemma there exists a small triangle whose vertices all have different colors. Repeating this process on the points inside this triangle infinitely gives a point that should be all three colors at the same time, which is possible only if that point is a fixed point (as the resulting vector is then not pointing towards any vertex). This demonstrates a fixed point exists for any continuous f from the simplex to the simplex, and since a simplex is homeomorphic to a closed ball, this demonstrates Brouwer's theorem.

8.7 Kakutani fixed point theorem

Generalization of Brouwer's fixed point theorem. Let S be a compact and convex (or is homeomorphic to convex) subset of Euclidean space $\mathbb{R}^n$. Let $\phi : S \rightarrow 2^S$ be a continuous set-valued function (=multivalued function or correspondence) on S which is (1) UHC, (2) convex-valued (so no 'holes' or crevices in the correspondence), and (3) non-empty, then it has a fixed point.

Counterexamples. If it's discontinuous, it's not UHC. If it's discontinuous with two $f(x)$ values at one x (a correspondence), it won't be a correspondence.

Proof. For most cases, can prove by simply taking a continuous function from

the correspondence and using Brouwer's fixed point theorem. Continuous functions are an element of correspondences. However, this doesn't work if can't draw a continuous function from a correspondence that otherwise works (e.g. a correspondence with horizontal + vertical lines). To address this, 'thicken' the vertical section with ϵ -neighborhoods around every point of the correspondence, so now there can be a 'near'-vertical continuous function at the vertical correspondence section. Formally, it uses the von Neumann approximation theorem - which requires convex-valuedness. So have 'thickened' points. Shrink ϵ -balls with Bolzano-Weierstrass to get convergence, so $(x, y) \rightarrow (x^*, x^*)$ so apply UHC to get the fixed point $x^* \in G(x^*)$.

8.8 Berge maximum theorem

General setup. $v(\theta) = \max_x f(x, \theta)$ subject to $x \in G(\theta)$. The solution correspondence is $\Gamma(\theta) = \{x \in G(\theta) | v(\theta) = f(x, \theta)\}$. So v is the value of the maximum, and Γ is the 'range' of the variables.

Maximum theorem. There is a result. Moreover, if f is a continuous function and G is a continuous correspondence (so both UHC and LHC), then v is continuous and Γ is UHC.

Proof. $f(x)$ is continuous so $v(\theta) = \max f(x, \theta)$ is continuous. Further, g is continuous (and therefore UHC) so $G(\theta)$ is compact. This means $x \in G(\theta)$ is compact. Get a sequence of (θ_n) that corresponds to (x_n) , which yields $(v(\theta_n) = f(x, \theta_n))$ and $(x_n \in G(\theta_n))$. Since $\theta_n \rightarrow \bar{\theta}$, $x \in G(\theta)$ with G being UHC means (x_n) is bounded, and by Bolzano-Weierstrass, the subsequence $(x_m) \rightarrow \bar{x}$ so the subsequence $\theta_m \rightarrow \bar{\theta}$. By UHC of G , $x \in G(\theta)$! Since θ converges, x converges, then $v = f(x, \theta)$ converges.

Also need to show $v(\theta) = f(x, \theta)$. Prove by contradiction. Assume $v(\theta) \neq f(x, \theta)$, then $f(x, \theta) < v(\theta)$ as it's not optimal. Let instead y be the optimal value for v . Then $f(x, \theta) < f(y, \theta)$ and $y \in G(\theta)$ as it must be feasible. Meanwhile, (θ_n) converges which has a corresponding (y_n) . So $f(y, \theta)$ must also converge. Use G 's LHC to construct (y_n) which corresponds to $G(\theta_n)$. But we already had (x_n) moving with (θ_n) so $f(x_n, \theta_n) = v(\theta_n) \geq f(y_n, \theta_n)$ means $f(x, \theta) \geq f(y, \theta)$.

Show Γ is UHC. As (θ_n) converges, $x \in G(\theta)$ converges and since G is UHC, G is bounded so (x_n) is bounded. As (θ_n) converges, $f(x_n, \theta_n) = v(\theta_n)$ (with v being continuous) converges so (x_n) converges with $x \in G(\theta)$ so $x \in \Gamma(\theta)$.

8.9 Nash existence theorem

Quasi-concavity condition. It makes no sense to impose a stronger condition, like concavity, on this function, because the only significant property of the

function is the character of its level curves (ordinality), not the specific numbers (cardinality) assigned to these curves. Player utilities are quasi-concave rather than quasi-convex because of diminishing marginal returns as with utility functions.

Normal-form game. Finite players, $A_i \in \mathcal{R}^k$ is the action set of player i (which are vectors). Every player selects $a_i \in A_i$ and payoff $u^i = A_1 \times A_2 \dots = \prod A_i$ manifested as $u^i(a_1, a_2 \dots)$. $(a_1^*, a_2^* \dots)$ is a NE if $\forall i, \forall N, u^i(a_1^*, a_2^* \dots a_i^*, \dots) \leq u^i(a_1^*, a_2^* \dots a_i^*, \dots)$.

Conditions for NE existence. A_i is compact and convex (from Kakutani). u^i is continuous and quasi-concave in own action (so if all other players' actions are fixed, then a_i is quasi-concave).

Berge's maximum shows that the solution correspondence exists (can use Weierstrass' theorem instead). Kakutani shows there is a fixed point for the NE, i.e. $a \in B(a)$.

Proof. Restrict to $N = 2$ wlog for simplicity. P1: $v^1(a_1) = \max u^1(a_1, a_2)$ (continuous) subject to $a_1 \in A_1$ (note: A_1 is like $G(\theta)$ from the maximum theorem, but in this case A_1 is independent of a_2). Further, A_1 is convex, which means a_2 is a convex correspondence. P2 best response correspondence: $B^1(a_2) = (a_1 \in A_1 | u^1(a_1, a_2) = v^1(a_2))$. By the maximum theorem, v is continuous and B^1 is UHC. Since a_2 is convex, $B^1(a_2)$ is convex. Define P2 analogously.

$u^1(a_1^*, a_2^*) \geq u^1(a_1, a_2^*)$ and analogously for P2. Then $a_1^* \in B^1(a_2^*)$ and $B^1 : A_2 \rightarrow A_1$ and similarly for P2. Then take Cartesian product $B : A_1 \times A_2 \rightarrow A_1 \times A_2$, so $B(a_1, a_2) = ((b_1, b_2) \in A_1 \times A_2 | b_1 \in B^1, b_2 \in B^2)$. $NE = (a_1^*, a_2^*) \in B(a_1^*, a_2^*)$. Then:

$$\begin{aligned} a_1^* &\in B^1(a_2^*) \\ a_2^* &\in B^2(a_1^*) \end{aligned}$$

We know $A_1 \times A_2$ and that each A_i is compact and convex. Turn each element into a vector (a_1, a_2) , with each element being bounded because they come from compact sets, and since the limiting vector will converge, the product is compact. With two 2D vectors, the first component will be a convex set and likewise for the second component. So $A_1 \times A_2$, the domain, is compact and convex.

Let $(b_1, b_2) \in B(a_1, a_2)$. Then B is UHC. Take sequence of $(a_{1,n}, a_{2,n}) \rightarrow (a_1, a_2)$ and likewise $(b_{1,n}, b_{2,n}) \in B(a_{1,n}, a_{2,n})$. Have to show the b 's are bounded and converges to a 's. First show $(b_{1,n}, b_{2,n})$ is bounded.

$$\begin{aligned} a_{1,n} &\rightarrow a_1 \\ a_{2,n} &\rightarrow a_2 \end{aligned}$$

$$b_{1,n} \in B^1(a_{2,n})$$

$$b_{2,n} \in B^2(a_{1,n})$$

Since B^2 is UHC, $(b_{2,n})$ is bounded. Likewise, B^1 is UHC, $(b_{1,n})$ is bounded. Then $(b_{1,n}, b_{2,n})$ is bounded. Next show convergence to a's.

$$a_{1,n} \rightarrow a_1$$

$$a_{2,n} \rightarrow a_2$$

$$b_{1,n} \in B^1(a_{2,n})$$

$$b_{2,n} \in B^2(a_{1,n})$$

$$b_{1,n} \rightarrow b_1$$

$$b_{2,n} \rightarrow b_2$$

By UHC of B^2 , $b_2 \in B^2(a_1)$ and similarly to get $b_1 \in B^1(a_2)$.

Now show B is convex. B^1 and B^2 are separately convex. Let (b_1, b_2) and (c_1, c_2) both $\in B(a_1, a_2)$.

$$b_1 \in B^1(a_2)$$

$$b_2 \in B^2(a_1)$$

$$c_1 \in B^1(a_2)$$

$$c_2 \in B^2(a_1)$$

The convex combinations

$$\alpha b_1 + (1 - \alpha)c_1 \in B^1(a_2)$$

$$\alpha b_2 + (1 - \alpha)c_2 \in B^2(a_1)$$

become $(\alpha b_1 + (1 - \alpha)c_1, \alpha b_2 + (1 - \alpha)c_2) \in B(a_1, a_2)$. This is equal in vector form to $\alpha \vec{b} + (1 - \alpha)\vec{c} \in B\vec{a}$.

8.10 Borsuk-Ulam

1-dimension. If $f : S^1 \rightarrow \mathcal{R}$ continuous, then $\exists x \in S^1$ s.t. $f(x) = f(-x)$. Let $g(x) = f(x) - f(-x)$. if $g(x_0) = 0$ then done, else $g(x) \neq 0 \forall x$. Assume $g(x) > 0$, then $g(-x) = f(-x) - f(x) = -(f(x) - f(-x)) = -g(x) < 0$. So $x \rightarrow -x$ means $g(x) > 0$ goes to $g(-x) < 0$, so by the intermediate value theorem, $\exists x_0$ $g(x_0) = 0$.

n-dimension. If $f : S^n \rightarrow \mathcal{R}^n$ continuous, then $\exists x \in S^n$ s.t. $f(x) = f(-x)$. Use vector spaces. Let $g(\vec{p}) = f(\vec{p}) - f(-\vec{p}) = -g(-\vec{p})$ where $g(\vec{p})$ and $g(-\vec{p})$ are 'opposite' rotations around the origin. Then shrink this 'loop' into a single point - at some $\vec{p}$, $g(\vec{p}) = \vec{0}$.

8.11 Euler characteristic

$\chi = V - E + F = 2$ for any convex polyhedra. χ is different for different shapes. Many proofs.

9 Graph Theory

Examples. Seven bridges of Konigsberg. Guarini's Knights on a chessboard. Represent as a circular graph and rotate four times = 16 moves.

Degree sum. The sum of all degrees (for vertices, also called nodes) in a graph is twice the number of edges E :

$$\sum_{v \in V} \deg(v) = 2|E|$$

Handshaking lemma. A graph has an even number of odd nodes.

Lower bound. An undirected graph $G(V,E)$ has at least $|V| - |E|$ connected components. If a graph is connected, then $|E| \geq |V| - 1$. Theorem not applicable if $|V| \leq |E|$.

A metagraph of connected components, where the connected components are each a node, is a directed acyclic graph.

9.1 Cycles

Eulerian cycle. Visits each EDGE exactly once. Works for both directed and undirected graphs. Start and end at same node.

- Connected undirected graphs. Eulerian cycle iff the degree of every node is even.
- Strongly connected directed graphs. Eulerian cycle iff every node is balanced (indegree=outdegree).

Hamiltonian cycle. Visits each NODE exactly once. No simple criteria known yet. No P-time algorithm.

Some problems can be expressed as either Eulerian and Hamiltonian cycles, and can get different non-unique solutions (i.e. cycles).

9.2 Graph types

9.2.1 Trees

Applicable to road repair problem, satellites, etc.. Trees don't have cycles.

Use Kruskal algorithm to find the minimum spanning tree, i.e. with minimum weights.

9.2.2 Bipartite graphs

Applicable to assignments of objects to different people. Can show how to assign, if matchings are possible based on preferences, etc.

A graph is bipartite iff it has no cycles of odd length.

Hall's Theorem: the left side has matchings if any subset has at least as many neighbors on the other side.

9.2.3 Planar graphs

No edges cross. Maps are naturally planar. Also called an embedding. E.g. subway, ICs.

Euler's theorem for planar graphs (originates in topology):

$$V - E + F = 2$$

Faces include the infinite-area face on the outside. Works for trees (only one infinite face). Removing an edge from a cycle means one less E and one less F, and repeat until there's a tree. Intuition:

- Start with 3 vertices and 3 edges resulting in 2 faces ($3-3+2=2$)
- Add 1 vertex and 2 edges resulting in 1 more face ($((3+1)-(3+2)+(2+1)=(3-3+2)+(1-2+1)=2+0=2)$)
- Repeat previous step

Faces. Let p be the number of (face, edge) pairs.

- Each face has at least 3 edges. $p \geq 3f$.
- Each edge belongs to 1 or 2 faces. $p \leq 2e$.

So $f \leq \frac{2e}{3}$ in a planar graph. Combine with Euler's theorem:

$$2 = v - e + f \leq v - e + \frac{2e}{3} = v - \frac{e}{3}$$

$$2 \leq v - \frac{e}{3}$$

$$e \leq 3v - 6$$

A connected planar graph has vertices with degrees of at most 5. If all vertex degrees were 6, then $e = 6v/2 = 3v$ which violates the equality above. So every planar graph has a vertex which has AT MOST 5 degrees.

Connected bipartite graphs don't have odd-length cycles. Each connected bipartite face has at least 4 edges. So $p \geq 4f$. Then $f \leq \frac{e}{2}$ and:

$$2 = v - e + f \leq v - e + \frac{e}{2} = v - \frac{e}{2}$$

$$2 \leq v - \frac{e}{2}$$

$$e \leq 2v - 4$$

9.3 Dual graphs

Connect planar graph faces to each other and to the outside face. Should get another planar graph. Dual graph of a dual graph returns the original graph.

If a graph is isomorphic to its dual graph, then it's self-dual.

9.4 Chromatic numbers

Four Color Theorem. May need another color for water, and countries have to be connected (no exclaves). Any map can be colored with at most 4 colors without adjacent countries using the same color. Every planar graph can be a map, and vice versa. Weaker versions of the theorem (6, 5 color) can be proven analytically, but 4-color theorem still uses a computer to prove all cases.

Chromatic numbers. For path graphs, 2. For full graphs, number of vertices. For cycles of even length, 2. For cycles of odd length, 3. For bipartite graphs, 2.

Greedy coloring. A graph of maximum degree Δ can be colored with $\Delta + 1$. This simply involves looking at the maximum number of neighbors. Stronger versions of the theorem exist.

Examples. Exam scheduling - each exam is a vertex, and if at least one student takes two courses, there is an edge between them. If there are no edges between vertices (color them with same color), those courses can be scheduled on the same day. Use for bandwidth allocation, assigning aircraft to flights, taxi scheduling.

9.5 Cliques

All vertices are connected to each other in a clique. The largest clique in a graph corresponds to the clique number $\omega(G)$.

None of the vertices are connected to each other in an independent set. The independent set is a complement of the clique. The largest independent set

corresponds to the independent set number $\alpha(\bar{G})$. Independent sets correspond to different colors from the four-color theorem.

$$(1) \omega(G) = \alpha(\bar{G})$$

(2) Since n vertices can be divided into $\chi(G)$ (the chromatic number) and these are independent sets with $\leq \alpha(G)$, then $\frac{n}{\chi(G)} \leq \alpha(G) \Rightarrow n \leq \chi(G)\alpha(G)$.

Mantel's theorem: If a graph has no triangles, then it has at most $\lfloor n^2/4 \rfloor$ edges. Partition the graph into a bipartite graph with $\lfloor n/2 \rfloor$ and $\lfloor n/2 \rfloor$ vertices which has no triangle. For bipartite graphs, the maximum number of edges comes with $n(k-n)$ which is maximized at $n^2/4$. Then the most edges it can have is $\lfloor n/2 \rfloor \lfloor n/2 \rfloor = \lfloor n^2/4 \rfloor$. Can be proved by induction.

Turan's theorem: generalization of Mantel's theorem. If a graph contains no K_{r+1} (clique of size $r+1$), then it has at most $(1 - \frac{1}{r}) \frac{n^2}{2}$ edges.

Extremal graph theory looks at interaction between global parameters and local graph structures. It includes Mantel's and Turan's theorems.

9.6 Ramsey theory

Ramsey numbers. $R(k, l)$ is the minimum number of vertices with a k -clique OR l -independent set. Difficult to find these numbers, as combinations can be huge.

- $R(3, 3) = 6$. Every 6-vertices graph contains either a 3-clique or 3-independent set.
- $R(4, 4) = 18$.
- $43 \leq R(5, 5) \leq 48$. Very big to calculate.

Ramsey's theorem: $R(k, l) \leq R(k-1, l) + R(k, l-1)$. So $R(k, l)$ upper bounds always exist. Proof:

- (1) Consider G on $R(k-1, l) + R(k, l-1)$ vertices.
- (2) Pick arbitrary vertex v . A is the set of neighbors, and B is the remainder. So v is connected to all vertices in A but not any in B .
- (3) So the number of vertices is:

$$|A| + |B| + 1 = R(k-1, l) + R(k, l-1)$$

- (4) One of following must hold:

$$|A| \leq R(k-1, l)$$

Then A has an l -independent set OR together with v has a k -clique.

$$|B| \leq R(k, l-1)$$

Then B has a k -clique OR together with v has an l -independent set.

9.7 Vertex covers

Set of vertices that is connected to every edge. The minimum vertex cover $(\beta(G)) +$ maximum independent set $(\alpha(G)) = G$.

The vertices of any maximal matching form a (not necessarily minimal) vertex cover. (A matching contains only disjoint or non-adjacent edges).

Konig's Theorem. In a bipartite graph, the number of EDGES in maximal matching equals the number of VERTICES in a minimum vertex cover. Prove by showing that the the number of edges in a maximal matching is upper-bounded by the number of vertices in a minimum vertex cover (intuitive) and vice versa (harder; involves Hall's theorem).

9.8 Other theorems

Menger's theorem. Let u and v be non-adjacent vertices. The size of a minimum u - v separating set is the maximum number of internally disjoint u - v paths in G .

9.9 Isomorphisms

Two graphs are the same if the vertices are the same and the edges are the same.

Two graphs are isomorphic if they can be made equal by relabelling the vertices (if they are labelled graphs) which means they have the same size and same order. Their degree sequences will also be the same.

9.10 Sperner's lemma

1D simplex (line). Each vertex has a different color. Then the number of edges with different colors will be odd. Think about each vertex of one color as a door with a doormat (but no doormats to the outer, infinite face).

2D simplex (triangle). Every end-vertex has a different color. Every boundary vertex can only be one of two colors of the end-vertices of that boundary. The inner vertices can be any of the three colors.

Theorem. In the 2D simplex, there exists an odd number ($n > 1$) of small triangles with 3 different vertex colors. Think of each node of two selected colors having a door with a doormat to the outside (except to the outer infinite face). There is an even number on the inside (as any edge of the selected two vertex colors will be one door and therefore two doormats), and an odd number on the boundary edge as it's only possible on one of the three boundary edges and the rest is as per the 1D case. So there are odd doormats. The only small triangles possible are those with no doormats, two doormats, and only one triangle with a single doormat (that has vertices with all different colors).

So there is an integer number of the last.

Can be generalized to higher n-simplices.

10 Fractals and Scaling

10.1 Fractals

Dimensions: property of object does not change when the scale is changed. They are topologically invariant.

10.1.1 Self-similarity dimension

Fractals are self-similar.

$$\text{Copies} = (\text{Magnification factor})^D$$

where D is the self-similarity dimension. Intuitively, it is the 'density' of the copies per magnification.

Lengths. Koch curve. Take line segment L and divide into thirds, replacing the middle third segment with two segments that are each (L/3) length. The total length is then (4L/3). Repeat for (16L/9) length at next level to $\frac{4^n L}{3^n}$. The number of segment lengths is increasing faster than the shrinking lengths.

Lengths and areas. Sierpinski triangle. Area gets progressively smaller and head to zero, but perimeter get progressively larger and heads to infinity.

Fractals are scale-free. A tomato, for instance, has a bounded range of sizes, but a fractal's range is not bounded.

10.1.2 Generating fractals

Fractal generating notation systems. Iterated functions (IFS; transformations usually represented by matrices) or L-systems (Lindenmayer systems; instructions using axiom, production rule and repeatedly nesting the production rule) to generate fractals.

Deterministic v stochastic fractal generation. Adds more variety and features to fractals. Two examples.

- Chaos game. Stochastic process leads to non-random outcome (e.g. Sierpinski triangle).
- Particles undergoing a random walk due to Brownian motion cluster together to form aggregates of such particles.

10.1.3 Box-counting dimension

Modify the self-similarity dimension to get:

$$N = c \left(\frac{1}{s} \right)^D$$

where N is the number of boxes, s is the side length of each box, D is the dimension and c is the length of the object. Defined as $s \rightarrow \infty$, so good approximations when s is sufficiently small but not too small that it reaches pixel limit of computer. Can be used for all shapes and gives same answer as self-similarity dimension for Sierpinski triangle, Cantor set, and others.

$$N = c \left(\frac{1}{s} \right)^D$$

$$\log(N) = \log(c) - D \log(s)$$

Collect data and use log-log regression to estimate D and c .

10.2 Power laws

Power laws are scale-free:

$$f(x) = Ax^\alpha$$

Let $y = kx$. Then,

$$\begin{aligned} f(y) &= A \left(\frac{y}{k} \right)^\alpha \\ &= \frac{A}{k^\alpha} y^\alpha \\ &= A' y^\alpha \end{aligned}$$

Renormalization group theory (from statistical mechanics) is used to calculate the critical exponent α .

10.2.1 Only power laws are scale-free

Normal distribution occurs when sample sizes or number of factors increase (central limit theorem), in contrast to power laws which are self-similar. Power law:

$$\begin{aligned} p(x) &= Ax^{-\alpha} \\ \frac{p(x)}{p(bx)} &= \frac{Ax^{-\alpha}}{A(bx)^{-\alpha}} = \frac{1}{(bx)^{-\alpha}} = \text{constant} \end{aligned}$$

Power laws are similar to exponential distributions, which are geometric (e.g. probability of scoring goal is $1/3$ while missing is $2/3$, then probability of scoring goal after possibly multiple tries is $(2/3)^{n-1}(1/3)$). Exponential distributions occur when waiting times between events occur with constant probability.

However, power laws are self-similar and scale-free, but exponentials are not self-similar and scale-free. Exponentials have the independent variable as the exponent rather than the base. For instance, $\frac{p(2)}{p(4)} = \frac{p(4)}{p(8)}$ for power laws, but $\frac{p(2)}{p(4)} \neq \frac{p(4)}{p(8)}$ for exponentials (and the differences get bigger over time, whereas it is constant for power laws). So power laws have longer tails as they take longer to decay than exponential distributions.

Only power laws are scale-free. Proof. If

$$\frac{p(x)}{p(bx)} = g(b)$$

then p is scale-free. For $x = 1$,

$$\begin{aligned}\frac{p(x)}{p(bx)} &= \frac{p(1)}{p(b)} = g(b) \\ p(bx) &= \frac{p(x)p(b)}{p(1)}\end{aligned}$$

Take derivative:

$$\begin{aligned}\frac{d}{db}p(bx) &= \frac{d}{db} \frac{p(x)p(b)}{p(1)} \\ x p'(bx) &= \frac{p(x)p'(b)}{p(1)}\end{aligned}$$

Set $b = 1$:

$$x p' = \frac{p'(1)}{p(1)} p(x) = a p(x)$$

which is a separable DE:

$$\begin{aligned}x \frac{dp}{dx} &= a p(x) \\ \int \frac{dp}{p} &= \int a \frac{dx}{x} \\ \ln(p) &= a \ln(x) + K\end{aligned}$$

Set $x = 1$ to get $K = \ln(p(1))$. Insert K back into equation and solve:

$$p(x) = p(1)x^a = Ax^{-\alpha}$$

10.2.2 Estimating power law parameters

Example. Plot number of words (y-axis) versus frequencies (i.e. how many times the word appears) (x-axis). Will be exponentially decreasing. Re-express plot as probability versus frequency, which is still exponential. Re-plot as log-probability versus log-frequency, which is linear and a power law, like dimensions. Estimate parameters to get $p(x) = Ax^{-D}$ where A is a parameter, x is

the number of words, and $p(x)$ is the probability that a word appears x times.

Sometimes averages don't exist, e.g. St. Petersburg game where winnings are 2^x , where x is the number of times to get T (tails) in coin tosses. Occasionally, there is a very big payoff, which raises the average but if there are no big payoffs, the average declines. So average continuously fluctuates but occasionally rises at rare events.

$$\begin{aligned} \text{Average} &= P(H)2^1 + P(TH)2^2 + P(TTH)2^3 + \dots \\ &= \frac{1}{2}2^1 + \left(\frac{1}{2}\right)^2 2^2 + \left(\frac{1}{2}\right)^3 2^3 + \dots \\ &= 1 + 1 + 1 \dots \\ &= \infty \end{aligned}$$

Because of long tails in $\alpha \leq 2$, this kind of power law has no average. If $\alpha > 2$ then tail declines fast enough for average (prove using calculus). If $\alpha \leq 3$ then no standard deviation (i.e. it is infinite). If $\alpha > 3$ then there is both average and standard deviation.

The PDF is $p(x) = Ax^{-\alpha}$ then CDF (defined as the probability being greater than x , so sloping downward and sometimes called the complementary CDF) by integration is $P(x) = \int_x^\infty p(x)dx = \frac{A}{\alpha-1}x^{-(\alpha-1)} = cx^{-(\alpha-1)}$. The CDF is equivalent to rank-frequency plots. Can convert word log-rank v word log-frequency CDF and this will be linear (and more linear than the log-log PDF plot). Many relationships in the world have linear CDFs.

Estimating α in $p(x) = Ax^{-\alpha}$ with MLE. Form $p(x|\alpha) = p(x_1|\alpha)p(x_2|\alpha)\dots p(x_n|\alpha)$ and choose α that maximizes the probability of the data $x_1, x_2, \dots, x_n$. Then estimator is (for continuous x ; discrete x estimator is different):

$$\hat{\alpha} = 1 + n \left(\sum_i^n \ln \frac{x_i}{\hat{x}_{min}} \right)^{-1}$$

where transient data $x < x_{min}$ is thrown out. To obtain $\hat{x}_{min}$, systematically try each x_{min} and then select best $\hat{\alpha}$ by minimizing distance (usually Kolmogorov-Smirnov) between $P_{model}(x)$ and $P_{data}(x)$:

$$p_{model}(x) = Ax^{-\hat{\alpha}} \rightarrow P_{model}(x)$$

Kolmogorov-Smirnov distance is $D = \max_{x \geq x_{min}} \|P_{model} - P_{data}\|$. Bootstrap to get error estimates (resample with replacement from data to get confidence intervals).

Goodness of fit. Generate synthetic data from model using $\hat{\alpha}, \hat{x}_{min}$, and estimate new α, x_{min} . Then calculate distances between the data. Repeat and

obtain multiple distances, and probability (p-value) $p(d_i \geq d)$. If p is large, then model is a good fit and evidence of a power law. Use goodness of fit to determine if data fits a power law or something related, like log-normal and/or exponential distributions. Can also use p-values, likelihood ratios.

10.2.3 Generating power laws

Power laws can arrived through stochastic processes and/or optimization. There are many mechanisms.

Network models. Rich-get-richer. Preferential attachment model. Assume an existing network graph. Attaching to random node with probability p, and attach to a node proportional to its in-links (directed edges ending at that node) with total probability 1-p. Fraction of nodes with k inlinks is $\approx ck^{-\alpha}$, $\alpha = 1 + \frac{1}{1-p}$ (derivation is mathematically involved and requires DEs; see academic literature). If p is large, then attachments are more random but if p is small then mostly preferential attachment and more network effects.

Combination of exponentials. Exponential growth $X(T) = e^{\mu T}$ combined with ages of those growths $p(T) = e^{-\nu T}$ lead to $p(X) = Ax^{-(\frac{\mu}{\nu}+1)}$ (a power law). Derivation is complex; see literature.

Multiplicative processes. Can lead to log-normal or power law relationships. For **log-normal**, start with:

$$x_t = \epsilon_t x_{t-1}$$

where ϵ is a random variable. Then,

$$\begin{aligned} x_t &= \epsilon_t \epsilon_{t-1} \dots \epsilon_1 x_0 \\ \ln(x_t) &= \ln(\epsilon_t \epsilon_{t-1} \dots \epsilon_1 x_0) \\ &= \ln(x_0) + \sum_{i=1}^t \ln(\epsilon_i) \end{aligned}$$

Since the second term is a sum of random variables (the $\ln(\epsilon_i)$), so it will converge to a normal distribution. As $t \rightarrow \infty$ or grows large, $\ln(x_0)$ becomes insignificant and $\ln(x_t)$ is normal, so x_t is log-normal. For **power law**, again start with:

$$x_t = \epsilon_t x_{t-1}$$

and impose lower threshold on x_t . This will force x_t to be larger, leading to a power law.

Optimization. In a **distribution network** like electrical grid or transportation routes. To minimize the number of stops and simultaneously minimize total network path distance, can have multiple configurations (e.g. hub-and-spoke, direct paths, hybrid of two). In **highly-optimized tolerance**, the best (most

robust to random disruptions) is a hybrid, which is fractal-like. In Information cost minimization, minimizing information cost c while maximizing information transmission h , means minimizing $\frac{c}{h}$. This explains word frequency. Optimization can be designed by humans but can also be from natural selection and energy minimization.

Phase transitions. Power law behavior seen at and very near transition points, e.g. fusion/freezing. Moreover, the phase transition exponents are the same for each universality class, which depend on the system's topological dimension and order parameters are the same or similar (the 'renormalization group'). From Quanta magazine (17 Sept 2020): 'At low temperatures, most atoms align, magnetizing the material. At high temperatures they grow disordered and the lattice demagnetizes. But at a critical transition point, islands of aligned atoms of all sizes coexist. Crucially, the ways in which certain quantities vary around this "critical point" appeared identical in the Ising model, in real magnets of varying materials, and even in unrelated systems such as a high-pressure transition where water becomes indistinguishable from steam. The discovery of this phenomenon, which theorists called universality, was as bizarre as finding that elephants and egrets move at precisely the same top speed.'

10.3 Scaling

10.3.1 Metabolic scaling

Kleiber's law. Log metabolic rate ($Y = \frac{\text{energy}}{\text{time}}$) versus log mass is linear for organisms. If surface area controlled metabolism, then since $S \propto L^2, M \propto V, V \propto L^3, L \propto M^{1/3}, S = kL^{2/3}$. However, empirically,

$$Y = kM^{3/4}$$

West-Brown-Enquist scaling theory. Explains Kleiber's law. Metabolic rate determined by properties of vascular network which is fractal as radius and lengths decrease at each level. Flow is optimized in the vascular network. Larger organisms have more levels but the capillary (where blood components enter into cells and remove waste) sizes are same, set by cell size which make up the vascular network. Networks fill the volume, and so each lower level volume must equal upper level volume. Metabolic rate is proportional to capillary volume flow. This explains the 3/4 power law.

Scaling of biological time. From Kleiber's law $Y = kM^{3/4}$, on a per mass basis:

$$\frac{Y}{M} = kM^{-1/4}$$

which fits number of mitochondria per cell, heart rate, and respiratory rate. Lifetime is reciprocal to the above, so they multiply to a constant

$$Time = kM^{1/4}$$

When multiplied, each organism has about 1.5 billion heartbeats, etc. (at least for vertebrates). This is about 10^{16} ATP molecules.

10.3.2 Urban scaling

Cities scale superlinearly (log quantity versus log population; usually with $\beta \approx 1.15$) for GDP, wages, crime, patents. Cities scale sublinearly (usually $\beta \approx 0.85$) for roads, electrical cables, gas stations. Cities scale linearly ($\beta \approx 1$) for housing, employment.

By analogy to vascular networks in metabolic scaling, urban infrastructure has roads, sidewalk widths, faucet sizes, etc.

10.4 Renormalization

Three steps

1. Coarse-grain
2. Rescale by substituting rescaled independent variables
3. Renormalize by multiplying the dependent variable by a renormalization factor

10.4.1 Coarse-graining

Renormalization is efficiently simplifying complexity. Merges microstates of the world into an 'equivalence class', irreversible mappings (onto but not 1-1), irreversible transformations (uncertainty of descriptions have declined), coarse-graining (where each pixel of an image 'takes a vote' or other rule and is mapped to one color). Data compression, clustering algorithm (vector space to clusters) or Gaussian mixture model.

Coarse-graining axiom:

$$H(A \text{ or } B \text{ or } C) = H(A \text{ or } (B \text{ or } C)) + (p_B + p_C)H(B \text{ or } C)$$

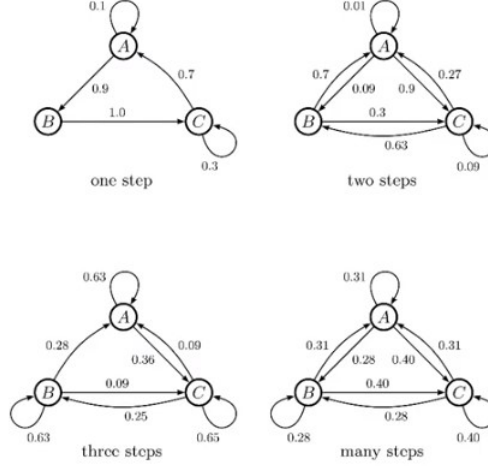
Renormalization group flow is how model parameters change when the data is coarse-grained.

Decimation is one way of coarse-graining, where one representative data point (instead of e.g. majority vote) is used from a period or a group of data points.

10.4.2 Markov chains

Coarse-grain complicated data to simpler data, then model will be renormalized. E.g. coarse-grain time series data to a lower resolution of time.

Do this with Markov chain, then at lower resolutions, the probabilities become fixed so there are fewer degrees of freedom, so are lower-dimensional. So theories change when coarse-graining data. Note: Markov chain has no memory.



With a 2-state (A and B) Markov chain which is symmetric, with $p(\text{remain}) = \epsilon$ and $p(\text{move}) = 1 - \epsilon$. In matrix notation:

$$\begin{pmatrix} p(A|A) & p(A|B) \\ p(B|A) & p(B|B) \end{pmatrix} \begin{pmatrix} p_A \\ p_B \end{pmatrix} = \begin{pmatrix} p(A|A)p(A) + p(A|B)p(B) \\ p(B|A)p(A) + p(B|B)p(B) \end{pmatrix}$$

where the first matrix is a stochastic matrix (columns add to 1).

$$T = \begin{pmatrix} \epsilon & 1 - \epsilon \\ 1 - \epsilon & \epsilon \end{pmatrix}$$

$$T^2 = \begin{pmatrix} \epsilon^2 + (1 - \epsilon)^2 & 2\epsilon(1 - \epsilon) \\ 2\epsilon(1 - \epsilon) & \epsilon^2 + (1 - \epsilon)^2 \end{pmatrix}$$

Since T is symmetric, T has two eigenvalues and eigenvectors, which allows diagonalization:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2\epsilon - 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2(2\epsilon - 1) \end{pmatrix}$$

where the left hand vector with α s is a probability matrix that sums to 1. Since the eigenvectors form a basis (and included in the basis-changing matrix often denoted P), the probability vector can be written as a linear combination of the basis eigenvectors: $p = \alpha_1 v_1 + \alpha_2 v_2$ where v's are the basis eigenvectors. (Note: any stochastic matrix will have as its largest eigenvalue being the absolute value of 1 (with irreducibility and ergodicity). Other eigenvalues are less than absolute value of unity.)

$$\hat{T} = \begin{pmatrix} 1 & 0 \\ 0 & 2\epsilon - 1 \end{pmatrix}$$

$$\hat{T}^n = \begin{pmatrix} 1 & 0 \\ 0 & (2\epsilon - 1)^n \end{pmatrix}^n$$

Since $2\epsilon - 1 < 1$, $(2\epsilon - 1)^n < 1$, and $(2\epsilon - 1)^n \rightarrow 0$ as $n \rightarrow \infty$. Then,

$$\lim_{n \rightarrow \infty} \hat{T}^n \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ 0 \end{pmatrix}$$

which is a fixed point for ending up always, regardless of starting p_A, p_B , at v_1 (the eigenvector for the largest eigenvalue): $p = \alpha_1 v_1 + \alpha_2 v_2 = \alpha_1 v_1 + 0 = \alpha_1 v_1$. Since $\alpha_1 + \alpha_2 = 1 \rightarrow \alpha_1 = 1$, so it maps every vector $vecp$ to v_1 (the stationary distribution).

Going in the reverse direction, if resolution is actually higher. Example with twice resolution. Solve $A^2 = T$. Set

$$A = \begin{pmatrix} a & 1-b \\ 1-b & a \end{pmatrix}$$

$$A^2 = \begin{pmatrix} a^2 + (1-b)(1-a) & a(1-b) + b(1-b) \\ a(1-a) + b(1-a) & (1-a)(1-b) + b^2 \end{pmatrix} = \begin{pmatrix} \epsilon & 1-\epsilon \\ 1-\epsilon & \epsilon \end{pmatrix} = T$$

For this to work, $a = b$. Then solve for a in terms of ϵ , from a quadratic equation. There are limits on ϵ s.t. it can't be complex. For values that are complex, $A^2 \neq T$ so T could not have come from coarse-graining a higher-resolution system, or it must have come from a higher-resolution system with more complexity, features, nodes, etc.

10.4.3 Cellular automata

Similar to Markov chain, has no memory. But cellular automata are more complicated than Markov chains as something far away horizontally will affect the outcome.

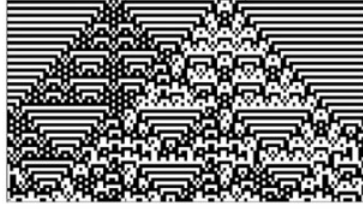
Cellular automata have 0 or 1 (white or black) on one horizontal line that determines deterministically what happens on the line below through triplets. Call this function f . There are $2^3 = 8$ triplets, and only 2 possibilities for each, so there are $2^8 = 256$ combinations of possible rules.

Each CA triplet affects the next period. So each quintuplet affects the line two periods into the future. When coarse-grain a CA by decimating one of the horizontal lines, then need a more complicated set of rules because it now requires quintuplets instead of triplets. However, a duet is affected by a quartet above and a sextet two periods before, which is just two triplets. So each duet (supercell) is affected by two supercells in the preceding level, and three supercells two periods before.

However, the supercells are no longer binary variables. Address this with a

projection operator P which maps the supercells $(x, y) \rightarrow 0$ or 1 , turning it into a binary variable once again. Then the rules are represented by function g instead of f , ultimately resulting in $f = PgP^{-1}$. So f and g both should have 256 rules. So f and g are in the same solution space, so each g will correspond to an f .

Rule 105, fine-grained level



Rule 150, coarse-grained level



So, $f^2 = Pg^2P^{-1}$ and if Rule x normalizes to Rule y normalizes to Rule z , then two normalizations of Rule x commutes with Rule z . Sometimes, there may be a fixed point, e.g. if Rule x normalizes to Rule y and vice versa. Exactly which rules normalize to which other rules and whether there are fixed points depends on the situation.

Rule 110 is Turing-complete as rules can encode logic gates (e.g. XOR) so encoding these into the horizontal line, the next line(s) will be the computed solution. However, not possible to find a projection (normalization or commutation) of Rule 110 that will speed up by coarse-graining, as the projections didn't keep all the data that is valuable. There is too much data that is lost.

Cannot implement a CA Rule and then coarse-grain the result (e.g. by decimation) then it becomes non-renormalizable (the f cannot relate simply to g). This is because some information in the previous level required for calculation is lost, and so no consistent rule can explain this. It would make the rules much more complicated and requires data from deeper in history ('non-locality').

10.4.4 Ising model

Describes atoms (nodes) in a lattice. Use a 2D lattice. Each node can be $+1$ or -1 . If a node is $+1$, then neighbor will tend to $+1$. So there should be larger 'patches' or regions of $+1$ or -1 . The probability (using Boltzmann distribution)

of node i σ_i being a particular value is:

$$p(\sigma_i) = \frac{e^{\beta \sum J_{ij} \sigma_i \sigma_j}}{Z(\beta)}$$

where J (the connection matrix) is 1 if i and j are neighbors and 0 otherwise. As β regulator (inverse temperature) gets larger, P gets larger and alignment between neighbors gets more important. σ are node values (+1 or -1). Z is a normalization factor ('partition function').

Isolate σ_A node, together with its neighbors connected through edges, $\sigma_1, \sigma_2, \sigma_3, \sigma_4$.

$$p(\sigma) = \frac{B e^{\beta(\sigma_A \sigma_1 + \sigma_A \sigma_2 + \sigma_A \sigma_3 + \sigma_A \sigma_4)}}{Z(\beta)}$$

where B is for nodes other than A :

$$B = e^{\beta \sum_{i,j \neq A} J_{ij} \sigma_i \sigma_j}$$

Coarse-grain by removing σ_A , by taking its average:

$$\begin{aligned} \sum_{\sigma_A=+1,-1} p(\sigma_i) &= (exp \beta(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4) + exp((- \beta)(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4))) \frac{B}{Z(\beta)} \\ &= 2 \cosh \beta(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4) \frac{B}{Z(\beta)} \\ &= exp(\ln \cosh \beta(\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4)) \frac{B}{Z'(\beta)} \end{aligned}$$

where the 2 is absorbed into $Z'(\beta)$. Note if there are 4 negative or 4 positive $\sigma_{1,2,3,4}$, then $\ln \cosh 4\beta$. If there are 3 negative and 1 positive or vice versa, $\ln \cosh 2\beta$. For 2 positive and 2 negative, $\ln \cosh 0 = \ln(1) = 0$. So,

$$\sum_{\sigma_A=+1,-1} p(\sigma_i) = exp(S_2(\sigma_1 \sigma_2 + \sigma_1 \sigma_3 + \sigma_1 \sigma_4 + \sigma_2 \sigma_3 + \sigma_2 \sigma_4 + \sigma_3 \sigma_4) + S_4 \sigma_1 \sigma_2 \sigma_3 \sigma_4) \frac{B}{Z'(\beta)}$$

where

$$\begin{aligned} S_2 &= \frac{1}{8} \ln \cosh 4\beta \\ S_4 &= S_2 - \frac{1}{2} \ln \cosh 2\beta = \frac{1}{8} \ln \cosh 4\beta - \frac{1}{2} \ln \cosh 2\beta \end{aligned}$$

where S_2 are new pairwise (diagonal) as well as 2-step vertical/horizontal interactions (generated by each node connecting to every other node around σ_A in the absence of σ_A), and S_4 are 'quartet' interactions connecting the nodes around σ_A . This is the rescaling. The new equation is more complicated than the original equation:

$$p'(\sigma_i) = \frac{e^{\beta \sum J'_{ij} \sigma_i \sigma_j + \sum K_{ijkl} \sigma_i \sigma_j \sigma_k \sigma_l}}{Z(\beta)}$$

Now extend this coarse-graining to all σ_A nodes in the infinite 2D lattice, so every other node in a lattice structure. There are now no original (1-step vertical or horizontal) connections. Instead, there are S_2, S_4 connections. Each diagonal S_2 is induced twice (one on each side), so $2S_2$. Each 2-step vertical/horizontal interaction S_2 is weaker than the diagonals:

$$2S_2 = \frac{1}{4} \ln \cosh 4\beta$$

$$S_2 = \frac{1}{8} \ln \cosh 4\beta$$

and S_4 remains as above. This is a renormalization. The original coupling strength was β but is now $2S_2 = \frac{1}{4} \ln \cosh 4\beta$, though there is still the quartet S_4 . Adjust the renormalization for simplicity: $\beta \rightarrow S = \frac{3}{8} \ln \cosh 4\beta$. This renormalization is an approximation, so repeated decimation leads to divergence.

Analogy to phase transitions. To visualize, graph S (a sigmoid curve) versus β , and superimpose with $\beta = \beta$ (a 45-degree) line. Then starting with any low β (below the intersection of the two graphs), decimation will lead to $\beta \rightarrow \frac{3}{8} \ln \cosh 4\beta$ (i.e. at a lower β). Repeating, β will end up at 0. So the system moves to a decoupled state, where each node is uncorrelated to its neighbors; they're just random.

Starting with any high β (above the intersection of the two graphs), repeated decimation leads β to infinity.

Between the 0 and infinity, there is an unstable fixed point (critical value) where $\beta = S$. It is scale-invariant (so a fractal) even after repeated decimation (and therefore repeated renormalizations). In an Ising model with +1 being black and -1 being white, the fixed point at 0 will give pure 'white' when zooming in or out. Fixed point at infinity will give 'grey'. At the critical fixed point, it will give patches of black and white when zooming in or out.

This is similar to phase transitions, as well as the tragedy of the commons where resource extracted is exactly equal to resource replenished.

10.4.5 Krohn-Rhodes theorem

Groups. With sequence 123, cycle perturbation operator leads to 312 and 231. This set A is closed and can't access 321, so introduce swap operator, which can covert these numbers respectively to 132, 321, 213. This new closed set B through cycle perturbations move in the opposite direction as the first closed set. A and B are equivalence classes and groups.

More finer-grained groups, e.g. with 123 and 132 as group 1, 312 and 321 in group 2, and 231 and 213 in group 3, have ambiguous group transformations.

For instance, a cycle perturbation on 1 will transform it to group 2 (if the element is 123) or group 3 (if the element is 132). This can be resolved at the more coarse-grained level which specifies the group element. The Jordan-Holder decomposition theorem says that there is a set of groups that are equivalent in composition and the number depends on the group, for reversible processes.

Renormalization group. From Quanta magazine (17 Sept 2020): ‘Wilson’s renormalization group describes a transformation from a theory of building blocks into a theory of structures. You start with a theory of small pieces, say the atoms in a billiard ball. Turn Wilson’s mathematical crank, and you get a related theory describing groups of those pieces — perhaps billiard ball molecules. As you keep cranking, you zoom out to increasingly larger groupings — clusters of billiard ball molecules, sectors of billiard balls, and so on. Eventually you’ll be able to calculate something interesting, such as the path of a whole billiard ball.’

Krohn-Rhodes. Krohn-Rhodes says can decompose for irreversible processes in semi-groups.

10.4.6 Quantum theories

As get closer and closer to object, such as electron (quantum electrodynamics), the scaling laws declining by $\propto \frac{1}{r^2}$ no longer work because there are new terms to address. There are fundamentally different forces at different scales so the equations are different at different scales.

From Quanta magazine (17 Sept 2020): ‘When a QED calculation leads to an infinite sum, cut it short. Stuff the part that wants to become infinite into a coefficient — a fixed number — in front of the sum. Replace that coefficient with a finite measurement from the lab. Finally, let the newly tamed sum go back to infinity... Mathematically cutting the sums short and shuffling the infinity around, physicists now understand, is the right way to do a calculation when your theory has a built-in minimum grid size. “The cutoff is absorbing our ignorance of what’s going on” at lower levels...’

10.4.7 Rate-distortion theory

Efficient v good representations. Basically utility function. If there is undiluted x information in the world and $\tilde{x}$ information wanted by the brain, then mutual information estimates the benefit $I(x, \tilde{x}) = H(\tilde{x}) - H(\tilde{x}|x)$. The eye maps $x \rightarrow \tilde{x}$ with $p(\tilde{x}|x)$. The less p is, the easier I is. Simultaneously, minimize penalty is $d(x, \tilde{x})$.

11 Number Theory

11.1 Numbers

$\mathcal{N}$: natural numbers: 1,2,3,4..... $\mathcal{N}_0$ includes 0. $\mathcal{Z}$: integers: minus infinity to positive infinity. $\mathcal{Q}$: rational numbers. $\mathcal{R}$: real numbers. $\mathcal{C}$: complex numbers.

Counting is a shift of $\mathcal{N}_0$ to $\mathcal{N}$, applying logic to set theory.

Addition is generalized counting. Subtraction arises as equivalence classes of Cartesian products of $\mathcal{N}$ and expanding numbers to $\mathcal{Z}$. Addition (together with subtraction) is an Abelian group, as the operations are associative, commutative, possess an identity (0), and inverses (negatives).

Multiplication is generalized addition. Division arises as equivalence classes of Cartesian products $\mathcal{Z}$ and expanding numbers to $\mathcal{Q}$. Multiplication (together with division) forms an Abelian group.

Fields. $\mathcal{Q}$ with addition (an Abelian group), multiplication (an Abelian group), and distributive laws form a field.

11.2 p-adics

$\mathcal{Q}$ converges to $\mathcal{R}$ or p-adics. p-adics are just re-based numbers with different metrics.

The size of a p-adic number is determined by the prevalence of p in its prime factorization. Numbers with more ps are smaller. For example, in the 3-adics, 486 is “small” because it has many 3s in its prime factorization ($486 = 2 \times 3 \times 3 \times 3 \times 3 \times 3$).

Another way to think about size is to think about which numbers are close to 0. In the p-adics, integers are closer together when they share a room at higher levels of the tower. The numbers 0 and 486 share a room up to the fifth level, whereas 0 and 6 share a room on only the first level — indicating that 0 is closer to 486 than to 6 (and thus 486 is smaller than 6).

Can be viewed as prime factorization towers. The p-adic towers accommodate fractions by expanding the tower belowground.

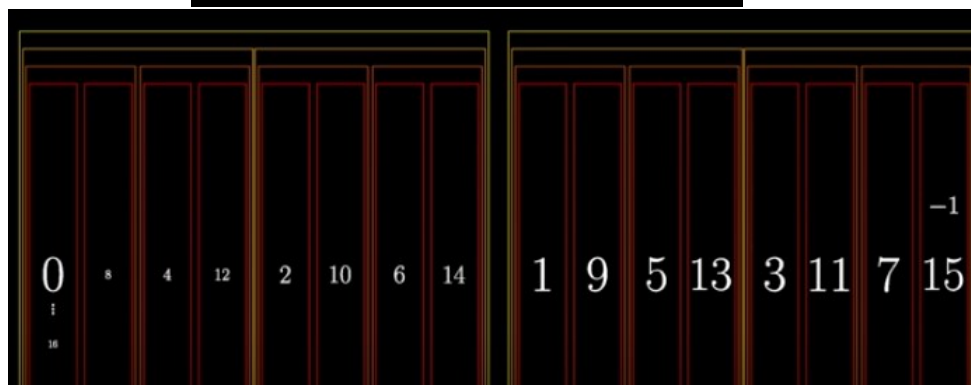
Infinite number behavior is counter-intuitive. Example. $\infty + 1 = \infty$, $\infty \times 2 = \infty$ but $\dots 3333 \times 2 = \dots 6666$ so they’re different infinities.

$1 + 2 + 4 + 8 + \dots = -1$ using the conventional formula. Use p(rime)-adics. Example with 3-adic. $\dots 22222 + 1 = \dots 00000$, so $\dots 222 = -1$. The -1 is related to the power from the distance metric when dividing $\dots 22222$.

Infinite sum formula works if in right base. Apply p-base to infinite sums: $1 + 2 + 4 + 8 + \dots$ in decimal is $1 \cdot 2^0 + 1 \cdot 2^1 + 2 \cdot 2^2 + \dots$ so $\dots 111111$ (coefficients) in base-2 or 2-adic. So $\dots 1111 + 1 = \dots 0000$ and $\dots 1111 = -1$ which is related to the power from the distance metric when dividing $\dots 1111$.

The p-adics take a different shape than the real number line: They form a fractal made of the infinitely nested rooms at the “top” of the p-adic tower. But this fractal has its own gaps. Mathematicians fill them by forming the “completion” of the p-adic rationals.

$$\begin{aligned}
 1 &= 1 \\
 3 &= 1 + 2 \\
 7 &= 1 + 2 + 2^2 \\
 15 &= 1 + 2 + 2^2 + 2^3 \\
 31 &= 1 + 2 + 2^2 + 2^3 + 2^4 \\
 63 &= 1 + 2 + 2^2 + 2^3 + 2^4 + 2^5 \\
 127 &= 1 + 2 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 \\
 255 &= 1 + 2 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 \\
 &\vdots
 \end{aligned}$$



p-adic numbers extend infinitely to left, real numbers extend infinitely to right. p-adics extend $\mathcal{Q}$ like $\mathcal{R}$ but in different directions. More 0's on left, smaller number in reals. More 0's on right, smaller number in reals. The number of 0's in right indicate how many times the base goes into the number.

Distance metric. Example with 2-adic system. $5 - 4$ goes into 2 zero times so $= \frac{1}{2^0} = 1$ but $8 - 4$ is $D = \frac{1}{2^2} = \frac{1}{4}$.

Ostrowski's theorem. Only 2 non-trivial distance metrics on $\mathcal{Q}$. Absolute val-

ues on reals and p-adic metric on p-adics. Truncated sums approach the infinite sums for both metrics.

11.3 Randomness and pseudo-randomness

Randomness. Human randomness v real randomness (e.g. flipping a coin). Both will have Bernoulli $p=0.5$, but the histogram of n -sequences will be flat for real randomness but not for human randomness. This is frequency stability.

Pseudo-randomness. Many kinds, all use seeds. von Neuman's algorithm: square the seed and output middle of the result and use this as the next seed and repeat (the middle squares method). However, it loops, with the size of the loop depending on the length of the seed. With a seed space much smaller, which shrinks the effective key space to the seed space. Share same seed and expand into random-looking one-time pad sequence.

There is always some randomness, even in a computer memory where heat, cosmic rays, radioactive elements can flip bits.

11.4 Primes

Prime number theorem. Number primes up to n is about $n \ln n$. The number of primes increases logarithmically ($\ln n$) so the density is the inverse, $\frac{1}{\ln n}$. Design of least resistance: logarithmic spiral, seen in galaxies and flowers.

Prime partitions. Every odd number is the sum of three (or more) primes. Goldbach conjecture: every even integer is the sum of two prime numbers.

Sieve of Eratosthenes to find primes. Start with 2, then eliminate all multiples of 2. Move to 3, eliminate all multiples of 3. For primes to 100, only need to do this to 11 as $11^2 > 100$. Inefficient because have to calculate lots of multiples. Can add starting primes, but this takes up enormous memory, and there is a time-space (space=storage) tradeoff.

Trial division (naive) algorithm. Try numbers iteratively up to square root of n .

Random primality test. Can divide by integers - if it divides, it's composite, and if not, then it increases conditional probability of being prime. Probability changes depending on number of digits to test and the number of integers to test. Can quantify confidence. But algorithm can be inefficient.

Fermat primality test. Look for rule that asymmetrically applies to primes. Number of steps doesn't scale with the digits unlike the naive algorithm. Use Fermat's little theorem with randomly generated a 's to test if the residue is 1 (or 0, depending on the form of FLT). However, due to pseudoprimes (shows false

positives), several a's should be tested. No false negatives (if it's composite, it's 100 percent composite). Even with repeats, more efficient algorithm.

Carmichael numbers. Some pseudoprimes will pass all Fermat tests but composite. There are infinite Carmichael numbers. But only small proportion of numbers are Carmichael numbers.

Other primality tests. Some use multiple stacked tests. AKS test is perfect and fast (though still slower than Fermat) - tests polynomial coefficients.

Prime factorization is difficult. No efficient algorithms for large numbers. Run-away effect as numbers get larger.

11.5 Divisibility

The number line is spaced by the divisor but shifted by the remainder. Dividing by negative number: -13 divided by 3 is -5 with remainder 2. Can have + or - remainders and shift them.

11.6 Binary

Each digit is 0 or 1 and 2 is the next digit over like 10 in decimal:

$$101101 = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2 + 1 = 45$$

Dividing by 2 (so whether the number is even or odd): the remainder is the last bit and the quotient is the rest:

$$101101 = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2 + 1 = 2(2^4 + 2^2 + 2) + 1$$

To represent 2^n in base-10 requires $n + 1$ bits. Example: 2^4 is 10000. 2^n is the smallest number that requires $n + 1$ bits, and anything else requires $< n + 1$ bits. Example: $2^4 - 1 = 2^3 + 2^2 + 2^1 + 2^0 = 1111$.

A positive integer n has b bits when $2^{b-1} \leq n \leq 2^b - 1$. Expressed differently, $b = \text{floor}(\log_2 n) + 1$.

Binary naturally fits physical computer processes, e.g. transistors and memory.

11.7 Hexadecimal

Base-16. Since 16 is a power of 2 ($2^4 = 16$), hexadecimal is more condensed than binary. Easier for people than binary to work with and to glance when looking at long strings of numbers. Uses 0-9 and A-F.

Can convert through decimal. To directly convert, break up into groups of 4 as each hexadecimal digit is represented by 4 binary digits.

11.8 Modular arithmetic and congruences

Notation: $a \bmod b$ = remainder or residue.

Two numbers a and b are congruent modulo m if they have the same remainder when divided by m :

$$a \equiv b \pmod{m}$$

$$a - b \equiv 0 \pmod{m}$$

a is congruent modulo m to its remainder:

$$a \equiv r \pmod{m}$$

$$a \equiv a + k \times m \pmod{m}$$

Congruence is preserved under addition and multiplication:

$$a + c \equiv b + c \pmod{m}$$

$$ac \equiv bc \pmod{m}$$

$$c \equiv d \pmod{m} \Rightarrow a + c \equiv b + d \pmod{m}$$

Example 1. $14 + 41 + 20 + 13 + 29 \equiv 2 + 1 + 0 + 1 + 1 \equiv 5 \equiv 1 \pmod{4}$, so the remainder is 1.

Example 2. $17(12 \times 19 + 5) - 23 \equiv 2(12 \times 19 + 5) - 2 \equiv 2(0 \times 1 + 2) - 2 \pmod{3}$. Shift remainders 0, 1, 2 to -1, 0, 1: $-1(0 \times 1 - 1) + 1 \equiv 2 \pmod{3}$.

Example 3. The last two digits of 99^{99} is the remainder after dividing by 100. Since $99 \equiv -1 \pmod{100}$, $99^{99} \equiv (-1)^{99} \equiv -1 \equiv 99 \pmod{100}$.

Example 4. To check whether 3457 is divisible by 3, split the number: $3 \times 10^4 + 4 \times 10^3 + 5 \times 10 + 7 = 3 + 4 + 5 + 7 = 19 \equiv 1 \pmod{3}$, so not divisible by 3. Note $10 \equiv 1 \pmod{3} \Rightarrow 10^k \equiv 1^k \pmod{3}$.

11.9 Euclidean algorithm

11.9.1 Greatest common divisors

Naive algorithm using modulo of lower of two numbers and look for 0 remainder, if not then try a number 1 lower. Inefficient.

Euclid's lemma. d divides a and b iff d divides $a-b$ and b (note: $a > b$).

Build more efficient algorithms to find gcd with Euclid's lemma. Example $\gcd(6, 22)$. $22 \bmod 6 = 4$ so 6 doesn't divide 22. So recurse the gcd problem by looking for $\gcd(6, 4)$ since this will divide 6 and 22: $6 \bmod 4 = 2$ and $4 \bmod 2 = 0$. Then gcd candidates drop by factor of 2. Lemma: $a \bmod b \leq a/2$. Can prove for case where $b > a/2$ and $b < a/2$ where $b < a$. So: $\log_2 a + \log_2 b$ max iterations.

11.9.2 Least common multiple

Reduces computational complexity when adding fractions.

Lemma: $lcm(a, b)gcd(a, b) = a \times b$. Intuitive. Prove by contradiction. Let $d = gcd(a, b)$ and $a = dp$ and $b = dq$. Then:

$$m = \frac{ab}{d} = \frac{dpdq}{d} = pdq = pb$$

$$m = \frac{ab}{d} = \frac{dpdq}{d} = dpq = aq$$

So m is a multiple of a and b . If there were a smaller multiple of a and b $\bar{m} < m$, then contradiction as $d \neq gcd(a, b)$:

$$\bar{d} = \frac{ab}{\bar{m}} > d$$

11.9.3 Extended Euclidean algorithm

To check whether $d = gcd(a, b)$, it is enough if (i) d divides a and b (so $d \leq gcd(a, b)$), and (ii) $d = ax + by$ (so $d \geq gcd(a, b)$ - naturally, gcd should divide the linear combination of a and b).

So look for q and p in Bezout's identity:

$$aq + bp = gcd(a, b)$$

Extended Euclid algorithm. Use Euclidean algorithm which recurses $gcd(a, b) = gcd(b, a \bmod b)$.

$$d = gcd(b, a \bmod b) = bp + (a \bmod b)q$$

Which replaces a in Bezout's identity with the remainder. Then:

$$d = bp + \left(a - \mathcal{Z} \left(\frac{a}{b}\right) b\right) q$$

$$d = aq + b \left(p - \mathcal{Z} \left(\frac{a}{b}\right) q\right)$$

Recurse the above. Let $x=q$ and y be the bracketed term of Bezout's identity:

$$ax + by = gcd(a, b)$$

Intuitively, the algorithm uses recursion within the Euclidean algorithm to find the gcd and then recursively backs out solutions to Bezout's identity.

This algorithm is also called the pulverizer.

11.9.4 Diophantine equations

Linear diophantine equations can have no solutions, 1 or more finite number of solutions, or infinitely many solutions.

Theorem. A linear diophantine equation $ax + by = gcd(a, b)$ has integer solutions. Dividing the left and right sides gives no remainders.

To solve a linear diophantine equation $ax + by = c$, find $gcd(a, b)$ first. For $10x + 6y = 14$:

$$gcd(6, 10) = 2 = 10(-1) + 6(2)$$

$$14 = 2(7) = 10(-1)(7) + 6(2)(7)$$

Euclid's lemma. If n divides ab and $gcd(a, n) = 1$ then n divides b .

All solutions of a linear diophantine equation. Let $gcd(a, b) = d$, $a = dp$, $b = dq$. If (x_0, y_0) is a solution, i.e.

$$ax_0 + by_0 = c$$

Then all solutions are:

$$a(x_0 + tq) + b(y_0 - tp) = c$$

Note:

$$a(x_0 + tq) + b(y_0 - tp) = c$$

$$ax + by + aqt - btp = ax + by + (dp)qt - (dq)tp = ax + by + dpqt^2 - dpqt^2 = ax + by$$

Example. For $3x + 5y = 22$, $x_0 = 9$, $y_0 = -1$, $a = 3$, $b = 5$, $d = gcd(a, b) = 1$. Since $a=dp$, $b=dq$, $p=3$, $q=5$:

$$x = x_0 + tq = 9 + 5t$$

$$y = y_0 - tp = -1 - 3t$$

If want conditions like $x \geq 0$ and $y \leq 0$, then solve inequality system.

Note: if $gcd(a, b) = 1 = ax + by$, then $1 \equiv ax \pmod{b}$ as b divides by completely - with $(\text{mod } a, \text{mod } b) = (0, 1)$. Further, $1 \equiv by \pmod{a}$ - with $(\text{mod } a, \text{mod } b) = (1, 0)$.

11.9.5 Modular division and inverses

Modular division is not possible in some cases. Example. $3x \equiv 1 \pmod{6}$ isn't possible because 6 divides multiples of 3 with remainder 0 or 3 only.

Can't arbitrary cancel terms in congruences. Example.

$$8 \cdot 2 \equiv 3 \cdot 2 \pmod{10}$$

Can't just cancel 2:

$$8 \not\equiv 3 \pmod{10}$$

To cancel, it must be that 10 and 2 are coprime.

Need a modular multiplicative inverse $\bar{a}$ such that $a \times \bar{a} \equiv 1 \pmod{n}$. Occurs iff $\gcd(a, n) = 1$ (they are coprime). Proof. Set up linear diophantine $sk + tn = 1$ and just let k' be s . Then:

$$sk + tn = 1$$

$$sk + tn \equiv 1 \pmod{n}$$

$$sk + t(0) \equiv 1 \pmod{n}$$

$$sk \equiv 1 \pmod{n}$$

So s is an inverse of k . Let $s=k'$. Now can cancel k in $ak \equiv bk \pmod{n}$ iff $\gcd(n, k) = 1$.

If a has a multiplicative inverse, then it is unique. If x and y are both multiplicative inverses of a :

$$x = x(ay) = (xa)y = y$$

Modular algebra example. $\gcd(9, 2) = 1$ so can compute $2x \equiv 7 \pmod{9}$. Extended Euclidean algorithm to solve the linear diophantine: $9(1) + 2(-4) = 1$ (alternatively, $2x \equiv 1 \Rightarrow x = -4$) so $-4 \equiv 5 \pmod{9}$ is the inverse of $2 \pmod{9}$. So $2x \cdot 5 \equiv 7 \cdot 5 \pmod{9} \Rightarrow x \equiv 35 \equiv 8 \pmod{9}$.

11.10 Integer factorization

Fundamental theorem of arithmetic. Every integer has a unique prime factorization.

Two numbers are coprime if $\gcd(a, b) = 1$. No known efficient algorithms for prime factorization. For gcd use Euclid's algorithm.

GCD and LCM can be calculated using prime factorizations, but prime factorizations are hard. So use Euclid's algorithm: $\text{lcm}(a, b)\gcd(a, b) = ab$.

11.11 Chinese remainder theorem

Lemma. If n_1 and n_2 have the same remainders modulo b and $a|b$, then n_1 and n_2 have the same remainders modulo a . Example. 27 and 31 have the same remainder (3) modulo 4. Since 2 divides 4, 27 and 31 have remainder (1) modulo 2.

If two numbers have $\gcd(a, b) > 1$, their remainders modulo a and b are dependent. Example with $a=6$, $b=4$. If n is odd, then $n \pmod{4}$ can't be determined by $n \pmod{6}$. $n = 1$, $n \equiv 1 \pmod{6}$ and $n \equiv 1 \pmod{4}$. If $n = 7$, $n \equiv 1 \pmod{6}$ and

$n \equiv 3 \pmod{4}$. So remainder mod 6 doesn't uniquely identify remainder mod 4. But if n is even, then remainders for both a and b are 0, 2, or 4.

If two numbers have $\gcd(a, b) = 1$, their remainders are independent.

Chinese remainder theorem. If a and b are coprimes then the system $n \equiv r_a \pmod{a}$ and $n \equiv r_b \pmod{b}$ has a solution for n as well as added multiples of ab (so $n_1 \equiv n_2 \pmod{ab}$ for $n_1, n_2 \in n$).

Chinese remainder theorem proof. Since $\gcd(a, b) = 1 = ax + by$ and $ax \equiv 1 \pmod{b}$ and $by \equiv 1 \pmod{a}$, there are remainders $(1, 0)$ and $(0, 1)$. So:

$$(r_a, r_b) = r_a(1, 0) + r_b(0, 1)$$

$$n = r_a by + r_b ax \equiv r_a by \equiv r_a \pmod{a}$$

$$n = r_a by + r_b ax \equiv r_b ax \equiv r_b \pmod{b}$$

If n_1 and n_2 correspond to the same remainder pair (r_a, r_b) then $n_1 \equiv n_2 \pmod{ab}$. Then:

$$n_1 \equiv r_a \equiv n_2 \pmod{a}$$

$$n_1 \equiv n_2 \pmod{a} \Rightarrow a | (n_1 - n_2)$$

Similarly for $b | (n_1 - n_2)$. a and b are coprime so both divide $n_1 - n_2$ so $ab | (n_1 - n_2)$.

Corollary 1. Each remainder modulo ab uniquely corresponds to pair (r_a, r_b) .

Corollary 2. If $r_a = r_b$, then $r \pmod{ab}$. Example: if $x \equiv 2 \pmod{5}$ and $x \equiv 2 \pmod{7}$, then $x \equiv 2 \pmod{35}$.

Intuition for Chinese remainder theorem. If one solution to $x \equiv a \pmod{p}$ is found, then can add p and still have a solution. Consider the system of congruences $x \equiv 5 \pmod{7}$, $x \equiv 4 \pmod{11}$, and $x \equiv 4 \pmod{8}$. The first congruence has solution $x = 5$. But (since it's a congruence mod 7), it will also have solutions $x = 12, 19, 26, 33$, etc. Look among these for a solution to the second congruence: we see $x = 26$ solves the first and second congruences. To keep this as a solution, we need to add multiples of $7 \times 11 = 77$: 7, because adding a multiple of 7 won't undo our solution to the first congruence; and 11, for the same reason (this is $n_1 \equiv n_2 \pmod{a}$). So solutions to the first two congruences are $x = 26, 103, 180$, and so on. We keep going until we find a solution to the third congruence: $x = 180$ solves the third congruence and, because of the way we've constructed it, it will also solve the other two.

More intuition for the Chinese remainder theorem. Two numbers a and b have different starting points r_a and r_b and cycle with period ab : $r_a + a(n_a)$ and $r_b + b(n_b)$. If $\gcd(a, b) \neq 1$ then $a-b$ is restricted to being a multiple of the gcd by Euclid's lemma. So non-coprime integers may not meet at the same n . Coprime integers aren't restricted so will meet at the same n in a cycle.

11.12 Modular exponentiation

Calculate $b^e \pmod{m}$.

Naive algorithm ($O(e)$ time) example with $c \leftarrow 7^4 \pmod{11}$:

$$c \leftarrow 1$$

$$c \leftarrow (cb) \equiv (1 \times 7) \equiv 7 \pmod{11} = 7$$

$$c \leftarrow (cb) \equiv (7 \times 7) \equiv 7 \pmod{11} = 5$$

$$c \leftarrow (cb) \equiv (5 \times 7) \equiv 7 \pmod{11} = 2$$

$$c \leftarrow (cb) \equiv (2 \times 7) \equiv 7 \pmod{11} = 3$$

Works because the remainder of a product is the product of the remainders.

Fast modular exponentiation ($O(k)$ time) example with $c \leftarrow 7^{2^7=128} \pmod{11}$.

$$7^2 \equiv 7 \times 7 \equiv 49 \pmod{11} = 5$$

$$7^4 \equiv 7^2 \times 7^2 \equiv 5 \times 5 \equiv 25 \pmod{11} = 3$$

$$7^8 \equiv 7^4 \times 7^4 \equiv 3 \times 3 \equiv 9 \pmod{11} = 9$$

...

$$7^{128} \equiv 7^{64} \times 7^{64} \equiv 3 \times 3 \equiv 9 \pmod{11} = 9$$

Time complexity at most $O(\log_2 e = \log_2 2^k = k)$. If e is not a power of 2, use a binary decomposition of k : $b^{13} = b^{8+4+1} = b^8 b^4 b^1 = b^{2^3} b^{2^2} b^{2^0}$ and use remainder of a product is the product of the remainders:

$$7^{2^0} \pmod{11} = 7$$

$$7^{2^2} \pmod{11} = 3$$

$$7^{2^3} \pmod{11} = 9$$

So $c = 7 \times 3 \times 9 = 189$.

11.13 Modular arithmetic on quadratics

Above addresses linear residues. Modular arithmetic goes further. Example. Can define quadratic residues, Legendre and Jacobi symbols, Euler's criterion, and finally quadratic reciprocity for quadratic polynomial residues.

11.13.1 Fermat's Little Theorem

For prime p that doesn't divide integer a , $a^p \equiv a \pmod{p}$. Equivalently, $a^{p-1} \equiv 1 \pmod{p}$.

Intuition. Analogize with p -length string or necklace and a colors - there are $a^p - a$ multi-colored strings or necklaces (a tuple minus the uni-colored objects) which form $\binom{k+n-1}{n-1} - n = \binom{p+a-1}{a-1} - n$.

Proof. Consider all remainders up to $p - 1$. Multiplying each by a is invertible since p doesn't divide a (so coprime), so the new remainder is non-zero. This cycles in a graph when repeatedly multiplying r by a to get $a^l \equiv 1 \pmod{p}$ where l is cycle length. There are c possible cycles, so $cl = p - 1$ (on a directed cyclic graph, these are different rotation combinations). Then $a^{p-1} = a^{cl} = (a^l)^c \equiv 1^c \equiv 1 \pmod{p}$.

Optimized fast modular exponentiation. If p doesn't divide a , then Fermat's little theorem: $a^{p-1} \equiv 1 \pmod{p}$. This means $a^n \equiv a^{n \bmod (p-1)} \pmod{p}$. If p divides a , $a^n \equiv 0 \equiv a^{n \bmod (p-1)} \pmod{p}$. So break up calculations using powers of $p - 1$.

11.14 Euler's Totient function

$\phi(n)$ counts integers between 0 and $n - 1$ which are coprime with n . $\phi(1) = 1$, $\phi(2) = 1$, $\phi(3) = 2$, $\phi(10) = 4$. If p is prime, $\phi(p) = p - 1$.

If p and q are primes (not just coprime), $\phi(pq) = \phi(p)\phi(q) = (p - 1)(q - 1)$. Proof. Consider all pq remainders r of modulo pq . By the Chinese remainder theorem, each r corresponds to a unique pair (r_p, r_q) . Since $\gcd(pq, r) = 1$, eliminate $r_p = 0$ and/or $r_q = 0$ (otherwise, p and/or q can factor the number and $r = 0$). Since the remainders of p and q can go up to $p - 1$ and $q - 1$ respectively, there are $(p - 1)(q - 1)$ remainders.

11.15 Euler's theorem

Generalization of Fermat's little theorem. If a is coprime with n , $a^{\phi(n)} \equiv 1 \pmod{n}$. When n is prime, $\phi(n) = p - 1$ which is Fermat's little theorem.

Proof similar for Fermat's little theorem. Consider all $\phi(n)$ remainders modulo n which are coprime with n . Multiplying a is invertible because $\gcd(a, n) = 1$, so new remainder is coprime with n . Again in a directed cyclic graph, $a^l \equiv 1 \pmod{n}$. Since $cl = \phi(n) = p - 1$, $a^{\phi(n)} = a^{cl} = (a^l)^c \equiv 1^c \equiv 1 \pmod{n}$.

11.16 Cryptography

11.16.1 Early ciphers

Caesar cipher just shifts alphabet. Broken by frequency analysis - look at shift of frequency of letters from its usual distribution. If the cipher shifts the alphabet by 3, encryption is $C = (M + 3) \bmod 26$ and decryption is $M = (C - 3) \bmod 26$.

Polyalphabetic ciphers use e.g. a word that is turned into numbers (e.g. a=1, b=2) and the sequence of numbers is repeatedly applied to the message, to shift letters by different numbers. Broken by looking for the length of the cipher by looking for repeated frequency distributions. Then break each frequency distribution and find the numbers.

11.16.2 One-time pad

Use a random number or letter for each letter in the messages. Now there are no information leaks as the frequency of the letters is random. The number of combinations is 26^l where l is the message length.

One-time pad implementation. Send message m with shared key k by sending encrypted message $c = m \oplus k$. XOR is:

$$x \oplus y = (x + y) \bmod 2$$

If $a = (a_1, a_2 \dots a_n)$ and $b = (b_1, b_2 \dots b_n)$ then:

$$a \oplus b = (a_1 + b_1 \bmod 2, a_2 + b_2 \bmod 2 \dots a_n + b_n \bmod 2)$$

Since $x \oplus x = 0$ decrypt with:

$$c \oplus k = m \oplus k \oplus k = a$$

Two-time pad.

$$c_1 = m_1 \oplus k$$

$$c_2 = m_2 \oplus k$$

Then:

$$\begin{aligned} c_1 \oplus c_2 &= m_1 \oplus k \oplus m_2 \oplus k \\ &= m_1 \oplus k \oplus k \oplus m_2 = m_1 \oplus m_2 \end{aligned}$$

And $m_1 \oplus m_2$ is partially decipherable. If know i -th character of m_1 , then can decipher i -th character of m_2 .

Enigma. Automated one-time pad. Big key space. Machines are deterministic and not truly random, so make huge loops of the key space. Enigma key settings were changed and distributed daily. But the Enigma settings weren't changed that much each time, so the effective key space was smaller and could be scanned with the help of crypts (words known to be in the code). Enigma letters did not route back to themselves, so the frequency distribution was uneven.

11.16.3 Diffie-Hellman key exchange

Encryption. A and B agree on a generator and prime number, e.g. 3 mod 17. A randomly generates a number, e.g. 15, applies it: $3^{15} \bmod 17 \equiv 6$. Likewise, B also applies own randomly generated number e.g. 13: $3^{13} \bmod 17 \equiv 12$ which is sent to A.

Decryption. A decrypts with $12^{15} \bmod 17 \equiv 10$. B decrypts with $6^{13} \bmod 17 \equiv 10$. They're the same calculation:

$$12^{15} = (3^{13} \bmod 17)^{15} \bmod 17 \equiv 6^{13} = (3^{15} \bmod 17)^{13} \bmod 17$$

11.16.4 RSA cipher

One of the earliest forms of public-key cryptography.

(1) Key generation. Receiver generates two prime numbers p and q , with $n = pq$, as well as exponent e which is coprime with $(p-1)(q-1)$. Receiver then makes the public keys (n, e) available.

(2) Encryption. Sender generates message m converted to bits and then a single integer. Encrypt: $c = m^e \bmod n$ (can use fast modular exponentiation). Send c to receiver.

(3) Decryption. Receiver decodes using d generated from the private key (p, q) which is the inverse of e : $c^d = (m^e)^d = m^{ed} \equiv m \bmod n$, generating m (Note: $m^{ed} \neq m^1$ because the modular inverse isn't a 'normal' inverse).

Generating e and d . Euler's theorem:

$$m^{\phi(n)} \equiv 1 \bmod n$$

If $\gcd(m, n) = 1$ (almost certain). Then:

$$m^{k\phi(n)} \equiv 1^k = 1 \bmod n$$

$$m \cdot m^{k\phi(n)} \equiv m \bmod n$$

$$m^{k\phi(n)+1} \equiv m \bmod n$$

Which should be equivalent to:

$$m^{ed} \equiv m \bmod n$$

So:

$$ed = k\phi(n) + 1$$

$$d = \frac{k\phi(n) + 1}{e}$$

Since $\phi(n)$ is extremely hard to determine without p and q , it is effectively a 'one-way' function. So hard to determine d without p and q .

Proof of d and e . $m^{ed} \equiv m \pmod{n}$, where $n = pq$ and p and q are coprime. By the Chinese remainder theorem:

$$m^{ed} \equiv m \pmod{p}$$

$$m^{ed} \equiv m \pmod{q}$$

By Fermat's little theorem, because p and q are coprime:

$$m^k \equiv m^{k \bmod (p-1)} \pmod{p}$$

Let $k = 1$, then:

$$m \equiv m^{\bmod (p-1)} \pmod{p}$$

This should be equivalent to:

$$m^{ed} \equiv m \pmod{n}$$

So require:

$$ed \equiv 1 \pmod{(p-1)}$$

$$ed \equiv 1 \pmod{(q-1)}$$

Which is fulfilled if:

$$ed \equiv 1 \pmod{(p-1)(q-1)}$$

If e is coprime with $(p-1)(q-1)$, then use Extended Euclidean algorithm to generate the modular multiplicative inverse, d .

11.16.5 Attacking RSA encryption

(1) If the message is very small, may be deciphered. Solution: pad the message with extra, random bits.

(2) If the private key is too small, can be deciphered. Solution: use a big private key.

(3) If difference between p and q are small, then easier to find these numbers. Solution: regenerate if p and q too close to each other.

(4) Insufficient random seeds. If there are two or more systems that are generating similar seeds, similar p and q . So take their public keys and $\gcd(n_1, n_2) = p$ and break the codes. Worked in about 0.4 percent of cases when tested. Solution: make sure seed is random.

(5) Hastad's broadcast attack. Broadcasting the same message repeatedly can make it easier to decipher. Solution: add different extra, random bits.

(6) If interact with server and measure how long it takes server to decipher, then can guess d better.

(7) Send intercepted encrypted message but change it and send it to server to get error that partially reveals the message.

(8) If understand how much power is consumed for deciphered, then can guess d better.

12 Real Analysis

12.1 Inequalities

12.1.1 Cauchy-Schwarz inequality

Generalization of the triangle inequality, which can be used to derive the triangle inequality and other relations. Multiplying vector lengths is larger than multiplying their components separately and then adding them.

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

Can be viewed as

$$\left| \left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle \right| \leq 1$$

The length of the projection of the unit vector is ≤ 1 . So projecting a vector can't make it longer.

Proof of generalization of triangle inequality:

$$\begin{aligned} \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} &\leq \sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2} \\ (x_1^2 - 2x_1y_1 + y_1^2) + (x_2^2 - 2x_2y_2 + y_2^2) &\leq (x_1^2 + x_2^2) + 2\sqrt{x_1^2 + x_2^2}\sqrt{y_1^2 + y_2^2} + (y_1^2 + y_2^2) \\ -2x_1y_1 - 2x_2y_2 &\leq 2\sqrt{x_1^2 + x_2^2}\sqrt{y_1^2 + y_2^2} \\ (x_1y_1 + x_2y_2)^2 &\leq (x_1^2 + x_2^2)(y_1^2 + y_2^2) \\ (x_1y_1 + x_2y_2) &\leq \sqrt{x_1^2 + x_2^2}\sqrt{y_1^2 + y_2^2} \\ |\langle x, y \rangle| &\leq \|x\| \|y\| \end{aligned}$$

12.1.2 Triangle inequality

Triangle inequality. One triangle side is longer than the sum of the other two sides. Provides (1) generalized notion that a straight line is the shortest distance between two points, and (2) transitivity to distances. Proof:

$$|a + b| \leq |a| + |b|$$

$$|a + b|^2 \leq (|a| + |b|)^2$$

$$|a + b|^2 \leq |a|^2 + 2|a||b| + |b|^2$$

$$|a + b|^2 = (a + b)^2 = a^2 + 2ab + b^2 \leq |a|^2 + 2|a||b| + |b|^2$$

The Cauchy-Schwarz inequality $|\langle x, y \rangle| \leq \|x\|\|y\|$ is more restrictive than the triangle inequality as it requires an inner product space. Generalizes multiplication and provides angle transitivity. So if there is a Cauchy-Schwarz inequality, the triangle inequality follows:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2 \end{aligned}$$

Reverse triangle inequality. One triangle side is longer than the difference of the other two sides.

$$||a| - |b|| \leq |a - b|$$

Can be written and proven in alternate form:

$$||a| - |b|| \leq |a + b|$$

$$(1) |a| = |a + b - b| = |(a + b) + (-b)| \leq |a + b| + |-b| = |a + b| + |b|$$

$$|a| - |b| \leq |a + b|$$

$$(2) |b| = |b + a - a| = |(a + b) + (-a)| \leq |a + b| + |-a| = |a + b| + |a|$$

$$|b| - |a| \leq |a + b|$$

Square of the sum of the products $\leq$ the product of the sum of the squares. Can generalize this to settings outside of inner products: <https://web.williams.edu/Mathematics/lg5/Cauchy-Schwarz.pdf>

12.1.3 Bernoulli's inequality

$$(1 + a)^n \geq 1 + na$$

For $a \geq -1$ and $n \in \mathbb{Z}^+$. Base case of $a = -1$:

$$(1 + a)^1 = 1 + a = 1 + (1)a$$

Induction:

$$(1 + a)^k \geq 1 + ka$$

$$(1 + a)^{k+1} \geq 1 + (k + 1)a$$

$$(1 + a)^k(1 + a) \geq 1 + a + ka + ka^2 = 1 + (k + 1)a + ka^2 \geq 1 + (k + 1)a$$

Because $a \geq -1 \rightarrow a^2 \geq 0$.

12.2 Metric spaces

12.2.1 Metric spaces and topology

With a metric space, can define an open ball because the radius is defined with the distance metric. Can define topology based on open sets using open balls (note: a topology is a set of open sets): $T = \{O \subset E | \forall x \in O, \exists r > 0, B_r(x) \subset O\}$. (i) The empty set and the entire set is an element of T, (ii) any union of elements is an element of T, and (iii) any finite intersection of elements is in T. A metric space will induce a Hausdorff space because the limit is unique in a metric space. A Hausdorff may or may not be metrizable. Example of non-metrizable topology is the trivial topology.

12.2.2 Metrization

A topological space is metrizable if and only if it is a paracompact and locally metrizable Hausdorff space.

Urysohn metrization theorem. Every Hausdorff second-countable regular space is metrizable. Regular spaces mean points and closed subsets can be separated by neighborhoods.

12.2.3 Isometry

Also called congruence or congruent transformation. A distance-preserving transformation between metric spaces. If X and Y are metric spaces and d_X, d_Y are distances, $f : X \rightarrow Y$ is an isometry if

$$d_X(a, b) = d_Y(f(a), f(b))$$

A global isometry, isometric isomorphism, or congruence mapping is a bijective isometry. Any translation, reflection, or rotation is an isometric isomorphism in Euclidean spaces.

12.2.4 Norms

L^1 norm 'ball' is a diamond, L^2 norm 'ball' is a disk, and L^∞ norm 'ball' is a square. For the unit circle in each of these metrics: $1 = (|x|^n + |y|^n)^{1/n}$, so $1 = 2|x|^n$ for 'corner' points' which is $(\frac{1}{2})^{1/n} = x$.

Strength of a norm. N_a is stronger than N_b if $N_b \leq C_a N_a(x)$ so that N_a balls fit into N_b up to a homothetic transformation. A stronger norm induces a stronger topology, which impacts convergence.

Norm equivalence. N_a and N_b are equivalent if they can be included in each other: $C_1 N_a(x) \leq N_b(x) \leq C_2 N_a(x)$. All finite-dimensional vector space norms are equivalent. So there is only one topology, the usual topology (i.e. all open sets).

12.2.5 Urysohn lemma

A topological space is normal iff any two disjoint closed subsets can be separated by a continuous function. Formally, a topological space X is normal iff for any two non-empty closed disjoint subsets A and B of X , there exists a continuous map $f : X \rightarrow [0, 1]$ s.t. $f(A) = 0$ and $f(B) = 1$.

12.2.6 Tietze extension theorem

Generalizes Urysohn's lemma. Continuous functions on a closed subset of a normal topological space can be extended to the entire space, preserving boundedness if necessary.

Example. Consider the $\mathcal{R}$ -subspace $x \in [1, 2]$ and $f : x \rightarrow \frac{1}{x}$. Then f is continuous and $[1, 2]$ is closed and $\mathcal{R}$ is normal so by Tietze extension, f can be extended to g on $\mathcal{R}$ s.t. g is continuous and $f(x) = g(x) \forall x \in [1, 2]$. An example extension is:

$$g(x) = \begin{cases} f(x), & x \in [1, 2] \\ 1, & x \in (-\infty, 1) \\ 1/2, & x \in (2, \infty) \end{cases}$$

12.3 Sequences

12.3.1 Limits

Limit of a sequence. $a_n \rightarrow L$ where L is a limit, $\forall \epsilon > 0, \exists N \in \mathbb{Z}^+$ s.t. $\forall n > N, |a_n - L| < \epsilon$.

The limit of a sequence is unique. If there were two limits, it wouldn't be a limit. Proof by contradiction. Suppose $a \neq b$ and $a_n \rightarrow a$ and $a_n \rightarrow b$. Then

$\epsilon = |a - b| > 0$. Since $a_n \rightarrow a$, $\exists N_1 \in \mathcal{Z}^+$ s.t. $\forall n > N_1$ $|a_n - a| < \frac{\epsilon}{2}$. Further, $a_n \rightarrow b$, $\exists N_2 \in \mathcal{Z}^+$ s.t. $\forall n > N_2$ $|a_n - b| < \frac{\epsilon}{2}$.

$$|a - b| = |a - a_n + a_n - b| \leq |a - a_n| + |a_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon = |a - b|$$

12.3.2 Convergent sequences

A convergent sequence has a limit.

Every convergent sequence is bounded. Proof. Suppose $a_n \rightarrow L$ as $n \rightarrow \infty$. $\exists N \in \mathcal{Z}^+$ s.t. $\forall n > N$ $|a_n - L| < \epsilon$. By the triangle inequality, $|a_n| \leq \max(|a_1|, |a_2|, \dots, |a_N|)$ for $n \leq N$. Now set $M = \max(|a_1|, |a_2|, \dots, |a_N|, \epsilon + |L|)$. Then $\forall n \in \mathcal{Z}^+$, $|a_n| \leq M$ and the sequence is bounded. Bounded sequences aren't necessarily convergent (e.g. $((-1)^n)$).

(x_n) is a sequence that converges to L if $\forall V \in \mathcal{V}(L)$ and $\exists N \in \mathcal{N}$ $n \geq N \rightarrow x_n \in V$. May converge to different L 's but in a Hausdorff space, it will be unique. If the topology is stronger, there will be more open sets, so it will be harder to converge.

12.3.3 Cauchy sequences

A problem with using limits to define convergent sequences is that the limit is required beforehand (i.e., extrinsic information). Further, showing convergence is difficult. To address this, use Cauchy sequences (which only require intrinsic information). Cauchy sequences also allow check of completeness.

For metric spaces, a Cauchy sequence is where $d(u_m, u_n) < \epsilon$.

(a_n) is Cauchy if $\forall \epsilon > 0$, $\exists N \in \mathcal{Z}^+$ s.t. for $\forall n, m > N$, $|a_n - a_m| < \epsilon$.

Every Cauchy sequence is bounded. Proof. Suppose (a_n) is Cauchy. Set $\epsilon = 1$. Then $\forall N \in \mathcal{Z}^+$ s.t. $\exists n, m > N$ and $|a_n - a_m| < 1$. Set $m = N + 1 > N$, $\exists n > N$. Then $|a_n| = |a_n - a_m + a_m| \leq |a_n - a_m| + |a_m| < 1 + |a_m| = 1 + |a_{N+1}|$. So, $|a_n| < 1 + |a_{N+1}|$ and $\exists n > N$. Then $M = \max(|a_1|, |a_2|, \dots, |a_N|, 1 + |a_{N+1}|)$ and then $\exists n \in \mathcal{Z}^+$ and $|a_n| \leq M$.

Every convergent sequence is Cauchy. Proof. Suppose $|a_n - L| < \frac{\epsilon}{2}$. Then $|a_n - a_m| = |a_n - L + L - a_m| \leq |a_n - L| + |a_m - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

12.3.4 Completeness

A Cauchy sequence isn't necessarily convergent. Example. A Cauchy sequence of $\mathcal{Q}$ may not have a limit $\in \mathcal{Q}$, so it may not be complete. The sequence $(3, 3.1, 3.14, 3.145, \dots) \in \mathcal{Q}$, but its limit is in $\mathcal{R}$. So $\mathcal{Q}$ is not complete as it has 'holes'. If every Cauchy sequence of a set converges, the set is complete. If

the set is complete, every Cauchy sequence in the set converges. So Cauchy sequence convergence defines completeness.

Cauchy sequence + completeness = convergent sequence.

12.3.5 Closedness

A set is closed if it contains all its limit points. So if all convergent sequences in the set have their limits in the set, the set is closed.

Open and closed sets can be equivalently defined with ϵ -balls. Set A is open if $\forall x \in A, \exists \epsilon > 0$ s.t. the ϵ -neighborhood $\subset A$. Set A is closed if $\neg A = \mathcal{R} \setminus A$ is open. $\emptyset$ and $\mathcal{R}$ are open and closed, $[-2, 2]$ is closed but not open, $(-2, 2)$ is open but not closed, and $(-2, 2]$ is neither.

12.3.6 Monotone convergence

Supremum is the lowest upper bound, works even for open intervals. Infimum is the highest lower bound, works even for open intervals.

If a sequence is monotonically decreasing and is bounded by an infimum, or monotonically increasing and bounded by a supremum, the sequence is convergent.

Prove:

$$a_n = \left(1 + \frac{1}{n}\right)^n$$

is convergent. Apply monotone convergence theorem in two parts. (i) Show it is monotonically increasing:

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n}\right)^n} = \left(1 + \frac{1}{n}\right) \frac{\left(1 + \frac{1}{n+1}\right)^{n+1}}{\left(1 + \frac{1}{n+1}\right)^{n+1}} \\ &= \left(1 + \frac{1}{n}\right) \left(\frac{\left(1 + \frac{1}{n+1}\right)n(n+1)}{\left(1 + \frac{1}{n+1}\right)n(n+1)}\right)^{n+1} = \left(1 + \frac{1}{n}\right) \left(\frac{n(n+1) + n + 1 - 1}{n(n+1) + n + 1}\right)^{n+1} \\ &= \left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n^2 + 2n + 1}\right)^{n+1} \geq \left(1 + \frac{1}{n}\right) \left(1 + (n+1) \left(-\frac{1}{n^2 + 2n + 1}\right)\right) = 1 \end{aligned}$$

Where the last step uses Bernoulli's inequality where $(1+x)^k \geq 1+kx$ for $k \in \mathbb{N}$ and $x \geq -1$ where $k = n+1$ and $x = -\frac{1}{(n+1)^2}$. So $\frac{a_{n+1}}{a_n} \geq 1$ and a_n is monotonically increasing. (ii) Show bounded by supremum:

$$a_n = \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} 1^{n-k} \left(\frac{1}{n}\right)^k$$

$$\begin{aligned}
&= \binom{n}{0} 1^n \left(\frac{1}{n}\right)^0 + \binom{n}{1} 1^{n-1} \left(\frac{1}{n}\right)^1 + \sum_{k=2}^n \binom{n}{k} \left(\frac{1}{n}\right)^k = 1 + 1 + \sum_{k=2}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \\
&= 2 + \sum_{k=2}^n \frac{n!}{(n-k)!k!} \left(\frac{1}{n}\right)^k = 2 + \sum_{k=2}^n \frac{n(n-1) + \dots + (n-k+1)}{n^k} \frac{1}{k!} \\
&= 2 + \sum_{k=2}^n \frac{1}{k(k-1)} = 2 + \sum_{k=2}^n \frac{1}{k-1} - \frac{1}{k} = 2 + 1 - \frac{1}{n} = 3 - \frac{1}{n} \leq 3
\end{aligned}$$

So a_n is convergent. It is e .

12.3.7 Subsequences

A subsequence (a_{n_k}) is a selection of a sequence (a_n) where the sequence order is preserved and subsequence contains infinitely many members. Sequences can be difficult to work with, subsequences are simpler and still preserve many properties of the parent sequences.

Every subsequence of a convergent sequence is convergent at the accumulation values (also called cluster points, limits points, partial limits).

A subsequence of a divergent sequence may be convergent at one or more accumulation values.

Expand definition of accumulation values with improper accumulation values: $+\infty$ or $-\infty$. A sequence can have 0, 1, or 2 improper accumulation values. A sequence lacking an upper bound will have a positive improper accumulation value and vice versa. Limit superiors and inferiors:

$$\begin{aligned}
\limsup_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \sup(a_n | n \geq m) \\
\liminf_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} \inf(a_n | n \geq m)
\end{aligned}$$

Properties:

$$\begin{aligned}
\limsup_{n \rightarrow \infty} (a_n + b_n) &\leq \limsup_{n \rightarrow \infty} (a_n) + \limsup_{n \rightarrow \infty} (b_n) \\
\limsup_{n \rightarrow \infty} (a_n b_n) &\leq \limsup_{n \rightarrow \infty} (a_n) \limsup_{n \rightarrow \infty} (b_n) \\
\liminf_{n \rightarrow \infty} (a_n + b_n) &\geq \liminf_{n \rightarrow \infty} (a_n) + \liminf_{n \rightarrow \infty} (b_n) \\
\liminf_{n \rightarrow \infty} (a_n b_n) &\geq \liminf_{n \rightarrow \infty} (a_n) \liminf_{n \rightarrow \infty} (b_n)
\end{aligned}$$

Monotone subsequence theorem. Every sequence has a monotonic subsequence. Assume there are infinite supremums or infimums, then there is a monotonic subsequence of these. Suppose there are no more supremums or infimums. Then the sequence must be flat, rise, or fall - so it has an accumulation value, or has more supremums or infimums.

Bolzano-Weierstrass theorem. A bounded sequence has an accumulation value, i.e. it has a convergent subsequence. Can prove with monotone subsequence theorem + bounded sequences. Can also prove by infinitely and recursively bisecting the bounded interval, selecting the or an interval with infinitely many sequence terms. By the sandwich theorem, there is an accumulation value.

12.3.8 Compactness

A set A is compact (or equivalently, sequentially compact) if $\forall a_n \in A$, it has an accumulation value in A . If the set contains one or more divergent sequences $(1, 2, 3, \dots, \infty)$, there are no accumulation values for its subsequences. But if the divergent sequence moves around in circles with radii that are infinitesimally larger each round, it is divergent but there are convergent subsequences and therefore accumulation values.

Compactness is a generalisation of finiteness. Finite sets tend to be easy to prove things about because we can work by exhaustion: each element can be individually inspected, if necessary, to prove or disprove a hypothesis. Compactness generalises that to infinite sets by giving us a way to reduce things to the finite case: if we can say something about an open set then compactness says we only need to inspect finitely many such open sets to obtain a result about whatever the set covers.

The standard definition of compactness: every open cover of a set has a finite open subcover. This is directly connected to sequences: construct an open set around each element of a sequence. Take those sets large enough, then there is an open cover of the space. If there is a finite subcover, then the space is compact. But a finite subcover is a subsequence. This has to be true for every sequence in the space. Hence sequential compactness.

Examples. $\emptyset$ is compact, (5) is compact, $\mathcal{R}$ is not compact, $[c, d]$ is compact because it is bounded so by Bolzano-Weierstrass has an accumulation value and that value must be within the closed set because the sequence is within the closed set.

Heine-Borel theorem. In Euclidean R^n space, a compact set is bounded and closed. Forward direction proof. A compact set's convergent sequences have limits, so the set must be closed. The compact set must be bounded because if it were not, convergent sequences in the set can't have accumulation values (contradiction). Reverse direction proof. If a set is bounded, it has an accumulation value (which may be a limit) by Bolzano-Weierstrass, and if the set is closed, that value must be within the set because the sequence is within the closed set.

12.4 Series

Note: the harmonic series is divergent.

12.4.1 Cauchy convergence test

$\sum_{k=1}^{\infty} a_k$ is convergent iff $\forall \epsilon > 0, \exists N \in \mathcal{N} \forall n \geq m \geq N$, and:

$$\left| \sum_{k=m}^n a_k \right| < \epsilon$$

Proof. S_n is convergent iff S_n is a Cauchy sequence, so $|S_n - S_m| < \epsilon$.

Example. $\sum_n (-1)^n$ diverges. Choose $n - m = 3$:

$$\left| \sum_{k=N}^{N+2} (-1)^k \right| = |1| = 1$$

regardless of whether N is even or odd. Since the above is not true that for ALL ϵ , including $\epsilon < 1$, it's divergent.

12.4.2 Alternating series test

For alternating series, if (1) $\lim_{n \rightarrow \infty} a_k = 0$ and (2) is monotonically decreasing, then

$$\sum_{k=1}^{\infty} (-1)^k a_k$$

is convergent.

Proof. Define:

$$S_n = \sum_{k=1}^n (-1)^k a_k$$

and $a_k \geq 0$. Even S are monotonically decreasing:

$$S_{2n+2} - S_{2n} = -a_{2n+1} + a_{2n+2} \leq 0$$

Odd S are monotonically increasing:

$$S_{2n+3} - S_{2n+1} = -a_{2n+3} + a_{2n+1} \geq 0$$

Since $S_{2n+1} - S_{2n} = -a_{2n+1} \leq 0$, $S_{2n+1} \leq S_{2n}$. Since S_{2n+1} (odd) is monotonically increasing, $S_3 \leq S_{2n+1}$ for $n > 1$. Since S_{2n} (even) is monotonically decreasing, $S_{2n} \leq S_2$ for $n > 1$. So S_{2n+1} and S_{2n} are bounded:

$$S_3 \leq S_{2n+1} \leq S_{2n} \leq S_2$$

Moreover, S_{2n} is monotonically decreasing and bounded, and S_{2n+1} is monotonically increasing and bounded. So each converges. Further,

$$\lim_{n \rightarrow \infty} S_{2n+1} - S_{2n} = -a_{2n+1} = 0$$

So

$$\lim_{n \rightarrow \infty} S_{2n+1} = \lim_{n \rightarrow \infty} S_{2n}$$

So

$$\lim_{n \rightarrow \infty} S_n = S$$

12.4.3 Comparison test

$$\sum_{k=1}^{\infty} a_k$$

is absolutely convergent if

$$\sum_{k=1}^{\infty} |a_k|$$

is convergent. For convergence, test for the majorant criterion by applying the Cauchy convergence test to b_k :

$$\left| \sum_{k=m}^n b_k \right| < \epsilon$$

where $|a_k| \leq b_k$ for all $k \geq n_0$. For divergence, test for the minorant criterion by testing for Cauchy on:

$$\left| \sum_{k=m}^n b_k \right|$$

which is divergent and $a_k \geq b_k$.

12.4.4 Ratio test

Apply comparison test with geometric series in the majorant criterion.

If $|a_k| < Cr^k$ and $r < 1$ beyond a certain n , then absolutely convergent. So test for:

$$\left| \frac{a_{k+1}}{a_k} \right| \leq r \leq 1$$

For $a_k \neq 0$ and $k \geq n_0$ (to exclude e.g. harmonic series divergence).

Proof.

$$|a_{k+1}| \leq r|a_k| \leq r^2|a_{k-1}| \leq \dots \leq r^{k+1-n_0}|a_{n_0}| = r^{k+1} \frac{|a_{n_0}|}{r^{n_0}}$$

So $|a_{k+1}| \leq Cr^{k+1}$.

12.4.5 Root test

As with the ratio test, apply the comparison test with geometric series. Since $|a_k| \leq r^k$ is absolutely convergent, test

$$|a_k|^{1/k} \leq r \Leftrightarrow \limsup_{k \rightarrow \infty} |a_k|^{1/k} < 1$$

If

$$\limsup_{k \rightarrow \infty} |a_k|^{1/k} > 1$$

then divergent. If

$$\limsup_{k \rightarrow \infty} |a_k|^{1/k} = 1$$

then could be absolutely convergent, convergent, or divergent.

12.4.6 Dirichlet's test

If (1) a_k is monotonically decreasing to 0, and (2) $|\sum_k^m b_k| \leq M$ for all m , then $\sum_k^\infty a_k b_k$ converges.

Example. To show $\frac{\sin(k)}{k}$ is convergent, use $a_k = \frac{1}{k}$ which monotonically converges to 0, and $b_k = \sin(k)$. To show $\sum_k^m \sin(k)$ is bounded, use $2\sin(a)\sin(b) = \cos(a-b) - \cos(a+b)$. Choose $2\sin(1)$. Then:

$$\begin{aligned} 2\sin(1) \sum_k^m \sin(k) &= \sum_k^m 2\sin(1)\sin(k) \\ &= \sum_k^m \cos(1-k) - \cos(1+k) = \sum_k^m \cos(k-1) - \cos(k+1) \\ &= (\cos(0) - \cos(2)) + (\cos(1) - \cos(3)) + (\cos(2) - \cos(4)) + \dots + (\cos(m-1) - \cos(m+1)) \\ &= 1 + \cos(1) - \cos(m) - \cos(m+1) \\ \left| \sum_k^m \sin(k) \right| &= \left| \frac{1 + \cos(1) - \cos(m) - \cos(m+1)}{2\sin(1)} \right| \leq \left| \frac{1 + 1 + 1 + 1}{2\sin(1)} \right| = M \end{aligned}$$

12.4.7 Re-ordering

Reordering doesn't change finite sums. However, it changes infinite sums, e.g.

$$\sum_0^\infty (-1)^k = 1 + (-1) + 1 + (-1) + 1 + \dots$$

isn't convergent and has two accumulation values: 0, 1. Reordering to $1 + 1 + (-1) + 1 + (-1) + \dots$, there are different accumulation values: 1, 2.

A reordering of $\sum_1^\infty a_k$ to $\sum_1^\infty a_{\tau(k)}$ where $\tau : \mathcal{N} \rightarrow \mathcal{N}$ in a bijective map.

If $\sum_1^\infty a_k$ is absolutely convergent, then $\sum a_{\tau(k)}$ is also absolutely convergent, AND:

$$\sum_1^\infty a_k = \sum_1^\infty a_{\tau(k)}$$

Proof. If $\sum_1^\infty a_k$ is absolutely convergent, of course $\sum_1^\infty |a_k|$ is convergent. So by the Cauchy test:

$$(*) \sum_m^n |a_k| < \epsilon$$

Now show, choosing $n > m > N_1$:

$$\begin{aligned} \left| \sum_1^\infty a_k - \sum_1^n a_{\tau(k)} \right| &= \left| \sum_1^\infty a_k - \sum_1^{N_1-1} a_k + \sum_1^{N_1-1} a_k - \sum_1^n a_{\tau(k)} \right| \\ &\leq \underbrace{\left| \sum_1^\infty a_k - \sum_1^{N_1-1} a_k \right|}_A + \underbrace{\left| \sum_1^{N_1-1} a_k - \sum_1^n a_{\tau(k)} \right|}_B \\ A &= \left| \sum_k^\infty a_k - \sum_1^{N_1-1} a_k \right| = \left| \sum_{N_1}^\infty a_k \right| = \lim_{n \rightarrow \infty} \left| \sum_{N_1}^n a_k \right| \leq \lim_{n \rightarrow \infty} |a_k| < \epsilon \end{aligned}$$

The last step uses result (*). For B , note that $n > N_1$:

$$B = \left| \sum_{\substack{? \\ ?}}^? a_{\tau(k)} \right| \leq \sum_{\substack{? \\ ?}}^? |a_{\tau(k)}| = \sum_{j=N_1}^\infty |a_j| < \epsilon$$

The last step gain uses result (*). The penultimate step is re-indexation after subtraction - it's unclear which terms are left. In sum:

$$\left| \sum_1^\infty a_k - \sum_1^n a_{\tau(k)} \right| < \epsilon + \epsilon = \epsilon'$$

12.4.8 Cauchy product

Motivation.

$$\begin{aligned} &(a_0 + a_1 + a_2)(b_0 + b_1 + b_2) \\ &= \underbrace{(a_0 b_0)}_0 + \underbrace{(a_1 b_0 + a_0 b_1)}_1 + \underbrace{(a_2 b_0 + a_1 b_1 + a_0 b_2)}_2 + \underbrace{(a_2 b_1 + a_1 b_2)}_3 + \underbrace{(a_2 b_2)}_4 \end{aligned}$$

The subscripts add as in the underbraces as the indices 0 to 2 for a and b can be placed perpendicular in a table, with the diagonals representing the underbrace indices.

Cauchy product.

$$\sum_0^{\infty} a_k \sum_0^{\infty} b_k = \sum_0^{\infty} c_k$$

$$c_k = \sum_{l=0}^k a_l b_{k-l}$$

where $l + k - l = k$ adds to a constant k .

If $\sum_0^{\infty} a_k$ and $\sum_0^{\infty} b_k$ are each convergent and at least one of these is absolutely convergent, the Cauchy product is absolutely convergent.

Example.

$$e^x = \sum_0^{\infty} \frac{x^k}{k!}$$

is absolutely convergent by the ratio test. For $e^x e^y$:

$$c_k = \sum_{l=0}^k \frac{x^l}{l!} \frac{y^{k-l}}{(k-l)!} = \frac{1}{k!} \sum_{l=0}^k \binom{k}{l} x^l y^{k-l} = \frac{1}{k!} (x+y)^k$$

So:

$$e^{x+y} = \sum_0^{\infty} \frac{1}{k!} (x+y)^k = \sum_0^{\infty} c_k = \left(\sum a_k \right) \left(\sum b_k \right) = e^x e^y$$

12.5 Taylor's theorem

Taylor polynomials for f with continuous derivatives:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(x_0)(x-x_0)^n}{n!}$$

However, can get a finite exact series with the Taylor theorem. Nth degree Taylor polynomial centered at x_0 :

$$f(x) = P_n + R_n$$

$$f(x) = \sum_{n=0}^N \frac{f^n(x_0)(x-x_0)^n}{n!} + \frac{f^{N+1}(\bar{x})(x-x_0)^{N+1}}{(N+1)!}$$

where $\bar{x}$ is between $[x_0, x]$. Note the error term is R_N :

$$\lim_{N \rightarrow \infty} R_N = 0$$

As N gets larger, P_N gets larger and R_N is smaller. If x and x_0 are very close to each other, then again R_N gets smaller.

Proof. Let $f(x) = P_n + M(x - x_0)^{n+1}$. Then $M = \frac{f(x) - P_n}{(x - x_0)^{n+1}}$. Also let $f(t) = P_t + M(t - x_0)^{n+1} + g(t)$. Then $g(t) = f(t) - P_t - M(t - x_0)^{n+1}$. Now differentiate $n + 1$ times: $g^{n+1}(t) = f^{n+1}(t) - 0 - M(n + 1)!$. Now look for c s.t. $g^{n+1}(t) = 0$ then proof is complete. For $k \leq n$, $g^k(x_0) = f^k(x_0) - P_n^k(x_0)$ but the Taylor polynomial is s.t. $f^k(x_0) = P_n^k(x_0)$ so $g^k(x_0) = f^k(x_0) - P_n^k(x_0) = 0$. Call $t \rightarrow x$. So to make $g(x) = 0$, i.e. $g(x) = f(x) - P_x - M(x - x_0)^{n+1} = 0$, need $M = \frac{f(x) - P_n}{(x - x_0)^{n+1}}$. Notice $g(x) = 0$ and $g(x_0) = 0$, so by Rolle's theorem to find $c_1 \in [x, x_0]$ to get $g'(c_1) = 0$. Apply Rolle's theorem again to g' to $[x, c_1]$ or $[c_1, x_0]$ and repeat for $c_2, c_3 \dots c_n$ so get $g^{n+1}(c) = 0$.

12.6 Functions and Limits

12.6.1 Archimedean property

For any number c , there exists a natural number that is greater than c .

For every positive number ϵ , $\exists N \in \mathcal{Z}^+$ s.t. $\frac{1}{n} < \epsilon$ for all $n \geq N$. Proof. Let $\epsilon > 0 \Rightarrow \frac{1}{\epsilon} > 0$, then $\exists N \in \mathcal{N}$ s.t. $N > \frac{1}{\epsilon}$. Since $n \geq N \Rightarrow \frac{1}{n} \leq \frac{1}{N}$ and $N > \frac{1}{\epsilon} \Rightarrow \frac{1}{N} < \epsilon$, $\frac{1}{n} \leq \epsilon$.

12.6.2 δ - ϵ proofs

Limit of a function. For:

$$\lim_{x \rightarrow a} f(x) = L$$

$\forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall x \in \mathcal{R}$ with $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

Prove:

$$\lim_{x \rightarrow 2} (3x + 5) = 11$$

$|3x + 5 - 11| < \epsilon \Rightarrow |3x - 6| = 3|x - 2| < \epsilon \Rightarrow |x - 2| < \frac{\epsilon}{3}$. So let $\delta = \frac{\epsilon}{3}$. Proof. Let $\delta = \frac{\epsilon}{3}$. If $|x - 2| < \delta$, then $|3x + 5 - 11| = 3|x - 2| = 3\frac{\epsilon}{3} = \epsilon$.

Prove:

$$\lim_{x \rightarrow 2} x^2 = 4$$

Note: $|x^2 - 4| = |x - 2||x + 2| = \delta|x + 2| < \epsilon$. Since $x \rightarrow 2$, set $|x + 2| < 5 \Rightarrow \delta|x + 2| < 5\delta = \epsilon \Rightarrow \delta = \frac{\epsilon}{5}$. Proof. Let $\delta = \frac{\epsilon}{5}$. If $|x - 2| < \delta$ then $|x^2 - 4| = |x - 2||x + 2| < 5\delta < \frac{\epsilon}{5}5 = \epsilon$.

12.6.3 Sandwich theorem

Sandwich or squeeze theorem. If (a_n) , (b_n) , and (c_n) are sequences and $a_n \leq c_n \leq b_n \forall n \in \mathcal{Z}^+$, if $a_n \rightarrow L$ and $b_n \rightarrow L$, then $c_n \rightarrow L$. Proof. Since $|a_n - L| < \epsilon \Rightarrow L - \epsilon < a_n < L + \epsilon$ and likewise for b_n , and:

$$L - \epsilon < a_n \leq c_n \leq b_n < L + \epsilon$$

Then $|c_n - L| < \epsilon$.

Example. Prove:

$$\lim_{x \rightarrow 1} (x^2 - 1)^3 \sin^3 \left(\frac{1}{x-1} \right) = 0$$

Proof:

$$\begin{aligned} -1 &\leq \sin^3 \left(\frac{1}{x-1} \right) \leq 1 \\ -(x^2 - 1)^3 &\leq (x^2 - 1)^3 \sin^3 \left(\frac{1}{x-1} \right) \leq (x^2 - 1)^3 \\ \lim_{x \rightarrow 1} -(x^2 - 1)^3 &\leq \lim_{x \rightarrow 1} (x^2 - 1)^3 \sin^3 \left(\frac{1}{x-1} \right) \leq \lim_{x \rightarrow 1} (x^2 - 1)^3 \\ 0 &\leq \lim_{x \rightarrow 1} (x^2 - 1)^3 \sin^3 \left(\frac{1}{x-1} \right) \leq 0 \end{aligned}$$

Prove if g is bounded and $\lim_{x \rightarrow c} f(x) = 0$ then:

$$\lim_{x \rightarrow c} f(x)g(x) = 0$$

Proof. Suppose g is bounded by M . Then:

$$0 \leq |f(x)g(x)| = |f(x)||g(x)| \leq |f(x)|M = 0 \cdot M = 0$$

12.6.4 Limit properties

Limit of sum is the sum of limits:

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= L \\ \lim_{x \rightarrow a} g(x) &= K \\ \lim_{x \rightarrow a} (f(x) + g(x)) &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L + K \end{aligned}$$

Proof. $\exists \delta_1 > 0$ s.t. $\forall x \in \mathcal{R}$ with $0 < |x - a| < \delta_1$ leads to $|f(x) - L| < \frac{\epsilon}{2}$. $\forall x \in \mathcal{R}$ with $0 < |x - a| < \delta_2$ leads to $|g(x) - K| < \frac{\epsilon}{2}$. Choose $\delta = \min(\delta_1, \delta_2)$. Then for $|x - a| < \delta$ we have $|f(x) + g(x) - (L + K)| = |(f(x) - L) + (g(x) - K)| \leq |f(x) - L| + |g(x) - K| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

Similarly:

$$\lim_{x \rightarrow a} f(x)g(x) = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x) = LK$$

And for subtraction and division.

Example. The limit of:

$$\lim_{n \rightarrow \infty} \frac{2x^2 + 5x - 1}{-5n^2 + n + 1} = \frac{\lim_{n \rightarrow \infty} 2 + (5/n) - (1/n^2)}{\lim_{n \rightarrow \infty} -5 + (n/n^2) + (1/n^2)} = -\frac{2}{5}$$

Monotonicity.

$$a_n \leq b_n \Rightarrow \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n$$

Sandwich theorem. If $(a_n) \leq (b_n) \leq (c_n)$, and

$$\lim_{n \rightarrow \infty} a_n = L$$

$$\lim_{n \rightarrow \infty} b_n = L$$

Then:

$$\lim_{n \rightarrow \infty} c_n = L$$

Determine the limit of $c_n = \sqrt{n^2 + 1} - n$. Note:

$$c_n = (\sqrt{n^2 + 1} - n) \frac{\sqrt{n^2 + 1} + n}{\sqrt{n^2 + 1} + n} = \frac{n^2 + 1 - n}{\sqrt{n^2 + 1} + n} = \frac{1}{\sqrt{n^2 + 1} + n} \leq \frac{1}{n}$$

So the limit of c_n is 0:

$$0 \leq \lim_{n \rightarrow \infty} c_n \leq \frac{1}{n} = 0$$

12.6.5 Multivariate limits

The multivariate limit:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$$

means $\forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall (x,y) \in \mathcal{R}^2$ with

$$\| (x,y) - (x_0,y_0) \| = \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$$

and $|f(x,y) - L| < \epsilon$.

Prove:

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{x^3 y}{x^2 + y^2} = 0$$

For $\sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$,

$$\left| \frac{x^3 y}{x^2 + y^2} \right| = |y| \left| \frac{x^3}{x^2 + y^2} \right| \leq |y| \left| \frac{x^3}{x^2} \right| = |y| |x| \leq (\sqrt{(x-x_0)^2 + (y-y_0)^2})^2 < \delta^2$$

So choose $(\sqrt{\epsilon})^2 = \epsilon$.

12.6.6 Miscellaneous proofs

Use $|x| \geq |\sin(x)|$ to prove:

$$\lim_{n \rightarrow \infty} \sin\left(\frac{1}{n}\right)$$

May require trigonometric identities for other proofs involving sin and cos.

To prove $(-1)^n$ is divergent, show odd and even n 's lead to contradictions. First look at evens. $|(-1)^{2n} - 1| < 1 \Rightarrow |1 - L| < 1 \Rightarrow -1 < 1 - L < 1 \Rightarrow -2 < -L < 0$. So even n leads to $(-2, 0)$. For odds: $|(-1)^{2n+1} - 1| < 1 \Rightarrow |-1 - L| < 1 \Rightarrow -1 < -1 - L < 1 \Rightarrow 0 < -L < 2$. So odds lead to $(0, 2)$. Divergence.

12.7 Continuity

Domain points close to each other map to the codomain (range) close to each other.

12.7.1 Pointwise continuity

A function f is continuous at $x = c$ if $\forall \epsilon > 0, \exists \delta > 0$ s.t. $\forall x \in \mathcal{R}, |x - c| < \delta, |f(x) - f(c)| < \epsilon$.

Prove $f(x) = \sqrt{x}$ is continuous over $[0, \infty)$. For $x = 0$, note $|x - 0| = |x| < \delta$ and $|\sqrt{x} - \sqrt{0}| = |\sqrt{x}| < \epsilon$, so apply $\delta < \epsilon^2$: $|\sqrt{x} - \sqrt{0}| = \sqrt{x} < \sqrt{\delta} = \sqrt{\epsilon^2} = \epsilon$. For $x > 0$, $|x - c| < \delta$ and $|\sqrt{x} - \sqrt{c}| < \epsilon$, and note:

$$\left| \sqrt{x} - \sqrt{c} \left(\frac{\sqrt{x} + \sqrt{c}}{\sqrt{x} + \sqrt{c}} \right) \right| = \left| \frac{x - c}{\sqrt{x} + \sqrt{c}} \right| < \frac{\delta}{\sqrt{x} + \sqrt{c}} \leq \frac{\delta}{\sqrt{c}} < \epsilon$$

So apply $\delta = \epsilon\sqrt{c}$. So:

$$|\sqrt{x} - \sqrt{c}| = \left| \frac{x - c}{\sqrt{x} + \sqrt{c}} \right| < \frac{\delta}{\sqrt{x} + \sqrt{c}} \leq \frac{\delta}{\sqrt{c}} = \epsilon$$

To disprove continuity for any function, use contradiction.

12.7.2 Uniform continuity

Generalization of continuity, where the same ϵ holds across the domain. In contrast to just continuity, ϵ depends only on δ , not c . f is uniformly continuous on $\mathcal{R}$ if $\forall \epsilon \exists \delta > 0$ s.t. $\forall x, y \in \mathcal{R}$ with $|x - y| < \delta, |f(x) - f(y)| < \epsilon$.

Prove

$$f(x) = \frac{1}{1 + x^2}$$

is uniformly continuous. Proof.

$$\left| \frac{1}{1 + x^2} - \frac{1}{1 + y^2} \right| = \frac{|x + y|}{|1 + x^2||1 + y^2|} |x - y|$$

$$\leq \left(\frac{|x|}{(1+x^2)(1+y^2)} - \frac{|y|}{(1+x^2)(1+y^2)} \right) |x-y| \leq \left(\frac{|x|}{(1+x^2)} - \frac{|y|}{(1+y^2)} \right) |x-y|$$

Note:

$$\frac{|x|}{|1+x^2|} \begin{cases} x \leq 1, & \frac{|x|}{|1+x^2|} \leq \frac{1}{1+x^2} \leq 1 \\ x > 1, & \frac{|x|}{|1+x^2|} \leq \frac{|x|}{x^2} = \frac{1}{|x|} < 1 \end{cases}$$

So:

$$\left| \frac{1}{1+x^2} - \frac{1}{1+y^2} \right| < (1+1)|x-y| = 2|x-y| < 2\delta$$

If $\delta = \frac{\epsilon}{2}$, then:

$$\left| \frac{1}{1+x^2} - \frac{1}{1+y^2} \right| < 2\delta = \epsilon$$

Prove:

$$f(x) = \frac{1}{x}$$

is not uniformly continuous on $x \in (0, 1)$. Proof by contradiction, so objection is to find $|x-y| < \delta$ and $\left| \frac{1}{x} - \frac{1}{y} \right| > 1$. Pick $y = \frac{x}{2}$:

$$|x-y| = \left| x - \frac{x}{2} \right| = |x|/2 = \frac{x}{2}$$

If $x < 2\delta$, then $|x-y| < \delta$. But:

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \left| \frac{1}{x} - \frac{2}{x} \right| = \left| -\frac{1}{x} \right| = \frac{1}{x}$$

Which is > 1 over the interval $(0, 1)$, so contradiction.

12.7.3 Continuity v uniform continuity

Continuity. The maximum distance between $f(x)$ and $f(y)$ may depend on x and y . So for all x , for all ϵ , there exist such a δ ...

Uniform continuity. A function f is uniformly continuous if $f(x)$ and $f(y)$ can be arbitrarily close to each other by requiring only that x and y be sufficiently close to each other. For all ϵ , there exists such a δ that for all x, y . For continuity, check one x at a time. Doing this on $f(x) = 1/x$ on $(0, 1)$, find a δ for every x and ϵ . However, if you fix ϵ , the values for δ that you need become arbitrarily small as x approaches 0. For uniform continuity, select an ϵ , then find a δ which is good for ALL x values in the set. For $f(x) = 1/x$, $a\delta$ does not exist.

Continuous functions can fail to be uniformly continuous if they are unbounded on a bounded domain, such as $f(x) = \frac{1}{x}$ on $(0, 1)$ at 0, or if their slopes become unbounded on an infinite domain, such as $f(x) = x^2$ on the real line.

12.7.4 Absolute continuity

There are two definitions of absolute continuity out there. One refers to an absolutely continuous function and the other to an absolutely continuous measure.

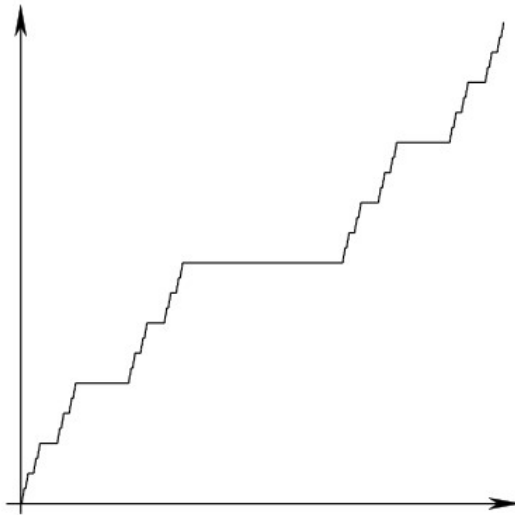
An absolutely continuous function is just a continuous function with extra constraints placed upon it so that it is guaranteed to have a derivative (almost everywhere) and its derivative can be integrated to return the original function.

More formally, a function $f : [a, b] \rightarrow [0, \infty]$ is absolutely continuous if $\forall \epsilon > 0$ and all $n > 0 \exists \delta > 0$ s.t. whenever $(a_i, b_i)_i^n \subset [a, b]$ is a pairwise disjoint collection of open intervals with $\sum_i^n b_i - a_i < \delta$ we have $\sum_i^n |f(b_i) - f(a_i)| < \epsilon$.

A measure f is absolutely continuous wrt to a measure g if all g -null sets are also f -null. But these definitions are basically the same, modulo some limits. Further, the fact that absolutely continuous functions have a derivative almost everywhere is the content of the Radon-Nikodym theorem.

Absolute continuity is similar to uniform continuity. Absolute continuity requires the δ - ϵ mapping to "work" even when the δ -sized intervals are divided up into finitely many disjoint "mini-intervals" whose lengths sum to $< \delta$. The corresponding "mini-epsilons" still must sum to $< \epsilon$.

The Cantor function (Devil's staircase) is uniformly continuous but not absolutely continuous. Its derivative is zero almost everywhere; all of its increasing is done on a null set.



Although the definitions of an absolutely continuous function and an absolutely continuous measure are different, they are intrinsically related, linked together

by *Lebesgue's Fundamental Theorem of Calculus*:

$$\int_a^x F' d\mu = F(x) - F(a)$$

where the integral is a Lebesgue integral. Absolute continuity characterizes which functions can be an antiderivative.

Let μ, ν be positive measures on a measure space (X, Σ) . Then ν is absolutely continuous with respect to μ (notationally $\mu \gg \nu$) if $\mu(A) = 0 \Rightarrow \nu(A) = 0, \forall A \in \Sigma$. Note that

$$\nu(A) < \delta \Rightarrow \int_A f d\mu < \epsilon$$

12.7.5 Hemicontinuity

Concept of continuity doesn't transfer well to correspondences.

Upper hemicontinuity (UHC). Start with limit at $(x, R(f(x)))$ (where R is the range which may have non-zero height). Then x-sequences and y-sequences must converge. Can't have open boundaries in any vertical angle, otherwise y-sequences might not converge.

Lower hemicontinuity (LHC). Start with any limit at $(x, y \in f(x))$ which must be a single point. Then x-sequences and y-sequences must converge to that single point. Limit can't have non-zero height; otherwise, x-sequences may not converge.

If a limit is both UHC and LHC then it's a continuous correspondence.

Often applicable for proofs. UHC useful for showing closed or boundedness. LHC useful for converging a sequence.

12.7.6 Semicontinuity

Useful for understanding max and min of discontinuous functions. Upper semi-continuity means the 'solid dot' is above the 'open dot' discontinuity. Lower semi-continuity means the reverse.

12.7.7 Lipschitz continuity

Describe subclasses of continuous functions. Used in existence and uniqueness of solutions of ODEs and SDEs.

Function f is Lipschitz continuous if there is a constant $K > 0$ s.t.

$$|f(x) - f(y)| \leq K|x - y|$$

The Lipschitz condition (or Lipschitz continuity) ensures that your function always remains entirely outside the white cone, so it cannot e.g. become infinitely steep at one point.

When moving along the function, there is a 'cone' of slopes

$$m_1 = \frac{f(x) - f(y)}{x - y}, m_2 = \frac{-(f(x) - f(y))}{x - y}$$

which the function does not touch. The 'cone' exists because there is a finite $K > 0$.

12.7.8 Holder continuity

Generalization of Lipschitz continuity. Again describes subclasses of continuous functions. Used in existence and uniqueness of solutions of ODEs and SDEs.

Function f is Holder continuous of order $0 < \alpha \leq 1$ if there is a constant $K > 0$ s.t.

$$|f(x) - f(y)| \leq K|x - y|^\alpha$$

Holder continuity indicates smoothness. The exponent is the lowest number (the flattest angle of the 'cones') at which the relationship holds. Moreover, if f is Holder continuous with $\alpha > 1$, then f is constant. If the Holder exponent is very low, then the function is very rough because steep cones are the flattest possible cones.

Note:

$$\begin{aligned} \text{Continuous differentiable} &\subset \text{Lipschitz continuous} \\ &\subset \text{Holder continuous} \subset \text{Uniformly continuous} \end{aligned}$$

12.8 Convergence

12.8.1 Pointwise convergence

Pointwise convergence. $f_n(x_0)$ converges pointwise if $\forall \epsilon > 0$ and for each specific $x_0 \in E$, $\exists N \in \mathbb{Z}^+$ s.t. $\forall n > N$, $|f_n(x_0) - L| < \epsilon$ (Limit function $L = f(x)$). Basically, ϵ can depend on x_0 . For a specific x_0 , looking for: $f(x_0) - \epsilon < f_n(x_0) < f(x_0) + \epsilon$.

Example. Prove $f_n(x) = x^n$ over $[0, 1]$ is pointwise convergent. For $x = 0$, $f_n(0) = 0^n = 0 \rightarrow 0$, so $|f_n(0) - 0| = |0 - 0| = 0 < \epsilon$. If $x = 1$, $f_n(1) = 1^n = 1 \rightarrow 1$, so $|f_n(1) - 1| = |1 - 1| = 0 < \epsilon$. For $0 < x < 1$, $f_n(x) = x^n \rightarrow 0$, and $|f_n(x) - 0| = |x^n - 0| = x^n < \epsilon$ so use $n \ln x < \ln \epsilon \Rightarrow n > \frac{\ln \epsilon}{\ln x}$.

12.8.2 Uniform convergence

Pointwise convergence doesn't necessarily preserve function properties. Example where pointwise convergence doesn't preserve continuity. $f_n(x) = x^n$ over $E \in [0, 1]$ with:

$$L = f(x) = \begin{cases} 1, & x = 1 \\ 0, & 0 \leq x < 1 \end{cases}$$

Then $f_n(x)$ converges pointwise over E , and each $f_n(x)$ is continuous BUT $L = f(x)$ is not continuous at $x = 1$. So need uniform convergence.

Uniform convergence. More general than pointwise. $f_n(x)$ converges uniformly if $\forall \epsilon > 0, \exists N \in \mathbb{Z}^+$ s.t. $\forall n > N$ and $\forall x \in E, |f_n(x) - L| < \epsilon$. Uniform convergence implies pointwise convergence but not vice versa. Preserves at least some properties upon convergence, including continuity. For any x , looking for: $f(x) + \epsilon < f_n(x) < f(x) + \epsilon$.

Example. Prove $f_n(x) = \frac{1}{1+x^n}$ over $[2, \infty)$ uniformly converges. $\left| \frac{1}{1+x^n} - 0 \right| = \frac{1}{1+x^n} \leq \frac{1}{x^n} \leq \frac{1}{2^n}$, so apply $n > \frac{\ln \epsilon}{\ln(1/2)}$.

12.8.3 Bernstein polynomials

Bernstein basis polynomials are

$$b_{\nu,n}(x) = \binom{n}{\nu} x^\nu (1-x)^{n-\nu}$$

and basis polynomials of degree n form a vector space basis. A linear combination with Bernstein or Bezier coefficients is the Bernstein polynomial:

$$B_n(x) = \sum_{\nu=0}^n \beta_\nu b_{\nu,n}(x)$$

Bernstein polynomials have many properties including always being positive, and

$$\begin{aligned} \sum_k \binom{n}{k} x^k (1-x)^{n-k} &= 1 \text{ probability} \\ \sum_k \frac{k}{n} \binom{n}{k} x^k (1-x)^{n-k} &= x \text{ mean} \\ \sum_k \left(x - \frac{k}{n}\right)^2 \binom{n}{k} x^k (1-x)^{n-k} &= \frac{x(1-x)}{n} \text{ variance} \end{aligned}$$

where the intuition for the probability comes from summing the k -constituents of n and the intuition for the mean comes from k/n .

12.8.4 Stone-Weierstrass

Any continuous $f : [0, 1] \rightarrow \mathbb{R}$ is the uniform limit of a sequence of polynomials.

Proof. Define

$$Z_n = \frac{\sum^n X_i}{n}$$

where $X_i \sim \text{Bern}(x)$. First prove $E(f(Z_n)) = B_n(x)$.

$$\begin{aligned} E(f(Z_n)) &= \sum_{k=0}^n f\left(\frac{k}{n}\right) P\left(Z_n = \frac{k}{n}\right) \\ &= \sum_{k=0}^n f\left(\frac{k}{n}\right) P(\text{Bin}(n, x) = k) \\ &= \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k} = B_n(x) \end{aligned}$$

Second prove $P(|Z_n - x| > \delta) \leq \frac{1}{4n\delta^2}, \delta > 0$.

$$\begin{aligned} E(Z_n) &= x \\ \text{Var}(Z_n) &= \frac{\text{Var}(\sum^n X_i)}{n^2} = \frac{nx(1-x)}{n^2} = \frac{x(1-x)}{n} \end{aligned}$$

where the variance is at a maximum when $x = 1/2$ so

$$\text{Var}(Z_n) \leq \frac{1}{4n}$$

Then

$$P(|Z_n - x| > \delta) = P(|Z_n - E(Z_n)| > \delta) \leq \frac{\text{Var}(Z_n)}{\delta^2} \leq \frac{1}{4n\delta^2}$$

Third:

$$|f(x) - B_n(x)| = |E(f(x) - f(Z_n))|$$

noting that $f(x)$ is deterministic so not a random variable, so $E(f(x)) = f(x)$. Further note: (a) continuity, $\forall \epsilon > 0, \exists \delta > 0$ s.t. $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$, and (b) boundedness, $M = \sup_{y \in [0,1]} |f(y)| < \infty$ because $[0,1]$ is closed and f is continuous so can't be asymptotic and must be bounded. By the triangle inequality,

$$|f(x) - B_n(x)| \leq E(|f(x) - f(Z_n)| \mathbb{1}_{|x - Z_n| < \delta}) + E(|f(x) - f(Z_n)| \mathbb{1}_{|x - Z_n| \geq \delta})$$

where the first right-side term $\leq \epsilon$ by continuity and the second term $|f(x) - f(Z_n)| \leq 2M$ by boundedness and $\mathbb{1}_{|x - Z_n| \geq \delta} \Rightarrow P(|Z_n - x| > \delta) \leq \frac{1}{4n\delta^2}$. So, the second term is $\leq \frac{2M}{4n\delta^2}$. As $n \rightarrow \infty$, the latter term shrinks to ϵ . So,

$$|f(x) - B_n(x)| \leq 2\epsilon$$

So,

$$\sup_{x \in [0,1]} |f(x) - B_n(x)| \rightarrow 0$$

12.9 Convexity and concavity

12.9.1 Quasi-concavity and quasi-convexity

Useful e.g. for ordinal utilities so there may not be smooth functions.

Convex function. Positive paraboloid. Quasi-convexity: $\max(f(x), f(y)) \geq f(\lambda x + (1 - \lambda)y)$, which is a looser form of convexity.

Quasi-concavity: $\min(f(x), f(y)) \leq f(\lambda x + (1 - \lambda)y)$, a looser form of concavity.

Quasi-convex and quasi-concave functions are defined on convex set domains.

12.10 Differentiation

A function differentiable at $x = c$ if it is continuous there. So:

$$\lim_{x \rightarrow c} f(x) = f(c)$$

$$\lim_{x \rightarrow c} (f(x) - f(c)) = 0$$

Proof. If f is differentiable at c , then its limit exists there:

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = f'(c)$$

$$\lim_{x \rightarrow c} (f(x) - f(c)) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} (x - c) = f'(c) \cdot 0 = 0$$

Then:

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} f(x) - f(c) + f(c) = \lim_{x \rightarrow c} f(x) - f(c) + \lim_{x \rightarrow c} f(c) = 0 + f(c) = f(c)$$

Prove $f(x) = |x|$ is not differentiable at $c = 0$. Proof:

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

is a piecewise function, so limit doesn't exist. So not differentiable.

Use

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(h)}{h} = f'(x)$$

to show that limits and derivatives exist.

Differentiation can be generalized to metric spaces with the metric derivative.

12.11 Existence theorems

12.11.1 Intermediate value theorem

A consequence of completeness. For any $\min(f(a), f(b)) < u < \max(f(a), f(b))$ then $c \in (a, b)$ s.t. $f(c) = u$.

Proof. Let S be the set of all $x \in [a, b]$ s.t. $f(x) \leq u$. This is a non-empty set. By completeness, $c = \sup S$. Claim that $u = f(c)$. Since f is continuous, $\exists \delta > 0$ s.t. $|f(x) - f(c)| < \epsilon$ whenever $|x - c| < \delta$.

$$f(x) - \epsilon < f(c) < f(x) + \epsilon$$

$\exists a^* \in (c - \delta, c]$ by the properties of sup, so

$$f(c) < f(a^*) + \epsilon \leq u + \epsilon$$

$\exists a^{**} \in (c, c + \delta)$ by the properties of sup, so

$$f(c) > f(a^{**}) - \epsilon > u - \epsilon$$

So,

$$u - \epsilon < f(c) < u + \epsilon$$

12.11.2 Extreme value theorem

Every continuous function $[a, b]$ has an absolute minimum and absolute maximum.

12.11.3 Mean value theorem

If f is a continuous function on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then there exists a point $c \in (a, b)$ s.t. the tangent at c is parallel to the secant line through the endpoints $(a, f(a))$ and $(b, f(b))$. So:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Rolle's theorem. If $f(a) = f(b)$ then $f'(c) = 0$. Usually prove this with EVT and then generalize to MVT.

Implications. (1) If $f'(x) = g'(x) \forall x \in (a, b)$ of the domain of these functions, then $f - g$ is constant or $f = g + c$ where c is a constant on (a, b) . (2) If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is $F(x) + c$.

12.12 Integration

12.12.1 Stieltjes integral

A generalization of the concept of the Riemann integral, realizing the notion of integrating a function f with respect to another function u . Let two functions f and u be defined and bounded on $[a, b]$ and let $a = x_0 < \dots < x_n = b$. The Stieltjes integral is

$$I = \int_a^b f(x) du(x)$$

the function u is said to be the integrating function.

12.12.2 Riemann integral

The Riemann integral is a particular case of the Stieltjes integral, when a function $x + C$ is taken as the integrating function u .

13 Measure Theory

For a long time the Riemann integral was considered the correct integral to handle bounded functions. Research continued on how best to do improper integrals but there wasn't much doubt that the Riemann integral did its job correctly on bounded functions. Until Volterra found a differentiable function with a bounded derivative that was not Riemann integrable. That doomed the Riemann integral.

Lebesgue proposed to generalize the Riemann integral to handle the Volterra example. He started by imagining that he could produce an integral that handled all bounded functions and that possessed the properties that one would require. He argued that if he could do that then, at the same time (for free) he would have also produced a nice measure theory for how to estimate the length.

This says the length of the set is equal to the sum of the lengths of its pieces, which is reasonable. A very nice feature was that the integration theory produced the measure theory (for free) and it also works backwards: if you have the measure theory then (almost free with a bit of work) you can produce the integration theory.

So Lebesgue set out to develop the measure theory (using ideas that Borel had already published). As it turns out you cannot quite do this: not for all sets, only for nice sets.

To find out which sets were the nice sets he turned to an ancient method that the Greeks used (the method of exhaustion): if you want a correct measurement of an area or a volume, take your best shot at an overestimate and call it the upper area or upper volume. Then take your best shot at an underestimate and

call it the lower area or lower volume. If the two agree then you have succeeded. If they disagree then either your methods are bad or else there really is no area or volume that makes sense.

So Lebesgue used this idea to produce an upper length $l_u(E)$ and also a lower length $l_d(E)$. If (just as the Greeks would have done) $l_u(E) = l_d(E)$ then he called the set measurable (i.e., nice).

This meant, however, a slight change in the integration theory. Now he couldn't integrate all bounded functions, only ones that were closely related to the nice measurable sets. He called these functions measurable (using the same name as for sets). Then instead of producing an integral that integrates all bounded functions, it integrates all bounded measurable functions. But, restricted to these functions, the integral has all the strong integration properties that he required.

So the outer measure of the problem really traces back to this story with a lot of credit to the Greeks for the upper/lower or outer/inner idea, and a lot of credit to Lebesgue who managed to connect integration theory with measure theory.

13.1 Non-measurable sets

Measure problem. Look for measure μ on $P(\mathcal{R})$ where (1) $\mu[a, b] = b - a$ (intuitive concept of length) and (2) $\mu(x + A) = \mu(A)$ (translation invariance). This would generalize length measure. But this doesn't exist.

Let $\mu(I = (0, 1]) < \infty$ (unit measure is finite) and satisfies (2). Then μ makes no sense, and doesn't satisfy (1). Proof. Define $I = (0, 1]$ with equivalence relation $x \sim y$ iff $x - y \in \mathcal{Q}$. So $[x] = (x + r : r \in \mathcal{Q}, x + r \in I)$. This is equivalent to $\mathcal{Q} \in (0, 1]$ shifted by irrational numbers. $A \in I$ with the property that for each $[x]$, there is an $a \in A$ with $a \in [x]$, which is equivalent to the Axiom of Choice (given a potentially infinite collection of non-empty sets, we can form a new set that contains one element from each set). Further define $A_n = r_n + A$ where $(r_n) \in \mathcal{Q}(-1, 1]$. Note that $A_n \cup A_m = \emptyset$ iff $n \neq m$ as the equivalence classes are disjoint. Further, note $(0, 1] \subseteq \bigcup A_n \subseteq (-1, 2]$ since $r_n \in (0, -1]$.

Deeper into the proof. Assume μ on $P(\mathcal{R})$ with $\mu((0, 1]) < \infty$ and (2). By (2), $\mu(x + A) = \mu(A)$, which when combined with $(0, 1] \subseteq \bigcup A_n \subseteq (-1, 2]$ leads to $\mu(0, 1] \leq \mu(\bigcup A_n) \leq \mu(-1, 2]$. Now let $\mu(0, 1] = 1$, then by σ -additivity, this is decomposable into $\mu((-1, 0] \cup (0, 1] \cup (1, 2]) = 3$. Then:

$$1 \leq \sum_n^{\infty} \mu(A_n) \leq 3$$

By (2) translation invariance,

$$1 \leq \sum_n^{\infty} \mu(A) \leq 3$$

which series will explode if $\mu(A) > 0$. Further, it can't be $\mu(A) = 0$ (otherwise, $1 \leq 0 \leq 3$). So μ is non-measurable.

Intuition. There are uncountably infinite irrationals in the real interval $[0, 1]$. So giving each number a measure leads to infinite length while 0 measure leads to 0 length. Theoretically, assigning measures may be possible with $\mathcal{Q}$ in the real interval but then the irrationals are missing so the length is incomplete.

The axiom of choice leads to non-measurable sets. To address this, reject the axiom of choice or accept non-measurable sets. Accept non-measurable sets and therefore avoid them with measurable sets such as Borel sets.

13.2 σ -algebras

Also called σ -fields. $P(X)$ is the power set of X . $\mathcal{A} \subseteq P(X)$ is a σ -algebra and measurable if (i) $\emptyset, X \in \mathcal{A}$, (ii) closed under complements so $A \in \mathcal{A}$ and $\neg A \in \mathcal{A}$, and (iii) closed under countable unions so $A_i \in \mathcal{A}$ for $i \in \mathcal{N}$ and $\bigcup_i^{\infty} A_i \in \mathcal{A}$.

Notes. (i) can be justified using (ii)+(iii) because if $x \in A$ then $x^c \in A$ and due to countable union closure $x \cup x^c = A$. Further, since the whole set $A \in \mathcal{A}$ $A^c = \emptyset$. Finally, closed under countable intersections: $\bigcap E_i = \bigcap (E_i^c)^c = (\bigcap E_i^c)^c \in \mathcal{A}$ by de Morgan's law and because of closure under complements.

$\mathcal{A} = (\emptyset, X)$ is the smallest σ -algebra. $\mathcal{A} = P(X)$ is the largest σ -algebra. Intersections of σ -algebras $\bigcap \mathcal{A}_i$ are also σ -algebras (i.e. they both have $\emptyset, X$, include complements and closed unions as they come as 'packs').

A σ -algebra is closed under countably infinite unions and under complementation. A topology is closed under any (including uncountably infinite) union and in general not closed under complementation (as a function space just needs to be defined in terms of what it is, not what it's not).

σ -algebras are closed under countable unions but not uncountable unions because intervals (which can be paired with $\mathcal{Z}$ or $\mathcal{N}$) lead to countable unions but points lead to uncountable unions.

Intuition. Let S be any subset of the possible outcomes. Naively, one would like to think that it is meaningful to ask what the probability is that the outcome is in S . The collection of sets S is meaningful. The axioms of a σ -algebra reflect the sorts of arguments we think we can make for meaningful questions. If we

can ask whether the outcome is in S , we ought also to be able to ask whether the outcome is not in S (closure under complementation). If we can ask whether one or more outcomes are in S , we ought also to ask whether whether the union (or-sets) are in S (closure under unions).

13.3 Borel sets

For $M \subseteq P(X)$, the smallest σ -algebra that contains M is

$$\sigma(M) = \bigcap_{\mathcal{A} \supseteq M} \mathcal{A}$$

Borel sets are formed from open or closed sets through the operations of countable union, countable intersection, and relative complement. It is a *countable* set; if it were uncountable, then non-measurable sets (like those from Banach-Tarski's paradox) would be possible.

Borel sets connect topology with measure. Topologies don't necessarily have measure, as they are closed under uncountable unions (as opposed to countable unions in σ -algebras). This can lead to non-measurable sets. So use Borel or other countable sets.

13.4 Measurable sets and Measure spaces

$(X, \mathcal{A})$ is a measurable space. $\mu : \mathcal{A} \rightarrow [0, \infty]$.

$\mu : \mathcal{A}_i$ is a measure if (i) $\mu(\emptyset) = 0$, and (ii) σ -additive so that $\mu(\bigcup_i^\infty A_i) = \sum_i^\infty \mu(A_i)$ and A_i are pairwise disjoint. $(X, \mathcal{A}, \mu)$ is a measure space.

Calculations: (i) $x + \infty = \infty$ for $x \in [0, \infty]$, (ii) $x \cdot \infty = \infty$ for $x \in (0, \infty]$, and (iii) $0 \cdot \infty = 0$.

If $\mathcal{A} = P(X)$, then:

(A) Counting measure (just counting number of elements):

$$\mu(A) = \begin{cases} \text{No. } A, & A \text{ has finite elements} \\ \infty, & \text{else} \end{cases}$$

(B) Dirac measure for $p \in X$ (concentrates the measure at a single point):

$$\delta_p(A) = \begin{cases} 1, & p \in A \\ 0, & \text{else} \end{cases}$$

(C) Look for $X = \mathcal{R}^n$:

(i) $\mu([0, 1]^n) = 1$ (unit cuboids)

(ii) $\mu(x + A) = \mu(A)$ (translation invariance).

13.5 Measurable maps (functions)

Map $f : \Omega_1 \rightarrow \Omega_2$ between measurable spaces $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ is measurable wrt $\mathcal{A}_1, \mathcal{A}_2$ if: $f^{-1}(A_2) \in \mathcal{A}_1, \forall A_2 \in \mathcal{A}_2$.

For $(\Omega, \mathcal{A})$ and $(\mathcal{R}, B(\mathcal{R}))$, if $f, g : \Omega \rightarrow \mathcal{R}$ are measurable, then $f + g, f - g, f \circ g$, and $|f|$ are all measurable.

Measurability is a generalization of continuity. Continuity implies measurability but not vice versa as measurability is broader. This leads to a broader class of integrable functions (Lebesgue integration versus Riemann integration).

A measurable function is a function between the underlying sets of two measurable spaces that preserves the structure of the spaces: the preimage of any measurable set is measurable. This is in direct analogy to the definition that a continuous function between topological spaces preserves the topological structure: the preimage of any open set is open. In real analysis, measurable functions are used in the definition of the Lebesgue integral. In probability theory, a measurable function on a probability space is known as a random variable.

13.6 Riemann integral

$f : \mathcal{R} \rightarrow \mathcal{R}$. Uses convergence of lower and upper rectangle sums. 3 problems.

First problem. Difficult to expand to higher dimensions. Adding just one dimension e.g. $f : \mathcal{R}^2 \rightarrow \mathcal{R}$ requires cuboids which is more complicated.

Second problem. Some continuity requirements - only finite discontinuities are allowed. Infinite discontinuities not Riemann-integrable.

Third problem. Limit and Riemann integral only commute if there is uniform convergence. But we know they should commute in wider contexts.

Contrast to Lebesgue integrals, where $f : \mathcal{R}^n(\Omega) \rightarrow \mathcal{R}$ still has a real y-axis so partition this to integrate instead of the x-axis. c_i can be upper or lower height.

13.7 Lebesgue integral

This is essentially a sideways Riemann integral with extra features, e.g. being able to use the Lebesgue measure. So it can integrate more functions than Riemann. Further, the set of Lebesgue-integrable functions forms a complete vector space, where convergence and other properties are more consistent. This in turn allows use of DCT, MCT, Fatou's lemma, etc. as integration rules. Finally, it is easier to integrate higher-dimensional integrals with Lebesgue versus Riemann.

In measure space $(X, \mathcal{A}, \mu)$, let $\chi_A : X \rightarrow \mathcal{R}$ be a step function (also: characteristic function, where the value is 1 if the preimage is $A \in \mathcal{A}$ and 0 if else). This is for simple functions, like step functions. Define integral $I(\chi_A) = \mu(A)$. Then:

$$f(x) = \sum_i^n c_i \cdot \chi_{A_i}(x)$$

where $c_i \in \mathcal{R}$ (the y-distance or height), is measurable. Then these are integrals:

$$I(f) = \sum_i^n c_i \mu(A_i)$$

But what about if it's infinity, e.g. $3 \cdot \infty - 2 \cdot \infty$. Limit constant $c > 0$. Why have closed sets with infinity? If measure is non-atomic, then no difference. If Dirac mass at ∞ , then there is a difference.

Definition. $\mathcal{S}^+ = \{f : X \rightarrow \mathcal{R} \mid f = \text{simple}, f \geq 0\}$ but f can be measurable and finite c . Can choose representations that are equivalent.

Lebesgue integral wrt μ .

$$I(f) = \int_x f \cdot d\mu = \int_x f(x) d\mu(x) = \sum_i^n c_i \mu(A_i) \in [0, \infty]$$

Properties. (1) Linear $I(af + bg) = aI(f) + bI(g)$, with $a, b \geq 0$. (2) Monotonic $f \leq g$ implies $I(f) \leq I(g)$.

General Lebesgue integral. Let $h = \sum c_i \chi_{A_i}$, which is always below the curve where the function range is ≥ 0 . Then

$$\int_x f d\mu = \sup(I(h) \in \mathcal{S}^+ \mid h \leq f)$$

and f is μ -integrable if $\int_x f d\mu < \infty$. Note that simple functions are still used to define the general Lebesgue integral.

More cumbersome for integrating simple functions (Riemann is easier) but can be done in multiple ways.

13.8 Convergence theorems

Determine when limits and integrals commute. The MCT and DCT tell us that if you place certain restrictions on both the f_n and f , then can interchange the limit and integral. Fatou's Lemma, on the other hand, says, "Here's the best you can do if you don't make any extra assumptions about the functions."

13.9 Monotone convergence

μ almost-everywhere: $\mu(x \in X | f(x) \neq g(x)) = 0$ so the functions can be slightly different (e.g. one or more discontinuities) that add to 0 so $\int f d\mu = \int g d\mu$. If $f \leq g$ and μ almost-everywhere then $\int f d\mu \leq \int g d\mu$.

Monotone convergence theorem. If $f_n : X \rightarrow [0, \infty)$ is a sequence of measurable functions on a measurable set X , $f_1 \leq f_2 \leq \dots \leq f_n$, μ (pointwise) almost-everywhere, and

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

Then:

$$\lim_{n \rightarrow \infty} \int_x f_n(x) d\mu = \int_x \lim_{n \rightarrow \infty} f_n(x) d\mu = \int_x f(x) d\mu$$

Proof. Involves converting

$$\begin{aligned} f_1 &\leq f_2 \leq \dots \leq f_n \\ \int f_1 &\leq \int f_2 \leq \dots \leq \int f_n \end{aligned}$$

13.10 Fatou's lemma

A type of convergence theorem. Measurable space $(X, \mathcal{A}, \mu)$ and non-negative measurable function $f : X \rightarrow [0, \infty]$. Then:

$$\int_x \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_x f_n d\mu$$

Proof. Let $g_n = \inf f_n$. Then:

$$\int_x \lim_{n \rightarrow \infty} g_n d\mu = \lim_{n \rightarrow \infty} \int_x g_n d\mu = \liminf_{n \rightarrow \infty} \int_x g_n d\mu$$

where the second equality uses the monotone convergence theorem, the third term just uses the inf twice (once outside the integral and once inside the integral, implicit in g_n), and the equality between the first and third terms is Fatou's lemma. Since we let $g_n = \inf f_n$, $g_n \leq f_n$. By monotone convergence, $\int_x g_n(x) d\mu \leq \int_x f_n(x) d\mu$. Apply this to the above to get:

$$\int_x \liminf_{n \rightarrow \infty} f_n d\mu = \liminf_{n \rightarrow \infty} \int_x g_n d\mu \leq \liminf_{n \rightarrow \infty} \int_x f_n d\mu$$

Illustration. $X = [0, 2]$ and piecewise function

$$f_n = \begin{cases} \chi_{(1,2]} & n = \text{even} \\ \chi_{[0,1]} & n = \text{odd} \end{cases}$$

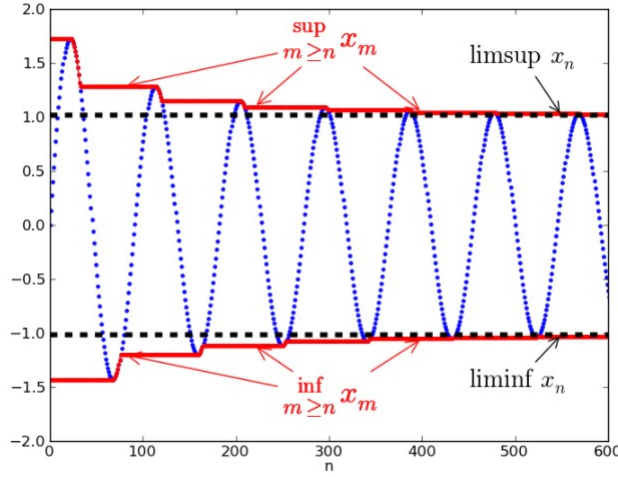
Then $\int f_n = 1$ because the function just switches between a line at $y = 1$ over $[0, 1]$ and over $[1, 2]$ and 0 elsewhere, depending on whether n is even or odd. Further, the limit inferior is 0 the entire time. So:

$$0 = \int_x \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_x f_n d\mu = 1$$

Intuition. The integral is a type of averaging, so the average of the lim inf is less than or equal to the lim inf of the average. In fact, Fatou's lemma for expectation convergences is precisely this. If $X_n \geq 0$ then

$$E(\liminf_n X_n) \leq \liminf_n E(X_n)$$

Visualize using below which also visually depicts liminf and limsup:



13.11 Dominated convergence theorem

Measurable space $(X, \mathcal{A}, \mu)$ and set of Lebesgue-integrable non-negative functions:

$$\mathcal{L}^1(\mu) = \left(f : X \rightarrow \mathcal{R} \mid \int_x |f|^1 d\mu < \infty \right)$$

For $f \in \mathcal{L}^1(\mu)$, $f = f^+ - f^-$ (≥ 0 and ≤ 0 respectively) where $f^+, f^- \geq 0$ because we have an absolute value in the set.

Lebesgue dominated convergence theorem. $f_n : X \rightarrow \mathcal{R}$ and μ (pointwise) almost everywhere. f_n converges to $f \in \mathcal{L}^1(\mu)$ while $|f_n| < g$ (so $f \leq g$) with $g \in \mathcal{L}^1(\mu)$. g is called the integrable majorant. Then:

$$\lim_{n \rightarrow \infty} \int_x f_n d\mu = \int_x \lim_{n \rightarrow \infty} f_n d\mu = \int_x f d\mu$$

Proof.

$$|f_n - f| \leq |f_n| + |f| \leq 2g$$

$$h_n = 2g - |f_n - f| \geq 0$$

So $h_n : X \rightarrow [0, \infty]$ measurable. Apply Fatou's lemma:

$$\int_x \liminf h_n d\mu \leq \liminf \int_x h_n d\mu$$

$$\int_x \liminf h_n d\mu = \int_x 2g$$

To get the smallest h_n need to subtract the largest $|f_n - f|$:

$$\liminf \int_x h_n d\mu = \int_x 2g d\mu - \limsup \int_x |f_n - f| d\mu$$

Then:

$$\int_x 2g d\mu \leq \int_x 2g d\mu - \limsup \int_x |f_n - f| d\mu$$

$$0 \leq - \limsup \int_x |f_n - f| d\mu$$

$$\limsup \int_x |f_n - f| d\mu \leq 0$$

But it must be that:

$$0 \leq \liminf \int_x |f_n - f| d\mu \leq \limsup \int_x |f_n - f| d\mu \leq 0$$

So the above must be equalities and all are equal to 0. So limit exists and:

$$\lim \int_x |f_n - f| d\mu = 0$$

This essentially proves Lebesgue's dominated convergence theorem.

Illustration. If f is not dominated by any function g , then the limit and integral may not commute. For instance, the area of the square is always 1 in the piecewise function which increases in height to infinity as n increases:

$$f_n(x) = \begin{cases} n & 0 < x \leq \frac{1}{n} \\ 0 & \text{else} \end{cases}$$

$$1 = \lim \int f_n(x) dx \neq \int \lim f_n(x) dx = 0$$

13.12 Lebesgue's FTC

Lebesgue's FTC: a function $F : [a, b] \rightarrow \mathcal{R}$ is absolutely continuous iff:

$$\int_a^x F' d\mu = F(b) - F(a)$$

where the integral is a Lebesgue integral. It characterizes which functions can be an antiderivative.

Let $f : [a, b] \rightarrow [0, \infty]$ be an integrable function and suppose

$$F(x) = \int_a^x f d\mu$$

Then f is absolutely continuous.

Lemma. For any measurable set $A \subset \mathcal{R}$ and any integrable function $f : [a, b] \rightarrow [0, \infty]$ for each $\epsilon > 0 \exists \delta > 0$ s.t. $\mu(A) > \delta$ implies

$$\int_A f d\mu > \epsilon$$

13.13 Measures

Ideally any measure has 4 desirable properties. (1) Defined for every subset of $\mathcal{R}$. (2) The measure of any interval is equal to its length. (3) The measure of a countable union of disjoint sets is equal to the sum of measures of each of the disjoint sets. (4) The measure should be translation invariant. It turns out that no matter how you attempt to define a measure, you can never satisfy these 4 ideas simultaneously. σ -algebras provide (2)-(4). (3) becomes uncountable unions instead of countable unions if (1) is included.

The outer measure of A is defined to be the smallest sum of lengths of intervals that cover A and it turns out that the outer measure does not have property 3.

13.14 Caratheodory's extension theorem

Let X be a set and $\mathcal{A} \subseteq P(X)$ which is not a σ -algebra but rather a semiring. $p : X \rightarrow [0, \infty]$ is a pre-measure, a map that is defined before defining measure. Then:

(A) Existence. p has an extension $\mu : \sigma(\mathcal{A}) \rightarrow [0, \infty]$ where $p(A) = \mu(A)$ where $A \in \mathcal{A}$.

(B) Uniqueness. μ is σ -finite (a generalization of finite measure). If there is a sequence of subsets (s_j) where $s_j \in \mathcal{A}$, $\bigcup s_j = X$, and $p(s_j) < \infty$, then μ is

unique.

Semi-ring has 3 properties. (1) $\emptyset \in \mathcal{A}$ as in σ -algebras. (2) $A, B \in \mathcal{A}$ and $A \cap B \in \mathcal{A}$ which is weaker than for σ -algebras. (3) There are pairwise disjoint sets (S_n) s.t. $\bigcup S_n = A/B$.

Example. Let $\mathcal{A} = [a, b)$ (i.e. only finite intervals) where $a, b \in \mathcal{R}$, which is not a σ -algebra because the entire set $\mathcal{R}$ is not an element of $\mathcal{A}$. But $\sigma(\mathcal{A}) = \mathcal{B}(\mathcal{R})$ (so generates a Borel σ -algebra).

Pre-measure. $p : \mathcal{A} \rightarrow [0, \infty]$. (1) $p(\emptyset) = 0$. (2) σ -additivity.

13.15 Outer measures

Use intervals that cover the set 'from the outside'. The outer measure is the smallest measurement over all interval combinations that have some part outside of our set. In contrast, an inner measure would be using the largest intervals that our set can cover. The inner measure is the largest measurement over all interval combinations that lie completely inside our set.

Doesn't have to be an actual measure. Don't need it for most measure theory but need it to prove Caratheodory extensions.

Map $\phi : P(X) \rightarrow [0, \infty]$ is an outer measure if it fulfills the following properties. (1) $\phi(\emptyset) = 0$. (2) Monotonicity: $A \subseteq B \subseteq X \rightarrow \phi(A) \leq \phi(B)$. (3) σ -subadditivity $\phi(\bigcup A_i) \leq \sum \phi(A_i)$.

Restrict sets to get σ -additivity. $A \in P(X)$ is ϕ -measurable if for all $Q \in P(X)$, $\phi(Q) = \phi(A \cap Q) + \phi(A^c \cap Q)$. The collection of A s that are ϕ -measurable forms a σ -algebra denoted $\mathcal{A}_\phi$. Then, $\mu : \mathcal{A}_\phi \rightarrow [0, \infty]$, $\mu(A) = \phi(A)$ is a measure.

Example 1: counting measure. $\phi : P(\mathcal{N}) \rightarrow [0, \infty]$, $\phi(A) = |A|$, the cardinality of A . This is an outer measure but not a measure.

Example 2. $I = [a, b)$, $\mu[a, b) = b - a$. $\phi : P(\mathcal{R}) \rightarrow [0, \infty]$.

$$\phi(A) = \inf \left(\sum_j \mu(I_j) : A \subseteq \bigcup_j I_j \right)$$

where I 's are coverings. Proof. (1) $\phi(\emptyset) = 0$. (2) Monotonicity because $A \subseteq B$ implies

$$\phi(B) = \inf \left(\sum_j \mu(I_j) : A \subseteq B \subseteq \bigcup_j I_j \right)$$

$$\geq \inf \left(\sum_j^\infty \mu(I_j) = A \subseteq \bigcup_j^\infty I_j \right) = \phi(A)$$

(3) σ -subadditivity: $\phi(\bigcup A_n) = \sum \phi(A_n)$ where $n \in \mathcal{N}$. This is intuitive since the left side contains potential intersections. Let $\epsilon > 0$ with $\sum \epsilon_n \geq \epsilon$. $\phi(A_n) = \sum \mu(I_{j,n}) - \epsilon_n$. Then $\bigcup A_n \subseteq \bigcup_n \bigcup_j I_{j,n} = \bigcup_{j,n} I_{j,n}$. So:

$$\phi(\bigcup A_n) \leq \phi(\bigcup I_{j,n})$$

by monotonicity, and $\leq \sum \mu(I_{j,n})$ because the left-hand side contains intersected intervals, and:

$$= \sum \left(\sum \mu(I_{j,n}) \right) \leq \sum (\phi(A_n) + \epsilon_n) = \sum (\phi(A_n) + \epsilon)$$

which means

$$\phi(\bigcup A_n) \leq \sum \phi(A_n)$$

13.16 Lebesgue measure

Lebesgue measure. Let $\mathcal{A} = [a, b)$ where $a, b \in \mathcal{R}$ and $p : \mathcal{A} \rightarrow [0, \infty]$. So $p([a, b)) = b - a$ is a pre-measure. By Caratheodory extension, there is a unique extension to $B(\mathcal{R})$ (a Borel σ -algebra), which is the Lebesgue measure (and which exists and is unique).

Lebesgue-Stieltjes measure. $F : \mathcal{R} \rightarrow \mathcal{R}$ monotonically increasing (so non-decreasing and may have lots of discontinuities). Can define the limits of $F(a)$ and $F(b)$ for $[a, b)$ and $(a, b]$ respectively using left or right limits. By Caratheodory extension, this is a Borel set $B(\mathcal{R}) \rightarrow [0, \infty]$.

Defining the integral. Let $F : \mathcal{R} \rightarrow \mathcal{R}$ and the derivative $F' : \mathcal{R} \rightarrow [0, \infty)$ if F is monotonically increasing and continuously differentiable everywhere. Then, $\mu[a, b) = F(b) - F(a)$, but by FTC this is $= \int_a^b F'(x) dx$ where $F'(x)$ is the density function.

The Lebesgue integral is the area under the graph, where the area is understood to be the Lebesgue measure, similarly to the Riemann-Darboux integral being the ordinary area (= the Jordan measure) under the graph.

13.17 Lebesgue decomposition

$(X, \mathcal{A}, \lambda) \rightarrow (\mathcal{R}, B(\mathcal{R}), \lambda)$ measure space. Lebesgue measure $\lambda[a, b) = b - a$ is the reference measure. Interested in another measure $\mu : B(\mathcal{R}) \rightarrow [0, \infty]$ in reference to λ .

Absolute continuity of a measure. Let μ and λ be measures on a measurable space $(X, \mathcal{A})$. μ is absolutely continuous wrt Lebesgue measure ($\mu \ll \lambda$) if

$$\lambda(A) = 0 \rightarrow \mu(A) = 0.$$

μ is singular wrt λ if there is a measurable set $N \in B(\mathcal{R})$ s.t. $\lambda(N) = 0$ and $\mu(N^c) = \mu(\mathcal{R}/N) = 0$. $\mu \perp \lambda$. Basically, disjoint sets that are orthogonal. Like the 'opposite' to absolute continuity.

Example. Dirac measure $\delta_0(0) = 1$, so $\delta_0 \perp \lambda$. So $N = \{0\}$.

Lebesgue decomposition theorem. $\mu : B(\mathcal{R}) \rightarrow [0, \infty]$ (σ -finite). There are 2 uniquely determined measures, μ_{ac} (absolutely continuous), μ_s (singular) with $\mu = \mu_{ac} + \mu_s$.

13.18 Radon-Nikodym theorem

Radon-Nikodym theorem. $\mu : B(\mathcal{R}) \rightarrow [0, \infty]$ (σ -finite). There is a measurable map $h : \mathcal{R} \rightarrow [0, \infty)$ with:

$$\mu_{ac}(A) = \int_A h d\lambda$$

for all Borel sets $A \in B(\mathcal{R})$. h is a density function.

13.19 Push-forward measure

Image measure = push-forward measure. Let $(X, \mathcal{A})$ equipped with measure μ and $(Y, \mathcal{C})$ equipped with no measure (but which will be inferred as $\bar{\mu}$) be two measurable spaces with measurable map $h : X \rightarrow Y$. Then for $C \in \mathcal{C}$, $\bar{\mu}(C) = \mu(h^{-1}(C))$ (also, without using $\mathcal{C}$: $\mu \circ h^{-1}$, pushing forward the measure to Y).

13.20 Substitution formula

Substitution formula = change of variables formula. Switch between $y = h(x)$. Following on from above, $g : Y \rightarrow \mathcal{R}$. Then:

$$\int_Y g d(\mu \circ h^{-1}) = \int_X g \circ h d\mu$$

Equivalently:

$$\int_Y g(y) d(\mu \circ h^{-1})(y) = \int_X g(h(x)) d\mu(x)$$

This can be used to derive:

$$\int g(y) dy = \int g(F(x)) F'(x) dx$$

13.21 Product measure

Let $(X_1, \mathcal{A}_1, \mu_1)$ and $(X_2, \mathcal{A}_2, \mu_2)$ be two measure spaces. $(X_1 \times X_2, \mathcal{A}, \mu)$ and μ is the product measure. Then combining length and width to get area $\mu_1(A_1 \times A_2) = \mu_1(A_1) \times \mu_2(A_2)$.

Product σ -algebra. $\mathcal{A}_1 \times \mathcal{A}_2$ is not a σ -algebra because offset rectangles won't form a rectangle and therefore be closed. But it's a semi-ring. So adjust: $\mathcal{A} = \sigma(\mathcal{A}_1 \times \mathcal{A}_2)$.

Product measure. Define $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$. Apply Caratheodory extension on the above semi-ring to get product measure. However, the product measure isn't in general unique.

Address uniqueness. If μ_1, μ_2 are σ -finite, then $\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$ is unique and is (and there is equivalence with Cavalieri's principle):

$$\mu(M) = \int_{X_2} \mu_1(M_y) d\mu_2(y) = \int_{X_1} \mu_2(M_x) d\mu_1(x)$$

13.22 Fubini's theorem

Generalization of Cavalier's principle.

Let μ_1, μ_2 be σ -finite, μ the product measure and measurable function $f : X_1 \times X_2 \rightarrow [0, \infty]$ (so $f \in \mathcal{L}^1(\mu)$). Then

$$\begin{aligned} \int_{X_1 \times X_2} f d\mu &= \int_{X_2} \int_{X_1} f(x, y) d\mu_1(x) d\mu_2(y) \\ &= \int_{X_1} \int_{X_2} f(x, y) d\mu_2(y) d\mu_1(x) \end{aligned}$$

14 Functional Analysis

14.1 Infimum and supremum

The *infimum* of a subset S of a partially ordered set P is a greatest element in P that is less than or equal to all elements of S , if such an element exists. It is the greatest lower bound.

The *supremum* is the least element in P that is greater than or equal to all elements of S , if such an element exists. It is the least upper bound.

Inf and sup are similar to min and max but more general as they apply even when there is no min or max. For instance, the set $\mathcal{R}^+$ does not have a minimum but an infimum (0) which is smaller than all the positive real numbers and greater than any other real number which could be used as a lower bound.

14.2 Spaces

14.2.1 Metric spaces

The mapping $d: X^n \rightarrow [0, \infty)$ is a metric space if (1) $d(x, y) = 0$, (2) $d(x, y) = d(y, x)$, and (3) $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).

The metric space is (X, d) . Examples. (1) Manhattan metric. $X = \mathcal{R}^n$ is a metric space with $d(x, y) = |y_1 - x_1| + |x_2 - x_1|$. (2) Euclidean or standard metric. $X = \mathcal{R}^n$ is a metric space with $d(x, y) = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$. (3) $X = \mathcal{R}^n$ with $d(x, y) = \max(|x_1 - y_1|, \dots, |x_n - y_n|)$. (4) $d(s_n, t_n)$ where s_n and t_n are bounded sequences, and d is the maximum distance between them. (5) Similar to (4), $d(f, g) = \max(|f(x) - g(x)|)$ for continuous functions that are bounded, or taking a metric over an interval. (6) Generalized Manhattan metric. $d(f, g) = \int_a^b |f(x) - g(x)| dx$, so the are between functions. (7) Generalized Euclidean metric. $d(f, g) = \sqrt{\int_a^b |f(x) - g(x)|^2}$. (8) $d(a, b) = \inf\{|a - b| | a \in A, b \in B\}$ so the shortest distance between two sets A and B. (9) Discrete metric. (X, d) where $X = (0, 3)$ and

$$d(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

Like a circle, or for the set $(1, 2, 3)$, an equilateral triangle.

14.2.2 Vector space

Set closed under addition and scalar multiplication as specified in detail requirements: commutativity, associativity, additive identity, additive inverse, associativity of scalar multiplication, distributivity, scalar multiplication identity. Vector spaces are groups (commutative groups) with scalars over a field that interact through distributive and associative properties.

Can represent functions into vectors.

Examples. (1) 2×3 real matrices are commutative form a commutative group, can add and scale, and fulfill all vector space properties, so is a vector space. (2) Real polynomials of degree 5 or less are a vector space. Can add them, and multiply by a real number and get another real polynomial of degree 5 or less. (3) $V = \{f(x) | f \text{ is continuous on } \mathcal{R}\}$. So $f(x)$ and $g(x)$ are continuous, and then so is $f(x) + g(x)$. Also, since $f(x)$ is continuous, $cf(x)$ is continuous. (4) All real polynomials, described using bases $(1, x, x^2, x^3, \dots, x^n)$ with a coefficient vector. (5) $x \in \mathcal{R}^n$, including $n = 1$ (i.e. the real number line) and $n = \infty$, (6) all real-valued functions $f: x \rightarrow \mathcal{R}$, (7) all bounded real-valued functions, (8) all continuous real-valued functions.

Maps BETWEEN vector spaces, i.e. transform vectors to another set, have

to maintain linearity: $T(ax + y) = aT(x) + T(y)$. Examples. (1) Matrix multiplication. (2) Derivatives of e.g. polynomials.

14.2.3 Normed vector spaces

Let $\mathcal{F} \in (\mathcal{R}, \mathcal{C})$. Let X be a $\mathcal{F}$ -vector space. $\|\cdot\|: X \rightarrow [0, \infty)$ is a norm if: (1) $\|x\| = 0 \Leftrightarrow x = 0$, (2) $\|\lambda x\| = |\lambda| \|x\|$, and (3) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality). Then, $(X, \|\cdot\|)$ is a normed space. If $\|\cdot\|$ is a norm for $\mathcal{F}$ -vector space X then $d_{\|\cdot\|}(x, y) = \|x - y\|$.

Examples. (1) For $x \in \mathcal{R}$, $\|x\| = |x|$ (so absolute value), (2) for $x \in \mathcal{R}^k$, Euclidean norm but there are others, (3) $x \in \mathcal{R}^\infty$, one example is the supremum of bounded values $\|x\| = \sup |x|$, and can extend this to all bounded functions, (4) for $C(0, 1) = (f : [0, 1] \rightarrow \mathcal{R})$ and $\|f\| = \int_0^1 |f(x)| dx$ but there are others.

14.2.4 Banach space

The normed vector space $(X, \|\cdot\|)$ where $(X, d_{\|\cdot\|})$ is complete. With completeness, convergence will lead to limits. So properties proved in Banach spaces will hold at limits.

Examples. $(\mathcal{R}^1, |\cdot|)$. $(X = (0), \|\cdot\|)$.

Define function $\phi : x \rightarrow \mathcal{R}_{++}$ and $\phi(x) > 0$. $f : x \rightarrow \mathcal{R}$ and $\|f\|_\phi = \sup \frac{|f(x)|}{\phi(x)}$. Let X be the set of all functions f s.t. $\|f\|_\phi$ exists and is $< M$ where $M \in \mathcal{R}$. So f is bounded in the ϕ -norm. If $\phi(x) = 1$ then it doesn't do anything. Depending on ϕ , characteristics change. Example. $f(x) = \alpha x + \beta$, $\phi(x) = |x| + 1$, then if $|\alpha| < M$ and $|\beta| < M$ (under the condition $\|f\|_\phi < M$) then f is bounded in the ϕ -norm. Can similarly ϕ -norm bound other types of functions, including quadratics and exponentials.

14.2.5 Sequences and convergence

Can have sequences and convergences of vectors as well as functions. Use ϵ - δ definitions. Some function sequences may be Cauchy but not converge to a continuous limit. Those that do are complete. Banach spaces are complete.

14.2.6 L^p space

$(l^p, \|\cdot\|_p)$ is a Banach space. $L^p = (\mathcal{N}, \mathcal{F})$ with $p \in [1, \infty)$ (note: $p = 1$ is the Manhattan metric, $p = 2$ is the Euclidean metric). $\|\cdot\|_p: l^p \rightarrow [0, \infty)$ with

$$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{\frac{1}{p}}$$

If all sequences $(x_n)_{n \in \mathcal{N}}$ in $\mathcal{F} \in (\mathcal{R}, \mathcal{C})$ s.t.

$$\sum_{n=1}^{\infty} |x_n|^p < \infty$$

(i.e. converges) then (x_n) is in L^p . Proof. L^p is a $\mathcal{F}$ -vector space as the sequence members are added and scaled component-wise and $\|\cdot\|_p$ is a norm. For completeness, let (x^k) be a collection of Cauchy sequences in L^p . Then any specific iteration across different sequences being less than the sum of the other differences across iterations:

$$|x_m^k - x_m^l|^p \leq \sum_{n=1}^{\infty} |x_n^k - x_n^l|^p = \|x^k - x^l\|_p^p$$

$\forall \epsilon > 0, \exists k \in \mathcal{N} \forall k, l \geq k, |x_m^k - x_m^l| \leq \|x^k - x^l\|_p < \epsilon$ and $\lim_{k \rightarrow \infty} x_m^k = \tilde{x} \in \mathcal{F}$. Choose $k \in \mathcal{N}$ s.t. $\forall k, l \geq K : \|x^k - x^l\|_p < \epsilon'$, and

$$\|x^k - \tilde{x}\|_p^p = \sum_{n=1}^{\infty} |x_n^k - \tilde{x}_n|^p = \lim_{N \rightarrow \infty} \sum_{n=1}^N |x_n^k - \tilde{x}_n|^p = \lim_{N \rightarrow \infty} \lim_{l \rightarrow \infty} \underbrace{\sum_{n=1}^N |x_n^k - x_n^l|^p}_{< (\epsilon')^p}$$

Therefore $\|x^k - \tilde{x}\|_p^p \leq (\epsilon')^p$, so $\|x^k - \tilde{x}\|_p \leq \epsilon' < \epsilon$ where ϵ can be chose appropriately. Now,

$$\tilde{x} = \underbrace{(\tilde{x} - x^k)}_{\in l^p} + \underbrace{x^k}_{\in l^p}$$

So $\tilde{x} \in l^p$.

14.2.7 Inner product spaces

$\langle \cdot, \cdot \rangle : X^2 \rightarrow \mathcal{F}$ is an inner product on X if: (i) $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0 \Leftrightarrow x = 0$, (ii) $\langle x, y \rangle = \langle y, x \rangle$ for $\mathcal{R}$ and $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for $\mathcal{C}$, and (iii) $\langle x, y_1 + y_2 \rangle = \langle x, y_1 \rangle + \langle x, y_2 \rangle$ and $\langle x, \lambda y \rangle = \lambda \langle x, y \rangle$. If $\langle \cdot, \cdot \rangle$ is an inner product, $\|x\|_{\langle \cdot, \cdot \rangle} = \sqrt{\langle x, x \rangle}$ defines a norm.

14.2.8 Hilbert spaces

An inner product space $(X, \langle \cdot, \cdot \rangle)$ where $(X, \|\cdot\|_{\langle \cdot, \cdot \rangle})$ is a Banach space. $\mathcal{R}^n$, $\mathcal{C}^n$, and L^2 (with $\langle x, y \rangle = \sum_i x_i y_i$) are all Hilbert spaces. To prove, show they are (i) Banach spaces, and (ii) inner product spaces.

14.2.9 Orthogonality

If $\langle x, y \rangle = 0$, x and y are orthogonal, or $x \perp y$. Vector subspaces can also be orthogonal. If $U, V \subseteq X$, $U \perp V$ if $x \perp y \forall x \in U$ and $\forall y \in V$.

Orthogonal complement. For $U \subseteq X$ the orthogonal complement of U is $U^\perp = \{x \in X \mid \langle x, u \rangle = 0, \forall u \in U\}$, and $U^\perp$ is always a subspace of X . Examples. (1) $(0)^\perp = X \Leftrightarrow X^\perp = (0)$. (2) $U \subseteq V$, then $U^\perp \supseteq V^\perp$. Proof. If $x \in V^\perp \Rightarrow \langle x, v \rangle = 0, \forall v \in V$ but since $U \subseteq V$, $\langle x, v \rangle = 0, \forall u \in U \Rightarrow x \in U^\perp$. (3) $x \perp y \Rightarrow \|x + y\|_{\langle \cdot, \cdot \rangle}^2 = \|x\|_{\langle \cdot, \cdot \rangle}^2 + \|y\|_{\langle \cdot, \cdot \rangle}^2$

14.3 Structure

14.3.1 Open and closed sets

ϵ -radius ball: $B_\epsilon(x) = \{y \in X \mid d(x, y) < \epsilon\}$. Boundary points on set A : $\partial A = \{x \in X \mid x \text{ is a boundary point for } A\}$.

Open set: $A \subseteq X$ is open if $x \in A$, $\exists B_\epsilon(x) \subseteq A$. Equivalently, $A \cap \partial A = \emptyset$. Closed set: A^c is open. Closure: $A \cup \partial A = \bar{A}$ (always closed).

Example. $X = (1, 3] \cup (4, \infty)$ and (X, d) is a metric space. $A = (1, 3] \subseteq X$ is an open set (as $B_{\epsilon < 1}(3)$ is one-sided and doesn't overlap with $A^c = (4, \infty)$) AND closed ($\epsilon \geq 1$ or A^c is open).

14.3.2 Dense sets

Let (X, d_x) be a metric space. Let $M \subseteq X$. M is dense in X if $\forall x \in X$, x is a limit point of M , i.e. $\forall r > 0 \exists m \in M$ s.t. $m \in B(x, r)$. So no matter how small the neighborhood of any point in X (can even be outside of M but still in X), there will be a part of m in the ϵ -neighborhood of that point. Example. $\mathcal{Q}$ is dense in $\mathcal{R}$.

14.3.3 Baire category theorem

If (X, d) is a complete metric space and (u_n) is a collection of open and dense sets, then $\bigcap_{n=1}^\infty u_n$ is dense. Intuitively, each dense set covers so much of the set that the intersection also does.

Proof. Show if $x \in X$ and $r > 0$, $B(x, r) \cap \bigcap u_n$, then $\exists y \in B(x, r)$ and $y \in u_n$. Fix x and $r > 0$, which must intersect with u_1 as u_n is dense. Then there must be a further $\overline{B(x_1, r_1)} \subseteq B(x, r) \cap u_1$ in the intersection space. Repeat this, so $\overline{B(x_2, r_2)} \subseteq B(x_1, r_1) \cap u_2$. More generally, $x_n \in B(x_n, r_n) \subseteq \overline{B(x_n, r_n)} \subseteq B(x_{n-1}, r_{n-1}) \cap u_n$. (x_n) is Cauchy convergent as the metric space is complete:

$$\lim_{n \rightarrow \infty} x_n \in \overline{B(x_m, r_m)} \subseteq B(x_{m-1}, r_{m-1}) \cap u_m \subseteq u_m$$

where $n > m$. Then $y \in \overline{B(x_1, r_1)} \subseteq B(x, r) \cap u_1 \subseteq B(x, r)$.

14.3.4 Completeness

Use sequences to describe metric spaces. (x_n) is convergent in metric space (X, d) if $\forall \epsilon > 0, \exists N \in \mathcal{N} \forall n \geq N d(x_n, \tilde{x}) < \epsilon$ (so $B_\epsilon(\tilde{x})$):

$$\lim_{n \rightarrow \infty} x_n = \tilde{x} \in X$$

$A \subseteq X$ is closed iff for every convergent sequence $(a_n) \subseteq A$,

$$\lim_{n \rightarrow \infty} a_n \in A$$

$(x_n) \subseteq X$ is a Cauchy sequence if $\forall \epsilon > 0, \exists N \in \mathcal{N} \forall n, m \geq N, d(x_n, x_m) < \epsilon$ (a generalization of convergent sequences, as it doesn't require the limit).

(X, d) is a complete (i.e. no holes) metric space if all Cauchy sequences converge.

Example. (X, d) where $X = (0, 3)$ and $d(x, y) = |x - y|$ is incomplete because $\lim_{n \rightarrow \infty} (x_n) = \lim_{n \rightarrow \infty} \frac{1}{n}$ isn't in $X = (0, 3)$.

Example. (X, d) where $X = [0, 3]$ and $d(x, y) = |x - y|$ is complete.

Example. The discrete metric is complete. Proof. Let $(x_n) \subseteq X$ be a Cauchy sequence. Choose $\epsilon = \frac{1}{2}$. Then $\exists N \in \mathcal{N}$ s.t. $\forall n, m \geq N, d(x_n, x_m) = 0 < \frac{1}{2}$, i.e. when $x_n = x_m$.

14.3.5 Compactness

Finiteness is for functions. Compactness is the analog for CONTINUOUS functions. If a set X is finite, $f : X \rightarrow \mathcal{R}^n$ is bounded. If a set X is compact, every continuous function $f : X \rightarrow \mathcal{R}^n$ is bounded.

Compactness in metric space (X, d) . $A \subseteq X$ is (sequentially compact) if for $(x_n) \subseteq A$ one finds a convergent subsequence (x_{n_k}) with $\tilde{x} = \lim_{k \rightarrow \infty} x_{n_k} \in A$.

If $A \subseteq X$ is compact, then A is closed and bounded (so $\exists x \in X$ s.t. $B_\epsilon(x) \supseteq A$). Prove by showing A is (1) closed and (2) bounded.

(1) A is closed. Let $(x_n) \subseteq A$ be a convergent sequence with limit $x_a \in X$. Since it's compact, there is convergent subsequence (x_{n_k}) with limit $x_b \in A$. Since the limit is unique, $x_a = x_b \in A$. So A is closed.

(2) A is bounded by contraposition. Assume A is not bounded. For $a \in A$ $\exists x_n \in A$ with $d(a, x_n) > n$. For any subsequence (x_{n_k}) and any point $b \in A$ $n < d(a, x_n) = d(a - b + b, x_n) \leq d(a, b) + d(b, x_n)$ by the triangle inequality. So, $n - d(a, b) \leq d(b, x_n)$. Then $d(b, x_n)$ can't be 0 $\forall b \in A$.

14.3.6 Arzela-Ascoli theorem

It can be shown that a compact set in Banach space of FINITE dimensions is equivalent to being closed and bounded. A compact set in Banach space of INFINITE dimensions induces a closed and bounded space, but being closed and bounded are necessary but not sufficient to induce a compact set. Examples. $A = \{f \in C[0, 1] \mid \|f\|_\infty \leq 1\}$ where the supremum norm is $\|f\|_\infty = \sup(|f(t)| \mid t \in [0, 1])$. There are sets of functions f that can't get close enough to each other in a δ - ϵ definition. Moreover, there are sequences that have no convergent subsequences, such as basis vectors in each of the infinite dimensions. However, with finite dimensions, there are convergent subsequences, as the sequences loop.

Uniform equicontinuity. Uniform continuity is for a single function. Uniform equicontinuity generalizes this to an entire SET of functions (i.e. sequence of functions), and takes pointwise convergence to uniform convergence. So one δ - ϵ is used for the entire set of functions.

Arzela-Ascoli theorem. Only applies to continuous functions. For any normed space, a set is compact if it is closed, bounded, and uniformly equicontinuous.

14.3.7 Contraction mapping

Banach space $C_\phi(A) = \{f : A \rightarrow B\}$ which is continuous and bounded. $T : X \rightarrow X$ is a contraction mapping. $\exists \beta \in [0, 1)$ and $\|T(x) - T(y)\| \leq \beta \|x - y\|$. So repeating, the norm $\rightarrow 0$ due to repeated shrinking.

14.3.8 Banach fixed point theorem

Contraction mapping $T : X \rightarrow X$ is a fixed point if $T(x) = x$.

Banach fixed point theorem: fixed point x is unique and $\|T^n x_0 - x\| \leq \beta^n \|x_0 - x\|$. Intuitively, maps have to contract to one point.

Proof. Take sequence $x_0, T(x_0) = x_1$, etc.

$$\|x_{n+1} - x_n\| \leq \beta \|x_n - x_{n-1}\| \leq \beta^n \|x_1 - x_0\|$$

by contraction mapping. By the triangle inequality:

$$\begin{aligned} \|x_n - x_m\| &\leq \|x_n - x_{n-1}\| + \|x_{n-1} - x_{n-2}\| + \dots + \|x_m - x_{m+1}\| \\ &\leq \beta^{n-1} \|x_1 - x_0\| + \beta^{n-2} \|x_1 - x_0\| + \dots + \beta^m \|x_1 - x_0\| = \|x_1 - x_0\| \sum_{m=0}^{n-1} \beta^m \\ &\leq \|x_1 - x_0\| \sum_{m=0}^{\infty} \beta^m \end{aligned}$$

Since this is a geometric series,

$$\lim \|x_1 - x_0\| \frac{\beta^m}{1 - \beta} \rightarrow 0$$

as $n, m \rightarrow \infty$. Since the space is Banach (complete), there is convergence: $x_n \rightarrow x$. Now show that $T(x) = x$:

$$\|T(x) - x\| \leq \|T(x) - T^n(x_0)\| + \|T^n(x_0) - x\|$$

by the triangle inequality. Then:

$$\leq \beta \|x - T^{n-1}(x_0)\| + \|T^n(x_0) - x\| = \beta \|x - x_{n-1}\| + \|x_n - x\| \rightarrow 0$$

The limit approaches 0 because of the limit proved above. It shows that $T(x) = x$. Prove uniqueness. Let $T(x) = x$ and $T(y) = y$,

$$\|x - y\| = \|T(x) - T(y)\| \leq \beta \|x - y\|$$

This can only be true if both sides are 0. So fixed point is unique.

Lemma. Let X' be a closed subset of X . So $x \in X'$ and $T(x) \in X'$. Then the unique fixed point of T contraction mapping is in X' . This is because T can't escape from the subset once inside.

Blackwell theorem. Conditions for contraction mapping. Under (1) monotonicity $f \leq g \rightarrow T(f) \leq T(g)$, (2) discounting $T(f + a\phi)(x) \leq Tf(x) + \beta a\phi(x)$.

14.3.9 Perron-Frobenius theorem

Perron Frobenius theorem: If all entries of a $n \times n$ matrix A are positive, it has a unique maximal eigenvalue. Its eigenvector has positive entries.

Proof. Consider the sphere $x_1^2 + x_2^2 + \dots + x_n^2 = 1$ (not the solid) with the non-negative 'quadrants' ($x_1 \geq 0, \dots, x_n \geq 0$) which is a closed, bounded set X . The matrix A defines a map $T(v) = \frac{Av}{|Av|}$ (normalized to 1) on X because the entries of the matrix are positive and therefore non-negative. Because they are positive, TX is contained in the interior of X . This map is a contraction, there exists $0 < k < 1$ s.t. $d(T(x), T(y)) \leq kd(x, y)$ where d is the geodesic sphere distance (the distance between two points on a sphere). Such a map has a unique fixed point v by Banach's fixed point theorem. This is the eigenvector $Av = \lambda v$ we were looking for. We have seen now that on X , there is only one eigenvector (because the Banach fixed point is unique). Every other eigenvector $Aw = \mu w$ must have a coordinate entry which is negative. Write $|w|$ for the vector with coordinates $|w_j|$. The computation

$$|\mu||w|_i = |\mu w_i| = \left| \sum_j A_{ij} w_j \right| \leq \sum_j |A_{ij}| |w_j| = \sum_j A_{ij} |w_j| = (A|w|)_i = \lambda |w|_i$$

shows that $|\mu|L \leq \lambda L$. So, $|\mu| \leq \lambda$.

Intuition. Consider an initial vector that has all components positive and real, and repeatedly apply the matrix to it. Since everything is positive, our iterations keep all components positive, indefinitely. However, iterating the vector like this is dominated by the largest magnitude eigenvalue.

14.4 Operators

14.4.1 Continuity

Continuity of functions mapping between metric spaces (X, d_x) and (Y, d_y) . $f : X \rightarrow Y$ is continuous if the preimage $f^{-1}(B)$ is open in X for all open sets $B \subseteq Y$. So an open X input maps to an open Y output in continuous functions. For metric spaces, equivalently there is sequential continuity: $\forall \tilde{x} \in X$ and $(x_n) \in X$ with limit $\tilde{x}$, $f(x_n) \rightarrow f(\tilde{x})$.

Examples. (1) If (X, d_x) is the discrete metric space and (Y, d_y) is any metric space, $f : X \rightarrow Y$ is continuous because the discrete metric space is open (a ball of $r, 1$, so $B_{r < 1}(x)$ is open; the discrete metric space is also closed because all complement sets are open). (3) (X, d_x) and (Y, d_y) are metric spaces and $y_0 \in Y$ is fixed. Then $f : X \rightarrow Y$ with $x \mapsto y_0$ is continuous. It is sequentially continuous because $f(x_n) \rightarrow f(\tilde{x})$ as $n \rightarrow m$.

Let $(X, \|\cdot\|)$ be a normed space and $Y = \mathcal{R}$ with standard metric. then $f : X \rightarrow \mathcal{R}$ with $x \mapsto \|x\|$ is continuous. Proof. Let $(x_n) \subseteq X$ be a sequence with limit $\tilde{x} \in X$. Then $f(x_n) = \|x_n\| = \|x_n - \tilde{x} + \tilde{x}\| \leq \|x_n - \tilde{x}\| + \|\tilde{x}\| = d(x_n, \tilde{x}) + f(\tilde{x})$ where $d(x_n, \tilde{x}) \rightarrow 0$ as $n \rightarrow m$. So:

$$\lim_{n \rightarrow \infty} f(x_n) \leq f(\tilde{x})$$

For the opposite direction, $f(\tilde{x}) = \|\tilde{x}\| = \|\tilde{x} - x_n + x_n\| \leq \|\tilde{x} - x_n\| + \|x_n\| = d(\tilde{x}, x_n) + f(x_n)$ where $d(x_n, \tilde{x}) \rightarrow 0$ as $n \rightarrow m$. So:

$$\lim_{n \rightarrow \infty} f(\tilde{x}) \leq f(x_n)$$

For the inner product space X , and $Y = \mathcal{C}$ with standard metric $x_0 \in X$ fixed, $f : X \rightarrow \mathcal{C}$ and $x \mapsto \langle x_0, x \rangle$ is continuous. Proof. Let $(x_n) \subseteq X$ be a sequence with limit $\tilde{x} \in X$. Then $|f(x_n) - f(\tilde{x})| = |\langle x_0, x_n \rangle - \langle x_0, \tilde{x} \rangle| = |\langle x_0, x_n - \tilde{x} \rangle|$ and by the Cauchy-Schwarz inequality, $\leq \|x_0\| \|x_n - \tilde{x}\| \rightarrow 0$ since the second term $\|x_n - \tilde{x}\| \rightarrow 0$ as $n \rightarrow \infty$. Analogously, $g : X \rightarrow \mathcal{C}$ and $x \mapsto \langle x, x_0 \rangle$ is continuous.

Orthogonal complements are always closed. Let $(X, \langle \cdot, \cdot \rangle)$ is an inner product space $U \subseteq X$. Then $U^\perp$ is closed. Proof. Let $(x_n) \subseteq U^\perp$ be a sequence with limit $\tilde{x} \in X$. Then $\langle x_n, u \rangle = 0 \forall u \in U$. Moreover, $\lim_{n \rightarrow \infty} \langle x_n, u \rangle = 0 \forall u \in U$. So, $\tilde{x} \in U^\perp$.

14.4.2 Linear operators

Linear operators between normed spaces: $T : X \rightarrow Y$ where the normed spaces are $(X, \|\cdot\|_x)$ and $(Y, \|\cdot\|_y)$. It is operator and not a function because it may map functions as inputs or outputs. T is linear if it (i) preserves linear structure (i.e. algebraic structure), and (ii) is continuous, i.e. preserves topological structure (so open sets). The derivative is a linear operator.

If X and Y are normed spaces, $T : X \rightarrow Y$ is linear if $T(x + \tilde{x}) = Tx + T\tilde{x}$ and $T(\lambda x) = \lambda Tx$. Further:

$$\|T\| = \|T\|_{X \rightarrow Y} = \sup \left(\frac{\|Tx\|_y}{\|x\|_x} \mid x \in X, x \neq 0 \right)$$

Instead of *max*, use *sup* as the quotient may be infinite in infinite-dimensions. T maps the length $\|x\|_x$ to length $\|Tx\|_y$. If $\|T\| < \infty$, then T is bounded. This concept of boundedness is different from bounded functions. T is not bounded only if X is infinite-dimensional.

$T : X \rightarrow Y$ between normed spaces X and Y . This is equivalent to: (A) T is continuous, (B) T is continuous at $x = 0$, and (C) T is bounded. (A) $\rightarrow$ (B): if T is continuous everywhere, it is certainly continuous at $x = 0$. (B) $\rightarrow$ (C): For sequence $(x_n) \subseteq X$ with $x_n \rightarrow 0$ and $Tx_n \rightarrow 0$, $\exists \delta > 0$ s.t. $\|Tx\|_y < 1 \forall x \in X$ with $\|x\|_x < \delta$. Proof. For all $n \in \mathbb{N}$, $x_n \in X$ with $\|x_n\| < \frac{1}{n}$ and $\|Tx_n\|_y \geq 1$ (so Tx converges to 0 but the image does not). Then:

$$\frac{\|Tx\|_y}{\|x\|_x} = \frac{\|Tx\|_y \frac{\delta}{2} \frac{1}{\|x\|_x}}{\|x\|_x \frac{\delta}{2} \frac{1}{\|x\|_x}} = \frac{\left\| T \left(\frac{\delta}{2} \frac{x}{\|x\|_x} \right) \right\|_y < 1}{\left\| \frac{\delta}{2} \frac{x}{\|x\|_x} \right\|_x = \frac{\delta}{2}} \leq \frac{2}{\delta}$$

Where the $\frac{\delta}{2}$ is used as $\|x\|_x < \delta$, and $\frac{x}{\|x\|_x}$ is a scaling to 1. Then:

$$\|T\| = \|T\|_{X \rightarrow Y} = \sup \left(\frac{\|Tx\|_y}{\|x\|_x} \mid x \in X, x \neq 0 \right) \leq \frac{2}{\delta} < \infty$$

(C) $\rightarrow$ (A). Let $(x_n) \subseteq X$ be convergent with limit $\tilde{x} \in X$. Then:

$$\|Tx_n - T\tilde{x}\|_y = \|T(x_n - \tilde{x})\|_y \leq \|T\| \|x_n - \tilde{x}\|_x \rightarrow 0$$

as $n \rightarrow \infty$. So Tx_n converges to $T\tilde{x}$.

Example. Characterize $T : X \rightarrow Y$ where $X = (C([0, 1], \mathcal{F}), \|\cdot\|_\infty)$ and $Y = (\mathcal{F}, |\cdot|)$ with $g \in X$ s.t. $g(t) \neq 0 \forall t \in [0, 1]$. $T_g(f) = \int_0^1 g(t)f(t)dt$ and

$$\|T_g\| = \sup \left(\frac{|T_g(f)|}{\|f\|_\infty} \mid f \in X, f \neq 0 \right)$$

$$\begin{aligned}
&= \sup(|T_g(f)| | f \in X, \|f\|_\infty = 1) \\
&= \sup \left(\left| \int_0^1 g(t)f(t)dt \right| | f \in X, \|f\|_\infty = 1 \right) \\
&= \sup \left(\left| \int_0^1 g(t)f(t)dt \right| \leq \int_0^1 |g(t)| \underbrace{|f(t)|}_{\leq \|f\|_\infty = 1} dt | f \in X, \|f\|_\infty = 1 \right) \\
&\leq \int_0^1 |g(t)| dt < \infty
\end{aligned}$$

So it's bounded. Now define

$$h(t) = \frac{\overline{g(t)}}{|g(t)|}$$

with $\|h\|_\infty = 1$. Then:

$$\|T_g\| \geq |T_g(h)| = \left| \int_0^1 g(t) \frac{\overline{g(t)}}{|g(t)|} dt \right| = \int_0^1 \frac{|g(t)|^2}{|g(t)|} dt = \int_0^1 |g(t)| dt$$

So:

$$\|T_g\| = \int_0^1 |g(t)| dt$$

14.4.3 Compact operators

A bounded linear operator mapping any finite-dimensioned normed spaces (often Banach spaces) $T : X \rightarrow Y$ is compact if the unit ball is closed: $\overline{T[B_1(0)]} \subseteq Y$.

Example. $T : C[0,1] \rightarrow C[0,1]$ for $k \in C[0,1]^2$ and f in the domain has the sup-norm $\|\cdot\|_\infty$ with:

$$(T_k f)(s) = \int_0^1 k(s,t)f(t)dt$$

with $s \in [0,1]$. k is uniformly continuous with $\forall \epsilon > 0 \exists \delta > 0 \forall (s_1, t_1)(s_2, t_2)$ with $\|(s_1, t_1) - (s_2, t_2)\| < \delta \rightarrow |k(s_1, t_1) - k(s_2, t_2)| < \epsilon$. $(Tf)(s)$ must be continuous, so

$$\begin{aligned}
|(T_k f)(s_1) - (T_k f)(s_2)| &= \left| \int_0^1 (k(s_1, t)f(t) - k(s_2, t)f(t))dt \right| \\
&\leq \int_0^1 \underbrace{|k(s_1, t) - k(s_2, t)|}_{< \epsilon} \underbrace{|f(t)|}_{\leq \|f\|_\infty} dt < \epsilon \|f\|_\infty
\end{aligned}$$

$\forall \epsilon > 0 \exists \delta > 0$ $s_1, s_2 \in [0,1]$ with $|s_1 - s_2| < \delta$. So it's continuous, and therefore bounded. Now need it to be uniformly equicontinuous and closed. For uniform

equicontinuity, with $A = T_k[B_1(0)]$, $\forall \epsilon > 0 \exists \delta > 0 \forall s_1, s_2 \in [0, 1] \forall g \in A$, $|s_1 - s_2| < \delta \rightarrow |g(s_1) - g(s_2)| < \epsilon$. So T_k is uniformly equicontinuous. To show closure, calculate the norm:

$$\begin{aligned} \|T_k\| &= \sup(\|T_k f\|_\infty \mid \|f\|_\infty = 1) \\ &= \sup(\underbrace{\sup_{s \in [0, 1]} \left| \int_0^1 k(s, t) f(t) dt \right|}_{\leq \|f\|_\infty}) \\ &\leq \sup(\underbrace{\sup_{s \in [0, 1]} \int_0^1 |k(s, t)|}_{\leq \|f\|_\infty = 1} \underbrace{|f(t)|}_{\leq \|f\|_\infty} dt \mid \|f\|_\infty) \\ &\leq \sup(\underbrace{\sup_{s \in [0, 1]} \int_0^1 |k(s, t)| dt}_{\leq \|k\|_\infty}) \end{aligned}$$

So it is bounded by the sup-norm of k . Unimportant what its exact value is, just as long as it is constant (which it is). By Arzela-Ascoli, $\overline{T_k[B_1(0)]}$ is compact, so T_k is a compact operator.

14.5 Operator theorems

The closed graph theorem, the open mapping theorem and the bounded inverse theorem are all equivalent.

14.5.1 Uniform boundedness

A sequence of functions f_n is pointwise bounded if $\forall x, |f_n(x)| \leq K(x)$. A sequence of functions is uniformly bounded if $|f_n(x)| \leq K$ where K applies to all x .

Banach-Steinhaus theorem. Let X and Y be normed spaces, with X being a Banach space (so complete). Define the linear bounded operator as $B(X, Y) = \{T : X \rightarrow Y \mid T \text{ is linear and bounded}\}$. Then for every subset $M \subseteq B(X, Y)$, M being bounded pointwise on X is equivalent to M being uniformly bounded. Equivalently, $\forall x \in X \exists C_x \geq 0 \forall T \in M \|Tx\|_Y \leq C_x$ is equivalent to $\exists C \geq 0 \forall T \in M \|T\|_{X \rightarrow Y} \leq C$. Further equivalently, $\sup \|Tx\|_Y < \infty \rightarrow \sup \|T\|_{B(X, Y)} < \infty$. Operator norms are hard to calculate, so if it's sufficient to use the norm of Tx , that's better.

Proposition: X and Y are normed spaces with X being a Banach space. Let $T_n \in B(X, Y) \forall n \in \mathcal{N}$ with $\lim_{n \rightarrow \infty} T_n x$ existing $\forall x \in X$. Then $T : X \rightarrow Y$ with $Tx = \lim_{n \rightarrow \infty} T_n x$ is linear and bounded (bounded implies continuity). Proof. Let $M = (T_n \mid n \in \mathcal{N})$ be bounded pointwise on X . Then $\exists C \geq 0$ with $\|T_n\| \leq C$ for all n . Then

$$\begin{aligned} \|T\|_{X \rightarrow Y} &= \sup(\underbrace{\|Tx\|_Y}_{\lim_{n \rightarrow \infty} \|T_n x\|_Y} \mid \|x\|_X = 1) \\ &= \lim_{n \rightarrow \infty} \sup(\|T_n x\|_Y \mid \|x\|_X = 1) = \lim_{n \rightarrow \infty} \|T_n\| \leq C \end{aligned}$$

14.5.2 Open mapping theorem

Open maps. Let (X, d_x) and (Y, d_y) be metric spaces. $f : X \rightarrow Y$ is open if $A \subseteq X$ is open in $X \rightarrow f(A) \subseteq Y$ open in Y . Examples. $f : \mathcal{R} \rightarrow \mathcal{R}$ with $x \mapsto x^3$ is open. $f : \mathcal{R} \rightarrow \mathcal{R}$ with $x \mapsto x^2$ is not open, as $A = (-2, 2) \rightarrow f(A) = [0, 4)$ (note: this isn't a linear operation).

Open mapping theorem (Banach-Schauder). Let X, Y be Banach spaces (and therefore, complete). For $T \in B(X, Y)$ (so linear and bounded), T being surjective is equivalent to T being an open map.

Intuition. If a linear map is open then it maps the open unit ball to an open ellipsoid centred at 0. By expanding the unit ball the ellipsoid increases with it (linearity of T) and eventually covers the whole space. The point of the theorem is that this is still valid in infinite dimensions. The strategy of most proofs is to show that the unit open ball maps to an open set that contains a small ball centered at 0. So by expanding them, the whole space is covered. An open map instead of a closed map is used as it generalizes better (e.g. topology).

Proof. Let $A : X \rightarrow Y$ be a surjective, continuous, and linear operator. Further, let $U = B_1^X(0)$ and $V = B_1^Y(0)$ be open unit balls around the origins. Then $X = \bigcup_{k \in \mathcal{N}} kU$. Prove surjectivity leads to open mapping, and open mapping leads to surjectivity.

Surjectivity leads to open mapping. Since A is surjective, $Y = A(X) = A(\bigcup_{k \in \mathcal{N}} kU) = \bigcup_{k \in \mathcal{N}} A(kU)$ due to linearity. Since Y is Banach (and therefore complete), apply the Baire category theorem: $\exists k \in \mathcal{N}, \overline{A(kU)} \neq \emptyset$. For $c \in Y$ and $r > 0$, $B_r^Y(c) \subseteq \overline{A(kU)} \neq \emptyset$. Let $v \in V$ then $c + rv \in B_r^Y(c) \subseteq \overline{A(kU)}$. By linearity, $rv \in \overline{A(kU)} - c \subseteq \overline{A(kU)} + \overline{A(kU)} \subseteq \overline{A(2kU)}$. Let $rL = 2k$ and with linearity again, $V \subseteq \overline{A(LU)}$. Then $\forall y \in Y$ and $\forall \epsilon > 0$, $\exists x \in X$ s.t. (1) $\|x\|_X \leq L\|y\|_Y$ and $\|y - Ax\|_Y < \epsilon$.

Open mapping leads to surjectivity. Show $V \subseteq A(2LU)$. By (1), $\exists x_1$ with $\|x_1\| < L$ and $\|y - Ax_1\| < 1/2$. Define (x_n) inductively. Assume (2) $\|x_n\| < \frac{L}{2^{n-1}}$ and $\|y - A(x_1 + x_2 + \dots + x_n)\| < \frac{1}{2^n}$. By (1), choose x_{n+1} s.t. $\|x_{n+1}\| < \frac{L}{2^n}$ and $\|y - A(x_1 + \dots + x_n) - A(x_{n+1})\| < \frac{1}{2^{n+1}}$. So (2) is satisfied for x_{n+1} . Let $s_n = x_1 + \dots + x_n$. From (2), (s_n) is Cauchy and since X is complete, it converges to some $x \in X$. By (2), $As_n \rightarrow y$ so $Ax = y$ by continuity of A . Also,

$$\|x\| = \lim_{n \rightarrow \infty} \|s_n\| \leq \sum_{n=1}^{\infty} \|x_n\| < 2L$$

where the inequality is from the triangle inequality and the $2L$ is from an infinite geometric series. This shows $y \in A(2LU)$ and that $V \subseteq A(2LU)$. So $A(B_1^X(x)) \supseteq B_{V/2L}^Y(y)$.

14.5.3 Bounded inverse theorem

Let X, Y be Banach spaces and $T \in B(X, Y)$. Then T being bijective means $T^{-1} \in B(Y, X)$ is continuous. Proof. T is bijective $\rightarrow T$ is surjective $\rightarrow T$ is an open map $\rightarrow T^{-1}$ is continuous.

Counterexample. Let $X = (C[0, 1], \|\cdot\|_\infty)$ and $Y = (f \in C'[0, 1], \|\cdot\|_\infty)$ (so Y is NOT complete) where $f(0) = 0$ for bijectivity. Then the operation:

$$(Tf)(t) = \int_0^t f(s)ds$$

is linear, bounded, and bijective.

$$\|Tf\|_\infty = \sup_{t \in [0, 1]} \left| \int_0^t f(s)ds \right| \leq \|f\|_\infty$$

Then $\|T\|_{X \rightarrow Y} \leq 1$. Let $f_k(t) = \sin(kt)$. Then $(Tf_k)(t) = \frac{1}{k}(1 - \cos(kt)) = g_k(t)$ with its maximum value being $\frac{2}{k}$. Now, $T^{-1}g_k = f_k$. So:

$$\|T^{-1}\|_{Y \rightarrow X} \geq \frac{\|T^{-1}g_k\|_\infty}{\|g_k\|_\infty} = \frac{\|f_k\|_\infty \leq 1}{\|g_k\|_\infty \leq \frac{2}{k}} \geq \frac{k}{2} \rightarrow \infty$$

14.5.4 Closed graph theorem

A linear operator with a closed graph will, under certain conditions, be continuous.

Graph of a map or function. $f : X \rightarrow Y$. $Gr f = \{(x, f(x)) \in X \times Y | y = f(x)\}$. X and Y can be topologies with $X \times Y$ being the product topology.

Closed graph. $D \subseteq X$ and $f : D \rightarrow Y$ then f has a closed graph in $X \times Y$ if $Gr f$ is a closed subset of $X \times Y$. Closed means it contains all limit points.

Closed graph theorem. If X and Y are Banach spaces, and $T : X \rightarrow Y$ is a linear operator, then T is continuous iff its graph is closed in the product topology $X \times Y$. Proof follows from the open mapping theorem.

14.5.5 Adjoints

Transposes are contravariant transformations: $AA' = A'A =$ multiplication of the scale changes, as the rotations cancel each other out. Conjugate transposes are required for complex vector spaces to ensure complex vector magnitudes multiply. Adjoints generalize conjugate transposes: $\langle Av, w \rangle = \langle v, A^*w \rangle$ where $A : v \rightarrow w$ (operator) and $A^* : w \rightarrow v$ (adjoint operator). If $A = A^*$, A is a self-adjoint operator.

14.6 Dual Spaces

14.6.1 Dual spaces

Vector space V will have a dual space $V^* = \mathcal{L}(V, \mathcal{R})$ where the latter is a linear functional. So it's a space of such linear transformations (maps) - and maps are called linear functionals. V and V^* is isomorphic. Can graph $\mathcal{L} = f$ against V to get a 2D graph.

Example 1. Take the $\mathcal{R}^2$ vector space. Define f_1 that takes any vector $V = (x, y)$ and returns the x-value and y-value: $f_1(V) = x$ and $f_2(V) = y$. f_1 and f_2 are themselves vectors. Similar to basis vectors in an alternate vector space, the dual space. Can add them together: $f_1 + f_2$ as a function that takes any vector and returns the sum of the x-coordinate and the y-coordinate. Can multiply them by numbers: $7 \times f_1$ as a function that takes any vector and returns the value of the x coordinate multiplied by 7. Can form any linear combination: $3.5 \times f_1 - 5 \times f_2$ as a function that takes any vector and returns the number 3.5 times the x coordinate minus 5 times the y coordinate.

Example 2. $V = \mathcal{R}^3$, $f : V \rightarrow \mathcal{R}$ where $f(x, y, z) = 2x + 3y - 5z$.

Other examples. Trace of a matrix, integration.

Non-examples. $f(x, y, z) = x^2 + y^2 + z^2$ because it's not linear (so no longer a vector space). $f(x, y) = (x + y, x - y)$ because the map is to $\mathcal{R}^2$.

Intuition. Can think of the elements of the vector space as column vectors, and the elements of the dual as row vectors. The dot product results in a map to $\mathcal{R}$ so each row vector gives a map from V to $\mathcal{R}$.

Purpose. There are lots of reasons for this; here is one: Remember how important it is, when you solve systems of linear equations, to add equations together, or to multiply them by scalars (here I am referring to all the steps you typically make when performing Gaussian elimination on a collection of simultaneous linear equations)? Well, under the dictionary above between linear equations and linear functionals, these processes correspond precisely to adding together linear functionals, or multiplying them by scalars. If you ponder this for a bit, you can hopefully convince yourself that making the set of linear functionals a vector space is a pretty natural thing to do.

14.6.2 Dual basis

Can graph $\mathcal{L} = f$ against V to get a 2D graph. From this, can back out the bases $f_i(v_{ij}) = 1$ (and 0 everywhere else) where $i = 3$ if $V \in \mathcal{R}^3$ (an indicator function). This is why the vector and its dual space are isomorphic.

$f \in V^*$ can be decomposed as $f = f(v_1)f_1 + f(v_2)f_2 + \dots + f(v_n)f_n$ where

f_i are the basis vectors.

14.6.3 Double dual space

It turns out that the isomorphism between the initial vector space V and its double dual is canonical as we shall see right away. Let $v \in V, \phi \in V^*$ and $\bar{v} \in V^{**}$. We can now define a linear map: $\bar{v}(\phi) = \phi(v)$. Mind that we are talking about finite dimensional vector spaces V , i.e. $\dim(V) < \infty$.

The dual is intuitively the space of "rulers" (or measurement-instruments) of our vector space. Its elements measure vectors. This is what makes the dual space and its relatives so important in Differential Geometry, for instance. This immediately motivates the study of the dual space. For motivations in other areas, the other answers are quite well-versed.

This also happens to explain intuitively some facts. For instance, the fact that there is no canonical isomorphism between a vector space and its dual can then be seen as a consequence of the fact that rulers need scaling, and there is no canonical way to provide one scaling for space. However, if we were to measure the measure-instruments, how could we proceed? Is there a canonical way to do so? Well, if we want to measure our measures, why not measure them by how they act on what they are supposed to measure? We need no bases for that. This justifies intuitively why there is a natural embedding of the space on its bidual. (Note, however, that this fails to justify why it is an isomorphism in the finite-dimensional case).

14.6.4 Homomorphism and isomorphism

Homomorphism. Map that preserves mathematical structure. Example 1. a linear map $f(\lambda x) = \lambda f(x)$ and $f(x + x') = f(x) + f(x')$ preserves vector space structure, so a homomorphism in vector spaces is a linear map. Example 2. (X, d_x) and (Y, d_y) are two metric spaces and $f : X \rightarrow Y$, then $d_x(x, x') \geq d_y(f(x), f(x'))$ or $d_x(x, x') \leq d_y(f(x), f(x'))$ are homomorphisms, as they preserve structure.

Isomorphism. A homomorphism which is also bidirectional. So: homomorphism + bijectivity, but also need + inverse also being homomorphic to ensure bijectivity. Example 1. Linear maps in vector spaces are also isomorphic if the inverse exists (if the two spaces are of the same dimension, the inverse will exist). Example 2. Polynomials of degree 2 or less and $\mathcal{R}^3$ are isomorphic. Example 3. (X, d_x) and (Y, d_y) are two metric spaces and $f : X \rightarrow Y$, then $d_x(x, x') = \lambda d_y(f(x), f(x'))$ is isomorphic. If $\lambda = 1$, it is isometric isomorphic. Example 4. A Banach space is isometric isomorphic if it is linear, bijective, and $\|f(x)\|_y = \|x\|_x$. Example 5. The right-shift operator $S_r : l^p(\mathcal{N}) \rightarrow l^p(\mathcal{N})$ which shifts $(x_1, x_2, \dots) \rightarrow (0, x_1, \dots)$ is not isomorphic as it's not bijective. However, $S : l^p(\mathcal{Z}) \rightarrow l^p(\mathcal{Z})$ which shifts $(\dots x_{-1}, x_0, x_1 \dots) \rightarrow (\dots x_0, x_1, x_2 \dots)$ is isomorphic

(and isometric isomorphic).

14.6.5 Riesz representation theorem

Every linear functional can be represented as an inner product. Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space. For each continuous linear map $l : X \rightarrow \mathcal{F}$ (a continuous linear functional), there is exactly one (so it's bijective) vector $x_l \in X$ s.t. $l(x) = \langle x_l, x \rangle \forall x \in X$ and operator norm $\|l\|_{X \rightarrow \mathcal{F}} = \|x_l\|_x$. Only holds in Hilbert spaces.

Prove: (1) existence, (2) uniqueness. Then (3) calculate operator norm.

(1) Existence. The kernel $\ker(l) = X$ is $x_l = 0$ as only the 0-vector sends x to $\mathcal{F} = 0$. $\ker(l) \neq X$ is $x_l \in \ker(l)^\perp$, and it doesn't subsume (0) because $\ker(l)$ is closed and orthogonal projections exist in Hilbert spaces. The kernel is the pre-image of the closed set (0) and with continuity, the kernel is closed. Choose $\hat{x} \in \ker(l)^\perp$ with normalization $\|\hat{x}\|_x = 1$ and set the scaling factor as $x_l = \overline{l(\hat{x})}\hat{x}$ (which will be validated below). Now, with linearity:

$$\begin{aligned} l(x) &= \left(x - \frac{l(x)}{l(\hat{x})}\hat{x} + \frac{l(x)}{l(\hat{x})}\hat{x} \right) = l \left(x - \frac{l(x)}{l(\hat{x})}\hat{x} \right) + l \left(\underbrace{\frac{l(x)}{l(\hat{x})}\hat{x}}_{\lambda} \right) \\ &= l(x) - \frac{l(x)}{l(\hat{x})}l(\hat{x}) + l(\lambda\hat{x}) = 0 + l(\lambda\hat{x}) \end{aligned}$$

This also means that the term in brackets is 0:

$$l \left(x - \frac{l(x)}{l(\hat{x})}\hat{x} \right) = 0 = l(0)$$

Now, due to normalization:

$$\begin{aligned} l(x) &= \lambda l(\hat{x}) \cdot 1 = \lambda l(\hat{x}) \langle \hat{x}, \hat{x} \rangle = \lambda \langle l(\hat{x})\hat{x}, \hat{x} \rangle \\ &= \langle x_l, \lambda\hat{x} \rangle = \langle x_l, \underbrace{\lambda\hat{x} - x}_{=0 \text{ as above}} + x \rangle = \langle x_l, x \rangle \end{aligned}$$

So $l(x) = \langle x_l, x \rangle$ exists.

(2) Uniqueness. Assume $x_l, \hat{x}_l \in X$. Then $l(x) = \langle x_l, x \rangle = \langle \hat{x}_l, x \rangle$ so $\langle x_l - \hat{x}_l, x \rangle = 0 \forall x \in X$ by factoring out x . So it gives a 0 vector no matter what x is. Then $\langle x_l - \hat{x}_l, x_l - \hat{x}_l \rangle = 0$. So, $x_l = \hat{x}_l$.

(3) Operator norm. use a unit vector to remove fractions:

$$\|l\| = \sup(|l(x)| \|x\|_x) = 1$$

$$\|l\| = \sup(\underbrace{|\langle x_l, x \rangle|}_{\leq \|x_l\|_x \|x\|_x = 1}) \leq \|x_l\|$$

Further, just add a unit vector to get the lower bound:

$$\|l\| \geq \left| l \left(\underbrace{\frac{x_l}{\|x_l\|}}_{=1} \right) \right| = \left| \langle x_l, \frac{x_l}{\|x_l\|} \rangle \right| = \|x_l\|$$

Where the last step is due to $\langle x_l, x_l \rangle = \|x_l\|^2$ as it is in Hilbert space.

14.6.6 Young's inequality

For $a, b > 0$,

$$a \cdot b \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

Proof. Jensen's inequality with $f : x \mapsto e^x$ and $\lambda = \frac{1}{p}$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

$$f\left(\frac{1}{p}\log(a^p) + \frac{1}{p'}\log(b^{p'})\right) \leq \frac{1}{p}f(\log(a^p)) + \frac{1}{p'}f(\log(b^{p'}))$$

$$f(\log a + \log b) \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

$$e^{\log ab} \leq \frac{e^{\log a^p}}{p} + \frac{e^{\log b^{p'}}}{p'}$$

$$a \cdot b \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

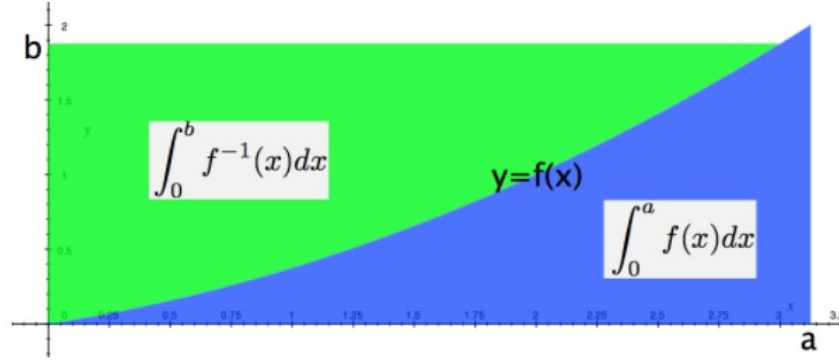
Intuition and alternate proof. For any increasing $f(x)$,

$$ab \leq \int_0^a f(x)dx + \int_0^b f^{-1}(y)dy$$

The inequality is an equality if $b = f(a)$. Young's inequality: $f(x) = x^{p-1}$. $\frac{1}{p} + \frac{1}{p'} = 1 \rightarrow \frac{1}{p-1} = p' - 1$. Then:

$$a \cdot b \leq \int_0^a x^{p-1}dx + \int_0^b y^{1/(p-1)=p'-1}dy$$

$$a \cdot b \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$



14.6.7 Holder's inequality

Generalization of Cauchy-Schwarz to L^p . For $x \in \mathcal{F}^n$ and $p, p' \in (1, \infty)$ (p' is the Holder conjugate) with $\frac{a^p}{p} + \frac{b^{p'}}{p'} = 1$, and

$$\|x\|_q = \left(\sum_{j=1}^n |x_j|^q \right)^{\frac{1}{q}}$$

where $q \in [1, \infty)$, and $(xy)' = (x_1 y_1, x_2 y_2, \dots, x_n y_n)$. Then Holder's inequality: $\|xy\|_{q=1} \leq \|x\|_p \|y\|_{p'}$. When $p = p' = 2$, it is Cauchy-Schwarz.

Proof. If $x = 0$ or $y = 0$, the inequality is satisfied. For other cases:

$$1 \geq \frac{1}{\|x\|_p \|y\|_{p'}} \|xy\|_1 = \frac{1}{\|x\|_p \|y\|_{p'}} \sum_{j=1}^n |x_j y_j| = \sum_{j=1}^n \frac{|x_j| |y_j|}{\|x\|_p \|y\|_{p'}}$$

Apply Young's inequality:

$$\leq \sum_{j=1}^n \frac{1}{p} \frac{|x_j|^p}{\|x\|_p^p} + \sum_{j=1}^n \frac{1}{p'} \frac{|y_j|^{p'}}{\|y\|_{p'}^{p'}} = \frac{1}{p} + \frac{1}{p'} = 1$$

14.6.8 Minkowski's inequality

Triangle inequality in L^p space:

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

$\forall x, y \in l^p(\mathcal{N})$ with $p \in [1, \infty)$. Proof. For $p = 1$, this is simply the triangle inequality. For $p > 1$,

$$\|x + y\|_p^p = \sum_{j=1}^{\infty} |x_j + y_j|^p = \lim_{n \rightarrow \infty} \sum_{j=1}^n |x_j + y_j|^p$$

Take the bracketed term:

$$(|x_j| + |y_j|)^p = (|x_j| + |y_j|)(|x_j| + |y_j|)^{p-1}$$

$$\underbrace{|x_j|}_{a_j} \underbrace{(|x_j| + |y_j|)^{p-1}}_{b_j} + \underbrace{|y_j|}_{c_j} \underbrace{(|x_j| + |y_j|)^{p-1}}_{b_j}$$

Apply Holder's inequality with the definition of p-norms:

$$\|b\|_{p'} = \left(\sum_{j=1}^n |b_j|^{p'} \right)^{\frac{1}{p'}} = \left(\sum_{j=1}^n (|x_j| + |y_j|)^{p-1} \right)^{\frac{1}{p'}}$$

Use Holder's inequality with $\frac{1}{p} + \frac{1}{p'} \Leftrightarrow \frac{p}{p-1} = p'$:

$$= \left(\sum_{j=1}^n (|x_j| + |y_j|)^p \right)^{\frac{1}{p'}}$$

Take the inner term:

$$\sum_{j=1}^n (|x_j| + |y_j|)^p \leq \|a\|_p \|b\|_{p'} + \|c\|_p \|b\|_{p'}$$

$$= (\|a\|_p + \|c\|_p) \left(\sum_{j=1}^n (|x_j| + |y_j|)^p \right)^{\frac{1}{p'}}$$

Divide both sides by the last term:

$$\left(\sum_{j=1}^n (|x_j| + |y_j|)^p \right)^{1 - \frac{1}{p'}} \leq \left(\sum_{j=1}^n |a_j x_j|^p \right)^{\frac{1}{p}} + \left(\sum_{j=1}^n |c_j y_j|^p \right)^{\frac{1}{p}}$$

As $n \rightarrow \infty$,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

14.7 Dual space of a normed space is a Banach space

The dual space of X is $X' = \{l : X \rightarrow \mathcal{F} \mid l \text{ is linear and bounded}\}$. For Hilbert spaces, from the Riesz representation theorem, there is a unique linear functional, and the operator norm $\|l\|_{X \rightarrow \mathcal{F}} = \|x_l\|_x$, so $X = X'$ and it is an isometric isomorphism. But this isn't true in general for non-Hilbert Banach spaces.

If X is a normed space, then $(X', \|\cdot\|_{X \rightarrow \mathcal{F}})$ is always a Banach space. Proof. Need to show completeness. Let $(l_k) \subseteq X'$ be a Cauchy sequence. This means $\forall \epsilon > 0, \exists N \in \mathcal{N}$, indices $\forall n, m \geq N$:

$$\|l_n - l_m\|_{X \rightarrow \mathcal{F}} < \epsilon$$

But since operator norms are defined by supremums:

$$\frac{1}{\|x\|_x} |l_n(x) - l_m(x)| \leq \|l_n - l_m\|_{X \rightarrow \mathcal{F}} < \epsilon$$

So $(l_k(x)) \subseteq \mathcal{F}$ (i.e, with the (x)) is a Cauchy in the Banach space. Since $\mathcal{F}$ is complete, $l(x)$ is complete:

$$\lim_{k \rightarrow \infty} l_k(x) = l(x)$$

To prove this, show (1) l is linear, (2) l is bounded, and (3) $\|l_k - l\|_{X \rightarrow \mathcal{F}} \rightarrow 0$ as $k \rightarrow \infty$. (1) l is linear as defined. (2) l is bounded as defined, and because it's Cauchy. Further:

$$\|l_n\|_{X \rightarrow \mathcal{F}} \leq \underbrace{\|l_n - l_N\|_{X \rightarrow \mathcal{F}}}_{< \epsilon} + \underbrace{\|l_N\|_{X \rightarrow \infty}}_{= \text{constant}} \leq \epsilon + \text{constant}$$

Can show that the sup-norm is bounded:

$$|l(x)| = \left| \lim_{k \rightarrow \infty} l_k(x) \right| = \lim_{k \rightarrow \infty} |l_k(x)| \leq \underbrace{\lim_{k \rightarrow \infty} \|l_k\|_{X \rightarrow \mathcal{F}}}_{\leq \epsilon + \text{constant}} \|x\|_x$$

So:

$$\frac{1}{\|x\|_x} |l(x)| \leq \lim_{k \rightarrow \infty} \|l_k\|_{X \rightarrow \mathcal{F}} \leq \epsilon + \text{constant}$$

$$\|l\|_{X \rightarrow \mathcal{F}} < \infty$$

(3) $\forall \epsilon > 0 \exists N \in \mathcal{N}$ s.t. $\forall n, m > N$,

$$\frac{1}{\|x\|_x} |l_n(x) - l_m(x)| < \epsilon$$

$$\frac{1}{\|x\|_x} \left| l_n(x) - \underbrace{\lim_{m \rightarrow \infty} l_m(x)}_{l(x)} \right| \leq \epsilon$$

Apply sup as it's an operator norm:

$$\sup \frac{1}{\|x\|_x} |l_n(x) - l(x)| < \epsilon$$

$$\|l_n - l\|_{X \rightarrow \mathcal{F}} < \epsilon$$

14.7.1 L^p dual space is isometric isomorphic

Let $X = l^p(\mathcal{N})$ with $p \in (1, \infty)$. Then its dual space $l^{p'}(\mathcal{N})$ with $p' \in (1, \infty)$ is isometric isomorphic (as it is a Banach space). The p' is a Holder conjugate: $\frac{1}{p} + \frac{1}{p'} = 1$, as the vector (Banach) space and dual space (also a Banach space) are combined and mapped to the scalar field.

Let $T : l^{p'} \rightarrow l^p$. So Tx maps x in the dual space to the vector space. Assume $y \in X$. Then $(Tx)(y)$ maps x from the dual space to the vector space and combined with y . This results in the inner product:

$$(Tx)(y) = \sum_{k=1}^{\infty} y_k x_k$$

Proof. Show that T is (1) well-defined, (2) linear, (3) bounded, (4) surjective, (5) isometric.

(1) T is well defined. Use the triangle inequality and then Holder's inequality:

$$|(Tx)(y)| \leq \lim_{n \rightarrow \infty} \sum_{k=1}^n |y_k x_k| \leq \|y\|_p \|x\|_{p'} < \infty$$

So $|(Tx)(y)| < \infty$ and therefore bounded.

(2) T is linear, as it's an inner product.

(3) T is bounded. See (1). Alternatively:

$$\|Tx\|_{l^p(\mathcal{N}) \rightarrow \mathcal{F}} = \sup(\underbrace{|(Tx)(y)|}_{\leq \|y\|_p \|x\|_{p'}} \mid \|y\|_p = 1) \leq \|x\|_{p'}$$

with $\|y\| = 1$ so $\|T\| \leq 1$.

(4) Let $y' \in l^p$ and the unit vector $e_k = (0, 0, 0, \dots, 0, 1, 0, 0, \dots)$ with the 1 at the k th element. Define $x_k = y'(e_k)$ and $x = (x_k)$. Is $x \in l^{p'}$ with $Tx = y'$?

$$\sum_{k=1}^n |x_k|^{p'} = \sum_{k=1}^n x_k t_k$$

Where

$$t_k = \begin{cases} \frac{|x_k|^{p'}}{x_k}, & x_k \neq 0 \\ 0, & x_k = 0 \end{cases}$$

Then:

$$= \sum_{k=1}^n t_k y'(e_k) = y'(\sum_{k=1}^n t_k e_k) \leq \|y'\| \left\| \sum_{k=1}^n t_k e_k \right\|_p$$

Where $t_k e_k = (t_1, t_2, \dots, t_k, 0, 0, \dots)$. So:

$$\begin{aligned} &= \|y'\| \left(\sum_{k=1}^n |t_k|^p \right)^{1/p} = \|y'\| \left(\sum_{k=1}^n \frac{|x_k|^{p'}}{|x_k|} \right)^{1/p} \\ &= \|y'\| \left(\sum_{k=1}^n |x_k|^{(p'-1)p} \right)^{1/p} = \|y'\| \left(\sum_{k=1}^n |x_k|^{p'} \right)^{1/p} \end{aligned}$$

So:

$$\|x\|_{p'} \leq \|y'\|_{l^p(\mathcal{N}) \rightarrow \mathcal{F}}$$

as $n \rightarrow \infty$. For $y \in l^p(\mathcal{N})$:

$$(Tx - y')(y) = (Tx - y') \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n y_k e_k \right) = \lim_{n \rightarrow \infty} (Tx - y') \left(\sum_{k=1}^n y_k e_k \right)$$

due to continuity. Further, due to linearity,

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n y_k (Tx - y')(e_k) = 0$$

as $Tx - y' = 0$ due to $x_k = y'(e_k)$. So it's surjective.

(5) T is isometric isomorphic. Using results from (3) and (4),

$$\|Tx\| \leq \|x\|_{p'} \leq \|y'\|_{l^p(\mathcal{N}) \rightarrow \mathcal{F}} = \|Tx\|_{l^p(\mathcal{N}) \rightarrow \mathcal{F}}$$

And since the first and last parts are the same, the inequalities are actually equalities. So it's isometric isomorphic.

14.7.2 Hahn-Banach

Addresses whether there are “enough” continuous linear functionals on X to separate the points of X . The answer is yes. This is a kind of analogue, for continuous linear functionals on a normed linear space X , of Urysohn's lemma for general continuous functions on an arbitrary metric space X). There is also an extension theorem for continuous linear functionals defined on a proper linear subspace of X . This is a kind of analogue of the Tietze extension theorem for general continuous functions defined on a proper closed subset of an arbitrary metric space X .

X' is large and therefore can use it to analyze X . Theorem. Let $U \subseteq X$ be a subspace, and an element of its dual space $u' : U \rightarrow \mathcal{F}$ a continuous linear functional. Then $\exists x' : X \rightarrow \mathcal{F}$ (another continuous linear functional) with $x'(u) = u'(u) \forall u \in U$ with $\|x'\|_{x'} = \|u'\|_{u'}$.

(A) Let $(X, \|\cdot\|_x)$ be normed space. Then $\forall x \in X, x \neq 0, \exists x' \in X'$ with $\|x'\|_{x'} = 1$ and $x'(x) = \|x\|_x$. Proof. Since x is just a vector, $u' : U \rightarrow \mathcal{F}$ with $U = \text{span}(X)$ (the 1D subspace), so $\lambda \cdot x \mapsto \lambda\|x\|$, a continuous linear functional. Then by Hahn-Banach, $x' : X \rightarrow \mathcal{F} \ x'(x) = u'(x) = \|x\|_x$ ($\lambda = 1$). It also follows that $\|x'\|_{x'} = \|u'\|_{u'} = 1$.

(B) X' separates points of X . For $x_1, x_2 \in X$ and $x_1 \neq x_2 \exists x' \in X'$ with $x'(x_1) \neq x'(x_2)$. Proof. Let $x = x_2 - x_1$. Then by (A), $x'(x) = \|x\|_x \neq 0$ and $x'(x_2) - x'(x_1) \neq 0 \rightarrow x'(x_2) \neq x'(x_1)$.

(C) $\forall x \in X \ \|x\|_x = \sup(|x'(x)| |x' \in X', \|x'\| = 1)$. Proof.

$$\|x'\|_{x'} \geq \frac{|x'(x)|}{\|x\|_x}$$

Apply supremum to both sides:

$$1 = \sup_{\|x'\|=1} \|x'\|_{x'} \geq \sup_{\|x'\|=1} \frac{|x'(x)|}{\|x\|_x}$$

Rearrange:

$$\|x\|_x \geq \sup_{\|x'\|=1} |x'(x)|$$

Apply $\|x\|_x = x'(x)$ from (A):

$$\|x\|_x \leq \sup_{\|x'\|=1} |x'(x)|$$

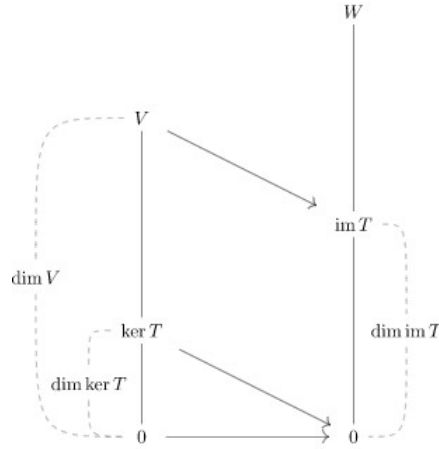
(D) Let $U \subseteq X$ be a closed subspace, $x \in X$ and $x \notin U$. $\exists x' \in X'$ with $x'(u) = 0$ and $x'(x) \neq 0$. Proof. $\frac{X}{U} = ([x] = x+U | x \in X, u \in U)$ is the quotient space (set of all affine subsets of X parallel to U). Then $\|[x]\|_{X/U} = \inf_{u \in U} \|x+u\|_x$ which is

a normed space $(\frac{X}{U}, \|\cdot\|_{X/U})$. Then by (A), $\exists y' \in (\frac{X}{U})'$ a linear functional with $y'([x]) \neq 0$. Define $x' \in X'$ by $x'(z) = y'([z])$ for $z \in X$. Then x' is a bounded linear functional, inputting any vector of u results in $= 0$, inputting x results in $\neq 0$.

14.8 Spectral theory

Generalization of eigenvalues. For $\lambda \in \mathbb{C}^n, \exists x \in \mathbb{C}^n \setminus \{0\} : Ax = \lambda x \Leftrightarrow (A - \lambda I)x = 0$. So $\ker(A - \lambda I) \neq 0 \Leftrightarrow \text{map } x \mapsto (A - \lambda I)x \text{ is not injective because the kernel is bigger than the 0-space (so excludes } \ker(A - \lambda I) \neq 0)$.

Rank-nullity theorem. If $M \in \mathbb{C}^n, \text{Rank} + \text{nullity} = n \rightarrow \dim(\text{Range}(M)) + \dim(\ker(M)) = n$.



Problematic that rank-nullity doesn't hold for infinite dimensions. So need new approach.

Let X be a complex Banach space. Then $T : X \rightarrow X$ be a bounded linear operator. The spectrum of T is all the eigenvalues: $\sigma(T) = \{\lambda \in \mathbb{C} \mid (T - \lambda I) \text{ is not bijective}\}$. The resolvent of T is $\rho(T) = \{\lambda \in \mathbb{C} \mid (T - \lambda I) \text{ is bijective and } (T - \lambda I)^{-1} \text{ is bounded}\}$, which is invertible. Since this is in Banach space, apply the bounded inverse theorem, s.t. $\sigma(T) = \mathbb{C} \setminus \rho(T)$ so that the spectrum is the complement set of the resolvent.

In finite dimensions, there is just the point spectrum. In infinite dimensions, there is this disjoint union: $\sigma(T) = \sigma_p(T) \cup \sigma_c(T) \cup \sigma_r(T)$ where the latter two are only present in infinite dimensions. The point spectrum $\sigma_p(T) = \{\lambda \in \mathbb{C} \mid (T - \lambda I) \text{ is not injective}\}$ is the non-trivial kernel, which produces eigenvalues (the classical case). The continuous spectrum is $\sigma_c(T) = \{\lambda \in \mathbb{C} \mid (T - \lambda I) \text{ is injective but not surjective with } \overline{\text{Range}(T - \lambda I)} = X\}$ (so is ALMOST surjective). The usually empty residual set is $\sigma_r(T) = \{\lambda \in \mathbb{C} \mid (T - \lambda I) \text{ is injective but not surjective with } \overline{\text{Range}(T - \lambda I)} \neq X\}$.

14.8.1 Linear algebra

Let A be an $n \times n$ matrix. Then its characteristic polynomial is calculated from:

$$\begin{aligned} p(\lambda) &= |A - \lambda I| = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_0) \\ &= (-1)^n (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n) \end{aligned}$$

Now set $\lambda = 0$

$$p(0) = |A| = (-1)^n (-1)^n \lambda_1 \lambda_2 \dots \lambda_n$$

So the determinant equals the product of the eigenvalues.

The matrix $\text{diag}(\lambda_j)$ has as its eigenvalues (λ_j) (just the basis vector stretches

and compressions). Algebraically, $\text{diag}(\lambda_j - \lambda)$ so $p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$ which are all the eigenvalues. So the diagonal matrix of the eigenvalues will produce the matrix spectrum.

14.8.2 Examples

X is a complex Banach space and T is a bounded linear operator. $\lambda \in \sigma(T)$ iff $(T - \lambda I) = (T - \lambda)$ is uninvertible.

Example 1. Finite-dimension Banach space. $X = \mathcal{C}^n$. $Tx = \text{diag}(\lambda_j)x_j$ so $\sigma(T) = (\lambda_j) = \sigma_p(T)$. The eigenvectors are unit basis vectors.

Example 2. Infinite-dimension Banach space. $X = l^p(\mathcal{N})$ where $p \in [1, \infty)$. $Tx = \text{diag}(\lambda_j)x$ with $j \in [1, \infty)$. To get a bounded operator, $\sup_{j \in \mathcal{N}} |\lambda_j| < \infty$.

$T : l^p(\mathcal{N}) \rightarrow l^p(\mathcal{N})$ $(Tx)_j = \lambda_j x_j$. Let $e_1 = (1, 0, 0, \dots)$ be an eigenvector with λ_1 , etc. Then $\sigma(T) \supseteq (\lambda_j) = \sigma_p(T)$ (it's not an equality because there are accumulation points in the infinite λ_j s which prevents 'completeness').

Let $\mu \in \mathcal{C}$ be a number with $\mu \notin (\lambda_j)$ but $\mu \in \overline{(\lambda_j)}$ (e.g. $\lambda_j = \frac{1}{j}$ then $\mu = 0$). With a Banach (complete) space, $T - \mu$ is always injective. Show that $T - \mu$ isn't surjective. Proof by contradiction. Assume $T - \mu$ is surjective. Then $T - \mu$ is bijective and by the bounded inverse theorem, $(T - \mu)^{-1}$ which is bounded. Then, $\|(T - \mu)^{-1}\| \geq \|(T - \mu)^{-1}e_j\|_{l^p} = \|(\lambda_j - \mu)^{-1}e_j\|_{l^p} = |(\lambda_j - \mu)^{-1}| = \frac{1}{|\lambda_j - \mu|} \rightarrow \infty$.

Result:

$$\sigma(T) = \underbrace{(\lambda_1 \dots \lambda_n)}_{\sigma_p(T)} \cup \underbrace{(\mu \in \mathcal{C} | \mu \notin (\lambda_j) \wedge \mu \in \overline{(\lambda_j)})}_{\sigma_c(T) \cup \sigma_r(T)}$$

14.8.3 Properties

(1) Closed and open sets. $\sigma(T) = (\lambda \in \mathcal{C} | (T - \lambda) \text{ not invertible})$ is closed, and $\rho(T) = (\lambda \in \mathcal{C} | (T - \lambda) \text{ invertible})$ is open.

(2) $\rho(T) \rightarrow B(X)$ where B is bounded. $\lambda \mapsto (T - \lambda)^{-1}$ is analytical (locally can be expressed as a Taylor series).

(3) $\lambda \in \rho(T) : \|(T - \lambda)^{-1}\| \geq \frac{1}{\text{dist}(\lambda, \sigma(T))}$. Closer to $\sigma(T)$, larger norm.

Proof. (1) Choose $\lambda_0 \in \rho(T)$ and set $C = \|(T - \lambda_0)^{-1}\|$, $\epsilon = \frac{1}{C}$. ϵ -ball and $\lambda_0 \in \rho(T)$ and show $\rho(T)$ is open. Take any $\lambda \in \mathcal{C}$ with $|\lambda - \lambda_0| < \epsilon$. Show that $|T - \lambda|$ is invertible $= T - \lambda_0 + \lambda_0 - \lambda = T - \lambda_0 - (\lambda - \lambda_0) = (T - \lambda_0)(I - \underbrace{(\lambda - \lambda_0)(T - \lambda_0)^{-1}}_S)$. Then, $\|S\| < \epsilon \cdot C = 1$. Neumann series:

$(I - S)$ with $\|S\| < 1$ invertible because like geometric series:

$$(I - S) \sum_{k=0}^n S^k = (I - S) \frac{I - S^{n+1}}{I - S} = (I - S^{n+1}) \rightarrow I$$

as $n \rightarrow \infty$ as $\|S\| < 1$.

$$(I - S)^{-1} = \sum_{k=0}^{\infty} S^k$$

$(T - \lambda)$ is invertible because product of 2 invertible operations $(= (T - \lambda_0)(I - S)^{-1})$. So $\lambda \in \rho(T)$ and $\rho(T)$ is open.

(2)

$$T - \lambda = (T - \lambda_0)(I - S)$$

and invert (which reverses order of RHS):

$$\begin{aligned} (T - \lambda)^{-1} &= (I - S)^{-1}(T - \lambda_0)^{-1} = \sum_{k=0}^{\infty} S^k (T - \lambda_0)^{-1} \\ &= \sum_{k=0}^{\infty} (\lambda - \lambda_0)^k (T - \lambda_0)^{-k} (T - \lambda_0)^{-1} = \sum_{k=0}^{\infty} (T - \lambda_0)^{-(k+1)} (\lambda - \lambda_0)^k \end{aligned}$$

which is a Taylor series and therefore analytic.

(3) $\lambda \in \rho(T)$ and from above, $|\lambda - \lambda_0| \geq \epsilon \frac{1}{|\lambda - \lambda_0|} \leq C$. $\|(T - \lambda)^{-1}\| = \frac{1}{\text{dist}(\lambda_0, \rho(T))} = \frac{1}{\inf |\lambda - \lambda_0|} = \sup \frac{1}{|\lambda - \lambda_0|} \leq \|(T - \lambda)^{-1}\|$

15 Algebra

15.1 Infinite arithmetic series

Instead of using Gauss's equation, consider numbers as having a 'width' of 1, so they are like barcharts, with 0x1, 1x1, 2x1, etc. blocks. Then just take the area. Gauss's proof is just stacking two of these bar charts so that a rectangle emerges and taking half the area of this.

15.2 Infinite geometric series

The formula for an infinite geometric series

$$1 + r + r^2 + r^3 + \dots$$

is

$$S = \frac{1}{1 - r}$$

The proof involves subtracting staggered series to get the sum of a finite geometric series and then taking the limit to get infinite sums. The first part of the proof - subtracting staggered sums - can be visualized, and then the second by removing the subtraction of 'higher number' portion of the sets as they go to infinity. This is an indirect intuition of the formula.

The direct intuition of an infinite geometric series is a rebasing of the number system by r . This is most intuitive with base 0.1 or 10. For instance, if $r = 0.1$, then

$$1 + 0.1 + 0.01 + 0.001 + \dots$$

If $r = 10$, then

$$1 + 10 + 100 + 1000 + \dots$$

In the case of $r = 0.1$,

$$S = \frac{1}{1 - 0.1} = \frac{1}{0.9} = 1.1111\dots$$

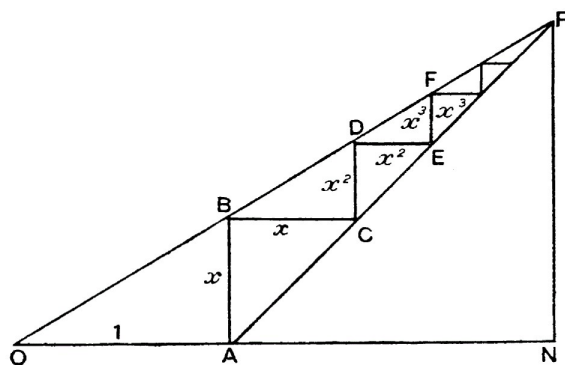
A division of 1 by a fraction such as 0.2 is about turning the 1 as 0.2 - so 5 units of the 1 are needed to make the 1 be 0.2 of the 5. A division of 1 by 0.9 means turning the 1 into a 0.9.

A division of 1 by 0.9 results in a value where the 1 is 0.9 of the resulting value. This is because the rest of the digits (0.1111....) need to form 0.1 of the resulting value after the 1 is now 0.9 of the resulting value.

A visual approach is to use the formula for a finite geometric series

$$S = \frac{r^{n+1} - 1}{r - 1}$$

which is about scaling triangles from this diagram:



And then taking the limit of $n \rightarrow \infty$, ending with

$$S = \frac{1}{1-r}$$

15.2.1 Euler's identity

Euler's identity is based on

$$e^{xi} = \lim_{n \rightarrow \infty} \left(1 + \frac{xi}{n}\right)^n$$

If $x = 1$ then there is a rotation of i . If this is divided by infinity, and the rotation is repeated infinitely, it will end up somewhere with an imaginary component. However, if $x = \pi$, which is divided by infinity and then rotated infinitely, it will result in -1 - this is like $\frac{\pi}{\infty} \times \infty = \pi$ arc, which arcs to -1 .

15.3 Euler's number

Related to rates. Rates are related to factorials. Factorials are related to probabilities. So e is related to rates, factorials, and probabilities.

15.3.1 Compound interest

$$e^r = \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n$$

15.3.2 Probability

If the probability of winning is 1 in a million,

$$p(win) = \frac{1}{1000000}$$

$$p(loss) = 0.999999$$

Then the probability of losing every time in a million trials is

$$e^{-1} = \left(1 - \frac{1}{1000000}\right)^{1000000} = \left(1 + \frac{-1}{1000000}\right)^{1000000}$$

15.3.3 Infinite series

Related to the above (i.e. compound interest) formula,

$$e = \sum \frac{1}{n!} = 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \dots$$

15.3.4 Gamma function

Start with

$$\int_0^{\infty} e^{ax} dx = \frac{-1}{a}$$

Differentiate n times,

$$\int_0^{\infty} x^n e^{ax} dx = \frac{(-1)^{n+1} n!}{a^{n+1}}$$

Substitute $a = -1$,

$$\Gamma(n+1) = \int_0^{\infty} x^n e^{-x} dx = n!$$

Re-writing,

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx = (n-1)!$$

Note

$$\begin{aligned} \Gamma(n+1) &= \int_0^{\infty} x^n e^{-x} dx \\ &= -x^n e^{-x} + \int_0^{\infty} x^{n-1} e^{-x} dx \\ &= n\Gamma(n) \end{aligned}$$

15.3.5 Stirling's approximation

Integration by parts:

$$\int \ln(x) = x \ln(x) - x + C$$

Stirling's approximation. An asymptotic approximation of factorials.

$$\ln(n!) = \sum_j^n \ln(j) \approx \int_1^n \ln(x) dx = n \ln(n) - n + 1$$

Since the approximation is valid for $n \rightarrow \infty$, ignore the $+1$. Then,

$$n! = e^{n \ln(n) - n} = e^{n \ln(n)} e^{-n} = \left(\frac{n}{e}\right)^n$$

Intuition. Consider all $n!$ permutations of n items. Can represent information (=uncertainty) in each of the $n!$ terms separately using $\log_2(n)$ bits. With n terms, the upper bound is $n \log_2(n)$. But the last term is fully determined by the earlier terms, so subtract 1 for each of the n terms: $n \log_2(n) - n$. The log base-2 can be converted using $\ln(n) = \frac{\log_2(n)}{\log_2(e)}$. The conversion won't change the expression much as $2 \approx e$. However, not necessary to convert. Instead, can use Shannon entropy unit 'nats' with natural logarithms (where the uncertainty is not divided by 2, but rather by Euler's number e).

16 Single Variable Calculus

16.1 Derivatives

16.2 Integration

16.3 Fundamental theorem of calculus

17 Linear Algebra

17.1 Products

17.1.1 Dot product

Multiplying vectors by multiplying components:

$$A \cdot B = a_1 b_1 + a_2 b_2$$

Take vector A along x-axis $a_1 = |A|\cos\theta = |A|\cos 90 = |A|$ and $a_2 = |A|\sin\theta = |A|\sin 0 = 0$ and vector B at an angle in the first quadrant $b_1 = |B|\cos\theta$ and $b_2 = |B|\sin\theta$. Multiplying the two:

$$A \cdot B = a_1 b_1 + a_2 b_2 = |A||B|\cos\theta + |A|\sin 0 |B|\sin\theta = |A||B|\cos\theta$$

Alternative explanation. The dot product intuitively is a multiplication of the projection of one vector $\vec{a}$ onto another $\vec{b}$ by $|\vec{a}|\cos\theta$ or the other way around would be $|\vec{b}|\cos\theta$. Either way, $|\vec{a}||\vec{b}|\cos\theta$. This is equivalent to the multiplication of components because

$$\vec{a} \cdot \vec{b} = (a_x \vec{i} + a_y \vec{j}) \cdot (b_x \vec{i} + b_y \vec{j}) = a_x b_x + a_y b_y$$

because $\vec{i} \cdot \vec{j} = 0$ as they are orthogonal.

17.1.2 Cross product

The cross-product is similar to the determinant. For instance, both give the parallelogram area below:

$$|\vec{a}||\vec{b}|\sin\theta = \vec{a} \times \vec{b} = \begin{vmatrix} a_x & a_y \\ b_x & b_y \end{vmatrix} = a_x b_y - b_x a_y$$

The cross-product takes advantage of the determinant formula to get the determinant in the perpendicular (orthogonal) direction:

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k} \end{aligned}$$

The determinant formula comes from generalizing areas and volumes. By convention, use right-hand rule for cross-product direction.

Intuitively, the area of the determinant in the perpendicular direction. Lay vector A on the x-axis so $a_1 = |A|$, $a_2|A|\sin 0 = 0$, $a_3 = 0$. Vector B is on the xy-plane but at an angle so $b_1 = |B|\cos\theta$, $b_2 = |B|\sin\theta$, $b_3 = 0$. The general formula collapses to

$$\mathbf{A} \times \mathbf{B} = a_1b_2 - a_2b_1 = |A||B|\sin\theta - 0 = |A||B|\sin\theta$$

17.1.3 Dot versus cross product

As vectors have direction as opposed to regular numbers, there are two 'ways' to multiply them. The dot product multiplies vectors in a 'parallel' fashion $|A||B|\cos\theta$, while the cross product multiplies vectors in a 'perpendicular' fashion $|A||B|\sin\theta$.

17.2 Analytic geometry

Parametric equation of a line in n-dimensions. Example with 3-dimensions. Measure distance from origin to line with two vectors r and r_0 . Then $r - r_0 = ut$ where u is the 'velocity'. Then $r = r_0 + ut$, and split the components:

$$x = x_0 + u_1t$$

$$y = y_0 + u_2t$$

$$z = z_0 + u_3t$$

$$t = \frac{x - x_0}{u_1} = \frac{y - y_0}{u_2} = \frac{z - z_0}{u_3}$$

Parametric equation of a plane in n-dimensions. Example with 3-dimensions. Start with r_1, r_2, r_3 and distances $s_1 = r_2 - r_1$, $s_2 = r_3 - r_2$, with normal N perpendicular to plane. Then:

$$N = S_1 \times S_2$$

$$N \cdot (r - r_1) = 0$$

where r is any (x, y, z) . This implies $N \cdot r = N \cdot r_1 = d$

$$r = xi + yi + zk$$

$$N = ai + bj + ck$$

$$N \cdot r_1 = d$$

So by the dot product,

$$N \cdot r = ax + by + cz = d$$

17.3 Notation

17.3.1 Kronecker delta

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Without Einstein notation, $\delta = I_n$. $\delta_{ij}A_j = \delta_{ii}A_i + \delta_{ij}A_j = A_i + 0 = A_i$. Example. For 3D, $\delta_{ij}A_j = \delta_{11}A_1 + \delta_{12}A_2 + \delta_{13}A_3 + \delta_{21}A_1 + \delta_{22}A_2 + \delta_{23}A_3 + \delta_{31}A_1 + \delta_{32}A_2 + \delta_{33}A_3$, but in Einstein notation, i is the free index ($\delta_{ij}A_j = \sum_j \delta_{ij}A_j$), so there are only 3 terms of which 2 are equal to 0: $\delta_{ij}A_j = \delta_{11}A_1 + \delta_{12}A_2 + \delta_{13}A_3 = \delta_{11}A_1 + 0 + 0 = \delta_{11}A_1$. Therefore, $\delta_{ij}A_j = A_i$.

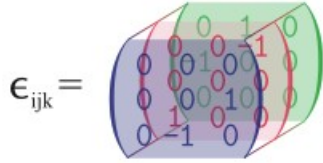
If two subscripts from different Kronecker deltas are the same, they can be combined. With Einstein notation, $\delta_{ik}\delta_{kj} = \delta_{ii}\delta_{ij} + \delta_{ij}\delta_{jj} + \delta_{ik}\delta_{kj} = \delta_{ij} + 0 + 0 = \delta_{ij}$. Since Kronecker delta order doesn't matter, $\delta_{ik}\delta_{kj} = \delta_{ik}\delta_{jk} = \delta_{ki}\delta_{kj}$.

Dot product with Kronecker delta and Einstein notation: $a \cdot b = \delta_{ij}a_i b_j$

17.3.2 Levi-Civita

Can be formulated in 2D and higher dimensions. For 3D there are 'clockwise' (+1), 'counter-clockwise' (-1), and other (0) rotations.

$$\epsilon_{ijk} = \begin{cases} +1, & (i, j, k) = (1, 2, 3), (j, k, i) = (2, 3, 1), (k, i, j) = (3, 1, 2) \\ -1, & (k, j, i) = (3, 2, 1), (j, i, k) = (2, 1, 3), (i, k, j) = (1, 3, 2) \\ 0, & \text{else (at least one of } i=j, j=k, k=i) \end{cases}$$



where i is the depth (blue: $i = 1$; red: $i = 2$; green: $i = 3$), j is the row and k is the column.

$$\begin{aligned} \epsilon_{ijk} &= \epsilon_{jki} = \epsilon_{kij} \\ \epsilon_{ijk} &= -\epsilon_{jik} \text{ and so forth.} \end{aligned}$$

Cross-product:

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \end{aligned}$$

Cross-product in terms of Levi-Civita:

$$\sum_i (A \times B)_i = \sum_j \sum_k \epsilon_{ijk} A_j B_k$$

In terms of Einstein notation

$$\sum_i \sum_j \sum_k \epsilon_{ijk} A_j B_k = \epsilon_{ijk} A_j B_k$$

$$(A \times B)_i = \epsilon_{111} A_1 B_1 + \epsilon_{112} A_1 B_2 + \epsilon_{113} A_1 B_3 + \epsilon_{121} A_2 B_1 \\ + \epsilon_{122} A_2 B_2 + \epsilon_{131} A_3 B_1 + \epsilon_{123} A_2 B_3 + \epsilon_{132} A_3 B_2 + \epsilon_{133} A_3 B_3$$

Most terms have duplicate subscript terms so are 0. Then:

$$= \epsilon_{123} A_2 B_3 + \epsilon_{132} A_3 B_2 = A_2 B_3 - A_3 B_2$$

Relation between Kronecker delta and Levi-Civita (Einstein notation is irrelevant because there are only free indices):

$$\epsilon_{ijk} \epsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix}$$

$$\epsilon_{ijk} \epsilon_{lmn} = \delta_{ii}(\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) - \delta_{im}(\delta_{jl}\delta_{kn} - \delta_{jn}\delta_{kl}) + \delta_{in}(\delta_{jl}\delta_{km} - \delta_{jl}\delta_{km})$$

Contraction identity ($i = l$) (note use of Einstein summation notation):

$$\epsilon_{ijk} \epsilon_{imn} = \delta_{ii}(\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) - \delta_{im}(\delta_{ji}\delta_{kn} - \delta_{jn}\delta_{ki}) + \delta_{in}(\delta_{ji}\delta_{km} - \delta_{ji}\delta_{km}) \\ = 3(\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) - (\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) + (\delta_{jn}\delta_{km} - \delta_{jn}\delta_{km}) \\ = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}$$

Note: $\epsilon_{ijk} \epsilon_{ijn} = 3\delta_{ii}\delta_{kn} - \delta_{kn} = 3\delta_{kn} - \delta_{kn} = 2\delta_{kn}$

17.3.3 Einstein summation notation

$$\sum_i^n \delta_{ii} \equiv \delta_{ii} = n$$

$$\epsilon_{ijk} \epsilon_{ijk} = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + (-1)(-1) + (-1)(-1) + (-1)(-1) = 6$$

$$\mathbf{A} \cdot \mathbf{B} = A_i B_i$$

$$(\mathbf{A} \times \mathbf{B})_i = \epsilon_{ijk} A_j B_k = \epsilon_{123} A_2 B_3 + \epsilon_{132} A_3 B_2 = A_2 B_3 - A_3 B_2$$

In Einstein notation, the free index only appears once in a term and is the 'reference' term (i below). The dummy index is repeated twice in a term and is summed over (j below). There are no terms that are repeated three times or more (wrong or can be reduced).

$$\delta_{ij} A_j = \sum_j \delta_{ij} A_j$$

17.4 Projections

If there is a vector $\vec{y}$ lying 'over' $\vec{x}$ and the latter is extended to $b\vec{x}$, then the dot product of the difference and $\vec{x}$ is 0:

$$\vec{x} \cdot (\vec{y} - b\vec{x}) = 0$$

Solving for b:

$$b = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}|^2}$$

The projection of $\vec{y}$ onto $\vec{x}$ is:

$$b\vec{x} = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}|^2} \vec{x}$$

17.5 Adjoints

Adjoints are generalized transposes. A dot product is

$$\langle u, v \rangle = \sum uv$$

Generalizing,

$$\langle f, g \rangle = \int f(x)g(x)dx$$

$$\langle Au, v \rangle = (Au)'v = (u'A')v = u'(A'v) = \langle u, A'v \rangle$$

Let u and v are vectors and u lie above v. If A rotates vector u clockwise (resulting in Au) closer to v then the dot product Au gets larger. To get the same effect operating on v, it must be that there is a linear transformation that rotates v in the opposite direction, i.e. counterclockwise, closer to u to get the same dot product. That linear transformation is the opposing rotation A' or the adjoint (the transpose in this simple case).

17.6 Quadratic forms

Positive definite matrix. Apply a linear transformation A to vector x: $y = Ax$. If A is positive definite, the the angle between x and y is less than 90 degrees: $x'Ax > 0$. If the angle were greater than 90 degrees, the projection of Ax on x would be in the opposite direction and so negative. Intuition. Positive definite matrices are the generalization of positive real numbers to the matrix case, and positive semidefinite matrices are the generalization of nonnegative numbers.

A covariance matrix is always PSD. Can always find a transformation of the covariance matrix to make it diagonal. On the diagonal are variances of transformed variables which are either zero or positive.

17.7 Singular value decomposition

Square matrices have eigenvalues and eigenvectors. For non-square matrices, have singular values and singular vectors. Every matrix (square or non-square) factors into $A = U\Sigma V^T$. U and V are square orthogonal matrices (both rotations). Σ is $m \times n$ aspect ratio matrix.

$$A^T A = (V\Sigma U)^T U \Sigma V^T = V \Sigma^T \Sigma V^T$$

A was rectangular, $A^T A$ is symmetric positive semi-definite. So eigenvalues of $A^T A$ are σ^2 , and the eigenvalues of A are σ . Moreover,

$$A A^T = U \Sigma V^T V \Sigma^T U^T = U \Sigma \Sigma^T U^T$$

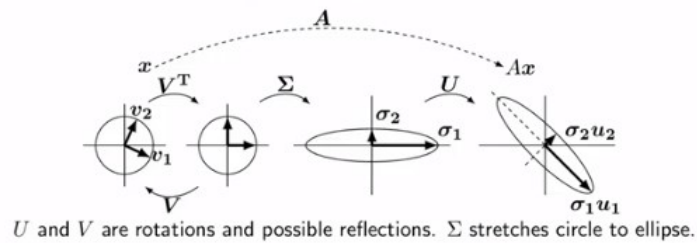
So $V^T V$ is now the identity. U is the eigenvector matrix for $A A^T$.

$$A = U \Sigma V^T \text{ with } U^T U = I \text{ and } V^T V = I$$

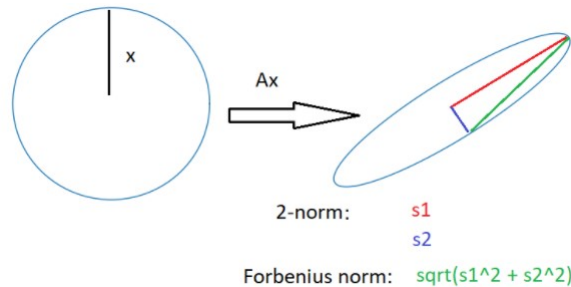
$AV = U\Sigma$ means

$$A \begin{bmatrix} v_1 & \cdots & v_r \end{bmatrix} = \begin{bmatrix} u_1 & \cdots & u_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \text{ and } Av_i = \sigma_i u_i$$

SINGULAR VALUES $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ $r = \text{rank of } A$



SVD is often used in PCA (for instance, the first entry a_{11} of a matrix can be decomposed into its components, $u_1 \sigma_1 v_1^T$). Using singular values, can calculate the L2 norms (σ_i) and Frobenius norm (the Pythagorean distance of the σ s):



17.8 Vector algebra

17.8.1 Scalar triple product

$$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B)$$

Proof.

$$A \cdot (B \times C) = A_i (B \times C)_i = B_j \epsilon_{jki} C_k A_i = B_j \cdot (C \times A)_j = B \cdot (C \times A)$$

$$B \cdot (C \times A) = B_i (C \times A)_i = C_j \epsilon_{jki} A_k B_i = C_j \cdot (A \times B)_j = C \cdot (A \times B)$$

Intuition. Parallelepiped volume.

$$A \cdot (B \times C) = A \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

17.8.2 Vector triple product

$$A \times (B \times C) = (A \cdot C)B - (A \cdot B)C$$

Proof.

$$\begin{aligned} (A \times (B \times C))_i &= \epsilon_{ijk} A_j (B \times C)_k = \epsilon_{ijk} A_j \epsilon_{klm} B_l C_m \\ &= \epsilon_{kij} \epsilon_{klm} A_j B_l C_m = (\delta_{il} \delta_{jm} - \delta_{jl} \delta_{im}) A_j B_l C_m \\ &= A_j (\delta_{il} B_l) (\delta_{jm} C_m) - A_j (\delta_{jl} B_l) (\delta_{im} C_m) \\ &= A_j B_i C_j - A_j B_j C_i = A_j C_j B_i - A_j B_j C_i = ((A \cdot C)B - (A \cdot B)C)_i \end{aligned}$$

Intuition. Since $B \times C$ is orthogonal to the plane, $A \times (B \times C)$ lies on the plane so must be a linear combination of B and C .

17.8.3 Jacobi identity

Jacobi identity. The cross-product is not associative but satisfies this identity, which is proved by decomposition with the vector triple product:

$$A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0$$

Intuitively, the vectors are all pointing equally away from the origin so add to 0.

17.8.4 Scalar quadruple product

Proved similarly as for previous vector identities:

$$(A \times B) \cdot (C \times D) = (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C)$$

$$(A \times B) \cdot (C \times D) = \begin{vmatrix} A \cdot C & A \cdot D \\ B \cdot C & B \cdot D \end{vmatrix}$$

It turns the dot product of cross products into a 2D area calculation. More intuition seen from the Lagrange identity:

$$|A \times B|^2 = (A \times B) \cdot (A \times B) = (A \times A) \cdot (B \times B) - (A \times B) \cdot (B \times A) = |A|^2 |B|^2 - (A \cdot B)^2$$

17.8.5 Vector quadruple product

$$(A \times B) \times (C \times D) = ((A \times B) \cdot D)C - ((A \times B) \cdot C)D$$

17.9 Intuition

17.9.1 Eigenvalues

Real eigenvalues indicate scalings. Imaginary eigenvalues indicate rotations. Complex eigenvalues indicate spirals.

A rotation matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Has eigenvalues $\lambda_1 = i, \lambda_2 = -i$. Complex eigenvalues always come in pairs. The corresponding eigenvectors are $(i, 1)'$ and $(-i, 1)'$ respectively. When the eigenvalues are repeatedly applied to these complex eigenvectors, the real part is a rotation corresponding to the matrix. Along this real rotation however are two rotations along the imaginary axis, counter-clockwise $\lambda_1 = i$ and clockwise $\lambda_2 = -i$, which oppose each other.

17.9.2 The matrix

A matrix is a linear transformation:

$$T(av, bw) = aT(v) + bT(w)$$

So it's a scaling by a in the v-direction, a scaling by b in the w-direction, and adding these.

A matrix has its (i) scaling components along the diagonal and (ii) rotation PLUS stretching in the off-diagonal. In a 2x2 matrix, the top left element scaled (alternatively, stretched) in the x-direction, and the bottom left element rotates in the y-direction followed by a compensatory scaling. The bottom right element scales in the y-direction, and the top right element rotates in the x-direction followed by a compensatory scaling.

Diagonal-similar matrices can be factored using diagonalization which is just a rotation, scaling, and another rotation. Symmetric real matrices can be factored using orthogonal diagonalization which is similar in that it's a rotation, scaling, and another rotation. SVD is also similar. So matrices are scalings, rotations, and further scalings.

17.9.3 Matrix multiplication

A matrix acts on a vector in matrix multiplication to transfer it by dot products to the matrix coordinate basis - this is the matrix multiplication algorithm. Matrix multiplication on a vector can also be seen as scaling the matrix coordinate

bases by the vector's components:

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} e \\ f \end{pmatrix} = e \begin{pmatrix} a \\ b \end{pmatrix} + f \begin{pmatrix} c \\ d \end{pmatrix}$$

The matrix has an 'x-vector' basis which is $(a, b)'$ and a 'y-vector' basis which is $(c, d)'$. Swapping the matrix columns is effectively 'flipping' the vector bases - i.e. flipping the x and y coordinates of the matrix. The vector $(e, f)'$ has x-scalar e and y-scalar f . The x-scalar scales the matrix x-vector which is added to the y-scaled matrix y-vector. If instead of multiplying a vector with a matrix, it were matrix multiplication, then this process is repeated for the 'second' vector in the right-hand matrix.

17.9.4 Matrix inverses

A matrix inverse algorithm can be explained by inverting its determinant, which is a scaling-down (squishing) since the determinant represents the 'volume' of the matrix. The matrix is then transposed, which is reversing the rotations - since the inverse matrix is a 'backward-operation' operator.

17.9.5 Transpose matrix

Transposition is a the regular arithmetic equivalent of a multiplication-division hybrid. Matrix multiplication is a rotation, stretch (scaling), and another rotation. Matrix inversion is a reverse rotation, reverse scaling, and another reverse rotation. Matrix transposition keeps the original scaling because the diagonal elements remain the same (so like matrix multiplication because it scales based on the diagonal elements), but the scaling is reversed because the off-diagonal elements flip places (so like matrix inversion).

With the dot product $\langle v, w \rangle$, as a way to talk about projection of vectors onto other vectors, and if A is a matrix, then its transpose is the unique matrix A' so that $\langle v, Aw \rangle = \langle A'v, w \rangle$ for every pair of vectors v, w . That is, v projected onto Aw is the same as w projected onto $A'v$. With a pair of vectors v and w , and you rotate w by some angle, then this compares to v in the same way the original w compares to v rotated by the same angle but in the opposite direction. The matrix transpose must have the reverse rotation in order for

$$Av \cdot w = v \cdot A'w$$

If A rotates v in one direction, then it must rotate w in the opposite direction for the angles to move in the same direction after transformation.

17.9.6 Matrix trace

The matrix trace is the sum of the main diagonal elements.

$$tr(A) = \sum a_{ii}$$

This is obvious for diagonal matrices. If the diagonal matrix is then warped (and therefore no longer diagonal), it still 'lives' in its 'own world' as a diagonal matrix. So its eigenvalues must still be the same sum - and the sum is a scalar.

The trace of a matrix transpose is equal to the trace of the matrix:

$$\text{tr}(A) = \text{tr}(A')$$

If you are just working in a finite-dimensional Euclidean space, then by using the fact that we can calculate the trace of A as $\sum_{j=1}^n \langle Ax, e_j \rangle$

for any choice of orthonormal basis $e_1, \dots, e_n$, one obtains

$$\text{tr}(A) = \int_{x \in B} \langle Ax, x \rangle dm(x)$$

where B is the Euclidean unit sphere, and m is the uniform measure on B normalised to have total mass 1.

The trace of the inner product is the same whether it is $A'B$ or $B'A$:

$$\text{tr}(A'B) = \text{tr}((A'B)') = \text{tr}(B'A)$$

which is intuitive, since they're both adding the same inner products. Further,

$$\text{tr}(AB) = \text{tr}(BA)$$

because matrix multiplication and the reverse multiplication have the same sum of diagonal elements:

$$AB = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} e & g \\ f & h \end{pmatrix} = \begin{pmatrix} ae + cf & ag + ch \\ be + df & bg + dh \end{pmatrix}$$

$$BA = \begin{pmatrix} e & g \\ f & h \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} ea + bg & ec + gd \\ fa + hb & fc + hd \end{pmatrix}$$

where

$$(ae + cf) + (bg + dh) = (ea + bg) + (fc + hd)$$

because the middle two terms on each side swap. So this is why the SUM of the diagonal elements is the trace. It is spreading the x-vector of the right-side matrix into the x-components of both the x-vector and y-vector of the left-side matrix and repeating with the right-side y-vector, and adding these. When the left and right-side matrices are flipped, it is vice versa. The trace is an abstract 'dimension', which is a sum of the components which make up the trace. Note the sum of off-diagonal elements do not equal:

$$(ag + ch) + (be + df) \neq (ec + gd) + (fa + hb)$$

Because

$$\text{tr}(AB) = \text{tr}(BA)$$

TWO elements can be swapped at a time,

$$\text{tr}(ABC) = \text{tr}((AB)C) = \text{tr}(C(AB)) = \text{tr}(CAB)$$

$$\text{tr}(A(BC)) = \text{tr}((BC)A) = \text{tr}(BCA)$$

since only TWO elements can be swapped at a time,

$$\text{tr}(ABC) \neq \text{tr}(ACB)$$

Note in orthogonal diagonalization:

$$\text{tr}(P'AP) = \text{tr}(APP') = \text{tr}(AI) = \text{tr}(A)$$

which helps connect the idea of the sum of eigenvalues to the sum of the diagonal elements:

$$\sum a_{ii} = \sum \lambda_i$$

The trace is constant even if the basis changes, *with respect to that new basis* - the trace will no longer necessarily be the same in the original basis. The trace is constant in whatever basis is used because the trace is equal to the sum of the eigenvalues.

Multiplying the eigenvalues gives the determinant and volume; adding the eigenvalues gives the dimension of the matrix in a different way. Suppose small changes are made in the edges of a parallelepiped. Then the main contribution to the change in volume of the parallelepiped is due to the change of each edge in its own direction, changes in the direction of the other edges making only a second-order contribution to the change in volume. So the sum of eigenvalues has an intuition.

Finally, a projection onto a plane keeps the vectors inside the plane the same ($Av = v$) and it completely kills the vectors perpendicular to the plane ($Av = 0$) (and a projection matrix has eigenvalues of only 1 or 0). Any set of linearly independent vectors that span the plane is a set of eigenvectors. All those eigenvectors have eigenvalue 1 because $Av = v$ for those vectors. Hence the sum of the eigenvalues is the dimension of that plane.

Another intuition. If A is diagonalizable, and x is the x-direction unit vector $(1, 0)'$ and y is the y-direction unit vector $(0, 1)'$,

$$x'Ax + y'Ay = \sum \lambda = (1, 1)D \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Then

$$\begin{aligned}
x'Ax + y'Ay &= (1, 1)D(1, 1)' \\
&= ((1, 0) + (0, 1))D((1, 0) + (0, 1))' \\
&= (x + y)'D(x + y) \\
&= (x + y)' \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} (x + y) \\
&= (x + y)' \begin{pmatrix} \frac{1}{\lambda_1} & 0 \\ 0 & \frac{1}{\lambda_2} \end{pmatrix}^{-1} (x + y)
\end{aligned}$$

The penultimate line is interpretable as the diagonal matrix containing the eigenvalues in a quadratic with the $(1, 1)'$ vector. So the sum of the eigenvalues is the L1 or taxicab metric distance, as is the LHS expression. So it traces the LHS distance in L1 along an L1 'circle' until it reaches the RHS distance. The last line is interpretable as the Mahalobis distance of the vector $(1, 1)'$.

The trace is analogous to the real part of a complex number. The trace divided by the matrix' rank is the "scalar part" of the matrix, while the other parts are 'rotational' like imaginary numbers.

17.9.7 Trace properties

The trace property for a given matrix A

$$tr(A) = \sum a_{ii} = \sum \lambda_i$$

where ii are the diagonal entries and λ_i are the eigenvalues, is proven in many sources. The intuition is that the diagonal elements of a matrix (which represent the stretches or compressions of the linear transformation) determine the stretch or compression factor of the eigenvectors, i.e. the eigenvalues. The off-diagonal matrix entries indicate rotations - but a rotation increases one eigenvalue exactly as it offsets another. For instance, in

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

the eigenvalue corresponding to the eigenvector spanning quadrants 1 and 3 grows with the rotations of 1 and 1 of both the i and j unit vectors - i and j 'swing in' and 'squeeze' the eigenvector, thereby increasing the eigenvalue. However, because this is a linear transformation, the increase of the eigenvector spanning quadrants 1 and 3 simultaneously leads to a decrease or a compression of the eigenvector spanning quadrants 2 and 4, thereby shrinking that eigenvalue by a corresponding, additive, amount. The off-diagonal elements are not relevant for determining the eigenvalues (even if they help in determining the eigenvectors) because the linear transformation ensures that an increase somewhere is exactly offset by a decrease somewhere else.

The trace property

$$\text{tr}(AB) = \text{tr}(BA)$$

can be proven algebraically, i.e. looking at which entries are multiplied with which other entries. The intuition is that again this is a linear transformation and even if matrix multiplication $AB \neq BA$ necessarily, the 'jaws' of both matrices (jaws creating by the space between the matrix unit vectors) are the same and that's where the eigenvalue comes from. The 'jaws' end up stretching or compressing the eigenvector by the same amount regardless of order, even if the eigenvectors are pointing in different directions.

Once the trace commutativity property above is known, the trace property

$$\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA) \neq \text{tr}(ACB) = \text{tr}(BAC) = \text{tr}(CBA)$$

can be proven easily:

$$\text{tr}(ABC) = \text{tr}((AB)C) = \text{tr}(C(AB)) = \text{tr}((CA)B) = \text{tr}(B(CA))$$

The intuition is that the sum of the eigenvalues, i.e. trace, of AB, BA are the same but they are not necessarily equal because matrix multiplication is not commutative. Even if the trace is the same, AB, BA will usually have different eigenvectors. If now matrix C (e.g. a stretch of the x-unit vector by 2) is multiplied to these combinations, it will 'skew' the eigenvector and the eigenvalue differently in ABC and CBA . The intuition then arises because the non-commutativity of matrix multiplication.

17.9.8 Idempotent and nilpotent matrices

Idempotent matrices:

$$A^2 = A$$

So, they collapse to themselves. The OLS projection matrix P is idempotent:

$$P = X(X'X)^{-1}X'$$

$$P^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = X(X'X)^{-1}X' = P$$

The projection matrix P takes a vector y and outputs $\hat{y}$ on the plane of X :

$$Py = X(X'X)^{-1}X'y = X\beta = \hat{y}$$

And so

$$P^2y = Py = \hat{y}$$

and

$$P\hat{y} = \hat{y}$$

because the idempotent matrix has already projected $y \rightarrow \hat{y}$.

Idempotent matrices have eigenvalues that are 0 or 1, because they do not stretch $\lambda = 1$ and do not project further or rotate further $\lambda = 0$.

Nilpotent matrices:

$$A^2 = 0$$

So they collapse onto themselves. They only have eigenvalues of 0.

17.9.9 Matrix factorization

A square matrix can be factored into a lower-triangular matrix and an upper-triangular matrix, each representing shears, using the LU decomposition algorithm. The algorithm basically takes the matrix A and embodies Gauss-Jordan elimination as the L -matrix to get $LA = U$, where U is the upper-triangular matrix. Then invert L to get $A = LU$. Note: inverse of a triangular matrix (a shear) is also triangular because it is the shear in reverse.

Other matrix decompositions often rely on the LU decomposition. If A is symmetric, then

$$A = LU = LDL'$$

using a simple algorithm. If D 's eigenvalues are non-negative, then,

$$A = LDL' = LD^{1/2}D^{1/2}L' = CC'$$

which is the Cholesky decomposition.

17.9.10 Vector space summation and direct sums

Vector spaces summation takes matrix A living in R^m space and matrix B living in R^n space and creates a matrix C in R^{m+n} space:

$$C = ((A, 0)', (0, B)')$$

So it's a diagonal block matrix as matrix A is missing the elements of R^n and matrix B is missing the elements of R^m .

A direct sum is related to a vector space sum, and also assumes the spaces of m and n are orthogonal. It can sum vectors and just re-arranges the elements differently in the sum. It can take two vectors A and B as:

$$C = (A, B)'$$

which is another vector.

A direct sum just adds components,

$$(1, 2, 3)' \oplus (4, 5)' = (1, 2, 3, 4, 5)'$$

17.9.11 Inner product

A dot product is essentially vector multiplication (a generalized version of multiplication, which is a precursor to matrix multiplication which is a combination of vector multiplications or dot products). Vector division cannot occur because a vector is uninvertible.

A vector projection is a dot product scaled by the square (indicating the Euclidean space) of the projection surface.

A correlation coefficient is a dot product scaled by X and Y, so its absolute value must be less than or equal to 1. This can also be viewed as a projection onto the XY-surface. A regression coefficient is vector projection onto X. A Fourier transform is a vector projection onto the complex exponential surface - conversely, a vector projection is a Fourier transform (a regression splits the dependent variable Y into its 'projected' components, while a Fourier transform splits the signal into its complex exponential components). They are conceptually equivalent to a degree.

The dot product helps determine the orientation of two vectors relative to each other. The cross-product helps determine the orientation of three or more vectors.

The inner product combines vectors by 'multiplying' them into a scalar - it collapses information. The outer product takes all combinations of the vector components, so retains all information.

17.9.12 Cauchy-Schwarz inequality

A generating function is the product of sums. A transform is the sum of products. They are only equal if the cross-terms are 0, i.e. the components are orthogonal, and the vectors are parallel. For this to happen however, the orthogonal components must be scalar multiples of each other.

In general, one can write a product of sums as a sum of a products:

$$\left(\sum x_i\right)\left(\sum y_j\right) = \sum x_i y_j$$

The reverse, a sum of products, is not generally true:

$$\sum x_k y_k$$

which is missing the cross-terms. They are related by

$$\sum x_k y_k \leq \sum x_k \sum y_k$$

because the left side is a dot product, which must be smaller than multiplying two parallel vectors. It is from the Cauchy-Schwartz inequality:

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

Cauchy-Schwartz inequality. A generalization of the triangle inequality which is itself a summary of the Euclidean metric. The square of the sum of products is less than or equal to the product of the sum of squares. Generalized to the Holder inequality.

A norm provides a consistent definition of distance. The Cauchy-Schwarz inequality shows how the distances are consistent with each other.

At first glance, the scalar product and the distance notion are largely independent entities. As a matter of fact, they are intimately related, and the Cauchy-Schwarz inequality provides one of the links. The Cauchy-Schwarz inequality is important because it shows how the notion of an inner product from linear algebra is related to the notion of length. In multivariable calculus, this inequality is used to say that the gradient vector of a function points in the direction of steepest ascent of the function. The Cauchy-Schwarz inequality just tells you that this cannot be greater than itself: a vector cannot be made longer by projecting it. Furthermore, equality holds only when a vector is being projected onto a parallel vector, because this leaves it, and hence its length, unchanged. (This is also part of the statement of the inequality.)

17.9.13 The Gram matrix $X'X$

The $X'X$ matrix is the Gram matrix. It is not an inner product. It is the multiplication of the transpose of a matrix with the matrix (read left to right). It is like the outer product of inner products:

$$X'X = \begin{pmatrix} v'_1 \\ v'_2 \end{pmatrix} (v_1 \ v_2) = \begin{pmatrix} v'_1 v_1 & v'_1 v_2 \\ v'_2 v_1 & v'_2 v_2 \end{pmatrix}$$

Since the Gram matrix is a product of a matrix transpose with the matrix, and the matrix transpose keeps the scaling of the matrix but reverses rotation, the Gram matrix has dot products of the basis vectors (=norms of the basis vectors) in the diagonal and rotation-reverse rotation products in the off-diagonals (where these off-diagonal elements are symmetric since they're the same dot products of the different basis vectors).

The Gram matrix contains information from different component combinations like an outer product. The Gram matrix is intuitively symmetric from the above. It is also positive semi-definite:

$$u' Au = u' X' X u = (Xu)' Xu = \langle Xu, Xu \rangle \geq 0$$

17.9.14 Positive (semi) definiteness

The vector u upon transformation Au can be $u' Au \geq 0$ (positive semidefinite, PSD) or $u' Au > 0$ (positive definite, PD).

Intuition is that the linear transformation A keeps the vector u on the 'same side', so that the angle between u and Au is ≤ 90 degrees (PSD) or < 90 degrees (PD). A PSD matrix is analogous to a non-negative number and a PD matrix is analogous to a positive number, because the dot product (from which the angle between u and Au can be calculated) which is like multiplication of vectors, is positive for PD and non-negative for PSD.

Mathematically, a diagonal matrix is PD if its diagonal elements are > 0 , so that Au keeps u 's diagonal signs the same and therefore remains in u 's 'quadrant'. A diagonal matrix is PSD if its diagonal elements are ≥ 0 , so that AU keeps u 's diagonal signs the same or 0 and therefore remains in u 's 'quadrant' or 0. However, if the matrix is not a diagonal matrix, the diagonal elements rule here does not work generally.

Instead, use eigenvalues. If the matrix eigenvalues are all positive, they will form a positive diagonal matrix upon diagonalization. If the matrix eigenvalues are all non-negative, they will form a non-negative diagonal matrix upon diagonalization. Mathematically,

$$Au = \lambda u$$

If the eigenvalue is positive, $\lambda > 0$, so Au remains in the same direction as u . If the eigenvalue is negative, $\lambda < 0$, so Au points in the opposite direction as u . Zero eigenvalues are not counted, but if it were, then $Au = 0$.

PD and PSD are useful in determining turning points with the multivariate Taylor expansion:

$$f(x) = f(a) + f'(a)(x - a) + \frac{1}{2}(x - a)'f''(a)(x - a) + \dots$$

If for instance $f'(a) = 0$ then there might be a turning point there, so check the Hessian $f''(a)$. If the Hessian is PD or PSD, then there's a local minimum there.

17.9.15 Diagonalization

Only square matrices are diagonalizable.

A matrix is diagonalizable if and only if for each eigenvalue the dimension of the eigenspace is equal to the multiplicity of the eigenvalue to constitute D:

$$A = PDP^{-1}$$

Further, P must be a basis set of eigenvectors and must be invertible.

A matrix is orthogonally diagonalizable if it is a square symmetric real matrix.

$$A = PDP^{-1} = PDP'$$

If the diagonalizable matrix is PSD, then its eigenvalues in the diagonal matrix can be square-rooted.

17.9.16 Symmetric PSD matrices are Gramian

Every symmetric PSD matrix is Gramian. Call the matrix A and since it's symmetric, it's orthogonally diagonalizable:

$$A = PDP'$$

where $PP' = P'P = I$. Since A is PSD,

$$D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

$$B^2 = B'B = D$$

$$B = \text{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n})$$

From the above,

$$A = PDP' = PB'BP' = PB'BP'PBP' = (PBP')'(PBP')$$

Since

$$(PBP')' = PB'P' = PBP'$$

$$A = (PBP')'(PBP') = X'X$$

Unlike the Gram matrix where the diagonal elements are norms of the basis vectors and the trace is the sum of these norms, XX' has diagonal elements comprising norms of the basis covectors and the trace is the sum of these norms. Since the covector basis norm sum equals the vector basis norm sum,

$$\text{tr}(XX') = \text{tr}(X'X)$$

which is used to show the cyclic permutation property of traces.

17.9.17 OLS derivation

$$Xb = y$$

$$X'Xb = X'y$$

$$b = (X'X)^{-1}X'y$$

where $X'X$ is like the Gram matrix in that it is like an outer product of inner products.

17.9.18 XX'

This matrix is the reverse of the Gram matrix. It is not an outer product, which would give the Kronecker product which is a higher-dimensional object. Instead, it is simply the multiplication of a matrix with its transpose (read from left to right). It is like the inner product of outer products:

$$XX' = (v_1 \ v_2) \begin{pmatrix} v'_1 \\ v'_2 \end{pmatrix} = v_1 v'_1 + v_2 v'_2$$

which is the sum of the outer products. Since the outer product is a collection of information, and the sum of outer products is a sum of collections of information, there does not appear to be concrete geometric intuition.

XX' is symmetric - though this is intuitive, given outer products are symmetric (their off-diagonal elements are dot products so are symmetric) and XX' is sum of outer products.

$$(XX')' = (X')'X' = XX'$$

It is also positive semidefinite:

$$u' Au = u'(XX')u = (X'u)(X'u) \geq 0$$

Since XX' is symmetric and PSD, it can be rewritten as a Gram matrix:

$$XX' = P\Lambda P' = Y'Y$$

This is obvious, since XX' is a matrix multiplied by its transpose, which viewed from a different perspective is just the matrix transpose multiplied by its transpose if we let $X = Y'$ and $X' = Y$, so that $XX' = Y'Y$. Note however $X'X$ and $Y'Y$ are very different since $X' \neq Y$, $X \neq Y'$.

17.9.19 Outer product

The inner product splits the components of the LEFT vector u by transposing it and turning it into two (for 2D) scalars, and multiplying these scalars with the right vector. So the right vector is being projected or collapsed onto the scalar components. This collapses the information and loses all directional information. It also happens to have an intuitive interpretation.

The outer product is the reverse of the inner product. It splits the components of the RIGHT vector and has the left vector multiply it. Knowing that matrix multiplication is not commutative, it is not the 'inverse' or 'matrix division', as it exists of its own accord. It is a matrix because the matrix 'warps' the vector with which it's multiplied to the left vector u .

There are two ways to multiply vectors, the inner product collapses the two multiplied vectors into a scalar so 'loses' its direction information; the outer

product preserves the direction of the left vector and uses the right vector as a 'scaling' element in the ROW SPACE - though if the matrix is turned, ie from the COLUMN SPACE, it is actually the right vector's direction scaled by the left vector's components. So the matrix preserves all scalars and directions because it's ultimately a matrix of the combinations of the original vector components.

For any vector x ,

$$uv'(x) = (v \cdot x)u$$

If u and v are unit vectors, $uv'(x)$ is the component of x in the v direction, taken into the u direction. So it results in a 'convolution' of the u and v vectors. While the outer product is a matrix, it is only 1-dimensional, in the direction of the left vector. Intuitively, v 's x -component is scaled into a 'miniature' u vector and the same for v 's y -component. So it forms a set of linearly dependent vectors in the outer product matrix. This interpretation is useful in SVD:

$$A = U\Sigma V' = \sum (u_k \otimes v_k) \sigma_k$$

17.9.20 Kronecker delta

Makes a matrix or tensor its identity by turning its diagonal elements to 1 or a multiple if there are coefficients and the cross-diagonal elements 0.

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

17.9.21 Kronecker products

The outer product is shaped differently from the Kronecker product because the outer product transposes the right vector, but they both have the same entries. They are both tensor products. The Kronecker product is like a generalization of the outer product, and the tensor product is a generalization of the Kronecker product.

The tensor product of two vector spaces is a way of multiplying vector spaces. For example, suppose you want to keep track of the positions of each of your hands, so you have two coordinate planes. Each coordinate plane has two directions, right and up. What is a basis for the combined system? The answer is all possible combinations of the basis for each hand: (left hand right, right hand right), (left hand up, right hand right), (left hand right, right hand up), (left hand up, right hand up). So now suppose we have two matrices, the first matrix transforms your left hand and the second matrix transforms your right hand. The Kronecker product of the two matrices gives you a new matrix, which describes how the combined system of your left hand + right hand transforms, according to the basis given above.

In quantum theory, there are vectors in $V \otimes W$ which cannot be written as a tensor product of any v from V and w from W . These are entangled states because the state of two entangled particles does not have a representation in V or W individually.

Another use is for creating projection operators. To project a vector v onto a 1D subspace spanned by the vector w , could use $proj(v, w) = dot(v, w) * w$. Or, you could use $proj(v, w) = (w \otimes w)v$. Here, $w \otimes w$ is a rank-1 matrix (because it is only specified by 1 vector).

A third use is information compression. If you have a very large matrix, for example a $100,000 \times 100,000$ matrix, then it has $(10^5 \times 10^5 = 10^{10})$ entries. Call this large matrix M . If there is a matrix A which is 103×103 and a matrix B which is 102×102 , then $A \otimes B$ is 105×105 (because $103 \times 102 = 105$). This compresses the information to $(103 \times 103) + (102 \times 102) = 1010000$ numbers. Can $A \otimes B = M$ if A and B are chosen correctly? In general no, and it can be really unequal. However, it's a trade-off.

17.9.22 Covectors, covariance, and contravariance

The transpose of a column vector is a row vector which is just two numbers. This is a covector which is a function. So in an outer product, the covector acts on the vector to produce a scalar, which is a dual space to the vector space in which the column vector resides. The covector space is like a space of contour lines, and the covector bases are basically the contour lines from the basis vectors. When the covector (row vector) (a, b) acts on the column vector $(x, y)'$, it forms the dot product $ax + by$ which is a series of those covector (contour) lines.

The basis covectors are just the covectors of the basis vectors.

A matrix will transform the old basis vectors to a new set of basis vectors. The matrix inverse is the backward transformation. Basis vectors are (1,0)-tensors (covariant).

Vectors are contravariant to the transformation of basis vectors. If a vector grows, this is the same as the vector bases shrinking. If the basis vectors grow, the vector components shrink. If the vector rotates CW, this is equivalent to the basis vectors rotating CCW. To get from an old vector to a new one, the backward transformation of the basis vectors (the matrix inverse) is therefore used. If the vector is distance, then an increase in distance from cm to m will lead to the vector's numerical value shrinking. Vectors are (0,1)-tensors (contravariant).

Covectors are covariant to basis vectors. If the basis vectors grow, the number of contours increases. Physically, covector units are the inverse of vector units (e.g. second versus per second or distance versus per distance). This is

like the change in the gradient increasing if the distance unit changes. If the distance unit increases from cm to m, then the change in the gradient from per cm to per m increases. Covectors are (1,0)-tensors.

Covector bases are also contravariant to the transformation of basis vectors. If a basis covector grows (e.g. the gradient gets steeper), so there are more contour lines, this is equivalent to the basis vector shrinking (e.g. going from m to cm). So the backward transformation of the vector basis is used. Basis covectors are (0,1)-tensors.

Linear maps are both contravariant and covariant, so (1,1)-tensors:

$$\tilde{L} = BLF$$

$$L = F\tilde{L}B$$

where the tilde indicates a new basis, vector, covector, etc.

17.9.23 Tensor products

Kronecker products are arrays of numbers in various shapes, when rows and/or columns are nested into each other. Tensor products are basically the same, except these numbers are basically created from vector and/or covectors so the numbers in the arrays are from the coefficients in the vectors and/or covectors and the shapes are determined by how many vector and/or covectors are used - this means tensor products have more physical meaning.

Tensor product example. A metric tensor is a collection of dot products that allows a new basis to act like it's a standard orthonormal basis, i.e. so that the pythagorean theorem works, even if the basis is not standard orthonormal.

A tensor with m contravariant components and n covariant components, it's an (m,n)-tensor.

Linear maps are combinations of basis vectors and basis covectors (dual basis vectors). Tensor products can be linear combinations of covector-vector pairs as bases. So they're combinations of vector and/or covectors.

17.9.24 Jordan form

Many matrices are not diagonalizable. However, all matrices can be 'approximately' diagonalizable as the Jordan form, which is a 'slightly distorted' version of a pure diagonal matrix. The Jordan form matrix has eigenvalues along its diagonal, and some 'leftover' 1's on off-diagonals.

At a 'lower resolution' or lower dimension, the diagonal and off-diagonal elements turn into blocks, so the entire matrix in lower dimensions is a diagonal matrix.

18 Vector and Matrix Calculus

18.1 Scalar and vector fields

Scalar field: $s(r, t) = s(x, y, z; t)$

Vector field: $v(r, t) = v_1(x, y, z; t)\mathbf{i} + v_2(x, y, z; t)\mathbf{j} + v_3(x, y, z; t)\mathbf{k}$

18.2 Partial derivatives

Definition:

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

Young's theorem (requires continuous functions):

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = f_{yx}$$

Multivariable Taylor series:

$$f(x, y) = f + xf_x + yf_y + \frac{1}{2}(x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}) + \dots$$

18.3 Chain rule

Total differential of $f(x, y)$ (where both variables are changing):

$$\begin{aligned} df &= f(x+dx, y+dy) - f(x, y) \\ &= f(x+dx, y+dy) - f(x, y+dy) + f(x, y+dy) - f(x, y) \\ &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \end{aligned}$$

Chain rule:

$$\begin{aligned} f &= f(x(t), y(t)) \\ \frac{df}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \end{aligned}$$

18.4 Triple product rule

$$\begin{aligned} f(x, y, z) &= 0 \\ x &= x(y, z) \\ z &= z(x, y) \\ dx &= \frac{\partial x}{\partial y} dy + \frac{\partial x}{\partial z} dz \\ dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \end{aligned}$$

$$dx = \frac{\partial x}{\partial y} dy + \frac{\partial x}{\partial z} \left(\frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \right)$$

$$\left(1 - \frac{\partial x}{\partial z} \frac{\partial z}{\partial x} \right) dx = \left(\frac{\partial x}{\partial y} + \frac{\partial x}{\partial z} \frac{\partial z}{\partial y} \right) dy$$

Left side:

$$\frac{\partial x}{\partial z} \frac{\partial z}{\partial x} = 1$$

Right side:

$$\frac{\partial x}{\partial y} = - \frac{\partial x}{\partial z} \frac{\partial z}{\partial y}$$

$$\frac{\partial x}{\partial y} \frac{\partial y}{\partial z} \frac{\partial z}{\partial x} = -1$$

Unlike single-variable derivatives d , can't cancel top and bottom as they mean different things (depends on the variable being held constant and changing).

Triple product rule intuition. If points are located at $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ then the slopes (the partial derivatives) along the coordinate planes (xy, yz, xz) are all -1. So the product is -1. It's negative because it's odd. The quadruple product rule is positive:

$$\frac{\partial x}{\partial y} \frac{\partial y}{\partial z} \frac{\partial z}{\partial t} \frac{\partial t}{\partial x} = 1$$

18.5 Vector derivatives

18.5.1 Gradient

A vector of partial derivatives.

$$f = f(x, y, z)$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= \left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \right) \cdot (dx \mathbf{i} + dy \mathbf{j} + dz \mathbf{k})$$

$$df = \nabla f \cdot d\mathbf{r}$$

where the differential operator is

$$\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$$

Intuition. ∇f is the gradient which is maximized along the direction of $\mathbf{r}$ (i.e., the direction of change).

18.5.2 Divergence

Describes net flow away or toward a point in a vector field. $Div > 0$ means away and $Div < 0$ means toward.

Vector field $u = u_1(x, y, z)\mathbf{i} + u_2(x, y, z)\mathbf{j} + u_3(x, y, z)\mathbf{k}$. The divergence is

$$\begin{aligned} Div u &= \nabla \cdot u = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k} \right) \cdot (u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}) \\ &= \frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} + \frac{\partial u_3}{\partial z} \end{aligned}$$

Intuition. If there is a positive velocity change (change in size of velocity vector) then there's more material flowing outward than inward.

18.5.3 Curl

Describes vorticity or swirl in the vector field. Counter-clockwise rotation is +, clockwise rotation is -. So it outputs a scalar in a 2D vector field at a certain location.

The 2D curl measures the change in partial derivatives of the gradients horizontally and vertically, and if > 0 then it's counter-clockwise and if < 0 then it's clockwise:

$$Curl u = \nabla \times u = \frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y}$$

Generalize 2D case to 3D and higher dimensions. In a 3D vector field, it outputs a vector with magnitude of the curl and direction being the rotation axis from the right-hand rule. Vector field $u = u_1(x, y, z)\mathbf{i} + u_2(x, y, z)\mathbf{j} + u_3(x, y, z)\mathbf{k}$. The curl is

$$\begin{aligned} Curl u &= \nabla \times u = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_1 & u_2 & u_3 \end{vmatrix} \\ &= \left(\frac{\partial u_3}{\partial y} - \frac{\partial u_2}{\partial z} \right) \mathbf{i} - \left(\frac{\partial u_3}{\partial x} - \frac{\partial u_1}{\partial z} \right) \mathbf{j} + \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) \mathbf{k} \end{aligned}$$

18.5.4 Laplacian

$$\nabla^2 = \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Acts on scalars:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Acts on vectors:

$$\nabla^2 u = \nabla^2 u_1 \mathbf{i} + \nabla^2 u_2 \mathbf{j} + \nabla^2 u_3 \mathbf{k}$$

Applied in PDEs.

18.6 Vector calculus

18.6.1 Identities

Curl of a gradient is 0 (a unidirectional arrow has no rotation):

$$\nabla \times (\nabla f) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} = 0$$

Divergence of a curl is 0 (a curl doesn't have net out or in flow):

$$\nabla \cdot (\nabla \times u) = 0$$

Curl of a curl, where the last term is the Laplacian:

$$\nabla \times (\nabla \times u) = \nabla(\nabla \cdot u) - \nabla^2 u$$

This is a scalar:

$$\nabla \cdot (fu) = u \cdot \nabla f + f \nabla \cdot u$$

Other identities:

$$\nabla \times (fu) = \nabla f \times u + f \nabla \times u$$

$$\nabla(u \cdot v) = (u \cdot \nabla)v + (v \cdot \nabla)u + u \times (\nabla \times v) + v \times (\nabla \times u)$$

$$\nabla \times (u \times v) = u(\nabla \cdot v) - v(\nabla \cdot u) + (v \cdot \nabla)u - (u \cdot \nabla)v$$

$$\nabla \cdot (u \times v) = v \cdot (\nabla \times u) - u \cdot (\nabla \times v)$$

Proof of $\nabla \cdot (u \times v) = v \cdot (\nabla \times u) - u \cdot (\nabla \times v)$. Use:

$$(\nabla \times u)_i = \epsilon_{ijk} \frac{\partial u_k}{\partial x_j}$$

$$\nabla \cdot u = \frac{\partial u_i}{\partial x_i}$$

So:

$$\nabla \cdot (u \times v) = \frac{\partial}{\partial x_i} (\epsilon_{ijk} u_j v_k)$$

Apply product rule:

$$= \epsilon_{ijk} \frac{\partial u_j}{\partial x_i} v_k + \epsilon_{ijk} u_j \frac{\partial v_k}{\partial x_i}$$

Rearrange to get closer to identity by moving v_k and u_j to front which requires changing the subscripts on the Levi-Civita symbols to match the above equations:

$$\begin{aligned} & v_k \epsilon_{kij} \frac{\partial u_j}{\partial x_i} - u_j \epsilon_{jik} \frac{\partial v_k}{\partial x_i} \\ &= v \cdot (\nabla \times u) - u \cdot (\nabla \times v) \end{aligned}$$

18.7 Double and triple integrals

Volumes

$$\int \int_A f(x, y) dx dy$$

Example 1. Volume above xy-plane of $f(x, y) = x^y$ above the unit square in the first quadrant formed from $(0,0), (1,0), (0,1), (1,1)$:

$$\int_0^1 \int_0^1 x^2 y dx dy = \int_0^1 \int_0^1 x^2 dy dx = \int_0^1 y dy \int_0^1 x^2 dx$$

Example 2. Volume above xy-plane of $f(x, y) = x^y$ above the triangle in the first quadrant formed from $(0,0), (1,0), (0,1)$:

$$\int_0^1 \int_0^{1-y} x^2 y dx dy = \int_0^1 \int_0^{1-x} x^2 y dy dx$$

Mass of 3D object

$$\int \int \int \rho(x, y, z) dx dy dz$$

18.8 Polar coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Unit vector of $\hat{r}$ at angle θ :

$$\hat{r} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$$

Unit vector of $\hat{\theta}$ is perpendicular to $\hat{r}$:

$$\hat{\theta} = -\sin \theta \mathbf{i} + \cos \theta \mathbf{j}$$

If θ increases by $d\theta$, then $\hat{r}$ moves in the direction of $\hat{\theta}$:

$$\frac{d\hat{r}}{d\theta} = \hat{\theta}$$

If θ increases by $d\theta$, then $\hat{\theta}$ is being pulled towards the origin like a centripetal force:

$$\frac{d\hat{\theta}}{d\theta} = -\sin \theta \mathbf{i} + \cos \theta \mathbf{j} = -\hat{r}$$

Gradient.

$$f = f(r, \theta)$$

$$df = \nabla f \cdot d\vec{r}$$

$$df = \frac{\partial f}{\partial r} dr + \frac{\partial f}{\partial \theta} d\theta = \nabla f \cdot d\vec{r}$$

where

$$d\vec{r} = d(r\hat{r}) = dr\hat{r} + r\frac{d\hat{r}}{d\theta}d\theta$$

due to the product and chain rules. From $\frac{d\hat{r}}{d\theta} = \hat{\theta}$,

$$= dr\hat{r} + rd\theta\hat{\theta}$$

Now,

$$df = \frac{\partial f}{\partial r}dr + \frac{\partial f}{\partial \theta}d\theta = \nabla f \cdot (dr\hat{r} + rd\theta\hat{\theta})$$

So ∇f (gradient) must be:

$$\nabla f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta}$$

because

$$\nabla f \cdot (dr\hat{r} + rd\theta\hat{\theta}) = \left(\frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} \right) \cdot (dr\hat{r} + rd\theta\hat{\theta})$$

and $\hat{r}$'s match and $\hat{\theta}$'s match. The $\frac{1}{r}$ term appears because the gradient is about length (the denominators are lengths) and θ is dimensionless - 'project' out θ to the 'circumference' by multiplying $r\partial\theta$.

Note:

$$r\hat{r} = r(\cos\theta\mathbf{i}) + r(\sin\theta\mathbf{j})$$

$$r\hat{\theta} = r(-\sin\theta\mathbf{i}) + r(\cos\theta\mathbf{j})$$

Additionally:

$$\frac{d\hat{r}}{dr} = 0$$

$$\frac{d\hat{\theta}}{dr} = 0$$

18.9 Divergence in polar coordinates

Start with

$$\vec{u} = u = u_r(r, \theta)\hat{r} + u_\theta(r, \theta)\hat{\theta}$$

$$\nabla = \hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta}$$

$$\frac{d\hat{r}}{d\theta} = \hat{\theta}, \frac{d\hat{\theta}}{d\theta} = -\hat{r}, \frac{d\hat{r}}{dr} = 0, \frac{d\hat{\theta}}{dr} = 0$$

$$\hat{r} \times \hat{\theta} = \hat{k}$$

$$(A + B) \times (C + D) = A \times C + A \times D + B \times C + B \times D$$

Divergence in polar coordinates

$$\begin{aligned}
\nabla \cdot u &= \left(\hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \cdot (u_r \hat{r} + u_\theta \hat{\theta}) \\
&= \hat{r} \frac{\partial}{\partial r} (u_r \hat{r}) + \hat{r} \frac{\partial}{\partial r} (u_\theta \hat{\theta}) + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} (u_r \hat{r}) + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} (u_\theta \hat{\theta}) = \\
&= \hat{r} \left(\frac{\partial u_r}{\partial r} \hat{r} + \frac{\partial}{\partial r} \hat{r} u_r \right) + \hat{r} \left(\frac{\partial u_\theta}{\partial r} \hat{\theta} + \frac{\partial \hat{\theta}}{\partial r} u_\theta \right) + \hat{\theta} \frac{1}{r} \left(\frac{\partial u_r}{\partial \theta} \hat{r} + \frac{\partial \hat{r}}{\partial \theta} u_r \right) + \hat{\theta} \frac{1}{r} \left(\frac{\partial u_\theta}{\partial \theta} \hat{\theta} + u_\theta \frac{\partial \hat{\theta}}{\partial \theta} \right) \\
&= \frac{\partial u_r}{\partial r} + \frac{u_r}{r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} = \frac{1}{r} \left(\frac{\partial}{\partial r} r u_r \right) + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} = \frac{1}{r} \left(\frac{\partial r u_r}{\partial r} + \frac{\partial u_\theta}{\partial \theta} \right)
\end{aligned}$$

where the ru_r is like r^2 to compensate for the $\frac{1}{r}$. Curl in polar coordinates is analogous:

$$\begin{aligned}
\nabla \times u &= \left(\hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \times (u_r \hat{r} + u_\theta \hat{\theta}) \\
&= \hat{r} \times \frac{\partial}{\partial r} (u_r \hat{r}) + \hat{r} \times \frac{\partial}{\partial r} (u_\theta \hat{\theta}) + \hat{\theta} \times \frac{1}{r} \frac{\partial}{\partial \theta} (u_r \hat{r}) + \hat{\theta} \times \frac{1}{r} \frac{\partial}{\partial \theta} (u_\theta \hat{\theta})
\end{aligned}$$

Use the product rule and $\hat{r} \times \hat{r} = \hat{\theta} \times \hat{\theta} = 0$:

$$\begin{aligned}
&(\hat{r} \times \hat{\theta}) \frac{\partial u_\theta}{\partial r} + (\hat{\theta} \times \hat{r}) \frac{1}{r} \frac{\partial u_r}{\partial \theta} - \frac{1}{r} u_\theta (\hat{\theta} \times \hat{r}) \\
&= \left(\frac{\partial u_\theta}{\partial r} + \frac{1}{r} u_\theta - \frac{1}{r} \frac{\partial u_r}{\partial \theta} \right) \mathbf{k} \\
&= \frac{1}{r} \left(\frac{\partial r u_\theta}{\partial r} - \frac{\partial u_r}{\partial \theta} \right) \mathbf{k}
\end{aligned}$$

18.10 Laplacian in polar coordinates

$$\begin{aligned}
\nabla^2 &= \left(\hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \cdot \left(\hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \\
&= \frac{\partial^2}{\partial r^2} + 0 + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}
\end{aligned}$$

where the second term involves $\frac{\partial \hat{\theta}}{\partial r} = 0$ and the third term involves $\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$ and $\hat{\theta} \cdot \hat{\theta} = 1$,

$$= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$$

18.11 Conservation of angular momentum

$\vec{r} = r\hat{r}$ and Newton's 2nd law $\vec{F} = F = m\frac{d^2\vec{r}}{dt^2} = m\ddot{\vec{r}} = m\ddot{r}$. With centripetal force,

$$\begin{aligned}\frac{d\hat{r}}{dt} &= \frac{d\hat{r}}{d\theta} \frac{d\theta}{dt} = \dot{\theta}\hat{\theta} \\ \frac{d\hat{\theta}}{dt} &= \frac{d\hat{\theta}}{d\theta} \frac{d\theta}{dt} = -\dot{\theta}\hat{r}\end{aligned}$$

Velocity:

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

Acceleration:

$$\vec{a} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + \ddot{\theta}r - r\dot{\theta}^2\hat{r} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Force:

$$\vec{F} = -f(r)\hat{r} + 0 = m(\ddot{r} - r\dot{\theta}^2)\hat{r} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

So,

$$\begin{aligned}m(\ddot{r} - r\dot{\theta}^2) &= -f(r) \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) &= 0\end{aligned}$$

Multiply second equation by r to get conservation of angular momentum:

$$\frac{d}{dt}(mr^2\dot{\theta}) = \frac{d}{dt}(mvr) = \frac{dL}{dt} = 0$$

Angular momentum itself is a vector formed from a cross-product (noting $\vec{r} = \mathbf{r} = r\hat{r}$):

$$\begin{aligned}\mathbf{l} &= \mathbf{r} \times \mathbf{p} = \mathbf{r} \times (m\dot{\mathbf{r}}) = \mathbf{r} \times \left(m\frac{d\mathbf{r}}{dt}\right) = \mathbf{r} \times \left(m\left(\dot{r}\hat{r} + r\frac{d\hat{r}}{dt}\right)\right) \\ &= \mathbf{r} \times \left(m\left(\dot{r}\hat{r} + r\frac{d\hat{r}}{d\theta}\frac{d\theta}{dt}\right)\right) = mr\hat{r} \times (\dot{r}\hat{r} + r\dot{\theta}\hat{\theta})\end{aligned}$$

Distribute:

$$(mr\hat{r} \times \dot{r}\hat{r}) + (mr\hat{r} \times r\dot{\theta}\hat{\theta}) = 0 + (mr^2\hat{r} \times \dot{\theta}\hat{\theta})$$

Collect scalars:

$$= mr^2\dot{\theta}(\hat{r} \times \hat{\theta})$$

and note $|\hat{r} \times \hat{\theta}| = 1$ as they're unit vectors.

18.12 Change of variables: substitution

$$I = \int_a^b f(x)dx$$

Let $x = u(s)$, so $dx = u'(s)ds$, so

$$I = \int_{u^{-1}(a)}^{u^{-1}(b)} f(u(s))u'(s)ds$$

18.13 Change of variables: Jacobian

$$I = \int \int_A f(x, y) dx dy$$

Let $x = x(s, t)$ and $y = y(s, t)$. Then

$$dx dy = \begin{vmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{vmatrix} ds dt$$

With local linearity, when the integrated base area changes, then want the area of $dx dy$ to also change. So for

$$\begin{bmatrix} f_1(x, y) \\ f_2(x, y) \end{bmatrix} = \begin{bmatrix} x + \sin(y) \\ y + \sin(x) \end{bmatrix}$$

the determinant (area) changes by $\frac{\partial f_1}{\partial x} \frac{\partial f_2}{\partial y} - \frac{\partial f_1}{\partial y} \frac{\partial f_2}{\partial x}$. So the Jacobian is

$$J = \begin{vmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{vmatrix}$$

Changing to polar coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Then, $\vec{r} = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j}$, so

$$\frac{\partial \vec{r}}{\partial r} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$$

$$\frac{\partial \vec{r}}{\partial \theta} = -r \sin \theta \mathbf{i} + r \cos \theta \mathbf{j}$$

So the Jacobian is

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

$$\int \int_A dx dy = \int_0^{2\pi} \int_0^R r dr d\theta$$

where the extra r is associated with the θ : $r d\theta$, to sweep a circle sector by the length dr and the arc $r d\theta$.

18.14 Gaussian integral

$$\begin{aligned}
 I &= \int_{-\infty}^{\infty} e^{-x^2} dx \\
 I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right)^2 \\
 &= \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\
 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta
 \end{aligned}$$

Let $u = r^2$ so $du = 2r dr$:

$$\begin{aligned}
 &= \int_0^{2\pi} \int_0^{\infty} e^{-u} \frac{du}{2} d\theta = \int_0^{2\pi} d\theta \int_0^{\infty} \frac{e^{-u}}{2} du \\
 &= \pi \int_0^{\infty} e^{-u} du = -\pi e^{-u} = \pi
 \end{aligned}$$

So,

$$I^2 = \pi \rightarrow I = \sqrt{\pi}$$

18.15 Cylindrical coordinates

$(x, y, z) \rightarrow (\rho, \phi, z)$ and $\vec{r} = \rho\hat{\rho} + z\hat{z}$ with $x = \rho\cos\phi$, $y = \rho\sin\phi$, $z = z$. Then $dx dy dz = \rho d\rho d\phi dz$ (dz doesn't add anything other than the vertical component).

$$\nabla = \hat{\rho} \frac{\partial}{\partial \rho} + \hat{\phi} \frac{1}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z}$$

with straightforward but tedious derivation of ∇^2 . Further, if $\vec{A} = A_\rho\hat{\rho} + A_\phi\hat{\phi} + A_z\hat{z}$, there are tedious mechanical derivations of $\nabla \cdot \vec{A}$, $\nabla \times \vec{A}$

18.16 Spherical coordinates

$(x, y, z) \rightarrow (r, \theta, \phi)$ and

$$\begin{aligned}
 x &= r \sin\theta \cos\phi \\
 y &= r \sin\theta \sin\phi \\
 z &= r \cos\theta \\
 dx dy dz &= r^2 \sin\theta dr d\theta d\phi
 \end{aligned}$$

In Cartesian coordinates:

$$\begin{aligned}
 \hat{r} &= \sin\theta \cos\phi \mathbf{i} + \sin\theta \sin\phi \mathbf{j} + \cos\theta \mathbf{k} \\
 \hat{\theta} &= \cos\theta \cos\phi \mathbf{i} + \cos\theta \sin\phi \mathbf{j} - \sin\theta \mathbf{k}
 \end{aligned}$$

$$\hat{\phi} = -\sin\phi \mathbf{i} + \cos\phi \mathbf{j}$$

and invert matrix for $\mathbf{i}, \mathbf{j}, \mathbf{k}$. Differentiate to get

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}, \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}, \frac{\partial \hat{\phi}}{\partial \theta} = 0$$

$$\frac{\partial \hat{r}}{\partial \phi} = \sin\hat{\phi}, \frac{\partial \hat{\theta}}{\partial \phi} = \cos\hat{\phi}, \frac{\partial \hat{\phi}}{\partial \phi} = -\sin\theta \hat{r} - \cos\theta \hat{\theta}$$

From

$$\nabla = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

can calculate $\nabla^2, \nabla \cdot A, \nabla \times A$.

18.17 Line integral on scalar field

$$\int_c f(\vec{r}) ds$$

$$ds = \sqrt{(dx)^2 + (dy)^2}$$

So:

$$\int_c f(\vec{r}) \sqrt{(dx)^2 + (dy)^2}$$

Convert to Cartesian coordinates:

$$y = y(x)$$

$$dy = y'(x) dx$$

$$ds = \sqrt{(dx)^2 + (y'(x) dx)^2} = \sqrt{1 + (y'(x))^2} dx$$

$$\int_c f(\vec{r}) \sqrt{(dx)^2 + (dy)^2} = \int_c f(\vec{r}) \sqrt{1 + (y'(x))^2} dx$$

Convert to parametric coordinates:

$$x = x(t) \rightarrow dx = \dot{x} dt$$

$$y = y(t) \rightarrow dy = \dot{y} dt$$

$$ds = \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2} dt$$

$$\int_c f(\vec{r}) \sqrt{(dx)^2 + (dy)^2} = \int_c f(\vec{r}) \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2} dt$$

18.18 Arc length

Just the distance (disregarding the value of the function at each point on the line). Line integral without $f(\vec{r})$ in the integral:

$$\text{perimeter} = p = \int_c ds$$

$$x = a \cos \theta \rightarrow dx = -a \sin \theta d\theta$$

$$y = b \sin \theta \rightarrow dy = b \cos \theta d\theta$$

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} d\theta$$

Ellipse:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

$$\text{eccentricity} = e = \sqrt{1 - \left(\frac{b}{a}\right)^2}$$

$$e^2 + \left(\frac{b}{a}\right)^2 = 1$$

$$\begin{aligned} ds &= a \sqrt{(1 - \cos^2 \theta) + \left(\frac{b}{a}\right)^2 \cos^2 \theta} d\theta = a \sqrt{1 - (1 - \left(\frac{b}{a}\right)^2) \cos^2 \theta} d\theta \\ &= a \sqrt{1 - e^2 \cos^2 \theta} d\theta \end{aligned}$$

Complete elliptic integral of the second kind:

$$\begin{aligned} p &= \int_c ds = a \int_0^{2\pi} \sqrt{1 - e^2 \cos^2 \theta} d\theta \\ &= 4a \int_0^{\pi/2} \sqrt{1 - e^2 \cos^2 \theta} d\theta = 4a \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 \theta} d\theta \end{aligned}$$

18.19 Line integral of a vector field

$$\int_c \vec{u} \cdot d\vec{r} = \int_c (\vec{u} \cdot \hat{t}) ds$$

$$d\vec{r} = \hat{t} ds$$

where $\vec{r}$ is along the line and $\hat{t}$ is the tangent unit vector. Now let $\vec{r} = \vec{r}(t)$, $d\vec{r} = \frac{d\vec{r}}{dt} dt$ then

$$\int_c \vec{u} \cdot d\vec{r} = \int_{t_0}^{t_f} \vec{r}'(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt} dt$$

Note: the line integrals of a diagonal to (1,1) from the origin and through (1,0) are the same!

18.20 Work-energy theorem

$$F = ma$$

$$\vec{F} = \frac{d\vec{v}}{dt}$$

$$\vec{v} \cdot \frac{d\vec{v}}{dt} = \frac{1}{2} \frac{d}{dt} (\vec{v} \cdot \vec{v}) = \frac{1}{2} \frac{d}{dt} |\vec{v}|^2$$

$$\frac{d}{dt} \frac{1}{2} m |\vec{v}|^2 = \vec{F} \cdot \vec{v}$$

Integrate both sides

$$\int \frac{d}{dt} \frac{1}{2} m |\vec{v}|^2 dt = \int \vec{F} \cdot \frac{d\vec{r}}{dt} dt$$

$$\frac{1}{2} m |\vec{v}_f|^2 - \frac{1}{2} m |\vec{v}_i|^2 = \int_c \vec{F} \cdot d\vec{r}$$

$$KE_f - KE_i = W$$

18.21 Surface integral of a scalar field

$$\int_S f(\vec{r}) dS$$

$$\vec{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$$

$$dS = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv$$

Now assume $z = z(x, y)$ and $\vec{r} = x\mathbf{i} + y\mathbf{j} + z(x, y)\mathbf{k}$. As there are only two actual variables:

$$\frac{\partial \vec{r}}{\partial x} = \mathbf{i} + \frac{\partial z}{\partial x} \mathbf{k}$$

$$\frac{\partial \vec{r}}{\partial y} = \mathbf{j} + \frac{\partial z}{\partial y} \mathbf{k}$$

Use the determinant of a cross-product to determine

$$\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} = -\frac{\partial z}{\partial x} \mathbf{i} - \frac{\partial z}{\partial y} \mathbf{j} + \mathbf{k}$$

By Pythagoras,

$$\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| = \sqrt{\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 1}$$

So the integral is:

$$\int_S f(\vec{r}) dS = \int \int_A f(x, y, z(x, y)) \sqrt{1 + \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2} dx dy$$

18.22 Surface integral of a sphere

$$x = R \sin \theta \cos \phi$$

$$y = R \sin \theta \sin \phi$$

$$z = R \cos \theta$$

$$\vec{r}(\theta, \phi) = R \sin \theta \cos \phi \mathbf{i} + R \sin \theta \sin \phi \mathbf{j} + R \cos \theta \mathbf{k}$$

$$\frac{\partial \vec{r}}{\partial \theta} = R \cos \theta \cos \phi \mathbf{i} + R \cos \theta \sin \phi \mathbf{j} - R \sin \theta \mathbf{k}$$

$$\frac{\partial \vec{r}}{\partial \phi} = -R \sin \theta \sin \phi \mathbf{i} + R \sin \theta \cos \phi \mathbf{j}$$

Compute the determinant:

$$\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} = R^2 \sin^2 \theta \cos \phi \mathbf{i} + R^2 \sin^2 \theta \sin \phi \mathbf{j} + R^2 \cos \theta \sin \theta \mathbf{k}$$

$$dS = \left| \frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right| = R^2 \sin \theta (\sin^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta)^{1/2} = R^2 \sin \theta$$

$$S = \int_S dS = \int_0^{2\pi} \int_0^\pi R^2 \sin \theta d\theta d\phi = R^2 \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta = 4\pi R^2$$

18.23 Surface integral of a vector field

$$\int_S \vec{u} \cdot d\vec{S}$$

Now look for $d\vec{S} = \hat{n} dS$ where $\hat{n}$ is the normal unit vector and dS is the small surface area. So

$$= \int_S (\vec{u} \cdot \hat{n}) dS$$

Example. Surface area of a sphere.

$$\vec{r} = R\hat{r}$$

$$d\vec{S} = \hat{r} dS$$

$$\oiint \vec{r} \cdot d\vec{S} = \oint_S \vec{r} \cdot d\vec{S} = \oint_S R dS = 4\pi R^3$$

18.24 Flux integrals

Example. Mass flux is mass per time through a surface (e.g. charge through a spherical surface)

$$\int_S \rho \vec{u} \cdot d\vec{S}$$

18.25 Fundamental theorem of calculus

$$f(x)h \approx A(x+h) - A(x)$$
$$f(x) \approx \frac{A(x+h) - A(x)}{h}$$

At the limit $h \rightarrow 0$,

$$f(x) = A'(x)$$

$$f(x) = \frac{d}{dx}A(x)$$

So,

$$\int f(x) = A(x)$$

and $f(x)$ is the anti-derivative.

18.26 Gradient theorem

$$d\phi = \nabla\phi \cdot d\vec{r}$$

Value just depends on end points:

$$\int_c \nabla\phi \cdot d\vec{r} = \int_c d\phi = \phi(\vec{r}_2) - \phi(\vec{r}_1)$$

So over a closed curve:

$$\oint_c \nabla\phi \cdot d\vec{r} = 0$$

FTC applied to gradients. Going around a closed loop, the gradient 'dot' direction will cancel each other out at opposing sides, so the sum is 0.

18.27 Conservative vector field

4 equivalent conditions.

(1) $\vec{r} = \nabla\phi$.

(2) Curl is 0. $\nabla \times \vec{u} = 0$ because $\nabla \times \vec{u} = \nabla \times (\nabla \times \phi)$.

(3) $\int_c \vec{u} \cdot d\vec{r} = \int_c \nabla\phi \cdot d\vec{r}$ is path independent (just state-dependent as per the gradient theorem).

(4) $\oint_c \vec{u} \cdot d\vec{r} = 0$.

18.28 Conservation of energy

$$\int_c \vec{F} \cdot d\vec{r} = KE_f - KE_i$$

Since $\vec{F}$ is a conservative vector field, $\vec{F} = -\nabla PE$. Then

$$\begin{aligned} &= - \int_c \nabla PE \cdot d\vec{r} = -(PE_f - PE_i) = KE_f - KE_i \\ &PE_i + KE_i = PE_f + KE_f \end{aligned}$$

18.29 Divergence theorem

Volume flow = surface change (inside the volume, there are cancellations):

$$\int_V \nabla \cdot \vec{u} dV = \oint_S \vec{u} \cdot d\vec{S}$$

Alternative integral notations:

$$\iiint \nabla \cdot \vec{u} dV = \int_V \nabla \cdot \vec{u} dV = \iint_S \vec{u} \cdot d\vec{S} = \oiint \vec{u} \cdot d\vec{S} = \oint_S \vec{u} \cdot d\vec{S}$$

Over a box with one corner in origin and other corner at $(x, y, z) = (a, b, c)$

$$\begin{aligned} &\int_V \nabla \cdot \vec{u} dV = \int_0^c \int_0^b \int_0^a \left(\frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} + \frac{\partial u_3}{\partial z} \right) dx dy dz \\ &= \int_0^c \int_0^b \int_0^a \frac{\partial u_1}{\partial x} dx dy dz + \int_0^c \int_0^a \int_0^b \frac{\partial u_2}{\partial y} dy dx dz + \int_0^b \int_0^a \int_0^c \frac{\partial u_3}{\partial z} dz dx dy \end{aligned}$$

which reduce to surface integrals (where each of the three integrals has two terms so indicate the difference between two opposing sides of the cube with $d\vec{S} = \pm i dy dz$, so $x = 0$ and $x = a$ for the 'x-faces'):

$$\begin{aligned} &= \int_0^c \int_0^b u_1(a, y, z) - u_1(0, y, z) dy dz \\ &+ \int_0^c \int_0^a u_2(x, b, z) - u_2(x, 0, z) dx dz \\ &+ \int_0^b \int_0^a u_3(x, y, c) - u_3(x, y, 0) dx dy \\ &= \oint_S \vec{u} \cdot d\vec{S} \end{aligned}$$

Can be proven in spherical coordinates too:

$$\begin{aligned} \int_V (\nabla \cdot \vec{u}) dV &= \int_0^{2\pi} \int_0^\pi \int_0^R \left(\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta u_\theta) + \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi} \right) r^2 \sin \theta dr d\theta d\phi \\ &= \int_0^{2\pi} \int_0^\pi u_r(R, \theta, \phi) R^2 \sin \theta d\theta d\phi = \oint_S \vec{u} \cdot d\vec{S} \end{aligned}$$

18.30 Continuity equation

Divergence theorem:

$$\int_V \nabla \cdot \vec{u} dV = \oint_S \vec{u} \cdot d\vec{S}$$

Convert to flux with fluid density ρ :

$$\oint_S \rho \vec{u} \cdot d\vec{S} = \int_V \nabla \cdot (\rho \vec{u}) dV$$

On the left side, with fluid exiting when mass changes,

$$\frac{dM}{dt} = - \oint_S \rho \vec{u} \cdot d\vec{S}$$

On the right side,

$$M = \int_V \rho dV$$

$$\frac{dM}{dt} = \int_V \frac{\partial \rho}{\partial t} dV$$

Combining,

$$\frac{dM}{dt} = \int_V \frac{\partial \rho}{\partial t} dV = - \oint_S \rho \vec{u} \cdot d\vec{S} = - \int_V \nabla \cdot (\rho \vec{u}) dV$$

$$\int_V \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) \right) dV = 0$$

So,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0$$

For incompressible fluids, $\rho = \text{constant}$ so this is

$$= 0 + \rho \nabla \cdot \vec{u} = 0$$

so $\nabla \cdot \vec{u} = 0$ which means no divergence, i.e. what flows in flows out because it's incompressible.

18.31 Green's theorem

Stoke's theorem in 2D. Let $\vec{u} = u_1(x, y)\mathbf{i} + u_2(x, y)\mathbf{j}$ be a vector field. Then

$$\int_S \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) dS = \oint_C (u_1 dx + u_2 dy)$$

where the left side is the 3rd component of the curl (i.e. the curl in 2D) and the right side is the line integral enclosing surface S. So, the curl of the surface is equal to the line integral enclosing it. The surface can be broke down into

much smaller line integrals, where the internal contacts cancel each other out, leaving just the boundary - the line integral. The curl is state-dependent and path-independent, as in the FTC.

Proof. Let $(x, y) = (a, c)$ be one corner of a square in 2D and (b, d) be the other corner, with counterclockwise rotation of the vector field. Converting to xy-coordinates:

$$\begin{aligned} \int_S \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) dS &= \int_c^d \int_a^b \left(\frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} \right) dx dy \\ &= \int_c^d (u_2(b, y) - u_2(a, y)) dy - \int_a^b (u_1(x, d) - u_1(x, c)) dx \end{aligned}$$

where each of the four terms corresponds to a side of the square.

$$= \oint_C (u_1 dx + u_2 dy)$$

18.32 Stoke's theorem

Generalize Green's theorem to 3D.

$$\begin{aligned} \frac{\partial u_2}{\partial x} - \frac{\partial u_1}{\partial y} &= (\nabla \times \vec{u}) \cdot \mathbf{k} \\ d\vec{S} &= \hat{k} dS \\ u_1 dx + u_2 dy &= \vec{u} \cdot d\vec{r} \\ \int_S (\nabla \times \vec{u}) \cdot d\vec{S} &= \oint_C \vec{u} \cdot d\vec{r} \end{aligned}$$

18.33 Coordinate-free definitions

Divergence theorem leads to divergence:

$$\int_V (\nabla \cdot \vec{u}) dV = \oint_S \vec{u} \cdot d\vec{S}$$

Shrink this to a point

$$\nabla \cdot \vec{u} = \lim_{V \rightarrow 0} \frac{1}{V} \oint_S \vec{u} \cdot d\vec{S}$$

Source: $\nabla \cdot \vec{u} > 0$. Sink: $\nabla \cdot \vec{u} < 0$.

Stokes' theorem leads to curl:

$$\int_S (\nabla \times \vec{u}) \cdot d\vec{S} = \oint_C \vec{u} \cdot d\vec{r}$$

Shrink this to a point:

$$(\nabla \times \vec{u}) \cdot \hat{n} = \lim_{S \rightarrow 0} \frac{1}{S} \oint_C \vec{u} \cdot d\vec{r}$$

19 Tensor Algebra

19.1 Tensor definitions

Multidimensional array of numbers. Scalar is rank 0 (dimensions of the 'matrix'), vector is rank 1, matrices are rank 2, tensors are rank 3.

Object that is invariant under coordinate change, components that change in a special, predictable way under the change. Different coordinate systems are equally valid.

Best definition: Collection of vectors and covectors combined by the tensor product.

19.2 Vectors and vector transformations

Vectors are invariant but vector components vary depending on coordinates. A vector is a member of a vector space with $(V, S, +, \cdot)$ where V is vector, S is scalar, + is addition, and $\cdot$ and scalar multiplication. Vector is the actual arrow object in reality, which can be described with different coordinate bases.

Forward and backward transformations between coordinate systems. Write new basis vectors in terms of old basis vectors, and express as a transformation matrix F, the forward transformation. Do the reverse, with matrix B, the backward transformation. Then $FB = I$, so $B = F^{-1}$. For the forward transformation (n=new basis, o=old basis):

$$e_o = \sum_n F_{no} e_n$$

For example:

$$(e_1 \ e_2) = (e_a \ e_b) \begin{pmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{pmatrix}$$

For the backward transformation:

$$e_n = \sum_o B_{on} e_o$$

For example:

$$(e_a \ e_b) = (e_1 \ e_2) \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

Combine F and B:

$$e_n = \sum_o B_{on} e_o = \sum_o B_{on} \sum_n F_{no} e_n = \sum_o \left(\sum_n F_{no} B_{on} \right) e_n$$

We want therefore

$$\sum_o F_{no} B_{on} = \begin{cases} 1 & o = n \\ 0 & o \neq n \end{cases} = I = \delta_{on} = \delta_{no}$$

which is the Kronecker delta. Note that

$$\mathbf{e}_{new} = B\mathbf{e}_{old}$$

$$\mathbf{e}_{old} = F\mathbf{e}_{new}$$

which appears counterintuitive. But this is correct. Example 1. If the coordinate basis rotates CW, it 'appears' the vector 'rotates' CCW. Example 2. If the coordinate basis is doubled, it grows but the vector 'appears' to 'shrink'. So the vector experiences opposing change to the transformation. So vector components are contravariant. So vectors are contravariant tensors.

The vector can be described in terms of the new and old bases: $\vec{v} = \sum_j^n v_j e_j = \sum_i^n u_i e_i$. Then combine the new basis description with the backward transformation to get:

$$\vec{v} = \sum_j^n v_j e_j = \sum_j^n v_j \left(\sum_i^n B_{ij} e_i \right) = \sum_i^n \left(\sum_j^n B_{ij} v_j \right) e_i$$

So it must be that

$$\sum_i^n \left(\sum_j^n B_{ij} v_j \right) e_i = \sum_i^n u_i e_i$$

19.3 Covectors

In an orthonormal basis only, a column vector is just a re-oriented row vector. In other bases, not equivalent.

Row vector (covector, dual vector) is a function of column vectors. Example:

$$[2, 1] \left(\begin{bmatrix} 3 \\ -4 \end{bmatrix} \right) = 2$$

So dot product:

$$a(\vec{v}) = \sum_i^n a_i v^i$$

So $a : V \rightarrow \mathcal{R}$, and it's a linear functional and linear map: $a(n\vec{v} + m\vec{w}) = na(\vec{v}) + ma(\vec{w})$.

Visualize covectors.

$$[2, 1] \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = 2x + 1y$$

$$\begin{aligned}
2x + 1y = 0 &\rightarrow y = -2x \\
2x + 1y = 1 &\rightarrow y = -2x + 1 \\
&\vdots \\
2x + 1y = n &\rightarrow y = -2x + n
\end{aligned}$$

which are parallel lines and $[2, 1]$ is perpendicular to the set of lines, and the dot product (scalar) is the number of lines crossed by the column vector input. To increase the scalar, make the covector smaller (more dense):

$$\begin{aligned}
a(\vec{v}) &= 2 \\
2a(\vec{v}) &= 4 \\
(2a)(\vec{v}) &= 4
\end{aligned}$$

which makes the covector 'smaller' but appears to make it 'grow' as the covector components get numerically larger. Can add covectors to 'compose' a column vector input. It adds linearly:

$$\begin{aligned}
\beta(\vec{v}) &= 3 \\
\gamma(\vec{v}) &= 2
\end{aligned}$$

Then: $(\beta + \gamma)(\vec{v}) = \beta(\vec{v}) + \gamma(\vec{v}) = 5$. So it forms the dual space with $(V^*, S, +^*, \cdot^*)$.

Note: if the covector basis increased (e.g. from $[2, 1]$ to $[4, 2]$ then $y = -2x + n/2$), the field of parallel lines becomes denser. So the covector basis is contravariant.

19.4 Covector bases

Covectors themselves are invariant (so the visualization of 'parallel lines' doesn't change). However, covector components are not invariant (they can be described in different bases).

Covector bases for 2D are $\epsilon^1, \epsilon^2 : V \rightarrow \mathcal{R}$. The coefficients on these bases determine the number of parallel lines of the covector are pierced. Then,

$$\begin{aligned}
\epsilon^1(e_1) &= 1 \\
\epsilon^2(e_1) &= 0 \\
\epsilon^1(e_2) &= 1 \\
\epsilon^2(e_2) &= 0
\end{aligned}$$

So $\epsilon^i(e_j) = \delta_{ij}$. Then:

$$\begin{aligned}
\epsilon^1(\vec{v}) &= \epsilon^1(v^1 e_1 + v^2 e_2) = v^1 \epsilon^1(e_1) + v^2 \epsilon^1(e_2) = v^1(1) + v^2(1) = v^1 + v^2 \\
\epsilon^2(\vec{v}) &= \epsilon^2(v^1 e_1 + v^2 e_2) = v^1 \epsilon^2(e_1) + v^2 \epsilon^2(e_2) = v^1(0) + v^2(1) = v^2
\end{aligned}$$

So $\epsilon^i(\vec{v}) = v^i$ and define the coefficients:

$$a(e_1) = a_1$$

$$a(e_2) = a_2$$

So $a = a_1\epsilon^1 + a_2\epsilon^2$.

Example. Change the covector $[2, 1]$ in orthonormal basis so $a = 2\epsilon^1 + 1\epsilon^2$ (so **i** pierces the covector 'parallel lines' twice and **j** pierces once) and therefore $a(e_1) = 2, a(e_2) = 1$. Change the basis s.t. $a(e_a) = 5, a(e_b) = -3/4$, so $\hat{a}_1 = a(e_a) = a(2e_1 + 1e_2) = 2a(e_1) + 1a(e_2) = 2(2) + 2(1) = 5, \hat{a}_2 = a(e_b) = a(-1/2e_1 + 1/4e_2) = -1/2a(e_1) + 1/4a(e_2) = -1/2(2) + 1/4(1) = -3/4$ and $a = \hat{a}_1\hat{\epsilon}^1 + \hat{a}_2\hat{\epsilon}^2$. Then $[2, 1]F = [5, -3/4]$ and $[5, -3/4]B = [2, 1]$ where F and B change between the bases. The equations are therefore the reverse of vectors:

$$B \begin{vmatrix} 2 \\ 1 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

$$F \begin{vmatrix} 1 \\ 0 \end{vmatrix} = \begin{vmatrix} 2 \\ 1 \end{vmatrix}$$

In the orthonormal basis, $\begin{vmatrix} 2 \\ 1 \end{vmatrix}$ and $[2, 1]$ look similar even though they're completely different objects (a covector 'field of parallel lines' versus a vector 'arrow').

19.5 Covariance and Contravariance

Basis vectors transform like this:

$$b_{new} = F b_{old}$$

$$b_{old} = B b_{new}$$

Vector components transform in the opposite way (e.g. if the bases grow, the vector 'shrinks') so are contravariant ((1,0)-tensors):

$$v_{new} = F v_{old}$$

$$v_{old} = B v_{new}$$

Covector bases transform in the opposite way to basis vectors too (increasing the basis means increasing 'density' of the 'field of parallel lines'), so are contravariant.

Covector components transform in the same way as basis vectors so are covariant ((0,1)-tensors). Visualize if the orthonormal basis vectors grow 2x, then the covector components (the number of times the parallel lines are pierced) is also 2x.

19.6 Notation

L_1^2 means 1 is row, 2 is column.

Under Einstein notation: $L(e_j) = \sum_k^n L_j^k e_k = L_j^k e_k$.

Kronecker deltas are like index cancellors: $MI = M$ so $(MI)_{ki} = (MI)_k^i = \sum_j M_j^i \delta_k^j = M_k^i \delta_k^i = M_k^i = M$; the Kronecker delta forces the columns=rows indices as it is effectively an identity matrix.

19.7 Linear maps

Linear maps are (1,1)-tensors as they involve a covariant element and a contravariant element.

$$L_i^q \tilde{e}_q = L(e_i)$$

with $\tilde{e}_i = F_i^j e_j$,

$$= F_i^j L(e_j)$$

with $e_k = B_k^l \tilde{e}_l$,

$$\tilde{L}_i^q \tilde{e}_q = F_i^j L_j^k B_k^l \tilde{e}_l$$

$$\tilde{L}_i^l \tilde{e}_l = F_i^j L_j^k B_k^l \tilde{e}_l$$

So like moving to 'other' basis, performing L , then coming back:

$$\tilde{L}_j^i = B_k^i L_l^k F_j^l$$

Backward derivation uses $B_y^z F_z^x = \delta_y^x$:

$$\tilde{L}_j^i = B_k^i L_l^k f_j^l$$

Multiply left sides by F_l^s and right sides by B_t^i :

$$F_l^s \tilde{L}_i^l B_t^i = F_l^s B_k^l L_j^k F_i^j B_t^i = L_t^s$$

$$F_k^s \tilde{L}_l^k B_t^l = L_j^i$$

19.8 Metric tensors calculate lengths

Pythagorean theorem only works in orthonormal bases, doesn't work in non-orthonormal bases (don't get right angle triangles and right 'scaling'). The generalization of the Pythagorean theorem is the dot product:

$$\|\vec{v}\|^2 = \vec{v} \cdot \vec{v} = (v^1 e_1 + v^2 e_2) \cdot (v^1 e_1 + v^2 e_2) = (v^1)^2 (e_1 \cdot e_1) + 2v^1 v^2 (e_1 \cdot e_2) + (v^2)^2 (e_2 \cdot e_2)$$

In an orthonormal basis,

$$e_1 \cdot e_2 = 0$$

because they're orthogonal (i.e., $e_i \cdot e_j = \delta_{ij}$), so:

$$\|\vec{v}\|^2 = (v^1)^2 + (v^2)^2 = \|\tilde{v}\|^2$$

If non-orthonormal, need to change basis:

$$\tilde{e}_1 = 2e_1 + 1e_2$$

$$\tilde{e}_2 = -1/2e_1 + 1/4e_2$$

Then expand in terms of e_1 and e_2 using the above basis change

$$\|\tilde{\tilde{v}}\|^2 = \tilde{\tilde{v}} \cdot \tilde{\tilde{v}} = (\tilde{v}^1 \tilde{e}_1 + \tilde{v}^2 \tilde{e}_2) \cdot (\tilde{v}^1 \tilde{e}_1 + \tilde{v}^2 \tilde{e}_2)$$

and the result will be $\|\vec{v}\| = \|\tilde{\tilde{v}}\|$, but note the latter will have a 'middle' term (3 terms instead of the 2 in orthonormal bases).

Express norms like this, with the metric tensors g in old and new bases:

$$\|\vec{v}\|^2 = [v^1 \ v^2] \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{g_{old}} \begin{bmatrix} v^1 \\ v^2 \end{bmatrix} = [\tilde{v}^1 \ \tilde{v}^2] \underbrace{\begin{bmatrix} 5 & -3/4 \\ -3/4 & 5/10 \end{bmatrix}}_{g_{new}} \begin{bmatrix} \tilde{v}^1 \\ \tilde{v}^2 \end{bmatrix} = \|\tilde{\tilde{v}}\|^2$$

The metric tensor can be expressed in terms of dot products - as an example for the old basis:

$$g_{old} = \begin{bmatrix} e_1 \cdot e_1 & e_1 \cdot e_2 \\ e_2 \cdot e_1 & e_2 \cdot e_2 \end{bmatrix}$$

Note the metric tensor is symmetric because $g_{ij} = e_i \cdot e_j = e_j \cdot e_i = g_{ji}$.

19.9 Metric tensors calculate angles

Rotate unit vector e_1 by θ to $\tilde{e}_2$. Then $\tilde{e}_2 = (\cos\theta)e_1 + (\sin\theta)e_2$. Then

$$\tilde{e}_1 \cdot \tilde{e}_1 = e_1 \cdot e_1 = 1$$

$$\tilde{e}_1 \cdot \tilde{e}_2 = \cos\theta$$

$$\tilde{e}_2 \cdot \tilde{e}_2 = 1$$

So

$$g_{old} = I$$

$$g_{new} = \begin{bmatrix} 1 & \cos\theta \\ \cos\theta & 1 \end{bmatrix}$$

Now,

$$\vec{v} \cdot \vec{w} = (a\tilde{e}_1) \cdot (b\tilde{e}_2) = ab(\tilde{e}_1 \cdot \tilde{e}_2) = ab \cos\theta = \|\vec{v}\| \|\vec{w}\| \cos\theta$$

and

$$\begin{aligned} \vec{v} \cdot \vec{w} &= (v^1 e_1 + v^2 e_2) \cdot (w^1 e_1 + w^2 e_2) \\ &= v^1 w^1 g_{11} + v^2 w^2 g_{22} + v^1 w^2 g_{12} + v^2 w^1 g_{21} = v^i w^j g_{ij} \end{aligned}$$

so

$$\vec{v} \cdot \vec{w} = v^i w^j g_{ij}$$

19.10 Metric tensors

Metric tensor is function $g : V \times V \rightarrow \mathcal{R}$. To get length, $g(\vec{v}, \vec{v}) = v^i v^i g_{ii} = \|\vec{v}\|^2$.
To get angles, $g(\vec{v}, \vec{w}) = v^i w^i g_{ij} = |\vec{v}| |\vec{w}| \cos \theta$.

Intuition. Like change of areas.

19.11 Metric tensor change of basis

Use $\tilde{e}_j = F_j^i e_i$ to get

$$\tilde{g}_{ij} = \tilde{e}_i \cdot \tilde{e}_j = (F_i^k e_k) \cdot (F_j^l e_l) = F_i^k F_j^l (e_k \cdot e_l) = F_i^k F_j^l g_{kl}$$

Use $e_j = B_j^i \tilde{e}_i$ to get

$$g_{kl} = B_k^i B_l^j \tilde{g}_{ij}$$

These are (0,2)-tensors as they have no contravariant elements and 2 covariant types. There are two elements in the change of basis because there are two vectors (in the dot product) to change.

Note: $\|\vec{v}\|^2 = \tilde{v}^i \tilde{v}^j \tilde{g}_{ij} = v^k v^l g_{kl}$

19.12 (m,n)-Tensors

Superscripts are m (contravariant elements) and subscripts are n (covariant elements):

$$\tilde{T}^{abc\dots}_{xyz\dots} = (B_i^a B_j^b B_k^c \dots) T_{rst\dots}^{ijk\dots} (F_x^r F_y^s F_z^t \dots)$$

$$T_{ijk\dots}^{rst\dots} = (F_a^i F_b^j F_c^k \dots) \tilde{T}_{xyz\dots}^{abc\dots} (B_r^x B_s^y B_t^z \dots)$$

19.13 Bilinear forms

Another type of tensor. The metric tensor is an example of a bilinear form.

Metric tensors:

$$g : V \times V \rightarrow \mathcal{R}$$

$$ag(v, w) = g(av, w) = g(v, aw)$$

$$g(v + u, w) = g(v, w) + g(u, w)$$

$$g(v, w + t) = g(v, w) + g(v, t)$$

$$g(v, w) = v^i w^j g_{ij}$$

Bilinear form:

$$B : V \times V \rightarrow \mathcal{R}$$

$$aB(v, w) = B(av, w) = B(v, aw)$$

$$B(v + u, w) = B(v, w) + B(u, w)$$

$$B(v, w + t) = B(v, w) + B(v, t)$$

$$B(v, w) = v^i w^j B_{ij}$$

Forms: $V \times V \times \dots \times V \rightarrow \mathcal{R}$. Covectors are linear forms or 1-forms because they're linear and take one V as input.

Bilinear forms are 2-forms with $V \times V \rightarrow \mathcal{R}$. Bilinear because if v or w is held constant, the other is like covectors.

Metric tensors are very specific bilinear forms with two extra properties (symmetric matrix + non-negative norms):

$$g(v, w) = v^i w^j g_{ij} = v^i w^j g_{ji} = g(w, v)$$

$$g(v, v) = \|v\|^2 \geq 0$$

Bilinear forms are (0,2)-tensors.

19.14 Linear maps from the outer product

Linear maps are linear combinations of vectors and covectors using the outer product (the Kronecker or tensor product).

Vectors are (1,0)-tensors, covectors are (0,1)-tensors, so everything else is a combination of vectors and covectors.

Covector times vector. Example. $[2, 1] \begin{vmatrix} 3 \\ -4 \end{vmatrix} = 2$ (a scalar, (0,0)-tensor) but $\begin{vmatrix} 3 \\ -4 \end{vmatrix} [2, 1] = \begin{vmatrix} 6 & 3 \\ -8 & -4 \end{vmatrix}$ (a matrix or linear map, (1,1)-tensor).

Pure matrices can be constructed from vectors and covectors in an outer product to form a matrix but are linearly dependent ($L_i^j = v^i a_j$). Impure matrices cannot be constructed from vectors and covectors as an outer product but are linearly independent ($L_i^j \neq v^i a_j$).

Note: all outer products are pure matrices.

Any linear map can be written using the vector bases $e_1 = (1, 0)'$, $e_2 = (0, 1)'$ and covector bases $\epsilon_1 = (1, 0)$, $\epsilon_2 = (0, 1)$:

$$e_1 \epsilon_1 = \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}$$

$$e_1 \epsilon_2 = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}$$

$$e_2 \epsilon_1 = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix}$$

$$e_2\epsilon_2 = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix}$$

and these four matrices can construct any matrix:

$$a \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + b \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} + c \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} + d \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

so $e_1\epsilon_1, e_1\epsilon_2, e_2\epsilon_1, e_2\epsilon_2$ is a basis for $V \rightarrow V$ (but they're not the only ones - there are an infinite number).

$$L = ae_1\epsilon_1 + be_1\epsilon_2 + ce_2\epsilon_1 + de_2\epsilon_2$$

So

$$L = L_j^i e_i \epsilon^j$$

which is a linear map. Note: $e_i \epsilon^j = e_i \otimes \epsilon^j$, the tensor (Kronecker) product of a vector and covector.

19.15 Advantages of linear maps as outer products

Expressing a linear maps as outer products.

(1) Can transform basis of linear maps more easily. Express as vectors and covectors, and then change the vector and covector separately.

$$L_j^i e_i \epsilon^j = L_j^i (e_i \otimes \epsilon^j)$$

To change from the old basis to the new basis, where the basis vector is covariant and the covector basis is contravariant, and the last step contains the actual linear transformation:

$$L_l^k e_k \epsilon^l = L_l^k (B_k^i \tilde{e}_i) (F_j^l \tilde{\epsilon}^j) = (B_k^i L_l^k F_j^l) \tilde{e}_i \tilde{\epsilon}^j = (B_k^i L_l^k F_j^l) \tilde{e}_i \tilde{\epsilon}^j = \tilde{L}_j^i \tilde{e}_i \tilde{\epsilon}^j$$

Reverse:

$$L = \tilde{L}_l^k \tilde{e}_k \tilde{\epsilon}^l = \tilde{L}_l^k (F_k^i e_i) (B_j^l \epsilon^j) = (F_j^i L_l^k B_j^l) e_i \epsilon^j = L_j^i e_i \epsilon^j$$

(2) Input a vector more easily. If the input is a vector $\vec{v} = v^k e_k$, then

$$\vec{w} = L(\vec{v}) = L_j^i e_i \epsilon^j (v^k e_k) = L_j^i v^k e_i \epsilon^j e_k = L_j^i e_i (v^k \delta_k^j) = L_j^i e_i v^j = L_j^i v^j e_i$$

so it outputs a vector.

(3) Tensor product gives correct array:

$$\begin{vmatrix} v^1 \\ v^2 \end{vmatrix} [a_1 \ a_2] = \begin{vmatrix} v^1 \\ v^2 \end{vmatrix} \otimes [a_1 \ a_2] = \begin{vmatrix} v^1 \\ v^2 \end{vmatrix} a_1 \begin{vmatrix} v^1 \\ v^2 \end{vmatrix} a_2 = \begin{vmatrix} v^1 a_1 & v^1 a_2 \\ v^2 a_1 & v^2 a_2 \end{vmatrix}$$

19.16 Bilinear forms

Bilinear forms are linear combinations of covector-covector pairs. Since each covector takes a vector as an input, the covector-covector pair inputs a pair of vectors = matrix.

$$B = B_{ij}\epsilon^i\epsilon^j = B_{ij}(\epsilon^i \otimes \epsilon^j)$$

(1) Bilinear transformation:

$$B = B_{kl}\epsilon^k\epsilon^l = B_{kl}(F_i^k\tilde{\epsilon}^i)(F_j^l\tilde{\epsilon}^j) = (F_i^kF_j^lB_{kl})\tilde{\epsilon}^i\tilde{\epsilon}^j = \tilde{B}_{ij}\tilde{\epsilon}^i\tilde{\epsilon}^j$$

and the reverse is:

$$B = \tilde{B}_{kl}\tilde{\epsilon}^k\tilde{\epsilon}^l = \tilde{B}_{kl}(B_i^k\epsilon^i)(B_j^l\epsilon^j) = (B_j^k\tilde{B}_{kl}B_i^l)\epsilon^i\epsilon^j = B_{ij}\epsilon^i\epsilon^j$$

(2) Component multiplication formula:

$$\begin{aligned} B(\vec{v}, \vec{w}) &= B_{ij}\epsilon^i\epsilon^j(v^ke_k, w^le_l) = B_{ij}\epsilon^i(v^ke_k)\epsilon^j(w^le_l) \\ &= B_{ij}v^kw^l\epsilon^i(e_k)\epsilon^j(e_l) = B_{ij}v^kw^l\delta_k^i\delta_l^j = B_{ij}v^iw^j \end{aligned}$$

(3) Tensor product gives correct form

$$\begin{aligned} &[a_1 \ a_2] \otimes [b_1 \ b_2] \\ &= [[a_1 \ a_2]b_1 \ [a_1 \ a_2]b_2] = [[a_1b_1 \ a_2b_1] \ [a_1b_2 \ a_2b_2]] \end{aligned}$$

which can be used in multiplication. The form

$$B(\vec{v}, \vec{w}) = B_{ij}v^iw^j = [v^1 \ v^2] \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} \begin{bmatrix} w^1 \\ w^2 \end{bmatrix}$$

isn't right as $\vec{v}$ is a vector, not a covector (row vector). But with the above tensor product,

$$\begin{aligned} B(\vec{v}, \vec{w}) &= [[B_{11} \ B_{12}] \ [B_{21} \ B_{22}]] \begin{bmatrix} v^1 \\ v^2 \end{bmatrix} \begin{bmatrix} w^1 \\ w^2 \end{bmatrix} \\ &= (B_{11}v^1 + B_{12}v^2)w^1 + (B_{21}v^1 + B_{22}v^2)w^2 \\ &= B_{11}v^1w^1 + B_{12}v^1w^2 + B_{21}v^2w^1 + B_{22}v^2w^2 = B_{ij}v^iw^j \end{aligned}$$

19.17 Higher-type tensors

Best definition of tensors: collection of vectors and covectors combined by the tensor product.

Can generalize tensor products to higher-type tensors. Examples: $D = D^{ab}e_ae_b$ which is a (2,0)-tensor and $Q = Q_{jk}^ie_i\epsilon^j\epsilon^k$ which is a (1,2)-tensor. Can determine change of bases and array shapes, but since the tensors become complicated, there are many ways of feeding in vectors and feeding in other tensors.

19.18 Tensor products and tensor product spaces

Tensor product:

$$n(\vec{v} \otimes a) = (n\vec{v}) \otimes a = \vec{v} \otimes (na)$$

$$\vec{v} \otimes a + \vec{v} \otimes b = \vec{v} \otimes (a + b)$$

$$\vec{v} \otimes a + \vec{v} \otimes a = (\vec{v} + \vec{v}) \otimes a$$

Since it's a linear map, it's a vector space. It's a tensor product of vector spaces, the tensor product space:

$$\vec{v}, \vec{w}, e_1, e_2 \in V$$

$$a, b, \epsilon^1, \epsilon^2 \in V^*$$

$$\vec{v}a, \vec{v}b, \vec{w}a, \vec{w}b, e_1\epsilon^2, L_j^i e_i \epsilon^j \in V \otimes V^*$$

What are the elements of the tensor product space? (1,1)-tensors. In the notation $L_j^i e_i \epsilon^j$, the higher indices are for vectors and lower indices are for covectors.

$$L_j^i v^j = w^i, V \rightarrow V$$

$$L_j^i a_i = b_j, V^* \rightarrow V^*$$

$$L_j^i v^j a_i = \text{scalar}, V \times V^* \rightarrow \mathcal{R}$$

$$L_j^i a_i v^j = \text{scalar}, V^* \times V \rightarrow \mathcal{R}$$

The tensor product space of (0,2)-tensors is $B_{ij} \epsilon^i \epsilon^j \in V^* \otimes V^*$:

$$B_{ij} v^i w^j = \text{scalar}, V \times V \rightarrow \mathcal{R}$$

$$B_{ij} v^i = a_j, V \rightarrow V^*$$

$$B_{ij} v^j = b_i, V^* \rightarrow V$$

Using vectors and covectors, can build any tensor product space: $V \otimes V : A^{ij}$, $V \otimes V^* : L_j^i$, $V^* \otimes V : L_j^i$, $V^* \otimes V^* : B_{ij}$, $V^* \otimes V \otimes V^* : T_{ik}^j$. Examples.

$$T_{ikl}^j v^i a_j w^k u^l, V \times V^* \times V \times V \rightarrow \mathcal{R}$$

$$T_{ikl}^j U^{ikl} b_j, (V \otimes V \otimes V) \times V^* \rightarrow \mathcal{R}$$

$$T_{ikl}^j a_j D^{kl}, V^* \times (V \otimes V) \rightarrow V^*$$

$$T_{ikl}^j L_j^i, (V \otimes V^*) \rightarrow V^* \otimes V^*$$

19.19 Tensor v Kronecker product

Tensor product is algebraic, while the Kronecker product is about arrays. Basically the same thing.

Example. Tensor product:

$$(e_i \otimes \epsilon^j)(\vec{v}) = e_i \otimes (\epsilon^j(\vec{v})) = e_i \otimes (\epsilon^j(v^k e_k)) = v^k e_i \otimes (\epsilon^j(e_k)) = v^k e_i \delta_k^j = v^j e_i$$

but the Kronecker product would just do this with arrays.

Example 2. Tensor product:

$$\vec{v} \otimes a = (v^i e_i) \otimes (a_j \epsilon^j) = v^i a_j (e_i \otimes \epsilon^j)$$

Analogous Kronecker product:

$$\begin{bmatrix} v^1 \\ v^2 \end{bmatrix} \otimes [a_1 \ a_2] = \begin{bmatrix} v^1 a_1 \\ v^2 a_1 \end{bmatrix} \begin{bmatrix} v^1 a_2 \\ v^2 a_2 \end{bmatrix}$$

19.20 Multilinear maps

Holding any all except one of the tensors constant, the tensor product should just reduce to a linear map. So a multilinear map has these properties:

$$T(x_1 \dots n x_i \dots x_n) = n T(x_1 \dots x_i \dots x_n)$$

$$T(x_1 \dots x_i + y_i \dots x_n) = T(x_1 \dots x_i \dots x_n) + T(x_1 \dots y_i \dots x_n)$$

19.21 Relating vectors and covectors

Vector: $\vec{v} = v^i e_i$. Covector: $a = a_j \epsilon^j$. How to relate V and V^* ? In other words, e_i with ϵ^i ? Problem: vector bases are covariant but covector bases are contravariant. So, if $\tilde{e}_i = 2e_i$ (so uses the forward transformation F) then $\tilde{\epsilon}^i = 1/2\epsilon^i$ (so uses the backward transformation B), so they're not easily related.

Dot product relates them. $\vec{v} \cdot - : V \rightarrow \mathcal{R} \in V^*$ because that's what a covector does (turn a vector into a scalar). Then,

$$\vec{v} \leftrightarrow \vec{v} \cdot -$$

$$2\vec{v} \leftrightarrow 2\vec{v} \cdot - = x_i \epsilon^i$$

which builds in terms of covectors. To do this, use the metric tensor:

$$\vec{v} \cdot \vec{w} = g(\vec{v}, \vec{w}) = g_{ij} \epsilon^i \epsilon^j (v^k e_k, w^l e_l)$$

The metric tensor is symmetric ($g(\vec{v}, \vec{w}) = g(\vec{w}, \vec{v})$), so the order doesn't matter:

$$= g_{ij} \epsilon^i (v^k e_k) \epsilon^j (w^l e_l) = g_{ij} v^k w^l \epsilon^i (e_k) \epsilon^j (e_l) = g_{ij} v^k w^l \delta_k^i \delta_l^j = g_{ij} v^i w^j$$

Now,

$$\vec{v} \cdot - = g(\vec{v}, -) = g_{ik} \epsilon^i \epsilon^k (v^j e_j) = g_{ik} v^j \epsilon^i \epsilon^k (e_j) = g_{ik} v^j \epsilon^i \delta_j^k = g_{ij} v^j \epsilon^i = v_i \epsilon^i$$

So, $v_i = g_{ij} v^j$ and the metric tensor 'lowers' the index. This is sometimes called the 'covariant metric tensor'. The metric tensor is

$$g : V \times V \rightarrow \mathcal{R}$$

$$g \in V^* \otimes V^*$$

But it's ALSO now

$$g : V \rightarrow V^*$$

The reverse, $V^* \rightarrow V$ is through the inverse metric tensor or $\mathfrak{g} : V \otimes V$. This is the contravariant metric tensor. Definition:

$$\mathfrak{g}^{ki} g_{ij} = \delta_j^k$$

So input $v_i = g_{ij} v^j$ to get

$$\mathfrak{g}^{ki} v_i = \mathfrak{g}^{ki} g_{ij} v^j = \delta_j^k v^j = v^k$$

so it raises the index.

Example.

$$\begin{aligned} Q &\in V \otimes V^* \otimes V \\ Q &= Q_{jk}^i e_i \epsilon^j \epsilon^k = Q_{jk}^i \mathfrak{g}^{jx} = Q_k^{ix} \\ Q' &= Q_k^{ix} e_i e_x \epsilon^k \end{aligned}$$

which is $Q' \in V \otimes V \otimes V^*$. So use the covariant and contravariant metric tensor to move between tensor product spaces.

19.22 Sharp and flat notation

Flat notation: turns vector into covector.

$$\vec{v} \cdot - = g(\vec{v}, -) = \flat \vec{v}$$

So:

$$\vec{v} = v^i e_i$$

$$\flat \vec{v} = v_i \epsilon^i$$

Sharp notation: turns covector into vector.

$$a = \vec{a} \cdot - = \sharp(a, -) = \sharp a$$

So:

$$a = a_i \epsilon^i$$

$$\sharp a = a^i e_i$$

20 Tensor Calculus and Differential Geometry

20.1 Basis vectors are partial derivatives

In Cartesian coordinates:

$$e_x = \lim_{h \rightarrow 0} \frac{\vec{R}_h(x+h, y) - \vec{R}(x, y)}{h} = \frac{\partial \vec{R}}{\partial x}$$

Similarly,

$$e_y = \frac{\partial \vec{R}}{\partial y}$$

In polar coordinates:

$$e_r = \frac{\partial \vec{R}}{\partial r}$$

$$e_\theta = \frac{\partial \vec{R}}{\partial \theta}$$

but note that the last is often normalized because it's bigger when further from the origin:

$$e_\theta = \frac{1}{r} \frac{\partial \vec{R}}{\partial \theta}$$

20.2 Jacobian and inverse Jacobian

Change from Cartesian to polar basis vectors. Use multivariate chain rule:

$$e_r = \frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y} \frac{\partial y}{\partial r}$$

$$e_\theta = \frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y} \frac{\partial y}{\partial \theta}$$

which is then

$$\begin{aligned} \frac{\partial \vec{R}}{\partial r} &= \frac{\partial \vec{R}}{\partial x} \cos \theta + \frac{\partial \vec{R}}{\partial y} \sin \theta \\ \frac{\partial \vec{R}}{\partial \theta} &= \frac{\partial \vec{R}}{\partial x} (-r \sin \theta) + \frac{\partial \vec{R}}{\partial y} r \cos \theta \end{aligned}$$

where the 'r' exists because the partial derivative wasn't normalized in polar coordinates for θ . Above is equivalent to:

$$\begin{aligned} \frac{\partial \vec{R}}{\partial r} &= \frac{\partial \vec{R}}{\partial x} \frac{x}{r} + \frac{\partial \vec{R}}{\partial y} \frac{y}{r} \\ \frac{\partial \vec{R}}{\partial \theta} &= \frac{\partial \vec{R}}{\partial x} (-y) + \frac{\partial \vec{R}}{\partial y} x \end{aligned}$$

Then

$$F = \begin{vmatrix} \frac{x}{r} & -y \\ \frac{y}{r} & x \end{vmatrix} = \begin{vmatrix} \cos\theta & -r \sin\theta \\ \sin\theta & r \cos\theta \end{vmatrix} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = J$$

Note: J is a general formula and its specific values depend on the specific point in Cartesian coordinates.

$$e_r = \frac{\partial x}{\partial r} e_x + \frac{\partial y}{\partial r} e_y$$

$$e_\theta = \frac{\partial x}{\partial \theta} e_x + \frac{\partial y}{\partial \theta} e_y$$

For polar to Cartesian basis vector change:

$$e_x = \frac{\partial \vec{R}}{\partial x} = \frac{\partial \vec{R}}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial \vec{R}}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$e_y = \frac{\partial \vec{R}}{\partial y} = \frac{\partial \vec{R}}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial \vec{R}}{\partial \theta} \frac{\partial \theta}{\partial y}$$

which is then, with $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$,

$$e_x = \frac{\partial \vec{R}}{\partial x} = \frac{\partial \vec{R}}{\partial r} \frac{x}{\sqrt{x^2 + y^2}} + \frac{\partial \vec{R}}{\partial \theta} \frac{-y}{x^2 + y^2}$$

$$e_y = \frac{\partial \vec{R}}{\partial y} = \frac{\partial \vec{R}}{\partial r} \frac{y}{\sqrt{x^2 + y^2}} + \frac{\partial \vec{R}}{\partial \theta} \frac{x}{x^2 + y^2}$$

So:

$$e_x = \frac{\partial \vec{R}}{\partial x} = \frac{\partial \vec{R}}{\partial r} \frac{x}{r} + \frac{\partial \vec{R}}{\partial \theta} \frac{-y}{r^2}$$

$$e_y = \frac{\partial \vec{R}}{\partial y} = \frac{\partial \vec{R}}{\partial r} \frac{y}{r} + \frac{\partial \vec{R}}{\partial \theta} \frac{x}{r^2}$$

Then:

$$B = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} \\ \frac{-y}{r^2} & \frac{x}{r^2} \end{vmatrix} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = J^{-1}$$

$$e_x = \frac{\partial r}{\partial x} e_r + \frac{\partial \theta}{\partial x} e_\theta$$

$$e_y = \frac{\partial r}{\partial y} e_r + \frac{\partial \theta}{\partial y} e_\theta$$

Note:

$$JJ^{-1} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial x}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial x}{\partial \theta} \frac{\partial \theta}{\partial x} & \frac{\partial x}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial x}{\partial \theta} \frac{\partial \theta}{\partial y} \\ \frac{\partial y}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial \theta}{\partial x} & \frac{\partial y}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial y}{\partial \theta} \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = I$$

To get $a_1 1$ entry, note

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial x}$$

but then replace f with x:

$$1 = \frac{\partial x}{\partial x} = \frac{\partial x}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial x}{\partial \theta} \frac{\partial \theta}{\partial x}$$

More generally, if

$$\begin{aligned} c_1 &= x, c_2 = y \\ p_1 &= r, p_2 = \theta \end{aligned}$$

then:

$$J J^{-1} = \frac{\partial c_i}{\partial p_j} \frac{\partial p_j}{\partial c_k} = \frac{dc_i}{dc_k} = \delta_k^i$$

20.3 Derivatives are vectors

Want to get vector components for vector fields along curves. So look at tangent vectors of $\vec{R}$ which is calculated ultimately from

$$\frac{\partial \vec{R}}{\partial \lambda} = \lim_{h \rightarrow 0} \frac{\vec{R}(\lambda + h) - \vec{R}(\lambda)}{h}$$

to get

$$\frac{\partial \vec{R}}{\partial \lambda} = \frac{\partial \vec{R}}{\partial x} \frac{dx}{d\lambda} + \frac{\partial \vec{R}}{\partial y} \frac{dy}{d\lambda} = \frac{dx}{d\lambda} e_x + \frac{dy}{d\lambda} e_y = \frac{dc^i}{d\lambda} \frac{\partial \vec{R}}{\partial c^i}$$

Similarly, in polar coordinates:

$$\frac{\partial \vec{R}}{\partial \lambda} = \frac{\partial \vec{R}}{\partial r} \frac{dr}{d\lambda} + \frac{\partial \vec{R}}{\partial \theta} \frac{d\theta}{d\lambda} = \frac{dr}{d\lambda} e_r + \frac{d\theta}{d\lambda} e_\theta = \frac{dp^i}{d\lambda} \frac{\partial \vec{R}}{\partial p^i}$$

So, this is analogous to:

$$\begin{aligned} \vec{v} &= v^1 e_1 + v^2 e_2 = v^i e_i \\ \vec{v} &= \tilde{v}^1 \tilde{e}_1 + \tilde{v}^2 \tilde{e}_2 = \tilde{v}^i \tilde{e}_i \end{aligned}$$

where any vector $\vec{v}$ can be written from different perspectives (different bases).

20.4 Vector field components are contravariant

Recall vector components are contravariant:

$$\tilde{v}^i = B_j^i v^j$$

Analogously, vector field components are contravariant. First note the basis vectors in the vector field are:

$$\frac{\partial \vec{R}}{\partial p^j} = \frac{\partial c^i}{\partial p^j} \frac{\partial \vec{R}}{\partial c^i} = F \frac{\partial \vec{R}}{\partial c^i} = J \frac{\partial \vec{R}}{\partial c^i}$$

$$\frac{\partial \vec{R}}{\partial c^j} = \frac{\partial p^i}{\partial c^j} \frac{\partial \vec{R}}{\partial p^i} = B \frac{\partial \vec{R}}{\partial p^i} = J \frac{\partial \vec{R}}{\partial p^i}$$

whose vector field components are:

$$\begin{aligned} \frac{dp^i}{d\lambda} &= \frac{\partial p^i}{\partial c^j} \frac{dc^j}{d\lambda} = B \frac{dc^j}{d\lambda} \\ \frac{dc^i}{d\lambda} &= \frac{\partial c^i}{\partial p^j} \frac{dp^j}{d\lambda} = F \frac{dp^j}{d\lambda} \end{aligned}$$

20.5 Derivative operators

Remove $\vec{R}$ from the equations, as it requires an origin and later on, for manifolds, there won't be an origin. So the basis vectors are derivative operators which can be added and scaled by scalars:

$$\begin{aligned} e_x &= \frac{\partial}{\partial x} \\ e_y &= \frac{\partial}{\partial y} \end{aligned}$$

so e_x is just differentiating in the x-direction, and e_y is differentiating in the y-direction. Then, $\frac{d}{d\lambda}$ is just differentiating along the parameter of the curve. Can break this 'angled' operator into its components using the chain rule:

$$\frac{d}{d\lambda} = \frac{dx}{d\lambda} \frac{\partial}{\partial x} + \frac{dy}{d\lambda} \frac{\partial}{\partial y}$$

20.6 Tangent vector space

Tangent vector space $T_p M$ is where

$$\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$$

live. As they add and scale, they really form a vector space. Most useful in INTRINSIC geometry, e.g. can't use e_x, e_y on curved surfaces within those surfaces, so use tangent vector spaces.

20.7 Differentials are covectors

Differentials dx are covectors.

d is an operator that takes a scalar field f and outputs a covector field df . The covector field is a set of contours of the rate of change of the scalar field. The scalar field f is called a '0-form' and the covector field df is a '1-form' or 'differential form'.

$dx, dy, dr, d\theta$ are all covector fields (dr is radial outward, while $d\theta$ is a radial-rotation).

When df (the covector field) acts on $\vec{v}$, it outputs a scalar. It indicates the number of covector 'stacks' 'pierced', but since the 'stacks' change at each point in the covector field, use the 'tangent' covector field at the origin of $\vec{v}$. In fact, $df(\vec{v})$ is the directional derivative:

$$\nabla_{\vec{v}} f = \frac{\partial f}{\partial \vec{v}} = df(\vec{v})$$

Note that

$$df(e_x) = df\left(\frac{\partial}{\partial x}\right) = \frac{\partial f}{\partial x}$$

$$df(e_y) = df\left(\frac{\partial}{\partial y}\right) = \frac{\partial f}{\partial y}$$

Using above:

$$dx(e_x) = dx\left(\frac{\partial}{\partial x}\right) = \frac{\partial x}{\partial x} = 1$$

$$dx(e_y) = dx\left(\frac{\partial}{\partial y}\right) = \frac{\partial x}{\partial y} = 0$$

$$dy(e_y) = 1, dy(e_x) = 0$$

which means:

$$dc^i\left(\frac{\partial}{\partial c^j}\right) = \frac{\partial c^i}{\partial c^j} = \delta_j^i$$

But this is just like the covector basis acting on basis vectors $\epsilon^1(e_1) = 1, \epsilon^2(e_1) = 0, \epsilon^1(e_2) = 0, \epsilon^2(e_2) = 1$ or $\epsilon^i e_j = \delta_j^i$. So differentials are covector bases. To see this algebraically:

$$\frac{df}{d\lambda} = \frac{\partial f}{\partial x} \frac{dx}{d\lambda} + \frac{\partial f}{\partial y} \frac{dy}{d\lambda}$$

$$df \frac{d}{d\lambda} = \frac{\partial f}{\partial x} dx \frac{d}{d\lambda} + \frac{\partial f}{\partial y} dy \frac{d}{d\lambda}$$

$$df\left(\frac{d}{d\lambda}\right) = \frac{\partial f}{\partial x} dx\left(\frac{d}{d\lambda}\right) + \frac{\partial f}{\partial y} dy\left(\frac{d}{d\lambda}\right)$$

Factoring out the bracketed term,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

So dx, dy are dual bases of differential forms (covector fields). So:

$$e_1 = \frac{\partial}{\partial x}, e_2 = \frac{\partial}{\partial y}$$

$$\epsilon^1 = dx, \epsilon^2 = dy$$

Can switch between bases:

$$\begin{aligned} df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \frac{\partial f}{\partial r} dr + \frac{\partial f}{\partial \theta} d\theta \\ &= \frac{\partial f}{\partial c^i} dc^i = \frac{\partial f}{\partial p^i} dp^i \end{aligned}$$

So geometric intuition of differentials (of planes) is the same as the geometric intuition of differential forms (of covector fields) which combine covector components.

20.8 Covector bases are contravariant

$$dp^i = \frac{\partial p^i}{\partial c^j} dc^j = J^{-1} dc^j$$

$$\left| \frac{dr}{d\theta} \right| = J^{-1} \left| \frac{dx}{dy} \right|$$

$$dc^i = \frac{\partial c^i}{\partial p^j} dp^j = J dp^j$$

$$\left| \frac{dx}{dy} \right| = J \left| \frac{dr}{d\theta} \right|$$

$$df = \frac{\partial f}{\partial c^i} dc^i = \frac{\partial f}{\partial p^i} p^i$$

So:

$$\frac{\partial f}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial f}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial f}{\partial y}$$

$$\frac{\partial f}{\partial \theta} = \frac{\partial x}{\partial \theta} \frac{\partial f}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial f}{\partial y}$$

which is

$$\frac{\partial f}{\partial p^j} = \frac{\partial c^i}{\partial p^j} \frac{\partial f}{\partial c^i} = J \frac{\partial f}{\partial c^i}$$

So:

$$\frac{\partial f}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta}$$

$$\frac{\partial f}{\partial y} = \frac{\partial r}{\partial y} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta}$$

which is

$$\frac{\partial f}{\partial c^i} = \frac{\partial p^j}{\partial c^i} \frac{\partial f}{\partial p^j} = J^{-1} \frac{\partial f}{\partial p^j}$$

20.9 Differential form interpretation of integrals

FTC:

$$\int_a^b dF = F(b) - F(a)$$

Fundamental theorem of line integrals = Gradient theorem:

$$\int_a^b \nabla F \cdot dr = F(b) - F(a)$$

where all the dot products are being summed and is path-independent. It results in only state-dependence. Easier to visualize as a path in a scalar field, but the contours of that scalar field can be represented as a covector field. The integral is path-independent because it's just about the initial and final states and how many covector field contours separate the two.

$$\int d\theta = \theta(b) - \theta(a)$$

$$\int_p f(x)dx$$

where p is the path and $f(x)dx$ is the covector field, with $f(x)$ being the type of covector field.

$$\int_p f(x)dx = \int_p \frac{dF}{dx} dx = \int dF = F(b) - F(a)$$

so F is the covector field and dF is a covector field.

Example 1. A 1D integral gives a 1D covector field.

$$\int (-6x + 4)dx$$

with

$$F(x) = -3x^2 + 4x + c$$

Example 2. A 2D integral gives a 2D covector field:

$$\int ydx + xdy = \int \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = F(b) - F(a)$$

where $F(x, y) = xy + c$.

Substitution (e.g. $u = 2x, v = 2y$) will simply re-base and re-scale the covector field and components but won't change the answer:

$$dF = \frac{dF}{dx} dx + \frac{dF}{dy} dy = \frac{dF}{du} du + \frac{dF}{dv} dv$$

20.10 Arc lengths

Arc lengths:

$$\begin{aligned}
 \int \left\| \frac{d\vec{R}}{dt} \right\| dt &= \int \sqrt{\frac{d\vec{R}}{dt} \cdot \frac{d\vec{R}}{dt}} dt = \int \sqrt{\left(\frac{\partial \vec{R}}{\partial x} \frac{dx}{dt} + \frac{\partial \vec{R}}{\partial y} \frac{dy}{dt} \right) + \left(\frac{\partial \vec{R}}{\partial x} \frac{dx}{dt} + \frac{\partial \vec{R}}{\partial y} \frac{dy}{dt} \right)} dt \\
 &= \int \sqrt{\left(\frac{dx}{dt} \right)^2 \left(\frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial x} \right) + 2 \frac{dx}{dt} \frac{dy}{dt} \left(\frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial y} \right) + \left(\frac{dy}{dt} \right)^2 \left(\frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial y} \right)} dt \\
 &= \int \sqrt{\left(\frac{dx}{dt} \right)^2 (1) + 2 \frac{dx}{dt} \frac{dy}{dt} (0) + \left(\frac{dy}{dt} \right)^2 (1)} dt \\
 &= \int \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt
 \end{aligned}$$

Magnitude of velocity vector tangent to curve:

$$\left\| \frac{d\vec{R}}{dt} \right\| = \sqrt{\sum_i \sum_j \frac{dq^i}{dt} \frac{dq^j}{dt} \left(\frac{\partial \vec{R}}{\partial q^i} \cdot \frac{\partial \vec{R}}{\partial q^j} \right)} = \sqrt{\frac{dq^i}{dt} \frac{dq^j}{dt} \left(\frac{\partial \vec{R}}{\partial q^i} \cdot \frac{\partial \vec{R}}{\partial q^j} \right)}$$

20.11 Metric tensors from tangent norms

$$\begin{aligned}
 \left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} = \frac{dc^i}{d\lambda} \frac{dc^j}{d\lambda} g_{ij} = \frac{dp^i}{d\lambda} \frac{dp^j}{d\lambda} \tilde{g}_{ij} \\
 \tilde{g}_{ij} &= \frac{\partial c^k}{\partial p^i} \frac{\partial c^l}{\partial p^j} g_{kl} \\
 g_{kl} &= \frac{\partial p^i}{\partial c^k} \frac{\partial p^i}{\partial c^l} \tilde{g}_{ij}
 \end{aligned}$$

20.12 Metric tensors in curved surfaces

Extrinsic geometry. Parametric equation of a sphere. Use a function f to move from flat (u, v) to curved surfaces like a sphere $(X(u, v), Y(u, v), Z(u, v))$:

$$X = \cos(v) \sin(u)$$

$$Y = \cos(v) \sin(v)$$

$$Z = \cos(u)$$

In this case,

$$\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 = \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda}$$

expanding out,

$$= \left(\frac{dx}{d\lambda}\right)^2 + \left(\frac{dy}{d\lambda}\right)^2 + \left(\frac{dz}{d\lambda}\right)^2$$

The metric tensor is 3x3 as there are three variables: X, Y, Z.

Intrinsic geometry.

$$\left\|\frac{d\vec{R}}{d\lambda}\right\|^2 = \begin{vmatrix} \frac{du}{d\lambda} & \frac{dv}{d\lambda} \end{vmatrix} g_{uv} \begin{vmatrix} \frac{du}{d\lambda} \\ \frac{dv}{d\lambda} \end{vmatrix}$$

expanding out,

$$g_{uv} = \begin{vmatrix} 1 & 0 \\ 0 & \sin^2 u \end{vmatrix}$$

So,

$$\left\|\frac{d\vec{R}}{d\lambda}\right\|^2 = \left(\frac{du}{d\lambda}\right)^2 + \sin^2 u \left(\frac{dv}{d\lambda}\right)^2$$

where $\sin^2 u$ shows that longitude (x=v) depends on latitude (y=u).

Along a Mercator projection, metric tensor changes everywhere. So it's a metric tensor field. To be truly intrinsic, remove $\vec{R}$ which depends on an extrinsic origin.

20.13 ∇ and d operators

f is a scalar field. Then ∇f is the gradient, so it forms a vector field. df is the differential form (comprised of covectors) so forms a covector field. So $\nabla f \in V$ and $df \in V^*$.

Review:

$$\vec{v} \cdot \vec{w} = v(v^i e_i) \cdot (w^j e_j) = v^i w^j (e_i \cdot e_j) = v^i w^j g_{ij}$$

and therefore

$$\vec{v} \cdot - = v^i g_{ij} \epsilon^j$$

with $\epsilon^j(e_i) = \delta_i^j$ the covector basis.

Since from above the directional derivative is $df(\vec{v})$,

$$df(\vec{v}) = \nabla f \cdot \vec{v} = ((\nabla f)^i e_i) \cdot (v^j e_j) = (\nabla f)^i v^j (e_i \cdot e_j) = (\nabla f)^i v^j g_{ij}$$

using the basis covectors $d\epsilon^j$,

$$df = \nabla f \cdot - = (\nabla f)^i g_{ij} d\epsilon^j$$

Proof using tensor product spaces. Generally, $\vec{v} \cdot \vec{w} = g(\vec{v}, \vec{w}) = (\epsilon^i \otimes \epsilon^j)(\vec{v}, \vec{w}) = v^i w^j g_{ij}$, where

$$g_{ij} = \begin{vmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{vmatrix} = g_{11} \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + g_{12} \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} + g_{21} \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} + g_{22} \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix}$$

$$\begin{aligned}
&= g_{11}(\epsilon^1 \otimes \epsilon^1) + g_{12}(\epsilon^1 \otimes \epsilon^2) + g_{21}(\epsilon^2 \otimes \epsilon^1) + g_{22}(\epsilon^2 \otimes \epsilon^2) \\
&= (g_{ij}(\epsilon^i \otimes \epsilon^j))(v^k e_k, w^l e_l) = g_{ij} \epsilon^i(v^k e_k) \epsilon^j(w^l e_l) \\
&= g_{ij} v^k w^l \epsilon^i(e_k) \epsilon^j(e_l) = g_{ij} v^k w^l \delta_k^i \delta_l^j = g_{ij} v^l w^j
\end{aligned}$$

So,

$$\begin{aligned}
\vec{v} \cdot - &= g(\vec{v}, -) = g_{ij}(\epsilon^i \otimes \epsilon^j)(v^k e_k) = g_{ij} \epsilon^i(v^k e_k) \epsilon^j \\
&= g_{ij} v^k \epsilon^i(e_k) \epsilon^j = g_{ij} v^k \delta_k^i \epsilon^j = g_{ij} v^i \epsilon^j
\end{aligned}$$

Analogously:

$$\begin{aligned}
g &= g_{ij}(dc^i \otimes dc^j) \\
dc^j \left(\frac{\partial}{\partial c^i} \right) &= \delta_i^j \\
\nabla \cdot \vec{v} &= g(\nabla f, \vec{v}) = (g_{ij}(dc^i \otimes dc^j))((\nabla f)^k \frac{\partial}{\partial c^k}, v^l \frac{\partial}{\partial c^l}) \\
&= g_{ij} dc^i \left((\nabla f)^k \frac{\partial}{\partial c^k} \right) dc^j \left(v^l \frac{\partial}{\partial c^l} \right) \\
&= g_{ij} (\nabla f)^k v^l dc^i \frac{\partial}{\partial c^k} dc^j \frac{\partial}{\partial c^l} \\
&= g_{ij} (\nabla f)^k v^l \delta_k^i \delta_l^j = g_{ij} (\nabla f)^i v^j
\end{aligned}$$

Then

$$\begin{aligned}
\nabla f \cdot - &= g(\nabla f, -) = (g_{ij}(dc^i \otimes dc^j))((\nabla f)^k \frac{\partial}{\partial c^k}) \\
&= g_{ij} dc^i ((\nabla f)^k \frac{\partial}{\partial c^k}) dc^j \\
&= g_{ij} (\nabla f)^k dc^i \frac{\partial}{\partial c^k} dc^j \\
&= g_{ij} (\nabla f)^k \delta_k^i dc^j = g_{ij} (\nabla f)^i dc^j
\end{aligned}$$

Differential form in Einstein notation:

$$df = \frac{\partial f}{\partial c^i} dc^i$$

Equate the two:

$$\begin{aligned}
df &= \frac{\partial f}{\partial c^j} dc^j = (\nabla f)^i g_{ij} dc^j \\
\frac{\partial f}{\partial c^j} &= (\nabla f)^i g_{ij}
\end{aligned}$$

Using $g_{ij} \mathfrak{g}^{jk} = \delta_i^k$,

$$\frac{\partial f}{\partial c^j} \mathfrak{g}^{jk} = (\nabla f)^i g_{ij} \mathfrak{g}^{jk} = (\nabla f)^i \delta_i^k = (\nabla f)^k$$

This means that

$$\begin{aligned}\nabla f &= (\nabla f)^k \frac{\partial}{\partial c^k} = \mathfrak{g}^{jk} \frac{\partial f}{\partial c^j} \frac{\partial}{\partial c^k} \\ \nabla f &= (\nabla f)^k e_k = \mathfrak{g}^{jk} \frac{\partial f}{\partial c^j} e_k\end{aligned}$$

In Cartesian coordinates,

$$\begin{aligned}g_{ij} &= \delta_{ij} = I \\ \mathfrak{g}^{jk} &= \delta^{jk} = I\end{aligned}$$

So,

$$\nabla f = \delta^{jk} \frac{\partial f}{\partial c^j} e_k = \delta^{jk} \frac{\partial f}{\partial c^j} e_k = \frac{\partial f}{\partial c^k} e_k = \frac{\partial f}{\partial x} e_x + \frac{\partial f}{\partial y} e_y$$

In polar coordinates,

$$\begin{aligned}\tilde{g}_{ij} &= \begin{vmatrix} 1 & 0 \\ 0 & r^2 \end{vmatrix} \\ \tilde{g}^{ij} &= \begin{vmatrix} 1 & 0 \\ 0 & 1/r^2 \end{vmatrix}\end{aligned}$$

so,

$$\nabla f = \tilde{g}^{ij} \frac{\partial f}{\partial p^j} e_i = \tilde{g}^{rr} \frac{\partial f}{\partial r} e_r + \tilde{g}^{\theta\theta} \frac{\partial f}{\partial \theta} e_\theta = \frac{\partial f}{\partial r} e_r + \frac{1}{r^2} \frac{\partial f}{\partial \theta} e_\theta$$

Summary:

$$\begin{aligned}(\nabla f)^k &= \mathfrak{g}^{jk} \frac{\partial f}{\partial c^j} \\ (\tilde{\nabla} f)^k &= \tilde{\mathfrak{g}}^{jk} \frac{\partial f}{\partial p^j}\end{aligned}$$

In Cartesian coordinates, $\nabla f = df$:

$$\begin{aligned}df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \\ \nabla f &= \frac{\partial f}{\partial x} e_x + \frac{\partial f}{\partial y} e_y\end{aligned}$$

In polar coordinates,

$$\begin{aligned}df &= \frac{\partial f}{\partial r} dr + \frac{\partial f}{\partial \theta} d\theta \\ \nabla f &= \frac{\partial f}{\partial r} e_r + \frac{1}{r^2} \frac{\partial f}{\partial \theta} e_\theta\end{aligned}$$

Spherical coordinates:

$$\begin{aligned}\mathfrak{g} &= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1/r^2 & 0 \\ 0 & 0 & 1/(r^2 \sin^2 \theta) \end{vmatrix} \\ \nabla f &= \frac{\partial f}{\partial r} e_r + \frac{1}{r^2} \frac{\partial f}{\partial \theta} e_\theta + \frac{1}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} e_\phi\end{aligned}$$

20.14 Geodesics and Christoffel symbols

Geodesics are straight paths having no tangential acceleration. However, they may possess normal acceleration.

The tangent vectors of a 2D surface (in 3D or higher dimensions) are:

$$\frac{\partial R}{\partial u} = \frac{\partial R}{\partial x}, \frac{\partial R}{\partial v} = \frac{\partial R}{\partial y}$$

Note: $R = \vec{R}$. The velocity vector of a surface is in tangent plane:

$$\frac{dR}{d\lambda} = \frac{du}{d\lambda} \frac{\partial R}{\partial u} + \frac{dv}{d\lambda} \frac{\partial R}{\partial v}$$

Acceleration vector is, using the chain rule:

$$\frac{d}{d\lambda} \left(\frac{dR}{d\lambda} \right) = \frac{d}{d\lambda} \left(\frac{du}{d\lambda} \frac{\partial R}{\partial u} + \frac{dv}{d\lambda} \frac{\partial R}{\partial v} \right) = \frac{d}{d\lambda} \left(\frac{du}{d\lambda} \frac{\partial R}{\partial u} \right) + \frac{d}{d\lambda} \left(\frac{dv}{d\lambda} \frac{\partial R}{\partial v} \right)$$

Expand again using the chain rule, re-arrange, and express using Einstein notation

$$\frac{d^2 R}{d\lambda^2} = \frac{d^2 u^i}{d\lambda^2} \frac{\partial R}{\partial u^i} + \frac{du^i}{d\lambda} \frac{\partial^2 R}{\partial \lambda \partial u^i \partial u^j}$$

where the last term is a 3D vector. The 3D basis is $\frac{\partial R}{\partial u^i}, \frac{\partial R}{\partial u^j}, \hat{n}$, where the first two are in the tangent plane. Expand the last term in this 3D basis:

$$\frac{\partial^2 R}{\partial u^i \partial u^j} = \Gamma_{ij}^1 \frac{\partial R}{\partial u^i} + \Gamma_{ij}^2 \frac{\partial R}{\partial u^j} + L_{ij} \hat{n}$$

Re-expressing in Einstein notation,

$$\frac{\partial^2 R}{\partial u^i \partial u^j} = \Gamma_{ij}^k \frac{\partial R}{\partial u^k} + L_{ij} \hat{n}$$

where Γ_{ij}^k are Christoffel symbols (which are for tangent components) and second fundamental form (which are for normal components). Dot product both sides with the tangent to get rid of the normal component (dot product of tangent with normal component is 0):

$$\frac{\partial^2 R}{\partial u^i \partial u^j} \cdot \frac{\partial R}{\partial u^l} = \left(\Gamma_{ij}^k \frac{\partial R}{\partial u^k} + L_{ij} \hat{n} \right) \cdot \frac{\partial R}{\partial u^l} = \Gamma_{ij}^k \frac{\partial R}{\partial u^k} \cdot \frac{\partial R}{\partial u^l} + 0 = \Gamma_{ij}^k g_{kl}$$

Using $g_{kl} \mathfrak{g}^{lm} = \delta_k^m$, solve for the Christoffel symbol:

$$\frac{\partial^2 R}{\partial u^i \partial u^j} \frac{\partial R}{\partial u^l} \mathfrak{g}^{lm} = \Gamma_{ij}^k g_{kl} \mathfrak{g}^{lm} = \Gamma_{ij}^k \delta_k^m = \Gamma_{ij}^m$$

Analogously, solve for the second fundamental form by adding the dot product to both sides:

$$\frac{\partial^2 R}{\partial u^i \partial u^j} \cdot \hat{n} = \left(\Gamma_{ij}^k \frac{\partial R}{\partial u^k} + L_{ij} \hat{n} \right) \cdot \hat{n} = L_{ij}$$

$$\frac{e_i \times e_j}{\|e_i \times e_j\|} = L_{ij}$$

Return to computing the acceleration vector:

$$\frac{d^2 R}{d\lambda^2} = \frac{d^2 u^i}{d\lambda^2} \frac{\partial R}{\partial u^i} + \frac{u^i}{d\lambda} \frac{u^j}{d\lambda} \frac{\partial^2 R}{\partial u^i \partial u^j}$$

by substituting:

$$\frac{\partial^2 R}{\partial u^i \partial u^j} = \Gamma_{ij}^k \frac{\partial R}{\partial u^k} + L_{ij} \hat{n}$$

Combine and re-arrange:

$$\frac{d^2 R}{d\lambda^2} = \underbrace{\left(\frac{d^2 u^k}{d\lambda^2} + \Gamma_{ij}^k \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \right)}_{Tangent} \frac{\partial R}{\partial u^k} + L_{ij} \underbrace{\left(\frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \right)}_{Normal} \hat{n}$$

Since a geodesic is where there is 0 tangential acceleration, the geodesic equation is

$$\frac{d^2 u^k}{d\lambda^2} + \Gamma_{ij}^k \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} = 0$$

20.15 Alternative derivation of geodesic

Vector being parallel transported to itself, from the intrinsic definition of the covariant derivative.

$$\begin{aligned} \nabla_{\frac{d}{d\lambda}} \frac{d}{d\lambda} &= \frac{d}{d\lambda} \left(\frac{d}{d\lambda} \right) = \frac{d}{d\lambda} \left(\frac{du^j}{d\lambda} \frac{\partial}{\partial u^j} \right) \\ &= \frac{d}{d\lambda} \left(\frac{du^j}{d\lambda} \right) \frac{\partial}{\partial u^j} + \frac{du^j}{d\lambda} \frac{d}{d\lambda} \left(\frac{\partial}{\partial u^j} \right) \\ &= \frac{d^2 u^j}{d\lambda^2} \frac{\partial}{\partial u^j} + \frac{du^j}{d\lambda} \frac{d}{d\lambda} \left(\frac{\partial}{\partial u^j} \right) \\ &= \frac{d^2 u^j}{d\lambda^2} \frac{\partial}{\partial u^j} + \frac{du^j}{d\lambda} \left(\frac{du^i}{d\lambda} \frac{\partial}{\partial u^i} \right) \left(\frac{\partial}{\partial u^j} \right) \\ &= \frac{d^2 u^j}{d\lambda^2} \frac{\partial}{\partial u^j} + \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \left(\frac{\partial^2}{\partial u^i \partial u^j} \right) \\ &= \frac{d^2 u^j}{d\lambda^2} e_j + \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \left(\frac{\partial e_j}{\partial u^i} \right) \\ &= \frac{d^2 u^k}{d\lambda^2} e_k + \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \Gamma_{ij}^k e_k \\ &= \left(\frac{d^2 u^k}{d\lambda^2} + \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} \Gamma_{ij}^k \right) e_k = 0 \end{aligned}$$

20.16 Christoffel symbols are not tensors

Conversion derivation of Christoffel symbols between different bases is tedious. Result:

$$\tilde{\Gamma}_{ij}^k = \frac{\partial p^k}{\partial c^b} \left(\frac{\partial c^a}{\partial p^i} \frac{\partial c^r}{\partial p^j} \Gamma_{ar}^b + \frac{\partial^2 c^b}{\partial p^i \partial p^j} \right)$$

where the second term is not part of tensor basis conversions. So Christoffel symbols are not tensors.

20.17 Geodesic examples

To find geodesics, (1) calculate the Christoffel symbols and (2) input into geodesic equation.

Example 1. Flat plane: $R(u, v) = \vec{p} + u\vec{a} + v\vec{b}$. Computing the Christoffel symbol, $\Gamma_{ij}^k = 0$ (nothing 'pops out' of the flat plane). Input this into geodesic equation:

$$\begin{aligned} \frac{d^2 u^k}{d\lambda^2} + \Gamma_{ij}^k \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} &= 0 \\ \frac{d^2 u^k}{d\lambda^2} + (0) \frac{du^i}{d\lambda} \frac{du^j}{d\lambda} &= 0 \\ \frac{d^2 u^k}{d\lambda^2} &= 0 \end{aligned}$$

which is in Einstein notation. Expanding out the set of differential equations:

$$\begin{aligned} \frac{d^2 u}{d\lambda^2} &= 0 \\ \frac{d^2 v}{d\lambda^2} &= 0 \end{aligned}$$

The solutions are:

$$\begin{aligned} u &= k_u \lambda + u_0 \\ v &= k_v \lambda + v_0 \end{aligned}$$

Substitute into $R(u, v)$:

$$R(\lambda) = (\vec{p} + u_0 \vec{a} + v_0 \vec{b}) + \lambda(k_u \vec{a} + k_v \vec{b}) = \vec{q} + \lambda \vec{r}$$

where $\vec{q}$ is the initial position and $\vec{r}$ is the initial velocity. Unsurprisingly, the geodesic is a straight line.

Example 2. Sphere. $X = \cos(u)\sin(v), Y = \sin(u)\sin(v), Z = \cos(v)$. First solve below using the chain rule:

$$\frac{\partial R}{\partial u}, \frac{\partial R}{\partial v}, \frac{\partial}{\partial u} \left(\frac{\partial R}{\partial u} \right), \frac{\partial}{\partial v} \left(\frac{\partial R}{\partial v} \right), \frac{\partial}{\partial u} \left(\frac{\partial R}{\partial v} \right)$$

Compute the metric tensor for a sphere:

$$g_{ij} = \begin{vmatrix} 1 & 0 \\ 0 & \sin^2 u \end{vmatrix}$$

Since the metric tensor matrix is diagonal, the inverse is simply the inverse of the diagonal elements:

$$\mathfrak{g}^{ij} = g_{ij}^{-1} = \begin{vmatrix} 1 & 0 \\ 0 & 1/\sin^2 u \end{vmatrix}$$

Solve for Christoffel symbols and geodesic equations. Geodesic equations are differential equations. For sphere, difficult to find general solutions. By trial and error, induction, and intuition:

$$u = \frac{\pi}{2}, v = k\lambda$$

Since $\frac{\pi}{2}$ lies on equator, this means the equator has no tangential acceleration and is therefore a geodesic. The acceleration vector points to the center of the equator which is the center of the sphere itself. Other great circles by rotational symmetry are also geodesics. Other 'latitudes' have tangential acceleration as acceleration vectors point to the center of that latitude (i.e. NOT to the center of the sphere), so are not geodesics.

20.18 Covariant derivative

Studies rate of change of tensor fields with changing basis vectors. There are multiple definitions of the covariant derivative, see below.

20.18.1 Flat space definition

In flat space, the covariant derivative is just the chain rule.

Cartesian coordinates.

$$\begin{aligned} \vec{v} &= v^x e_x + v^y e_y = v^i e_i \\ \frac{\partial}{\partial x} \vec{v} &= \frac{\partial}{\partial x} (v^x e_x + v^y e_y) = \frac{\partial}{\partial x} (v^x e_x) + \frac{\partial}{\partial x} (v^y e_y) \\ &= \frac{\partial v^x}{\partial x} e_x + v^x \frac{\partial e_x}{\partial x} + \frac{\partial v^y}{\partial x} e_y + v^y \frac{\partial e_y}{\partial x} \end{aligned}$$

This is repeated for $\frac{\partial}{\partial y} \vec{v}$. More generally,

$$\frac{\partial}{\partial c^i} \vec{v} = \frac{\partial}{\partial c^i} (v^j e_j) = \frac{\partial v^j}{\partial c^i} e_j + v^j \frac{\partial e_j}{\partial c^i}$$

but since

$$\frac{\partial e_j}{\partial c^i} = \Gamma_{ij}^1 e_1 + \Gamma_{ij}^2 e_2 = \Gamma_{ij}^k e_k$$

can write the general equation as

$$\frac{\partial}{\partial c^i} \vec{v} = \frac{\partial}{\partial c^i} (v^j e_j) = \frac{\partial v^j}{\partial c^i} e_j + v^j \frac{\partial e_j}{\partial c^i} = \left(\frac{\partial v^k}{\partial c^i} + v^j \Gamma_{ij}^k \right) e_k$$

In flat spaces, the basis vectors don't change in the x-direction, so

$$= \frac{\partial v^x}{\partial x} e_x + v^x \cdot 0 + \frac{\partial v^y}{\partial x} e_y + v^y \cdot 0$$

If the coefficients of $\vec{v}$ are constants like $\vec{v} = k_1 e_x + k_2 e_y$, then

$$= 0 \cdot e_x + 0 \cdot e_y = 0$$

Can alternatively compute using $\Gamma_{ij}^k = 0$.

Polar coordinates.

$$\frac{\partial}{\partial \theta} \vec{v} = \frac{\partial v^r}{\partial \theta} e_r + \frac{\partial v^\theta}{\partial \theta} e_\theta + v^r \frac{\partial e_r}{\partial \theta} + v^\theta \frac{\partial e_\theta}{\partial \theta}$$

$$\frac{\partial}{\partial r} \vec{v} = \frac{\partial v^r}{\partial r} e_r + \frac{\partial v^\theta}{\partial r} e_\theta + v^r \frac{\partial e_r}{\partial r} + v^\theta \frac{\partial e_\theta}{\partial r}$$

Compute partial derivatives, which all have geometric intuition:

$$\frac{\partial e_r}{\partial r} = 0, \frac{\partial e_\theta}{\partial r} = \frac{\partial e_r}{\partial \theta} = \frac{1}{r} e_\theta, \frac{\partial e_\theta}{\partial \theta} = -r e_r$$

Substituting,

$$\frac{\partial}{\partial r} \vec{v} = \frac{\partial v^r}{\partial r} e_r + \left(\frac{\partial v^\theta}{\partial r} + v^\theta \frac{1}{r} \right) e_\theta$$

$$\frac{\partial}{\partial \theta} \vec{v} = \left(\frac{\partial v^r}{\partial \theta} - r v^\theta \right) e_r + \left(\frac{\partial v^\theta}{\partial \theta} + v^r \frac{1}{r} \right) e_\theta$$

In matrix notation,

$$\frac{\partial}{\partial r} \vec{v} = \begin{bmatrix} \frac{\partial v^r}{\partial r} \\ \frac{\partial v^\theta}{\partial r} + v^\theta \frac{1}{r} \end{bmatrix}$$

$$\frac{\partial}{\partial \theta} \vec{v} = \begin{bmatrix} \frac{\partial v^r}{\partial \theta} - r v^\theta \\ \frac{\partial v^\theta}{\partial \theta} + v^r \frac{1}{r} \end{bmatrix}$$

where the partial derivative terms describe how the vector components are changing and the non-derivative terms describe how the basis vectors are changing.

20.18.2 Extrinsic definition

Covariant derivative is the rate of change of a vector $\vec{v}$ in a given direction $\vec{w}$ with the normal component subtracted.

Constant vectors on curved surfaces are impossible to define. Example: 'constant' vector in a loop on a spherical surface. Parallel transport of vector is as constant as possible, but still suffers in loops. It keeps the vector the same length. Parallel transport is where acceleration vector points to the center of the sphere for spherical surfaces, so the normal vector $\vec{n}$ points from the surface to the center for forward velocities:

$$\frac{d\vec{v}}{d\lambda} = \vec{n}$$

$$\frac{d\vec{v}}{d\lambda} - \vec{n} = 0$$

Covariant derivative $\nabla_{\vec{w}}\vec{v}$ helps parallel transport, whether tangent to the surface or not.

$$\nabla_{\frac{d\lambda}{du^i}}\vec{v} = \frac{d\vec{v}}{d\lambda} - \vec{n} = 0$$

where in flat spaces, $\vec{n} = 0$.

$$\begin{aligned}\nabla_{\frac{d}{du^i}}\vec{v} &= \frac{d\vec{v}}{du^i} - \vec{n} = \frac{\partial}{\partial u^i}(v^1 e_1 + v^2 e_2) - \vec{n} = \frac{\partial}{\partial u^i}(v^j e_j) - \vec{n} \\ &= \left(\frac{\partial v^j}{\partial u^i} e_j + v^j \frac{\partial e_j}{\partial u^i} \right) - \vec{n}\end{aligned}$$

From earlier,

$$\frac{\partial e_j}{\partial u^i} = \Gamma_{ij}^k e_k + L_{ij} \hat{n}$$

Substituting,

$$\begin{aligned}\nabla_{\frac{d}{du^i}}\vec{v} &= \left(\frac{\partial v^j}{\partial u^i} e_j + v^j (\Gamma_{ij}^k e_k + L_{ij} \hat{n}) \right) - \vec{n} \\ &= \left(\frac{\partial v^j}{\partial u^i} e_j + v^j \Gamma_{ij}^k e_k \right) + L_{ij} \hat{n} - \vec{n} \\ &= \underbrace{\left(\frac{\partial v^k}{\partial u^i} + v^j \Gamma_{ij}^k \right) e_k}_{Tangent} + \underbrace{L_{ij} \hat{n} - \vec{n}}_{Normal=0}\end{aligned}$$

The normal is 0 because of parallel transport. Now solve for Christoffel symbols. From earlier,

$$\frac{\partial e_j}{\partial u^i} = (\Gamma_{ij}^k e_k + L_{ij} \hat{n})$$

Dot product both sides:

$$\frac{\partial e_j}{\partial u^i} \cdot e_l = (\Gamma_{ij}^k e_k + L_{ij} \hat{n}) \cdot e_l$$

$$= \Gamma_{ij}^k e_k \cdot e_l = \Gamma_{ij}^k g_{kl}$$

Using $g_k l \mathfrak{g}^{lm} = \delta_k^m$, dot product both sides:

$$\frac{\partial e_j}{\partial u^i} \cdot e_l \mathfrak{g}^{lm} = \Gamma_{ij}^k g_{kl} \mathfrak{g}^{lm} = \Gamma_{ij}^k \delta_k^m = \Gamma_{ij}^m$$

So,

$$\Gamma_{ij}^k = \left(\frac{\partial e_j}{\partial u^i} \cdot e_l \right) \mathfrak{g}^{lk}$$

Can use covariant derivative to determine whether a path is parallel transported (=0) or how a vector should change directions along a given path to be parallel transported.

20.18.3 Intrinsic definition

In extrinsic geometry,

$$\Gamma_{ij}^k = \left(\frac{\partial e_j}{\partial u^i} \cdot e_l \right) \mathfrak{g}^{lk}$$

but without extrinsic geometry, can't calculate dot products as they can exist outside intrinsic geometry. Intrinsic geometry has no normal, so the parallel transport is:

$$\begin{aligned} \nabla_{\frac{\partial}{\partial u^i}} \vec{v} &= \frac{\partial \vec{v}}{\partial u^i} - \vec{n} = \frac{\partial \vec{v}}{\partial u^i} \\ &= \frac{\partial}{\partial u^i} (v^j e_j) = \frac{\partial v^j}{\partial u^i} e_j + v^j \frac{\partial e_j}{\partial u^i} \end{aligned}$$

and using the vector decomposition

$$\frac{\partial e_j}{\partial u^i} = \Gamma_{ij}^k e_k + L_{ij} \hat{n} = \Gamma_{ij}^k e_k + 0 = \left(\frac{\partial v^k}{\partial u^i} + v^j \Gamma_{ij}^k \right) e_k$$

There is a $T_p M$ or tangent vector space at every point where the intrinsic Christoffel symbols exist. Need to find intrinsic geometry Christoffel symbol. Use the 'metric compatibility' property, a combination of dot product and product rule, to expand derivative of metric tensor:

$$\begin{aligned} \frac{\partial}{\partial u^k} g_{ij} &= \frac{\partial}{\partial u^k} e_i \cdot e_j = \frac{\partial e_i}{\partial u^k} \cdot e_j + e_i \cdot \frac{\partial e_j}{\partial u^k} = (\Gamma_{ik}^l) \cdot e_j + e_i \cdot (\Gamma_{jk}^l e_l) \\ &= \Gamma_{ij}^l (e_l \cdot e_j) + \Gamma_{jk}^l (e_i \cdot e_l) = \Gamma_{ik}^l g_{jl} + \Gamma_{jk}^l g_{il} \end{aligned}$$

Use this to calculate all metric tensors:

$$\frac{\partial g_{ij}}{\partial u^k} = \Gamma_{ik}^l g_{jl} + \Gamma_{jk}^l g_{il}$$

$$\frac{\partial g_{ik}}{\partial u^j} = \Gamma_{kj}^l g_{il} + \Gamma_{ij}^l g_{kl}$$

$$\frac{\partial g_{jk}}{\partial u^i} = \Gamma_{ij}^l g_{kl} + \Gamma_{ki}^l g_{jl}$$

Combining and cancelling terms,

$$\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i} = \Gamma_{jk}^l g_{li} + \Gamma_{jk}^l g_{li} = 2\Gamma_{jk}^l g_{li}$$

Multiply both sides by $\mathfrak{g}^{im}$,

$$\mathfrak{g}^{im} \left(\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i} \right) = 2\Gamma_{jk}^l g_{li} \mathfrak{g}^{im} = 2\Gamma_{jk}^l \delta_l^m = 2\Gamma_{jk}^m$$

$$\Gamma_{jk}^m = \frac{\mathfrak{g}^{kl}}{2} \left(\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i} \right)$$

With some manipulation, it can be shown that this equivalent to the extrinsic equation

$$\Gamma_{ij}^k = \left(\frac{\partial e_j}{\partial u^i} \cdot e_l \right) \mathfrak{g}^{kl}$$

20.18.4 Abstract definition of covariant derivative

Define in terms of properties, like a vector space. covariant derivative is a linear map. Notation: $\frac{\partial}{\partial u_1} = \partial_1$.

(1) Additive property 1.

$$\nabla_{a\partial_1 + b\partial_2} \vec{v} = (a\partial_1 + b\partial_2) \vec{v} = a(\partial_1 \vec{v}) + b(\partial_2 \vec{v})$$

So,

$$\nabla_{a\vec{w} + b\vec{t}} \vec{v} = a\nabla_{\vec{w}} \vec{v} + b\nabla_{\vec{t}} \vec{v}$$

(2) Additive property 2.

$$\nabla_{\partial_i} (\vec{v} + \vec{w}) = \partial_i (\vec{v} + \vec{w}) = \partial_i \vec{v} + \partial_i \vec{w} = \nabla_{\partial_i} \vec{v} + \nabla_{\partial_i} \vec{w}$$

(3) Product rule.

$$\nabla_{\partial_i} (a\vec{v}) = \frac{\partial}{\partial u^i} (a\vec{v}) = \frac{\partial a}{\partial u^i} \vec{v} + a \frac{\partial \vec{v}}{\partial u^i} = (\nabla_{\partial_i} a) \vec{v} + a(\nabla_{\partial_i} \vec{v})$$

(4) Directional derivative generalization.

$$\nabla_{\partial_i} (a) = \frac{\partial}{\partial u^i} (a) = \frac{\partial a}{\partial u^i}$$

20.18.5 Covariant derivatives in engineering notation

$$\begin{aligned}\frac{\partial}{\partial c^i} \vec{v} &= \frac{dc^i}{d\lambda} \frac{\partial}{\partial c^i} (v^j e_j) = \frac{dc^i}{d\lambda} \left(\frac{\partial v^j}{\partial c^i} e_j + v^j \frac{\partial e_j}{\partial c^i} \right) \\ &= \frac{dc^i}{d\lambda} \underbrace{\left(\frac{\partial v^k}{\partial c^i} + v^j \Gamma_{ij}^k \right)}_{\nabla_i v^k} e_k\end{aligned}$$

In engineering notation,

$$\nabla_i v^k = v_{;i}^k = \left(\frac{\partial v^k}{\partial c^i} + v^j \Gamma_{ij}^k \right)$$

Substituting,

$$\frac{\partial}{\partial \lambda} \vec{v} = (v_{;i}^k) e_k$$

In matrix notation,

$$= \begin{vmatrix} v_{;1}^1 & v_{;1}^2 \\ v_{;2}^1 & v_{;2}^2 \end{vmatrix}$$

To convert bases,

$$\tilde{v}_{;i}^k = \frac{\partial p^k}{\partial c^b} \frac{\partial c^a}{\partial p^i} (v_{;a}^b) = J^{-1} J(v_{;a}^b)$$

20.19 Levi-Civita connection

The covariant derivative takes a direction and input field and outputs an output field (a rate of change):

$$\nabla_{\text{direction}}(\text{input}) = \text{output}$$

(Note: in a geodesic, $\nabla_{\vec{v}} \vec{v} = 0$, as it is the covariant derivative of the input in the input direction.)

Previously, Christoffel symbols were defined through linear combinations of the partial derivatives of basis vectors:

$$\frac{\partial}{\partial u^i} e_j = \Gamma_{ij}^k e_k$$

and the abstract definition is then

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k$$

where the Christoffel symbols are also called 'connection coefficients'. On a surface, the covariant derivative parallel transports a vector from one tangent vector space $T_p S$ to another $T_q S$. It connects the tangent vector spaces. So it is a 'connection'.

Christoffel symbols aren't unique in general, so covariant derivatives aren't unique in general. However, they can be made unique by adding two properties: the Torsion-free property and metric compatibility.

Torsion-free property: $\nabla_{e_i} e_j = \nabla_{e_j} e_i$. First consider the Lie bracket:

$$\nabla_{\vec{w}} \vec{v} - \nabla_{\vec{v}} \vec{w} = [\vec{v}, \vec{w}] = \vec{v}\vec{w} - \vec{w}\vec{v}$$

Since $[\partial_i, \partial_j] = \partial_i \partial_j - \partial_j \partial_i = 0$,

$$\nabla_{\vec{w}} \vec{v} = \nabla_{\vec{v}} \vec{w}$$

So, like Young's theorem for partial derivatives:

$$\nabla_{e_i} e_j = \nabla_{e_j} e_i$$

$$\nabla_{e_i} e_j = \Gamma_{ij}^k$$

$$\nabla_{e_j} e_i = \Gamma_{ji}^k$$

$$\Gamma_{ij}^k = \Gamma_{ji}^k$$

Metric compatibility. Parallel transport two vectors while keeping their dot product the same. So angle and vector lengths remain the same.

$$\nabla_{\partial_i} (\vec{v} \cdot \vec{u}) = (\nabla_{\partial_i} \vec{v}) \cdot \vec{u} + \vec{v} \cdot (\nabla_{\partial_i} \vec{u})$$

$$\nabla_{e_k} (e_i \cdot e_j) = (\nabla_{e_k} e_i) \cdot e_j + e_i \cdot (\nabla_{e_k} e_j)$$

$$\partial_k (e_i \cdot e_j) = (\Gamma_{ik}^l e_l) \cdot e_j + e_i \cdot (\Gamma_{jk}^l e_l)$$

$$\partial_k g_{ij} = \Gamma_{ik}^l g_{jl} + \Gamma_{kj}^l g_{il}$$

Combine torsion-free and metric compatibility properties to get a result similar to intrinsic geometry definition.

$$\partial_k g_{ij} = \Gamma_{ik}^l g_{jl} + \Gamma_{jk}^l g_{il}$$

$$\partial_j g_{ki} = \Gamma_{kj}^l g_{il} + \Gamma_{ij}^l g_{kl}$$

$$\partial_i g_{jk} = \Gamma_{ji}^l g_{kl} + \Gamma_{ki}^l g_{jl}$$

Then,

$$\partial_k g_{ij} + \partial_j g_{ki} - \partial_i g_{jk} = 2\Gamma_{jk}^l g_{il}$$

$$\Gamma_{jk}^m = \frac{1}{2} \mathfrak{g}^{im} (\partial_k g_{ij} + \partial_j g_{ki} - \partial_i g_{jk})$$

This is the Levi-Civita connection. The fundamental theorem of Riemannian geometry: in a Riemannian manifold (curved space with a metric), the Christoffel symbols (connection coefficients) are unique if they possess the torsion-free and metric compatibility properties.

The Levi-Civita connection is the most important abstract definition of covariant derivatives. Another is

$$\Gamma_{ij}^k = 0$$

which leads to:

$$\nabla_{\frac{\partial}{\partial u^i}} \vec{v} = \left(\frac{\partial v^k}{\partial u^i} + v^j \Gamma_{ij}^k \right) e_k = \left(\frac{\partial v^k}{\partial u^i} + v^j(0) \right) e_k$$

which leads to a different parallel transport mechanism. For instance, on a spherical surface, it will parallel transport a vector on the equator pointing downwards to another point on the equator with the vector still pointing down. In contrast, the Levi-Civita connection would turn or spin the vector direction.

Covariant derivative of a covector.

$$\nabla_{\partial_i}(a) = \nabla_{\partial_i}(a_j \epsilon^j)$$

Using the 'product rule' of covariant derivatives,

$$\begin{aligned} &= (\nabla_{\partial_i} a_j) \epsilon^j + a_j (\nabla_{\partial_i} \epsilon^j) = \frac{\partial a_j}{\partial u^i} \epsilon^j + a_j (\Lambda_{ik}^j \epsilon^k) = \left(\frac{\partial a_k}{\partial u^i} + a_j \Lambda_{ik}^j \right) \epsilon^k \\ &= \left(\frac{\partial a_k}{\partial u^i} - a_j \Gamma_{ik}^j \right) \epsilon^k \end{aligned}$$

where Λ is the Christoffel symbol for covectors, and is defined from:

$$\begin{aligned} \nabla_{\partial_i}(a(\vec{v})) &= \nabla_{\partial_i}(\vec{a} \cdot \vec{v}) \\ \partial_i(a(\vec{v})) &= (\nabla_{\partial_i} \vec{a}) \cdot \vec{v} + \vec{a} \cdot (\nabla_{\partial_i} \vec{v}) \\ \partial_i(a_k \epsilon^k (v^r e_r)) &= \left(\frac{\partial a_k}{\partial u^i} + a_j \Lambda_{ik}^j \right) \epsilon^k (\vec{v}) + \left(\frac{\partial v^k}{\partial u^i} + v^j \Gamma_{ij}^k \right) a(e_k) \\ &= \left(\frac{\partial a_k}{\partial u^i} + a_j \Lambda_{ik}^j \right) v^k + \left(\frac{\partial v^k}{\partial u^i} + v^j \Gamma_{ij}^k \right) a_k \\ 0 &= a_j v^k \Lambda_{ik}^j + a_k v^j \Gamma_{ij}^k \\ 0 &= a_k v^j \Lambda_{ij}^k + a_{ik} v^j \Gamma_{ij}^k \\ 0 &= \Lambda_{ij}^k + \Gamma_{ij}^k \\ \Lambda_{ij}^k &= -\Gamma_{ij}^k \end{aligned}$$

Covariant derivative of a tensor. Use this property:

$$\begin{aligned} \nabla_{\vec{w}}(T \otimes S) &= (\nabla_{\vec{w}} T) \otimes S + T \otimes (\nabla_{\vec{w}} S) \\ \nabla_{\text{partial}_i}(g) &= \nabla_{\partial_i}(g_{rs} \epsilon^r \otimes \epsilon^s) = \nabla_{\partial_i}(g_{rs})(\epsilon^r \otimes \epsilon^s) + g_{rs} \nabla_{\partial_i}(\epsilon^r \otimes \epsilon^s) \\ &= \partial_i(g_{rs})(\epsilon^r \otimes \epsilon^s) + g_{rs}((\nabla_{\partial_i} \epsilon^r) \otimes \epsilon^s + \epsilon^r \otimes (\nabla_{\partial_i} \epsilon^s)) \end{aligned}$$

$$\begin{aligned}
&= \partial_i(g_{rs})(\epsilon^r \otimes \epsilon^s) - (g_{rs}\Gamma_{ik}^r(\epsilon^k \otimes \epsilon^s) + g_{rs}\Gamma_{ik}^s(\epsilon^r \otimes \epsilon^k)) \\
&= \partial_i(g_{rs})(\epsilon^r \otimes \epsilon^s) - (g_{ks}\Gamma_{ir}^k(\epsilon^r \otimes \epsilon^s) + g_{rk}\Gamma_{is}^k(\epsilon^r \otimes \epsilon^s)) \\
\nabla_{\partial_i}(g) &= \left(\frac{\partial g_{rs}}{\partial u^i} - g_{ks}\Gamma_{ir}^k - g_{rk}\Gamma_{is}^k \right) (\epsilon^r \otimes \epsilon^s)
\end{aligned}$$

Summary. Covariant derivatives of vectors are (0,1)-tensors as they have one covariant element and so a Christoffel symbol is added:

$$\nabla_{\partial_i}\vec{v} = \left(\frac{\partial v^k}{\partial u^i} + v^j\Gamma_{ij}^k \right) e_k$$

Covariant derivatives of covectors are (1,0)-tensors as they have one contra-variant element and so a Christoffel symbol is subtracted:

$$\nabla_{\partial_i}a = \left(\frac{\partial a_k}{\partial u^i} - a_j\Gamma_{ij}^k \right) e_k$$

Covariant derivatives of tensors are (2,0)-tensors as they have two contra-variant elements and so two Christoffel symbols are subtracted:

$$\nabla_{\partial_i}g = \left(\frac{\partial g_{rs}}{\partial u^i} - g_{ks}\Gamma_{ir}^k - g_{rk}\Gamma_{is}^k \right) (\epsilon^r \otimes \epsilon^s)$$

For a covariant derivative which is an (m,n)-tensor, add a Christoffel symbol for each covariant element and subtract one for each contravariant element.

Metric compatibility. With metric compatibility,

$$\begin{aligned}
\frac{\partial g_{rs}}{\partial u^i} &= g_{ks}\Gamma_{ir}^k + g_{rk}\Gamma_{is}^k \\
\nabla_{\partial_i}g &= \left(\frac{\partial g_{rs}}{\partial u^i} - g_{ks}\Gamma_{ir}^k - g_{rk}\Gamma_{is}^k \right) (\epsilon^r \otimes \epsilon^s) = (0)(\epsilon^r \otimes \epsilon^s) = 0
\end{aligned}$$

So $\nabla(g) = 0$. The covariant derivative of the metric tensor on any curved surface is 0.

20.20 Lie bracket

Lie bracket or commutator calculates the 'gap' between flow curves in a vector field when operations are non-commutative. The gap is the separation vector. If it's 0, then it commutes.

$$[u, v] = u(v) - v(u)$$

where $u = \vec{u}, v = \vec{v}$ and the first terms are like derivative operators. So $u(v)$ is the derivative of v in the direction of u .

Example. $u = 1e_x, v = xe_y$.

$$\begin{aligned}
v(u) &= v^i e_i(u^j e_j) = v^i \partial_i(u^j \partial_j) = v^i (\partial_i u^j) \partial_j + u^j (\partial_i \partial_j) \\
&= v^i (\partial_i u^j) \partial_j + v^i u^j (\partial_i \partial_j) = v^y (\partial_y u^x) \partial_x + v^y u^x (\partial_y \partial_x) \\
&= x (\partial_y 1) \partial_x + x(1) (\partial_y \partial_x) = x \partial_y \partial_x \\
&= x (\partial_y e_x) = x \frac{\partial x}{\partial y} = x(0) = 0
\end{aligned}$$

in Cartesian coordinates because the basis vectors are orthogonal. That $v(u) = 0$ means when moving in the v-direction, u isn't changing. For the other half of the commutator:

$$\begin{aligned}
u(v) &= u^i e_i(v^j e_j) = u^x (\partial_x v^y) \partial_y + u^x v^y (\partial_x \partial_y) = (1) (\partial_x x) \partial_y + 1x (\partial_x \partial_y) \\
&= \partial_y + 0 = \partial_y = e_y
\end{aligned}$$

so when moving in the u-direction, there is a change in the v-direction.

Coordinate lines are flow curves of basis vectors, so they must close: $[e_i, e_j] = 0, [\partial_i, \partial_j] = 0$.

Lie bracket is not linear:

$$\begin{aligned}
[ar, s] &\neq [a[r, s]] \\
[ar, s] &= ar(s) - s(ar) = ar(s) - s(a)r - as(r) \\
&= a(r(s) - s(rs)) - s(a)r = a[r, s] - s(a)r = a[r, s] - s(a)(r)
\end{aligned}$$

20.21 Torsion

Recall the covariant derivative:

$$\nabla_{\vec{u}} \vec{v} = \nabla_u v$$

where the connection ∇ acts on a vector in the v-direction in the u-direction. Expanding it out,

$$\nabla_u v = \nabla_{u^i \partial_i} (v^j \partial_j) = u^i \nabla_{\partial_i} (v^j \partial_j) = u^i [\nabla_{\partial_i} (v^j) \partial_j + v^j \nabla_{\partial_i} \partial_j]$$

Using $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$,

$$= u^i [(\partial_i v^k) \partial_k + v^j \Gamma_{ij}^k \partial_k] = u^i (\partial_i v^k + v^j \Gamma_{ij}^k) \partial_k$$

Many ways to define Γ such as Levi-Civita and $\Gamma = 0$ which provide different parallel transport instructions:

$$\nabla_u(v) = 0, \nabla_v(u) = 0$$

If $\nabla_u(v) \neq 0, \nabla_v(u) \neq 0$ then deviates from parallel transport by that value, e.g., $\nabla_u(v) = k$ since $\nabla_u(v) - 0(P.T.) = k - 0 = k$ where P.T.=parallel transport.

If the parallel transport is 'closed', then

$$\nabla_u(v) - \nabla_v(u) = 0$$

$$\nabla_u(v) = \nabla_v(u)$$

If there is a gap, then Lie bracket is non-zero: $[u, v] \neq 0$. Define the torsion tensor:

$$T(u, v) = \nabla_u(v) - [u, v] - \nabla_v(u)$$

and if $T(u, v) = 0$, then it's 'torsion-free', and means that:

$$\nabla_u(v) - \nabla_v(u) = [u, v]$$

The torsion-free property is a property of connections, not vector fields. The expression can be simplified:

$$\begin{aligned} &= \nabla_u v - \nabla_v u - [u(v) - v(u)] \\ &= u^i(\partial_i v^k + v^j \Gamma_{ij}^k) \partial_k - v^i(\partial_i u^k + u^j \Gamma_{ij}^k) \partial_k - [[u^i \partial_i (v^j \partial_j)] - [v^i \partial_i (u^j \partial_j)]] \\ &= u^i(\partial_i v^k + v^j \Gamma_{ij}^k) \partial_k - v^i(\partial_i u^k + u^j \Gamma_{ij}^k) \partial_k - [[u^i(\partial_i v^j) \partial_j] + u^i v^j (\partial_i \partial_j)] - [v^i(\partial_i u^j) \partial_j] + v^i u^j (\partial_i \partial_j)] \\ &= u^i(\partial_i v^k + v^j \Gamma_{ij}^k) \partial_k - v^i(\partial_i u^k + u^j \Gamma_{ij}^k) \partial_k - [[u^i(\partial_i v^j) \partial_j] + u^i v^j (\partial_i \partial_j)] - [v^i(\partial_i u^j) \partial_j] + v^i u^j (\partial_i \partial_j)] \\ &\text{where } v^i u^j (\partial_i \partial_j) = v^j u^i (\partial_j \partial_i) = u^i v^j (\partial_i \partial_j), \end{aligned}$$

$$\begin{aligned} &= u^i(\partial_i v^k + v^j \Gamma_{ij}^k) \partial_k - v^i(\partial_i u^k + u^j \Gamma_{ij}^k) \partial_k - [u^i(\partial_i v^j) \partial_j - v^i(\partial_i u^j) \partial_j] \\ &= (u^i(\partial_i v^k) \partial_k + u^i v^j \Gamma_{ij}^k \partial_k) - (v^i(\partial_i u^k) \partial_k + v^i u^j \Gamma_{ij}^k \partial_k) - [u^i(\partial_i v^j) \partial_j - v^i(\partial_i u^j) \partial_j] \end{aligned}$$

Change some of the notation: $j \rightarrow k$

$$\begin{aligned} &= (u^i(\partial_i v^k) \partial_k + u^i v^j \Gamma_{ij}^k \partial_k) - (v^i(\partial_i u^k) \partial_k + v^i u^j \Gamma_{ij}^k \partial_k) - [u^i(\partial_i v^k) \partial_k - v^i(\partial_i u^k) \partial_k] \\ &= u^i v^j \Gamma_{ij}^k \partial_k - v^i u^j \Gamma_{ij}^k \partial_k = u^i v^j \Gamma_{ij}^k \partial_k - v^j u^i \Gamma_{ji}^k \partial_k \\ &= u^i v^j \underbrace{(\Gamma_{ij}^k - \Gamma_{ji}^k)}_{T_{ij}^k} \partial_k \end{aligned}$$

So if torsion-free, $T(u, v) = T_{ij}^k = 0$, then

$$\Gamma_{ij}^k = \Gamma_{ji}^k$$

20.22 Riemann curvature tensor

How to determine whether a surface is curved or flat?

Often just have basis vectors and metric tensor g . From metric tensor, can derive Γ s. In Cartesian coordinates, $g_{ij} = I, \Gamma_{ij}^k = 0$, but this doesn't necessarily mean the space is flat because polar coordinates won't give this result even though it's on a flat surface. While polar coordinates can be converted to $g_{ij} = I, \Gamma_{ij}^k = 0$, so can points on curved surfaces. A point on the equator of a spherical surface has $g_{ij} = I, \Gamma_{ij}^k = 0$, and any non-equatorial point can be made this way by re-orienting the sphere so that it lies on the equator of the re-oriented sphere (the Riemann normal coordinate or 'local inertial frame').

Use Riemann curvature tensor to detect curvature:

$$R(u, v)w = \nabla_u \nabla_v w - \nabla_v \nabla_u w - \nabla_{[u, v]} w$$

where $u = \vec{u}$, etc. It is a (1,3)-tensor R_{abc}^d with one contravariant element and three covariant elements. The Riemann curvature tensor can be used to detect curvature in two ways: (1) holonomy, (2) geodesic deviation.

20.22.1 Holonomy

When a vector is parallel transported in a loop, if there's a gap between the final and initial vectors, then the surface is curved.

The Riemann curvature tensor R can be derived by using a parallelogram loop with sides A, B, C, D, and parallel transporting vector w :

$$\begin{aligned} R(\vec{u}, \vec{v})\vec{w} &= \lim_{x, y \rightarrow 0} \frac{w - DCBAw}{xy} \\ &= \lim_{x, y \rightarrow 0} \frac{1}{xy} (DCC^{-1}D^{-1}w - DCBAw) \\ &= \lim_{x, y \rightarrow 0} \frac{1}{xy} DC(C^{-1}D^{-1}w - BA w) \\ &= \lim_{x, y \rightarrow 0} \frac{1}{xy} DC(C^{-1}D^{-1}w - C^{-1} + C^{-1}w - w - BA w + Bw - Bw + w) \\ &= \lim_{x, y \rightarrow 0} \frac{1}{xy} DC[(C^{-1}(D^{-1}w - w) + (C^{-1}w - w) - B(Aw - w) + (Bw - w))] \\ &= \lim_{x, y \rightarrow 0} DC \left[\frac{C^{-1}}{x} \frac{(D^{-1}w - w)}{y} + \frac{1}{y} \frac{(C^{-1}w - w)}{x} - \frac{B}{y} \frac{(Aw - w)}{x} + \frac{1}{x} \frac{(Bw - w)}{y} \right] \\ &= \lim_{x, y \rightarrow 0} DC \left[\frac{C^{-1}}{x} \nabla_v w + \frac{1}{y} \nabla_u w - \frac{B}{y} \nabla_u w + \frac{1}{x} \nabla_v w \right] \end{aligned}$$

At the limit, $DC = II = I$ because they're so small, so

$$= \nabla_u \nabla_v w - \nabla_v \nabla_u w$$

if $[u, v] = 0$, so add Lie bracket for the general formula:

$$R(u, v)w = \nabla_u \nabla_v w - \nabla_v \nabla_u w - \nabla_{[u, v]} w$$

which is a multilinear map, so is linear in each of u, v, w .

20.22.2 Geodesic deviation

On a flat surface, geodesic rays (straight lines) originating from a single point will move apart from each other at a constant rate c :

$$\nabla_{\vec{v}} \vec{s} = c$$

So, $\nabla_v \nabla_v s = 0$. In a curved space, the geodesic rays will move apart from each other until the equator and then converge at the other side. So the rate of separation isn't constant:

$$\nabla_v \nabla_s \neq 0$$

Assuming the connection is torsion-free,

$$[v, s] = \nabla_v s - \nabla_s v = 0$$

$$\nabla_v s = \nabla_s v$$

On a geodesic,

$$\nabla_v v = 0$$

because there is no change in the forward vector moving in the forward direction along a geodesic. So $\nabla_s \nabla_v v = 0$.

$$\underbrace{\nabla_s \nabla_v v - \nabla_v \nabla_s v}_{R(s, v)v, [v, s]=0} + \nabla_v \underbrace{\nabla_s v}_{Torsion-free} = 0$$

$$R(s, v)v + \nabla_v \nabla_v s = 0$$

$$-R(s, v)v = \nabla_v \nabla_v s$$

So if left side is 0, then flat surface. Note:

$$-R(s, v)v = R(v, s)v$$

Moreover, by convention sometimes the negative is a positive in some references.

20.22.3 Riemann tensor components

Riemann tensor R is multilinear, so

$$R(u, v)w = u^i v^j w^k R(e_i, e_j)e_k$$

Take the basis component:

$$R(e_a, e_b)e_c = \nabla_{e_a} \nabla_{e_b} e_c - \nabla_{e_b} \nabla_{e_a} e_c - \nabla_{[e_a, e_b]} e_c$$

Since Lie bracket of basis vectors is zero ($[e_i, e_j] = 0$, no basis vector flow gaps),

$$e_i(e_j) - e_j(e_i) = 0$$

$$\frac{\partial^2}{\partial x^i \partial x^j} = \frac{\partial^2}{\partial x^j \partial x^i}$$

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k$$

So,

$$= \nabla_{e_a} \nabla_{e_b} e_c - \nabla_{e_b} \nabla_{e_a} e_c - 0$$

Recall definition of connection coefficients:

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k$$

Then,

$$R(e_a, e_b)e_c = \nabla_{e_a}(\Gamma_{bc}^i e_i) - \nabla_{e_b}(\Gamma_{ac}^j e_j)$$

Product rule:

$$= \nabla_{e_a}(\Gamma_{bc}^i e_i) + \Gamma_{bc}^i \nabla_{e_a} e_i - \nabla_{e_b}(\Gamma_{ac}^j e_j) - \Gamma_{ac}^j \nabla_{e_b} e_j$$

$$= \partial_a(\Gamma_{bc}^i) e_i - \partial_b(\Gamma_{ac}^j) e_j + \Gamma_{bc}^i \Gamma_{ai}^k e_k - \Gamma_{ac}^j \Gamma_{bj}^l e_l$$

Change basis vector indices to e_d :

$$= \partial_a(\Gamma_{bc}^d) e_d - \partial_b(\Gamma_{ac}^d) e_d + \Gamma_{bc}^i \Gamma_{ai}^d e_d - \Gamma_{ac}^j \Gamma_{bj}^d e_d$$

$$= [\partial_a(\Gamma_{bc}^d) - \partial_b(\Gamma_{ac}^d) + \Gamma_{bc}^i \Gamma_{ai}^d - \Gamma_{ac}^j \Gamma_{bj}^d] e_d$$

$$= R_{cab}^d e_d$$

So

$$R_{cab}^d = \partial_a(\Gamma_{bc}^d) - \partial_b(\Gamma_{ac}^d) + \Gamma_{bc}^i \Gamma_{ai}^d - \Gamma_{ac}^j \Gamma_{bj}^d$$

which is ugly. Derive symmetry rules that can help:

(1) 34-symmetry. Start with intuitive equation:

$$R(e_b, e_a) = -R(e_a, e_b)$$

So:

$$R_{cba}^d = -R_{cab}^d$$

Equivalently (can use metric tensor g to lower indices):

$$R_{abdc} = -R_{abcd}$$

(2) Bianchi identity (requires torsion-free property). Start with:

$$R(e_a, e_b)e_c + R(e_c, e_a)e_b + R(e_b, e_c)e_a$$

Expand, use torsion-free connection, cancel terms, and using the fact that the Lie bracket of basis vectors is 0,

$$R(e_a, e_b)e_c + R(e_c, e_a)e_b + R(e_b, e_c)e_a = 0$$

So:

$$R_{cab}^d + R_{bca}^d + R_{abc}^d = 0$$

Equivalently (can use metric tensor g to lower indices):

$$R_{dcab} + R_{dbca} + R_{dabc} = 0$$

(3) 12-symmetry (requires metric compatibility), skip long proof:

$$R_{bacd} = -R_{abcd}$$

(4) Flip (derived from the other three identities), skip proof:

$$R_{abcd} = R_{cdab}$$

In 2D, there are 2^4 Riemann tensors R_{cab}^d to compute because each of a,b,c,d can be 1 or 2. In 3D, it is $3^4 = 81$, etc. The symmetry rules reduce the number of tensor computations. In 2D, all R_{cab}^d except $R_{1212} = -R_{1221} = R_{2121} = -R_{2112}$ are 0, and computing one of these 4 gives the other 3, there's only one actual computation.

Example 1. 2D polar coordinates.

$$\begin{aligned} R_{\theta r \theta}^r &= \partial_r(\Gamma_{\theta\theta}^r) - \partial_\theta(\Gamma_{r\theta}^r) + \Gamma_{\theta\theta}^i \Gamma_{ri}^r - \Gamma_{r\theta}^j \Gamma_{\theta j}^r \\ &= \partial_r(\Gamma_{\theta\theta}^r) - \Gamma_{r\theta}^\theta \Gamma_{\theta\theta}^r = \partial_r(-r) - (-r) \frac{1}{r} = -1 + 1 = 0 \end{aligned}$$

By symmetry, the other '3' R_{1212} -type computations are 0. Since the other 12 computations also 0 in 2D, $R_{cab}^d = 0$ in 2D polar coordinates. Surface is flat.

Example 2. Sphere.

$$\begin{aligned} R_{212}^1 &= \partial_1(\Gamma_{22}^1) - \partial_2(\Gamma_{12}^1) + \Gamma_{22}^i \Gamma_{1i}^1 - \Gamma_{12}^j \Gamma_{2j}^1 \\ &= \partial_1(\Gamma_{22}^1) - \Gamma_{22}^1 \Gamma_{11}^1 + \Gamma_{22}^2 \Gamma_{12}^1 - \Gamma_{12}^1 \Gamma_{21}^1 - \Gamma_{12}^2 \Gamma_{22}^1 \\ &= \partial_1(\Gamma_{22}^1) - \Gamma_{12}^2 \Gamma_{22}^1 \end{aligned}$$

After computing the connection coefficients,

$$= \sin^2(u^1)$$

which is $\neq 0$ so surface is curved.

20.23 Ricci curvature tensor and Ricci scalar

Tracks volume changes along geodesics, which measures curvature.

20.23.1 Geodesic deviation

Recall in geodesic deviation, if the separation vector $\vec{s}$ is constant (so in flat space) or constantly changing (also in flat space), then

$$\nabla_{\vec{v}}\vec{s} = \nabla_v s = \text{constant}$$

So,

$$\nabla_v \nabla_v s = 0$$

In curved space, if s is growing smaller, i.e. converging, then

$$\nabla_v \nabla_v s < 0$$

and

$$[R(s, v)v] \cdot s = [\nabla_v \nabla_v s] \cdot \vec{s} < 0$$

In curved space, if s is growing larger, i.e. diverging, then

$$\nabla_v \nabla_v s > 0$$

and

$$[R(s, v)v] \cdot s = [\nabla_v \nabla_v s] \cdot \vec{s} > 0$$

Make above is normalized for size and made directionless with the square of the parallelogram area. Area of parallelogram A:

$$A = bh = \|u\|\|w\|\sin\theta = \|u \times w\|$$

$$\begin{aligned} A^2 &= \|u\|^2 \|w\|^2 \sin^2\theta = \|u\|^2 \|w\|^2 (1 - \cos^2\theta) \\ &= \|u\|^2 \|w\|^2 - (\|u\|\|w\|\cos\theta)^2 \\ &= (u \cdot w)(w \cdot w) - (u \cdot w)^2 \end{aligned}$$

20.23.2 Sectional curvature

Normalize the geodesic deviation with the square of the parallelogram area to get the sectional curvature:

$$K(u, w) = \frac{[R(u, w)w] \cdot u}{(u \cdot u)(w \cdot w) - (u \cdot w)^2}$$

which is the same even if a different set of u and w is chosen ($u = au + bw, w = cu + dw$). If $K > 0$ then it converges and if $K < 0$ it diverges.

20.23.3 Ricci curvature

Only works in orthonormal basis.

In an orthonormal basis $e_1, e_2 \dots e_n$, choose a particular direction and call it $v = \vec{v} = e_n$. It's orthonormal to the other directions. Define Ricci tensor as the sum of sectional curvatures:

$$Ric(v) = \sum_{i=1}^n K(e_i, v) = \sum_i \frac{[R(e_i, v)v] \cdot e_i}{(e_i \cdot e_i)(v \cdot v) - (e_i \cdot v)^2}$$

If $= 0$ then area of a circle remains the same, though the dimensions may be distorted.

Since v is orthonormal,

$$\begin{aligned} \frac{[R(e_i, v)v] \cdot e_i}{(1)(1) - 0} &= [R(e_i, v)v] \cdot e_i \\ &= [R(e_i, v^j e_j)v^k e_k] \cdot e_i = v^j v^k [R(e_i, e_j)e_k] \cdot e_i = v^j v^k R_{kij}^i \end{aligned}$$

Since summing over i's,

$$R_{ab} = \sum_i R_{aib}^i$$

this is the same as the 'generic' form: $= v^j v^k R_{kj}$ As it's a (3,1)-tensor, the change of basis is:

$$R_{abc}^{\tilde{b}} = F_a^k F_b^i F_c^j R_{kij}^i B_i^b$$

where in generic form:

$$Ric(v) = R_{ab} v^a v^b$$

Example. Spherical surface. Calculate Ricci tensor components $R_{11}, R_{12}, R_{21}, R_{22}$ of spherical surface, use Riemann tensor components. For 2D, there's one calculation, for R_{212}^1 and the metric tensor g_{ij} .

$$R_{11} = \mathfrak{g}^{22} R_{2121}^2 = \mathfrak{g}^{22} R_{1212} = \mathfrak{g}^{22} g_{11} R_{212}^1$$

Final result: $Ric(v) > 0$, so there is positive curvature.

20.23.4 Volume element derivative

In contrast to Ricci curvature, works even if not orthonormal basis.

$\omega(a, b)$ is area, $\omega(x, y, z)$ is volume. Note $\omega(e_1, e_2 \dots e_n) = 1$. If (u, v) aren't orthonormal, then change:

$$u = u^1 e_1 + u^2 e_2$$

$$w = w^1 e_1 + w^2 e_2$$

$$w(u, v) = \begin{vmatrix} u^1 & w^1 \\ u^2 & w^2 \end{vmatrix} = u^1 w^2 - u^2 w^1 = \epsilon_{ij} u^i w^j$$

where ϵ is the Levi-Civita symbol.

$$\begin{aligned} &= \epsilon_{11} u^1 w^1 + \epsilon_{12} u^1 w^2 + \epsilon_{21} u^2 w^1 + \epsilon_{22} u^2 w^2 \\ &= 0 + (1)u^1 w^2 + (-1)u^2 w^1 + 0 \end{aligned}$$

Similarly,

$$\omega(u, w, t) = \epsilon_{ijk} u^i w^j t^k$$

but note that this formula only works for orthonormal bases.

If ω is in a non-orthonormal basis, then convert to orthonormal basis:

$$\tilde{e}_1 = F_1^1 e_1 + F_1^2 e_2$$

$$\tilde{e}_2 = F_2^1 e_1 + F_2^2 e_2$$

So $\omega(\tilde{e}_1, \tilde{e}_2) = \det F$. Can specify this further. Take metric tensor in new basis:

$$\begin{aligned} \tilde{g}_{ij} &= \tilde{e}_i \cdot \tilde{e}_j \\ &= (F_i^a e_a) \cdot (F_j^b e_b) \\ &= F_i^a F_j^b g_{ab} \end{aligned}$$

Take determinants of both sides,

$$\det \tilde{g} = (\det F)(\det F)(\det g) = (\det F)^2 (I) = (\det F)^2$$

Previously, $V = \det F$. But now, $\det F = \sqrt{\det \tilde{g}}$, so $V = \det F = \sqrt{\det \tilde{g}}$. So if vectors are defined in terms of non-orthonormal bases,

$$u = u^1 \tilde{e}_1 + u^2 \tilde{e}_2$$

$$w = w^1 \tilde{e}_1 + w^2 \tilde{e}_2$$

So $\omega(\tilde{e}_1, \tilde{e}_2) = \det F = \sqrt{\det \tilde{g}}$. This means,

$$\omega(u, v) = \sqrt{\det \tilde{g}} \epsilon_{ij} u^i w^j$$

Similarly, can convert to orthonormal partial derivative bases (note: $\frac{\partial}{\partial x^1} = e_1, \frac{\partial}{\partial x^2} = e_2$):

$$\begin{aligned} \frac{\partial}{\partial \tilde{x}^1} &= \frac{\partial x^1}{\partial \tilde{x}^1} \frac{\partial}{\partial x^1} + \frac{\partial x^2}{\partial \tilde{x}^1} \frac{\partial}{\partial x^2} \\ \frac{\partial}{\partial \tilde{x}^2} &= \frac{\partial x^1}{\partial \tilde{x}^2} \frac{\partial}{\partial x^1} + \frac{\partial x^2}{\partial \tilde{x}^2} \frac{\partial}{\partial x^2} \end{aligned}$$

Then $\omega\left(\frac{\partial}{\partial \tilde{x}^1}, \frac{\partial}{\partial \tilde{x}^2}\right) = \det J$, the Jacobian matrix. Then:

$$\det \tilde{g} = (\det J)^2 (\det g) = (\det J)^2$$

so basis volume $V = \det J = \sqrt{\det \tilde{g}}$, and a similar result follows.

20.23.5 Volumes tracing geodesics

Recall $\nabla_v \omega = 0$ along a geodesic. With the Levi-Civita connection, which assumes metric compatibility, parallel transporting a vector preserves lengths and angles. This means it also preserves volumes:

$$\nabla_v V = 0$$

Then,

$$\nabla_v [\omega(a, b, c)] = 0$$

By the product rule,

$$\nabla_v \omega(a, b, c) + \omega(\nabla_v a, b, c) + \omega(a, \nabla_v b, c) + \omega(a, b, \nabla_v c) = 0$$

Parallel transporting vectors leads to covariant derivatives being zero:

$$\nabla_v \omega(a, b, c) + 0 + 0 + 0 = 0$$

$$\nabla_v \omega = \frac{d}{d\lambda} \sqrt{\det g} \epsilon_{ijk} = 0$$

If a parallelepiped is moving in 3D space,

$$V = \sqrt{\det g} \epsilon_{ijk} u^i w^j t^k$$

and equating the sides of the parallelepiped as being the separation vectors and changing notation,

$$V = \sqrt{\det g} \epsilon_{\mu_1, \mu_2, \mu_3} s_1^{\mu_1} s_2^{\mu_2} s_3^{\mu_3}$$

Generalizing,

$$V = \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} \prod_{i=1}^D s_i^{\mu_i}$$

Since volume is a scalar, with λ being the path parameter,

$$\nabla_v \nabla_v V = \frac{\partial^2}{\partial \lambda^2} V$$

which is

$$= \frac{d}{d\lambda} \frac{d}{d\lambda} \left(\sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} \prod_{i=1}^D s_i^{\mu_i} \right) = \frac{d}{d\lambda} \frac{d}{d\lambda} \left(\prod_{i=1}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D}$$

Applying the product rule once,

$$= \frac{d}{d\lambda} \left(\dot{s}_j^{\mu_j} \prod_{i=1, i \neq j}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D}$$

Applying the product rule a second time,

$$= \left(\ddot{s}_j^{\mu_j} \prod_{i=1, i \neq j}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} + \left(\dot{s}_j^{\mu_j} \dot{s}_k^{\mu_k} \prod_{i=1, i \neq j, k}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D}$$

Recall that

$$\nabla_v \nabla_v s = -R(s, v)v$$

which is equivalent to:

$$\ddot{s}_j^{\mu_j} = -R_{xyz}^{\mu_j} s_j^y v^z v^x$$

substituting,

$$= \left(-R_{xyz}^{\mu_j} s_j^y v^z v^x \prod_{i=1, i \neq j}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} + \left(\dot{s}_j^{\mu_j} \dot{s}_k^{\mu_k} \prod_{i=1, i \neq j, k}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D}$$

The first term is non-zero only when $y = \mu_j$. Every other time, ϵ will lead to 0 as there will be a repetitive term. So,

$$= \left(-R_{x\mu_j z}^{\mu_j} s_{\mu_j}^y v^z v^x \prod_{i=1, i \neq j}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} + \left(\dot{s}_j^{\mu_j} \dot{s}_k^{\mu_k} \prod_{i=1, i \neq j, k}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D}$$

Since $i \neq j$, it's removed in the first term,

$$\begin{aligned} &= -R_{x\mu_j z}^{\mu_j} v^z v^x \underbrace{\left(\prod_{i=1}^D s_i^{\mu_i} \right)}_V \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} + \left(\dot{s}_j^{\mu_j} \dot{s}_k^{\mu_k} \prod_{i=1, i \neq j, k}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} \\ &= -R_{x\mu_j z}^{\mu_j} v^z v^x V + \left(\dot{s}_j^{\mu_j} \dot{s}_k^{\mu_k} \prod_{i=1, i \neq j, k}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D} \end{aligned}$$

Since $R_{x\mu_j z}^{\mu_j} = R_{xz}$,

$$= -\underbrace{R_{xz} v^z v^x}_{Ric(v)} V + \left(\dot{s}_j^{\mu_j} \dot{s}_k^{\mu_k} \prod_{i=1, i \neq j, k}^D s_i^{\mu_i} \right) \sqrt{\det g} \epsilon_{\mu_i, \dots, \mu_D}$$

First term. So the second derivative has a term that is proportional to V, the volume. The Ricci curvature is the constant of proportionality. So the first term is about the effect of curved space. Second term is about how flat space changes.

20.23.6 Ricci scalar

Compares how much more area there is in curved space versus flat space for a given perimeter or boundary. Compares how size of ball deviates in curved space compared to flat space.

The radius r from the 'north pole' of a surface, ρ is the radius on a flat-plane slice, the radius from the center of the sphere being $\mathfrak{R}$ is related by

$$\rho = \mathfrak{R} \sin \theta$$

$$r = \mathfrak{R} \phi$$

The surface area of a portion of the sphere is

$$SA = \int dA = \int 2\pi \rho ds = \int 2\pi \rho \mathfrak{R} d\theta$$

where $2\pi \rho$ is the perimeter.

$$= \int_0^\phi 2\pi (\mathfrak{R} \sin \theta) \mathfrak{R} d\theta = 2\pi \mathfrak{R}^2 (1 - \cos \phi)$$

with $r = \mathfrak{R} \rightarrow \phi = \frac{r}{\mathfrak{R}}$ s.t.

$$= 2\pi \mathfrak{R}^2 (1 - \cos \left(\frac{r}{\mathfrak{R}} \right))$$

Compare areas of curved surface with flat surface as a ratio

$$\frac{A_{curved}}{A_{flat}} = \frac{2\pi \mathfrak{R}^2 (1 - \cos \left(\frac{r}{\mathfrak{R}} \right))}{\pi r^2}$$

Expand numerator using Taylor series:

$$= 1 - \frac{1}{24} \frac{2}{\mathfrak{R}^2} r^2 + O(r^4)$$

Metric tensor of a sphere is

$$g^{ij} = \begin{vmatrix} 1/(\mathfrak{R}^2) & 0 \\ 0 & 1/(\mathfrak{R} \sin^2 u^1) \end{vmatrix}$$

The Ricci tensor of a sphere is

$$R_{ij} = \begin{vmatrix} 1 & 0 \\ 0 & \sin^2 u^1 \end{vmatrix}$$

The Ricci scalar is

$$\begin{aligned} R &= R_i^i = g^{ij} R_{ij} \\ &= g^{11} R_{11} + g^{12} R_{12} + g^{21} R_{21} + g^{22} R_{22} \end{aligned}$$

$$= \frac{1}{\Re^2}(1) + \frac{1}{\Re^2 \sin^2 u^1} \sin^2 u^1$$

$$R = \frac{2}{\Re^2}$$

Substituting the last into the Taylor expansion of the area ratio:

$$\frac{A_{curve}}{A_{flat}} = 1 - \underbrace{\frac{1}{24}(R)r^2}_{<1} + O(r^4) > 0$$

More generally in D-dimensions,

$$\frac{V_{curve}}{V_{flat}} = 1 - \frac{R}{6(D+2)}r^2 + O(r^4)$$

For the same circumference, curvature leads to MORE area. For the same radius, curvature leads to LESS area.

20.23.7 Semicolon derivative

$$\begin{aligned} \nabla_{e_i} \vec{v} &= \nabla_{e_i} (v^j e_j) = \nabla_{e_i} v_j e_j + v^j \nabla_{e_i} e_j \\ &= (\partial_i v^j) e_j + v^j \Gamma_{ij}^k e_k \\ &= (\partial_i v^k + v^j \Gamma_{ij}^k) e_k \\ &= v_{;i}^k e_k \end{aligned}$$

where i is the direction of the covariant derivative. Can do similar with metric tensor:

$$\nabla_{e_i} g = \nabla_{e_i} (g_{mn} \epsilon^m \otimes \epsilon^n)$$

noting that $\epsilon^i = dx^i$ and $\epsilon^i(e_j) = dx^i(\frac{\partial}{\partial x^j}) = \delta_j^i$,

$$\begin{aligned} &= (\partial_i g_{mn} - g_{mx} \Gamma_{in}^x - g_{xn} \Gamma_{im}^x) \epsilon^m \otimes \epsilon^n \\ &= g_{mn;i} \epsilon^m \otimes \epsilon^n \end{aligned}$$

Riemann tensor:

$$\begin{aligned} \nabla_{e_i} R &= \nabla_{e_i} (R_{cab}^d e_d \otimes \epsilon^c \otimes \epsilon^a \otimes \epsilon^b) \\ &= (\partial_i R_{cab}^d + R_{cab}^x \Gamma_{ix}^d - R_{xab}^d \Gamma_{ic}^x - R_{cxb}^d \Gamma_{ia}^x - R_{cax}^d \Gamma_{ib}^x) \end{aligned}$$

where each of the components is itself expanded ($R_{cab}^d = \partial_a \Gamma_{bc}^d - \partial_b \Gamma_{ac}^d + \Gamma_{bc}^k \Gamma_{ak}^d - \Gamma_{ac}^j \Gamma_{bj}^d$),

$$= R_{cab;i}^d$$

20.23.8 Properties of the Ricci tensor and scalar

Recall contraction of Ricci tensors:

$$R_{ab} = \sum_i R_{aib}^i$$

as well as the Ricci scalar:

$$R = \sum_i R_i^i$$

(1) Which Riemann tensor contractions are interesting?

$$R_{iab}^i = g^{ij} R_{jiab} = -g^{ji} R_{ijab} = -R_{jab}^j$$

using one of the Riemann tensor properties and the fact that the metric tensor is symmetric. Since

$$R_{iab}^i = -R_{jab}^j$$

then both are equal to zero. Further,

$$R_{aib}^i = -R_{abi}^i$$

which is just the negative of R_{abi}^i , which is the only Riemann tensor contraction that is interesting, i.e. the Ricci tensor.

(2) Ricci tensor is symmetric. Using Bianchi identity,

$$R_{bc} = R_{cb}$$

so the matrix is symmetric.

(3) Second Bianchi identity, which has a very long proof:

$$R_{cab;i}^d + R_{cia;b}^d + R_{cbi;a}^d = 0$$

(4) Contracted Bianchi identity

$$R_{;n}^{mn} - \frac{1}{2} g^{mn} R_{;n} = 0$$

sometimes written as

$$\underbrace{\nabla_n (R^{mn} - \frac{1}{2} g^{mn} R)}_{G^{mn}} = 0$$

$$\nabla_n G^{mn} = 0$$

where G^{mn} is the Einstein tensor. The contracted Bianchi identity means this:

$$\nabla_{e_i} (R^{mn} (e_m \otimes e_n) - \frac{1}{2} g^{mn} (e_m \otimes e_n) R) = 0$$

(5) Einstein field equations. T^{mn} is the energy-momentum tensor while $\nabla_n T^{mn} = 0$ is conservation of energy-momentum. So, $\nabla_n G^{mn} = 0$. Therefore,

$$G^{mn} \propto T^{mn}$$

$$G^{mn} = \frac{8\pi G}{c^4} T^{mn}$$

which comprises 16 equations, but with mn elements being symmetric and therefore equal to nm elements, there are 6 fewer equations, so 16-6=10 unique equations. Sometimes, the EFQ is expanded to:

$$G^{mn} + \Lambda g^{mn} = \frac{8\pi G}{c^4} T^{mn}$$

where Λ is the cosmological constant.

21 Relativity

21.1 Relativity math

Invariance. Tensor is same in all coordinate bases. Time dilation is a change of coordinates in 4D spacetime.

Covariance and contravariance.

$$\begin{aligned} \vec{v} &= 1e_x + 1e_y \\ &= [e_x, e_y] \begin{vmatrix} 1 \\ 1 \end{vmatrix} = [e_x, e_y] I \begin{vmatrix} 1 \\ 1 \end{vmatrix} \\ &= [e_x, e_y] \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \begin{vmatrix} 1 \\ 1 \end{vmatrix} = [e_x, e_y] \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix}^{-1} \begin{vmatrix} 1 \\ 1 \end{vmatrix} \\ &= [e_x, e_y] \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} \begin{vmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{vmatrix} \begin{vmatrix} 1 \\ 1 \end{vmatrix} = [\tilde{e}_x, \tilde{e}_y] \begin{vmatrix} 1/2 \\ 1/2 \end{vmatrix} \end{aligned}$$

So in general,

$$\vec{v} = [e_x, e_y] \underbrace{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}_{\text{basis change}} \underbrace{\begin{vmatrix} a & b \\ c & d \end{vmatrix}^{-1}}_{\text{component change}} \begin{vmatrix} x \\ y \end{vmatrix}$$

Galilean relativity matrix:

$$\vec{v} = [e_x, e_y] \begin{vmatrix} 1 & 0 \\ v & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ -v & 1 \end{vmatrix} \begin{vmatrix} t \\ x \end{vmatrix}$$

Special relativity (Lorentz) matrix:

$$\vec{v} = [e_x, e_y] \begin{vmatrix} \gamma & \gamma \frac{v}{c} \\ \gamma \frac{v}{c} & \gamma \end{vmatrix} \begin{vmatrix} \gamma & -\gamma \frac{v}{c} \\ -\gamma \frac{v}{c} & \gamma \end{vmatrix} \begin{vmatrix} t \\ x \end{vmatrix}$$

21.2 Galilean relativity

In inertial (non-accelerating) reference frames, all observers agree on the laws of motion. In non-inertial frames, $F = ma + F_{extra}$ where the extra force is due to the extent and direction of acceleration.

In Galilean relativity, any inertial reference frame is as valid as any other. There is no 'special' reference frame.

Maxwell's equations of EM, when combined, say that the speed of light is constant in vacuum. Sound waves are relative to air, water waves are relative to water, but EM waves are not relative to anything.

21.2.1 Spacetime diagrams and separation vector

On a spacetime, inertial reference frame is the vertical worldline, but can so that it's from another inertial reference frame. More horizontal worldlines mean faster velocities.

All agree on time but disagree on position of events. Spacetime separation vectors are 'distances' across spacetime diagrams. So $\vec{s}$ is invariant, but depending on reference frame, the coordinate bases e_t, e_x are different.

21.2.2 Galilean transform

Change of bases in one direction:

$$\tilde{e}_t = e_t + v e_x$$

$$\tilde{e}_x = e_x$$

Change of bases in the reverse direction:

$$e_t = \tilde{e}_t - v \tilde{e}_x$$

$$e_x = \tilde{e}_x$$

So, the separation vector is, by combining the above,

$$\vec{s} = s = t e_t + x e_x = t \tilde{e}_t + (-vt + x) \tilde{e}_x = \tilde{t} \tilde{e}_t + \tilde{x} \tilde{e}_x$$

In matrix notation,

$$\begin{aligned} s &= t e_t + x e_x = [e_t, e_x] \begin{vmatrix} t \\ x \end{vmatrix} = [e_t, e_x] \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \begin{vmatrix} t \\ x \end{vmatrix} \\ &= [e_t, e_x] \begin{vmatrix} t & x \end{vmatrix} = [e_t, e_x] \begin{vmatrix} 1 & 0 \\ v & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ v & 1 \end{vmatrix}^{-1} \begin{vmatrix} t \\ x \end{vmatrix} \\ &= [e_t, e_x] \begin{vmatrix} t & x \end{vmatrix} = [e_t, e_x] \begin{vmatrix} 1 & 0 \\ v & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 \\ -v & 1 \end{vmatrix} \begin{vmatrix} t \\ x \end{vmatrix} \end{aligned}$$

$$= [\tilde{e}_t, \tilde{e}_x] \left| \begin{array}{c} \tilde{t} \\ \tilde{x} \end{array} \right| = \tilde{t}\tilde{e}_t + \tilde{x}\tilde{e}_x = s$$

In the other direction,

$$\begin{aligned} s &= \tilde{t}\tilde{e}_t + \tilde{x}\tilde{e}_x = [\tilde{e}_t, \tilde{e}_x] \left| \begin{array}{c} \tilde{t} \\ \tilde{x} \end{array} \right| \\ &= [\tilde{e}_t, \tilde{e}_x] \left| \begin{array}{cc} 1 & 0 \\ -v & 1 \end{array} \right| \left| \begin{array}{cc} 1 & 0 \\ v & 1 \end{array} \right| \left| \begin{array}{c} \tilde{t} \\ \tilde{x} \end{array} \right| = [e_t, e_x] \left| \begin{array}{c} t \\ x \end{array} \right| = te_t + xe_x = s \end{aligned}$$

21.2.3 Metric tensor

Pythagorean theorem can't be used in other (even orthonormal) bases other than the standard basis. So use dot product. Example:

$$\begin{aligned} \|R\|^2 &= R \cdot R = (xe_x + ye_y) \cdot (xe_x + ye_y) \\ &= x^2(e_x \cdot e_x) + xy(e_x \cdot e_y) + yx(e_y \cdot e_x) + y^2(e_y \cdot e_y) = x^2 + y^2 \\ &= [x, y]g \left| \begin{array}{c} x \\ y \end{array} \right| \end{aligned}$$

Another example:

$$\|R\|^2 = 2\hat{x}^2 + 2\hat{x}\hat{y} + \hat{y}^2 = [\hat{x}, \hat{y}]g_{\hat{x}\hat{y}} \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right|$$

To convert between bases and metric tensors g:

$$\begin{aligned} \|R\|^2 &= \left| \begin{array}{c} x \\ y \end{array} \right|' g \left| \begin{array}{c} x \\ y \end{array} \right| \\ &= \left(\left| \begin{array}{cc} a & b \\ c & d \end{array} \right| \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right| \right)' g \left| \begin{array}{cc} a & b \\ c & d \end{array} \right| \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right| \\ &= \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right|' \left| \begin{array}{cc} a & b \\ c & d \end{array} \right|' g \left| \begin{array}{cc} a & b \\ c & d \end{array} \right| \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right| = \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right|' g \left| \begin{array}{c} \hat{x} \\ \hat{y} \end{array} \right| \end{aligned}$$

so, the metric tensor is 2-covariant:

$$\hat{g} = \left| \begin{array}{cc} a & b \\ c & d \end{array} \right|' g \left| \begin{array}{cc} a & b \\ c & d \end{array} \right|$$

Note:

$$g(R, T) = R \cdot T = [R_x, R_y]g \left| \begin{array}{c} T_x \\ T_y \end{array} \right| = [\hat{R}_x, \hat{R}_y]\hat{g} \left| \begin{array}{c} \hat{T}_x \\ \hat{T}_y \end{array} \right|$$

21.2.4 Problems with Galilean relativity

In Galilean relativity, time is the same for everyone.

Newton's laws of motion.

$$F = ma = m \left(\frac{d^2 x(t)}{dt^2} \right)$$

if $x(t) + k$ or $x(t) + vt$

$$= m \frac{d^2}{dt^2} (x(t) + k) = m \left(\frac{d^2 x(t)}{dt^2} \right) = ma$$

but if in $\frac{1}{2}\alpha t^2$,

$$= m \frac{d^2}{dt^2} (x(t) + k) = m \left(\frac{d^2 x(t)}{dt^2} \right) + m\alpha = ma + m\alpha$$

in a non-inertial reference frame. So the laws of motion are the same in inertial frames but not in non-inertial frames.

Electromagnetism. The Lorentz force combined with Maxwell's equations leads to a contradiction. A proton near a charged wire will be pushed away from an outsider's perspective but from the proton's perspective it will remain stationary. If the Lorentz contraction is introduced, which contracts the protons in the wire which from the proton's perspective are 'moving', then the reference frames agree. This suggests lengths are not invariant. Further, solving Maxwell's equations, the second derivative with respect to space is proportional to the second derivative with respect to time for both electricity and magnetism:

$$\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2}$$

so they're wave equations. Moreover, they're the same thing, viewed from different reference frames (outsider and proton). Finally, light speed c is a constant from solving Maxwell's equations. This suggests c is invariant.

21.3 Special relativity

Einstein accepted Galilean relativity but expanded it to include Maxwell's equations and therefore treated c as a law of physics that does not depend on the reference frame. So every observer, regardless of reference frame, will agree on c . But this is internally inconsistent, as an observer with a light should be seen by another observer at speed $v+c$, but this should be c .

To resolve this inconsistency, make time and space relative as well. Previously, it was energy, momentum, etc. that were relative but time and space were all agreed upon by all observers. If time and space become relative but c is constant regardless of the reference frame, then pick a reference frame (a 'moving' frame) where the time that passes for light to travel is

$$\Delta t = \frac{d}{c}$$

and from a 'stationary' reference frame, where the same light travels $c\tilde{\Delta}t$ across the distance d but moving at speed v and therefore has moved $v\tilde{\Delta}t$. Then

$$\begin{aligned}(v\tilde{\Delta}t)^2 + d^2 &= (c\tilde{\Delta}t)^2 \\ \frac{d^2}{c^2} + \frac{v^2}{c^2}(\tilde{\Delta}t)^2 &= (\tilde{\Delta}t)^2 \\ &= (\tilde{\Delta}t)^2 \left(1 - \frac{v^2}{c^2}\right) = \frac{d^2}{c^2} = \Delta t^2\end{aligned}$$

So,

$$(\tilde{\Delta}t)^2 = \Delta t \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \right) = \Delta t \gamma_v$$

where $\gamma_v \geq 1$ and is the Lorentz factor. Since light moves at c , $\Delta t = 0$ so light experiences no time, and so there is no concept of time for light. Time is 'vertically flat'. Light speed c is what is constant in the universe, where the universe or space-time is the medium. If Galilean relativity held for c , then for one observer cause occurs before effect but the reverse can occur for another observer as the effect can arrive first - so time is essentially moving backwards and observers cannot agree on the order of cause and effect.

21.3.1 Lorentz transformation

In Galilean transforms, c is different depending on the reference frame. It only shifts e_x but all time is the same for all observers. Geometrically, the basis 'unit squares' aren't rhombuses; they're just parallelograms.

In Lorentz transformations, the basis 'unit squares' are rhombuses as both time and position basis vectors tilt. The time axis tilts as in Galilean transforms. The position axis tilts because simultaneity is now at angle. It is tilted as light moving in a different inertial reference frame appears distorted to a different inertial frame. In this Lorentz transform, c is constant in all reference frames.

Vertical axis: ct (light-time). Horizontal axis: x . Since light travels at c , for each unit of time t , $x = ct$ and therefore the slope is 1. In a Lorentz transformation, the vertical slope is $x = \beta ct$ where $\beta = \frac{v}{c}$ and the horizontal slope is,

since the transformation forms a rhombus, $x = \frac{1}{\beta}ct$. The Galilean transforms were:

$$\tilde{t} = t$$

$$\tilde{x} = -vt + x$$

in the reverse direction,

$$t = \tilde{t}$$

$$x = +v\tilde{t} + \tilde{x}$$

analogously, the Lorentz transformation is

$$c\tilde{t} = ct - \beta x$$

$$\tilde{x} = -\beta ct + x$$

in the reverse direction,

$$ct = c\tilde{t} - \beta \tilde{x}$$

$$x = +\beta c\tilde{t} + \tilde{x}$$

and scaling by the Lorentz factor γ ,

$$ct = \gamma(c\tilde{t} - \beta \tilde{x})$$

$$x = \gamma(+\beta c\tilde{t} + \tilde{x})$$

Then:

$$\tilde{x}x = \gamma(-\beta ct + x)\gamma(+\beta c\tilde{t} + \tilde{x})$$

Using $x = ct$ and $\tilde{x} = c\tilde{t}$,

$$(c\tilde{t})(ct) = \gamma ct(-\beta + 1)\gamma c\tilde{t}(\beta + 1)$$

$$1 = \gamma^2(-\beta + 1)(\beta + 1)$$

$$1 = \gamma^2(1 - \beta)^2$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}$$

In matrix notation,

$$\begin{vmatrix} c\tilde{t} \\ \tilde{x} \end{vmatrix} = \gamma \underbrace{\begin{vmatrix} 1 & -\beta \\ -\beta & 1 \end{vmatrix}}_{\text{Lorentz matrix}} \begin{vmatrix} ct \\ x \end{vmatrix}$$

Since the Lorentz transform is like a Jacobian, a basis change, it reflects the area change of unit shapes (squares, rhombuses):

$$\begin{vmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{vmatrix} = \gamma^2 - \gamma^2\beta^2 = \gamma^2(1 - \beta^2) = \left(\frac{1}{\sqrt{1 - \beta^2}}\right)^2 (1 - \beta^2) = 1$$

So no change in area. From the above, can derive:

$$\tilde{e}_t = \gamma(e_t + \beta e_x)$$

$$\tilde{e}_x = \gamma(\beta e_t + e_x)$$

Time dilation. Just time component of spacetime vector in different bases. Time between 2 events is always shortest in the reference where it occurs, with the observer needing time for light to reach them.

Length contraction. Objects are longest where they are, with observers seeing them shorter as they move while the light from the end of the object arrives.

Note the Lorentz transformation doesn't allow faster-than-light speeds which would effectively reverse the axes. The Lorentz transformation also forms light cones, and while simultaneity is relative, cause-and-effect is not reversible within the light cones. Outside the light cones, objects cannot be causally connected so their orders can be reversed, depending on the Lorentz transformation.

21.3.2 Velocity addition in special relativity

To add velocities in special relativity, with one reference frame being:

$$c\tilde{t} = \gamma_v(ct - \beta_v x)$$

$$\tilde{x} = \gamma_v(-\beta_v ct + x)$$

and the other reference frame:

$$c\hat{t} = \gamma_u(c\tilde{t} - \beta_u \tilde{x})$$

$$\hat{x} = \gamma_u(-\beta_u c\tilde{t} + \tilde{x})$$

Combining,

$$c\hat{t} = \gamma_u \gamma_v ((1 + \beta_u \beta_v) ct - (\beta_v + \beta_u) x)$$

$$\hat{x} = \gamma_u \gamma_v (-(\beta_u + \beta_v) ct + (1 + \beta_u \beta_v) x)$$

In matrix notation,

$$\begin{pmatrix} c\hat{t} \\ \hat{x} \end{pmatrix} = \underbrace{\gamma_u \gamma_v \begin{pmatrix} 1 + \beta_u \beta_v & -(\beta_u + \beta_v) \\ -(\beta_u + \beta_v) & 1 + \beta_u \beta_v \end{pmatrix}}_{Transformation} \begin{pmatrix} ct \\ x \end{pmatrix}$$

The transformation can be mathematically re-arranged to:

$$\frac{1}{\sqrt{1 - \left(\frac{\beta_u + \beta_v}{1 + \beta_u \beta_v}\right)^2}} \begin{pmatrix} 1 & -\left(\frac{\beta_u + \beta_v}{1 + \beta_u \beta_v}\right)^2 \\ -\left(\frac{\beta_u + \beta_v}{1 + \beta_u \beta_v}\right)^2 & 1 \end{pmatrix}$$

The Lorentz transformation is

$$\frac{1}{\sqrt{1-\beta^2}} \begin{vmatrix} 1 & -\beta \\ -\beta & 1 \end{vmatrix}$$

So the new velocity, i.e. after addition, is

$$0 < \frac{\beta_u + \beta_v}{1 + \beta_u \beta_v} < 1$$

21.3.3 Minkowski geometry

Euclidean geometry. The Euclidean metric tensor is:

$$\begin{vmatrix} e_x \cdot e_x & e_x \cdot e_y \\ e_y \cdot e_x & e_y \cdot e_y \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = I$$

which leads to all observers agreeing on Euclidean distance:

$$L^2 = [x, y] I [x, y]' = x^2 + y^2$$

Minkowski geometry. Spacetime distance is:

$$\|\vec{S}\|^2 = S \cdot S = s^2 = (ct)^2 - x^2$$

which all observers agree on. It is constructed using the orthonormal Minkowski metric tensor:

$$\begin{vmatrix} e_t \cdot e_t & e_t \cdot e_x \\ e_x \cdot e_t & e_x \cdot e_x \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$$

So:

$$s^2 = [ct, x] \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} [ct, x]' = (ct)^2 - x^2$$

Note: Metric tensors work regardless of coordinate basis change. So s^2 is the same in all inertial reference frames.

$$s^2 = (c\tilde{t})^2 - (\tilde{x})^2$$

when $c\tilde{t} = \gamma(ct - \beta x)$ and $\tilde{x} = \gamma(-\beta ct + x)$ are substituted in,

$$s^2 = (c\tilde{t})^2 - (\tilde{x})^2 = (ct)^2 - x^2$$

In contrast to $L^2 \geq 0$, s^2 can be anything, including $s^2 < 0$. Case (i) $s^2 > 0$ ('time-like') then it must be that $ct > x \rightarrow \|\beta\| = \frac{x}{ct} < 1$. At $x = 0$, $s = ct = c\tau$ where τ is the proper time (shortest time possible as other frames are time-dilated). Case (ii) $s^2 = 0$ ('light-like') $s = 0 \rightarrow x = ct$ so it's light and forms the edge of light-cones. Case (iii) $s^2 < 0$ ('space-like') where $ct < x \rightarrow \|\beta\| = \frac{x}{ct} > 1$ and $s^2 = -x^2 \rightarrow \sqrt{-s^2} = x = L_0$, the proper length, the object's length in its own reference frame (longest possible length as other frames see it length-contracted).

Minkowski metric intuition. Like 'reversed' Pythagorean calculation, as the moving spacetime vector receives light from the stationary observer in the future (consistent with time dilation).

21.3.4 Hyperbolic geometry

In Euclidean geometry, use a unit circle $x^2 + y^2 = 1$ is represented by $x = \cos\theta$ and $y = \sin\theta$, and combining, get trigonometric identity $\sin^2\theta + \cos^2\theta = 1$ with $Area = \frac{\theta}{2}$.

In Minkowski geometry, use a unit hyperbola $y^2 - x^2 = \pm 1$ (positive for 'vertical' hyperbolas, negative for 'horizontal' hyperbolas; in the unit hyperbola $r^2 = 1$) represented by $x = \sinh\phi$ and $y = \cosh\phi$. When combined, $\cosh^2\phi - \sinh^2\phi = \pm 1$. The $Area = \frac{\phi}{2}$. From Minkowski geometry, $(ct)^2 - x^2 = s^2$ where $s^2 > 0$ is a vertical hyperbola, $s^2 < 0$ is a horizontal hyperbola, and $s^2 = s = 0$ are asymptotes.

The unit orthonormal metric is a circular rotation, where $A = \theta/2 = \text{constant}$ (so the determinant is constant) and the r^2 of $x^2 + y^2 = r^2$ is also constant:

$$[\hat{e}_x, \hat{e}_y] = [e_x, e_y] \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$$

The unit hyperbolic metric is a hyperbolic rotation, where $A = \phi/2 = \text{constant}$ (determinant is constant; also, rectangles 'boxed' in by the vertical and horizontal hyperbolas form rectangles) and the lengths and widths change. The r^2 is also constant:

$$[\hat{e}_x, \hat{e}_y] = [e_x, e_y] \begin{vmatrix} \cosh\phi & \sinh\phi \\ \sinh\phi & \cosh\phi \end{vmatrix}$$

which is equivalent to the Lorentz transformation:

$$\gamma = \cosh\phi = \frac{1}{\sqrt{1 - \beta^2}}$$

$$\cosh^2\phi = \frac{1}{1 - \beta^2}$$

$$(1 - \beta^2)\cosh^2\phi = 1$$

and eventually,

$$\beta = \frac{\sinh\phi}{\cosh\phi} = \tanh\phi$$

which can be substituted into

$$\gamma\beta = \cosh\phi \frac{\sinh\phi}{\cosh\phi} = \sinh\phi$$

So substituting $\gamma = \cosh\phi$ and $\gamma\beta = \sinh\phi$ into the Lorentz transformation matrix,

$$\begin{vmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{vmatrix} = \begin{vmatrix} \cosh\phi & \sinh\phi \\ \sinh\phi & \cosh\phi \end{vmatrix}$$

so the Lorentz transformation is a hyperbolic rotation. The Lorentz transformation that forms 'rhombuses' are consistent with hyperbolic rotations.

21.3.5 Special relativity physics

Euclidean vectors like

$$\vec{r} = xe_x + ye_y + ze_z$$

$$\vec{v} = \frac{dr}{dt}$$

and momentum, acceleration, and force are 3-vectors. They depend on the reference frame. They don't include time. In contrast, Minkowski vectors like

$$\vec{S} = cte_t + xe_x + ye_y + ze_z = cte_t + ie_i$$

$$\vec{V} = \frac{dS}{d\tau}$$

and corresponding p, a, and F are 4-vectors which all observers agree on, as it is defined with respect to proper time τ .

4-velocity.

$$\frac{dS}{d\tau} = \frac{d}{d\tau}(cte_t + ie_i)$$

In inertial frames, derivatives of basis vectors are 0, so after the product rule:

$$= c \frac{dt}{d\tau} e_t + \frac{di}{d\tau} e_i$$

For the first term,

$$ct = \gamma(c\tilde{t} + \beta\tilde{x})$$

since $\tilde{x} = 0$ when $t = \tau$,

$$ct = \gamma c\tau \rightarrow t = \gamma\tau$$

so,

$$c \frac{dt}{d\tau} e_t + \frac{di}{d\tau} e_i = c\gamma e_t + \frac{di}{d\tau} e_i$$

Use chain rule on second term,

$$\frac{dx}{d\tau} = \frac{dx}{dt} \frac{dt}{d\tau} = u^x \gamma$$

so,

$$c\gamma e_t + \frac{di}{d\tau} e_i = c\gamma e_t + u^i e_i =$$

$$\vec{U} = U = cU^t e_t + U^i e_i$$

which is invariant in all coordinate bases (e.g. $[e_t, e_x]$, $[\tilde{e}_t, \tilde{e}_x]$). U 's are relativistic velocities. Since for the 'rest' reference frame, position vectors are 0, so $U = c\gamma + 0 = c(1) = c$ (the Minkowski length). All other reference frames should be the same, since the 4-velocity is invariant.

4-momentum. $\vec{P} = P = mU$. Using U from above,

$$P = mc\gamma e_t + mu^i e_i$$

Expand the first term $P^t = mc\gamma$ using Taylor series:

$$\gamma = 1 + \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + \frac{5}{16}\beta^6 + \dots$$

Since $\beta = \frac{u}{c}$,

$$\begin{aligned}\gamma &= 1 + \frac{1}{2}\frac{u^2}{c^2} + \frac{3}{8}\frac{u^4}{c^4} + \frac{5}{16}\frac{u^6}{c^6} + \dots \\ cP^t &= mc^2 \left(1 + \frac{1}{2}\frac{u^2}{c^2} + \frac{3}{8}\frac{u^4}{c^4} + \frac{5}{16}\frac{u^6}{c^6} + \dots \right) \\ &= mc^2 + \frac{1}{2}mu^2 + mc^2 \left(\frac{3}{8}\frac{u^4}{c^4} + \frac{5}{16}\frac{u^6}{c^6} + \dots \right)\end{aligned}$$

where the first term is rest mass (KE when not moving), the second is KE, and the higher order terms are further KE terms that Newton was unaware of. So,

$$E = cP^t \rightarrow P^t = \frac{E}{c}$$

So,

$$\begin{aligned}P &= \frac{E}{c}e_t + \gamma p^i e_i \\ P \cdot P &= (mU) \cdot (mU) = m^2(U \cdot U) = m^2c^2 \\ P \cdot P &= (P^t)^2 - (P^i)^2 = \frac{E^2}{c^2} - \gamma^2(p^i)^2 \\ m^2c^2 &= \frac{E^2}{c^2} - \gamma^2p^2\end{aligned}$$

which results in Einstein's energy-mass-momentum relation:

$$E^2 = m^2c^4 + \gamma^2p^2c^2$$

and for rest masses, $E = mc^2$. For massless objects like light, set $P = 0 \rightarrow P^t = P^i \rightarrow \frac{E}{c} = P \rightarrow E = cP$ and from QM, $E = hf$.

Example of 4-momentum. In the 'rest' reference frame, it's just rest mass, $P = mc \rightarrow \frac{E}{c} = mc \rightarrow E = mc^2$. From a moving observer's frame, there is rest mass which is larger but which is subtracted by KE.

4-momentum is conserved. In Newtonian physics, total momentum vectors add to 0. With 4-momentum, even if an object is at rest, there is momentum in the time dimension (related to the rest energy from $E = mc^2$).

21.4 Acceleration in Special Relativity

Assume constant acceleration for simplicity. In 2D spacetime, acceleration is not parabolic (as in Newtonian physics, as this will mean eventually can go faster than c). Instead, it's hyperbolic with $x = ct$, the edge of the light-cone, being asymptotic.

The Lorentz transformation can't adjust to non-inertial reference frames, but it can adjust to specific points, where $e_t = \vec{U}$ (which is tangent to the worldline and $U = ce_t + 0e_i$) and $e_x = \vec{A}$ (which is orthogonal to the worldline and $A = 0e_t + a^i e_i$).

Minkowski orthogonality is orthogonality in Lorentz frames:

$$\vec{U} \cdot \vec{A} = U \cdot A = 0$$

So 4-velocity and 4-acceleration are Minkowski orthogonal. Proof. Since $\frac{d}{d\tau}(U \cdot U) = \frac{d}{d\tau}(c^2) = 0$ and $\frac{d}{d\tau}(U \cdot U) = \frac{d}{d\tau} \cdot U + U \cdot \frac{d}{d\tau} = 2(U \cdot A) = 0$, so $U \cdot A = 0$.

With $U = U^t e_t + U^x e_x$ and $A = A^t e_t + A^x e_x$ and the Minkowski metric tensor,

$$U \cdot U = c^2 = (U^t)^2 - (U^x)^2$$

$$U \cdot A = 0 = (U^t)(A^t) - U^x A^x$$

$$A \cdot A = -a^2 = (A^t)^2 - (A^x)^2$$

Combine these equations to get

$$A^t = \frac{a}{c} U^x$$

$$A^x = \frac{a}{c} U^t$$

Combine with

$$A = \frac{dU}{d\tau} = A^t e_t + A^x e_x$$

to get

$$A^t = \frac{a}{c} U^x = \frac{dU^t}{d\tau}$$

$$A^x = \frac{a}{c} U^t = \frac{dU^x}{d\tau}$$

and combine further:

$$\frac{a^2}{c^2} U^x = \frac{d^2 U^x}{d\tau^2}$$

Solve the DE at boundary conditions $U^x = 0, \tau = 0$:

$$U^t(\tau) = c \frac{1}{2} (e^{(a/c)\tau} + e^{-(a/c)\tau})$$

$$U^x(\tau) = c \frac{1}{2} (e^{(a/c)\tau} - e^{-(a/c)\tau})$$

Since

$$\cosh \phi = \frac{1}{2} (e^\phi + e^{-\phi})$$

$$\sinh \phi = \frac{1}{2} (e^\phi - e^{-\phi})$$

$$\frac{d}{d\phi} \cosh \phi = \sinh \phi$$

$$\frac{d}{d\phi} \sinh \phi = \cosh \phi$$

Then:

$$U^t(\tau) = c \frac{1}{2} (e^{(a/c)\tau} + e^{-(a/c)\tau}) = c \cosh \left(\frac{a}{c} \tau \right)$$

$$U^x(\tau) = c \frac{1}{2} (e^{(a/c)\tau} - e^{-(a/c)\tau}) = c \sinh \left(\frac{a}{c} \tau \right)$$

So:

$$U^t = \frac{d(ct)}{d\tau} = c \cosh \left(\frac{a}{c} \tau \right)$$

$$U^x = \frac{dx}{d\tau} = c \sinh \left(\frac{a}{c} \tau \right)$$

It follows that:

$$A^t = \frac{dU^t}{d\tau} = a \sinh \left(\frac{a}{c} \tau \right)$$

$$A^x = \frac{dU^x}{d\tau} = a \cosh \left(\frac{a}{c} \tau \right)$$

And solving the DEs,

$$ct = \int U^t d\tau = \frac{c^2}{a} \sinh \left(\frac{a}{c} \tau \right)$$

$$x = \int U^x d\tau = \frac{c^2}{a} \cosh \left(\frac{a}{c} \tau \right)$$

So,

$$S \cdot S = - \left(\frac{c^2}{a} \right)^2$$

21.4.1 Rindler horizon

An accelerating observer will approach the light-cone asymptote. Above the asymptote, it is darkness to the observer as this light will not reach them. This occurs at the Rindler horizon, where the stationary object appears at a constant distance to the accelerating observer at:

$$L_0 = \sqrt{-S \cdot S} = \frac{c^2}{a}$$

The accelerating observer is outrunning the stationary source's light, so the distance remains constant. Further, time dilates to the point where the stationary object appears frozen in time $\tilde{t} = \gamma t = (0)t = 0$ to the accelerating observer.

21.4.2 Rindler coordinates

Bell's spaceship paradox. If two ships accelerate at same speed, distance will change. The ship behind has to accelerate faster to keep distance same due to length contraction. This can be viewed using angled position basis vectors as with Lorentz transformations. Generalizing this motion leads to Born Rigid Motion. Generalizing this to curvilinear coordinates, leading to Rindler coordinates. Rindler coordinates are bound by the Rindler horizons.

Using $D = \frac{c^2}{a}$ inside the brackets (to maintain a constant distance for the coordinate system) but $\tilde{x} = \frac{c^2}{a}$ for an 'initial' or chosen x, and $\tau = \tilde{t}$:

$$ct = \frac{c^2}{a} \sinh\left(\frac{a}{c^2} c\tau\right) = \tilde{x} \sinh\left(\frac{1}{D} c\tau\right)$$

$$x = \frac{c^2}{a} \cosh\left(\frac{a}{c^2} c\tau\right) = \tilde{x} \cosh\left(\frac{1}{D} c\tau\right)$$

Then if $D \rightarrow 2D$ then $a \rightarrow \frac{a}{2}$, etc. Moreover, as

$$ct = 2D \sinh\left(\frac{c\tau}{2D}\right) = \tilde{x} \sinh\left(\frac{c\tilde{t}}{D}\right)$$

$$x = 2D \cosh\left(\frac{c\tau}{2D}\right) = \tilde{x} \cosh\left(\frac{c\tilde{t}}{D}\right)$$

So,

$$\frac{c\tilde{t}}{D} = \frac{c\tau}{2D} \rightarrow 2\tilde{t} = \tau$$

$$\tilde{x} = 2D \rightarrow 2 = \frac{\tilde{x}}{D}$$

Combining, $\frac{\tilde{x}\tilde{t}}{D} = \tau$. This means along a line of simultaneity, proper time τ is different. It also means acceleration must be different between different positions to maintain constant distances:

$$\tau = \frac{\tilde{x}\tilde{t}}{D} \rightarrow \frac{d\tau}{d\tilde{t}} = \frac{\tilde{x}}{D}$$

which is combined with

$$ct = \tilde{x} \sinh\left(\frac{c\tilde{t}}{D}\right)$$

to get U^t, A^t, A^x and

$$\sqrt{-A \cdot A} = \frac{c^2}{\tilde{x}} = \frac{aD}{\tilde{x}}$$

So if $\tilde{x}$ increases (farther from origin) then a decreases.

Coordinate conversions. From above, one direction of the conversion is:

$$ct = \tilde{x} \sinh\left(\frac{1}{D} c\tau\right)$$

$$x = \tilde{x} \cosh\left(\frac{1}{D} c\tau\right)$$

The other direction through algebraic manipulation is:

$$\tilde{t} = \frac{c^2}{a} \operatorname{arctanh}\left(\frac{1}{D} c\tau\right)$$

Using $s^2 = (ct)^2 - x^2 = -\tilde{x}^2$, can get

$$\tilde{x} = \sqrt{x^2 - (ct)^2}$$

Then with $x = ct + x_0$ where x_0 is the initial position,

$$\frac{d\tilde{x}}{d\tilde{t}} = \frac{cx_0}{D}$$

and at $x_0 = D \rightarrow \frac{d\tilde{x}}{d\tilde{t}} = c$ $x_0 = 2D \rightarrow \frac{d\tilde{x}}{d\tilde{t}} = 2c$ and etc. This happens as it is measuring light using NON-LOCAL clocks.

21.4.3 Rindler coordinate basis changes

$$[e_t, e_x] J J^{-1} [U^t, U^x]'$$

where the Jacobian and its inverse are derived essentially from the multivariable chain rule.

$$\tilde{e}_t = -e_t + -e_x$$

$$\tilde{e}_x = -e_t + -e_x$$

$$\frac{\partial S}{\partial \tilde{ct}} = -\frac{\partial S}{\partial ct} + -\frac{\partial S}{\partial x}$$

$$\frac{\partial S}{\partial x} = -\frac{\partial S}{\partial ct} + -\frac{\partial S}{\partial x}$$

$$\frac{\partial S}{\partial \tilde{ct}} = \frac{\partial ct}{\partial \tilde{ct}} \frac{\partial S}{\partial ct} + \frac{\partial x}{\partial \tilde{ct}} \frac{\partial S}{\partial x}$$

$$\frac{\partial S}{\partial \tilde{x}} = \frac{\partial ct}{\partial \tilde{x}} \frac{\partial S}{\partial ct} + \frac{\partial x}{\partial \tilde{x}} \frac{\partial S}{\partial x}$$

In matrix notation,

$$\begin{bmatrix} \frac{\partial S}{\partial ct}, \frac{\partial S}{\partial \tilde{x}} \end{bmatrix} = \begin{bmatrix} \frac{\partial S}{\partial ct}, \frac{\partial S}{\partial x} \end{bmatrix} \begin{bmatrix} \frac{\partial ct}{\partial ct}, \frac{\partial ct}{\partial \tilde{x}} \\ \frac{\partial x}{\partial ct}, \frac{\partial x}{\partial \tilde{x}} \end{bmatrix}$$

Using

$$\begin{aligned} ct &= \tilde{x} \sinh\left(\frac{a}{c^2} c\tilde{t}\right) \\ x &= \tilde{x} \cosh\left(\frac{a}{c^2} c\tilde{t}\right) \end{aligned}$$

Calculate $\frac{\partial S}{\partial ct}$:

$$\begin{aligned} \frac{\partial S}{\partial ct} &= \left(\frac{a}{c^2} \tilde{x} \cosh\left(\frac{a}{c^2} c\tilde{t}\right)\right) \frac{\partial S}{\partial ct} + \left(\frac{a}{c^2} \tilde{x} \sinh\left(\frac{a}{c^2} c\tilde{t}\right)\right) \frac{\partial S}{\partial x} \\ \frac{\partial S}{\partial ct} &= \frac{a}{c^2} x \frac{\partial S}{\partial ct} + \frac{a}{c^2} ct \frac{\partial S}{\partial x} \end{aligned}$$

Calculate $\frac{\partial S}{\partial \tilde{x}}$:

$$\begin{aligned} \frac{\partial S}{\partial \tilde{x}} &= \left(\sinh\left(\frac{a}{c^2} c\tilde{t}\right)\right) \frac{\partial S}{\partial ct} + \left(\cosh\left(\frac{a}{c^2} c\tilde{t}\right)\right) \frac{\partial S}{\partial x} \\ \frac{\partial S}{\partial \tilde{x}} &= \frac{ct}{\tilde{x}} \frac{\partial S}{\partial ct} + \frac{x}{\tilde{x}} \frac{\partial S}{\partial x} \end{aligned}$$

So the Jacobian is:

$$J = \begin{bmatrix} \frac{a}{c^2} x & \frac{ct}{\tilde{x}} \\ \frac{a}{c^2} ct & \frac{x}{\tilde{x}} \end{bmatrix}$$

Using

$$\begin{aligned} c\tilde{t} &= \frac{c^2}{a} \operatorname{arctanh}\left(\frac{ct}{x}\right) \\ \tilde{x} &= \sqrt{x^2 - (ct)^2} \\ J^{-1} &= \begin{bmatrix} \frac{c^2 x}{ax^2} & -\frac{c^2 ct}{ax^2} \\ -\frac{ct}{\tilde{x}} & \frac{x}{\tilde{x}} \end{bmatrix} \end{aligned}$$

J, J^{-1} depends on position and time, so they're not constant in Rindler coordinates.

21.4.4 Twin paradox

Twin paradox. (1) If one frame (earth or spaceship) is inertial, and they meet back up in the future, then the other must be non-inertial. So can't say they're the same. (2) In spacetime, the triangle inequality is reversed: the straight path takes the LONGEST spacetime, rather than the shortest distance in other standard systems.

21.4.5 Sum of spacetime paths and Rindler metric

To sum complicated spacetime paths,

$$\sum c\tau = \int \left\| \frac{dS}{d\lambda} \right\| d\lambda$$

Note: Minkowski length $\left\| \frac{dS}{d\tau} \right\| = c$ is different. This is for proper time of the inertial reference frame only. Summing a complex spacetime path will involve non-inertial reference frames.

To sum spacetime vectors, take standard bases $ct(\lambda), x(\lambda)$ and using the multivariate chain rule, calculate $\frac{dS}{d\lambda}$. From this, calculate:

$$c\tau = \int \left\| \frac{dS}{d\lambda} \right\| d\lambda = \int \sqrt{\frac{dS}{d\lambda} \cdot \frac{dS}{d\lambda}} d\lambda = \text{Constant}(\lambda_f - \lambda_i)$$

This can be calculated in standard basis or Rindler basis $(\tilde{ct}, \tilde{x})$. Use the Rindler metric tensor to change bases:

$$J = \begin{bmatrix} \left(\frac{\tilde{x}}{\tilde{D}}\right)^2 & 0 \\ 0 & -1 \end{bmatrix}$$

which is calculated from the Rindler basis vectors $\tilde{e}_t(e_t, e_x)$ and $\tilde{e}_x(e_t, e_x)$.

21.4.6 Covariant derivatives in Rindler coordinates

In inertial reference frames, basis vectors are constant so their derivatives are zero. In Rindler coordinates, they are changing so more complicated. In addition to using the chain rule as usual, need to calculate these as well. This results in covariant derivatives (covariant derivatives are derivatives which all coordinate systems agree on - they consider changes not only of vector components, but the bases as well). To calculate the basis vector changes, use Christoffel symbols (where Γ_{ab}^c , c indicates which basis vector Γ is associated with, a is the starting basis and b is the basis to change to; with 3 indices with 2 choices each (time and position), there are $2^3 = 8$ Christoffel symbols). To shorten expressions, use Einstein notation, and note some Christoffel symbols are the same as others by symmetry.

$$\begin{aligned} \frac{\partial \tilde{e}_t}{\partial \tilde{ct}} &= 0\tilde{e}_t + \frac{a}{c^2} \tilde{x}\tilde{e}_x = \tilde{\Gamma}_{tt}^t \tilde{e}_t + \tilde{\Gamma}_{tt}^x \tilde{e}_x \\ \frac{\partial \tilde{e}_t}{\partial \tilde{x}} &= \frac{1}{\tilde{x}} \tilde{e}_t + 0\tilde{e}_x = \tilde{\Gamma}_{tx}^t \tilde{e}_t + \tilde{\Gamma}_{tx}^x \tilde{e}_x \\ \frac{\partial \tilde{e}_x}{\partial \tilde{ct}} &= \frac{1}{\tilde{x}} \tilde{e}_t + 0\tilde{e}_x = \tilde{\Gamma}_{xt}^t \tilde{e}_t + \tilde{\Gamma}_{xt}^x \tilde{e}_x \\ \frac{\partial \tilde{e}_x}{\partial \tilde{x}} &= 0\tilde{e}_t + 0\tilde{e}_x = \tilde{\Gamma}_{xx}^t \tilde{e}_t + \tilde{\Gamma}_{xx}^x \tilde{e}_x \end{aligned}$$

To calculate the 'basis-less' 4-vectors like $\vec{U}, \vec{A}$, i.e. the covariant derivatives, use the product rule to obtain the change in vector component and change in basis component. Then substitute in the above basis vector changes.

Below calculation results involved calculating basis vector derivatives and using the chain rule as usual:

$$\vec{U} = \frac{dS}{d\tau} = \frac{d\tilde{x}}{d\tau} \tilde{e}_x + \frac{dct}{d\tau} \tilde{e}_t$$

$$\vec{A} = \frac{dU}{d\tau} = \left(\frac{d^2 ct}{d\tau^2} + 2 \frac{dct}{d\tau} \frac{d\tilde{x}}{d\tau} \frac{1}{\tilde{x}} \right) e_t + \left(\frac{d^2 \tilde{x}}{d\tau^2} + \left(\frac{dct}{d\tau} \right)^2 \left(\frac{a}{c^2} \right)^2 \tilde{x} \right) e_x$$

21.4.7 Geodesics

In inertial frames, straight lines are geodesics but not in Rindler coordinates of non-inertial frames. To compute geodesic paths, solve for the geodesic equation:

$$\frac{d^2 S}{d\lambda^2} = 0$$

Inertial frames. Linear worldlines as geodesics.

$$S = ct e_t + x e_x$$

Since basis vectors aren't changing,

$$\frac{de_t}{d\lambda} = \frac{de_x}{d\lambda} = 0$$

When S is differentiated twice,

$$\frac{d^2 S}{d\lambda^2} = \frac{d^2 ct}{d\lambda^2} e_t + \frac{d^2 x}{d\lambda^2} e_x = 0$$

So set

$$\frac{d^2 ct}{d\lambda^2} = 0$$

$$\frac{d^2 x}{d\lambda^2} = 0$$

Solve the DE to get a linear system:

$$ct(\lambda) = d^t \lambda + k^t$$

$$x(\lambda) = d^x \lambda + k^x$$

where the d -component is the initial direction vector and k -component is initial position. Choose values for these and substitute back into $S = ct e_t + x e_x$ to generate inertial geodesics, which are straight lines.

Non-inertial frames. In Rindler coordinates,

$$ct = \tilde{x} \sinh\left(\frac{c\tilde{t}}{D}\right)$$

$$x = \tilde{x} \cosh\left(\frac{c\tilde{t}}{D}\right)$$

If $x = x_0$,

$$\tilde{x} = x_0 \frac{1}{\cosh\left(\frac{c\tilde{t}}{D}\right)} = x_0 \operatorname{sech}\left(\frac{c\tilde{t}}{D}\right)$$

$$ct = \frac{x_0}{\cosh\left(\frac{c\tilde{t}}{D}\right)} \sinh\left(\frac{c\tilde{t}}{D}\right) = x_0 \tanh\left(\frac{c\tilde{t}}{D}\right)$$

Since $-1 < \tanh(x) < 1$, $ct \in (-x_0, x_0)$ which is the Rindler horizon.

In Rindler coordinates, light has exponential worldlines. Light is $x = ct + x_0$, so using

$$ct = \tilde{x} \sinh\left(\frac{c\tilde{t}}{D}\right)$$

$$x = \tilde{x} \cosh\left(\frac{c\tilde{t}}{D}\right)$$

and after algebra,

$$\tilde{x} = x_0 e^{c\tilde{t}} D$$

Geodesics in Rindler coordinates.

$$\frac{d^2 S}{d\lambda^2} = 0\tilde{e}_t + 0\tilde{e}_x$$

$$= \left(\frac{d^2 c\tilde{t}}{d\tau^2} + 2 \frac{dc\tilde{t}}{d\tau} \frac{d\tilde{x}}{d\tau} \frac{1}{\tilde{x}} \right) \tilde{e}_t + \left(\frac{d^2 \tilde{x}}{d\tau^2} + \left(\frac{dc\tilde{t}}{d\tau} \right)^2 \left(\frac{a}{c^2} \right)^2 \tilde{x} \right) \tilde{e}_x$$

Setting parenthetical terms to zero:

$$\frac{d^2 c\tilde{t}}{d\tau^2} + 2 \frac{dc\tilde{t}}{d\tau} \frac{d\tilde{x}}{d\tau} \frac{1}{\tilde{x}} = 0$$

$$\frac{d^2 \tilde{x}}{d\tau^2} + \left(\frac{dc\tilde{t}}{d\tau} \right)^2 \left(\frac{a}{c^2} \right)^2 \tilde{x} = 0$$

This is a non-linear coupled DE. After lots of math, the solution is:

$$\tilde{x}(c\tilde{t}) = \frac{1}{C_1 \cosh\left(\frac{c\tilde{t}}{D}\right) + C_2 \sinh\left(\frac{c\tilde{t}}{D}\right)}$$

Generate different geodesics depending on choices of C_1, C_2 . If $C_1 = \frac{1}{x_0}, C_2 = 0$ then $\tilde{x}(c\tilde{t}) = x_0 \operatorname{sech}\left(\frac{c\tilde{t}}{D}\right)$ which is an inertial worldline. If $C_1 = \frac{1}{x_0} = C_2$, then $\tilde{x}(c\tilde{t}) = x_0 e^{c\tilde{t}} D$ which is exponential worldline of light.

21.5 Waves and Doppler effect

Vector indicates direction and size. Covectors indicate direction and density.

Wavelength is λ , wavenumber is $\frac{1}{\lambda}$, period is T , frequency is $f = \frac{1}{T}$, wave covector is κ , velocity is $v = \frac{f}{\kappa_x}$.

Galilean relativity Doppler effect. The Galilean transform is

$$\tilde{e}_t = 1e_t + ve_x$$

$$\tilde{e}_x = 0e_t + 1e_x$$

So

$$\tilde{f} = \kappa(\tilde{e}_t) = \kappa(1e_t + ve_x)$$

using linearity of covectors,

$$= 1\kappa(e_t) + v\kappa(e_x) = f + v\kappa_x$$

Analogously,

$$\tilde{\kappa}_x = \kappa(\tilde{e}_x) = \kappa(0e_t + 1e_x) = 0f + 1\kappa_x$$

So the Galilean Doppler shift is

$$\tilde{f} = f + v\kappa_x$$

combined with $v_w = \frac{f}{\kappa_x}$,

$$\tilde{f} = f \left(\frac{v_w + v}{v_w} \right)$$

Special relativity Doppler effect. The Lorentz transform is:

$$\tilde{e}_t = \gamma(e_t + \beta e_x)$$

$$\tilde{e}_x = \gamma(\beta e_t + e_x)$$

where $\beta = \frac{v}{c}$. Note for light:

$$v = c = \frac{f}{\kappa_x}, \kappa_x = \kappa_t = \frac{f}{c}$$

So:

$$\tilde{\kappa}_t = \kappa(\tilde{e}_t) = \gamma(\kappa_t + \beta\kappa_x)$$

$$\tilde{\kappa}_x = \kappa(\tilde{e}_x) = \gamma(\beta\kappa_t + \kappa_x)$$

So:

$$\tilde{\kappa}_t = \kappa_t \gamma(1 + \beta) = \kappa_t \sqrt{\frac{1 + \beta}{1 - \beta}}$$

$$\tilde{f} = f \sqrt{\frac{1 + \beta}{1 - \beta}}$$

Accelerating reference frames. In Rindler coordinates.

$$\tilde{\kappa}_t = \kappa_t e^{c\tilde{t}/D} \in (0, \kappa_t, \infty)$$

21.6 General Relativity

21.6.1 Equivalence principle

In small regions of spacetime, local gravitation is equivalent to acceleration.

In Newtonian physics, $F = ma$ and $F = m\frac{GM}{r^2}$ and the masses are equivalent, so $a = \frac{GM}{r^2}$. This is different from electric forces like $F = \frac{kQq}{r^2}$. So gravity is different from other forces.

Special relativity which assumes flat spacetime, can't explain gravity. But curved spacetime as in GR can.

Proper acceleration α is a tensor, as it is invariant and everyone will agree on it. In an accelerating frame where the force is generated by a rocket or magnetic or other external force, $\alpha > 0$. This is equivalent not to gravitational free-fall ($\alpha = 0$) but when there is a force like the ground opposing gravity.

In Newtonian physics, an object at rest on the ground has $\sum F = F_{up} - g = 0$ but in GR, $\sum F = F_{up} > 0$ as gravity is not a force but rather curved spacetime. The object naturally moves along the 4D geodesic which is gravity unless there is a force that opposes it.

21.6.2 Manifolds

Small spacetime regions are flat as in special relativity while larger regions are curved as in general relativity. So spacetime is a manifold, a curved space that is locally flat. As there is a distance metric on the manifold, it is a Riemannian manifold.

$$g(u, v) = u \cdot v$$

$$g(e_i, e_j) = e_i \cdot e_j = g_{ij}$$

$$g(r, r) = \|r\|^2$$

In relativity:

$$g(v, v) = \|v\|^2 > 0, \text{Time-like so get proper time}$$

$$g(v, v) = \|v\|^2 = 0, \text{Light-like}$$

$$g(v, v) = \|v\|^2 < 0, \text{Space-like so get proper length}$$

Since in relativity distances can be negative, it is a pseudo-Riemannian manifold. The length of a path in a manifold is still

$$L = \int \left\| \frac{dS}{d\lambda} \right\| d\lambda$$

and to calculate, need tangent vector $\frac{d}{d\lambda}$ which exists in a different $T_p M$ at each point in space. While the tangent vectors will partly exist outside the manifold

(so can be viewed fully extrinsically), in intrinsic geometry can only see part of it - where they exist in the manifold dimensions. This is sufficient.

Between different $T_p M$ s at different manifold points, use the covariant derivative (which is a vector in vector field) to compute the difference. In flat spaces, the covariant derivative is

$$\frac{dV}{d\lambda} = \frac{dx^\mu}{d\lambda} \left(\frac{\partial V^\sigma}{\partial x^\mu} + V^\nu \Gamma_{\mu\nu}^\sigma \right) e_\sigma$$

Notation:

$$\frac{dV}{d\lambda} = \nabla_{\frac{d}{d\lambda}} V = \nabla_W V$$

And:

$$\frac{dV}{d\lambda} = \frac{dx^\mu}{d\lambda} \nabla_{e_\mu} V = \frac{dx^\mu}{d\lambda} (V_{;\mu}^\sigma + V^\nu \Gamma_{\mu\nu}^\sigma) e_\sigma = \frac{dx^\mu}{d\lambda} V_{;\mu}^\sigma e_\sigma = \frac{dx^\mu}{d\lambda} \nabla_\mu V^\sigma e_\sigma$$

Use parallel transport between different $T_p M$ s at different manifold points:

$$\nabla_{\frac{d}{d\lambda}} V = 0$$

but parallel transport alone is insufficient to completely define a path. So can get different Γ and therefore $T_p M$ and covariant derivatives. To get unique Γ values, add two conditions. First, ensure length and angles of tangent vectors remain constant during parallel transport (metric compatibility), which means the product rule applies:

$$\nabla_W (V_1 \cdot V_2) = \nabla_W V_1 \cdot V_2 + V_1 \cdot \nabla_W V_2$$

so when combined with parallel transport $\nabla_W V_1 = \nabla_W V_2 = 0$, $\nabla_W (V_1 \cdot V_2) = 0$. The second condition is that the vectors are torsion-free. This means some Christoffel symbols will be symmetric. The two conditions together give the Levi-Civita connection, which uniquely connects $T_p M$ s.

The geodesic equation is essentially a no-acceleration condition:

$$\frac{d^2 S}{d\lambda^2} = 0$$

alternatively:

$$\frac{d^2 x^\sigma}{d\lambda^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0$$

21.6.3 Curvature

Classical differential geometry. Gaussian curvature K describes only 2D curvature. $K > 0$ is positive curvature (two crossing parabolas), $K = 0$ is neutral curvature (at least one straight line), and $K < 0$ is negative curvature (saddle). To generalize curvature concept to higher dimensions, use Riemann curvature

tensor.

Riemann curvature. Conceptualize by (1) holonomy or (2) geodesic deviation.

(1) Holonomy. Change in vector $\vec{T}$ after parallel transporting in a loop. Can create parallelogram loop (generalization of basis vectors in 2D space) with $\vec{V}, \vec{W}$. If $R(V, W)T \neq 0$ then there is curvature. The R is the difference between the starting $\vec{T}$ and end $\vec{T}$. Since R is linear,

$$R(V, W)T = R(V^\mu e_\mu W^\nu e_\nu)T^\sigma e_\sigma = V^\mu W^\nu T^\sigma R(e_\mu, e_\nu)e_\sigma = V^\mu W^\nu T^\sigma R^\rho_{\sigma\mu\nu} e_\rho$$

where e_ρ is the direction of the difference and $R^\rho_{\sigma\mu\nu}$ is the magnitude of the difference. To calculate R , need Γ s.

(2) Geodesic deviation. Change in spread between geodesics with common origin. Along any geodesic, $\nabla_V V = 0$, and after algebraic manipulation, $\nabla_V \nabla_V T = -R(T, V)V$.

Ricci curvature tensor. Like a summary or matrix trace of the Riemann tensor. So tracks volume changes along geodesics. Detects changes in volume but not shape.

$$R_{\alpha\beta} = R^\rho_{\alpha\rho\beta} = R^0_{\alpha 0\beta} + R^1_{\alpha 1\beta} + R^2_{\alpha 2\beta} + R^3_{\alpha 3\beta}$$

where each of the four terms on the right is a 4x4 matrix, so the sum is another 4x4 tensor, the Ricci curvature tensor.

Ricci scalar. $Ric(V, W)$ and is a scalar. $R = R(V, V) = V^\alpha V^\beta R_{\alpha\beta}$, which is Ricci curvature. Compute the Ricci scalar from the Ricci tensor: $R = g^{\alpha\beta} R_{\alpha\beta} = R^\alpha_\alpha$.

To calculate the Ricci scalar, compute (1) the metric tensor, (2) Christoffel symbols using the Levi-Civita connection, (3) Riemann tensor, (4) Ricci tensor, (5) Ricci scalar.

21.6.4 Newton-Cartan theory

Gravity explained by spacetime curvature. Positive divergence $\nabla \cdot V > 0$ means vectors are leaving (a vector source), and negative divergence (convergence) $\nabla \cdot V < 0$ means vectors are entering (a sink). See gravitation as a vector field $\nabla \cdot g = -4\pi G\rho$ where ρ is the mass density. If ϕ is the gravitational potential as a scalar, then:

$$\begin{aligned} g &= \frac{GM}{r^2} = -\nabla\phi \\ \nabla \cdot (-\nabla\phi) &= -4\pi G\rho \\ \nabla^2\phi &= 4\pi G\rho \end{aligned}$$

which is the Poisson equation which detects 'bumps' (sinks or sources of vectors) so divergence or convergence. Since $\rho \geq 0$, $\nabla^2 \phi \geq 0$ so the vectors point inward (convergence) and it's a well.

$$F = ma = mg$$

$$a = g$$

$$\frac{d^2 x}{dt^2} = -\nabla \phi$$

$$\frac{d^2 x^i}{dt^2} = -\frac{\partial \phi}{\partial x^i}$$

$$\frac{d^2 x^i}{dt^2} + \frac{\partial \phi}{\partial x^i} = 0$$

which is similar to the geodesic equation:

$$\frac{d^2 x^\sigma}{dt^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} = 0$$

To make it match, let $\lambda \rightarrow t$ as in Galilean relativity, where time is not relative. Further, let $x^0 = t$. So for time:

$$\frac{d^2 x^0}{dt^2} + \Gamma_{\mu\nu}^0 \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} = 0$$

$$0 + \Gamma_{\mu\nu}^0 \frac{dx^\mu}{dt} \frac{dx^\nu}{dt} = 0$$

So it must be that $\Gamma_{\mu\nu}^0 = 0$ (time components are unchanging). Now look at spatial components. $\Gamma_{\mu\nu}^i \neq 0$ possible (spatial components changing), but since $\frac{dx^i}{dt} = 0$ (it doesn't appear in the geodesic equation so must zero), the only Christoffel symbol that is unchanging must be $\Gamma_{00}^i \neq 0$ and others must be zero. So in the geodesic equation for space:

$$\frac{d^2 x^0}{dt^2} + \Gamma_{00}^i \frac{dx^0}{dt} \frac{dx^0}{dt} = 0$$

$$\frac{d^2 x^0}{dt^2} + \Gamma_{00}^i \frac{dt}{dt} \frac{dt}{dt} = 0$$

$$\frac{d^2 x^0}{dt^2} + \Gamma_{00}^i = 0$$

Comparing with the geodesic equation, it must be that

$$\Gamma_{00}^i = \frac{\partial \phi}{\partial x^i} = \partial_i \phi$$

and the geodesic equation for Newtonian gravity is therefore

$$\vec{a} + \nabla \phi = 0$$

The Riemann curvature of Newtonian gravity is

$$R_{\sigma\mu\nu}^{\rho} = \partial_{\mu}(\Gamma_{\nu\sigma}^{\rho}) - \partial_{\nu}(\Gamma_{\mu\sigma}^{\rho}) + \Gamma_{\nu\sigma}^{\alpha}\Gamma_{\mu\alpha}^{\rho} - \Gamma_{\mu\sigma}^{\beta}\Gamma_{\nu\beta}^{\rho}$$

$$R_{0\mu\nu}^i = \partial_{\mu}(\Gamma_{\nu 0}^i) - \partial_{\nu}(\Gamma_{\mu 0}^i)$$

which when calculated for each of the spatial dimensions i in 3D space gives a 3x3 tensor $= \partial_j \partial_i \phi$, where the geodesic deviation leads to the r -direction being elongated and the other dimensions being compressed. This is the tidal force, $\Phi_{ij} = \partial_j \partial_i \phi$. The general Ricci tensor is

$$R_{\alpha\beta} = R_{\alpha\mu\beta}^{\mu}$$

and the corresponding Ricci tensor for Newtonian gravity will be

$$R_{00} = R_{0\mu 0}^{\mu} = R_{000}^0 + R_{010}^1 + R_{020}^2 + R_{030}^3$$

$$= 0 + \partial_1 \partial_1 \phi + \partial_2 \partial_2 \phi + \partial_3 \partial_3 \phi$$

$$= \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \right) = \nabla^2 \phi \geq 0$$

$$R_{00} = R_{tt} = \nabla^2 \phi = 4\pi G \rho \geq 0$$

Since $R > 0$, there is convergence in the time direction, so space is shrinking.

21.6.5 Energy-momentum tensor

Newtown-Cartan's theory uses ρ which is not invariant. To make it invariant, use the energy-momentum tensor.

Number-flux vectors

$$\vec{N} = N = [N^t, N^x, N^y, N^z]'$$

where the component indicates the number of objects in a unit spacetime box where that component's dimension is held constant:

$$N^t = \frac{Number}{\Delta x \Delta y \Delta z}$$

$$N^x = \frac{Number}{\Delta t \Delta y \Delta z}$$

$$N^y = \frac{Number}{\Delta x \Delta t \Delta z}$$

$$N^z = \frac{Number}{\Delta x \Delta y \Delta t}$$

With $N^t = n$ as the rest mass number density, above can be rewritten as

$$N^t = \frac{\Delta t}{\Delta t} \frac{Number}{\Delta x \Delta y \Delta z} = U^t n$$

$$N^x = \frac{\Delta x}{\Delta t} \frac{\text{Number}}{\Delta x \Delta y \Delta z} = U^x n$$

$$N^y = \frac{\Delta y}{\Delta t} \frac{\text{Number}}{\Delta x \Delta y \Delta z} = U^y n$$

$$N^z = \frac{\Delta z}{\Delta t} \frac{\text{Number}}{\Delta x \Delta y \Delta z} = U^z n$$

Now,

$$\vec{N} = n[U^t, U^x, U^y, U^z]'$$

Time-flux changes in Galilean relativity and time- and space-flux change in relativity. Galilean relativity can be handled within the above framework $\vec{N} = n\vec{U}$. In special relativity,

$$S = [ct, x, y, z]'$$

$$N = n[c, U^x, U^y, U^z]\gamma$$

Divergence of spacetime number-flux should be zero as the divergence is zero:

$$\frac{\partial N^\mu}{\partial t} = 0 \rightarrow \partial_\mu N^\mu = 0$$

which is the 'Continuity Equation'.

Momentum. $T^{\alpha\beta}$ is the flux of α -momentum through a unit spacetime box of β .

$$T^{tt} = \frac{P^t}{\Delta x \Delta y \Delta z}$$

$$T^{it} = \frac{P^i}{\Delta x \Delta y \Delta z}$$

which comprises the outer product of $\vec{P}$ (column vector) and $\vec{N}$ (row vector):

$$T = \begin{bmatrix} T^{tt} & T^{tx} & T^{ty} & T^{tz} \\ T^{xt} & T^{xx} & T^{xy} & T^{xz} \\ T^{yt} & T^{yx} & T^{yy} & T^{yz} \\ T^{zt} & T^{zx} & T^{zy} & T^{zz} \end{bmatrix}$$

where T^{tt} is the energy density, the first row is energy flux, the first column is spatial momentum density, and the remainder is spatial momentum fluxes in various directions relative to various directions (and within this remainder, the diagonal elements are pressure $T^{ii} = \frac{P^i}{\Delta t \Delta y \Delta z} = \frac{F^x}{\Delta y \Delta z}$ and the remaining reminder elements are shears).

Since $T^{\alpha\beta} = T(e_\alpha, e_\beta)$ then $T(V, W)$ is the momentum in the v-direction inside a unit spacetime box perpendicular to w-direction.

Special relativity. $\Gamma = 0$ as basis vectors don't change in flat spacetime. Conservation of energy: energy flux (first row of energy-momentum tensor) divergence is zero:

$$\frac{\partial T^{ti}}{\partial i} = 0$$

$$\frac{\partial \rho_{energy}}{\partial ct} + \nabla \cdot T^{ti} = 0$$

and the remaining rows of the tensor correspond to momentum conservation by setting their divergences to zero:

$$\frac{\partial T^{ti}}{\partial i} = 0$$

The energy-momentum conservation equations can be compacted to:

$$\partial_\nu T^{\mu\nu} = 0$$

or $\nabla \cdot T = 0$.

General relativity. Γ not necessarily zero.

$$\begin{aligned} \nabla_{e_\sigma} T &= \nabla_{e_\sigma} (T^{\mu\nu} e_\mu \otimes e_\nu) \\ &= \nabla_{e_\sigma} (T^{\mu\nu}) e_\mu \otimes e_\nu + T^{\mu\nu} \nabla_{e_\sigma} (e_\mu) \otimes e_\nu + T^{\mu\nu} e_\mu \otimes \nabla_{e_\sigma} e_\nu \\ &= (\partial_\sigma T^{\mu\nu} + T^{\alpha\nu} \Gamma_{\alpha\sigma}^\mu + T^{\mu\alpha} \Gamma_{\alpha\sigma}^\nu) e_\mu \otimes e_\nu \\ &= \nabla_\sigma T^{\mu\nu} e_\mu \otimes e_\nu \end{aligned}$$

So energy-momentum conservation (locally) is

$$\nabla_\nu T^{\mu\nu} = T^{\mu\nu}_{;\nu} = 0$$

21.6.6 Einstein field equations

Einstein tensor is proportionate to mass-energy-momentum tensor:

$$G^{\mu\nu} = \kappa T^{\mu\nu}$$

$$R^{\mu\nu} - \frac{1}{2} R g^{\mu\nu} = \kappa T^{\mu\nu}$$

where the Ricci tensor is like the 'first derivative' of the Riemann tensor and the Ricci scalar is like the 'second derivative'. Can add the cosmological constant:

$$R^{\mu\nu} - \frac{1}{2} R g^{\mu\nu} - \Lambda g^{\mu\nu} = \kappa T^{\mu\nu}$$

So curvature of spacetime is proportionate to mass-energy-momentum. Taking the divergence of both sides,

$$\nabla_\mu (G^{\mu\nu}) = 0$$

$$\nabla_\mu(\kappa T^{\mu\nu}) = 0$$

So along a geodesic in curved spacetime, mass-energy-momentum are conserved.

Einstein tensor. Derived from the contracted Bianchi identity, which uses Riemann tensor symmetries. The Ricci curvature tensor indicates how much spacetime is curved, and the Ricci scalar about the magnitude of the change.

Cosmological constant and $g^{\mu\nu}$. These are derived from $\nabla G^{\mu\nu} = 0$ and the covariant derivative of metric tensors being zero, $\nabla g = 0$ (which requires metric compatibility). The cosmological constant indicates changes in the size of the universe.

Constant κ is derived from assuming low velocities and weak gravitation, essentially Poisson's equation $\nabla^2\phi = 4\pi G\rho$.

22 Variational Calculus

Calculus of functionals (function of functions). Example. Finding the FUNCTION that minimizes the path f between two points, which will be a stationary point of the functional.

22.1 Catenary rope and Euler's derivation

Gravitational potential energy is:

$$E_p \propto m, g, h \rightarrow E_p = mgh$$

For a catenary rope,

$$E_p = (m_1h_1 + m_2h_2 + \dots + m_nh_n)g$$

but only one curve will give the lowest potential energy. Each small section of the catenary rope can be derived using Pythagoras:

$$\begin{aligned} ds^2 &= dx^2 + dy^2 \\ \frac{ds^2}{dx^2} &= 1 + \frac{dy^2}{dx^2} \\ \frac{ds}{dx} &= \sqrt{1 + \frac{dy^2}{dx^2}} \\ ds &= \sqrt{1 + \frac{dy^2}{dx^2}} dx \\ ds &= \sqrt{1 + y'(x)^2} dx \end{aligned}$$

Rope mass M with length L leads to rope density $d = \frac{M}{L}$. Each rope section has mass $m = \frac{M}{L}s$, and shrinking this infinitesimally, $dm = \frac{M}{L}ds$. So,

$$dm = \frac{M}{L}ds$$

$$ds = dm \frac{L}{M}$$

Substituting into the above,

$$dm = \frac{M}{L} \sqrt{1 + y'(x)^2} dx$$

Substituting into $E_p = mgh$,

$$E_p = \int_0^L \frac{M}{L} \sqrt{1 + y'(x)^2} gy(x) dx$$

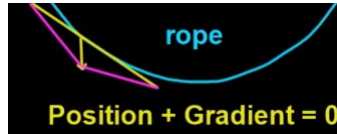
Let $c = \frac{M}{L}g = 1$ be a collection of constants that is normalized. Then,

$$E_p = \int_0^L \sqrt{1 + y'(x)^2} y(x) dx$$

Generalizing,

$$S = \int f(x, y(x), y'(x)) dx$$

which is a functional (note: there could be an explicit dependence on just x , though this doesn't happen with E_p). Trying to find $y(x)$ that minimizes S . A functional maps a function to another function instead of a variation to a function. In equilibrium,



which is the intuition for the Euler-Lagrange equation.

$$\frac{df}{dx} \delta x = \Delta y$$

which can be re-expressed as

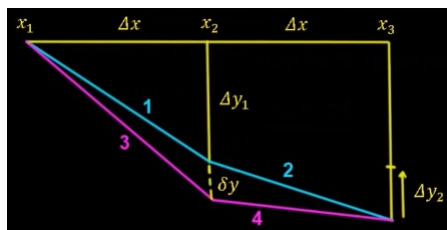
$$\frac{df}{dy} \delta y = \Delta x$$

So, for a general system,

$$E_p = \sqrt{1 + y'(x)^2} y(x)$$

becomes

$$\delta E_p = \delta S = \frac{\partial f}{\partial y(x)} \Delta y$$



Segment gradients:

$$1. \quad y'(x) \Big|_{x_1} = \frac{\Delta y_1}{\Delta x}$$

$$2. \quad y'(x) \Big|_{x_2} = \frac{\Delta y_2}{\Delta x}$$

$$3. \quad y'(x) \Big|_{x_1} = \frac{\Delta y_1 + \delta y}{\Delta x}$$

$$4. \quad y'(x) \Big|_{x_1} = \frac{\Delta y_2 - \delta y}{\Delta x}$$

Subtract (3)-(1): $\frac{\delta y}{\Delta x}$.

Subtract (4)-(2): $-\frac{\delta y}{\Delta x}$.

Use both of these differences. Note that $x_2 = x_1 + \Delta x$.

$$\begin{aligned} \delta S_2 &= \frac{\partial f}{\partial y'} \Big|_{x_1} \Delta y'(x) + \frac{\partial f}{\partial y'} \Big|_{x_2} \Delta y'(x) \\ &= \frac{\partial f}{\partial y'} \Big|_{x_1} \frac{\delta y}{\Delta x} - \frac{\partial f}{\partial y'} \Big|_{x_2} \frac{\delta y}{\Delta x} \\ &= \frac{\delta y}{\Delta x} \left(\frac{\partial f}{\partial y'} \Big|_{x_1} - \frac{\partial f}{\partial y'} \Big|_{x_2} \right) \\ &= \frac{\left(\frac{\partial f(x_1 + \Delta x)}{\partial y'} - \frac{\partial f(x_1)}{\partial y'} \right) \delta y}{\Delta x} \end{aligned}$$

Take the limit:

$$\delta S_2 = \lim_{\Delta x \rightarrow 0} \frac{\left(\frac{\partial f(x_1 + \Delta x)}{\partial y'} - \frac{\partial f(x_1)}{\partial y'} \right) \delta y}{\Delta x} = -\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \delta y$$

Combine:

$$\begin{aligned} \delta S_1 + \delta S_2 &= \frac{\partial f}{\partial y} \delta y - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \delta y \\ &= \delta y \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right) \end{aligned}$$

As $\delta y \rightarrow 0$, but $y \neq 0$ as the rope has to move vertically. So,

$$\boxed{\underbrace{\frac{\partial f}{\partial y}}_{\text{position}} - \underbrace{\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right)}_{\text{gradient}} = 0}$$

which is the **Euler-Lagrange equation**. This can be simplified using the Beltrami identity.

22.2 Lagrange's derivation

Instead of $y(x)$, use

$$Y(x) = y(x) + \epsilon \eta(x)$$

$$Y'(x) = y'(x) + \epsilon \eta'(x)$$

where $\epsilon \eta(x)$ is the 'variation' and $\eta(x)$ is 0 at endpoints. As $\epsilon \rightarrow 0$, $Y(x) \rightarrow y(x)$. Since

$$I = \int f(x, y, y') dx$$

Then

$$I + \delta I = \int f(x, Y, Y') dx$$

and interested in

$$\frac{dI}{d\epsilon} = \frac{d}{d\epsilon} \left(\int f(x, Y, Y') dx \right) = 0$$

x is not a function of ϵ but Y and Y' are. So,

$$I = \int f(x, \underbrace{y(x) + \epsilon \eta(x)}_Y, \underbrace{y'(x) + \epsilon \eta'(x)}_{Y'}) dx$$

Using the chain rule,

$$\begin{aligned} \frac{dI}{d\epsilon} &= \int \left(\frac{\partial f}{\partial x} \frac{dx}{d\epsilon} + \frac{\partial f}{\partial Y} \frac{dY}{d\epsilon} + \frac{\partial f}{\partial Y'} \frac{dY'}{d\epsilon} \right) dx \\ &= \int \left(\frac{\partial f}{\partial x} \cdot 0 + \frac{\partial f}{\partial Y} \frac{dY}{d\epsilon} + \frac{\partial f}{\partial Y'} \frac{dY'}{d\epsilon} \right) dx \\ &= \int \left(\frac{\partial f}{\partial Y} \frac{dY}{d\epsilon} + \frac{\partial f}{\partial Y'} \frac{dY'}{d\epsilon} \right) dx \\ &= \int \left(\frac{\partial f}{\partial Y} \eta(x) + \frac{\partial f}{\partial Y'} \eta'(x) \right) dx \\ &= \int \left(\frac{\partial f}{\partial y} \eta(x) + \frac{\partial f}{\partial y'} \eta'(x) \right) dx \\ &= \int \frac{\partial f}{\partial y} \eta(x) dx + \int \frac{\partial f}{\partial y'} \eta'(x) dx = 0 \end{aligned}$$

where $\epsilon \rightarrow 0$ in the penultimate step. Integrate by parts the second term:

$$\begin{aligned} & \int \frac{\partial f}{\partial y'} \eta'(x) dx \\ & \int u dv = uv - \int v du \\ & dv = \eta'(x), v = \eta(x), u = \frac{\partial f}{\partial y'}, du = \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \\ & \frac{\partial f}{\partial y'} \eta(x) - \int \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \eta(x) dx, \eta(a) = \eta(b) = 0 \\ & 0 - \int \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \eta(x) dx \end{aligned}$$

Returning to the derivation,

$$\begin{aligned} \frac{dI}{d\epsilon} &= \int \left(\frac{\partial f}{\partial y} \eta(x) + \frac{\partial f}{\partial y'} \eta'(x) \right) dx \\ &= \int \left(\frac{\partial f}{\partial y} \eta(x) - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \eta(x) \right) dx \\ &= \int \eta(x) \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right) dx = 0 \end{aligned}$$

Since $\eta(x) \neq 0$, it must be that the bracketed term is zero, which is the Euler-Lagrange equation. See fundamental lemma below for explanation.

Fundamental lemma of variational calculus. If $\int u(x)\eta(x)dx = 0$ and $\eta(x) \neq 0$ then $u(x) = 0$. Proof by contradiction, assuming $u(x) \neq 0$. Choose $\eta(x) > 0$ function: $\eta(x) = (x-x_0)(x_1-x)$ with $x_0 < x < x_1$. This is the product of two lines (one positive slope, one negative slope) which is a parabola. So,

$$\int_{x_0}^{x_1} u(x)\eta(x)dx > 0$$

If $u(x) < 0$, then

$$\int_{x_0}^{x_1} u(x)\eta(x)dx < 0$$

So for the integral to be zero, it must be that $u(x) = 0$.

22.3 Beltrami identity

Explicit dependence on x , y , and y' . Use the Euler-Lagrange equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

No explicit dependence on x. Example:

$$f = (1 + y'^2)^{1/2} y$$

The Euler-Lagrange equation form to be used is the Beltrami identity:

$$f - y' \frac{\partial f}{\partial y'} = c$$

Proof. Begin by applying the chain rule, and with no dependence on x,

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial f}{\partial y'} \frac{\partial y'}{\partial x} \\ &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \end{aligned}$$

The Euler-Lagrange equation is

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y'} \right)$$

Substituting,

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y'} \right) y' + \frac{\partial f}{\partial y'} y''$$

The product rule is

$$d(uv) = u dv + v du$$

and with $u = y', v = \frac{\partial f}{\partial y'}$, the above expression shrinks via the product rule to:

$$\begin{aligned} \frac{df}{dx} &= \frac{d}{dx} \left(y' \frac{\partial f}{\partial y'} \right) \\ \frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) &= 0 = \frac{d}{dx} c \end{aligned}$$

So,

$$f - y' \frac{\partial f}{\partial y'} = c$$

which is the Beltrami identity.

No explicit dependence on x and y.

$$f = (1 + y'^2)^{1/2}$$

Begin with the Euler-Lagrange equation and simplify:

$$\begin{aligned} \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) &= 0 \\ -\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) &= 0 \\ \frac{\partial f}{\partial y'} &= c \end{aligned}$$

22.4 Differentiation

The Euler-Lagrange equation is

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y'} \right) = 0$$

where $f = f(x, y(x), y'(x))$. If $g(x, y(x), y'(x)) = g = \frac{\partial f}{\partial y'}$, the total derivative is

$$\begin{aligned} \frac{\partial f}{\partial y} - \left(\frac{\partial g}{\partial x} \frac{dy}{dx} + \frac{\partial g}{\partial y} \frac{dy}{dx} + \frac{\partial g}{\partial y'} \frac{dy'}{dx} \right) &= 0 \\ \frac{\partial f}{\partial y} - \left(\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} y' + \frac{\partial g}{\partial y'} y'' \right) &= 0 \\ \frac{\partial f}{\partial y} - \left(\frac{\partial}{\partial x} \frac{\partial f}{\partial y'} + \frac{\partial}{\partial y} \frac{\partial f}{\partial y'} y' + \frac{\partial}{\partial y'} \frac{\partial f}{\partial y'} y'' \right) &= 0 \\ \frac{\partial f}{\partial y} - \frac{\partial^2 f}{\partial x \partial y'} - \frac{\partial^2 f}{\partial y \partial y'} y' - \frac{\partial^2 f}{\partial y'^2} y'' &= 0 \end{aligned}$$

22.5 Zero identity

If the Euler-Lagrange equation equals 0, the functional is a function of only the endpoints. Example.

$$\int (y^2 + 2xyy') dx$$

Using the Euler-Lagrange equation,

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y'} \right) = 2y + 2xy' - (2y + 2xy') = 0$$

But using $\frac{d}{dx}(xy^2) = y^2 + 2xyy'$.

$$\int (y^2 + 2xyy') dx = xy^2 \Big|_a^b$$

Proof is involved and omitted.

22.6 Applications

22.6.1 Catenary

Returning to the catenary rope,

$$E_p = \int_0^L \sqrt{1 + y'(x)^2} y(x) dx$$

The function f is

$$f = \sqrt{1 + y'(x)^2} y(x)$$

substituting this into Euler-Lagrange using the Beltrami identity

$$f - y' \frac{\partial f}{\partial y'} = c$$

the result is, using the chain rule,

$$y(1 + y'^2)^{1/2} - y' y \frac{1}{2} (1 + y'^2)^{-1/2} 2y' = c$$

$$y(1 + y'^2)^{1/2} - \frac{y'^2 y}{(1 + y'^2)^{1/2}} = c$$

$$\frac{y(1 + y'^2)^{1/2} (1 + y'^2)^{1/2} - y y'^2}{(1 + y'^2)^{1/2}} = c$$

$$\frac{y + y y'^2 - y y'^2}{(1 + y'^2)^{1/2}} = c$$

$$\frac{y}{(1 + y'^2)^{1/2}} = c$$

$$\frac{y^2}{(1 + y'^2)} = c^2$$

$$c^2 (1 + y'^2) = y^2$$

$$c^2 + c^2 y'^2 = y^2$$

$$1 + y'^2 = \frac{y^2}{c^2}$$

$$y' = \frac{dy}{dx} = \sqrt{\frac{y^2}{c^2} - 1}$$

$$dx = \frac{dy}{\sqrt{\frac{y^2}{c^2} - 1}}$$

Use the integral identity

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \left(\frac{x}{a} \right) + C$$

Integrate both sides,

$$\int dx = \int \frac{dy}{\sqrt{\frac{y^2}{c^2} - 1}}$$

$$x(y) = \cosh^{-1} \left(\frac{y}{c} \right) + C$$

$$y(x) = c \cosh \left(\frac{x + b}{c} \right)$$

which is the catenary.

22.6.2 Shortest distance in 2D between two points

The shortest distance between two points in 2D will be $y = mx + b$ using Euler-Lagrange by minimizing the path between two points. The path integral is

$$S = \int_a^b ds$$

where again

$$ds^2 = dx^2 + dy^2$$

$$\frac{ds^2}{dx^2} = 1 + y'^2$$

$$ds = (1 + y'^2)^{1/2} dx$$

again using the Beltrami identity, the calculation leads to $y = mx + b$.

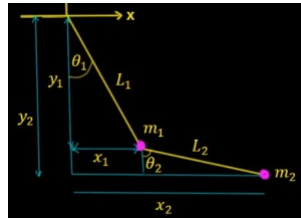
22.6.3 Simple pendulum

Minimize

$$\text{Lagrangian} = L = KE - PE$$

from Lagrangian mechanics. It minimizes energy differences and amount of energy transfer. The result will be the acceleration of the angle of the pendulum swings.

22.6.4 Double pendulum



$$x_1 = L_1 \sin \theta_1$$

$$x_2 = x_1 + L_2 \sin \theta_2$$

$$y_1 = -L_1 \cos \theta_1$$

$$y_2 = y_1 - L_2 \cos \theta_2$$

Differentiating,

$$\dot{x}_1 = (L_1 \cos \theta_1) \dot{\theta}_1$$

$$\dot{x}_2 = (L_1 \cos \theta_1) \dot{\theta}_1 + (L_2 \cos \theta_2) \dot{\theta}_2$$

$$\dot{y}_1 = (L_1 \sin \theta_1) \dot{\theta}_1$$

$$\dot{y}_2 = (L_1 \sin \theta_1) \dot{\theta}_1 + (L_2 \sin \theta_2) \dot{\theta}_2$$

Lagrangian is $L = KE - PE$. Start with KE

$$\begin{aligned}
KE &= \frac{1}{2}mv^2 = \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2) \\
&= \frac{1}{2}m_1((L_1^2\cos^2\theta_1)\dot{\theta}_1^2 + (L_1^2\sin^2\theta_1)\dot{\theta}_1^2) \\
&\quad + \frac{1}{2}m_2(((L_1\cos\theta_1)\dot{\theta}_1 + (L_2\cos\theta_2)\dot{\theta}_2)^2 + (L_1\sin\theta_1)\dot{\theta}_1 + (L_2\sin\theta_2)\dot{\theta}_2)^2) \\
&= \frac{1}{2}(m_1 + m_2)L_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2L_2^2\dot{\theta}_2^2 + m_2L_1L_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)
\end{aligned}$$

where the trig identity $\cos(\theta_1 - \theta_2) = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2$ is used. The PE is

$$\begin{aligned}
PE &= m_1gy_1 + m_2gy_2 \\
&= g(-m_1L_1\cos\theta_1 + m_2(-L_2\cos\theta_1 - L_2\cos\theta_2)) \\
&= -g((m_1 + m_2)L_1\cos\theta_1 + m_2\cos\theta_2)
\end{aligned}$$

Combining KE and PE,

$$\begin{aligned}
L &= KE - PE \\
&= \frac{1}{2}(m_1 + m_2)L_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2L_2^2\dot{\theta}_2^2 + m_2L_1L_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) + g((m_1 + m_2)L_1\cos\theta_1 + m_2\cos\theta_2)
\end{aligned}$$

Euler-Lagrange:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

where there are two degrees of freedom: $q_1 = \theta_1, q_2 = \theta_2$. For θ_1 :

$$(m_1 + m_2)L_1^2\ddot{\theta}_1 + m_2L_1L_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) + m_2L_1L_2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + L_1g(m_1 + m_2)\sin\theta_1 = 0$$

For θ_2 :

$$m_2L_2\ddot{\theta}_2^2 + m_2L_1\ddot{\theta}_1^2\cos(\theta_1 - \theta_2) - m_2L_1\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + m_2g\sin\theta_2 = 0$$

The above are coupled second-order DEs. Convert to coupled first-order DEs by $\dot{\theta} = \omega = 2\pi f = 2\frac{\pi}{t}, \theta_1 - \theta_2 = \Delta\theta$:

$$(m_1 + m_2)L_1^2\dot{\omega}_1 + m_2L_1L_2\dot{\omega}_2\cos(\Delta\theta) + m_2L_1L_2\omega_2^2\sin(\Delta\theta) + L_1g(m_1 + m_2)\sin\theta_1 = 0$$

$$m_2L_2\dot{\omega}_2^2 + m_2L_1\dot{\omega}_1^2\cos(\Delta\theta) - m_2L_1\omega_1^2\sin(\Delta\theta) + m_2g\sin\theta_2 = 0$$

Simplify further with $a = (m_1 + m_2)L_1, b = m_2L_2\cos(\Delta\theta), c = m_2L_1\cos(\Delta\theta), d = m_2L_2, e = -m_2L_2\omega_2^2\sin(\Delta\theta) - (m_1 + m_2)g\sin\theta_1, f = m_2L_1\omega_1^2\sin(\Delta\theta) - m_2g\sin\theta_2$:

$$a\dot{\omega}_1 + b\dot{\omega}_2 = e$$

$$c\dot{\omega}_1 + d\dot{\omega}_2 = f$$

which is a simultaneous system of equations. Solve:

$$\dot{\omega}_1 = \frac{ed - bf}{ad - bc}$$

$$\dot{\omega}_2 = \frac{af - ce}{ad - bc}$$

22.6.5 Other applications

- Shortest distance in polar form
- Geodesic on a cylinder (a helix)
- Geodesic on a sphere (a plane)
- Projectile motion (a parabola). See principle of least action.

22.7 Second variation

The Taylor expansion of the functional is

$$f(x, Y, Y') - f(x, y, y') = \epsilon \eta(x) + O(\epsilon^2)$$

where the first variation (in Lagrange's derivation) solution assumed $O(\epsilon^2) \rightarrow 0$ and

$$\lim_{\epsilon \rightarrow 0} \eta(x) = \frac{f(x, Y, Y') - f(x, y, y')}{\epsilon}$$

which is satisfied by the Euler-Lagrange equation:

$$\frac{df}{dy} - \frac{d}{dx} \left(\frac{df}{dy'} \right) = 0$$

which can be satisfied if f (position) and f' (gradient) CHANGE are both set to 0. The Euler-Lagrange equation finds the function f which is a stationary point of the functional (the function which is a function of f). However, it does not indicate whether f is a max, min, or inflection. To address this, use the second variation. Can have third, fourth, etc. variations by including higher-order Taylor expansion terms.

Include the second term in the Taylor expansion of the functional for the second variation:

$$f(x, Y, Y') - f(x, y, y') = \epsilon \eta(x) + \frac{\epsilon^2}{2!} \eta'(x) + O(\epsilon^3)$$

As in Lagrange's derivation,

$$Y(x) = y(x) + \epsilon \eta(x)$$

$$Y'(x) = y'(x) + \epsilon \eta'(x)$$

$$\frac{dY}{d\epsilon} = \eta(x)$$

$$\frac{dY'}{d\epsilon} = \eta'(x)$$

$$I = \int f(x, \underbrace{y(x) + \epsilon \eta(x)}_Y, \underbrace{y'(x) + \epsilon \eta'(x)}_{Y'}) dx$$

Differentiating both sides in Lagrange's derivation:

$$\frac{dI}{d\epsilon} = \int \left(\frac{\partial f}{\partial y} \eta(x) + \frac{\partial f}{\partial y'} \eta'(x) \right) dx$$

Differentiating a second time, the integral disappears:

$$\frac{d^2 I}{d\epsilon^2} = \frac{\partial}{\partial Y} \left(\frac{dI}{d\epsilon} \right) \frac{\partial Y}{\partial \epsilon} + \frac{\partial}{\partial Y'} \left(\frac{dI}{d\epsilon} \right) \frac{\partial Y'}{\partial \epsilon}$$

Evaluate the first left term:

$$\frac{\partial}{\partial Y} \left(\frac{dI}{d\epsilon} \right) \frac{\partial Y}{\partial \epsilon} = \left(\frac{\partial^2 f}{\partial Y^2} \eta + \frac{\partial \eta}{\partial Y} \frac{\partial f}{\partial Y} \right) \frac{\partial Y}{\partial \epsilon} + \left(\frac{\partial^2 f}{\partial Y \partial Y'} \eta' + \frac{\partial \eta'}{\partial Y} \frac{\partial f}{\partial Y'} \right) \frac{\partial Y}{\partial \epsilon}$$

Since $\frac{\partial \eta(x)}{\partial Y} = 0$, $\frac{\partial \eta'(x)}{\partial Y} = 0$, the above simplifies to:

$$\frac{\partial}{\partial Y} \left(\frac{dI}{d\epsilon} \right) \frac{\partial Y}{\partial \epsilon} = \frac{\partial^2 f}{\partial Y^2} \eta^2 + \frac{\partial^2 f}{\partial Y \partial Y'} \eta' \eta$$

Evaluate the second left term:

$$\frac{\partial}{\partial Y'} \left(\frac{dI}{d\epsilon} \right) \frac{\partial Y'}{\partial \epsilon} = \left(\frac{\partial^2 f}{\partial Y \partial Y'} \eta + \frac{\partial \eta}{\partial Y'} \frac{\partial f}{\partial Y} \right) \frac{\partial Y'}{\partial \epsilon} + \left(\frac{\partial^2 f}{\partial Y'^2} \eta' + \frac{\partial \eta'}{\partial Y'} \frac{\partial f}{\partial Y'} \right) \frac{\partial Y'}{\partial \epsilon}$$

Since $\frac{\partial \eta(x)}{\partial Y'} = 0$, $\frac{\partial \eta'(x)}{\partial Y'} = 0$, the above simplifies to:

$$\frac{\partial}{\partial Y'} \left(\frac{dI}{d\epsilon} \right) \frac{\partial Y'}{\partial \epsilon} = \frac{\partial^2 f}{\partial Y \partial Y'} \eta \eta' + \frac{\partial^2 f}{\partial Y'^2} \eta'^2$$

Combining the left and right terms, then $\epsilon \rightarrow 0$ s.t. $Y(x) \rightarrow y(x)$, $Y'(x) \rightarrow y'(x)$:

$$\begin{aligned} \frac{d^2 I}{d\epsilon^2} &= \frac{\partial^2 f}{\partial Y^2} \eta^2 + \frac{\partial^2 f}{\partial Y \partial Y'} \eta' \eta + \frac{\partial^2 f}{\partial Y \partial Y'} \eta \eta' + \frac{\partial^2 f}{\partial Y'^2} \eta'^2 \\ &= \frac{\partial^2 f}{\partial y^2} \eta^2 + 2 \frac{\partial^2 f}{\partial y \partial y'} \eta' \eta + \frac{\partial^2 f}{\partial y'^2} \eta'^2 \end{aligned}$$

Since

$$\begin{aligned} I &= \int f(x, y, y') dx \\ I + \delta I &= \int f(x, Y, Y') dx \end{aligned}$$

Then

$$\delta I = \int (f(x, Y, Y') - f(x, y, y')) dx = \delta I_1 + \delta I_2 + \dots$$

depending on the degree of precision. From the first variation,

$$\delta I_1 = \epsilon \int \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) dx$$

From above, the second variation leads to

$$\delta I_2 = \frac{\epsilon^2}{2!} \int \left(\frac{\partial^2 f}{\partial y^2} \eta^2 + 2 \frac{\partial^2 f}{\partial y \partial y'} \eta' \eta + \frac{\partial^2 f}{\partial y'^2} \eta'^2 \right) dx$$

22.7.1 Jacobi accessory equation

First derivation. Re-express δI_2 :

$$\begin{aligned}
\delta I_2 &= \frac{\epsilon^2}{2!} \int_a^b \left(\frac{\partial^2 f}{\partial y^2} \eta^2 + 2 \frac{\partial^2 f}{\partial y \partial y'} \eta' \eta + \frac{\partial^2 f}{\partial y'^2} \eta'^2 \right) dx \\
&= \frac{\epsilon^2}{2!} \int_a^b (F_{00} \eta^2 + 2F_{01} \eta' \eta + F_{11} \eta'^2) dx \\
&= \frac{\epsilon^2}{2!} \int_a^b \left(F_{00} \eta^2 + F_{01} \frac{d}{dx} \eta^2 + F_{11} \eta'^2 \right) dx \\
&= \frac{\epsilon^2}{2!} \int_a^b \left(F_{00} \eta^2 - \eta^2 \frac{d}{dx} F_{01} + F_{11} \eta'^2 \right) dx \\
&= \frac{\epsilon^2}{2!} \int_a^b \left(\eta^2 \left(F_{00} - \frac{d}{dx} F_{01} \right) + F_{11} \eta'^2 \right) dx \\
&= \frac{\epsilon^2}{2!} \int_a^b \left(\eta^2 \left(F_{00} - \frac{d}{dx} F_{01} \right) - \eta \frac{d}{dx} (F_{11} \eta') \right) dx \\
&= \frac{\epsilon^2}{2!} \int_a^b \left(\eta \left(\eta \left(F_{00} - \frac{d}{dx} F_{01} \right) - \frac{d}{dx} (F_{11} \eta') \right) \right) dx = 0
\end{aligned}$$

Note 1: Fourth step uses integration by parts with $u = F_{01}, v = \eta(x)$ and $\eta(a) = \eta(b) = 0$:

$$\begin{aligned}
\int_a^b F_{01} \frac{d}{dx} \eta'(x) dx &= F_{01} \eta(x) \Big|_a^b - \int_a^b \eta(x) \frac{d}{dx} F_{01} dx \\
&= 0 - \int_a^b \eta(x) \frac{d}{dx} F_{01} dx
\end{aligned}$$

So,

$$F_{01} \frac{d}{dx} \eta'(x) dx = -\eta(x) \frac{d}{dx} F_{01} dx$$

Note 2: Sixth step uses (together with integration by parts similar to step four):

$$-\eta \frac{d}{dx} (F_{11} \eta') = - \left(-F_{11} \eta' \frac{d}{dx} \eta \right) = F_{11} \eta'^2$$

Using the fundamental lemma of variational calculus, $\eta(x) \neq 0$ so the inner bracket term of the last equation is 0. But η is still in

$$\eta \left(F_{00} - \frac{d}{dx} F_{01} \right) - \frac{d}{dx} (F_{11} \eta') = 0$$

So call use a substitute function $u(x)$, which forms the basis for a different 'family' of variations than $\eta(x)$:

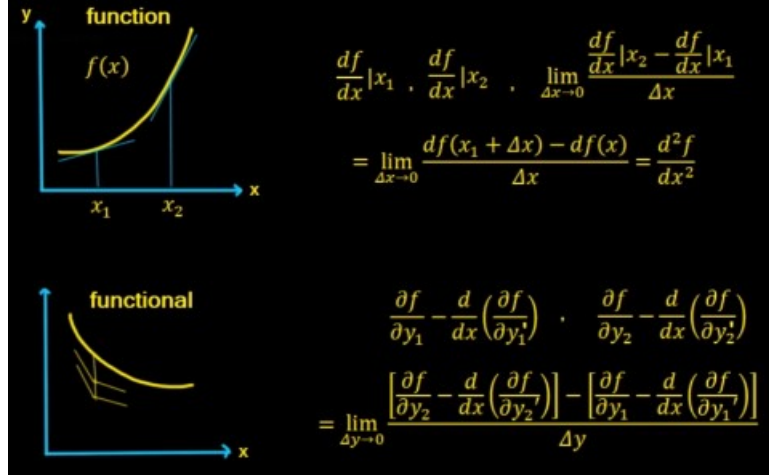
$$u \left(F_{00} - \frac{d}{dx} F_{01} \right) - \frac{d}{dx} (F_{11} u') = 0$$

where:

$$Y_1(x) = y(x) + \epsilon \eta(x)$$

$$Y_2(x) = y(x) + \epsilon u(x)$$

Second derivation. More intuitive and builds on first derivation.



Just as the second derivative from regular calculus can be understood as

$$\lim_{\Delta x \rightarrow 0} f(x) = \frac{\frac{df}{dx}|_{x_2} - \frac{df}{dx}|_{x_1}}{\Delta x} = f''(x)$$

The second variation can be derived from

$$\lim_{\Delta y \rightarrow 0} f(x) = \frac{\left(\frac{\partial f}{\partial y_2} - \frac{d}{dx} \left(\frac{\partial f}{\partial y_2'} \right) \right) - \left(\frac{\partial f}{\partial y_1} - \frac{d}{dx} \left(\frac{\partial f}{\partial y_1'} \right) \right)}{\Delta y}$$

Using two different bases for variations $\eta(x), u(x)$ from the first derivation,

$$\begin{aligned} \lim_{\Delta \epsilon \rightarrow 0} f(x) &= \frac{\left(\frac{\partial f(x, Y_2, Y_2')}{\partial y_2} - \frac{d}{dx} \left(\frac{\partial f(x, Y_2, Y_2')}{\partial y_2'} \right) \right) - \left(\frac{\partial f(x, Y_1, Y_1')}{\partial y_1} - \frac{d}{dx} \left(\frac{\partial f(x, Y_1, Y_1')}{\partial y_1'} \right) \right)}{\epsilon} \\ &= \frac{\left(\frac{\partial f(x, y + \epsilon u, y' + \epsilon u')}{\partial y_2} - \frac{d}{dx} \left(\frac{\partial f(x, y + \epsilon u, y' + \epsilon u')}{\partial y_2'} \right) \right) - \left(\frac{\partial f(x, y + \epsilon \eta, y' + \epsilon \eta')}{\partial y_1} - \frac{d}{dx} \left(\frac{\partial f(x, y + \epsilon \eta, y' + \epsilon \eta')}{\partial y_1'} \right) \right)}{\epsilon} \end{aligned}$$

Since the difference between first derivatives is the second derivative multiplied by the horizontal distance (when $\epsilon \rightarrow 0$),

$$\begin{aligned}
&= \frac{\partial^2 f}{\partial y^2} u + \frac{\partial^2 f}{\partial y \partial y'} u' - \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y \partial y'} + \frac{\partial^2 f}{\partial y'^2} u' \right) \\
&= \frac{\partial^2 f}{\partial y^2} u + \frac{\partial^2 f}{\partial y \partial y'} u' - u' \frac{\partial^2 f}{\partial y \partial y'} - u \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y \partial y'} \right) - \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y'^2} u' \right) \\
&= u \frac{\partial^2 f}{\partial y^2} - u \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y \partial y'} \right) - \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y'^2} u' \right) \\
&= u \left(F_{00} - \frac{d}{dx} F_{01} \right) - \frac{d}{dx} (F_{11} u') = 0
\end{aligned}$$

which is the Jacobi accessory equation. This is the second derivative equivalent in variational calculus.

22.7.2 Solutions to the Jacobi accessory

With $F_0 = \frac{\partial f}{\partial y}$, $F_1 = \frac{\partial f}{\partial y'}$, Euler-Lagrange (i.e. first variation) is

$$F_0 - \frac{d}{dx} F_1 = 0$$

where the general solution to the functional $f(x, y, y')$ is the function $y = y(x, c_1, c_2)$, where c_1, c_2 are parameters.

The Jacobi accessory equation (i.e. the second variation) is

$$u \left(F_{00} - \frac{d}{dx} F_{01} \right) - \frac{d}{dx} (F_{11} u') = 0$$

and the solutions are

$$u_1 = \frac{\partial y}{\partial c_1}, u_2 = \frac{\partial y}{\partial c_2}$$

Proof (for u_1 only; u_2 is analogous). Begin with the Euler-Lagrange equation,

$$F_0 - \frac{d}{dx} F_1 = 0$$

Differentiating,

$$\frac{\partial F_0}{\partial y} \frac{\partial y}{\partial c_1} + \frac{\partial F_0}{\partial y'} \frac{\partial y'}{\partial c_1} - \frac{d}{dx} \left(\frac{\partial F_1}{\partial y} \frac{\partial y}{\partial c_1} + \frac{\partial F_1}{\partial y'} \frac{\partial y'}{\partial c_1} \right) = 0$$

Letting $u = \frac{\partial y}{\partial c_1}$, $u' = \frac{\partial y'}{\partial c_1}$,

$$\frac{\partial F_0}{\partial y} u + \frac{\partial F_0}{\partial y'} u' - \frac{d}{dx} \left(\frac{\partial F_1}{\partial y} u + \frac{\partial F_1}{\partial y'} u' \right) = 0$$

Working through the calculus and algebra,

$$F_{00}u + F_{01}u' - u \frac{d}{dx} F_{01} - u' F_{01} - u' \frac{d}{dx} F_{11} - u'' F_{11} = 0$$

$$u \left(F_{00} - \frac{d}{dx} F_{01} \right) - \left(\frac{d}{dx} (F_{11}u') \right) = 0$$

which is the Jacobi accessory. So, the solution for $u = u_1$ in the Jacobi accessory is $u = \frac{\partial y}{\partial c_1}, u' = \frac{\partial y'}{\partial c_1}$.

If u_1, u_2 are independent solutions, then

$$u_1 \left(F_{00} - \frac{d}{dx} F_{01} \right) - \left(\frac{d}{dx} (F_{11}u'_1) \right) = 0$$

$$u_2 \left(F_{00} - \frac{d}{dx} F_{01} \right) - \left(\frac{d}{dx} (F_{11}u'_2) \right) = 0$$

So,

$$u_1 u_2 \left(F_{00} - \frac{d}{dx} F_{01} \right) - u_2 \left(\frac{d}{dx} (F_{11}u'_1) \right) = 0$$

$$u_2 u_1 \left(F_{00} - \frac{d}{dx} F_{01} \right) - u_1 \left(\frac{d}{dx} (F_{11}u'_2) \right) = 0$$

Combining,

$$u_2 \frac{d}{dx} (F_{11}u'_1) = u_1 \frac{d}{dx} (F_{11}u'_2)$$

$$u_2 \frac{d}{dx} (F_{11}u'_1) - u_1 \frac{d}{dx} (F_{11}u'_2) = 0$$

$$\frac{d}{dx} (F_{11}(u_2 u'_1 - u_1 u'_2)) = 0$$

$$\int d(F_{11}(u_2 u'_1 - u_1 u'_2)) = \int 0 dx$$

$$F_{11}(u_2 u'_1 - u_1 u'_2) = C$$

$$u_2 u'_1 - u_1 u'_2 = \frac{C}{F_{11}}$$

Assuming $F_{11} \neq \infty$, then cannot have repeated roots (i.e. $u_1 \neq u_2$). Moreover, know that this cannot occur: $u_i = \frac{du_i}{dx} = 0$. Further, rewriting the above using the quotient rule,

$$\frac{d}{dx} \frac{u_1}{u_2} = \frac{B}{F_{11}u_1^2}$$

and if F_{11} is consistent + or - over the interval (a, b) , then $\frac{u_1}{u_2}$ is strictly increasing or decreasing respectively.

22.7.3 Conjugate points

Express $u(x)$ as a linear combination:

$$u(x) = \beta_1 u_1(x) + \beta_2 u_2(x)$$

$$0 = \beta_1 u_1(a) + \beta_2 u_2(a)$$

$$0 = \beta_1 u_1(a') + \beta_2 u_2(a')$$

$$\frac{u_1(a)}{u_2(a)} = \frac{u_1(a')}{u_2(a')} = -\frac{\beta_2}{\beta_1}$$

So to find all conjugate points a' , look for x 's which satisfy the above:

$$\frac{u_1(a)}{u_2(a)} = \frac{u_1(x)}{u_2(x)}$$

$$\text{As } u_1 = \frac{\partial y}{\partial c_1}, u_2 = \frac{\partial y}{\partial c_2},$$

$$\frac{\frac{\partial y}{\partial c_1}}{\frac{\partial y}{\partial c_2}} = \left(\frac{\frac{\partial y}{\partial c_1}}{\frac{\partial y}{\partial c_2}} \right) \bigg|_{x=a}$$

Conjugate points are intuitively indicate the 'maximum usable distance' (e.g. extreme distance envelopes in parabolic launches or the other side of a sphere's great arc) of a curve for variational calculus. So want the integration limits to lie **within**, and have no part lying outside (a, a') .

22.7.4 Legendre test

Tests whether an extremum a max or min by using

$$\delta I_2 = \frac{\epsilon^2}{2!} \int_a^b \left(\frac{\partial^2 f}{\partial y^2} \eta^2 + 2 \frac{\partial^2 f}{\partial y \partial y'} \eta' \eta + \frac{\partial^2 f}{\partial y'^2} \eta'^2 \right) dx$$

Test. (1) Euler-Lagrange equation is satisfied. (2) Integration limits are sufficiently small. (3) $\frac{\partial^2 f}{\partial y'^2}$ does not change sign over (a, b) . Then,

$$\frac{\partial^2 f}{\partial y'^2} < 0 \rightarrow f \text{ is max}$$

$$\frac{\partial^2 f}{\partial y'^2} > 0 \rightarrow f \text{ is min}$$

Proof. Simplify

$$\begin{aligned}
\delta I_2 &= \frac{\epsilon^2}{2!} \int \left(\frac{\partial^2 f}{\partial y^2} \eta^2 + 2 \frac{\partial^2 f}{\partial y \partial y'} \eta' \eta + \frac{\partial^2 f}{\partial y'^2} \eta'^2 \right) dx \\
&= \frac{\epsilon^2}{2!} \int \left(F_{00} \eta^2 + F_{01} \frac{d}{dx} \eta^2 + F_{11} \eta'^2 \right) dx \\
&= \frac{\epsilon^2}{2!} \int \left(F_{00} \eta^2 - \eta^2 \frac{d}{dx} F_{01} - \eta \frac{d}{dx} (F_{11} \eta') \right) dx \\
&= \frac{\epsilon^2}{2!} \int \left(\eta^2 \left(F_{00} - \frac{d}{dx} F_{01} \right) - \eta \frac{d}{dx} (F_{11} \eta') \right) dx \\
&= \frac{\epsilon^2}{2!} \int \left(\eta^2 \left(\frac{1}{u} \frac{d}{dx} (F_{11} u') \right) - \eta \frac{d}{dx} (F_{11} \eta') \right) dx \\
&= \frac{\epsilon^2}{2!} \int \left(\frac{\eta}{u} \left(\eta \left(\frac{d}{dx} (F_{11} u') \right) - u \frac{d}{dx} (F_{11} \eta') \right) \right) dx \\
&= \frac{\epsilon^2}{2!} \int \left(\frac{\eta}{u} \left(\frac{d}{dx} (F_{11} (\eta u' - u \eta')) \right) \right) dx \\
&= -\frac{\epsilon^2}{2!} \int \left(F_{11} (\eta u' - u \eta') \frac{d}{dx} \frac{\eta}{u} \right) dx \\
&= -\frac{\epsilon^2}{2!} \int \left(F_{11} (\eta u' - u \eta') \frac{\eta' u - \eta u'}{u^2} \right) dx \\
&\dots \text{algebraic manipulation} \dots \\
&= \underbrace{\frac{\epsilon^2}{2!}}_{+} \int F_{11} \underbrace{\left(\eta' - \frac{\eta u'}{u} \right)^2}_{\text{Jacobi: } +} dx
\end{aligned}$$

So

$$\begin{aligned}
F_{11} > 0 &\rightarrow \min \\
F_{11} < 0 &\rightarrow \max
\end{aligned}$$

Notes: The third step uses integration by parts on the left second and left third terms with $\eta(a) = \eta(b) = 0$:

$$\frac{d}{dx} \left(\int_a^b F_{01} \frac{d}{dx} \eta^2 dx = \underbrace{F_{01} \eta^2 \Big|_a^b}_{=0} - \int_a^b \eta^2 \frac{d}{dx} F_{01} dx \right)$$

and

$$\frac{d}{dx} \left(\int_a^b (F_{11} \eta') \frac{d}{dx} \eta dx = \underbrace{(F_{11} \eta') \eta \Big|_a^b}_{=0} - \int_a^b \eta \frac{d}{dx} (F_{11} \eta') dx \right)$$

The fifth step substitutes the Jacobi accessory equation

$$u \left(F_{00} - \frac{d}{dx} F_{01} \right) - \frac{d}{dx} (F_{11} u') = 0$$

$$F_{00} - \frac{d}{dx} F_{01} = \frac{1}{u} \frac{d}{dx} (F_{11} u')$$

The eighth step uses similar integration by parts as the third step.

22.7.5 Jacobi test

Adds extra condition to Legendre test. If

$$\left(\eta' - \frac{\eta u'}{u} \right)^2 = 0$$

then

$$\eta(x) = \alpha u(x)$$

$$\alpha u'(x) - \alpha u(x) \frac{u'(x)}{u(x)} = 0$$

Jacobi test is the same as the Legendre test, but it further ensures that

$$\left(\eta' - \frac{\eta u'}{u} \right)^2 \neq 0$$

by

$$\eta(x) \neq \alpha u(x)$$

which occurs if endpoints (a, b) are not the same for $\eta(x), u(x)$. More specifically that the conjugate point $a' \geq b$ (if $a' < b$ then length between a', b will be 0 and not possible to determine whether max, min, or inflection).

22.8 Generalized functionals

Above, the functional form has been

$$I = \int f(x, y(x), y'(x)) dx$$

22.8.1 Multiple coordinates

The functional can be generalized to

$$I = \int f(x, y_1(x), y_2(x), y_1'(x), y_2'(x)) dx$$

using different variable names,

$$I = \int f(t, q_1(t), q_2(t), q_1'(t), q_2'(t)) dt$$

Then,

$$\begin{aligned}\delta I_1 &= \int \left(\frac{\partial f}{\partial q_1} \frac{\partial q_1}{\partial \epsilon_1} + \frac{\partial f}{\partial q'_1} \frac{\partial q'_1}{\partial \epsilon_1} \right) dt = \int \left(\frac{\partial f}{\partial q_1} \eta_1 + \frac{\partial f}{\partial q'_1} \eta'_1 \right) dt \\ \delta I_2 &= \int \left(\frac{\partial f}{\partial q_2} \frac{\partial q_2}{\partial \epsilon_2} + \frac{\partial f}{\partial q'_2} \frac{\partial q'_2}{\partial \epsilon_2} \right) dt = \int \left(\frac{\partial f}{\partial q_2} \eta_2 + \frac{\partial f}{\partial q'_2} \eta'_2 \right) dt\end{aligned}$$

$\epsilon_1 = \epsilon_2$

The first variation is

$$\begin{aligned}\frac{\partial f}{\partial q_1} - \frac{d}{dt} \left(\frac{\partial f}{\partial q'_1} \right) &= 0 \\ \frac{\partial f}{\partial q_2} - \frac{d}{dt} \left(\frac{\partial f}{\partial q'_2} \right) &= 0\end{aligned}$$

which might be used for xy-coordinates, where the variable is time t , with $x(t)$ and $y(t)$ as position. The second variation, letting $q_1 = 1, q_2 = 2, q'_1 = 3, q'_2 = 4$,

$$\begin{aligned}\delta I &= \frac{\epsilon^2}{2!} \int ((F_{11}\eta_1^2 + F_{22}\eta_1'^2 + F_{33}\eta_2^2 + F_{44}\eta_2'^2) + (2F_{13}\eta_1\eta'_1 + 2F_{12}\eta_1\eta_2 + 2F_{14}\eta_1\eta'_2) \\ &\quad + (2F_{23}\eta_1'\eta_2 + 2F_{34}\eta_1'\eta_2' + 2F_{34}\eta_1'\eta_2' + 2F_{24}\eta_2\eta_2'))dt\end{aligned}$$

and the Legendre test is: (1) the integration 'distance' is small, (2) F_{22}, F_{44} has a constant sign over the integration limits, (3) $F_{22}F_{44} > F_{24}^2$, then

$$F_{22} > 0 \rightarrow \min$$

$$F_{22} < 0 \rightarrow \max$$

and this can be generalized to more variables. Finding the conjugate points involves a complicated matrix.

22.8.2 Higher derivatives

Reminder:

$$\begin{aligned}Y(x) &= y(x) + \epsilon\eta(x) \\ Y'(x) &= y'(x) + \epsilon\eta'(x) \\ \frac{dY}{d\epsilon} &= \eta \\ \frac{dY'}{d\epsilon} &= \eta'\end{aligned}$$

The functional can be generalized to

$$\begin{aligned}I &= \int f(x, y, y', y'' \dots y^n) dx \\ \delta I &= \int f(x, Y, Y', Y'') - f(x, y, y', y'') dx\end{aligned}$$

$$\begin{aligned}
I &= \epsilon \int \left(\frac{\partial f}{\partial Y} \frac{\partial Y}{\partial \epsilon} + \frac{\partial f}{\partial Y'} \frac{\partial Y'}{\partial \epsilon} + \frac{\partial f}{\partial Y''} \frac{\partial Y''}{\partial \epsilon} \right) dx + O(\epsilon^2) \\
&= \epsilon \int \left(\frac{\partial f}{\partial Y} \eta + \frac{\partial f}{\partial Y'} \eta' + \frac{\partial f}{\partial Y''} \eta'' \right) dx + O(\epsilon^2)
\end{aligned}$$

After integration by parts on the second and third terms,

$$= \int \eta \left(\frac{\partial f}{\partial Y} - \frac{d}{dx} \left(\frac{\partial f}{\partial Y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial f}{\partial Y''} \right) \right) dx$$

With $\eta(x) \neq 0$ and $\epsilon \rightarrow 0$ (so $Y(x) \rightarrow y(x)$),

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial f}{\partial y''} \right) = 0$$

which is analogous to the Euler-Lagrange equation. If it's satisfied, then the Legendre test applies to the nature of the stationary point.

22.8.3 Multiple independent variables

The functional can be generalized to

$$I = \int \int f(x, y, z, p, q) dx dy$$

where x and y are the independent variables, $z(x, y)$ is a parameter, and $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$. Instead of $\eta(x)$, there is $\eta(x, y)$.

$$\delta I = \int \int f(x, y, z(x, y) + \epsilon \eta, p + \epsilon \frac{d\eta}{dx}, q + \epsilon \frac{d\eta}{dy}) dx dy - \int \int f(x, y, z, p, q) dx dy$$

After significant algebra and calculus,

$$= \int \int \left(\eta \frac{\partial f}{\partial z} + \frac{d\eta}{dx} \frac{\partial f}{\partial p} + \frac{d\eta}{dy} \frac{\partial f}{\partial q} \right) dx dy = 0$$

To separate η , use **Green's theorem**:

$$\int \int \left(\frac{\partial \phi}{\partial x} + \frac{\partial \psi}{\partial y} \right) dx dy = \int (\phi dy - \psi dx)$$

which is a re-expression of the FTC turning a surface integral to a line integral (all internal paths of the surface cancel each other out as they move in opposition directions, leaving only the outside):

$$\int_a^b \frac{d\phi}{dx} dx = \phi_b - \phi_a$$

Apply Green's theorem using $\phi = \eta \frac{\partial f}{\partial p}$, $\psi = \eta \frac{\partial f}{\partial q}$ and after algebra and calculus,

$$\int \int \eta \left(\frac{\partial f}{\partial s} - \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial p} \right) - \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial q} \right) \right) dx dy = 0$$

With $\eta \neq 0$ and applying the fundamental theorem of variational calculus,

$$\frac{\partial f}{\partial s} - \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial p} \right) - \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial q} \right) = 0$$

which is the Euler-Lagrange equivalent for a multivariable functional.

22.9 Noether's theorem

A specific case of Euler-Lagrange.

For every continuous symmetry of a Lagrangian, there is a conserved quantity along trajectories that satisfy Newton's equations of motion. Symmetry of translation leads to conserved momentum, symmetry of rotation leads to conserved angular momentum, symmetry of time translation leads to conserved energy. There are others, including conserved charges in quantum field theory.

22.9.1 Intuition

Time translation symmetry gives conservation of energy; space translation symmetry gives conservation of momentum; rotation symmetry gives conservation of angular momentum, etc.

Suppose Lagrangian L has a symmetry so doesn't change when sending q to some new position $q(s)$. Then

$$\frac{d}{ds} L(q(s), \dot{q}(s)) = 0$$

and in this case, the claim is that

$$Constant = C = p \frac{dq(s)}{ds} \equiv p$$

Proof. Use product rule,

$$\begin{aligned} \dot{C} &= \dot{p} \frac{dq(s)}{ds} + p \frac{\dot{q}(s)}{ds} \\ \dot{C} &= \frac{\partial L}{\partial q} \frac{dq(s)}{ds} + \frac{\partial L}{\partial \dot{q}} \frac{\dot{q}(s)}{ds} \end{aligned}$$

Reversing the chain rule,

$$\frac{d}{ds} L(q(s), \dot{q}(s)) = 0$$

22.9.2 Spatial translation

Intuition. Lagrangeian $L = L'$ because a spatial translation is a constant where the derivative is zero: velocity in KE is derivative of position, and PE's gravity is just the difference in distances, which doesn't change with spatial translations.

$$S(q(t)) = \int_{t_1}^{t_2} L(q, \dot{q}) dt$$

If $q(t)$ is stationary (i.e. satisfies least action) then $\delta S = 0$. With U as potential energy,

$$\begin{aligned}
S(q(t) + \epsilon(t)) &= S(q(t)) + \delta S + O(\epsilon^2) \\
&= \int_{t_1}^{t_2} \left(\frac{1}{2} m_1 (\dot{q}_1 + \dot{\epsilon})^2 + \frac{1}{2} m_2 (\dot{q}_2 + \dot{\epsilon})^2 - U(q_1 + \epsilon - q_2 - \epsilon) \right) dt \\
&= \int_{t_1}^{t_2} \left(\frac{1}{2} m_1 (\dot{q}_1^2 + 2\dot{q}_1\dot{\epsilon} + \dot{\epsilon}^2) + \frac{1}{2} m_2 (\dot{q}_2^2 + 2\dot{q}_2\dot{\epsilon} + \dot{\epsilon}^2) - U(q_1 - q_2) \right) dt \\
&= \underbrace{\int_{t_1}^{t_2} \left(\frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 - U(q_1 - q_2) \right) dt}_S + \underbrace{\int_{t_1}^{t_2} (m_1 \dot{q}_1 \dot{\epsilon} + m_2 \dot{q}_2 \dot{\epsilon}) dt}_{\delta S} + O(\epsilon^2)
\end{aligned}$$

With $\epsilon(t_1) = \epsilon(t_2) = 0$ and integration by parts as in Euler-Lagrange work,

$$\delta S = \underbrace{(m_1 \dot{q}_1 \epsilon + m_2 \dot{q}_2 \epsilon) \Big|_{t_1}^{t_2}}_{=0} - \int_{t_1}^{t_2} \epsilon \underbrace{(m_1 \ddot{q}_1 + m_2 \ddot{q}_2)}_{=0} dt = 0$$

for any ϵ . From the second part of the last line,

$$\begin{aligned}
m_1 \ddot{q}_1 + m_2 \ddot{q}_2 &= 0 \\
\frac{d}{dt} (m_1 \dot{q}_1 + m_2 \dot{q}_2) &= \text{constant} \\
\frac{d}{dt} p &= \text{constant}
\end{aligned}$$

So momentum is conserved.

22.9.3 Rotation translation

Analogous to above, but for rotational translation being symmetric and angular momentum being conserved. If there is a disturbance to the rotational symmetry, there is an extra term. The extra term is torque.

22.9.4 Noether procedure

1. Symmetry transformation $q \rightarrow q + \delta q$, where δq depends on small constant parameter *epsilon*. This sends Lagrangian $L \rightarrow L$ (i.e. no change).
2. Make ϵ be time-dependent, so $\epsilon(t)$, which sends $L \rightarrow L + \dot{\epsilon} B$.
3. In solutions to Newton's equations of motion, $\delta S = 0$ (principle of least action) if $\epsilon(t_1) = \epsilon(t_2) = 0$. Therefore, $0 = \delta S = \int_{t_1}^{t_2} \dot{\epsilon} B dt = \int_{t_1}^{t_2} \epsilon \dot{B} dt$ because of integration by parts (as in Euler-Lagrange work).

Proof. If ϵ is constant, then $L \rightarrow L + \epsilon A$, where the transformation is proportional to A . If $\epsilon(t)$, then $L \rightarrow L + \epsilon A + \dot{\epsilon} B$. If $q \rightarrow q + \delta q$ is a symmetry transformation, then $L \rightarrow L$ so $A = 0$ and so $L \rightarrow L + \dot{\epsilon} B$, and solutions to Newton's equations of motions lead to $0 = \delta S = \int_{t_1}^{t_2} \dot{\epsilon} B dt = \int_{t_1}^{t_2} \epsilon \dot{B} dt$ and $\dot{B} = 0$, so B is conserved.

23 Mechanics

23.1 Newtonian mechanics

Based on Newton's three laws of motion. Complicated calculations for all but trivial cases.

23.2 Least action principle

Action. Definition:

$$I = S = \int L dt = \int (KE - PE) dt$$

Principle of least action (aka stationary action principle). I is a stationary point. So, an object travels in a path s.t. I is optimized. E.g. movement between any two points is characterized by the action of those two points which finds the best 'balance' between KE and PE (where the difference is smallest). Intuition: nature moves in optimal paths (e.g. where the KE-PE energy differences are minimized so that energy imbalances are in turn minimized). Take above (Lagrangian) and combine with energy conservation:

$$Constant = TE = KE + PE, L = KE - PE$$

$$L = KE - (TE - KE) = 2KE - TE$$

which results in

$$I = S = \int L dt = \int 2KE - TE dt = 2 \int KE dt - \int TE dt = \underbrace{2 \int KE dt}_{\text{Least action}} - constant$$

So minimizing

$$2 \int KE dt = S + constant$$

will produce results. Note: this is Hamilton's formulation of the least action principle, so sometimes called Hamilton's principle.

23.2.1 Brachistochrone problem

Path of shortest time.

$$\begin{aligned} \Delta PE &= \Delta KE \\ mgy(x) &= \frac{1}{2} m \left(\frac{ds}{dt} \right)^2 \\ 2gy(x) &= \left(\frac{ds}{dt} \right)^2 \end{aligned}$$

$$\sqrt{2gy} = \left(\frac{ds}{dt} \right)$$

$$dt = \frac{ds}{\sqrt{2gy}}$$

Since

$$ds^2 = dx^2 + dy^2$$

$$ds = \sqrt{1 + y'^2} dx$$

Then

$$dt = \frac{\sqrt{1 + y'^2} dx}{\sqrt{2gy}}$$

Total time is

$$\frac{1}{\sqrt{2g}} \int \frac{\sqrt{1 + y'^2} dx}{\sqrt{y}}$$

Using the Beltrami identity to solve this, and renaming and standard substituting using $2b = A = y(1 + y'^2)$, where $\phi = 2\theta$ and θ is the angle of fall, then

$$x = b(1 - \sin\phi), y = b(1 - \cos\phi)$$

which is the equation of a cycloid.

23.3 Lagrangian mechanics

Applies the least action principle. Equivalent to Newton's second law $F = ma$. The Lagrangian is $L = KE(\dot{q}) - PE(q)$, then from Euler-Lagrange

$$\begin{aligned} \frac{\partial L}{\partial q} &= \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \\ \frac{\partial}{\partial q} KE(\dot{q}) - PE(q) &= \frac{d}{dt} p \\ \frac{\partial}{\partial q} KE(\dot{q}) - PE(q) &= \frac{d}{dt} \frac{\partial}{\partial \dot{q}} KE(\dot{q}) - PE(q) \\ -\frac{\partial}{\partial q} PE(q) &= \frac{d}{dt} \frac{\partial}{\partial \dot{q}} KE(\dot{q}) \\ F &= \frac{d}{dt} \frac{\partial}{\partial \dot{q}} \frac{1}{2} m \dot{q}^2 \\ F &= m \ddot{q} = ma \end{aligned}$$

Note: Newton's laws of motion are large-scale approximations of quantum laws, and analogously Lagrangian mechanics are large-scale approximations of Feynman path integrals.

23.4 Legendre transformation

Maps one function $F(x)$ to another $G(x)$. F gradient becomes x -coordinate of G , which is $F'(x)$. F 's negative y -intercept becomes y -coordinate of G , which is $F'(x)x - F(x)$. So G coordinates are $(F'(x), F'(x)x - F(x))$. Now call $s = s(x) = F'(x)$. Then the inverse is $x(s)$. So G coordinates can be re-expressed as $(s, s \cdot x(s) - F(x(s)))$. So,

$$G(s) = sx(s) - F(x(s))$$

Example. $F(x) = x^2$ so $s = F'(x) = 2x \rightarrow x = \frac{s}{2}$. Then

$$G(s) = sx(s) - F(x(s)) = s \frac{s}{2} - F\left(\frac{s}{2}\right) = \frac{s^2}{2} - \left(\frac{s}{2}\right)^2 = \frac{s^2}{4}$$

Other examples.

$$F(x) = c \rightarrow (0, -c)$$

$$F(x) = ax + b \rightarrow (a, -b)$$

$$F(x) = a + bx + cx^2 \rightarrow G(s) = \frac{1}{4c}(s - b)^2 - a$$

$$F(x) = e^x \rightarrow G(s) = s \ln s - s$$

Inverse. Legendre transform of a Legendre transform.

$$y(s) = \frac{dG}{ds} = G'(s)$$

where the inverse is $s(y)$. Then follow steps as before:

$$H(y) = ys(y) - G(s(y))$$

But this means

$$G(s(y)) = ys(y) - H(y)$$

which means $H(y) \equiv F(x)$. So the Legendre transform of a Legendre transform is just an inverse. The Legendre transformation is therefore an involution: $f(f(x)) = x$. Note: analogous to inverse functions requiring monotonic F , inverse Legendre transforms require convex F , i.e. F' is monotonic (so that each point on F has a unique slope). Otherwise, will have two values per x in the transformed function G .

Canonical conjugate pairs. Since

$$G(s) = sx - F(x)$$

Then

$$G(s) + F(x) = sx$$

so s and x depend on each other.

Differentials

$$\begin{aligned}
 G(s) &= sx - F(x) \\
 dG &= sdx + xds - dF \\
 dG &= sdx + xds - \frac{dF}{dx}dx \\
 dG &= \frac{dF}{dx}dx + xds - \frac{dF}{dx}dx \\
 dG &= xds \\
 x &= \frac{dG}{ds}
 \end{aligned}$$

or just differentiate $G(s)$ with respect to variations of s . Repeat for variations of x to by differentiating $F(x) = sx - G(s)$ to get $s = \frac{dF}{dx}$. So x and s indicate sensitivity of G and S , respectively. It also provides more intuition for inverse transformations.

Multiple independent variables. Surface instead of line: $F(x_1, x_2)$.

$$\begin{aligned}
 dF &= \frac{\partial F}{\partial x_1}dx_1 + \frac{\partial F}{\partial x_2}dx_2 \\
 dF &= s_1dx_1 + s_2dx_2
 \end{aligned}$$

Function spaces are now $(F, x_1, x_2) \leftrightarrow (G, s_1, s_2) \leftrightarrow (G_2, x_1, s_2) \leftrightarrow (G_3, s_1, x_2)$.

$$\begin{aligned}
 dG &= \frac{\partial G}{\partial s_1}ds_1 + \frac{\partial G}{\partial s_2}ds_2 \\
 dG &= x_1ds_1 + x_2ds_2
 \end{aligned}$$

Combining,

$$\begin{aligned}
 d(F + G) &= (s_1dx_1 + s_2dx_2) + (x_1ds_1 + x_2ds_2) = d(x_1s_1 + x_2s_2) \\
 F(x_1, x_2) + G(s_1, s_2) &= x_1s_1 + x_2s_2 \\
 G(s_1, s_2) &= x_1s_1 + x_2s_2 - F(x_1, x_2)
 \end{aligned}$$

which is analogous to the 1D case of $G(s) = sx - F(x)$. This can be generalized to higher dimensions.

$$\begin{aligned}
 G(s) &= \left(\sum_i^N s_i x_i \right) - F(x(s)) \\
 s_i &= \frac{\partial F}{\partial x_i}
 \end{aligned}$$

23.5 Hamiltonian mechanics

23.5.1 Derivation from the Lagrangian

The Lagrangian is $L = KE - PE = \frac{1}{2}m\dot{q}^2 - mgq = L(q, \dot{q}, t)$. The total differential of the Lagrangian is

$$\frac{dL}{dt} = \frac{\partial L}{\partial q} \frac{dq}{dt} + \frac{\partial L}{\partial \dot{q}} \frac{d\dot{q}}{dt} + \frac{\partial L}{\partial t}$$

The Euler-Lagrange equation in this context is

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q}$$

Substitute in the Euler-Lagrange equation into the total Lagrangian differential:

$$\frac{dL}{dt} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \dot{q} + \frac{\partial L}{\partial \ddot{q}} \ddot{q} + \frac{\partial L}{\partial t}$$

Note that

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \dot{q} + \frac{\partial L}{\partial \ddot{q}} \ddot{q} = \frac{d}{dt} \left(\dot{q} \frac{\partial L}{\partial \dot{q}} \right)$$

So,

$$\begin{aligned} \frac{dL}{dt} &= \frac{d}{dt} \left(\dot{q} \frac{\partial L}{\partial \dot{q}} \right) + \frac{\partial L}{\partial t} \\ 0 &= \frac{d}{dt} \left(\dot{q} \frac{\partial L}{\partial \dot{q}} \right) - \frac{dL}{dt} + \frac{\partial L}{\partial t} \\ 0 &= \frac{d}{dt} \left(\dot{q} \frac{\partial L}{\partial \dot{q}} - L \right) + \frac{\partial L}{\partial t} \end{aligned}$$

Since $KE = \frac{1}{2}m\dot{q}^2$, $L = KE - PE$, and PE is not a function of q , $p = \frac{\partial KE}{\partial \dot{q}} = \frac{\partial L}{\partial \dot{q}}$. This is the generalized momentum. Substituting this in,

$$0 = \frac{d}{dt} \underbrace{(\dot{q}p - L)}_H + \frac{\partial L}{\partial t}$$

So,

$$H = \dot{q}p - L(q, \dot{q}, t)$$

and since $\frac{\partial L}{\partial t} = 0$ for the least (or stationary) action, this is also the conversion between the Hamiltonian and Lagrangian.

23.5.2 Conversion is a Legendre transformation

Abbreviating for simplicity $L(q, \dot{q}, t) \rightarrow L(\dot{q})$ and $H(q, p, t) \rightarrow H(p)$,

$$H(p) = p\dot{q} - L(\dot{q})$$

which is of the same form as

$$G(s) = sx - F(x)$$

which is the Legendre transformation, where

$$\dot{q} \rightarrow p = \frac{\partial L}{\partial \dot{q}}$$

the conjugate pair, and more specifically the conjugate momentum in this case. The inverse conversion, from Hamiltonian back to the Lagrangian is an inverse Legendre transformation

$$L(\dot{q}) = p\dot{q} - H(p)$$

23.5.3 Hamiltonian as energy conservation

As $L = KE - PE = \frac{1}{2}m\dot{q}^2 - mgq$,

$$\begin{aligned} H &= \dot{q}p - L(q, \dot{q}, t) \\ &= m\dot{q}\dot{q} - \frac{1}{2}m\dot{q}^2 + mgq \\ &= \frac{1}{2}m\dot{q}^2 + mgq \\ &= KE + PE \end{aligned}$$

which is energy conservation.

23.5.4 Derivation of Hamilton's equations

Hamiltonian analog of Euler-Lagrange equations. Begin with

$$H(q, p, t) = p\dot{q} - L(q, \dot{q}, t)$$

$$\frac{\partial H}{\partial q}dq + \frac{\partial H}{\partial p}dp + \frac{\partial H}{\partial t}dt = p d\dot{q} + \dot{q}dp - \frac{\partial L}{\partial q}dq - \frac{\partial L}{\partial \dot{q}}d\dot{q} - \frac{\partial L}{\partial t}dt$$

Substitute in Euler-Lagrange: $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = \frac{d}{dt}p = \dot{p}$ and $\frac{\partial L}{\partial \dot{q}} = p$:

$$\frac{\partial H}{\partial q}dq + \frac{\partial H}{\partial p}dp + \frac{\partial H}{\partial t}dt = p d\dot{q} + \dot{q}dp - \dot{p}dq - p d\dot{q} - \frac{\partial L}{\partial t}dt$$

$$\frac{\partial H}{\partial q}dq + \frac{\partial H}{\partial p}dp + \frac{\partial H}{\partial t}dt = \dot{q}dp - \dot{p}dq - \frac{\partial L}{\partial t}dt$$

So,

$$\frac{\partial H}{\partial q} = -\dot{p}$$

$$\frac{\partial H}{\partial p} = \dot{q}$$

Instead of one second-order PDE (Euler-Lagrange), there are two first-order ODEs. There is additionally

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \rightarrow dH = -dL \rightarrow \frac{\partial H}{\partial L} = -1$$

but this is a restatement of

$$H(q, p, t) = \dot{q}p - L(q, \dot{q}, t)$$

23.5.5 Intuition of Hamilton's equations

Hamilton's equations

$$\begin{aligned} \frac{dx}{dt} &= \frac{\partial H}{\partial p} \Big|_{x,p} \\ \frac{dp}{dt} &= -\frac{\partial H}{\partial x} \Big|_{x,p} \end{aligned}$$

This can be re-expressed as

$$\begin{aligned} \frac{dx}{dt} &= S_x(x, p) = \frac{\partial H}{\partial p} \\ \frac{dp}{dt} &= S_p(x, p) = -\frac{\partial H}{\partial x} \end{aligned}$$

where $S(x, p) = (S_x, S_p)$ is a vector field on a p versus x plane. Including time as a parameter, $S(x(t), p(t))$. This traces a trajectory in the vector field, so $S(x(t), p(t)) = (S_x, S_p) = \left(\frac{x(t)}{dt}, \frac{p(t)}{dt}\right)$. $H(x, p)$ are contour lines tracing $S(x(t), p(t))$. Taking its gradient, $\nabla H(x, p) = \left(\frac{\partial H}{\partial x}, \frac{\partial H}{\partial p}\right)$. But $\nabla H(x, p)$ is perpendicular to $S(x(t), p(t))$. So rotate $\nabla H(x, p)$ clockwise 90 degrees (which is a transformation taking $+x \rightarrow -y, +y \rightarrow +x$): $\nabla H(x, p) = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial x}\right)$. Equating the two forms results in Hamilton's equations.

23.5.6 Zero divergence and flux

$$S = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial x}\right)'$$

is equivalent to

$$\text{div}(S) = \frac{\partial S_x}{\partial x} + \frac{\partial S_p}{\partial p}$$

23.5.7 Determinism and reversibility

Determinism is

$$q(t_f) = f(q(t_i))$$

$$|\Sigma(t_f)| \leq |\Sigma(t_i)|$$

and reversibility is

$$x(t_i) = \hat{f}(x(t_f))$$

$$|\Sigma(t_i)| \leq |\Sigma(t_f)|$$

and both if

$$\hat{f} = f^{-1}$$

$$|\Sigma(t_i)| = |\Sigma(t_f)|$$

where

$$q = (x, p)'$$

$$\Sigma = \begin{pmatrix} \sigma_x^2 & \sigma_x \sigma_p \\ \sigma_x \sigma_p & \sigma_p^2 \end{pmatrix}$$

For

$$|\Sigma(t_f)| = |\Sigma(t_i)|$$

$$|\Sigma(t + dt)| = |\Sigma(t)|$$

where $|\Sigma(t + dt)|$ can be made equivalent to $|J\Sigma(t + dt)J'|$ with a suitable J. That J is

$$|\Sigma(t + dt)| = |J\Sigma(t + dt)J'| = |J||\Sigma(t + dt)||J'| = |\Sigma(t)|$$

s.t. $|J| = 1$ (which preserves area). So,

$$|J| = \begin{vmatrix} \frac{\partial x(t+dt)}{\partial x} & \frac{\partial x(t+dt)}{\partial p} \\ \frac{\partial p(t+dt)}{\partial x} & \frac{\partial p(t+dt)}{\partial p} \end{vmatrix} = \begin{vmatrix} \frac{\partial(x+S_x dt)}{\partial x} & \frac{\partial(x+S_x dt)}{\partial p} \\ \frac{\partial(p+S_p dt)}{\partial x} & \frac{\partial(p+S_p dt)}{\partial p} \end{vmatrix}$$

$$= \begin{vmatrix} 1 + \frac{\partial S_x}{\partial x} dt & \frac{\partial S_x}{\partial p} dt \\ \frac{\partial S_p}{\partial x} dt & 1 + \frac{\partial S_p}{\partial p} dt \end{vmatrix} = 1 + \left(\frac{\partial S_x}{\partial x} + \frac{\partial S_p}{\partial p} \right) dt + O(dt^2) = 1$$

So,

$$div(S) = \frac{\partial S_x}{\partial x} + \frac{\partial S_p}{\partial p} = 0$$

which is zero divergence and flux.

23.5.8 Thermodynamic conservation

Since the Hamiltonian is deterministic and reversible, there is no energy added or removed. So,

$$\begin{aligned}\Delta E &= \oint dE = 0 \\ &= \oint dE_{KE} + dE_{work} = \oint (vdp - Fdx) = \oint \left(\frac{dx}{dt} dp - \frac{dp}{dt} dx \right) \\ &= \oint (S_x dp - S_p dx) = \oint \vec{S} \cdot d\vec{n} = \int \int \text{div}(S) dA = 0\end{aligned}$$

So,

$$\text{div}(S) = \frac{\partial S_x}{\partial x} + \frac{\partial S_p}{\partial p} = 0$$

again. Further, energy is path-independent and so is the Hamiltonian:

$$\Delta E = H(q_2) - H(q_1) = \oint \vec{S} \cdot d\vec{n}$$

23.5.9 Information conservation

Discrete Shannon entropy is $I = -\int \rho \log(\rho)$ but using the continuous analog leads to a Jacobian: $I = -\int \rho \log(|J|)$. Similar to thermodynamics, since the Hamiltonian is deterministic and reversible, there is no information added or removed (alternatively, can make the same argument noting that information is energy):

$$\begin{aligned}I(t_f) &= I(t_i) \\ I(t + dt) &= I(t) \\ I(t + dt) &= I(t) + \int \rho \log(|J|) = I(t)\end{aligned}$$

So it must be again that $|J| = 1$ as above, which is a transformation where the area is preserved.

23.5.10 Mapping

The Hamiltonian is just a bijective map in the discrete case, which in zero divergence in the continuous case.

24 Complex analysis

24.1 Algebra

24.1.1 Cartesian coordinates

Modulus is $|z| = \sqrt{x^2 + y^2}$. Multiplication is $(x + iy)(u + iv) = (xu - yv) + i(xv + yu)$. Division uses complex conjugate:

$$\frac{z}{w} = \frac{x + iy}{u + iv} \frac{u - iv}{u - iv}$$

$$\begin{aligned}
z\bar{z} &= (x+iy)(x-iy) = x^2 + y^2 = |z|^2 \\
\operatorname{Re} z &= \frac{z+\bar{z}}{2}, \operatorname{Im} z = \frac{z-\bar{z}}{2i} \\
\left(\frac{\bar{z}}{w}\right) &= \frac{\bar{z}}{\bar{w}}
\end{aligned}$$

Note. If $z = x + iy$, $\bar{z} = x - iy$:

$$\begin{aligned}
\operatorname{Re}(z) &= \frac{z+\bar{z}}{2} = x \\
\operatorname{Im}(z) &= \frac{z-\bar{z}}{2i} = y
\end{aligned}$$

24.1.2 Euler's formula and polar coordinates

$$\begin{aligned}
e^x &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{nx} \\
&= \lim_{n \rightarrow \infty} \left(1 + \frac{x}{nx}\right)^{nx} \\
&= \lim_{M \rightarrow \infty} \left(1 + \frac{x}{M}\right)^M
\end{aligned}$$

So,

$$e^i = \lim_{n \rightarrow \infty} \left(1 + i\frac{1}{n}\right)^n$$

which is a repeated multiplication of a complex number $a + ib$ and therefore a CCW rotation on the complex plane. This also leads to the Euler identity $e^{i\pi} = -1$.

By Taylor expansion,

$$\begin{aligned}
e^{ix} &= 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \dots \\
&= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \dots \\
&= \left(-\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots\right) \\
&= \cos(x) + i \sin(x)
\end{aligned}$$

So, $e^{i\theta} = \cos\theta + i \sin\theta$. Further:

$$\begin{aligned}
z &= r(\cos\theta + i \sin\theta) = r e^{i\theta} \\
e^{i(\theta+2\pi k)} &= e^{i\theta}
\end{aligned}$$

Intuition. Compound interest being rotated.

$$e^i = \lim_{n \rightarrow \infty} \left(1 + \frac{i}{n}\right)^n$$

Argument properties.

$$\arg(\bar{z}) = -\arg(z)$$

$$\arg\left(\frac{1}{z}\right) = -\arg(z)$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

where the last works up to modulo 2π and can be seen from $e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}$. With this result, $z_1 z_2 = r_1 e^{i\theta} r_2 e^{i\phi} = (r_1 r_2) e^{i(\theta+\phi)}$.

De Moivre's theorem.

$$(e^{i\theta})^n = (e^{in\theta}) = (\cos\theta + i \sin\theta)^2 = \cos(n\theta) + i \sin(n\theta)$$

24.1.3 Complex roots

If not already, convert to polar form: $re^{i\theta+2k\pi}$. Take n-th root: $\rho = r^{1/n}$ and $\phi = \frac{\theta}{n} + \frac{2k\pi}{n}$ where $k=0,1,\dots,n-1$ (just repeats after this). So compute the three cube roots of 8 from $8^{1/3}e^{i(\frac{0+2k\pi}{3})}$. The n-th roots of unity are computed from $1 = 1e^{i0}$, which is expanded to

$$= 1^{1/n} e^{i(\frac{0}{n} + \frac{2k\pi}{n})} = e^{i(\frac{2k\pi}{n})}$$

24.1.4 Analysis

A disk centered at z_0 with radius r is $|z - z_0| < r$, where $z_0 = x_0 + iy_0$.

A open set is where the entire set is interior, and a closed set contains all its boundary points. A set can be open, closed, neither, or both. A closure will add all boundaries if not already in the set. A separated set is where there are disjoint open sets.

24.2 Complex functions and iterations

Complex functions. For instance, $f(z) = iz + 2$ where $z = x + iy$. So it's essentially a composition. Intuition. If $f(z) = z^2$, then it's a rotation. If $f(z) = z^2 + c$ then it's a rotation and then adding c .

Iterations. Example. $f^2(z) = f(f(z))$, so not differentiation nor powers. Example. $f^2(z) = f(f(z))$ with $f(z) = z + 1$ and $f^n(z) = z + n$ for the n-th iteration.

24.2.1 Sequences and limits

Usual analysis concepts apply, including $\epsilon - \delta$ definitions of limits, limit addition, squeeze theorem, monotone convergence of bounded functions. Further:

(1) $\lim_{n \rightarrow \infty} z^n = 0$ for $|z| < 1$.

(2) If $\lim_{n \rightarrow \infty} |s_n| \rightarrow 0$, then $\lim_{n \rightarrow \infty} s_n \rightarrow 0$.

(3) $\lim_{z \rightarrow i} \text{Arg}(z) = \frac{\pi}{2}$, $\lim_{z \rightarrow 1} \text{Arg}(z) = 0$, $\lim_{z \rightarrow -1} \text{Arg}(z)$ doesn't exist $\in -\pi < \text{Arg}(z) \leq \pi$.

24.2.2 Iteration of quadratics

The quadratic $f(z) = z^2 + c$ behaves equivalently under iteration to $p(z) = az^2 + bz + d$ if $c = ad + \frac{b}{2} - \left(\frac{b}{2}\right)^2$ and $\phi(z) = az + \frac{b}{2}$, using $p(z) = \phi^{-1}(f(\phi(z))) = \phi^{-1} \circ f \circ \phi$. This is because

$$p \circ p = (\phi^{-1} \circ f \circ \phi) \circ (\phi^{-1} \circ f \circ \phi) = \phi^{-1} \circ f \circ f \circ \phi$$

$$p^2 = \phi^{-1} \circ f^2 \circ \phi$$

$$p^n = \phi^{-1} \circ f^n \circ \phi$$

Example 1. $f(z) = z^2 \rightarrow f^n(z) = z^{2^n}$, so $z = re^{i\theta} \rightarrow f^n(z) = r^{2^n} e^{i2^n\theta}$. If $|z| < 1 \rightarrow |f^n(z)| = |z|^{2^n} \rightarrow 0$ as $n \rightarrow \infty$, if $|z| > 1 \rightarrow |f^n(z)| \rightarrow \infty$, and if $|z| = 1 \rightarrow z = e^{i\theta}$ since $r = 1$ so $|f^n(z)| = 1$ for all n . So $|z| = 1$ leads to chaotic behavior and is a Julia set.

Julia set: $J(f) = \{z : |z| = 1\}$ which is closed and bounded. Fatou set: $F(f) = \{z : |z| > 1 \cup |z| < 1\}$ which is open and unbounded. The Julia and Fatou sets are invariant: $f(J) = J$ and $f(F) = F$.

Example 2. $f(z) = z^2 - 2$. Let $\phi = w + \frac{1}{w}$. Note:

$$f(\phi) = f(z) = z^2 - 2 = \left(w + \frac{1}{w}\right)^2 - 2 = w^2 + \frac{1}{w^2} = \phi(w^2)$$

So,

$$\phi^{-1}(f(\phi)) = \phi^{-1}(\phi(w^2)) = w^2$$

$$f(z) = \phi(g(\phi^{-1}(w)))$$

where $g(w) = w^2$ and since this iterates to ∞ , $f^n(z) \rightarrow \infty$. This means $J(f) = [-2, 2]$ and the basin of attraction to infinity $A(\infty) \in \mathcal{C}$ excluding $[-2, 2]$.

24.2.3 Julia and Mandelbrot sets

Julia sets are $f(z) = z^2 + c$ where c is chosen and z is the variable, and the filled-in set is where the iterates don't explode. Mandelbrots are where $z=0$ and c is the parameter where the Julia set is connected.

Theorem. Julia set is connected if $|f^n(0)| \leq 2$.

Misiurewicz point. A preperiodic point in a Mandelbrot set.

24.3 Complex derivative

For a limit to exist at $f(z_0)$, its limits from both the imaginary axis and the real axis must be equal and exist.

Complex functions are differentiable just like functions in $\mathcal{R}$. So product and quotient rules, for instance, are equally applicable.

If a point is differentiable, it is continuous. Proof:

$$\lim_{z \rightarrow z_0} f(z) - f(z_0) = \lim_{z \rightarrow z_0} \left(\frac{f(z) - f(z_0)}{z - z_0} (z - z_0) \right) = f'(z_0) \cdot 0 = 0$$

A function is analytic if it is complex differentiable at each point. If it's differentiable everywhere on $\mathcal{C}$, then it's entire. Polynomials are analytic and entire functions, while $f(z) = |z|$ is not analytic (while it's differentiable at $z=0$, that's the only point and one point isn't enough to be analytic). Further, $f(z) = \text{Re}(z)$, $f(z) = \text{Im}(z)$ are not analytic.

24.4 Cauchy-Riemann equations

If $f(z) = z^2$ where $z = x + iy$ and so $f(z) = (x^2 - y^2) + i2xy = u(x, y) + iv(x, y)$. Then $f'(z) = 2z = 2x + i2y = u_x + iv_x = f_x(z) = -i(u_y + iv_y) = if_y(z)$. So:

$$u_x = v_y$$

$$u_y = -v_x$$

which are the Cauchy-Riemann equations. If $u_x \neq v_y$ at a point, then f is not differentiable there. If the Cauchy-Riemann equations are satisfied at a point AND the first partial derivatives are continuous (e.g. there can't be discontinuities), then the function is analytic at that point.

Matrix $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ is pure scaling and rotation (it does not scale different directions differently, nor does it shear). This is exactly the type of operation that is represented by multiplication by a complex number. In fact, $A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ with $(a + bi)(x + yi)$, where $\begin{bmatrix} x \\ y \end{bmatrix} \leftrightarrow x + iy$. Note further that $\begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.

So the Cauchy-Riemann equations indicate the Jacobian of the functions $u(x, y)$ and $v(x, y)$ is of this form (the diagonal elements are equal, the off-diagonal elements are exactly opposite) which means that it can be represented as multiplication by some complex number. This is needed in order to have the derivative of your complex function at a point be another complex number rather than a 2×2 matrix.

More intuition for Cauchy-Riemann. In a rotation, which is complex numbers do, $\frac{du}{dx} = \frac{dv}{dy}$ because a stretch of the radius in the u-direction should also result in a stretch in the v-direction. Further, $\frac{du}{dy} = -\frac{dv}{dx}$ is a rotational stretch so if the u-direction rotates upward (the positive y-direction), the v-direction rotates in the negative x-direction.

24.4.1 Complex exponentials and logarithms

The complex exponential is a spiral. It is e^z where $z = x + iy$:

$$e^x \cos y + ie^x \sin y = e^x (\cos y + i \sin y) = e^x e^{iy} = e^{x+iy} = e^z$$

Properties: (1) $|e^z| = |e^x| |e^{iy}| = |e^x| = e^x$. note that $e^{iy} = \cos y + i \sin y$ is a circle of radius 1.

(2) $\arg(e^z) = \arg(e^x e^{iy}) = y (+2k\pi)$.

(3) $e^{\bar{z}} = e^{x-iy} = e^x e^{-iy} = \bar{e^z}$, which is a horizontal reflection.

(4) If $e^z = e^w \rightarrow e^{z-w} = 1 \rightarrow z = w + 2k\pi$.

(5) Suppose $w = e^z$ and $z, w \in \mathcal{C}$, which maps w to z. With $w = x + iy_0$, a series of horizontal lines in w becomes rays from the origin in z. With $w = x_0 + iy$, a series of vertical lines in z becomes circles in z.

Complex log. When is $e^w = z$? If $w = u + iv$, $z = |z|e^{i\theta}$, then

$$e^w = e^u e^{iv} = |z|e^{i\theta}$$

So $v = \theta + 2k\pi$, and

$$u = \ln|z|$$

So $w = \ln|z| + i \arg(z)$. The complex log is the inverse of the complex exponential. With $e^{i\pi} = -1$,

$$\ln(-1) = \ln(e^{i\pi}) = i\pi$$

$$\ln(z) = \ln(re^{i\theta}) = \ln(r) + i\theta$$

where $-\pi < \theta \leq \pi$.

24.4.2 Complex trigonometry

As

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta$$

Then

$$e^{i\theta} + e^{-i\theta} = 2\cos\theta \rightarrow \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$e^{i\theta} - e^{-i\theta} = 2i\sin\theta \rightarrow \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Complex hyperbolic trigonometry:

$$\begin{aligned}\sin(z) &= \sin(x + iy) = \sin(x)\cos(iy) + \cos(x)\sin(iy) \\ &= \sin(x)\frac{e^y + e^{-y}}{2} + i\cos(x)\frac{e^y - e^{-y}}{2} \\ &= \sin(x)\cosh(y) + i\cos(x)\sinh(y)\end{aligned}$$

Can similarly show that:

$$\cos(z) = \cos(x)\cosh(y) - i\sin(x)\sinh(y)$$

24.4.3 Analytic function properties

If $f(z)$ is analytic on domain D (must be connected), and if $f'(z) = 0$ for all $z \in D$, then $f(z)$ is constant. Note, for $f'(z) = 0$, it's not enough for $u_x = 0$, as this only part of the requirement for $f'(z) = 0$.

If $f(z) = u + iv$ is analytic in D and u is a constant in D , AND the above condition is satisfied, then f must be constant in D . If u is constant, then $u_x = 0$, and since f must be constant, then $u_y = 0 = v_x = v_y$. So if u is constant, v must also be constant.

If $|f|^2$ is constant, then $u^2(z) + v^2(z)$ (Pythagorean or Euclidean distance) is constant in D . There is an algebraic proof for this.

24.4.4 Inverse functions

As $z = |z|e^{i\theta}$, $\text{Log}(z) = \ln|z| + i\text{Arg}(z)$ which is the principal branch of the log, while $\log(z) = \ln|z| + i\arg(z)$ is the multivalued function. Other branches lie on different complex 'planes'. So $\text{Log}(z) + 2k\pi i$. Further, note that $|i| = 1$, $\text{Arg}(i) = \frac{\pi}{2}$, and that for complex logs, there is a discontinuity on the negative real axis as the argument is defined continuously over $-\pi < \theta \leq \pi$. To calculate complex logs:

$$\begin{aligned}e^{\text{Log}(z)} &= z \\ e^{\text{Log}(z)} \frac{d}{dz} \text{Log}(z) &= 1 \\ \frac{d}{dz} \text{Log}(z) &= \frac{1}{z} = z^{-1}\end{aligned}$$

Inverse function theorem. If $f : U \rightarrow \mathcal{C}$ is analytic and $g : D \rightarrow U$, and $f(g(z)) = z$, then g is analytic in D and, using the product rule:

$$g'(z) = \frac{1}{f'(g(z))}$$

Example. $f(z) = z^2$, so $g(z) = \sqrt{z^2} = \pm\sqrt{z}e^{i(\frac{\text{Arg}(z)}{2} + \pi)}$. Square roots as approach the negative x-axis (i.e., -1) map to $\sqrt{-1} = +\sqrt{z^2}$. If pass -1, then $-\sqrt{z} = -\sqrt{-1} = -i$.

24.5 Conformal mapping

A function g is conformal if it preserves angles, specifically tangents, between curves at intersection points. To find tangent, differentiate with respect to parameter, e.g. t in $\gamma(t) = x(t) + iy(t)$.

If $\gamma_1 \angle \gamma_2 \rightarrow g(\gamma_1) \angle g(\gamma_2)$. Note: $g'(z) \neq 0$ must hold. If g is analytic and $g'(z_0) \neq 0$, then it's conformal. If $\gamma_1(t_0) = \gamma_2(t_1)$ intersect,

$$(g \circ \gamma_1)'(t_0) = g'(\gamma_1(t_0))\gamma_1'(t_0) = g'(z_0)\gamma_1'(t_0)$$

$$(g \circ \gamma_2)'(t_1) = g'(\gamma_2(t_1))\gamma_2'(t_1) = g'(z_0)\gamma_2'(t_1)$$

If g is further bijective, then it's a conformal mapping.

24.5.1 Mobius transformations

Also called fractional linear transformations. If $\hat{\mathcal{C}} = \mathcal{C} \cup \infty$ is the extended complex plane, then Mobius transformations are conformal mappings (in fact, the only mappings) of $\hat{\mathcal{C}} \rightarrow \hat{\mathcal{C}}$.

$$f(z) = \frac{az + b}{cz + d}$$

Note: $ad - bc \neq 0$ since it must be that $f'(z) \neq 0$:

$$f'(z) = \frac{ad - bc}{(cz + d)^2}$$

Example 1. Bijection from $z \in \hat{\mathcal{C}} \rightarrow w \in \hat{\mathcal{C}}$,

$$f(z) = w$$

$$az + b = w(cz + d)$$

$$\frac{wd - b}{-wc + a}$$

Example 2. $c = 0, d = 1$, so $f(z) = az + b$, an affine transformation where a is a rotation and stretch, and $a = |a|e^{i\theta}$, $\infty \rightarrow \infty, \mathcal{C} \rightarrow \mathcal{C}$.

Example 3. Inversion, $a = 0, b = 1, c = 1, d = 0$. So $f(z) = \frac{1}{z}$. So $z = re^{i\theta} \rightarrow \frac{1}{z} = \frac{1}{r}e^{-i\theta}$. Exchanges outside and inside of unit circle. $|z - 3| = 1$ becomes $|w - \frac{3}{8}| = \frac{1}{8}$, another circle.

Mobius transformations are transformations of circles and lines to circles and lines. Lines are equivalent to infinite circles.

Compositions of Mobius transformations are also Mobius transformations. If $g(z) = \frac{ez+f}{gz+h}$ then

$$f(z) = \frac{az + b}{cz + d}$$

$$f(g(z)) = \frac{ag(z) + b}{cg(z) + d} = \frac{z(\cdot) + (\cot)}{z(\cdot) + (\cdot)}$$

Inverses of Mobius transformations are also Mobius transformations. If $w = \frac{az+b}{cz+d}$, then solve for $z = g(w)$ which will be in the form of another Mobius transformation.

Mobius transformations are only of 3 types: rotations/stretches, translations, and inversions. All Mobius transformations can be constructed from these operations or decomposed algebraically and conceptually into these. This is why Mobius transformations map circles and lines to circles and lines.

The Mobius transformation:

$$f(z) = \frac{z - z_1}{z - z_0} \frac{z_2 - z_3}{z_2 - z_1}$$

maps $f(z_1) = 0, f(z_2) = 1, f(z_3) = \infty$. Using this, can calculate for instance a Mobius transformation from $z_1, z_2, z_3 \rightarrow (w_1, w_2, w_3)$ by (1) calculating $f_1(z)$ with $z_1, z_2, z_3 \rightarrow (0, 1, \infty)$, (2) calculating $f_2(z)$ with $w_1, w_2, w_3 \rightarrow (0, 1, \infty)$, (3) solving for f_2^{-1} , (4) calculating $f = f_2^{-1} \circ f_1$. Another way to calculate this is to use $f(z_1) = w_1, f(z_2) = w_2, f(z_3) = w_3$.

To map $\hat{\mathcal{C}} \rightarrow \hat{\mathcal{C}}$, $\frac{az+b}{cz+d}$ because if $f(\infty) = \infty$ then $c = 0$ and if $f(\infty) \neq \infty$ then $c \neq 0$. To map $\mathcal{C} \rightarrow \mathcal{C}$, $az + b$ as $c = 0$. There is no conformal mapping from a subset of $\mathcal{C}$ or $\hat{\mathcal{C}}$ - for this, use the Riemann mapping theorem.

24.5.2 Riemann mapping

Riemann mapping theorem. If D is a simply connected domain (no holes, open, and connected) and a subset of $\mathcal{C}$, there is a conformal mapping $D \rightarrow \mathcal{D}$ of the unit disk. D is then conformally equivalent to $\mathcal{D}$. Proof uses pure math arguments (from topology, group theory, etc.).

24.6 Complex integration

Basically a path integral. In $\mathcal{C}$, integrate over curves. For $\gamma(t), t \in [a, b]$, sum the tangents with the value of the function:

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

Example 1. Alternative integrations exist:

$$\begin{aligned} \int_0^{\pi} e^{it} dt &= \int_0^{\pi} (\cos(t) + i \sin(t)) dt = 2i \\ \int_0^{\pi} e^{it} dt &= -ie^{it} = 2i \end{aligned}$$

Example 2. For $z = e^{it}, 0 \leq t \leq 2\pi$:

$$\int_{|z|=1} \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{\gamma(t)} \gamma'(t) dt$$

Example 3. If $\gamma(t)$ is defined over $0 \leq t < 1$ and $f(z) = \operatorname{Re}(z)$, and

$$\int_{\gamma} f(z) dz = \int_0^1 \operatorname{Re}(\gamma(t)) \gamma'(t) dt$$

Example 4. If $f(z) = \bar{z}$,

$$\int_{\gamma} f(z) dz = \int \overline{\gamma(t)} \gamma'(t) dt$$

Generally,

$$\int_{|z|=1} z^m dz = \begin{cases} 2\pi i, & m = -1 \\ 0, & \text{Otherwise} \end{cases}$$

Substitution is same as real integration. Moreover, the final result as in real integration is independent of the parameterization.

24.6.1 Reverse paths

Define $\gamma : [a, b] \rightarrow \mathcal{C}$. Then $(-\gamma) : [a, b] \rightarrow \mathcal{C}$. Then

$$(-\gamma)(t) = \gamma(a + b - t)$$

which then leads to $(-\gamma)(a) = \gamma(b), (-\gamma)(b) = \gamma(a)$.

$$\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz$$

24.6.2 Arc lengths

Arc lengths. Add tangents.

$$\int_a^b |\gamma'(t)| dt$$

Note $|e^{it}| = 1, |i| = 1, |1 + i| = \sqrt{2}$, etc.

24.6.3 Integration with respect to arc length

Integrate the function's value over the arc length.

$$\int_{\gamma} f(z) |dz| = \int_a^b f(\gamma(t)) |\gamma'(t)| dt$$

$$\text{Length} = \int_{\gamma} |dz| = \int_{\gamma} |d|$$

ML-estimate. Maximum-length estimate. Provides upper bound inequality.

$$\left| \int_{\gamma} f(z) dz \right| \leq \int_{\gamma} |f(z)| |dz|$$

which is similar to the triangle inequality. If $|f(z)| \leq M$ on γ ,

$$\left| \int_{\gamma} f(z) dz \right| \leq M \cdot L$$

$$\left| \int_{\gamma} f(z) dz \right| \leq \int |\gamma(t)| |\gamma'(t)| dt$$

where M is the maximum and L is the length. Both are upper bounds.

24.6.4 FTC of analytic functions

Analogous to FTC for reals. The complex analogue of antiderivatives in the primitive. If f is continuous on a domain D and if f has a primitive F in D , then for any curve $\gamma : [a, b] \rightarrow D$, then $\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a))$.

Note: if the function's path endpoints of integration are undefined, then use limit to approach it. For instance, $F(z) = \text{Log}(z)$ is not defined on the negative real axis $(-\infty, 0]$. So,

$$\int_{\gamma} \frac{1}{z} dz = \int_{\tilde{\gamma}} \frac{1}{z} dz = \text{Log}(\tilde{\gamma}(b)) - \text{Log}(\tilde{\gamma}(a))$$

Goursat's theorem. Let D be a simply connected domain in $\mathcal{C}$ and let f be analytic in D . Then f has a primitive in D . Moreover, a primitive is given explicitly by picking $z_0 \in D$ and letting $F(z) = \int_{z_0}^z f(w) dw$ where the integral is taken over an arbitrary curve in D . So f has a primitive under these conditions.

Cauchy theorem for triangles. Let D be an open set in $\mathcal{C}$ and let f be analytic in D . Let T be a triangle that fits into D (including its boundary) and let ∂T be its boundary, oriented positively. Then $\int_{\partial T} f(z) dz = 0$.

Morera's theorem. If f is continuous on a simply connected domain D , and if $\int_{\gamma} f(z) dz = 0$ for any triangular curve in D , then f has a primitive in D .

24.7 Cauchy theorems

Cauchy's theorem for simply connected domains. Let D be a simply connected domain in $\mathcal{C}$, f be analytic in D , $\gamma : [a, b] \rightarrow D$ be a piecewise smooth, closed curve in D (so $\gamma(a) = \gamma(b)$). Then:

$$\int_{\gamma} f(z) dz = 0$$

Proof idea. Shrink loop to a point, since there are no holes. Then integral is equal to zero.

Corollary. If γ_1, γ_2 are simple closed curves (so don't intersect themselves) oriented counter-clockwise, with γ_2 inside γ_1 . If f is analytic in the domain, that contains both curves and the 'area' between them. Then,

$$\int_{\gamma_1} f(z)dz = \int_{\gamma_2} f(z)dz$$

Note that holes are possible, particularly inside γ_1 .

Cauchy's integral formula. Let D be simply connected, bounded by a piecewise smooth curve γ and let f be analytic in a set U that contains the closure of D (in other words, contains D and γ). Then:

$$f(w) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-w} dz$$

for all $w \in D$. So can calculate the f -value of w using only γ , the boundary. Note if $w \notin D$ but f is analytic in D , use the theorem for simply connected domains and its corollary above. Proof of Cauchy's integral formula. Let $z = \gamma(t) = w + \epsilon e^{it}$ where ϵ is the ball radius around a point z . Then,

$$\begin{aligned} \frac{1}{2\pi} \int_{\partial Ball} \frac{f(z)}{z-w} dz &= \frac{1}{2\pi} \int_0^{2\pi} \frac{f(\gamma(t))}{\gamma(t)-w} \gamma'(t) dt \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(\gamma(t))}{\epsilon e^{it}} \epsilon i e^{it} dt = \frac{1}{2\pi} \int_0^{2\pi} f(w + \epsilon e^{it}) dt \end{aligned}$$

and as $\epsilon \rightarrow \infty$, $\frac{1}{2\pi} \int_0^{2\pi} f(w) dt$.

Theorem (consequence of Cauchy's integral formula). If f is analytic on an open set U , then its derivative f' is also analytic in U . This can be repeated for further derivatives. So once f is found analytic, then all derivatives are analytic. Proof idea. Using

$$\lim_{h \rightarrow 0} \frac{f(w+h) - f(w)}{h}$$

or the quotient rule,

$$f'(w) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-w)^2} dz$$

and show it's analytic. The Cauchy formula for derivatives follows.

Cauchy formula for derivatives. If D is a simply connected domain, bounded by a piecewise smooth curve γ , f is analytic in U that contains closure of D (so includes both D and γ). Then,

$$f^k(w) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-w)^{k+1}} dz$$

24.7.1 Cauchy estimate

If f is analytic in an open set that contains closed ball $B_r(z_0)$ centered at z_0 and having radius r , $|f(z)| \leq m$, then

$$|f^k(z_0)| \leq \frac{k!m}{r^k}$$

Proof uses Cauchy's integral formula.

$$|f^k(z_0)| = \frac{k!}{2\pi} \left| \int_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{k+1}} dz \right| \leq \frac{k!}{2\pi} \int_{|z-z_0|=r} \frac{|f(z)|}{|z-z_0|^{k+1}} |dz|$$

which uses the reverse triangle inequality.

$$\leq \frac{k!m}{2\pi r^{k+1}} 2\pi r = \frac{k!m}{r^k}$$

Note:

$$\int_{|z-z_0|=r} |dz| = 2\pi r$$

24.7.2 Liouville's theorem

If f is analytic in $\mathcal{C}$ and is an entire function. If f is bounded then f must be constant.

This is different from $\mathcal{R}$, where for instance $y = \sin(x)$ is bounded yet isn't constant. For $\mathcal{C}$, $\sin(z) = \frac{e^{ix} - e^{-ix}}{2i}$ then $x = ni$ makes the function unbounded.

Proof of Liouville's theorem. Suppose $|f(z)| \leq m$ for all $z \in \mathcal{C}$. Pick z_0 . Since $\mathcal{C}$ contains closed ball $B_r(z_0)$, Cauchy's estimate with $k = 1$ leads to $|f'(z_0)| \leq \frac{m}{r}$. Let $r \rightarrow \infty$ then $f'(z_0) = 0$.

Application. If $f = u + iv$ has $u(z) \leq 0$, then f is constant. Consider $g(z) = e^{f(z)}$ then $|g(z)| = e^{\operatorname{Re}(f(z))} = e^{u(z)} \leq e^0 = 1$. So it's bounded, so g is constant, so f is constant: $g'(z) = e^{f(z)} f'(z)$ leads to $0 = e^{f(z)} f'(z)$ so it must be that $f'(z) = 0$.

Note: $|e^z| = e^{\operatorname{Re}(z)} = e^x$.

24.7.3 Fundamental theorem of algebra

Any polynomial $p(z) = a_0 + a_1z + \dots + a_nz^n$ where $a_n \neq 0$ has a zero in $\mathcal{C}$, there exists $z_0 \in \mathcal{C}$ s.t. $p(z_0) = 0$.

Proof. Suppose to the contrary there exists $p(z)$ as in the theorem that has no zeros. Then, $f(z) = \frac{1}{p(z)}$ is an entire function. Since $p(z) \neq 0$ then $f(z)$ is a bounded function. But since it's bounded, f is constant. But then p is constant. But p is a polynomial with a variables in it so not constant. So there's a contradiction.

24.7.4 Maximum modulus principle

Let f be analytic in D and suppose there exists $z_0 \in D$ s.t. $|f(z)| \leq |f(z_0)|$ for all $z \in D$. Then f is constant in D .

If f is analytic and continuous in D , $|f|$ must be maximum on ∂D . This is the inverse of the above statement.

24.8 Series

24.8.1 Infinite series

$$\sum_{k=0}^{\infty} z^k = \frac{1}{1-z}$$

If $|z| \geq 1$, then it diverges for $z \in \mathcal{C}$.

If $z = re^{i\theta}$, then $z^k = r^k e^{ik\theta} = r^k \cos(k\theta) + ir^k \sin(k\theta)$,

$$\begin{aligned} \sum z^k &= \sum r^k \cos(k\theta) + i \sum r^k \sin(k\theta) \\ \frac{1}{1-z} &= \frac{1}{1-re^{i\theta}} = \frac{1-re^{-i\theta}}{(1-re^{i\theta})(1-re^{-i\theta})} \\ &= \frac{1-r\cos\theta + ir\sin\theta}{1-2r\cos\theta + r^2} = \frac{1-r\cos\theta}{1-2r\cos\theta + r^2} + \frac{ir\sin\theta}{1-2r\cos\theta + r^2} \end{aligned}$$

Harmonic series. $\sum \left| \frac{z^k}{k} \right| = \sum \frac{1}{k}$ diverges:

$$\sum \frac{1}{k} = 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4} \right)}_{\geq 1/2} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right)}_{\geq 1/2} + \underbrace{\left(\frac{1}{9} + \dots + \frac{1}{16} \right)}_{\geq 1/2} + \dots$$

So there are $n \rightarrow \infty$ of $\frac{n}{2}$ so diverges.

Alternating harmonic series. $\sum_{k=1}^{\infty} \frac{i^k}{k} = \sum_{n=1}^{\infty} \frac{i^{2n}}{2n} + \sum_{n=1}^{\infty} \frac{i^{2n+1}}{2n+1}$ when $k = 2n \rightarrow i^k = (-1)^n$ is real. So,

$$= \frac{1}{2} \sum \frac{(-1)^n}{n} + i \sum \frac{(-1)^n}{2n+1}$$

converges because

$$\sum \frac{(-1)^n}{n} = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \dots$$

becomes smaller and smaller as a net sum.

Absolute convergence. $\sum a_k$ converges absolutely if $\sum |a_k|$ converges. Absolute convergence means $\left| \sum a_k \right| \leq \sum |a_k|$. Further,

$$\left| \sum z^k \right| \leq \sum |z|^k \rightarrow \left| \frac{1}{1-z} \right| \leq \frac{1}{1-|z|}$$

24.8.2 Power series

A power series is

$$\sum_0^{\infty} a_k (z - z_0)^k$$

The radius of convergence theorem. In a power series, there exists an R (convergence radius) where the series absolutely converges for $|z - z_0| < R$ and diverges in $|z - z_0| > R$. Uniform convergence in $|z - z_0| \leq r$. R can be 0 in some cases, and it can be ∞ in others.

Theorem. In a power series with $R > 0$,

$$f(z) = \sum_0^{\infty} a_k (z - z_0)^k$$

is analytic in $|z - z_0| < R$. Then can differentiate term by term:

$$f'(z) = \sum_1^{\infty} a_k k (z - z_0)^{k-1}$$

$$f''(z) = \sum_2^{\infty} a_k k(k-1)(z - z_0)^{k-2}$$

and etc. In particular,

$$f^k(z_0) = a_k k!$$

$$a_k = \frac{f^k(z_0)}{k!}$$

Power series integration. For any power series with $|w - z_0| < R$,

$$\int_{z_0}^w \sum_0^{\infty} a_k (z - z_0)^k dz = \sum_0^{\infty} a_k \int_{z_0}^w (z - z_0)^k dz = \sum_0^{\infty} a_k \frac{1}{k+1} (w - z_0)^{k+1}$$

Integral is taken over any curve in disk $|z - z_0| < R$ from $z_0 \rightarrow w$.

24.8.3 Radius of convergence

Recall the power series is analytic in the convergence radius R .

Ratio test. If the sequence $\left\{ \left| \frac{a_k}{a_{k+1}} \right| \right\}$ has a limit as $k \rightarrow \infty$ then this limit is R , the convergence radius, of $\sum_0^{\infty} a_k (z - z_0)^k$. So if $\sum_0^{\infty} z^k$ then $z_0 = 0, a_k = 1$ and $\left| \frac{a_k}{a_{k+1}} \right| \rightarrow 1$ as $k \rightarrow \infty$, and so $R = 1$.

Note: $a_k = \left| \frac{1}{k^k} \right| \rightarrow \infty$.

Root test. If $\{|a_k|^k\}$ has a limit as $k \rightarrow \infty$, then

$$R = \frac{1}{\lim_{k \rightarrow \infty} \{|a_k|^k\}}$$

Note: the ratio and root tests don't work if there are consistently fluctuating values, i.e. no convergence.

Note: $|(-1)^k| = 1$ as the radius is consistently 1. Further, $|k^{\frac{1}{k}}| = 1$.

Cauchy-Hadamard criterion. This equation always works, even if the ratio and root tests don't work, to calculate R:

$$R = \frac{1}{\limsup_{k \rightarrow \infty} |a_k|^k}$$

where *limsup* is the largest accumulation point. If the sequence is $\{0, 0.5, 0, 0.5, 0, \dots\}$, *limsup* = 0.5.

Theorem. Let $f : U \rightarrow \mathcal{C}$ be analytic and let $|z - z_0| < r \in U$ (note: $R > r$). Within this disk, f has power series representation

$$f(z) = \sum_0^{\infty} a_k (z - z_0)^k$$

where

$$a_k = \frac{f^k(z_0)}{k!}$$

Note: a_k 's equation must be not just for the first derivative, but for the k-derivative.

Note: a_k can be expanded, particularly for 'cyclic' series like sin and cos.

$$\sin(z) = \sum_0^{\infty} \frac{(-1)^k}{(2k+1)!} z^{2k+1}$$

$$\cos(z) = \frac{d}{dz} \sin(z) = \sum_0^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}$$

Corollary. If f and g are analytic in $|z - z_0| < r$ and $f^k(z_0) = g^k(z_0)$, then $f(z) = g(z)$.

24.9 Riemann zeta function and hypothesis

Euler showed, using the ζ function, that $\sum_{p=\text{prime}} \frac{1}{p}$ diverges. It diverges because prime numbers don't have big gaps between them.

Recall the harmonic series $\sum_1^\infty \frac{1}{n}$ diverges but $\sum_1^\infty \frac{1}{n^s}$ converges if $s > 1$. The series can be viewed as the area under $f(x) = \frac{1}{x^s}$ as rectangles starting at $x = 1$. So,

$$\sum_1^\infty \frac{1}{n^s} \leq 1 + \int_1^\infty \frac{1}{x^s} dx = \frac{s}{s-1}$$

so it's finite and bounded.

Riemann zeta function. If $s \in \mathcal{C}$ with $Re(s) > 1$,

$$\zeta(s) = \sum_1^\infty \frac{1}{n^s}$$

$n^s = e^{\ln n^s} = e^{s \ln n} = n^s$. Further, $|n^s| = |e^{s \ln n}| = e^{Re(s \ln n)} = n^{Re(s)}$.

$$\sum_1^\infty \left| \frac{1}{n^s} \right| = \sum \frac{1}{n^{Re(s)}}$$

So $Re(s) > 1$ converges. So, $\sum \frac{1}{n^s}$ converges absolutely over $Re(s) > 1$. In fact, ζ is analytic over $Re(s) > 1$.

Theorem. ζ has an analytic continuation into $\mathcal{C}$ excluding 1 and this continuation satisfies that $\zeta(s) \rightarrow \infty$ as $s \rightarrow 1$. Proof is difficult. But consider proof for more limited case over $Re(s) > 0$:

$$\sum_1^N \frac{1}{n^s} = \int_1^{N+1} \frac{1}{x^s} dx + \sum_1^N \delta_n(s)$$

where $\delta_n(s) = \frac{1}{n^s} - \int_n^{n+1} \frac{1}{x^s} dx$. The second right term contains pieces above the curve $\frac{1}{x^s}$. Then,

$$\zeta(s) = \frac{1}{s-1} + H(s)$$

where $H(s)$ is analytic over $Re(s) > 0$.

Theorem. The only zeros of $\zeta(s)$ are outside the strip $0 \leq Re(s) \leq 1$ are at negative even integers (-2,-4,-6, etc.). There are no zeros on $Re(s) = 1$ nor $Re(s) = 0$. There is an unknown between these regions, but Riemann hypothesized that in this region, all zeros lie on $Re(s) = \frac{1}{2}$.

24.9.1 Prime number theorem

$\pi(x)$ is the prime counting function which is equal to the number of prime numbers below x . Example. $\pi(6) = 3$.

Prime number theorem. $\pi(x) \sim \frac{x}{\ln x}$, or

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{x/\ln x} = 1$$

Euler showed that

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

for $s > 1, \operatorname{Re}(s) > 1$. Proof.

$$\begin{aligned} \zeta(s) &= \frac{1}{1^s} + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \frac{1}{5^s} + \frac{1}{6^s} + \dots \\ &= \left(1 + \frac{1}{2^s} + \frac{1}{4^s} + \dots\right) \left(1 + \frac{1}{3^s} + \frac{1}{9^s} + \dots\right) \left(1 + \frac{1}{5^s} + \frac{1}{25^s} + \dots\right) \dots \\ &= \prod_p \sum_{k=0}^{\infty} \frac{1}{p^{ks}} = \prod_p \sum_{k=0}^{\infty} (p^{-s})^k = \prod_p \frac{1}{1-p^{-s}} = \prod_p \frac{1}{1-p^{-s}} \end{aligned}$$

which shows that $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$ since $\frac{1}{1-p^{-s}} \neq 0$.

Prime number theorem further leads to $\pi(x) \sim \operatorname{Li}(x)$,

$$\operatorname{Li}(x) = \int_2^x \frac{1}{\ln t} dt$$

which is the offset log integral function. Further, it was shown that $\pi(x) = \operatorname{Li}(x) + \text{error}$ where the error term grows to infinity at a controlled rate. It was still further shown that $|\pi(x) - \operatorname{li}(x)| < \frac{\sqrt{x} \ln x}{8\pi}$ is equivalent to the Riemann hypothesis. Note $\operatorname{li}(x) = \int_0^x \frac{1}{\ln t} dt$ where $\operatorname{Li}(x) = \operatorname{li}(x) - \operatorname{li}(2)$.

24.10 Laurent series

There is a power series representation if f is analytic in a disk. But what if there is no such disk, and just an annulus?

If $f : U \rightarrow \mathcal{C}$ is analytic and $r < |z - z_0| < R \in U$ then f has a Laurent series:

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z-z_0)^k = \dots + \frac{a_{-2}}{(z-z_0)^2} + \frac{a_{-1}}{(z-z_0)} + a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \dots$$

that converges at each point of the annulus and converges absolutely in each sub annulus $s \leq |z - z_0| \leq t$ where $r < s < t < R$. The annulus has inner radius r and outer radius R .

To use Laurent series, can find various annuli but the f function will be the same. Rewrite the power series to fit that particular annulus so that $|a_k| < 1$. Note: to find a_{-1} can look for the coefficient of $\frac{1}{z}$.

For Taylor series $f(z) = \sum_0^\infty a_k(z - z_0)^k$,

$$a_k = \frac{f^k(z_0)}{k!}$$

For Laurent series $f(z) = \sum_{-\infty}^\infty a_k(z - z_0)^k$, if f is analytic in $r < |z - z_0| < R$,

$$a_k = \frac{1}{2\pi} \int_{|z-z_0|=s} \frac{f(z)}{(z - z_0)^{k+1}} dz$$

for any $r < s < R$.

24.10.1 Isolated singularities

z_0 is an isolated singularity if f is analytic in a punctured disk $0 < |z - z_0| < r$. Here, can use Laurent series. The Laurent series is divided into the principal terms (a_{-n}) and analytic terms (a_n).

Laurent series can have no ISs, finite ISs, and infinite ISs. So the singularity is: (1) Removable if it becomes a Taylor series $a_0 + a_1(z - z_0) + \dots$, (2) N-order pole if $\frac{a_{-N}}{(z - z_0)^N} + \dots + \frac{a_{-1}}{(z - z_0)} + a_0 + a_1(z - z_0) + \dots$, (3) Simple pole $\frac{a_{-1}}{(z - z_0)} + a_0 + a_1(z - z_0) + \dots$, (4) Essential singularity $\dots + \frac{a_{-1}}{(z - z_0)} + a_0 + a_1(z - z_0) + \dots$

Riemann's theorem. z_0 is removable singularity iff f is bounded near z_0 .

Poles. z_0 is a pole iff

$$\lim_{z \rightarrow z_0} |f(z)| \rightarrow \infty$$

Casorati-Weierstrass. If z_0 is an essential singularity, $z_n \rightarrow z_0$ sequence s.t. $f(z_n) \rightarrow w_0$ as $n \rightarrow \infty$. Example. $e^{\frac{1}{z_n}} = w_0 = 1 + \sqrt{3}i$, where w_0 is chosen. Then,

$$\frac{1}{z_n} = \log(1 + \sqrt{3}i) = \ln|n| + i \arg(z)$$

$z_n \rightarrow 0$ as $n \rightarrow \infty$. $f(z_n) = w_0$.

Picard's theorem. If z_0 is an essential singularity of f , there is a sequence z_n such that $f(z_n) = w_0$, where there is at most one exception (e.g. $e^{\frac{1}{z_n}} \neq 0$).

24.10.2 Residue theorem

If

$$f(z)dz = \sum_{-\infty}^\infty (z - z_0)^k$$

then integrating both sides,

$$\int_{|z-z_0|=\rho} f(z)dz = \sum_{-\infty}^{\infty} a_k \int_{|z-z_0|=\rho} (z-z_0)^k dz$$

To solve this integral, take the case where $k \neq -1$. Then,

$$\frac{1}{k+1}(z-z_0)^{k+1} = 0$$

because it's a closed curve. Now take the case where $k = -1$. Then

$$\int_{|z-z_0|=\rho} \frac{1}{z-z_0} dz$$

and to solve this, can use Cauchy's integral formula, but can also solve by noting that

$$\begin{aligned} \int_{\gamma} f(z)dz &= \int_a^b f(\gamma(t))\gamma'(t)dt \\ \gamma(t) &= z_0 + \rho e^{it}, \gamma'(t) = \rho i e^{it} \end{aligned}$$

So,

$$\int_{|z-z_0|=\rho} \frac{1}{z-z_0} dz = \int_0^{2\pi} i dt = 2\pi i$$

Combining the $k \neq -1$ and $k = -1$ cases,

$$\int_{|z-z_0|=\rho} f(z)dz = \sum_{-\infty}^{\infty} a_k \int_{|z-z_0|=\rho} (z-z_0)^k dz = \sum_{-\infty}^{\infty} a_{-1} \int_{|z-z_0|=\rho} (z-z_0)^{-1} dz = 2\pi i a_{-1}$$

If z_0 is an isolated singularity, then the residue of f at z_0 is $Res(f, z_0) = a_{-1}$. Can do this by finding Laurent series expression and reading off a_{-1} . If not, then 0. Can ignore the analytic, or Taylor series, portion.

Residue theorem. If f is analytic in D except isolated singularities, and C is a simple closed curve in D (oriented CCW) not passing through the isolated singularities, and let $z_1, z_2, \dots, z_n$ be the isolated singularities lying inside C , then

$$\int_C f(z)dz = 2\pi i \sum_{k=1}^n Res(f, z_k)$$

This is a generalization of the Laurent series in a punctured disk.

24.10.3 Residues

Simple pole.

$$f(z) = \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \dots$$

$$f(z)(z - z_0) = a_{-1} + a_0(z - z_0) + \dots$$

$$Res(f, z_0) = a_{-1} = \lim_{z \rightarrow z_0} ((z - z_0)f(z))$$

Double pole.

$$f(z) = \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \dots$$

$$f(z)(z - z_0)^2 = a_{-2} + a_1(z - z_0) + \dots$$

$$\frac{d}{dz} f(z)(z - z_0)^2 = \frac{d}{dz} (a_{-2} + a_1(z - z_0) + \dots)$$

$$Res(f, z_0) = a_{-1} = \lim_{z \rightarrow z_0} \frac{d}{dz} ((z - z_0)^2 f(z))$$

For the n-pole.

$$Res(f, z_0) = a_{-1} = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \frac{d^{n-1}}{dz^{n-1}} ((z - z_0)^n f(z))$$

Remark. If g and h are analytic near z_0 , h is zero at z_0 , then

$$Res(f(z), z_0) = \frac{g(z_0)}{h'(z_0)}$$

24.10.4 Terminology

A function is analytic if it has a complex derivative at every point of its domain, and in consequence possessing derivatives of all orders and agreeing with its Taylor series locally. Can be applied to real and complex functions.

Complex analysis frequently involves both Taylor series and Laurent series. For the latter, it's important to distinguish whether the number of negative powers is finite or infinite. Meromorphic allows poles (i.e., finitely many negative powers in the Laurent series), while holomorphic does not.

24.11 Quaternions

Vectors and matrices lead to gimbal lock when using Euler rotations, and it's not commutative at all. However, quaternions better address these issues:

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = -ji = k$$

$$ki = -ik = j$$

$$jk = -kj = i$$

$$\cos\theta + \sin\theta(i + j + k)$$

Use conjugation to rotate the unit sphere ($q \cdot p \cdot q^{-1}$): $q \cdot p \cdot q^{-1}$. The unit sphere in 3D i,j,k is just a projection of a hypersphere. A hypersphere 'rotation' leads to projections that are not unit spheres, so to 'fix' this, use the inverse, q^{-1} to isolate the rotation desired.

Higher complex numbers: octonions and other hypercomplex numbers. Lose degrees of freedom in various ways in higher-dimensions so become less useful.

25 Integral Transforms

25.1 Background math

(1) For any integer m:

$$\int_0^{2\pi} \sin(mt) dt = -\frac{1}{m} \cos(mt) \Big|_0^{2\pi} = 0$$

(2) For any integer m > 0:

$$\int_0^{2\pi} \cos(mt) dt = \frac{1}{m} \sin(mt) \Big|_0^{2\pi} = 0$$

(3) For any integers m and n:

$$\int_0^{2\pi} \sin(mt) \cos(nt) dt = \int_0^{2\pi} \frac{1}{2} [\sin((m+n)t) + \sin((m-n)t)] dt$$

using a trigonometric identity, and

$$= \frac{1}{2} \int_0^{2\pi} \underbrace{\sin((m+n)t)}_{integer} dt + \frac{1}{2} \int_0^{2\pi} \underbrace{\sin((m-n)t)}_{integer} dt = 0 + 0 = 0$$

(4) Use a trig identity to establish:

$$\begin{aligned} \int_0^{2\pi} \sin^2(mt) dt &= \int_0^{2\pi} \frac{1}{2} [\cos((m-n)t) - \cos((m+n)t)] dt \\ &= \frac{1}{2} \int_0^{2\pi} \underbrace{\cos((m-n)t)}_{integer} dt - \frac{1}{2} \int_0^{2\pi} \underbrace{\cos((m+n)t)}_{integer} dt = 0 + 0 = 0 \end{aligned}$$

Using above, when integers $m \neq n$ or $m \neq -n$:

$$\int_0^{2\pi} \sin(mt) \sin(nt) dt = 0$$

Using above, when m=n:

$$\int_0^{2\pi} \sin(mt) \sin(nt) dt$$

$$= \frac{1}{2} \int_0^{2\pi} \underbrace{\cos((m-n)t)}_0 dt - \frac{1}{2} \int_0^{2\pi} \underbrace{\cos((m-n)t)}_{integer} dt = \frac{1}{2} \int_0^{2\pi} 1 dt + 0 = \pi$$

(5) When $m=n$ and $m \neq 0$:

$$\begin{aligned} & \int_0^{2\pi} \frac{1}{2} [\cos((m-n)t) + \cos((m-n)t)] dt \\ &= \frac{1}{2} \int_0^{2\pi} \underbrace{\cos((m-n)t)}_0 dt + \frac{1}{2} \int_0^{2\pi} \underbrace{\cos((m-n)t)}_{2m} dt \\ &= \frac{1}{2} \int_0^{2\pi} 1 dt + \frac{1}{2} \int_0^{2\pi} \cos(2mt) dt = \pi + 0 = \pi \end{aligned}$$

25.2 Fourier series derivation

(1) Decompose any function into weighted sines and cosines:

$$\begin{aligned} f(t) = & \underbrace{a_0}_{a_0 \cos(0t)} + a_1 \cos(t) + a_2 \cos(2t) + \dots + a_n \cos(nt) \\ & + b_1 \sin(t) + b_2 \sin(2t) + \dots + b_n \sin(nt) \end{aligned}$$

Integrate everything:

$$\begin{aligned} \int_0^{2\pi} f(t) dt = & \int_0^{2\pi} a_0 dt + \int_0^{2\pi} a_1 \cos(t) dt + \int_0^{2\pi} a_2 \cos(2t) dt + \dots + \int_0^{2\pi} a_n \cos(nt) dt \\ & + \int_0^{2\pi} b_1 \sin(t) dt + \int_0^{2\pi} b_2 \sin(2t) dt + \dots + \int_0^{2\pi} b_n \sin(nt) dt \end{aligned}$$

Using the background,

$$\int_0^{2\pi} f(t) dt = \int_0^{2\pi} a_0 dt + 0 + 0 + \dots = a_0(2\pi)$$

Therefore,

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(t) dt$$

Note a_0 is the average value of the function

(2) Decompose any function into weighted sines and cosines:

$$\begin{aligned} f(t) = & \underbrace{a_0}_{a_0 \cos(0t)} + a_1 \cos(t) + a_2 \cos(2t) + \dots + a_n \cos(nt) \\ & + b_1 \sin(t) + b_2 \sin(2t) + \dots + b_n \sin(nt) \end{aligned}$$

Integrate everything and multiply by $\cos(nt)$:

$$\begin{aligned} & \int_0^{2\pi} f(t)\cos(nt)dt \\ &= \int_0^{2\pi} a_0\cos(nt)dt + \int_0^{2\pi} a_1\cos(t)\cos(nt)dt + \int_0^{2\pi} a_2\cos(2t)\cos(nt)dt + \dots + \int_0^{2\pi} a_n\cos(nt)\cos(nt)dt \\ &+ \int_0^{2\pi} b_1\sin(t)\cos(nt)dt + \int_0^{2\pi} b_2\sin(2t)\cos(nt)dt + \dots + \int_0^{2\pi} b_n\sin(nt)\cos(nt)dt \end{aligned}$$

Using the background:

$$\int_0^{2\pi} f(t)\cos(nt)dt = 0 + \dots + \int_0^{2\pi} a_n\cos(nt)\cos(nt)dt + 0 + \dots = \int_0^{2\pi} a_n\cos^2(nt)dt = a_n\pi$$

Therefore,

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(t)\cos(nt)dt$$

(3) Decompose any function into weighted sines and cosines:

$$\begin{aligned} f(t) = & \underbrace{a_0}_{a_0\cos(0t)} + a_1\cos(t) + a_2\cos(2t) + \dots + a_n\cos(nt) \\ & + b_1\sin(t) + b_2\sin(2t) + \dots + b_n\sin(nt) \end{aligned}$$

Integrate everything and multiply by $\sin(nt)$:

$$\begin{aligned} & \int_0^{2\pi} f(t)\sin(nt)dt \\ &= \int_0^{2\pi} a_0\sin(nt)dt + \int_0^{2\pi} a_1\cos(t)\sin(nt)dt + \int_0^{2\pi} a_2\cos(2t)\sin(nt)dt + \dots + \int_0^{2\pi} a_n\cos(nt)\sin(nt)dt \\ &+ \int_0^{2\pi} b_1\sin(t)\sin(nt)dt + \int_0^{2\pi} b_2\sin(2t)\sin(nt)dt + \dots + \int_0^{2\pi} b_n\sin(nt)\sin(nt)dt \end{aligned}$$

Using the background:

$$\int_0^{2\pi} f(t)\sin(nt)dt = 0 + \dots + \int_0^{2\pi} b_n\sin(nt)\sin(nt)dt + 0 + \dots = \int_0^{2\pi} b_n\sin^2(nt)dt = b_n\pi$$

Therefore,

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(t)\sin(nt)dt$$

(4) We now have

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_0^{2\pi} f(t)dt \\ a_n &= \frac{1}{\pi} \int_0^{2\pi} f(t)\cos(nt)dt \\ b_n &= \frac{1}{\pi} \int_0^{2\pi} f(t)\sin(nt)dt \end{aligned}$$

25.3 Fourier transform derivation

(5) Turn real Fourier series into complex exponential form:

$$f(t) = a_0 + \sum_i^{\infty} [a_n \cos(wt) + b_n \sin(wt)]$$

where $w = \frac{n\pi}{T} = n\pi f$. Use the trig and complex identities:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

So,

$$\begin{aligned} f(t) &= a_0 + \sum_i^{\infty} \left(a_n \left(\frac{e^{iwt} + e^{-iwt}}{2} \right) + b_n \left(\frac{e^{iwt} - e^{-iwt}}{2i} \right) \right) \\ &= a_0 + \frac{1}{2} \sum_i^{\infty} \left(a_n (e^{iwt} + e^{-iwt}) \left(\frac{i}{i} \right) + b_n \left(\frac{e^{iwt} - e^{-iwt}}{i} \right) \left(\frac{i}{i} \right) \right) \\ &= a_0 + \frac{1}{2} \sum_i^{\infty} [(a_n - ib_n)e^{iwt} + (a_n + ib_n)e^{-iwt}] \end{aligned}$$

Since $w = \frac{n\pi}{T}$, if n becomes $-n$, then w becomes $-w$:

$$= a_0 + \frac{1}{2} \sum_i^{\infty} (a_n - ib_n)e^{iwt} + \frac{1}{2} \sum_{-1}^{-\infty} (a_{-n} + ib_{-n})e^{-i(-w)t}$$

(6) Alter (4) without requiring a Jacobian transformation since $w = \frac{n\pi}{T}$ and not, e.g. $w = \frac{n2\pi}{T}$:

$$a_n = \frac{1}{T} \int_{-T}^T f(t) \cos(wt) dt$$

$$b_n = \frac{1}{T} \int_{-T}^T f(t) \sin(wt) dt$$

Rename a_0 so that it no longer includes the '2' in the denominator:

$$a_0 = \frac{1}{T} \int_{-T}^T f(t) dt$$

(7) Call $\frac{1}{2}(a_n - ib_n) = c_n$ so when n is positive and apply (6) to (5):

$$c_n = \frac{1}{2} \left[\frac{1}{T} \int_{-T}^T f(t) \cos(wt) dt - i \frac{1}{T} \int_{-T}^T f(t) \sin(wt) dt \right]$$

$$= \frac{1}{2T} \int_{-T}^T [f(t)\cos(wt) - i f(t)\sin(wt)]dt$$

Use Euler's identity

$$e^{-i\theta} = \cos\theta - i \sin\theta$$

So,

$$c_n = \frac{1}{2T} \int_{-T}^T f(t)e^{-iwt}dt$$

And call it $\hat{f}(w)$:

$$\hat{f}(w) = \frac{1}{2T} \int_{-T}^T f(t)e^{-iwt}dt$$

(8) Call $\frac{1}{2}(a_n + ib_n) = c_{-n}$ so when n is negative and apply (6) to (5):

$$\begin{aligned} c_n &= \frac{1}{2} \left[\frac{1}{T} \int_{-T}^T f(t)\cos(wt)dt + i \frac{1}{T} \int_{-T}^T f(t)\sin(wt)dt \right] \\ &= \frac{1}{2T} \int_{-T}^T [f(t)\cos(wt) + i f(t)\sin(wt)]dt \end{aligned}$$

Use Euler's identity

$$e^{i\theta} = \cos\theta + i \sin\theta$$

So, with n being negative and therefore w being negative

$$c_n = \frac{1}{2T} \int_{-T}^T f(t)e^{i(-w)t}dt$$

And call it $\hat{f}(w)$ as in (7):

$$\hat{f}(w) = \frac{1}{2T} \int_{-T}^T f(t)e^{-iwt}dt$$

(9) Call $c_n = \frac{a_n}{2}$ and therefore, as in (7) and (8):

$$\hat{f}(w) = \frac{1}{2T} \int_{-T}^T f(t)e^{-iwt}dt$$

(10) Combine (6) to (9):

$$f(t) = a_0 + \sum_i^{\infty} [a_n \cos(wt) + b_n \sin(wt)] = \sum_{n=-\infty}^{\infty} c_n e^{iwt} = \sum_{n=-\infty}^{\infty} \hat{f}(w) e^{iwt}$$

(11) Take

$$\lim_{T \rightarrow \infty} w = \lim_{T \rightarrow \infty} \left(\frac{n\pi}{T} \right)$$

(12) We take

$$f(t) = \sum_{n=-\infty}^{\infty} \hat{f}(w) e^{iwt} = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2T} \int_{-T}^T f(t) e^{-iwt} dt \right) e^{iwt}$$

And take the limit from (11):

$$\begin{aligned} \mathcal{F}^{-1}\{f(t)\} &= \lim_{T \rightarrow \infty} f(t) = \lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} \left(\frac{1}{2T} \int_{-T}^T f(t) e^{-iwt} dt \right) e^{iwt} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\frac{\pi}{dw}} \int_{-T}^T f(t) e^{-iwt} dt \right) e^{iwt} = \int_{-\infty}^{\infty} \underbrace{\left(\frac{1}{2\pi} \int_{-T}^T f(t) e^{-iwt} dt \right)}_{\hat{f}(w)} e^{iwt} dw \end{aligned}$$

(13) FT and inverse FT:

$$\begin{aligned} \mathcal{F}\{f(t)\} &= \hat{f}(w) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-iwt} dw \\ \mathcal{F}^{-1}\{\hat{f}(w)\} &= f(t) = \int_{-\infty}^{\infty} \hat{f}(w) e^{iwt} dw \end{aligned}$$

(14) Alternatively:

$$\begin{aligned} \mathcal{F}\{f(t)\} &= \hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-iwt} dw \\ \mathcal{F}^{-1}\{\hat{f}(w)\} &= f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w) e^{iwt} dw \end{aligned}$$

25.4 Convolution theorem for Fourier transforms

Use the FT form of (14) in the convolution theorem:

$$\begin{aligned} \mathcal{F}\{(f * g)(t)\} &= \sqrt{2\pi} \hat{f}(w) \hat{g}(w) \\ \mathcal{F}^{-1}\{\hat{f}(w) \hat{g}(w)\} &= \frac{1}{\sqrt{2\pi}} (f * g)(t) \end{aligned}$$

Definition of a convolution:

$$(f * g)(x) = \int_{-\infty}^{\infty} f(p) g(x - p) dp$$

Proof (of first form, inverse proof is analogous):

$$\begin{aligned} \mathcal{F}\{(f * g)(t)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (f * g)(t) e^{-iwt} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(p) g(x - p) dp \right] e^{-iwt} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p) g(x - p) e^{-iwt} dt dp \quad (\text{Fubini}) \end{aligned}$$

Substitute

$$q = x - p$$

$$dq = dp$$

$$x = p + q$$

Finish the proof:

$$\begin{aligned}\mathcal{F}\{(f * g)(t)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)g(q)e^{-iw(p+q)} dp dq \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(p)e^{-iwp} dp \int_{-\infty}^{\infty} g(q)e^{-iwq} dq \\ &= \hat{f}(w)\hat{g}(w)\sqrt{2\pi}\end{aligned}$$

So, a Fourier transform of a convolution is an integral inside another integral, which is re-arranged as a product of those two integrals.

Intuition and analogy. When two polynomials are multiplied, the terms in the expanded product are convolutions of the terms in the two factors. However when the product is evaluated at specific arguments, it is of course the product of values of the two factors at those arguments. This is related to combinatorics.

25.5 Discrete Fourier Transform (DFT)

This is actually discretization of the Fourier SERIES.

$$\begin{aligned}\hat{f}_k &= \sum_{j=0}^{n-1} f_j e^{\frac{-i2\pi jk}{n}} \\ f_k &= \frac{1}{n} \sum_{j=0}^{n-1} \hat{f}_j e^{\frac{i2\pi jk}{n}} \\ &\quad e^{\frac{-i2\pi jk}{n}}\end{aligned}$$

Turn DFT into matrix form:

$$\begin{aligned}\hat{f} &= F_n f \\ \begin{pmatrix} \hat{f}_0 \\ \hat{f}_1 \\ \hat{f}_2 \\ \dots \\ \hat{f}_{n-1} \end{pmatrix} &= \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & w & w^2 & w^3 & \dots & w^{n-1} \\ 1 & w^2 & w^4 & w^6 & \dots & w^{2(n-1)} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & w^{n-1} & w^{2(n-1)} & w^{3(n-1)} & \dots & w^{(n-1)(n-1)} \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \\ f_2 \\ \dots \\ f_{n-1} \end{pmatrix} \\ w &= e^{\frac{i2\pi}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\end{aligned}$$

The Vandermonde matrix F is orthogonal (column inner spaces, which involve multiplying complex conjugates, $= 0$). It is also Hermitian as $F^H F = \bar{F}^T F = I$. Note $(w_6 4)^2 = w_3 2$ for example.

25.6 Fast Fourier Transform (FFT)

Speeds up the DFT.

Decompose F :

$$F_n = \begin{pmatrix} I_{n/2} & D_{n/2} \\ I_{n/2} & -D_{n/2} \end{pmatrix} \begin{pmatrix} F_{n/2} & 0 \\ 0 & F_{n/2} \end{pmatrix} (P)$$

Where P is the permutation matrix such that:

$$(P) \begin{pmatrix} f_0 \\ f_1 \\ f_2 \\ \dots \\ f_{n-1} \end{pmatrix} = \begin{pmatrix} f_{even} \\ f_{odd} \end{pmatrix}$$

Moreover:

$$D_{n/2} = \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & w & 0 & 0 & \dots & 0 \\ 0 & 0 & w^2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & w^{(n-1)} \end{pmatrix}$$

Each $F_{n/2}$ is recursively decomposed into $F_{n/4}$, etc.

If $n=64$ for example, each $F_{n/2}$ results in 32^2 operations, and there are 2 of them $= 2(32)^2$. Then add 64 for the 2 diagonal matrices D and $-D = 2(32)^2 + 64$. Recursively decompose:

$$\begin{aligned} 2(32)^2 + 64 &\Rightarrow 2(2(16)^2 + 32) + 64 \Rightarrow 2(2(2(8)^2 + 16) + 32) + 64 \\ &\Rightarrow 2(2(2(2(4)^2 + 8) + 16) + 32) + 64 \Rightarrow 2(2(2(2(2(2)^2 + 4) + 8) + 16) + 32) + 64 \\ &\Rightarrow 2(2(2(2(2(2(1)^2 + 2) + 4) + 8) + 16) + 32) + 64 \\ &= 64(6) = n \log_2 n \end{aligned}$$

The normal DFT is $O(n^2)$ but the FFT is $O(n \log n)$.

25.7 Laplace Transform

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

Where $s = a + iw$. It is $\mathcal{F}$ when $a=0$, with $\mathcal{L}$ generalizing $\mathcal{F}$ beyond that one 'slice' of $a=0$. It is also the $\mathcal{F}$ of $f(t)e^{-at}$:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-(a+iw)t} dt = \int_0^{\infty} f(t)e^{-iwt}e^{-at} dt = \mathcal{F}\{f(t)e^{-at}\}$$

Since the e^{-at} is damping or damping potential, $\mathcal{L}$ is just $\mathcal{F}$ of a system with damping or damping potential.

25.8 $\mathcal{L}$ Derivation from power series

The power series is:

$$\sum_0^{\infty} a_n x^n$$

Example 1. If $a_n = 1$, then:

$$1 + x + x^2 + \dots = \frac{1}{1-x} \text{ where } -1 < x < 1$$

Example 2. If $a_n = \frac{1}{n!}$, then it's e^x

The continuous analog is:

$$\int_0^{\infty} a_t x^t dt$$

Make cosmetic changes. With $0 < x < 1$, $\ln x < 0$:

$$\int_0^{\infty} a_t e^{t \ln x} dt = \int_0^{\infty} f(t) e^{-st} dt = \mathcal{L}\{f(t)\} = F(s)$$

So $\mathcal{L}$ is the continuous analog of the power series.

25.9 $\mathcal{L}$ Derivation from $\mathcal{F}$

$\mathcal{F}$ is more complicated with functions which don't tail off to 0 at infinity and negative infinity, like e^{-at} and the Heaviside function.

$\mathcal{L}$ is a one-sided $\mathcal{F}$. Multiply $f(t)$ by e^{-at} so that:

$$\lim_{t \rightarrow \infty} f(t)e^{-at} \rightarrow 0$$

Then multiply by the Heaviside function:

$$F(t) = f(t)e^{-at}H(t) = \begin{cases} 0 & t < 0 \\ f(t)e^{-at} & t \geq 0 \end{cases}$$

Take $\mathcal{F}$:

$$\begin{aligned}\mathcal{F}\{F(t)\} &= \hat{F}(w) = \int_{-\infty}^{\infty} F(t)e^{-iwt} dt = \int_0^{\infty} f(t)e^{-at}e^{-iwt} dt \\ &= \int_0^{\infty} f(t)e^{-(a+iw)t} dt = \int_0^{\infty} f(t)e^{-st} dt = F(s)\end{aligned}$$

25.10 $\mathcal{L}^{-1}$

Find $\mathcal{L}^{-1}$:

$$\begin{aligned}F(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(w)e^{iwt} dw \\ e^{at}F(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{at}\hat{F}(w)e^{iwt} dw \\ f(t) = e^{at}F(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{at}\hat{F}(w)e^{iwt} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s)e^{(a+iw)t}\frac{ds}{i}\end{aligned}$$

Since $s = a + iw$, $ds = i dw \Rightarrow dw = \frac{ds}{i}$:

$$= \frac{1}{2\pi i} \int_{-\infty}^{\infty} F(s)e^{st} ds$$

Can use the above inverse equation, but this is difficult.

Alternatively, the inverse $\mathcal{L}$ can be calculated by mathematically manipulating $F(s)$ so that it looks like one of the $\mathcal{L}$ already known.

25.11 $\mathcal{L}$ Properties

(1) $\mathcal{L}$ is a linear operator:

$$\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 \mathcal{L}\{f(t)\} + c_2 \mathcal{L}\{g(t)\}$$

(2) $\mathcal{L}$ of a derivative: use integration by parts

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} f'(t)e^{-st} dt = s\mathcal{L}\{f(t)\} - f(0)$$

$$\mathcal{L}\{f''(t)\} = s\mathcal{L}\{f'(t)\} - f'(0) = s(s\mathcal{L}\{f(t)\} - f'(0)) - f'(0) = s^2\mathcal{L}\{f(t)\} - sf'(0) - f'(0)$$

Laplace transform of higher derivatives:

$$\mathcal{L}\{y^4\} = s^4\mathcal{L}\{y\} - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)$$

Note:

$$\begin{aligned}\mathcal{L}\{f'\} &= s\mathcal{L}\{f\} - f(0) \\ \mathcal{L}\{f\} &= \frac{1}{s}\{\mathcal{L}\{f'\} + f(0)\}\end{aligned}$$

(3) Powers. We know using integration by parts that:

$$\mathcal{L}\{1\} = \frac{1}{s}$$

So:

$$\begin{aligned}\mathcal{L}\{t\} &= \frac{1}{s}\mathcal{L}\{1\} = \frac{1}{s^2} \\ \mathcal{L}\{t^2\} &= \frac{1}{s}\mathcal{L}\{2t\} = \frac{2}{s}\{t\} = \frac{2}{s}\frac{1}{s^2} = \frac{2}{s^3} \\ \mathcal{L}\{t^3\} &= \frac{3(2)}{s^4} \\ \mathcal{L}\{t^n\} &= \frac{n!}{s^{n+1}}\end{aligned}$$

(4) Periodic functions. Use integration by parts to get:

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$$

Use this to calculate $\mathcal{L}\{\cos at\}$.

(5) Translations.

$$\begin{aligned}\mathcal{L}\{f(t)\} &= F(s) \\ \mathcal{L}\{e^{at}f(t)\} &= \int_0^\infty f(t)e^{at}e^{-st}dt = \int_0^\infty f(t)e^{-t(s-a)}dt = F(s-a)\end{aligned}$$

So:

$$\begin{aligned}\mathcal{L}\{\cos(2t)\} &= F(s) = \frac{s}{s^2 + 4} \\ \mathcal{L}\{e^{3t}\cos(2t)\} &= F(s-3) = \frac{s-3}{(s-3)^2 + 4}\end{aligned}$$

25.12 $\mathcal{L}$ of Discontinuous Functions

The Heaviside (unit step) function is:

$$u_c(t) = \begin{cases} 0 & t < c \\ 1 & t \geq c \end{cases}$$

Can alter to make different kinds of step functions. Example:

$$f(t) = 2 - 2u_\pi(t)$$

Two steps, one at π and another at 2π :

$$f(t) = 2 - 2u_\pi(t) + u_{2\pi}(t)$$

Shift $g(t)$ to the right by c and then 'start' the function:

$$g(t) = u_c(t)f(t - c)$$

$$\mathcal{L}\{u_c(t)f(t - c)\} = \int_0^\infty u_c(t)f(t - c)e^{-st}dt$$

Change left boundary:

$$= \int_c^\infty \underbrace{u_c(t)}_{=1} f(t - c)e^{-st}dt = \int_t^\infty e^{-st}f(t - c)dt$$

Let $x = t - c \Rightarrow t = x + c \Rightarrow dx = dt$.

$$\begin{aligned} &= \int_0^\infty e^{-s(x+c)}f(x)dx = \int_0^\infty e^{-sx}e^{-sc}f(x)dx \\ &= e^{-sc} \int_0^\infty e^{-sx}f(x)dx = e^{-sc}\mathcal{L}\{f(t)\} \end{aligned}$$

So:

$$\mathcal{L}\{u_c(t)f(t - c)\} = e^{-sc}\mathcal{L}\{f(t)\}$$

Application example:

$$\mathcal{L}\{u_\pi(t)\sin(t - \pi)\} = e^{-\pi s}\mathcal{L}\{\sin(t)\} = \frac{e^{-\pi s}}{s^2 + 1}$$

25.13 Dirac δ -function

Apply as an impulse or instantaneous change:

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

And define:

$$\int_{-\infty}^\infty \delta(x)dx = 1$$

For intuition, define:

$$d(t) = \begin{cases} \frac{1}{2\tau} & -\tau < t < \tau \\ 0 & \text{everywhere else} \end{cases}$$

$$\int_{-\infty}^\infty d(t)dt = \int_{-\tau}^\tau \frac{1}{2\tau}dt = 1$$

$$\lim_{\tau \rightarrow 0} d(t) = 1$$

For:

$$\delta(t-c) = \begin{cases} \infty & t = c \\ 0 & t \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t-c) dt = 1$$

$$\int_{-\infty}^{\infty} 2\delta(t-c) dt = 2$$

So $2\delta(t)$ is twice as high as $\delta(t)$!

25.14 $\mathcal{L}$ of the Dirac δ -function

Multiplying a normal function $f(t)$ with $\delta(t-c)$:

$$\mathcal{L}\{f(t)\delta(t-c)\} = \int_0^{\infty} e^{-st} f(t) \delta(t-c) dt = \int_0^{\infty} e^{-st} \underbrace{f(c)}_{\text{notice}} \delta(t-c) dt$$

Because the multiplication of the functions is just an impulse now at $x=c$.

$$= e^{-sc} f(c) \underbrace{\int_0^{\infty} \delta(t-c) dt}_{=1}$$

So:

$$\mathcal{L}\{\delta(t-c)f(t)\} = e^{-sc} f(c)$$

So:

$$\mathcal{L}\{\delta(t-c)(1)\} = \mathcal{L}\{\delta(t-c)\} = e^{-sc}(1) = e^{-sc}$$

25.15 z -Transform and the DTFT

The discrete analog of $\mathcal{L}$:

$$X(z) = \sum_{-\infty}^{\infty} x(n) z^{-n}$$

Where $z = r e^{iw}$. If $r=1$, then it is the $X(e^{iw}) = DTFT$. On the z -plane (the complex plane), DTFT is the complex unit circle.

Generalize:

$$X(z) = \sum_{-\infty}^{\infty} \underbrace{x(n)r^{-n}}_{*} e^{-iwn}$$

Where $*$ is the DTFT of $x(n)\frac{1}{r^n}$. The z -plane complex circle will have radius or modulus r . Bigger r = bigger radius. If $x(n)$ signal has lots of energy, can cancel it with a bigger r as r^{-n} is a signal decay. This forms the region of convergence where it is possible to take the DTFT - i.e. outside the circle with radius r .

25.16 Fourier series as inputs in LTI systems

An LTI system takes $x(t)$ as an input which is convolved with the impulse response $h(t)$ to produce $y(t)$:

$$y(t) = h(t) * x(t)$$

If $x(t) = w^t$, it's an eigenfunction of convolution using the z-transform:

$$y(t) = \sum_{\tau=-\infty}^{\infty} h(\tau)x(t-\tau) = \sum_{\tau=-\infty}^{\infty} h(\tau)w^{t-\tau} = \sum_{\tau=-\infty}^{\infty} h(\tau)w^{-\tau}w^t = H(w)w^t = H(w)x(t)$$

Usually $x(t)$ is more complicated, like this:

$$\begin{aligned} x(t) &= \mathcal{F}^{-1}\{x[t]\} = e^{iwt} \\ \mathcal{O}[x(t)] &= \mathcal{O}[\mathcal{F}^{-1}\{\hat{x}(w)\}] = \mathcal{O}\left[\sum_{w=-\infty}^{\infty} \hat{x}(w)e^{iwt}\right] = \sum_{w=-\infty}^{\infty} \hat{x}(w)\mathcal{O}[e^{iwt}] = \sum_{w=-\infty}^{\infty} \hat{x}(w)\mathcal{O}[g(t)] \\ &= \sum_{w=-\infty}^{\infty} \hat{x}(w)\left[\sum_{\tau=-\infty}^{\infty} h(\tau)g(t-\tau)\right] = \sum_{w=-\infty}^{\infty} \hat{x}(w)\left[\sum_{\tau=-\infty}^{\infty} h(\tau)e^{-i w \tau}e^{i w t}\right] = \sum_{w=-\infty}^{\infty} \hat{x}(w)H(w)e^{i w t} \end{aligned}$$

Where $H(w)$ is the eigenvalue and e^{iwt} is the eigenfunction.

$$\mathcal{O}^2[x(t)] = \mathcal{O}\left[\sum_{w=-\infty}^{\infty} \hat{x}(w)H(w)e^{iwt}\right] = \sum_{w=-\infty}^{\infty} \hat{x}(w)H(w)\mathcal{O}[e^{iwt}] = \sum_{w=-\infty}^{\infty} \hat{x}(w)H^2(w)e^{iwt}$$

$$\mathcal{O}^n[x(t)] = \sum_{w=-\infty}^{\infty} \hat{x}(w)H^n(w)e^{iwt} = y(t)$$

So the convolution in the time domain:

$$y(t) = h(t) * x(t)$$

is a multiplication in the frequency domain:

$$\hat{x}(w)H^n(w)$$

which are the weights of the complex exponentials.

25.17 Intuition: Poles

With periodic functions like $f(t) = \cos(wt)$, the frequency (w) domain will have two 'poles' at $-w$ and w .

$$f(t) = \cos(wt) = \frac{1}{2}(e^{-iwt} + e^{iwt}) = \frac{1}{2}(e^{i(-w)t} + e^{i(w)t})$$

On the Im v Re graph, these indicate rotation directions (w in the ccw direction and -w in the cw direction). The -w and w cancel each other out in the Im axis, so there is no imaginary component. The -w and w add to each other in superposition, leading to a double-amplitude cosine wave, which is halved to get back the original cosine wave.

25.18 $\mathcal{F}$ Intuition

The frequency (w) domain has amplitudes = modulus on the Im v Re 'rotation plot'. When w=0, the amplitude (so the y-intercept) in the frequency (w) domain is the area under f(t) with domain t:

$$\hat{f}(w) = \int_{-\infty}^{\infty} f(t)e^{-iwt} dt$$

$$\hat{f}(0) = \int_{-\infty}^{\infty} f(t)e^{-i(0)t} dt = \int_{-\infty}^{\infty} f(t) dt = \text{Area under curve of } f(t)$$

If the $\mathcal{F}$ complex exponential is decomposed to its periodic components:

$$\mathcal{F}\{f(t)\} = \hat{f}(w) = \int_{-\infty}^{\infty} f(t)e^{-iwt} dt$$

$$= \int_{-\infty}^{\infty} f(t)\cos(wt) dt - i \int_{-\infty}^{\infty} f(t)\sin(wt) dt$$

For each w in the complex unit circle (Im v Re graph), we are taking the radius or modulus. With the above decomposition, we can just taken all complex unit circles 'added together'. The decomposition above also shows that f(t) is just being multiplied by cosine and sine waves to get the amplitude on the frequency (w) domain. This can also be used to 'scan' for cos and sin frequencies and how much - and if there are 'pure' frequencies, there will be δ -functions because areas will become infinite when $\cos^2(t)$ and/or $\sin^2(t)$ will lead to all signals j , 0 - these are also the poles in $\mathcal{L}$ intuition.

25.19 $\mathcal{L}$ Intuition

The frequency (w) domain has amplitudes = modulus on the Im v Re 'rotation plot'. The w is then part of the complex plane a+iw, with amplitudes being the heights of the complex plane. When a=0, $\mathcal{L}=\mathcal{F}$. When a=0 and w=0, the amplitude (so the y-intercept) in the frequency (w) domain is the area under f(t) with domain t.

On the complex s domain, there are 2 'poles' which indicate the boundary of convergence. In the function $f(t) = e^{-t}\sin(t)$, the poles are at a=1, because $e^{at} = e^t$ cancels e^{-t} so that f(t) becomes the pure sine signal. The Im-coordinates are -1 and 1, which are -w and w of $\sin(t)$. So the poles are at $a + iw = 1 + i(\pm w)$. The a and w found using $\mathcal{L}$ is therefore from

$$f(t) = e^{-at} \sin(\omega t).$$

At the poles, the amplitudes are infinite, as they are δ -functions. Approaching these poles, the $\mathcal{L}$ on the frequency (ω) domain approaches the pure frequencies. Depending on $f(t)$, one side will be the region of divergence (reverse of a damped system or explosive growth). On the other side of the poles, the region of convergence, $f(t)$ becomes more damped.

25.20 Fourier versus Laplace transform applications

Fourier transform (FT) - (roughly) a tool to visualize ANY signal as a sum of sinusoids.

Laplace transform (LT) - a tool to analyze the stability of systems.

FT cannot analyze systems as it cannot handle exponentially growing signals. That in turn is because a signal needs to be absolutely integrable for it to have a FT.

25.21 Convolutions and integral transforms

Total variation is the trace of a covariance matrix

$$TV = \text{trace}(\Sigma) = \sigma_1^2 + \sigma_2^2 + \dots$$

Generalized variation is the determinant of a covariance matrix

$$GV = |\Sigma|$$

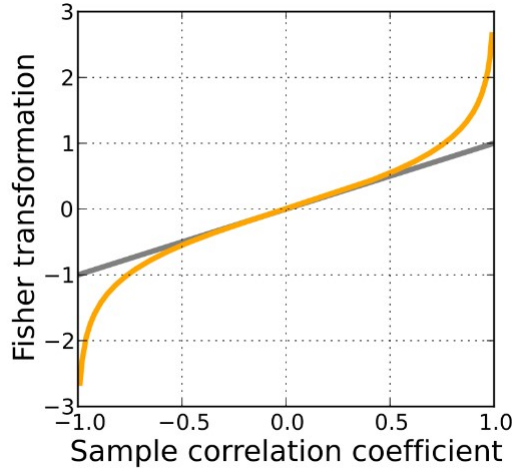
The larger these variances, the greater the data dispersion. Note for a bivariate covariance matrix,

$$|\Sigma| = \sigma_1^2 \sigma_2^2 (1 - \rho^2)$$

The variance of a linear combination Y is a quadratic

$$\text{Var}(Y) = \text{Var}(a_1 x_1 + a_2 x_2 + \dots) = \text{Var}(a'X) = a' \Sigma a$$

Can take the covariance and correlation of linear combinations. Fisher Z-Transformation. A way to transform the sampling distribution of Pearson's r (i.e. the correlation coefficient) so that it becomes normally distributed. The "z" in Fisher Z stands for a z-score.



A graph of the transformation (in orange). The untransformed sample correlation coefficient is plotted on the horizontal axis, and the transformed coefficient is plotted on the vertical axis. The identity function (gray) is also shown for comparison.

By the CLT, taking the mean of groups of sample variances from the chi-squared distribution, correlation coefficients from the non-Fisher transformed distribution (+Fisher-transformed) will lead to a normal distribution.

25.21.1 Convolution

To add independent discrete random variables X and Y as $Z = X + Y$, look for a $g(X, Y)$. Example (note: X is the random variable, x is just a value that X , Y , or Z can be, e.g. $Z = 3$ in the example):

$$\begin{aligned}
 p_Z(X + Y = 3) &= p_Z(3) \\
 &= \dots + p_Z(X = 0, Y = 3) + p_Z(X = 1, Y = 2) + p_Z(X = 2, Y = 1) + \dots \\
 &\quad \dots + p_Z(X = 3, Y = 0) + p_Z(X = 4, Y = -1) + \dots \\
 &= \dots + p_X(X = 0)p_Y(Y = 3) + p_X(X = 1)p_Y(Y = 2) + \dots \\
 &= \dots + p_X(X = 0)p_Y(3 - X) + p_X(X = 1)p_Y(3 - X) + \dots \\
 &= \dots + p_X(x)p_Y(3 - x) + p_X(x)p_Y(3 - x) + \dots \\
 &= \sum p_X(x)p_Y(z - x)
 \end{aligned}$$

which is a convolution where the Y PMF $p_Y(x)$ is vertically flipped and shifted to the right by $Z = z$ units to be multiplied by $p_X(x)$. The continuous analog is

$$f_Z(z) = \int f_X(x)f_Y(z - x)dx$$

This is actually just the marginal distribution, since,

$$f_{Z|X}(z|x) = f_{Y+x|X}(z|x) = f_{Y+x}(z) = f_Y(z - x)$$

so,

$$f_Z(z) = \int f_X(x)f_Y(z-x)dx = \int f_X(x)f_{Z|X}(z|x)dx = \int f_{X,Z}(x,z)dx$$

A convolution can also be conceptualized as a Toeplitz matrix.

25.21.2 Convolution with Dirac delta

Dirac delta (impulse) function

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

The delta function convolved with a function leaves the function as it is. So, the delta function is the identity in convolution. The sifting property shows this:

$$\int f(\tau)\delta(t-\tau)d\tau = \int f(t-\tau)\delta(\tau)d\tau = f(t)$$

where in the first integral the user sweeps along the function f until the delta function is found, and in the second integral the user sweeps the delta function until the value at $f(t)$ is found.

25.21.3 Fourier transform

Both the convolution and Fourier transforms are inner products, and the Fourier transform is a specific kind of convolution.

The Fourier transform is a convolution and a complex inner product so has an imaginary number. It does not have to use an imaginary number because it can be performed using real sines and cosines, as Fourier originally did. However, using complex numbers packages the sine and cosine into one compact notation. It is possible to derive this compact notation - the standard notation - from the sine and cosine waves in a few lines. The complex number notation takes advantage of the imaginary dimension, compactifying the notation by placing the sine into the imaginary dimension. However, the imaginary number will disappear as we take the norm - and the norm in complex vector spaces multiplies complex conjugates. Intuitively, the norm is just the distance from the origin, and by using the complex plane we can still get the real norm but compactify the space by placing one of the basis vectors (the sine) into the imaginary dimension.

The imaginary number can be positive, though it is more naturally negative for intuition and because the math is more natural - when the inner product norm is calculated.

The inverse Fourier transform of a Fourier transform is the original function. The proof uses the Dirac delta function.

The convolution theorem is a dot product of a dot product. Intuition is that probabilities (decomposed into amplitudes) are multiplied - so the Fourier transforms, i.e. amplitudes, in the frequency domain must be multiplied.

25.21.4 LTI systems

A system is linear time-invariant if the transfer function responds to inputs in a linear and time-invariant manner. So the signal can be scaled and added as in linear transformations, and the output will be scaled and added. Horizontally-shifted inputs become horizontally-shifted outputs because of time invariance.

The input is viewed, at least discretely, as a series of impulse responses:

$$x(n) = \sum x(i)\delta(n-i)$$

Since the output is

$$y(n) = T(x(n)) = T(\sum x(i)\delta(n-i))$$

and because of linearity and time invariance,

$$T(\sum x(i)\delta(n-i)) = \sum x(i)T(\delta(n-i)) = \sum x(i)h(n-i)$$

So this is a convolution. This means the input may be fed into the transfer function from the right side first, but it will be processed from the left side first if n starts low and increases. It will read and processed from the right side first if n starts high and declines

Since convolution integrals are difficult to calculate, use the Laplace or Fourier transform.

25.21.5 Laplace transform

A Laplace transform is a generalization of the Fourier transform. The Fourier transform has sines and cosines as basis vectors, whereas the Laplace transform has sines, cosines, and exponentials. The Fourier transform can calculate spring oscillations of masses, but the Laplace transform can express this ($s=0$) and also exponential decays of such oscillations. This is also why sometimes Fourier transforms have no answer whereas Laplace transforms can solve all situations.

Fourier transforms are 'slices' of Laplace transforms where there is no exponential decay. The Laplace transform can also be viewed as the Fourier transform of

$$f(t)e^{-\alpha t}$$

By changing α , we can change the Fourier transform 'slices'. So the Laplace transform can be viewed on the complex plane, with the poles (which come in

pairs) as being the Dirac deltas where there is no damping from the exponential term or where the exponential term is cancelled out. So the α is the x-axis, whereas the imaginary axis on the complex plane is the frequency from the Fourier transform. If the α of the pole is very negative, then there is a lot of damping. If the ω frequency is very high, there are very high-frequency waves in the signal. The pole of a system indicates how much damping and how much oscillation is in a system.

Laplace transforms (and Fourier transforms, for a limited class of problems) are useful in solving differential equations because they can turn them into Laplace transforms - which turns calculus into algebra. The same is for convolutions and therefore LTI systems.

A power series is

$$\sum f(t)x^t = F(x)$$

in the region of convergence for infinite series, $0 < x < 1$. The continuous version is

$$\int f(t)x^t dt = F(x)$$

To clean this up,

$$\int f(t)e^{ln(x)t} dt$$

but the region of convergence is $-1 < ln(x) < 0$, which means $ln(x)$ is always negative. To work with positive numbers,

$$\int f(t)e^{-ln(x)t} dt = F(x)$$

Let $s = ln(x)$:

$$\int f(t)e^{-st} dt = F(s)$$

The Fourier transform is very similar, so the $ln(x)$ is just sine and cosine wave bases, whereas for Laplace transforms it is sines, cosines, and exponentials. Fourier and Laplace transforms are just different types of power series.

Laplace transforms can solve LTI systems because they are convolutions, using the convolution theorem. Laplace transforms can also solve DEs by turning the problem from one of calculus to algebra - this is intuitive because the Laplace transform converts the time-domain DE to s-domain LT.

25.21.6 Power series

A polynomial is like an integer as it has 'clumps' (see section on abstract algebra and rings). Polynomials are therefore easy to work with. A power series is a polynomial. 'Hanging' the coefficients on a power series allows a lot of simple algebraic manipulation.

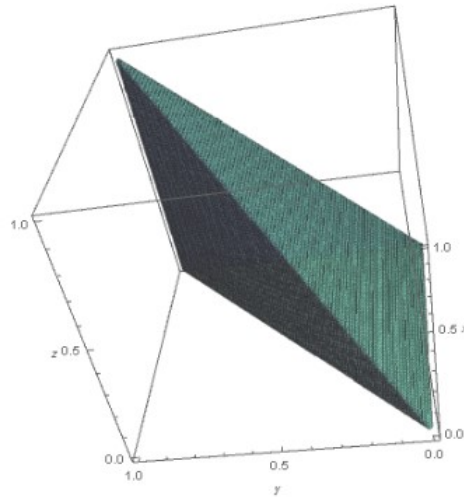
25.21.7 Simplices

A simplex is a polytope made from flat sides. Simplices are related to factorials, perhaps most intuitive using probability simplices:

$$\sum x_n = 1$$

$$x_n \geq 0$$

where n is the number of dimensions of the simplex. The 0D-simplex (a dot) corresponds to a probability simplex fraction of $0! = 1$, the 1D-simplex (a line) corresponds to a simplex fraction of $1! = 1$, the 2D simplex (area) corresponds to a simplex fraction of $2! = 2$, 3D-simplex (volume) of $3! = 6$, and so forth.



The proof of $n!$ is that the set of all points $(x_1, x_2, \dots, x_n)$ with $0 \leq x_1 \leq x_2 \leq \dots \leq x_n \leq 1$. The volume measures the probability that a random sequence of n real numbers is in sorted order, and there are $n!$ ways to permute a set of n values, and only one of them is sorted.

25.21.8 Taylor series

A Taylor series is a power series, as are Fourier and Laplace series, and these are all related.

A Taylor series has coefficients that are higher and higher derivatives because the function is being approximated at higher and higher detail - which is actually going into higher and higher dimensions. The power series arises because higher and higher dimensions require greater powers of x . The inverted factorials appear because higher and higher dimensions mean the simplex in those higher dimensions becomes a smaller proportion of those dimensions.

A first-degree Taylor expansion is the y-intercept (technically, it is in the 0D simplex, so divided by $0! = 1$).

In a second-degree expansion the y-intercept plus the slope multiplied by x , in a 1D simplex so is divided by $1! = 1$.

A third-degree expansion views the function $f(x)$ as an area under the curve $f_{area}(x)$, and the change in that area is the y-intercept plus the area change $f'_{area}(x)(x - a)$ plus $f'_{area}(x)$ multiplied by $x - a$, and bringing this back to the original function $f(x)$, is $f'_{area}(x)(x - a) = f''(x)(x - a)^2$. The third term represents a triangle, which is technically in the 2D simplex, so divide by $2! = 2$.

A fourth-degree expansion views the function one dimension higher, and this takes place in the 3D simplex, so the fourth term is divided by $3! = 6$. The fourth term is the 'triangle' in the 3D simplex.

Euler's number

$$e^x = \frac{1}{0!}x^0 + \frac{1}{1!}x^1 + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots$$

is just taking simplices at higher and higher dimensions.

Fourier, Laplace series, as well as moment generating functions, can be understood as related to Taylor series. A Taylor series approximates a function at higher and higher details at higher and higher dimensions while Laplace, Fourier series approximate using higher and higher frequencies. MGFs approximate at higher and higher moments.

25.21.9 Generating functions

Z-transform is the discrete analog of Laplace transform.

A polynomial is a finite series or a dot product.

Use a generating function if a simpler method does not work.

It is basically a polynomial, so a dot product with powers as basis vectors.

A generating function can solve recurrence relations by multiplying both sides by the generalized basis vector x^n , i.e. transfer it to a polynomial in vector space. Then solve for the generating function form

$$A(x) = F(x) = \sum a_n x^n$$

which can then be expanded out to (often) obtain an explicit formula for a_n .

Cauchy product is the discrete convolution of two infinite series.

Euler's number appears in probability in many places because it is related to factorials which relate to probability problems.

The t in MGFs is like the 'dimension'. MGFs are like integral transforms, and they are related by Euler's identity: MGFs use the series expansion and integral transforms use the sine and cosine form. Since MGFs are like integral transforms, they are unique to the $f(t) = PDF$.

Although in the end we set $t = 0$, for the integral, it must be that t works for any value. But it might diverge for some t 's, which is why the MGF does not always exist.

25.21.10 Moments

The first moment is average, the second variance, third is skew, fourth is kurtosis. Fifth is more skew, sixth is more kurtosis, seventh is even more skew, eighth is even more kurtosis, etc.

If the moment generating function $MX(t) = E^{etX}$ of the random variable X exists (for t in some open interval containing zero), then all the moments of X exist. So one way to show that t distributions do not have moment generating functions is to show that not all moments exist.

But it is well known that the t -distribution with ν degrees of freedom only have moments up to order $\nu-1$, so the MGF do not exist.

25.21.11 Characteristic function

An MGF may not always exist because moments may not exist as e^{tX} is not bounded. Mathematically, however, it will always exist - 'exist' for PDFs means convergent, not divergent. In contrast, the characteristic function of a PDF always exists as $e^{itX} = 1$. But MGFs may have moments that spiral out of control, leading to divergent series.

While both Laplace and Fourier transforms can diverge, for PDFs, the t is not bounded so Laplace transforms can diverge while Fourier transforms with PDFs will always converge.

From a probabilistic perspective, moment generating functions are essentially interesting only when they converge in a real neighborhood of 0. (Indeed, in that case they converge in the corresponding complex strip where the imaginary part is allowed to be arbitrary.) To say the moment generating function converges on the imaginary axis is no more than to say the characteristic function exists, which is of course fundamental, but since it's always true it's in some

sense uninteresting.

If X is a random variable, then what it means for the characteristic function of X to exist is that the random variable $|e^{itX}|$ must have finite expectation for every t . This is automatically true because $|e^{itX}| = 1$ for all t , hence $E(|e^{itX}|) = 1 < \infty$. This is true regardless of whether or not the moments exist. The characteristic function can be Taylor expanded in terms of the moments only if the moments actually exist, but in general they will not.

25.21.12 Probability generating function

The PGF is

$$G(z) = E(z^X) = \sum_{x=0}^{\infty} p(x)z^x$$

and produces the probability masses. The MGF, which is a generalization of the PGF, is

$$M(t) = E(e^{tX})$$

Intuitively, the MGF is like the derivative of the PGF, since taking the derivative of it gives you the derivative values, as in Taylor series. They are related by logarithms, which are generalizations of powers and factorials, so has a relationship to the inverted factorials in the Euler-Taylor expansion.

Now define

$$\log(z) = t \Rightarrow z = e^t$$

Then

$$G(z) = E(z^X) = E(e^t)^X = E(e^{tX}) = M(t) = M(\log(z))$$

The relationship is

$$G(z) = M(\log(z))$$

An alternative way to look at probability generating functions, let $G(z)$ instead be called $f(x)$:

$$f(x) = p_0 + p_1x + p_2x^2 + p_3x^3 + \dots$$

It is like the frequency domain, because the probabilities note the frequencies of events happening. Tossing a coin:

$$f(x) = \frac{1}{2} + \frac{1}{2}x$$

Rolling a die:

$$f(x) = \frac{1}{6} + \frac{1}{6}x + \frac{1}{6}x^2 + \frac{1}{6}x^3 + \dots + \frac{1}{6}x^6$$

Tossing two coins:

$$f(x) = \frac{1}{4} + \frac{2}{4}x + \frac{1}{4}x^2 = \left(\frac{1}{2} + \frac{1}{2}x\right)^2$$

Generally, if $f(x)$ and $g(x)$ are the probability generating functions of two independent random variables X and Y , then the generating function, corresponding to the sum $X + Y$ equals the product $f(x)g(x)$. This is like Fourier transforms which transform convolutions into arithmetic.

25.21.13 Abstract algebra

A group is closed under one operation ('addition'), a ring is closed under two operations ('addition' and 'multiplication'), and a field is a commutative division ring. Rings that are commutative for multiplication are commutative rings, and rings with multiplicative inverses are division rings. Polynomials are rings because a polynomial multiplied by a polynomial is another polynomial. Moreover, polynomials are commutative so form a commutative ring. However, the set of all polynomials is not a division ring. While many polynomials are divisible, many others are not.

A ring is intuitively like 'clumps' that are irreducible (read: indivisible), and so they are like natural numbers where prime numbers are indivisible numbers and so are like indivisible polynomials. Another type of 'clump' are lego pieces - some objects are divisible into smaller lego pieces while some pieces are not - but they are all legos so form a closed set.

Rings are like whole numbers as some are divisible by prime numbers while others are indivisible because they are prime numbers. To obtain divisibility, whole numbers need to become real numbers - so the number line has to expand. To obtain divisibility, polynomials have to somehow expand but they cannot. So they remain rings.

Groups, rings, fields, vector spaces, etc. are all 'spaces' where the rules differ. A polynomial fits in a vector space, and a vector space allows addition of vectors but not multiplication of vectors - a vector space only has multiplication by scalars. Multiplication of vectors makes no sense in vector spaces - it is like putting a giant car in a room and asking it to move quickly - the space does not allow it so the question is meaningless.

Polynomials are vectors in vector spaces as long as we assume that $x, x^2, x^3 \dots$ are orthogonal to each other. However, as soon as polynomials are multiplied, they are no longer orthogonal because $x^2 \times x^2 = x^4$ which means x^2, x^4 are not orthogonal: it is like $x = 1$ so the orthogonality collapses and so polynomials are just basically scalars now (which can be multiplied, even in vector spaces).

Square matrices form a ring. But vectors do not form a ring, because vectors cannot be 'multiplied' with each other.

A vector space is a combination of a field over scalars and group over vector addition. Because vectors cannot be multiplied, dot products or inner products

were created, which are analogous to multiplication.

A polynomial can be a vector, but since vectors are not rings, polynomials naturally cannot be multiplied under vector representation. They can be multiplied using Toeplitz or Toeplitz-like matrices, but this is not naturally vector 'multiplication'. Or it might be something resembling a cross-product. Polynomial vectors cannot be multiplied as vectors cannot be multiplied, but a matrix can be multiplied by a vector to output another vector - so convolution or polynomial multiplication can be represented by a matrix multiplied with a vector. More specifically, as polynomial multiplication is a convolution, use a Toeplitz matrix. Intuitively, because a Toeplitz matrix has a 'shifted' symmetry - symmetry across diagonals, it will be a 'shifted' version of the coefficients across the polynomial vector space, which allows convolution (i.e. matrix multiplication).

26 Differential Equations

26.1 Euler method

Step 1. Initial condition $y(x_0) = y_0$.

Step 2. Move a small amount along slope $\frac{dy}{dx} = f(x, y)$. Get equation $y_1 = y_0 + \Delta x f(x_0, y_0)$.

Step 3. Repeat. Will obtain a curve.

26.2 Runge-Kutta methods

The Euler method can be re-expressed as $y_{n+1} = y_n + k_1$ where $k_1 = \Delta x f(x_n, y_n)$, which is the first-order RK method. The second order method is $y_{n+1} = y_n + ak_1 + bk_2$ where $k_2 = \Delta x f(x_n + \alpha \Delta x, y_n + \beta k_1)$. There are further constraints: $a + b = 1, \alpha b = \beta b = \frac{1}{2}$.

26.3 Separable first-order

$$g(y) \frac{dy}{dx} = f(x), y(x_0) = y_0$$

$$\int_{y_0}^y g(y) dy = \int_{x_0}^x f(x) dx$$

which is equivalent to

$$\int_{x_0}^x g(y(x)) y'(x) dx = \int_{x_0}^x f(x) dx$$

through substitution of $u = y(x), du = y'(x) dx$.

26.4 Linear first-order ODE

$$\frac{dy}{dx} + p(x)y = g(x), y(x_0) = y_0$$

Let $\mu(x)$ be an integrating factor:

$$\mu(x) \left(\frac{dy}{dx} + p(x)y \right) = \mu(x)g(x)$$

Let the LHS be:

$$\mu \frac{dy}{dx} + \mu p y = \frac{d}{dx}(\mu y)$$

Then

$$\mu \frac{dy}{dx} + \mu p y = \mu \frac{dy}{dx} + y \frac{d\mu}{dx}$$

Cancelling terms out,

$$\begin{aligned} \frac{d\mu}{dx} &= p\mu \\ \int_{\mu_0}^{\mu} \frac{d\mu}{\mu} &= \int_{x_0}^x p dx \\ \ln \mu - \ln \mu_0 &= \int_{x_0}^x p dx \end{aligned}$$

Here, let $\mu_0 = 1$ as it doesn't matter what it is,

$$\mu(x) = \exp \left(\int_{x_0}^x p(x) dx \right)$$

So equate LHS with RHS

$$\begin{aligned} \frac{d}{dx}(\mu y) &= \mu g \\ \int_{x_0}^x d(\mu y) &= \int_{x_0}^x \mu g dx \\ \mu(x)y(x) \Big|_{x_0}^x &= \int_{x_0}^x \mu g dx \\ \mu(x)y(x) - \mu(x_0)y(x_0) &= \int_{x_0}^x \mu g dx \\ \mu(x)y(x) - e^0 y(x_0) &= \int_{x_0}^x \mu g dx \\ \mu(x)y(x) - y(x_0) &= \int_{x_0}^x \mu g dx \\ y(x) &= \frac{1}{\mu(x)} \left(y_0 + \int_{x_0}^x \mu(x)g(x) dx \right) \end{aligned}$$

26.5 Application: compound interest

$$S(t + \Delta t) = S(t) + r\Delta t S(t) + k\Delta t$$

where k is the deposit. Then,

$$\lim_{\Delta t \rightarrow 0} \frac{S(t + \Delta t) - S(t)}{\Delta t} = rS(t) + k$$

$$\frac{dS}{dt} = rS + k$$

Let $S(0) = S_0$. Then

$$\frac{dS}{dt} - rS = k$$

$$\mu(t) = e^{-rt}$$

$$S(t) = e^{rt} \left(S_0 + \int_0^t e^{-rt} k dt \right) = S_0 e^{rt} + \frac{k}{r} e^{rt} (1 - e^{-rt})$$

Note by l'Hopital's rule,

$$\lim_{r \rightarrow 0} S(t) = S_0 + kt$$

26.6 Application: terminal velocity

$$F = mg + kv = ma$$

$$m \frac{dv}{dt} = -mg - kv$$

$$\frac{dv}{dt} + \frac{k}{m}v = -g, v(0) = 0$$

At equilibrium, $\frac{dv}{dt} = 0, v \rightarrow v_\infty$. So,

$$v_\infty = \frac{-mg}{k}$$

Solving the linear 1st-order ODE,

$$\mu(t) = e^{-kt/m}$$

$$\begin{aligned} v(t) &= e^{-kt/m} \int_0^t e^{kt/m} (-g) dt \\ &= -\frac{mg}{k} (1 - e^{-kt/m}) \\ &= v_\infty (1 - e^{-kt/m}) \end{aligned}$$

26.7 Application: RC circuit

Ohm's law is $V_R = iR$. Capacitor equation is $q = V_C C$, where $i = \frac{dq}{dt}$. Kirchoff's law is $\epsilon = V_R + V_C$. Then,

$$V_R = iR = \frac{dq}{dt}R = \frac{dV_C}{dt}R = RC \frac{dV_C}{dt}$$

Charging:

$$\begin{aligned} RC \frac{dV_C}{dt} + V_C &= \epsilon \\ \frac{dV_C}{dt} + \frac{1}{RC} V_C &= \frac{\epsilon}{RC} \\ \mu(t) &= e^{t/RC} \\ V_C &= e^{-t/RC} \int_0^t \frac{e^{t/RC}}{RC} dt = \epsilon(1 - e^{-t/RC}) \end{aligned}$$

Discharging:

$$\begin{aligned} \frac{dV_C}{dt} + \frac{1}{RC} V_C &= 0 \\ V_C(t) &= \epsilon e^{-t/RC} \end{aligned}$$

26.8 Application: SIR epidemic model

S is susceptible, I is infected, and R is removed. β is degree of contagiousness. Further, $\beta\Delta t$ is the probability of I infecting S in Δt . γ is the rate of removal. So $\gamma\Delta t$ is the probability of moving from I to R in Δt . Structural equations:

$$\begin{aligned} S(t + \Delta t) &= S(t) - \beta\Delta t S(t)I(t) \\ I(t + \Delta t) &= I(t) + \beta\Delta t S(t)I(t) - \gamma\Delta t I(t) \\ R(t + \Delta t) &= R(t) + \gamma\Delta t I(t) \end{aligned}$$

Re-express:

$$\begin{aligned} \frac{S(t + \Delta t) - S(t)}{\Delta t} &= -\beta S(t)I(t) \\ \frac{I(t + \Delta t) - I(t)}{\Delta t} &= \beta S(t)I(t) - \gamma I(t) \\ \frac{R(t + \Delta t) - R(t)}{\Delta t} &= \gamma I(t) \end{aligned}$$

The limit as $\Delta t \rightarrow 0$.

$$\begin{aligned} \frac{dS}{dt} &= -\beta SI, \frac{dI}{dt} = \beta SI - \gamma I, \frac{dR}{dt} = \gamma I \\ S(0) &= N - n, I(0) = n, R(0) = 0 \end{aligned}$$

Solving with $S(t) \approx N, I \ll N$,

$$\frac{dI}{dt} = (\beta S(t) - \gamma)I = (\beta N - \gamma)I$$

Solving as a first-order linear ODE,

$$\frac{dI}{dt} - (\beta N - \gamma)I = 0$$

$$I(t) = I(0)e^{(\beta N - \gamma)t}$$

where there is an epidemic when $\beta N - \gamma > 0$, or when $\frac{\beta N}{\gamma} > 1 \rightarrow R_0 > 1$. Graph normalized populations, $S/N, I/N, R/N$ versus time. Flatten curve by reducing R_0 .

26.9 Second-order Euler method

Generalization of first-order Euler method to second-order DEs. Define $\ddot{x} = f(t, x, \dot{x})$ and $\dot{x} = u$. Then, $\dot{u} = \ddot{x} = f(t, x, u)$.

Let initial conditions be $x(t_0) = x_0, u(t_0) = u_0$. If these are graphed on Cartesian coordinates: (1) x v. t slope is u_0 , (2) u v. t slope is $\dot{u}_0 = f(0, x_0, u_0)$. Then, each 'Euler step' is: (1) $x_1 = x_0 + \Delta t u_0$, and (2) $u_1 = u_0 + \Delta t f(0, x_0, u_0)$.

Use this method to write a second-order ODE as a system of two equations: (1) $\dot{x} = u$ and (2) converting the second-order ODE into a first-order ODE by substituting (1).

The second-order Runge-Kutta modified Euler method (predictor-corrector method) for the following system of two first-order ODEs $\dot{x} = f(t, x, y)$ and $\dot{y} = g(t, x, y)$ are:

$$k_1 = \Delta t f(t_n, x_n, y_n)$$

$$k_2 = \Delta t f(t_n + \Delta t, x_n + k_1, y_n + l_1)$$

$$l_1 = \Delta t g(t_n, x_n, y_n)$$

$$l_2 = \Delta t g(t_n + \Delta t, x_n + k_1, y_n + l_1)$$

and

$$x_{n+1} = x_n + \frac{1}{2}(k_1 + k_2)$$

$$y_{n+1} = y_n + \frac{1}{2}(l_1 + l_2)$$

26.10 Principle of superposition

For 2nd-order linear homogenous ODE.

$$\ddot{x} + p(t)\dot{x} + q(t)x = 0$$

Suppose $x = x_1(t)$ and $x = x_2(t)$ are solutions. Then by the principle of superposition

$$x = c_1x_1 + c_2x_2$$

is also a solution. Proof.

$$\ddot{x} + p\dot{x} + qx = 0$$

$$c_1\ddot{x}_1 + c_2\ddot{x}_2 + p(c_1\dot{x}_1 + c_2\dot{x}_2) + q(c_1x_1 + c_2x_2) = 0$$

$$c_1(\ddot{x}_1 + p\dot{x}_1 + qx_1) + c_2(\ddot{x}_2 + p\dot{x}_2 + qx_2) = 0$$

$$c_1(0) + c_2(0) = 0$$

26.11 Wronskian

Leaving out the time parameter (so $x = x(t)$), if a 2nd-order linear homogenous equation is $\ddot{x} + p\dot{x} + qx = 0$, and $x = c_1x_1 + c_2x_2$ is a solution and want to satisfy the initial conditions

$$x(t_0) = x_0$$

$$\dot{x}(t_0) = u_0$$

then get two equations after combining:

$$c_1x_1(t_0) + c_2x_2(t_0) = x_0$$

$$c_1x_1(t_0) + c_2x_2(t_0) = u_0$$

use the Wronskian, which is a determinant:

$$W = \begin{vmatrix} x_1(t_0) & x_2(t_0) \\ \dot{x}_1(t_0) & \dot{x}_2(t_0) \end{vmatrix} \neq 0$$

gives a unique solution if the determinant is non-zero:

$$= x_1(t_0)\dot{x}_2(t_0) - x_2(t_0)\dot{x}_1(t_0)$$

There are many solutions. Example solutions are:

$$x_1(t) = \sin\omega t$$

$$x_2(t) = \cos\omega t$$

$$W = -\omega\sin^2\omega t - \omega\cos^2\omega t = -\omega$$

which is non-zero when $\omega \neq 0$.

26.12 Homogenous 2nd-order ODE with constant coefficients

$$a\ddot{x} + b\dot{x} + cx = 0$$

where a, b, c are constant. Try $x(t) = e^{rt}$. then

$$ar^2e^{rt} + bre^{rt} + ce^{rt} = 0$$

$$ar^2 + br + c = 0$$

is the characteristic equation (related to the eigenvalue problem). It has 2 real roots, complex conjugate roots, or 1 real root.

26.12.1 Case 1: distinct real roots

Example. Solve $\ddot{x} + 5\dot{x} + 6x = 0$ with initial conditions $x(0) = 2, \dot{x}(0) = 3$. Try $x = e^{rt}$ to get characteristic equation $r^2 + 5r + 6 = 0$ which leads to solutions $r = -2, -3$. So by the superposition principle,

$$x(t) = c_1e^{-2t} + c_2e^{-3t}$$

where its *Wronskian* $\neq 0$. Differentiating,

$$\dot{x}(t) = -2c_1e^{-2t} - 3c_2e^{-3t}$$

Use the two initial conditions:

$$x(0) = c_1e^{-2(0)} + c_2e^{-3(0)}$$

$$\dot{x}(0) = -2c_1e^{-2(0)} - 3c_2e^{-3(0)}$$

which reduce to a system of equations:

$$c_1 + c_2 = 3$$

$$-2c_1 - 3c_2 = 3$$

where $c_1 = 0, c_2 = -7$.

26.12.2 Case 2: complex-conjugate roots

Suppose the complex conjugate solutions are:

$$r = \lambda + i\mu$$

$$\bar{r} = \lambda - i\mu$$

Plugging into the ansatz equations,

$$z = e^{\lambda t}e^{i\mu t}$$

$$\bar{z} = e^{\lambda t} e^{-i\mu t}$$

By definition and the superposition principle, there are two solutions:

$$\frac{z + \bar{z}}{2} = \operatorname{Re}(z) = x_1$$

$$\frac{z - \bar{z}}{2i} = \operatorname{Im}(z) = x_2$$

So from $z = e^{\lambda t} e^{i\mu t}$, and recalling Euler's formula $e^{i\mu t} = \cos(\mu t) + i \sin(\mu t)$, then

$$x_1(t) = \frac{z + \bar{z}}{2} = \operatorname{Re}(z) = e^{\lambda t} \cos(\mu t)$$

$$x_2(t) = \frac{z - \bar{z}}{2i} = \operatorname{Im}(z) = e^{\lambda t} \sin(\mu t)$$

Example. $\ddot{x} + \dot{x} + x = 0$ with $x(0) = 1, \dot{x}(0) = 1$. With $x = e^{rt}$, the characteristic equation is $r^2 + r + 1 = 0$, so $r = \frac{-1 \pm i\sqrt{3}}{2}$. The general solution is then

$$x(t) = \exp\left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i\right)t$$

$$x(t) = e^{-t/2} e^{\pm \frac{\sqrt{3}}{2}it}$$

So the general solution is:

$$x(t) = e^{-t/2} (Ax_1(t) \pm Bx_2(t))$$

$$x(t) = e^{-t/2} \left(A \cos \frac{\sqrt{3}}{2}t + B \sin \frac{\sqrt{3}}{2}t \right)$$

and the derivative is

$$\dot{x}(t) = -\frac{1}{2}e^{-t/2} \left(A \cos \frac{\sqrt{3}}{2}t + B \sin \frac{\sqrt{3}}{2}t \right) + \frac{\sqrt{3}}{2}e^{-t/2} \left(-A \sin \frac{\sqrt{3}}{2}t + B \cos \frac{\sqrt{3}}{2}t \right)$$

Plugging in initial conditions,

$$\begin{aligned} A &= 1 \\ -\frac{1}{2}A + \frac{\sqrt{3}}{2}B &= 0 \\ B &= \frac{1}{\sqrt{3}} \end{aligned}$$

And these can be used to obtain the general solution.

26.12.3 Case 3: repeated roots

The general roots are $r = \lambda \pm i\mu$. The general solution is

$$x(t) = e^{\lambda t}(A \cos(\mu t) + B \sin(\mu t))$$

With repeated roots, $\mu = 0$ which leads to a missing solution as the second term disappears. So it must be that $B \rightarrow \infty \propto \frac{1}{\mu}$. With initial conditions $x(0) = x_0, \dot{x}(0) = u_0$, and using $x(t), \dot{x}(t)$,

$$A = x_0$$

$$\lambda A + \mu B = u_0$$

$$B = \frac{u_0 - \lambda x_0}{\mu}$$

So the general solution is now

$$x(t) = e^{\lambda t}(x_0 \cos(\mu t) + \frac{u_0 - \lambda x_0}{\mu} \sin(\mu t))$$

$$\lim_{\mu \rightarrow 0} x(t, \mu) = e^{\lambda t}(x_0 + (u_0 - \lambda x_0) \lim_{\mu \rightarrow 0} \frac{\sin(\mu t)}{\mu})$$

Using l'Hopital's rule,

$$= e^{\lambda t}(x_0 + (u_0 - \lambda x_0)t) = e^{\lambda t}(c_1 + c_2 t)$$

26.13 Inhomogenous 2nd-order ODE

$$\ddot{x} + p(t)\dot{x} + q(t)x = g(t)$$

$$x(0) = x_0$$

$$\dot{x}(0) = u_0$$

(1) Solve the homogenous equation

$$x_h(t) = c_1 x_1(t) + c_2 x_2(t)$$

(2) Find the particular solution, $x_p(t)$ (3) Combine (1) and (2), $x(t) = x_h(t) + x_p(t)$. Determine c_1, c_2 from initial conditions.

$$\ddot{x} + p(t)\dot{x} + q(t)x = g(t)$$

$$= \ddot{x}_h + \ddot{x}_p + p(\dot{x}_h + \dot{x}_p) + q(x_h + x_p)$$

$$= \ddot{x}_h + p\dot{x}_h + qx_h + \ddot{x}_p + p\dot{x}_p + qx_p = 0 + g(t)$$

So,

$$\ddot{x}_h + p\dot{x}_h + qx_h = 0$$

$$\ddot{x}_p + p\dot{x}_p + qx_p = g(t)$$

26.14 Exponential inhomogenous terms

Example. $\ddot{x} - 3\dot{x} - 4x = 3e^{2t}$ with initial conditions are $x(0) = 1, \dot{x}(0) = 0$.

(1) Homogenous equation. try $x = e^{rt}$. Then $r^2 - 3r - 4 = 0 \rightarrow x_h(t) = c_1e^{4t} + c_2e^{-t}$.

(2) Particular solution. Try $x = Ae^{2t}$. $4A - 6A - 4A = 3, A = \frac{-1}{2}$.

(3) $x(t) = x_h + x_p = c_1e^{4t} + c_2e^{-t} - (1/2)e^{2t}$. Use the set of initial conditions to get

$$\begin{aligned}c_1 + c_2 - 1/2 &= 0 \\4c_1 - c_2 - 1 &= 0 \\c_1 &= 1/2, c_2 = 1 \\x(t) &= \frac{1}{2}e^{4t} - \frac{1}{2}e^{2t} + e^{-t}\end{aligned}$$

26.15 Trigonometric inhomogenous terms

First method. Uses system of equations.

$$\ddot{x} - 3\dot{x} - 4x = 2\sin t$$

Try $x = A\cos t + B\sin t$. Then,

$$\begin{aligned}-A\cos t - B\sin t + 3A\sin t - 3B\cos t - 4A\cos t - 4B\sin t &= 2\sin t \\-\underbrace{(5A + 3B)}_0 \cos t + \underbrace{(3A - 5B)}_2 \sin t &= 2\sin t \\A = \frac{3}{17}, B = \frac{-5}{17} \\x(t) &= \frac{1}{17}(3\cos t - 5\sin t)\end{aligned}$$

Second method. Uses complex numbers.

$$\ddot{x} - 3\dot{x} - 4x = 2\sin t$$

Particular solution: $z \in \mathcal{C}$. Since the inhomogenous term is a sine, use the imaginary part (if it were cosine, use the real part):

$$\ddot{z} - 3\dot{z} - 4z = 2e^{it}$$

So, $x = \text{Im}(z)$ as it's a sine. Try $z = Ce^{it}$:

$$\begin{aligned}-C - 3Ci - 4C &= 2 \\-C(5 + 3i) &= 2 \\C &= \frac{1}{17}(-5 + 3i) \\x_p &= \frac{1}{17}\text{Im}((-5 + 3i)(\cos t + i\sin t)) = \frac{1}{17}(3\cos t - 5\sin t)\end{aligned}$$

26.16 Polynomial inhomogeneous term

Particular solution. Example. $\ddot{x} + \dot{x} - 2x = t^2$. Try $x = At^2 + Bt + C$:

$$2A + 2At + B - 2At^2 - 2Bt - 2C = t^2$$

$$A = \frac{-1}{2}, B = \frac{-1}{2}, C = \frac{-3}{4}$$

26.17 Resonance

$$\ddot{x} + w_0^2 x = f \cos(wt)$$

where w_0 is the natural frequency and w is the added or applied frequency. When

$$\lim_{t \rightarrow \infty} w = w_0$$

there is resonance. Now,

$$x_h(t) = c_1 \cos(w_0 t) + c_2 \sin(w_0 t)$$

Try $x_p(t) = A \cos(wt)$,

$$-w^2 A + w_0^2 A = f$$

$$A = \frac{f}{w_0^2 - w^2}$$

$$x(t) = c_1 \cos(w_0 t) + c_2 \sin(w_0 t) + \frac{f}{w_0^2 - w^2} \cos(wt)$$

Initial conditions are $x(0) = \dot{x}(0) = 0$. Then,

$$c_1 + \frac{f}{w_0^2 - w^2} = 0 \rightarrow c_1 = \frac{-f}{w_0^2 - w^2}$$

$$w_0 c_2 = 0 \rightarrow c_2 = 0$$

$$x(t) = \frac{-f}{w_0^2 - w^2} \cos(w_0 t) + 0 + \frac{f}{w_0^2 - w^2} \cos(wt)$$

$$= \frac{f}{w_0^2 - w^2} (\cos(wt) - \cos(w_0 t))$$

At resonance,

$$x(t) = f \lim_{w \rightarrow w_0} \frac{\cos(wt) - \cos(w_0 t)}{w_0^2 - w^2}$$

Apply l'Hopital's rule,

$$= f \lim_{w \rightarrow w_0} \frac{-t \sin(wt)}{-2w} = tf \frac{\sin(w_0 t)}{2w_0}$$

where the amplitude increases linearly with time, tf .

26.18 Nondimensionalization

This equation has 6 units:

$$ml\ddot{\theta} + cl\dot{\theta} + mgsin\theta = F_0cos(wt)$$

Turn this into a unitless equation:

$$\ddot{\theta} + \frac{c}{m}\dot{\theta} + \frac{g}{l}sin\theta = \frac{F_0}{ml}cos(wt)$$

$$w_0 = \sqrt{\frac{g}{l}}$$

$$\tau = w_0 t$$

$$\frac{d\theta}{dt} = \frac{d\theta}{d\tau} \frac{d\tau}{dt} = w_0 \frac{d\theta}{d\tau}$$

$$\ddot{\theta} = w_0^2 \frac{d^2\theta}{d\tau^2}$$

$$w_0^2 \frac{d^2\theta}{d\tau^2} + \frac{cw_0}{m} \frac{d\theta}{d\tau} + w_0^2 sin\theta = \frac{F_0}{ml} cos \frac{w}{w_0} \tau$$

$$\frac{d^2\theta}{d\tau^2} + \frac{c}{mw_0} \frac{d\theta}{d\tau} + sin\theta = \frac{F_0}{mlw_0^2} cos \frac{w}{w_0} \tau$$

$$\frac{d^2\theta}{d\tau^2} + \alpha \frac{d\theta}{d\tau} + sin\theta = \gamma cos(\beta\tau)$$

which is a unitless equation. There were 6 units to begin, 3 fundamental units (mass, length, time) while the other units are composite units (like gravity, m/s^2) and there are now 3 units. By the Buckingham pie theorem, there are 6-3=3 units.

26.19 Application: RLC circuit

$$V_R = iR, V_C = \frac{q}{C}, V_L = \frac{di}{dt}L, i = \frac{dq}{dt}$$

Kirchoff's law: $E(t) = V_L + V_R + V_C = E_0cos(wt)$.

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E_0cos(wt)$$

This is a 2nd-order linear inhomogenous ODE with constant coefficients. Now make it unitless.

$$L \frac{d^2q}{dt^2} = -\frac{q}{C}$$

$$\frac{d^2q}{dt^2} = -\frac{q}{LC} = \frac{1}{s^2} = w_0^2$$

Then, $w_0 = \frac{1}{\sqrt{LC}}$. Define $\tau = w_0 t$ so that when τ spans 0 to 1 then t spans 0 to $1/w_0$. Define the unitless charge as

$$Q = \frac{w_0^2 L}{E_0} q = \left(\frac{1}{LC} \right) \frac{L}{E_0} q = \frac{q}{CE_0}$$

since the units of $E \approx \frac{q}{C}$. Then the unitless equation is

$$\frac{d^2 Q}{d\tau^2} + \alpha \frac{dQ}{d\tau} + Q = \cos(\beta\tau)$$

with $\alpha = \frac{R}{Lw_0}$, $\beta = \frac{w}{w_0}$.

26.20 Application: mass on a spring

Newton's 2nd law: $F = ma = m \frac{d^2 x}{dt^2}$. Hooke's law $F_x = -kx$. Frictional force $F - f = -c \frac{dx}{dt}$. External force $F_e = F_0 \cos(\omega t)$. Together,

$$m \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx = F_0 \cos(\omega t)$$

Define $w_0 = \sqrt{\frac{k}{m}}$, $\tau = w_0 t$, $X = \frac{mw_0^2 x}{F_0}$:

$$\frac{d^2 X}{d\tau^2} + \alpha \frac{dX}{d\tau} + X = \cos(\beta\tau)$$

where $\alpha = \frac{c}{mw_0}$ is the damping coefficient. Further, $\beta = \frac{w}{w_0}$.

26.21 Application: pendulum

Arc length is $s = l\theta$ with velocity $v = l\dot{\theta}$ and acceleration $a = l\ddot{\theta}$. The forces are $F_g = -mg \sin\theta$, $F_f = -c \frac{ds}{dt} = -cl\dot{\theta}$, $F_e = F_0 \cos(\omega t)$. So the DE is

$$ml\ddot{\theta} + cl\dot{\theta} + mg \sin\theta = F_0 \cos(\omega t)$$

Let $w_0 = \sqrt{\frac{g}{l}}$ and $\tau = w_0 t$. Then:

$$\frac{d^2 \theta}{d\tau^2} + \alpha \frac{d\theta}{d\tau} + \sin\theta = \gamma \cos(\beta\tau)$$

where

$$\alpha = \frac{c}{mw_0}, \beta = \frac{w}{w_0}, \gamma = \frac{F_0}{mlw_0^2}$$

This is a non-linear equation as it contains $\sin\theta$. To linearize, $\sin\theta \approx \theta$ and define $\Theta = \frac{\theta}{\gamma}$. Then

$$\frac{d^2 \Theta}{d\tau^2} + \alpha \frac{d\Theta}{d\tau} + \Theta = \gamma \cos(\beta\tau)$$

26.22 Damped resonance

$$\ddot{x} + \alpha\dot{x} + x = \cos(\beta t)$$

where $\alpha > 0$ is the damping coefficient and attached to velocity. For the homogenous equation, ansatz $x = e^{rt}$, so

$$r^2 + \alpha r + 1 = 0$$

$$r = \frac{-\alpha \pm \sqrt{\alpha^2 - 4ac}}{2}$$

so $\operatorname{Re}(z) < 0$ and

$$\lim_{t \rightarrow \infty} x_h(t) = 0$$

Use complex method

$$\ddot{z} + \alpha\dot{z} + z = e^{i\beta t}$$

where $x = \operatorname{Re}(z)$. Ansatz $z_p(t) = Ae^{i\beta t}$.

$$-\beta^2 A + i\alpha\beta A + A = 1$$

$$((1 - \beta^2) + i\alpha\beta)A = 1$$

$$A = \frac{1}{1 - \beta^2 + i\alpha\beta}$$

Resonance: $\beta = 1$. Then, $A = \frac{1}{i\alpha}$.

$$x_p = \operatorname{Re}\left(\frac{1}{i\alpha}e^{it}\right) = R\frac{\sin t}{\alpha}$$

when α is small, the friction is low and the amplitude x_p becomes large. Sine and cosine are out of phase by 90 degrees.

26.23 Laplace transform method

Most useful when inhomogeneous term has a discontinuity or impulse.

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$$

$\mathcal{L}$ transforms a function from t-space to s-space. It is linear: $\mathcal{L}\{c_1 f_1(t) + c_2 f_2(t)\} = c_1 \mathcal{L}\{f_1(t)\} + c_2 \mathcal{L}\{f_2(t)\}$. The inverse transforms from s-space to t-space:

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-(s-a)t} dt = -\frac{1}{s-a} e^{-(s-a)t} \Big|_0^\infty = \frac{1}{s-a}$$

with $s > a$.

26.24 LT of constant coefficient ODE

$$a\ddot{x} + b\dot{x} + cx = g(t)$$

$$a\mathcal{L}\{\ddot{x}\} + b\mathcal{L}\{\dot{x}\} + c\mathcal{L}\{x\} = \mathcal{L}\{g(t)\}$$

To solve

$$\mathcal{L}\{\dot{x}\} = \int_0^\infty e^{-st} \dot{x}(t) dt$$

use integration by parts with $u = e^{-st}$, $du = -se^{-st}dt$, $dv = \dot{x}(t)dt$, $v = x(t)$.

$$\mathcal{L}\{\dot{x}(t)\} = e^{-st}x(t) \Big|_0^\infty + s \int_0^\infty e^{-st}x(t)dt = 0 - x(0) + sX(s) = sX(s) - x(0)$$

To solve

$$\mathcal{L}\{\ddot{x}(t)\} = s^2X(s) - sx(0) - \dot{x}(0)$$

integrate by parts and recycle result from first derivative.

$$a(s^2X - sx_0 - u_0) + b(sX - x_0) + cX = G$$

which is now linear. Now solve for X.

26.25 Example

$$\ddot{x} + x = \sin(2t)$$

$$x(0) = 2, \dot{x}(0) = 1$$

$$\mathcal{L}\{\ddot{x}\} + \mathcal{L}\{x\} = \mathcal{L}\{\sin(2t)\}$$

$$s^2X - x(0)s - \dot{x}(0) + X = \frac{b}{s^2 + b^2}$$

$$s^2X - 2s - 1 + X = \frac{2}{s^2 + 4}$$

$$X = \frac{2s + 1}{s^2 + 1} + \frac{2}{(s^2 + 1)(s^2 + 4)}$$

Invert. Use partial fractions first, to turn solution into a form which is easier to invert.

$$\begin{aligned} X &= 2 \frac{s}{s^2 + 1} + \frac{1}{s^2 + 1} + \frac{2}{3} \frac{1}{(s^2 + 1)} - \frac{1}{3} \frac{2}{s^2 + 4} \\ &= 2\cos t + \frac{5}{3}\sin t - \frac{1}{3}\sin(2t) \end{aligned}$$

26.26 Heaviside step function

On a graph of x versus t

$$x = u_c(t) = \begin{cases} 0 & t < c \\ 1 & t \geq c \end{cases}$$

Laplace transform:

$$\mathcal{L}\{u_c(t)\} = \int_0^\infty e^{-st} u_c(t) dt = \int_c^\infty e^{-st} dt = \frac{e^{-cs}}{s}$$

Example. Shift $f(t)$ to the right by c while ensuring $f(t) = 0$ where $t < c$: $u_c(t)f(t-c)$. Use the Laplace transform table to transform (note variable change $t' = t - c$):

$$\begin{aligned} \mathcal{L}\{u_c(t)f(t-c)\} &= \int_0^\infty e^{-st} u_c(t) f(t-c) dt \\ &= \int_c^\infty e^{-st} f(t-c) dt = \int_0^\infty e^{-s(t'+c)} f(t') dt' = e^{-cs} \int_0^\infty e^{-st'} f(t') dt' = e^{-cs} F(s) \end{aligned}$$

Example.

$$f(t) = \begin{cases} f_1(t) & t < c \\ f_2(t) & t \geq c \end{cases} = f_1(t) + (f_2(t) - f_1(t))u_c(t)$$

Example. Heaviside step-down function.

$$1 - u_c = \begin{cases} 1 & t < c \\ 0 & t \geq c \end{cases}$$

Example. Step-up and step-down function.

$$u_a - u_b = \begin{cases} 0 & t < a \\ 1 & a \leq t \leq b \\ 0 & t > b \end{cases}$$

26.27 Dirac delta

Not a function but rather a limit of a function. Limit of Heaviside step up and step down function. Area is 1, with rectangle from $c - \epsilon$, $c + \epsilon$ so length is ϵ and height is $\frac{1}{2\epsilon}$.

$$\delta(t-c) = \begin{cases} 0 & t \neq c \\ \infty & t = c \end{cases} = \lim_{\epsilon \rightarrow 0} \frac{1}{2\epsilon} (u_{c-\epsilon}(t) - u_{c+\epsilon}(t))$$

$$\int_0^\infty f(t) \delta(t-c) dt = f(c) \int_0^\infty \delta(t-c) dt = f(c) \mathcal{L}\{\delta(t-c)\} = \int_0^\infty e^{-st} \delta(t-c) dt = e^{-cs}$$

Example. Prove $\delta(ax) = \frac{1}{|a|}\delta(x)$. For $a > 0$,

$$\int_{-\infty}^{\infty} f(x)\delta(ax)dx = \frac{1}{a} \int_{-\infty}^{\infty} f\left(\frac{x'}{a}\right)\delta(x')dx' = \frac{1}{a}f(0)$$

but

$$\frac{1}{a}f(0) = \int_{-\infty}^{\infty} f(x)\delta(x)dx$$

so it must be that $\delta(ax) = -\frac{1}{a}\delta(x)$. For $a < 0$, $\delta(ax) = \frac{1}{|a|}\delta(x)$.

26.28 Example: discontinuous inhomogenous term

$$\ddot{x} + 3\dot{x} + 2x = 1 - u_c(t)$$

$$x(0) = \dot{x}(0) = 0$$

$$s^2X + 3sX + 2X = \frac{1}{s}(1 - e^{-s})$$

$$X = \frac{1 - e^{-s}}{s(s+1)(s+2)}$$

By defining $F(s) = \frac{1}{s(s+1)(s+2)}$,

$$X(s) = F(s) - e^{-s}F(s)$$

Using the Laplace transform table,

$$u_c(t)f(t-c) = e^{-cs}F(s)$$

$$x(t) = f(t) - u_1(t)f(t-1)$$

For $\mathcal{L}^{-1}\{F(s)\}$:

$$F(s) = \frac{1}{s(s+1)(s+2)} = \frac{1/2}{s} + \frac{-1}{s+1} + \frac{1/2}{s+2}$$

$$f(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

26.29 Example: impulsive inhomogeneous term

$$\ddot{x} + 3\dot{x} + 2x = \delta(t)$$

$$x(0) = \dot{x}(0) = 0$$

$$s^2X + 3sX + 2X = 1$$

$$X = \frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

$$x(t) = e^{-t} - e^{-2t} = e^{-t}(1 - e^{-t})$$

But there's a velocity discontinuity from the impulse:

$$\dot{x}(t) = e^{-t} + 2e^{-2t}$$

$$\dot{x}(0^-) = 0$$

$$\dot{x}(0^+) = 0$$

26.30 Series solution

$$y'' + y = 0$$

Ansatz:

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

Plug into DE:

$$\sum_{n=2}^{\infty} a_n x^{n-2} + \sum_0^{\infty} a_n x^n = 0$$

Re-index:

$$\sum_0^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_0^{\infty} a_n x^n = \sum_0^{\infty} ((n+2)(n+1)a_{n+2} + a_n)x^n = 0$$

where the bracketed term must be zero. So,

$$a_{n+2} = \frac{-a_0}{(n+1)(n+2)}$$

which is a recursion relation. Taken the even powers:

$$a_0, a_2 = \frac{-a_0}{2 \cdot 1}, a_4 = \frac{-a_2}{4 \cdot 3} = \frac{-a_0}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{-a_0}{4!}, \dots$$

Take the odd powers next:

$$a_1, a_3 = \frac{-a_1}{3 \cdot 2}, a_5 = \frac{-a_3}{5 \cdot 4} = \frac{-a_1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{-a_1}{5!}, \dots$$

$$\begin{aligned} y(x) &= a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} \dots \right) \\ &= a_0 \cos(x) + a_1 \sin(x) \end{aligned}$$

A different form:

$$y'' - y = 0$$

Ansatz:

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

Plug into DE:

$$\sum_{n=2}^{\infty} a_n x^{n-2} - \sum_0^{\infty} a_n x^n = 0$$

Re-index:

$$\sum_0^{\infty} (n+2)(n+1)a_{n+2}x^n - \sum_0^{\infty} a_n x^n = \sum_0^{\infty} ((n+2)(n+1)a_{n+2} - a_n)x^n = 0$$

where the bracketed term must be zero. So,

$$a_{n+2} = \frac{a_0}{(n+1)(n+2)}$$

which is a recursion relation. Taken the even powers:

$$a_0, a_2 = \frac{a_0}{2 \cdot 1}, a_4 = \frac{a_2}{4 \cdot 3} = \frac{a_0}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{-a_0}{4!}, \dots$$

Take the odd powers next:

$$a_1, a_3 = \frac{a_1}{3 \cdot 2}, a_5 = \frac{a_3}{5 \cdot 4} = \frac{a_1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{a_1}{5!}, \dots$$

$$\begin{aligned} y(x) &= a_0 \left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} \dots \right) + a_1 \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} \dots \right) \\ &= a_0 \cosh(x) + a_1 \sinh(x) \end{aligned}$$

26.31 Airy's equation

$$y'' - xy = 0$$

Ansatz:

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

Plug into DE:

$$\sum_{n=2}^{\infty} a_n x^{n-2} - \sum_0^{\infty} a_n x^n = 0$$

Re-index:

$$\sum_{-1}^{\infty} (n+3)(n+2)a_{n+3}x^{n+1} - \sum_0^{\infty} a_n x^{n+1}$$

where the first term has a constant but the second doesn't. So isolate the constant from the first term:

$$= 2a_2 + \sum_0^{\infty} ((n+3)(n+2)a_{n+3} - a_n)x^{n+1} = 0$$

So,

$$\begin{aligned} a_2 &= 0 \\ a_{n+3} &= \frac{a_n}{(n+3)(n+2)} \end{aligned}$$

which is a recursion relation. Start at $n = 0$:

$$a_0, a_3 = \frac{a_0}{3 \cdot 2}, a_6 = \frac{a_3}{6 \cdot 5} = \frac{a_0}{6 \cdot 5 \cdot 3 \cdot 2}, \dots$$

Take $n = 1$:

$$a_1, a_4 = \frac{a_1}{4 \cdot 3}, a_7 = \frac{a_4}{7 \cdot 6} = \frac{a_1}{7 \cdot 6 \cdot 4 \cdot 3}, \dots$$

$$y(x) = a_0 \left(1 + \frac{x^3}{3 \cdot 2} + \frac{x^6}{6 \cdot 5 \cdot 3 \cdot 2} \dots \right) + a_1 \left(x + \frac{x^4}{4 \cdot 3} + \frac{x^7}{7 \cdot 6 \cdot 4 \cdot 3} \dots \right)$$

$$= a_0 y_0(x) + a_1 y_1(x)$$

Note: $y_0(0) = 0, y'_0(0) = 0, y_1(0) = 0, y'_1(0) = 1$. For $x < 0$, the function is oscillatory as the equation becomes $y'' + xy = 0$ which resembles the oscillatory ODE with cosine/sine solutions $y'' + y = 0$. For $x > 0$, the function remains $y'' - xy = 0$ which resembles the ODE with exponential solutions $y'' - y = 0$.

26.32 Matrix algebra review

$$Ax = \lambda x$$

$$\det \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = (a - \lambda)(d - \lambda) - bc$$

$$= \lambda^2 - (a + d)\lambda + ad - bc = \lambda^2 - \text{Tr}(A)\lambda + \det A = 0$$

26.33 Systems of linear 1st-order ODEs

$$\dot{x}_1 = ax_1 + bx_2$$

$$\dot{x}_2 = cx_1 + dx_2$$

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\dot{x} = Ax$$

Ansatz $x(t) = ve^{\lambda t}$

$$\lambda ve^{\lambda t} = Ave^{\lambda t}$$

Eigenvalue problem:

$$Av = \lambda v$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Try (i) distinct real roots, (ii) complex conjugate roots, and (iii) repeated roots. Last will not be covered here.

Example. Distinct real roots.

$$\dot{x}_1 = x_1 + x_2$$

$$\dot{x}_2 = 4x_1 + x_2$$

Ansatz $x(t) = ve^{\lambda t}$. Solving for the eigenvalues, $\lambda_1 = -1, \lambda_2 = 3$. For $\lambda_1 = -1$,

$$(A - \lambda_1 I)v_1 = 0$$

$$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{21} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2v_{11} + v_{21} = 0$$

$$v_1 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

For $\lambda_1 = 3$,

$$(A - \lambda_2 I)v_2 = 0$$

$$\begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} v_{12} \\ v_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$2v_{12} + v_{22} = 0$$

$$v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

General solution

$$x(t) = c_1 \begin{pmatrix} -1 \\ 2 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}$$

Re-express the general solution:

$$x_1 = c_1 e^{-t} + c_2 e^{3t}$$

$$x_2 = -2c_1 e^{-t} + 2c_2 e^{3t}$$

Example. Complex conjugate roots.

$$\dot{x}_1 = -\frac{1}{2}x_1 + x_2$$

$$\dot{x}_2 = -x_1 - \frac{1}{2}x_2$$

Ansatz $x(t) = ve^{\lambda t}$. Solving for the eigenvalues, $\lambda = -\frac{1}{2} + i, \bar{\lambda} = -\frac{1}{2} - i$. For λ ,

$$(A - \lambda I)v_1 = 0$$

$$\begin{pmatrix} -i & 1 \\ -1 & -i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-iv_1 + v_2 = 0$$

$$v = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

For $\bar{\lambda}$,

$$\bar{v} = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Use superposition of the real and imaginary parts (see above) for general solution:

$$\begin{aligned} x(t) &= c_1 \operatorname{Re}(ve^{\lambda t}) + c_2 \operatorname{Im}(ve^{\lambda t}) \\ &= c_1 e^{-t/2} \operatorname{Re} \left(\begin{pmatrix} 1 \\ i \end{pmatrix} (\cos t + i \sin t) \right) + c_2 e^{-t/2} \operatorname{Im} \left(\begin{pmatrix} 1 \\ i \end{pmatrix} (\cos t + i \sin t) \right) \end{aligned}$$

So the general solution is

$$= c_1 e^{-t/2} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + c_2 e^{-t/2} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

Review. Superposition of complex results. Suppose the complex conjugate solutions are:

$$r = \lambda + i\mu$$

$$\bar{r} = \lambda - i\mu$$

Plugging into the ansatz equations,

$$z = e^{\lambda t} e^{i\mu t}$$

$$\bar{z} = e^{\lambda t} e^{-i\mu t}$$

By definition and the superposition principle, there are two solutions:

$$\frac{z + \bar{z}}{2} = \operatorname{Re}(z) = x_1$$

$$\frac{z - \bar{z}}{2i} = \operatorname{Im}(z) = x_2$$

So from $z = e^{\lambda t} e^{i\mu t}$, and recalling Euler's formula $e^{i\mu t} = \cos(\mu t) + i \sin(\mu t)$, then

$$x_1(t) = \frac{z + \bar{z}}{2} = \operatorname{Re}(z) = e^{\lambda t} \cos(\mu t)$$

$$x_2(t) = \frac{z - \bar{z}}{2i} = \operatorname{Im}(z) = e^{\lambda t} \sin(\mu t)$$

26.34 Phase portrait

Visualize $\dot{x} = Ax$ on a graph of x_2 versus x_1 , where the origin is a fixed point (equilibrium). The equilibrium can be stable or unstable. The classification is

Eigenvalues	Fixed point
$\mathcal{R} < 0 + \mathcal{R} < 0$	Stable node
$\mathcal{R} > 0 + \mathcal{R} > 0$	Unstable node
$\mathcal{R} > 0 + \mathcal{R} < 0$	Saddle point
$\mathcal{C} + \mathcal{R} < 0$	Stable spiral
$\mathcal{C} + \mathcal{R} > 0$	Unstable spiral

Example.

$$\dot{x}_1 = -3x_1 + \sqrt{2}x_2$$

$$\begin{aligned}\dot{x}_2 &= \sqrt{2}x_1 - 2x_2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= c_1 \begin{pmatrix} 1 \\ -\sqrt{2}/2 \end{pmatrix} e^{-4t} + c_2 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{-t} \\ c_2 = 0 &\rightarrow x_2 = \frac{-\sqrt{2}}{2}x_1 \\ c_1 = 0 &\rightarrow x_2 = \sqrt{2}x_1\end{aligned}$$

So these are axes and the movement with time is inward on both of these axes as the eigenvalues are both negative. However, the axis with eigenvalue -4 will move faster than the axis with eigenvalue -1.

Phase portraits for ODE systems with complex conjugate eigenvalues. Real part will determine whether the spiral is stable or unstable, and look on the x_2 axis where $x_1 = 0$ to see whether $\dot{x}_1 < 0$ (CCW) or $\dot{x}_1 > 0$ (CW).

The characteristic equation for DEs and characteristic polynomial from linear algebra give the eigensolutions which are 'characteristic' of the system - characterize these to characterize the system.

26.35 Application: coupled oscillators

Two walls, two masses connected to the walls by Hooke's constant k and each other by Hooke's constant K . Use $F = ma$, $F = -kx$.

$$\begin{aligned}m\ddot{x}_1 &= -kx_1 - K(x_1 - x_2) \\ m\ddot{x}_2 &= -kx_2 - K(x_2 - x_1) \\ m \frac{d^2}{dt^2} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} -(k+K) & K \\ K & -(k+K) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ m\ddot{x} &= Ax\end{aligned}$$

Ansatz $x(t) = ve^{rt}$, where v is the constant vector like A in preceding sections.

$$\begin{aligned}mr^2ve^{rt} &= Ave^{rt} \\ mr^2v &= Av \\ \lambda v &= Av\end{aligned}$$

where A here is the matrix with k and K - the transition matrix. This is an eigenvalue problem.

$$\begin{aligned}\det(A - \lambda I) &= 0 \\ (-(k+K) - \lambda)^2 - K^2 &= 0 \\ (k+K + \lambda)^2 &= K \\ \lambda_1 = -k, \lambda_2 &= -(k+2K)\end{aligned}$$

Since $\lambda = mr^2$, $r = \pm\sqrt{\frac{\lambda}{m}}$.

$$r_1 = \pm\sqrt{\frac{-k}{m}} = \pm i\sqrt{\frac{k}{m}}$$

$$r_2 = \pm\sqrt{\frac{-(k+2K)}{m}} = \pm i\sqrt{\frac{k+2K}{m}}$$

Using the superposition principle, combine the complex results to get sines and cosines

$$\omega_1 = \sqrt{\frac{k}{m}}, \omega_2 = \sqrt{\frac{k+2K}{m}}$$

Normal mode 1: $\lambda_1 = -k$.

$$(A - \lambda_1 I)v_1 = 0$$

$$\begin{pmatrix} -K & K \\ K & -k \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{21} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

which is associated with $\omega_1 = \sqrt{\frac{k}{m}}$. The two masses are moving in tandem so K has no effect and does not appear in ω_1 .

Normal mode 2: $\lambda_2 = -k - 2K$:

$$(A - \lambda_2 I)v_2 = 0$$

$$\begin{pmatrix} -K & K \\ K & -k \end{pmatrix} \begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

which is associated with $\omega_2 = \sqrt{\frac{k+2K}{m}}$. The two masses move in opposite direction, so each of compression and expansion have double the effect of K.

The actual movement of the spring is a superposition of the normal modes.

Review. Superposition of complex results. Suppose the complex conjugate solutions are:

$$r = \lambda + i\mu$$

$$\bar{r} = \lambda - i\mu$$

Plugging into the ansatz equations,

$$z = e^{\lambda t} e^{i\mu t}$$

$$\bar{z} = e^{\lambda t} e^{-i\mu t}$$

By definition and the superposition principle, there are two solutions:

$$\frac{z + \bar{z}}{2} = \operatorname{Re}(z) = x_1$$

$$\frac{z - \bar{z}}{2i} = \operatorname{Im}(z) = x_2$$

So from $z = e^{\lambda t} e^{i\mu t}$, and recalling Euler's formula $e^{i\mu t} = \cos(\mu t) + i \sin(\mu t)$, then

$$x_1(t) = \frac{z + \bar{z}}{2} = \operatorname{Re}(z) = e^{\lambda t} \cos(\mu t)$$

$$x_2(t) = \frac{z - \bar{z}}{2i} = \operatorname{Im}(z) = e^{\lambda t} \sin(\mu t)$$

26.36 PDEs: the heat equation

$$\frac{\partial T(x, t)}{\partial t} = \alpha \frac{\partial^2 T(x, t)}{\partial x^2} = \alpha \nabla^2$$

where ∇^2 is the Laplacian, which indicates how different a point is from its neighbors. It is also the divergence of the gradient. The heat equation can be derived from discretely considering neighboring points. A higher-dimensional version of the heat equation is

$$\frac{\partial T(x, y, z, t)}{\partial t} = \alpha \left(\frac{\partial^2 T(x, y, z, t)}{\partial x^2} + \frac{\partial^2 T(x, y, z, t)}{\partial y^2} + \frac{\partial^2 T(x, y, z, t)}{\partial z^2} \right) = \alpha \nabla^2$$

Solve:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} = -\alpha T$$

The solution combines spatial and temporal dimensions:

$$T(x, t) = \underbrace{\sin(x)}_{\text{spatial}} \underbrace{e^{-\alpha t}}_{\text{temporal}}$$

There are different rules at the boundary, so the solution must fit the boundary conditions. In the heat equation, the edges will act differently from the interior, so

$$\frac{\partial T(0, t)}{\partial x} = \frac{\partial T(L, t)}{\partial x} = 0$$

Further, how the boundary acts (fixed or flexible) determines whether the solution is sine or cosine.

26.37 Complex exponentials in DEs

Real trig functions can be decomposed into their complex parts, which are opposite direction rotations:

$$2\cos\theta = e^{i\theta} + e^{-i\theta}$$

Note for exponential solutions Ae^{rt} that $A = e^{\pi/4}$ would be an initial rotation of 45 degrees: $Ae^{rt} = e^{\pi/4+rt}$.

Exponential solutions.

$$\frac{d}{dt}e^t = e^t$$

which denotes accelerating linear movement

$$\frac{d}{dt}e^{xt} = xe^{xt}$$

which denotes even faster accelerating linear movement

$$\frac{d}{dt}e^{it} = ie^{it}$$

which denotes rotational movement

26.38 Rotational solutions in DEs

The system

$$\frac{dx}{dt} = -y$$

$$\frac{dy}{dt} = x$$

can be viewed as separate movements along two completely separate 1D axes. It can also be combined into a 2D plane with the movement being $(x, y)'$, so the above can be re-expressed as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

where the transition matrix is a rotational matrix. The above is re-expressed as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

(which also indicates $e^{ix} = \cos(x) + i\sin(x)$). Both equations can be re-expressed as

$$\frac{d}{dt}v = Mv$$

Just like the problem

$$\frac{d}{dt}x = rx$$

has the solution

$$\frac{d}{dt}e^{rt} = re^{rt}$$

and with initial value

$$\frac{d}{dt}e^{rt} = re^{rt}x_0$$

The solution to

$$\frac{d}{dt}v = Mv$$

is

$$\begin{pmatrix} x \\ y \end{pmatrix} = e^{Mt} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

so,

$$\frac{d}{dt}e^{Mt} = Me^{Mt}$$

26.39 Matrix exponentials in DEs

$$e^{At}$$

where A is a matrix can be re-expressed using a Taylor expansion:

$$e^x = x^0 + x^1 + \frac{1}{2}x^2 + \dots$$

so

$$e^A = A^0 + A^1 + \frac{1}{2}A^2 + \dots$$

and if A is a rotation matrix, then the sum of the terms will approach that rotation.

26.40 Fourier series

Let $f(x)$ be a periodic function with period $2L$. Express it as:

$$f(x) = \frac{a_0}{2} + \sum_1^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

The sums are for superposition. Linearity is the basis for superposition and is reflected in the Fourier series. Use orthogonality conditions to derive the coefficients above

$$\int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = L\delta_{nm}$$

$$\int_{-L}^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = L\delta_{nm}$$

$$\int_{-L}^L \cos \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = 0$$

where δ_{nm} is the Kronecker delta with

$$\delta_{nm} = \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}$$

So,

$$\begin{aligned} \int_{-L}^L f(x) \sin \frac{m\pi x}{L} dx &= \sum_{m=1}^{\infty} b_m \int_{-L}^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx \\ &= \sum_{m=1}^{\infty} b_m L \delta_{nm} = L b_n \end{aligned}$$

So the coefficient formulas are:

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \end{aligned}$$

26.41 Fourier cosine and sine series

Even function ($f(-x) = f(x)$) multiplied by an odd function ($f(-x) = -f(x)$) results in an odd function. Integration of an odd function is 0.

Can eliminate sine or cosine if function is even or odd.

Fourier cosine series. When $f(x)$ is even.

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} \\ a_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx \end{aligned}$$

Fourier sine series. When $f(x)$ is odd.

$$\begin{aligned} f(x) &= \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} \\ b_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx \end{aligned}$$

Fourier coefficients are derived in an alternative expression from:

$$c_n = \int_0^1 f(t) e^{-2\pi i n t} dt$$

which holds each frequency ($n2\pi = \omega$) as a constant while setting all other terms to 0 and then integrating to get the constant.

26.42 Diffusion equation

$$u(x, t) = \text{density} = \frac{\text{mass}}{\text{length}} = \frac{M}{x}$$

$$M = u(x, t)\Delta x$$

Differentiate mass with respect to time:

$$\frac{dM}{dt} = \text{influx} - \text{outflux} = J(x, t) - J(x + \Delta x, t)$$

Fick's law of diffusion:

$$J(x, t) = -Du_x(x, t)$$

where D is the diffusion constant in units of $\text{length}^2/\text{time}$. Combine:

$$\begin{aligned}\frac{dM}{dt} &= \text{influx} - \text{outflux} = J(x, t) - J(x + \Delta x, t) \\ &= -Du_x(x, t) + Du_x(x + \Delta x, t)\end{aligned}$$

Back to differentiating mass with respect to time:

$$u_t(x, t)\Delta x = D(u_x(x + \Delta x, t) - u_x(x, t))$$

Rearrange:

$$u_t(x, t) = \frac{D(u_x(x + \Delta x, t) - u_x(x, t))}{\Delta x}$$

Limit as $\Delta x \rightarrow 0$:

$$u_t = Du_{xx}$$

which is the 1-dimensional diffusion equation.

26.43 Solving the diffusion equation

$$u_t = Du_{xx}$$

Ansatz $u(x, t) = X(x)T(t)$

$$XT' = DX''T$$

Divide by XT:

$$\frac{T'}{T} = D\frac{X''}{X}$$

So the equations are separated. The left side is independent of X and the right side is independent of T, so both sides must be a constant. Call the separation constant $-\lambda$.

$$\frac{T'}{T} = D\frac{X''}{X} = -\lambda$$

From this, generate two separate equations:

$$X'' + \lambda X = 0$$

$$T' + \lambda DT = 0$$

Boundary conditions. No concentration of dye at either end of the pipe - like an infinite fluid reservoir with no dye.

$$u(0, t) = 0 = X(0)T(t)$$

$$u(L, t) = 0 = X(L)T(t)$$

So,

$$X(0) = 0, X(L) = 0$$

which are homogeneous boundary conditions. Since the homogeneity is on the function itself, it's a homogenous Dirichlet boundary condition set.

$$u(x, t) = X(x)T(t)$$

$$X'' + \lambda X = 0$$

$$T' + \lambda DT = 0$$

$$X(0) = X(L) = 0$$

Let $\lambda = \mu^2$, so

$$X'' + \mu^2 X = 0$$

Since the solution of X must cycle twice, it is trigonometric and since the eigenvalue is μ^2 , it must be with period μ :

$$X(x) = A\cos\mu x + B\sin\mu x$$

With the boundary conditions,

$$X(0) = A = 0$$

$$X(L) = B\sin(\mu L) = 0$$

If B=0, uninteresting (trivial) solutions. So just set B=1 for simplicity and let $\sin(\mu L) = 0$. Solve this:

$$\mu_n = \frac{n\pi}{L}$$

where $n \in \mathbb{Z}$.

$$\lambda_n = \mu_n^2 = \frac{n^2\pi^2}{L^2}$$

Then:

$$X_n = \sin\frac{n\pi x}{L}$$

NOTE: $\lambda \geq 0$ must hold. Otherwise,

$$X'' - \mu^2 X = 0$$

so the characteristic equation solutions are $\pm\mu$.

$$X(x) = Ae^{\mu x} + Be^{-\mu x}$$

$$X(0) = 0 \rightarrow A + B = 0 \rightarrow B = -A$$

$$X(L) = 0 \rightarrow Ae^{\mu L} + Be^{-\mu L} = 0$$

Combining,

$$A(e^{\mu L} - e^{-\mu L}) = 0$$

for which the only nontrivial solution is $A=0$.

Fourier series.

$$T' + \lambda_n DT = 0$$

Solution T will be s.t. derivative is negative of $\lambda_n DT$, which is $e^{<0}$:

$$T_n(t) = e^{-(n^2 \pi^2 D/L^2)t}$$

$$u_n(x, t) = X(x)T(t) = \sin \frac{n\pi x}{L} e^{-(n^2 \pi^2 D/L^2)t}$$

General solution is superposition of all solutions:

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-(n^2 \pi^2 D/L^2)t}$$

Initial conditions with $t=0$:

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

which is a Fourier sine series, so

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

Example. Applies the results from the diffusion equation above. Start with dye at single pipe cross-section middle $\frac{L}{2}$. Then, model it with Dirac delta:

$$f(x) = M_0 \delta \left(x - \frac{L}{2} \right)$$

When integrating over the Dirac delta, pick out $x = \frac{L}{2}$:

$$b_n = \frac{2M_0}{L} \int_0^L \delta \left(x - \frac{L}{2} \right) \sin \frac{n\pi x}{L} dx = \frac{2M_0}{L} \sin \frac{n\pi L}{2L} = \frac{2M_0}{L} \sin \frac{n\pi}{2}$$

In the equation

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-(n^2 \pi^2 D/L^2)t}$$

$n=1$ is the most influential term, as the higher-order terms decay much faster. So approximate the solution with just $n=1$,

$$b_1 = \frac{2M_0}{L}$$

$$u(x, t) \approx \frac{2M_0}{L} \sin \frac{\pi x}{L} e^{-(\pi^2 D t)/L^2}$$

So the maximum is at $x = \frac{L}{2}$ so highest dye concentration in pipe middle with less at ends. The graph would be a half of a sine function with the max at the middle. This is the Fourier sine series with $n=1$.

26.43.1 Picard's existence and uniqueness theorem

Theorem for first-order ODEs. Suppose f is bounded

$$|f(t, x)| \leq M$$

$\forall x, t$ and f is Lipschitz,

$$|f(t, x) - f(t, y)| \leq L|x - y|$$

$\forall x, t$. Then the DE

$$\frac{dx}{dt} = f(t, x), x(t_0) = x_0$$

has solution $x(t)$ which is defined on some interval $t_0 - \epsilon < t < t_0 + \epsilon$.

Another way to express this theorem. Consider the initial value problem

$$y' = f(x, y), y(x_0) = y_0$$

Suppose $f(x, y), \frac{\partial f}{\partial y}(x, y)$ are contained in some open rectangle $R = [(x, y) : a < x < b, c < y < d]$ that contains the point x_0, y_0 . Then the problem has a unique solution in $x_0 - h, x_0 + h$. Moreover, the the Picard iteration

$$y_{n+1}(x) = y_0 + \int_{x_0}^x f(t, y_n(t)) dt$$

produces a sequence of functions that converges to the solution uniformly.

Heuristic proof. Define operator T

$$T[y](x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$$

Then $T[y] = y$ would be a fixed point and a solution by the Banach fixed point theorem for operators. To calculate this, use Picard iteration. This will lead to uniform convergence (where the norm decreases to 0). Use boundedness and Lipschitz conditions for the rectangle of solutions.

27 Dynamical Systems and Chaos

27.1 Iterated functions

Deterministic versus stochastic functions. Deterministic functions always give the same output from a given input. Stochastic functions do not.

Functions versus time series (iterated orbits). For instance, $y = 3x$ plots a line for a function but $y = 3^n x$ so a geometric progression in a time series.

Dynamical system: System that evolves in time through a well-defined unchanging rule. Iterated functions are dynamical systems.

Phase line for $y = x^2$. 0 and 1 are fixed points $f(x) = x$. 1 is an unstable fixed point (repeller or source) as numbers >1 move to infinity and <1 move to 0, while 0 is a stable fixed point (attractor or sink). Solve for fixed points by letting $f(x) = x$.

27.2 Differential equations

Iterated function is $x_{n+1} = f(x_n)$ and once x_0 is specified, the orbit is specified. Differential equation is $\frac{dx}{dt} = f(x)$ and once $x_0 = x(0)$ is specified, then the solutions $x = x(t)$ are specified. So differential equations are another, different, type of dynamical system.

Numerical method of differential equation: Euler's method. Start with initial condition, then assume constant change over some Δt and add it to initial condition.

$$x(t + \Delta t) = x(t) + \left(\frac{dx}{dt} \Delta t \right)$$

Then at new level value, repeat. Gets more accurate as $\Delta t \rightarrow 0$. This is a way of moving from indirect information (change rate) to direct information (level values).

Better numerical methods: (1) Runge-Kutta methods where several change rates are averaged over each Δt , (2) Adaptive stepsizes where Δt is smaller where the change rates are faster.

Qualitative methods. Example: Newton's law of cooling

$$\frac{dT}{dt} = 0.2(20 - T)$$

Plotting $\frac{dT}{dt}$ versus T , downward straight line. Stable fixed point at room temperature. From this, get phase line, and then graph level values. Generally, if $\frac{dx}{dt} = f(x)$, graph this versus x then find fixed points at $f(x) = \frac{dx}{dt} = 0$ and find phase line, stability, and graph level values.

Analytic methods. Use calculus to solve exactly. However, most non-linear DEs cannot be solved analytically and even if possible, not intuitive.

Note 1: if $\frac{dx}{dt} = f(x)$ then the change rate is only a function of x . So each value of x can only get one change rate, so the time series cannot 'double back' on itself - so never oscillatory solutions.

Note 2: Autonomous systems don't depend on time $\frac{dx}{dt} = f(x)$ whereas non-autonomous systems depend on time $\frac{dx}{dt} = f(x, t)$.

27.3 Chaos and the butterfly effect

27.3.1 Population models

$$f(P) = rP$$

$$P_{n+1} = rP_n$$

where r is the growth rate. If $r > 1$, population explodes. If $0 < r < 1$ then population shrinks to 0. If $r = 1$, population is fixed.

27.3.2 Logistic function

$$f(P) = rP \left(1 - \frac{P}{A} \right)$$

where A is the annihilation population, where if $P = A$ then $f(P) = 0$. If $P \ll A$, then $f(P) \approx rP$. Graphing $f(P)$ versus P is an inverted parabola with the zeros at 0 and A . Then divide both sides by A

$$\frac{P_{n+1}}{A} = r \frac{P_n}{A} \left(1 - \frac{P_n}{A} \right)$$

and let $x = \frac{P}{A}$

$$x_{n+1} = rx_n(1 - x_n)$$

$$f(x) = rx(1 - x) = rx - rx^2$$

which is the inverted parabola. Solve for stability:

$$x = rx(1 - x)$$

$$x_F = \frac{r - 1}{r}$$

For $0 \leq r \leq 1$, there is only fixed point at 0 because $x_F < 0$ from the fixed point solution equation above. Moreover, it is a stable fixed point. For $1 < r \leq 3$, there are two fixed points, 0 and x_F . However, 0 is an unstable fixed point (repeller), so when drawing bifurcation diagrams, don't use the 0 by seeding with $x_0 \neq 0$. For $r > 3$, there continue to be two fixed points, 0 and x_F , but both

are unstable (repellers). So seed values other than 0 and x_F in the bifurcation diagram. Further, when $r > 3$, some seeds give very low-cycle orbits, so avoid those exact seeds - but with SDIC, can select seeds very close to those values as they're unstable (see 'density' of orbits).

From above, apparent that fixed point stability changes with r .

For $r=4$, logistic equation is proved aperiodic mathematically.

27.3.3 Butterfly effect

Sensitive dependence on initial conditions (SDIC). Need multiple orders of magnitudes more accuracy for only slightly better accuracy. For fixed points, not sensitive because everything just converges - but for aperiodic systems, matters more. Tiny changes are magnified quickly even if the system is predictable. Long-term forecasting is impossible.

Formally, if f is a function and x_0, y_0 are seeds, f has SDIC if $|x_0 - y_0| < \epsilon \rightarrow |x_n - y_n| > \delta$ after n iterations for ANY initial conditions. Note: with some initial conditions, can end up out of phase for some systems (those with periodicity) but then can find other initial conditions for these systems that are in-phase.

SDIC may be so sensitive that physically impossible to be that precise - e.g. so precise that it has to be much more precise than normal deviations in that quantity.

A dynamical system is chaotic if: (1) it's deterministic (e.g. iterated functions and DEs), (2) orbits are bounded (and so can't go to ∞), (3) orbits are aperiodic, and (4) system has SDIC.

Shadowing lemma. Computed orbit by a computer, because of rounding errors, leads to an orbit that is not the true orbit but is nevertheless close to some other orbit because there are so many other orbit possibilities.

Lyapunov exponent. Expresses degree of SDIC. Let $D_0 = |x_0 - y_0|$ and $D_t = |x_t - y_t|$. For many systems, $D_t \approx D_0 2^{\lambda t}$ (can use different bases other than 2), where $\lambda > 0$ means SDIC and larger values indicate greater sensitivity, and $\lambda < 0$ means convergence to an attractor.

27.3.4 Randomness

(1) Deterministic sources of randomness. Important for math generally (to have a random number generator for Monte Carlo simulations, cryptography, random trials) and for chaos.

Symbolic dynamics. Convert an orbit into a symbol sequence (like RLLRRL-

RLRLRRRR).

Example. $x_t < 0.5 \rightarrow L$ and $x_t \geq 0.5 \rightarrow R$, which is a 'generating partition'. Only certain partitions are generating partitions, which preserve the properties of the original orbit. Since $P(R) = P(L) = 0.5$, which is the same as coin tossing, they're statistically the same.

Properties become more accurate as sequence goes to infinity. Easier to work with symbols sequences than numerical orbits.

(2) Incompressibility of information. Number of algorithms is countably infinite, but the number of sequences is uncountably infinite. So can't find compressing algorithms for all sequences - there are a lot of sequences which are random, and it is almost certain that if a sequence is chosen arbitrary, that there will be no algorithm for it. The logistic equation produces all random sequences but the initial conditions are generally irrational so still results in no information compression.

27.4 Logistic differential equations

Logistic differential equation is a change rate:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$$

where K is the 'carrying capacity'. Logistic equation is levels:

$$f(P) = rP \left(1 - \frac{P}{A}\right)$$

They are both inverted parabolas but with different meaning.

The phase line of the logistic differential equation is generally just an attractor at the carrying capacity, for normal values. The logistic differential equation is continuous, unlike the logistic equation. Moreover, the logistic differential equation cannot have cycles - if it did, the graph of P versus t would have multiple change rates for a single value of P, which is not possible from the equation. Since there are no cycles, there is no chaos in logistic differential equations.

To solve for fixed points in a logistic equation: $P = rP \left(1 - \frac{P}{A}\right)$. to solve for fixed points in a logistic differential equation, $0 = rP \left(1 - \frac{P}{K}\right)$.

Logistic differential equation with harvest.

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right) - h$$

where h is the harvest, which is a downward vertical translation in the $\frac{dP}{dt}$ versus P graph (note: K is the attractor and lies at $\frac{dP}{dt} = 0$ on the graph). If $h > 0$, then

below the lower critical value (moved UP from 0 when $h=0$) there is extinction. As h increases, there will be a point at which inverted parabola top will touch x-axis at one point, a 'semi-stable' fixed point. As h rises further, there is only extinction as $\frac{dP}{dt} < 0$ everywhere.

27.4.1 Bifurcation

A sudden qualitative change in a dynamical system as a parameter is varied. Bifurcation map graphs x versus parameter like r or h .

Hysteresis. If the system becomes even slightly complicated and contains bifurcations, the system can develop hysteresis (path dependence) because approaching from left or right at equilibrium (i.e. fixed points) the bifurcation map indicates different movement when these points are at the same parameter value.

$$\frac{dx}{dt} = rx + x^3 - x^5$$

27.5 Universality

Other maps are similar in shape, e.g.

$$f(x) = rx^2(1-x)$$

$$f(x) = r \sin\left(\frac{\pi}{2}x\right)$$

27.5.1 Feigenbaum constant

Regardless of which map is used,

$$\delta = \lim_{n \rightarrow \infty} \frac{r_{n+1} - r_n}{r_{n+2} - r_{n+1}} = \lim_{n \rightarrow \infty} \frac{\Delta_n}{\Delta_{n+1}} = 4.669...$$

which is universal for all iterated functions that map an interval to itself and has a single quadratic maximum. Note: cubic function has a single local quadratic max. Can't have more than one local maximum.

27.5.2 Renormalization

Change length scale. Any smooth curve, when magnified, resembles a straight line. Magnifying a straight line also ends up with a straight line. In fact, any smooth curve will end up at a straight line, which is a 'universal curve'. The line is a 'universal compound' function. By renormalization, each bifurcation will take the function from f_1 to f_2 to f_4 etc. and ends up at a line that has δ (which is a repetition, and therefore like another dynamical system). The bifurcation diagram is a fractal, as the period doubling keeps occurring at a constant rate. So it's universal for 1D cases.

There is also (semi)-group of operations, the renormalization group, as two magnifications are the same as one magnification.

As multiple different starting points (e.g. curve, straight line) end up with a straight line, which is a fixed point. So it's also an attractor.

27.6 Phase spaces

27.6.1 2D differential equations

Lotka-Volterra predator-prey models (foxes and rabbits) are coupled differential equations:

$$\begin{aligned}\frac{dR}{dt} &= f(R, F) = aR - bRF = R - \frac{1}{4}RF \\ \frac{dF}{dt} &= g(R, F) = cRF - dF = 0.2RF - 0.6F\end{aligned}$$

where a is measure of how fast rabbits can multiply and b is how deadly foxes are (note if rabbits increase then bR means easier to hunt rabbits), c is how easy foxes can hunt rabbits and more rabbits means easier to hunt, and d is death rate of foxes. Example initial conditions: $R(0) = 10, F(0) = 6$.

Plot $x(t)$ versus t , $y(t)$ versus t separately. Then combine into phase plane.

Phase portraits and trajectories. 2D version of the final state diagram. Plotting one variable against the other, like F versus R , gives phase diagram (phase portrait). Different initial conditions lead to different phase portraits. If the phase trajectories are closed cycles, can move between them by external changes, e.g. adding/subtracting R and/or F . Neutral or fixed point in the middle of a closed trajectory. Trajectories follows the vectors in the vector field. Note vector field means the phase path cannot cross. Types of fixed points: (1) direct attractor, (2) spiral attractor, (3) direct repeller, (4) spiral repeller.

Nullclines. If

$$\begin{aligned}\frac{dR}{dt} &= R \left(1 - \frac{1}{4}F \right) \\ \frac{dF}{dt} &= F(0.2R - 0.6)\end{aligned}$$

Solutions are $(0,0)$ and $(4,3)$. For second solution, if $F=4$, $\frac{dR}{dt} = 0$ and if $R=3$, $\frac{dF}{dt} = 0$, both of which are straight lines. On phase plane, plot $(4,3)$ which is center of two perpendicular nulleline straight lines, dividing into 4 quadrants each with combinations of $R' > 0, R' < 0, F' > 0, F' < 0$. This will allow vector field determinations (e.g. by diagonal arrows). Can then graphically represent solution, instead of using numerical solutions.

van der Pol equations.

$$\begin{aligned}\frac{dX}{dt} &= Y \\ \frac{dY}{dt} &= -X + (1 - X^2)Y\end{aligned}$$

It has an attractor cycle, no matter where the initial conditions. Sometimes called a 'limit cycle'.

Poincare-Bendixson theorem. Bounded aperiodic behavior is impossible for 2D DEs of the van der Pol equations. Bounded - e.g. boxed in a phase diagram, and aperiodic means trajectories can't cross. So chaos is impossible - need higher dimensions than 2D.

27.6.2 2D iterated functions

Henon map.

$$\begin{aligned}x_{n+1} &= y_n + 1 - ax_n^2 \\ y_{n+1} &= bx_n\end{aligned}$$

Example $a = 0.9, b = 0.3, x_0 = 0.2, y_0 = 0.4$. Then $x_1 = 1.364$

27.6.3 3D differential equations

Phase plane will become phase space. Lorenz equations are coupled differential equations:

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

Plot $x(t)$ versus t , $y(t)$ versus t , $z(t)$ versus t separately. Then combine into phase space.

Let $\sigma = 10, \rho = 160, \beta = 8/3$. $x(t), y(t), z(t)$ are periodic. Like a butterfly, closed and periodic.

Let $\sigma = 10, \rho = 28, \beta = 8/3$. $x(t), y(t), z(t)$ are aperiodic.

27.7 Strange attractor

Strange attractor is: (1) attractor and (2) chaotic. Globally stable (attractor shape; statistically predictable) but locally unstable (motion on attractor is aperiodic). So long-term locally unpredictable but long-term statistically predictable.

Henon strange attractor. With aperiodic $x(t)$ and $y(t)$, a side U-shaped aperiodic attractor. SDIC and $x(t)$ and $y(t)$ curves diverge but the exact same shape - so it's an attractor. So bounded (to attractor shape) and aperiodic, so chaotic. Can keep changing a parameter (a or b) like r in the 1D case, and this will lead to period-doubling following δ until chaos on the Henon strange attractor. It also has fractal features.

Lorenz strange attractor.

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

In the spiral centers are unstable fixed points that set the derivatives above each to 0.

Rossler attractor. Simplest or one of the simplest strange attractors. Circle with part of it being stretched and folded onto itself. Stretching leads to SDIC. Folding keeps orbits bounded.

$$\begin{aligned}\frac{dx}{dt} &= -y - z \\ \frac{dy}{dt} &= x + ay \\ \frac{dz}{dt} &= b + (x - c)\end{aligned}$$

Note: stretching and folding also occur in the 1D logistic equation (inverted parabola is a folding of a straight line with positive slope), which is then projected onto the y -axis (y_{n+1} which is then projected onto the previous time period y_n).

27.8 Pattern formation

Chaos can also produce patterns.

Diffusion. Concentration of A is $u(x,y)$ and concentration of B is $v(x,y)$ in 2D.

$$\begin{aligned}\frac{du}{dt} &= A\nabla^2 u \\ \frac{dv}{dt} &= B\nabla^2 v\end{aligned}$$

Reaction-diffusion. A and B interact. Inhibitor diffuses faster than activator $B > A$.

$$\begin{aligned}\frac{du}{dt} &= A\nabla u^2 + f(u, v) \\ \frac{dv}{dt} &= B\nabla v^2 + g(u, v) \\ f(u, v) &= u(1 - v) - 12 \\ g(u, v) &= 16 - uv\end{aligned}$$

Doesn't have to be chemicals - e.g. rabbits can be activator and rabbits are inhibitor. Now, there can be local concentrations of rabbits (rabbits make more rabbits), attracting foxes, and then rabbits run away and foxes chase them but are faster. So foxes create a ring and trap the rabbits. This creates spatial structures depending on A, B, f, g. Sometimes they are time stable, sometimes they are oscillations. This is counter-intuitive because diffusion wants to make concentrations uniform. See also Belousov-Zhabotinsky reaction. Patterns and structures can form spontaneously from homogeneity.

28 Complexity

28.1 What is complexity?

Properties: (1) simple units or agents, (2) nonlinear interactions means the whole isn't the sum of the parts, (3) no central control, (4) emergent behaviors and properties (evolution, dynamics, etc.).

28.2 Dynamics and chaos

Linear dynamical system means the whole is the sum of the parts. E.g. exponential population growth, as $n_0 = 2$ and $n_0 = 10$ will give same results at n_t scaled by 5. Nonlinear system means whole is not the sum of parts.

Population model, simple:

$$n_{t+1} = rn_t$$

Population model, with limitations (deaths):

$$n_{t+1} = r(n_t - \text{offspring deaths}) = rn_t - dn_t$$

Assume offspring deaths:

$$\text{offspring deaths} = \frac{n_t^2}{\text{max pop}}$$

Extended model (logistic model):

$$n_{t+1} = (\text{birth rate} - \text{death rate}) \left(n_t - \frac{n_t^2}{\text{max pop}} \right)$$

This is no longer a linear system. Re-express:

$$n_{t+1} = R \left(n_t - \frac{n_t^2}{K} \right)$$

$$\frac{n_{t+1}}{K} = R \left(\frac{n_t}{K} - \frac{n_t^2}{K^2} \right)$$

$$x_{t+1} = R(x_t - x_t^2) = Rx_t(1 - x_t)$$

R is confined to a range $0 \leq R \leq 4$ because the maximum of $x(1 - x) = 0.25$ with $x = 0.5$. So, $Rx(1 - x) \leq 1$ as $x_{t+1} \leq 1$. Solving for the maximum: $R = 4$. Inputting R values outside of the range always leads to negative infinity.

If $R = 1$, because $0 < x_t < 1$, $0 \leq x(1 - x) \leq 1$ so is continuously shrinking.

28.3 Fractals

Beyond Euclidean geometry.

Self-similar. In nature however, objects are only fractal-like and not mathematically fractal (perfect self-similarity).

Koch curve. Take line segment L and divide into thirds, replacing the middle third segment with two segments that are each (L/3) length. The total length is then (4L/3). Repeat for (16L/9) length at next level to $\frac{4^n L}{3^n}$. The number of segment lengths is increasing faster than the shrinking lengths.

Dimensions. When a dimension is bisected on each side of its dimensions, it results in $\frac{1}{2^D}$ copies where D is the dimension. For m-sections, it is $\frac{1}{M^D}$ copies. Call the copies N. Hausdorff dimension is $N = M^D \rightarrow D = \frac{\log N}{\log M}$. For Koch curve, $D = \frac{\log 4}{\log 3} \approx 1.26$, a fractional dimension of self-similarity, the object's degree of jaggedness or dense it is.

Other examples. Sierpinski triangle. Cantor set length shrinks so dimension is less than 1. Jackson Pollock's paintings.

Box-counting dimension. At a resolution of the size of the box and boxes that include the figure. Then, $D = \frac{\log(\text{number of boxes})}{\log(1/\text{box side length})}$. Plot $\log(\text{number of boxes}) = D \log(1/\text{box side length})$ and calculate slope D.

28.4 Information entropy

Maxwell's demon was found not to be a paradox: the demon's information was entropy. This led to information entropy. Thermodynamics sees entropy as

heat loss. Statistical mechanics see entropy as the number of microstates. Microstates are position, velocity, etc. of every particle. More ordered state has fewer microstates. Macrostates are collections of microstates.

Boltzmann entropy:

$$S = k \ln W$$

S is entropy (Joules per Kelvin), k is the Boltzmann constant, W is the number of microstates corresponding to the macrostate. Higher entropy is more probable. Second law of thermodynamics: the universe will tend to the macrostate that is most probable.

Shannon information entropy. Based on Boltzmann entropy.

$$H = \log_2 M$$

M is the number of messages possible. Units are bits.

$$H = - \sum_i^M p_i \log_2(p_i) = - \sum_i^M p_i \log_2(N_i) + \log_2(M) = \log_2(M) - \sum_i^M p_i \log_2(N_i)$$

where the negative is present because log probability is negative and want $H \geq 0$ and $p_i = N_i/M$. The first p_i is a weight over the different possibilities for M, which will sum to 1. The second is the information content of each of those possibilities. The final expression shows H is just as before but subtracts repeated portions of the messages (i.e. $N_i > 1$).

28.5 Genetic algorithms

Evolve a strategy, with actions for all possible situations. Many actions are bad, so genetically evolve. Initialize with random strategies for individuals. For each individual, calculate fitness (average reward minus penalties). Then select most fit individuals and cross them by mixing actions together with some random mutations. Old generation dies out. Repeat.

Mechanisms. Natural selection, punctuated equilibrium, exaptation (the process by which features acquire functions for which they were not originally adapted or selected).

Genetic programming. Instead of strategies represented by matching situations to actions, use strategy trees. Generate random population of trees and then enforce logical constraints (e.g. ifelse syntax). Then fit individuals cross and exchange subtrees. Introduce random mutations that replace subtrees.

28.6 Cellular automata

Idealized models of complex systems. Game of Life can converge to fixed points, attractors (including multi-period attractors). So can move from high entropy

to low entropy.

Elementary CA. 1D and only 256 possibilities. Wolfram divided into 4 classes. Can view logistic map as elementary CA. Langton hypothesis: $0 \leq \lambda \leq 1$ values which encode proportion of 1's in the binary class with values around 0.5 being more complex.

Genetic algorithm can lead to CAs where the trajectories through iterations are 'particles' that collide like in real-life and lead to other particles or annihilate each other.

28.7 Self-organization and emergence

Can be approximated by very few simple rules. Biological processes can be viewed as computers with emergent behavior.

28.7.1 Social systems

Traditional economics use analytic models and equilibrium dynamics. Involves perfect rationality, perfect knowledge, and deductive reasoning.

Complexity economics use non-analytic models and agent-based models which continually adapt. Involves bounded rationality, imperfect knowledge, and inductive reasoning.

Prisoner's dilemma. Repeated games, best strategy is tit-for-tat. Always defect or rat wins early but eventually tit-for-tat is best. Can decompose into kernel (does nothing), strategic, and externality components. Externality is about the other side being able to influence you. If you act strategically, then it imposes an externality on the other side.

		Player 2	
		Cooperate	Defect
Player 1	Cooperate	3, 3	-1, 5
	Defect	5, -1	1, 1

		"Kernel"		"Nash" or "Strategic"		"Behavioral"	
		C	D			C	D
C		2, 2	2, 2	C		2, 2	-2, 2
D		2, 2	2, 2	D		2, -2	-2, -2
		No info		Selfish info		Externalities	

		"Nash" or "Strategic"	
		A	B
A		a, c	b, -c
B		-a, d	-b, -d
		Selfish info	

		"Behavioral"	
		A	B
A		m, n	-m, n
B		m, -n	-m, -n
		Externalities	

		"Kernel"	
		A	B
A		j, k	j, k
B		j, k	j, k
		No info	

See 'Strategic and behavioral decomposition of games' Jessie and Saari 2015.

El Farol problem. 60 people fit comfortably in a bar but 100 want to go. Each agent only knows (perfectly) how many people attended last time. Solution. Let N be the number of strategies and M the weeks of attendance known. t is

the time, with $t-1$, $t-2$, etc. with $A(t)$ being attendance at time t . Each strategy has the following form:

$$S(t) = 100(w_1A(t-1) + w_2A(t-2) + \dots + w_mA(t-M) + c)$$

where $w_i \in [-1, 1]$. Each person has N such strategies with different weights (some weights can be 0), and each person's strategies are different. Each person has a best strategy S^* , s.t. $S^*(t) > \text{overcrowding threshold}$ then don't go. Each person is initialized with random weights and initial history with $M \in [0, 99]$. Each person has errors

$$\text{Error}(S) = |S(t) - A(t)| + |S(t-1) - A(t-1)| + \dots + |S(t-M) - A(t-M)|$$

The best strategy minimizes this error. This also maximizes overall utility as the bar gets close to being full but not overcrowded much.

28.7.2 Network

Terminology. Uses graph theory edges (directed, undirected) and nodes. Clustering is the fraction of neighbors that are connected to each other with 1 neighbor = 0. Clustering coefficient is the average clustering across all nodes in the network, and indicates how well-connected the network is (e.g. how fast information will be transmitted across the network, how vulnerable the network is to one node being removed).

Degrees of separation. Milgram's experiment connecting Boston stockbrokers and Nebraskan farmers showed average was six degrees of separation.

Real-world networks. Prototype networks are either low average distances and low average clustering (highly ordered networks), or high average distances and high average clustering (random networks). Comparing different networks from different fields, find real-world networks have low average distances between nodes but high average clustering. These real-world networks can be built starting with regular network (each node has the same pattern of connections to other nodes), and randomly rewiring.

Scale-free networks. A random network is Poisson-distributed and symmetrical about a positive value. Scale-free networks instead have their highest value at or near 0 and exponentially decline. They are long-tailed distributions. The declination shape is fractal-like and follow power laws $k \propto \frac{1}{k^2}$, where k is the connectivity degree. Hubs branch into sub-hubs which branch to further sub-hubs, and finally end at single or often single nodes, so are fractal-like.

Most real-world networks have long-tailed degree distributions and many of these are fractal-like. Google takes advantage of long-tailed distributions (easier to rank pages) rather than random connections in a Poisson-distribution (harder to rank pages). Preferential attachment model: scale-free networks

arise as already-more cited pages get cited even more. Long-tailed distributions have larger tails than normal distributions so have more extreme values, so more likely to incur catastrophic risk.

Cascading or domino effect. Many networks display this property.

28.8 Scaling in biology and society

Power law of scaling:

$$attribute \propto (scale)^\alpha$$

where α is constant and is the power. Plotting $y = cx^\alpha$ is exponential but plotting its transformed version $\log y = \alpha \log x + \log c$ is a straight line on a log-log plot. Gutenberg-Richter law on earthquake frequencies v magnitudes, metabolic rate v body size, crime, income, etc v city population are all straight lines.

28.8.1 Metabolic rate

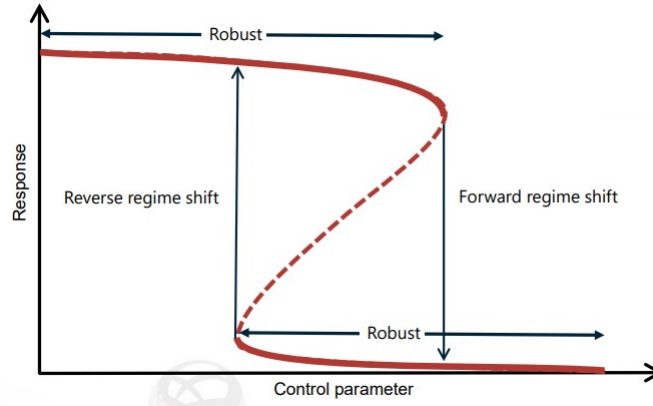
With power law, might assume volume scales with surface area as $V \propto r^3$ and $SA \propto r^2$. As heat is emitted through $\frac{SA}{V}$ and volume grows much faster, a large animal can't emit much heat so this appears incorrect. However, since $SA \propto V^{2/3}$ perhaps metabolic rate $\propto (\text{body mass})^{2/3}$ (surface hypothesis), but experiments show the scaling factor is 3/4 (Kleiber's law) so evolution has led to more efficiency than expected. Many other biological scales have 1/4 or 3/4 scales (quarter-power scaling laws) because the space-filling aspects (e.g. lung bronchi) are fractal-like so increase dimensions from 3D to nearly 4D.

28.9 Nonlinear relationships

In a graph, two nodes connected to each other and to no others leads to a linear relationship. A change in one node is proportionally transmitted to the other node. If three nodes are connected to each other (in a triangle), this can lead to non-linear relationships, with a strong change in one node leading to feedback, a weak node leading to dissipation, and a fixed point. The more complex system is more robust as a small change in one node may not to much effect.

28.10 Regime shifts

Regime shifts may be path-dependent (i.e. subject to hysteresis).



This is seen in the Tavoni-Schluter-Levin model, and depends on the strength of the connectivity (more hysteresis effect with higher connectivity; if there is no connectivity, there is no hysteresis and just a sigmoid-type curve). There can be more than two regimes, and they can be different in nature.

28.10.1 Tipping points and universality

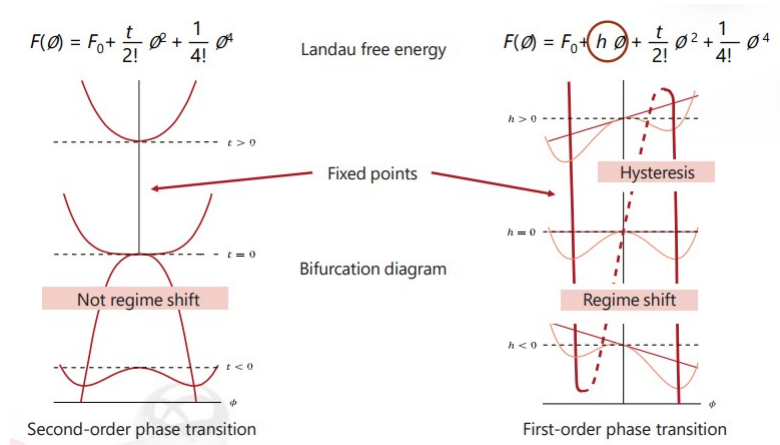
In physical systems, tipping points are phase transitions. All materials have phase transitions having the same critical exponents and are therefore in the same universality class. In social systems, regime switching. Feigenbaum's constant is applicable for a certain universality class.

28.10.2 Landau theory

Begin with Landau free energy:

$$F(\phi) = F_0 + \frac{t}{2!}\phi^2 + \frac{1}{4!}\phi^4$$

which is a Taylor expansion about $\phi = 0$, the order parameter. In physical interpretations, t is the scale temperature. The linear term disappears because the Landau free energy assumes it is at the lowest free-energy state (the parabolic minimum). Then cubic term disappears because it is assumed that this is a symmetric scenario.



The global minimum or minima is/are the fixed points, the points of lowest free energy. If $t > 0$, then F is parabolic+quartic (but parabolic dominates) with $\phi = 0$ is the minimum. If $t = 0$, then quartic, and again $\phi = 0$ is the minimum. If $t < 0$, then quartic minus parabolic, but quartic will dominate because of its higher power (a double-well potential, a w-shape). For $t < 0$, the optimal points are $\phi = 0$, one positive value, and one negative value. But $\phi = 0$ is a maximum, while the positive and negative values are minima (which are described by the t values, which are attached to the parabolic term of F). So the $t = 0 \rightarrow t < 0$ progression leads to a doubling of fixed points. This is a bifurcation. In physics, this is a second-order phase transition (to a supercritical fluid), but in other fields, it is often not called a regime shift in other fields.

First-order phase transition. Add linear term:

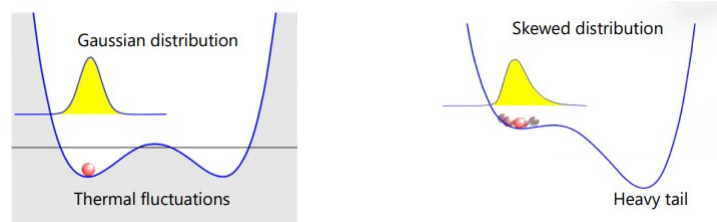
$$F(\phi) = F_0 + h\phi + \frac{t}{2!} \phi^2 + \frac{1}{4!} \phi^4$$

where h is like the magnetization or magnetic field. At $h = 0$, it is identical to the previous scenario. However, $h \neq 0$ will 'tilt' the double-well shape, so only one, instead of two, global minimum will be left. Further however, beginning in the left well, if h is reduced, then it remains there even though the right well is the actual global minimum, because of an energy barrier. As h is reduced further, the system will suddenly transition. Between these paths, there is a path of unstable fixed points indicated by a dotted path in the diagram. This path dependency is hysteresis.

Equilibrium versus critical transitions. Equilibrium transitions: movement between two wells away from critical temperature. It is a 'natural' fluctuation by climbing over the energy barrier between the two wells. Critical transition: an equilibrium transition near the critical temperature.

28.10.3 Forecasting tipping points

Landau theory indicates there are early warning signs before regime shifts. System behavior changes just before regime shift.



As the system approaches regime shifts, the energy barrier between the two wells decreases. The fluctuation (the variance) grows to infinity. The restoring force is weaker, and it takes longer to restore ('spectral reddening', as the frequency of fluctuations shifts to lower frequencies). So the pattern changes slow down significantly. The distribution will also become skewed. At the critical temperature, for most power laws (critical exponent less than 3), the variance becomes infinite.

However, forecasting techniques using e.g. neural networks and autoregressive techniques do not include tipping point equations or modelling. They include memory or other mechanisms instead. So difficult to forecast tipping points with these kinds of techniques.

28.10.4 Tipping point forecasting techniques

Log periodic power law (LPPL). Bethe lattice has discrete scale invariance (scale invariance at only discrete scales) so should see log-periodic power laws. Before market crashes, see a rise in prices. As crash proceeds, prices will fluctuate more and more.

Soup-of-groups. Maps phenomena such as earthquakes and financial markets to groups or clusters of objects that move through time and when they approach a power law, the groups or clusters swell into giants and break apart, which is a regime shift.

28.11 Resilience

Resilience is when a system responds from a shock quickly and returns to original state. Unless 100 percent resilient, the system will behave differently.

Sustainability. Tragedy of the commons. Let S be grass grow at rate α , cows L eating grass at rate β , carrying capacity K , death rate δ , conversion factor ϕ .

Using logistic growth of S:

$$\begin{aligned}\frac{dS}{dt} &= \alpha S \left(1 - \frac{S}{K}\right) - \beta LS \\ \frac{dL}{dt} &= L(\delta + \phi\beta S)\end{aligned}$$

Fixed points:

$$\frac{dS}{dt} = 0 = \frac{dL}{dt}$$

The trivial fixed points are $(S, L) = (0, 0)$ and $(S, L) = (K, 0)$, which are both unstable because a small change (introducing a small amount of grass or cows will change the equilibrium). The non-trivial fixed point, which is stable, is:

$$(S, L) = \left(-\frac{\delta}{\phi\beta}, \frac{\alpha}{\beta} \left(1 + \frac{\delta}{\phi\beta K}\right)\right)$$

28.12 Evolution

Evolutionary game theory. Will have continually shifting dominant strategies when some evolutionary games are repeated or played for many iterations.

28.13 Swarms

Must be decentralized. Cannot have centralized control. Combines network and information theories.

Requires three parameters: moving in same direction, stay close to neighbors, avoid collisions. These parameters also depend on another parameter, connectivity (how far each agent is looking). The connectivity helps determines regime stability and resilience after it is attacked.

Use information theory. Channel information capacity from Shannon-Hartley:

$$C = B \log_2 \left(1 + \frac{Signal}{Noise}\right)$$

where B is the bandwidth. Use min-cut max-flow algorithm (Ford-Fulkerson) to find information bottlenecks. Find that swarm unsurprisingly requires minimum information channel capacity.

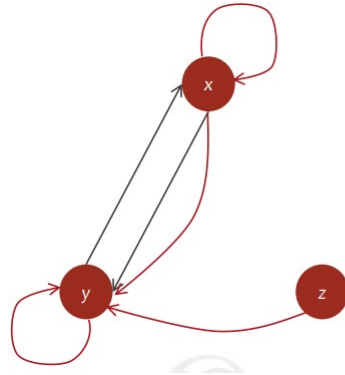
Damaging one node will lead to network adjustment, so it continues to function, albeit differently.

28.14 Agent-based modelling

Can apply to swarms, but used more for complex agents that may be heterogeneous. May need to be supplemented with social science (e.g. interviews, surveys).

28.15 Network theory

Behind every complex system, there is a network.



Lorenz Equation:

$$\frac{dx}{dt} = \sigma(y - x)$$

$$\frac{dy}{dt} = x(p - z) - y$$

$$\frac{dz}{dt} = xy = \beta z$$

Equation is unknown.

Can still understand a lot!

- Clustering coefficient. How much neighbors are connected to each other. Specify average for network.
- Assortativity. Defined globally over network. If $r > 0$ then high-degree nodes tend to be connected to high-degree nodes, and same for low-degree nodes. If $r < 0$ then opposite.
- Path length. If average is low, well-connected. Vice versa.
- Network diameter. Longest shortest path between any two nodes.
- Other measures include degree, betweenness, closeness, and others.

Random networks. Generate random networks from a given set of nodes. With N nodes, probability of connection p , probability of no connection $(1-p)$, and k connections, one node's connection being $p^k(1-p)^{N-1-k}$. For any k connections, it is a binominal distribution:

$$p(k) = \binom{N-1}{k} p^k (1-p)^{(N-1)-k}$$

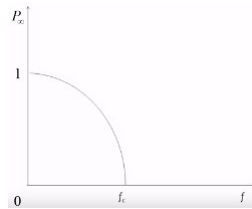
The random networks are form a family and plotted. Fraction of nodes in the giant (main) component is f . When N is large and p small, this becomes a Poisson distribution. However, random networks are not realistic.

Small-world network. Watts-Strogatz model. Rewire regular lattice, which reduces diameter considerably.

Scale-free networks. Degree distributions are power-laws. Preferential attachment model explain power laws.

28.15.1 Percolation

Attacking a network by removing nodes leads to percolation, where networks break apart into pieces (subgraphs). An attack will lead to very rapid reductions in largest subgraphs. So fraction of nodes in giant component decreases quickly.



Scale-free networks are more robust to random attacks but more susceptible to targeted attacks (as they attack high-degree nodes). Reverse for random networks.

28.15.2 Code

Use Python NetworkX and Gephi packages.

28.16 Emergence

Complexity at different scales. No general mathematical theory, just descriptions and classifications, as well as specific examples. Might be approached using agent-based modelling, renormalization, scaling.

Different scales (integrative) levels interact. Lower levels comprise higher level but higher level direction causes lower level to react.

Variable space can differ between scale theories, e.g. atoms described by velocities and coordinates but gases described by pressure, temperature, etc. So the number of variables and degrees of freedom changes. If the variables map 1-1, then fine-graining and if map many-to-1, then coarse-graining.

Many dimensions of comparisons. Homeostructural theories (theories that are similar in nature) versus heterostructural theories (theories that are very different in nature). Fine-graining versus coarse-graining. Weak emergence (microtheories can be used to transition to macro theories) and strong emergence (no known good examples, only hypothetical theories such as random photos that make a mosaic).

Generally, macrotheories can't exactly specify microtheories but microtheories exactly specify macrotheories.

29 Numerical methods

29.1 Binary numbers

Convert from binary to decimal with powers of 2, positive powers for integer part and negative powers after the radix.

Convert from decimal to binary. Think about decimal position system which is based on integer part being successively divided by 10 and decimals being multiplied by 10. Binary. Take integer part and keep dividing by 2, moving from rightmost position to leftmost position. Take decimal part and keep multiplying by 2, moving from leftmost point (next to radix) and further away.

Shortcut that sometimes works for converting decimal to binary. $7 = 2^3 - 1 = 100 - 001 = 011$.

29.2 IEEE floating point standards

Single (32-bit) and double (64-bit) precision standards. In both cases, represent a very wide range of numbers with 4-byte or 8-byte representations. First bit is sign (negative or positive), next bits are exponent ($2^{exponent}$ and the final bits are significands (provide precision). For double precision, the equation is:

$$(-1)^{sign} \times 2^{exponent-1023} \times 1.f$$

Since the significand is multiplied by the exponent part, the floating point precision depends on the scale (i.e. exponent) of the number.

REALMAX= 1.7977×10^{308} , REALMIN= 2.225×10^{-308} . EPS=Machine epsilon is the spacing between numbers.

29.3 Logistic map

29.4 Logistic map: intuition

1D map. Simplest negative feedback loop:

$$x_{n+1} = rx_n(1 - x_n)$$

Graphing x_{n+1} versus x_n is an inverted parabola, so the lower the current population, the lower the population next year as well on the left side of the inverted parabola. But on the right side, the higher the population this year, the LOWER the population next year!

When graphing x_n versus r , it is 0 until $r=1$ and then rises for the left half of the inverted parabola. After, it splits as there is an oscillation - the oscillation between high and low populations between years, so the paths bifurcate. Going further, each of these bifurcates further. Between 4- and 5-period cycles,

bifurcations are chaotic. Going even further, there are periods of more stability.

This is a fractal, which is the vertical part of the Mandelbrot set. The main cardioid of the Mandelbrot set are stable are one value, and then smaller cardioids are for bifurcations - even smaller cardioids have more bifurcations. The Mandelbrot needle is chaotic after the Feigenbaum number. The Mandelbrot medallion in the needle are the stable islands amidst chaos. Bifurcation only exists on real line as the logistic map exists only on the real line. The Mandelbrot 'bulbs' which are complex are oscillating but not on the logistic map - they oscillate up and down as well therefore.

Feigenbaum δ is ratio of increase to r . Works for other equations that have a single hump negative feedback as well, example:

$$x_{n+1} = rx_n \sin(x_n)$$

29.4.1 Logistic map: fixed points

$$x_{n+1} = f(x_n)$$

Fixed point solution is x_* :

$$x_* = f(x_*)$$

Is the fixed point stable? How stable is it? Answer with linear stability analysis (linear because it restricts the solution to the linear part of the Taylor expansion):

$$x_n = x_* + \epsilon_n$$

$$x_{n+1} = x_* + \epsilon_{n+1}$$

which is stable if $\epsilon_{n+1} < \epsilon_n$ and unstable if $\epsilon_{n+1} > \epsilon_n$. Inserting the above two equations into $x_{n+1} = f(x_n)$:

$$x_* + \epsilon_{n+1} = f(x_* + \epsilon_n)$$

Apply linear stability analysis by keeping the linear term in the Taylor expansion:

$$f(\epsilon_n + x_*) = f(x_*) + \epsilon_n f'(x_*) + \dots$$

so drop the quadratic term and terms after. Since $x_* = f(x_*)$:

$$x_* + \epsilon_n f'(x_*)$$

$$\left| \frac{\epsilon_{n+1}}{\epsilon_n} \right| = |f'(x_*)|$$

where x_* is stable if $|f'(x_*)| < 1|$ and unstable if $|f'(x_*)| > 1|$.

29.4.2 Logistic map: quadratic map

Let the logistic map be quadratic:

$$x_{n+1} = \mu x_n(1 - x_n)$$

and restrict it to $0 < \mu < 4$. Chart x versus μ . The fixed points are:

$$x_* = \mu x_*(1 - x_*)$$

so, the first fixed solution is:

$$x_* = 0$$

and the second is:

$$1 = \mu(1 - x_*)$$
$$x_* = 1 - \frac{1}{\mu}$$

The quadratic logistic map is changing at rate:

$$f'(x) = \mu - 2\mu x$$

So with the two fixed points above,

$$f'(0) = \mu$$

which is stable for $0 < \mu < 1$ which converges to 0 because $\mu < 1$. Further:

$$f'\left(1 - \frac{1}{\mu}\right) = 2 - \mu$$

which is stable for $1 < \mu < 3$.

Bifurcations. Splits into two fixed points:

$$x_{**} = f(x_*)$$

$$x_* = f(x_{**})$$

The quadratic map bifurcates at $\mu = 3$. Take $x = f(f(x))$ and combine with $f(x) = \mu x(1 - x)$ to get:

$$\mu^3 x^4 - 2\mu^3 x^3 + \mu^2(1 + \mu)x^2 + (1 - \mu^2)x = 0$$

Since $x_* = 0, x_* = 1 - \frac{1}{\mu}$ are known roots, factor these out. The remainder is

$$\mu^2 x^2 - \mu(\mu + 1)x + (\mu + 1) = 0$$

Solve:

$$x = \frac{1}{2\mu} \left((\mu + 1) \pm \sqrt{(\mu + 1)(\mu - 3)} \right)$$

which is valid for $\mu \geq 3$. If $\mu = 3$, then $x = \frac{2}{3}$ which is the same as $x_* = 0, x_* = 1 - \frac{1}{\mu}$.

29.5 Root finding

29.5.1 Bisection method

Slow but always converges. For $f(x) = 0$, bracket the root $x = r$ with x_0, x_1 . Then calculate their bisection $\frac{x_0+x_1}{2} = x_1 - \left(\frac{x_1-x_0}{2}\right)$. Test $f(x)$ to see which bracket (in this case, left or right) to use, and repeat.

29.5.2 Newton-Raphson method

Fast but initial guess has to be good (otherwise might not converge) and must be able to take analytical derivative. On Cartesian coordinates, graph $f(x)$ and take tangent at (x_n, y_n) , extending the tangent $f'(x)$ down to the x-axis at point $(x_{n+1}, 0)$. Then,

$$\begin{aligned}y - y_n &= f'(x_n)(x - x_n) \\0 - f(x_n) &= f'(x_n)(x_{n+1} - x_n) \\x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)}\end{aligned}$$

Example. $\sqrt{2}$. Use $f(x) = x^2 - 2$ and $f'(x) = 2x$. Then

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n^2 + 2}{2x_n}$$

29.5.3 Secant method

Problem with Newton's method: must have analytical derivative. So adapt Newton's method of

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

by using an approximation to the derivative:

$$f'(x_n) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

Convergence is relatively fast but not as fast as Newton but faster than bisection.

29.5.4 Order of convergence

With $f(x) = 0, x = r, \epsilon_n = r - x_n$, if

$$|\epsilon_{n+1}| = k|\epsilon_n|^p$$

where p is the convergence order. For bisection,

$$|\epsilon_{n+1}| = \frac{1}{2}|\epsilon_n|^1$$

so is order 1. Newton is $p = 2$ (the best possible) and secant is $p \approx 1.6$.

Newton method convergence order.

$$\begin{aligned}
 x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\
 \epsilon_n &= r - x_n \\
 r - x_{n+1} &= r - x_n + \frac{f(x_n)}{f'(x_n)} \\
 \epsilon_{n+1} &= \epsilon_n + \frac{f(r - \epsilon_n)}{f'(r - \epsilon_n)} \\
 \epsilon_{n+1} &= \epsilon_n + \frac{f(r) - \epsilon_n f'(r) + \frac{\epsilon_n^2}{2} f''(r) + \dots}{f'(r) - \epsilon_n f''(r) + \dots}
 \end{aligned}$$

Divide both sides by $f'(r)$ and noting $f(r) = 0$ as it is the root,

$$\epsilon_{n+1} = \epsilon_n + \frac{0 - \epsilon_n + \frac{\epsilon_n^2}{2} \frac{f''(r)}{f'(r)} + \dots}{1 - \epsilon_n \frac{f''(r)}{f'(r)} + \dots}$$

Since $\frac{1}{1-\epsilon} = 1 + \epsilon + \epsilon^2 + \dots$,

$$\begin{aligned}
 \epsilon_{n+1} &= \epsilon_n + \left(-\epsilon_n + \frac{\epsilon_n^2}{2} \frac{f''(r)}{f'(r)} + \dots \right) \left(1 - \epsilon_n \frac{f''(r)}{f'(r)} + \dots \right) \\
 \epsilon_{n+1} &= \epsilon_n + \left(-\epsilon_n + \frac{\epsilon_n^2}{2} \frac{f''(r)}{f'(r)} - \epsilon_n^2 \frac{f''(r)}{f'(r)} + \dots \right) = -\frac{f''(r)}{2f'(r)} \epsilon_n^2
 \end{aligned}$$

So,

$$|\epsilon_{n+1}| = k |\epsilon_n|^2$$

noting that if $f'(r) = 0$, it is problematic (the derivative is 0 at the root).
Second method convergence order.

$$x_{n+1} = x_n - \frac{(x_n - x_{n-1})f(x_n)}{f(x_n) - f(x_{n-1})}$$

Using $\epsilon_n = r - x_n$ and subtracting r ,

$$\epsilon_{n+1} = \epsilon_n + \frac{(\epsilon_{n-1} - \epsilon_n)f(r - \epsilon_n)}{f(r - \epsilon_n) - f(r - \epsilon_{n-1})}$$

Taylor expand $f(r - \epsilon_n)$ and $f(r - \epsilon_{n-1})$ with $f(r) = 0$,

$$\epsilon_{n+1} = \epsilon_n + \frac{-\epsilon_n f'(r) + \frac{1}{2} \epsilon_n^2 f''(r) + \dots}{f'(r) - \frac{1}{2} (\epsilon_{n-1} + \epsilon_n) f''(r) + \dots}$$

and dividing by $f'(r)$,

$$\epsilon_{n+1} = \epsilon_n + \frac{-\epsilon_n + \frac{1}{2}\epsilon_n^2 \frac{f''(r)}{f'(r)} + \dots}{1 - \frac{1}{2}(\epsilon_{n-1} + \epsilon_n) \frac{f''(r)}{f'(r)} + \dots}$$

Since $\frac{1}{1-\epsilon} = 1 + \epsilon + \epsilon^2 + \dots$,

$$\epsilon_{n+1} = \epsilon_n + \left(-\epsilon_n + \frac{1}{2}\epsilon_n^2 \frac{f''(r)}{f'(r)} + \dots\right) \left(1 + \frac{1}{2}(\epsilon_{n-1} + \epsilon_n) \frac{f''(r)}{f'(r)} + \dots\right)$$

$$\epsilon_{n+1} = -\frac{1}{2}(\epsilon_{n-1}\epsilon_n) \frac{f''(r)}{f'(r)}$$

$$|\epsilon_{n+1}| = \left|\frac{1}{2} \frac{f''(r)}{f'(r)}\right| |\epsilon_{n-1}| |\epsilon_n|$$

With

$$|\epsilon_{n+1}| = k |\epsilon_n|^p$$

$$|\epsilon_n| = k |\epsilon_{n-1}|^p$$

Then,

$$k^{p+1} |\epsilon_{n-1}|^{p^2} = \frac{k}{2} \left| \frac{f''(r)}{f'(r)} \right| |\epsilon_{n-1}|^{p+1}$$

$$p^2 = p + 1$$

So,

$$p = \frac{\sqrt{5} + 1}{2}$$

which is the golden ratio.

29.6 Newton Fractal

Cube roots of unity:

$$f(z) = z^3 - 1$$

$$f'(z) = 3z^2$$

$$z_{n+1} = z_n - \frac{f(z_n)}{f'(z_n)}$$

If points on the complex plane are each run through Newton's method, they will converge to one of the cube roots of unity.

29.7 Feigenbaum δ

Let μ be bifurcation points,

$$\mu_n : 2^n \rightarrow 2^{n+1}$$

However, at bifurcation points, there are many transient points. It takes a long time to converge. So use 'superstable points' instead.

Orbit. $N = 2^n$, where cycles repeat:

$$x_0, x_1, x_2, \dots, x_{N-1}, x_0, x_1, \dots$$

Superstable cycle. A theoretical result says if it contains $x = \frac{1}{2}$ in the orbit, then the number of iterations to convergence is lowest. Transient number is low, so convergence is fast. So the orbit is:

$$x_0, x_1, x_2, \dots, x_{N-1}, x_0, x_1, \dots$$

$$\frac{1}{2}, x_1, x_2, \dots, x_{N-1}, \frac{1}{2}, x_1, \dots$$

Compute δ by finding μ where the orbit contains $x_0 = \frac{1}{2}$. Deefine m_n as period 2^n cycle with $x_0 = \frac{1}{2}$ in orbit.

$$m_0 : \frac{1}{2} = \frac{1}{2}\mu \left(1 - \frac{1}{2}\right) \rightarrow \mu = 2$$

$$m_1 : x_1 = \mu x_0 (1 - x_0)$$

$$x_0 = \mu x_1 (1 - x_1)$$

Combine with $x_0 = \frac{1}{2}$,

$$\mu^3 - 4\mu^2 + 8 = 0$$

We know $\mu = 2$ is a root and using long division of $\mu - 2$, obtain $\mu^2 - 2\mu - 4 = 0$, so the positive root is

$$m_1 = 1 + \sqrt{5}$$

For m_2 and beyond, use numerical methods. To obtain the superstable point from

$$x_0, x_1, x_2, \dots, x_{N-1}, x_0, x_1, \dots$$

$$\frac{1}{2}, x_1, x_2, \dots, x_{N-1}, \frac{1}{2}, x_1, \dots$$

solve this equation for its root:

$$g(\mu) = x_N - \frac{1}{2}$$

given $x_0 = \frac{1}{2}$. The root will be $x_N = \frac{1}{2}$. Use the fastest algorithm to solve for the root, Newton-Raphson. So need derivative:

$$g'(\mu) = x'_N$$

where $x'_0 = \frac{1}{2}$. Use the product rule since $\mu = \mu(x_i)$:

$$x_{i+1} = \mu x_i (1 - x_i)$$

$$x'_{i+1} = x_i(1 - x_i) + \mu x'_i(1 - 2x_i)$$

Newton-Raphson:

$$\mu^{j+1} = \mu^j - \frac{x_N - \frac{1}{2}}{x'_N}$$

Need an initial guess for $m_n = \mu^0$ by using the original equation for the Feigenbaum δ :

$$\delta = \lim_{n \rightarrow \infty} \frac{m_{n-1} - m_{n-2}}{m_n - m_{n-1}}$$

$$\delta(m_n - m_{n-1}) = m_{n-1} - m_{n-2}$$

$$m_n - m_{n-1} = \frac{m_{n-1} - m_{n-2}}{\delta}$$

$$m_n \approx m_{n-1} + \frac{m_{n-1} - m_{n-2}}{\delta_{n-1}}$$

To find m_2 , start at perhaps δ of 4-5.

29.8 Matrix algebra

29.8.1 Roundoff errors in Gaussian elimination

Let ϵ be smaller than eps. Example.

$$2 + \epsilon = 2$$

$$4 - \epsilon = 4$$

Example. Solve:

$$\epsilon x_1 + 2x_2 = 4$$

$$x_1 - x_2 = 1$$

So:

$$\begin{pmatrix} \epsilon & 2 & 4 \\ 1 & -1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} \epsilon & 2 & 4 \\ 0 & -2/\epsilon & -4/\epsilon + 1 \end{pmatrix}$$

where the solutions are approximately $x_2 \approx 2, x_1 \approx 3$. Using Gaussian elimination,

$$\epsilon x_1 + 2x_2 = 4$$

$$(\epsilon + 2)x_2 = 4 - \epsilon$$

$$x_2 = 2$$

but:

$$\epsilon x_1 = 4 - 2x_2 = 0$$

$$x_1 = 0$$

which is wrong. To fix, use partial pivoting.

29.8.2 Partial pivoting

Partial pivoting = interchange rows. Full pivoting = interchange rows and columns, but this is usually not necessary. Example. Solve:

$$\epsilon x_1 + 2x_2 = 4$$

$$x_1 - x_2 = 1$$

So:

$$\begin{pmatrix} \epsilon & 2 & 4 \\ 1 & -1 & 1 \end{pmatrix}$$

Use a row interchange, comparing the first columns (the potential pivots) since $1 \gg \epsilon$:

$$\begin{pmatrix} 1 & -1 & 1 \\ \epsilon & 2 & 4 \end{pmatrix}$$

Then solve:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 + \epsilon & 4 - \epsilon \end{pmatrix}$$

Using Gaussian elimination,

$$\epsilon x_1 + 2x_2 = 4$$

$$(\epsilon + 2)x_2 = 4 - \epsilon$$

$$x_2 = 2$$

and:

$$x_1 - x_2 = 1$$

$$x_1 = 3$$

which is better.

29.8.3 LU decomposition

Encode Gaussian elimination of matrix A into elementary matrices M_1, M_2, M_3 for instance. Then the upper matrix U is:

$$M_3 M_2 M_1 A = U$$

$$A = M_1^{-1} M_2^{-2} M_3^{-3}$$

where for instance if

$$M_1 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

then

$$M_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Invert other elementary matrices similarly. Then,

$$M_1^{-1} M_2^{-2} M_3^{-3} = L$$

So,

$$A = LU$$

Use this to repeatedly solve $Ax = b$. Let $y = Ux$. Then (1) solve $Ly = b$ for y and (2) solve $Ux = y$ for x .

29.8.4 LU decomposition with partial pivoting

Example.

$$A = \begin{pmatrix} -2 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -8 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 6 & -6 & 7 \\ -2 & 2 & -1 \\ 3 & -8 & 4 \end{pmatrix} = P_{12}A$$

$$\begin{pmatrix} 6 & -6 & 7 \\ 0 & 0 & 4/3 \\ 0 & -5 & 1/2 \end{pmatrix} = M_2 M_1 P_{12} A$$

$$\begin{pmatrix} 6 & -6 & 7 \\ 0 & -5 & 1/2 \\ 0 & 0 & 4/3 \end{pmatrix} = P_{23} M_2 M_1 P_{12} A = U$$

$P_{23} = P_{23}^{-1}$ but note that multiplication from left will interchange rows but multiplication on right will interchange columns so result of $P_{23} X P_{23}$ will be an interchange of a_{21} and a_{31} :

$$P_{23} M_2 M_1 P_{23} P_{12} A = U$$

$$\begin{aligned}
(P_{23}M_2M_1P_{23})P_{23}P_{12}A &= U \\
P_{23}P_{12}A &= (P_{23}M_2M_1P_{23})^{-1}U = LU \\
A &= (P_{23}P_{12}L)U
\end{aligned}$$

29.8.5 Cholesky decomposition

Cholesky decomposition is roughly analogous to taking a square root in multiple dimensions. For example, in the normal distribution, the standard deviation is the square root of the variance and used to draw samples from the distribution. For the multivariate normal distribution, the variance is a matrix (unfortunately named the covariance matrix), so use the Cholesky decomposition in place of the square root to obtain the equivalent of a standard deviation.

For a Hermite positive definition matrix (unless it's positive definite, can't 'square root' the matrix):

$$A = LL'$$

where L is a triangular matrix.

29.9 Operation counts

Example. Matrix multiplication. $A \times B = AB$. Per element, there are n multiplications and $n-1$ additions. Since they are $n \times n$ matrices, there are n^2 elements, so there are a total of n^3 multiplications and $n^2(n-1)$ additions. So AB scales at $O(n^3)$, throwing away the constants and lower-orders. So $T_n = kn^3$. If the matrix sizes are doubled, then $2n \times 2n$ and $T_{2n} = k(2n)^3 = 8(kn^3) = 8T_n$.

Example. One isolated step of Gaussian elimination of $n \times n$ matrix at pivot position (i,i) . Then, there are $(n-i)$ multiplications and additions and over $n-i$ rows equals $(n-i)^2$. The total number of operations is then

$$\sum_{i=1}^{n-1} (n-i)^2 = (n-1)^2 + (n-2)^2 + \dots + 1^2$$

$$\sum_{k=1}^{n-1} k^2 = \frac{1}{6}(n-1)(2n-1)n$$

Since $\sum_{k=1}^n k^2 = \frac{1}{6}n(2n+1)(n+1)$.

$$= \frac{1}{3}n^3 + \dots = O(n^3)$$

Example. LU backward substitution (so for U or L matrices):

$$x_{ii} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=i+1}^n a_{ij}x_j \right)$$

Since there are $n - i + 1$ multiplications and additions with n rows,

$$\begin{aligned}\sum i &= 1^n(n - i + 1) = n + (n - 1) + \dots + 1 \\ &= \sum_{k=1}^n k = \frac{1}{2}n(n + 1) \approx \frac{1}{2}n^2\end{aligned}$$

LU forward substitution is similar, though the equation is

$$x_{ii} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=i}^n a_{ij}x_j \right)$$

and also results in $O(n^2)$. So for forward + backward substitution, it is double or $O(n^2)$, which is much faster than Gaussian elimination.

29.10 Eigenvalue power method

To find the largest eigenvalue λ_1 and associated eigenvector e . $A_{n \times n}$ matrix with n real distinct λ s. Any column vector x_0 is a combination of linearly independent eigenvectors:

$$\begin{aligned}x_0 &= c_1e_1 + c_2e_2 + \dots + c_ne_n = \sum_i^n c_ie_i \\ x_1 &= Ax_0 = A \sum_i^n c_ie_i = \sum_i^n c_iAe_i = \sum_i^n c_i\lambda_ie_i\end{aligned}$$

Using powers, the difference between the largest and other eigenvalues grows:

$$x_p = A^p x_0 = \sum_i^n c_i \lambda_i^p e_i = c_1 \lambda_1^p e_1 + \sum_{i=2}^n c_i \lambda_i^p e_i = \lambda_1^p \left(c_1 e_1 + \sum_{i=2}^n c_i \left(\frac{\lambda_i}{\lambda_1} \right)^p e_i \right) \approx \lambda_1^p c_1 e_1$$

where λ_1 is the dominant eigenvalue.

$$x_{p+1} \approx \lambda_1^{p+1} c_1 e_1$$

To calculate the dominant eigenvalue

$$\lambda_1 = \frac{x'_p x_{p+1}}{x'_p x_p}$$

$$x_p = \lambda_1^p c_1 e_1$$

Since eigenvectors are unique up to a constant and $\lambda_1 c_1$, $e_1 = x_p$. To prevent underflow or overflow, normalize:

$$x_p \rightarrow \frac{x_p}{(x'_p x_p)^{1/2}}$$

Example.

$$A = \begin{pmatrix} 6 & 5 \\ 4 & 5 \end{pmatrix}$$

where $\lambda_1 = 10, v_1 = \begin{pmatrix} 5/4 \\ 1 \end{pmatrix}$ and $\lambda_2 = 1, v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Choose arbitrary vector to begin:

$$x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$x_1 = A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

$$x_2 = A \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 56 \\ 44 \end{pmatrix}$$

$$x_3 = A \begin{pmatrix} 56 \\ 44 \end{pmatrix} = \begin{pmatrix} 556 \\ 444 \end{pmatrix}$$

and there is a pattern.

$$\lambda_1 = \frac{x'_4 x_5}{x'_4 x_4} \approx 10$$

$$e_1 = \begin{pmatrix} 556/444 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.25 \\ 1 \end{pmatrix}$$

29.11 Systems of nonlinear equations

Non-linear equation system:

$$f(x, y) = 0, g(x, y) = 0$$

Compute by starting with initial guess (x_0, y_0) and then computing (x_1, y_1) , etc. until convergence.

$$f(x_{n+1}, y_{n+1}) = f(x_n, y_n) + (x_{n+1} - x_n)f_x(x_n, y_n) + (y_{n+1} - y_n)f_y(x_n, y_n) + \dots$$

$$g(x_{n+1}, y_{n+1}) = g(x_n, y_n) + (x_{n+1} - x_n)g_x(x_n, y_n) + (y_{n+1} - y_n)g_y(x_n, y_n) + \dots$$

Equivalently,

$$0 = f(x_n, y_n) + \Delta x f_x(x_n, y_n) + \Delta y f_y(x_n, y_n) + \dots$$

$$0 = g(x_n, y_n) + \Delta x g_x(x_n, y_n) + \Delta y g_y(x_n, y_n) + \dots$$

Further:

$$-f(x_n, y_n) = \Delta x f_x(x_n, y_n) + \Delta y f_y(x_n, y_n) + \dots$$

$$-g(x_n, y_n) = \Delta x g_x(x_n, y_n) + \Delta y g_y(x_n, y_n) + \dots$$

which is a matrix equation

$$\begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} \begin{pmatrix} \Delta x_n \\ \Delta y_n \end{pmatrix} = - \begin{pmatrix} f \\ g \end{pmatrix}$$

$$J \begin{pmatrix} \Delta x_n \\ \Delta y_n \end{pmatrix} = - \begin{pmatrix} f \\ g \end{pmatrix}$$

where J and the RHS are evaluated at (x_n, y_n) so must be re-computed each iteration.

29.11.1 Non-linear system example: Lorentz equations

$$\dot{x} = \sigma(y - x)$$

$$\dot{y} = x(r - z) - y$$

$$\dot{z} = xy - \beta z$$

Fixed point: $\dot{x} = \dot{y} = \dot{z} = 0$ because the derivative is 0 there and remains there.
Rename the system:

$$\dot{x} = \sigma(y - x) = 0 = f(x, y, z)$$

$$\dot{y} = x(r - z) - y = 0 = g(x, y, z)$$

$$\dot{z} = xy - \beta z = 0 = h(x, y, z)$$

and calculate the Jacobian:

$$J = \begin{pmatrix} -\sigma & \sigma & 0 \\ r - z & -1 & -x \\ y & x & -\beta \end{pmatrix}$$

$$J \begin{pmatrix} \Delta x_n \\ \Delta y_n \\ \Delta z_n \end{pmatrix} = - \begin{pmatrix} \sigma(y_n - x_n) \\ x_n(r - z_n) - y_n \\ x_n y_n - \beta z_n \end{pmatrix}$$

To solve, begin with initial values $(x_0, y_0, z_0)'$, enter into the system of non-linear equations above, and solve for $(\Delta x_n, \Delta y_n, \Delta z_n)$, which is used to calculate

$$x_1 = x_0 + \Delta x_n$$

$$y_1 = y_0 + \Delta y_n$$

$$z_1 = z_0 + \Delta z_n$$

29.12 Quadrature and interpolation

29.12.1 Midpoint rule

$$I_h = \int_0^h f(x) dx$$

where

$$f(x) = f\left(\frac{h}{2}\right) + (x - h/2)f'\left(\frac{h}{2}\right) + \frac{(x - h/2)^2}{2}f''\left(\frac{h}{2}\right) + \frac{(x - h/2)^3}{6}f'''\left(\frac{h}{2}\right) + \frac{(x - h/2)^4}{24}f^{(4)}\left(\frac{h}{2}\right) + \dots$$

Change variables in $I_h = \int_0^h f(x)dx$ by $y = x - h/2$:

$$I_h = \int_{-h/2}^{h/2} f(y)dy$$

Due to symmetry, double the even powers of y and set the odd powers of y to 0:

$$I_h = \int_{-h/2}^{h/2} f(y)dy = 2 \int_0^{h/2} \left(f\left(\frac{h}{2}\right) + 0 + \frac{y^2}{2} f''\left(\frac{h}{2}\right) + 0 + \frac{y^4}{24} f^{(4)}\left(\frac{h}{2}\right) \right) dy$$

$$\int_0^{h/2} y^2 dy = \frac{h^3}{24}, \int_0^{h/2} y^4 dy = \frac{h^5}{160}$$

$$I_h = hf\left(\frac{h}{2}\right) + \frac{h^3}{24} f''\left(\frac{h}{2}\right) + \frac{h^5}{1920} f^{(4)}\left(\frac{h}{2}\right) = hf\left(\frac{h}{2}\right) + O(h^3)$$

Intuition.

$$\int_0^{h/2} f(x)dx = \int_0^{h/2} (a + bx + cx^2)dx = h \left(a + \frac{bh}{2} + \frac{ch^2}{3} \right)$$

$$f\left(\frac{h}{2}\right) = a + \frac{bh}{2} + \frac{ch^2}{4}$$

$$f''\left(\frac{h}{2}\right) = 2c$$

So

$$\begin{aligned} \int_0^h f(x)dx &= h \left(a + \frac{bh}{2} + \frac{ch^2}{3} \right) \\ &= h \left(a + \frac{bh}{2} + \frac{ch^2}{4} \right) + \left(\frac{ch^2}{3} - \frac{ch^2}{4} \right) \\ &= hf\left(\frac{h}{2}\right) + h \left(\frac{ch^2}{12} \right) \\ &= hf\left(\frac{h}{2}\right) + \frac{h^3}{24} f''\left(\frac{h}{2}\right) \end{aligned}$$

29.12.2 Trapezoidal rule

Recall

$$f(x) = f\left(\frac{h}{2}\right) + (x-h/2)f'\left(\frac{h}{2}\right) + \frac{(x-h/2)^2}{2}f''\left(\frac{h}{2}\right) + \frac{(x-h/2)^3}{6}f''' \left(\frac{h}{2}\right) + \frac{(x-h/2)^4}{24}f^{(4)}\left(\frac{h}{2}\right) + \dots$$

If $x = 0$,

$$f(0) = f\left(\frac{h}{2}\right) - \frac{h}{2}f'\left(\frac{h}{2}\right) + \frac{h^2}{8}f''\left(\frac{h}{2}\right) - \frac{h^3}{48}f''' \left(\frac{h}{2}\right) + \frac{h^4}{384}f^{(4)}\left(\frac{h}{2}\right) + \dots$$

If $x = h$, the above negative terms become positive:

$$f(h) = f\left(\frac{h}{2}\right) + \frac{h}{2}f'\left(\frac{h}{2}\right) + \frac{h^2}{8}f''\left(\frac{h}{2}\right) + \frac{h^3}{48}f'''\left(\frac{h}{2}\right) + \frac{h^4}{384}f^{(4)}\left(\frac{h}{2}\right) + \dots$$

The trapezoidal rule is

$$\frac{h}{2}(f(0) + f(h)) = hf\left(\frac{h}{2}\right) + \frac{h^3}{8}f''\left(\frac{h}{2}\right) + \frac{h^5}{384}f^{(4)}\left(\frac{h}{2}\right) + \dots$$

Combine the midpoint with the trapezoidal rule:

$$I_h = \frac{h}{2}(f(0) + f(h)) - \frac{h^3}{12}f''\left(\frac{h}{2}\right) - \frac{h^5}{480}f^{(4)}\left(\frac{h}{2}\right) + \dots$$

Note that the trapezoidal rule can also be derived by

$$\begin{aligned} f(x) &\approx f(0) + \frac{f(h) - f(0)}{h}x \\ \int_0^h f(x)dx &= \int_0^h \left(f(0) + \frac{f(h) - f(0)}{h}x\right) dx \\ &= f(0)h + \frac{1}{2}(f(h) - f(0))h \\ &= \frac{h}{2}(f(0) + f(h)) \end{aligned}$$

29.12.3 Simpson's rule

Midpoint rule:

$$I_h = hf\left(\frac{h}{2}\right) + \frac{h^3}{24}f''\left(\frac{h}{2}\right) + \frac{h^5}{1920}f^{(4)}\left(\frac{h}{2}\right) = hf\left(\frac{h}{2}\right) + O(h^3)$$

Trapezoidal rule:

$$\frac{h}{2}(f(0) + f(h)) = hf\left(\frac{h}{2}\right) + \frac{h^3}{8}f''\left(\frac{h}{2}\right) + \frac{h^5}{384}f^{(4)}\left(\frac{h}{2}\right) + O(h^7)$$

Combine in Simpson's rule:

$$3I_h = \frac{h}{2}(f(0) + f(h)) + 2hf\left(\frac{h}{2}\right) + O(h^5)$$

$$I_h = \frac{h}{6}(f(0) + 4f\left(\frac{h}{2}\right) + f(h)) + O(h^5)$$

$$I_{2h} = \int_0^{2h} f(x)dx = \frac{h}{3}(f(0) + 4f(h) + f(2h)) + O(h^5)$$

29.12.4 Composite quadrature rule

$$I = \int_a^b f(x)dx$$

where $a = x_0, x_1, \dots, x_n = b$.

$$\int_a^b f(x)dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x)dx$$

Substitution: let $s = x - x_i$ so $x = s + x_i$ and $ds = dx$. Further, define $h_i = x_{i+1} - x_i$ and $f_i = f(x_i)$. Then,

$$\int_a^b f(x)dx = \sum_{i=0}^{n-1} \int_0^{h_i} f(x_i + s)ds$$

$$\int_a^b f(x)dx = \sum_{i=0}^{n-1} \frac{h_i}{2} (f_i + f_{i+1})$$

Assuming equally spaced points such that $h_i = h$ as a constant,

$$\int_a^b f(x)dx = \frac{h}{2} (f_0 + 2f_1 + 2f_2 + \dots + 2f_{n-1} + f_n) + O(h^2)$$

because the 'middle points' (overlapping points) get doubled. Derive Simpson's rule by adding all separate intervals:

$$\begin{aligned} \int_a^b f(x)dx &= \frac{h}{3} (f_0 + 4f_1 + f_2) + \frac{h}{3} (f_2 + 4f_3 + f_4) + \dots + O(h^4) \\ &= \frac{h}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + \dots + 2f_{n-2} + 4f_{n-1} + f_n) + O(h^4) \end{aligned}$$

29.13 Gaussian quadrature

Very different from other quadrature techniques.

$$\int_a^b W(x)f(x)dx = \sum_{i=1}^n w_i f(x_i)$$

where $W(x)$ is the weight function, w_i are weights, and x_i are nodes. There are several Gaussian quadrature methods:

$$\text{Chebyshev} = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} f(x)dx$$

$$\text{Laguerre} = \int_0^\infty e^{-x} f(x)dx$$

$$\text{Hermite} = \int_{-\infty}^\infty e^{-x^2} f(x)dx$$

$$\text{Legendre} = \int_{-1}^1 f(x)dx$$

Example. Two-point Legendre.

$$\int_{-1}^1 f(x)dx = w_1 f(x_1) + w_2 f(x_2)$$

For low-degree polynomials, with 4 parameters then exact for $f(x) = 1 (= x^0), x (= x^1), x^2, x^3$:

$$f(x) = 1 : 2 = w_1 + w_2$$

$$f(x) = x : 0 = w_1 x_1 + w_2 x_2$$

$$f(x) = x^2 : \frac{2}{3} = w_1 x_1^2 + w_2 x_2^2$$

$$f(x) = x^3 : 0 = w_1 x_1^3 + w_2 x_2^3$$

Solution. Expect symmetry about $x=0$, so (1) $x_1 = -x_2$ and (2) $w_1 = w_2$. So based on above equations, $w_1 = w_2 = 1$ and $x_1 = -\frac{1}{\sqrt{3}}, x_2 = \frac{1}{\sqrt{3}}$.

29.14 Adaptive quadrature

$$I = \int_a^b f(x)dx$$

'Level 1' over the interval from a to b with $h = b - a$. Use trapezoidal rule:

$$I = \frac{h}{2}(f(a) + f(b)) - \frac{h^3}{12}f''(\psi)$$

Compare this to the integral over (a, b) but with c in the middle: (a, c, b) with trapezoidal rule:

$$I = \frac{h}{4}(f(a) + 2f(c) + f(b)) - \frac{(h/2)^3}{12}f''(\psi_l) - \frac{(h/2)^3}{12}f''(\psi_r)$$

Define

$$s_1 = \frac{h}{2}(f(a) + f(b))$$

$$s_2 = \frac{h}{4}(f(a) + 2f(c) + f(b))$$

$$E_1 = -\frac{h^3}{12}f''(\psi)$$

$$E_2 = -\frac{1}{4}\frac{h^3}{12}f''(\psi)$$

For E_2 , assume $\psi_l = \psi_r$ as the intervals shrink with more intervals, the approximation gets better so these errors ψ get smaller. From above,

$$E_1 = 4E_2$$

$$S_1 + E_1 = S_2 + E_2$$

$$S_1 + 4E_2 = S_2 + E_2$$

$$|E_2| = \frac{|S_2 - S_1|}{3}$$

If $|E_2| > \textit{tolerance}$ then increase level (i.e. separate the interval (a, b) further. Different parts of the integration (where $f(x)$ changes a lot) will be integrated at higher levels whereas more flat areas where be integrated at lower levels.

29.15 Interpolation

Three types:

(1) Polynomial. Draw lines between endpoints. Global because one function goes through all points. Not used much beyond 2 or 3 points.

(2) Piecewise linear. Connect points by straight lines.

(3) Cubic spline. Smooths out piecewise linear by using cubic piecewise polynomials.

29.15.1 Polynomial

Example. If a function passes through $(0,0)$, $(1,1)$, and $(2,1)$, approximating by a quadratic $f(x) = ax^2 + bx + c$,

$$c = 0$$

$$a + b + c = 1$$

$$4a + 2b + c = 1$$

Solve for these, and enter into $f(x)$.

29.15.2 Piecewise linear

Interpolating function is $y = g(x) = g_0 + g_1 + \dots$, which spans the entire part of the function to be integrated. Then,

$$g_i(x) = a_i(x - x_i) + b_i$$

but these are known values:

$$g_i(x_i) = y_i$$

$$g_i(x_{i+1}) = y_{i+1}$$

So,

$$b_i = y_i$$

$$a_i = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}$$

29.15.3 Cubic spline

Connect graph points using a cubic form:

$$g_i(x) = a_i(x - x_i)^3 + b_i(x - x_i)^2 + c_i(x - x_i) + d_i \quad (0)$$

over $(0, n - 1)$ and $x_i \leq x \leq x_{i+1}$. With n points, there are $4n$ unknowns. Constraints to help solve the coefficients. We know:

$$g_i(x_i) = y_i \quad (1)$$

$$g_i(x_{i+1}) = y_{i+1} \quad (2)$$

which are $2n$ constraints. Then want to smooth out transition points so that first and second derivatives are the same (since this is a cubic function, third derivatives won't mean much). First derivative:

$$g'_i(x_{i+1}) = g'_{i+1}(x_{i+1}) \quad (3)$$

which are $(n-1)$ constraints. Second derivative:

$$g''_i(x_{i+1}) = g''_{i+1}(x_{i+1}) \quad (4)$$

which is another $(n-1)$ constraints. So in total so far there have been $2n + 2(n - 1) = 4n - 2$ constraints.

Define $h_i = x_{i+1} - x_i$ and $\eta = y_{i+1} - y_i$ for notational simplicity. Then using (0) and (1) above,

$$d_i = y_i \quad (a)$$

From (0) and (2),

$$a_i h_i^3 + b_i h_i^2 + c_i h_i = \eta_i \quad (b)$$

From (3),

$$3a_i h_i^2 + 2b_i h_i + c_i = c_{i+1} \quad (c)$$

From (4),

$$6a_i h_i + 2b_i = 2b_{i+1} \quad (d)$$

Solve (c) for a_i in terms of b_i :

$$a_i = \frac{1}{3h_i}(b_{i+1} - b_i)$$

Solve (d) for c_i in terms of b_i :

$$c_i = \frac{1}{h_i}(\eta_i - a_i h_i^3 - b_i h_i^2) = \frac{\eta_i}{h_i} - \frac{1}{3}h_i(b_{i+1} - 2b_i)$$

Then (d) in terms of b_i becomes

$$\frac{1}{3}h_i b_i + \frac{2}{3}(h_i + h_{i+1})b_{i+1} + \frac{1}{3}h_{i+1}b_{i+2} = \frac{\eta_{i+1}}{h_{i+1}} - \frac{\eta_i}{h_i}$$

which is a matrix (with top and bottom rows missing) to solve for b_i . To complete the matrix, use the 'not-a-knot' condition:

$$g_0(x) = g_1(x)$$

which means the first and second approximations are the same function. Because the first and second derivatives are already known, this condition is equivalent to:

$$g_0'''(x_1) = g_1'''(x_1)$$

and

$$a_0 = a_1$$

which means:

$$\begin{aligned} a_0 &= \frac{1}{3h_0}(b_1 - b_0) = \frac{1}{3h_1}(b_2 - b_1) = a_1 \\ \frac{b_1 - b_0}{h_0} &= \frac{b_2 - b_1}{h_1} \\ h_1 b_0 - (h_0 + h_1)b_1 + h_0 b_2 &= 0 \end{aligned}$$

The not-a-knot condition will also lead to:

$$g_{n-2}(x) = g_{n-1}(x)$$

$$a_{n-2} = a_{n-1}$$

analogous to the above, this leads to:

$$h_{n-1}b_{n-2} - (h_{n-2} + h_{n-1})b_{n-1} + h_{n-2}b_n = 0$$

When the above conditions complete the matrix, solve for b_i and subsequently a_i, c_i . Instead of the not-a-knot condition, if $g_0'(x), g_n'(x)$ are known, can also solve. An example 4x4 matrix is:

$$\begin{pmatrix} h_1 & -(h_0 + h_1) & h_0 & 0 \\ \frac{1}{3}h_0 & \frac{2}{3}(h_0 + h_1) & \frac{1}{3}h_1 & 0 \\ 0 & \frac{1}{3}h_1 & \frac{2}{3}(h_1 + h_2) & \frac{1}{3}h_2 \\ 0 & h_2 & -(h_1 + h_2) & h_1 \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{\eta_1}{h_1} - \frac{\eta_0}{h_0} \\ \frac{\eta_1}{h_1} - \frac{\eta_0}{h_0} \\ 0 \end{pmatrix}$$

29.16 ODEs

29.17 First-order Euler method

Let a system be defined by:

$$\dot{x} = f(t, x)$$

$$x(t_0) = x_0$$

Use a first-order approximation:

$$x_1 = x_0 + \Delta t f(t_0, x_0)$$

$$x_2 = x_1 + \Delta t f(t_1, x_1)$$

...

$$x_{n+1} = x_n + \Delta t f(t_n, x_n)$$

29.18 Second-order Euler method

Again, $\dot{x} = f(t, x)$. To increase accuracy, try taking average of slopes:

$$x_{n+1} = x_n + \frac{\Delta t}{2}(f(t_n, x_n) + f(t_n + \Delta t, \underbrace{x_{n+\Delta t}}_{\text{unknown}}))$$

but this is circular due to the last term. Instead, use the predictor-corrector method:

$$\begin{aligned} x_{n+1}^{pred} &= x_n + \Delta t f(t_n, x_n) \\ x_{n+1} &= x_n + \frac{\Delta t}{2}[f(t_n, x_n) + f(t_n + \Delta t, x_{n+1}^p)] \end{aligned}$$

which is second-order Runge-Kutta (RK2). Define

$$\begin{aligned} K_1 &= \Delta t f(t_n, x_n) \\ K_2 &= \Delta t f(t_n + \Delta t, x_n + K_1) \\ x_{n+1} &= x_n + \frac{1}{2}(K_1 + K_2) \end{aligned}$$

which is the average of the slopes.

29.18.1 Runge-Kutta methods

Begin with

$$\dot{x} = f(t, x)$$

then,

$$\ddot{x} = f_t(t_n, x_n) + \dot{x}(t_n)f_x(t_n, x_n) = f_t(t_n, x_n) + f(t_n, x_n)f_x(t_n, x_n)$$

A second-order Taylor expansion is

$$x_{n+1} = x(t_n + \Delta t) = x(t_n) + \Delta t \dot{x}(t_n) + \frac{(\Delta t)^2}{2!} \ddot{x}(t_n) + O(\Delta t)^3$$

$$x_{n+1} = x_n(= x(t_n)) + \Delta t f(t_n, x_n) + \frac{1}{2}(\Delta t)^2(f_t(t_n, x_n) + f(t_n, x_n)f_x(t_n, x_n)) \quad (1)$$

Find a second Taylor expansion and equate. For the second Taylor expansion, define $k_1 = \Delta t f(t_n, x_n)$, $k_2 = \Delta t f(t_n + \alpha \Delta t, x_n + \beta k_1)$, and $x_{n+1} = x_n + a k_1 + b k_2$. The last is a type of weighted average of slopes. Then,

$$x_{n+1} = x_n + \underbrace{a \Delta t f(t_n, x_n)}_{k_1} + \underbrace{b \Delta t f(t_n + \alpha \Delta t, x_n + \beta k_1)}_{k_2} + O(\Delta t)^3 \quad (2a)$$

Taylor expand $f(t_n + \alpha \Delta t, x_n + \beta \Delta t f(t_n, x_n))$:

$$f(t_n + \alpha \Delta t, x_n + \beta \Delta t f(t_n, x_n)) = f(t_n, x_n) + \alpha \Delta t f_t(t_n, x_n)$$

$$+\beta\Delta t f(t_n, x_n) f_x(t_n, x_n) + O(\Delta t)^2 \quad (2b)$$

Substitute (2b) into (2a):

$$\begin{aligned} x_{n+1} &= x_n + a\Delta t f(t_n, x_n) + b\Delta t (f(t_n, x_n) + \alpha\Delta t f_t(t_n, x_n) \\ &\quad + \beta\Delta t f(t_n, x_n) f_x(t_n, x_n) + O(\Delta t)^2) + O(\Delta t)^3 \end{aligned} \quad (2d)$$

which reduces to

$$= x_n + (a+b)\Delta t f(t_n, x_n) + (\Delta t)^2(\alpha\beta f_t(t_n, x_n) + \beta b f(t_n, x_n) f_x(t_n, x_n) + O(\Delta t^3)) \quad (2e)$$

Comparing (1) and (2e),

$$a+b=1, \alpha b = \frac{1}{2}, \beta b = \frac{1}{2}$$

29.18.2 Family of Runge-Kutta methods

$$x_{n+1} = x_n + ak_1 + bk_2$$

Modified Euler:

$$a=b=\frac{1}{2} \rightarrow \alpha=\beta=1$$

$$K_1 = \Delta t f(t_n, x_n)$$

$$K_2 = \Delta t f(t_n + \Delta t, x_n + K_1)$$

$$x_{n+1} = x_n + \frac{1}{2}(K_1 + K_2)$$

Midpoint method: choose middle of interval

$$\alpha = \frac{1}{2} \rightarrow b=1, \beta = \frac{1}{2}, a=0$$

$$K_1 = \Delta t f(t_n, x_n)$$

$$K_2 = \Delta t f(t_n + \frac{1}{2}\Delta t, x_n + \frac{1}{2}K_1)$$

$$x_{n+1} = x_n + K_2$$

Second-order Runge-Kutta methods don't differ significantly in accuracy.

29.18.3 Runge-Kutta methods related to quadrature

If $\dot{y} = f(x)$ and $y(0)$ is the initial value, then by separation of variables,

$$dy = f(x)dx \rightarrow \int_{y(0)}^{y(h)} dy = \int_0^h f(x)dx = y(h) - y(0)$$

With the midpoint rule,

$$k_1 = hf(0), k_2 = hf(h/2), y(h) = y(0) + hf(h/2)$$

Then,

$$\int_0^h f(x)dx = hf(h/2)$$

As another example, modified Euler:

$$k_1 = hf(0), k_2 = hf(h), y(h) = y(0) + \frac{1}{2}h(f(0) + f(h))$$

Then,

$$\int_0^h f(x)dx = \frac{1}{2}h(f(0) + f(h))$$

29.18.4 Higher-order Runge-Kutta methods

Third-order RK:

$$\dot{x} = f(t, x)$$

Using Taylor expansions,

$$k_1 = \Delta t f(t_n, x_n)$$

$$k_2 = \Delta t f(t_n + \alpha \Delta t, x_n + \beta k_1)$$

$$k_3 = \Delta t f(t_n + \gamma \Delta t, x_n + \delta k_1 + \epsilon k_2)$$

$$x_{n+1} = x_n + ak_1 + bk_2 + ck_3$$

Order	2	3	4	5	6	7	8
Stages	2	3	4	6	7	9	11

So RK4 is appealing as the next one involves a lot more work. RK4 has two 'midpoints' in the interval $(t_n, t_n + \Delta t)$:

$$k_1 = \Delta t f(t_n, x_n)$$

$$k_2 = \Delta t f(t_n + (1/2)\Delta t, x_n + (1/2)k_1)$$

$$k_3 = \Delta t f(t_n + (1/2)\Delta t, x_n + (1/2)k_2)$$

$$k_4 = \Delta t f(t_n + \Delta t, x_n + k_3)$$

$$x_{n+1} = x_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

29.19 Higher-order ODEs and systems

$$\ddot{x} = f(t, x, \dot{x})$$

Let

$$u = \dot{x}$$

$$\dot{x} = u$$

$$\dot{u} = \ddot{x} = f(t, x, u)$$

If there is a system, then couple the integrals:

$$\dot{x} = f(t, x, y)$$

$$\dot{y} = g(t, x, y)$$

Modified Euler method:

$$k_1 = \Delta t f(t_n, x_n, y_n)$$

$$l_1 = \Delta t g(t_n, x_n, y_n)$$

$$k_2 = \Delta t f(t_n + \Delta t, x_n + k_1, y_n + l_1)$$

$$l_2 = \Delta t g(t_n + \Delta t, x_n + k_1, y_n + l_1)$$

$$x_{n+1} = x_n + \frac{1}{2}(k_1 + k_2)$$

$$y_{n+1} = y_n + \frac{1}{2}(l_1 + l_2)$$

29.20 Adaptive Runge-Kutta

MATLAB uses Dormand-Prince (ode45.m), which is RK6. Below RK6 there are RK4 and RK5. RK4 is $X_{n+1} + O(\Delta t)^5$ and RK5 is $x_{n+1} + O(\Delta t)^6$. The actual error is then

$$e = |x_{n+1} - X_{n+1}| \approx O(\Delta t)^5$$

where ϵ is the allowed error or tolerance is ϵ .

$$\frac{e}{\epsilon} = \frac{(\delta t)^5}{(\Delta \tau)^5}$$

$$\Delta \tau = (s) \Delta t \left(\frac{\epsilon}{e} \right)^{1/5}$$

where $\Delta \tau$ is the timestep that should be used and Δt is the timestep that was actually used. The s is the safety factor, usually 0.9. Want $e < \epsilon$, otherwise reject data.

29.21 Lotka-Volterra equations

Predator-prey model

$$\begin{aligned}\dot{x} &= rx(1-x) \\ \dot{y} &= \frac{1}{r}y(x-1)\end{aligned}$$

where x is prey and y is predator.

29.22 Boundary value problems: shooting method

$$\frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

Initial values are $y(x_0) = y_0, y(x_f) = y_f$. This makes it a 2-point boundary problem. Convert the above into a system of two ODEs:

$$\frac{dy}{dx} = z \quad (1)$$

$$\frac{dz}{dx} = f(x, y, z) \quad (2)$$

where z is a new variable. Initial value $y(x_0) = y_0$ is still known. Use $y(x_f) = y_f$ to obtain $y'(x_0) = z(x_0) = \eta$, the second initial value. Define $F = F(z(x_0)) = F(\eta) = y(x_f) - y_f$ which is a root finding problem now for $F(\eta) = 0$. This works because η is now part of each side of the complicated equation $y(x_f) = y_f$. Guess some initial value of η , calculate $F(\eta)$. If $F > 0$ then reduce and vice versa.

29.23 Partial differential equations

29.23.1 Boundary and initial value problems

Boundary problem: boundaries are given and solve for all interior points. Usually for steady state or equilibrium.

Interior value problem: boundaries given and initial conditions ($t=0$) are also given. Solution evolves over time.

29.23.2 Central difference approximation

1D solution, $y = y(x)$.

$$y(x+h) = y(x) + hy'(x) + \frac{h^2}{2}h''(x) + \frac{h^3}{6}y'''(x) + \frac{h^4}{24}y^{(4)}(x) + \dots \quad (1)$$

$$y(x-h) = y(x) - hy'(x) + \frac{h^2}{2}h''(x) - \frac{h^3}{6}y'''(x) + \frac{h^4}{24}y^{(4)}(x) + \dots \quad (2)$$

Subtract: (1) minus (2)

$$y(x+h) - y(x-h) = 2hy'(x) + \frac{h^3}{3}y'''(x) + \dots$$

$$y'(x) = \frac{y(x+h) - y(x-h)}{2h} + O(h^2)$$

Add: (1) plus (2)

$$y(x+h) + y(x-h) = 2y(x) + h^2 y''(x) + \frac{h^4}{12} y^{(4)}(x) + \dots$$

$$y''(x) = \frac{y(x+h) - 2y(x) + y(x-h)}{h^2} + O(h^2)$$

29.23.3 Boundary problem: Direct method and discrete Laplace equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \Phi = 0$$

$$x_i = (i-1)\Delta x, y_j = (j-1)\Delta y$$

where $i=[1, n_x]$ and $j=[1, n_y]$. Note that $\Phi(x_i, y_j) = \Phi_{ij}$. Apply central difference approximation:

$$\frac{\partial^2 \Phi}{\partial x^2} = \frac{\Phi_{i+1,j} - 2\Phi_{ij} + \Phi_{i-1,j}}{(\Delta x)^2}$$

$$\frac{\partial^2 \Phi}{\partial y^2} = \frac{\Phi_{i,j+1} - 2\Phi_{ij} + \Phi_{i,j-1}}{(\Delta y)^2}$$

For simplicity, assume $\Delta x = \Delta y$. Discrete Laplace equation is then

$$4\Phi_{ij} - \Phi_{i+1,j} - \Phi_{i-1,j} - \Phi_{i,j+1} - \Phi_{i,j-1} = 0$$

Natural ordering. Convert 2D grid to 1D.

$$(i, j) \rightarrow k = i + (j-1)n_x$$

$$\Phi = (\Phi_{11}, \Phi_{21}, \dots, \Phi_{n_x,1}, \Phi_{12}, \Phi_{22}, \dots, \Phi_{n_x,2}, \Phi_{31}, \dots, \Phi_{n_x, n_y})'$$

Then, the discrete Laplace equation

$$4\Phi_{ij} - \Phi_{i+1,j} - \Phi_{i-1,j} - \Phi_{i,j+1} - \Phi_{i,j-1} = 0$$

becomes

$$4\Phi_k - \Phi_{k+1} - \Phi_{k-1} - \Phi_{k+n_x} - \Phi_{k-n_x} = 0$$

The boundary conditions then become

$$\text{Bottom: } j=1 \text{ Left: } i=1, \text{ Top: } j=n_y, \text{ Right: } i=n_x$$

Formulate into matrix $A\Phi = b$. Use the grid example of

$$\begin{vmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix}$$

so that the interior point (subject to the discrete Laplace equation) is 5, while all others are boundary conditions. Then, $k = 5, n_x = n_y = 3$, and

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \\ \Phi_4 \\ \Phi_5 \\ \Phi_6 \\ \Phi_7 \\ \Phi_8 \\ \Phi_9 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ 0 \\ b_6 \\ b_7 \\ b_8 \\ b_9 \end{pmatrix}$$

29.23.4 Boundary problems: iterative methods

Derive iterative methods by re-expressing Laplace equation:

$$4\Phi_{ij} - \Phi_{i+1,j} - \Phi_{i-1,j} - \Phi_{i,j+1} - \Phi_{i,j-1} = 0$$

Jacobi. Most common now.

$$\Phi_{ij}^{(n+1)} = \frac{1}{4}(\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n)})$$

where the superscript is the iteration number. An older version of Jacobi is Gauss-Seidel, which is different due to FORTRAN-style looping:

$$\Phi_{ij}^{(n+1)} = \frac{1}{4}(\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n+1)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n+1)})$$

Gauss-Seidel isn't used a lot anymore. SOR (successive over-relaxation) is also not as common as Jacobi, but can choose λ to determine convergence speed $\lambda > 1$ is over-relaxation and is faster convergence while $\lambda < 1$ is under-relaxation and is slower but possibly more accurate:

$$\Phi_{ij}^{(n+1)} = \Phi_{ij}^{(n)} + \frac{\lambda}{4}(\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n)} - 4\Phi_{ij}^{(n)})$$

where the final term is to compensate for the addition of the bracketed term (which is itself Jacobi). Re-express SOR like this:

$$\Phi_{ij}^{(n+1)} = (1 - \lambda)\Phi_{ij}^{(n)} + \frac{\lambda}{4}(\Phi_{i+1,j}^{(n)} + \Phi_{i-1,j}^{(n)} + \Phi_{i,j+1}^{(n)} + \Phi_{i,j-1}^{(n)})$$

29.24 Time-stepping methods for initial value problems

29.24.1 Explicit methods for solving the diffusion equation

Explicit because $t+1$ values only depend on t values.

$$u = u(x, t)$$

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$$

where D is the diffusion constant. Boundary conditions:

$$u(-L_x, t) = u(L_x, t) = 0$$

Initial value conditions:

$$u(x, 0) = u_0(x)$$

Notation is $u(x_j, t_l) = u_j^l$.

Forward-time centered space (FTCS).

$$\frac{u_j^{l+1} - u_j^l}{\Delta t} = D \frac{u_{j-1}^l - 2u_j^l + u_{j+1}^l}{\Delta x^2}$$

where the left side is the Euler method and the right side is the finite difference approximation. Solving for u_j^{l+1} ,

$$u_j^{l+1} = u_j^l + \frac{D\Delta t}{\Delta x^2}(u_{j-1}^l - 2u_j^l + u_{j+1}^l)$$

29.24.2 Von Neumon stability analysis

FTCS:

$$u_j^{l+1} = u_j^l + \frac{D\Delta t}{\Delta x^2}(u_{j-1}^l - 2u_j^l + u_{j+1}^l)$$

if Δt is too large, unstable because it diverges. To show this, ansatz $u_j^l = \xi^l e^{ikj\Delta x}$ to determine stability. If $|\xi| > 1$ for any k, then unstable. Substitute:

$$\xi^{l+1} e^{ikj\Delta x} = \xi^l e^{ikj\Delta x} + \frac{D\Delta t}{\Delta x^2} (\xi^l e^{ik(j+1)\Delta x} - 2\xi^l e^{ikj\Delta x} + \xi^l e^{ik(j-1)\Delta x})$$

Divide by $\xi^l e^{ikj\Delta x}$

$$\begin{aligned} \xi &= 1 + \frac{D\Delta t}{\Delta x^2} (e^{ik\Delta x} - 2 + e^{-ik\Delta x}) \\ &= 1 + \frac{2D\Delta t}{\Delta x^2} (\cos(k\Delta x) - 1) \end{aligned}$$

When $k = 1 \rightarrow \xi = 1$ and $k = -1 \rightarrow \xi < 1$. Note $\Delta x = 1$. Most dangerous situation is when $\cos(k\Delta x) = -1$ as ξ may fall below -1 and therefore become unstable. When does $\xi < -1$?

$$\begin{aligned} 1 - \frac{4\Delta t D}{\Delta x^2} &< -1 \\ \Delta t &> \frac{\Delta x^2}{2D} \end{aligned}$$

So if the time step Δt is too large, which depends on Δx and D , then it can become unstable.

29.24.3 Implicit methods for solving the diffusion equation

In the explicit method, the 2nd derivative was taken at the previous time step, t . In the implicit method, take 2nd derivative at current time step, $t+1$.

$$u_j^{l+1} = u_j^l + \frac{\Delta t D}{(\Delta x)^2} (u_{j+1}^{l+1} - 2u_j^{l+1} + u_{j-1}^{l+1})$$

Ansatz $u_j^l = \xi^l e^{ikj\Delta x}$ again and eventually get

$$\xi = 1 + \frac{2\Delta t D}{(\Delta x)^2} \xi (\cos(k\Delta x) - 1)$$

where the ξ on the right-hand side is the difference between the explicit and implicit methods. Solving,

$$\xi = \frac{1}{1 + \frac{2\Delta t D}{(\Delta x)^2} (1 - \cos(k\Delta x))}$$

As $0 < \xi \leq 1$, unconditionally stable.

29.24.4 Crank-Nicolson method

Average of explicit and implicit methods, like $l + \frac{1}{2}$, so both time and space-stepping become second order. Advantage: unconditionally stable. Disadvantage: must solve matrix equation each iteration.

Apply to diffusion equation.

$$u_j^{l+1} = u_j^l + \frac{\Delta t D}{2(\Delta x)^2} [(u_{j+1}^l - 2u_j^l + u_{j-1}^l) + (u_{j+1}^{l+1} - 2u_j^{l+1} + u_{j-1}^{l+1})]$$

which is still implicit because u^{l+1} is still on the right-hand side. Let $a = \frac{\Delta t D}{(\Delta x)^2}$ and multiply both sides by 2:

$$-au_{j+1}^{l+1} + 2(1-a)u_j^{l+1} - au_{j-1}^{l+1} = au_{j+1}^l + 2(1-a)u_j^l + au_{j-1}^l$$

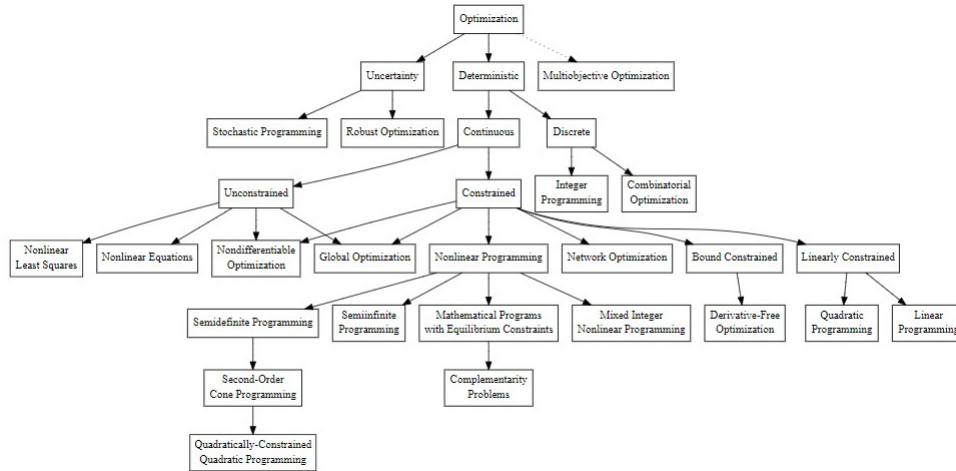
Matrix example with 4 iterations:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ -a & 2(1+a) & -a & 0 \\ 0 & -a & 2(1+a) & -a \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1^{l+1} \\ u_2^{l+1} \\ u_3^{l+1} \\ u_4^{l+1} \end{pmatrix} = \begin{pmatrix} 0 \\ au_1^l + 2(1-a)u_2^l + au_3^l \\ au_2^l + 2(1-a)u_3^l + au_4^l \\ 0 \end{pmatrix}$$

30 Optimization

30.1 Introduction

30.1.1 Optimization problem classification



30.1.2 Math background

Directional derivative of f in direction of $\vec{p}$ where $\|\vec{p}\| = 1$:

$$\nabla_p f(x) = D(f, \vec{p}) = \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon \vec{p}) - f(x)}{\epsilon}$$

Gradient of f at x :

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \Big|_x \\ \vdots \\ \frac{\partial f}{\partial x_n} \Big|_x \end{pmatrix}$$

If the system is dynamic, use chain rule

$$\frac{\partial f}{\partial x_i} \Big|_{x(t)} = \nabla f(x)^T \frac{dx}{dt} \Big|_t = \sum_{t=1}^n \frac{\partial f}{\partial x_i} \Big|_{x(t)} \frac{\partial x_i}{\partial t} \Big|_t$$

The directional derivative is related to the gradient:

$$\nabla_p f(x) = \nabla f(x)^T p$$

The Hessian

$$H(x) = \nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} \Big|_x & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \Big|_x \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} \Big|_x & \cdots & \frac{\partial^2 f}{\partial x_n^2} \Big|_x \end{pmatrix}$$

1D Taylor expansion:

$$f(x) = f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2 + O(x^3)$$

where $f'(a) = 0, f''(a) > 0$ is 'positive definite' (concave up), $f'(a) = 0, f''(a) < 0$ is 'negative definite' (concave down), etc.

2D Taylor expansion:

$$f(x) = f(a) + \nabla f(a)(x - a) + \frac{1}{2}(x - a)^T \nabla^2 f(a)(x - a) + O(x^3)$$

Expanding the Hessian in the 2D Taylor expansion,

$$(x, y) \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{\partial^2 f}{\partial x^2} x^2 + 2 \frac{\partial^2 f}{\partial x \partial y} xy + \frac{\partial^2 f}{\partial y^2} y^2$$

which is the second derivative in all directions in the xy-plane multiplied by the distance moved in the xy-plane. So the Hessian indicates concavity as with the 1D case. Since Hessians are symmetric, their eigenvalues are always real and exist. By the spectral theorem, a symmetric square matrix can be factored with a diagonal matrix in the basis of the Q matrices as

$$A = Q^T \Lambda Q = Q^T \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} Q$$

If the eigenvalues are positive, any independent variable x values will lead to positive second derivative values, so this is concave up. If eigenvalues are negative then concave down, if mixed then saddle point, if 0 then must investigate further (e.g. graphically).

Multivariate Taylor expansion with vectors around x wrt $\vec{p}$. Since

$$\begin{aligned} \nabla_p f(x) &= D(f, \vec{p}) = \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon \vec{p}) - f(x)}{\epsilon} \\ f(x + \epsilon p) &= f(x) + \epsilon \nabla_p f(x) + O(\epsilon^2) \\ &= f(x) + \epsilon \nabla f(x)^T \vec{p} + O(\epsilon^2) \end{aligned}$$

Specify 2nd order Taylor expansion terms, noting the quadratic form $p^T \nabla^2 f(x) p$:

$$= f(x) + \epsilon \nabla f(x)^T \vec{p} + \frac{1}{2} \epsilon^2 \vec{p}^T \nabla^2 f(x) \vec{p} + O(\epsilon^3)$$

Derivative of quadratic form. Two proofs.

$$f(x) = x^T A x \rightarrow f'(x) = 2Ax$$

Proof 1. $f(x)$ is a scalar, $g(x)^T$ is a row vector, and $h(x)$ is a column vector. Decompose:

$$f(x) = g(x)^T h(x)$$

$$\nabla f(x)^T = f'(x) = (g(x)^T)' h(x) + g(x)^T h'(x)$$

Re-express

$$= h(x)^T g'(x) + g(x)^T h'(x)$$

$$\nabla f(x) = \nabla g(x) h(x) + \nabla h(x) g(x)$$

Let $g(x) = x, h(x) = Ax$. Then, $f'(x) = 2Ax$. Proof 2. Let $f(x) = x^T Ax$. Then

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^T A(x+h) - x^T Ax}{h} = \lim_{h \rightarrow 0} \frac{x^T A(x+h) + h^T A(x+h) - x^T Ax}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^T Ax + x^T Ah + h^T Ax + h^T Ah - x^T Ax}{h}$$

$$hf'(x) = x^T Ah + h^T Ax + h^T Ah$$

Drop last term which is a higher order term and divide both sides by h: $f'(x) = x^T A + Ax = 2Ax$.

Matrix norms. Vector norms are in L1, L2, etc.

$$\|u\|_1 = |u_x| + |u_y|$$

$$\|u\|_2 = \sqrt{u_x^2 + u_y^2}$$

By analogy, construct matrix norms. These are the largest value when fed a unit vector u. For L_p :

$$\|A\|_p = \|u\|_p = \max \|Au\|_p$$

Application.

$$\|A^k\|_p = \|u\|_p = \max \|A^k u\|_p$$

Let λ_i be eigenvalue(s) of A. Using the triangle equality

$$\|AU\| = |\lambda_i| \|U_i\| \leq \|A\| \|U_i\|$$

$$|\lambda_i| \leq \|A\|$$

$$\max |\lambda_i|^k \leq \|A^k\|$$

So the largest eigenvalue provides a lower bound on the powers of A.

Polynomial interpolation. Example. Four ordered pairs as data for $f(x) = a+bx+cx^2+dx^3$. This can be rewritten as $Ax = b$ where the basis is $(1, x, x^2, x^3)$. This matrix of powers is more specifically, a Vandermonde matrix as it has

unique properties. The 'Runge phenomenon' where a polynomial interpolation leads to bumps where a function should be perfectly horizontal, can be addressed by adjusting the polynomial to $f(x) = a + bx + cx^2 + d\frac{1}{x}$. Another, unrelated polynomial interpolation is to use Lagrange (or atomic) polynomials where the basis will be (p_1, p_2, p_3, p_4) :

$$p_1(x) = \frac{(x-2)(x-3)(x-4)}{(1-2)(1-3)(1-4)} = 1$$

$$p_2(x) = \frac{(x-1)(x-3)(x-4)}{(2-1)(2-3)(2-4)} = 1$$

$$p_3(x) = \frac{(x-1)(x-2)(x-4)}{(3-1)(3-2)(3-4)} = 1$$

$$p_4(x) = \frac{(x-1)(x-2)(x-3)}{(4-1)(4-2)(4-3)} = 1$$

so $p_i = 1$ but 0 elsewhere. So these polynomials are 'atomic'. To construct $f(x) = a + bx + cx^2 + dx^3$, take $ap_1 + bp_2 + cp_3 + dp_4$.

30.2 Unconstrained optimization

30.2.1 Smooth unconstrained optimality

First-order necessary conditions (but not sufficient as it can be maximum or saddle point) for x' as minimizer:

$$\nabla f(x') = 0$$

Proof. Since x' is a minimizer,

$$f(x' + \epsilon \vec{p}) \geq f(x')$$

Further,

$$\nabla_p f(x') = (x')^T p \geq 0 \quad (1)$$

If choose $p = -\nabla f(x')$,

$$\nabla f(x')^T p = -(x')^T \nabla f(x') = -\|\nabla f(x')\|^2 \leq 0 \quad (2)$$

By (1) and (2), $\nabla f(x') = 0$.

Second-order necessary conditions

$$\nabla f(x') = 0$$

$\nabla^2 f(x')$ is positive semi-definite

where f is twice continuously differentiable, then x' is a local minimizer. Note that even if $H(x) = 0$, such as for $y = x^3$, it is not a minimum. Proof by contradiction, so it is negative definite:

$$p^T \nabla^2 f(x') p < 0$$

So Taylor expansion at x' :

$$f(x' + \epsilon p) = \underbrace{f(x') + \epsilon \nabla f(x')^T p}_{f(x')=0} + \underbrace{\frac{1}{2} \epsilon^2 p^T \nabla^2 f(x') p}_{\text{dominates}} + O(\epsilon^3)$$

Since $p^T \nabla^2 f(x') p < 0$, $f(x' + \epsilon p) < f(x')$ so x' is not a minimizer due to the contradiction. So $\nabla^2 f(x')$ must be positive semi-definite.

Second-order sufficient conditions (but not necessary, e.g. $y = x^4$)

$$\begin{aligned} \nabla f(x') &= 0 \\ \nabla^2 f(x') &\text{ is positive definite} \end{aligned}$$

where f is twice continuously differentiable, then x' is a STRICT local minimizer.

If second-order necessary conditions are satisfied but second-order sufficient conditions are NOT satisfied, must look at the function from various cross-sections.

30.2.2 Convexity

Convex optimization is not difficult, but non-convex optimization is difficult. Note: if given objective functions are non-convex but the methods for unconstrained optimization can only find a local optimum, then if we change the initial point, it is possible that the solver will identify a different optimum than before.

Convex optimization if the objective function and feasible set are convex. If the domain is convex, the inequalities are convex, and the equalities are linear, then the feasible set is convex. If any inequality is non-convex and/or any equality is non-linear, then feasible set is non-convex.

Sets. Convex set if $x_1, x_2 \in \Omega$ with $\alpha \in [0,1]$ with $\alpha x_1 + (1 - \alpha)x_2 \in \Omega$.

Functions. Convex function if $f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2)$ (as in the interior of the parabola). Can be strictly convex. If f is convex, $-f$ is concave. Affine linear functions are convex and concave but not strictly so. Can have functions like $y = x^3$ which are neither convex nor concave.

Theorem. If f is convex, then Hessian is PSD. If Hessian is PD, f is strictly convex.

30.2.3 Direct v indirect methods

Note: nomenclature of direct and indirect are inconsistent in optimization field.

Indirect methods. Set up optimality equations and constraints and analytically or numerically solve.

Direct methods. Use iterative descent (line search, trust region, etc.) to find a convergent sequence of the objective function. For minimization, $f^{(k+1)}(x) < f^{(k)}(x)$ and $\lim_{k \rightarrow \infty} x^{(k)} = x'$.

Iterative descent convergence rate. $\exists c \in (0, 1)$ s.t.

$$\|x^{(k+1)} - x'\| \leq c \|x^{(k)} - x'\|^p$$

where p is the convergence order and $p = 1$ is linear, $p > 1$ means convergence accelerates. If c is a sequence that converges to 0, $\lim_{k \rightarrow \infty} c_k = 0$.

30.2.4 Direct method: line search

Line search algorithm for minimization. (1) Choose descent direction $\vec{p}$ at $x^{(k)}$ s.t. $\nabla f(x^{(k)})^T \vec{p}^{(k)} < 0$ which will give a negative slope for the minimization. (2) Determine step length α_k . (3) Set $x^{(k+1)} = x^{(k)} + \alpha_k \vec{p}^{(k)}$.

Line search: step length

Want to minimize $\Phi(\alpha) = f(x^{(k)} + \alpha \vec{p}^{(k)})$ efficiently. Finding global or even local step lengths α are expensive. Armijo's condition states that $\exists \alpha > 0$ s.t. $\Phi(\alpha) \leq \Phi(0) + \alpha c_1 \Phi'(0)$ where $c_1 \in (0, 1)$. This means $\Phi(\alpha) < \Phi(0)$ which leads to descent. If $c_1 \approx 1$ then step length α is small, leading to slow convergence. If $c_1 \approx 0$ then $\Phi(\alpha)$ and therefore f decrease slowly, so need lots of steps (which also takes a lot of time). How to optimize?

Progressively-shrinking α algorithm. (1) Choose $\alpha_1 > 0$ and $c_1, \rho \in (0, 1)$. (2) Set $\alpha = \alpha_1$. (3) $\alpha \leftarrow \rho \alpha$ until $\Phi(\alpha) \leq \Phi(0) + \alpha c_1 \Phi'(0)$.

A better algorithm. (1) Choose α_0 and $c_1 \in (0, 1)$. If $\Phi(\alpha_0) \leq \Phi(0) + \alpha_0 c_1 \Phi'(0)$ satisfied, stop. Else, find better $\alpha \in (0, \alpha_0)$ through quadratic interpolation:

$$f(x) = ax^2 + bx + c$$

$$\Phi(0) = c$$

$$\Phi'(\alpha) = 2a\alpha + b \rightarrow \Phi'(0) = b$$

$$\Phi(\alpha_0) = a\alpha_0^2 + \Phi'(0)\alpha_0 + \Phi(0) \rightarrow a = \frac{\Phi(\alpha_0) - \Phi'(0)\alpha_0 - \Phi(0)}{\alpha_0^2}$$

$$\Phi'(\alpha_1) = 0 = 2\alpha_1 \cdot a + \Phi'(0) \rightarrow \alpha_1 = \frac{-\Phi'(0)\alpha_0}{2(\Phi(\alpha_0) - \Phi'(0)\alpha_0 - \Phi(0))}$$

If $\Phi(\alpha_0) \leq \Phi(0) + \alpha_1 c_1 \Phi'(0)$, stop. Else, find α_2 through cubic interpolation and repeat cubic interpolation until $\Phi(\alpha_0) \leq \Phi(0) + \alpha_n c_1 \Phi'(0)$.

Wolfe: an even better algorithm. (1) Choose $c_1 \in (0, 1)$ and $c_2 \in (c_1, 1)$. Wolfe states $\exists \alpha > 0$ s.t. $\Phi(\alpha) \leq \Phi(0) + \alpha c_1 \Phi'(0)$ and the slope condition $\Phi'(\alpha) \geq c_2 \Phi'(0)$. This effectively reduces the feasible range for the minimum step length to a 'reasonable' range.

Line search: direction

Steepest descent. Start with Taylor expansion about $x^{(k)}$ so differentiate wrt $\vec{p}$:

$$f(x^{(k)} + \alpha \vec{p}^{(k)}) = f(x^{(k)}) + \nabla f(x^{(k)})^T \vec{p}^{(k)} + O(\alpha^2)$$

To first-degree (linear) approximation, steepest descent direction is where $f(x^{(k)})$ is lowest.

$$\nabla f(x^{(k)})^T \vec{p}^{(k)} = \|\nabla f(x^{(k)})\| \|\vec{p}\| \cos \theta$$

Since $\|\vec{p}\| = 1$, to minimize, set $\cos \theta = -1 \rightarrow \theta = \pi$. So,

$$\begin{aligned} \vec{p} &= -\frac{\|\nabla f(x^{(k)})\|}{\|\nabla f(x^{(k)})\|} = -\frac{\|\nabla f(x^{(k)})\|}{\|\nabla f(x^{(k)})\|} \times \frac{\nabla f(x^{(k)})}{\|\nabla f(x^{(k)})\|} \\ &= -\frac{\|\nabla f(x^{(k)})\| \times \nabla f(x^{(k)})}{\|\nabla f(x^{(k)})\|^2} = -\frac{\nabla f(x^{(k)})}{\|\nabla f(x^{(k)})\|} \end{aligned}$$

Steepest descent algorithm. Choose x_0 . (1) $\vec{p} = -\nabla f(x^{(k)})$, (2) determine step length (e.g. Armijo rule), (3) $x^{(k+1)} = x^{(k)} + \alpha \vec{p}$, (4) while $\|\nabla f(x^{(k)})\| > \epsilon$, repeat.

Newton's descent.

$$f(x^{(k+1)}) = f(x^{(k)}) + \nabla f(x^{(k)})^T (x^{(k+1)} - x^{(k)}) + \dots$$

Differentiate

$$\begin{aligned} 0 &= \nabla f(x^{(k)}) + \nabla^2 f(x^{(k)}) (x^{(k+1)} - x^{(k)}) \\ x^{(k+1)} - x^{(k)} &= -\frac{\nabla f(x^{(k)})}{\nabla^2 f(x^{(k)})} = \vec{p} \end{aligned}$$

Use $\alpha_k = 1$. Newton descent algorithm. Choose x_0 . (1) $\vec{p} = -\frac{\nabla f(x^{(k)})}{\nabla^2 f(x^{(k)})}$, (2) $x^{(k+1)} = x^{(k)} + \vec{p}$ (since $\alpha = 1$), (3) while $\|\nabla f(x^{(k)})\| > \epsilon$, repeat. Advantage is that it is local quadratic approximation, disadvantage is that second derivative calculation and matrix inversion are time-consuming. Can combine with line search.

Line search: complexity

Analytical complexity (number of operations to get answer) for ϵ , which is the norm of the gradient (distance to convergence) for steepest descent is $O\left(\frac{1}{\epsilon^2}\right)$. For Newton's method, $O\left(\frac{1}{\epsilon^{2-\tau}}\right)$ where $0 < \tau < 1$.

Variants of Newton's method: Inexact method

Let f be the function, g the gradient, and H the Hessian. From Newton's method:

$$\begin{aligned}x^{(k+1)} &= x^{(k)} + \vec{p}^{(k)} \\ H^{(k)} \vec{p}^{(k)} &= -g^{(k)}\end{aligned}$$

Assuming the above second condition is positive definite, it is solved numerically by the conjugate gradient method.

Algorithm. Choose x_0 . (1) Calculate g and H , (2) solve the system of linear equations $H^{(k)} \vec{p}^{(k)} = -g^{(k)}$ by conjugate gradients, (3) $x^{(k+1)} = x^{(k)} + \alpha_k \vec{p}^{(k)}$ (though can set $\alpha = 1$), (4) while $\nabla f(x^{(k)}) > \epsilon$, repeat.

Variants of Newton's method: Modified method

If H is singular or not positive definite. Same as inexact method but replace H with an approximation $B^{(k)} = H^{(k)} + E^{(k)}$, where $E^{(k)} = \tau_R I, \tau_R \geq 0$. Choose τ_R by judgment and trial-and-error. I is the identity (a constant).

Variants of Newton's method: Quasi-Newton method

Instead of using the constant E , calculate B from last iteration information with a quadratic approximation of f at $x^{(k)}$:

$$m^{(k)}(\vec{p}) = f^{(k)} + g^{(k)T} p + \frac{1}{2} p^T B^{(k)} p$$

B must be positive symmetric definite. Algorithm. (1) B must also be s.t. it is consistent with the first-order optimality condition $p^{(k)} = -B^{(k)-1} g^{(k)}$, (2) then minimize m , (3) set $x^{(k+1)} = x^{(k)} + \alpha_k \vec{p}^{(k)}$, (4) repeat to get:

$$m^{(k+1)}(\vec{p}) = f^{(k+1)} + g^{(k+1)T} p + \frac{1}{2} p^T B^{(k+1)} p$$

where $\nabla m^{(k+1)} = \nabla f(x^{(k+1)}) = \nabla f(x^{(k+1)})$ is desired. Differentiating the above,

$$\nabla m^{(k+1)}(\vec{p}) = g^{(k+1)} + B^{(k+1)} \vec{p}$$

where at $x^{(k+1)}, p = 0$, would like B s.t. $\nabla m^{(k+1)}(0) = g^{(k+1)}$. To get there, at $x = x^{(k)}, \vec{p} = -\alpha_k \vec{p}^{(k)}$, want $\nabla m^{(k+1)}(-\alpha_k \vec{p}^{(k)}) = g^{(k)}$, so solve:

$$\begin{aligned}\nabla m^{(k+1)}(\vec{p}) &= g^{(k+1)} + B^{(k+1)} \vec{p} \\ g^{(k)} &= g^{(k+1)} - \alpha_k \vec{p}^{(k+1)} B^{(k+1)} \\ g^{(k+1)} - g^{(k)} &= B^{(k+1)} \alpha_k \vec{p}^{(k)}\end{aligned}$$

where $\alpha_k \vec{p}^{(k)} = x^{(k+1)} - x^{(k)}$. Re-express the above as

$$B^{(k+1)} s^{(k)} = y^{(k)}$$

Since B is symmetric positive definite,

$$s^{(k)T} B^{(k+1)} s^{(k)} = s^{(k)T} y^{(k)} > 0$$

However, this will give multiple solutions. To get a unique solution, make $B^{(k+1)}$ as close to $B^{(k)}$ as possible:

$$\min_B \|B - B^{(k)}\|_w$$

which is the weighted Frobenius norm. Solve above s.t. $B^T = B$ for symmetry and $B s^{(k)} = y^{(k)}$. There is more than one way to solve this, and the solutions are complicated iterative formulas of the form $B^{(k+1)} = f(B^{(k)}, y^{(k)}, s^{(k)})$.

Line search direction: parameter estimation

Example. Finding parameters of regression. If the concentration in a batch reactor is represented by DEs and the theoretical solution is $c_A(t) = c_A e^{-kt}$ which will differ from measured:

$$\begin{aligned} c_{A,theoretical}(t_j) &= \Phi(\vec{x}, t_j) = x_1 e^{-x_j t_j} \\ c_{A,measured}(t) &= y_j \end{aligned}$$

Define the residual error to be

$$\epsilon_j = y_j - \Phi(\vec{x}, t_j)$$

With (non-linear) least-squares,

$$\min_x \frac{1}{2} \|\epsilon(x)\|_2^2 = \frac{1}{2} \sum_j (y_j - \Phi(\vec{x}, t_j))^2$$

Use Gauss-Newton method to solve

$$\min_x f(x) = \min_x \frac{1}{2} \|\epsilon(x)\|_2^2 = \min_x \frac{1}{2} \epsilon(x)^T \epsilon(x)$$

Take the Jacobian of $\epsilon(x)$, resulting in a non-square matrix

$$J(x) = \nabla \epsilon(x)$$

Take the derivatives of $f(x)$ using $f(x) = x^T A x \rightarrow f'(x) = 2Ax$:

$$\begin{aligned} \nabla f(x) &= J(x)^T \epsilon(x) \\ \nabla^2 f(x) &= J(x)^T J(x) + \underbrace{\sum_{j=1}^m \epsilon_j(x) \nabla^2 \epsilon_j(x)}_{=0} \end{aligned}$$

where the second term is zero in the Gauss-Newton method because (i) $\nabla^2 \epsilon(x) = 0$ in almost-linear cases, and/or (ii) set $\epsilon(x) \approx 0$ (small values) as starting values when initializing with a good guess for x . Now use Newton's direction:

$$\nabla^2 f(x^{(k)}) p_k = -\nabla f(x^{(k)})$$

and re-express as (using the above):

$$J^{(k)T} J^{(k)} p^{(k)} = -J^{(k)T} \epsilon^{(k)}$$

which is a system of equations.

30.2.5 Direct methods: trust region methods

Approximate $f(x^{(k)})$ by a quadratic model:

$$m^{(k)}(p) = f^{(k)} + g^{(k)T} p + \frac{1}{2} p^T B^{(k)} p$$

where g is the gradient and B is symmetric. If p is selected as a very small step, the approximation error is small (this is a Taylor series - so taking a small step away from the pivot means the approximation error is small). So find the minimum within a 'trust region radius' $\Delta^{(k)}$. Solve the minimization problem

$$\min_p m^{(k)}(p)$$

within $\|p\| \leq \Delta^{(k)}$ and set $p^{(k)}$ as the solution, calculate $x^{(k+1)} = x^{(k)} + p^{(k)}$ and repeat.

Updating $\Delta^{(k)}$. Compare agreement between model m and function f at previous iterations using the contraction rate:

$$\rho_k = \frac{f(x^{(k)}) - f(x^{(k)} + p^{(k)})}{m^{(k)}(0) - m^{(k)}(p^{(k)})} = \frac{\text{actual reduction}}{\text{predicted reduction}}$$

The denominator is positive because this is a minimization. If ρ is near 1 then $\Delta^{(k)}$ should be increased due to good reductions. If ρ is near 0 then $\Delta^{(k)}$ should be decreased due to poor reductions. If $\rho < 0$ then it is ascending so reject the step. Repeat. Codify this into algorithm using judgment for ρ thresholds. Several specific strategies.

30.3 Constrained optimization

Let I be all inequalities (usually physical and economic limitations) and E be equalities (usually mass balance, energy balance, $\leq 0, \geq 0$).

30.3.1 Equality constraints

$$\min_x f(x) \text{ s.t. } c(x) = 0$$

A point on the function f can be minimized (more generally, optimized) if feasibility on the equality constraint $c(x) = 0$ (linear or non-linear) is maintained while decreasing the function. So a decrease in f while p in $\nabla f(x)^T p < 0$ points to a descent direction. Since the equality constraint has to retain feasibility,

$$\begin{aligned} c(x + p) &= c(x) + \nabla c(x)^T p + \dots = 0 \\ \nabla c(x)^T p &= 0 \end{aligned}$$

At the optimum there should NOT exist p s.t. $\nabla f(x)^T p < 0$ and $\nabla c(x)^T p = 0$. So $\nabla c(x)$ and $\nabla f(x)$ must be parallel or anti-parallel, so

$$L(x, \lambda) = \nabla f(x) + \lambda \nabla c(x) = 0$$

which is the Lagrangian.

$$\nabla L(x, \lambda) = \begin{pmatrix} L_x(x, \lambda) \\ L_\lambda(x, \lambda) \end{pmatrix} = 0$$

This results in

$$\begin{aligned} \nabla L_x(x, \lambda) &= \nabla f(x) + \lambda \nabla c(x) = 0 \\ \nabla L_\lambda(x, \lambda) &= \nabla c(x) = 0 \end{aligned}$$

where the first is stationarity and the second is primal feasibility. Even if optimization problem is to minimize objective function, get multiple λ_i but can't tell whether it is max or min as can just multiply $c(x) = 0 \rightarrow -c(x) = 0$ which will change the signs of λ_i . So must check each λ_i . If there are multiple equality constraints, then just add to the Lagrangian.

30.3.2 Inequality constraints

$$\min_x f(x) \text{ s.t. } c(x) \leq 0$$

Optimality conditions:

1. Stationarity: $\nabla f(x) + \lambda \nabla c(x) = 0$
2. Primal feasibility: $c(x) \leq 0$
3. Complementary slackness: $\lambda c(x) = 0$
4. Dual feasibility: $\lambda \geq 0$

Dual feasibility requires $\lambda \geq 0$. If optimal point lies on $c(x) = 0$ and f is maximized, then ∇f and ∇c must point in opposite directions, so $\nabla f = -\lambda \nabla c$ with $\lambda > 0$. If optimal point lies on $c(x) = 0$ and f is minimized, then ∇f and ∇c must point in the same direction, so $\nabla f = \lambda \nabla c$ with $\lambda > 0$. If optimal point does not lie on $c(x) = 0$, regardless of whether f is minimized or maximized, can ignore it as in unconstrained optimization so $\lambda = 0 \rightarrow \nabla f = 0$. This is related to complementary slackness:

$$c(x) = 0 \rightarrow \lambda \geq 0$$

$$c(x) \leq 0 \text{ (or } c(x) \geq 0) \rightarrow \lambda = 0$$

where the second is essentially unconstrained optimization. If an inequality constraint is inactive (essentially unconstrained optimization), then ignore (Lagrange multipliers will be zero anyways). If active, set as boundary $c(x) = 0$ as in constrained optimization.

30.3.3 KKT optimality conditions

For any combination of inequality and equality constraints, proceed with equality constraints and active inequality constraints in a Lagrangian.

KKT conditions arise from differentiating the Lagrangian $L(x, \lambda_i) = f(x) + \lambda_i(c(x))$:

1. Stationarity: $\nabla f(x') + \sum \lambda_i \nabla c_i(x') = 0$
2. Primal feasibility, equality: $c(x') = 0$.
3. Primal feasibility, inequality: $c(x') \leq 0, c(x') \geq 0$.
4. Dual feasibility: $\lambda_i \geq 0$.
5. Complementary slackness: $\lambda_i c_i(x') = 0$. Trivially satisfied for equalities.

KKT conditions good for solving for optimality and inform about sensitivity using stationarity.

$$\nabla f(x') + \sum \lambda_i \nabla c_i(x') = 0$$

$$\nabla f(x') = - \sum \lambda_i \nabla c_i(x')$$

together with

$$f(x) - f(x') \approx (x - x')^T \nabla f(x') = - \sum \lambda_i (x - x')^T \nabla c_i(x') \approx - \sum \lambda_i (c_i(x) - c_i(x'))$$

So larger multipliers indicate importance of constraint:

$$\Delta f(x) \approx - \sum \lambda_i \Delta c_i$$

Notes. (1) Lagrange multiplier values indicate the degree of 'projection' onto the objective function gradient. If $|LM| > 0$, the constraint (equality or inequality)

is active, so $LM \neq 0$ for equality constraints. (2) Complementary slackness only has real meaning for inequalities, as equalities produce $|LM| > 0$ so the second half of CS must be 0. (3) LM signs for equality constraints don't have much meaning. However, they can be rewritten by multiplying by -1 which simply results in the sign of LM changing.

30.4 Linear programming

All linear programs are convex because only convex feasible sets are possible geometrically. So any local solution is a global solution. Because it's convex, can have infinitely many solutions, solutions that go to infinity (unbounded), no solution (if constraints contradict each other), or unique solution.

Change linear inequality constraints to equalities by introducing slack variables:

$$c(x) \geq 0 \rightarrow c(x) - v = 0, v \geq 0$$

$$c(x) \leq 0 \rightarrow c(x) + v = 0, v \geq 0$$

30.4.1 Simplex method

Classic LP solution method.

$$\min_x d^T x$$

s.t. $Ax = b, x \geq 0$. Form Lagrangian (second term is from non-negativity inequality constraint):

$$L = d^T x + \lambda_E(b - Ax) - \lambda_I x$$

$$\nabla L_x = 0 = d^T - \lambda_E A - \lambda_I$$

and the remaining constraints are:

$$Ax = b, x \geq 0, \lambda_i \geq 0, x\lambda = 0$$

Split the variables into basic (slack variables to convert inequalities to equalities) and non-basic (original) variables:

$$(B, N)^T \lambda_E + [\lambda_{I,B}^T, \lambda_{I,N}^T]^T = [d_B^T, d_N^T]^T$$

$$Bx_B + Nx_N = b$$

Choose $x_N = 0$ and $\lambda_{I,B} = 0$ (complementary slackness satisfied). Solving yields:

$$\lambda_E = (B^T)^{-1} d_B \geq 0$$

$$\lambda_{I,N} = d_N - N^T \lambda_E \geq 0$$

$$x_B = B^{-1} b \geq 0$$

Algorithm. Solutions will be corner solutions. Start with basic 'feasible' (corner) point x . Move to next corner point. Do this by creating matrix, and solving for either the basic or non-basic variables. Several types of algorithm strategies. Algorithm terminates when the Lagrange multipliers of all inequality constraints are greater than or equal to 0.

30.4.2 Duality

Motivation. To solve a primal maximization means to solve the dual minimization. For instance, the upper bound of a primal maximization is the dual minimization value.

(1a) Primal maximization:

$$\begin{aligned} \max \quad & c_1x_1 + c_2x_2 + c_3x_3 \\ \text{s.t.} \quad & A_{11}x_1 + A_{12}x_2 + A_{13}x_3 \geq b_1 \quad (\times y_1) \\ & A_{21}x_1 + A_{22}x_2 + A_{23}x_3 \leq b_2 \quad (\times y_2) \\ & A_{31}x_1 + A_{32}x_2 + A_{33}x_3 = b_3 \quad (\times y_3) \\ & x_1 \geq 0, x_2 \leq 0, x_3 \in \mathcal{R} \end{aligned}$$

(1b) Dual minimization:

$$\begin{aligned} \min \quad & b_1y_1 + b_2y_2 + b_3y_3 \\ \text{s.t.} \quad & A_{11}y_1 + A_{21}y_2 + A_{31}y_3 \geq c_1 \quad (\times x_1) \\ & A_{12}y_1 + A_{22}y_2 + A_{32}y_3 \leq c_1 \quad (\times x_2) \\ & A_{13}y_1 + A_{23}y_2 + A_{33}y_3 = c_1 \quad (\times x_3) \\ & y_1 \leq 0, y_2 \geq 0, y_3 \in \mathcal{R} \end{aligned}$$

(2a) In standard form, primal maximization is

$$\begin{aligned} \max \quad & c_1x_1 + c_2x_2 + c_3x_3 \\ \text{s.t.} \quad & A_{11}x_1 + A_{12}x_2 + A_{13}x_3 = b_1 \quad (\times y_1) \\ & A_{21}x_1 + A_{22}x_2 + A_{23}x_3 = b_2 \quad (\times y_2) \\ & A_{31}x_1 + A_{32}x_2 + A_{33}x_3 = b_3 \quad (\times y_3) \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0 \end{aligned}$$

(2b) Standard form dual minimization:

$$\begin{aligned} \min \quad & b_1y_1 + b_2y_2 + b_3y_3 \\ \text{s.t.} \quad & A_{11}y_1 + A_{21}y_2 + A_{31}y_3 \geq c_1 \quad (\times x_1) \\ & A_{12}y_1 + A_{22}y_2 + A_{32}y_3 \geq c_1 \quad (\times x_2) \\ & A_{13}y_1 + A_{23}y_2 + A_{33}y_3 \geq c_1 \quad (\times x_3) \end{aligned}$$

Standard form matrix representation for primal is $\max c^T x$ s.t. $Ax = b, x \geq 0$, and for dual is $\min y^T b$ s.t. $y^T A \geq c^T$.

LP weak duality. If x and y are dual feasible (i.e. simultaneously feasible), then $c^T x \leq y^T b$. Proof.

$$c^T x \leq y^T A x = y^T b$$

because $x \geq 0$ and $y^T A \geq c^T$ (from the dual minimization constraint) and $Ax = b$.

If $c^T x = y^T b$ then the solution is optimal. There is an intuitive proof for this theory.

Difference between primal and dual optimal values is the duality gap. Duality gap is zero for strong duality (convex problems like LP that satisfy strict feasibility).

(3a) Primal minimization:

$$\begin{aligned} \min \quad & c_1 x_1 + c_2 x_2 + c_3 x_3 \\ \text{s.t.} \quad & A_{11} x_1 + A_{12} x_2 + A_{13} x_3 \geq r_1 \quad (\times y_1) \\ & A_{21} x_1 + A_{22} x_2 + A_{23} x_3 \leq r_2 \quad (\times y_2) \\ & A_{31} x_1 + A_{32} x_2 + A_{33} x_3 = r_3 \quad (\times y_3) \\ & x_1 \geq 0, x_2 \leq 0, x_3 \in \mathcal{R} \end{aligned}$$

(3b) Dual maximization:

$$\begin{aligned} \max \quad & r_1 y_1 + r_2 y_2 + r_3 y_3 \\ \text{s.t.} \quad & A_{11} y_1 + A_{21} y_2 + A_{31} y_3 \geq c_1 \quad (\times x_1) \\ & A_{12} y_1 + A_{22} y_2 + A_{32} y_3 \leq c_2 \quad (\times x_2) \\ & A_{13} y_1 + A_{23} y_2 + A_{33} y_3 = c_3 \quad (\times x_3) \\ & y_1 \geq 0, y_2 \leq 0, y_3 \in \mathcal{R} \end{aligned}$$

The primal LP problem is

$$\begin{aligned} \max_x \quad & d^T x \\ & Ax = b, x \geq 0 \end{aligned}$$

Strong duality (optimal objective values of primal and dual are same)

$$\begin{aligned} d^T x' &= (A^T \lambda_E + \lambda_I) x' \\ &= \lambda_E^T A x' + \lambda_I^T x' \\ &= \lambda_E^T A x' \\ &= b^T \lambda_E \end{aligned}$$

So b, d are dual and x, λ_E are dual. The dual LP problem, which is the same problem viewed differently, is:

$$\max_{\lambda_E} b^T \lambda_E \leftrightarrow \min_{\lambda_E} -b^T \lambda_E$$

$$A^T \lambda_E \leq d$$

and the Lagrangian for the dual problem is:

$$L(x, \lambda_E) = -b^T \lambda_E + x^T (A^T \lambda_E - d)$$

where x are the multipliers. Since $A^T \lambda_E \leq d$ is an inequality, $x \geq 0$ (and this is the reason why the dual LP problem constraint is an inequality, though the standard LP problem constraint is an equality). From the Lagrangian, obtain the dual LP problem KKT conditions:

$$\begin{aligned} L_{\lambda_E} &= 0 \rightarrow Ax = b \\ A^T \lambda_E &\leq d \\ x &\geq 0 \\ x(A\lambda_E - d) &= 0 \end{aligned}$$

Depending on the problem, it may be easier to solve the dual LP problem. For any primal LP, the dual LP is unique. The dual of the dual of the primal is the primal.

Shadow price. For an LP optimal solution, the shadow price of a constraint is the change in the objective function value when the constraint is marginally increased, assuming the optimality conditions are still satisfied. It is the net loss/benefit from marginal additional labor, budget, etc. and the rate of improvement. It is the Lagrange multiplier. The sign of the shadow price depends on the type of problem (max or min) and direction of constraint ($\geq, \leq, =$). The shadow price is zero if a non-binding constraint is changed.

30.4.3 Interior point methods

Primal-dual method (PDM) uses a variation of Newton's method to solve (1)-(3) equalities of the KKT conditions:

1. $A^T \lambda_E + \lambda_I = d$
2. $Ax = b$
3. $x_i \lambda_{I,i}$
4. $x \lambda_I \geq 0$

Define

$$F(x, \lambda_E, \lambda_I) = \begin{pmatrix} A^T \lambda_E + \lambda_I - d \\ Ax - b \\ X \Lambda_I e \end{pmatrix} = 0$$

where

$$X = \begin{bmatrix} x_1 & & \\ & \ddots & \\ & & x_n \end{bmatrix}, \Lambda_I = \begin{bmatrix} \lambda_{I,1} & & \\ & \ddots & \\ & & \lambda_{I,n} \end{bmatrix}, e = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

PDM inputs F at current position to find direction $(\delta x, \delta \lambda_E, \delta \lambda_I)$. PDM 'relaxes' complementary slackness as constraints may be violated when moving in a new direction. Do this by setting $x \lambda_I \geq \tau, \tau > 0$ and then shrinking τ . This involves a complicated matrix equation. Solve to find new direction. Fewer steps than simplex method because start in interior and move directly to optimal corner.

30.5 Nonlinear and integer programming

Solvers can be compared by plotting along Dolan-More performance profiles. Often tradeoff between time and breadth.

30.5.1 Quadratic programming

QP is quadratic objective function and linear constraints. QCQP is QP with quadratic constraints. KKT are necessary but not sufficient conditions. Use NLP more generally to solve.

30.5.2 Nonlinear programming

Difficult. For instance, nonlinear objective f and nonlinear equality constraint s.t. the system is non-convex (so can't use above techniques). However, solve for one variable in the equality constraint and plug into objective function. The result may become convex (as checked by examining Hessian of combined objective function)!

Solve simply by (1) eliminate variables (to convert to unconstrained optimization), (2) approximate as series of unconstrained problems, (3) approximate as series of simpler constrained problems.

Elimination of variables. n variables and m equality constraints. Solve m equality constraints and plug results back into original formulation to get the 'reduced-space formulation' with $n-m$ variables.

30.5.3 Penalty and barrier methods

Approximate NLP as series of unconstrained problems.

Quadratic penalty method (QPM). For the problem

$$\begin{aligned} \min_x & f(x) \\ \text{s.t. } & c_E(x) = 0 \\ & c_I(x) \leq 0 \end{aligned}$$

Add quadratic penalty terms to the objective function that iteratively increase, leading to better approximations.

$$Q(x, \mu) = f(x) + \frac{1}{2\mu} \sum c_E(x)^2 + \frac{1}{2\mu} \sum \max(0, c_I(x))^2$$

where μ is user-specified. As $\mu \rightarrow 0$, the Hessian becomes more and more difficult to calculate. To alleviate, use ALM.

Augmented Lagrangian method (ALM). For the problem

$$\begin{aligned} \min_x & f(x) \\ \text{s.t. } & c_i(x) = 0 \end{aligned}$$

Augment the Lagrangian with a quadratic penalty:

$$L(x, \lambda, \mu) = f(x) + \sum \lambda_i c_i(x) + \frac{1}{2\mu} \sum c_i(x)^2$$

Fewer iterations than QPM as the Lagrangian has more information than the objective function.

A variation of ALM is linearly-constrained Lagrangian method (LCLM), which uses a linear approximation of the constraint and a modified objective (which is an augmented Lagrangian with the linearly-approximated constraint).

Log-barrier method (LBM). For the problem

$$\begin{aligned} \min_x & f(x) \\ \text{s.t. } & c_i(x) \leq 0 \end{aligned}$$

Add a barrier term to the objective

$$P(x, \mu) = f(x) - \mu \sum \log(-c_i(x))$$

where μ is the barrier parameter. To add equality constraints, unfortunately can only add quadratic penalty:

$$P(x, \mu) = f(x) + \frac{1}{2\mu} \sum c_i(x)^2 - \mu \sum \log(-c_i(x))$$

30.5.4 Sequential quadratic programming

Approximate as series of simpler constrained problems. For the QP

$$\begin{aligned} \min_x & f(x) \\ \text{s.t. } & c_i(x) = 0 \end{aligned}$$

Approximate the QP at each iteration using Taylor expansions:

$$\begin{aligned} \min_p & \frac{1}{2} p^T \nabla^2 f(x^{(k)}) + p(\nabla f(x^{(k)}))^T p \\ \text{s.t. } & (\nabla c(x^{(k)}))^T p + c(x^{(k)}) = 0 \end{aligned}$$

Can be interpreted as Newton's method to solve KKT conditions.

30.5.5 Integer programming

Not convex because of gaps between integers.

General formulation for mixed-integer program:

$$\begin{aligned} \min_x & f(x, y) \\ \text{s.t. } & c_i(x, y) = 0 \\ & c_i(x, y) \leq 0 \end{aligned}$$

where $x \in \mathcal{R}^n, y \in \mathcal{Z}^n$. Many integer programs have integers contribute linearly:

$$\begin{aligned} \min_x & f(x) + d^T y \\ \text{s.t. } & c_i(x) = 0 \\ & c_i(x) + a^T y \leq 0 \end{aligned}$$

Often used for yes/no 0-1 processes (e.g. $y = 1$ for yes, $y = 0$ for no). Then include in the constraints, e.g. max plant capacity $c \leq 2y$.

Solve by relaxing the integer requirement, by allowing decimal solutions.

Branch and bound algorithm. Branch using 0-1 as branches to find lower and upper bounds and keep fathoming until lower and upper bounds are equal. Minimization algorithm manually initialized with global lower bound, and if not integer solution then choosing a non-integer x to branch. If the branch doesn't lead to integer solution, then local bound, and if it does lead to integer solution, then (global=local) upper bound. Repeat for other branches and branch further. Repeat until lower bound of branches exceeds upper bound.

30.6 Global optimization

30.6.1 Branch and bound

For any (convex or non-convex) objective function, generate a convex relaxation (a function that is lower than the objective function everywhere). Find the minimum of the convex relaxation. Divide the objective function into two at the convex relaxation minimum. Repeat for both sides, eliminating any sides where that piece's convex relaxation minimum is higher than other pieces' lower objective function values.

30.6.2 Relaxations of non-convex functions

Natural interval extensions. Simplify non-convex function through a sequence of modifications and evaluate the simplified function at endpoints, resulting in a lower and upper bound of the non-convex function. Works but exaggerates function and linear convergence so not efficient.

α BB-relaxation. Add a quadratic term to the objective function, adding curvature to it. Better approximation and quadratic convergence.

McCormick auxiliary variable relaxation. Replace independent variables with more variables, convexify, and linearize. Used in most global optimization codes now.

Relaxation convergence rates. Generate upper and lower relaxations of the objective function over a range, which are mirror images: $f^l(x) \leq f(x) \leq f^u(x)$. The pointwise convergence rate γ is

$$\sup_x (f(x) - f^l(x), f^u(x) - f(x)) \leq C(\delta(x))^\gamma$$

and the Hausdorff convergence rate β is

$$\max(\inf_x f(x) - \inf_x f^l(x), \inf_x f^u(x) - \inf_x f(x)) \leq C(\delta(x))^\beta$$

Global solvers obtain tolerance ϵ to bound the global optimum, not the global optimum itself.

30.6.3 Full and reduced space

Full space includes in the optimization problem all variables, degree of freedom variables and state variables. Reduced space optimization problems exclude the state variables.

30.6.4 Black box methods

Black box methods. Numerically solve functions without gradients. Find better solutions iteratively without gradients. Advantage because can avoid local suboptimal regions, no derivatives required. Disadvantage is slow because sample the solution space, and only get upper bound (instead of upper and lower bound).

Stochastic methods, genetic algorithms, simulation optimization are example of solution classes. Specific examples are gradient-free simplex methods (not same as LP simplex method), gradient-free univariate search (search in one direction and step if better, and repeat for other dimensions, and repeat whole process).

30.6.5 Stochastic optimization

Random search. Randomly find a potential solution. Evaluate function. Repeat by comparing with previous best and keep best so far.

Hybrid stochastic-deterministic solutions. Example: multi-start algorithm. Try different initialization values stochastically and then use deterministic methods after.

Genetic algorithms. Start with initial population. Use a merit function to keep the best (a deterministic process). Mutate and/or cross the survivors (a stochastic process). Can hybridize with deterministic solvers to look for local solution each round.

30.7 Dynamic optimization

Complication as including time means there are infinite variables.

$$\begin{aligned} \min_x & \Phi(x(t)) \\ \text{s.t. } & \dot{x}(t) = f(x(t), u(t)) \\ & g_p(x(t), u(t)) \leq 0 \\ & x(t_0) = x_0 \\ & g_T(x(t)) \leq 0 \end{aligned}$$

where x is the state variable (determined endogenously), u is the control variable (which has the degrees of freedom and so is determined exogenously). The first constraint above is the state equation (model), the second is the path constraint, the third is the initial condition, and the fourth is the terminal condition.

Indirect methods. Write Hamiltonian by adjoining the constraints, which is similar to constructing the Lagrangian.

Direct methods: Simultaneous method. Discretize time $0 = t_0, t_1, \dots, t_n$ and therefore the state and control variables: $y^T = (u_0, x_1, u_1, \dots, u_{M-1}, x_M)$. Note the control variable begins at $t=0$ and ends at $t=M-1$ while state variable begins at $t=1$ and ends at $t=M$. This approximates the dynamic problem as a nonlinear program. Further,

$$\dot{x}(t) = \frac{x_{k+1} - x_k}{t_{k+1} - t_k}$$

so its solution is also discretized, specifically, using numerical methods (e.g. modified Euler, Runge-Kutta). For instance using Euler:

$$c_k(y) = x_{k+1} - x_k - \Delta t f(x_k, u_k) = 0$$

So the simultaneous method turns the original problem where the objective function is only at the final time period (and simplified to remove the path constraint which is difficult):

$$\begin{aligned} \min_x & \Phi(x(t_{final})) \\ \text{s.t. } & \dot{x}(t) = f(x(t), u(t)) \\ & x(t_0) = x_0 \end{aligned}$$

into

$$\begin{aligned} \min_y \quad & \hat{F}(y) \\ \text{s.t.} \quad & c_k(y) = 0 \end{aligned}$$

The Lagrangian is then

$$\begin{aligned} L(y, \lambda_0, \lambda_1, \dots, \lambda_k) &= \hat{F}(y) + \sum_{k=0}^{M-1} \lambda_k^T c_k(y) \\ &= \Phi(x_M) + \sum_{k=0}^{M-1} \lambda_k^T (x_k + \Delta t f(x_k - u_k) - x_{k+1}) \end{aligned}$$

and the KKT conditions are:

$$\begin{aligned} \frac{\partial L}{\partial x_k} &= (\lambda_k - \lambda_{k-1}) + \Delta t \left(\frac{\partial f}{\partial x_k} \right)^T \lambda_k = 0 \\ \frac{\partial L}{\partial x_M} &= -\lambda_{M-1} + \frac{\partial \Phi}{\partial x_M} = 0 \\ \frac{\partial L}{\partial u_k} &= \Delta t \left(\frac{\partial f}{\partial u_k} \right)^T \lambda_k = 0 \\ \frac{\partial L}{\partial \lambda_k} &= x_{k+1} - x_k - \Delta t f(x_k, u_k) = 0 \end{aligned}$$

which is equivalent to

$$\nabla L(y, \lambda_i) = \begin{pmatrix} \nabla_y \hat{F}(y) + \sum_{k=0}^{M-1} (\nabla_y c_k(y))^T \lambda_k \\ c_1(y) \\ c_2(y) \\ \vdots \\ c_k(y) \end{pmatrix} = 0$$

The more time is discretized into smaller Δt , the better the NLP approximation to the dynamic program. Note that the NLP approximation is very sparse so often requires specialized solvers.

Using the modified Euler method to solve the ODE is slow (slow convergence order). The technique of **collocation** is faster but is much more involved mathematically. In collocation,:

$$\begin{aligned} \dot{x}(t) &= f(t, x(t)) \\ x(t_0) &= x_0 \end{aligned}$$

Assume an s-degree polynomial

$$P_s(t) = (t - t_0)^s \alpha_s + (t - t_0)^{s-1} \alpha_{s-1} + \dots + (t - t_0) \alpha_1 + \alpha_0$$

so for n-states, there are $n(s+1)$ coefficients here. Want to find polynomial at s time points that matches the ODE exactly:

$$P_s(t_0) = x_0$$

$$\dot{P}_s(t_0 + c_i \Delta t) = f(t_0 + c_i \Delta t, \underbrace{P_s(t_0 + c_i \Delta t)}_{x(t_0 + c_i \Delta t)})$$

where there are now $n(s+1)$ nonlinear equations. The state and control variables x and u are represented by Lagrange polynomials for s collocation points:

$$u_k(\sigma) = P_u(t_{k-1} + \sigma \Delta t) = \sum_{j=1}^s L_j(t) u_{k,j}$$

$$x_k(\sigma) = P_x(t_{k-1} + \sigma \Delta t) = \sum_{j=0}^s L_j(t) x_{k,j}$$

Impose continuity of state variables. Imposing continuity on controls is optimal as it doesn't have to be a continuous process. Solve ODE at each collocation point c_i :

$$\sum_{j=0}^s \dot{L}_j(c_i) \underbrace{x_{k,j}}_{P_x(t_{k-1} + c_j \Delta t)} = P_x(t_{k-1} + c_i \Delta t) = f\left(\underbrace{x_{k,i}}_{P_x(t_{k-1} + c_i \Delta t)}, \underbrace{u_{k,i}}_{P_u(t_{k-1} + c_i \Delta t)}\right)$$

which solves the nonlinear system of equations for $x_{k,j}$ for fixed $u_{k,i}$ and fixed c_i values. Now, the optimization problem is

$$\begin{array}{ll} \min_x & \Phi(x_{k+1,0}) \\ \text{ODE} & x_{k+1,0} = \sum_{j=0}^s L_j(1) x_{k,j} \\ \text{Initial} & x_{1,0} = x_0 \\ \text{Final} & x_{k+1,0} = \sum_{j=0}^s L_j(1) x_{k,j} \\ \text{Path constraint} & u_l \leq u \leq u_u \end{array}$$

Direct methods: Single shooting method (sequential method). First discretize control variables $u(t)$ by a series of orthonormal basis functions $\sigma_{i,k}(t)$ and corresponding coefficients $p_{i,k}$:

$$u_i(t) \approx \sum_{k=1}^s p_{i,k} \sigma_{i,k}(t), i = 1, \dots, n_u$$

so $u(t)$ is now parameterized by p . This will also parameterize $x(t)$ at the same time. The original problem was

$$\begin{aligned} \min_{x,u} & \Phi(x(t_{final})) \\ \text{s.t. } & \dot{x}(t) = f(x(t), u(t)) \\ & x(t_0) = x_0 \\ & u_l \leq u \leq u_u \end{aligned}$$

is now

$$\begin{aligned} \min_p & \phi(p) \\ \text{s.t. } & c(p) \leq 0 \\ & x(t_0) = x_0(p) \end{aligned}$$

where $x(t)$ solves $\dot{x}(t) = f(x(t), u(t))$, $x(t_0) = x_0$. Further,

$$\min_{x,u} \Phi(x(t_{final}))$$

has become

$$\min_p \phi(p)$$

Still further,

$$u_l \leq u \leq u_u$$

has become

$$c(p) \leq 0$$

To solve:

1. Calculate $\nabla\phi(p)$ and then p by optimization.
2. Input p to output $x_0, u(t)$.
3. Input $x_0, u(t)$ into the differential equations and solve for $x(t)$.
4. Input $x(t)$ into $\phi(p) = \Phi(x(t_f))$. Repeat.

To calculate $\nabla\phi(p)$, use finite differences:

$$\nabla\phi(p) = \left(\frac{\partial\phi}{\partial p_1}, \dots, \frac{\partial\phi}{\partial p_j} \right)'$$

One p at a time perturb:

$$\frac{\partial\phi}{\partial p_j} = \frac{\phi(p + e_j\Delta) - \phi(p)}{\Delta}, e_j = (0, \dots, 0, 1, 0, \dots)'$$

where $e_j = 1$ at the j -th position. Further, using the chain rule,

$$\frac{\partial\phi}{\partial p_j} = \sum_i \left(\frac{\partial\Phi}{\partial x_i} \cdot \frac{\partial x_i}{\partial p_j} \right)$$

where the first term in brackets is derived from Φ and $x(t)$, and the second term is the sensitivity $S_{i,j}(t)$ to p . Re-express:

$$\nabla \phi(p) = \left(\sum_i \left(\frac{\partial \Phi}{\partial x_i} \cdot S_{i,1}(t_f) \right), \dots, \sum_i \left(\frac{\partial \Phi}{\partial x_i} \cdot S_{i,j}(t_f) \right), \dots \right)'$$

Now,

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)), x(t_0) = x_0 \\ \frac{d\dot{x}}{dp} &= \frac{df}{dp} = \frac{\partial f}{\partial x} \frac{dx}{dp} + \frac{\partial f}{\partial u} \frac{du}{dp} \end{aligned}$$

where the derivatives of f are from manually or computationally differentiating f . To calculate $S(t)$:

$$\dot{S}(t) = \frac{d\dot{x}}{dp} = \frac{\partial f}{\partial x} S(t) + \frac{\partial f}{\partial u} \frac{du}{dp}$$

where the last term is from parameterization of u using p , derivatives of f are from differentiating f , and the initial condition is $S(t_0) = \frac{d\dot{x}_0}{dp}$.

Direct methods: Multiple shooting method. The single shooting method is very sensitive to initial values because the dynamical system blows up, e.g. exponentially. Multiple shooting addresses this by splitting the domain up and introducing new initial values for each domain segment. This reduces the sensitivity to the initial value.

30.7.1 Path constraints and switching structures

$$\begin{aligned} \min_{x,u} & \Phi(x(t_{final})) \\ \text{s.t. } & \dot{x}(t) = f(x(t), u(t)) \\ & x(t_0) = x_0 \\ & g_p(x(t), u(t)) \leq 0, t \in (t_0, t_f) \end{aligned}$$

where g_p is the path constraint. Discretize g_p , but as in the Runge phenomenon (from polynomial interpolation), it violates between the discrete points in time. Could try tweaking the constraints, e.g. $g_p \leq -0.01$ but this ruins optimality. Instead, can use semi-infinite programming (SIP) or approximate and parameterize the path control using 'switching structures'.

30.7.2 Model predictive control

Comprises MPC (model predictive control) and trajectory optimization. Allows control and optimization of process real-time (online). Can be linear or non-linear.

Can be used in multiple-input, multiple-output (MIMO) systems where the

inputs and outputs may interact (e.g. $(x_1, x_2) \rightarrow (y_1, y_2)$ where $x_1 \rightarrow (y_1, y_2)$ and $x_2 \rightarrow (y_1, y_2)$). MPC can also handle constraints. This is in contrast to individual PID controllers.

However, it requires a lot of memory and computational resources.

30.8 Machine learning

30.8.1 Topologies

Example. If x_1 is within a certain narrow range (similar to optimization inequality constraint) and x_2 is also within a different, certain narrow range. This is a mechanistic model, i.e. the inequality constraints based on mechanisms we know, where extrapolation outside these inequality ranges is invalid. So a neural network inputting both variables x_1, x_2 to obtain one output y is likely invalid as it will extrapolate beyond the inequality ranges. So include the mechanistic component in the topology, e.g. one neural network (NN) for x_1 and a separate one for x_2 , and then combine these NN outputs in a mechanistic component that results in y .

30.8.2 Neutral network optimization

Recall full space is

$$\begin{aligned} \min_{x,z} f(x, z) \\ \text{s.t. } g(x, z) \leq 0 \\ h(x, z) = 0 \end{aligned}$$

Usually, $\dim(z) \gg \dim(x)$, so solve $h(x, z) = 0$ for z to obtain the reduced space:

$$\begin{aligned} \min_x \hat{f}(x, z) \\ \text{s.t. } \hat{g}(x, z) \leq 0 \end{aligned}$$

Formulate neural networks as optimizations. Example with two inputs $x = (x_1, x_2)$ and one hidden layer. Full space is

$$\begin{aligned}
& \min_{x,z} y_{ANN} \\
& \text{s.t. } y_{ANN} = (z_1 + 1) \frac{y^{train} - y}{z} + y \\
& z_1 = \sum_1^3 w_{j,1} z_j + b_1 \\
& z_j = \tanh\left(\sum_1^3 w_{j,1} z_j + b_i\right) \\
& z_j = \frac{x_j - x_j^L}{x_j^U - x_j^L} 2 - 1
\end{aligned}$$

However, this is a complicated NLP with long run-times. Solve equalities and substitute into objective function to obtain the reduced space:

$$\min_x y_{ANN}$$

30.8.3 Gaussian processes (Kriging) and Bayesian optimization

Nonlinear interpolation which provides prediction as well as variance (uncertainty). Model-free optimization (guides where to get next point).

Full space

$$\begin{aligned}
& \min_{x,k,K,r,y} y_* \\
& \text{s.t. } y_* = K K^{-1} y_{train} \\
& K = \begin{pmatrix} k(x_1, x_1) & \dots & k(x_1, x_N) \\ \dots & \dots & \dots \\ k(x_N, x_1) & \dots & k(x_N, x_N) \end{pmatrix} \\
& K_* = (k(x_*, x_1), \dots, k(x_*, x_N)) \\
& k(x_i, x_j) = \sigma_f^2 e^{(-r_{ij}/l)}, k \in [k^l, k^u] \\
& r_{ij} = \|x_i - x_j\|, r_{ij} \in [r_{ij}^l, r_{ij}^u], i, j \in [1, N] \\
& x \in [x^l, x^u]
\end{aligned}$$

where K is the covariance matrix and k is the covariance (function), there are D dimensions of training points, and N training points total. This is a difficult problem, so convert to reduced space by solving equalities and propagating McCormick relaxations and subgradients.

Reduced space

$$\begin{aligned} \min_x & \text{Gaussian}(x_*) \\ \text{s.t. } & x_* \in [x^l, x^u] \end{aligned}$$

30.9 Optimization under uncertainty

If one of the parameters (y) is a result of a stochastic process, it affects the objective function. If it were fixed instead of stochastic, then can use NLP. However, for stochastic parameter(s), optimize via:

1. Parametric. Consider parameter in terms of variable.
2. Stochastic. Consider probability measures over all uncertainties.
3. Robust. Consider worst-case uncertainty.

30.9.1 Parametric optimization

Solves parameter(s) y in terms of the variable x . Then can solve for objective function depending on the $(x, y(x))$, so it's deterministic.

$$\begin{aligned} \min_x & f(x, y) \\ \text{s.t. } & c(x, y) \leq 0 \end{aligned}$$

Example.

$$\begin{aligned} \min_x & f(x, y) = (x - y)^2 + y \\ \text{s.t. } & x^*(y) = y \end{aligned}$$

where the optima lie along the line $y = x$. Note: $f^*(y) = f(x^*(y), y) = (y - y)^2 + y = y$.

30.9.2 Stochastic optimization

Know the probabilities $p(y)$ of the parameter.

$$\begin{aligned} \min_x & \mathbb{E}_y(f(x, y)) \\ \text{s.t. } & \mathbb{P}_y(c(x, y) \leq 0) \end{aligned}$$

2-stage stochastic optimization. The first stage is

$$\begin{aligned} \min_x & f(x) + \mathbb{E}_y(f(x, y)) \\ \text{s.t. } & c(x, y) \leq 0 \end{aligned}$$

The second stage is

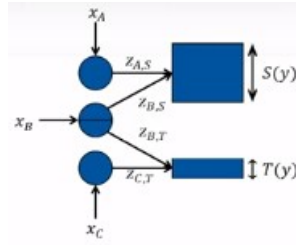
$$\begin{aligned} F(x, y) &= \min_z f(y, z) \\ \text{s.t. } & c(x, y, z) \leq 0 \end{aligned}$$

The two stages can be combined into a single-stage:

$$\begin{aligned} \min_{x,z} & f(x) + \sum p f(y, z) \\ \text{s.t. } & c(x, y) \leq 0 \\ & c(x, y, z) \leq 0 \end{aligned}$$

which can be estimated through discretization and scenario analysis if the probability distribution is calculation-intensive.

Example. Economy with inputs x , controls for actual production z , and demand S and T .



$$\begin{aligned} \min_x & d_A x_A + d_B x_B + d_C x_C + \mathbb{E}_y(F(x, y)) \\ \text{s.t. } & x_A + x_B + x_C \leq x_{max} \\ & x_A, x_B \geq 0 \end{aligned}$$

where the objective function is the cost, and $\sum x_i \leq x_{max}$ is the budget constraint. The second stage is

$$\begin{aligned} F(x, y) = \min_z & -d_S(z_{A,S} + z_{B,S}) - d_T(z_{B,T} + z_{C,T}) \\ \text{s.t. } & z_{A,S} \leq x_A \\ & z_{B,S} + z_{B,T} \leq x_B \\ & z_{C,T} \leq x_C \\ & z_{A,S} + z_{B,S} \leq S(y) \\ & z_{B,T} + z_{C,T} \leq T(y) \end{aligned}$$

30.9.3 Robust optimization and semi-infinite programming

The parameter y is bounded to the worst-case.

$$\begin{aligned} \min_x \max_y & f(x, y) \\ \text{s.t. } & c(x, y) \leq 0 \end{aligned}$$

Semi-infinite programming (SIP). Has finite variables x and infinite constraints due to infinite y . Useful for solving robust optimization problems.

$$\begin{aligned} \min_x & f(x) \\ \text{s.t. } & \min_y c(x, y) \leq 0 \end{aligned}$$

where the constraint is the **lower-level problem** which must be solved globally (not just locally). Example. Fit a circle $(z_1 - x_1)^2 + (z_2 - x_2)^2 = r^2$ in a shape defined by parameter y s.t. $z_1 = h_1(y), z_2 = h_2(y)$. Then, maximize r in $(h_1(y) - x_1)^2 + (h_2(y) - x_2)^2 = r^2$.

SIP can't be solved using global optimization as the constraint is not differentiable. KKT conditions can't be used, as it only solves for local solutions, and one of the local solutions might not be feasible within $c(x, y) \leq 0$.

SIP solution methods. Often involve estimating a lower bound and upper bound (e.g. converting $c(x, y) \leq 0 \rightarrow c(x, y) \leq -\epsilon$), and iterating.

1. Local reduction. Find all KKT points. Use NLP solution methods.
2. Discretization. Finite approximation of y . Use NLP solution methods.

31 Dynamic programming

31.1 Bellman equation

The value function v with discount rate β , payoff or utility F , state x :

$$v(x_0) = \max_{a_0, a_1, \dots} \sum_{t=0}^{\infty} \beta^t F(x_t, a_t)$$

where $a_t \in \Gamma(x_t)$ (a correspondence), the law of motion is $x_{t+1} = r(x_t, a_t)$, and x_0 is given. Lagrangian is impractical with infinite a 's. However, with limited time horizon (one period, from $t=0$ to $T=0$),

$$v(x_0) = F(x_0, a_0)$$

Further,

$$v(x_1) = \max_{a_1, a_2, \dots} \sum_{t=1}^{\infty} \beta^{t-1} F(x_t, a_t)$$

Combining,

$$v(x_0) = \max_{a_0} (F(x_0, a_0) + \beta v(x_1))$$

which is the Bellman equation. It is an implicit equation. To solve, can guess $v_0(k)$, substituting for $v(x_1) \rightarrow v_1(x_0)$ and repeating $v_1(x_0) \rightarrow v(x_1) \rightarrow v_2(x_0) \rightarrow v(x_1) \rightarrow v_2(x_0) \dots$, which is value function iteration.

31.1.1 Vector spaces

Closed under addition ($x, y \in V \rightarrow x + y \in V$) and scalar multiplication ($a \in \mathcal{R}, x \in V \rightarrow ax \in V$). Properties: commutative, associative, and distributive.

Norm. Can be absolute value, Euclidean norm, etc. For infinite-dimensional spaces, can use weighted average, or supremum $\|x\| = \sup_i |x_i|$ which satisfies the definition and properties (e.g. Cauchy-Schwartz) of norms. A vector space with a norm is a normed vector space.

Example 1.

$$B(A) = (f : A \rightarrow \mathcal{R}, f \text{ is bounded})$$

Can attach a norm $\|f\| = \sup_x |f(x)|$. (B) Norm might additionally specify an additional function $\phi(x) > 0$ and where $\|f\|_\phi = \sup \frac{|f(x)|}{\phi(x)} < M$, which satisfies the definition and properties (e.g. Cauchy-Schwartz) of norms. Those f bounded in the ϕ -norm might be a subset of the normed vector space. Just substitute $f(x), \phi(x)$ and solve the inequality for parameters that bound f .

Example 2.

$$C(A) = (f : A \rightarrow \mathcal{R}, f \text{ is continuous})$$

where an example is polynomials that can be added and scaled. Can attach a norm $\|f\| = \int |f(x)| dx$ (area between $f(x)$ and x-axis), which satisfies the definition and properties (e.g. Cauchy-Schwartz) of norms.

31.1.2 Banach spaces

Complete normed vector space. Complete means every Cauchy sequence converges (not true that every Cauchy sequence converges in general).

Theorem. $C_\phi(A) = (f : A \rightarrow \mathcal{R}, f \text{ is continuous and bounded in the } \phi\text{-norm})$, where $\phi(x) \in \mathcal{R}$, is a Banach space. Proof omitted.

31.1.3 Contraction mappings

Use Banach space $C_\phi(A) = (f : A \rightarrow B)$ which is continuous and bounded. Then T is as an operator s.t. $T : X \rightarrow X$ is a contraction mapping. $\exists \beta \in [0, 1)$ and $\|T(x) - T(y)\| \leq \beta \|x - y\|$. So repeating, the $\|\cdot\| \rightarrow 0$ due to repeated shrinking. Basically, it shrinks the space in all directions.

Banach fixed point theorem (contraction mapping theorem). Contraction mapping $T : X \rightarrow X$ is a fixed point if $T(x) = x$. Banach fixed point theorem: fixed point x is unique and $\|T^n x_0 - x\| \leq \beta^n \|x_0 - x\|$. Intuitively, maps have to contract to one point. See Functional Analysis chapter for proof.

Lemma. Let X' be a closed subset of X . So $x \in X'$ and $T(x) \in X'$. Then the unique fixed point of T contraction mapping is in X' . This is because T can't escape from the subset once inside.

So there is a contraction mapping of a function in Banach space to another function in Banach space:

$$T : C_\phi(A) \rightarrow C_\phi(A)$$

where the contraction-mapped function is $Tf = T(f)$.

Blackwell theorem. Sufficient conditions for contraction mapping: (1) monotonicity $f \leq g \rightarrow T(f) \leq T(g)$, (2) discounting $T(f + a\phi)(x) \leq Tf(x) + \beta a\phi(x)$.

31.1.4 Maximization theorems

Theory of the maximum. $\max f(x)$ s.t. $x \in G$. However, $f(x) \rightarrow \infty$ is possible. To prevent this, ensure f is continuous and G is compact (closed and bounded, where closed means convergence and bounded means $\|x\| \leq M$).

Berge's maximization theorem. A generalization of the theory of the maximum. The (optimal) value function is $v(x) = \max_a f(x, a)$ s.t. $a \in G(x)$. To make sure $v(x)$ is continuous, ensure f is continuous and G is both UHC and LHC (analogous to compactness). Further, the solution correspondence (all the solutions):

$$\Gamma(x) = \{a \in G(x) : f(x, a) = v(x)\}$$

would then also be UHC. Additionally, if f is strictly strictly quasi-concave in a and G is convex-valued, then $G(x)$ is a convex set and $\Gamma(x)$ is unique and continuous.

31.1.5 Optimization theorem

Applying Berge's theorem and associated conditions (e.g. G is UHC and LHC) to a series of value functions and objective functions. Then the solutions to

$$\gamma_n(x) = \max_a f_n(x)$$

$$\gamma(x) = \max_a f(x)$$

will be unique. However, establish that the assumptions

$$\lim_{n \rightarrow \infty} f_n = f$$

$$\lim_{n \rightarrow \infty} \sup_{x, a} \frac{|f_n(x, a) - f(x, a)|}{\phi(x)} = 0$$

result in convergence:

$$\lim_{n \rightarrow \infty} \gamma_n(x) = \gamma(x)$$

To do this,

$$\begin{aligned}
0 &\leq f(x, \gamma(x)) - f(x, \gamma_n(x)) \\
&\leq f(x, \gamma(x)) - f(x, \gamma_n(x)) + \underbrace{f_n(x, \gamma_n(x)) - f_n(x, \gamma(x))}_{\geq 0} \\
&= \phi(x) \frac{f(x, \gamma(x)) - f_n(x, \gamma(x))}{\phi(x)} + \phi(x) \frac{f_n(x, \gamma_n(x)) - f(x, \gamma_n(x))}{\phi(x)}
\end{aligned}$$

So,

$$\begin{aligned}
f(x, \gamma(x)) - f(x, \gamma_n(x)) &\leq \phi(x) \frac{f(x, \gamma(x)) - f_n(x, \gamma(x))}{\phi(x)} + \phi(x) \frac{f_n(x, \gamma_n(x)) - f(x, \gamma_n(x))}{\phi(x)} \\
\frac{f(x, \gamma(x)) - f(x, \gamma_n(x))}{\phi(x)} &\leq \frac{f(x, \gamma(x)) - f_n(x, \gamma(x))}{\phi(x)} + \frac{f_n(x, \gamma_n(x)) - f(x, \gamma_n(x))}{\phi(x)}
\end{aligned}$$

Taking limsup of both sides, both RHS terms are zero based on

$$\lim_{n \rightarrow \infty} \sup_{x, a} \frac{|f_n(x, a) - f(x, a)|}{\phi(x)} = 0$$

above, so

$$\lim_{n \rightarrow \infty} \sup_{x, a} \frac{|f(x, \gamma(x)) - f(x, \gamma_n(x))|}{\phi(x)} = 0$$

Now assume (in a proof by contradiction) that

$$\lim_{n \rightarrow \infty} \|\gamma_n(x) - \gamma(x)\| \neq 0$$

and assume that x is compact. So,

$$\lim_{n \rightarrow \infty} \|\gamma_n(x) - \gamma(x)\| \geq \epsilon$$

where $\epsilon > 0$. Define the neighborhood

$$B_\epsilon = \{a \in G(x) : \|a - \gamma(x)\| \geq \epsilon\}$$

Then with $\epsilon > 0$, $\delta(x)B_\epsilon$ but $\delta_n(x) \in B_\epsilon$. Further, the neighborhood is compact (closed because as $a_n \rightarrow a$, they always satisfy both the conditions of the neighborhood; bounded because $B_\epsilon \in G(x)$). However, if the neighborhood is compact, it contradicts $\delta(x)B_\epsilon$. So it must be that

$$\lim_{n \rightarrow \infty} \|\gamma_n(x) - \gamma(x)\| = 0$$

Through an analogous contradiction proof can show that the above is the same as the below. Can also see that it is intuitively equivalent:

$$\lim_{n \rightarrow \infty} \|\gamma_n - \gamma\|_\phi = \lim_{n \rightarrow \infty} \sup \frac{\|\gamma_n(x) - \gamma(x)\|}{\phi(x)} = 0$$

31.1.6 Bellman operator

Bellman equation:

$$v(x) = \max_{a \in G(x)} (F(x, a) + \beta v(r(x, a)))$$

From above, the contraction mapping operator,

$$T : C_\phi(X) \rightarrow C_\phi(X)$$

If can show that

$$Tv(x) = \max_{a \in G(x)} (F(x, a) + \beta v(r(x, a)))$$

then can solve the Bellman equation because $Tv(x) = v(x)$ by the Banach fixed point theorem. The T is now called the Bellman operator.

Regularity conditions. F is continuous, G is UHC and LHC, r is continuous,

$$\max_{a \in G(x)} |F(x, a)| \leq M\phi(x)$$

$$\beta \max_{a \in G(x)} \phi(r(x, a)) \leq \theta\phi(x)$$

where $\theta \in [0, 1)$.

Since F and r are continuous, v is continuous. So by Berge's maximization theorem, Tv is also continuous. Now show that Tv is bounded in the ϕ -norm using the regularity conditions above:

$$\begin{aligned} Tv(x) &= \max_{a \in G(x)} (F(x, a) + \beta v(r(x, a))) \\ &\leq \max_{a \in G(x)} (|F(x, a)| + \beta |v(r(x, a))|) \\ &\leq M\phi(x) + \beta \max_{a \in G(x)} \phi(r(x, a)) \underbrace{\frac{|v(r(x, a))|}{\phi(r(x, a))}}_{\leq \|v\|_\phi} \\ &\leq M\phi(x) + \|v\|_\phi \beta \max_{a \in G(x)} \phi(r(x, a)) \\ &\leq M\phi(x) + \|v\|_\phi \theta\phi(x) \end{aligned}$$

$$\|Tv\|_\phi = \sup \frac{Tv(x)}{\phi(x)} \leq \sup (M + \|v\|_\phi \theta)$$

so Tv is indeed bounded. Now apply Blackwell's theorem, which requires monotonicity and discounting. For monotonicity,

$$v \leq w$$

So,

$$\begin{aligned}
Tv(x) &= \max_{a \in G(x)} (F(x, a) + \beta v(r(x, a))) \\
&\leq \max_{a \in G(x)} (F(x, a) + \beta w(r(x, a))) \\
&= Tw(x)
\end{aligned}$$

For discounting, show $T(v + a\phi) \leq Tv + \theta b\phi$:

$$\begin{aligned}
T(v + a\phi)(x) &\leq \max_{a \in G(x)} (F(x, a) + \beta(v(r(x, a)) + b\phi(r(x, a)))) \\
&\leq \max_{a \in G(x)} (F(x, a) + \beta v(r(x, a))) + \beta \max_{a \in G(x)} b\phi(r(x, a)) \\
&\leq Tv(x) + b\theta\phi(x)
\end{aligned}$$

So this is a contraction mapping

$$T : C_\phi(X) \rightarrow C_\phi(X)$$

and $Tv = v$.

31.1.7 Infinite horizon

Show that $w_p = \beta^t F(x_t, a_t)$ or the sequence $w_0, w_1, w_2 \dots$ is Cauchy and therefore converges (so the metric space is complete, as in a Banach space):

$$\begin{aligned}
|w_n - w_m| &= \left| \sum_{t=0}^n \beta^t F(x_t, a_t) - \sum_{t=0}^m \beta^t F(x_t, a_t) \right| \\
&= \left| \sum_{t=m+1}^n \beta^t F(x_t, a_t) \right| \\
&= \sum_{t=m+1}^n \beta^t |F(x_t, a_t)| \\
&= \sum_{t=m+1}^n \beta^t M\phi(x_t) \\
&= M \sum_{t=m+1}^n \beta^t \phi(x_t) \\
\phi(x_t) &= \phi(r(x_{t-1}, a_{t-1}))
\end{aligned}$$

The last regularity condition is

$$\phi(r(x, a)) \leq \frac{\theta\phi(x)}{\beta}$$

So,

$$\begin{aligned}
\phi(x_t) &= \phi(r(x_{t-1}, a_{t-1})) \\
&\leq \frac{\theta}{\beta} \phi(x_{t-1}) \\
&= \frac{\theta}{\beta} \phi(r(x_{t-2}, a_{t-2})) \\
&\leq \left(\frac{\theta}{\beta}\right)^2 \phi(x_{t-2}) \\
&\dots \\
&= \left(\frac{\theta}{\beta}\right)^t \phi(x_0)
\end{aligned}$$

so this puts bounds on $\phi(x)$. Continuing with the above derivation,

$$\begin{aligned}
|w_n - w_m| &= \left| \sum_{t=0}^n \beta^t F(x_t, a_t) - \sum_{t=0}^m \beta^t F(x_t, a_t) \right| \\
&\dots \\
&= M \sum_{t=m+1}^n \beta^t \phi(x_t) \\
&= M \sum_{t=m+1}^n \beta^t \left(\frac{\theta}{\beta}\right)^t \phi(x_0) \\
&= M \sum_{t=m+1}^n \theta^t \phi(x_0) = M \phi(x_0) \sum_{t=m+1}^n \theta^t \\
&\leq M \phi(x_0) \sum_{t=m+1}^{\infty} \theta^t = M \phi(x_0) \frac{\theta^{m+1}}{1-\theta} = 0
\end{aligned}$$

where the last line follows from $0 < \theta < 1$. So w is Cauchy and converges.

Show that $w_p \leq v(x_0)$.

$$\begin{aligned}
v(x_0) &= \max_{a \in G(x_0)} (F(x_0, a) + \beta v(r(x_0, a))) \\
&\geq F(x_0, a_0) + \beta v(r(x_0, a_0)) \\
&= F(x_0, a_0) + \beta v(x_1) \\
&= F(x_0, a_0) + \beta \max_{a \in G(x_1)} (F(x_1, a) + \beta v(r(x_1, a))) \\
&\dots \\
&= \sum_{t=0}^T \beta^t F(x_t, a_t) + \lim_{T \rightarrow \infty} \beta^{T+1} v(x_{T+1}) \\
&= w_p + \lim_{T \rightarrow \infty} \beta^T v(x_T)
\end{aligned}$$

$$v(x_T) \leq \frac{|v(x_T)|}{\phi(x_T)} \frac{\phi(x_T)}{\phi(x_{T-1})} \frac{\phi(x_{T-1})}{\phi(x_{T-2})} \dots \frac{\phi(x_1)}{\phi(x_0)} \phi(x_0)$$

Applying the last regularity condition,

$$\frac{\phi(x_t)}{\phi(x_{t-1})} = \frac{\phi(r(x_{t-1}, a_{t-1}))}{\phi(x_{t-1})} \leq \frac{\theta}{\beta}$$

Applying this,

$$\begin{aligned} v(x_T) &\leq \frac{|v(x_T)|}{\phi(x_T)} \left(\frac{\theta}{\beta}\right)^{T-1} \phi(x_0) \\ &= \|v\|_{\phi} \left(\frac{\theta}{\beta}\right)^{T-1} \phi(x_0) \end{aligned}$$

$$\beta^T v(x_T) \leq \|v\|_{\phi} \beta \theta^{T-1} \phi(x_0)$$

$$\lim_{T \rightarrow \infty} \beta^T v(x_T) \leq \lim_{T \rightarrow \infty} \|v\|_{\phi} \beta \theta^{T-1} \phi(x_0) = 0$$

So,

$$v(x_0) \geq w_p + \lim_{T \rightarrow \infty} \beta^T v(x_T) = w_p$$

To show that for the optimal path p^* , show $v(x_0) = w_{p^*}$, which simply involves recycling the above with equalities:

$$\begin{aligned} v(x_0) &= \max_{a \in G(x_0)} (F(x_0, a) + \beta v(r(x_0, a))) \\ &= F(x_0, a_0) + \beta v(x_1) \\ &= F(x_0, a_0) + \beta \max_{a \in G(x_1)} (F(x_1, a) + \beta v(r(x_1, a))) \\ &= F(x_0, a_0) + \beta F(x_1, a_1) + \beta^2 v(x_2) \\ &= \sum_{t=0}^T \beta^t F(x_t, a_t) + \beta^{T+1} v(x_{T+1}) \\ &= w_{p^*} + 0 \end{aligned}$$

31.1.8 Properties

From regularity conditions and contraction mapping lemma above, can prove that the Bellman operator T (proofs omitted) is:

1. Monotonic. If $x \leq x'$, and F (function) and G (correspondence) are monotone, then the optimal value function $v^*(x)$ is monotone.
2. Strictly concave. If F (function) is strictly concave and G (correspondence) is convex, the optimal value function $v^*(x)$ is strictly concave.

31.1.9 Finishing the proof

Given F is strictly concave, G is convex, and the regularity conditions, value function v is strictly concave from above. Further,

$$\gamma_n(x) = \max_{a \in G(x)} (F(x) + \beta v_n(a))$$

will pointwise converge to $\gamma(x)$ and if x is compact, it will uniformly converge $\|\gamma_n(x) - \gamma(x)\|_\phi \rightarrow 0$. It was shown above that the assumptions

$$\begin{aligned} \lim_{n \rightarrow \infty} f_n &= f \\ \lim_{n \rightarrow \infty} \sup_{x, a} \frac{|f_n(x, a) - f(x, a)|}{\phi(x)} &= 0 \end{aligned}$$

result in convergence:

$$\lim_{n \rightarrow \infty} \gamma_n(x) = \gamma(x)$$

Proof of assumptions. Let

$$\begin{aligned} f_n(x, a) &= F(x, a) + \beta v_n(a) \\ f(x, a) &= F(x, a) + \beta v(a) \end{aligned}$$

Then,

$$\begin{aligned} |f_n(x, a) - f(x, a)| &= |\beta| |v_n(a) - v(a)| \\ &= \beta \frac{|v_n(a) - v(a)|}{\phi(a)} \phi(a) \\ &= \beta \phi(a) \|v_n - v\|_\phi \\ &\leq \beta \max_{a \in G(x)} \phi(a) \|v_n - v\|_\phi \\ &\leq \beta \max_{a \in G(x)} \phi(r(x, a)) \|v_n - v\|_\phi \\ &\leq \theta \phi(x) \|v_n - v\|_\phi \end{aligned}$$

where $\beta, \theta \in [0, 1)$ and the penultimate step is from the regularity conditions. Note $\|v_n - v\|_\phi$ is just a number. Take suprema of both sides above,

$$\sup \frac{|f_n(x, a) - f(x, a)|}{\phi(x)} \leq \sup \|v_n - v\|_\phi \rightarrow 0$$

where the RHS is a scalar that approaches zero because T is a contraction mapping so $v_n \rightarrow v$ as a fixed point. This proves the optimization theorem above.

31.1.10 Differentiability

Benveniste-Scheinkman lemma. If x is convex, v is continuous and concave, and w is continuous, concave and $w'(x_0) = c$. Further, $w(x_0) = v(x_0)$ and $w(x) \leq v(x)$. Then $w'(x_0) = v'(x_0)$ because of geometric intuition (so proof omitted).

Envelope theorem. Differentiability of the Bellman value function.

$$v(x) = \max_a (F(x, a) + \beta v(a))$$

Assume regularity, F strictly concave, G convex. Then v is strictly concave and extracted policy function $\gamma^*(x)$ is continuous. Now assume additionally $\gamma(x_0)$ is in the interior of $G(x_0)$ and F is differentiable. Then v is differentiable at x_0

$$v'(x_0) = F'(\underbrace{x_0}_{diff}, \underbrace{\gamma(x_0)}_{fixed})$$

which relies on Benveniste-Scheinkman (but proof omitted).

Example.

$$v(k_t) = \max_{k_{t+1}} u(Ak_t - k_{t+1}) + \beta v(k_{t+1}) \quad (0)$$

Differentiate wrt k_t :

$$v'(k_t) = u'(Ak_t - k_{t+1}) + \beta v'(k_{t+1}) \quad (1)$$

which is the same as the version moved forward in time

$$v'(k_{t+1}) = u'(Ak_{t+1} - k_{t+2}) + \beta v'(k_{t+2}) \quad (1a)$$

Also differentiate equation (0) directly wrt k_{t+1} :

$$-u'(Ak_t - k_{t+1}) + \beta v'(k_{t+1}) = 0 \quad (2)$$

Combine equations (1a) and (2):

$$v'(k_{t+1}) = u'(Ak_{t+1} - k_{t+1})A - u'(Ak_t - k_{t+1}) + \beta v'(Ak_{t+1} - k_{t+1})A = 0$$

$$u'(ct) = \beta A u'(c_{t+1})$$

which is a DE.

31.1.11 Euler equation

Generalize Bellman value function differentiability.

$$\max_{a_0, a_1, \dots} \sum \beta^t F(x_t, x_{t+1})$$

is (where prime indicates optimal, not derivative)

$$F(x'_0, x'_1) + \dots + \beta^{t-1} F(x'_{t-1}, x_t) + \beta^t F(x_t, x'_{t+1}) + \beta^{t+1} F(x_{t+1}, x_{t+2}) + \dots$$

Differentiate wrt x_t and since F is strictly concave and differentiable, the derivative will lead to optimality:

$$\beta^{t-1} F_{x_t}(x'_{t-1}, x'_t) + \beta^t F_{x_t}(x'_t, x'_{t+1}) = 0$$

$$F_{x_t}(x'_{t-1}, x'_t) = -\beta F_{x_t}(x'_t, x'_{t+1}) = 0$$

which is the Euler equation.

31.1.12 Iterations

Value function iteration. Initialize value function, find an improved value function in an iterative process, until reaching the optimal value function. Then extract (via argmax) the optimal policy from that optimal value function.

$$v_1(x) = Tv_0(x) = \max_a (F(x, a) + \beta v_0(r(x, a)))$$

$$v_2(x) = Tv_1(x) = \max_a (F(x, a) + \beta v_1(r(x, a)))$$

etc. This converges to a stable fixed point. Intuition is a 45-degree line and the value function trajectory. Then extract the optimal policy:

$$\gamma(x) = \underset{a}{\operatorname{argmax}} (F(x, a) + \beta v(r(x, a)))$$

Policy function iteration ('Howard improvement'). Initialize a policy, find the corresponding value function, then find an improved policy based on the previous value function until an optimal policy is found. Repeat steps (1) and (2):

$$Hv_n(x) = (F(x, \gamma_n(x)) + \beta v(\gamma_n(x))) \quad (1)$$

$$\gamma_{n+1}(x) = \underset{a}{\operatorname{argmax}} (F(x, a) + \beta v_{n+1}(a)) \quad (2)$$

$$Hv_{n+1}(x) = (F(x, \gamma_{n+1}(x)) + \beta v(\gamma_{n+1}(x))) \quad (1)$$

$$\gamma_{n+2}(x) = \underset{a}{\operatorname{argmax}} (F(x, a) + \beta v_{n+2}(a)) \quad (2)$$

where H is the Howard operator, also a contraction mapping. Policy function iteration can be faster than value function iteration because there is no maximum (there is no 'max' in equation (1)) required.

Howard operator is bounded (solution must convergence to same result as in value function iteration). Proof that Howard operator iterations must increase.

$$\gamma_{n+1}(x) = \underset{a}{\operatorname{argmax}} (F(x, a) + \beta v_n(a))$$

$$\begin{aligned}
v_{n+1}(x) &= F(x, \gamma_{n+1}(x)) + \beta v(\gamma_{n+1}(x)) \\
H_{n+1}v_n(x) &= F(x, \gamma_{n+1}(x)) + \beta v_n(\gamma_{n+1}(x)) \\
&= \max_a (F(x, a) + \beta v_n(a)) = Tv_n(x) \\
Tv_n(x) &\geq \underbrace{F(x, \gamma_n(x)) + \beta v_n(\gamma_n(x))}_{\text{fixed point of } H_n} = v_n(x)
\end{aligned}$$

where $\gamma_n(x)$ are not fixed points (so suboptimal).

$$Tv_n = H_{n+1}v_n \leq H_{n+1}^2v_n \leq H_{n+1}^3v_n \dots \leq v_{n+1}$$

So,

$$\begin{aligned}
H_{n+1}^2v_n(x) - H_{n+1}v_n(x) &= F(x, \gamma_{n+1}(x)) + \beta H_{n+1}v_n(\gamma_{n+1}(x)) \\
&\quad - F(x, \gamma_{n+1}(x)) - \beta v_n(\gamma_{n+1}(x)) \\
&= \beta (H_{n+1}v_n(\gamma_{n+1}(x)) - v_n(\gamma_{n+1}(x))) \geq 0 \\
H_{n+1}^k v_n - H_{n+1}^{k+1} v_n &= F(x, \gamma_{n+1}(x)) + \beta H_{n+1}^k v_n(\gamma_{n+1}(x)) \\
&\quad - F(x, \gamma_{n+1}(x)) - \beta H_{n+1}^{k-1} v_n(\gamma_{n+1}(x)) \\
&= \beta (H_{n+1}^k v_n(\gamma_{n+1}(x)) - H_{n+1}^{k-1} v_n(\gamma_{n+1}(x))) \geq 0
\end{aligned}$$

etc. in a proof by induction.

31.2 Pontryagin's maximum principle

Describes a necessary condition of an optimal trajectory satisfying (1) the co-state equation:

$$\begin{aligned}
-\dot{\lambda}(t) &= H_x(\lambda(t), x(t), u(t)) \\
&= -\dot{\lambda}(t) \underbrace{f_x(x(t), u(t))}_{\text{Jacobian}} + \underbrace{L_x(x(t), u(t))}_{\text{Gradient}}
\end{aligned}$$

and the (2) terminal condition

$$\lambda'(T) = \Psi_x(x(T))$$

31.2.1 Control Hamiltonian

Consider a dynamical system of n first-order DEs:

$$\dot{x}(t) = f(x(t), u(t), t)$$

where x is a vector of state variables and u is a vector of control variables. Once initial conditions are specified, the solution is the trajectory $x(t, x_0, t_0)$.

In optimal control, find $u(t)$ that maximizes or minimizes an objective function. Specifically, optimize a performance index $I(x(t), u(t), t)$ at each point in time.

$$\max_{u(t)} J = \int_0^T I(x(t), u(t), t) dt$$

The solution method involves defining an ancillary function, the control Hamiltonian:

$$\begin{aligned} H(x(t), u(t), \lambda(t), t) &= I(x(t), u(t), t) + \lambda'(t) \dot{x}(t) \\ &= I(x(t), u(t), t) + \lambda'(t) f(x(t), u(t), t) \end{aligned}$$

which combines the objective function and the state equations like a Lagrangian in static optimization, but $\lambda'(t)$ (the costate variables) are functions of time rather than constants.

The goal is to find an optimal control policy function $u^*(t)$ and, with it, an optimal trajectory of the state variable $x^*(t)$, which by Pontryagin's maximum principle are the arguments that maximize the Hamiltonian.

31.2.2 Proof

The system is described by $\dot{x} = f(x(t), u(t))$ and the cost or control functional is

$$J = J(u(t)) = \underbrace{\Psi(x(T))}_{\text{terminal}} + \int_0^T \underbrace{L(x(t), u(t))}_{\text{running cost}} dt$$

where $u(t) = u_{\text{optimal}}(t)$. Perturb $u \rightarrow \delta u \rightarrow \delta J$:

$$\bar{J} = J - \int_0^T \lambda'(t) \underbrace{(\dot{x}(t) - f(x(t), u(t)))}_{\text{from } \dot{x}(t)=f(x,u)} dt$$

where $\lambda'(t)$ is a derivative vector of Lagrange multipliers. Define the control Hamiltonian as

$$H(\lambda, \dot{x}, u) = \lambda' f(x, u) + L(x, u)$$

Then,

$$\begin{aligned} \bar{J} &= J - \int_0^T \lambda'(t) (\dot{x}(t) - f(x(t), u(t))) dt \\ &= \Psi(x(T)) + \int_0^T L(x(t), u(t)) dt - \int_0^T \lambda'(t) (\dot{x}(t) - f(x(t), u(t))) dt \\ &= \Psi(x(T)) + \int_0^T (L(x, u) + \lambda' f(x, u) - \lambda' \dot{x}) dt \\ &= \Psi(x(T)) + \int_0^T (H(\lambda, x, u) - \lambda' \dot{x}) dt \end{aligned}$$

Perturb $u_{optimal}$:

$$\int_0^T |u(t) - v(t)| dt < \epsilon$$

Then

$$\delta \bar{J} = \bar{J}(v) - \bar{J}(u)$$

$$\delta \bar{J} = \Psi(x(T) + \delta x(T)) - \Psi(x(T)) + \int_0^T (H(\lambda, x + \delta x, v) - H(\lambda, x, u) - \lambda' \delta \dot{x}) dt$$

Integrate by parts the last term with $a = \delta x, b = \lambda'$:

$$\int_0^T \lambda' \delta \dot{x} dt = \lambda'(T) \delta x(T) - \underbrace{\lambda'(0) \delta x(0)}_0 - \int_0^T \dot{\lambda}' \delta x dt$$

since $\delta x(0) = 0$ as the perturbation doesn't change the initial state. Then,

$$\begin{aligned} \delta \bar{J} &= \Psi(x(T) + \delta x(T)) - \Psi(x(T)) + \int_0^T (H(\lambda, x + \delta x, v) - H(\lambda, x, u) - \lambda' \delta \dot{x}) dt \\ &= \underbrace{\Psi(x(T) + \delta x(T)) - \Psi(x(T))}_{(1)} - \lambda'(T) \delta x(T) + \int_0^T \underbrace{(H(\lambda, x + \delta x, v) - H(\lambda, x, u))}_{(2)} - \dot{\lambda}' \delta x dt \end{aligned}$$

where (1) and (2) are Taylor expanded:

$$\Psi(x(T) + \delta x(T)) \approx \Psi(x(T)) + \Psi_x(x(T)) \delta x(T)$$

$$H(\lambda, x + \delta x, v) \approx H(\lambda, x, v) + H_x(\lambda, x, v) \delta x$$

Then,

$$\delta \bar{J} = (\Psi_x(x(T)) - \lambda'(T)) \delta x(T) + \int_0^T (H_x(\lambda, x, v) + \dot{\lambda}') \delta x dt + \int_0^T (H(\lambda, x, v) - H(\lambda, x, u)) dt$$

Now simplify this equation. Until now, λ was not restricted in any way. So can select λ . Select it s.t.

$$\begin{aligned} -\dot{\lambda}(t) &= H_x(\lambda(t), x(t), u(t)) \\ &= -\dot{\lambda}(t) \underbrace{f_x(x(t), u(t))}_{\text{Jacobian}} + \underbrace{L_x(x(t), u(t))}_{\text{Gradient}} \end{aligned}$$

Further, impose the terminal condition

$$\lambda'(T) = \Psi_x(x(T))$$

Then,

$$\delta \bar{J} = \int_0^T (H(\lambda, x, v) - H(\lambda, x, u)) dt$$

The Hamiltonian is always a max (and not a min). So want above to be < 0 .
At every point in time, want

$$H(\lambda(t), x(t), v(t)) \leq H(\lambda(t), x(t), u(t))$$

Proof by contradiction. Assume there exists a function ϕ s.t.

$$H(\lambda(t), x(t), \phi(t)) > H(\lambda(t), x(t), u(t))$$

Since H is continuous, there exists an interval $(t_1 - \sigma, t_1 + \sigma)$ where this holds. Let $v(t) = u(t)$ except in this interval, where $v(t) = \phi(t)$. For the perturbed function

$$\delta \bar{J} = \int_{t_1 - \sigma}^{t_1 + \sigma} H((\lambda, x, v) - H(\lambda, x, u)) dt > 0$$

where the inequality results from the integrand always being greater than zero. Contradiction.

31.3 Travelling salesperson person using computation

Dynamic programming through the travelling salesperson problem (TSP).

31.3.1 Problem types

TSP decision problem (given graph G , is there a path with path shorter than b) is NP-complete. If NP complexity is extended to optimization problems, TSP optimization is an NP-hard problem (given graph G with budget b , find the shortest path).

Can be depicted on a directed or undirected (symmetric or asymmetric) graph. Applications: travel routes, circuit board design, order of reconfiguration steps.

Euclidean TSP: just get points on Cartesian coordinates. Complete graph. Symmetric. Satisfies triangle inequality.

Shortest common superstring. E.g. Shortest common string of BCD, CDA, and ABC is ABCDA. Often applied in genome assembly.

31.3.2 Simple algorithms

Naive solution. Envision as shortest common superstring problem. Just concatenate, and look for permutation that minimizes overlap. Make directed graph of overlaps and find the maximum path, which is the shortest TSP path.

Brute force algorithm. Try all permutations and select best path. Cycles mean $n \cdot n!$ permutations, but if fix start and end at same node, $n!$. Infeasible for large n .

Random permutations. Expected path = $\frac{1}{n-1} \sum w$ where w is edge weight.

Nearest neighbor heuristic. Often produces reasonable result. However, sometimes gets trapped by disproportionately heavy edges in non-Euclidean graphs. Can also be suboptimal even in Euclidean scenarios.

Branch and bound. Can take a long time in some cases but exact. Create a tree graph showing the permutations with the start node at the root and branch tips. Cease calculating branches that exceed shortest result to that point. Can calculate various lower bounds, like:

$$\frac{1}{2} \sum_{v \in V} \text{two min edges to } v$$

Minimum spanning tree ($O(n)$) by removing one edge. Must be smaller because an edge is missing.

31.3.3 Dynamic programming solutions

Exact and powerful. Break into small subproblems, and solve one by one (generally from the end result, but can be bottom-up), recycling intermediate results (which are memoized). Gradually increase the subproblem sizes. Example. Used to return Fibonacci numbers.

TSP recurrence relation:

$$C(i, S) = \min(C(j, S - [i],) + w(j, i))$$

over all $j \in S$ and where $j \neq i$. Start at the end, and look for penultimate set S . Then look for lowest C . Iterate.

Number of sets S is 2^n . Binary has property of natural correspondence between integers in range 0 to $2^n - 1$. E.g. in 3-digit binaries, 5 is 101 which in turn is 0,2 because it has a 1 in the 0's digit and a 1 in the leftmost digit. Flip particular bits to get permutations in set S so use binary shift operator $<<$. E.g. $k(1 << j)$. Use XOR with 000010000 to flip 0 or 1 in j 's position.

Dynamic programming solution is $O(n^2 2^n)$. i goes from 0 to n and j from 0 to n . Better than $n!$ but still slow.

31.3.4 Linear programming

With $i \neq j$, $1 \leq i, j \leq N$, with whether to use an edge $x_{ij} = 1$ or not $x_{ij} = 0$, minimize

$$\sum_i \sum_j w_{ij} x_{ij}$$

Subject to each vertex having one edge out:

$$\sum_i^N x_{ij} = 1$$

And each vertex having one edge in:

$$\sum_j^N x_{ij} = 1$$

Further conditions can be added to prevent multiple loops.

31.3.5 Approximate algorithms

Exact algorithms are not solvable in polynomial time. Approximate solution guaranteed within almost-linear time but 2x longer than best solution. Many variations.

2-approximation. Only for metric version - graph edges are symmetric. Satisfies triangle inequality. Euclidean TSP is a specific instance of a metric TSP.

If G is an undirected graph, then the minimum spanning tree MST is less than the TSP: $MST(G) \leq TSP(G)$. Note removing an edge from the TSP results in a spanning tree (though might not be the minimum spanning tree). Steps:

- $T \leftarrow MST(G)$
- $D \leftarrow T$ with each edge doubled
- $D \leftarrow$ Eulerian cycle C in D . Doubling edges means all vertex degrees are even, allowing Eulerian paths.
- Find a cycle that visits each node just once. Avoid repeating a node by skipping previously-visited nodes - this will shorten the cycle.

Proof. The total length of $MST(G)$ is at most $OPT(G)$ because of the double-edged minimum spanning tree, which is further shortened by avoiding repeating nodes.

Local search. Start with initial solution, try to improve. Finds local optimum, not necessarily a global optimum. The more edges can be simultaneously changed, the better the result but longer running time. More complicated local searches sometimes increase weights of cycles, e.g. simulated annealing.

32 Game theory

Instead of a single agent, use multiple agents. Strategies are like MDPs. Maps state to action.

In a 2-player, zero-sum deterministic perfect information game, minimax=maximin is the optimal strategy. So there always exists an optimal pure strategy for each player. Where the payoff matrix 'crosses' for both players.

Works for non-deterministic (stochastic) games as they can be mapped into deterministic payoff matrices.

Imperfect information: minimax does not equal maximin. Use mixed strategy equilibria.

Nash equilibrium. $\forall i, s_i^* = \arg \max_{s_i} u_i(s_i, i = 1 \dots n)$.

Variations: cooperative and altruism.

32.1 Repeated games

Tit-for-tat is a famous strategy and both players playing this is a deterministic Nash equilibrium. This is like an MDP. Folk theorem: in repeated games, using threats and retaliation leads to cooperation. Grim trigger not credible.

32.2 Subgame perfection

Subgame perfect: take best response independent of history.

32.3 Stochastic games

Stochastic games generalize MDPs.

Zero-sum stochastic games. The basic form

$$Q_i^*(s, (a, b)) = R_i(s, (a, b)) + \gamma \sum_{s'} T(s, (a, b), s') \max_{a', b'} Q_i^*(s'(a', b'))$$

is optimistically delusional as it assumes the joint action (a,b) benefits player i only. The form

$$Q_i^*(s, (a, b)) = R_i(s, (a, b)) + \gamma \sum_{s'} T(s, (a, b), s') \min_{a', b'} \max_{a', b'} Q_i^*(s'(a', b'))$$

is the minimax-Q-learning, which is an improvement but may be unstable. The form

$$Q_i^*(s, (a, b)) = R_i(s, (a, b)) + \gamma \sum_{s'} T(s, (a, b), s') \text{Nash}_{a', b'} Q_i^*(s'(a', b'))$$

is Nash-Q-learning, which is stable.

32.4 Solution concepts

Nash, correlated equilibria, Stackelberg, trembling hand, subgame perfection, etc.

32.5 Correlated equilibrium

Extra information (e.g. referee, traffic lights for drivers) that guides when there are many NEs. Referee can provide incentives that can boost payoffs for everyone, and doesn't show the worst outcome (e.g. dare, dare).

Can find in P-time, unlike possibly NEs generally. All mixed NE are correlated equilibria, and all convex combinations of mixed NEs are correlated.

32.6 Coco values

Cooperative-competitive values. Can reveal side payments will raise utilities. The Coco value is

$$Coco(U, \bar{U}) = \underbrace{\max_{\max} \left(\frac{U + \bar{U}}{2} \right)}_{\text{cooperative}} + \underbrace{\min_{\max} \left(\frac{U - \bar{U}}{2} \right)}_{\text{competitive}}$$

where U is player i and $\bar{U}$ is the other player. Both maxmax (max value of payoff matrix) and minmax (min value of differences in payoffs) are computable in polynomial time. Side payments are

$$p = Coco(U, \bar{U}) - U(*)$$

$$\bar{p} = Coco(U, \bar{U}) - \bar{U}(*)$$

33 Control Theory

Control systems are mechanism that alter the future state of a system. Control theory is a strategy to select inputs.

33.1 Open and closed loop control systems

Open-loop. One way and has no feedback: $Input \rightarrow PLANT \rightarrow Output$.

Closed-loop. The output y is connected to a sensor H , which compares the signal with a reference signal V , generating a difference $E = V - yH$, which is subsequently fed into a controller D . The controller compensates for E and creates an input signal for the plant G .

Transfer function. Convert closed-loop system to an open-loop system. Combine D and G $\rightarrow D \cdot G$. Then,

$$y = EDG \rightarrow E = \frac{y}{DG}$$

$$E = V - yH = \frac{y}{DG}$$

$$y = \frac{DGV}{1 + DGH} = V \frac{DG}{1 + DGH}$$

$$\text{Output} = (\text{Reference} = \text{Input}) \frac{DG}{1 + DGH}$$

and the system is open-loop. Can conceptualize the entire closed loop system as one whole, and that whole is an open loop system.

33.2 Time and frequency domains

Distance is a function of time: $d = f(t) = vt$. Can plot distance versus time. Motion of spring is also a function of time analogously. Motion of a spring is $m\ddot{x} = -kx \rightarrow m\ddot{x} + kx = 0$ which is an ODE with general solution

$$x(t) = A \sin(\omega t + \phi)$$

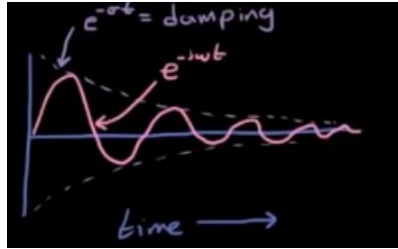
Can plot $x(t)$ (amplitude) against the time domain, which is sinusoidal. Can alternatively plot $x(t)$ against ω , i.e. the frequency domain. In this case, it will just be a Dirac delta function (if the phase shifts are ignored).

If there is $\omega_1 \neq \omega_2$, there is a superposition that is messy on the time domain. However, it is just two Dirac delta spikes in the frequency domain.

A complicated periodic function (with period T) can be represented by a Fourier series, but the frequencies in this case are multiplies of the first harmonic by integers. This can be easily visualized on the frequency domain with discrete Dirac deltas. But as $T \rightarrow \infty$, the harmonics become more densely packed together (as the first harmonic approaches 0) and eventually become continuous. This is the Fourier transform:

$$\mathcal{F} = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

Laplace transform. Transforms from the time domain to the S-plane, $S = \sigma + i\omega$ complex plane. The S-plane includes the dampening of a spring σ coefficient, whereas the Fourier transform only characterizes the sinusoidal action through ω coefficient.



$$\mathcal{L} = \int_{-\infty}^{\infty} f(t)e^{-\sigma t}e^{-i\omega t}dt = \int_{-\infty}^{\infty} f(t)e^{(-\sigma+i\omega)t}dt = \int_{-\infty}^{\infty} f(t)e^{st}dt$$

33.3 Laplace transforms

33.3.1 Laplace transforms and Dirac deltas

For any partially continuous function $f : \mathcal{R} \rightarrow \mathcal{R}$, the Dirac delta function has the nice property $f(t) = \int_{-\infty}^{\infty} f(s)\delta(ts)ds$. So, any partially continuous function can be written as a sum of Dirac delta functions. This is particularly useful for LTI systems. If we know the impulse response of a LTI system, we can calculate its output for a specific input function using the above property. In fact, it is called the "convolution integral".

33.3.2 Laplace transforms and transfer functions

The Laplace transform of an impulse response $\frac{y(t)}{x(t)}$ is the transfer function $\frac{Y(s)}{X(s)}$.

Zeros of a transfer function are s-values where the numerator becomes zero (and so makes the transfer function itself become zero). Poles of a transfer function are s-values where the denominator becomes zero.

Example. Find transfer function of the DE:

$$\ddot{y}(t) + 7\dot{y}(t) + 12y(t) = \dot{x}(t) + 2x(t)$$

where $x(t)$ is input and $y(t)$ is output.

$$s^2Y(s) + 7sY(s) + 12Y(s) = sX(s) + 2X(s)$$

$$Y(s)(s^2 + 7s + 12) = X(s)(s + 2)$$

$$\frac{Y(s)}{X(s)} = \frac{s + 2}{s^2 + 7s + 12} = \frac{s + 2}{(s + 3)(s + 4)}$$

Intuition. Zero gives $Y(s) = 0$, where the integral or summation of the oscillating signal (i.e. the Laplace transform) is zero (the + and - portions cancel each other out). So the output in LT space is 0. Pole gives $X(s) = 0$, where the integral or summation (i.e. the LT) is infinite (makes the signal's oscillations are in each oscillation on net +, e.g. resonance as the two frequencies are

identical so squared so are completely positive). The $0 \cdot \infty = \frac{0}{0} = \frac{Y(s)}{X(s)}$ is the transfer function of the system. Take inverse LT of the transfer function to get the impulse response (i.e. the transfer function in the time domain).

33.3.3 Poles and stability

In the transfer function $\frac{Y(s)}{X(s)}$, the characteristic equation is $X(s) = 0$, and the poles are the roots of the equation. If the real components of the poles are negative, the system is stable.

Example.

$$\frac{Y(s)}{X(s)} = \frac{1}{(s+1)(s+2)}$$

then the poles are at $s = -1, -2$. Using partial fractions,

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

Using the inverse Laplace transform, this becomes $Ae^{-t} + Be^{-2t}$ which is exponentially decreasing.

Poles with negative real values on the s-plane are stable. Poles with positive real values are unstable. Poles on the imaginary axis are oscillatory. A pole on the origin is stable (just a constant).

33.3.4 Steady states

Occurs when all transients are gone, i.e. when $\sigma = 0$ s.t. $s = \sigma + i\omega = 0 + i\omega = i\omega$.

Example. LTIs with $\cdot 2$ and $\cdot \frac{1}{2}$. Then, output is $y(t) = 2u(t) + \frac{u(t)}{2}$,

$$\frac{y(t)}{u(t)} = \frac{2s+1}{s} = \frac{2(i\omega)+1}{i\omega} = 2 - \frac{1}{\omega}i$$

since $s = i\omega$. The real part of s is 2, and the imaginary part is $\frac{1}{\omega}$. On the s-plane, the amplitude (gain) is $length = \sqrt{Re^2 + Im^2}$ and the angle to the horizontal is the phase $\phi = \arctan(Re, Im)$. Sweeping through frequencies ω , the s-graph forms a vertical line on the real component 2.

33.3.5 Transfer function types

All transfer functions are one of four subtypes: (1) constants, (2) poles and/or zeros at origin, (3) real poles and/or zeros, (4) complex poles and/or zeros. Transfer functions are combinations of these. These can be represented on Bode plots.

33.4 LTI systems

33.4.1 Properties

All linear and time invariance (LTI) systems have (1) homogeneity, (2) superposition, and (3) time invariance. (1) and (2) are a result of linearity.

1. Homogeneity. I/O is scaled. If input is doubled, output is correspondingly doubled.
2. Superposition (additivity). Sum of inputs = sum of outputs.
3. Time invariance. Translating input by time $+t$ will shift output by time $+t$.

33.4.2 Why use LTI systems?

Most real systems aren't LTI (usually not time invariant), but LTI is simple to solve.

If the impulses are continuous and increasing (positive straight slope), then the velocities of the object move faster with time. The infinite impulse responses are superimposed. This is a convolution in the time domain of input * impulse response. This is a difficult integration. But if both input and impulse response are converted to the S-domain using the Laplace transform, then the convolution becomes multiplication. Normally, if $u(t)$ is the input and $g(t)$ is the LTI impulse response, then the output is the convolution integral

$$y(t) = \int u(t)g(t)dt$$

However, converting to the S-domain through the Laplace transform s.t. $U(s)$ is the input and $G(s)$ is the LTI impulse response,

$$Y(s) = U(s)G(s)$$

where $G(s)$ is the **transfer function** and the system is easier to solve. This is particularly effective if there are multiple LTIs (e.g. actuators) in series. This is a convolution in the time domain but just multiply the transfer functions in the s-domain. (Definition of transfer function: Laplace transform of the impulse response of the LTI system when system initial conditions are 0.) A Fourier transform is not enough to represent phase shift ϕ , but Laplace transforms will include them as the angle ϕ . In both Fourier and Laplace transforms, amplitude is the distance from the origin.

LTI approximations are good in their 'operating region', like a spring which is not linear (according to Hooke's law) if it is stretched too far or compressed too much.

33.4.3 Sinusoidal solutions

There are limited mathematical operations on an LTI system: (1) $f(t) \cdot a$, (2) $\frac{df(t)}{dt}$, (3) $\int f(t)dt$, (4) $f_1(t) + f_2(t)$. A sinusoidal $f(t)$ is the only function for which frequency doesn't change in an LTI impulse response, though the amplitude and phase can change.

Example 1. LTI with two impulse responses, one with $\cdot 2$ and the other with $\int dt$. Then, input $x(t) = \sin(\omega t)$ leads to output $y(t) = 2\sin(\omega t) + \int \sin(\omega t)dt$. Through mathematical manipulation, including use of complicated trigonometric identities, this reduces to

$$y(t) = \sqrt{8}\sin(\omega t - 0.785)$$

where frequency remains ω .

33.5 Stability

BIBO, bounded input leads to bounded output. A ball in a valley is stable, a ball at a top of a hill is unstable. Want to build **margin** into the system, but this can't be so stable that it's too rigid to be practical.

Look at poles in the s-plane. If they have negative real values, then they're stable. The imaginary component only indicates frequency of oscillation, so isn't relevant to stability. Only the s-plane poles.

33.6 Final value and steady state error

Calculating the final value (i.e. when $t \rightarrow \infty$) can be difficult for complex systems, e.g.

$$\ddot{x}(t) + 3\dot{x}(t) + 2x(t) = \delta(0)$$

so convert using Laplace transform:

$$s^2X(s) + 3sX(s) + 2X(s) = 1$$

which is simpler as it's just algebra. Analogously, use this for $t \rightarrow \infty$.

Final value theorem.

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

However, this only works where the poles are on the origin (constant) and/or with negative real values (0). If the pole has positive real value, then the final value is infinite, and if on the imaginary axis (but not at the origin) then it's oscillatory.

Steady state error. If the input is $u(s)$, output $E(s)$, and sensor $G(s)$ that feeds $y(s)$ to the input, then

$$E(s) = u(s) - y(s)$$

$$y(s) = G(s) - E(s)$$

$$E(s) = \frac{u(s)}{1 + G(s)}$$

and the steady state error is

$$\lim_{s \rightarrow 0} \frac{u(s)}{1 + G(s)}$$

33.7 Controllers

Difference between reference (desired) signal and sensor output is the error. Controller input is error and output is a signal to make the plant act s.t. that the error moves to zero.

33.7.1 PID controller

Proportional-integral-derivative controller. Error input is split into at most three streams: (1) proportional, where the error is multiplied by a constant K_P , (2) integral, where the error multiplied by K_I and then integrated, so the total process is $K_I \cdot \frac{1}{s}$ in s-space, and (3) derivative, where the error is multiplied by K_D and then differentiated, so the total process is $K_D \cdot s$ in s-space. Then the outputs are combined.

A P controller reduces the error by multiplying the error by a proportion (constant). However, it may lead to instability if the signal is changed too rapidly for the actuator to act. This can be addressed using a PD controller.

A PD controller reduces the error by multiplying by a proportion and differentiating. As the error declines, the slope of the error changes, so the proportion declines. However, if the error doesn't change, nothing happens. This can be addressed using a PID controller.

A PID controller will additionally respond to the size of the error by integrating it. If the error is large, it will increase the proportion and if declines, it will decrease the proportion.

33.8 Classical control techniques

33.8.1 Bode plot

Two graphs. The first has gain (dB) which is logarithmic versus log frequency. The second is phase ϕ versus the same log frequency as the first graph. The two

graphs indicate how gain and phase evolve as they are swept across frequencies ω .

33.8.2 Routh-Hurwitz stability criterion

If there is even one unstable pole (i.e. has a positive real value in the s-plane), then the entire system is unstable, no matter how many stable poles it has. To quickly determine whether a system is stable or unstable, look at the denominator of the transfer function. The Routh-Hurwitz criterion determines whether the denominator has unstable poles, e.g. if the denominator polynomial in the transfer function doesn't have all positive or negative signs, then it will be unstable (if it has negative real values, the linear factors of the polynomial will all be positive so all signs should be positive; multiplying the entire polynomial by -1 will yield all negative signs). However, even if all signs are the same in the characteristic equation polynomial, it might not be stable. Use Routh array to determine whether a system is stable or not, and what parameters would lead to gain and/or phase margin (optimal values).

33.8.3 Root locus method

In the poles, there may be one or more parameters k that are unknown. Example.

$$TF = \frac{s^2 + s + 1}{s^3 + 4s^2 + ks + 1}$$

when k is varied, it traces out a locus on the s-plane, which helps determine which k -values are stable and which have which characteristics (e.g. oscillation). There are many short-cut rules to drawing and analyzing the root loci.

33.8.4 Nyquist stability criterion

Transform or map s-plane coordinates to w-plane by inputting s-coordinates into the transfer function. A different set of rules then can be used to determine zeros, poles, and therefore stability, such as how many loops are produced in the w-plane.

33.9 State space representations

A state-space representation is a mathematical model of a physical system as a set of input, output and state variables related by first-order differential equations or difference equations. State variables are variables whose values evolve over time in a way that depends on the values they have at any given time and on the externally imposed values of input variables. Output variables' values depend on the values of the state variables.

State variables are the minimum number of variables to describe the state,

$x(t)$. This is described in a set of ODEs, such as

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

Can change $x(t) \leftrightarrow z(t)$, which will change the matrices A, B, C, and D. This is just a coordinate transformation, but the number of state variables (the dimension) will remain the same (not degenerate).

A closed loop system can be represented as a DE in state space, where the state space is a multi-dimensional (multi-DOF) space, with state x :

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}$$

where y is the measurement of x after passing the input through C (which will be a linear combination of x). The u is the actuator or controller. If $u = -Kx$, then the system becomes

$$\begin{aligned}\dot{x} &= Ax + Bu = Ax - BKx = (A - BK)x \\ y &= Cx\end{aligned}$$

and the eigenvalues (and therefore the stability) change.

33.9.1 Eigenvalues

If the DE were just $\dot{x} = Ax$, it is stable if $A < 0$ and unstable if $A > 0$ because the solution is $x(t) = x(0)e^{At}$. A is a matrix, so e^{At} is expanded using Taylor series.

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

but computations are complicated. So transform and represent in more convenient coordinates. The eigendecomposition of A is

$$\begin{aligned}AT &= TD \\ T^{-1}AT &= D\end{aligned}$$

where D is a matrix with eigenvalues along its diagonal. Let there be a coordinate transformation

$$x = Tz$$

where z is the new coordinate system. Then,

$$\dot{x} = T\dot{z}$$

but from above, without the controller, $\dot{x} = Ax$. So,

$$\dot{x} = T\dot{z} = Ax$$

$$T\dot{z} = ATz$$

$$\boxed{\dot{z} = T^{-1}ATz = Dz}$$

which means the dimensional components of z are independent of each other (i.e. linearly independent).

$$\begin{pmatrix} \dot{z}_1 \\ \vdots \\ \dot{z}_n \end{pmatrix} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} = \begin{pmatrix} \lambda_1 z_1 \\ \vdots \\ \lambda_n z_n \end{pmatrix}$$

So,

$$z(t) = e^{Dt}z(0)$$

$$\begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} = \begin{pmatrix} e^{\lambda_1 t} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & e^{\lambda_n t} \end{pmatrix} \begin{pmatrix} z_1(0) \\ \vdots \\ z_n(0) \end{pmatrix}$$

To convert back to x -space, transform with $A = TDT^{-1}$.

$$\begin{aligned} e^{At} &= I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots \\ &= TT^{-1} + TDT^{-1}t + \frac{(TD^2T^{-1})t^2}{2!} + \frac{(TD^3T^{-1})t^3}{3!} + \dots \\ &= T \left(I + Dt + \frac{D^2 t^2}{2!} + \frac{D^3 t^3}{3!} + \dots \right) T^{-1} \\ &= Te^{Dt}T^{-1} \end{aligned}$$

So,

$$x(t) = x(0)e^{At} = x(0)Te^{Dt}T^{-1}$$

But since $x = Tz$, $T^{-1}x(0) = z(0)$ and $e^{Dt}T^{-1}x(0) = z(t)$ and $x(t) = Tz(t)$.

33.9.2 Continuous stability

If there is a complex eigenvalue, they will come in pairs and when returning to x -space, they will be real,

$$\lambda = a \pm ib$$

$$e^{\lambda t} = e^t(\cos(bt) \pm i\sin(bt))$$

where a indicates stability. If $a < 0$ then stable, if $a > 0$ is unstable, as with Laplace transforms. If unstable, then use Bu to drive the eigenvalues to stability (left side of complex plane).

33.9.3 Discrete stability

$$x_{k+1} = \tilde{A}x_k$$

where $x_k = x(k\Delta t)$. Further, $\tilde{A} = e^{A\Delta t}$.

$$x_{k+1} = \tilde{A}x_k$$

$$x_1 = \tilde{A}x_0 = (\tilde{T}\tilde{D}\tilde{T}^{-1})x_0 \rightarrow \lambda$$

$$x_2 = \tilde{A}^2x_0 = (\tilde{T}\tilde{D}\tilde{T}^{-1})x_0 \rightarrow \lambda^2$$

$$x_n = \tilde{A}^n x_0 = (\tilde{T}\tilde{D}\tilde{T}^{-1})x_0 \rightarrow \lambda^n$$

where λ is the eigenvalue from above. Since

$$\lambda = a \pm ib = Re^{i\theta}$$

which can be depicted in the complex plane. Further,

$$\lambda^2 = R^2 e^{i2\theta}$$

$$\lambda^n = R^n e^{in\theta}$$

So in complex plane, if the $R > 1$ then unstable. So the unit circle in the complex plane is the region of stability, and outside the unit circle unstable. This circle conformally maps to the left side (negative real values) of the complex plane.

33.10 Equivalent representations

There are three equivalent representations of a system.

State space representation.

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

Time domain. The impulse response is the convolution integral (convolution of the impulse response function h with the impulse u):

$$y(t) = \int_0^t h(t-\tau)u(\tau)d\tau$$

Frequency domain. The transfer function is

$$G(s) = \frac{B}{C(sI - A)}$$

Inputting a sinusoidal wave ($\sin(\omega t)$) into an LTI will maintain the frequency (though amplitude and phase can change $A \sin(\omega t + \phi)$). Then, with transfer function $G(s) = \frac{Y(s)}{U(s)}$,

$$A = |G(i\omega)|$$

$$\phi = \angle G(i\omega)$$

Example. Laplace transform a system with $\bar{x}$ being the solution,

$$\begin{aligned}\ddot{x} + \dot{x} + x &= u \\ s^2 \bar{x} + s\bar{x} + \bar{x} &= \bar{u} \\ (s^2 + s + 1)\bar{x} &= \bar{u} \\ \frac{\bar{x}}{\bar{u}} &= \frac{1}{(s^2 + s + 1)} = G(s)\end{aligned}$$

The frequency domain can be represented on a Bode plot, indicating its amplitude and phase over different frequencies.

The Laplace transform of a derivative is

$$\mathcal{L}(\dot{x}) = s\bar{x}(s) - x(0)$$

So for the system

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx \\ s\bar{x}(s) - x(0) &= A\bar{x} + B\bar{u} \\ (sI - A)\bar{x} &= B\bar{u} + x(0) \\ \bar{y}(s) &= C\bar{x}(s)\end{aligned}$$

And

$$G(s) = \frac{\bar{y}(s)}{\bar{u}(s)} = \frac{B}{C(sI - A)}$$

33.10.1 Impulse response

Graph of amplitude over time. In the Laplace form

$$G(s) = \frac{\bar{y}(s)}{\bar{u}(s)} = \frac{B}{C(sI - A)}$$

$u(s) = 1$ in the impulse response, so

$$y(t) = \mathcal{L}^{-1}(G(s))$$

33.11 Linearizing non-linear systems

If $\dot{x} = f(x)$ is not linear, convert to $\dot{x} = Ax$. Example.

$$\begin{aligned}\dot{x}_1 &= f(x_1, x_2) = x_1 x_2 \\ \dot{x}_2 &= f(x_1, x_2) = x_1^2 + x_2^2\end{aligned}$$

Find fixed points $\bar{x}$ s.t. $f(\bar{x}) = 0$ and then linearize about $\bar{x}$ using the Jacobian:

$$J = \begin{pmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{pmatrix}$$

At the fixed point,

$$\dot{x} = f(x) = f(\bar{x}) + J(x - \bar{x}) + J^2(x - \bar{x})^2 + \dots$$

Choosing the region around the fixed point to be very small, the higher-order terms disappear. Moreover, $f(\bar{x}) = 0$. So,

$$\dot{x} \approx J(x - \bar{x}) = \frac{Df}{Dx} \Delta x$$

which is approximated as

$$\Delta \dot{x} = A \Delta x$$

which is a linear system. Valid near fixed point, which is good control. Note this only works if the real part of the eigenvalues are NOT zero (i.e. not purely imaginary), as the approximation breaks down.

Example. Pendulum. Calculate Jacobian from equations of motion (an ODE system, with $\theta, \dot{\theta}$). For small oscillations at fixed points (down vertical or up vertical positions), calculate J. Eigenvalues of J (J=A) in down vertical are stable. However, one of the eigenvalues of J (J=A) of up vertical is unstable.

33.12 Controllability

Add feedback control Bu:

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

Then, $u = -Kx$ is the optimal solution. Can control K to do anything (including steering in unstable states). Some systems are controllable (alter K to get poles with negative real values in complex plane) and others are not. Usually A and B are set.

Example. This system is uncontrollable because u cannot affect x_1 :

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

Example. This system is controllable:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} u$$

If systems are coupled then it's controllable.

Iff the control matrix has full column rank

$$\mathcal{C} = (B, AB, A^2B, \dots, A^{n-1}B)$$

then it's controllable. The control matrix intuition seen from the discrete case:

$$x_{k+1} = Ax_k + Bu_k$$

where there is an impulse from $u_0 = 1$. Then

$$x_0 = 0, x_1 = B, x_2 = AB, x_3 = A^2B, \dots, x_n = A^{n-1}B$$

So full rank is required because the impulse must 'touch' all parts of the system.

Iff controllable, then poles (=eigenvalues) can be placed ANYWHERE by adjusting K. Moreover, there is **reachability**: If controllable, then there are $u(t)$ can drive $x(t)$ anywhere.

33.12.1 Degree of controllability

Taking the $U\Sigma V$ singular value decomposition (SVD) of the control matrix $\mathcal{C}$, the largest eigenvectors in U are the easiest directions or the most controllable. This is intuitive. Mathematically, begin with

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

Controllability Gramian:

$$W_t = \int_0^t e^{A\tau}BB'e^{A'\tau}d\tau \in \mathcal{R}^{n \times n}$$

$$W_t \approx \mathcal{C}\mathcal{C}'$$

$$W_tX = \lambda X$$

The eigenvectors corresponding to the largest eigenvalues λ are the easiest to control.

33.12.2 Popov-Belevitch-Hautus

Popov-Belevitch-Hautus (PBH) test for controllability is alternative to using control matrix $\mathcal{C}$. A and B system is controllable iff

$$\text{rank}((A - \lambda I)B) = n$$

Note $\text{rank}(A - \lambda I) = n$ except at eigenvalues. Moreover, B needs to have some component in each eigenvector direction to compensate for rank deficiency when λ is an eigenvalue (this is intuitive - want control in directions of eigenvectors which determine stability). There's a high probability that this will happen. So controllability is likely.

33.12.3 Cayley-Hamilton

Every square matrix A satisfies its own characteristic equation. Intuition: matrices are rescalings in appropriate eigenbases, and as eigenvalues satisfy the characteristic equation, the scaling is 0 in each base direction.

$$\det(A - \lambda I) = 0$$

$$\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_2\lambda^2 + a_1\lambda + a_0 = 0$$

Then,

$$A^n + a_{n-1}A^{n-1} + \dots + a_2A^2 + a_1A + a_0I = 0$$

Then A^n is a linear combination of 'basis' matrices (like adding 'linearly independent' matrices to create the matrix A^n , or like adding eigenvectors as bases):

$$A^n = -a_{n-1}A^{n-1} - \dots - a_2A^2 - a_1A - a_0I$$

And any A^w where $w \geq n$ can be rewritten in terms where the highest power is A^{n-1} . For A^{n+1} :

$$A^{n+1} = -a_{n-1} \underbrace{A^n}_{*} - \dots - a_2A^3 - a_1A^2 - a_0A$$

where the (*) term can be rewritten with the highest power being A^{n-1} as in the original equation. Similarly, for A^{n+2} :

$$A^{n+2} = -a_{n-1} \underbrace{A^{n+1}}_{*} - a_{n-2} \underbrace{A^n}_{*} - \dots - a_2A^4 - a_1A^3 - a_0A^2$$

and again where the (*) terms can be rewritten with the highest power being A^{n-1} as in the original equation.

The above is summarized as

$$A^{\geq n} = \sum_j^{n-1} a_j A^j$$

Cayley-Hamilton can be used to re-express e^{At} . Before, it was written with infinite terms:

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \dots$$

Applying Cayley-Hamilton,

$$e^{At} = \phi_0(t)I + \phi_1(t)A + \phi_2(t)A^2 + \dots + \phi_{n-1}(t)A^{n-1}$$

which has a finite number of terms. However, $\phi(t)$ can be difficult to compute.

33.12.4 Controllability leads to reachability

Reachability. If a vector ξ in state space is reachable, then there is an input $u(t)$ s.t.

$$\xi = \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau$$

which is a convolution integral (difficult to compute). From above,

$$e^{At} = \phi_0(t)I + \phi_1(t)A + \phi_2(t)A^2 + \dots + \phi_{n-1}(t)A^{n-1}$$

So,

$$\begin{aligned} \xi &= \int_0^t (\phi_0(t-\tau)IBu(\tau) + \phi_1(t-\tau)ABu(\tau) + \dots + \phi_{n-1}(t-\tau)A^{n-1}Bu(\tau)) d\tau \\ &= B \int_0^t (\phi_0(t-\tau)u(\tau) d\tau + AB \int_0^t \phi_1(t-\tau)u(\tau) d\tau + \dots + A^{n-1}B \int_0^t \phi_{n-1}(t-\tau)u(\tau) d\tau) \\ &= \underbrace{(B, AB, \dots, A^{n-1}B)}_{\mathcal{C}} \begin{pmatrix} \int_0^t \phi_0(t-\tau)u(\tau) d\tau \\ \int_0^t \phi_1(t-\tau)u(\tau) d\tau \\ \vdots \\ \int_0^t \phi_{n-1}(t-\tau)u(\tau) d\tau \end{pmatrix} \end{aligned}$$

If $\mathcal{C}$ has full rank, then any ξ can be reached as the value in each column of $\mathcal{C}$ is controllable. Moreover, there are an infinite $u(t)$'s to get to ξ as there are many paths.

33.12.5 Pendulum example

In the linearization equation

$$\dot{x} = Ax + Bu$$

at the fixed points, check controllability using A and B.

33.13 Observability

There is duality between controllability (A,B) and observability $\mathcal{O}(A, C)$.

$$\mathcal{O} = (C, AC, A^2C, \dots, A^{n-1}C)'$$

System is observable (can estimate x from y) if full rank: $rank(\mathcal{O}) = n$.

Analogous to controllability, SVD (U, Σ, V) of $\mathcal{O}$ indicates order of less noisy (i.e. higher signal-to-noise) observability directions (in the matrix V). Can also do this by calculating determinant of the observability Gramian, which indicates dimensions of observability in the state space. Specifically, isolate each x_i in $y = Cx$ as the measurement (so $y_1 = x_1$ for instance and $y_{i \neq 1} = 0$)

and calculate the Gramian. Those x-variables producing larger Gramian determinants lead to more observability. Usually, the observability here applies to all x-variables, not just the x_1 variable, but for due diligence check the size of eigenvalues to see what the dimensions of the determinants are.

33.13.1 Full-state estimation

Often, the full state data isn't available (e.g. suboptimal and/or expensive sensors). Usually less: $y = Cx$. So the system (PLANT) is

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

Take measurements y and estimate the full state of the system $\hat{x}$ (not exactly x because there can be noise, etc.). The full-state estimator (e.g. Kalman filter) has inputs both u and y and is a linear dynamical system:

$$\dot{\hat{x}} = f(u, y)$$

Then

$$u = -K\hat{x}$$

which can use LQR.

For the full-state estimator, which inputs u, y and outputs $\hat{x}$ and is a linear dynamical system,

$$\frac{d}{dt}\hat{x} = A\hat{x} + Bu + K_f(y - \hat{y})$$

where K_f is the gain matrix (a constant matrix), $\hat{y} = C\hat{x}$. The $y - \hat{y}$ is a correction update for the new $\hat{x}$.

$$\begin{aligned}\frac{d}{dt}\hat{x} &= A\hat{x} + Bu + K_f(y - \hat{y}) \\ &= A\hat{x} + Bu + K_f y - K_f C\hat{x} \\ &= (A - K_f C)\hat{x} + (B, K_f) \begin{pmatrix} u \\ y \end{pmatrix}\end{aligned}$$

Error $\epsilon = x - \hat{x}$.

$$\begin{aligned}\frac{d}{dt}\epsilon &= \frac{d}{dt}x - \frac{d}{dt}\hat{x} \\ &= Ax + Bu - A\hat{x} + K_f C\hat{x} - K_f y - Bu \\ &= Ax - A\hat{x} + K_f C\hat{x} - K_f y \\ &= Ax - A\hat{x} + K_f C\hat{x} - K_f Cx \\ &= A(x - \hat{x}) + K_f C(x - \hat{x}) \\ &= (A - K_f C)(x - \hat{x}) \\ &= (A - K_f C)\epsilon\end{aligned}$$

If (A,C) is observable, then can place eigenvalues by choosing K_f . So eigenvalues will be stable, and drive $x \rightarrow \dot{x}$ and thereby $\epsilon \rightarrow 0$. Then $(A - K_f C)\epsilon \rightarrow 0$. Based on this, eigenvalues should be as negative as possible. However, noise and disturbances can be added to the system

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

to result in

$$\dot{x} = Ax + Bu + w_{disturbance}$$

$$y = Cx + w_{noise}$$

So Kalman filter gain K_f may need to be optimized s.t. eigenvalues may need to be moderated.

33.14 Optimal control

33.14.1 Control speed

More negative eigenvalues (poles) lead to faster convergence to steady state. But this requires very powerful actuators and a lot of energy. To get a better balance, use LQR.

33.14.2 Linear quadratic regulator

Impose cost on energy of actuator and cost on time to convergence.

$$Cost = \int_0^t (x'Qx + u'Ru)dt$$

where Q is the penalty for times and R is penalty for energy expended by the actuator(s). Because of the quadratic nature, there is a best control law that minimizes the cost function above. The LQR will tune K to get the optimal eigenvalues (poles).

Example. Inverted pendulum on a cart has 4 variables $x = (x, \dot{x}, \theta, \dot{\theta})$:

$$Q = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 100 \end{pmatrix}$$

where the first two 1's are for $x, \dot{x}$ whereas the other larger numbers are penalties for $\theta, \dot{\theta}$.

33.14.3 Linear quadratic estimator: Kalman filter

Also called linear quadratic estimator (LQE).

Assume $w_{disturbance} = w_d$ is Gaussian with variance v_d . Assume $w_{noise} = w_n$ is also Gaussian and variance v_n . If $v_d \gg v_n$ then trust measurement $y = Cx + w_n$ more and if $v_d \ll v_n$ then trust model $\dot{x} = Ax + Bu + w_d$ more.

Aim to reduce error as fast as possible

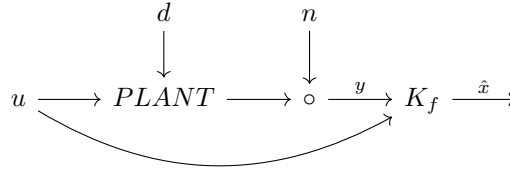
$$\epsilon = x - \hat{x}$$

$$\dot{\epsilon} = (A - K_f C)\epsilon$$

but need to balance the eigenvalues based on v_d, v_n and which are analogous to Q and R cost matrices in LQR. The cost function is

$$Cost = \mathbb{E}\epsilon' \epsilon = \mathbb{E}(x - \hat{x})'(x - \hat{x})$$

where minimizing this cost function by adjusting K_f is analogous to LQR and the math is also analogous based on the duality of controllability (A,B) and observability (A,C).



So the system now includes disturbance d and noise n:

$$\begin{aligned}\dot{x} &= Ax + Bu + V_d \cdot d + 0 \cdot n \\ y &= Cx + \underbrace{0 \cdot d + V_n \cdot n}_{D \cdot n = (0, 0, V_n)(u, d, n)'}\end{aligned}$$

The Kalman filter K_f can, based on just noisy y and u, estimate x with $\hat{x}$ with high accuracy.

33.14.4 Kalman filter math

Assume a linear state equation:

$$x_t = c_1 x_{1,t-1} + c_2 x_{2,t-1} + c_3 x_{3,t-1} + \dots c_{n+1}$$

With the update matrix being A_t , actuator effect matrix B_t , and multivariate Gaussian state error $\epsilon_t \sim \mathcal{N}(0, Q_t)$ (resulting from calculations to estimate the state),

$$x_t = A_t x_{t-1} + B_t u_t + \epsilon_t$$

which is a probability distribution $p(x_t|x_{t-1}, u_t) \sim \mathcal{N}$. With the sensor matrix C_t and multivariate Gaussian sensor error $\delta_t \sim \mathcal{N}(0, R_t)$,

$$z_t = C_t x_t + \delta_t$$

which is a probability distribution $p(z_t|x_t) \sim \mathcal{N}$.

The *a priori* estimates for the predicted state and predicted state covariance respectively are

$$\begin{aligned}\hat{x}_{t|t-1} &= A_t \hat{x}_{t-1|t-1} + B_t u_t \\ \Sigma_{t|t-1} &= A_t \Sigma_{t-1|t-1} A_t' + Q_t\end{aligned}$$

The *pre-fit residual* (or innovation) $\tilde{y}$ and associated covariance Σ respectively are

$$\begin{aligned}\tilde{y}_t &= z_t - C_t \hat{x}_{t|t-1} \\ S_t &= C_t \Sigma_{t|t-1} C_t' + R_t\end{aligned}$$

where the second equation adds the covariances of the state and sensor.

The *a posteriori* estimates for the predicted state and predicted state covariance respectively are

$$\begin{aligned}\hat{x}_{t|t} &= \hat{x}_{t|t-1} + K_t \hat{y}_t \\ \Sigma_{t|t} &= (I - K_t C_t) \Sigma_{t|t-1}\end{aligned}$$

where the first equation is a definition (and therefore defines the Kalman gain K_t), and the second is derived, and where the optimal Kalman gain is

$$K_t = \Sigma_{t|t-1} C_t' S_t^{-1} = \Sigma_{t|t-1} C_t' (C_t \Sigma_{t|t-1} C_t' + R_t)^{-1}$$

So if the sensor covariance (sensor error) R_t is high (relative to C_t and Σ where the latter depends on Q_t from above), then K_t depends more on predicted state calculations (i.e. $x_{t|t-1}$) and is low. If the sensor error is low then K_t depends less on predicted state calculations and more on sensor measurements and is high because $K_t \hat{y}_t \gg x_{t|t-1}$.

Derivation of $\Sigma_{t|t-1}$:

$$\begin{aligned}
\Sigma_{t|t} &= \text{Cov}(x_t - \hat{x}_{t|t}) \\
&= \text{Cov}(x_t - (\hat{x}_{t|t-1} + K_t \tilde{y}_t)) \\
&= \text{Cov}(x_t - (\hat{x}_{t|t-1} + K_t(z_t - C_t \hat{x}_{t|t-1}))) \\
&= \text{Cov}(x_t - (\hat{x}_{t|t-1} + K_t(C_t x_t + \delta_t - C_t \hat{x}_{t|t-1}))) \\
&= \text{Cov}((I - K_t C_t)(x_t - \hat{x}_{t|t-1}) - K_t \delta_t) \\
&= \text{Cov}((I - K_t C_t)(x_t - \hat{x}_{t|t-1})) + \text{Cov}(K_t \delta_t); \delta \text{ is uncorrelated} \\
&= (I - K_t C_t) \text{Cov}(x_t - \hat{x}_{t|t-1}) (I - K_t C_t)' + K_t \text{Cov}(\delta_t) K_t' \\
&= (I - K_t C_t) \Sigma_{t|t-1} (I - K_t C_t)' + K_t R_t K_t' \\
&= (I - K_t C_t) \Sigma_{t|t-1} (I' - K_t' C_t') + K_t R_t K_t' \\
&= (I - K_t C_t) \Sigma_{t|t-1} (I' - K_t' C_t') + K_t R_t K_t' \\
&= (\Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}) (I' - K_t' C_t') + K_t R_t K_t' \\
&= \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} - \Sigma_{t|t-1} K_t' C_t' + K_t (C_t \Sigma_{t|t-1} C_t' + R_t) K_t' \\
&= \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} - \Sigma_{t|t-1} K_t' C_t' + K_t S_t K_t' \\
&= \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} - K_t S_t K_t' + K_t S_t K_t' \\
&= \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}
\end{aligned}$$

where the penultimate step arises from the optimal Kalman gain

$$\begin{aligned}
K_t &= \Sigma_{t|t-1} C_t' S_t^{-1} \\
K_t S_t K_t' &= \Sigma_{t|t-1} C_t' S_t^{-1} S_t K_t' \\
K_t S_t K_t' &= \Sigma_{t|t-1} C_t' K_t'
\end{aligned}$$

and the optimal Kalman gain is derived from matrix differentiation of the quadratic

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1} - \Sigma_{t|t-1} K_t' C_t' + K_t S_t K_t'$$

which is:

$$\begin{aligned}
\frac{\partial \Sigma_{t|t-1}}{\partial K_t} &= -C_t \Sigma_{t|t-1} - \Sigma_{t|t-1} C_t' + 2K_t S_t \\
&= -2(C_t \Sigma_{t|t-1})' + 2K_t S_t = 0 \\
K_t S_t &= (C_t \Sigma_{t|t-1})' \\
K_t S_t &= \Sigma_{t|t-1} C_t' \\
K_t &= \Sigma_{t|t-1} C_t' S_t^{-1}
\end{aligned}$$

The Kalman filter is only for linear dynamic systems with Gaussian errors. For non-linear (but locally linear) dynamic systems with Gaussian errors, use the extended Kalman filter which calculates local Jacobians. This adds to computational cost. For non-linear and non-locally linear systems with Gaussian errors,

use the unscented Kalman filter which approximate the Gaussian using 'sigma points' instead (the minimal set of sample points s.t. mean and covariance are same as the Gaussian). Each sigma point is propagated through the dynamical system. For non-linear systems with non-Gaussian errors, use particle filters which generalize the concept of sigma points ('particle points') and consumes the most computational resources.

33.14.5 Linear quadratic Gaussian

Combines LQR and Kalman filter (LQE). System with $u = -K_r \hat{x}$:

$$\begin{aligned}\dot{x} &= Ax + Bu + w_d \\ &= Ax - K_r \hat{x} + w_d \\ &= Ax - K_r(x - (x - \hat{x})) + w_d\end{aligned}$$

And

$$\begin{aligned}\epsilon &= x - \hat{x} \\ \frac{d}{dt}\epsilon &= \frac{d}{dt}x - \frac{d}{dt}\hat{x} \\ &= Ax + Bu + w_d - A\hat{x} + K_f C \hat{x} - K_f y - Bu \\ &= Ax - A\hat{x} + K_f C \hat{x} - K_f y + w_d \\ &= Ax - A\hat{x} + K_f C \hat{x} - K_f (Cx + w_n) + w_d \\ &= A(x - \hat{x}) + K_f C(x - \hat{x}) + w_d - K_f w_n \\ &= (A - K_f C)(x - \hat{x}) + w_d - K_f w_n \\ &= (A - K_f C)\epsilon + w_d - K_f w_n\end{aligned}$$

So,

$$\frac{d}{dt} \begin{pmatrix} x \\ \epsilon \end{pmatrix} = \begin{pmatrix} (A - BK_r) & BK_r \\ 0 & (A - K_f C) \end{pmatrix} \begin{pmatrix} x \\ \epsilon \end{pmatrix} + \begin{pmatrix} I & 0 \\ I & -K_f \end{pmatrix} \begin{pmatrix} w_d \\ w_n \end{pmatrix}$$

Since the eigenvalues of a triangular matrix are the diagonal elements, the first matrix's eigenvalues remain $A - BK_r$ (the LQR) and $A - K_f C$ (the LQE, Kalman filter). This is the **separation principle**, so the LQR and LQE act independently.

33.15 Robust control

LQG has high performance but not necessarily robust. Use loop transfer functions to illustrate fundamental limits on the robustness of any system. Applies Nyquist stability criterion.

34 Combinatorics

34.1 Permutations and combinations

A permutation is a bijection or linear ordering. Permutations with repetitions are just tuples. k-permutations are injective maps.

For n types of objects that can be repeated in a group of k, use (n-1) delimiters dividing n spaces representing object types:

$$\binom{k+n-1}{n-1}$$

The below shows different objects can be arranged (arrangement of all objects, no choosing):

	With repetitions	Without repetitions
Ordered	Tuples n^k	k-permutations $\frac{n!}{(n-k)!}$
Unordered	Combinations with repetitions $\binom{k+n-1}{n-1}$	Combinations $\binom{n}{k}$

34.1.1 Circular arrangements

Number of trains with n colors and p carriages is n^p . If this is a merry-go-round instead, start with $\frac{n^p}{p}$, which 'cuts' the merry-go-round to trains p times. However, this doesn't work if they are all the same color, though this works if there are at least 2 colors and p is prime (as the rotations around the merry-go-round won't go back to themselves with prime numbers). Then add back n for unicolor train. So, it's $\frac{n^p-n}{p} + n$. It's not obvious that $\frac{n^p-n}{p}$ is an integer (it must be, since it's a combinatorics problem). This is Fermat's little theorem.

34.1.2 Multisets

A multiset function $\mu : X \rightarrow \mathbb{Z}_{\geq 0}$. The size of the multiset is $\sum_x \mu(x)$. Example k balls in n boxes, each has no maximum capacity. This corresponds to the number of multisets. Line up k balls and (n-1) separators, so

$$\binom{n+k-1}{n-1} = \binom{n+k-1}{k}$$

ways, or the number of k-element multisets.

34.1.3 Integer composition

A presentation of n as an ordered sum.

$$n = n_1 + n_2 + \dots + n_k$$

Example.

1 1 combination

$2 = 1 + 1$ 2 combinations

$3 = 2 + 1 = 1 + 2 = 1 + 1 + 1$ 4 combinations

$4 = \dots$ 8 combinations

1. The number of compositions of n is 2^{n-1} .
2. The number of compositions with exactly k summands is $\binom{n-1}{k-1}$.

Proof of (2). Compositions of n with k summands corresponds to configurations of n balls inside k boxes s.t. each box is non-empty. This is the same as placing one ball in each box and then adding the rest, which is the number of configurations of $n-k$ balls in k ,

$$\binom{n-k+k-1}{k-1} = \binom{n-1}{k-1}$$

Proof of (1). The total number of compositions

$$\binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{n-1} = 2^{n-1}$$

34.1.4 Inclusion-Exclusion

$$\begin{aligned} |A_1 \cup \dots \cup A_n| &= |A_1| + \dots + |A_n| \\ &- \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots \\ &+ (-1)^{n-1} |A_1 \cap \dots \cap A_n| \end{aligned}$$

34.1.5 Derangement

Example 1. A kid has 'N' letters labeled Letter1, Letter2, ..., LetterN and the corresponding 'N' addressed envelopes labeled Envelope1, Envelope2, ..., EnvelopeN. Then how many ways can the kid put the letters in the wrong envelope? (Letter1 should not go to Envelope1, Letter2 should not go to Envelope2, ..., LetterN should not go to EnvelopeN.)

Example 1. Secret santa with 8 people. Then want to find

$$|A| - |A_1 \cup \dots \cup A_n|$$

i.e. excluding the 'fixed points' where someone receives their own gift. For the first person *not* to receive their own gift:

$$|A_i| = 7!$$

For *any* two people not to receive their own gifts:

$$|A_i \cap A_j| = 6!$$

Repeating with the inclusion-exclusion formula,

$$|A| - |A_1 \cup \dots \cup A_n| = 8! - \binom{8}{1}7! + \binom{8}{2}6! - \dots + \binom{8}{8}1!$$

General case. How many permutations have no fixed points?

$$\begin{aligned} |A| - |A_1 \cup \dots \cup A_n| &= n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \dots + \binom{n}{n}1! \\ &= \underbrace{n!}_{total} \underbrace{\left(1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!}\right)}_{p(n)} \end{aligned}$$

Since

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \dots \\ \lim_{n \rightarrow \infty} p(n) &= 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \dots = e^{-1} = 0.37\dots \end{aligned}$$

34.2 Binomial expansion

(1) Binomial Theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

34.2.1 Combination Properties and Pascal's Triangle

(1) Binomial coefficients are the elements of Pascal's Triangle.

(2) Symmetry. Choosing k from n is the same as choosing $(n-k)$ from n :

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k}$$

Pascal's Triangle is symmetric.

(3) Power set. The number of combinations of sets from $1, 2, \dots, n$ is:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}$$

$$1 + n + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \dots + \frac{n!}{n!}$$

This is mathematically difficult. There are 2 choices for the first element (y/n) and repeat for each element. So it's 2^n :

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

This is adding the terms of a row in Pascal's Triangle. The sum of each row is twice the previous row because each of the previous row elements are used twice to make the lower row.

Another proof applies the binomial theorem. If $a = b = 1$ in $(a + b)^n$, then:

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

(4) Reduction. Taking one arbitrary element out of n , can either (i) include the arbitrary element in the set and therefore choose $k-1$ from $n-1$, or (ii) exclude the arbitrary element and choose k from $n-1$:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Intuition. Split k -element set into two subsets, one with a specific element (call it '1' for instance) and another without that specific element. The first subset corresponds to the left RHS term and the second subset corresponds to the right RHS term. In Pascal's triangle, summing two terms on the higher row results in a lower row element, and this corresponds to the above formula where the RHS terms are the higher row and the LHS term is on a lower row.

(5) Galton Board. A visual demonstration of Pascal's Triangle and binomial coefficients divided by 2^n for the row to get probabilities. Galton Board assumes decision independence at each level (e.g. no momentum or memory from prior level).

34.3 Linear recurrence relations

A solution to a recurrence relation gives the value of x_n in terms of n and does not require the value of any previous terms.

Definition. A linear recurrence relation of order k is a sequence defined by

$$A(n) = \sum_i^k c_i A(n-i)$$

Example. Pascal's triangle is a linear recurrence of order 1.

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

Example. Fibonacci sequence is a linear recurrence of order 2.

$$F(n) = F(n-1) + F(n-2)$$

Example. Nickels, dimes, and quarters that add to 1 dollar is a linear recurrence of order 5.

$$A(n) = A(n-1) + A(n-2) + A(n-5)$$

where n=20 for 1 dollar.

A sequence defined by a linear recurrence relation is uniquely determined by its first k terms $A(0), A(1), \dots, A(k-1)$. Suppose $a_0, a_1, \dots$ are solutions and $a'_0, a'_1, \dots$ are also solutions. Then $a_0 + a'_0, a_1 + a'_1, \dots$ are also solutions as well as $ca_0, ca_1, \dots$ (which are $a_0 + a_0 + \dots, a_1 + a_1 + \dots$). So the solutions are linear transformations. So the solutions form a vector space.

For k=1,

$$A(n) = cA(n-1)$$

So,

$$a_n = c^n a_0$$

which is a geometric progression.

34.3.1 Characteristic equations

If a geometric progression $a_0, \lambda a_0, \lambda^2 a_0, \dots$ satisfies the linear recurrence relation, then

$$A(n) = c_1 A(n-1) + c_2 A(n-2) + \dots$$

with

$$A(n) = \lambda^n a_0$$

Then,

$$a_0 \lambda^n = c_1 a_0 \lambda^{n-1} + c_2 a_0 \lambda^{n-2} + \dots$$

Dividing by a_0 ,

$$\lambda^n = c_1 \lambda^{n-1} + c_2 \lambda^{n-2} + \dots$$

Then λ is a polynomial root of the characteristic equation

$$t^k = c_1 t^{k-1} + c_2 t^{k-2} + \dots + c_k$$

For k=2,

$$A(n) = c_1 A(n-1) + c_2 A(n-2) + \dots$$

$$t^2 = c_1 t + c_2$$

Suppose λ, μ are roots with sequences

$$1, \lambda, \lambda^2, \dots$$

$$1, \mu, \mu^2, \dots$$

Then

$$a_n = C\lambda^n + D\mu^n$$

is also a solution (as solutions are linear transformations since they are part of a vector space). In fact, it is the general solution. There are no other solutions.

$$a_0 = C + D$$

$$a_1 = C\lambda + D\mu$$

Solving the linear system,

$$C = \frac{a_1 - a_0\mu}{-\mu}$$

$$D = \frac{a_0\lambda - \mu}{\lambda - \mu}$$

34.3.2 Binet formula

Fibonacci sequence is $F(n) = F(n-1) + F(n-2)$. So characteristic equation is

$$t^2 = t + 1 \Rightarrow t^2 - t - 1 = 0$$

Solutions are

$$\lambda = \frac{1 + \sqrt{5}}{2}$$

$$\mu = \frac{1 - \sqrt{5}}{2}$$

So,

$$f_n = C\lambda^n + D\mu^n$$

With 'initial conditions' $f_0 = 0, f_1 = 1$, find coefficients C and D:

$$0 = C + D$$

$$1 = C\lambda + D\mu$$

Solving, $C = \frac{1}{\sqrt{5}}, D = \frac{-1}{\sqrt{5}}$. Then, the explicit form is

$$\begin{aligned} f_n &= \frac{1}{\sqrt{5}}\lambda^n - \frac{1}{\sqrt{5}}\mu^n \\ &= \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right) \end{aligned}$$

which is the Binet formula. Though there are irrational numbers in the formula, the answer is always integer.

34.3.3 Golden ratio

From the Binet formula,

$$\lambda = \frac{1 + \sqrt{5}}{2} = 1.61...$$

$$\mu = \frac{1 - \sqrt{5}}{2} = \frac{-1}{\lambda} = 1 - \lambda$$

The Fibonacci sequence is

$$F(n) = 1, 1, 2, 3, 5, 8, 13...$$

$$\frac{F(n)}{F(n-1)} = 1, 2, 1.5, 1.66...1, 6, 1.625...$$

Since

$$\lim_{n \rightarrow \infty} f_n = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n$$

Then

$$\lim_{n \rightarrow \infty} \frac{F(n)}{F(n-1)} = \lambda$$

The intuition of the golden ratio is of a rectangle with sides a and a+b. Then,

$$\lambda = \frac{a}{b} = \frac{a+b}{a}$$

34.3.4 Linear recurrences of order k

For

$$A(n) = \sum_i^k c_i A(n-i)$$

Suppose there are k distinct roots,

$$a_n = c_1 \lambda_1^n + c_2 \lambda_2^n + .. + c_k \lambda_k^n$$

Suppose given $a_0, ... a_{k-1}$, want a solution with these initial conditions,

$$a_0 = c_1 + c_2 + ... + c_k$$

$$a_1 = c_1 \lambda_1 + c_2 \lambda_2 + ... + c_k \lambda_k$$

$$a_{k-1} = c_1 \lambda_1^{k-1} + ...$$

Using linear algebra, the determinant is Vandermonde

$$\begin{vmatrix} 1 & ... & 1 \\ \lambda_1 & ... & \lambda_k \\ \lambda_1^2 & ... & \lambda_k^2 \\ \vdots & \ddots & \vdots \\ \lambda_1^{k-1} & ... & \lambda_k^{k-1} \end{vmatrix} = \prod_{1 \leq j < i \leq k} (\lambda_i - \lambda_j)$$

which exists for distinct roots.

Repeated roots (a multiplicity). Take $k=2$,

$$A(n) = c_1 A(n-1) + c_2 A(n-2)$$

$$t^2 - c_1 t - c_2 = 0$$

λ is a unique (double) root when

$$(t - \lambda)^2 = 0 \Rightarrow t^2 - 2\lambda t + \lambda^2 = 0$$

In other words, when

$$c_1 = 2\lambda$$

$$-c_2 = \lambda^2$$

In this case, since $(t - \lambda)^2$, one solution is

$$1, \lambda, \lambda^2, \dots \lambda^n$$

But need another solution (2 solutions are needed for a linear transformation in vector space). Claim that

$$0, \lambda, 2\lambda^2, 3\lambda^3, \dots n\lambda^n$$

is another solution. Proof for $k=2$.

$$n\lambda^2 = c_1(n-1)\lambda^{n-1} + c_2(n-2)\lambda^{n-2}$$

which takes the last two solutions of the sequence above. Then since $c_1 = 2\lambda$ and $-c_2 = \lambda^2$ from above,

$$n\lambda^2 = 2\lambda(n-1)\lambda^{n-1} + (-\lambda)^2(n-2)\lambda^{n-2}$$

which dividing by λ^n ,

$$n = 2(n-1) - (n-2) = n$$

So the general solution is now

$$a_n = c_1 \lambda^n + c_2 n \lambda^n$$

for $k=2$. For arbitrary k ,

$$0, 1, \lambda, \lambda^2 \dots$$

$$0, \lambda, 2\lambda^2, \dots n\lambda^n$$

$$0, \lambda, 4\lambda^2, \lambda^3, \dots n^2\lambda^n$$

$$0, \lambda, 2^{r-1}\lambda^2, \dots n^{r-1}\lambda^n$$

34.4 Catalan numbers

34.4.1 $(n+1)$ -gons

In a convex $(n+2)$ -gon, how many triangles can be formed by splitting using $n-1$ non-crossing diagonals? The Catalan number C_n is the number of triangulations.

$$C_0 = 1, C_1 = 1, C_2 = 2, C_3 = 5, C_4 = 14$$

Recurrence relation derived by splitting $(n+1)$ -gon into a triangle and two other triangles with C_{k-1} and C_{n-k} and multiplying the two. So,

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + C_2 C_{n-3} + \dots + C_{n-1} C_0 = \sum_{k=0}^n C_{k-1} C_{n-k}$$

which is not linear and instead quadratic.

34.4.2 Binary trees

Binary trees have one 'starting' root (node) that has one edge to another node. Then each node branches into two edges each after that or end as a 'leaf'.

Theorem. The number of binary trees with $n+1$ leaves is equal to C_n . it is a bijection to the middle of edges and triangulations in an $(n+1)$ -gon.

34.4.3 Dyck paths

Catalan numbers also appear with $2n$ people in a lineup where half have 5-dollar bills and the other half have 10-dollar bills. There is a lineup to buy something that is 5-dollars but the cashier has no change. The number of line-ups that 'work' (cashier has enough change) is 1 for $n=1$, 2 for $n=2$, 3 for $n=3$, etc. in a Catalan sequence.

Dyck paths represent recurrence relations on a graph with the x-axis from 0 to $2n$. If the sequence contacts the x-axis at $2k$ (must be even) after moving up, then there becomes a natural break and the number of Dyck paths is then $C_{k-1} C_{n-k}$. Before arriving at $2k$, the Dyck path must move around $2k-2$. It then moves $2n-2k$ afterwards. These correspond to $k-1$ and $n-k$, respectively, in the number of Dyck paths. The total number of Dyck paths with different $2k$ values is the Catalan number

$$C_n = \sum_{k=1}^n C_{k-1} C_{n-k}$$

34.4.4 Formula derivation

The total number of Dyck paths from $(0,0)$ to $(2n,0)$ is

$$\binom{2n}{n}$$

where the path is $2n$ steps horizontally and there are n steps vertically up (and correspondingly, n steps down). However, this formula includes unacceptable Dyck paths that fall below the x-axis. Each of these 'bad' paths bijectively map to a Dyck path that ends at $(2n, -2)$ by reflecting the 'bad' path across $y = -1$ as it will at one point cross that horizontal line. The number of bad paths must therefore be

$$\binom{2n}{n-1} = \binom{2n}{n+1}$$

since there will be $n-1$ ups and $n+1$ downs in these bijectively mapped bad paths. The number of 'good' Dyck paths is then

$$\begin{aligned} C_n &= \binom{2n}{n} - \binom{2n}{n-1} \\ &= \frac{1}{n+1} \binom{2n}{n} \end{aligned}$$

34.5 Formal power series

An infinite polynomial.

$$A(q) = a_0 + a_1q + a_2q^2 + \dots$$

$$B(q) = b_0 + b_1q + b_2q^2 + \dots$$

Addition:

$$A(q) + B(q) = (a_0 + b_0) + (a_1 + b_1)q + \dots$$

Multiplication:

$$\begin{aligned} A(q)B(q) &= (a_0 + a_1q + a_2q^2 + \dots)(b_0 + b_1q + b_2q^2 + \dots) \\ &= a_0b_0 + (a_0b_1 + a_1b_0)q + \dots \end{aligned}$$

34.5.1 Inverse

Inverse:

$$A(q)^{-1} = 1 - q$$

$$(1 + q + q^2 + \dots)(1 - q) = 1 + q + q^2 + \dots - q - q^2 - q^3 - \dots = 1$$

Let $A(q)B(q) = 1$ and so $A(q) = B(q)^{-1}$. Then,

$$a_0b_0 = 1 \Rightarrow b_0 = \frac{1}{a_0}$$

$$(a_0b_1 + a_1b_0)q = 0 \Rightarrow b_1 = \frac{-a_1b_0}{a_0}$$

$$(a_0b_2 + a_1b_1 + a_2b_0)q^2 = 0 \Rightarrow b_2 = \frac{-a_1b_1 + a_2b_0}{a_0}$$

So can find inverse power series this way.

34.5.2 Compositions

Compositions.

$$A(B(q)) = a_0 + a_1 B(q) + a_2 (B(q))^2 + \dots = \sum_{n \geq 0} a_n (B(q))^n$$

The constant term of $A(B(q))$ is $a_0 + a_1 b_0 + a_2 b_0^2 + \dots$, which might be divergent. It is only meaningful if $b_0 = 0$. Note if $A(q) = 1 + q + q^2 + \dots = \frac{1}{1-q}$ and $B(q) = q^2$ then $A(B(q)) = 1 + q^2 + q^4 + \dots$

34.5.3 Calculus

Differentiation and integration. Let

$$A(q) = a_0 + a_1 q + a_2 q^2 + \dots$$

Then

$$A'(q) = a_1 + 2a_2 q + 3a_3 q^2 + \dots + na_n q^{n-1} + \dots$$

and the ordinary differentiation rules (e.g. product rule) apply. The chain rule also holds:

$$A(B(q))' = A'(B(q)) \cdot B'(q)$$

Corollary. Inverse function theorem. If $A(B(q)) = q$, then $A'(t) = \frac{1}{B'(q)}$, where $t = B(q)$. Proof.

$$A'(B(q))B'(q) = 1$$

So,

$$A'(B(q)) = \frac{1}{B'(q)}$$

For integration,

$$\int A(q) = a_0 q + \frac{a_1}{2} q^2 + \frac{a_2}{3} q^3 + \dots + \frac{a_n}{n+1} q^{n+1} + \dots$$

When integration the derivative, since the above integral result doesn't have a constant, there's no constant below:

$$\int A'(q) = a_1 q + a_2 q^2 + a_3 q^3 + \dots = A(q) - a_0 = A(q) - A(0)$$

which is the Newton-Leibniz formula.

34.5.4 Differential equations

Find all $F(q)$ satisfying

$$F'(q) = F(q)$$

If $F(q) = f_0 + f_1q + f_2q^2 + \dots$ then $F'(q) = f_1 + 2f_2q + 3f_3q^2$, and these must be equal. Comparing these two equations, $f_n = (n+1)f_{n+1}$. So, $f_n = \frac{f_0}{n!}$. Then

$$F(q) = f_0 \left(1 + \frac{1}{1!}q + \frac{1}{2!}q^2 + \dots \right)$$

But recall

$$e^q = 1 + \frac{1}{1!}q + \frac{1}{2!}q^2 + \dots$$

So,

$$F(q) = f_0 e^q$$

34.5.5 Logarithm

Note $\ln(e^q) = q$. To calculate $\ln(1+t)$,

$$\ln(1+t)' = \frac{1}{1+t} = 1 - t + t^2 - t^3 + \dots$$

So using Newton-Leibniz,

$$\ln(1+t) = \int (\ln(1+t))' = t - \frac{t^2}{2} + \frac{t^3}{3} - \dots = \sum \frac{(-1)^{n+1}t^n}{n}$$

Logarithmic derivative. Suppose

$$B(q) = 1 + b_1q + b_2q^2 + \dots$$

Then,

$$(\ln(B(q)))' = \frac{B'(q)}{B(q)}$$

This is used to help prove Newton's binomial theorem.

34.5.6 Newton's binomial theorem

$$(1+q)^\alpha \stackrel{def}{=} e^{\alpha \ln(1+q)}$$

Theorem. Let $\alpha \in \mathbb{Q}$.

$$(1+q)^\alpha = 1 + \alpha q + \frac{\alpha(\alpha-1)}{2!}q^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}q^3 + \dots = \sum \binom{\alpha}{n} q^n$$

where $\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}$. Proof.

$$(1+q)^\alpha = e^{\alpha \ln(1+q)}$$

$$(\ln(e^{\alpha \ln(1+q)}))' = \frac{(e^{\alpha \ln(1+q)})'}{e^{\alpha \ln(1+q)}} = \frac{((1+q)^\alpha)'}{(1+q)^\alpha}$$

Note that

$$(\alpha \ln(1+q))' = \frac{\alpha}{1+q}$$

So,

$$\frac{((1+q)^\alpha)' }{(1+q)^\alpha} = \frac{\alpha}{1+q}$$

And,

$$\begin{aligned}(1+q)^\alpha &= 1 + a_1q + a_2q^2 + \dots \\ ((1+q)^\alpha)' &= a_1 + 2a_2q + 3a_3q^2 + \dots\end{aligned}$$

So,

$$\begin{aligned}\alpha(1 + a_1q + a_2q^2 + \dots) &= (a_1 + 2a_2q + 3a_3q^2 + \dots)(1+q) \\ ka_k + (k-1)a_{k-1} &= \alpha a_{k-1} \\ \alpha_k = \frac{\alpha - (k-1)}{k} a_{k-1} &= \frac{(\alpha - (k-1))(\alpha - (k-2))}{k(k-1)} a_{k-2} = \dots = \binom{\alpha}{k}\end{aligned}$$

34.6 Generating functions

Working with a continuous function is sometimes much easier than working with a sequence. For example, in the analysis of functions, calculus is very useful. However, the discrete nature of sequences prevents us from using calculus on sequences. A generating function is a continuous function associated with a given sequence. For this reason, generating functions are very useful in analyzing discrete problems involving sequences of numbers or sequences of functions.

34.6.1 Definition

A generating function of sequence $a_0, a_1, \dots$ is the formal power series,

$$A(q) = a_0 + a_1q + a_2q^2 + \dots + a_nq^n + \dots = \sum_{n \geq 0} a_nq^n$$

Example 1. Sequence $(a_0, a_1, \dots) = (1, 1, \dots)$

$$A(q) = 1 + q + q^2 + \dots q^n + \dots = \frac{1}{1-q}$$

Example 2. Sequence $(a_0, a_1, \dots)$ is the m -th row of Pascal's triangle:

$$A(q) = \binom{m}{0} + \binom{m}{1}q + \binom{m}{2}q^2 + \dots = (1+q)^m$$

34.6.2 Negative integer exponents

$$\frac{1}{(1-q)^k} = \sum_{n \geq 0} \binom{n+k-1}{n} q^n$$

Proof.

$$\frac{1}{1-q} = 1 + q + q^2 + \dots$$

$$\frac{1}{(1-q)^k} = (1 + q + q^2 + \dots)^k = \sum q^{\sum d_i} = \sum_{n \geq 0} a_n q^n$$

where a_n is the number of ways to present $\sum d_i$ (integer sums) can be presented. This is the 'balls and boxes' scenario where boxes can be empty.

$$a_n = \binom{n+k-1}{n}$$

34.6.3 Fibonacci Binet formula

$$f_{n+2} = f_{n+1} + f_n, f_0 = 0, f_1 = 1$$

The Fibonacci sequence is 1, 1, 2, 3, 5, 8, 13... so the generating function is

$$\Psi(q) = q + q^2 + 2q^3 + 3q^4 + 5q^5 + \dots$$

And multiplying by q and q^2 ,

$$q\Psi(q) = q^2 + q^3 + 2q^4 + 3q^5 + 5q^6 + \dots$$

$$q^2\Psi(q) = q^3 + q^4 + 2q^5 + 3q^6 + 5q^7 + \dots$$

The characteristic polynomial is

$$\Psi(q)(1 - q - q^2) = q$$

$$\Psi(q) = \frac{q}{1 - q - q^2}$$

$$1 - q - q^2 = (1 - \phi q)(1 - \psi q)$$

where ϕ, ψ are inverses to the roots of $1 - q - q^2$:

$$\phi = \frac{1 + \sqrt{5}}{2}, \psi = \frac{1 - \sqrt{5}}{2}$$

where

$$\Psi(q) = \frac{C}{q - \phi q} + \frac{D}{1 - \psi q} = \frac{q}{1 - q - q^2}$$

Solving,

$$C = \frac{1}{\sqrt{5}}, D = \frac{-1}{\sqrt{5}}$$

Combining,

$$\begin{aligned}\Psi(q) &= \frac{1}{\sqrt{5}} \left(\frac{1}{1-\phi q} - \frac{1}{1-\psi q} \right) \\ &= \frac{1}{\sqrt{5}} (1 + \phi q + \phi^2 q^2 + \dots) - \frac{1}{\sqrt{5}} (1 + \psi q + \psi^2 q^2 + \dots)\end{aligned}$$

So,

$$f_n = \frac{1}{\sqrt{5}} (\phi^n - \psi^n)$$

34.6.4 Linear recurrence GFs are rational

Generalizes Fibonacci Binet formula.

Theorem. Let a_n be defined by a linear recurrence relation of order k :

$$a_{n+k} = c_1 a_{n+k-1} + c_2 a_{n+k-2} + \dots + c_k a_n$$

Then,

$$A(q) = a_0 + a_1 q + a_2 q^2 + \dots$$

is rational

$$A(q) = \frac{P(q)}{Q(q)}$$

where the polynomial degree of $Q(q)$ is k in the linear recurrent relation. Proof omitted but a further example is provided instead. For the sequence

$$a_n = a_{n-1} + 6a_{n-2}$$

with $a_0 = 0, a_1 = 1$, the sequence is 0, 1, 1, 7, 13, 55, 133.... So the generating function (formal power series) is

$$\begin{aligned}A(x) &= 0(1) + 1(x) + 1(x^2) + 7(x^3) + \dots \\ &= x + x^2 + 7x^3 + \dots \\ &= x + (x^2 + x^3 + 7x^4 + 13x^5 + \dots) + 6(x^3 + x^4 + 7x^5 + \dots) \\ &= x + xA(x) + 6x^2A(x) \\ &= \frac{x}{1-x-6x^2} \text{ denominator is characteristic polynomial (CP)} \\ &= \frac{1/5}{1-3x} - \frac{1/5}{1+2x} \text{ find CP roots and partial fractions}\end{aligned}$$

If $A(q)$ is rational, then a_n 's satisfy a linear recurrence relation of order of $Q(q)$.

Theorem. Let $A(q) = \frac{P(q)}{Q(q)}$ be a rational function. Then $A(q)$ can be uniquely presented as the linear combination of fractions of the form $\frac{1}{(1-\lambda_i q)^{k_i}}$ where $\frac{1}{\lambda_i}$'s are the complex roots of $Q(q)$ and $k_i \leq$ multiplicity of the root $\frac{1}{\lambda_i}$. Corollary. If $Q(q)$ doesn't have repeated roots then

$$A(q) = \frac{c_1}{1-\lambda_1 q} + \dots + \frac{c_k}{1-\lambda_k q}$$

34.6.5 Catalan GF

$$Cat(q) = \sum c_n q^n = 1 + q + 2q^2 + 5q^3 + \dots$$

Square and multiply by q ,

$$q(Cat(q))^2 = c_0^2 q + (c_0 c_1 + c_1 c_0) q^2 + (c_0 c_2 + c_1 c_1 + c_2 c_0) q^3 + \dots = Cat(q) - 1$$

Turn above into a quadratic,

$$q(Cat(q))^2 - Cat(q) + 1 = 0$$

$$Cat(q) = \frac{1 - \sqrt{1 - 4q}}{2q}$$

Taking the square root term and applying the binomial formula,

$$\sqrt{1 - 4q} = (1 - 4q)^{1/2} = \sum \binom{1/2}{n} (-1)^n 4^n q^n$$

Since

$$\begin{aligned} \binom{1/2}{n} &= \frac{1}{n!} \frac{1}{2} \frac{-1}{2} \frac{-3}{2} \dots \frac{-2n+3}{2} = \frac{(-1)^{n-1}}{2^n n!} \cdot 1 \cdot 3 \cdot \dots \cdot (2n-3) \\ &= \frac{(-1)^n}{2^n n!} \frac{1}{2 \cdot 4 \cdot \dots \cdot (2n-2)(2n-1)} (2n)! = \frac{(-1)^{n-1} (2n)!}{4^n (n!)^2} = \frac{(-1)^{n-1}}{4^n (2n-1)} \binom{2n}{n} \end{aligned}$$

So,

$$(1 - 4q)^{1/2} = - \sum \frac{1}{2n-1} \binom{2n}{n} q^n$$

Now insert this into $\frac{1 - \sqrt{1-4q}}{2q}$ to get $Cat(q)$. From $Cat(q)$, get c_n :

$$c_n = \frac{1}{2} \frac{1}{2n+1} \binom{2n+2}{n+1} = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{n!(n+1)!}$$

34.6.6 Combinatoric GFs

$$\begin{aligned} (x^0 + x^1)^n &= (x + 1)^n \\ &= \binom{n}{0} x^0 + \binom{n}{1} x^1 + \dots + \binom{n}{r} x^r + \dots + \binom{n}{n} x^n \end{aligned}$$

Example 1. Choosing 2 objects from 2 identical apples, 1 bannana, 1 pear, 1 papaya.

$$\begin{aligned} &\underbrace{(x^0 + x^1 + x^2)}_{0, 1, \text{ or } 2 \text{ apples}} \underbrace{(x^0 + x^1)}_{\text{bannanas}} \underbrace{(x^0 + x^1)}_{\text{pears}} \underbrace{(x^0 + x^1)}_{\text{papayas}} \\ &= 1 + 4x + 7x^2 + 7x^3 + 4x^4 + x^5 \end{aligned}$$

where the coefficient on x^2 , 7, is the answer. Can check this manually as the number is small, using tables or trees. Example 2. Choosing 4 pastries from 3 cheese, 2 cherry, and 4 raspberries.

$$(x^0 + x^1 + x^2 + x^3)(x^0 + x^1 + x^2)(x^0 + x^1 + x^2 + x^3 + x^4)$$

and check the x^4 coefficient. If raspberries only come in boxes of 2,

$$(x^0 + x^1 + x^2 + x^3)(x^0 + x^1 + x^2)(x^0x^2 + x^4)$$

and check the x^4 coefficient.

34.6.7 Probability GFs

$$G(x) = \sum_{k=0}^{\infty} a_k x^k$$

Alternatively,

$$G_X(s) = E(s^X) = \sum_{x=0}^{\infty} = p_0 + p_1 s + p_2 s^2 + \dots$$

Example. Rolling a die. Only $k = 1 \rightarrow 6$ have non-zero probabilities and therefore coefficients:

$$G(x) = p_1 x + p_2 x^2 + p_3 x^3 + p_4 x^4 + p_5 x^5 + p_6 x^6$$

Can recover the probabilities as coefficients:

$$G'(0) = p_1, G''(0) = p_2, \dots$$

With two dice:

$$G(x) \times G(x) = G^2(x) = \underbrace{p_1 p_1}_{\text{dice sum} = 2} x^2 + \underbrace{(p_1 p_2 + p_2 p_1)}_{\text{dice sum} = 3} x^3 + \underbrace{(p_1 p_3 + p_2 p_2 + p_3 p_1)}_{\text{dice sum} = 4} x^4 + \dots$$

Expectation of generating function.

$$\begin{aligned} \Phi(s) &= \sum p(x) s^x \\ &= p(0) + p(1)s + p(2)s^2 + p(3)s^3 + \dots \end{aligned}$$

Take the derivative:

$$\Phi'(s) = p(1) + 2p(2)s + 3p(3)s^2 + \dots$$

And let $s = 1$:

$$\Phi'(1) = p(1) + 2p(2) + 3p(3) + \dots$$

34.7 Partitions

34.7.1 Definition

A partition of $n \in \mathbb{N}$ is a presentation

$$n = \lambda_1 + \lambda_2 + \dots + \lambda_k$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k \geq 1$. Denote this as

$$n \vdash \lambda = (\lambda_1, \dots, \lambda_k)$$

Note that 3+2 and 2+3 are different *compositions* but same *partition*. The number of partitions is $p(n)$. So $p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5$. There is no explicit formula for $p(n)$. Partitions can be represented by *Young diagrams*.

34.7.2 Partition GF

$$P(q) = \sum p(n)q^n = 1 + q + 2q^2 + 3q^3 + 5q^4 + \dots$$

Let $p(n, k)$ be the number of partitions of n into summands not exceeding k , where $k \geq \lambda_1 \geq \dots \geq \lambda_i$.

$$P_k(q) = \sum p(n, k)q^n$$

For $k=1$,

$$p(n, 1) = 1$$

$$P_1(q) = 1 + q + q^2 + q^3 + \dots = \frac{1}{1-q}$$

For $k=2$, i.e. how many ways are there to represent n as a sum of 2's. 0 if n is odd, and 1 if n is even. The GF is

$$1 + q^2 + q^4 + \dots = \frac{1}{1-q^2}$$

Proposition.

$$P_2(q) = (1 + q + q^2 + \dots)(1 + q^2 + q^4 + \dots) = \frac{1}{(1-q)(1-q^2)}$$

Proof. Need to show the coefficient of q^n in RHS is $p(n, 2)$.

$$(1 + q + q^2 + \dots)(1 + q^2 + q^4 + \dots) = 1 \cdot 1 + q \cdot 1 + q^2 \cdot 1 + 1 \cdot q^2 + q \cdot q^2 + q^3 \cdot 1 + \dots$$

where the individual terms are $q^k q^{2l}$ where $k + 2l = n$. We can take

$$n = \underbrace{(1 + \dots + 1)}_k + \underbrace{(2 + \dots + 2)}_l$$

Proposition.

$$\begin{aligned}
 P_k(q) &= (1 + q + q^2 + \dots)(1 + q^2 + q^4 + \dots) \dots (1 + q^k + q^{2k} + \dots) \\
 &= \frac{1}{(1 - q)(1 - q^2) \dots (1 - q^k)} \\
 &= \prod_{j=1}^k \frac{1}{1 - q^j}
 \end{aligned}$$

Theorem (from Euler).

$$P(q) = \sum p(n)q^n = \prod_{j=1}^{\infty} \frac{1}{1 - q^j}$$

To compute each coefficient of $P(q)$, need only finitely many operations. For instance, if want 100th term, don't need to calculate partition polynomial beyond that point.

34.7.3 Odd and distinct summands

Theorem. Number of partitions into odd summands $p_o(n)$ is equal to the number of partitions into pairwise distinct summands $p_d(n)$, so $p_o(n) = p_d(n)$. Proof. The GF of $p_o(n)$ is

$$\begin{aligned}
 P_o(q) &= \sum p_o(n)q^n = (1 + q + q^2 + \dots)(1 + q^3 + q^6 + \dots)(1 + q^5 + q^{10} + \dots) \dots \\
 &= \prod_{k=1}^{\infty} \frac{1}{1 - q^{2k-1}} \\
 &= \frac{1}{1 - q} \frac{1}{1 - q^3} \frac{1}{1 - q^5} \dots \\
 &= \frac{1}{1 - q} \frac{1 - q^2}{1 - q^2} \frac{1}{1 - q^3} \frac{1 - q^4}{1 - q^4} \frac{1}{1 - q^5} \dots \\
 &= \frac{(1 - q^2)(1 - q^4)(1 - q^6) \dots}{(1 - q)(1 - q^2)(1 - q^3) \dots} \\
 &= (1 + q)(1 + q^2)(1 + q^3) \dots
 \end{aligned}$$

The GF of $p_d(n)$ is

$$P_d(q) = \sum p_d(n)q^n = (1 + q)(1 + q^2)(1 + q^3) \dots = P_o(q)$$

The theorem can alternatively be proved combinatorially using Sylvester's bijection.

34.7.4 Euler's pentagonal theorem

$$Q(q) = P(q)^{-1} = (1 - q)(1 - q^2)(1 - q^3) \dots = \prod_{k=1}^{\infty} (1 - q^k)$$

$$Q(q) = 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + q^{22} + q^{26} - \dots$$

Properties:

1. All coefficients are 0, -1, or +1.
2. The coefficients are 1,-1,-1,1,1,-1,-1,... (they alternate as pairs)
3. The differences of exponents are equal to 1,2,3,... ((1,2), (5,7), (12,15), (22,26))
4. From the preceding point, 1,5,12,22... are special numbers: the pentagonal numbers. (1, 3, 6, 10.... can be formed into pyramids, 1, 4, 9, 16.... can be formed into squares, and 1, 5, 12, 22 form pentagons). Pentagonal numbers are $\frac{n(3n-1)}{2} + n = \frac{n(3n+1)}{2}$.

Euler's pentagonal theorem.

$$Q(q) = 1 + \sum (-1)^n \left(q^{\frac{n(3n-1)}{2}} + q^{\frac{n(3n+1)}{2}} \right)$$

Proof.

$$Q(q) = (1 - q)(1 - q^2)(1 - q^3) \dots$$

After expanding the above, the coefficients are +1 for q^{2n} (even) and -1 for q^{2n+1} (odd) when q 's (not 1's) are multiplied (which corresponds to even and odd numbers of rows in a Young diagram). So the coefficient in front of q^n equals (number of diagrams n boxes and an even number of rows of distinct length)-(number of diagrams n boxes and an odd number of rows of distinct length). Need to show this is equal to +1 or -1 if $n = \frac{k(3k-1)}{2}$ or $n = \frac{k(3k+1)}{2}$ and 0 otherwise. Shift boxes in Young diagrams around until one of two patterns emerges, each being a uniformly 'stepped' box:

$$n = k + (k + 1) + (k + 2) + \dots + (2k - 1) = \frac{3k^2 - k}{2}$$

$$n = (k + 1) + (k + 2) + \dots + 2k = \frac{3k^2 + k}{2}$$

34.7.5 Pentagonal theorem application

How to compute $p(n)$?

$$P(q) = 1 + p(1)q + p(2)q^2 + \dots$$

$$P(q)^{-1} = Q(q) = 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + \dots$$

So,

$$P(q)Q(q) = 1$$

$$\left(\sum p(n)q^n \right) (1 - q - q^2 + q^5 + q^7 + \dots) = 1$$

Expanding and selecting only the coefficients of q^n ,

$$p(n) - p(n-1) - p(n-2) + p(n-5) + p(n-7) - p(n-12) - p(n-15) \dots = 0$$

For instance, for $p(6)$ there is of course a $p(6)q^6$ term, but also $-p(5)q^5q$, $-p(4)q^4q^2$, $+p(1)qq^5$ and no $p(3), p(2)$ terms with q^6 . Now start with $p(0) = 1$ and recursively calculate:

$$\begin{aligned} p(1) &= p(0) = 1 \\ p(2) &= p(1) + p(0) = 2 \\ p(3) &= p(2) + p(1) = 3 \\ p(4) &= p(3) + p(2) = 5 \\ p(5) &= p(4) + p(3) - p(0) = 7 \\ p(6) &= p(5) + p(4) - p(1) = 11 \end{aligned}$$

34.8 Gaussian binomial coefficients

34.8.1 Definition

The number of paths from one corner to the opposite corner of an $m \times n$ grid is

$$\binom{m+n}{n} = \binom{m+n}{m}$$

which also corresponds to the number of Young diagrams occupying one corner of the grid. If each Young diagram is denoted by q^n where n is the number of blocks, then

$$1 + q + 2q^2 + q^3 + q^4 = \left[\begin{matrix} 4 \\ 2 \end{matrix} \right]_q$$

where the q^2 coefficient of 2 is from a 1×2 Young diagram and a 2×1 Young diagram. Generalizing,

$$\left[\begin{matrix} m+n \\ n \end{matrix} \right]_q = \sum_{\lambda \subset m \times n} q^{|\lambda|}$$

which are q -analogs of binomial coefficients (q -binomial or Gaussian binomial coefficients). A q -analog is a generalized function that when $q \rightarrow 1$, the function becomes a number. q -Binomial coefficients are the number of subspaces of dimension n of a vector space of dimension $m+n$ over a field with q elements.

34.8.2 Properties

1. The degree of a q -binomial is $\deg \left(\left[\begin{matrix} m+n \\ n \end{matrix} \right]_q \right) = mn$

2. The q-binomial coefficients of $q^0, q^1, q^{mn-1}, q^{mn}$ are all 1. Coefficients of q^0, q^{mn} correspond to Young diagrams with 0 blocks and mn (i.e. entire) blocks. Coefficients of q^1, q^{mn-1} correspond to Young diagrams with 1 block and 1 block missing. So, $1 + q + \dots + q^{mn-1} + q^{mn}$.
3. A Young diagram for a $1 \times n$ block is $\begin{bmatrix} n+1 \\ n \end{bmatrix} = 1 + q + q^2 + \dots + q^n = \frac{1-q^{n+1}}{1-q}$.
4. $\begin{bmatrix} m+n \\ n \end{bmatrix}_{q=1} = \binom{m+n}{n}$, i.e. each Young diagram counts as 1, so this is the sum of all Young diagrams.
5. q-Binomials are symmetric: $\begin{bmatrix} m+n \\ n \end{bmatrix}_q = \begin{bmatrix} m+n \\ m \end{bmatrix}_q$. Intuitively, a reflection across the main diagonal of a Young diagram.

Proposition (can be seen by modifying Pascal's triangle):

$$\begin{bmatrix} m+n \\ m \end{bmatrix} = \begin{bmatrix} m+n-1 \\ m-1 \end{bmatrix} + q^m \begin{bmatrix} m+n-1 \\ m \end{bmatrix}$$

For $q=1$,

$$\binom{m+n}{m} = \binom{m+n-1}{m-1} + \binom{m+n-1}{m}$$

In an $m \times n$ Young diagram (vertical m and horizontal n), the first RHS term is 'moving up' 1 unit from the bottom left corner. Then the remaining number of Young diagrams is $m-1 \times n$, or selecting $m-1$ upward movements from $m-1+n$ moves. The second RHS term is 'moving right' 1 unit from the bottom left corner. Then the remaining number of Young diagrams is $m \times n-1$, which is selecting m upward moves from $m+(n-1)$ total moves.

34.8.3 q-Integers

For a $1 \times n$ Young diagram,

$$[n] \stackrel{def}{=} \begin{bmatrix} n \\ 1 \end{bmatrix} = 1 + q + q^2 + \dots + q^{n-1} = \frac{1-q^n}{1-q}$$

Further,

$$[n]_{q=1} = n$$

In general,

$$[m]_q + [n]_q \neq [m+n]_q$$

(though true if $p=1$) but

$$[m]_q + q^m [n]_q = [n]_q + q^n [m]_q = [m+n]_q$$

Factorials.

$$n! = 1 \cdot 2 \cdot 3 \dots n$$

$$[n]! \stackrel{def}{=} [1][2][3][4]\dots[n]$$

where $[0]! = 1$.

Proposition.

$$\begin{bmatrix} m+n \\ m \end{bmatrix} = \frac{[m+n]!}{[m]![n]!}$$

Proof.

$$\begin{aligned} \begin{bmatrix} m+n \\ m \end{bmatrix} &= \begin{bmatrix} m+n-1 \\ m-1 \end{bmatrix} + q^m \begin{bmatrix} m+n-1 \\ m \end{bmatrix} \\ &= \frac{[m+n-1]!}{[m-1]![n-1]!} + q^m \frac{[m+n-1]!}{[m]![n-1]!} \\ &= \frac{[m+n-1]!}{[m-1]![n-1]!} \left(\frac{1}{[n]} + \frac{q^m}{[m]} \right) \\ &= \frac{[m+n-1]!}{[m-1]![n-1]!} \left(\frac{[m] + q^m[n]}{[m][n]} \right) \\ &= \frac{[m+n]!}{[m]![n]!} \end{aligned}$$

Proposition

$$\begin{bmatrix} m+n \\ n \end{bmatrix} = \frac{(1-q^{m+1})(1-q^{m+2})\dots(1-q^{m+n})}{(1-q)(1-q^2)\dots(1-q^n)}$$

Proof.

$$\begin{aligned} \begin{bmatrix} m+n \\ n \end{bmatrix} &= \frac{[m+n]!}{[m]![n]!} \\ &= \frac{[m+1][m+2]\dots[m+n]}{[n]!} \\ &= \frac{\frac{1-q^{m+1}}{1-q} \frac{1-q^{m+2}}{1-q} \dots \frac{1-q^{m+n}}{1-q}}{\frac{1-q}{1-q} \frac{1-q^2}{1-q} \dots \frac{1-q^n}{1-q}} \\ &= \frac{(1-q^{m+1})\dots(1-q^{m+n})}{(1-q)\dots(1-q^n)} \end{aligned}$$

which completes the proof. Now take the limit:

$$\lim_{m \rightarrow \infty} \begin{bmatrix} m+n \\ n \end{bmatrix} = \frac{1}{(1-q)(1-q^2)\dots(1-q^n)} = P_n(q)$$

which is the generating function for partitions with parts not exceeding n (Young diagrams made from $n \times \infty$). Generalizing to infinity,

$$\lim_{m,n \rightarrow \infty} \begin{bmatrix} m+n \\ n \end{bmatrix} = \prod_{k=1}^{\infty} \frac{1}{(1-q^k)} = P(q)$$

Note. The above is convergence for formal power series, so the topology is discrete. The coefficient at each power of q should be eventually constant for the limit to exist. Indeed,

$$\begin{aligned} \begin{bmatrix} N \\ r \end{bmatrix} &= \frac{(1-q^N)(1-q^{N-1})\dots(1-q^{N-r+1})}{(1-q)(1-q^2)\dots(1-q^r)} \\ &= \frac{1}{(1-q)(1-q^2)\dots(1-q^r)} \cdot (1-q^N r + 1 + \text{higher powers of } q) \end{aligned}$$

so the power series for $\begin{bmatrix} N \\ r \end{bmatrix}$ and for $\frac{1}{(1-q)(1-q^2)\dots(1-q^r)}$ coincide up to power $q^N - r$. Letting $N \rightarrow \infty$, the desired limit appears.

34.8.4 q-Binomial theorem

The regular binomial theorem is

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

The q -binomial theorem is

$$(1+xq)(1+xq^2)\dots(1+xq^n) = \sum \begin{bmatrix} n \\ k \end{bmatrix} q^{k(k+1)/2} x^k$$

Proof. Make a Young diagram of a partition, with exactly k summands (i.e. terms), with each summand $\leq n$. This makes a triangle in one corner of the Young diagram. The leftover of the Young diagram will be $\begin{bmatrix} n \\ k \end{bmatrix}$. The aforementioned triangle will be $q^{1+2+3+\dots}$ so multiplying the two completes the proof.

34.9 Applications

34.9.1 Chessboard

(1) How many ways are there to get from $(0,0)$ to $(5,3)$ on a chessboard if the moves are limited to moving right or up? $\binom{8}{3} = \binom{8}{5} = 56$ or use Pascal's Triangle:

Solution Using Pascal's Triangle

1							
1							
1							
1							
1							
1							
1	7	21	35	35	21	7	1
1	1	1	1	1	1	1	1

(2) If there are n assignments and k people, this is a tuple of n^k arrangements.

(3) Split 12 people into groups of two - adjust for repetitive group combinations:

$$\frac{\binom{12}{2}\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2}}{6!}$$

34.9.2 Numbers and digits

(1) How many non-negative integers with at most four digits in strictly increasing order are there? There are $\binom{10}{4} = 210$ combinations of 4-digit numbers (no duplicates). Each combination can be arranged in only one way - since each number is unique - in increasing order. So there are 210 numbers.

(2) Number of integers less than 10,000 where the consecutive digit is non-increasing. Combination with repetitions which can be arranged in only one combination. $\binom{4+(10-1)}{10-1}$. Choose 9 delimiters with 4 total spaces between them, where the spaces represent numbers.

(3) The number of integers less than 10,000 where digits sum to 9. Assume nine 1's and split them (order doesn't matter) with 3 delimiters. $\binom{12}{3} = 220$. But if the sum is to 10, then $\binom{13}{3} - 4 = 282$, the subtraction is for adjusting when all ten 1's assigned to one digit.

34.10 Probability

(1) To maximize probability of choosing a white ball from one of two randomly chosen boxes, place one white ball and no black balls in one box and all other balls in the other box. The probability of choosing a white ball approaches 75 percent. This is a maximization problem, maximizing probabilities of the boxes by moving balls around.

(2) Birthday problem. Assume uniform distribution of birthdays. What is the probability of at least one pair of birthdays being the same? Note dependence

of one pair on another: if pair (a,b) doesn't have the same birthday, this isn't independent from (c,d). The probability of NO pairs being the same is:

$$\frac{365 \times 364 \times 363 \dots \times (365 - n + 1)}{365^n} = \frac{\binom{365}{n} \times n!}{365^n}$$

So the answer is the complement.

(2a) Birthday problem variation. Assume uniform distribution of birthdays. The probability that two randomly chosen people have the same birthday is $\frac{365}{365 \times 365} = \frac{1}{365}$. There are $\binom{n}{2}$ unordered pairs of people. So the expectation of the number of pairs with the same birthday WHEN CHOSEN THIS WAY is $\binom{n}{2} \times \frac{1}{365}$.

(3) Markov's inequality and algorithm stopping times. If an algorithm runs an average n^2 , and it always stops in $10^3 n^2$, then it makes mistakes with at most 10^{-3} probability:

$$p(f \geq 10^3 n^2) \leq \frac{E(f)}{10^3 n^2} = \frac{n^2}{10^3 n^2} = 10^{-3}$$

35 Probability Theory

35.1 Riemann integrability

To calculate the proportion of rationals in $[0,1]$, use the Dirichlet function

$$p(x \in \mathcal{Q}) = \int_{\mathcal{Q}} \mathbb{1}_{(0,1)}(x) dx = \int_0^1 \mathbb{1}_{\mathcal{Q}}(x) dx$$

which is not well-defined. It is not Riemann-integrable. To be Riemann-integrable, the x-coordinate points chosen for the rectangular sums must be arbitrary, e.g. left rectangular sums, right rectangular sums, etc. must result in the same integration. However, this doesn't work for the Dirichlet function. Let

$$f(x) = \begin{cases} 0 & x \in \mathcal{Q} \\ 1 & \text{otherwise} \end{cases}$$

Let $0 = x_0 < x_1 \dots < x_n = 1$. The Riemann upper sum is

$$\begin{aligned} U_x &= \sum_{i=1}^n (x_i - x_{i-1}) \sup_{x_{i-1} < x < x_i} f(x) \\ &= \sum_{i=1}^n (x_i - x_{i-1}) \cdot 1 = 1 \end{aligned}$$

But the Riemann lower sum is

$$\begin{aligned} L_x &= \sum_{i=1}^n (x_{i-1} - x_i) \inf_{x_{i-1} < x < x_i} f(x) \\ &= \sum_{i=1}^n (x_{i-1} - x_i) \cdot 0 = 0 \end{aligned}$$

So they will not integrate to the same value. So the Riemann integral is not defined.

35.2 Measure theory

35.2.1 σ -algebras

A collection $\mathcal{F}$ of subsets of Ω is a σ -algebra, used to model probability outcome spaces:

1. $\mathcal{F}$ is non-empty
2. Stability by complement: if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$
3. Stability by countable union: if $(A_n) \in \mathcal{F}$ (A_n is a sequence: $A_1, A_2, \dots$) then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

A measurable space $(\Omega, \mathcal{F})$ comprises a set and a σ -algebra. The *measurable space* defines the subsets that will be measured. A *measurable function* is $A \in \mathcal{F}$.

35.2.2 Borel σ -algebras

To define a probability measure on even an uncountable set, can assign a probability to the sub-intervals of $[0,1]$ which can be closed, open, open-closed, etc., which is a σ -algebra. But the smallest one, which is the Borel σ -algebra, provides the highest resolution. The subsets (Ω) are the Borel sets in the Borel σ -algebra.

Borel σ -algebra importance. When developing a theory of integration, one wants (roughly speaking) a somehow 'friendly' collection of sets over which a notion of 'size' (read: measure) can be defined. The first naive guess at what sets we can assign such a 'size' to is simply all sets in the space, but there are various sets that one can construct such that any 'nice' description of their 'size' will yield contradictions (see Banach-Tarski's paradox using the axiom of choice from ZFC set theory).

The second naive guess would be the topology in the space (i.e. open intervals/closed intervals, arbitrary (finite) unions of open (closed) intervals arbitrary (finite) intersections of closed (open) intervals etc). This however is too restrictive. There are all sorts of sets that ones' intuition suggests have a well defined size that are not open or closed. For example, one simply knows that

$[0,1]$ should have 'size' equal 1, but if one is only considering open and closed sets, then this set will not be a defined 'size'.

So one needs something between all sets, and the topological sets. The Borel sets sit in between these two naive guesses, and are the first step towards defining the right notion of sets with a legitimate 'size'.

The real numbers form a Borel set.

$$((-\infty, x) \cup (x, \infty))^c = \{x\} = B(\mathcal{R})$$

where x is a singleton.

$$\mathcal{Q} \in \bigcup_{x \in \mathcal{Q}} x \in B(\mathcal{R})$$

So $\mathcal{R}$ is a Borel set. The Borel set is consistent with stability under countable unions despite having uncountably infinite elements because, e.g. the rationals (countably infinite) are stable and can be used to deduce the sum of the uncountably infinite elements as a whole.

35.2.3 Measurable function

A function between the underlying sets of two measurable spaces that preserves the structure of the spaces: the preimage of any measurable set is measurable. Analogous to continuous functions between topological spaces, which preserve the topological structure: the preimage of any open set is open. Let

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{R}, B(\mathcal{R}))$$

be a *measurable function* if $\forall B \in B(\mathcal{R})$, the preimage is

$$X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F}$$

where $\omega \in \Omega$. X can be a random variable in probability.

35.2.4 Measures and σ -additivity

Let the positive measure μ be

$$\mu : \mathcal{F} \rightarrow [0, \infty]$$

($[0,1]$ and $\mu = p$ for probability spaces) s.t.

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n)$$

for non-overlapping A_i 's. The last property is σ -additivity.

The *Lebesgue measure* is measure space $(\mathcal{R}, B(\mathcal{R}), \mu)$ comprising $\Omega = \mathcal{R}$ and

Borel σ -algebra $A = B(\mathcal{R})$, with measure $\mu((a, b)) = b - a, a < b$.

Probability measures add to 1.

Banach-Tarski and Vitali sets are non-measurable because there is no measure that can describe them. This happens because these sets use the axiom of choice of *uncountably* infinite elements. These and other pathological sets are non-measurable because they are 'amorphously' defined sets, i.e. sets that have a selected an 'unknown' amount of uncountably infinite things (perhaps the axiom of choice may be unnecessary for countably infinite cases, as these are theoretically finite - and therefore 'fit' in the present universe). This number is beyond the finite universe (countable = theoretically finite) so there is not enough 'time' to specify all of them. With uncountable infinities, there are infinite universes stuffed into Banach-Tarski's 'starting sphere'.

35.2.5 Lebesgue integration

Integration is just a measure of a function.

From above, let X be the measurable function

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{R}, B(\mathcal{R}))$$

The Lebesgue integral is then

$$\int X d\mu = \mu(A)$$

Let X be a linear combination of simple (indicator) functions

$$X = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$$

where $a_i \in \mathcal{R}$. Then,

$$\int X d\mu = \sum_{i=1}^n a_i \int \mathbb{1}_{A_i} d\mu = \underbrace{\sum_{i=1}^n a_i \mu(A_i)}_{\text{vertical sum} = \text{horizontal sum}}$$

The last term can be viewed either as a 'vertical sum' (Riemann) or 'horizontal sum' (Lebesgue). Where Riemann-integrable, the Riemann and Lebesgue integrals should equal.

If

$$\int |X| d\mu < \infty$$

then this forms the $L^1(\Omega, \mathcal{F}, \mu)$ (Lebesgue) space.

To calculate the proportion of rationals in $[0,1]$,

$$p(X \in \mathcal{Q}) = \int_0^1 \mathbb{1}_{\mathcal{Q}} d\lambda$$

where λ is the Lebesgue measure, $\mathbb{1}_{\mathcal{Q}} = \lambda(\mathcal{Q} \cap (0,1))$ is a Borel set and therefore measurable. So the integral exists, unlike the Riemann integral. Note: $\mathcal{Q} \cap (0,1) = (x_1, x_2 \dots x_n \dots), x_i \neq x_j$.

$$\begin{aligned} \lambda(\mathcal{Q} \cap (0,1)) &= \lambda \left(\bigcup_{i=1}^{\infty} \underbrace{x_i}_{\text{singleton}} \right) \\ &= \sum_{i=1}^{\infty} \underbrace{\lambda(x_i)}_0 = 0 \end{aligned}$$

since the rationals are countably infinite but the irrationals are uncountably infinite.

The Lebesgue integral can be seen as an abstraction of the above example, from

$$\int_0^1 f(x) d\mu = 0 \cdot \underbrace{\mu(A_0)}_{\text{irrational}} + 1 \cdot \underbrace{\mu(A_1)}_{\text{rational}}$$

to

$$\int_a^b f(x) d\mu = \sum_{i=1}^n y_i \cdot \mu(A_i)$$

and taking infinite sums,

$$\int_a^b f(x) d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^n y_i \cdot \mu(A_i)$$

$$\int_a^b f(x) d\mu = \lim_{n \rightarrow \infty} \int f_n(x) d\mu$$

35.2.6 Monotone convergence

Let (X_n) be a non-decreasing sequence of positive measurable functions with pointwise limit X . Then X is measurable and

$$\lim_{n \rightarrow \infty} \int X_n d\mu = \int \lim_{n \rightarrow \infty} X_n d\mu$$

35.2.7 Dominated convergence

Stronger result than MCT. DCT implies MCT. Let (X_n) be a sequence of measurable functions dominated by an integrable function and with pointwise limit X . Then X is integrable and

$$\lim_{n \rightarrow \infty} \int |X_n - X| d\mu = 0$$

$$\lim_{n \rightarrow \infty} \int X_n d\mu = \int \lim_{n \rightarrow \infty} X_n d\mu$$

35.2.8 Fatou's lemma

If there are no limits on X_n , then the best is Fatou's lemma. Let (X_n) be positive measurable. Then,

$$\int \liminf_{n \rightarrow \infty} X_n d\mu \leq \liminf_{n \rightarrow \infty} \int X_n d\mu$$

35.2.9 Fubini's theorem

Let $(S, \mathcal{F}_S, \mu_S), (T, \mathcal{F}_T, \mu_T)$ are measure spaces, the Cartesian product being $\Omega = S \times T$, the σ -algebra $\mathcal{F} = \sigma(A \times B : A \in \mathcal{F}_S, B \in \mathcal{F}_T)$, the product measure $\mu(A \times B) = \mu_S(A)\mu_T(B)$.

Fubini's theorem. For every $X : (\Omega, \mathcal{F}, \mu) \rightarrow (\mathcal{R}, B)$ positive measurable or integrable on a σ -finite (i.e. the set Ω is a countable union of measurable sets s.t. that each set has a finite measure) measure space

$$\int_{\Omega} X d\mu = \int_S \int_T X d\mu_T d\mu_S = \int_T \int_S X d\mu_S d\mu_T$$

35.3 Inclusion-Exclusion

Inclusion-exclusion general formula:

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{i_1 < \dots < i_k} P(A_{i_1} \cap \dots \cap A_{i_k})$$

which can be proven with the binomial theorem. Examples.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

35.4 Likelihood

Conditional probability:

$$L(\theta|x) = p(x|\theta)$$

Not same as probability distributions because doesn't sum to 1. Probability is varying y fixing θ while likelihood is fixing y and varying θ .

Random sample = independent sample.

Exchangeable sample = can change order but might not be independent (e.g. drawing balls out of a container without replacement). De Finetti showed that exchangeable samples are asymptotically random samples.

35.5 Conditioning

35.5.1 Radon-Nikodym theorem and derivative

Let $\phi : (\Omega, \mathcal{F}, \mu) \rightarrow (\mathcal{R}, B)$ be positive and measurable. Define the measure

$$\nu(A) = \int_A \phi d\mu = \int \phi \mathbb{1}_A d\mu$$

s.t. $A \in \mathcal{F}, A_i \cap A_j = \emptyset$.

Absolute continuity. Let μ, ν be measures. Then ν is absolutely continuous with respect to μ , i.e. $\nu \ll \mu$ if $\mu(A) = 0 \rightarrow \nu(A) = 0, \forall A \in \mathcal{F}$.

Radon-Nikodym theorem. Let $\nu \ll \mu$ be σ -finite measures. Then there is a unique $\phi : (\Omega, \mathcal{F}) \rightarrow (\mathcal{R}, B)$ positive measurable, called the Radon-Nikodym derivative of ν with respect to μ s.t.

$$\nu(A) = \int_A \phi d\mu, \forall A \in \mathcal{F}$$

where $\mathcal{F}$ is like 'perfect' information. Can view ν as the new measure being defined for any measurable subset A and the function ϕ being the density at a given point. The integral is with respect to an existing measure μ , which may be the canonical Lebesgue measure. For example, if ϕ represented mass density and μ was the Lebesgue measure, then $\nu(A)$ is the total mass in a spatial region A .

The ϕ is the Radon-Nikodym derivative:

$$\phi = \frac{d\nu}{d\mu}$$

35.5.2 Conditional expectation

Conditional expectation $Z = E(X|\mathcal{G})$ can be understood as the 'best guess' for X when only partial information ($\mathcal{G} \in \mathcal{F}$, former is imperfect information, latter is perfect information) is given. The definition of a conditional expectation:

$$\int_{\mathcal{G}} X dP = \int_{\mathcal{G}} E(X|\mathcal{G}) dP = \int_{\mathcal{G}} Z dP$$

Let $X \in L'(\Omega, \mathcal{F}, P)$ and $\mathcal{G} \subset \mathcal{F}$ be a σ -subalgebra. Then $Z = E(X|\mathcal{G})$ is a random variable s.t.

1. Z is $\mathcal{G}$ -measurable
2. $E(Z\mathbb{1}_A) = E(X\mathbb{1}_A), \forall A \in \mathcal{G}$

Proof. $\exists! Z$ (exists and is unique). Let $\nu : \mathcal{G} \rightarrow [0, \infty]$. Then,

$$\nu(A) = \int_A X dP = E(X\mathbb{1}_A)$$

$P = 0 \rightarrow \nu(A) = 0$ so $\nu \gg P$. Then by Radon-Nikodym, $\exists! Z \geq 0$ s.t.

$$\nu(A) = \int_A Z dP = E(Z\mathbb{1}_A)$$

Note that

$$E(X|Y) = E(X|\sigma(Y))$$

and that the conditional probability of an event given a σ -algebra:

$$P(A|\mathcal{G}) = E(\mathbb{1}_A|\mathcal{G})$$

35.5.3 Conditional probability formulas

(1) For all $X \in L'(\Omega, \mathcal{F}, P)$,

$$E(E(X|\mathcal{G})) = E(X)$$

Measure theory proof. $\forall A \in \mathcal{G}, E(Z\mathbb{1}_A) = E(X\mathbb{1}_A)$. Since $\mathcal{G}$ is a σ -algebra, it will contain at least the empty set and the entire set Ω . So let $A = \Omega$. Then, $E(Z) = E(X)$. Finally, $Z = E(X|\mathcal{G})$ is well defined by Radon-Nikodym.

(2) When $\mathcal{G}$ is generated by a finite or countably infinite partition (B_n) ,

$$E(X|\mathcal{G}) = \sum_n E(X|B_n) \mathbb{1}_{B_n}$$

Measure theory proof. Since Z is $\mathcal{G}$ -measurable,

$$Z = \sum_n^{\infty} a_n \mathbb{1}_{B_n}$$

Then,

$$E(Z \mathbb{1}_{B_n}) = E(X \mathbb{1}_{B_n}) = E(a_n \mathbb{1}_{B_n}) = a_n P(B_n)$$

$Z = E(X|\mathcal{G})$ is well defined by Radon-Nikodym. So,

$$Z = E(X|\mathcal{G}) = \sum_{n=1}^{\infty} \frac{E(X \mathbb{1}_{B_n})}{P(B_n)} \mathbb{1}_{B_n} = \sum_{n=1}^{\infty} E(X|B_n) \mathbb{1}_{B_n}$$

(3) For all $A \in \mathcal{F}$,

$$P(A) = \sum_n P(A|B_n)P(B_n)$$

Measure theory proof.

$$\begin{aligned} P(A) &= E(\mathbb{1}_A) = E(E(\mathbb{1}_A|\mathcal{G})) \\ &= E\left(\sum_n E(\mathbb{1}_A|B_n) \mathbb{1}_{B_n}\right) \\ &= E\left(\sum_n P(A|B_n) \mathbb{1}_{B_n}\right) \\ &\stackrel{MCT}{=} \sum_n P(A|B_n)E(\mathbb{1}_{B_n}) \\ &= \sum_n P(A|B_n)P(B_n) \end{aligned}$$

(4) For all discrete random variables Y ,

$$E(X) = \sum_y E(X|Y=y)P(Y=y)$$

Measure theory proof.

$$\mathcal{G} = \sigma(Y) = \sigma((Y=y) : y \in \text{range of } Y)$$

$$\begin{aligned} E(X) &= E(E(X|Y)) \\ &= E\left(\sum_y E(X|Y=y) \mathbb{1}_{Y=y}\right) \\ &= \sum_y E(X|Y=y)P(Y=y) \end{aligned}$$

35.5.4 Properties

1. Monotonic, linear, MCT, DCT as conditional expectations are similar to unconditional expectations
2. Perfect information: $E(X|\mathcal{G})$ if X is $\mathcal{G}$ -measurable.

3. No information: $E(X|\mathcal{G}) = E(X)$ when X and $\mathcal{G}$ are independent. Definition of independence: two σ -algebras $\mathcal{G}, \mathcal{H}$ are independent if $\forall A \in \mathcal{G}$ and $\forall B \in \mathcal{H}$, $P(A \cap B) = P(A) \cdot P(B)$.
4. Assuming X is $\mathcal{G}$ -measurable, (1) $E(X + Y|\mathcal{G}) = X + E(Y|\mathcal{G})$ and (2) $E(XY|\mathcal{G}) = X \cdot E(Y|\mathcal{G})$.

In stochastic processes, $\mathcal{G}$ is the past, X is the present, and Y is a random element independent of the past so conditioning on the past is removed: (1) $E(X + Y) = X + E(Y)$ and (2) $E(XY) = X \cdot E(Y)$.

35.5.5 Pairwise versus global independence

Events can be pairwise independent but not globally independent. This can happen if one of the events 'links' the others. Example.

- A_1 : blue die 1
- A_2 : red die 6
- A_3 : blue=red

These events are pairwise independent:

$$P(A_1) = P(A_2) = P(A_3) = \frac{1}{6}$$

$$P(A_1 \cap A_2) = P(A_1 \cap A_3) = P(A_2 \cap A_3) = \frac{1}{36}$$

But they are not globally independent:

$$P(A_1 \cap A_2 \cap A_3) = 0 \neq P(A_1)P(A_2)P(A_3)$$

35.6 Convergence

There are multiple modes of convergence for random variables as there are several types of 'closeness' that can be measured. As random variables are functions, it is the 'distance between functions' from functional analysis.

Convergence in mean and convergence almost surely are stronger than convergence in probability, which in turn is stronger than convergence in distribution.

Use graphical intuition for cases below: probability on y-axis and random variable X on x-axis. In the random variable sequence, let n be the sample size.

35.6.1 Convergence in distribution

Weakest form of convergence. $X_n \xrightarrow{d} X$ if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

where F_X is continuous. F is a CDF. Examples: Markov chains, central limit theorem (any normalized average of iid random variables converges in distribution to a standard normal).

35.6.2 Convergence in probability

Also called stochastic convergence.

The probability of an “unusual” outcome becomes smaller and smaller as the sequence progresses. Each manifestation of X_∞ must be arbitrarily close to X . On a p v X graph, the two must be the same for convergence in probability.

$X_n \xrightarrow{p} X$ if

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$$

convergence in probability is stronger than convergence in distribution. Proof.

$$\begin{aligned} P(Y \leq y) &\leq P(Z \leq y + \epsilon) + P(Z \geq y + \epsilon) \\ &= P(Z \leq y + \epsilon) + P(|Z - Y| \geq \epsilon) \end{aligned}$$

Then,

$$F_Y(y) \leq F_Z(y + \epsilon) + P(|Z - Y| \geq \epsilon)$$

Using the above and letting $Y \rightarrow X_n, y \rightarrow x, Z \rightarrow X$,

$$F_{X_n}(x) \leq F_X(x + \epsilon) + P(|X_n - X| \geq \epsilon)$$

Further letting $Y \rightarrow X, y \rightarrow x - \epsilon, Z \rightarrow X_n$,

$$F_X(x - \epsilon) \leq F_{X_n}(x) + P(|X_n - X| \geq \epsilon)$$

As $n \rightarrow \infty$,

$$F_{X_n}(x) \leq F_X(x + \epsilon) + 0$$

$$F_X(x - \epsilon) \leq F_{X_n}(x) + 0$$

Assume $X_n \xrightarrow{p} X$, then

$$F_{X_n}(x - \epsilon) \leq \lim_{n \rightarrow \infty} F_{X_n}(x) \leq F_X(x + \epsilon)$$

If F_X is continuous at x ,

$$F_X(x) = \lim_{\epsilon \rightarrow 0} F_X(x - \epsilon) \leq \lim_{n \rightarrow \infty} F_{X_n}(x) \leq \lim_{\epsilon \rightarrow 0} F_X(x + \epsilon) = F_X(x)$$

By the envelope theorem,

$$F_X(x) = \lim_{n \rightarrow \infty} F_{X_n}(x)$$

So

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

The converse is not necessarily true. For example, let $X_1, X_2, X_3, \dots$ be a sequence of i.i.d. $Bernoulli(1/2)$ random variables. Let also $X \sim Bernoulli(1/2)$ be independent from the X_i 's. Then, $X_n \xrightarrow{d} X$. However, X_n does not converge in probability to X , since $|X_n X|$ is in fact also a $Bernoulli(1/2)$ random variable and

$$P(|X_n X| \geq \epsilon) = \frac{1}{2}$$

A special case in which the converse is true is when $X_n \xrightarrow{d} c$, where c is a constant. In this case, convergence in distribution implies convergence in probability.

35.6.3 Convergence almost surely

Almost pointwise convergence of a probability space, with the exception of some edge or boundary cases. Almost sure convergence is: $X_n \xrightarrow{as} X$ if

$$P((\omega \in \Omega : \lim_{n \rightarrow \infty} \underbrace{X_n(\omega)}_{\mathcal{R}} = X(\omega))) = 1$$

If this applied to all ω , then it is pointwise convergence. Example: Martingales.

One thing that could happen is that for EVERY SINGLE (possibly uncountably many) points in the domain, the sequence could converge to some particular function. That is sure (i.e. pointwise) convergence and is overkill. A few outcome points in the domain having probability zero are unimportant. It's enough to know that the convergence happens at "almost" all outcomes (where almost means we can leave out a few that have zero probability).

Almost sure convergence of our sequence means it's possible to find a significant selection of outcomes (the set of outcomes the event, E which is a subset of the set of ALL possible outcomes) with desirable properties. First, the probability of E is one. Second, at EVERY SINGLE, point $\omega \in E$, $X_n(\omega) \rightarrow X(\omega)$. E may not include EVERY outcome, as it's not "sure convergence" but contains "enough" outcomes that the probability is still 1.

Proof that almost sure convergence implies convergence in probability. Let

$$A_n = (|X_n - X| \geq \epsilon)$$

and fix $\epsilon > 0$. Then, let

$$B_n = \bigcup_{k \geq n} A_k = \bigcup_{k \geq n} (|X_k - X| \geq \epsilon)$$

The sequence of sets is monotonically decreasing:

$$B_n \supseteq B_{n+1} \supseteq B_{n+2} \dots$$

So,

$$B_\infty = \bigcap_{n \geq 1} B_n = \bigcap_{n \geq 1} \bigcup_{k \geq n} A_k$$

We want to find

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} P(A_n)$$

Since $A_n \subseteq B_n$,

$$\lim_{n \rightarrow \infty} P(A_n) \leq \lim_{n \rightarrow \infty} P(B_n) \stackrel{MCT}{=} P\left(\lim_{n \rightarrow \infty} B_n\right) = 0$$

35.6.4 Convergence a.s. v. p

Assume you have some device, that improves with time. So, every time you use the device the probability of it failing is less than before.

Convergence in probability says that the chance of failure goes to zero as the number of usages goes to infinity. So, after using the device a large number of times, you can be very confident of it working correctly, it still might fail, it's just very unlikely.

Convergence almost surely is a bit stronger. It says that the total number of failures is finite. That is, if you count the number of failures as the number of usages goes to infinity, you will get a finite number. As you use the device more and more, you will, after some finite number of usages, exhaust all failures. From then on the device will work perfectly.

It's not possible to actually know when all failures have been exhausted, so from a purely practical point of view, there is not much difference between the two modes of convergence.

35.6.5 Convergence in mean or L^p

Convergence in $L^p(\Omega, \mathcal{F}, P), p \in \mathcal{R}: X_n \xrightarrow{L^p} X$ if

$$\lim_{n \rightarrow \infty} \|X_n - X\|_p = \lim_{n \rightarrow \infty} (E(|X_n - X|^p))^{1/p} = 0$$

If $p = 1$ then *mean convergence*, and $p = 2$ then *mean-square convergence*. Example: Martingales.

The p-th mean (L^p convergence implies convergence in probability. Jensen's equality:

$$f(E(X)) \leq E(f(X))$$

if f is convex (positive-parabolic). Fix $p > q$ so $f(x) = x^{p/q}$ is convex (exponent is greater than 1). Assume p-th mean convergence:

$$X_n \xrightarrow{L^p} X$$

Then,

$$\begin{aligned} \|X_n - X\|_q &\stackrel{def}{=} (E|X_n - X|^q)^{1/q} \\ &= (f(E|X_n - X|^q))^{1/p} \\ &= (E(f(|X_n - X|^q)))^{1/p} \text{ by Jensen's inequality} \\ &\stackrel{def}{=} (E|X_n - X|^p)^{1/p} \\ &= \|X_n - X\|_p \rightarrow 0 \end{aligned}$$

where the last step follows from the initial assumption of p-th mean convergence. Then, p-th mean convergence implies q-th mean convergence where $p > q$. Next step of proof uses Markov's inequality

$$P(X \geq \epsilon) \leq \frac{E(X)}{\epsilon}$$

Now assume p-th mean convergence for some $p \geq 1$. Set $p = 1$ and fix $\epsilon > 0$. Then,

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{E|X_n - X|}{\epsilon} = 0$$

where the last step follows from L^1 mean convergence.

35.6.6 Limits supremum and infimum

With limsup, take the biggest value past a certain N in a sequence s_n . As N increases, there are fewer values to choose from, so limsup decreases (or stays constant).

With liminf, take the smallest value past a certain N in a sequence. As N increases, there are fewer values to choose from, so liminf increases (or stays constant).

Example.

$$s_n = (4, -4, 3, -3, 2, -2, 1, -1, 0, 0)$$

At $N = 1$ the largest value past or at N is 4 and the smallest is 4. At $N = 4$, the largest value past or at N is 2 and the smallest is now 3.

35.6.7 Borel-Cantelli lemmas

Limsup of a sequence of events or sets:

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n \geq 0} \bigcup_{k \geq n} A_k = \bigcap_n \bigcup_{k=n}^{\infty} A_k$$

so that the limsup of A_n is infinite.

Borel-Cantelli lemma 1.

$$\sum_0^{\infty} P(A_n) < \infty \Rightarrow P\left(\limsup_{n \rightarrow \infty} A_n\right) = 0$$

Proof.

$$\begin{aligned} N &= \sum_0^{\infty} \mathbb{1}_{A_n} \\ E(N) &= E\left(\sum_0^{\infty} \mathbb{1}_{A_n}\right) \text{ monotonic} \\ &\stackrel{MCT}{=} \sum_0^{\infty} E(\mathbb{1}_{A_n}) \\ &= \sum_0^{\infty} P(A_n) < \infty \end{aligned}$$

Then,

$$E(N) < \infty \Rightarrow P(N = \infty) = 0$$

So,

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = P(N = \infty) = 0$$

Borel-Cantelli lemma 2. Converse of lemma 1.

$$\sum_0^{\infty} P(A_n) = \infty \Rightarrow P\left(\limsup_{n \rightarrow \infty} A_n\right) = 1$$

and A_n is independent. Proof.

$$\begin{aligned} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) &= P\left(\lim_{m \rightarrow \infty} \bigcap_{k=n}^m A_k^c\right) \text{ monotonically decreasing} \\ &\stackrel{MCT}{=} \lim_{m \rightarrow \infty} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) \\ &= \lim_{m \rightarrow \infty} \prod_{k=n}^{\infty} (1 - P(A_k)) \\ &\leq \lim_{m \rightarrow \infty} \prod_{k=n}^{\infty} e^{-P(A_k)} \end{aligned}$$

where the last line follows from $1 - x \leq e^{-x}$ which is just $y = e^{-x}$ and the tangent at (0,1). Then,

$$\begin{aligned} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) &\leq \lim_{m \rightarrow \infty} \exp\left(-\sum_{k=n}^m P(A_k)\right) \\ &= \exp\left(\lim_{m \rightarrow \infty} -\sum_{k=n}^m P(A_k)\right) \text{ by continuity} \\ &= \exp(-\infty) = 0 \end{aligned}$$

So,

$$P\left(\limsup_{n \rightarrow \infty} A_n\right) = P\left(\bigcap_{k=n}^{\infty} A_k^c\right) \stackrel{MCT}{=} \lim_{m \rightarrow \infty} P\left(\bigcup_{n=k}^{\infty} A_k\right) = 1$$

35.6.8 Convergence almost surely of a subsequence

Convergence in probability implies almost sure convergence of a subsequence.
Proof. Assume convergence in probability:

$$\lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) = 0$$

Define the n-subsequence

$$n_k = \min\left(n > n_{k-1} : P\left(|X_n - X| \geq \left(\frac{1}{2}\right)^k\right) \leq \left(\frac{1}{2}\right)^k\right)$$

Let

$$\begin{aligned} A_k &= \left(|X_{n_k} - X| \geq \left(\frac{1}{2}\right)^k\right) \\ \sum_k^{\infty} P(A_k) &\leq \sum_k^{\infty} \left(\frac{1}{2}\right)^k < \infty \end{aligned}$$

By Borel-Cantelli lemma 1,

$$P\left(\limsup_{k \rightarrow \infty} A_k\right) = 0$$

So,

$$P\left(\limsup_{k \rightarrow \infty} X_{n_k} \neq X\right) \leq P\left(\limsup_{k \rightarrow \infty} A_k\right) = 0$$

35.7 Law of large numbers

35.7.1 Markov's inequality

$$P(X \geq \epsilon) \leq \frac{E(X)}{\epsilon}$$

Proof.

$$\begin{aligned} E(X) &= E(X \mathbb{1}_{X \geq \epsilon}) + E(X \mathbb{1}_{X < \epsilon}) \text{ by linearity} \\ &\geq E(X \mathbb{1}_{X \geq \epsilon}) \\ &\geq \epsilon E(\mathbb{1}_{X \geq \epsilon}) = \epsilon P(X \geq \epsilon) \end{aligned}$$

Note: the second line is due to the RHS adding values $X > \epsilon$ divided by the total number (not just those greater than ϵ) of samples.

35.7.2 Chebyshev's inequality

Given $E(X), Var(X) < \infty$,

$$P(|X - E(X)| \geq \epsilon) \leq \frac{Var(X)}{\epsilon^2}$$

Proof.

$$\begin{aligned} (X - E(X))^2 &\geq 0 \\ P(|X - E(X)| \geq \epsilon) &= P((X - E(X))^2 \geq \epsilon^2) \\ &\leq \frac{E((X - E(X))^2)}{\epsilon^2} \text{ by Markov's inequality} \end{aligned}$$

35.7.3 Weak law of large numbers

Let X_i be iid and $E(X_i) = \mu < \infty$. Then,

$$\frac{1}{n} \sum_i^n \xrightarrow{p} \mu$$

The variance doesn't have to be bounded. However, proving weak LLN in this case requires Martingale theory. Instead, prove weak LLN with finite variance. Re-express weak LLN:

$$S_n = \sum_i^n X_i, \frac{S_n}{n} \xrightarrow{as} \mu$$

Start with finite mean and variance:

$$\begin{aligned} E\left(\frac{S_n}{n}\right) &= \mu < \infty, Var(X_i) = \sigma^2 < \infty \\ Var\left(\frac{S_n}{n}\right) &= \frac{Var(\sum_i^n X_i)}{n^2} = \frac{\sigma^2}{n} \end{aligned}$$

Use Chebyshev's inequality,

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2} = 0$$

The variance of the sample mean approaches 0 because the numerator adds variance terms (which are squares or rectangles) divided by the total samples squared. The denominator outpaces the numerator. The denominator is n^2 because variance is defined using squares. This is also linked to the CLT. The intuition for approaching 0 is due to larger sample sizes' samples cancelling each other's changes and having fewer or more predictable big changes.

35.7.4 Strong law of large numbers

Let X_i be iid and $E(X_i) = \mu < \infty$. Then,

$$\frac{1}{n} \sum_i^n \xrightarrow{as} \mu$$

Again, the variance doesn't have to be bounded. However, proving strong LLN in this case requires Martingale theory. Instead, prove strong LLN with finite variance. Re-express strong LLN:

$$S_n = \sum_i^n X_i, \frac{S_n}{n} \xrightarrow{p} \mu$$

Begin with

$$Var\left(\frac{S_n}{n}\right) = \frac{n^2 Var(X_i)}{n^4} = \frac{\sigma^2}{n^2}$$

By Chebyshev's inequality,

$$P\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \leq \left(\frac{\sigma}{n\epsilon}\right)^2$$

Then,

$$\sum_{n=1}^{\infty} P\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \leq \sum_{n=1}^{\infty} \left(\frac{\sigma}{n\epsilon}\right)^2 < \infty$$

By Borel-Cantelli lemma 1,

$$P\left(\limsup_{n \rightarrow \infty} \left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) = P\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon \text{ infinitely often}\right) = 0$$

Since

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} \neq \mu\right)$$

implies $\exists \epsilon > 0$ s.t.

$$\left|\frac{S_n}{n} - \mu\right| \geq \epsilon$$

then

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} \neq \mu\right) \leq \sum_{\epsilon} P\left(\limsup_{n \rightarrow \infty} \left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) = 0$$

where the right-hand side is from the result applying Borel-Cantelli lemma 1 above. So,

$$\frac{S_{n^2}}{n^2} \xrightarrow{as} \mu$$

Fix $k \in \mathbb{N}$ and let $n^2 \leq k < (n+1)^2$ (so the difference between n^2 and k is at most $(n+1)^2 - n^2 = 2n+1 \approx 2n$). By the triangle inequality,

$$\left| \frac{S_{n^2}}{n^2} - \frac{S_k}{k} \right| \leq \left| \frac{S_{n^2}}{n^2} - \frac{S_{n^2}}{k} \right| + \left| \frac{S_{n^2}}{k} - \frac{S_k}{k} \right|$$

where

$$\left| \frac{S_{n^2}}{n^2} - \frac{S_{n^2}}{k} \right| \rightarrow \left| \frac{1}{n^2} - \frac{1}{k} \right| = \frac{k - n^2}{kn^2} \leq \mathcal{O}\left(\frac{1}{nk}\right) \rightarrow \mathcal{O}\left(\frac{1}{n^2}\right)$$

$$\left| \frac{S_{n^2}}{k} - \frac{S_k}{k} \right| \rightarrow \frac{S_{n^2} - S_k}{k} \leq \mathcal{O}\left(\frac{2n+1}{n^2}\right) \rightarrow \mathcal{O}\left(\frac{2n}{n^2}\right) \rightarrow \mathcal{O}\left(\frac{n}{n^2}\right) \rightarrow \mathcal{O}\left(\frac{1}{n}\right)$$

So,

$$\left| \frac{S_{n^2}}{n^2} - \frac{S_k}{k} \right| \leq \mathcal{O}\left(\frac{1}{n}\right)$$

and so

$$\left| \frac{S_{n^2}}{n^2} - \frac{S_k}{k} \right| \xrightarrow{as} 0$$

Assembling above results with the triangle inequality,

$$\left| \frac{S_k}{k} - \mu \right| \leq \left| \frac{S_k}{k} - \frac{S_{n^2}}{n^2} \right| + \left| \frac{S_{n^2}}{n^2} - \mu \right| \xrightarrow{as} 0$$

which completes the proof.

35.7.5 Weak versus strong LLN

The weak law states that for a specified large n , the average $\bar{X}_n$ is likely to be near μ . Thus, it leaves open the possibility that $|X_n - \mu| > \epsilon$ happens an infinite number of times, although at infrequent intervals. (Not necessarily $|X_n - \mu| \neq 0$ for all n).

The strong law shows that this almost surely will not occur. It implies that with probability 1, for any $\epsilon > 0$ the inequality $|X_n - \mu| < \epsilon$ holds for all large enough n .

With SLLN, a scientific experiment to obtain, say, the speed of light, is justified in taking averages. At least in theory, after obtaining enough data, you can get arbitrarily close to the true speed of light. There won't be any failures (however improbable) in the averaging process. Choose some $\delta > 0$ arbitrarily

small. You obtain n estimates $X_1, X_2, \dots, X_n$ of the speed of light (or some other quantity) that has some 'true' value, say μ . You compute the average

$$S_n = \frac{1}{n} \sum_{k=1}^n X_k$$

As we obtain more data (n increases) we can compute S_n for each $n = 1, 2, \dots$. The weak law says (under some assumptions about the X_n) that the probability

$$P(|S_n - \mu| > \delta) \rightarrow 0$$

as n goes to ∞ . The strong law says that the number of times that $|S_n - \mu|$ is larger than δ is finite (with probability 1). That is, if we define the indicator function $I(|S_n - \mu| > \delta)$ that returns one when $|S_n - \mu| > \delta$ and zero otherwise, then

$$\sum_{n=1}^{\infty} I(|S_n - \mu| > \delta)$$

converges. This gives considerable confidence in the value of S_n , because it guarantees (i.e. with probability 1) the existence of some finite n_0 such that $|S_n - \mu| < \delta$ for all $n > n_0$ (i.e. the average never fails for $n > n_0$). Note that the weak law gives no such guarantee.

Example. Consider the interval $\Omega = [0, 1]$ as a sample space. Use a uniform probability. Define the random variables $X_n : [0, 1] \rightarrow [0, 1]$ as follows: for $\omega \in [0, 1] = \Omega$, $X_n(\omega) = 1$ if $\omega \in A_n$, where A_n is a shifting or changing interval otherwise it is 0.

This shifting interval will start at $A_1 = [0, 0.1]$, $A_2 = [0.1, 0.2]$ and will shift right keeping its length until it reaches the right limit, that is, $A_{10} = [0.9, 1]$. Then its size will halve and will move to the left boundary, $A_{11} = [0, 0.05]$ and repeat. $P(X_n = 1) = P(A_n)$ is the length of A_n . So by construction $P(X_n = 1) \rightarrow 0$ that is, $P(X_n = 0) \rightarrow 1$. This is weak convergence (convergence in probability). For any outcome of the sample space, that is $\forall x \in [0, 1]$, the problematic interval A_n will hit to your outcome x infinitely many times. Therefore, for any outcome of your experiment, the value of X_n will never converge to 0. The "errors" ($X_n = 1$) will occur with decreasing frequency but they will always occur.

The strong law of large numbers is different. In an experiment (sample from the sample space), waiting long enough, the value of X_n will converge to 0, unless in a case which is extremely unlucky and sampled from a probability 0 event, e.g. if $A_n = [0, 1/n]$: for all outcomes except $x = 0$ the sequence X_n will converge to 0 eventually. With the weak law of large numbers, in any single experiment, however long the wait, convergence may not occur.

35.7.6 Ergodic theorem

Ergodic theorem. Let $X \in L^1(\Omega, \mathcal{F}, P)$ and $f: \Omega \rightarrow \Omega$. It must be that f is (1) measure-preserving, i.e. $P(f^{-1}(A)) = P(A), \forall A \in \mathcal{F}$, and (2) ergodic, i.e. $f^{-1}(A) = A \Rightarrow P(A) = 0$ or $P(A) = 1$ (like a fixed point). Then,

$$\frac{1}{n} \sum_{i=1}^n X(f^i(\omega)) \xrightarrow{as} E(X)$$

where f^i is f composed of itself i times. Note that f is measure-preserving and ergodic s.t. $X_i(\omega) = X(f^i(\omega)), \forall \omega \in \Omega$ iff the sequence (X_i) is *stationary*.

Stationarity is when

$$(X_1, X_2, \dots, X_k) \stackrel{d}{=} (X_n, X_{n+1}, \dots, X_{n+k-1})$$

where the equality is 'equal in distribution'. iid implies stationarity but not necessarily the reverse. So the strong LLN is stronger than the ergodic theorem.

35.7.7 Central limit theorem

The central limit theorem (CLT) establishes that, in many situations, when independent random variables are *summed up*, their properly normalized sum tends toward a normal distribution even if the original variables themselves are not normally distributed. The random variables have to be summed up; so sample means follow the CLT, but other sample statistics such as sample variances do not (they follow a distribution with a chi-square in it).

The CLT is the natural result to study after LLNs. Let X_i be iid. Further, assume $\mu = E(X_i) < \infty$ and $\sigma^2 = Var(X_i) < \infty$. Then,

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

which is equivalent to

$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Intuition. With a group of perpendicular vectors $x_1, \dots, x_n$ of length l , then $\frac{S_n = x_1 + \dots + x_n}{\sqrt{n}}$ is again of length l . The denominator is a renormalization factor. There is a deep connection between independent random variables and orthogonal vectors. When random variables are independent, that basically means that they are orthogonal vectors in a vector space of functions. Now just horizontally shift by μ and:

$$Z = \frac{X - \mu}{\sqrt{n}}$$

Alternative consideration of CLT. Since

$$\bar{X} = \frac{S_n}{n}$$

leads to a variance of the sample of 0 as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \text{Var} \left(\frac{S_n}{n} \right) = \lim_{n \rightarrow \infty} \frac{n\sigma^2}{n^2} = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$$

If instead X is normalized by multiplying by $\sqrt{n}$:

$$\sqrt{n}\bar{X} = \frac{S_n}{\sqrt{n}}$$

then the variance of the function is constant

$$\lim_{n \rightarrow \infty} \text{Var} \left(\frac{S_n}{\sqrt{n}} \right) = \lim_{n \rightarrow \infty} \frac{n\sigma^2}{n} = \sigma^2$$

which is equivalent to starting with $n\text{Var}(\bar{X})$. The distribution is a normal distribution (and after adjusting for the mean and variance, it's a standard normal distribution because it is centered at 0 and has variance 1):

$$Z = \sqrt{n} \frac{S_n - n\mu}{n\sigma} = \frac{S_n - n\mu}{\sqrt{n}\sigma} = \frac{\sqrt{n}(\bar{X} - \mu)}{n\sigma}$$

So as per the CLT:

$$\sqrt{n}(\bar{X} - \mu) \rightarrow \mathcal{N}(0, 1)$$

So the normalization or scaling factor $\sqrt{n}$ ultimately originates from the squared nature of variance, i.e. originates from our almost arbitrary definition of variance using squared terms.

To prove that the sum of random variables becomes a normal distribution, rather than another type of distribution, use moment generating functions.

There are multiple types of CLTs, depending on how the random variable acts. The simplest is the Lindeberg-Levy for iid random variables. If the random variables are less than iid, then other CLTs might apply.

35.7.8 Large deviation estimates

Indicates how fast the average is heading to the mean. In other words, the speed of

$$P \left(\frac{S_n}{n} \geq \mu + \epsilon \right) \rightarrow 0$$

Assume the moment-generating function of X_i is defined as

$$M(\theta) = E(e^{\theta X_i}) < \infty$$

$$M(0) = 1$$

Then, $\forall \epsilon > 0, \exists \gamma = \gamma(\epsilon) > 0$ s.t.

$$P\left(\frac{S_n}{n} \leq \mu - \epsilon\right) \leq e^{-\gamma\mu}$$

$$P\left(\frac{S_n}{n} \geq \mu + \epsilon\right) \leq e^{-\gamma\mu}$$

which is exponential decay.

35.7.9 Stirling's formula

Derive using CLT. Stirling's formula is

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

as $n \rightarrow \infty$. Let X_i be a sequence of independent Poisson random variables with parameter 1 ($\lambda = 1$). $S_n = X_1 + X_2 + \dots + X_n$ where $X_i \sim \text{Poisson}(1)$ leads, using CLT, to

$$\frac{S_n - n}{(1)\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

$$P(S_n = n) = P(n \leq S_n < n + 1)$$

$$= P\left(0 \leq \frac{S_n - n}{\sqrt{n}} < \frac{1}{\sqrt{n}}\right)$$

$$\sim \int_0^{1/\sqrt{n}} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

$$= \frac{1}{\sqrt{2\pi n}}$$

Since the sum of $\text{Poisson}(1)$ n times is $\text{Poisson}(n)$,

$$\sqrt{2\pi n} P(S_n = n) = \sqrt{2\pi n} \left(\frac{n^n}{n!} e^{-n}\right) \rightarrow 1$$

which completes the derivation.

36 Probability

36.1 Generating functions

Moment-generating functions (MGFs) are Laplace transforms and characteristic functions are Fourier transforms. MGFs do not exist for a PDF if there are undefined moments as these are like Laplace transform poles. Characteristic functions always exist for PDFs. Both MGFs (where it exists) and characteristic functions fully characterize PDFs as they are 1-1 mappings (which is intuitive based on Fourier and Laplace transforms). It is easier to extract moments from MGFs than characteristic functions (in the latter case, the calculation is only slightly more involved).

36.1.1 MGFs

The MGF defines the moments of a random variable.

The moment generating function of a random variable X is defined as the expected value of e^{tX} . By LOTUS,

$$M_X(t) \equiv E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$$

M_X is the MGF of X because its n -th derivative, evaluated at 0, gives the n -th moment of X , i.e. the expected value of X^n .

Flipping the sign on t in the integral above results in the two-sided Laplace transform of f_X . The MGF of X at t is the two-sided Laplace transform of f_X at $-t$. If the density function is zero for negative values, then the two-sided Laplace transform reduces to the more common (one-sided) Laplace transform.

$$\text{Two-sided: } \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$$

$$\text{One-sided: } \int_0^{\infty} e^{tx} f_X(x) dx$$

MGF of normal distribution.

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

So,

$$\begin{aligned} m_X(t) &= \int e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\ &= \int \frac{1}{\sqrt{2\pi}} e^{tx - \frac{1}{2}x^2} dx \\ &= \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} e^{\frac{1}{2}t^2} dx \\ &= e^{\frac{1}{2}t^2} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx \text{ latter term is } N(t,1) \\ &= e^{\frac{1}{2}t^2} (1) \end{aligned}$$

More generally,

$$m_X(t) = e^{E(X) + \frac{1}{2}V(X)}$$

where $E(X), V(X)$ are the first and second moments.

MGF of Brownian motion. Since $B(t) \sim \mathcal{N}(0, t)$ and $zB(t) \sim \mathcal{N}(0, z^2t)$,

$$m_B(z) = E(e^{zB(t)}) = e^{\frac{1}{2}tz^2}$$

36.1.2 Characteristic function

The characteristic function completely defines the PDF of a random variable.

The characteristic function of a random variable is a variation of the MGF. Rather than use $E(e^{tX})$, it uses $E(e^{itX})$. So the characteristic function of a random variable is the Fourier transform of its density function.

The right tail of the PDF f_X has to go to zero faster than e^{tX} , which isn't the case for fat-tailed distributions. That's not a problem for the characteristic function because the Fourier transform exists for any density function. The MGF doesn't need to exist because in particular it requires the existence of moments of any order.

36.1.3 Cumulants

Also called characteristic exponents. Log of MGF and can be mathematically useful because e.g. means of iid rv's = means of individual rv's:

$$\begin{aligned}K_{X+Y}(t) &= \log E(e^{t(X+Y)}) \\&= \log(E(e^{tX})E(e^{tY})) \\&= \log E(e^{tX}) + \log E(e^{tY}) \\&= K_X(t) + K_Y(t)\end{aligned}$$

36.2 Sum of independent random variables

Calculating $Z = X + Y$ where X and Y are independent random variables.

Discrete case. Example.

$$\begin{aligned}p_Z(e) &= p(X = 0, Y = 3) + P(X = 1, Y = 2) + \dots \\&= p_X(0)p_Y(3) + p_X(1)p_Y(2) + \dots \text{ by independence}\end{aligned}$$

Generalizing,

$$\begin{aligned}p_Z(z) &= \sum_x p(X = x, Y = z - x) \\&= \sum_x p_X(x)p_Y(z - x)\end{aligned}$$

which is a convolution. Intuition for $p_Y(z - x)$ is to flip the x-values across the y-axis (i.e. -x) and add shift horizontally to the right by z. Then multiply the two PDFs.

Continuous case. Can derive by analogy,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)dx$$

but this can be derived also using conditioning

$$\begin{aligned} f_{Z|X}(z|x) &= f_{X+Y|X}(z|x) \\ &= f_{X+Y}(z) \text{ assuming } X \text{ is known} \\ &= f_Y(z-x) \text{ by horizontal translation, e.g. } f_{X+b}(x) = f_X(x-b) \end{aligned}$$

So,

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_{X,Z}(x, z) \\ &= \int_{-\infty}^{\infty} f_X(x) f_{Z|X}(z|x) \\ &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) \end{aligned}$$

Normal distributions. If $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent then $\alpha X_1 + \beta X_2 \sim \mathcal{N}(\alpha\mu_1 + \beta\mu_2, \alpha^2\sigma_1^2 + \beta^2\sigma_2^2)$. So the means add (which is intuitive) and the variances add because they're independent (i.e. the covariance term is 0).

36.3 Infinitely divisible distributions

An infinitely divisible probability distribution can be expressed as the probability distribution of the sum of an arbitrary number of iid random variables. So ξ is an infinitely divisible distribution if

$$\xi \stackrel{d}{=} Y_1 + \dots + Y_n$$

The characteristic functions are related:

$$\Phi_\xi(u) = E(e^{iu\xi}) = E(e^{iu(nY_i)}) = E((e^{iu(Y_i)})^n) = (\Phi_{Y_i}(u))^n$$

So,

$$\begin{aligned} \Phi_\xi(u) &= (\Phi_{Y_i}(u))^n \\ (\Phi_\xi(u))^{1/n} &= \Phi_{Y_i}(u) \end{aligned}$$

must exist. Further, infinitely divisible distributions (1) have NO real solutions to $\Phi(u) = 0$; they're all complex solutions; and (2) the probability support $\mu = (x : \forall \text{ open sets } G \text{ containing } x : \mu(G) > 0)$ is unbounded.

Infinitely divisible distributions can be 'scaled' and look kind of same when their random variables are added. Examples of infinitely divisible distributions: normal, cauchy, gamma, exponential, negative binomial, geometric, Poisson, and compound Poisson. Most distributions are infinitely divisible. Uniform and Bernoulli distributions are the most common distributions that aren't infinitely divisible.

Example 1. For a normal distribution $\mathcal{N}(\mu, \sigma^2)$,

$$\Phi_\xi(u) = e^{i\mu u - \frac{1}{2}\sigma^2 u^2}$$

But for the sum

$$\xi \stackrel{d}{=} Y_1 + \dots + Y_n, Y_i \sim iid$$

The characteristic function is

$$(\Phi_\xi(u))^{1/n} = e^{i\frac{\mu}{n}u - \frac{1}{2}\frac{\sigma^2}{n}u^2}$$

which exists, so is infinitely divisible. So,

$$\xi \sim \mathcal{N}\left(\frac{\mu}{n}, \frac{\sigma^2}{n}\right)$$

Example 2. The Cauchy distribution is

$$p(x) = -\frac{1}{\pi\gamma\left(\frac{1+(x-x_0)^2}{\gamma^2}\right)}$$

where an expectation doesn't exist. It has a characteristic function

$$\Phi(u) = e^{x_0 iu - \gamma|u|}$$

But for the sum

$$\xi \stackrel{d}{=} Y_1 + \dots + Y_n, Y_i \sim iid$$

The characteristic function is

$$(\Phi_\xi(u))^{1/n} = e^{\frac{x_0}{n}iu - \frac{\gamma}{n}|u|}$$

which exists, so is infinitely divisible.

Example 3. The Gamma (Γ) distribution is

$$p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x > 0$$

with characteristic function

$$\Phi(u) = \left(1 - \frac{iu}{\beta}\right)^{-\alpha}$$

But for the sum

$$\xi \stackrel{d}{=} Y_1 + \dots + Y_n, Y_i \sim iid$$

The characteristic function is

$$(\Phi_\xi(u))^{1/n} = \left(1 - \frac{iu}{\beta}\right)^{-\alpha/n} \sim \Gamma\left(\frac{\alpha}{n}, \beta\right)$$

which exists, so is infinitely divisible.

Example 4. Uniform distribution $\xi \sim Unif(a, b)$ is not infinitely divisible since $\Phi(u) = 0$ with real solutions:

$$\begin{aligned}\Phi_\xi(u) &= \int_a^b e^{iux} \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \frac{e^{iua}}{iu} (e^{iu(b-a)} - 1)\end{aligned}$$

where $\Phi = 0$ iff $(e^{iu(b-a)} - 1) = 0 \Rightarrow u = \frac{2\pi k}{b-a}, k \in \mathbb{Z}/\{0\}$ from the unit circle. So the uniform distribution isn't infinitely divisible, which matches intuition.

Example 5. Bernoulli distribution isn't infinitely divisible because its support, $\{0, 1\}$, is bounded (i.e. finite domain). It is intuitive that it's not possible to infinitely divide with a finite support.

36.4 Stable distributions

A subclass of infinitely divisible distributions and that are specified by location (e.g. mean) and scale (e.g. variance). If $\xi = \xi_i \sim iid$, then

$$\sum \xi_i \stackrel{d}{=} a_n \xi + b_n$$

where a_n, b_n are deterministic (i.e. not stochastic). Examples of stable distributions: normal, Cauchy. This is intuitive because, e.g. a normal distribution is by the central limit theorem is many infinitely divided distributions added together.

A stable distribution is infinitely divisible but not vice versa. The location and particularly scale provide a sense of self-similarity.

36.5 Joint probabilities and expectations

If X, Y are independent,

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

Then using indicator functions,

$$E(I(X \leq x)I(Y \leq y)) = E(I(X \leq x))E(I(Y \leq y))$$

$$E(f(X)g(Y)) = E(f(X))E(g(Y))$$

36.6 Transition probabilities

The one-step transition probability from state i to j is

$$p_{ij} = p(X_{k+1} = j | X_k = i)$$

Markov processes are memoryless. If this is a Markov process, the transition probability from state i to j through n steps is the addition of products:

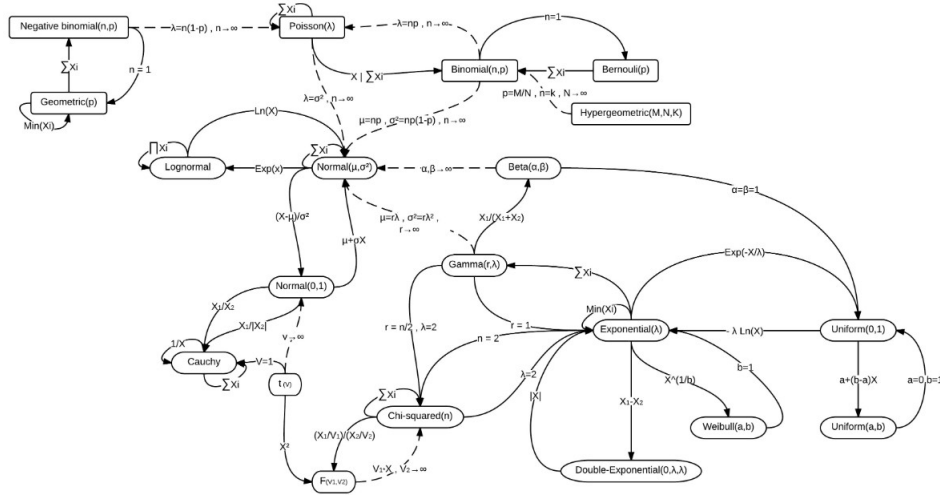
$$\begin{aligned} p_{ij} &= p_{ij}(X_{k+n} = j | X_k = i) \\ &= \sum p(X_{k+n} = j | X_u = r, X_k = i) p(X_u = r | X_k = i) \end{aligned}$$

through given n steps. This is generalized to the Chapman-Kolmogorov equation:

$$p(y, t, x, s) = \int p(y, t, z, u) p(dz, u, x, s)$$

37 Probability distributions

37.1 Exponential distribution family



37.2 Negative binomial

Relationship to binomial distribution. Event $X \geq r$ is the set of outcomes that satisfy "n trials led to r successes or more", which is equivalent to "r-th success happened at n-th trial or before", which is in turn equivalent to "n trials or less were required to get r successes", and that is it.

$$\begin{aligned} p(X \geq r) &= P(\text{at least } r \text{ successes in } n \text{ trials}) \\ &= P(r\text{-th success in } n\text{-th trial or before}) \\ &= P(n \text{ or fewer trials to get } r \text{ successes}) \\ &= P(Y \leq n) \end{aligned}$$

37.3 Log normal

Very useful if want normal distribution but not negative values: log normal and then convert to normal scale.

$$X = e^Y$$

where X is log-normally distributed and $Y \sim \mathcal{N}(\mu, \sigma^2)$. So,

$$\ln(X) = Y$$

The potential returns of a stock can be graphed in a normal distribution. The prices of the stock, however, can be graphed in a log-normal distribution. The log-normal distribution curve can therefore be used to help better identify the compound return that the stock can expect to achieve over a period of time.

37.3.1 PDF

The PDF is

$$\begin{aligned} \ln(X) &\sim \mathcal{N}(\mu, \sigma^2) \\ f(x) &= \frac{d}{dx} p(X \leq x) = \frac{d}{dx} p(\ln(X) \leq \ln(x)) \\ &= \frac{d}{dx} \Phi\left(\frac{\ln(x) - \mu}{\sigma}\right) \\ &= \phi\left(\frac{\ln(x) - \mu}{\sigma}\right) \frac{d}{dx} \left(\frac{\ln(x) - \mu}{\sigma}\right) \\ &= \phi\left(\frac{\ln(x) - \mu}{\sigma}\right) \frac{1}{\sigma x} \\ &= \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\ln(x) - \mu}{\sigma}\right)^2} \end{aligned}$$

37.3.2 Expected value

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_0^{\infty} x \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{\ln(x) - \mu}{\sigma}\right)^2} dx \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_0^{\infty} e^{-\frac{1}{2}\left(\frac{\ln(x) - \mu}{\sigma}\right)^2} dx \end{aligned}$$

Let $t = \frac{\ln(x) - \mu}{\sigma}$. Then $dt = \frac{1}{\sigma x} dx \Rightarrow \sigma x dt = dx \Rightarrow \sigma e^{\mu + \sigma t} dt = dx$. Then $x = 0 \rightarrow t = -\infty$ and $x = \infty \rightarrow t = \infty$. Continuing from above,

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}t^2} \sigma e^{\mu + \sigma t} dt \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(t^2 - 2\sigma t + \sigma^2)} \sigma e^{\mu + \frac{1}{2}\sigma^2} dt \\
&= e^{\mu + \frac{1}{2}\sigma^2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(t^2 - \sigma^2)} dt \\
&= e^{\mu + \frac{1}{2}\sigma^2} (1)
\end{aligned}$$

38 Statistics

38.1 Basics

38.1.1 Variance

The formulas for variance and standard deviation fall out from the geometry of the statistics. Almost all scatter plots are in variable space, for example, x axis is "height", and y axis is "weight." But a fundamentally different approach to plotting data is "subject space," where for example, the x axis is Subject 1; the y axis is Subject 2; and the z-axis is Subject 3. Each variable is then plotted as a vector.

If the vector is centered, its length is proportional to its standard deviation. The length of the vector is calculated by the Pythagorean theorem, which uses squares; therefore, the formula for calculating the SD and variance uses squares or square roots.

The correlation between variables is the cosine of the angle between vectors (in subject space, each vector represents a variable), which leads directly to the formula for calculating the Pearson correlation coefficient.

Calculation of p-values also is explained by this framework.

The sample variance is unbiased:

$$E(S^2) = \sigma^2$$

38.1.2 Standard deviation

The sample standard deviation is biased:

$$E(S) \neq \sigma$$

because of Jensen's inequality:

$$E(\sqrt{s^2}) = \sqrt{E(s^2)} = \sigma$$

However, the sample standard deviation is consistent:

$$\lim_{n \rightarrow \infty} \frac{\sigma^2}{S^2} = \lim_{n \rightarrow \infty} \frac{n-1}{n} = 1$$

The variance is unbiased, so $E(S^2) = \sigma^2$, but it does not have a normal distribution. It is *chi*²-squared according to Cochran's derivation. So if samples are drawn from that distribution, which is asymmetric, the samples drawn will be asymmetric but the expectation will still be unbiased. In small samples, there can be a few points drawn from the skewed part of the distribution, which will be shrunk a lot in a square root operation to get to the sample standard deviation. This will lead to a biased sample standard deviation.

38.1.3 Student t-distribution

That the sample deviation is consistent is why the t-distribution, which is a ratio of the standard normal distribution to the square root of the mean-square of standard normal distributions (i.e. chi-squared distribution divided by the degrees of freedom), converges to the normal distribution when the sample size increases. The chi-squared sum divided by the DOF, i.e. mean-square of standard deviations, is the sample variance, i.e.

$$\begin{aligned} S^2 &= \frac{1}{DOF} \sum^n (X - \bar{X})^2 \\ &= \frac{\sigma^2}{DOF} \sum^n \frac{(X - \bar{X})^2}{\sigma^2} \\ &= \frac{\sigma^2}{DOF} n \sum Z^2 \\ &= \frac{\sigma^2}{DOF} n \chi^2 \end{aligned}$$

So,

$$\begin{aligned} t &= \frac{Z}{\sqrt{\frac{\chi^2}{DOF}}} \\ &= \frac{Z}{\sqrt{S^2/n\sigma^2}} \\ &= \frac{Z}{S/\sqrt{n}\sigma} \\ &= \frac{(X - \mu)/\sigma}{S/\sqrt{n}\sigma} \\ &= \frac{(X - \mu)}{S/\sqrt{n}} \end{aligned}$$

38.1.4 Chi-squared tests

There are two general types of chi-squared tests. Uses

$$\chi^2 \sim \sum \frac{(O - E)^2}{E}$$

where O is observed value and E is expected value.

Chi-squared goodness-of-fit test. Used to determine whether or not a categorical variable follows a hypothesized distribution. Examples. (1) We want to know if a die is fair, so we roll it 50 times and record the number of times it lands on each number. (2) We want to know if an equal number of people come into a shop each day of the week, so we count the number of people who come in each day during a random week.

Chi-squared test of independence. Used to determine whether or not there is a significant association between two categorical variables. Uses marginal distributions to calculate the theoretical independent distributions and compares that to the observed value. (1) We want to know if gender is associated with political party preference so we survey 500 voters and record their gender and political party preference. (2) We want to know if a person's favorite color is associated with their favorite sport so we survey 100 people and ask them about their preferences for both.

38.2 Multivariate statistics

38.2.1 Basic Concepts

Calculations. Law of Iterated Expectations:

$$E(X) = E(E(X|Y)) = E_Y(E_X(X|Y))$$

Law of The Unconscious Statistician (LOTUS):

$$E[g(X)] = \sum_x g(x)f_X(x)$$

$$E[g(X)] = \sum_{-\infty}^{\infty} g(x)f_X(x)dx$$

$$E[g(X, Y)] = \sum_y \sum_x g(x, y)f(x, y)$$

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y)dx$$

Independence, Mean Independence, and Correlation:

(1) Independence applies to ALL MOMENTS:

$$p(y|x) = p(y)$$

(2) Mean independence only applies to the FIRST MOMENT:

$$E(y|x) = E(y)$$

So y still depends on x for e.g. variance. Independence implies mean independence:

$$E(y|x) = \int y p(y|x) dy = \int y p(y) dy = E(y)$$

However, mean independence doesn't necessarily imply independence. (3) Correlation. Mean independence implies non-correlation. Uncorrelatedness doesn't necessarily imply mean independence or independence EXCEPT for Gaussians.

Linear Combinations of Random Variables. If $Z = X + Y$, then:

$$f_{X+Y}(z) = f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$$

If X and Y are mutually independent, then joint PDF is a convolution:

$$f_Z(z) \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

Mahalanobis Distance.

$$MD = \sqrt{(x - \mu)' \Sigma^{-1} (x - \mu)} \sim \chi^2$$

MD measures Euclidean distance between a data point and a distribution (represented by its centroid) in multivariate analysis, similar to the standardized normal: $\frac{x - \mu}{\sigma}$. In fact, it is present in the Gaussian PDF. The ² statistic tests for outliers beyond the cluster of distribution points.

38.3 Non-parametric estimation

Nonparametric estimation is a statistical method that allows the functional form of a fit to data to be obtained in the absence of any guidance or constraints from theory. As a result, the procedures of nonparametric estimation have no meaningful associated parameters. Two types of nonparametric techniques are artificial neural networks and kernel estimation.

38.4 Multivariate Distributions

38.4.1 Multivariate Normal

A random variable X is multivariate normal if ANY linear combination is univariate normal. So the X's can be correlated, even highly correlated, e.g. $X+X+X=3X$ which is just a stretch of the support.

Can be degenerate, resulting in the Dirac delta function. But can't be degenerate in more than one x. So the covariance matrix Σ can be:

- Positive semi-definite: potentially degenerate
- Positive definite: no degeneracy

Like all density functions, must satisfy probability axioms where joint PDF sums to 1 and all probabilities are ≥ 0 .

$$f(x) = \frac{1}{\sqrt{|2\pi\Sigma|}} \exp\left(-\frac{1}{2}(x - \mu)' \Sigma^{-1}(x - \mu)\right)$$

Σ is PSD, so eigenvalues are ≥ 0 .

Proof. Let $Z_1, Z_2 \sim N(0, 1)$ and $X = \mu_1 + \sigma_1 Z_1$. Then:

$$E(X) = \mu_1 + \sigma_1 E(Z_1) = \mu_1$$

$$Var(X) = Var(\mu_1 + \sigma_1 Z_1) = \sigma_1^2 Var(Z_1) = \sigma_1^2$$

So $X \sim N(\mu_1, \sigma_1^2)$. Let $Y = \mu_2 + \sigma_2 \rho Z_1 + \sqrt{1 - \rho^2} \sigma_2 Z_2$. Then:

$$E(Y) = \mu_2 + \sigma_2 \rho E(Z_1) + \sqrt{1 - \rho^2} \sigma_2 E(Z_2) = \mu_2$$

$$Var(Y) = \sigma_2^2 \rho^2 Var(Z_1) + (1 - \rho^2) \sigma_2^2 Var(Z_2) = \sigma_2^2$$

So $Y \sim N(\mu_2, \sigma_2^2)$. then:

$$Cov(X, Y) = Cov(\mu_1 + \sigma_1 Z_1, \mu_2 + \sigma_2 \rho Z_1 + \sqrt{1 - \rho^2} \sigma_2 Z_2)$$

$$= Cov(\sigma_1 Z_1, \sigma_2 \rho Z_1 + \sqrt{1 - \rho^2} \sigma_2 Z_2)$$

$$= Cov(\sigma_1 Z_1, \sigma_2 \rho Z_1) + Cov(\sigma_1 Z_1, \sqrt{1 - \rho^2} \sigma_2 Z_2)$$

$$= \sigma_1 \sigma_2 \rho Cov(Z_1, Z_2) = \sigma_1 \sigma_2 \rho$$

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{\sigma_1 \sigma_2 \rho}{\sigma_1 \sigma_2} = \rho$$

$$f(Z_1, Z_2) = f(Z_1)f(Z_2) = \frac{e^{-\frac{Z_1^2}{2}}}{\sqrt{2\pi}} \frac{e^{-\frac{Z_2^2}{2}}}{\sqrt{2\pi}}$$

$$X = \mu_1 + \sigma_1 Z_1 \Rightarrow Z_1 = \frac{X - \mu_1}{\sigma_1}$$

$$Y = \mu_2 + \sigma_2 \rho Z_1 + \sqrt{1 - \rho^2} \sigma_2 Z_2 \Rightarrow Z_2 = \frac{1}{\sqrt{1 - \rho^2}} \left(\frac{Y - \mu_2}{\sigma_2} \right) - \frac{\rho}{\sqrt{1 - \rho^2}} \left(\frac{X - \mu_1}{\sigma_1} \right)$$

A Jacobian is required for $(Z_1, Z_2) \rightarrow (X, Y)$:

$$|J| = \left| \frac{\partial Z_1}{\partial X} \quad \frac{\partial Z_1}{\partial Y} \quad \frac{\partial Z_2}{\partial X} \quad \frac{\partial Z_2}{\partial Y} \right| = \left| \frac{1}{\sigma_1} \quad 0 \quad \frac{\rho}{\sigma_1 \sqrt{1 - \rho^2}} \quad \frac{1}{\sigma_2 \sqrt{1 - \rho^2}} \right| = \frac{1}{\sigma_1 \sigma_2 \sqrt{1 - \rho^2}}$$

$$\begin{aligned}
f(X, Y) &= f_1\left(\frac{X - \mu_1}{\sigma_1}\right) f_2\left(\frac{Y - \mu_2}{\sigma_2\sqrt{1 - \rho^2}} - \frac{\rho}{\sqrt{1 - \rho^2}}\left(\frac{X - \mu_1}{\sigma_1}\right)\right) |J| \\
&= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X - \mu_1}{\sigma_1}\right)^2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{Y - \mu_2}{\sigma_2\sqrt{1 - \rho^2}} - \frac{\rho}{\sqrt{1 - \rho^2}}\left(\frac{X - \mu_1}{\sigma_1}\right)\right)^2} \frac{1}{\sigma_1\sigma_2\sqrt{1 - \rho^2}} \\
&= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho^2}} \\
&\exp\left(-\frac{1}{2}\left(\frac{X - \mu_1}{\sigma_1}\right)^2 - \frac{1}{2(1 - \rho^2)}\left[\left(\frac{Y - \mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{Y - \mu_2}{\sigma_2}\right)\left(\frac{X - \mu_1}{\sigma_1}\right) + \rho^2\left(\frac{X - \mu_1}{\sigma_1}\right)^2\right]\right) \\
&= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1 - \rho^2}} \exp\left(-\frac{1}{2(1 - \rho^2)}\right) \left[\left(\frac{X - \mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{Y - \mu_2}{\sigma_2}\right)\left(\frac{X - \mu_1}{\sigma_1}\right) + \left(\frac{Y - \mu_2}{\sigma_2}\right)^2\right]
\end{aligned}$$

The above reduces to:

$$f(x) = \frac{1}{\sqrt{|2\pi\Sigma|}} \exp\left(-\frac{1}{2}(x - \mu)' \Sigma^{-1}(x - \mu)\right)$$

Where $x = (X, Y)$ and:

$$\begin{aligned}
\Sigma &= \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \\
\Sigma^{-1} &= \frac{1}{\sigma_1^2\sigma_2^2(1 - \rho^2)} \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}
\end{aligned}$$

Properties

(1) Any affine transformation of a Gaussian is Gaussian. If $X \sim N(\mu, \Sigma)$ then $AX + b \sim N(A\mu + b, A\Sigma A')$.

(2) $Z = X + Y$ where X and Y are standard normals:

$$X + Y \sim N(\mu_x + \mu_y, \text{Var}(X + Y) = \sigma_x^2 + \sigma_y^2 + \rho\sigma_x\sigma_y)$$

(3) Construction (de-sphering). If each of $X_1, X_2, \dots, X_n \sim N(0, 1)$ are mutually independent, $X \sim N(0, I) \Rightarrow AX + \gamma \sim N(\gamma, AA')$

(4) Sphering. If Σ is PD then $y \sim N(\mu, \Sigma) \Rightarrow A^{-1}(y - \mu) \sim N(0, I)$ where $\Sigma = AA'$. It can't be PSD because if eigenvalue = 0, then A is uninvertible.

Geometric intuition Let $X \sim N(0, I)$. Since Σ is symmetric,

$$\Sigma = U\lambda U' = U\lambda^{1/2}\lambda^{1/2}U' = (U\lambda^{1/2})(\lambda^{1/2}U') = AA'$$

Where U is an orthogonal matrix of eigenvectors (since Σ is symmetric), and

$$\lambda = \begin{pmatrix} \lambda_1 & 0 & \dots \\ 0 & \lambda_2 & \dots \\ \dots & \dots & \dots \end{pmatrix} = \begin{pmatrix} \sqrt{\lambda_1} & 0 & \dots \\ 0 & \sqrt{\lambda_2} & \dots \\ \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} \sqrt{\lambda_1} & 0 & \dots \\ 0 & \sqrt{\lambda_2} & \dots \\ \dots & \dots & \dots \end{pmatrix}$$

Where $U = (u_1 \ u_2 \ \dots)$. Since U is orthogonal, it's a reflection or rotation. Also, the eigenvectors are all mutually orthogonal. Now let

$$y = AX + \gamma = U\lambda^{1/2}X + \gamma \Rightarrow y \sim N(\gamma, \Sigma)$$

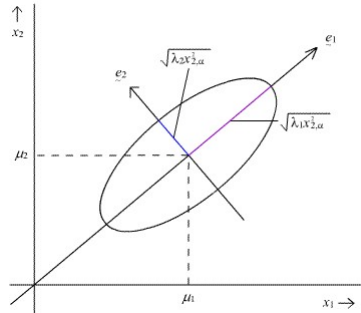
In the normal quadratic

$$(x - \mu)' \Sigma^{-1} (x - \mu) = (x - \mu)' U \lambda^{-1} U' (x - \mu)$$

Where $\lambda^{-1/2}$ is a scaling, U is a rotation, and γ is a translation. A small eigenvalue will lead to a larger quadratic:

$$\lambda^{-1} = \begin{pmatrix} \lambda_1^{-1} & 0 & 0 \\ 0 & \lambda_2^{-1} & 0 \\ 0 & 0 & \lambda_3^{-1} \end{pmatrix}$$

Geometry. The geometry of the multivariate normal distribution can be investigated by considering the orientation, and shape of the prediction ellipse.



Because of the χ^2 relationship, the $(1 - \alpha) \times 100\text{percent}$ prediction ellipse above is centered on the population means. The half-lengths of the ellipse axes are the square roots of eigenvalues times the chi-squared values.

38.4.2 Multivariate t

There may be several multivariate t distributions possible, but the one most cited is from elliptical distribution theory:

$$f(x) = \frac{1}{(\nu\pi)^{k/2}} \frac{\Gamma(\frac{\nu+k}{2})}{\Gamma(\frac{\nu}{2}) |\Sigma|^{\frac{1}{2}}} \left(1 + \frac{1}{\nu} (x - \mu)' |\Sigma|^{-1} (x - \mu) \right)^{\frac{\nu+k}{2}}$$

Where k is the number of dimensions and ν is the DOF. The joint distribution has an elliptical probability density (=elliptical copula).

Unlike the multivariate normal and like most other multivariate distributions, zero correlation does not lead to statistical independence. Set $k=2$, $A=I$, $\mu = 0$ to illustrate that the resulting expression won't factor:

$$f(x) = \left(\frac{1}{\nu\pi} \right) \frac{\Gamma\left(\frac{\nu}{2} + 1\right)}{\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{1}{\nu}(x_1^2 + x_2^2) \right)$$

If statistically independent univariate t distributions were instead multiplied, the joint distribution instead has a diamond-shaped probability density. Why? The normal distribution has steeper tails than the t distribution. As $\nu < 1$, $f(x)f(y) < f(x)$ and $f(y)$ where $f(x)$ and $f(y)$ are independent. Take $x=y$, then $p(x, y) = p(x)p(y) < p(x) = p(y)$. With steeper tails, the independent multivariate normal takes a shorter distance d for $f(x)f(y) = f(x+d) = f(y+d)$ leading to an elliptical arc whereas the multivariate takes disproportionately longer leading to a diamond shape.

38.5 Stochastic Dependence: Copulas

Transform univariate support $[0,1]$ to $[0, \infty]$, combine the probabilities, and then inverse-transform them back to $[0,1]$.

38.5.1 Generalized CDF Inverses

For flat (horizontal) areas and/or discontinuities (vertical). The usual case (leftmost X):

$$F_X^{-1}(p) = \inf\{x \in R | F_X(x) \geq p\}$$

Rightmost X:

$$F_X^{-1+}(p) = \sup\{x \in R | F_X(x) \leq p\}$$

For

$$\alpha \in [0, 1]$$

$$F_X^{-1(\alpha)}(p) = \alpha F_X^{-1}(p) + (1 - \alpha) F_X^{-1+}(p)$$

For strictly monotonic CDFs:

$$F_X^{-1(\alpha)}(p) = F_X^{-1}(p)$$

Properties of generalized CDF inverses: With right vertical discontinuous jumps:

$$F_X(F_X^{-1}(p)) \geq p$$

$$F_X^{-1}(p) \leq x \Leftrightarrow p \leq F_X(x)$$

Boundaries:

$$F_X^{-1+}(0) \text{ is the smallest possible value of } X$$

$F_X^{-1}(1)$ is the largest possible value of X

$$P[X \in [F_X^{-1+}(0), F_X^{-1}(1)]] = 1$$

So,

$$0 < F_X(x) < 1$$

Transformations If $h(X)$ is non-decreasing and left continuous then

$$F_{h(X)}^{-1}(p) = h(F_X^{-1}(p))$$

If h is non-decreasing and right-continuous then

$$F_{h(X)}^{-1+}(p) = h(F_X^{-1+}(p))$$

For any x in $(F_X^{-1+}(0), F_X^{-1}(1))$

$$\alpha_x = \begin{cases} \frac{F_X^{-1+}(F_X(x)) - x}{F_X^{-1+}(F_X(x)) - F_X^{-1}(F_X(x))}, & \text{if } F_X^{-1+}(F_X(x))F_X^{-1}(F_X(x)) \\ 1, & \text{otherwise} \end{cases}$$

38.5.2 Probability Integral Transform

Let X be a random variable and Y be the CDF of X :

$$Y = F_X(X)$$

Then $Y \sim U[0, 1]$:

$$U = F_X(X) \Rightarrow X = F_X^{-1}(U)$$

Proof:

$$F_Y(y) = P(Y \leq y) = P(F_X(X) \leq y) = P(X \leq F_X^{-1}(y)) = F_X^{-1}(F_X^{-1}(y)) = y$$

38.5.3 Copula Definition

A bivariate copula $C: [0,1] \times [0,1] \rightarrow [0,1]$ is a joint CDF for a bivariate random vector (U_1, U_2) with uniform marginal densities.

$$C(u_1, u_2) = P[U_1 \leq u_1, U_2 \leq u_2], (u_1, u_2) \in [0, 1]^2$$

It describes the correlation structure between the uniform marginal densities.

38.5.4 Sklar's Theorem

A copula C and its marginals defines a multivariate CDF of X (uniquely defines for continuous distributions). Every multivariate CDF of X can be expressed in terms of its marginal densities $F_n(x_n)$ and a copula C .

By the definition of a copula:

$$C(U_1, U_2, \dots, U_n) = C(F_1(x_1), F_2(x_2), \dots, F_n(x_n)) = F(x_1, x_2, \dots, x_n)$$

Rewrite:

$$F(x_1, x_2, \dots, x_n) = C(F_1(x_1), F_2(x_2), \dots, F_n(x_n))$$

Differentiate wrt $x_1, x_2, \dots, x_n$:

$$f(x_1, x_2, \dots, x_n) = c(F_1(x_1), F_2(x_2), \dots, F_n(x_n))f_1(x_1)f_2(x_2)\dots f_n(x_n)$$

38.5.5 Density Calculations

The copula density function c is:

$$c(F_1(x_1), F_2(x_2), \dots, F_n(x_n)) = \frac{f(x_1, x_2, \dots, x_n)}{f_1(x_1)f_2(x_2)\dots f_n(x_n)}$$

Then double-integrate c to get C . Double-differentiate C to get c :

$$c(u, v) = \frac{\partial^2}{\partial u \partial v} C(u, v)$$

The above calculates the COPULA DENSITY. To calculate the JOINT DENSITY:

$$c(F_1(x_1), F_2(x_2)) = \frac{\partial^2}{\partial x_1 \partial x_2} C(F_1(x_1), F_2(x_2))$$

Divide by appropriate marginal densities to get conditional densities. If the joint density is:

$$f(x_1, x_2) = c_{12}(F_1(x_1), F_2(x_2))f_1(x_1)f_2(x_2)$$

Then:

$$f_{2|1}(x_2|x_1) = \frac{c_{12}(F_1(x_1), F_2(x_2))f_1(x_1)f_2(x_2)}{f_1(x_1)} = c_{12}(F_1(x_1), F_2(x_2))f_2(x_2)$$

38.5.6 Copula Entropy

Recall the copula density is the quotient of the joint PDF and individual marginal densities:

$$c(F_1(x_1), F_2(x_2), \dots, F_n(x_n)) = c(x) = c(u_1, u_2, \dots, u_n) = \frac{f(x_1, x_2, \dots, x_n)}{f_1(x_1)f_2(x_2)\dots f_n(x_n)}$$

$$c(x) = c(u) = \frac{f(x)}{f_1(x_1)f_2(x_2)\dots f_n(x_n)}$$

Note the marginals are INDEPENDENT marginals. Recall entropy is:

$$H(x) = \int c(u) \log c(u) du$$

So:

$$H[c(x)] = H[f(x)] - H[\Pi f_n(x_n)]$$

$$H(\text{copula density}) = H(\text{joint PDF}) - H(\text{marginals})$$

Rearrange:

$$H[f(x)] = H[c(x)] + H[\Pi f_n(x_n)]$$

$$H(\text{joint PDF}) = H(\text{copula density}) + H(\text{marginals})$$

The 'total correlation' quantifies information redundancy:

$$-H[c(x)] = H[\Pi f_n(x_n)] - H[f(x)]$$

So, the total correlation of X is equal to the negative entropy of the copula density.

Since the entropy of a joint PDF is always less than the sum of the entropies of its marginals, $H[c(x)] \leq 0$. So, the total correlation is always ≥ 0 . The total correlation is 0 when the joint PDF is the product of the marginals, i.e. when the marginals are mutually independent:

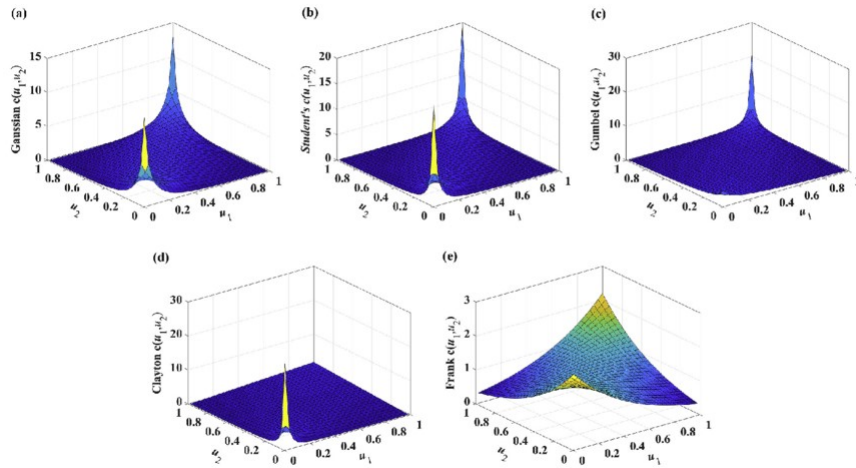
$$H[\Pi f_n(x_n)] = H[f(x)]$$

So the independence copula is the maximum entropy copula.

When modelling, select the lowest-entropy copula, i.e. the one that has the fewest restrictions.

38.5.7 Visualization

Some example copula densities $c(F(x), F(y))$:



monotonic Copulas. Extreme positive correlation:

$$(X_1, X_2) = (F_X^{-1}(U), F_{X_2}^{-1}(U))$$

Co-

Note the same U above. This is equivalent to:

$$F_{\vec{X}}(\vec{X}) = \min\{F_{X_1}(x_1), F_{X_2}(x_2)\}$$

Proof:

$$\begin{aligned} (X_1, X_2) &= (F_X^{-1}(U), F_{X_2}^{-1}(U)) \\ F_{\vec{X}}(\vec{X}) &= P(X_1 \leq x_1, X_2 \leq x_2) = P(F_{X_1}^{-1}(U) \leq x_1, F_{X_2}^{-1}(U) \leq x_2) \\ &= P(U \leq F_{X_1}(x_1), U \leq F_{X_2}(x_2)) = P(U \leq \min[\underbrace{F_{X_1}(x_1)}_{\in[0,1]}, \underbrace{F_{X_2}(x_2)}_{\in[0,1]}]) \\ &= \min[F_{X_1}(x_1), F_{X_2}(x_2)] \end{aligned}$$

The co-monotonic copula is the copula C_U such that

$$F_{\vec{X}}(\vec{X}) = C_U\{F_{X_1}(x_1), F_{X_2}(x_2)\}$$

Where C_U is:

$$C_U(u, v) = \min\{u, v\}$$

38.5.8 Counter-monotonic Copulas

Extreme negative correlation:

$$(X_1, X_2) = (F_{X_1}^{-1}(U), F_{X_2}^{-1}(1 - U))$$

This is equivalent to:

$$\begin{aligned} F_{\vec{X}}(\vec{x}) &= C_L(F_{X_1}(x_1), F_{X_2}(x_2)) \\ C_L(u, v) &= \max\{u + v - 1, 0\} \end{aligned}$$

Proof:

$$\begin{aligned} F_{\vec{X}}(x_1, x_2) &= P(X_1 \leq x_1, X_2 \leq x_2) = P(F_{X_1}^{-1}(U) \leq x_1, F_{X_2}^{-1}(1 - U) \leq x_2) \\ &= P(U \leq F_{X_1}(x_1), 1 - U \leq F_{X_2}(x_2)) \\ &= P(U \leq F_{X_1}(x_1), U \geq 1 - F_{X_2}(x_2)) \\ &= P(1 - F_{X_2}(x_2) \leq U \leq F_{X_1}(x_1)) \\ &= \begin{cases} 0 & F_{X_1}(x_1) \leq 1 - F_{X_2}(x_2) \\ F_{X_1}(x_1) + F_{X_2}(x_2) - 1 & F_{X_1}(x_1) > 1 - F_{X_2}(x_2) \end{cases} \\ &= \max\{F_{X_1}(x_1) + F_{X_2}(x_2) - 1, 0\} \end{aligned}$$

38.5.9 Frechet-Hoeffding Bounds

Copulas are bounded:

$$\underbrace{C_L(u_1, u_2)}_{\text{counter-monotonic}} \leq C(u_1, u_2) \leq \underbrace{C_U(u_1, u_2)}_{\text{co-monotonic}}$$

Upper bound proof:

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2)$$

So

$$C(u_1, u_2) \leq P(U_2 \leq u_2) = u_2$$

$$C(u_1, u_2) \leq P(U_1 \leq u_1) = u_1$$

Add the above two:

$$C(u_1, u_2) \leq \min(u_1, u_2) = C_U(u_1, u_2)$$

Lower bound proof:

$$P[(u_1, u_2) \in A] \geq 0$$

Where A is a rectangular area within the domain bounded by the marginal densities.

$$P[(u_1, u_2) \in A] = P(U_1 \leq 1, U_2 \leq 1) - \underbrace{P(U_1 \leq u_1)}_{u_1} - \underbrace{P(U_2 \leq u_2)}_{u_2} + \underbrace{P(U_1 \leq u_1, U_2 \leq u_2)}_{C(u_1, u_2)}$$

Combine above:

$$1 - u_1 + u_2 + C(u_1, u_2) \geq 0$$

$$C(u_1, u_2) \geq u_1 + u_2 - 1$$

And since

$$C(u_1, u_2) \geq 0$$

Then

$$C_L(u_1, u_2) = \max\{u_1 + u_2 - 1, 0\}$$

38.5.10 Independent Copulas

If $P(A) = 0.6$ and $P(B) = 0.7$, $P(A) + P(B) = 1.3$ so invalid as probability.

However:

$$-\ln P(A) = 0.51$$

$$-\ln P(B) = 0.36$$

$$-\ln P(A) - \ln P(B) = 0.87$$

$$e^{-0.87} = 0.42$$

Which is $= P(A)P(B) = (0.6)(0.7) = 0.42$.

This can be more derived:

$$p(x)|a = p(x|a) = p(x)^a \text{ where } a \text{ will be } 1$$

$$p(x) = E_a(p(x)|a) = E_a(p(x|a)) = E_a(p(x)^a) = E_a(e^{a \ln p(x)}) = \mathcal{L}\{-\ln p(x)\}$$

Where $\mathcal{L}$ is the Laplace transform. So:

$$\mathcal{L}\{-\ln p(x)\} = p(x)$$

$$\mathcal{L}^{-1}\{p(x)\} = -\ln p(x)$$

So:

$$p(x_1, x_2) = \mathcal{L}\{-\ln p_1(x_1) - \ln p_2(x_2)\} = \mathcal{L}\{-\ln p_1(x_1)p_2(x_2)\}$$

$$\mathcal{L}^{-1}\{p(x_1, x_2)\} = \mathcal{L}^{-1}\{\mathcal{L}\{-\ln p_1(x_1)p_2(x_2)\}\} = p_1(x_1)p_2(x_2)$$

The above is a type of independent copula:

$$C(F_X(X), F_Y(Y)) = F_X(X)F_Y(Y)$$

There are other independent copulas.

38.5.11 Bivariate Gaussian Copulas

Let $u = F_1(x_1)$ and $v = F_2(x_2)$.

$$c(u, v) = \frac{f(x_1, x_2)}{f_1(x_1)f_2(x_2)}$$

Use the joint normal distribution (bivariate Gaussian):

$$c(u, v) = \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2} \left(\frac{1}{1-\rho^2}\right) (\Phi^{-1}(u)^2 + \Phi^{-1}(v)^2) + \frac{\rho}{1-\rho^2} \Phi^{-1}(u)\Phi^{-1}(v)\right)$$

Suppose $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. Then:

$$\Phi^{-1}(u) = \Phi^{-1}(F_1(x_1)) = \Phi^{-1}\left(\Phi\left(\frac{x_1 - \mu_1}{\sigma_1}\right)\right)$$

$$\Phi^{-1}(v) = \Phi^{-1}(F_2(x_2)) = \Phi^{-1}\left(\Phi\left(\frac{x_2 - \mu_2}{\sigma_2}\right)\right)$$

38.5.12 Gaussian copula versus multivariate Gaussian

A Gaussian copula uses any marginal, but if it used normally-distributed marginals, it would be a multivariate Gaussian. So a multivariate Gaussian is a Gaussian copula with normally-distributed marginals. A Gaussian copula with other marginals looks very different.

38.5.13 Archimedean Copulas

Most popular copula class which includes Gaussian, Clayton, Frank, Gumbel, and others.

$$F(x)|a = p(X < x|a) = F(x)^a$$

Where $a \neq 1$ indicates correlation and $a = 1$ means uncorrelated. The a can be seen as frailty or the regression coefficient from Cox Proportional Hazards models.

$$F(x) = E_a(F(x)|a) = E_a(p(X < x|a)) = E_a(F(x)^a) = E_a(e^{a \ln F(x)}) = \mathcal{L}\{-\ln F(x)\}$$

Where $\mathcal{L}$ is the Laplace transform. So:

$$\Psi\{-\ln F(x)\} = F(x)$$

$$\Psi^{-1}\{F(x)\} = -\ln F(x)$$

Notation elsewhere may flip the Ψ and Ψ^{-1} . Above, the Ψ^{-1} is the generator function (and Ψ is the generator if notation is flipped). With more than one CDF:

$$\begin{aligned} P(F_1(x_1), F_2(x_2)) &= E_a(P(F_1(x_1), F_2(x_2))|a) = E_a(p(X_1 < x_1, X_2 < x_2|a)) \\ &= E_a(e^{a \ln F_1(x_1) + a \ln F_2(x_2)}) = \mathcal{L}\{-\ln F_1(x_1) - \ln F_2(x_2)\} \end{aligned}$$

Where $\mathcal{L}$ is the Laplace transform but now call it Ψ . So the Archimedean copula form is:

$$C(F_1(x_1), F_2(x_2)) = \Psi\{-\ln F_1(x_1) - \ln F_2(x_2)\} = \Psi\{\Psi^{-1}(F_1(x_1)) + \Psi^{-1}(F_2(x_2))\}$$

Since $\Psi\{-\ln F(x)\} = F(x) = u$:

$$C(u_1, u_2) = \Psi\{\Psi^{-1}(u_1) + \Psi^{-1}(u_2)\}$$

Archimedean copulas are exchangeable, so all marginals have the same dependence structure.

38.5.14 Tail dependence

There are different ways of measuring tail dependence.

One way: (a) Lower-tail dependence:

$$\lambda_L = \lim_{t \rightarrow 0} P(X_2 \leq F_2^{-1}(t) | X_1 \leq F_1^{-1}(t))$$

(b) Upper-tail dependence:

$$\lambda_U = \lim_{t \rightarrow 1} P(X_2 > F_2^{-1}(t) | X_1 > F_1^{-1}(t))$$

Another way: (a) Lower-tail dependence:

$$\lambda_L = \lim_{t \rightarrow 0} \frac{C(t, t)}{t}$$

(b) Upper-tail dependence:

$$\lambda_U = 2 - \lim_{t \rightarrow 1} \frac{1 - C(t, t)}{1 - t}$$

38.5.15 Clayton Copula

Let the frailty be:

$$a \sim \Gamma\left(\frac{1}{\theta}, 1\right)$$

This indicates the wait time to $\frac{1}{\theta}$ events. So a low θ means high frailty and waiting for many events, which takes a long time. Then:

$$\begin{aligned} \Psi(s) &= \mathcal{L}\{-\ln F(x)\} = F(s) = u = \int_0^\infty e^{-sa} f_\Gamma(\theta) da \\ &= \int_0^\infty e^{-sa} \frac{a^{\frac{1}{\theta}-1} e^{-a}}{\Gamma(\frac{1}{\theta})} da = (1+s)^{-\frac{1}{\theta}} \end{aligned}$$

Since:

$$\begin{aligned} \Psi(s) &= u = (1+s)^{-\frac{1}{\theta}} \\ u^{-\theta} &= 1+s \end{aligned}$$

So the generator function is:

$$s = u^{-\theta} - 1$$

So:

$$C(u) = (u_1^{-\theta} + u_2^{-\theta} + \dots + u_n^{-\theta} - n + 1)^{-\frac{1}{\theta}}$$

As $\theta \rightarrow 0$, the Clayton copula becomes the independence copula. At $\theta = -1$, it is the counter-monotonic copula, and as $\theta \rightarrow \infty$, it becomes the co-monotonic copula.

The contours are asymmetric owing to the asymmetry of the Γ distribution. The Γ also results in more dependence in the negative than positive tail.

The θ is related to Kendall's τ by:

$$\tau = \frac{\theta}{\theta + 2} \Leftrightarrow \theta = \frac{2\tau}{1 - \tau}$$

38.5.16 Other Copulas

Gumbel copulas are Archimedean and model the frailty as a stable distribution. It shows more positive than negative tail dependence.

$$\Psi(s) = e^{-\frac{s}{\theta}}$$

$$\Psi^{-1}(s) = (-\log S)^\theta$$

$$C(u) = \exp[-\{(-\log S_1)^\theta + (-\log S_2)^\theta\}^{\frac{1}{\theta}}]$$

Joe copulas also have more positive tail dependence and less negative tail dependence than Gumbel copulas.

Frank copulas are also Archimedean and model LESS positive AND negative tail dependence, so they have more square-like contours.

t copulas are Archimedean and have more positive, negative, and lateral dependence than Gaussian copulas.

Elliptical copulas trace out elliptical contours, as with Gaussian and t copulas.

Vine copulas are used at higher dimensions since standard copulas, such as Gaussian, Frank, and Clayton copulas, become too restrictive in their relationships. Vine copulas offer more flexibility in specifying dependency relationships between random variables. Use graph theory to specify the dependency structure. Example:

$$f(x_1, x_2, x_3) = f_{3|12}f_{2|1}f_1$$

With:

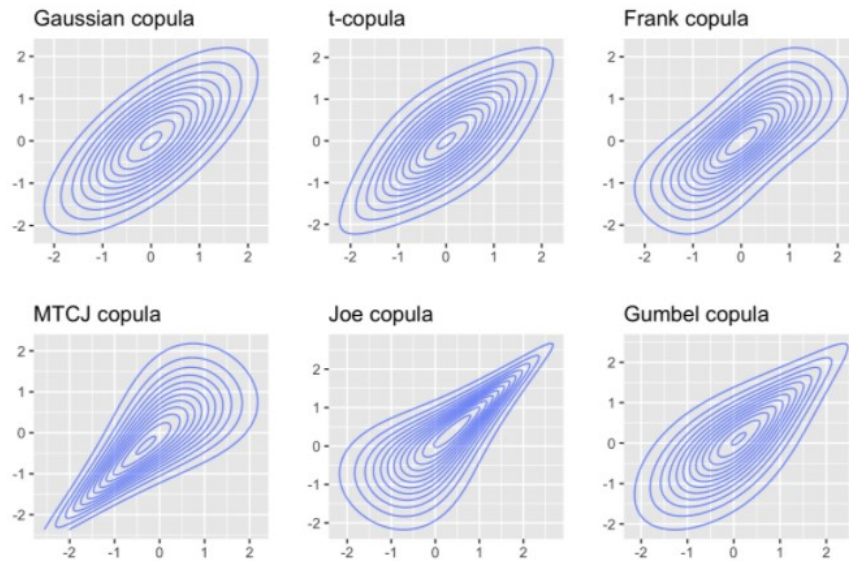
$$f_{3|12} = c_{13|2}f_{3|2}$$

$$f_{1|1} = c_{12}f_2$$

$$f_{3|2} = c_{23}f_3$$

These are generally re-organized and structured to = (pair copula densities) x (marginal densities). Use Bayesian estimate or MLE to determine the probability distributions.

There are many other copulas, including Ali-Mikhail-Haq and Galambos.

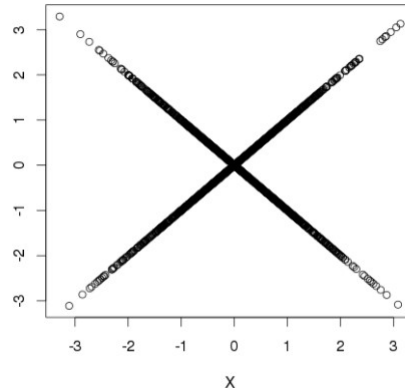


38.6 Correlation

Correlation tries to capture dependence between two random variables in a single, intuitive number. However, the dependence can be complicated, and correlation is insufficient to capture the full dependence structure. The multivariate normal is an exception: correlation is sufficient to capture dependence.

38.6.1 Non-correlation and Independence

If two random variables are normal (but not JOINTLY normal) and they're uncorrelated, they might not be independent. The marginals of these normal distributions can interact in ways s.t. that the final distribution has a shape that is clearly not independent (for an independent distribution, viewed from above, the 3D distribution is a 'cloud' of points for symmetrically-distributed random variables). For instance, suppose X has a normal distribution with expected value 0 and variance 1. Let W have a Rademacher distribution, so that $W=1$ or $W=-1$, each with probability $1/2$, and assume W is independent of X . Let $Y = WX$. Then X and Y are uncorrelated; both have the same normal distribution; and X and Y are not independent. The below is uncorrelated because the slope of a line through the points is 0, but it's not independent because it's not a 'cloud' of points (for symmetrically-distributed random variables).



If two random variables are JOINTLY normal and they're uncorrelated, they will form a 3D Gaussian, so the points are independent (viewed from above, the 3D distribution is 'cloud' of points). Algebraically, this can be shown by setting $\rho = 0$ in the multivariate normal and factoring will lead to independent marginal densities.

Other random variables where uncorrelated $\rightarrow$ independence are Bernoulli and uniform distributions.

If two random variables are not jointly normal and uncorrelated, they are not necessarily independent. Example:

$$Y = \begin{cases} X & |X| < c \\ -X & |X| > c \end{cases}$$

X and Y are each normal and are uncorrelated but aren't jointly normal because the joint density function isn't continuous. So X and Y aren't mutually independent $\rightarrow$ X can be independent, but Y depends on X. As another example, X and -X are normal and uncorrelated but dependent.

Even if random variables are jointly distributed, uncorrelated, and contour plots show no angled tendencies, they are not necessarily independent. See for instance the multivariate t distribution (joint distribution) where the marginals are mutually uncorrelated with no angled contour plots. But the random variables are not independent.

38.6.2 Pearson's r

Pearson's r measures linear correlation. If $r=1$, then $a > 0$ and (x,y) are co-monotonic. If $r=-1$ then $a < 0$ and (x,y) are counter-monotonic.

$$y = ax + b, a > 0$$

$$x = F_x^{-1}(U)$$

$$\begin{aligned}
y &= aF_x^{-1}(U) + b \\
F_{h(x)}(p) &= h(F_X^{-1}(p)) \\
y &= F_{ax+b}^{-1}(U) \\
y &= F_y^{-1}(U)
\end{aligned}$$

So they're co-monotonic:

$$(x, y) = (F_x^{-1}(U), F_y^{-1}(U))$$

r is bounded by Frechet-Hoeffding:

$$r(F_X^{-1}(U), F_Y^{-1}(1-U)) \leq r(X, Y) \leq r(F_X^{-1}(U), F_Y^{-1}(U))$$

r has numerous disadvantages:

(1) Changes in marginal distributions lead to changes in r even if the correlation hasn't changed. Pearson's r can be re-expressed by decomposing the covariance numerator into its CDF components:

$$\begin{aligned}
r &= \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{1}{\sigma_x \sigma_y} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (F_{xy}(x, y) - F_x(x)F_y(y)) dx dy \\
&= \frac{1}{\sigma_x \sigma_y} \int_0^1 \int_0^1 (C(u_x, u_y) - (u_x, u_y)) dF_x^{-1}(u_x) dF_y^{-1}(u_y)
\end{aligned}$$

So changes to the marginal distributions x and y lead to changes to $dx dy$ and $dF_x^{-1} dF_y^{-1}$, even if the correlation between them represented by the copula doesn't change. r is not strictly invariant under strictly increasing transformations:

$$r(X_1, X_2) \neq r(t_1(X_1), t_2(X_2))$$

(2) r doesn't capture non-linear dependence.

Due to numerous disadvantages, use instead rank correlation such as Spearman's ρ and/or Kendall's τ . r, ρ , and τ are popular to different degrees depending on use, usually result in slightly different values, can be used in different ways (e.g. as test statistics), have different properties (e.g. asymptotics), but τ is more popular for joint distributions.

38.6.3 Spearman's ρ

Ordinal/rank counterpart to r. Rank the X and Y values and use the Pearson correlation equation, which can be reduced to:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Will give perfect correlations (+1, -1) for monotonic relationships.

38.6.4 Kendall's τ

For a bivariate random variable $(x_i, y_i) \sim f(x, y)$,

(1) Concordance between (x_i, y_i) and (x_j, y_j) :

$$p_c = p(\text{concordance}) = p((x_i > x_j) \cap (y_i > y_j)) + p((x_i < x_j) \cap (y_i < y_j))$$

(2) Discordance

$$p_d = p(\text{discordance}) = p((x_i > x_j) \cap (y_i < y_j)) + p((x_i < x_j) \cap (y_i > y_j))$$

(3) Kendall's τ

$$\tau = p_c - p_d$$

Perfect concordance $\rightarrow \tau = 1$

Perfect discordance $\rightarrow \tau = -1$

$$\begin{aligned} p_c &= p(x_i > x_j) - p[(x_i > x_j) \cap (y_i < y_j)] \\ &\quad + p(x_i < x_j) - p[(x_i < x_j) \cap (y_i > y_j)] \\ &= p(x_i > x_j) + p(x_i < x_j) - p_d = 1 - p_d \\ p_c &= 1 - p_d \\ \Rightarrow \tau &= 2p_c - 1 = 1 - 2p_d \end{aligned}$$

(4) Independence

Assume x and y are independent:

$$\begin{aligned} p_c &= p(x_i < x_j) \cap (y_i < y_j) + p(x_i > x_j) p(y_i > y_j) \\ &= p(x_i > x_j) p(y_i < y_j) + p(x_i, x_j) p(y_i > y_j) = p_d \\ \Rightarrow \tau &= p_c - p_d = 0 \end{aligned}$$

In general, reverse is not true as with Pearson's correlation.

(5) Sample estimate of Kendall's τ

$$\begin{aligned} p_c &= p[9x_i - x_j)(y_i - y_j) > 0] \\ p_d &= p[(x_i - x_j)(y_i - y_j) < 0] \\ p_{tie} &= p(x_i = x_j) + p(y_i = y_j) \end{aligned}$$

Let $h_{ij} = \text{sign}(x_j - x_i) \text{sign}(y_j - y_i)$. Then:

$$E(h_{ij}) = (1)p_c + (0)p_{tie} + (-1)p_d = p_c - p_d = \tau$$

So h_{ij} is an unbiased estimator of τ

Let

$$T = \binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} \sum [\text{sign}(x_j - x_i)] [\text{sign}(y_j - y_i)]$$

Sample estimate:

$$E(T) = \binom{n}{2}^{-1} \underbrace{\sum \sum}_{1 \leq i < j \leq n} E(h_{ij}) = \binom{n}{2}^{-1} \binom{n}{2} \tau = \tau$$

(6) τ isn't limited to linear dependence. Further, it doesn't change if the marginal distributions change but the correlation doesn't change:

$$\tau = \frac{1}{\sigma_x \sigma_y} \int_0^1 \int_0^1 (C(u_x, u_y) - (u_x, u_y)) dF_x^{-1}(u_x) dF_y^{-1}(u_y) = E(C(U_1, U_2)) - 1$$

And invariant under strictly increasing transformations:

$$\tau(X_1, X_2) \neq \tau(t_1(X_1), t_2(X_2))$$

38.6.5 Kendall's τ of Copulas

Kendall's τ is (C=concordance, D=disconcordance):

$$\tau = \frac{C - D}{C + D} = \frac{C - D}{\binom{n}{2}}$$

$$= P[(X_1 - X_2)(Y_1 - Y_2) > 0] - P[(X_1 - X_2)(Y_1 - Y_2) < 0]$$

Let (X_1, Y_1) and (X_2, Y_2) be independent vectors of random variables with joint distributions H_1 and H_2 , respectively. Further, let $F(x)$ be the marginals for both X's and $G(y)$ be the marginals for both Y's. Let C_1 be the copula of (X_1, Y_1) and C_2 be the copula of (X_2, Y_2) . Then:

$$H_1(x, y) = C_1(F(x), G(y))$$

$$H_2(x, y) = C_2(F(x), G(y))$$

Let $Q = Q(C_1, C_2)$ and moreover:

$$Q = P[(X_1 - X_2)(Y_1 - Y_2) > 0] - P[(X_1 - X_2)(Y_1 - Y_2) < 0]$$

Since

$$= P[(X_1 - X_2)(Y_1 - Y_2) < 0] = 1 - P[(X_1 - X_2)(Y_1 - Y_2) > 0]$$

Then:

$$Q = 2 \underbrace{P[(X_1 - X_2)(Y_1 - Y_2) > 0]}_{\text{Decompose}} - 1$$

$$P[(X_1 - X_2)(Y_1 - Y_2) > 0] = \underbrace{P[X_1 > X_2, Y_1 > Y_2]}_{\text{Decompose}} + \underbrace{P[X_1 < X_2, Y_1 < Y_2]}_{\text{Decompose}}$$

For the first decomposition:

$$P[X_1 > X_2, Y_1 > Y_2] = P[X_2 < X_1, Y_2 < Y_1]$$

Marginalize out (X_1, Y_1) :

$$\begin{aligned} &= \int \int P[X_2 < x, Y_2 < y] dC_1(F(x), G(y)) \\ &= \int \int C_2(F(x), G(y)) dC_1(F(x), G(y)) \end{aligned}$$

Let $u=F(x)$ and $v=G(y)$:

$$= \int \int C_2(u, v) dC_1(u, v)$$

Repeat for the second decomposition:

$$P[X_1 < X_2, Y_1 < Y_2] = P[X_2 > X_1, Y_2 > Y_1]$$

Marginalize out (X_1, Y_1) again:

$$= \int \int P[X_2 > x, Y_2 > y] dC_1(F(x), G(y))$$

The integration area lies in the 'top right' so after subtracting $F(x)$ and $G(y)$, need to add back $C_2(F(x), G(y))$:

$$\begin{aligned} &= \int \int C_2(1 - F(x) - G(y) + C_2(F(x), G(y))) dC_1(F(x), G(y)) \\ &= \int \int (1 - u - v + C_2(u, v)) dC_1(u, v) \end{aligned}$$

Since u and v are uniform over $C_1(u, v)$, $E(U) = E(V) = \frac{1}{2}$:

$$= 1 - \frac{1}{2} - \frac{1}{2} \int \int C_2(u, v) dC_1(u, v)$$

So:

$$P[(X_1 - X_2)(Y_1 - Y_2) > 0] = 2 \int \int C_2(u, v) dC_1(u, v)$$

And:

$$Q(C_1, C_2) = 2 \left(2 \int \int C_2(u, v) dC_1(u, v) \right) - 1$$

Let $C_1 = C_2 = C$ and let $Q=\tau$:

$$\tau = 4 \int \int C(u, v) dC(u, v) - 1$$

38.6.6 Kendall Distribution Function of Archimedean

Let T be an Archimedean copula:

$$T = C(x, y) = \Psi^{-1}[\Psi(x) + \Psi(y)]$$

Differentiate:

$$(1) \Psi'(T) \frac{\partial T}{\partial x} = \Psi'(x) \Rightarrow \frac{\partial T}{\partial x} = \frac{\Psi'(x)}{\Psi'(T)}$$

$$(2) \Psi'(T) \frac{\partial T}{\partial y} = \Psi'(y) \Rightarrow \frac{\partial T}{\partial y} = \frac{\Psi'(y)}{\Psi'(T)}$$

$$(3) \Psi''(T) \frac{\partial T}{\partial x} \frac{\partial T}{\partial y} + \Psi'(T) \frac{\partial^2 T}{\partial x \partial y} = 0$$

Rearrange (3) and substitute in (1) and (2):

$$\frac{\partial^2 T}{\partial x \partial y} = t(x, y) = -\frac{\Psi''(T) \frac{\partial T}{\partial x} \frac{\partial T}{\partial y}}{\Psi'(T)} = -\frac{\Psi''(T) \Psi'(x) \Psi'(y)}{[\Psi'(T)]^3}$$

Introduce the Kendall distribution function $K(t)$, where t is a particular value of T :

$$K(t) = \int \int_{[0, t]^2} t(x, y) dx dy = - \int \int_{[0, t]^2} \frac{\Psi''(T) \Psi'(x) \Psi'(y)}{[\Psi'(T)]^3} dx dy$$

Rearrange (2):

$$\frac{\partial T}{\partial y} = \frac{\Psi'(y)}{\Psi'(T)} \Rightarrow dy = \frac{\Psi'(T)}{\Psi'(y)} dT$$

The Jacobian for the $dx dy \rightarrow dx dT$ transformation is:

$$J = \begin{vmatrix} \frac{\partial T}{\partial x} & \frac{\partial T}{\partial y} \\ \frac{\partial x}{\partial x} & \frac{\partial x}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{\partial T}{\partial x} & \frac{\partial T}{\partial y} \\ 1 & 0 \end{vmatrix} = -\frac{\partial T}{\partial y} = -\frac{\Psi'(T)}{\Psi'(y)}$$

Substitute above into $K(t)$:

$$K(t) = - \int \int_{[0, t]^2} \frac{\Psi''(T) \Psi'(x) \Psi'(y)}{[\Psi'(T)]^3} dx \underbrace{(-1)}_J \left(\frac{\Psi'(T)}{\Psi'(y)} dT \right) = \int \int_{[0, t]^2} \frac{\Psi''(T)}{[\Psi'(T)]^2} \underbrace{\Psi'(x) dx}_{\partial(\Psi(T))} dT$$

Since

$$T = C(x, y) = \Psi^{-1}[\Psi(x) + \Psi(y)]$$

$$\Psi(T) = [\Psi(x) + \Psi(y)]$$

$$\frac{\partial \Psi(T)}{\partial x} = \Psi'(x)$$

$$\partial(\Psi(T)) = \Psi'(x) \partial x$$

The Jacobian for the $dx dT \rightarrow d(\Psi(T)) dT$ transformation is:

$$J = \begin{vmatrix} \frac{\partial \Psi(T)}{\partial x} & \frac{\partial \Psi(T)}{\partial T} \\ \frac{\partial T}{\partial x} & \frac{\partial T}{\partial T} \end{vmatrix} = \begin{vmatrix} \frac{\partial \Psi(T)}{\partial x} & \frac{\partial \Psi(T)}{\partial T} \\ 0 & 1 \end{vmatrix} = \frac{\partial \Psi(T)}{\partial T}$$

Substitute above and take antiderivative of $(1)\partial(\Psi(T))$:

$$K(t) = \int \int_{[0,t]^2} \frac{\Psi''(T)}{[\Psi'(T)]^2} \underbrace{(1)}_J \partial(\Psi(T)) \partial T = \int_0^t \frac{\Psi''(T)}{[\Psi'(T)]^2} \Psi(T) \partial T$$

With integration by parts:

$$K(t) = t - \frac{\Psi(t)}{\Psi'(t)}$$

Check by differentiation:

$$\frac{d}{dt} \left(t - \frac{\Psi(t)}{\Psi'(t)} \right) = 1 - \frac{[\Psi'(t)]^2 - \Psi'(t)\Psi''(t)}{[\Psi'(t)]^2} = \frac{\Psi''(t)}{[\Psi'(t)]^2} \Psi(t) \partial T$$

38.6.7 Kendall's τ of Archimedean

Recall:

$$\tau = 4 \underbrace{\int \int C(u, v) dC(u, v)}_{E[C(u, v)]} - 1$$

$$E[C(u, v)] = E(T) = \int_0^1 t dK(t) = t K(t)|_0^1 - \int_0^1 K(t) dt = 1 - \int_0^1 K(t) dt$$

Substitute in Kendall's distribution function for Archimedean:

$$\begin{aligned} \tau &= 4 \left(1 - \int_0^1 K(t) dt \right) - 1 = 3 - 4 \int_0^1 K(t) dt = 3 - 4 \int_0^1 \left(t - \frac{\Psi(t)}{\Psi'(t)} \right) dt \\ &= 1 + 4 \int_0^1 \frac{\Psi(t)}{\Psi'(t)} dt \end{aligned}$$

38.6.8 Kendall's τ of the Clayton Copula

Recall the Clayton copula generating function:

$$\Psi(t) = \frac{1}{\theta} (t^{-\theta} - 1)$$

So:

$$\int_0^1 \frac{\Psi(t)}{\Psi'(t)} dt = \int_0^1 \frac{t^{-\theta} - 1}{-\theta t^{-\theta-1}} dt = -\frac{1}{\theta} \int_0^1 \frac{t^{-\theta} - 1}{t^{-\theta-1}} dt$$

$$\begin{aligned}
&= -\frac{1}{\theta} \int_0^1 \left(\frac{t^{-\theta}}{t^{-\theta-1}} - t^{\theta+1} \right) dt = -\frac{1}{\theta} \int_0^1 (t - t^{\theta+1}) dt = -\frac{1}{2\theta} + \frac{1}{\theta(\theta+2)} \\
&= \frac{-(\theta+2) + 2}{2\theta(\theta+2)} = -\frac{1}{2(\theta+2)}
\end{aligned}$$

So:

$$\tau = 1 + 4 \int_0^1 \frac{\Psi(t)}{\Psi'(t)} dt = 1 - 4 \frac{1}{2(\theta+2)} = \frac{\theta}{\theta+2}$$

38.7 ANOVA

One-way ANOVA evaluates the impact of one factor on a response variable. Two-way ANOVA has two independent variables. MANOVA (multivariate ANOVA) differs from ANOVA as the former tests for multiple dependent variables simultaneously while the latter assesses only one dependent variable at a time.

38.7.1 One-way ANOVA

Used to compare three or more groups. For two groups, just use a population comparison or t-test.

One-way ANOVA null hypothesis is that all means are the same: $\mu_1 = \mu_2 = \dots = \mu_T$. The alternate hypothesis is if one or more of the μ are different. To test this, compute the test statistic (F-statistic) that compares the between sample variability to within sample variability.

$$\begin{aligned}
F &= \frac{MS_{treatment}}{MS_{error}} \\
&= \frac{\frac{1}{T-1} SS_{treatment}}{\frac{1}{N-T} SS_{error}} \\
&= \frac{\frac{1}{T-1} \sum n_i (\bar{Y}_i - \bar{Y})^2}{\frac{1}{N-T} \sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2} \\
&= \frac{\frac{1}{T-1} \sum n_i (\bar{Y}_i - \bar{Y})^2}{\frac{1}{N-T} (\sum_i \sum_j (Y_{ij} - \bar{Y})^2 - \sum n_i (\bar{Y}_i - \bar{Y})^2)} \\
&= \frac{\frac{1}{T-1} SS_{treatment}}{\frac{1}{N-T} (SS_{total} - SS_{treatment})}
\end{aligned}$$

38.7.2 Pairwise comparisons

If the null hypothesis is rejected, use pairwise comparisons between treatments to determine which specifically are different. However, cannot simply use successive pairwise t-tests. For one t-test, the type I error is

$$\alpha = 1 - 0.95 = 0.05$$

For two t-tests, the type I error grows:

$$\alpha = 1 - (0.95 \times 0.95) = 0.0975$$

So, need specialized tests that address this problem. One is the Tukey procedure which has special critical values. Other pairwise comparison tests are Fisher's Least Protected Significant Difference, Bonferroni, Scheffe, and Dunnett.

38.7.3 Contrast analysis

Paired comparisons are made only between treatment mean pairs. The contrast analysis procedure can be used to carry out comparisons of multiple groups or even trends.

A linear contrast is

$$A = \sum_i a_i \bar{y}_i$$

which estimates

$$\sum_i a_i \mu_i$$

and where

$$\sum_i a_i = 0$$

By choosing a_i appropriately, linear contrasts can be used to make different comparisons among groups of treatment means but not limited to only mean pairs.

A pair of contrasts is orthogonal if

$$A = \sum_i a_i \bar{y}_i$$

$$B = \sum_i b_i \bar{y}_i$$

$$\sum_i a_i b_i = 0$$

so that $Cov(A, B) = 0$. The linear contrast can be used to formulate hypothesis:

$$H_0 : \sum_i a_i \mu_i = 0$$

$$H_a : \sum_i a_i \mu_i \neq 0$$

which can be tested with a specific t-statistic or F-statistic, where the latter is just the square of the former.

38.7.4 ANOVA as a linear model

ANOVA can be viewed as a linear model

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

where the τ are the treatment level effects that sum to 0. The ϵ are errors of the treatment averages. Then,

$$H_0 : Y_{ij} = \mu + \epsilon_{ij}$$

$$H_a : Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

The test statistic is

$$F = \frac{\frac{SSE(0) - SSE(a)}{DF(0) - DF(a)}}{\frac{SSE(a)}{DF(a)}} = \frac{\frac{SSE_{treatment}}{DF_{treatment}}}{\frac{SSE_{error}}{DF_{error}}} = \frac{MS_{treatment}}{MS_{error}}$$

Undertake diagnostics on the error and use Box-Cox to adjust if there are issues e.g. heteroskedasticity.

The linear model can take various configurations, depending on how the study is modelled. The overall means model is

$$Y_{ij} = \mu + \epsilon_{ij}$$

The cell means model is

$$Y_{ij} = \mu_i + \epsilon_{ij}$$

The dummy variable regression is

$$Y_{ij} = \beta_0 + \beta_1 + \beta_2 + \beta_{r-1} + \epsilon_{ij}$$

The effects (i.e. level means deviations from overall mean) model is

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

The linear models are just instances of the general linear model (GLM): $Y = X\beta + \epsilon$ where X is the 'design' matrix. This is intuitively $Data = model + error$.

38.7.5 Power analysis

If ANOVA or any other statistical procedure shows no differences but suspect there are still differences that are hidden by small sample size, etc. then use power analysis to determine the statistical power of the test. If low power, may want to improve experimental design.

38.7.6 Multi-factor ANOVA

MANOVA hypothesis is 'For each response variable, the means of all groups are equal'. This might appear a bit confusing because the null hypothesis of MANOVA is exactly the same as the combination of null hypotheses for a collection of univariate ANOVAs, but at the same time we know that doing MANOVA is not equivalent to doing univariate ANOVAs and then somehow "combining" the results (one could come up with various ways of combining). This is because running all univariate ANOVAs, even though would test the same null hypothesis, will have less power. There is a pairwise error rate when pooling hypotheses.

Factorial designs. Every combination of factors is tested or observed. Factors are said to be crossed. The combined effect is an interaction effect. In a single-factor model,

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

In a two-factor factorial model with interaction,

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

The means is partitioned as follows

$$\bar{Y}_{ij.} - \bar{Y}_{...} = (\bar{Y}_{i..} - \bar{Y}_{...}) + (\bar{Y}_{.j.} - \bar{Y}_{...}) + (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})$$

So this is the deviation around the overall mean equals the first main effect plus the second main effect plus the interaction effect.

Nested designs. Some factors are wholly within other factors. In a two-factor nested model with interaction,

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \epsilon_{ijk}$$

The means are partitioned as follows

$$\bar{Y}_{ijk} - \bar{Y}_{...} = (\bar{Y}_{i..} - \bar{Y}_{...}) + (\bar{Y}_{ij.} - \bar{Y}_{i..}) + (\bar{Y}_{ijk} - \bar{Y}_{ij.})$$

The deviation around the overall mean equals the first main effect plus the specific effect of the second effect at the i-th level, plus the residual.

With a cross-nested model,

$$Y_{ijkl} = \mu + \alpha_i + \gamma_k + (\alpha\gamma)_{ik} + (\beta\gamma)_{j(i)k} + \epsilon_{ijkl}$$

38.7.7 Mixed models

ANOVA factors can be fixed or random effects. A mixed model has a combination.

Fixed effects. Those controlled in the study or focus specifically on the chosen

factors. Example. Comparing the performance of the specific brands chosen for a study.

Random effects. Not controlled in study or general factor. Example. The overall variation in battery life, so that the results would be applicable to all brands of batteries, then, they may have chosen (presumably with a random sampling process) a sample of 4 of the many brands available and test 4 batteries of each of these brands.

Multi-factor ANOVA mixed models are different for each study combination, with each combination of crossed/nested factor interacting with whether it is fixed/random. So if there are 2 factors, then there are 6 combinations (crossed: fixed/fixed, fixed/random, random/random; nested: fixed/fixed, fixed/random, random/random). Each combination has specific tests that depend on the unique circumstance.

38.7.8 Experimental design

Randomization. Can randomize the experimental units. If there is a suspected micro-environmental gradient, use a randomized block design with one treatment level in each block, with the block oriented perpendicular to the gradient. The block would be a random effect and the means between the blocks should be netted out. Can have blocking in more than 1D. Use latin squares to build the block design.

Split-plot designs. By dividing experimental levels, can make experimental design more efficient (e.g. use less resources, reduce number of experimental units).

38.7.9 ANCOVA

Used when there is a mix of categorical predictors as in ANOVA and quantitative continuous predictors as in regression.

$$Y_{ij} = \mu + \tau_i + \gamma(X_{ij} - \bar{X}_{..}) + \epsilon_{ij}$$

which is a GLM.

Fitting a polynomial to express the impact of the quantitative predictor on the response is also called trend analysis and helps to evaluate the separate contributions of linear and nonlinear components of the polynomial. The polynomial with successive powers (up to the degrees of freedom) is inserted into the ANCOVA equation.

38.7.10 Repeated measures

If there is repeated measuring of an experimental unit, the errors or residuals will be correlated (autocorrelation). Using a 'free' autocorrelation structure

means too many variances to determine. Instead, fit to one of the common structures such as AR (autoregressive), variance components (i.e. 0 covariance, nonzero variances), and others. Use an information criterion like AIC, BIC to fit the covariance structure.

Cross-over repeated measures. Carry-over effects occur if the sequence of the treatments leads to difference covariance structures. There is a method of coding to address this issue.

39 Geometry of statistics

39.1 Variable and subject spaces

In variable spaces, the N observations are each plotted as points in scatterplots with the number of variables S (=number of subjects) being the number of dimensions. In subject spaces, the S subjects are plotted as vectors, with the number of observations N being the number of dimensions. Example. If there are 10 observations of 2 individuals X and Y , this results in a 2-dimensional scatterplot with 10 points or a 10-dimensional vector plot with 2 vectors.

Centering. An instance of variable X_i can be split into its average and deviation around the mean:

$$X_i = \bar{X} + x_i$$

The X_i is a vector that is projected onto $\bar{X}$ and x_i is the difference between the two vectors, which is orthogonal to $\bar{X}$ as it contains different information from the average. The average lies in the **1** axis, because

$$\mathbf{1} \cdot x_i = \sum x_i = \sum (X_i - \bar{X}) = 0$$

39.2 Vectors in subject spaces

Vector lengths are proportional to standard deviations because they are square roots of squared sums.

$$s_x = \sqrt{\frac{\sum x_i^2}{N-1}}$$

$$|\vec{x}| = \sqrt{N-1} s_x$$

Angles between vectors are related to correlations. Specifically, the cosine of the angle between the vectors is the Pearson correlation coefficient. The dot product is

$$|\vec{x}||\vec{y}|\cos\theta = \vec{x} \cdot \vec{y}$$

$$\cos\theta = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}||\vec{y}|}$$

So,

$$r_{xy} = \cos \theta = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}| |\vec{y}|} = \frac{\sum x_i y_i}{\sqrt{(\sum x_i^2)(\sum y_i^2)}} = \frac{\text{cov}(\vec{x}, \vec{y})}{\sqrt{\text{var}(\vec{x}) \cdot \text{var}(\vec{y})}}$$

39.3 Regression

From

$$\vec{Y} = a + b\vec{X} + e$$

the regression equation is

$$\hat{Y} = a + b\vec{X}$$

where the coefficients a and b are chosen to minimize the discrepancy which is the orthogonal error $\vec{e} = \vec{Y} - \hat{Y}$. This is the shortest distance, and which makes the error vector orthogonal to the projection $b\vec{x}$. The variables are centered to get

$$\hat{y} = b\vec{x}$$

Then,

$$\vec{x} \cdot (\vec{y} - b\vec{x}) = 0$$

$$\vec{x} \cdot \vec{y} - b(\vec{x} \cdot \vec{x}) = 0$$

$$\vec{x} \cdot \vec{y} - b|\vec{x}|^2 = 0$$

$$b = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}|^2} = \frac{\sum x_i y_i}{\sum x_i^2}$$

A projection of vector $\vec{y}$ onto $\vec{x}$ is

$$\vec{y}' = \left(\frac{\vec{y} \cdot \vec{x}}{|\vec{x}|^2} \right) \vec{x}$$

So, the regression coefficient is the length of the extension of $\vec{x}$ to get it as close as possible to $\vec{y}$. In contrast, the Pearson correlation coefficient is related to the cosine of the angle between the vectors. The regression coefficient and Pearson correlation coefficient are related by the ratio of the lengths of $\vec{y}, \vec{x}$:

$$r_{xy} \frac{\vec{y}}{\vec{x}} = b$$

With an intercept,

$$\begin{aligned} \hat{Y} &= a\vec{1} + b\vec{X} \\ &= a\vec{1} + b(\vec{X} + x) \\ &= (a + b\vec{X}) + bx \end{aligned}$$

Match up $(a + b\vec{X})\vec{1}$ with $\vec{Y}\vec{1}$ to get

$$a = \bar{Y} - b\bar{X}$$

39.4 Multiple regression

If there are two predictors $\vec{x}_1, \vec{x}_2$ and an outcome variable $\vec{y}$,

$$\vec{x}_1 \cdot \vec{e} = 0$$

$$\vec{x}_1 \cdot (\vec{y} - \hat{\vec{y}}) = 0$$

$$\vec{x}_1 \cdot (\vec{y} - b_1\vec{x}_1 + b_2\vec{x}_2) = 0$$

$$\vec{x}_1\vec{y} - b_1(\vec{x}_1\vec{x}_1) - b_2(\vec{x}_1\vec{x}_2) = 0$$

A companion equation is derived from

$$\vec{x}_2 \cdot \vec{e} = 0$$

$$\vec{x}_2\vec{y} - b_1(\vec{x}_2\vec{x}_1) - b_2(\vec{x}_2\vec{x}_2) = 0$$

The above equations are used to solve for the regression coefficients

$$b_1 = \frac{(\vec{x}_1 \cdot \vec{y})(\vec{x}_2 \cdot \vec{x}_2) - (\vec{x}_2 \cdot \vec{y})(\vec{x}_1 \cdot \vec{x}_2)}{(\vec{x}_1 \cdot \vec{x}_1)(\vec{x}_2 \cdot \vec{x}_2) - (\vec{x}_1 \cdot \vec{x}_2)^2}$$

$$b_2 = \frac{(\vec{x}_2 \cdot \vec{y})(\vec{x}_1 \cdot \vec{x}_1) - (\vec{x}_1 \cdot \vec{y})(\vec{x}_1 \cdot \vec{x}_2)}{(\vec{x}_1 \cdot \vec{x}_1)(\vec{x}_2 \cdot \vec{x}_2) - (\vec{x}_1 \cdot \vec{x}_2)^2}$$

39.5 Interpreting multiple regression coefficients

Examine coefficients of a multiple regression as a whole and not individually. They form a linear combination and should be viewed as part of that entire linear combination.

If the coefficients are all the same sign (positive or negative) and about the same size, then there is some kind of averaging effect. If the variables have different directions then there's a contrast effect.

Examine the angle between the predictor vector and $\hat{\vec{y}}$. The closer the angle, the more substantive it is. However, multicollinearity can create problems. For instance, if $\hat{\vec{y}} = 5\vec{x}_1 + 3\vec{x}_2 + 0.001\vec{x}_3$, it could be that $\vec{x}_3$ is actually the most important if the angle between $\vec{y}, \vec{x}_3$ is actually the closest. This can happen if the closest $\hat{\vec{y}}$ is constructed using $\vec{x}_1, \vec{x}_2$ and excludes or almost excludes $\vec{x}_3$.

39.5.1 Perfect collinearity

If there is perfect collinearity, remove variables or aggregate (or combine them somehow, e.g. by taking their differences) them. It cannot be solved because there is an infinite number of choices to get the regression vector from the equivalent predictor vectors.

39.5.2 Near multicollinearity

If there is near multicollinearity (ill-conditioning), problem arises because there are correlated variable(s) and uncertainty because exact measurements are impossible. If there is perfect measurement, correlated (but not perfectly correlated) variables can be solved for a unique regression vector that minimizes the error. However, this will still result in one or more of the coefficients of the correlated variables being very large in order to position the regression vector. Moreover, if one of the highly correlated vectors moves slightly off-plane, the regression plane can change dramatically, so the error and coefficients will change dramatically. So the uncertainty in the coefficients will be very large. Even a small change in-plane leads to big changes in the coefficients. Problems don't arise if only one of the correlated predictors is used; the problems arise when two or more of them are used together.

To measure near multicollinearity, can look at correlation between regression vector $\hat{y}$ using all predictors except x_i . If there is a high correlation between the two, the latter may be spanned by the former. A second approach is to take the volume spanned by the predictor variables, and if it is small compared to the number of dimensions then there may be near multicollinearity. A third approach is to use PCA.

To address near multicollinearity, could collect more samples to increase sample size and reduce error. Can also eliminate or aggregate variables, or use a selection criteria other than error vector minimization, like ridge regression.

39.5.3 Orthogonal predictors

Much more stable than with collinear or near collinear predictors. If there are two orthogonal or near orthogonal predictors x_1, x_2 , their errors e_1, e_2 will be larger than the error from the combining them to get the regression vector e . As these predictors are orthogonal, they're unrelated, so it's almost like having two separate regressions, though error shrinks upon combining them.

39.5.4 Suppressors

A variable that has little or no effect by itself suddenly has a significant effect when combined with another predictor. The suppressor variable is or nearly orthogonal to the outcome vector, so has no effect by itself. But if another predictor has an effect, the suppressor can help improve the regression vector and reducing the error.

39.6 Coefficient of determination

39.6.1 Simple regression

A regression model explains some variation in the dependent variable:

$$y_i = \hat{y}_i + e_i$$

$$Var(y) = Var(\hat{y}) + Var(e) + 2Cov(\hat{y}, e)$$

but $Cov(\hat{y}, e) = 0$ because the bracketed terms are not correlated.

$$\frac{1}{N} \sum (y_i - \bar{y})^2 = \frac{1}{N} \sum (\hat{y}_i - \bar{y})^2 + \frac{1}{N} \sum (e_i - \bar{e})^2$$

where $\bar{e} = 0$.

$$TSS = ESS + SSR$$

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}$$

This is equivalent to the squared Pearson correlation.

Note that

$$R = \cos \theta = \frac{|\hat{\vec{y}}|}{|\vec{y}|}$$

$$|\hat{\vec{y}}| = R|\vec{y}|$$

and

$$|\hat{\vec{y}}|^2 + |\vec{e}|^2 = |\vec{y}|^2$$

$$SS_{regression} + SS_{residual} = SS_{total}$$

$$SS_{regression} = |\hat{\vec{y}}|^2 = R^2 |\vec{y}|^2 = R^2 SS_{total}$$

$$SS_{residual} = |\vec{e}|^2 = (1 - R^2) SS_{total}$$

39.6.2 Multiple regression

Again,

$$R = \cos \theta = \frac{|\hat{\vec{y}}|}{|\vec{y}|}$$

$$\begin{aligned} \hat{\vec{y}} \cdot \hat{\vec{y}} &= (b_1 \vec{x}_1 + b_2 \vec{x}_2 + \dots + b_p \vec{x}_p) \cdot \hat{\vec{y}} \\ &= b_1 (\vec{x}_1 \cdot \hat{\vec{y}}) + b_2 (\vec{x}_2 \cdot \hat{\vec{y}}) + \dots + b_p (\vec{x}_p \cdot \hat{\vec{y}}) \end{aligned}$$

Note that

$$\vec{x}_j \cdot \vec{e} = \vec{x}_j \cdot (\vec{y} - \hat{\vec{y}}) = 0$$

$$\vec{x}_j \cdot \hat{\vec{y}} = \vec{x}_j \cdot \vec{y}$$

So,

$$|\hat{\vec{y}}|^2 = b_1 (\vec{x}_1 \cdot \vec{y}) + b_2 (\vec{x}_2 \cdot \vec{y}) + \dots + b_p (\vec{x}_p \cdot \vec{y})$$

Then,

$$R^2 = \frac{|\hat{y}|^2}{|\vec{y}|^2} = \frac{b_1(\vec{x}_1 \cdot \vec{y}) + b_2(\vec{x}_2 \cdot \vec{y}) + \dots + b_p(\vec{x}_p \cdot \vec{y})}{\vec{y} \cdot \vec{y}}$$

And again,

$$R = \cos \theta = \frac{|\hat{y}|}{|\vec{y}|}$$

$$|\hat{y}| = R|\vec{y}|$$

and

$$|\hat{y}|^2 + |\vec{e}|^2 = |\vec{y}|^2$$

$$SS_{regression} + SS_{residual} = SS_{total}$$

$$SS_{regression} = |\hat{y}|^2 = R^2|\vec{y}|^2 = R^2 SS_{total}$$

$$SS_{residual} = |\vec{e}|^2 = (1 - R^2) SS_{total}$$

The multiple regression model is

$$E(Y_i) = \mu_y + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}$$

which is an approximation of the ideal model. When the actual outcome variable is measured,

$$Y_i = \mu_y + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

As a vector model,

$$\vec{Y} = \mu_y \vec{1} + \beta_1 \vec{x}_1 + \beta_2 \vec{x}_2 + \dots + \beta_p \vec{x}_p + \vec{e}$$

The error vector $\vec{e}$ is random and is formed from N independent components $\epsilon_1, \epsilon_2, \dots$, each component having an independent normal distribution with mean 0. So it's spherically symmetric.

$$f(\vec{e}) = \left(\frac{1}{\sqrt{2\pi}} e^{-\epsilon_1^2/2} \right) \left(\frac{1}{\sqrt{2\pi}} e^{-\epsilon_2^2/2} \right) \dots \left(\frac{1}{\sqrt{2\pi}} e^{-\epsilon_N^2/2} \right) = \frac{1}{\sqrt{2\pi}} e^{-|\vec{e}|^2/2}$$

Since there's no angle in the equation, the error vector is uniformly spherical.

39.7 Statistical tests

39.7.1 Degrees of freedom

An outcome vector can be divided into its systematic component and error component. If the systematic component (which is proportional to its standard deviation) is larger than the error component, then the predictors are statistically significant.

In subject space, there are S-vectors (=p-vectors) in N-dimensional space. The degrees of freedom is therefore N, which is divided further. There is 1 DOF consumed for centering (i.e. the intercept), and p DOFs for each of the subjects. So the error space which is orthogonal to both the intercept and predictor (i.e. regression) spaces is $N - 1 - p$ which is the error DOF.

39.7.2 Statistical tests

To determine whether a multiple regression is helpful, test the null hypothesis that the explanatory part is 0, or more specifically that $\beta_1 = \beta_2 = \dots = \beta_p = 0$. When this is true, $\vec{y} = \vec{\epsilon}$, which still means $\hat{y} > 0$ and that there is still $\vec{\epsilon} > 0$. If the null hypothesis is false, then $\vec{\epsilon}$ points off the tip of the explanatory part of the multiple regression (instead of at 0, for the null hypothesis). This also means $\vec{\epsilon}$ is a small compared to the explanatory part of the multiple regression.

The error is

$$E(|\vec{\epsilon}|^2) = E(|\vec{\epsilon}_1|^2) + E(|\vec{\epsilon}_2|^2) + \dots + E(|\vec{\epsilon}_N|^2)$$

and if the average squared length of EACH component is M,

$$E(|\vec{\epsilon}|^2) = NM$$

So,

$$M(\vec{\epsilon}) = \frac{|\vec{\epsilon}|^2}{\dim(N)}$$

Under the null hypothesis,

$$|\vec{y}|^2 = |\vec{\epsilon}|^2 = |\vec{\epsilon}|^2 + |\hat{y}|^2$$

so

$$M(\hat{y}) \approx M(\vec{\epsilon})$$

but for the alternative hypothesis,

$$M(\hat{y}) > M(\vec{\epsilon})$$

which can be compared using the F ratio

$$F = \frac{M(\hat{y})}{M(\vec{\epsilon})} = \frac{\dim(e)|\hat{y}|^2}{\dim(\text{regression})|\vec{\epsilon}|^2}$$

When the null hypothesis is true, $F \approx 1$ and otherwise if the null hypothesis will be rejected. The F ratio can be re-expressed using R . Recall

$$|\hat{y}|^2 = R^2|\vec{y}|^2$$

$$|\vec{\epsilon}|^2 = (1 - R^2)|\vec{y}|^2$$

So,

$$M(\hat{y}) = \frac{R^2|\vec{y}|^2}{p}$$

$$M(\vec{\epsilon}) = \frac{(1 - R^2)|\vec{y}|^2}{N - p - 1}$$

So,

$$F = \frac{(N - p - 1)R^2}{p(1 - R^2)}$$

39.8 Conditional relationships

39.8.1 Partial correlation

If there are three vectors $\vec{x}$, $\vec{y}$, and $\vec{z}$. Even if $\vec{y}$ and $\vec{z}$ are separated by an acute angle (and so are positively correlated), removing the effect of $\vec{x}$ from each (resulting in $\vec{y}_{\perp x}$ and $\vec{z}_{\perp x}$) could lead them to be separated by an obtuse angle (and therefore negatively correlated). The regressions of $\vec{y}$ and $\vec{z}$ on $\vec{x}$ are

$$\vec{y}_x = b_{y \cdot x} \vec{x} \Rightarrow b_{y \cdot x} = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}|^2}$$

$$\vec{z}_x = b_{z \cdot x} \vec{x} \Rightarrow b_{z \cdot x} = \frac{\vec{x} \cdot \vec{z}}{|\vec{x}|^2}$$

Since

$$\vec{y}_{\perp x} = \vec{y} - \vec{y}_x = \vec{y} - b_{y \cdot x} \vec{x}$$

$$\vec{z}_{\perp x} = \vec{z} - \vec{z}_x = \vec{z} - b_{z \cdot x} \vec{x}$$

The partial correlation is

$$\begin{aligned} r_{yz \cdot x} &= \frac{\vec{y}_{\perp x} \cdot \vec{z}_{\perp x}}{|\vec{y}_{\perp x}| |\vec{z}_{\perp x}|} \\ &= \frac{(\vec{y} - b_{y \cdot x} \vec{x}) \cdot (\vec{z} - b_{z \cdot x} \vec{x})}{|\vec{y}_{\perp x}| |\vec{z}_{\perp x}|} \\ &= \frac{\vec{y} \cdot \vec{z} - b_{z \cdot x} (\vec{y} \cdot \vec{x}) - b_{y \cdot x} (\vec{z} \cdot \vec{x}) + b_{y \cdot x} b_{z \cdot x} (\vec{x} \cdot \vec{x})}{|\vec{y}_{\perp x}| |\vec{z}_{\perp x}|} \end{aligned}$$

Substituting the above b values and after algebra,

$$r_{yz \cdot x} = \frac{\vec{y} \cdot \vec{z} - \frac{(\vec{y} \cdot \vec{x})(\vec{z} \cdot \vec{x})}{|\vec{x}|^2}}{|\vec{y}_{\perp x}| |\vec{z}_{\perp x}|}$$

Since

$$\vec{y} \cdot \vec{z} = |\vec{y}| |\vec{z}| r_{yz}$$

$$|\vec{y}_{\perp x}|^2 = (1 - r_{xy}^2) |\vec{y}|^2$$

$$r_{yz \cdot x} = \frac{r_{yz} - r_{xy} r_{xz}}{\sqrt{1 - r_{xy}^2} \sqrt{1 - r_{xz}^2}}$$

To determine the additional effect of $\vec{z}$ beyond $\vec{x}$ on $\vec{y}$, remove the effect of $\vec{x}$ from $\vec{z}$.

$$|\vec{y}|^2 = |\hat{y}_x|^2 + |\hat{y}_{z \perp x}|^2 + |\vec{e}_{xz}|^2$$

So,

$$SS_y = SS_{y \cdot x} + SS_{y \cdot z | x} + SS_{error}$$

$$SS_{y \cdot x} = |\hat{y}_x|^2 = R_{y \cdot x}^2 |\vec{y}|^2$$

$$SS_{y \cdot xz} = |\hat{y}_{xz}|^2 = R_{y \cdot xz}^2 |\vec{y}|^2$$

$$SS_{error} = |\hat{e}_{xz}|^2 = (1 - R_{y \cdot xz}^2) |\vec{y}|^2$$

Then,

$$SS_{y \cdot xz} = |\vec{y}_{z \perp x}|^2 = |\hat{y}_{xz}|^2 - |\hat{y}_x|^2 = (R_{y \cdot xz}^2 - R_{y \cdot x}^2) |\vec{y}|^2$$

To test whether unconditional X and conditional Z are statistically significant, use F-tests. These F-tests require first calculating the mean-square lengths, which are the average per-dimension squared lengths:

$$M(\hat{y}_x) = \frac{|\vec{\hat{y}}_x|^2}{\dim(x)} = \frac{R_{y \cdot x}^2 |\vec{y}|^2}{p}$$

$$M(\hat{y}_{z \perp x}) = \frac{|\vec{\hat{y}}_{z \perp x}|^2}{\dim(z \perp x)} = \frac{(R_{y \cdot xz}^2 - R_{y \cdot x}^2) |\vec{y}|^2}{q}$$

$$M(\vec{e}) = \frac{|\vec{e}|^2}{\dim(e)} = \frac{(1 - R_{y \cdot xz}^2) |\vec{y}|^2}{N - p - q - 1}$$

Combine above for F-tests

$$F_X = \frac{M(\hat{y}_x)}{M(\vec{e})} = \frac{\frac{R_{y \cdot x}^2 |\vec{y}|^2}{p}}{\frac{(1 - R_{y \cdot xz}^2) |\vec{y}|^2}{N - p - q - 1}} = \frac{\frac{R_{y \cdot x}^2}{p}}{\frac{(1 - R_{y \cdot xz}^2)}{N - p - q - 1}}$$

$$F_{Z|X} = \frac{M(\hat{y}_{z \perp x})}{M(\vec{e})} = \frac{\frac{(R_{y \cdot xz}^2 - R_{y \cdot x}^2) |\vec{y}|^2}{q}}{\frac{(1 - R_{y \cdot xz}^2) |\vec{y}|^2}{N - p - q - 1}} = \frac{\frac{(R_{y \cdot xz}^2 - R_{y \cdot x}^2)}{q}}{\frac{(1 - R_{y \cdot xz}^2)}{N - p - q - 1}}$$

Part or semipartial correlation

$$|\vec{y}_{z \perp x}| = |\vec{y}| = \cos(\vec{y}, \hat{y}_{z \perp x})$$

So,

$$r_{y(z \cdot x)}^2 = \cos^2(\vec{y}, \hat{y}_{z \perp x}) = \frac{|\hat{y}_{z \perp x}|^2}{|\vec{y}|^2} = \frac{|\hat{y}_{xz}|^2 - |\hat{y}_x|^2}{|\vec{y}|^2} = R_{y \cdot xz}^2 - R_{y \cdot x}^2$$

39.9 Multivariate statistics

39.9.1 Standard error

The standard error (SE) of a statistic (usually an estimate of a parameter) is the standard deviation of its sampling distribution or an estimate of that standard deviation. If the statistic is the sample mean, it is called the standard error of the mean (SEM).

For the univariate setting,

$$Var(\bar{x}) = \frac{\sigma^2}{n}$$

and its estimate is

$$\widehat{Var}(\bar{x}) = \frac{s^2}{n}$$

and the square root is the standard error of the the mean. In the multivariate setting,

$$Var(\bar{x}) = \frac{1}{n}\Sigma$$

$$\widehat{Var}(\bar{x}) = \frac{1}{n}S$$

39.9.2 Correlation coefficient SE

The standard error of Pearson's correlation coefficient is

$$SE = \sqrt{\frac{1 - r^2}{n - 2}}$$

because

39.10 Principal Components Analysis

Principal components are new variables that are constructed as linear combinations of the initial variables. These new variables are uncorrelated and most of the information within the initial variables is stored into the first components and thereby reducing the dimensionality of the dataset without suffering a great penalty in terms of information loss.

The first principal component accounts for the largest possible variance; the second component will, intuitively, account for the second largest variance and so forth.

PCA can be used to select a few orthogonal variables instead of many correlated ones.

39.10.1 PCA versus regression

PCA is an unsupervised method (only takes in data, no dependent variables) and linear regression (in general) is a supervised learning method. If there is a dependent variable, a supervised method would be suited. To capture most of the variation in the data, PCA is a useful tool.

OLS regression minimizes squared distances (i.e. the squared error) along the dependent variable dimension. PCA ends up minimizing squared distances between the points and the component u because it maximizes the squared dot or inner products or alternatively the projections.

39.10.2 Covariance matrix

The covariance matrix defines both the spread (variance) and the orientation (covariance) of our data. A simple covariance matrix is

$$C = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}$$

Since $C = S^2$, where S is the standard deviation matrix,

$$S = \begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_y \end{pmatrix}$$

which indicates how much spread there is on the x-axis and the y-axis. The S matrix indicates how much the cloud of points falls within one standard deviation. 2S indicates how much it falls within two standard deviations and so forth.

39.10.3 Eigendecomposition

Place all eigenvectors into the columns of a Matrix V and all eigenvalues as the entries of a diagonal matrix L. Then

$$CV = VL$$

which means the covariance matrix multiplied by the eigenvector matrix can be restated as the eigenvector matrix multiplied by a diagonal matrix (similar to a scaling matrix).

$$C = VLV^{-1}$$

Since $C = S^2$,

$$C = VSSV^{-1} = TT'$$

Since

$$T = VS$$

T represents the rotation matrix V (i.e. the orthogonal eigenvectors matrix) and the scaling or standard deviation matrix S. It is the 'amount of correlation' in the data. Its inverse T^{-1} is used to whiten the data.

Singular value decomposition can also be used to obtain more general results.

39.10.4 Application: Mahalanobis distance

Mahalanobis distance is an effective multivariate distance metric that measures the distance between a point (vector) and a distribution. If the dimensions (dataset columns) are correlated to one another, which is typically the case in real-world datasets, the Euclidean distance between a point and the center of the points (distribution) can give little or misleading information about how close a point really is to the cluster. So, it cannot be used to really judge how

close a point actually is to a distribution of points. What we need here is a more robust distance metric that is an accurate representation of how distant a point is from a distribution.

The Mahalanobis distance measures multivariate distances with covariance. It calculates the uncorrelated distance between a point x to a multivariate normal distribution with

$$D(x) = \sqrt{(x - \mu)'C^{-1}(x - \mu)}$$

C is the covariance of the multivariate normal distribution (the set of points assumed to be normal distributed). So it removes the correlation before calculating distance. This is essentially a multivariate equivalent of the regular standardization ($z = (x - \mu)/\sigma$). For any constant $D(x) = c$, it is the same probability of that distribution (it is a contour).

Mahalanobis distances can be used to detect outliers in datasets and classification problems.

Note that the squared Mahalanobis distance is chi-squared distributed, so can be used to determine how much of a distribution lies within a multivariate normal distribution by

$$P(\sqrt{(x - \mu)'C^{-1}(x - \mu)} \leq \chi_{\alpha}^2) = 1 - \alpha$$

39.10.5 Normal distribution formula intuition

The formula

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

can be decomposed into the normalization factor

$$\frac{1}{\sqrt{2\pi\sigma^2}}$$

calculated from the Gaussian integral; the Mahalanobis distance

$$-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$$

and Euler's number which helps create an analytic continuation of factorials in the binomial distribution and which can be seen in Stirling's approximation and the Gamma function. Euler's number is exponential but the Mahalanobis distance is an inverted quadratic which suppresses the 'top' part of the normal distribution near the center, while the exponential 'overwhelms' the quadratic in the tails.

39.10.6 Optimization

PCA looks for a component u that is a linear combination of the proper axes:

$$u = a_1\vec{x}_1 + a_2\vec{x}_2 + \dots$$

To hold the new component u to the same scale as the original set,

$$a_1^2 + a_2^2 + \dots = 1$$

PCA can be derived from an optimization perspective, which is equivalent to the linear algebra representation. Data is projected to a proposed component u :

$$x_{proj} = (x'u)u$$

Where $(x'u)$ indicates the direction and amount of information after the data is projected onto the proposed component u . Now square this quantity so that $(x'u)^2$ which is still the amount of information after projection. To find u ,

$$\max \sum (x'u)^2$$

subject to

$$u'u = 1$$

Use Lagrange multipliers method to solve

$$\max \sum (x'u)^2 = \max \sum u'xx'u = \max u' \sum xx'u = u'Cu$$

where C is the covariance matrix because

$$C = \frac{1}{n} \sum xx'$$

Form the Lagrangian:

$$u'Cu - \lambda(u'u - 1) = 0$$

Differentiate with respect to u

$$2Cu - 2\lambda u = 0$$

This reduces to the eigenvector and eigenvalue equation:

$$Cu = \lambda u$$

So the direction u that maximizes the information is the eigenvector direction. The amount of information preserved after projection onto u is the eigenvalue,

$$\lambda = u'Cu$$

So the best direction to choose to get the most information is the one with the largest eigenvalues.

To find the next eigenvector, find one that is orthogonal to the preceding eigenvector by imposing this additional condition

$$\vec{u}_i\vec{u}_j = a_{1i}a_{1j} + a_{2i}a_{2j} + \dots = 0$$

39.11 Factor analysis

Factor analysis is an inversion of principal components. In factor analysis we model the observed variables as linear functions of the “factors.” In principal components, we create new variables that are linear combinations of the observed variables. In both PCA and FA, the dimension of the data is reduced.

Unlike regression, independent variables are unobservable. They are instead latent.

$$x_{i,m} - \mu_i = l_{i,1}f_{1,m} + \cdots + l_{i,k}f_{k,m} + \epsilon_{i,m}$$

where L is the loading matrix and F is the factor matrix. The X 's are observed variables and the F 's are the unobservable latent factors. The X , F , and ϵ are different for each individual, but the L is the same for all individuals. In matrix notation,

$$X = M + LF + \epsilon$$

$$X - M = LF + \epsilon$$

Assume (1) F, ϵ are independent, (2) $E(F) = 0$, and (3) $Cov(F) = I$ so that the factors are uncorrelated. Taking the covariances of both sides,

$$Cov(X - M) = Cov(LF + \epsilon)$$

$$\Sigma = L Cov(F) L' + Cov(\epsilon) = LL' + \Psi$$

where Ψ is a diagonal matrix.

The proportion of variance of the i th variable contributed by the m common factors is obtained by taking the variance of both sides

$$\sigma_{ii} = l_{i1}^2 + \dots + l_{im}^2 + \Psi_i$$

So the variance of X = communality + specific variance. The i th communality is the sum of squares of loadings of the i th variable on the m common factor.

The matrix equation is solved by MLE on computer. The solution then usually requires rotation and/or oblique transformation because the dimension axes should be adjusted so that not all factors load heavily on one axis and nearly nothing on the remaining axes. The axes should align with the factors. If they should be orthogonal, then this criteria can be imposed; otherwise, the axes can be correlated with each other (i.e. the factors can be correlated with each other). There are several algorithms for the rotation or other transformation. Varimax is easiest to compute by hand but is an oblique rotation so not recommended. Oblimin is recommended, as it usually does not make sense to impose orthogonal factors when they may be correlated.

40 Bayesian Statistics

40.1 Inferential Methods

40.1.1 Frequentists, hypotheses, and confidence intervals

The state of the world is fixed while observations are stochastic. Scrutinize models on how well they fit the data. There is only ‘yes’ or ‘no’ to each possible state of the world (e.g. loaded coin or fair coin). There are no probabilities between the two (and no priors) so can’t normalize conditional probability equation.

Select null and alternate hypotheses of the states of the world and use MLE to determine highest likelihood of data for hypothesis selection (e.g. loaded coin with $p=0.7$ and fair coin with $p=0.5$ and 44H+56T). So frequentist examine the likelihood of the data point given a model.

Use p from hypothesis to calculate confidence intervals. If $X \sim \text{Bernoulli}(p)$ then by CLT $\sum 100X_i \sim N(100p, 100p(1-p))$ so CI is $100p \pm 1.96\sqrt{100p(1-p)}$.

Hypothesis testing uses $p(\theta|x)$ to accept or reject hypotheses. CI is based on $p(x|\theta)$ by assuming a distribution (normal, t, etc.).

40.1.2 Bayesians and credible intervals

Posterior or credible intervals are generated from the posterior distribution. So Bayesians examine the likelihood of the model given the data point. Bayesian analogue of confidence intervals.

Select a prior distribution. Consider (1) equal-tailed probabilities and/or (2) highest posterior density HPD (narrowest interval equalling a specified p).

Bayesian inference involves:

- Prior knowledge - Bayesian priors
- Maximize likelihood - Bayesian likelihood
- Reduce assumptions (Occam’s Razor) - Information criteria

40.1.3 Confidence versus credible intervals

Confidence interval. If the experiment were run repeatedly and CIs calculated for each one, the true parameter value(s) lie within the ensemble of CIs 95 percent of the time. It can’t say whether any PARTICULAR CI contains the true value(s) or not - that is what a credible interval does. So a CI is arguably useless - e.g. a particular coin flip isn’t informative of its probability.

Credible interval. Given the observed data, there is a 95 percent probability

that the true parameter value(s) fall within the interval.

Confidence intervals focus on and fix parameters, while credible intervals focus on and fix the data.

There could be observations known that point to parameter estimates outside the confidence intervals, but frequentists might argue that it's beyond the confidence interval.

There could be parameters that aren't adequately covered by credible intervals that may appear likely, but Bayesians might argue that's beyond the observed data.

The confidence interval is often interpreted to be the credible interval, and this works for simple problems only because the two coincide.

40.1.4 Predictive distributions

Predictive distributions are generated from fixing θ and varying x . Bayesian analogue of frequentist prediction intervals. There are two types: prior and posterior predictive distributions.

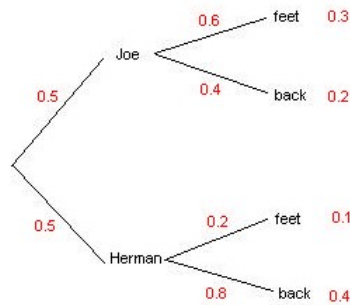
Prior predictive distributions are $f(y)$. Probability distribution of data (y) we expect before seeing the data, based on selected prior distribution. Same as marginal probability based on prior distribution - integrate across opposing variable.

Posterior predictive distributions are probability distribution of data y AFTER seeing some data y_1 . If this is repeated, then should get close to true value $\rightarrow$ Bayesian updating $\rightarrow$ what sampling attempts to replicate. Note: Bayesian updating is data order invariant. Use posterior predictive distribution to determine e.g. how much (e.g. person 1) is more likely to be taller than (e.g. person 2). Used for forecasting by sampling posterior $p(\theta|y_1)$ and then $p(y|\theta)$.

40.1.5 Bayesian inference and updating example

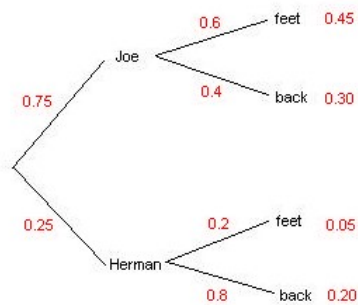
Joe and Herman are the only two frogs. Joe lands on feet 60 percent of the time and Herman lands on feet 20 percent of the time. A randomly chosen frog lands on its feet. The probability it is Joe involves a conditional probability calculation:

$$p(\text{Joe}|\text{feet}) = \frac{p(\text{feet}|\text{Joe})p(\text{Joe})}{p(\text{feet})} = \frac{p(\text{feet}, \text{Joe})}{p(\text{feet})} = \frac{0.3}{0.1 + 0.3} = 0.75$$



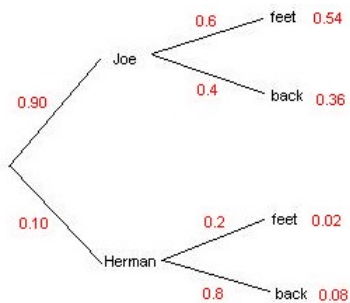
To get more information, test the chosen frog more. It lands on feet again. The probability it is Joe given (feet, feet) is:

$$p(\text{Joe}|\text{feet}) = \frac{p(\text{feet}, \text{Joe})}{p(\text{feet})} = \frac{0.45}{0.45 + 0.05} = 0.90$$



If the chosen frog lands on its back in the third trial:

$$p(\text{Joe}|\text{feet}) = \frac{p(\text{feet}, \text{Joe})}{p(\text{feet})} = \frac{0.36}{0.36 + 0.08} = 0.82$$

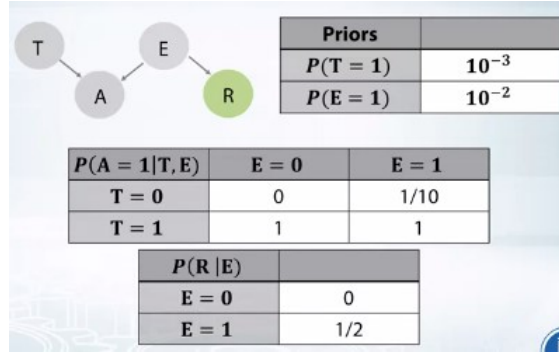


40.2 Networks

40.2.1 Bayesian Networks

In a graph, nodes are random variables and edges are direct impacts. If $A \rightarrow B$, $A \rightarrow C$, and $B \rightarrow C$, then the joint probability is $p(C|B, A)p(B|A)p(A)$.

Example.



$$P(T|A) = \frac{P(T, A)}{P(A)} = \frac{P(T, A, E) + P(T, A, \bar{E})}{P(T, A, E) + P(T, A, \bar{E}) + P(\bar{T}, A, E) + P(\bar{T}, A, \bar{E})}$$

$$= \frac{P(A|T, E)P(T|E)P(E) + P(A|T, \bar{E})P(T|\bar{E})P(\bar{E})}{\text{Numerator} + P(A|\bar{T}, E)P(\bar{T}|E)P(\bar{T})P(E) + P(A|\bar{T}, \bar{E})P(\bar{T}|\bar{E})P(\bar{E})}$$

Assume T and E are independent, so $P(T|E) = P(T)$ for instance:

$$= \frac{P(A|T, E)P(T)P(E) + P(A|T, \bar{E})P(T)P(\bar{E})}{\text{Numerator} + P(A|\bar{T}, E)P(\bar{T})P(\bar{T})P(E) + P(A|\bar{T}, \bar{E})P(\bar{T})P(\bar{E})}$$

And then evaluate = 0.48.

$$P(T|A, R) = \frac{P(T, A, R)}{P(A, R)} = \frac{P(T, A, R, E) + P(T, A, R, \bar{E})}{\text{Numerator} + P(A, R, \bar{T}, E) + P(A, R, \bar{T}, \bar{E})}$$

Simplify the above using, e.g.:

$$P(T, A, R, E) = \underbrace{P(A|T, R, E)}_{\text{R ind. of T(?)}} \underbrace{P(T|R, E)}_{\text{T is exogenous}} P(R|E)P(E)$$

$$= P(A|T, E)P(T)P(R|E)P(E)$$

Graphs can't have directed cycles, i.e. interdependent variables. Otherwise, need to joint random variables or use Markov random fields.

40.2.2 Markov network

A Bayesian network is a directed acyclic graph. So it represents directional dependencies.

A Markov network (Markov random field) is an undirected, possibly cyclic, graph with the Markov property (memoryless). So it can represent cyclic dependencies but not directional dependencies. Properties:

- Non-adjacent vertices are conditionally independent given all other variables.
- A vertex is conditionally independent of all other variables given its neighbors.
- Subsets of variables are conditionally independent given a separating subset.

The state of a random field is influenced by its neighbors in a Markov network. Use Bayes' to determine $p(f|data)$.

40.3 Curse of Dimensionality and Sparsity

Denominator and posterior, and their moments, are usually analytically intractable at high dimensions typical in Bayesian inference. Can discretize integrals or use numerical methods like Gaussian quadrature, but complexity rises exponentially with increasing dimensions (curse of dimensionality). For instance, if there are 10 calculations for 1D, then 100 in 2D, 10^N where N = number of dimensions.

Even if we ignore the denominator, discretizing at high dimension means posterior summary calculations like the below are difficult:

$$\sum_{\theta_1} \sum_{\theta_2} \dots \sum_{\theta_n} \theta_i p(\theta|x)$$

Even if this were calculable, priors are generally diffuse and likelihoods are generally concentrated, so only a small portion of priors are used. So making inferences with very small values.

40.4 Priors

40.4.1 Uninformative Priors

Make Bayesian statistics more objective by minimizing influence of prior.

40.4.2 Jeffreys priors

Jeffreys priors (1) align with likelihoods (uninformative) and (2) lead to posteriors with same statistical inferences (invariance).

Even if just increasing sample size, transforming a prior distribution will lead to different prior distributions which will lead to different posterior distributions. It will 'help' the likelihood to different relative degrees. E.g. $U[0, 1]$ is uninformative for $n = 1$ being positive but with $n = 2$ being positive it is no longer uniform - it's downward sloping.

It is generally impossible to have non-linear transformation-invariant prior distributions. However, prior distributions that differ only by a Jacobian transformation will lead to posterior distributions that also only differ by Jacobian transformations. This is the Jeffreys prior.

Jeffreys prior distribution is proportional to the square root of the Fisher information matrix of the LIKELIHOOD:

$$\begin{aligned} p(\psi) &\propto \sqrt{I(\theta)} \\ p(\psi) &= [-E[\frac{\partial^2 \log(p(y|\psi))}{\partial \psi^2}]]^{\frac{1}{2}} = [-E[\frac{\partial^2 \log(p(y|\theta))}{\partial \theta^2}]](\frac{d\theta}{d\psi})^2]^{\frac{1}{2}} \\ &= [-E[\frac{\partial^2 \log(p(y|\theta))}{\partial \theta^2}]](\frac{d\theta}{d\psi}) = p(\theta) \left| \frac{d\theta}{d\psi} \right| \end{aligned}$$

Example. Bernoulli likelihood Jeffreys prior $\sim \text{Beta}(0.5, 0.5)$. Then any $\text{Beta}(0.5, 0.5)$ distribution (e.g. with $\theta = 0.2$, $\theta = 0.4$, $\theta = 0.6$, etc.) will be a Jeffreys prior. Note: $\text{Beta}(0.5, 0.5)$ has a low effective sample size (=1) and graph is extreme at $\theta = 0$ and $\theta = 1$ because it is less likely that data here arises by chance (=more information). Jeffreys prior aligns with likelihoods.

Example. Normal likelihood with $Y \sim N(\mu, \sigma^2)$ leads to Jeffreys priors with $f(\mu) \propto 1$ and $f(\sigma^2) \propto \frac{1}{\sigma^2}$.

Difference between Jeffreys prior and vague priors is often negligible, can be complex, can be improper priors (so posteriors can be invalid). So Jeffreys priors not that useful in practice.

Fisher's information is different for different parameterizations, but that's because it is on a different scale: the curvature may be different, but so is the distance between points. The net result is invariance under a transformation.

To provide minimal help to data for the posterior, give less weight to points that are hard to find evidence against, and more weight to points that are easy to find evidence against. So take the prior distribution proportional to the expected curvature of the likelihood, which is the square root of Fisher information.

40.4.3 Improper priors

E.g. $U[0, 1] = \text{Beta}(1, 1)$ Effective sample size of 2 with 1H and 1T so not completely uninformative. So what about $\text{Beta}(0.1, 0.1)$ where effective sample size is very small. $\text{Beta}(0, 0) = f(\theta) \propto \theta^{-1}(1 - \theta)^{-1}$ (Haldane prior) which integrates to infinity. Can't normalize $\rightarrow$ improper prior (will lead to improper posterior). E.g.

$$f(\theta|y) \propto f(y|\theta)f(\theta) = \theta^k(1 - \theta)^{1-k}\theta^{-1}(1 - \theta)^{-1} = \theta - 1$$

for $k=1$ which integrates to infinity. Can still use improper priors: $f(\theta|y) \propto \theta^{y-1}(1 - \theta)^{n-y-1} \sim \text{Beta}(y, n - y)$ but there are restrictions: need at least 1H and 1T. Mean = $y/n \rightarrow$ same as for MLE (same as frequentist).

40.4.4 Vague priors

E.g. $Y \sim N(\mu, \sigma^2)$. If $\mu \sim (0, \infty^2)$ then $f(\mu) \propto 1$ (constant density) then this is an improper prior. If σ^2 is known:

$$f(\mu|x) \propto f(x|\mu)f(\mu) = \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right) (1) = \exp\left(-\frac{1}{2\sigma^2}(\mu - \bar{x})^2\right)$$

So $\mu|x \sim N(x, 2n)$, and same as MLE and frequentist approach. If σ^2 is unknown, $\Gamma^{-1} \sim (0, 0)$ which is an improper prior. Then $f(\sigma^2) = \frac{1}{\sigma^2}$ (Jacobian transformation with $x = \log \sigma^2$ will lead to uniform distribution). So $\mu| \sim \Gamma^{-1}(\frac{n}{2}, \frac{1}{2}\Sigma(y_i - y)^2)$ from formula above.

40.4.5 Reference priors

Maximize KL divergence between posterior to prior. Maximize influence of data on posterior:

$$KL = \text{argmax} \int_{\theta} p(\theta|t) \log\left[\frac{p(\theta|t)}{p(\theta)}\right] d\theta$$

Take expectation BEFORE data:

$$\begin{aligned} E\left(\int_{\theta} p(\theta|t) \log\left[\frac{p(\theta|t)}{p(\theta)}\right] d\theta\right) &= \int_t p(t) \int_{\theta} p(\theta|t) \log\left[\frac{p(\theta|t)}{p(\theta)}\right] d\theta dt \\ &= \int_t \int_{\theta} p(\theta, t) \log\left[\frac{p(\theta, t)}{p(\theta)p(t)}\right] d\theta dt \end{aligned}$$

Mutual information between t and θ which indicates dependency, or how much extra information in joint density versus only marginal densities.

Disadvantages. Calculating $p(\theta)$ is normally very difficult and requires variational calculus so not practical in many cases. In many cases, a large enough sample size (more than 100) yields negligible difference to weakly informative priors.

40.5 Empirical Bayes

Use empirical observations to determine prior. E.g. use mean of empirical observations, which yields more reasonable posterior but invalidates Bayesian methodology (but doesn't invalidate frequentist methodology).

Prior will be a function of some unknown parameters which can be calculated from the data (e.g. using method of moments, a frequentist technique).

40.5.1 Weakly informative priors

These priors fall between purely uninformative priors and priors constructed based on all pre-data knowledge. In practice, relax variance of prior, pick smooth distributions (e.g. long-tailed Cauchy instead of uniform). Most popular and useful.

40.6 Conjugacy

40.6.1 Γ^{-1} Variance Conjugate to $\mathcal{N}$

Define the precision as $\gamma = \frac{1}{\sigma^2}$, the inverse of variance. Let γ be Γ distributed:

$$\Gamma(\gamma|a, b) = \frac{b^a}{\Gamma(a)} \gamma^{a-1} e^{-b\gamma}$$

Where $\Gamma(n) = (n-1)!$ So:

$$p(\gamma) \propto \gamma^{a-1} e^{-b\gamma}$$

This also means that $\sigma^2 \sim \Gamma^{-1}$. Next, define $\mathcal{N}$ in terms of precision:

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \frac{\sqrt{\gamma}}{\sqrt{2\pi}} e^{-\gamma \frac{(x-\mu)^2}{2}} \propto \gamma^{1/2} e^{-\gamma \frac{(x-\mu)^2}{2}}$$

So

$$p(x|\gamma) \propto \gamma^{1/2} e^{-\gamma \frac{(x-\mu)^2}{2}}$$

Then:

$$p(\gamma|x) \propto p(x|\gamma)p(\gamma) = \left(\gamma^{1/2} e^{-\gamma \frac{(x-\mu)^2}{2}} \right) \gamma^{a-1} e^{-b\gamma} = \gamma^{\frac{1}{2}+a-1} e^{-\gamma(b + \frac{(x-\mu)^2}{2})}$$

So:

$$\gamma|x \sim \Gamma\left(a + \frac{1}{2}, b + \frac{(x-\mu)^2}{2}\right)$$

40.6.2 Dirichlet Conjugate to Multinomial

Dirichlet prior:

$$p(\theta) = \text{Dir}(\theta|\alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^K \theta_k^{\alpha_k - 1}$$

Multinomial likelihood:

$$p(x|\theta) = \frac{n!}{x_1! \dots x_k!} \theta_1^{x_1} \dots \theta_k^{x_k}$$

Then:

$$p(\theta|x) \propto \prod_{k=1}^K \theta_k^{\alpha_k + x_k - 1}$$

$$\theta|x \sim Dir(\theta|\alpha_k + x_k)$$

40.7 Variational Inference

Finds an exact solution to an approximation of the posterior. In contrast, MCMC finds an approximation to the exact posterior. The EM algorithm (i.e. variational EM) only results in the MAP estimator. Variational inference (i.e. variational Bayes) results in an estimated distribution:

(1) Mean field approximation. Approximate $p(\theta|x)$ with $q(\theta|x) = q(\theta)$ which is factorizable:

$$q(\theta) = q(\theta_1, \theta_2) = q_1(\theta_1)q_2(\theta_2)$$

For example, approximate a Gaussian-Inverse Gamma Bayesian posterior:

$$p(\mu, \sigma^2|x) \approx q(\mu, \sigma^2|x) = q(\mu, \sigma^2) = q_\mu(\mu)q_{\sigma^2}(\sigma^2)$$

More generally:

$$q(\theta) = \prod_i q_i(\theta_i)$$

(2) KL divergence minimization.

$$p(\theta|x) = \frac{p(x, \theta)}{p(x)}$$

Minimize distance between $p(\theta|x)$ and $q(\theta)$ by minimizing KL divergence:

$$\begin{aligned} KL(q(\theta) \parallel p(\theta|x)) &= \int q(\theta) \log \frac{q(\theta)p(x)}{p(x, \theta)} dz \\ &= \int q(\theta) \log \frac{q(\theta)}{p(x, \theta)} d\theta + \underbrace{\int q(\theta) \log p(x) d\theta}_{=1} \\ &= KL(q(\theta) \parallel p(x, \theta)) + \underbrace{\log p(x)}_{Constant} \end{aligned}$$

So minimize the first term.

(3) Variational calculus. Show that the solution to the KL divergence minimization is:

$$q_i(\theta_i) = \frac{\exp(E_{-i} \log p(x, \theta))}{\int \exp(E_{-i} \log p(x, \theta)) d\theta}$$

So:

$$q_i(\theta_i)p(x) = E_{-i} \log p(x, \theta)$$

$$\log q_i = E_{q_{-i}} \log p + \text{Constant}$$

Solve by minimizing q_i while holding q_{-i} constant. Example with $q_1(\theta_1)$ and $q_2(\theta_2)$. Solve for q_1 :

$$\begin{aligned} KL(q(\theta) \parallel p(x, \theta)) &= \int q_1 q_2 \log \frac{q_1 q_2}{p} d\theta \\ &= \int q_1 q_2 \log q_1 q_2 d\theta - \int q_1 q_2 \log p d\theta \\ &= \int q_1 q_2 \log q_1 d\theta + \int q_1 q_2 \log q_2 d\theta_1 d\theta_2 - \int q_1 q_2 \log p d\theta \\ &= \int q_1 \log q_1 \left(\underbrace{\int q_2 dz_2}_{=1} \right) d\theta_1 + \underbrace{\int q_1 d\theta_1}_{=1} q_2 \log q_2 d\theta_2 - \int q_1 q_2 \log p d\theta \\ &= \int q_1 \log q_1 d\theta_1 + \text{Constant}_0 - \int q_1 q_2 \log p d\theta \\ &= \int q_1 \log q_1 d\theta_1 - \int q_1 q_2 \log p d\theta_1 d\theta_2 + \text{Constant}_0 \\ &= \int q_1 \left(\log q_1 - \int q_2 \log p d\theta_2 \right) d\theta_1 + \text{Constant}_0 \\ &= \int q_1 \left(\log \frac{q_1}{E_{q_2}[p]} \right) d\theta_1 + \text{Constant}_0 \\ &= KL(q_1 \parallel E_{q_2}[p]) + \text{Constant}_0 \end{aligned}$$

So minimize KL divergence with:

$$\log q_1 = E_{q_2} \log p + \text{Constant}_1$$

Where the constant is from the KL divergence integral.

(4) Coordinate ascent. Repeat for q_{-i} . So in each step:

$$\log q_i = E_{q_{-i}} \log p + \text{Constant}$$

40.8 Sampling

Cuts through the curse of dimensionality as approximation doesn't depend on mathematical complexity. Instead of analytically solving for the entire posterior, make specific calculations one-at-a-time. Two types of sampling:

- Independent: get more information from the distribution but analytically intractable and/or too inefficient
- Dependent: each sample is related to the previous sample, so its information indicates the distribution but is diluted by the previous sample → samples can be locally tethered or stuck → effective sample size is smaller

40.8.1 Monte Carlo

If $x_1, \dots, x_n \sim p$ iid, then $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n f(x_i)$ is a basic Monte Carlo estimator of $E_p f(x)$, which is unbiased, consistent by WLLN if variance is finite $\sigma_{\hat{\mu}}^2 = \frac{1}{n} \sigma_{f(x)}^2 \rightarrow 0$, and converges to 0 at rate $\frac{1}{\sqrt{n}}$. Can approximate areas of intractable equations. Can easily use programs to calculate standard distributions but harder with non-standard posterior distributions.

Use indicator functions to determine probabilities:

$$p(x < \theta) = \int 1_{x < \theta} p(x) dx$$

For hierarchical model sampling, sample the prior and then sample the conditional likelihood. This amounts to joint sampling. Can also directly sample the joint distribution. Can discard all variables except the one of interest to obtain the marginal (=prior predictive) distribution.

40.8.2 Independent sampling

Given all else is the same, independent sampling is superior to dependent sampling as the effective sample size is higher.

40.8.3 Rejection Sampling

Pick x from proposal distribution, then y from uniform distribution. There is (A) uniform and (B) non-uniform rejection sampling, depending on the proposal distribution along the domain, but y is always chosen from the uniform distribution.

It is advantageous because it's independent sampling. It is disadvantageous as it's inefficient - it rejects a lot of points, and this problem grows exponentially with higher dimensions. So rejection sampling is generally only used for very simple cases.

- (A) Uniform rejection sampling. If PDF is known:
- (1) Generate random points from $U[0, x_{max}]$
 - (2) Choose y from $U[0, y_{max}]$ and accept if y -value $<$ PDF.

(B) Non-uniform rejection sampling.

- (1) Select proposal distribution $g(s)$, which sums to 1 as per a normal probability distribution. Select $g(s)$ such that it is 'close' to $p(s)$ and easy to sample from.
- (2) Scale $g(s)$ by M to get $Mg(s)$, so that it's higher than $p(s)$ everywhere ($p(s) \leq Mg(s)$). Note $Mg(s)$ sums to M .
- (3) Sample s from $g(s)$.
- (4) Pick y from $U[0, Mg(s)]$ and accept if $y < p(s)$, which is equivalent to accepting s with probability $\left\lfloor \frac{p(s)}{Mg(s)} \right\rfloor$. This is equivalent to sampling from $p(s)$. Since selection is from $p(s)$ with area 1 and $Mg(s)$ with area M , $\frac{1}{M}$ of the points will be selected, so $\left\lfloor \frac{p(s)}{Mg(s)} \right\rfloor$.

(C) Non-uniform rejection sampling - a variation useful for Bayes. Generally used when normalizing constant NC is intractable: $p(s) = \frac{f(s)}{NC}$.

- (1) Select proposal distribution $g(s)$, which sums to 1 as per a normal probability distribution. Select $g(s)$ such that it is 'close' to $f(s)$ and easy to sample from.
- (2) Scale $g(s)$ by $M' (= (NC)(W))$ to get $M'g(s)$, so that it's higher than $f(s)$ everywhere. Rationale:

$$p(s) = \frac{f(s)}{NC} \leq Wg(s)$$

$$f(s) \leq (NC)(W)g(s)$$

$$f(s) \leq M'g(s)$$

- (3) Sample s from $g(s)$.

- (4) Pick y from $U[0, M'g(s)]$ and accept if $y < f(s)$, which is equivalent to accepting s with probability $\left\lfloor \frac{f(s)}{M'g(s)} \right\rfloor$. This is equivalent to sampling from $p(s)$ because:

$$p(s) = \frac{f(s)}{NC} \leq Wg(s)$$

$$p(s) = \frac{f(s)}{(NC)(W)g(s)} \leq 1$$

$$p(s) = \frac{f(s)}{M'g(s)}$$

Rejection sampling suffers from the dimensionality curse. At higher dimensions, the NC has to be very high because the area (or volume measure) grows exponentially. So M also grows exponentially, and the upper bound is very large. It then takes many cycles to get an acceptance.

40.8.4 Importance Sampling

Hard to choose good proposal distributions, so want an alternative. Importance sampling is similar to rejection sampling with no rejections. However, it doesn't approximate the entire distribution - it only estimates the MEAN (=EXPECTED VALUE).

$$\sum xg(x) = E_g(x)$$

$$\sum xg(x) = \sum x \frac{g(x)}{f(x)} f(x) = \frac{1}{n} \sum x \frac{g(x)}{f(x)} = E_f \left(x \frac{g(x)}{f(x)} \right)$$

Sample from $f(x)$, and by knowing the ratio = weights = importances = $\frac{g(x)}{f(x)}$, can get $xg(x)$ What if $g(x)$ is unknown? For instance:

$$g(x) = \frac{h(x)}{z}$$

$$E_g(1) = 1 = \sum \frac{h(x)}{z} \Rightarrow z = \sum h(x) = \sum \frac{h(x)}{f(x)} f(x) \approx \frac{1}{n} \sum \frac{h(x)}{f(x)}$$

So,

$$\hat{z} \approx \frac{1}{n} \sum \frac{h(x)}{f(x)} = E(w) = \bar{w}$$

$$E_g(x) = \sum x \frac{h(x)}{z} = \sum x \frac{h(x)}{f(x)\hat{z}} f(x) \approx \frac{1}{\hat{z}} \frac{1}{n} \sum x \frac{h(x)}{f(x)} = \frac{\overline{xw}}{\bar{w}}$$

Proposal distribution should be closer to $g(x)$ to reduce variance of expected value. E.g. if using a flat uniform distribution, then expected value variance will be high because x samples will be all over the place.

40.8.5 Inverse Transform Sampling (Smirnov Sampling)

Uses the probability integral transform. Sample y from $U[0, 1]$ of inverse-CDF to generate x .

Advantageous because it's independent sampling and doesn't reject points. Disadvantageous because must know CDF to get inverse-CDF. This is even more difficult at higher dimensions. So use inverse transform sampling only for simple cases.

40.9 Markov Chain Monte Carlo

40.9.1 Markov Chains

The probability of a sequence $x_1, x_2, \dots, x_n$ can be decomposed:

$$p(x_1, x_2, \dots, x_n) = p(x_1)p(x_2|x_1)p(x_3|x_2, x_1)\dots p(x_n|x_{n-1}, \dots, x_1)$$

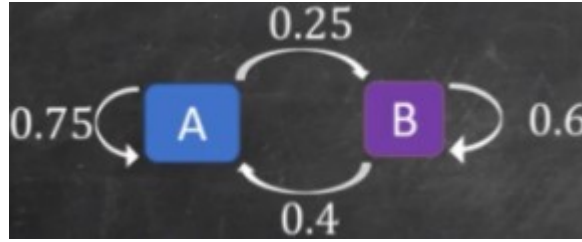
The Markov assumption is $p(x_n|x_{n-1}, \dots, x_1) = p(x_n|x_{n-1})$. Therefore:

$$p(x_1, x_2, \dots, x_n) = p(x_1)p(x_2|x_1)p(x_3|x_2)\dots p(x_n|x_{n-1})$$

A random walk can be expressed as a Markov process:

$$p(x_{t+1}|x_t) = \mathcal{N}(x_t, 1)$$

Encode a Markov process into a graph:



$$T = \begin{pmatrix} 0.75 & 0.4 \\ 0.25 & 0.6 \end{pmatrix}$$

$$p_1 = Tp_0$$

$$\begin{pmatrix} 0.75 \\ 0.25 \end{pmatrix} = \begin{pmatrix} 0.75 & 0.4 \\ 0.25 & 0.6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$p_2 = Tp_1 = T^2p_0$$

$$p_n = T^n p_0$$

Theorem: Any ergodic Markov process possesses a unique stationary distribution: $pT^n = p$ where T is the transition matrix, p is the position vector, and n is large.

Detailed Balance (or Reversibility) is stricter than stationarity:

$$p(x)T(y|x) = p(y)T(x|y)$$

$$\sum_x p(x)T(y|x) = \sum_x p(y)T(x|y)$$

$$\sum_x p(x)T(y|x) = p(y) \underbrace{\sum_x T(x|y)}_{=1}$$

$$\sum_x p(x)T(y|x) = p(y)$$

Where T is the stationary transition matrix ($= T^n$ with n being large).

Reversibility balances the probability flux of one region to another, i.e. the probability of going from one place = reverse to maintain stationarity. Reversibility requires ergodicity (irreducibility) and aperiodicity (non-periodic movement through the chain).

40.9.2 Bayes Factors

Want to sample at this ratio:

$$\frac{p(\theta_A|data)}{p(\theta_B|data)} = \frac{p(data|\theta_A)p(\theta_A)}{p(data|\theta_B)p(\theta_B)}$$

Don't need to calculate Bayes' denominator. But need to sample each data point with every other point to generate the ratios → computationally intense So, use dependent sampling: Make repeated steps based on current parameter, and compare you go, and map out parameter space Steps determined probabilistically - based on probabilities (depending on the dependent sampling technique) → steps determined by Markov chain and transition map

40.9.3 MCMC Sampling

Repeated sampling to arrive at the stationary Markov distribution after a period of 'burn-in' (i.e not yet at the stationary distribution). During the burn-in, the chain could get stuck or tethered somewhere for awhile depending on initialization. MCMC Sampling requires detailed balance condition.

Disadvantageous because it's dependent sampling so the effective sample size is relatively small. So the sampling is inefficient.

40.9.4 Metropolis-Hastings

Apply rejection sampling to Markov Chains. Pick proposal distribution Q that is close to p and is easy to sample. Draw candidate θ and accept with probability:

$$\alpha = \min \left(1, \frac{p(\theta_{t+1}|data)}{p(\theta_t|data)} \frac{Q(\theta_t|\theta_{t+1})}{Q(\theta_{t+1}|\theta_t)} \right)$$

Use Markov detailed balance:

$$p(a) = \frac{f(a)}{NC}$$

$$p(a)T(a \rightarrow b) = p(b)T(b \rightarrow a)$$

$$\frac{f(a)}{NC} Q(b|a) A(a \rightarrow b) = \frac{f(b)}{NC} Q(a|b) A(b \rightarrow a)$$

$$\frac{A(a \rightarrow b)}{A(b \rightarrow a)} = \frac{f(b)}{f(a)} \frac{Q(a|b)}{Q(b|a)} = \frac{\rho}{1} = \rho$$

To fulfill Markov detailed balance condition for stationary distribution:

$$A(a \rightarrow b) = \min\{1, \rho\}$$

ρ is a ratio of ratios: the ratio of conditional distributions f to the ratio of proposal probabilities:

$$\frac{A(a \rightarrow b)}{A(b \rightarrow a)} = \frac{\frac{f(b)/NC}{f(a)/NC}}{\frac{Q(a|b)}{Q(b|a)}} = \rho$$

In the case of plain Metropolis, Q is symmetric

$$\begin{aligned} \frac{f(a)}{NC} Q(b|a) A(a \rightarrow b) &= \frac{f(b)}{NC} Q(a|b) A(b \rightarrow a) \\ \frac{A(a \rightarrow b)}{A(b \rightarrow a)} &= \frac{f(b)}{f(a)} \frac{Q(a|b)}{Q(b|a)} = \frac{f(b)}{f(a)} = \frac{\rho_1}{1} = \rho_1 \end{aligned}$$

So:

$$A(a \rightarrow b) = \min\{1, \rho_1\}$$

40.9.5 Gibbs

Gibbs is MCMC with multiple parameters, a special case of Metropolis-Hastings. Can do MCMC with all parameters simultaneously $\rightarrow$ can do but complicated. Advantages:

- Multiple parameters, one at a time
- Accepts all proposals so more efficient at exploring sample space
- Independently sample from conditional posterior density

Disadvantages:

- Must analytically derive the conditional probabilities. This is tractable when $p(\theta)$ is conjugate to the posterior.
- For highly correlated parameters, Gibbs is slow and the effective sample size is low. Can move faster if Gibbs sample two parameters at a time but calculating the conditional probabilities becomes even more difficult. Can rotate the axes $\rightarrow$ differential evolution MCMC.

Sample from θ and ϕ alternatively in

$$p(\theta, \phi|y) = p(\phi|y)p(\theta|\phi, y)$$

No θ in first term so make it part of NC. Gibbs sampling:

$$P(\theta|X) = P(\theta_d, \theta_{-d}|X) = P(\theta_d|X, \theta_{-d})P(\theta_{-d}|X)$$

$$\alpha(\theta^*|\theta) = \min \left(1, \frac{P(\theta^*|X)P(\theta_d|X, \theta_{-d})}{P(\theta|X)P(\theta_d^*|X, \theta_{-d}^*)} \right)$$

$$= \min \left(1, \frac{P(\theta_d^*|X, \theta_{-d}^*)P(\theta_{-d}^*|X)P(\theta_d|X, \theta_{-d})}{P(\theta_d|X, \theta_{-d})P(\theta_{-d}^*|X)P(\theta_d^*|X, \theta_{-d}^*)} \right) = 1$$

... some terms cancel as they're the residual parameters after the * term is selected. Note intuitively:

$$\frac{P(\theta^*|X)P(\theta_d|X, \theta_{-d})}{P(\theta|X)P(\theta_d^*|X, \theta_{-d}^*)} = \frac{\frac{P(\theta^*|X)}{P(\theta|X)}}{\frac{P(\theta_d^*|X, \theta_{-d}^*)}{P(\theta_d|X, \theta_{-d})}}$$

Gibbs is guaranteed to asymptotically converge to posterior distribution. Proof:

$$\begin{aligned} & \sum_{x,y,z} p(x'|y, z)p(y'|x', z)p(z'|x', y')p(x, y, z) \\ &= \underbrace{p(z'|x', y')}_{\text{Ind. of } x,y,z} \sum_{y,z} [p(x'|y, z)p(y'|x', z) \underbrace{\sum_x p(x, y, z)}_{p(y,z)}] \end{aligned}$$

Combine $p(x'|y, z)$ and $p(y, z)$:

$$\begin{aligned} &= p(z'|x', y') \sum_z p(y'|x', z) \underbrace{\sum_y p(x', y, z)}_{p(x', z)} \\ &= p(z'|x', y') \underbrace{\sum_z p(x', y', z)}_{p(y', z')} = p(x', y', z') \end{aligned}$$

This means the Markov chain is stationary. Can split this algorithm and make it parallel and then use rejection sampling - with high rate of acceptance as probability distribution is 'close' to desirable already.

Example:

$$\begin{aligned} y_i | \mu, \sigma^2 &\sim N(\mu, \sigma^2) \\ \mu &\sim N(\mu_0, \sigma_0^2) \\ \sigma^2 &\sim IG(\nu_0, \beta_0) \\ p(\mu, \sigma^2 | y_1, y_2, \dots, y_n) &\propto p(y_1, y_2, \dots, y_n | \mu, \sigma^2) p(\mu) p(\sigma^2) \\ &= \prod_i^n N(y_i | \mu, \sigma^2) N(\mu | \mu_0, \sigma_0^2) IG(\sigma^2 | \nu_0, \beta_0) \\ &= \prod_i^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mu)^2}{2\sigma^2}\right) \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right) \frac{\beta_0^{\nu_0}}{\Gamma(\nu_0)} (\sigma^2)^{-(\nu_0+1)} \exp\left(-\frac{\beta_0}{\sigma^2}\right) I_{\sigma^2 > 0}(\sigma^2) \\ &\propto (\sigma^2)^{-n/2} \exp\left(-\frac{\sum_i^n (y_i - \mu)^2}{2\sigma^2}\right) \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right) (\sigma^2)^{-(\nu_0+1)} \exp\left(-\frac{\beta_0}{\sigma^2}\right) I_{\sigma^2 > 0}(\sigma^2) \end{aligned}$$

$$\begin{aligned}
p(\mu|\sigma^2, y_1, y_2, \dots, y_n) &\propto p(\mu, \sigma^2|y_1, y_2, \dots, y_n) \\
&\propto \exp\left(-\frac{\sum_i^n (y_i - \mu)^2}{2\sigma^2}\right) \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right) \\
&\propto \exp\left(-\frac{1}{2}\left(\frac{\sum_i^n (y_i - \mu)^2}{2\sigma^2} + \frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)\right) \\
&\propto N\left(\mu \mid \frac{n\bar{y}/\sigma^2 + \mu_0/\sigma_0^2}{n/\sigma^2 + 1/\sigma_0^2}, \frac{1}{n/\sigma^2 + 1/\sigma_0^2}\right) \\
p(\sigma^2|\mu, y_1, y_2, \dots, y_n) &\propto p(\mu, \sigma^2|y_1, y_2, \dots, y_n) \\
&\propto (\sigma^2)^{-n/2} \exp\left(-\frac{\sum_i^n (y_i - \mu)^2}{2\sigma^2}\right) (\sigma^2)^{-(\nu_0+1)} \exp\left(-\frac{\beta_0}{\sigma^2}\right) I_{\sigma^2>0}(\sigma^2) \\
&\propto (\sigma^2)^{-(\nu_0+n/2+1)} \exp\left(-\frac{1}{\sigma^2}\left(\beta_0 + \frac{\sum_i^n (y_i - \mu)^2}{2}\right)\right) I_{\sigma^2>0}(\sigma^2) \\
&\propto IG\left(\sigma^2 \mid \nu_0 + \frac{n}{2}, \beta + 0 + \frac{\sum_i^n (y_i - \mu)^2}{2}\right)
\end{aligned}$$

There are variations of Gibbs sampling. In 'Collapsed Gibbs' sampling, one or more variables are analytically marginalized out to result in a simpler scheme. This can be applied to LDA, for instance.

40.9.6 Hamiltonian Monte Carlo

Previous MC methods are problematic because the points are correlated. An auxiliary variable like momentum can produce uncorrelated points with a high degree of acceptance.

Vertically flip posterior density heights using $-\log[p(\text{data}|\theta)p(\theta)]$ so high densities become low and vice versa. Use log to reduce height differences.

Randomly draw momentum using independent sampling from a proposal distribution e.g. multivariate N. After predefined time T, accept or reject (so HMC is type of rejection sampling).

$$Hamiltonian = PE + KE = U(\theta) + KE(m) = -\log[p(\text{data}|\theta)p(\theta)] + KE(m)$$

Algorithm needs to discretize steps. Divergent iterations are possible around highly-changing posterior densities, so use symplectic (e.g. leapfrog) integrator algorithm which doesn't accept significant divergences. The problem is reduced if the steps are smaller but is more computationally intense. However, this biases the posterior distribution estimation.

Due to the bias, use rejection sampling:

$$r = \frac{p(X_{\theta_{t+1}})p(\theta_{t+1})}{p(X|\theta_t)p(\theta_t)} \frac{q(m')}{q(m)}$$

And compare to $u \sim U[0, 1]$.

Optimize T according to local terrain. If very 'hilly' then want small T (as it will make lots of U-turns which are inefficient) and if flat, then use a long T. Use No-U-Turn Sampler (NUTS) which adjusts T depending on starting and current position, while addressing detailed balance.

Derivation

$$r = \frac{p(x|\theta^*)p(\theta^*)N(m^*|0, 1)}{p(x|\theta)p(\theta)N(m|0, 1)} \frac{p(\theta, m|\theta^*, m^*)}{p(\theta^*, m^*|\theta, m)}$$

First term is Metropolis algorithm with momentum (like hierarchical modelling) and a Hastings term for asymmetry. Newton's laws are deterministic, not stochastic, and these determine the path from (θ, m) to (θ^*, m^*) , so $p(\theta^*, m^*|\theta, m) = 1$. Highly unlikely for the reverse process to occur, (θ, m^*) to (θ, m) , so $p(\theta, m|\theta^*, m^*) = 0$. But, this means the Hastings term = 0 so $r=0$. To address this, flip the m^* . Then, $p(\theta, m|\theta^*, -m^*) = 1$, and $p(\theta^*, -m^*|\theta, m) = 1$ which unitizes the Hastings term.

Divergent iterations occur if step size is too large and/or if there is high posterior curvature. If step sizes are too small, NUTS will hit max tree depth. May need to reparameterize likelihood and/or change prior weights.

40.9.7 Assessing MCMC convergence

When has Markov chain converged to stationary distribution? Generally several hundred to several thousand independent iterations. For posterior intervals, may need much higher - use Raftery-Louis diagnostic to determine how many iterations required for accuracy of posterior interval.

First, visually inspect the trace plot for convergence.

Second, calculate the Gelman-Rubin diagnostic, which compares variability within chain v between chains. $\hat{R}$ chains into two to address lack of intra-chain mixing. Discard burn-in portions which affect $\hat{R}$. Within-chain variance is:

$$W = \frac{1}{m} \sum_j s_j^2 = \frac{1}{m} \sum_j \left(\frac{1}{n-1} \sum_i (\theta_{ij} - \bar{\theta}_j)^2 \right)$$

$$B = \frac{n}{m-1} \sum_i (\bar{\theta}_j - \bar{\theta})^2$$

$$Var(\hat{\theta}|data) = \frac{n-1}{n} W + \frac{1}{n} B = W + \frac{1}{n} (B - W)$$

So as $n \rightarrow \infty$, $B \rightarrow W$

$$\hat{R} = \sqrt{\frac{W + \frac{1}{n}(B - W)}{W}} \rightarrow 1$$

Generally use $\hat{R} < 1.1$.

Autocorrelation reduces effective sample size. The effective sample size (ESS) is the number of independent samples - number of samples divided by number of lags to get 0 correlation in the ACF. Can reduce ACF also by increasing standard deviation of candidate draws. Calculate effective sample size from dependent samples:

$$ESS = \frac{mT}{1 + 2 \sum_J \hat{\rho}}$$

Where:

- m = number of Markov chains
- T = sample size
- T' = first integer of T where $\hat{\rho}_{T'+1} + \hat{\rho}_{T'+2} < 0$

With each autocorrelation $\hat{\rho}_J$ calculated by regressing θ_t against θ_{t-1} .

Proof of ESS formula. Posterior mean estimator

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T \theta_t$$

$$T^2 \text{var}(\hat{\mu}) = \text{var} \sum_{t=1}^T \theta_t = \sum_{t=1}^T \text{var}(\theta_t | \text{data}) + \sum_{t=1}^T \sum_{t=1}^T \text{cov}(\theta_t, \theta_{t-1})$$

Since MCMC produces autocorrelated samples,

$$\text{cov}(\theta_t, \theta_{t-\tau}) = \rho_t \text{var}(\theta | \text{data})$$

Substituting this into the above,

$$T \text{var}(\hat{\mu}) = \left(1 + 2 \sum_{t=1}^T \rho_t \right) \text{var}(\theta | \text{data})$$

For m chains,

$$mT \text{var}(\hat{\mu}) = \left(1 + 2 \sum_{t=1}^{mT} \rho_t \right) \text{var}(\theta | \text{data})$$

Since

$$\eta_{\text{effective}} = mT$$

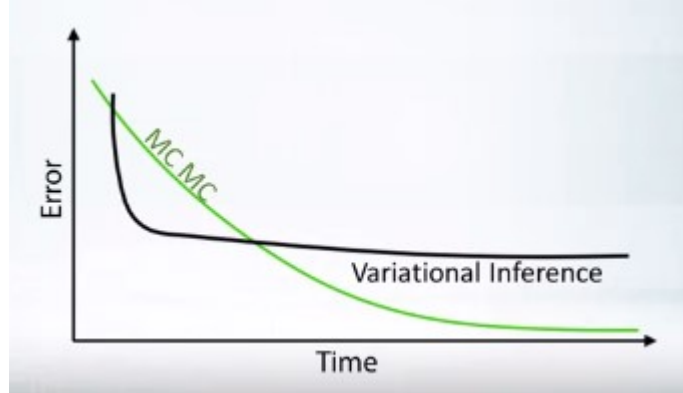
,

$$\eta_{\text{effective}} \text{var}(\hat{\mu}) = \text{var}(\theta | \text{data})$$

$$\eta_{\text{effective}} = \frac{mT}{1 + 2 \sum_{t=1}^{mT} \rho_t}$$

40.9.8 MCMC and Variational Inference

Generally, MCMC takes longer but provides a better approximation to the posterior:



40.10 Bayesian regression

As the data collected becomes infinite, the prior becomes negligible, and the Bayesian estimates approach OLS estimates. An advantage of Bayesian regression is reduced overfitting (though regularization can address this).

Frequentist generally only valid on large data sets, whereas Bayesian regression can be valid on small and large data sets.

Under the reference prior, posterior intervals of regression coefficient estimates are identical to frequentist confidence intervals.

<https://statswithr.github.io/book/introduction-to-bayesian-regression.html>

40.10.1 Estimators Under Loss Functions

Loss functions compare θ with $\hat{\theta}$. Most common:

- (i) Absolute loss $L(\theta|\hat{\theta}) = |\theta - \hat{\theta}|$.
- (ii) Squared loss $L(\theta|\hat{\theta}) = (\theta - \hat{\theta})^2$
- (iii) 0-1 loss $L(\theta|\hat{\theta}) = I(\theta \neq \hat{\theta}) = \begin{cases} 1 & \theta \neq \hat{\theta} \\ 0 & \theta = \hat{\theta} \end{cases}$

However, θ is unknown. So calculate over all θ . The mean loss (or risk, cost, etc.) is:

$$R(\hat{\theta}) = E[L(\theta|\hat{\theta})] = \int L(\theta|\hat{\theta})f(\theta|x)d\theta$$

The Bayesian estimator minimizes $\hat{\theta}$:

$$\hat{\theta} = \operatorname{argmin} R(\hat{\theta})$$

(i) Absolute loss

$$\begin{aligned}
R(\hat{\theta}) &= E[|\theta - \hat{\theta}|] = \int_{-\infty}^{\infty} |\theta - \hat{\theta}| f(\theta|x) d\theta \\
&= \int_{\hat{\theta}}^{\infty} (\theta - \hat{\theta}) f(\theta|x) d\theta + \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) f(\theta|x) d\theta \\
&= \int_{\hat{\theta}}^{\infty} \theta f(\theta|x) d\theta - \hat{\theta}(1 - F(\hat{\theta})) + \hat{\theta}F(\hat{\theta}) - \int_{-\infty}^{\hat{\theta}} \theta f(\theta|x) d\theta
\end{aligned}$$

Minimize with FOC:

$$\frac{d}{d\hat{\theta}} R(\hat{\theta}) = -\hat{\theta}f(\hat{\theta}|x) - 1 + F(\hat{\theta}|x) + \hat{\theta}f(\hat{\theta}|x) + F(\hat{\theta}|x) + \hat{\theta}f(\hat{\theta}|x) - \hat{\theta}f(\hat{\theta}|x)$$

$$2F(\hat{\theta}|x) - 1 = 0$$

$$F(\hat{\theta}|x) = \frac{1}{2}$$

So $\hat{\theta}$ is the median of $f(\theta|x)$. Note:

$$\frac{d^2 R(\hat{\theta})}{d^2 \hat{\theta}} = 2f(\hat{\theta}) > 0$$

(ii) Squared loss

$$\begin{aligned}
R(\hat{\theta}) &= E[(\theta - \hat{\theta})^2] = \int_{-\infty}^{\infty} (\theta - \hat{\theta})^2 f(\theta|x) d\theta \\
&= \hat{\theta}^2 \underbrace{\int_{-\infty}^{\infty} f(\theta|x) d\theta}_{=1} - 2\hat{\theta} \int_{-\infty}^{\infty} \theta f(\theta|x) d\theta + \int_{-\infty}^{\infty} \theta^2 f(\theta|x) d\theta \\
&= \hat{\theta}^2 - 2E[\theta]\hat{\theta} + E[\theta^2]
\end{aligned}$$

Minimize with FOC:

$$\frac{d}{d\hat{\theta}} R(\hat{\theta}) = 2\hat{\theta} - 2E[\theta] = 0$$

$$\frac{d^2 R(\hat{\theta})}{d^2 \hat{\theta}} = 2 > 0$$

So $\hat{\theta} = E[\theta]$.

(iii) 1-0 loss

$$\begin{aligned}
R(\hat{\theta}) &= E[I(\theta \neq \hat{\theta})] \\
&= \int_{-\infty}^{\hat{\theta}} (1) f(\theta|x) d\theta + \underbrace{\int_{\hat{\theta}}^{\hat{\theta}} (0) f(\theta|x) d\theta}_{=0} + \int_{\hat{\theta}}^{\infty} (1) f(\theta|x) d\theta
\end{aligned}$$

$$= \lim_{\Delta \rightarrow 0} \left(\int_{-\infty}^{\hat{\theta}+\Delta} f(\theta|x) d\theta + \int_{\hat{\theta}+\Delta}^{\infty} f(\theta|x) d\theta \right)$$

There is now a gap of 2Δ . Re-express:

$$\begin{aligned} &= \lim_{\Delta \rightarrow 0} \left(1 - \int_{\hat{\theta}-\Delta}^{\hat{\theta}+\Delta} f(\theta|x) d\theta \right) \\ &= 1 - \lim_{\Delta \rightarrow 0} \int_{\hat{\theta}-\Delta}^{\hat{\theta}+\Delta} f(\theta|x) d\theta = 1 - f(\hat{\theta}|x) \end{aligned}$$

To minimize $R(\hat{\theta})$, maximize $f(\theta|x)$. So, $\hat{\theta}$ is the mode of $f(\theta|x)$

40.10.2 Regression Coefficient Posterior with Normal Prior

Begin with the model:

$$y = x\beta + e$$

$$e \sim N(0, \sigma^2 I)$$

$$\beta \sim N(0, \tau^2 I)$$

Assume σ^2 and τ^2 are known. The above mean that:

$$y|\beta \sim N(x\beta, \sigma^2 I)$$

The posterior distribution of β is:

$$\begin{aligned} f(\beta|y) &= \frac{f(\beta, y)}{f(y)} = \frac{f(y|\beta)f(\beta)}{f(y)} \propto f(y|\beta)f(\beta) \\ &= \exp \left(-\frac{1}{2\sigma^2} (y - x\beta)'(y - x\beta) \right) \exp \left(-\frac{1}{2\tau^2} (\beta - 0)'(\beta - 0) \right) \\ &= \exp \left(\left(-\frac{1}{2\sigma^2} (y'y - y'x\beta - \beta'xiy - \beta'xix\beta) + \left(-\frac{\sigma^2}{2\sigma^2\tau^2} \beta'\beta \right) \right) \right) \end{aligned}$$

Call $\frac{\sigma^2}{\tau^2} = K$:

$$= \exp \left(-\frac{1}{2\sigma^2} (\beta'(x'x + KI)\beta - \beta'(x'y) - (x'y)'\beta + y'y) \right)$$

Since $y'y$ contains no β , remove it. Further, add $(x'y)'(x'x + KI)^{-1}(x'y)$ as a constant:

$$\propto \exp \left(-\frac{1}{2\sigma^2} (\beta(x'x + KI)\beta - \beta'(x'y) - (x'y)'(x'x + KI)^{-1}(x'x + KI)\beta + (x'y)'(x'x + KI)^{-1}(x'y)) \right)$$

Factorize:

$$= \exp \left(-\frac{1}{2\sigma^2} (\beta - (x'x + KI)^{-1}x'y)'((x'x + KI)\beta - xi'y)) \right)$$

Rearrange:

$$= \exp \left(-\frac{1}{2} (\beta - \underbrace{(x'x + KI)^{-1}x'y}_{mean})' \underbrace{\frac{(x'x + KI)}{\sigma^2}}_{\Sigma^{-1}} (\beta - \underbrace{(x'x + KI)^{-1}x'y}_{mean}) \right)$$

This is a multivariate normal:

$$f(\beta|y) \sim N((x'x + KI)^{-1}x'y, \sigma^2(x'x + KI)^{-1})$$

With $K = \frac{\sigma^2}{\tau^2}$. So the posterior distribution is normal. As it is normal, the absolute loss (median), squared loss (mean) and 0-1 loss (mode) are the same. So,

$$\hat{\beta}_{Bayes} = \hat{\beta}_{median} = \hat{\beta}_{mean} = \hat{\beta}_{mode} = (x'x + KI)^{-1}x'y$$

This is equivalent to L2 regularization (ridge regression). If $\tau \rightarrow \infty$ (vague prior) then $\hat{\beta}_{Bayes} \rightarrow \hat{\beta}_{OLS}$. So, the prior acts as a regularization term!

Further, generally with more data points, $x'x \gg KI$, so the likelihood dominates the prior, again leading to $\hat{\beta}_{Bayes} \rightarrow \hat{\beta}_{OLS}$.

A more intuitive approach:

$$f(\beta|y, X) = \frac{f(y, X, \beta)}{f(y, X)} = \frac{f(y|X, \beta)f(\beta|X)f(X)}{f(y|X)f(X)}$$

So can cross out the f(X)'s or even better, assume X is non-stochastic:

$$f(\beta|y) = \frac{f(y, \beta)}{f(y)}$$

$$\log f(\beta|y) = \log f(y|\beta) + \log f(\beta)$$

In MLE, maximize $f(y|\beta)$. Instead, proceed after defining the priors:

$$y|\beta \sim N(x\beta, \sigma^2 I)$$

$$\beta \sim N(0, \tau^2 I)$$

So:

$$\log f(\beta|y) = \log \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}}}_{K_1} \exp \left(-\frac{1}{2\sigma^2} (y - x\beta)'(y - x\beta) \right) + \log \underbrace{\frac{1}{\sqrt{2\pi\tau^2}}}_{K_2} \left(-\frac{1}{2\tau^2} (\beta - 0)'(\beta - 0) \right)$$

$$\begin{aligned}
\log f(\beta|y) &= \log K_1 - \frac{1}{2\sigma^2}(y - x\beta)'(y - x\beta) + \log K_2 - \frac{1}{2\tau^2}(\beta - 0)'(\beta - 0) \\
\frac{1}{2\sigma^2}(y - x\beta)'(y - x\beta) + \frac{1}{2\tau^2}\beta'\beta &= \log K_1 K_2 - \log f(\beta|y) \\
(y - x\beta)'(y - x\beta) + \frac{\sigma^2}{2\tau^2}\beta'\beta &= 2\sigma^2 \log K_1 K_2 - 2\sigma^2 \log f(\beta|y) \\
\|y - x\beta\|^2 + \frac{\sigma^2}{2\tau^2} \|\beta\|^2 &= 2\sigma^2 \log K_1 K_2 - 2\sigma^2 \log f(\beta|y)
\end{aligned}$$

Maximize $f(\beta|y)$ by maximizing:

$$\|y - x\beta\|^2 + \frac{\sigma^2}{2\tau^2} \|\beta\|^2$$

40.10.3 Variance Posterior with Vague Prior

Let $X_i \sim N(\mu, \sigma^2)$ be an iid variable. The μ is known so remains passive in the derivation. The likelihood is:

$$\begin{aligned}
f(x_i|\sigma^2, \mu) &= \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right) \\
&= L(\sigma^2, \mu|x) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_i^n (x_i - \mu)^2\right) \\
&\propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \left(\sum_i^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2\right)\right) \\
&\propto (\sigma^2)^{-n/2} \exp\left(-\frac{(n-1)S^2}{2\sigma^2}\right) \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{x})^2\right)
\end{aligned}$$

Use a uniform prior:

$$f(\sigma^2, \mu) \propto \frac{1}{\sigma^2}$$

Then the posterior distribution is:

$$\begin{aligned}
f(\sigma^2, \mu|x) &\propto L(\sigma^2, \mu|x) f(\sigma^2, \mu) \\
&\propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{(n-1)S^2}{2\sigma^2}\right) \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{x})^2\right)
\end{aligned}$$

Where the -1 in the σ^2 exponent comes from the uniform prior. Marginal posterior distribution for σ^2 :

$$\begin{aligned}
f(\sigma^2|x) &= \int_{-\infty}^{\infty} f(\sigma^2, \mu|x) d\mu \\
&\propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{(n-1)S^2}{2\sigma^2}\right) \underbrace{\int_{-\infty}^{\infty} \sqrt{\frac{2\pi\sigma^2}{n}} \sqrt{\frac{n}{2\pi\sigma^2}} \exp\left(-\frac{n}{2\sigma^2}(\mu - \bar{x})^2\right) d\mu}_{\text{Standard N}}
\end{aligned}$$

Where $\exp\left(\frac{-(n-1)S^2}{2\sigma^2}\right)$ does not depend on μ . So:

$$\propto \left(\frac{1}{\sigma^2}\right)^{\frac{n-1}{2}+1} \exp\left(\frac{-(n-1)S^2}{2\sigma^2}\right)$$

This is an Γ^{-1} :

$$f(\sigma^2|x) \sim \Gamma^{-1}\left(\frac{n-1}{2}, \frac{2}{(n-1)S^2}\right)$$

41 Stochastic Processes

T is time. A stochastic or random function is

$$X : T \times \Omega \rightarrow \mathbb{R}$$

where $X(t) = X_t$ is a random variable on the probability space $(\Omega, \mathcal{F}, P)$. If $T = \mathbb{R}$, then it is a continuous stochastic process, if $T = \mathbb{N}$ it is a discrete stochastic process, and if $T = \mathbb{R}^n$ then it is a random field.

The trajectory of a stochastic process is the path $X_t = X_1, X_2, \dots$ is when $\omega \in \Omega$ is fixed. As with time series, stochastic processes have X_t being dependent on each other unlike probability distributions where X_t can be independent.

41.1 Renewal theory

41.1.1 Renewal and counting processes

A renewal process is

$$\begin{aligned} S_0 &= 0 \\ S_n &= S_{n-1} + \xi_n \end{aligned}$$

where $\xi_i \sim iid$ and $\xi_i \stackrel{as}{>} 0$ so $P(\xi_i > 0) = 1 \Leftrightarrow F(0) = 0$.

A counting process is

$$N_t = \arg \max_k (S_k \leq t)$$

A counting process represents the total number of occurrences or events that have happened up to and including time t .

Graph the counting process N_t (a step function) on the y-axis and the renewal times S_k on the x-axis. The distances between S_k 's are ξ (inter-arrival times) so that $S_n = \sum_i^n \xi_i$. Note:

$$S_n > t \Leftrightarrow N_t < n$$

Renewal process examples. Number of customers in store. Reliability theory, with a sub-example being replacing failing cables in a multiple-cable system.

41.1.2 Convolutions

If $X \perp\!\!\!\perp Y, X \sim F_X, Y \sim F_Y$, then CDF convolution as a Stieltjes integral is

$$F_{X+Y}(x) = \int_{\mathbb{R}} F_X(x-y) dF(y)$$

which is $F_X * F_Y$. The PDF convolution assuming $X \perp\!\!\!\perp Y, X \sim p_X, Y \sim p_Y$

$$p_{X+Y}(x) = \int_{\mathbb{R}} p_X(x-y) p_Y(y) dy$$

which is $p_X * p_Y$.

Convolution properties of

$$F^{n*} = \underbrace{F * \dots * F}_n$$

First,

$$F^{n*}(x) \leq F^n(x)$$

if $F(0) = 0$ and $\xi_1, \dots, \xi_n \sim F(iid)$ because

$$(\xi_1 + \dots + \xi_n \leq x) \subset (\xi_1 \leq x, \dots, \xi_n \leq x)$$

by $xi \stackrel{as}{>} 0$. Now,

$$P(\xi_1 + \dots + \xi_n \leq x) \leq P(\xi_1 \leq x, \dots, \xi_n \leq x)$$

$$P(\xi_1 + \dots + \xi_n \leq x) \leq \bigcap_{k=1}^n P(\xi_k \leq x)$$

$$F^{n*}(x) \leq F^n(x)$$

Second,

$$F^{n*}(x) \geq F^{(n+1)*}(x)$$

which follows from

$$(\xi_1 + \dots + \xi_n \leq x) \subset (\xi_1 + \dots + \xi_{n+1} \leq x)$$

41.1.3 Convergence and counting expectation

Theorem. Given $S_n = S_{n-1} + \xi_n$, $\xi_i \sim F(iid)$ and $\xi_i \stackrel{as}{>} 0$ as $F(0) = 0$,

1. The sum of sum-CDFs converge. $U(t) = \sum_n^\infty F^{n*}(t) < \infty$.
2. $E(N_t) = U(t)$

The first is intuitive and the technical proof is omitted. Proof of second item.

$$\begin{aligned}
E(N_t) &= U(t) \\
&= E(\text{Count}(n : S_n \leq t)) \\
&= E\left(\sum_n^\infty \mathbb{1}(S_n \leq t)\right) \\
&= \sum_n^\infty E(\mathbb{1}(S_n \leq t)) \\
&= \sum_n^\infty p(S_n \leq t) = \sum_n^\infty F_t^{n*}
\end{aligned}$$

41.1.4 Laplace transforms

With PDF $f : \mathbb{R}_+ \rightarrow \mathbb{R}$,

$$\mathcal{L}_t(s) = \int_0^\infty e^{-sx} dx$$

Then,

1. If f is a PDF of ξ , $\mathcal{L}_t(s) = E(e^{-sx})$.
2. If f_1, f_2 are PDFs, $\mathcal{L}_{f_1 * f_2}(s) = \mathcal{L}_{f_1}(s) \cdot \mathcal{L}_{f_2}(s)$.
3. If F is a CDF with $F(0) = 0$ so almost surely > 0 and $p = F'$, $\mathcal{L}_F(s) = \frac{\mathcal{L}_p(s)}{s}$.

Proof of third property. Using integration by parts,

$$\begin{aligned}
\mathcal{L}_F(s) &= \int_{\mathbb{R}_+} F(x) e^{-sx} dx = \int_{\mathbb{R}_+} F(x) \frac{d(e^{-sx})}{s} \\
&= \underbrace{-\frac{F(x)e^{-sx}}{s}}_0 \Big|_0^\infty + \frac{1}{s} \int_{\mathbb{R}_+} p(x) e^{-sx} dx = \frac{\mathcal{L}_p(s)}{s}
\end{aligned}$$

which in the first line uses the substitution

$$\begin{aligned}
u &= e^{-sx} \Rightarrow du = e^{-sx}(-s)dx \\
-\frac{du}{s} &= -\frac{d(e^{-sx})}{s} = e^{-sx}
\end{aligned}$$

Two useful particular Laplace transform results. First,

$$\begin{aligned}
\mathcal{L}_{x^n}(s) &= \int_{\mathbb{R}_+} x^n \frac{d(e^{-sx})}{s} \\
&= \frac{n}{s} \int_{\mathbb{R}_+} x^{n-1} e^{-sx} dx \\
&= \frac{n}{s} \frac{(n-1)}{s} \frac{(n-2)}{s} \dots \frac{1}{s} \int_{\mathbb{R}_+} e^{-sx} dx \\
&= \frac{n!}{s^{n+1}}
\end{aligned}$$

Second,

$$\mathcal{L}_{e^{ax}}(s) = \frac{1}{s-a}$$

if $a < s$.

From above,

$$\begin{aligned}
E(N_t) = U_t &= \sum_1^\infty F^{n*}(t) \\
U_t &= \sum_1^\infty F^{n*}(t) \\
&= F(t) + F(t) * F(t) + F(t) * F(t) * F(t) + \dots \\
&= F(t) + F(t)(F(t) + F(t) * F(t) + \dots) \text{ by associativity} \\
&= F(t) + \underbrace{\left(\sum_1^\infty F^{n*}(t) \right)}_{U_t} * F(t) \\
&= F + U * F
\end{aligned}$$

where the convolution is for CDFs. Note however that

$$U = F + U * F = F + U * p$$

even though $F' = p$, not $F = p$, because

$$\int_{\mathbb{R}} U(x-y) dF(y) = \int_{\mathbb{R}} U(x-y) p(y) dy$$

which intuitively occurs as PDFs are different in nature than CDFs. Now,

$$U = F + U * p$$

Taking the Laplace transform of both sides,

$$\mathcal{L}_U(s) = \mathcal{L}_F(s) + \mathcal{L}_U(s) \cdot \mathcal{L}_p(s)$$

Using $\mathcal{L}_F(s) = \frac{\mathcal{L}_p(s)}{s}$,

$$\mathcal{L}_U(s) = \frac{\mathcal{L}_p(s)}{s} + \mathcal{L}_U(s) \cdot \mathcal{L}_p(s)$$

So,

$$\mathcal{L}_U(s) = \frac{\mathcal{L}_p(s)}{s(1 - \mathcal{L}_p(s))}$$

The above indicates how to calculate U_t from the distribution of ξ :

1. Use F or p to calculate $\mathcal{L}_p$.
2. Use $\mathcal{L}_p$ to calculate $\mathcal{L}_U$ using $\mathcal{L}_U(s) = \frac{\mathcal{L}_p(s)}{s(1 - \mathcal{L}_p(s))}$.
3. Inverse Laplace transform. Use $\mathcal{L}_U$ to calculate U .

However, the last step is difficult, involving Bromwich integrals. So in most cases, it's ad-hoc, often using the Laplace transform and inverse tables (which in turn often involves partial fractions). Note there is a 1-1 correspondence using the Laplace transform.

Example. From $S_n = S_{n-1} + \xi_n, \xi \sim iid$ where $p(x) = \frac{e^{-x}}{2} + e^{-2x}$.

1. $p \rightarrow \mathcal{L}_p(s) = \frac{3s+4}{2(s+1)(s+2)}$
2. $\mathcal{L}_p \rightarrow \mathcal{L}_U(s) = \frac{3s+4}{s^2(2s+3)}$
3. $\mathcal{L}_U \rightarrow U(t) = \frac{4t}{3} + \frac{1}{9} - \frac{e^{-1.5t}}{9}$

41.1.5 Limit theorems

Theorem 1. If $\mu = E(\xi_i) < \infty$ (i.e. finite) then

$$\frac{N_t}{t} \xrightarrow{as} \frac{1}{\mu}$$

which is an analogue of the strong LLN:

$$\frac{\sum \xi_i}{n} \xrightarrow{as} \mu$$

Proof. Begin with $S_{N_t} \leq t \leq S_{N_{t+1}}$, which is the x-axis of the step-function graph mentioned above (where N_t is the y-axis). Then,

$$\begin{aligned} \frac{N_t}{S_{N_{t+1}}} &\leq \frac{N_t}{t} \leq \frac{N_t}{S_{N_t}} \\ \lim_{t \rightarrow \infty} \frac{N_t}{S_{N_t}} &= \lim_{n \rightarrow \infty} \frac{n}{S_n} \xrightarrow{SLLN} \frac{1}{\mu} \end{aligned}$$

$$\lim_{t \rightarrow \infty} \frac{N_t}{S_{N_{t+1}}} = \lim_{t \rightarrow \infty} \underbrace{\frac{N_t}{N_{t+1}}}_1 \cdot \lim_{t \rightarrow \infty} \underbrace{\frac{N_{t+1}}{S_{N_{t+1}}}}_{1/\mu} = \frac{1}{\mu}$$

and by the envelope theorem, the limit is established.

Theorem 2. If $\sigma^2(\xi_i) < \infty$ (i.e. finite) then

$$Z_t = \frac{N_t - t/\mu}{\sigma\sqrt{t}/(\mu^{3/2})} \xrightarrow{d} \mathcal{N}(0, 1)$$

which is an analog of the CLT:

$$\frac{\sum \xi_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Note that

$$P(Z_t \leq x) \rightarrow \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = \Phi(x)$$

Proof.

$$P\left(\frac{S_n - n/\mu}{\sigma\sqrt{n}} \leq x\right) \xrightarrow{d} \Phi(x), x \in \mathbb{R}$$

$$P(S_n \leq n\mu + \sigma\sqrt{n}x) \rightarrow \Phi(x)$$

Let $t = n\mu + \sigma\sqrt{n}x$. Rearranging,

$$n = \frac{t}{\mu} - \frac{\sigma\sqrt{n}}{\mu}x$$

Letting the second RHS term be small such that $n \approx t/\mu$,

$$n = \frac{t}{\mu} - \frac{\sigma\sqrt{t}}{\mu^{3/2}}x$$

$$\frac{n - t/\mu}{\sigma\sqrt{t}/(\mu^{3/2})} = -x$$

The LHS is Z_t . So,

$$P(Z_t \geq -X) \rightarrow \Phi(x)$$

$$P(Z_t \leq X) = 1 - P(-Z_t > -X) = 1 - \Phi(-x) = \Phi(x)$$

41.2 Poisson processes

41.2.1 Definitions

Definition 1. A Poisson process is a specific renewal process s.t. $\xi_i \sim p(x) = \lambda e^{-\lambda x}$. λ is the intensity or rate; the average number of points per some unit of extent such as length, area, volume, or time, depending on the underlying mathematical space, and also called the mean density or mean rate.

Definition 2. N_t is a Poisson process if

1. $N_s = 0$ almost surely.
2. N_t has independent increments. $N_{t_1} - N_{t_0}, \dots, N_{t_n} - N_{t_{n-1}}$ are independent.
3. N_t has stationary increments. $N_t - N_s \stackrel{d}{=} N_{t-s}$.
4. $N_t - N_s \sim \text{Poisson}(\lambda(t-s))$.

Definition 3. N_t is a Poisson process if

1. $N_s = 0$ almost surely.
2. N_t has independent increments. $N_{t_1} - N_{t_0}, \dots, N_{t_n} - N_{t_{n-1}}$ are independent.
3. N_t has stationary increments. $N_t - N_s \stackrel{d}{=} N_{t-s}$.
4. $\lim_{h \rightarrow 0} \frac{p(N_{t+h} - N_t \geq 2)}{p(N_{t+h} - N_t = 1)} = 0$.

where the last definition follows from the below.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{1 - p(N_{t+h} - N_t = 0)}{h} &= \lim_{h \rightarrow 0} \frac{1 - e^{-\lambda h}}{h} \text{ i.e. let } n=0 \\ &= \lim_{h \rightarrow 0} \frac{\lambda e^{-\lambda h}}{1} \text{ by l'Hopital} \\ &= \lambda \end{aligned}$$

So,

$$\begin{aligned} \frac{1 - p(N_{t+h} - N_t = 0)}{h} - \lambda &= \bar{o}(1) \\ p(N_{t+h} - N_t = 0) &= 1 - \lambda h + \bar{o}(h) \end{aligned}$$

where

$$\lim_{h \rightarrow 0} \frac{o(1)h}{h} = 0$$

which follows from the definition

$$\bar{o}(g(h)) = \lim_{h \rightarrow 0} \frac{f(h)}{g(h)}$$

Then,

$$\begin{aligned} p(N_{t+h} - N_t = 0) &= 1 - \lambda h + \bar{o}(h) \\ p(N_{t+h} - N_t = 1) &= \lambda h + \bar{o}(h) \\ p(N_{t+h} - N_t \geq 2) &= \bar{o}(h) \end{aligned}$$

41.2.2 Theorems

Theorem. A Poisson process is characterized by

1. $p_{S_n} = \lambda \frac{(\lambda x)^{n-1}}{(n-1)!} e^{-\lambda x}$. This means that the CDF is $F_{S_n} = 1 - e^{-\lambda x} \sum_{k=0}^{n-1} \frac{(\lambda x)^k}{k!}$.
2. $p(N_t = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$ and $N_t \sim \text{Poisson}(\lambda t)$

Proof of 1. Proof by induction. For $n = 1$, $p_{S_1}(x) = \lambda e^{-\lambda x}$, which matches the definition. For $n \rightarrow n + 1$, calculate the convolution of $S_{n+1} = S_n + \xi_{n+1}$:

$$\begin{aligned}
 p_{S_{n+1}}(x) &= \int_0^x p_{S_n}(x-y) p_{\xi_{n+1}}(y) dy \\
 &= \int_0^x \left(\frac{\lambda^n (x-y)^{n-1}}{(n-1)!} e^{-\lambda(x-y)} \right) (\lambda e^{-\lambda y}) dy \\
 &= \frac{\lambda^{n+1}}{(n-1)!} e^{-\lambda x} \int_0^x (x-y)^{n-1} dy \\
 &= \frac{\lambda^{n+1}}{(n-1)!} e^{-\lambda x} \frac{(x-y)^n}{n} \Big|_0^x \\
 &= \frac{\lambda^{n+1}}{(n-1)!} e^{-\lambda x} \frac{x^n}{n} \\
 &= \frac{\lambda \cdot \lambda^n \cdot x^n}{n(n-1)!} = \lambda \frac{(\lambda x)^n}{n!} e^{-\lambda x}
 \end{aligned}$$

which matches 1. above because

$$p_{S_{n+1}} = \lambda \frac{(\lambda x)^{n+1-1}}{(n+1-1)!} e^{-\lambda x} = \frac{\lambda \cdot \lambda^n \cdot x^n}{n(n-1)!} = \lambda \frac{(\lambda x)^n}{n!} e^{-\lambda x}$$

Proof of 2. Since

$$(N_t = n) = \underbrace{(S_n \leq t)}_A \cap \underbrace{(S_{n+1} > t)}_B$$

and here $A \cap B = A \setminus B^c$ and $B^c \subset A$,

$$p(A \cap B) = p(A) - p(B)$$

So,

$$p(N_t = n) = p(S_n \leq t) - p(S_{n+1} \leq t)$$

and using the CDF from above,

$$\begin{aligned}
 p(N_t = n) &= p(S_n \leq t) - p(S_{n+1} \leq t) \\
 &= \left(1 - e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!} \right) - \left(1 - e^{-\lambda t} \sum_{k=0}^n \frac{(\lambda t)^k}{k!} \right) \\
 &= e^{-\lambda t} \frac{(\lambda t)^n}{n!}
 \end{aligned}$$

41.2.3 Memoryless property

A random variable X possesses the memoryless property iff $p(X > u + v) = p(X > u)p(X > v)$. Then, $p(X > u + v | X > v) = p(X > u)$. Note that both conditions must be satisfied. Example. If $p(X = -1) = 0.5, p(X \in U[0, 1]) = 0.5$, it's not memoryless because $p(X > 13 | X > 7) = 0, p(X > 6) = 0.5 - 0.3 = 0.2$.

Theorem 2. Random variable X is memoryless iff it has an exponential distribution, $p(x) = \lambda e^{-\lambda x}$.

41.2.4 Inhomogeneous Poisson processes

In homogeneous Poisson processes, $N_t \sim Pois(\lambda t) \Rightarrow E(N_t) = \lambda t$. In other words, $\Lambda = \lambda \cdot \nu$ where λ is a constant intensity or rate while ν is a Lebesgue measure like area, volume, or time.

Non-homogeneous (inhomogeneous) Poisson processes. Let $\Lambda(t)$ be a differentiable increasing function s.t. $\Lambda(0) = 0$. Example of inhomogeneous process. Arrival rate of customers is larger during lunch time compared to the late afternoon.

$X_t = N_t$ is a non-homogeneous Poisson process if

1. $N_0 = 0$
2. N_t has independent increments
3. $N_t - N_s \sim Pois(\Lambda(t) - \Lambda(s))$

where $\lambda(t) = \Lambda'(t)$, the intensity function.

Properties of non-homogeneous Poisson processes.

1. $E(N_t) = \Lambda(t)$ from (3) above. Example. $\Lambda(t) = \alpha t^\beta$.
2. If $\lambda(t) = \text{const.} \Rightarrow \Lambda(t) = \text{const.} \cdot t$.
3. $\Lambda(t)$ is differentiable $\Rightarrow \Lambda(t)$ is continuous. Moreover, it is defined to be increasing, so is invertible ($\Lambda^{-1}(t)$) with image $\Lambda(t) = \mathbb{R}$. So $N_{\Lambda^{-1}(t)}$ can be a homogeneous Poisson process but only with specific transformations (e.g. inverse sample transforms). This shows a potential connection between homogeneous and inhomogeneous Poisson processes.

41.2.5 Inhomogeneity and iid

Is $\xi_n = S_n - S_{n-1} \sim iid$?

Prove two relations first

1. $p_{\xi_1}(x) = \lambda(x)e^{-\Lambda(x)}$
2. $p_{\xi_2|\xi_1}(t|s) = \lambda(t+s)e^{-\Lambda(t+s)+\Lambda(s)}$

Proof of 1.

$$\begin{aligned}
p(\xi_1 \leq x) &= p(S_1 \leq x) \\
&= p(N_x \geq 1) \text{ since } (S_n > t) = (N_t < n) \\
&= 1 - p(N_x = 0) = 1 - e^{-\Lambda(x)}
\end{aligned}$$

Then taking derivatives of both sides, and recalling $\lambda(t) = \Lambda'(t)$, $p_{\xi_1}(x) = \lambda(x)e^{-\Lambda(x)}$.

Proof of 2.

$$\begin{aligned}
F_{\xi_1, \xi_2}(s, t) &= p(\xi_1 \leq s, \xi_2 \leq t) \\
&= \int_0^s p(\xi_1 \leq s, \xi_2 \leq t | \xi_1 = y) p_{\xi_1}(y) dy \\
&= \int_0^s p(N_{t+y} - N_y | \xi_1 = y) p_{\xi_1}(y) dy \\
&= \int_0^s p(N_{t+y} - N_y) p_{\xi_1}(y) dy \text{ independent increments} \\
&= \int_0^s (1 - e^{-\Lambda(t+y)-\Lambda(y)}) \lambda(y) e^{-\Lambda(y)} dy
\end{aligned}$$

Then,

$$\begin{aligned}
p_{\xi_1, \xi_2}(s, t) &= \frac{\partial}{\partial t} \left(\frac{\partial}{\partial s} F_{\xi_1, \xi_2}(s, t) \right) \\
&= \frac{\partial}{\partial t} \left((1 - e^{-\Lambda(t+s)-\Lambda(s)}) \underbrace{\lambda(s)e^{-\Lambda(s)}}_{p_{\xi_1}(s)} \right)
\end{aligned}$$

and dividing both sides by $p_{\xi_1}(s)$ completes the proof.

To determine whether $\xi_i \sim iid$, determine whether the following holds

$$p_{\xi_1}(t) = p_{\xi_2|\xi_1}(t|s)$$

To determine this,

$$\begin{aligned}
\lambda(t)e^{-\Lambda(t)} &= \lambda(t+s)e^{-\Lambda(t+s)+\Lambda(s)} \\
\underbrace{e^{-\Lambda(0)}}_{e^0} - e^{-\Lambda(T)} &= e^0 - e^{-\Lambda(T+s)+\Lambda(s)} \\
-e^{-\Lambda(T)} &= -e^{-\Lambda(T+s)+\Lambda(s)} \\
\Lambda(T) &= \Lambda(T+s) - \Lambda(s)
\end{aligned}$$

Since $\Lambda(t)$ was defined as monotonically increasing, it follows that only when $\Lambda(t) = \lambda t$ is $\xi_i \sim iid$. This means that only when N_t is a *homogeneous* process.

41.2.6 Queuing theory

Queuing theory often involves capacity utilisation, which is $\frac{\lambda}{n\mu}$, where λ is the arrivals, n is the number of servers, and μ is the service time. When the numerator is larger than the denominator, the queue gets longer.

There are numerous ways of modelling queues (a branch of operations research). Example. A Poisson queue is a queuing model in which the number of arrivals per unit of time and the number of completions of service per unit of time, when there are customers waiting, both have the Poisson distribution. Applicable for homogeneous (λ) or inhomogeneous ($\lambda(t)$) Poisson processes where applicable.

In Kendall's notation, A/S/c where A is arrival time process, S is service time process, and c is number of servers. *Arrival process*. M=Markovian/memoryless, so Poisson. D=degenerate or deterministic. G=general or any arbitrary distribution. *Service process*. M, D, G. *Number of servers*. Integers, can be infinity. If infinity, then arrivals are immediately served without wait.

M/G/ ∞ . Assume there is a homogenous N_t of arrivals that is decomposed into $N_1(t)$, $\lambda_1(t) = \lambda(1 - G(\tau - t))$, an inhomogeneous Poisson process of arrivals still being served at $t = \tau$, and $N_2(t)$, $\lambda_2(t) = \lambda(G(\tau - t))$ of arrivals already served by $t = \tau$. Then,

$$\begin{aligned} p(N_1(t + \delta) - N_1(t) = 1) &= p(N(t + \delta) - N(t) = 1) \underbrace{p(Y > \tau - t) + \bar{o}(\delta)}_{\text{customer}} \\ &= (\delta\lambda + \bar{o}(\delta))(1 - G(\tau - t) + \bar{o}(\delta)) \\ &= \delta\lambda(1 - G(t - \tau) + \bar{o}(\delta)) \end{aligned}$$

Now,

$$\begin{aligned} &p(N_1(t) = n_1, N_2(t) = n_2) \\ &= p(N_1(t) = n_1, N_2(t) = n_2 | N(t) = n_1 + n_2) \underbrace{p(N(t) = n_1 + n_2)}_{\sim \text{Pois}(\lambda t)} \\ &= \binom{n_1}{n_1 + n_2} (1 - G(\tau - t))^{n_1} (G(\tau - t))^{n_2} e^{-\lambda t} \frac{(\lambda t)^{n_1 + n_2}}{(n_1 + n_2)!} \\ &= \frac{\lambda t (1 - G(\tau - t))^{n_1}}{n_1!} e^{-\lambda(1 - G(\tau - t))} \lambda t (G(t - \tau))^{n_2} e^{-\lambda G(\tau - t)} \\ &= p(N_1(t) = n_1) p(N_2(t) = n_2) \end{aligned}$$

since it is a product of probabilities, they are independent. This means $N_1(t)$, $N_2(t)$ are independent even though they are from one singular process $N(t)$.

41.2.7 Compound Poisson process

$$X_t = \sum_{k=1}^{N_t} \xi_k$$

where $\xi_i \sim iid, N_t \sim Pois(\lambda)$. Example. ξ_i are insurance claim sizes, N_t are claim volumes, and X_t is the total claim value.

Use the probability-generating function with $\xi \in \mathbb{Z} \geq 0$. The general PGF is

$$G_X(s) = E(s^X) = \sum_{x=0}^{\infty} = p_0 + p_1 s + p_2 s^2 + \dots$$

and in this case is

$$PGF = \Phi_{\xi}(u) = E(u^{\xi})$$

where $|u| < 1$ to prevent series divergence. If $\xi_1 \perp\!\!\!\perp \xi_2$ then

$$\Phi_{\xi_1+\xi_2}(u) = \Phi_{\xi_1}(u) + \Phi_{\xi_2}(u)$$

Use also the MGF (two-sided Laplace transform):

$$MGF = \mathcal{L}_{\xi}(u) = E(e^{-u\xi})$$

In the case where ξ is not limited to $\mathbb{Z}$ and is arbitrary, use characteristic functions (why?)

$$\xi : \Phi_{\xi}(u) = E(e^{iu\xi}), \Phi : \mathbb{R} \rightarrow \mathbb{C}$$

$$\xi_1 \perp\!\!\!\perp \xi_2 \Rightarrow \Phi_{\xi_1+\xi_2}(u) = \Phi_{\xi_1}(u)\Phi_{\xi_2}(u)$$

Theorem.

$$\Phi_{X_t-X_s}(u) = e^{-\lambda(t-s)(\Phi_{\xi_1}(u)-1)}$$

Proof.

$$\begin{aligned} \Phi_{X_t-X_s}(u) &= E(e^{iu(X_t-X_s)}) \\ &= \sum_{k=0}^{\infty} E(e^{iu(X_t-X_s)} | N_t - N_s = k) p(N_t - N_s = k) \\ &= \sum_{k=0}^{\infty} E(e^{iu(X_t-X_s)}) \underbrace{p(N_t - N_s = k)}_{\sim Pois(\lambda(t-s))} \text{ as } \xi \perp\!\!\!\perp N_t \\ &= \sum_{k=0}^{\infty} E(e^{iuk\xi}) e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!} \text{ as } X_t - X_s \stackrel{d}{=} \xi_1 + \xi_2 + \dots \xi_k \\ &= \sum_{k=0}^{\infty} (\Phi_{\xi_1}(u))^k e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!} \\ &= e^{-\lambda(t-s)} \sum_{k=0}^{\infty} \frac{(\Phi_{\xi_1}(u)\lambda(t-s))^k}{k!} \\ &= e^{-\lambda(t-s)} e^{-\Phi_{\xi_1}(u)\lambda(t-s)} \\ &= e^{-\lambda(t-s)(\Phi_{\xi_1}(u)-1)} \end{aligned}$$

Corollaries.

1. $E(X_t) = \lambda t E(\xi_1)$
2. $Var(X_t) = \lambda t E(\xi_1^2)$

Proof of 1. First note $E(\xi^r) < \infty \Rightarrow \Phi(u)$ is r -times differentiable at 0. So,

$$\Phi^{(r)}(u) = i^r E(\xi^r) = E((i\xi)^r e^{iu\xi})$$

by linearity. So with $u = 0$,

$$\Phi^{(r)}(0) = E(i^r \xi^r e^{i0\xi}) = i^r E(\xi^r)$$

which also holds for X_t . So applying the above results with X_t instead of ξ ,

$$E(X_t) = \frac{\Phi'_{X_t}(0)}{i}$$

Then from above, $\Phi_{X_t}^{(r)}(u) = e^{\lambda t(\Phi_{\xi_1}(u)-1)}$, so

$$\Phi'_{X_t}(u) = \lambda t \Phi'_{\xi_1}(u) e^{\lambda t(\Phi_{\xi_1}(u)-1)} = \lambda t \Phi'_{\xi_1}(u) \Phi'_{X_t}(u)$$

Returning a few lines up,

$$E(X_t) = \frac{\Phi'_{X_t}(0)}{i} = \frac{\lambda t \Phi'_{\xi}(0) \Phi'_{X_t}(0)}{i} = \lambda t E(\xi)$$

where

$$\begin{aligned} \Phi'_{X_t}(0) &= 1 \\ \frac{\Phi'_{\xi}(0)}{i} &= E(\xi) \end{aligned}$$

Proof of 2. Follows from 1.

41.3 Markov processes

Definition. Markov chain S_n in a countable state space S which satisfies the Markov property

$$p(S_n = j | S_{n-1} = i_{n-1}, \dots, S_0 = i_0) = p(S_n = j | S_{n-1} = i_{n-1})$$

where $i_0, \dots, i_{n-1}, j \in S$ and $p(S_{n-1} = i_{n-1}, \dots, S_0 = i_0) \neq 0$. If the Markov property is satisfied,

$$\begin{aligned} &p(S_n = i_n, S_{n-1} = i_{n-1}, \dots, S_0 = i_0) \\ &= p(S_n = i_n | S_{n-1} = i_{n-1}) p(S_{n-1} = i_{n-1}, \dots, S_0 = i_0) \\ &= p(S_n = i_n | S_{n-1} = i_{n-1}) p(S_{n-1} = i_{n-1} | S_{n-2} = i_{n-2}) \dots p(S_1 = i_1 | S_0 = i_0) p(S_0 = i_0) \end{aligned}$$

If S_n 's are independent, then the conditioning is redundant, so there is a kind of 'pairwise independence'.

41.3.1 Random walk

$$\begin{aligned} S_0 &= 0 \\ S_n &= S_{n-1} + \xi_n \end{aligned}$$

where

$$\xi_i \sim iid \begin{cases} 1 & p \\ -1 & 1-p \end{cases}$$

Random walks are Markov chains:

$$p(S_n = j | S_{n-1} = i_{n-1}) = \begin{cases} p & j = i_{n-1} + 1 \\ 1-p & j = i_{n-1} - 1 \\ 0 & \text{otherwise} \end{cases}$$

Any renewal process is also a Markov chain but not interesting as the renewal process is unstructured so can take on any value. Random walks are not renewal processes because they can take negative values.

41.3.2 Matrix representation

Let the state space $S = (1, 2, \dots, M)$ be finite and

$$p(X_n = j | X_{n-1} = i) = p_{ij}$$

(doesn't depend on n) be a *homogeneous* Markov chain. Then $p = (p_{ij})_{i,j=1}^M$ is the transition matrix (e.g. p_{12} is the 1st row and 2nd column with $p_{12} = p(X_n = 2 | X_{n-1} = 1)$). Further, if

$$\sum_{j=1}^M p_{ij} = 1$$

and $p_{ij} > 0$, then it is a *stochastic matrix*.

Define

$$p_{ij}^{(m)} p(X_{n+m} = j | X_n = i)$$

and the *m-step transition matrix* as

$$p^{(m)} = (p_{ij}^{(m)})$$

$$p^m = (p(X_n = j | X_{n-1} = i))^m$$

Theorem. $p^{(m)} = p^m$. Proof.

$$\begin{aligned} p_{ij}^{(m)} &= \sum_{k=1}^M p(X_{n+m} = j | X_{n+m-1} = k) p(X_{n+m-1} = k | X_n = i) \\ &= \sum_{k=1}^M p_{kj} p_{ik}^{(m-1)} \\ &= p p^{(m-1)} = p^m \end{aligned}$$

The discrete PDF is

$$p(X_k = j) = \pi_j^{(k)}$$

$$\vec{\pi}^{(k)} = (\pi_1^{(k)}, \pi_2^{(k)}, \dots, \pi_M^{(k)})$$

with

$$\begin{aligned}\pi_j^{(k)} &= \sum_{i=1}^M p(X_k = j | X_{k-1} = i) p(X_{k-1} = i) \\ &= \sum_{i=1}^M p_{ij} \pi_i^{(k-1)} = \vec{\pi}^{(k)} \\ &= \vec{\pi}^{(k-1)} \cdot p = \dots = \vec{\pi}^{(0)} \cdot p^n\end{aligned}$$

where

$$\begin{aligned}&p_{1j} \pi_1^{(k-1)} + p_{2j} \pi_2^{(k-1)} + \dots + p_{Mj} \pi_M^{(k-1)} \\ &= (\pi_1^{(k-1)}, \pi_2^{(k-1)}, \dots, \pi_M^{(k-1)}) = \begin{pmatrix} p_{1j} \\ p_{2j} \\ \vdots \\ p_{Mj} \end{pmatrix}\end{aligned}$$

and

$$\begin{aligned}&(\pi_1^{(k)}, \pi_2^{(k)}, \dots, \pi_M^{(k)}) \\ &= (\pi_1^{(k-1)}, \pi_2^{(k-1)}, \dots, \pi_M^{(k-1)}) = \begin{pmatrix} p_{11} & \dots & p_{1M} \\ \vdots & \ddots & \vdots \\ p_{M1} & \dots & p_{MM} \end{pmatrix} \\ &\pi^{(k)} = \pi^{(k-1)} \cdot p\end{aligned}$$

and $\vec{\pi}^* \cdot p = \vec{\pi}^*$ is the stationary distribution.

$i \rightarrow j$ is accessibility if there exists a walk or path from i to j . If ij then i and j communicate.

41.3.3 Equivalence classes and relations

Y-set with equivalence relation $\sim$ has 3 properties

1. $a \sim a, a \in Y$ is reflexivity
2. $a \sim b \Rightarrow b \sim a, \forall a, b \in Y$ is symmetry
3. $a \sim b, b \sim c \Rightarrow a \sim c, \forall a, b \in Y$ is transitivity

Let B_i be equivalence classes. Then,

$$\forall j \in B_i, \forall k \in S \begin{cases} k \in B_i & k \leftrightarrow j \\ k \notin B_i & k \not\leftrightarrow j \end{cases}$$

Theorem. All states are either recurrent or transient equivalence classes. i is recurrent $\forall j : i \rightarrow j \Rightarrow j \rightarrow i$ (self-contained Markov chain). i is transient if $\exists j : i \rightarrow j, j \not\rightarrow i$ (a node with arrows away but no arrows towards; indicated by a column of 0's in a transition matrix). Proof by contradiction. Let k be a transient state. Then, $\exists j : k \rightarrow j, j \not\rightarrow k$. Let $i, k \in 1$ equivalence class s.t. $i \rightarrow k$. Since $k \rightarrow j$ from above, $i \rightarrow k \rightarrow j$. If it were the case that $j \not\rightarrow i$ then $j \rightarrow i \rightarrow k$ which is a contradiction.

41.3.4 Aperiodicity

Aperiodicity allows the Markov chain to move around instead of getting stuck somewhere.

The period of a state (node) i in a Markov chain is $\gcd(n : p_{ii}(n) \neq 0) = d(i)$, where ii denotes paths returning to the same state which cannot be 0. If $d(i) = 1$ then i is aperiodic.

Theorem. All elements in one equivalence class have the same period. So transients have the same period and recurrences have the same period. This happens because equivalence classes are connected to each other and self-contained.

41.3.5 Ergodic theorem

Ergodicity intuition. A process in which every sequence or sizable sample is equally representative of the whole. The average outcome of the group is the same as the average outcome of the individual over time. The time and ensemble (i.e. states) averages are the same. Trivial sampling is ergodic, and AR(1) processes (weak dependence, so forgets initial conditions after some time) are ergodic after transient states. Strongly path-dependent processes like Polya's urn are non-ergodic processes.

Birchhoff's equality:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(\omega(t)) dt = \int_{\omega} f(\omega) P(\omega) d\omega$$

This is what allows studying many atoms emitting radioactivity to be used to determine the half-life of a single atom. It equates states with time. If however a system is non-ergodic, then cannot use many atoms to determine the half-life of any single atom. The situation is equivalent to whether an 80 percent effective vaccine is 80 percent effective for 100 percent of the population (ergodic) or 100 percent effective for 80 percent of the population (non-ergodic).

Ergodic Markov chains are (i) limited to one equivalence class, (ii) are recurrent, and (iii) aperiodic ($d(i) = 1$). A Markov chain is ergodic iff $\exists m \in \mathbb{N}$ s.t. $p_{ij}(m) \neq 0, \forall i, j \in S$. Further, $\forall m \geq (M-1)^2 + 1$.

Ergodic theorem. Let X_t be an ergodic Markov chain. Then

$$\exists \lim_{n \rightarrow \infty} p_{ij}(n) = \pi_j^* > 0$$

recalling that $p(X_k = j) = \pi_j^{(k)}$ (the probability distribution that the k -th sequence of the Markov chain is at state j). Note that

$$\sum_{j=1}^n \pi_j^* = 1, \bar{\pi}^* = (\pi_1^*, \dots, \pi_M^*)$$

Corollaries.

1. $\bar{\pi}^*$ is a stationary distribution, i.e. $\bar{\pi}^* p = \bar{\pi}^*$.
2. $\lim_{n \rightarrow \infty} p(X_n = j) = \pi_j^*$.

Proof of (1). Let $i = 1 \dots M$. Then,

$$(\bar{\pi}^* p)_i = \sum_{j=1}^M \pi_j^* p_{ji} = \sum_{j=1}^M \lim_{n \rightarrow \infty} p_{kj}(n) p_{ji}$$

(which could be expressed in vector dot-product form). The second equality is from the theorem. Then,

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \sum_{j=1}^M p_{kj}(n) p_{ji} = \lim_{n \rightarrow \infty} \sum_{j=1}^M p^{(n)} p = \lim_{n \rightarrow \infty} \sum_{j=1}^M p^{(n+1)} \\ &= \lim_{n \rightarrow \infty} \sum_{j=1}^M (p^{(n+1)})_{ki} = \lim_{n \rightarrow \infty} \sum_{j=1}^M p_{ki}(n+1) = \lim_{n \rightarrow \infty} \sum_{j=1}^M p_{ki}(n) = \pi_i^* \end{aligned}$$

where an element ki is located in the transition matrix on the k -th row and the i -th column.

Proof of (2).

$$\begin{aligned} \lim_{n \rightarrow \infty} \pi_j^{(n)} &= \lim_{n \rightarrow \infty} \sum_{k=1}^M \pi_k^{(0)} p_{kj}^{(n)} \\ &= \sum_{k=1}^M \pi_k^{(0)} \lim_{n \rightarrow \infty} p_{kj}^{(n)} \\ &= \pi_j^* \sum_{k=1}^M \pi_k^{(0)} = \pi_j^* \end{aligned}$$

Example. For the Markov chain with transition matrix

$$p = \begin{pmatrix} 0.2 & 0.8 \\ 0.6 & 0.4 \end{pmatrix}$$

Let $\pi^* = (a, b)$ and solve $\pi^* p = \pi^* \Rightarrow a = \frac{3}{7}, b = \frac{4}{7}$. So, $p(X_n = 1) \rightarrow \frac{3}{7}, p(X_n = 2) \rightarrow \frac{4}{7}$.

41.4 Gaussian processes

Recall for $\mathcal{N}(\mu, \sigma^2)$, the PDF is

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

and degenerate case

$$p(X \stackrel{as}{=} \mu) = 1 \Rightarrow \sigma = 0$$

and its characteristic function is

$$\Phi(u) = e^{iu\mu - \frac{1}{2}\mu^2\sigma^2}$$

Definition. A *Gaussian vector* (for most or all purposes, a multivariate normal distribution). A random variable $\vec{X} = (X_1, \dots, X_n)$ is Gaussian (i.e. jointly normal) iff

$$\sum_{k=1}^n \lambda_k X_k \sim \mathcal{N}, \forall \lambda_i$$

Theorem. $\vec{X}$ is Gaussian if any of the following conditions hold.

1. $\Phi_{\vec{X}}(\vec{u}) = E(e^{i\langle \vec{u}, \vec{X} \rangle}) = \exp\left(i\langle \vec{u}, \vec{\mu} \rangle - \frac{1}{2}\vec{u}' C \vec{u}\right)$.
2. $\vec{X} = \vec{X}^0 + \vec{\mu}$.

where $\vec{X}^0$ is the standard normal vector, $\vec{\mu} = (E(X_1), \dots, E(X_n))$, and C is the covariance matrix (and so symmetric and positive semi-definite).

$$C = Cov = (c_{jk})_{j,k=1}^n, c_{jk} = Cov(X_j, X_k)$$

It is PSD because

$$\sum_{k,j=1}^n u_k c_{kj} u_j \geq 0 \Rightarrow \vec{u}' C \vec{u} \geq 0$$

$$\sum_{k,j=1}^n u_k Cov(x_j, x_k) u_j = cov \left(\sum_j u_j x_j, \sum_k u_k x_k \right) = Var \left(\sum_j u_j x_j \right) \geq 0$$

Further, $A = C^{1/2} \Rightarrow AA = C$. With diagonalization, $C = U' diag(d_1, \dots, d_n) U$ with $A = U' diag(\sqrt{d_1}, \dots, \sqrt{d_n}) U$.

Proof (*Def.* $\Leftrightarrow$ (i) $\Leftrightarrow$ (ii)). Start with *Def* $\Leftrightarrow$ (i). Let $\xi = \langle \vec{u}, \vec{X} \rangle \sim \mathcal{N}$. Then

$$\begin{aligned}\Phi_{\vec{X}}(\vec{u}) &= E(e^{i\langle \vec{u}, \vec{X} \rangle}) = E(e^{i\xi \cdot 1}) = E(e^{i\langle 1, \xi \rangle}) = \Phi_{\xi}(1) \\ &= e^{i\langle 1, \mu_{\xi} \rangle - (1/2)\sigma_{\xi}^2} = e^{i\langle 1, \mu_{\xi} \rangle - (1/2)(1)\sigma_{\xi}^2(1)}\end{aligned}$$

because $u \rightarrow 1, X \rightarrow \xi$, and μ_{ξ}, σ_{ξ} are needed to calculate the above:

$$\begin{aligned}\mu_{\xi} &= E\left(\sum_{k=1}^n u_k X_k\right) = \sum_{k=1}^n u_k \mu_k = \langle \vec{u}, \vec{\mu} \rangle \\ \sigma_{\xi}^2 &= \text{cov}\left(\sum_{k=1}^n u_k X_k, \sum_{j=1}^n u_j X_j\right) = \sum_k^n \sum_j^n u_k \text{Cov}(X_k, X_j) u_j = \vec{u}' C \vec{u}\end{aligned}$$

For (i) $\Rightarrow$ *Def.*, there's a 1-1 correspondence between the PDF and characteristic function. For (ii) $\Rightarrow$ (i), $\vec{X}^0$ is Gaussian.

$$\Phi_{\vec{X}^0}(\vec{u}) = \exp\left(-\frac{1}{2}\vec{u}'\vec{u}\right)$$

$$\begin{aligned}\Phi_{\vec{X}}(\vec{u}) &= E(e^{i\langle \vec{u}, A\vec{X}^0 + \mu \rangle}) = e^{i\langle \vec{u}, \vec{\mu} \rangle} \Phi_{\vec{X}^0}(A'\vec{u}) \\ &= e^{i\langle \vec{u}, \vec{\mu} \rangle} e^{-\frac{1}{2}\vec{u} A A' \vec{u}} = e^{i\langle \vec{u}, \vec{\mu} \rangle} e^{-\frac{1}{2}\vec{u} C \vec{u}}\end{aligned}$$

For (i) $\Rightarrow$ (ii), $A = C^{1/2}$.

41.4.1 Correlation and independence

Theorem. Let $X_1, X_2 \sim \mathcal{N}(0, 1)$ and $\text{Corr}(X_1, X_2) = 0$. Then $X_1 \perp\!\!\!\perp X_2$ iff (X_1, X_2) is a Gaussian vector (multivariate normal).

Proof. Let $\lambda_1 X_1 + \lambda_2 X_2 \sim \mathcal{N} \Rightarrow (X_1, X_2)$ is Gaussian (multivariate normal).

$$C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = C^{1/2}$$

From (ii) above,

$$\begin{aligned}\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} X_1^0 \\ X_2^0 \end{pmatrix}}_{\text{std. N vector}} + \underbrace{\begin{pmatrix} 0 \\ 0 \end{pmatrix}}_{\mu} \\ &\stackrel{d}{=} \begin{pmatrix} X_1^0 \\ X_2^0 \end{pmatrix}\end{aligned}$$

Since the standard normal vector is independent, i.e $X_1^0 \perp\!\!\!\perp X_2^0 \Rightarrow X_1 \perp\!\!\!\perp X_2$.

41.4.2 Gaussian processes

A collection of random variables is a Gaussian process if the joint distribution (multivariate normal) of its members is Gaussian. So the mean and variance remain constant throughout time. Mean is the same, with affine translation depending on where the process begins.

A Gaussian process is where X_t is a stochastic process, where $X_{t_1}, \dots, X_{t_n}$ is a Gaussian vector, and (1) expectation $m(t) = E(X_t)$ and (2) covariance function $K(t, s) = Cov(X_t, X_s)$. K is PSD and symmetric in argument $K(t, s) = K(s, t)$. It is PSD because for any $t_i, u_i \in \mathbb{R}$,

$$\sum_k^n \sum_j^n u_k u_j K(t_k, t_j) \geq 0$$

$$Cov \left(\sum_k^n u_k X_{t_k}, \sum_j^n u_j X_{t_j} \right) = Var \left(\sum_k^n u_k X_{t_k} \right) \geq 0$$

Gaussian random variables are specified by μ, σ , Gaussian vectors (MVN) are specified by $\vec{\mu}, Cov$, and Gaussian processes are specified by m and K .

Theorem. If K is symmetric and PSD, there exists a Gaussian process X_t s.t. $E(X_t) = m(t), Cov(X_t, X_s) = K(t, s)$.

Example. $K(t, s) = \min(t, s) \Rightarrow PSD$.

$$\sum_{j,k}^n u_j u_k \min(t_j, t_k) \geq 0$$

where u are random $\mathbb{R}$. Let $f_t(x) = \mathbb{1}_{x \in [0, t]}$, then

$$\int_0^\infty f_t(x) f_s(x) dx = \min(t, s)$$

where $f_t(x) f_s(x) = 1$ iff $x \in [0, t] \cap [0, s]$ (which fulfills the min function). Then,

$$\begin{aligned} \sum_{j,k}^n u_j u_k \min(t_j, t_k) &= \sum_{j,k}^n u_j u_k \int_{\mathbb{R}^+} f_{t_j}(x) f_{t_k}(x) dx \\ &= \int_{\mathbb{R}^+} \sum_{j,k}^n u_j u_k f_{t_j}(x) f_{t_k}(x) dx \\ &= \int_{\mathbb{R}^+} \sum_j^n u_j f_{t_j}(x) \sum_k^n u_k f_{t_k}(x) dx \\ &= \int_{\mathbb{R}^+} \left(\sum_k^n u_k f_{t_k}(x) \right)^2 dx \geq 0 \end{aligned}$$

Gaussian processes may or may not be stationary, depending on the species.

41.5 Brownian motion

Basic model for the cumulative effect of pure noise. If $B(t)$ denotes the position of a particle at time t , then the displacement $B(t) - B(0)$ is the effect of the purely random bombardment by the molecules of the fluid, or the effect of noise over time t .

Random walk is a discrete-space (integers) and discrete-time model, while Brownian Motion is a continuous-space and continuous-time model.

Synonymous with the Wiener process so $B_t = W_t$ (formally, Brownian motion is the physical phenomenon while the Wiener process is the mathematical description). Two equivalent definitions.

Definition 1. B_t is a Gaussian process if (1) $m(t) = 0$ and (2) $K(t, s) = \min(t, s)$.

Definition 2.

1. $B_0 = 0$ almost surely
2. B_t independent increments (this means: $B(t+s) = B(s) + (B(t+s) - B(s))$ so the additional increment is independent of the starting position.
3. $B_t - B_s \sim \mathcal{N}(0, t - s), t > s \geq 0$

Definitions are equivalent. Show Definition 1 leads to Definition 2. For $B_0 = 0$:

$$\begin{aligned} E(B_0) &= m(0) = 0 \\ \text{Var}(B_0) &= K(t, s) = K(0, 0) = 0 \\ B_t &\stackrel{as}{=} 0 \end{aligned}$$

For independent increments. Let $a < b < c < d$. Then increments (a, b) and (c, d) are uncorrelated:

$$\begin{aligned} & \text{Cov}(B_b - B_a, B_d - B_c) \\ &= \underbrace{\text{Cov}(B_b, B_d)}_b - \underbrace{\text{Cov}(B_a, B_d)}_a - \underbrace{\text{Cov}(B_b, B_c)}_b + \underbrace{\text{Cov}(B_a, B_c)}_a = 0 \end{aligned}$$

Further, the linear combination is Gaussian (so it's a Gaussian vector, or MVN):

$$\begin{aligned} & \lambda_1(B_b - B_a) + \lambda_2(B_d - B_c) \\ &= \lambda_1 B_b - \lambda_1 B_a + \lambda_2 B_d - \lambda_2 B_c \sim \mathcal{N} \end{aligned}$$

since each individual term is Gaussian. Since the increments are uncorrelated and Gaussian, they're independent.

For $B_t - B_s \sim \mathcal{N}$, since (B_t, B_s) is a Gaussian vector so a linear combination like $B_t - B_s \sim \mathcal{N}$. Further,

$$E(B_t - B_s) = E(B_t) - E(B_s) = m(t) - m(s) = 0$$

$$\begin{aligned}
\text{Var}(B_t - B_s) &= \text{Cov}(B_t - B_s, B_t - B_s) \\
&= \text{Var}(B_t) - 2\text{Cov}(B_t, B_s) + \text{Var}(B_s) \\
&= t - 2s + s = t - s
\end{aligned}$$

Now show Definition 2 leads to Definition 1. Let $t_1 < t_2 \dots < t_n$. Go backward in time:

$$\begin{aligned}
\sum_{k=1}^n \lambda_k B_{t_k} &= \lambda_n (B_{t_n} - B_{t_{n-1}}) + (\lambda_n + \lambda_{n-1}) B_{t_{n-1}} + \sum_{k=1}^{n-2} \lambda_k B_{t_k} \\
&= \sum_{k=1}^n d_k (B_{t_n} - B_{t_{n-1}}) \\
t_0 &= 0
\end{aligned}$$

Since the linear combination of increments (each Gaussian) so the sum is normal. So $B_{t_1}, \dots, B_{t_n}$ is a Gaussian vector. So B_t is a Gaussian process. Next, with $B_t \sim \mathcal{N}(0, t)$,

$$m(t) = E(B_t) = 0$$

$$\begin{aligned}
K(t, s) &= \text{Cov}(B_t, B_s) \\
&= \text{Cov}(B_t - B_s + B_s, B_s) \\
&= \text{Cov}(B_t - B_s, B_s - B_0 (= 0)) + \text{Cov}(B_s, B_s) \\
&= 0 + s = s
\end{aligned}$$

41.5.1 Arithmetic versus geometric

Arithmetic random walks are modeled as sums of random terms while geometric random walks are modeled as products of random factors.

Two examples are Brownian Motion and Geometric Brownian Motion. Brownian Motion has independent, identically distributed increments while the geometric version has independent, identically distributed ratios between successive factors.

41.5.2 Reflection principle

Let $B(t)$ be a Brownian motion started at x , and $\hat{B}(t) = -B(t)$. Then $\hat{B}(t)$ is a Brownian motion started at $-x$. The Brownian motion is reflected about the horizontal axis.

More generally, reflect about the horizontal axis passing through $(T, B(T))$. This is also a Brownian motion.

For $t \geq T$ the reflected path is

$$\hat{B}(t) - B(T) = -(B(t) - B(T))$$

so

$$\hat{B}(t) = 2B(T) - B(t)$$

Reflection principle. $\hat{B}(t)$ is a Brownian motion.

41.5.3 Joint distribution of Brownian motion with maximum

The joint distribution of $(B(t), M(t))$ has the density

$$f_{B,M}(a, b) = \sqrt{\frac{2}{\pi}} \frac{(2b-a)}{t^{3/2}} e^{-\frac{(2b-a)^2}{2t}}$$

Proof. Let for y -values $b > 0, b > a$ $\hat{B}(t)$ be $B(t)$ reflected at the vertical value $y = T_b$ (the time at which $B(t)$ reaches $y = b$). Then

$$\begin{aligned} P(B(t) \leq a, M(t) \geq b) &= P(T_b \leq t, B(t) \leq a) \text{ since } (M(t) \geq b) \rightarrow (T_b \leq t) \\ &= P(T_b \leq t, \hat{B}(t) \geq 2b - a) \text{ since } \hat{B}(t) = 2b - B(t) \\ &= P(T_b \leq t, B(t) \geq 2b - a) \text{ since } T_b \text{ is the same for } B \text{ and } \hat{B} \\ &= P(B(t) \geq 2b - a) \text{ since } b - a > 0 \rightarrow 2b - a > 0 \\ &= 1 - P(B(t) \leq 2b - a) \\ &= 1 - \Phi\left(\frac{2b-a}{\sqrt{t}}\right) \end{aligned}$$

The first step says if the maximum is above the Brownian motion value at t , then the maximum must have been reached before t , i.e. $T_b \leq t$. The last step follows from Brownian motion being normally distributed. Recall

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

(which integral is intractable). In this case,

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi t}} e^{-z^2/2t} dz$$

where $z = 2b - a$. Since $f_{B,M}(a, b)$ is a joint distribution, need to differentiate the above twice, once by a and the second time by b :

$$\frac{\partial^2}{\partial z^2} \left(1 - \int_{-\infty}^z \frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)} dz \right) \frac{\partial z}{\partial a} \frac{\partial z}{\partial b}$$

So compute step by step:

$$\frac{\partial}{\partial z} \left(1 - \int_{-\infty}^z \frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)} dz \right) \frac{\partial z}{\partial a} = -\frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)} \underbrace{\frac{\partial z}{\partial a}}_{-1} = \frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)}$$

$$\frac{\partial}{\partial z} \left(\frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)} dz \right) \frac{\partial z}{\partial b} = -\frac{1}{\sqrt{2\pi t}} \frac{2 \cdot 2(2b-a)}{2t} e^{-\frac{(2b-a)^2}{2t}} = \sqrt{\frac{2}{\pi}} \frac{(2b-a)}{t^{3/2}} e^{-\frac{(2b-a)^2}{2t}}$$

41.5.4 Brownian motion as the limit of a random walk

Brownian motion can be derived from a random walk. If $t = 0, 1, 2, \dots$ and

$$X = \begin{cases} 1 & p = 0.5 \\ -1 & p = 0.5 \end{cases}$$

Then,

$$E(X) = 0, \text{Var}(X) = 1$$

Scale the random walk infinitesimally small, which then approaches Brownian motion $W(t)$ with

$$E(W(t)) = \sum E(X_t) = 0$$

$$\text{Var}(W(t)) = \sum \text{Var}(X_t) = t \text{Var}(X_t) = t \cdot 1 = t$$

Note. The random walk is to the binomial distribution what the normal distribution is to Gaussian distribution and Brownian motion.

41.5.5 Brownian motion as a Gaussian process

Review. Let $X(t)$ be a process with independent Gaussian increments, that is, for any $s < t$, $X(t) - X(s)$ has a normal distribution, and is independent of the values $X(u)$, $u \leq s$ (the σ -field $\mathcal{F}_s$ generated by the process up to time s). Then $X(t)$ is a Gaussian process.

Theorem. A Brownian motion started at 0 is a Gaussian process with 0-mean function, and covariance function $\min(t, s)$. Conversely, a Gaussian process with 0-mean function, and covariance function $\min(t, s)$ is a Brownian motion.

41.5.6 Kolmogorov continuity theorem

Definition. X_t, y_t are *stochastically equivalent* if

$$p(X_t = y_t) = 1, \forall t \geq 0$$

Note that stochastically equivalent random variables can have very different properties. Example. $X_t = 0, \forall t \in [0, 1]$ and $y_t = \mathbb{1}(\tau = t), \tau \sim \text{Unif}(0, 1)$.

$$p(X_t = y_t) = p(0 = y_t) = p(t \neq \tau) = 1$$

because y_t will be 1 at exactly one point over the domain $[0, 1]$ and 0 everywhere else so will be discontinuous and therefore different from X_t which will be 0 everywhere.

Theorem. There exists $C, \alpha, \beta > 0$ if

$$E(|X_t - X_s|^\alpha) \leq C|t - s|^{1+\beta}$$

$\forall t, s \in [a, b]$. Then there exists y_t stochastically equivalent to X_t s.t. y_t has continuous trajectories. The Kolmogorov continuity theorem guarantees that a stochastic process that satisfies certain constraints on the moments of its increments will be continuous.

Example.

$$E(|B_t - B_s|^4) = (t - s)^2 E(\xi^4) = 3(t - s)^2$$

where $B_t - B_s \sim \mathcal{N}(0, t - s)$ and $B_t - B_s \stackrel{d}{=} \sqrt{t - s} \xi \sim (0, 1)$, and $E(\xi^4) = 3$. So $C = 3, \alpha = 4, \beta = 2$.

41.5.7 Quadratic variation

Variation is the structure of the codomain of a function or a measure. For a real-valued function, the total variation of a function on the interval is a measure of the one-dimensional arclength of the curve.

Divide time into partitions $t_0 < t_1 < \dots < t_n$. Can vary the number of partitions, e.g. $n = 10, n = 500$. The quadratic variation is finite:

$$S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n (B_{t_k} - B_{t_{k-1}})^2 = t$$

where $n \rightarrow \infty$ means $\max |t_i - t_{i-1}| \rightarrow 0$. Intuition:

$$\begin{aligned} E(S_n) &= E \sum_i |B(t_i^n) - B(t_{i-1}^n)|^2 \\ &= E \sum_i (B(t_i^n) - B(t_{i-1}^n))^2 \\ &= \text{Var}(B(t_i^n) - B(t_{i-1}^n)) \text{ with } E(B_i^n) = E(B_{i-1}^n) = 0 \\ &= \sum_i (t_i^n - t_{i-1}^n) \\ &= t - 0 = t \end{aligned}$$

Note the above implies

$$\lim_{n \rightarrow \infty} E(S_n - t)^2 = 0$$

The above is also used in the proof of

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n |B_{t_k} - B_{t_{k-1}}| = \infty$$

41.5.8 Continuity and differentiability

B_t is everywhere (stochastically) continuous but nowhere differentiable.

$$\lim_{h \rightarrow 0} B_{t+h} \xrightarrow{p} B_t, \forall t \geq 0$$

Note 1. Weierstrass function is a famous example of nowhere differentiable function, formulated as a Fourier series:

$$f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

which is a fractal curve, $0 < a < 1, ab > 1 + \frac{3}{2}\pi$.

Note 2. Stochastic continuity is

$$\lim_{t \rightarrow s} p(|X_t - X_s| > a) = 0$$

$\forall a > 0, \forall s \geq 0$.

41.5.9 Law of iterated logarithms

The (strong) LLN denominator is too high:

$$\lim_{t \rightarrow \infty} \frac{B_t}{t} \stackrel{as}{=} 0$$

The CLT denominator is too low:

$$\lim_{t \rightarrow \infty} \frac{B_t}{\sqrt{t}} \stackrel{as}{=} \infty$$

Law of iterated logarithms' denominator is 'just right':

$$\lim_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \cdot \log(\log t)}} = 1$$

Intuitively, $\frac{1}{\sqrt{2\pi \ln(n)}}$ 'mimics' the normal PDF. So equate the two:

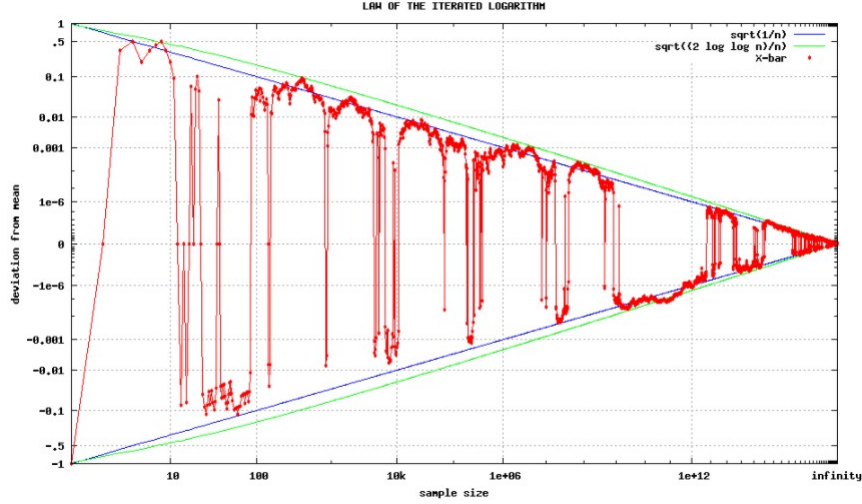
$$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = \frac{1}{\sqrt{2\pi \ln(n)}}$$

$$e^{-\frac{x^2}{2}} = \frac{1}{\ln(n)}$$

$$\ln(n) = e^{\frac{x^2}{2}}$$

$$\ln(\ln(n)) = \frac{x^2}{2}$$

$$\sqrt{2\ln(\ln(n))} = x$$



41.5.10 Properties of Brownian paths

$B(t)$ has these properties:

1. A continuous function of t
2. Not monotone in any interval, no matter how small the interval
3. Not differentiable at any point
4. Has infinite variation on any interval, no matter how small
5. Has quadratic variation on $[0, t]$ equal to t , for any t

Properties (1) and (3) of Brownian motion paths state that although any realization $B(t)$ is a continuous function of t , it has increments $\Delta B(t)$ over an interval of length Δt much larger than Δt as $\Delta \rightarrow 0$. Since $E(B(t + \Delta t) - B(t))^2 = \Delta t$, it suggests the increment is roughly like $\sqrt{\Delta t}$, which is made precise by property (5). If a continuous function has finite variation then its quadratic variation is zero. This implies a positive quadratic variation implies infinite variation so (4) flows from (5). Since a monotone function has finite variation, (2) follows from (4).

$B(t)$ is both a martingale and Markovian. A martingale is when the current expectation is the future expectation (as the forecast is based on current information):

$$E(g(B(t + s) - B(t)) | \mathcal{F}_t) = E(g(B(t + s) - B(t)))$$

$B(t)$ is a martingale because

$$\begin{aligned} E(B(t + s) | \mathcal{F}_t) &= E(B(t) + (B(t + s) - B(t)) | \mathcal{F}_t) \\ &= E(B(t) | \mathcal{F}_t) + E(B(t + s) - B(t) | \mathcal{F}_t) \\ &= B(t) + E(B(t + s) - B(t)) = B(t) \end{aligned}$$

The Markov property is 'memorylessness' (like a discrete probability analog of the martingale):

$$P(X(t+s) \leq y | \mathcal{F}_t) = P(X(t+s) \leq y | X(t))$$

$B(t)$ is Markovian. Proof uses generating functions.

41.6 Noise

41.6.1 White noise

White noise is the derivative of Brownian motion

$$WN = \frac{dB(t)}{dt} = B'(t)$$

It doesn't exist as a function because $B(t)$ is nowhere differentiable. It is called white noise because its Fourier transform gives all frequencies in its spectrum.

41.6.2 Gaussian noise

Statistical noise having a probability density function (PDF) equal to that of the normal distribution. The values that the noise can take on are Gaussian-distributed.

41.6.3 Gaussian white noise

White Gaussian noise. Values at any pair of times are iid. It is often incorrectly assumed that Gaussian noise (i.e., noise with a Gaussian amplitude distribution – see normal distribution) necessarily refers to white noise, yet neither property implies the other.

41.7 Random walk

The analogue of $B(t)$ in discrete time. However, there is no 'binomial or multinomial' process analogous to Gaussian processes because binomial and multinomial distributions are not stable distributions (because their domain of supports are finite). $B(t)$ can be constructed as the limit of random walks when step sizes get smaller and smaller.

$$\begin{aligned} S_0 &= 0 \\ S_n &= S_{n-1} + \xi_n \end{aligned}$$

where

$$\xi_i \sim iid \begin{cases} 1 & p \\ -1 & 1-p \end{cases}$$

41.8 Stationarity

41.8.1 Definitions

Strict stationarity definition. X_t is strictly stationary if

$$(X_{t_1+h}, \dots, X_{t_n+h}) \stackrel{d}{=} (X_{t_1}, \dots, X_{t_n})$$

Weak stationarity definition. X_t is weakly (second-order, wide-sense, covariance) stationary if

1. $m(t) = E(X_t) = \text{constant}$
2. $K(t, s) = \text{Cov}(X_t, X_s) = K(t+h, s+h)$

and the autocovariance function (ACF) is $\gamma : K(t, s) = \gamma(t-s)$. ACF properties:

1. $\gamma(0) = \text{Var}(X_t) \geq 0$
2. $|\gamma(t)| \leq \gamma(0)$. Proof uses Cauchy-Schwartz: $|\text{Cov}(X_t, X_0)| \leq \sqrt{\text{Var}(X_t)}\sqrt{\text{Var}(X_0)} = \sqrt{\gamma(0)}\sqrt{\gamma(0)} = \gamma(0)$.
3. γ is even: $\gamma(t) = \text{Cov}(X_t, X_0) = \text{Cov}(X_0, X - t) = \gamma(-t)$.

Notes

1. Strict stationarity implies weak stationarity
2. If $X_t \sim \mathcal{N}$ then strict stationarity $\leftrightarrow$ weak stationarity

41.8.2 Examples

White noise processes. $WN(0, \sigma^2)$.

1. $E(X_t) = 0, \text{Var}(X_t) = \sigma^2$.
2. $K(t, s) = 0, t \neq s$. $K(t, s) = \sigma^2 \mathbb{1}(t = s) = \gamma(t - s)$. $\gamma(x) = \sigma^2 \mathbb{1}(x = 0)$.

So it's weakly stationary. It would be strictly stationary if (list not exhaustive)

1. $X_i \sim iid$ (iid noise)
2. $X_t \sim \mathcal{N}$

Random walk.

$$\begin{aligned} S_0 &= 0 \\ S_n &= S_{n-1} + \xi_n \end{aligned}$$

where

$$\xi_i \sim iid \begin{cases} 1 & p \\ -1 & 1-p \end{cases}$$

$$S_n = \xi_1 + \xi_2 + \dots$$

$E(S_n) = nE(\xi_1) = n((1)p + (-1)(1-p)) = n(2p-1)$ and if $p \neq 1/2$ then $E(S_n) \propto n$ so not weakly stationary. So concentrate on case $p = 1/2$, and

$$\begin{aligned} K(n, m) &= \text{Cov}(S_m + \xi_{m+1} + \dots + \xi_n, S_m), n > m \\ &= \text{Cov}(S_m, S - m) + \text{Cov}(\underbrace{\xi_{m+1} + \dots + \xi_n}_{\text{Forward in time from } S_m}, S_m) \\ &= \text{Cov}(S_m, S - m) + 0 \\ &= \text{Var}(S_m) = m\text{Var}(\xi_1) \\ &= \min(m, n)\text{Var}(\xi_1) \end{aligned}$$

where the last line follows as $m < n$ and only the overlapping value is used (i.e. the minimum). Since $K(n, m)$ is not in closed form (i.e. not equal to $\text{Cov}(X_m, X_n)$), not weakly stationary. So random walks aren't stationary.

Brownian motion. $E(B_t) = 0, B_t - B_s \sim \mathcal{N}(0, t-s)$. With $s = 0, B_t \sim \mathcal{N}(0, t) \Rightarrow \text{Var}(B_t) = t$, so not weakly stationary.

Moving average, MA(q).

$$Y_t = X_t + a_1 X_{t-1} + \dots + a_q X_{t-q}$$

with

$$X_t \sim WN(0, \sigma^2)$$

Then,

$$\begin{aligned} E(Y_t) &= 0 \\ K(t, s) &= \text{Cov}\left(\sum_j^q a_j X_{t-j}, \sum_k^q a_k X_{s-k}\right) \\ &= \sum_j^q \sum_k^q a_j a_k \sigma^2 \mathbb{1}(t-s=j-k) \end{aligned}$$

ACF can be expressed in closed-form. MA is weakly stationary.

Autoregressive model, AR(p).

$$Y_t = b_1 Y_{t-1} + \dots + b_p Y_{t-p} + \xi_t$$

$$\xi \sim WN(0, \sigma^2)$$

$$\text{Cov}(\xi_t, Y_s) = 0, t > s$$

For AR(1)

$$Y_t - bY_{t-1} = \xi_t$$

$$Y_t = \sum_j^{\infty} b^j \xi_{t-j}$$

$$K(t, s) = \sum_{j,k}^{\infty} b^{j+k} \sigma^2 \mathbb{1}(t-s = j-k)$$

$$K(t, t) = \sum_k^{\infty} b^{2k} < \infty \Rightarrow |b| < 1$$

Then AR(1) is stationary.

41.8.3 Spectral density

Bochner-Khintchine theorem. If $\Phi(u) : \mathbb{R} \rightarrow \mathbb{C}$ is a characteristic function, then

1. Φ is continuous
2. Φ is PSD, $\sum_{j,k=1}^n z_j \bar{z}_k \Phi(u_j - u_k) \geq 0, \forall z_i \in \mathbb{C}, \forall u_i \in \mathbb{R}$.
3. $\Phi(0) = 1$

where

$$\Phi(u) = E(e^{iu\xi})$$

If (1)+(2), then there exists a measure $\mu : \Phi(u) = \int e^{iux} \mu(dx)$.

If (1)+(2) and $\int |\Phi(u)| du < \infty \Rightarrow \Phi(u) = \int e^{iux} s(x) dx$.

Assuming X_t is weakly stationary, $K(t, s) = \gamma(t - s)$ and if γ is continuous and $\int |\gamma(u)| du < \infty$ (finite) then by Bochner-Khintchine, there exists

$$g(x) : \gamma(u) = \int e^{iux} g(x) dx = \mathcal{F}[g](u)$$

which is a Fourier transform. To back out $g(x)$, use the inverse Fourier transform

$$g(x) = \frac{1}{2\pi} \int e^{-iux} \gamma(u) du$$

which is the *spectral density* of a weakly stationary process in continuous time. It is a decomposition of the ACF. The discrete time analog is

$$\frac{1}{2\pi} \sum_{-\infty}^{\infty} e^{-ihx} \gamma(h)$$

Example 1. White noise $WN(0, \sigma^2)$. Then, $\gamma(u) = \sigma^2 \mathbb{1}(u = 0)$ and the inverse FT gives $g(s) = \frac{\sigma^2}{2\pi}$. Example 2. MA(1).

$$\gamma(u) = \begin{cases} 0 & |u| > 0 \\ a\sigma^2 & |u| = 1 \\ (1 + a^2)\sigma^2 & u = 0 \end{cases}$$

$$g(x) = \frac{\sigma^2}{2\pi}(1 + a^2 + a2\cos(x))$$

Proposition. A real-valued function $g(x) \in [-\pi, \pi]$ is a spectral density of a stochastic density of a stochastic process X_t means

1. $g(x) \geq 0$
2. g is even
3. $\int_{-\infty}^{\infty} g(x)dx < \infty$

41.8.4 Stochastic integration

$$\int_a^b X_t dt = \lim \sum_{k=1}^k X_{t_{k-1}}(t_k - t_{k-1})$$

where the limit is s.t. $\max_i |t_i - t_{i-1}| \rightarrow 0$.

$$E \left(\sum - \int_a^b X_t dt \right)^2 \rightarrow 0$$

Theorem. Let $X_t : E(X_t^2) < \infty$, with $m(t), K(t, s)$ being continuous functions. Then, there exists

$$\int_a^b X_t dt$$

$K(t, s), \forall(t_0, s_0)$ is continuous iff $K(t, s), \forall(t_0, t_0)$ is continuous (i.e. the diagonal). If $K(t, s), \forall(t_0, t_0)$ is continuous then X_t is continuous at t_0 in the mean-squared sense: $E(X_t - X_{t_0}) \rightarrow 0$ as $t \rightarrow t_0$. Then X_t is continuous in t_0, s_0 , so $K(t_0, s_0)$ is continuous.

Properties

1. $E \left(\int X_t dt \right) = \int E(X_t) dt$ by Fubini (since E is also an integral, this is actually a double integral)
2. $E \left(\left(\int X_t dt \right)^2 \right) = \int \int E(X_t X_s) dt ds = \int \int X_t X_s dt ds$
3. $Var \int X_t dt = \int_a^b \int_a^b K(t, s) dt ds = 2 \int_a^b \int_a^s K(t, s) dt ds$ since K is symmetric

41.8.5 Filters

A transformation of one stochastic process into another: $X_t \rightarrow Y_t$. Examples:

$$Y_t = a_0 X_0 + a_1 X_{t-1} + \dots + a_n X_{t-n}$$

$$Y_t = \int_{\mathbb{R}} e^{-\beta(t-s)} X_s ds$$

Filters have two properties:

1. Linearity. If $X_t^{(1)} \rightarrow Y_t^{(1)}, X_t^{(2)} \rightarrow Y_t^{(2)} \Rightarrow c_1 X_1 + x_2 X_2 \rightarrow c_1 Y_1^{(1)} + c_2 Y_t^{(2)}$ with the same coefficients c.
2. Time invariance. $X_t \rightarrow Y_t \Rightarrow X_{t+h} \rightarrow Y_{t+h}$. For the second example above, $\int_{\mathbb{R}} e^{-\beta(t-s)} X_{s+h} ds = \int_{\mathbb{R}} e^{-\beta((t+h)-(s+h))} X_{s+h} ds = Y_{t+h}$.

General filter forms for continuous and discrete times respectively:

$$Y_t = \int_{\mathbb{R}} \rho(s) X_{t-s} ds$$

$$Y_t = \sum_{-\infty}^{\infty} \rho(h) X_{t-h}$$

Theorem. Let X_t be weakly stationary with $E(X_t) = 0$ and spectral density g_X , and $Y_t = \int_{\mathbb{R}} \rho(s) X_{t-s} ds$. Then,

1. Y_t is also a weakly stationary process
2. $g_Y(x) = g_X(s) \left| \mathcal{F}[\rho](x) \right|^2$ where $\mathcal{F}[\rho](x) = \int_{\mathbb{R}} e^{iux} \rho(u) du$

Proof. For (i) expectations,

$$E(Y_t) = E \left(\int_{\mathbb{R}} \rho(s) X_{t-s} ds \right) = \int_{\mathbb{R}} \rho(s) \underbrace{E(X_{t-s})}_0 ds = 0$$

For (i) ACF,

$$\begin{aligned} K_Y(t_1, t_2) &= E(Y_{t_1} Y_{t_2}) - E(Y_{t_1}) E(Y_{t_2}) \\ &= E(Y_{t_1} Y_{t_2}) - 0 \\ &= E \left(\int_{\mathbb{R}} \rho(s_1) X_{t_1-s_1} ds_1 \cdot \int_{\mathbb{R}} \rho(s_2) X_{t_2-s_2} ds_2 \right) \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \rho(s_1) \rho(s_2) E(X_{t_1-s_1} X_{t_2-s_2}) ds_1 ds_2 \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \rho(s_1) \rho(s_2) \underbrace{Cov(X_{t_1-s_1}, X_{t_2-s_2})}_{ACF=\gamma} ds_1 ds_2 \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \rho(s_1) \rho(s_2) \gamma((t_2 - s_2) - (t_1 - s_1)) ds_1 ds_2 \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \rho(s_1) \rho(s_2) \gamma(t_2 - t_1 - (s_2 - s_1)) ds_1 ds_2 \end{aligned}$$

Re-writing, the covariance $K(t, s)$ only depends on the difference between t_1, t_2 :

$$\gamma(x) = \int_{\mathbb{R}} \int_{\mathbb{R}} \rho(s_1) \rho(s_2) \gamma(x - (s_2 - s_1)) ds_1 ds_2$$

For (ii),

$$\begin{aligned}
\gamma_Y(x) &= \int \rho(s_1) \int \gamma_X(x + s_1 - s_2) \rho(s_2) ds_2 ds_1 \\
&= \int \rho(s_1) (\gamma_X * \rho)(x + s_1) ds_1 \text{ convolution} \\
&= \int_{\mathbb{R}} (\gamma_X * \rho)(x + s_1) \rho(s_1) ds_1 \\
&= \int_{\mathbb{R}} (\gamma_X * \rho)(x - s_1) \rho(-s_1) ds_1 \\
&= \int_{\mathbb{R}} (\gamma_X * \rho)(x - s_1) \rho^o(s_1) ds_1 \text{ let } \rho^o(s_1) = \rho(-s_1) \\
&= (\gamma_X * \rho * \rho^o)(x - s_1)
\end{aligned}$$

where a convolution is

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

From the spectral density definition,

$$\begin{aligned}
g_Y(x) &= \frac{1}{2\pi} \int e^{iux} \mathcal{F}[\gamma_Y](x) dx \\
&= \frac{1}{2\pi} \int e^{iux} \mathcal{F}[\gamma_X](x) \cdot \mathcal{F}[\rho](x) \cdot \mathcal{F}[\rho^o](x) dx \\
&= g_X(x) \cdot \left| \mathcal{F}[\rho](x) \right|^2
\end{aligned}$$

41.8.6 MA filter

Approximation of a weakly stationary process as an MA would allow use of tools for analysing MA processes

$$Y_n = a_1 X_{n-1} + a_2 X_{n-2}$$

How to find a_1, a_2 s.t. $\min E(X_n - Y_n)^2$ (the mean-square minimum)? Use

$$Z_n = X_n - Y_n = X_n - a_1 X_{n-1} - a_2 X_{n-2}$$

which is a linear filter $X_n \rightarrow Z_n$ so is also weakly stationary and

$$g_Z(s) = g_X(s) \left| \mathcal{F}[\rho](x) \right|^2$$

Using ρ the filtering function,

$$\rho(x) = \mathbb{1}(x = 0) - a_1 \mathbb{1}(x = 1) - a_2 \mathbb{1}(x = 2)$$

$$\mathcal{F}_\rho(x) = 1 - a_1 e^{ix} - a_2 e^{2ix}$$

To minimize $Var(Z_n)$, note again that Z_n is weakly stationary so

$$ACF = K_Z(n, n) = \gamma_Z(0) = \int_{\mathbb{R}} e^{i0x} g_Z(x) dx$$

where ACF is the Fourier transform of the spectral density so

$$= \int_{\mathbb{R}} g_X(x) \left| 1 - a_1 e^{ix} - a_2 e^{2ix} \right|^2 dx$$

Using $|Z^2| = Z\bar{Z}$,

$$\begin{aligned} &= \int_{\mathbb{R}} g_X(x) (1 - a_1 e^{ix} - a_2 e^{2ix})(1 - a_1 e^{-ix} - a_2 e^{-2ix}) dx \\ &= \sum_{i,j=1}^2 B_{ij} a_i a_j + \sum_{i=1}^2 c_i a_i + D \end{aligned}$$

Minimizing the quadratic polynomial over a_1, a_2 is to take the partial derivatives and turn this into a system of linear equations.

41.9 Ergodicity

Law of large numbers for $\xi_i \sim iid$:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \xi_n \rightarrow E(\xi_i)$$

where there is the weak, Khintchine, strong, etc. LLNs. Ergodicity extends LLNs to stochastic processes (and so adds the temporal dimension):

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T X_t \xrightarrow{p} c$$

41.9.1 Examples

Example 1. $X_t = \xi \sim \mathcal{N}(0, 1)$. So choose a point stochastically from the normal distribution, and for all time $t_0 \rightarrow t_\infty$ it is that constant value.

1. $m(t) = 0$
2. $K(t, s) = Var(\xi) = 1$
3. $\frac{1}{T} \sum_{t=1}^T X_t = \xi \neq c$

(1)+(2) indicate weak stationarity, but (3) indicates non-ergodicity. the initial stochastic selection determines the value, which is different for each trial. So not constant across trials.

Example 2. $X_t = \xi_t + a \cos\left(\frac{\pi t}{6}\right)$, $a \neq 0$, with $\xi \sim iid\mathcal{N}(0, 1)$. It has a cycle period of 12 (like 12 months). Since $m(t) = a \cos\left(\frac{\pi t}{6}\right) \neq c$, it is not stationary. However, it is ergodic:

$$\frac{1}{T} \sum_t^T X_t \sim \mathcal{N}\left(\frac{a}{T} \sum_t^T \cos\left(\frac{\pi t}{6}\right), \frac{1}{T}\right)$$

$$\left| \frac{a}{T} \sum_t^T \cos\left(\frac{\pi t}{6}\right) \right| \leq \frac{a}{T} \cdot 3 \rightarrow 0$$

where the 3 is the maximum non-cancelled oscillations leftover. Since the last converges to 0, it is ergodic.

41.9.2 Ergodicity criteria

Proposition. Let X_t be a discrete stochastic process which has a bounded variance $|K(s, t)| = |\gamma = ACF| \leq \alpha$. Further, $C(T) = Cov(X_T, M_T)$ where $M_T = \frac{1}{T} \sum_t^T X_t$. Then,

$$\lim_{T \rightarrow \infty} Var(M_T) \rightarrow 0 \Leftrightarrow \lim_{T \rightarrow \infty} C(T) \rightarrow 0$$

Corollary. If X_t is weakly stationary,

1. $\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{r=0}^{T-1} \gamma(r) \rightarrow 0$, then X_t is ergodic.
2. $\lim_{r \rightarrow \infty} \gamma(r) \rightarrow 0$, then X_t is ergodic.

Proof of (1). $E(X_t) = c \Rightarrow E(M_T) = c$ If $Var(M_T) = E((M_T - c)^2) \rightarrow 0$ then $M_T \xrightarrow{L^2} c \Rightarrow M_T \xrightarrow{p} c$ and therefore X_t is ergodic. To prove this,

$$\begin{aligned} C(T) &= Cov\left(X_T, \frac{1}{T} \sum_t^T X_t\right) \\ &= \frac{1}{T} \sum_t^T Cov(X_T, X_t) \\ &= \frac{1}{T} \sum_t^T \gamma(T - t) \\ &= \frac{1}{T} \sum_{r=0}^{T-1} \gamma(r) \\ &= 0 \text{ by the corollary above} \end{aligned}$$

Proof of (2). Stolz-Cesaro theorem: let a_n, b_n be sequences and b_n be strictly increasing and unbounded. Then,

$$\lim_{n \rightarrow \infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = a \Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = q$$

In this case, let

$$a_n = \sum_{r=0}^{n-1} \gamma(r), b_n = n$$

Then,

$$\lim_{n \rightarrow \infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = \frac{\gamma(n-1)}{1} \rightarrow 0 (= q)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \gamma(r) \rightarrow 0$$

which leads to corollary (1) so X_t is ergodic.

If X_t is not stationary, use heuristic arguments to show ergodicity.

41.10 Levy processes

A general stochastic process that includes Poisson processes and Brownian motion.

41.10.1 Properties

Properties:

1. $L_0 = 0$
2. Independent increments
3. Stationary increments $X_{t+h} - X_{s+h} \stackrel{d}{=} X_t - X_s$ (but Levy process itself is not generally stationary)
4. $L_t - L_s \sim p(t-s)$, where p is an infinitely divisible distribution (Poisson: $\sim \text{Pois}(\lambda(t-s))$; BM: $\sim \mathcal{N}(0, t-s)$)
5. Stochastic continuity

Properties (1)-(3) are shared with Poisson and BM processes. Property (4) is a generalization. Property (5) is:

$$\lim_{h \rightarrow 0} L_{t+h} \xrightarrow{p} L_t$$

$$\forall \epsilon > 0 : \lim_{h \rightarrow 0} p(|L_{t+h} - L_t| > \epsilon) = 0$$

so the process can't have jumps. But it can be modified by modelling trajectories as *cadlag processes* (left-limited, right-continuous processes). If f is a cadlag function,

$$\exists \lim_{s \rightarrow t^-} f(s) = f(t-)$$

$$\exists \lim_{s \rightarrow t^+} f(s) = f(t+) = f(t)$$

Notes:

1. Any Levy process at any time has an infinitely divisible distribution
2. For any infinitely divisible distribution, there is a corresponding Levy process

41.10.2 Characteristic exponents

Proposition. For any Levy process, there exists a characteristic exponent $\Psi : \mathbb{R} \rightarrow \mathbb{C}$ s.t.

$$\Phi_{L_t}(u) = E(e^{iuL_t}) = e^{t\Psi(u)}$$

Example 1.

$$L_t = \mu t \Rightarrow \Phi_{L_t}(u) = e^{iu\mu t} \Rightarrow \Psi(u) = iu\mu$$

Example 2. Brownian motion.

$$L_t = \sigma W_t \Rightarrow \Psi(u) = -\frac{1}{2}\sigma^2 u^2$$

Example 3. Compound Poisson.

$$L_t = \sum_{k=1}^{N_t} \xi_k \Rightarrow \Psi(u) = \lambda(\Phi_{\xi_1}(u) - 1) = \int (e^{iux} - 1) \lambda \underbrace{F_\xi}_{=p_\xi(x)} dx$$

where F_ξ is a Stieltjes integral.

Example 4. BM + Compound Poisson (CPP).

$$L_t = \mu t + \sigma W_t + CPP \Rightarrow \Psi(u) = iu\mu - \frac{1}{2}\mu^2\sigma^2 + \int (e^{iux} - 1)F_\xi$$

$\overline{dx}$ Corollaries.

1. The distribution of L_t is determined by L_1 (or any other time point): $\Phi_{L_1}(u) = e^{(t=1)\Psi(u)} \Rightarrow (1)\Psi(u) = \Psi(u)$ which determines the characteristic function, which in turn completely determines the distribution.
2. Since $E(e^{iuL_t}) = e^{t\Psi(u)} \Rightarrow E(e^{iuL_2}) = e^{2\Psi(u)} = e^{\Psi(u)}e^{\Psi(u)}$, so $E(L_t) = t \cdot E(L_1)$, $Var(L_t) = t \cdot Var(L_1)$, $K(t, s) = \min(t, s) \cdot Var(L_1)$. The last indicates L_t is not stationary.

41.10.3 Levy measures

Characterizes Levy jumps.

$$\nu(B) = E(t \in [0, 1] : \Delta X_t \in B)$$

where $B \subset \mathbb{R}/\{0\}$ and $\Delta X_t = X_t - X_{t-}$.

Example 1. Brown motion, $\nu(B) = 0$ as it has no jumps.

Example 2. Poisson process, all jumps of size 1 as it's a counting process.

$$\nu(B) = \begin{cases} 0 & 1 \notin B \\ E(N_1) = \lambda & 1 \in B \end{cases} = \lambda \cdot \mathbb{1}(1 \in B)$$

Example 3. Compound Poisson process, $\sum \xi_k, \xi_k \sim iid$.

$$\nu(B) = \lambda \cdot p(\xi_1 \in B)$$

If ξ_1 has a density p_ξ

$$\nu(B) = \int_B \underbrace{\lambda p_\xi(x)}_{=s(x)} dx$$

where $s(x)$ is the Levy density.

Any Levy measure has this property:

$$\int_{\mathbb{R}} (1 \wedge x^2) \nu(dx) = \begin{cases} \int_{|x| < 1} x^2 \nu(dx) < \infty \\ \int_{|x| \geq 1} \nu(dx) < \infty \end{cases}$$

and any measure with this property is a Levy measure for some Levy process.

41.10.4 Levy-Khintchine representation

The characteristic exponent of any L_t is

$$\Psi(u) = iu\mu - \frac{1}{2}\mu^2\sigma^2 + \int_{\mathbb{R}} \underbrace{(e^{iux} - 1)}_{\text{CPP}} \underbrace{- iux \mathbb{1}(|x| < 1)}_{\text{Compensated CPP}} \nu(dx)$$

where $\mu \in \mathbb{R}, \sigma \geq 0$. The compensated CPP corrects for the drift introduced by the CPP jump (like a net vertical translation) by subtracting the average, and the average of a Poisson process is $-\lambda x$ where here $\lambda = iu$.

(μ, σ, ν) is the Levy triplet (linear drift, stochasticity, and jump respectively) which completely determines the Levy process. The characteristic function is then

$$\Phi_{X_t}(u) = \exp \left(t \left(iu\mu - \frac{1}{2}\mu^2\sigma^2 + \int_{\mathbb{R}} (e^{iux} - 1 - iux \mathbb{1}(|x| < 1)) \nu(dx) \right) \right)$$

The random variable is

$$X_t = \underbrace{\mu t + \sigma W_t}_{\text{continuous}} + \underbrace{J_t}_{\text{discontinuous}}$$

and is decomposed into continuous and discontinuous parts. Note the only continuous Levy process is Brownian motion (e.g. Poisson process is a jump process)

so isn't continuous). The jump process J_t is decomposed into two compound Poisson processes:

$$J_t \approx \underbrace{\sum_{0 < s < t} \Delta X_s}_{|\Delta X_s| > 1} + \lim_{\epsilon \rightarrow 0} \underbrace{\sum_{0 < s < t} \Delta X_s}_{\epsilon < |\Delta X_s| \leq 1}$$

Moreover,

$$J_t = \int_{|x| < 1} \underbrace{(e^{iux} - 1 - iux)}_{< \infty} \nu(dx) + \int_{|x| > 1} \underbrace{(e^{iux} - 1)}_{\text{complex modulus } \leq 2\nu(dx) < \infty}$$

The Blumenthal-Gettoor index is another way of characterizing jump activity.

Example 1. X_t is a stochastic process of bounded variation, i.e.

$$\sum_{k=1}^n |X_{t_k} - X_{t_{k-1}}| \rightarrow \exists$$

as $\max |t_i - t_{i-1}| \rightarrow 0$ iff

1. $\sigma = 0$ (no diffusion)
2. $\int_{|x| < 1} x \nu(dx) < \infty$

Then

$$\begin{aligned} \Phi_{X_t}(u) &= \exp \left(t \left(iu\mu - \frac{1}{2}\mu^2\sigma^2 + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}(|x| < 1))\nu(dx) \right) \right) \\ &= \exp \left(t \left(iu\mu - 0 + \int_{\mathbb{R}} (e^{iux} - 1 - iux\mathbb{1}(|x| < 1))\nu(dx) \right) \right) \\ &= \exp \left(t \left(iu\tilde{\mu} + \int_{\mathbb{R}} (e^{iux} - 1)\nu(dx) \right) \right) \end{aligned}$$

where $\tilde{\mu} = \mu - \int_{|x| < 1} x \nu(dx)$.

Example 2. X_t is a CPP. Then,

1. $\sigma = 0$
2. $\int_{\mathbb{R}} \nu(dx) = \nu(\mathbb{R}) < \infty$

Then,

$$\int_{\mathbb{R}} (1 \wedge x^2) \nu(dx) < \infty$$

because of (2) above and x^2 for $|x| < 1$ is finite. Then as in example 1,

$$\Phi_{X_t}(u) = \exp \left(t \left(iu\tilde{\mu} + \int_{\mathbb{R}} (e^{iux} - 1)\nu(dx) \right) \right)$$

Example 3. X_t is a subordinator. A subordinator is a stochastic process within another stochastic process, the subordinated stochastic process. To be a subordinator, $X_t \geq 0$ almost surely. Since Levy processes have independent increments which are now also non-negative $X_t - X_s \stackrel{d}{=} X_{t-s} \geq 0$, combined with $X_t \geq 0$, the result is: $X_t \geq X_s$ almost surely (monotonically increasing) for all $t > s$. X_t is a subordinator iff

1. $\sigma = 0$ (no diffusion)
2. $\nu(\mathbb{R}) = 0$
3. $\int_0^1 x\nu(dx) < \infty$

Then as in example 1,

$$\Phi_{X_t}(u) = \exp \left(t \left(iu\tilde{\mu} + \int_{\mathbb{R}} (e^{iux} - 1)\nu(dx) \right) \right)$$

41.10.5 Levy measure estimation

Let X_t be a Levy process $X_\Delta, X_{2\Delta}, \dots, X_{n\Delta}$. It is useful to estimate the Levy measure ν as it completely characterizes jump structure.

Let X_t be a process of bounded variation

$$\begin{aligned} \Phi_{X_\Delta}(u) &= \exp \left(\Delta \left(iu\tilde{\mu} + \int_{\mathbb{R}} (e^{iux} - 1)s(x)(dx) \right) \right) \\ \Phi'_{X_\Delta}(u) &= \Phi_{X_\Delta}(u) \Delta i \left(\tilde{\mu} + \int_{\mathbb{R}} (e^{iux} - 1)s(x)(dx) \right) \\ \Phi''_{X_\Delta}(u) &= \Phi'_{X_\Delta}(u) \frac{\Phi'_{X_\Delta}(u)}{\Phi_{X_\Delta}(u)} + \Phi_{X_\Delta}(u) \underbrace{\left(- \int_{\mathbb{R}} e^{iux} x^2 s(x)(dx) \right)}_{\mathcal{F}(x^2 s(x))(u)} \end{aligned}$$

where the last term is a Fourier transform. Re-arranging the last equation,

$$\mathcal{F}(x^2 s(x))(u) = -\frac{1}{\Delta} \left(\frac{\Phi''_{X_\Delta}(u)}{\Phi_{X_\Delta}(u)} - \left(\frac{\Phi'_{X_\Delta}(u)}{\Phi_{X_\Delta}(u)} \right)^2 \right)$$

So for $X_\Delta, X_{2\Delta}, \dots, X_{n\Delta}$, can find Φ, Φ', Φ'' . For instance, use non-parametric estimator

$$\hat{\Phi}(u) = \frac{1}{n} \sum_{k=1}^n e^{iu(X_{K\Delta} - X_{(K-1)\Delta})}$$

and then calculate Φ, Φ', Φ'' as well as $\mathcal{F}(x^2 s(x))(u)$. Then use the inverse Fourier transform to back out $s(x)$, the estimate of the Levy measure.

41.10.6 Levy-based models

Stochastic time change models account for differences in tempo or activity. For instance, during a certain time period there may be more transactions and therefore more stochasticity than other times. To account for this, in the Levy process X_t , make time itself another (subordinated) Levy process $T(s)$. Then $X_{T(s)}$ can be modelled through the characteristic function (derivation omitted)

$$\Phi_{X_{T(s)}}(u) = \mathcal{L}_{T(s)}(-\Psi(u))$$

Stochastic volatility models account for changing σ . $\sigma = V_t$, a stochastic process random variable. In the *Cox-Ingersoll-Ross* SDE,

$$dV_t = (a - bV_t)dt + c\sqrt{V_t}dW_t$$

41.11 Martingales

41.11.1 Definition

A process X_n wrt $\mathcal{G}_n$ is a Martingale if

1. X_n is $\mathcal{G}_n$ -adapted (omit for natural filtrations)
2. $E(X_{n+1}|\mathcal{G}_n) = X_n$ almost surely. This is a fair game. A supermartingale is unfavorable: $E(X_{n+1}|\mathcal{G}_n) \leq X_n$. A submartingale is very favorable: $E(X_{n+1}|\mathcal{G}_n) \geq X_n$.

This is in contrast to Markov chains, where

$$p(X_{n+1} \in B|\mathcal{F}_n) = p(X_{n+1} \in B|X_n)$$

almost surely.

41.11.2 Compare to Markov chains

The Markov property states that a stochastic process essentially has "no memory". This means that the conditional probability distribution of the future states of the process are independent of any previous state, with the exception of the current state. The Martingale property states that the future expectation of a stochastic process is equal to the current value, given all known information about the prior events.

A random walk coin flip ($p(W_n(H) = 1) = p$; $p(W_n(T) = -1) = q$) is not a Martingale:

$$E(W_{n+1}|\mathcal{F}_n) = E(W_{n+1}) = p - q \neq W_n$$

though the player's fortune is a Martingale. In contrast, this is a Markov chain:

$$p(W_{n+1} = 1|\mathcal{F}_n) = p(W_{n+1} = 1|W_n)$$

A random walk coin flip looking at fortunes is both a Martingale and Markov chain.

A random coin flip where a player bets each time is a Martingale but not a Markov chain. The winnings are

$$Y_n = B_1W_1 + \dots + B_nW_n = \sum B_iW_i$$

and the bet is made before, not after, the coin flip:

$$B_n \in \mathcal{F}_{n-1}$$

So,

$$\begin{aligned} E(Y_{n+1}|\mathcal{F}_n) &= E(Y_n + B_{n+1}W_{n+1}|\mathcal{F}_n) \\ &= Y_n + B_{n+1}E(W_{n+1}|\mathcal{F}_n) \\ &= Y_n + (p - q)E(W_{n+1}) \end{aligned}$$

which is a Martingale for a fair coin $p - q = 0$ but can be a supermartingale or submartingale depending on p and q .

41.11.3 Optional stopping theorem

Also called Doob's optional sampling theorem.

Let X_n be a supermartingale wrt filtration $\mathcal{G}_n$.

$$E(X_n) = E(E(X_n|\mathcal{G}_{n-1})) \leq E(X_{n-1}) \leq \dots \leq E(X_0)$$

This is true for a random stopping (hitting) time. Let $T : \Omega \rightarrow \bar{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$ be a stopping time wrt filtration $\mathcal{G}_n$ if $(T = n) \in \mathcal{G}_n$. The decision to stop depends on the past, not the future. An example of a stopping time is $T = \inf(n : X_n \in B)$.

The process X_n^T stopped at T is a supermartingale with respect to filtration $\mathcal{G}_n$, where

$$X_n^T = \begin{cases} X_n & T \geq n \\ X_T & T < n \end{cases}$$

Proof.

$$\begin{aligned} E(X_{n+t}^T|\mathcal{G}_n) - X_n &= E((X_{n+1}^T - X_n^T)|\mathcal{G}_n) \\ &= E((X_{n+1}^T - X_n^T)\mathbb{1}(T > n)|\mathcal{G}_n) \\ &= \mathbb{1}(T > n)E(X_{n+1} - X_n|\mathcal{G}_n) \leq 0 \end{aligned}$$

where the last line follows as $E(X_{n+1} - X_n|\mathcal{G}_n)$ is a supermartingale wrt $\mathcal{G}_n$.

Let X_n be a supermartingale. If $\exists c < \infty$ (bounded) s.t. $|X_n^T| \leq c$ and

$P(T < \infty) = 1$ then $E(X_T) \leq E(X_0)$. Alternatively, $\exists c < \infty$ (bounded) s.t. $E(|X_{n+t}^T - X_n^T| | \mathcal{G}_n) \leq c$ and $E(T) < \infty$ then $E(X_T) \leq E(X_0)$. Proof.

$$\begin{aligned} E(X_T) &= E(\lim_{n \rightarrow \infty} X_n^T) \\ &= \lim_{n \rightarrow \infty} E(X_n^T) \text{ by DCT} \\ &\leq \lim_{n \rightarrow \infty} E(X_0^T) = E(X_0^T) = E(X_0) \end{aligned}$$

DCT = dominated Lebesgue's theorem, due to $|X_n^T| \leq c < \infty$ which makes it integrable. This condition is equivalent to $E(|X_{n+t}^T - X_n^T| | \mathcal{G}_n) \leq c$.

$$\begin{aligned} |X_n^T| &\leq |X_0| + \sum_{k=0}^{\infty} |X_{k+1}^T - X_k^T| \\ E(|X_n^T|) &\leq E|X_0| + \sum_{k=0}^{\infty} E|X_{k+1}^T - X_k^T| \\ &= E|X_0| + \sum_{k=0}^{\infty} E(E|X_{k+1}^T - X_k^T| \mathbb{1}(T > k) | \mathcal{G}_k)) \\ &= E|X_0| + \sum_{k=0}^{\infty} E(\mathbb{1}(T > k) E|X_{k+1}^T - X_k^T| | \mathcal{G}_k)) \\ &= E|X_0| + \sum_{k=0}^{\infty} E(\mathbb{1}(T > k) d) \text{ from } E(|X_{n+t}^T - X_n^T| | \mathcal{G}_n) \leq c \\ &= E|X_0| + d \sum_{k=0}^{\infty} E(\mathbb{1}(T > k)) \\ &= E|X_0| + d \sum_{k=0}^{\infty} p(T > k) \\ &= E|X_0| + d(0p(0) + 1p(1) + 2p(2) + \dots) \\ &= E|X_0| + dE(T) < \infty \end{aligned}$$

Since the expectation is finite, it must be that $|X_n^T| \leq c < \infty$.

Note: the Martingale betting strategy of doubling bets will lead to $E(X_T) = 1 \neq 0 = E(X_0)$, seemingly violating the optional stopping theorem. However, this is because the bets are not bounded increments; they're unbounded. If bounded, it should follow OST. If unbounded, the expected value is higher than the fair game 'should' allow, but the expectation takes into account near-infinite cases. For instance, if the game were bounded in time, it would also follow OST under modified conditions because many of the resulting 1's would take a very long time ('near-infinite').

41.11.4 Upcrossing inequality

Lemma. For all $k \in \mathbb{N}$ and supermartingale X_n defined at time ∞ ,

$$p(u(a, b) > k) \leq \frac{1}{b-a} E(X_\infty - a)^- \mathbb{1}(u(a, b) = k)$$

where $u(a, b)$ is the number of upcrossings for $X_{T_{2k+1}} \leq a < b \leq X_{T_{2k+2}}$ where the T 's are stopping times. Alternatively, $u(a, b) = \sup(k \geq 0 : T_{2k} < \infty)$ (the largest k s.t. $T_{2k} < \infty$).

- $T_0 = 0$
- Even: $T_{2k+1} = \inf(n > T_{2k} : X_n \leq a)$
- Odd: $T_{2k+2} = \inf(n > T_{2k} : X_n \geq b)$

Proof. Note $(X_n - a)$ is just a translation of the supermartingale so is also a supermartingale.

$$(u, (a, b) > k) = (\sigma_2 < \infty) \subset (\sigma_1 < \infty) \cap (X_{\sigma_2} \geq b)$$

Prove wlog $a = 0$:

$$\begin{aligned} p(u(a, b) > k) &= E(\mathbb{1}(u(a, b) > k)) \\ &= \underbrace{\frac{1}{b} E(X_{\sigma_2}^+ \mathbb{1}(u(a, b) > k))}_{p(X_{\sigma_2} \geq b)} \\ &= \frac{1}{b} E(X_{\sigma_2}^+ \mathbb{1}(\sigma_1 < \infty)) \\ &\leq \frac{1}{b} E(X_{\sigma_1} \mathbb{1}(\sigma_1 < \infty)) \text{ supermartingale and OST} \\ &\leq 0 \end{aligned}$$

Since the last result means it is less than 0,

$$\begin{aligned} \frac{1}{b} E(X_{\sigma_2}^+ \mathbb{1}(\sigma_1 < \infty)) &\leq \frac{1}{b} E(X_{\sigma_2}^- \mathbb{1}(\sigma_1 < \infty)) \\ &= \frac{1}{b} E(X_{\sigma_1} \mathbb{1}(\sigma_1 < \infty, \underbrace{X_{\sigma_2} \leq 0}_{\sigma_2 = \infty})) \end{aligned}$$

where the underbrace results from X_n being a supermartingale and as a consequence of OST. So $\sigma_2 = \infty$. Add back a to get

$$p(u(a, b) > k) \leq \frac{1}{b-a} E(X_\infty - a)^- \mathbb{1}(u(a, b) = k)$$

41.11.5 Martingale convergence

Let X_n be a supermartingale bounded in $L^1(\Omega, \mathcal{F}, p)$, i.e. $\sup_n E|X_n| < \infty$. Then

$$X_n \xrightarrow{as} X_\infty$$

where $X_\infty \in L^1(\Omega, \mathcal{F}, p)$.

Proof.

$$\begin{aligned} E(u_n(a, b)) &= \sum_{k=0}^{\infty} p(u_n(a, b) > k) \\ &\leq \frac{1}{b-a} \sum_{k=0}^{\infty} E((X_n - a)^- \mathbb{1}(u_n(a, b) = k)) \text{ Upcrossing inequality} \\ &= \frac{1}{b-a} E((X_n - a)^- \sum_{k=0}^{\infty} \mathbb{1}(u_n(a, b) = k)) \text{ Fubini or MCT} \\ &\leq \frac{1}{b-a} E((X_n - a)^-(1)) \text{ indicator f's span infinity} \\ &\leq \frac{1}{b-a} E|X_n| + |a| \end{aligned}$$

So,

$$\begin{aligned} E(u(a, b)) &= E(\lim_{n \rightarrow \infty} u_n(a, b)) \\ &= \lim_{n \rightarrow \infty} E(u_n(a, b)) \text{ MCT} \\ &\leq \frac{1}{b-a} (\sup_n E|X_n| + |a|) < \infty \text{ a.s.} \end{aligned}$$

So, $E(u(a, b)) < \infty$ almost surely $\Rightarrow p(u(a, b) = \infty) = 0$ almost surely $X_n \xrightarrow{as} X_\infty$.

41.11.6 Galton-Watson branching processes

Models population dynamics with iid number of offspring. Let X_n be the number of individuals at time or generation n . Let Y_{n-1} be the probability mass function with random variable PMF $P(Y_{n,i} = k) = p_k$, with i being individual.

$$X_{n+1} = Y_{n,1} + Y_{n,2} + \dots + Y_{n,X_n}$$

X_n is a discrete-time Markov chain, 0 is an absorbing state (no offspring once it reaches 0). It is also a martingale:

$$\begin{aligned} E(X_{n+1} | \mathcal{F}_n) &= E(Y_{n,1} + Y_{n,2} + \dots + Y_{n,X_n} | \mathcal{F}_n) \\ &= X_n E(Y_{n,i} | \mathcal{F}_n) \text{ No. of r.v.'s, memoryless} \\ &= X_n E(Y_{n,i}) \\ &= X_n \mu, \mu = \sum_{k=0}^{\infty} k p_k \end{aligned}$$

$$E\left(\frac{X_{n+1}}{\mu^{n+1}} \middle| \mathcal{F}_n\right) = \frac{\mu X_n}{\mu^{n+1}} = \frac{X_n}{\mu^n}$$

so it is a martingale. Moreover,

$$E(X_n) = \mu^n$$

and intuitively if $\mu < 1$ it's a supermartingale and subcritical (leading to extinction), if $\mu = 1$ it's a martingale and critical (eventually leading to extinction almost surely), $\mu > 1$ it's a submartingale and supercritical (possibility of survival). More specifically,

1. $\mu = E(Y_{n,i}) < 1 \Rightarrow P(X_n \neq 0) \leq \mu^n \rightarrow 0$ exponentially fast
2. $(\mu = E(Y_{n,i}) = 1, p_1 \neq 1 \Rightarrow P(T = \infty) = 0, E(T) = \infty$
3. $\mu = E(Y_{n,i}) > 1$, then $P(X_n > 0, \forall n > 0) > 0$

42 Stochastic Calculus

42.1 Stopping times

Also called optional stopping times.

In the study of a stochastic process it is often useful to consider some properties of the process that hold up to a random time. A natural question is for instance: How long is the process less than a given constant?

At any time, know whether an event has occurred or not, e.g. whether the process has reached a y-value of 5. Don't need the whole process to find out, e.g. whether the process has hit its maximum or minimum. So, a stopping time is $\mathcal{F}_t$ -measurable.

Stopping times are random variables - e.g. how long it takes to hit a certain y-value is a random variable.

Intuitive examples.

Parking (not a stopping time): Suppose I am driving along a very long road, and that I'm looking for the parking spot which is furthest towards the other end of the road (call this "the last parking spot"). I pass by available spots along the way, but at any time, I never know if I have passed the last free parking spot. Why? I could just have passed some empty spot, but I cannot see if there are more empty spots later on, and I wouldn't know if the spot that I just passed was the last one or not.

Birthday this year (a stopping time): This is a deterministic stopping time.

At any time, I know whether or not my birthday has occurred this year. In fact, I know exactly when my birthday occurs, which makes this a non-typical stopping time in the sense that it is deterministic.

42.2 Variation of a function

42.2.1 Variation

$$V_g([a, b]) = \lim_{\delta \rightarrow 0} \sum_{i=1}^n |g(t_i^n) - g(t_{i-1}^n)|$$

where g is a function of a real variable.

A finite variation function can have no more than countably many discontinuities. moreover, all discontinuities are jumps.

42.2.2 Quadratic variation

If g is a function of a real variable,

$$[g](t) = \lim_{\delta \rightarrow 0} \sum_{i=1}^n (g(t_i^n) - g(t_{i-1}^n))^2$$

Quadratic variation is important in stochastic calculus but not standard calculus as smooth functions usually have zero quadratic variation.

Theorem. If g is continuous and of finite variation then its quadratic variation is zero. Proof.

$$\begin{aligned} [g](t) &= \lim_{\delta \rightarrow 0} \sum_{i=0}^{n-1} (g(t_{i+1}^n) - g(t_i^n))^2 \\ &\leq \lim_{\delta \rightarrow 0} \max_i |g(t_{i+1}^n) - g(t_i^n)| \sum_{i=0}^{n-1} |g(t_{i+1}^n) - g(t_i^n)| \\ &\leq \lim_{\delta \rightarrow 0} \max_i |g(t_{i+1}^n) - g(t_i^n)| V_g(t) \end{aligned}$$

Since g is continuous, it is uniformly continuous on $[0, t]$, hence

$$\lim_{\delta \rightarrow 0} \max_i |g(t_{i+1}^n) - g(t_i^n)| = 0$$

which completes the proof.

42.2.3 Quadratic covariation

Define the covariation as

$$[f, g](t) = \lim_{\delta \rightarrow 0} \sum_{i=0}^{n-1} (f(t_{i+1}^n) - f(t_i^n))(g(t_{i+1}^n) - g(t_i^n))$$

The polarization identity is

$$[f, g](t) = \frac{1}{2}([f + g, f + g](t) - [f, f](t) - [g, g](t))$$

Theorem. If f is continuous and g is of finite variation, then $[f, g](t) = 0$.

Corollary. If X and Y are Ito processes and X is of finite variation, then $[X, Y](t) = 0$.

42.3 Stochastic differentiation

Let X_t be a stochastic process is differentiable at $t = t_0$ if the following limit exists in the mean-squared sense:

$$\lim_{h \rightarrow 0} \frac{X_{t_0+h} - X_{t_0}}{h} \stackrel{L^2}{=} \eta = X'_{t_0}$$

That is,

$$\lim_{h \rightarrow 0} E \left(\frac{X_{t_0+h} - X_{t_0}}{h} - \eta \right)^2 = 0$$

Proposition. If $E(X_t^2) < \infty$ (finite) then X_t is differentiable iff

1. $m(t) = E(X_t)$ is differentiable at $t = t_0$
2. $\frac{\partial^2}{\partial t \partial s} K(t, s)$ exists at $t = t_0$

Most deterministic processes, represented by real-valued functions, are differentiable. But not stochastic processes.

Example 1. X_t is weakly stationary. Then $m(t) = \text{constant}$ and so differentiable. The ACF is $K(t, s) = \gamma(t - s)$ and must exist at 0:

$$\left. \frac{\partial^2 K}{\partial t \partial s} \right|_{(t_0, t_0)} = -\gamma''(0)$$

As a subexample, $\gamma(r) = e^{-\alpha|r|}$ is not differentiable because the first and second derivatives at 0 ($\gamma'(0), \gamma''(0)$) don't exist. As another subexample, $\gamma(r) = \text{Cos}(\alpha r) \Rightarrow X_t$ is differentiable.

Example 2. Brownian motion. Not differentiable at any point. ACF is $K(t, s) = \min(t, s)$

$$\frac{K(t_0 + h, t_0) - K(t_0, t_0)}{h} = \frac{\min(t_0, t_0 + h) - t_0}{h} = \begin{cases} 0 & h > 0 \\ 1 & h < 0 \end{cases} \xrightarrow{h \rightarrow 0} \text{does not exist}$$

Example 3. Any stochastic process with independent increments and $X_0 = 0$. Then,

$$\begin{aligned}
K(t, s) &= \text{Var}(X_{\min(t, s)}) \\
&= \text{Cov}(X_t, X_s) \quad t > s \\
&= \text{Cov}(X_t - X_s, X_s) + \text{Cov}(X_s, X_s) \\
&= 0 + \text{Var}(X_s) = \text{Var}(X_{\min(t, s)})
\end{aligned}$$

where the last line follows as t and s only overlap over the minimum (s in this case because $t > s$).

42.4 Continuity in L^2

A stochastic process X_t is mean-squared continuous $X_t \xrightarrow{L^2} X_{t_0}$ iff $E(X_t - X_{t_0})^2 \rightarrow 0$ as $t \rightarrow t_0$.

Proposition. If $E(X_t) = 0$, then

1. If $K(t, s)$ is continuous at (t_0, t_0) then X_t is mean-squared continuous at $t = t_0$.
2. If X_t is mean-squared continuous at (t_0, s_0) then $K(t, s)$ is continuous at (t_0, s_0) .

Proof of (1)

$$\begin{aligned}
E(X_t - X_{t_0})^2 &= E(X_t^2) - 2E(X_t X_{t_0}) + E(X_{t_0})^2 \\
&= K(t, t) - 2K(t, t_0) + K(t_0, t_0) \\
&= K(t_0, t_0) - 2K(t_0, t_0) + K(t_0, t_0) \text{ as } t \rightarrow t_0 \\
&= 0
\end{aligned}$$

Proof of (2)

$$\begin{aligned}
K(t, s) - K(t_0, s_0) &= K(t, s) + K(t_0, s) - K(t_0, s) - K(t_0, s_0) \\
&= (K(t, s) - K(t_0, s)) + (K(t_0, s) - K(t_0, s_0))
\end{aligned}$$

For the first expression,

$$\begin{aligned}
|K(t, s) - K(t_0, s)| &= |E(X_t X_s) - E(X_{t_0} X_s)| \\
&= |E(X_t X_s) - 0 - E(X_{t_0} X_s) + 0| \\
&= |E((X_t - X_{t_0}) X_s)| \\
&\leq \sqrt{E(X_t - X_{t_0})^2} \sqrt{E(X_s)^2} \\
&= 0 \text{ as } t \rightarrow t_0
\end{aligned}$$

The second expression also equals 0.

42.5 Stochastic integration

Stochastic integrals are themselves random variables.

Four types in order of increasing generality:

1. $\int X_t dt$ stochastic Riemann or Lebesgue
2. $\int f(t) dW_t$ Wiener integral
3. $\int X_t dW_t$
4. $\int X_t dH_t$ Ito integral, the most general

Integral type (1). $X_t : \Omega \times \mathbb{R}_+ \rightarrow \mathbb{R}$, where Ω is the probability space with realizations ω and $\mathbb{R}_+$ is time. With a concrete realization ω , this becomes a Riemann integral

$$\int_a^b X_t(\omega) dt = \lim \sum_{k=1}^n X_{t_{k-1}}(t_k - t_{k-1})$$

where the limit is a mean-squared limit

$$\max |t_k - t_{k-1}| \rightarrow 0, E \left(\sum - \int X_t dt \right)^2 \rightarrow 0$$

Theorem. If $m(t)$, $K(t, s)$ are continuous, then $\int_a^b X - t dt$ exists. Moreover, by Fubini,

$$\begin{aligned} E \int X_t dt &= \int E(X_t) dt \\ Var \int X_t dt &= \int \int K(t, s) dt ds \end{aligned}$$

42.5.1 Wiener integral

Wiener integral:

$$\int f(t) dW_t$$

where $f \in L^2([a, b])$. This is in Hilbert space, where the inner product is

$$\langle f, g \rangle = \int f(x)g(x)dx$$

which is symmetric, linear, and where $\langle f, f \rangle \geq 0$. The inner product is an analog of the scalar product in Euclidean space. The function is from the Hilbert space iff $\int f^2(x)dx < \infty$ and that $f_n \xrightarrow{L^2} f \Leftrightarrow \langle f_n - f, f_n - f \rangle \xrightarrow{n \rightarrow \infty} 0$ or in other words

$$\lim_{n \rightarrow \infty} \int (f_n(x) - f(x))^2 dx = 0$$

Step functions are

$$f(x) = \sum_{i=1}^n \alpha_i \mathbb{1}(t_{i-1} \leq x < t_i)$$

where the Wiener integral is defined on step functions as

$$\int f(t) dW_t = \sum_{i=1}^n \alpha_i (W_{t_i} - W_{t_{i-1}})$$

Theorem. If f is a step function, then

$$\begin{aligned} \int f(t) dW_t &= I(f) \\ I(f) &\sim \mathcal{N}(0, \int f^2(x) dx) \end{aligned}$$

Proof.

$$\begin{aligned} E(I(f)) &= \sum_{i=1}^n \alpha_i \underbrace{E(W_{t_i} - W_{t_{i-1}})}_{0-0=0} = 0 \\ \text{Var}(I(f)) &= \sum_{i=1}^n \alpha_i^2 \underbrace{\text{Var}(W_{t_i} - W_{t_{i-1}})}_{t_i - t_{i-1}} = \int f^2(x) dx \end{aligned}$$

where the results are from Brownian motion $E(W_t) = 0$ and independent increments.

Step functions f_n can mean-square converge to any function (even non-step functions) $f : f_n \xrightarrow{L^2} f$, that is, $\int (f_n(t) - f(t))^2 dt \xrightarrow{n \rightarrow \infty} 0$. So,

$$I(f) = \lim_{n \rightarrow \infty} I(f_n) = \lim_{n \rightarrow \infty} \int f_n(t) dW_t$$

as mean-square convergence, i.e. $\lim_{n \rightarrow \infty} E((I(f_n) - I(f))^2) = 0$. Moreover, $I(f)$ doesn't depend on f_n because of the below result.

Theorem. If $f_n, \tilde{f}_n$ are two sequences of step functions that both mean-square converge to an arbitrary (even non-step) function f , then

$$\lim_{n \rightarrow \infty} I(f_n) = \lim_{n \rightarrow \infty} I(\tilde{f}_n)$$

Proof.

$$\lim_{n \rightarrow \infty} I(f_n) - I(\tilde{f}_n) = \lim_{n \rightarrow \infty} I(\underbrace{f_n - \tilde{f}_n}_{\text{also a step f}})$$

Since the difference is a step function, use the results of step functions from above for variance,

$$\begin{aligned} \text{Var}(I(f)) &= \int f^2(x)dx \\ &= \int (f_n(x) - \tilde{f}_n(x))^2 dx \\ &= E((I(f_n) - I(f))^2) \\ &= 0 \text{ as } n \rightarrow \infty \end{aligned}$$

due to mean-square convergence.

Theorem. $I(f) \sim \mathcal{N}(0, \int f^2(x)dx)$. Proof.

$$I(f) = \lim_{n \rightarrow \infty} I(f_n) \sim \mathcal{N}(0, \int f_n^2(x)dx)$$

So,

$$I(f) \sim \mathcal{N}(0, \lim_{n \rightarrow \infty} \sum f_n^2(x)dx)$$

42.5.2 Filtrations

A *filtration* is a sequence of σ -algebras $\mathcal{F}_t$ on $(\Omega, \mathcal{F}, p)$ s.t. $\mathcal{F}_t \subset \mathcal{F}_s, \forall t < s$. As time passes, an observer knows more and more information, and finer partitions of Ω .

$(\Omega, \mathcal{F}, \mathbb{F}, p)$ is a *filtered probability space*. A stochastic process on this space is *adapted* if for all t , the process is $\mathcal{F}_t$ -measurable, that is, if for any t , $\mathcal{F}_t$ contains all information about the stochastic process.

42.5.3 Stochastic process integral

Integrals of type

$$I(X_t) = \int X_t dW_t$$

Let $L_{adapted}^2([a, b], \Omega)$ be the space of stochastic processes with properties:

1. X_t is $\mathcal{F}_t$ -adapted, i.e. $\mathcal{F}_t$ -measurable, which means $(X_t \in B) \subset \mathcal{F}_t, \forall t, \forall B \subset B(\mathbb{R})$.
2. $\int E(X_t^2)dt < \infty$.

W_t is $\mathcal{F}_t$ -Brownian motion if

1. W_t is $\mathcal{F}_t$ -adapted
2. $W_t - W_s \perp \mathcal{F}_s, \forall t > s$

Define for

1. Step processes: $X_t = \sum \xi \mathbb{1}(t_{i-1} \leq t < t_i)$, i.e. divided into subintervals. Based on this, define $I(X_t) = \sum_{i=1}^n \xi_{i-1}(W_{t_i} - W_{t_{i-1}})$.
2. $X_t \in L_{adapted}^2$ where can find a sequence of X_t^n step processes which mean-square converge $\lim_{n \rightarrow \infty} \int_a^b E(X_t^n - X_t)^2 dt = 0$. If the sequence satisfies these criteria, then $I(X_t) = \lim_{n \rightarrow \infty} I(X_t^n)$ mean-squared.

Theorem. Let $m(t), K(t, s)$ be continuous. Then, define

$$X_t^n = \sum_{i=1}^n X_{t_{i-1}} \mathbb{1}(t_{i-1} \leq t < t_i)$$

$$X_t^n \rightarrow X_t : \lim_{n \rightarrow \infty} \int_a^b E(X_t^n - X_t)^2 dt = 0$$

Proof.

$$\begin{aligned} E(X_t - X_s)^2 &= E(X_t^2) - 2E(X_t X_s) + E(X_s^2) \\ &= (K(t, t) + m^2(t)) - 2(K(t, s) + m(t)m(s)) + (K(s, s) + m^2(s)) \\ &= 0 \text{ as } s \rightarrow t \end{aligned}$$

So, there is mean-squared convergence:

$$\lim_{n \rightarrow \infty} X_t^n \rightarrow X_t$$

Lebesgue's DCT:

$$\int \lim_{n \rightarrow \infty} f(n, t) dt = \lim_{n \rightarrow \infty} \int f(n, t) dt$$

if $\exists M(t) : |f(n, t)| \leq M(t), \int M(t) dt, \infty$. Since $(a - b)^2 \leq 2a^2 + 2b^2$,

$$E(X_t^n - X - t) \leq 2E((X_t^n)^2) + 2E(X_t)^2$$

Then,

$$2E(X_t^n)^2 + 2E(X_t)^2 \leq 4 \max_{t \in [a, b]} E(X_t^2) = 4 \max_{t \in [a, b]} (K(t, t) + m^2(t))$$

where the last term is finite because any continuous function on a closed interval is finite. Apply Lebesgue's DCT:

$$\lim_{n \rightarrow \infty} \int_a^b E(X_t^n - X_t)^2 dt = \int_a^b \lim_{n \rightarrow \infty} E(X_t^n - X_t)^2 dt = 0$$

42.5.4 Ito integral

The most general stochastic integral:

$$\int_a^b X_t dH_t$$

where H is a general stochastic process called an Ito process, which is a drift-diffusion process, a linear combination of stochastic processes (one for trend and one for stochasticity). The Ito process can be represented in at least two equivalent ways:

$$\begin{aligned} H_t &= H_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s \\ dH_t &= b_t dt + \sigma_t dW_t \end{aligned}$$

Then

$$\int_a^b X_t dH_t = \int_a^b b_s X_s ds + \int_a^b \sigma_s X_s dW_s$$

by inserting $dH_t = b_t dt + \sigma_t dW_t$. There is an $\mathcal{F}_t$ filtration, and b_s, σ_s are $\mathcal{F}_t$ -adapted stochastic processes. Further, W_t is $\mathcal{F}_t$ -BM and H_0 is $\mathcal{F}_0$ -measurable. X_t is defined wrt

$$\int_a^b (|X_s b_s| + X_s^2 b_s^2) ds < \infty$$

Theorem. Let H_t be an Ito process, $f(t, x)$ is twice continuously-differentiable. Then,

$$f(t, H_t) = f(0, H_0) + \int_0^t f'_1(s, H_s) ds + \int_0^t f'_2(s, H_s) dH_s + \frac{1}{2} \int_0^t f''_{22}(s, H_s) \sigma_s^2 ds$$

which is the *Ito formula*. f is the antiderivative of g wrt the second argument, $f'_2 = g$. An equivalent alternative representation (as an SDE) is

$$df(t, H_t) = f'_1(t, H_t) dt + f'_2(t, H_t) dH_t + \frac{1}{2} f''_{22}(t, H_t) \sigma_t^2 dt$$

Example. $\int_0^t g(s, W_s) dW_s$. Then

$$f(t, W_t) = f(0, 0) + \int_0^t f'_1(s, W_s) ds + \int_0^t g(s, W_s) dW_s + \frac{1}{2} \int_0^t g'_2(s, W_s) ds$$

Rearranging,

$$\int_0^t g(s, W_s) dW_s = f(t, W_t) - f(0, 0) - \int_0^t (f'_1(s, W_s) + \frac{1}{2} g'_2(s, W_s)) ds$$

which converts the LHS dW to the RHS ds .

42.5.5 Ito integral derivation

This is a sketch of the proof. Let X_t be a drift-diffusion Ito process and B_t be Brownian motion.

$$dX_t = \mu_t dt + \sigma_t dB_t$$

If $f(t, x)$ is a twice-differentiable function, its Taylor expansion is

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dx + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}dx^2 + \dots$$

Substituting,

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}(\mu_t dt + \sigma_t dB_t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(\mu_t dt + 2\mu_t \sigma_t dt B_t + \sigma_t dB_t^2) + \dots$$

As $dt \rightarrow 0$, $dt^2 \rightarrow 0$ and $dt dB_t \rightarrow 0$ faster than dB_t^2 , which is $O(dt)$. Set $dt^2 = dt dB_t = 0$ and substitute $dt = dB_t^2$ from quadratic variation of Brownian motion:

$$\begin{aligned} df &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}(\mu_t dt + \sigma_t dB_t) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(0 + 0 + \sigma_t dB_t^2) + \dots \\ &= \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2} \right) dt + \frac{\partial f}{\partial x} dB_t + \dots \end{aligned}$$

42.5.6 Black-Scholes

Black-Scholes is modelled as an SDE

$$dX_t = X_t \mu dt + X_t \sigma dW_t, \sigma > 0$$

which is equivalent to

$$X_t = X_0 + \mu \int_0^t X_s ds + \sigma \int_0^t X_s dW_s$$

Let $f(t, x) = \ln(x)$, $H_t = X_t$. Then use Ito's formula:

$$\begin{aligned} d(\ln X_t) &= 0 \cdot dt + \frac{1}{X_t} dX_t - \frac{1}{X_t^2} \frac{X_t^2 \sigma^2}{2} dt \\ &= \frac{1}{X_t} (X_t \mu dt + X_t \sigma dW_t) - \frac{\sigma^2}{2} dt \\ &= \mu dt + \sigma dW_t - \frac{\sigma^2}{2} dt \\ &= \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t \end{aligned}$$

where the first line's RHS zero term is from differentiating f wrt time. Convert from SDE to integral representation:

$$\ln(X_t) = \ln(X_0) + \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t$$

So,

$$X_t = X_0 e^{\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t}$$

42.5.7 Vasicek model

A mean-reverting SDE:

$$dX_t = (a - bX_t)dt + c dW_t$$

where $b, c > 0$, and the model is mean-reverting because

$$b \left(\frac{a}{b} - X_t \right) dt = \begin{cases} < 0 & X_t < \frac{a}{b} \\ > 0 & X_t > \frac{a}{b} \end{cases}$$

So, b is the speed of reversion.

Use solution $f(t, x) = xe^{bt}$ (this model is simple and so doesn't require Ito; get same result whether using the total differential or Ito's formula):

$$\begin{aligned} d(X_t e^{bt}) &= bX_t e^{bt} dt + e^{bt} dX_t \\ &= bX_t e^{bt} dt + e^{bt} ((a - bX_t)dt + c dW_t) \\ &= ae^{bt} dt + ce^{bt} dW_t \end{aligned}$$

Can now convert to integral representation and solve. Or move through detailed steps. Integrating both sides:

$$\begin{aligned} X_t e^{bt} - X_0 &= \frac{a}{b} e^{bt} - \frac{a}{b} + c \int_0^t e^{bs} dW_s \\ X_t e^{bt} &= X_0 + \frac{a}{b} (e^{bt} - 1) + c \int_0^t e^{bs} dW_s \end{aligned}$$

Multiplying both sides by e^{-bt} ,

$$X_t = X_0 e^{-bt} + \frac{a}{b} (1 - e^{-bt}) + c \int_0^t e^{b(s-t)} dW_s$$

42.5.8 Ornstein-Uhlenbeck process

Models stochastic movement of particle V_t in Brownian fluid. Einstein proposed the free particle movement version as

$$m dV_t = dW_t$$

but this was later modified to include friction with friction coefficient λ :

$$m dV_t = dW_t - \lambda V_t dt$$

Using the solution is $f(t, x) = xe^{\frac{\lambda}{m}t}$ with the Ito formula (and skipping the math):

$$V_t = e^{-\frac{\lambda}{m}t} \left(V_0 + \frac{1}{m} \int_0^t e^{\frac{\lambda}{m}s} dW_s \right)$$

If it is assumed that

$$V_0 \sim \mathcal{N}\left(0, \frac{1}{\lambda m}\right) \perp\!\!\!\perp W_t$$

Then V_t is a Gaussian process with mean 0 and

$$K(t, s) = \frac{m}{2\lambda} e^{-\frac{\lambda}{m}|t-s|}$$

so it's stationary. And since it's Gaussian, it's both weak and strong stationary.

42.6 Ito integrals

For a process,

$$X(t) = \xi_0 I_0(t) + \sum_{i=0}^{n-1} \xi_i I_{(t_i, t_{i+1}]}(t)$$

The stochastic integral

$$\int_0^T X(t) dB(t)$$

is a random variable:

$$Y(t) = \int_0^t X(s) dB(s)$$

which is continuous but nowhere differentiable like Brownian motion. The Ito integral is itself a process that depends on Brownian motion. This can be interpreted as the wealth from a martingale $B(t)$ of bets $X(t)$.

42.6.1 Mean and variance

$$\int_0^T X(t) dB(t) = \sum_{i=0}^{n-1} \xi_i (B(t_{i+1}) - B(t_i))$$

The ξ_i 's can be stochastic. Assume below variance of the stochastic integral is finite.

By Brownian motion independent increments,

$$E\left(\int_0^T X(t) dB(t)\right) = E\sum_{i=0}^{n-1} \xi_i (B(t_{i+1}) - B(t_i)) = 0$$

and the variance is

$$\begin{aligned} Var\left(\int_0^T X(t) dB(t)\right) &= Var\left(\sum_{i=0}^{n-1} \xi_i (B(t_{i+1}) - B(t_i))\right) \\ &= \sum Var(\xi_i (B(t_{i+1}) - B(t_i))) \\ &= \sum \xi_i^2 (t_{i+1} - t_i) \end{aligned}$$

42.6.2 Isometry

The Ito isometry property is

$$E \left(\int_0^T X(t) dB(t) \right)^2 = \int_0^T E(X^2(t)) dt$$

Proof.

$$\begin{aligned} & E \sum \xi(B(t_{i+1}) - B(t_i))^2 \\ &= \sum E(\xi^2(B(t_{i+1}) - B(t_i))^2) + 2 \sum_{i < j} E(\xi_i \xi_j (B(t_{i+1}) - B(t_i))(B(t_{j+1}) - B(t_j))) \end{aligned}$$

Using the martingale property of Brownian motion,

$$\begin{aligned} \sum_E (\xi_i^2 (B(t_{i+1}) - B(t_i))^2) &= \sum E E(\xi_i^2 (B(t_{i+1}) - B(t_i))^2 | \mathcal{F}_{t_i}) \\ &= \sum E(\xi_i^2 E((B(t_{i+1}) - B(t_i))^2 | \mathcal{F}_{t_i})) \\ &= \sum E(\xi_i^2)(t_{i+1} - t_i) \end{aligned}$$

where the last sum is $\int_0^T E(X^2(t)) dt$ since $X(t) = \xi_i$ on (t_i, t_{i+1}) . By conditioning in a similar way for $i < j$,

$$E(\xi_i \xi_j (B(t_{i+1}) - B(t_i))(B(t_{j+1}) - B(t_j))) = 0$$

so that the sum $\sum_{i < j}$ vanishes.

A consequence of Ito isometry is

$$E \left(\int_0^T X(t) dB(t) \int_0^T Y(t) dB(t) \right) = \int_0^T E(X(t)Y(t)) dt$$

Proof. Denote the Ito integrals

$$\begin{aligned} I_1 &= \int_0^T X(t) dB(t) \\ I_2 &= \int_0^T Y(t) dB(t) \end{aligned}$$

Write their product by using the identity

$$I_1 I_2 = (I_1 + I_2)^2 / 2 - I_1^2 / 2 - I_2^2 / 2$$

Then use isometry.

42.6.3 Martingale property

It can be shown that

$$E \left(\int_s^t X(u) dB(u) | \mathcal{F}_s \right) = 0$$

So,

$$\begin{aligned} E(Y(t) | \mathcal{F}_s) &= E \left(\int_0^t X(u) dB(u) | \mathcal{F}_s \right) \\ &= \int_0^s X(u) dB(u) + E \left(\int_s^t X(u) dB(u) | \mathcal{F}_s \right) \\ &= \int_0^s X(u) dB(u) \\ &= Y(s) \end{aligned}$$

42.6.4 Square integrability

A martingale is *square integrable* if its second moments are bounded.

42.6.5 Quadratic variation and covariation

Quadratic variation of Ito integral $Y(t)$:

$$\begin{aligned} [Y, Y](t) &= \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} (Y(t_{i+1}^n) - Y(t_i^n))^2 \\ \left[\int_0^t X(s) dB(s), \int_0^t X(s) dB(s) \right] (t) &= \int_0^t X^2(s) ds \end{aligned}$$

Quadratic covariation of Ito integrals $Y_1(t), Y_2(t)$:

$$\begin{aligned} [Y_1, Y_2](t) &= \lim_{n \rightarrow \infty} \sum (Y_1(t_{i+1}^n) - Y_1(t_i^n))(Y_2(t_{i+1}^n) - Y_2(t_i^n)) \\ &= \frac{1}{2}([Y_1 + Y_2, Y_1 + Y_2](t) - [Y_1, Y_1](t) - [Y_2, Y_2](t)) \\ &= \int_0^t X_1(s) X_2(s) ds \end{aligned}$$

42.6.6 Covariance and Gaussianity

If $X(t)$ is non-random s.t. $\int_0^T X^2(s) ds < \infty$ then its Ito integral $Y(t) = \int_0^t X(s) dB(s)$ is a Gaussian process with zero mean and covariance function

$$\text{Cov}(Y(t), Y(t+u)) = \int_0^t X^2(s) ds$$

Gaussianity arises follows from independence and normality of increments $M(t) - M(s) = \int_s^t X(u)dB(u)$ with $M(t) = \int_0^t X(s)dB(s)$. Proof of covariance function. Write $\int_0^{t+u}$ as $\int_0^t + \int_t^{t+u}$ and use the martingale property of $Y(t)$:

$$E \left(\int_0^t X(s)dB(s) E \left(\int_t^{t+u} X(s)dB(s) | \mathcal{F}_t \right) \right) = 0$$

Hence

$$\begin{aligned} Cov(Y(t), Y(t+u)) &= E \left(\int_0^t X(s)dB(s) \int_0^{t+u} X(s)dB(s) \right) \\ &= E \left(\int_0^t X(s)dB(s) \right)^2 \\ &= \int_0^t EX^2(s)ds \\ &= \int_0^t X^2(s)ds \end{aligned}$$

If $X(t)$ is a regular non-random function with $\int_0^t X^2(t, s)ds < \infty$ then $Y(t) = \int_0^t X(t, s)dB(s)$ is a Gaussian process with mean zero and covariance function

$$Cov(Y(t), Y(t+u)) = \int_0^t X(t, s)X(t+u, s)ds$$

Proof of covariance function.

$$Y(t+u) = \int_0^t X(t+u, s)dB(s) + \int_t^{t+u} X(t+u, s)dB(s)$$

Since $X(t+u, s)$ is non-random, the Ito integral

$$\int_t^{t+u} X(t+u, s)dB(s)$$

is independent of $\mathcal{F}_t$. Therefore,

$$E \left(\int_0^t X(t, s)dB(s) \int_t^{t+u} X(t+u, s)dB(s) \right) = 0$$

and

$$\begin{aligned} Cov(Y(t), Y(t+u)) &= E(Y(t)Y(t+u)) \\ &= E \left(\int_0^t X(t, s)dB(s) \int_0^t X(t+u, s)dB(s) \right) \\ &= \int_0^t X(t, s)X(t+u, s)ds \end{aligned}$$

42.6.7 Ito formula

Also known as the change of variable formula and chain rule. If $B(t)$ is Brownian motion and $f(x)$ is twice continuously differentiable,

$$f(B(t)) = f(0) + \int_0^t f'(B(s))dB(s) + \frac{1}{2} \int_0^t f''(B(s))ds$$

Proof.

$$f(B(t)) = f(0) + \sum_{i=0}^{n-1} (f(B(t_{i+1}^n)) - f(B(t_i^n)))$$

Taylor expand the difference

$$f(B(t_{i+1}^n)) - f(B(t_i^n)) = f'(B(t_i^n))(B(t_{i+1}^n) - B(t_i^n)) + \frac{1}{2} f''(B(t_i^n))(B(t_{i+1}^n) - B(t_i^n))^2 + \dots$$

$$f(B(t)) = f(0) + \sum_{i=0}^{n-1} f'(B(t_i^n))(B(t_{i+1}^n) - B(t_i^n)) + \frac{1}{2} \sum_{i=0}^{n-1} f''(B(t_i^n))(B(t_{i+1}^n) - B(t_i^n))^2 + \dots$$

Taking limits results in the Ito formula.

Example. Taking $f(x) = e^x$,

$$e^{B(t)} = 1 + \int_0^t e^{B(s)} dB(s) + \frac{1}{2} \int_0^t e^{B(s)} ds$$

42.7 Ito processes

Let $Y(t)$ be an Ito process (a drift-diffusion process);

$$Y(t) = Y(0) + \int_0^t \mu(s)ds + \int_0^t \sigma(s)dB(s)$$

which is the stochastic differential

$$dY(t) = \mu(t)dt + \sigma(t)dB(t)$$

The Ito formula becomes

$$d(f(B(t))) = f'(B(t))dB(t) + \frac{1}{2} f''(B(t))dt$$

42.7.1 Quadratic variation

Let $Y(t)$ be an Ito process.

$$\begin{aligned}
[Y](t) &= [Y, Y]([0, t]) \\
&= \lim_{\delta \rightarrow 0} \sum_{i=0}^{n-1} (Y(t_{i+1}^n) - Y(t_i^n))^2 \\
&= \left[\int_0^t \mu(s) ds + \int_0^t \sigma(s) dB(s) \right] (t) \\
&= \left[\int_0^t \mu(s) ds \right] (t) + 2 \left[\int_0^t \mu(s) ds, \int_0^t \sigma(s) dB(s) \right] (t) + \left[\int_0^t \sigma(s) dB(s) \right] (t) \\
&= 0 + 0 + \left[\int_0^t \sigma(s) dB(s) \right] (t) \text{ Covariation with finite variation f is 0} \\
&= \int_0^t \sigma^2(s) ds \text{ Quadratic variation of Ito integral}
\end{aligned}$$

Note

$$\begin{aligned}
d[X, Y](t) &= dY(t)dX(t) \\
d[Y, Y](t) &= (dY(t))^2
\end{aligned}$$

Since $X(t)$ is a continuous function of finite variation and $Y(t) = B(t)$ is continuous with quadratic variation t ,

$$dB(t)dt = 0, (dt)^2 = 0$$

but

$$(dB(t))^2 = d[B, B](t) = dt$$

42.7.2 Ito formula

If $X(t)$ is defined with

$$dX(t) = \mu(t)dt + \sigma(t)dB(t)$$

then $Y(t) = f(X(t))$ is

$$\begin{aligned}
df(X(t)) &= f'(X(t))dX(t) + \frac{1}{2}f''(X(t))d[X, X](t) + \dots \\
&= f'(X(t))dX(t) + \frac{1}{2}f''(X(t))\sigma^2(t)dt
\end{aligned}$$

where the second step uses $d[X, X](t) = \sigma^2(t)dt$ where a quadratic covariation with respect to a finite variation function is zero. Integrating both sides,

$$f(X(t)) - f(X(0)) = \int_0^t f'(X(s))dX(s) + \frac{1}{2} \int_0^t f''(X(s))\sigma^2(s)ds$$

Example 1. Geometric Brownian motion. Let $X(t)$ have the SDE

$$dX(t) = X(t)dB(t) + \frac{1}{2}X(t)dt$$

Solve the SDE by finding a process $X(t)$ satisfying the SDE. There may be more than one solution. Look for a positive solution. Use $Y(t) = \ln X(t)$. Then, differentiate as above:

$$\begin{aligned} d\ln X(t) &= \frac{1}{X(t)}dX(t) - \frac{1}{2X^2(t)}X^2(t)dt + \dots \\ &= \frac{1}{X(t)}(X(t)dB(t) + \frac{1}{2}X(t)dt) + \frac{1}{2}dt \\ &= dB(t) + \frac{1}{2}dt - \frac{1}{2}dt \\ &= dB(t) \end{aligned}$$

So,

$$d\ln X(t) = dB(t)$$

Integrating both sides,

$$\begin{aligned} \int_0^t d\ln X(t) &= \int_0^t dB(t) \\ \ln X(t) - \ln X(0) &= B(t) - B(0) = B(t) - 0 = B(t) \\ X(t) &= X(0)e^{B(t)} \end{aligned}$$

Example 2. More general geometric Brownian motion. Let $X(t)$ have the SDE

$$dX(t) = \mu X(t)dB(t) + \sigma X(t)dt$$

Look for a positive solution. Use $Y(t) = \ln X(t)$.

$$\begin{aligned} d\ln X(t) &= \frac{1}{X(t)}dX(t) + \frac{1}{2}\left(-\frac{1}{X^2(t)}\right)\sigma^2 X^2(t)dt + \dots \\ &= \frac{1}{X(t)}(\mu X(t)dB(t) + \sigma X(t)dt) - \frac{1}{2}\sigma^2 dt \\ &= (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB(t) \end{aligned}$$

So,

$$d\ln X(t) = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB(t)$$

Integrating both sides,

$$X(t) = X(0)e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma B(t)}$$

42.7.3 Integration by parts

If $X(t), Y(t)$ are Ito processes, quadratic covariation is

$$\begin{aligned}
[X, Y](t) &= \lim_{\delta \rightarrow 0} \sum (X(t_{i+1}) - X(t_i))(Y(t_{i+1}) - Y(t_i)) \\
&= \sum (X(t_{i+1})Y(t_{i+1}) - X(t_i)Y(t_i)) - \sum X(t_i)(Y(t_{i+1}) - Y(t_i)) - \sum Y(t_i)(X(t_{i+1}) - X(t_i)) \\
&= X(t)Y(t) - X(0)Y(0) - \sum X(t_i)(Y(t_{i+1}) - Y(t_i)) - \sum Y(t_i)(X(t_{i+1}) - X(t_i)) \\
&= X(t)Y(t) - X(0)Y(0) - \int_0^t X(s)dY(s) - \int_0^t Y(s)dX(s)
\end{aligned}$$

So, the integration by parts or stochastic product rule formula is

$$X(t)Y(t) - X(0)Y(0) = \int_0^t X(s)dY(s) + \int_0^t Y(s)dX(s)$$

Alternatively,

$$d(X(t)Y(t)) = X(t)dY(t) + Y(t)dX(t) + d[X, Y](t)$$

If

$$dX(t) = \mu_X(t)dt + \sigma_X(t)dB(t)$$

$$dY(t) = \mu_Y(t)dt + \sigma_Y(t)dB(t)$$

then using

$$\begin{aligned}
d[X, Y](t) &= dX(t)dY(t) \\
&= \sigma_X(t)\sigma_Y(t)(dB(t))^2 \\
&= \sigma_X(t)\sigma_Y(t)dt
\end{aligned}$$

and therefore

$$d(X(t)Y(t)) = X(t)dY(t) + Y(t)dX(t) + \sigma_X(t)\sigma_Y(t)dt$$

If one of the processes is of finite variation, then the covariance term is zero. For such processes, the stochastic product rule is the same as standard calculus.

42.8 Stochastic differential equations

Let $\xi(t)$ be white noise,

$$\xi(t) = \frac{dB(t)}{dt} = B'(t)$$

SDEs arise when the coefficients of ODEs are perturbed by white noise.

42.8.1 Black-Scholes-Merton model

Growth with uncertain rate of return, $r + \sigma\xi(t)$. The SDE is then

$$\frac{dX(t)}{dt} = (r + \sigma\xi(t))X(t)$$

$$dX(t) = rX(t)dt + \sigma X(t)dB(t)$$

because white noise from above is $dB(t) = \xi(t)dt$. If $\sigma = 0$ then no noise and it's deterministic and therefore an ODE with solution $x(t) = x(0)e^{rt}$. If $\sigma \neq 0$ then there's noise and the solution is a geometric Brownian motion as above,

$$X(t) = X(0)e^{r-\sigma^2/2)t+\sigma B(t)}$$

42.8.2 Stochastic exponential

The solution to

$$dU(t) = U(t)dX(t)$$

Then U is called the stochastic exponential of X , and is noted by $\Xi(X)$. If $X(t)$ is of finite variation, then

$$\frac{dU(t)}{U(t)} = dX(t)$$

$$\int \frac{dU(t)}{U(t)} = \int dX(t)$$

$$\ln U(t) = X(t)$$

$$U(t) = e^{X(t)}$$

For Ito processes, the only solution is

$$U(t) = e^{X(t)-X(0)-1/2[X,X](t)} = e^{V(t)}$$

Proof. Direct verification. Differentiate as above:

$$dU(t) = d(e^{V(t)}) = e^{V(t)}dV(t) + \frac{1}{2}e^{V(t)}d[V,V](t)$$

Since $[X, X](t)$ is of finite variation, and $X(t)$ is continuous, $[X, [X, X]](t) = 0$ and therefore $[V, V](t) = [X, X](t)$. Since $V(t) = X(t) - X(0) - \frac{1}{2}[X, X](t)$, plug in $dV(t) = dX(t) - \frac{1}{2}[X, X](t)$,

$$d\Xi(X)(t) = e^{V(t)}dX(t) - \frac{1}{2}e^{V(t)}d[X, X](t) + \frac{1}{2}e^{V(t)}d[X, X](t) = e^{V(t)}dX(t)$$

So,

$$U(t) = U(0)\Xi(X)(t)$$

Example. Let $S(t)$ be a stock price and assume an Ito process and

$$dR(t) = \frac{dS(t)}{S(t)}$$

$$dS(t) = S(t)dR(t)$$

So the stock price is the stochastic exponential of the return. If $R(t) = \mu t + \sigma B(t)$, then

$$\begin{aligned} S(t) &= S(0)\Xi(R(t)) = S(0)e^{R(t)-R(0)-\frac{1}{2}[R,R](t)} \\ &= S(0)e^{(\mu-\frac{1}{2}\sigma^2)t+\sigma B(t)} \end{aligned}$$

42.8.3 Stochastic logarithm

If $U = \Xi(X)$ then X is the stochastic log of U . This is the inverse operator of the stochastic exponential. For instance, the stochastic exponential of $B(t)$ is $e^{B(t)-\frac{1}{2}t}$, so $B(t)$ is the stochastic log of $e^{B(t)-\frac{1}{2}t}$. If $U = \Xi(X)$, then

$$\begin{aligned} dU(t) &= U(t)dX(t) \\ dX(t) &= \frac{dU(t)}{U(t)} \end{aligned}$$

So,

$$X(t) = \ln \left(\frac{U(t)}{U(0)} \right) + \int_0^t \frac{d[U, U](S)}{2U^2(S)}$$

Example. Let $U(t) = e^{B(t)}$. Then $dU(t) = e^{B(t)}dB(t) + \frac{1}{2}e^{B(t)}dt$. Can find stochastic log directly by

$$dX(t) = \frac{dU(t)}{U(t)} = \frac{e^{B(t)}dB(t) + \frac{1}{2}e^{B(t)}dt}{e^{B(t)}} = dB(t) + \frac{1}{2}dt$$

or can use the above formula, which will give the same answer.

42.8.4 Stochastic exponential SDEs

Solve the SDE

$$dU(t) = \beta(t)U(t)dt + \delta(t)U(t)dB(t)$$

which can be re-expressed as

$$\begin{aligned} dU(t) &= U(t)(\beta(t)dt + \delta(t)dB(t)) \\ &= U(t)dY(t) \end{aligned}$$

Use the above stochastic exponential result

$$\begin{aligned} U(t) &= U(0)\Xi(Y)(t) \\ &= U(0)\exp \left(Y(t) - Y(0) - \frac{1}{2}[Y, Y](t) \right) \\ &= U(0)\exp \left(\int \beta(t)dt + \int \delta(t)dB(t) - \frac{1}{2} \int \delta^2(t)ds \right) \\ &= U(0)\exp \left(\int (\beta(t) - \frac{1}{2}\delta^2(t))dt + \int \delta(t)dB(t) \right) \end{aligned}$$

where $[Y, Y](t)$ is obtained from $d[Y, Y](t) = dY(t)dY(t) = \delta^2(t)dt$.

42.8.5 General linear SDEs

Solve the SDE

$$dX(t) = (\alpha(t) + \beta(t)X(t))dt + (\gamma(t) + \delta(t)X(t))dB(t)$$

Use a solution of the form

$$X(t) = U(t)V(t)$$

where

$$\begin{aligned} dU(t) &= \beta(t)U(t)dt + \delta(t)U(t)dB(t) \\ dV(t) &= a(t)dt + b(t)dB(t) \end{aligned}$$

Take the LHS of the SDE:

$$\begin{aligned} dX(t) &= d(U(t)V(t)) \\ &= U(t)dV(t) + V(t)dU(t) + dU(t)dV(t) \\ &= U(t)dV(t) + V(t)dU(t) + (0 + 0 + 0 + \delta(t)U(t)b(t)dB^2(t)) \\ &= U(t)dV(t) + V(t)dU(t) + \delta(t)U(t)b(t)dt \\ &= U(t)(a(t)dt + b(t)dB(t)) + V(t)(\beta(t)U(t)dt + \delta(t)U(t)dB(t)) + \delta(t)U(t)b(t)dt \end{aligned}$$

where the third step involves

$$\begin{aligned} \alpha(t)\beta(t)U(t)dt^2 &= 0 \text{ since } dt^2 = 0 \\ \alpha(t)\delta(t)U(t)dtdB(t) &= 0 \text{ since } dtdB = 0 \\ b(t)\beta(t)U(t)dtdB(t) &= 0 \text{ since } dtdB = 0 \end{aligned}$$

because t is a continuous function of finite variation and $B(t)$ is continuous with quadratic variation, $dB(t)dt = 0$, $dt^2 = 0$. Next take the RHS of the SDE:

$$\begin{aligned} &(\alpha(t) + \beta(t)U(t)V(t))dt + (\gamma(t) + \delta(t)U(t)V(t))dB(t) \\ &= \alpha(t)dt + \beta(t)U(t)V(t)dt + \gamma(t)dB(t) + \delta(t)U(t)V(t)dB(t) \end{aligned}$$

Equating LHS=RHS,

$$\begin{aligned} b(t)U(t) &= \gamma(t) \\ a(t)U(t) &= \alpha(t) - \delta(t)\gamma(t) \end{aligned}$$

Now solve for V :

$$dV(t) = a(t)dt + b(t)dB(t)$$

Integrating both sides,

$$V(t) - V(0) = \int a(t)dt + \int b(t)dB(t)$$

So,

$$\begin{aligned} V(t) &= V(0) + \int a(t)dt + \int b(t)dB(t) \\ &= V(0) + \int \frac{\alpha(t) - \delta(t)\gamma(t)}{U(t)}dt + \int \frac{\gamma(t)}{U(t)}dB(t) \end{aligned}$$

So,

$$X(t) = U(t)V(t) = U(t) \left(V(0) + \int \frac{\alpha(t) - \delta(t)\gamma(t)}{U(t)}dt + \int \frac{\gamma(t)}{U(t)}dB(t) \right)$$

where U is a stochastic exponential SDE solution from above:

$$U(t) = U(0)\exp\left(\int(\beta(t) - \frac{1}{2}\delta^2(t))dt + \int\delta(t)dB(t)\right)$$

42.8.6 Langevin-type SDEs

$$dX(t) = a(t)X(t)dt + dB(t)$$

Use the above formula for general linear SDEs which is

$$dX(t) = (\alpha(t) + \beta(t)X(t))dt + (\gamma(t) + \delta(t)X(t))dB(t)$$

with $\beta(t) = a(t), \gamma(t) = 1, \alpha(t) = \delta(t) = 0$. Further,

$$dU(t) = a(t)U(t)dt$$

which gives

$$U(t) = \exp\left(\int_0^t a(s)ds\right)$$

So the above equation

$$X(t) = U(t)V(t) = U(t) \left(V(0) + \int \frac{\alpha(t) - \delta(t)\gamma(t)}{U(t)}dt + \int \frac{\gamma(t)}{U(t)}dB(t) \right)$$

becomes

$$X(t) = \exp\left(\int_0^t a(s)ds\right)(X(0) + 0 + \int_0^t \exp\left(-\int_0^s a(u)du\right)dB(u))$$

42.8.7 Brownian bridge

The Brownian bridge is used when the end point is pinned, in addition to the starting point. The SDE linearly subtracts the end result so that the rest can be stochastic. Used for stochastic (Brownian motion) interpolation.

$$dX(t) = \frac{b - X(t)}{T - t}dt + dB(t)$$

This is just a linear SDE with

$$\alpha(t) = \frac{b}{T-t}, \beta(t) = \frac{-1}{T-t}$$

So,

$$X(t) = a(1 - \frac{t}{T}) + b\frac{t}{T} + (T-t) \int \frac{1}{T-s} dB(s)$$

42.8.8 Strong and weak solutions

The main difference between weak and strong solutions is indeed that for strong solutions we are given a Brownian motion on a given probability space whereas for weak solutions we are free to choose the Brownian motion and the probability space.

Theorem. If

$$dX(t) = \mu(X(t), t)dt + \sigma(X(t), t)dB(t)$$

then if there exists a unique strong solution $X(t)$ if the following 3 conditions are fulfilled: 1. Coefficients are locally Lipschitz in x uniformly in t , i.e., for every T and N there is a constant K depending only on T and N , s.t. for all $|x|, |y| \leq N$ and $0 \leq t \leq T$,

$$|\mu(x, t) - \mu(y, t)| + |\sigma(x, t) - \sigma(y, t)| < K|x - y|$$

2. Coefficients satisfy the linear growth condition

$$|\mu(x, t)| + |\sigma(x, t)| \leq K(1 + |x|)$$

3. $X(0)$ is independent of $B(t), 0 \leq t \leq T$ and $EX^2(0) < \infty$.

SDE strong solutions can be shown to exist similar to ODEs (Picard's theorem).

Let $W_t \geq 0$ be a standard Brownian motion on a probability space $(\Omega, \mathcal{F}, p)$ and let $F_t \geq 0$ be the completion of the minimal filtration by null sets. A strong solution of the SDE

$$dX_t = \mu(X_t)dt + \omega(X_t)dW_t$$

with initial condition $x \in \mathbb{R}$ is an adapted process X_t with continuous paths such that for all $t \geq 0$,

$$X_t = x + \int_0^t \mu(X_s)ds + \int_0^t \omega(X_s)dW_s, \text{ a.s.}$$

A weak solution of the SDE with initial condition x is a continuous stochastic process X_t defined on some probability space $(\Omega, \mathcal{F}, p)$ s.t. for *some* Wiener process W_t and *some* admissible filtration $\mathcal{F}$ the process $X(t)$ is adapted and satisfies the stochastic integral equation.

As a consequence of these definitions, consider different notions of uniqueness. For strong solutions, typically looking for pathwise unique solutions, i.e. if $(X_t^{(1)})_{t \geq 0}$ and $(X_t^{(2)})_{t \geq 0}$ are strong solutions with the same initial condition, then pathwise uniqueness means

$$P \left(\sup_{t \geq 0} |X_t^{(1)} - X_t^{(2)}| = 0 \right) = 1$$

As the following simple example shows it doesn't make sense to talk about pathwise uniqueness of weak solutions.

Example. Let $(W_t^{(1)})_{t \geq 0}$ and $(W_t^{(2)})_{t \geq 0}$ be two Brownian motions (possibly defined on different probability spaces), then both $X_t^{(1)} := W_t^{(1)}$ and $X_t^{(2)} := W_t^{(2)}$ are weak solutions to the SDE

$$dX_t = dB_t, X_0 = 0$$

According to the definition, can freely choose the driving Brownian motion, so can set $B_t^{(1)} := W_t^{(1)}$ and $B_t^{(2)} := W_t^{(2)}$, respectively, and then

$$dX_t^{(i)} = dB_t^{(i)}$$

Example. The SDE

$$dX_t = \text{sign}(X_t)dB_t, X_0 = 0$$

has a weak solution but no strong solution.

42.9 Stratonovich calculus

Alternative to Ito calculus. Basic rules such as chain rule and integration by parts are same as in standard calculus. Stratonovich integrals can be reduced to Ito integrals to theory is equivalent.

Stratonovich integral is

$$\int_0^t Y(s) \partial X(s)$$

is a limit in mean square L^2 of approximating sums

$$\sum_{i=0}^{n-1} \frac{1}{2} (Y(t_{i+1}^n) + Y(t_i^n)) (X(t_{i+1}^n) - X(t_i^n))$$

where the $\frac{1}{2}$ and $+$ term appear as it takes the average value of the interval is taken in contrast to Ito which takes the left-most value. Convert between Stratonovich and Ito integrals:

$$\int_0^t Y(s) \partial X(s) = \int_0^t Y(s) dX(s) + \frac{1}{2} [Y, X](t)$$

$$Y(t)\partial X(t) = Y(t)dX(t) + \frac{1}{2}d[Y, X](t)$$

Proof. Take the left-most 'height' of Y in Ito calculus.

$$\begin{aligned} Y\partial X &= \frac{1}{2}(Y_t + Y_{t+1})(X_{t+1} - X_t) \\ &= \frac{1}{2}(2Y_t - Y_t + Y_{t+1})dX \\ &= (Y_t + \frac{1}{2}(Y_{t+1} - Y_t))dX \\ &= (Y + \frac{1}{2}dY)dX \\ &= YdX + \frac{1}{2}dYdX \\ &= YdX + \frac{1}{2}d[X, Y] \end{aligned}$$

42.9.1 Integration by parts and product rule

Integration by parts:

$$X(t)Y(t) - X(0)Y(0) = \int_0^t X(s)\partial Y(s) + \int_0^t Y(s)\partial X(s)$$

and from this derive the product rule:

$$\partial(X(t)Y(t)) = X(t)\partial Y(t) + Y(t)\partial X(t)$$

Proof of integration by parts uses product rule:

$$\begin{aligned} d(X(t)Y(t)) &= X(t)dY(t) + Y(t)dX(t) + d[X, Y](t) \\ &= X(t)dY(t) + \frac{1}{2}[X, Y](t) + Y(t)dX(t) + \frac{1}{2}[X, Y](t) \\ &= X(t)\partial Y(t) + Y(t)\partial X(t) \end{aligned}$$

42.9.2 Chain rule

Let X be continuous and f three times continuously differentiable, then

$$\begin{aligned} f(X(t)) - f(X(0)) &= \int_0^t f'(X(s))\partial X(s) \\ \partial f(X(t)) &= f'(X(t))\partial X(t) \end{aligned}$$

Example. If $B(t)$ is Brownian motion, then its Stratonovich differential is

$$\partial B^2(t) = 2B(t)\partial B(t)$$

which can be solved using the chain rule. Alternatively, use the product rule

$$\partial B(t)\partial B(t) = B(t)\partial B(t) + \partial B(t)B(t)$$

Compare this to the Ito differential

$$\begin{aligned} dB^2(t) &= B(t)dB(t) + dB(t)B(t) + [B(t)]^2 \\ &= 2B(t)dB(t) + dt \end{aligned}$$

42.9.3 Converting SDEs

Suppose $X(t)$ satisfies the Stratonovich SDE

$$dX(t) = \mu(X(t))dt + \sigma(X(t))\partial B(t)$$

Then $X(t)$ satisfies the Ito SDE

$$dX(t) = \left(\mu(X(t)) + \frac{1}{2}\sigma'(X(t))\sigma(X(t)) \right) dt + \sigma(X(t))dB(t)$$

Proof. Since

$$Y(t)\partial X(t) = Y(t)dX(t) + \frac{1}{2}d[Y, X](t)$$

Then

$$dX(t) = \mu(X(t))dt + \sigma(X(t))dB(t) + \frac{1}{2}d[\sigma(X), B](t)$$

Compute the last term separately:

$$d[\sigma(X), B](t) = d\sigma(X(t))dB(t)$$

Apply Ito's formula to $\sigma(X)$:

$$d\sigma(X(t)) = \sigma'(X(t))dX(t) + \frac{1}{2}\sigma''(X(t))d[X, X](t)$$

Calculate this term using the second equation of this proof:

$$d[X, B](t) = dX(t)dB(t) = 0 + \sigma(X(t))dt + 0 = \sigma(X(t))$$

Go back to calculating $d[\sigma(X), B](t)$:

$$\begin{aligned} d[\sigma(X), B](t) &= d\sigma(X(t))dB(t) \\ &= \sigma'(X(t))dX(t)dB(t) \\ &= \sigma'(X(t))\sigma(X(t))dt \\ &= \sigma'(X(t))\sigma(X(t))dt \end{aligned}$$

42.9.4 Ito or Stratonovich

The Ito integral takes the left-most function value, which is independent of the ΔB increment:

$$\int f(x, B_t, t)dB_t = \lim_{N \rightarrow \infty} \sum f(x_{t_k}, B_{t_k}, t_k) \underbrace{(B_{t_{k+1}} - B_{t_k})}_{\Delta B}$$

The Ito integral is therefore a martingale, and this property eases subsequent proofs of stochastic calculus. Moreover, the independence is intuitive; you can't know the function value ahead of the Brownian motion noise.

The Stratonovich integral takes the function value in the middle of the Brownian motion process. This is actually intuitive however for more realistic processes, which occur in finite time. There is some correlation time between the function value which changes slowly and the Brownian motion which changes faster but still at a finite speed. So the Stratonovich integral also has physical meaning. When this is distilled into an equation, the result is:

$$dX_t = f dt + \sigma dB_t$$

where the function value f is not uncorrelated to B . This is a differential equation, so Stratonovich calculus obeys the product and chain rules like ordinary calculus just as DEs do. However, it loses its martingale property.

42.10 Diffusion

42.10.1 Fokker-Planck equation

Describes dynamical probability, $p(x, t)$. Also called the Kolmogorov forward equation.

Fokker-Planck for standard Brownian motion. $B_t - B_s \sim N(0, t - s)$. Further, $X_t = B_t \rightarrow f(X_t)$, $dX_t = dB_t$, where f is arbitrary. Use Ito's lemma

$$df(x) = f'(x)dX_t + \frac{1}{2}f''(x)dX_t^2$$

with Brownian motion $dX_t = dB_t$, $dX_t^2 = dt$:

$$df(x) = f'(x)dB_t + \frac{1}{2}f''(x)dt$$

$$\begin{aligned} E(df(x)) &= E\left(f'(x)dB_t + \frac{1}{2}f''(x)dt\right) \\ &= \frac{1}{2}E(f''(x)dt), \text{ since } E(dB_t) = 0 \\ \frac{d}{dt}E(f(x)) &= \frac{1}{2}E(f''(x)) \end{aligned}$$

with $p(x, t)$:

$$\begin{aligned}
\frac{d}{dt} \int_{-\infty}^{\infty} f(x) p(x, t) dx &= \frac{1}{2} \int_{-\infty}^{\infty} f''(x) p(x, t) dx \\
&= \frac{1}{2} \left(f'(x) p(x, t) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x) \frac{\partial p(x, t)}{\partial x} dx \right) \text{ integration by parts} \\
&= 0 - \frac{1}{2} \int_{-\infty}^{\infty} f'(x) \frac{\partial p(x, t)}{\partial x} dx \\
&= -\frac{1}{2} \left(\underbrace{f(x) \frac{\partial p(x, t)}{\partial x} \Big|_{-\infty}^{\infty}}_{=0} - \int_{-\infty}^{\infty} f(x) \frac{\partial^2 p(x, t)}{\partial x^2} dx \right) \\
\frac{d}{dt} \int_{-\infty}^{\infty} f(x) p(x, t) dx &= \frac{1}{2} \int_{-\infty}^{\infty} f(x) \frac{\partial^2 p(x, t)}{\partial x^2} dx \\
\int_{-\infty}^{\infty} f(x) \frac{\partial p(x, t)}{\partial t} dx &= \frac{1}{2} \int_{-\infty}^{\infty} f(x) \frac{\partial^2 p(x, t)}{\partial x^2} dx \\
\int_{-\infty}^{\infty} f(x) \underbrace{\left(\frac{\partial p(x, t)}{\partial t} - \frac{1}{2} \frac{\partial^2 p(x, t)}{\partial x^2} \right)}_{\text{Fokker-Planck for BM}} dx &= 0
\end{aligned}$$

where the Fokker-Planck equation is just the diffusion equation.

Fokker-Planck for SDE

$$dX_t = \mu(x_t, t)dt + \sigma(x_t, t)dB_t$$

Apply Ito's lemma

$$df(x) = f'(x)dX_t + \frac{1}{2}f''(x)dX_t^2$$

to the SDE,

$$\begin{aligned}
df(x) &= \left(\mu f'(x) + \frac{1}{2} \sigma^2 f''(x) \right) dX_t + f'(x) \sigma dB_t \\
E(df) &= E \left(\left(\mu f'(x) + \frac{1}{2} \sigma^2 f''(x) \right) dt + f'(x) \sigma dB_t \right) \\
&= E \left(\mu f'(x) + \frac{1}{2} \sigma^2 f''(x) \right) dt \\
\frac{d}{dt} E(f(x)) &= E \left(\mu f'(x) + \frac{1}{2} \sigma^2 f''(x) \right) \\
\frac{d}{dt} \int_{-\infty}^{\infty} f(x) p(x, t) dx &= \int_{-\infty}^{\infty} \mu(x, t) f'(x) + \frac{1}{2} \sigma^2(x, t) f''(x) p(x, t) dx \\
&= - \int_{-\infty}^{\infty} f(x) \frac{\partial}{\partial x} (\mu(x, t) p(x, t)) dx - \frac{1}{2} \int_{-\infty}^{\infty} f'(x) \frac{\partial}{\partial x} \sigma^2 p(x, t) dx
\end{aligned}$$

where the second step is the result of integration by parts twice. Now,

$$\frac{d}{dt} \int_{-\infty}^{\infty} f(x)p(x,t)dx = \int_{-\infty}^{\infty} f(x) \frac{\partial p(x,t)}{\partial t} dt$$

So,

$$\int_{-\infty}^{\infty} f(x) \left(\frac{\partial p(x,t)}{\partial t} + \frac{\partial}{\partial x} (\mu(x,t)p(x,t)) - \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2(x,t)p(x,t)) \right) dx = 0$$

So the SDE Fokker-Planck is

$$\frac{\partial p(x,t)}{\partial t} + \frac{\partial}{\partial x} (\mu(x,t)p(x,t)) - \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2(x,t)p(x,t)) = 0$$

42.10.2 Kolmogorov backward equation

In contrast to the Kolmogorov forward equation which calculates probabilities at the present location and time, x, t , the backward equation considers how the probabilities change at the initial or previous x_0, t_0 . So,

$$\lim_{h \rightarrow 0} \frac{p(x, t|x_0, t_0 - h) - p(x, t|x_0, t_0)}{h} = - \frac{\partial p(x, t|x_0, t_0)}{\partial t_0}$$

To calculate the first term, start with the Chapman-Kolmogorov transition probability

$$p(x, t|x_0, t_0) = \int p(x, t|y, t)p(dy, t|x_0, t_0)$$

and lag this by time h and assume for now that x is discrete (i.e. moves up and down by 1 unit only)

$$p(x, t|x_{-1}, t_0 - h) = \int p(x, t|y, t)p(dy, t|x_{-1}, t_0 - h)$$

Since it's arbitrary to consider where the initial location is, x_{-1} can be renamed x_0 :

$$p(x, t|x_0, t_0 - h) = \int p(x, t|y, t)p(dy, t|x_0, t_0 - h)$$

Returning to the above equation,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{p(x, t|x_0, t_0 - h) - p(x, t|x_0, t_0)}{h} \\ &= \frac{1}{h} \left(\int p(x, t|y, t_0)p(dy, t_0|x_0, t_0 - h) \right) - \frac{1}{h} p(x, t|x_0, t_0) \end{aligned}$$

The latter term is

$$p(x, t|x_0, t_0) = p(x, t|x_0, t_0) \underbrace{\int p(dy, t_0|x_0, t_0 - h)}_1$$

So,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{p(x, t|x_0, t_0 - h) - p(x, t|x_0, t_0)}{h} \\ &= \frac{1}{h} \int (p(x, t|y, t_0) - p(x, t|x_0, t_0)) p(dy, t_0|x_0, t_0 - h) \end{aligned}$$

where the bracketed difference is the difference in probabilities when the initial locations are different at the same starting time but end up in the same x, t . So it is nothing more than $\|y - x_0\|$, which can be Taylor expanded

$$\begin{aligned} & p(x, t|x_0, t_0 - h) - p(x, t|x_0, t_0) \Big|_{\|y - x_0\|} \\ &= \frac{\partial p(x, t|x_0, t_0)}{\partial x_0} (y - x_0) + \frac{1}{2} \frac{\partial^2 p(x, t|x_0, t_0)}{\partial x_0^2} (y - x_0)^2 + \dots \end{aligned}$$

Ignore higher terms. Then,

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{p(x, t|x_0, t_0 - h) - p(x, t|x_0, t_0)}{h} \\ &= \frac{1}{h} \int (p(x, t|y, t_0) - p(x, t|x_0, t_0)) p(dy, t_0|x_0, t_0 - h) \\ &= \frac{1}{h} \int \left(\frac{\partial p(x, t|x_0, t_0)}{\partial x_0} (y - x_0) + \frac{1}{2} \frac{\partial^2 p(x, t|x_0, t_0)}{\partial x_0^2} (y - x_0)^2 \right) p(dy, t_0|x_0, t_0 - h) \end{aligned}$$

The first term is

$$\begin{aligned} &= \frac{1}{h} \int \left(\frac{\partial p(x, t|x_0, t_0)}{\partial x_0} (y - x_0) \right) p(dy, t_0|x_0, t_0 - h) \\ &= \frac{\partial p(x, t|x_0, t_0)}{\partial x_0} \lim_{h \rightarrow 0} \frac{1}{h} \int (y - x_0) p(dy, t_0|x_0, t_0 - h) \\ &= \frac{\partial p(x, t|x_0, t_0)}{\partial x_0} \mu(x_0, t_0) \end{aligned}$$

The second term is

$$\begin{aligned} &= \frac{1}{h} \int \left(\frac{1}{2} \frac{\partial^2 p(x, t|x_0, t_0)}{\partial x_0^2} (y - x_0)^2 \right) p(dy, t_0|x_0, t_0 - h) \\ &= \frac{1}{2} \frac{\partial^2 p(x, t|x_0, t_0)}{\partial x_0^2} \lim_{h \rightarrow 0} \frac{1}{h} \int (y - x_0)^2 p(dy, t_0|x_0, t_0 - h) \\ &= \frac{1}{2} \frac{\partial^2 p(x, t|x_0, t_0)}{\partial x_0^2} \sigma^2(x_0, t_0) \end{aligned}$$

Combining the above two terms,

$$-\frac{\partial p(x, t|x_0, t_0)}{\partial t_0} = \frac{\partial p(x, t|x_0, t_0)}{\partial x_0} \mu(x_0, t_0) + \frac{1}{2} \frac{\partial^2 p(x, t|x_0, t_0)}{\partial x_0^2} \sigma^2(x_0, t_0)$$

42.10.3 Kolmogorov forward versus backward

Kolmogorov forward (Fokker-Planck) and backward equations are adjoints of each other. If time dependence is removed, the forward equation operator is

$$L^*(x) = \mu(x) \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2(x) \frac{\partial^2}{\partial x^2}$$

The backward operator is

$$L(x) = -\frac{\partial}{\partial x} \mu(x) + \frac{1}{2} \frac{\partial^2}{\partial x^2} \sigma^2(x)$$

The operators indicate the small change in probability densities over different positions. In the forward equation, x_0, t_0 are constant. In the backward equation, x, t are constant. However, both equations are written with x, t confusingly. Adjoints are generalized transposes. A dot product is

$$\langle u, v \rangle = \sum uv$$

Generalizing,

$$\langle f, g \rangle = \int f(x)g(x)dx$$

$$\langle Au, v \rangle = (Au)'v = (u'A')v = u'(A'v) = \langle u, A'v \rangle$$

Let u and v be vectors and u lie above v . If A rotates vector u clockwise (resulting in Au) closer to v then the dot product Au gets larger. To get the same effect operating on v , it must be that there is a linear transformation that rotates v in the opposite direction, i.e. counterclockwise, closer to u to get the same dot product. That linear transformation is the opposing rotation A' or the adjoint (the transpose in this simple case).

The Kolmogorov forward and backward equations are adjoints of each other:

$$\langle Lf, g \rangle = \langle f, L^*g \rangle$$

The former takes probability f back in time with L and then multiplies it with the spatial function g . The latter takes the probability and multiplies it with the spatial function taken forward in time by L^* . This can be confirmed by straightforward algebra.

42.10.4 Generators

Let $X(t)$ satisfy

$$dX(t) = \mu(X(t), t)dt + \sigma(X(t), t)dB(t)$$

and L_t is the generator of $X(t)$, i.e.

$$L_t f(x, t) = (L_t f)(x, t) = \frac{1}{2} \sigma^2(x, t) \frac{\partial^2 f}{\partial x^2}(x, t) + \mu(x, t) \frac{\partial f}{\partial x}(x, t)$$

then

$$df(X(t), t) = \frac{\partial f}{\partial x}(X(t), t)dX(t) + \frac{\partial f}{\partial t}(X(t), t)dt + \frac{1}{2}\sigma^2(t)\frac{\partial^2 f}{\partial x^2}(X(t), t)dt$$

and replacing $dX(t)$ with the above,

$$df(X(t), t) = \left(L_t f(X(t), t) + \frac{\partial f}{\partial t}(X(t), t) \right) dt + \frac{\partial f}{\partial x}(X(t), t)\sigma(X(t), t)dB(t)$$

Note the generator of Brownian motion

$$L = \frac{1}{2} \frac{d^2}{dx^2} \cdot \Delta = \frac{d^2}{dx^2}$$

is the Laplacian.

The generator of a time-homogenous Markov process is a linear operator

$$Lf(x) = \lim_{t \rightarrow 0} \frac{E(f(X(t)) | X_0 = x) - f(x)}{t}$$

If the limit exists, f is said to be in the domain of the generator.

If the generator is assumed 0 (i.e. no local distortions), then we can get a martingale. First,

$$Lf = 0$$

$$\frac{1}{2}\sigma^2(x)f''(x) + \mu(x)f'(x) = 0$$

With $h = f'$,

$$\frac{h'}{h} = -\frac{2\mu(x)}{\sigma^2(x)}$$

so,

$$f'(x) = e^{-\int \frac{2\mu(y)}{\sigma^2(y)} dy}$$

and integrating,

$$f(x) = \frac{C}{\sigma^2(x)} e^{-\int \frac{2\mu(y)}{\sigma^2(y)} dy}$$

where C is found from

$$\int f(x)dx = 1$$

42.10.5 Martingales

If f is bounded, and Ito's formula is

$$f(B(t)) = f(0) + \int_0^t \frac{1}{2} f''(B(s))ds + \int_0^t f'(B(s))dB(s)$$

then the last term is a martingale $M(t)$. Then,

$$f(B(t)) = f(0) + \int_0^t \frac{1}{2} f''(B(s)) ds + M(t)$$

and

$$M(t) = f(B(t)) - f(0) - \int_0^t \frac{1}{2} f''(B(s)) ds$$

is also a martingale. So, the above equation

$$df(X(t), t) = \left(L_t f(X(t), t) + \frac{\partial f}{\partial t}(X(t), t) \right) dt + \frac{\partial f}{\partial x}(X(t), t) \sigma(X(t), t) dB(t)$$

becomes

$$f(X(t), t) = \int_0^t \left(L_u f + \frac{\partial f}{\partial t} \right) (X(u), u) du + M(t)$$

and so

$$M(t) = f(X(t), t) - \int_0^t \left(L_u f + \frac{\partial f}{\partial t} \right) (X(u), u) du$$

42.10.6 Dynkin's formula

In the above process where f is bounded,

$$Ef(X(t), t) = f(X(0), 0) + E \int_0^t \left(L_u f + \frac{\partial f}{\partial t} \right) (X(u), u) du$$

42.10.7 Solutions to Kolmogorov backward equation

Let $f(x, t)$ solve the backward equation, with L_t given by

$$L_t f(x, t) + \frac{\partial f}{\partial t}(x, t) = 0$$

with $f(x, T) = g(x)$. If $f(x, t)$ is a martingale, then

$$f(x, t) = E(g(X(T)) | X(t) = x)$$

Proof. The martingale property gives

$$E(f(X(T), T) | \mathcal{F}_t) = f(X(t), t)$$

On the boundary $f(x, T) = g(x)$, so that $f(X(T), T) = g(X(T))$ and

$$f(X(t), t) = E(g(X(T)) | \mathcal{F}_t)$$

According to the Markov property of $X(t)$,

$$f(X(t), t) = E(g(X(T)) | X(t))$$

which completes the proof.

42.10.8 Feynman-Kac

Generalizes the above result on solutions to the Kolmogorov backward equation.

Links PDEs and stochastic processes. For given bounded functions $r(x, t), g(x)$, let

$$C(x, t) = E \left(e^{-\int_t^T r(X(u), u) du} g(X(T)) \mid X(t) = x \right)$$

Assume that there is a solution to

$$\frac{\partial f}{\partial t}(x, t) + L_t f(x, t) = r(x, t)f(x, t)$$

with

$$f(x, T) = g(x)$$

Then the solution is unique and $C(x, t)$ is that solution.

Proof. If f is a solution, use Ito's formula

$$df(X(t), t) = \left(\frac{\partial f}{\partial t}(X(t), t) + L_t f(X(t), t) \right) dt + \frac{\partial f}{\partial x}(X(t), t) \sigma(X(t), t) dB(t)$$

The last term is a martingale, so rewrite as $dM(t)$. Use

$$\frac{\partial f}{\partial t}(x, t) + L_t f(x, t) = r(x, t)f(x, t)$$

to get

$$df(X(t), t) = r(X(t), t)f(X(t), t)dt + dM(t)$$

which is a Langevin-type SDE, so use the above result where $B(t)$ is replaced by $M(t)$.

$$f(X(T), T) = f(X(t), t)e^{\int_t^T r(X(u), u) du} + e^{\int_t^T r(X(u), u) du} \int_t^T e^{\int_t^s r(X(u), u) du} dM(s)$$

Since

$$f(x, T) = g(x)$$

then

$$f(X(T), T) = g(X(T))$$

So, rearranging the above,

$$g(X(T))e^{-\int_t^T r(X(u), u) du} = f(X(t), t) + \int_t^T e^{\int_t^s r(X(u), u) du} dM(s)$$

As the last term is in the integral of a bounded function with respect to the martingale, it is itself a martingale with zero mean. Taking expectation given $X(t) = x$, $C(x, \cdot) = f(x, \cdot)$.

42.10.9 Exit times from an interval

For the time-homogenous SDE

$$df(X(t)) = \mu(X(t))dt + \sigma(X(t))dB(t)$$

there is no separate t term, i.e. it is not

$$dX(t) = \mu(X(t), t)dt + \sigma(X(t), t)dB(t)$$

So,

$$df(X(t)) = Lf(X(t))dt + f'(X(t))\sigma(X(t))dB(t)$$

Since the integral of the second term is a martingale,

$$f(X(t)) = \int Lf(X(t))dt + M(t)$$

$$M(t) = f(X(t)) - \int Lf(X(t))dt$$

and

$$f(x) = \left(f(X(t)) - \int Lf(X(t))dt \right)$$

but assuming

$$Lf = 0$$

as in the generator is 0, then

$$Ef(X(t)) = f(x)$$

So then

$$p(\tau_b < \tau_a) = \frac{f(x) - f(a)}{f(b) - f(a)}$$

Define τ to be the first time the diffusion exits the interval (a, b) . $\tau = \min(T_a, T_b) = T_a \wedge T_b$.

42.10.10 Recurrence and transience

Recurrence. A point x is recurrent for diffusion $X(t)$ if the probability of the process coming back to x infinitely often is one, that is,

$$p(X(t) = x \text{ for a sequence of } t\text{'s increasing to infinity}) = 1$$

Transience. A point x is transient for diffusion $X(t)$ if

$$p(\lim_{t \rightarrow \infty} |X(t)| = \infty) = 1$$

If all points of a diffusion are recurrent, the diffusion is recurrent. If all points of a diffusion are transient, the diffusion is transient.

Theorem. Let $X(t)$ be a diffusion satisfying existence and uniqueness conditions, i.e. $\mu(x)$ and $\sigma(x)$ are bounded on finite intervals, $\sigma(x)$ is continuous and positive, and $\mu(x)$, $\sigma(x)$ satisfy the linear growth condition. Then, (1) if there is one recurrent point, then all points are recurrent, and (2) if there are no recurrent points, then the diffusion is transient.

42.10.11 Stationary distributions and ergodicity

For the diffusion process given by the SDE

$$dX(t) = \mu(X(t))dt + \sigma(X(t))dB(t)$$

with $X(0)$ having a distribution

$$\nu(x) = p(X(t) \leq x)$$

This distribution is *stationary* or *invariant* for the diffusion process $X(t)$ if for any t the distribution of $X(t)$ is the same as $\nu(x)$. If $p(t, x, y) = p(X(t) \leq y | X(0) = x)$ denotes the transition probability function of $X(t)$, then

$$p(X(0) \leq y) = p(X(t) \leq y) = \int p(X(t) \leq y | X(0) = x) d\nu(x)$$

and

$$\nu(y) = \int p(x, t, y) d\nu(x)$$

Stationarity means finite-dimensional distributions do not change with shifts in time. for Gaussian processes stationarity is equivalent to the covariance function to be a function $|t - s|$ only, i.e. $Cov(X(t), X(s)) = h(|t - s|)$.

If the generator is assumed 0, then

$$Lf = 0$$

$$\frac{1}{2}\sigma^2(x)f''(x) + \mu(x)f'(x) = 0$$

With $h = f'$,

$$\frac{h'}{h} = -\frac{2\mu(x)}{\sigma^2(x)}$$

so,

$$f'(x) = e^{-\int \frac{2\mu(y)}{\sigma^2(y)} dy}$$

and integrating,

$$f(x) = \frac{C}{\sigma^2(x)} e^{-\int \frac{2\mu(y)}{\sigma^2(y)} dy}$$

where C is found from

$$\int f(x) dx = 1$$

which indicates stationary distributions with 0 generators are Gaussian-like.

Stationary distributions are finite invariant measures, and they may exist only if a diffusion is recurrent. A diffusion is recurrent and admits a stationary distribution if, and only if, an expected hitting time is finite. When this property holds, diffusion is also called ergodic or positive recurrent. Brownian motion is *not* stationary as it drifts around. The Ornstein-Uhlenbeck process is the *only* process that is Gaussian, Markov, and stationary.

42.11 Martingales

42.11.1 Local time

Tanaka's formula. Let $X(t)$ be a continuous semimartingale. then for any a there exists a continuous non-decreasing adapted process $L^a(t)$, the local time at a of X , such that

$$|X(t) - a| = |X(0) - a| + \int_0^t \text{sign}(X(s) - a) dX(s) + L^a(t)$$

Local time is formally defined as

$$\begin{aligned} L^a(t) &= \int_0^t \delta(a - B(s)) d[B, B](s) \\ &= \lim_{\epsilon \rightarrow 0} \frac{1}{2\epsilon} \int_0^t 1_{a-\epsilon < B(s) < a+\epsilon} d[B, B](s) \\ &= \int_0^t 1_a X(s) ds \end{aligned}$$

Rewriting Tanaka,

$$L^a(t) = |X(t) - a| - |X(0) - a| - \int_0^t (1_{\text{positive}}(X(t) - a) - 1_{\text{negative}}(X(t) - a)) dX$$

Intuition. Taking $A = (a, a + da)$ and $g(x) = I_{(a, a+da)}(x)$ its indicator, $L^a(t)da$ is the time B spends in $(a, a + da)$ up to time t . Local time is the amount of time B spends near a .

Occupation time.

$$\begin{aligned} \int_0^t g(X(s)) d[X, X](s) &= \int_{-\infty}^{\infty} g(a) L^a(t) da \\ [X, X](t) &= \int_{-\infty}^{\infty} L^a(t) da \end{aligned}$$

42.12 Change of measure

42.12.1 Definitions

Equivalence. Let $\Omega = (\omega_1, \omega_2)$ with probability measure P as $P(\omega_1) = p, P(\omega_2) = 1 - p$, and probability measure Q as $Q(\omega_1) = q, Q(\omega_2) = 1 - q$. Q is equivalent to P ($Q \sim P$) if they have the same null sets, i.e. $Q(A) = 0$ iff $P(A) = 0$.

Absolute continuity. Q is absolutely continuous with respect to P $Q \ll P$ if $Q(A) = 0$ whenever $P(A) = 0$. P and Q are equivalent if $Q \ll P$ and $P \ll Q$.

42.12.2 Normal random variables

Let μ be any real number, $f_\mu(x)$ denote the probability density of $\mathcal{N}(\mu, 1)$ distribution, and P_μ the $\mathcal{N}(\mu, 1)$ probability measure. Then,

$$f_\mu(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\mu)^2} = f_0(x) e^{\mu x - \frac{\mu^2}{2}}$$

So,

$$f_\mu = f_0 e^{\frac{1}{2}(x-\mu)^2} = f_0 \Lambda(x)$$

where

$$\Lambda(x) = e^{\mu x - \frac{\mu^2}{2}}$$

Since

$$dP = P(dx) = f(x)dx$$

$$P_\mu(dx) = \Lambda(x)P_0(dx)$$

and

$$\Lambda(x) = \frac{dP_\mu}{dP_0}(x) = e^{\mu x - \frac{\mu^2}{2}}$$

42.12.3 Removing the mean or drift

Let $X \sim \mathcal{N}(0, 1)$ under probability measure P and $Y = X + \mu$. Then there is an equivalent probability $Q \sim P$, s.t. $Y \sim \mathcal{N}(0, 1)$ under probability measure Q .

$$\frac{dQ}{dP}(X) = \Lambda(X) = e^{-\mu X - \frac{\mu^2}{2}}$$

42.12.4 Change of normal probability measures

Let $X \sim N(\mu_1, \sigma_1^2)$ under probability measure P . Define Q by

$$\frac{dQ}{dP}(X) = \Lambda(X) = \frac{\sigma_1}{\sigma_2} e^{\frac{(X-\mu_1)^2}{2\sigma_1^2} - \frac{(X-\mu_2)^2}{2\sigma_2^2}}$$

Then $X \sim N(\mu_2, \sigma_2^2)$ under probability measure Q .

42.12.5 Radon-Nikodym

If $Q \ll P$ then there exists a random variable Λ s.t.

$$Q(A) = E_P(\Lambda I(A)) = \int_A \Lambda dP$$

Conversely, if Q is defined as above, then it is a probability measure and $Q \ll P$.

42.12.6 Girsanov's theorems

Girsanov's theorem for Brownian motion. Let $B(t)$ be a Brownian motion $0 \leq t \leq T$ under probability measure P . Consider $W(t) = B(t) + \mu t$. Define measure Q by

$$\Lambda = \frac{dQ}{dP}(B(T)) = e^{-\mu B(T) - \frac{1}{2}\mu^2 T}$$

Then Q is equivalent to P , and $W(t)$ is a Q -Brownian motion.

$$\Lambda = \frac{dQ}{dP}(W(T)) = e^{-\mu(W(T) - \mu T) - \frac{1}{2}\mu^2 T} = e^{-\mu W(T) + \frac{1}{2}\mu^2 T}$$

$$\frac{1}{\Lambda} = \frac{dP}{dQ}(W(T)) = e^{\mu W(T) - \frac{1}{2}\mu^2 T}$$

Intuition.

$$X_n = \sum_{k=1}^N \eta_k$$

where $\eta \sim N(0, 1)$. Then,

$$p(x_1, x_2, \dots, x_N) = C e^{-\frac{1}{2}(x_1^2 + (x_2 - x_1)^2 + \dots + (x_N - x_{N-1})^2)}$$

now let $\eta \sim N(\mu_n, 1)$. So the average is moving with each step n . Then,

$$p(x_1, x_2, \dots, x_N) = C e^{-\frac{1}{2}((x_1 - \mu_1)^2 + (x_2 - x_1 - \mu_2)^2 + \dots + (x_N - x_{N-1} - \mu_N)^2)}$$

Then by Radon-Nikodym,

$$\begin{aligned} \frac{dQ}{dP} &= e^{-(\mu_1 x_1 - \mu_2(x_2 - x_1) - \dots - \mu_N(x_N - x_{N-1})) + \frac{1}{2}(\mu_1^2 + \mu_2^2 + \dots + \mu_N^2)} \\ &= e^{-\sum \mu_k(x_k - x_{k-1}) - \frac{1}{2}\sum \mu_k^2} \\ &= e^{-\mu B(t) - \frac{1}{2}\mu^2 t} \end{aligned}$$

where the last step takes $n \rightarrow \infty$ up to time t .

Girsanov's theorem for removing drift. Let $B(t)$ be a P -Brownian motion and $H(t)$ s.t.

$$X(t) = - \int_0^t H(s) dB(s)$$

Define equivalent probability measure Q by

$$\Lambda = \frac{dQ}{dP}(B) = e^{-\int_0^T H(s) dB(s) - \frac{1}{2} \int_0^T H^2(s) ds}$$

Then the process

$$W(t) = B(t) + \int_0^t H(s) ds$$

is a Q-Brownian motion.

Change diffusion drift. Let

$$dX(t) = \mu_1(X(t), t)dt + \sigma(X(t), t)dB(t)$$

$$H(t) = \frac{\mu_1(X(t), t) - \mu_2(X(t), t)}{\sigma(X(t), t)}$$

$$\Lambda = \frac{dQ}{dP} = e^{-\int_0^T H(t)dB(t) - \frac{1}{2}\int_0^T H^2(t)dt}$$

Then

$$\begin{aligned} dW(t) &= dB(t) + H(t)dt \\ &= dB(t) + \frac{\mu_1(X(t), t) - \mu_2(X(t), t)}{\sigma(X(t), t)}dt \end{aligned}$$

and so

$$dX(t) = \mu_2(X(t), t)dt + \sigma(X(t), t)dW(t)$$

So the change in probability measure results in a change in drift.

42.12.7 Likelihood functions

When observations X are made from competing models, the likelihood is

$$\Lambda = \frac{Q(X = x)}{P(X = x)}$$

Example.

$$H_0 : dX(t) = dB(t)$$

$$H_a : \text{signal+noise} = dX(t) = h(t)dt + dB(t)$$

So $\mu_1(x) = 0, \mu_2(x) = h(t), \sigma(x) = 1$. The likelihood is

$$\Lambda(X) = \frac{dQ}{dP} = e^{\int_0^T h(t)dX(t) - \frac{1}{2}\int_0^T h^2(t)dt}$$

which leads to the likelihood ratio test

$$P(e^{\int_0^T h(t)dX(t) - \frac{1}{2}\int_0^T h^2(t)dt} \geq k) = \alpha$$

Remark. If X is the same except $\sigma_1 \neq \sigma_2$, then the processes are singular and by looking at them long enough can decide precisely which equation from which it came.

43 Econometrics

43.1 Variable transformations

Variables that appear in a proportion or percent form, such as the unemployment rate, the participation rate in a pension plan, the percentage of students passing a standardized exam, and the arrest rate on reported crimes - can appear in either the original or logarithmic form, although there is a tendency to use them in level forms. This is because any regression coefficients involving the original variable - whether it is the dependent or the independent variable - will have a percentage point change interpretation. If we use, say, $\log(\text{unem})$ in a regression, where unem is the percentage of unemployed individuals, we must be very careful to distinguish between a percentage point change and a percentage change. Remember, if unem goes from 8 to 9, this is an increase of one percentage point, but a 12.5 percent increase from the initial unemployment level. Using the log means that we are looking at the percentage change in the unemployment rate: $\log(9) - \log(8) = 0.118$ or 11.8 percent, which is the logarithmic approximation to the actual 12.5 percent increase.

43.2 Gauss-Markov Theorem

OLS provides BLUE coefficients under these assumptions:

- (1) True model is linear - LINEARITY
- (2) Non-singular design matrix - ESTIMATION IS POSSIBLE
- (3) Mean independence - UNBIASEDNESS - this still allows higher-moment dependence between the regressor and the error, though the second moment is addressed with (4) no heteroskedasticity
- (4) No autocorrelation or heteroskedasticity - BEST, MOST EFFICIENT

(1) is argued, (2) is just a solution limit, (3) is tested through exogeneity and endogeneity, and (4) is tested.

43.3 Quantile Regression

Useful in small samples where they can be proportionally more outliers. Least squares is sensitive to outliers. Also useful if error is not distributed symmetrically or Gaussian, or where more flexible inference is desirable.

43.4 Least Absolute Deviation

Originated before and supplanted by least squares. The LAD estimator is:

$$\underset{\beta(\tau)}{\operatorname{argmin}} \sum_i^n |y_i - x\beta(\tau)|$$

Finds the median estimate through the data and $\tau = 0.5$ (the 50th quantile or percentile).

LAD can be generalized with the quantile regression. Minimize:

$$\underset{\beta(\tau)}{\operatorname{argmin}} \sum_i^n \rho(y_i - x\beta(\tau))$$

Where $\tau \in [0, 1]$. Solve with:

$$\rho = \tau \max(u, 0) + (1 - \tau) \max(-u, 0) \geq 0$$

Which biases the weights - e.g. for a 10th quantile regression, want 90 percent of errors to be positive (above the quantile line) and 10 percent negative. So bias ρ so that positive errors have 0.1 weight and negative errors have 0.9 weight.

44 Time Series

44.1 Autocorrelation

Positive autocorrelation is more common and manifests as slower oscillations around the trend. Negative autocorrelation is less common and manifests as faster oscillations around the trend. Autocorrelation is more likely to be a problem with shorter time intervals. With longer time intervals, there is more time for autocorrelation effects to decline.

A lagged dependent variable will lead to biased coefficient estimates in the presence of autocorrelation.

$$y_t = b_0 + b_1 x_t + b_2 y_{t-1} + u_t$$

$$y_{t-1} = b_0 + b_1 x_{t-1} + b_2 y_{t-2} + u_{t-1}$$

If the error is AR(1):

$$u_t = \rho u_{t-1} + e_t$$

Then both y_t and u_t depend on u_{t-1} :

$$\operatorname{Cov}(y_t, u_t) \neq 0$$

If there is no lagged dependent variable, regression coefficients remain unbiased because it's just the oscillation frequency changing around the trend. But the standard errors will be wrong as they will be biased downwards. However, OLS will no longer be BLUE because there are alternative regression techniques that are more efficient - that have lower variance coefficient estimators. OLS will be LUE.

44.1.1 Durbin-Watson Test

44.1.2 Breusch-Godfrey Test

44.1.3 Other Tests

44.2 Stationarity and Dependence

44.2.1 Stationarity

A time series is strongly stationary if the joint probability distribution of any pair of observations differ only by their distance and not their particular times. So the joint distribution $(x_1, x_2, \dots, x_t)$ is the same as the joint distribution of $(x_{1+h}, x_{2+h}, \dots, x_{t+h})$. Applies to all moments of the joint distribution.

A time series is weakly stationary (or covariance stationary) if the mean is constant and the autocovariance only differs by the distance in time, not particular times. Limited to first two moments, so the distribution can be changing through time.

Strict stationarity doesn't necessarily imply weak stationarity. Strict stationarity doesn't require finite variance, whereas weak stationarity does because autocovariance must be defined. When the autocovariance is just the variance (so the distance in time = 0), the variance must exist. So e.g. iid Cauchy is strictly stationary but not weakly stationary.

Weak stationarity usually doesn't imply strict stationarity except for Gaussian white noise because the distribution is fully characterized by the first two moments.

44.2.2 Dickey-Fuller and ADF

44.2.3 KPSS

44.2.4 Weak Dependence

<https://www.youtube.com/watch?v=7xJ8mT6gZb4>

44.3 Exogeneity and Causality

Exogeneity of a variable is with respect to its coefficient. Exogeneity and causality helps determine which variables can be analyzed separately or before others. They determine whether the joint distribution is factorized into conditional and marginal distributions.

44.3.1 Strict exogeneity

This is conditional mean independence, when u_t is independent of past, present, and future X s. Gauss-Markov requires strict exogeneity for OLS estimators to

be BLUE. Specifically, strict exogeneity leads OLS estimators to be unbiased.

$$\hat{\beta} = (X'X)^{-1}X'y = (X'X)^{-1}X'(X\beta + \epsilon) = \beta + (X'X)^{-1}X'\epsilon$$

With strict exogeneity $E(\epsilon|X) = 0$ and $E(\epsilon) = 0$ (i.e. an appropriate intercept due to the Gauss-Markov linearity assumption):

$$E(\hat{\beta}) = \beta + E((X'X)^{-1}X'\epsilon) = \beta + (X'X)^{-1}X'E(\epsilon) = \beta + 0 = \beta$$

Strict exogeneity is more restrictive than weak exogeneity.

44.3.2 Weak exogeneity

Weak exogeneity is when u_t is independent of X_t only. Weak exogeneity can't guarantee unbiasedness but can guarantee consistent OLS estimators. A biased estimator may be consistent and therefore still allow estimation at large sample sizes. Using $Cov(x_i, \epsilon_i) = 0 \Rightarrow E(x_i\epsilon_i) - E(x_i)E(\epsilon_i) = E(x_i\epsilon_i) - E(x_i)0 = E(x_i\epsilon_i)$:

$$plim \hat{\beta} = \beta + plim(X'X)^{-1}X'\epsilon = \beta + plim\left(\frac{X'X}{n}\right)^{-1} plim\frac{X'\epsilon}{n}$$

$$= \beta + (\text{Non-singular matrix})E(x_i\epsilon_i) = \beta + (\text{Non-singular matrix})0 = \beta$$

Weakly exogenous drivers can be used for coefficient estimation at large sample sizes as all potentially feed-forwarding effects (reverse-causation forward in time) are accounted.

Weak exogeneity is necessary but insufficient to forecast x before y . Because of potential feed-forwarding effects, x and y generally need to be forecasted simultaneously.

44.3.3 Feed-forwarding

This is reverse causality moving forward in time. If:

$$y_t = a_0 + a_1x_t + u_t$$

But $x_t = b_0 + b_1y_{t-1} + e_t$, then any impulse on x_t will have the following effect:

$$x_t \rightarrow y_t \rightarrow x_{t+1} \rightarrow y_{t+1}$$

So, coefficient a_1 will be upwardly biased as it will include positive forward feedback. u_t now contains a dependency on x_{t-1} - a hidden effect. A lagged dependent variable also causes forward feedback.

Even if errors are iid, a lagged dependent variable will lead e_t to effectively contain the effect of y_{t-2} during feed-forwarding.

$$y_t = \alpha y_{t-1} + e_t$$

$$y_{t-1} = \alpha y_{t-2} + e_{t-1}$$

Then $Cov(y_{t-1}, e_t) = 0$ in the first equation but $Cov(y_{t-1}, e_{t-1}) \neq 0$ from the second equation.

44.3.4 Endogeneity

Endogeneity is when weak exogeneity fails. It can arise from:

- Reverse causation
- Lagged dependent variable + autocorrelation
- Measurement error
- Reverse causality
- Omitted variable or uncontrolled confounder
- Simultaneous equations bias

44.3.5 Sequential exogeneity

Variables determined prior to the current period. The current period error term is uncorrelated with current and lagged values of the predetermined variable but may be correlated with future values. Also called predetermined variables.

$$E(u_{it}|x_{it}) \neq 0$$

Where $s < t$. Often used in panel methods.

44.3.6 Granger non-causality

Y_{t-1} does not Granger cause $z_t|X_{t-1}$ iff:

$$p(z_t|X_{t-1}, \theta) = p(z_t|Z_{t-1}, Y_0, \theta)$$

Put differently:

$$z_t \parallel Y_{t-1}|Z_{t-1}, Y_0$$

44.3.7 Strong exogeneity

Strong exogeneity is weak exogeneity + Granger non-causality. The coefficients are independent of the VALUES of x. Strongly exogenous drivers can be used for both estimation AND forecasting. They can be forecasted before, e.g. dependent variables, because there are no feed-forward effects. There may be feed-forwarding from lagged dependent variables, but these are still dependent variables forecasted after the strongly exogenous variables are forecasted.

44.3.8 Super exogeneity

Required for simulation and policy analysis. Related to the Lucas critique. Strong exogeneity allows forecasting but if policy changes, e.g. change in money growth supply rule, then the coefficients will change. To address this, super exogenous variables, which are more fundamental than strongly exogenous variables, are required. Super exogenous regressors have coefficients independent of the DATA GENERATING PROCESS of x .

Weak exogeneity + coefficient is invariant to marginal density of x .

44.4 Orthogonality

44.4.1 Orthogonal regression

If $y \sim X$ contains a constant and the X variables are uncorrelated to each other, the coefficients are the same as the coefficients in the individual simple regressions of $y \sim x_i$ in turn.

44.4.2 Orthogonality condition

Generates the least squares estimator:

$$E(x_i \epsilon_i) = 0$$

44.5 Outliers

- AO: Additive outlier. Isolated spike.
- LS: Level shift. Sudden change in mean, may be seasonal or not.
- TC: Temporary change. Spike that disappears after a few periods.
- IO: Intervention outlier. Innovation shock.

44.6 Dummies

Level and/or difference dummies can be added in level and/or difference regressions. The interpretations are different. Level dummies added to difference regressions are about slope changes for differences. Differenced dummies are pulses where level dummies are consecutive - they are +1 pulses at the beginning of the consecutive period and -1 at the end.

When interacting dummies with differenced variables, ensure the correct time period.

In statistical programming, if the '0' dummies must be treated as 'n/a', NOT 0's - otherwise, parameter estimates WILL BE AFFECTED. Dummies listed as

'n/a' cannot be differenced if there are an odd number of such entries. Nevertheless, the statistical program may treat '0' as 'n/a' depending on the situation. If there is any value other than '0' or '1' in the series, it will automatically not be a dummy series.

44.6.1 Cointegration

An exception to spurious regressions.

Macroeconomic time series of x_t and y_t usually contain non-stationary data because variables like GDP and population grow over time. So their distributions are changing over time. They tend to be I(1) or higher integration order. So $u_t = y_t - x_t\beta$ also tends to be I(1) or higher integration order. OLS is then no longer BLUE:

- u_t may have a time trend and/or be heteroskedastic - so the u_t distribution is changing over time. Errors are then often autocorrelated.
- Usually $Cov(x_t, u_t) \neq 0$ because two non-stationary variables are often correlated.

If however u_t is stationary, y_t and x_t are cointegrated. There is a long-run relationship. Test using ADF or other tests.

Despite cointegration, the error may be autocorrelated. The coefficient estimates remain unbiased as there is no lagged dependent variable. Moreover, there may be times where the error is persistently high or low, there may be omitted variables, etc. so the t-distribution is unknown. Get a more useful equation with ECMs.

44.6.2 ECMs

If the error is I(0) and y_t and x_t are I(1), there is cointegration:

$$u_t = y_t - a - bx_t$$

Transform the variables (but not the whole equation):

$$y_t = a + bx_t + u_t$$

$$y_t - y_{t-1} + y_{t-1} = a + bx_t - x_{t-1} + x_{t-1} + u_t$$

$$y_t - y_{t-1} = a + bx_t - x_{t-1} + x_{t-1} - y_{t-1} + u_t$$

$$y_t - y_{t-1} = a + bx_t - bx_{t-1} + bx_{t-1} - y_{t-1} + u_t$$

$$\Delta y_t = a + b\Delta x_t + (bx_{t-1} - y_{t-1}) + u_t$$

$$\Delta y_t = a - c + b\Delta x_t + (c + bx_{t-1} - y_{t-1}) + u_t$$

$$\Delta y_t = (a - c) + b\Delta x_t + (-y_{t-1} + c + bx_{t-1}) + u_t$$

$$\Delta y_t = \gamma + b\Delta x_t + \underbrace{(-1)(y_{t-1} - c - bx_{t-1})}_{\text{Amalgamated error}} + u_t$$

So the amalgamated error is decomposed. Since Δy_t and Δx_t are both stationary, the amalgamated error should also be stationary. However, u_t will remain autocorrelated. So try an ARDL:

$$\begin{aligned} y_t &= a + bx_t + cy_{t-1} + dx_{t-1} + u_t \\ y_t - y_{t-1} &= a + bx_t + bx_{t-1} - bx_{t-1} + cy_{t-1} - y_{t-1} + dx_{t-1} + u_t \\ \Delta y_t &= a - f + b\Delta x_t + f + (b + d)x_{t-1} + (c - 1)y_{t-1} + u_t \\ \Delta y_t &= (a - f) + b\Delta x_t + (c - 1)y_{t-1} + f + (b + d)x_{t-1} + u_t \\ \Delta y_t &= \gamma + b\Delta x_t + (c - 1) \left(y_{t-1} - \frac{-f}{c - 1} - \frac{-(b + d)}{c - 1} x_{t-1} \right) + u_t \\ \Delta y_t &= \gamma + b\Delta x_t + \underbrace{\theta(y_{t-1} - \alpha_0 - \alpha_1 x_{t-1})}_{\text{Amalgamated error}} + u_t \end{aligned}$$

If u_t is not autocorrelated, then OLS can be BLUE. Further, we keep separately the cointegrated $v_t = y_t - \alpha_0 - \alpha_1 x_{t-1}$ which is the long-run component:

$$\Delta y_t = \underbrace{b\Delta x_t}_{\text{Short-run}} + \underbrace{\theta(y_{t-1} - \alpha_0 - \alpha_1 x_{t-1})}_{\text{Long-run } v_t} + u_t$$

If $v_t > 0$ then y_{t-1} is above its long-run equilibrium value and should decline and vice versa - so the error v_t is 'corrected'. Therefore, $\theta < 0$. Further, $-1 < \theta < 0$ to prevent divergence. If $|\theta|$ is small, the correction is slow.

44.6.3 ECM with additional regressors

If z_t is I(1), then $a_2 = 0$ at equilibrium:

$$\begin{aligned} y_t &= a_0 + a_1 x_t + a_2 z_t + u_t \\ u_t &= y_t - a_0 - a_1 x_t - a_2 z_t \\ u_t &= y_t - a_0 - a_1 x_t \end{aligned}$$

As before. Now transform:

$$\begin{aligned} y_t &= a + bx_t + cy_{t-1} + dx_{t-1} + \beta z_t + u_t \\ y_t - y_{t-1} &= a + bx_t + bx_{t-1} - bx_{t-1} + cy_{t-1} - y_{t-1} + dx_{t-1} + \beta z_t - \beta z_{t-1} + \beta z_{t-1} + u_t \\ \Delta y_t &= a - f + b\Delta x_t + f + (b + d)x_{t-1} + (c - 1)y_{t-1} + \beta \Delta z_t + \beta z_{t-1} + u_t \\ \Delta y_t &= (a - f) + b\Delta x_t + (c - 1)y_{t-1} + f + (b + d)x_{t-1} + \beta \Delta z_t + \beta z_{t-1} + u_t \\ \Delta y_t &= \gamma + b\Delta x_t + \beta \Delta z_t + (c - 1) \left(y_{t-1} - \frac{-f}{c - 1} - \frac{-(b + d)}{c - 1} x_{t-1} - \frac{-\beta}{c - 1} z_{t-1} \right) + u_t \end{aligned}$$

$$\Delta y_t = \gamma + b\Delta x_t + \beta\Delta z_t + \theta(y_{t-1} - a_0 - a_1x_{t-1} - a_2z_{t-1}) + u_t$$

If z_t is just a short-term regressor, then $a_2 = 0$ and:

$$\Delta y_t = \gamma + b\Delta x_t + \beta\Delta z_t + \theta(y_{t-1} - a_0 - a_1x_{t-1}) + u_t$$

So short-term regressors of proper integration order can be simply added to the ECM.

44.6.4 ECM with more lags

Derive an ECM with 2 lags each on x and y :

$$y_t = a + bx_t + cy_{t-1} + dx_{t-1} + b_0y_{t-2} + b_1x_{t-2} + u_t$$

$$y_t - y_{t-1} = a + bx_t + bx_{t-1} - bx_{t-1} + cy_{t-1} - y_{t-1} + dx_{t-1} + b_0y_{t-2} - b_0y_{t-1} + b_0y_{t-1} + b_1x_{t-2} - b_1x_{t-1} + b_1x_{t-1} + u_t$$

$$\Delta y_t = a - f + b\Delta x_t + f + (b+d)x_{t-1} + (c-1)y_{t-1} - b_0\Delta y_{t-1} + b_0y_{t-1} - b_1\Delta x_{t-1} + b_1x_{t-1} + u_t$$

$$\Delta y_t = (a-f) + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} + (c-1+b_0)y_{t-1} + f + (b+d+b_1)x_{t-1} + u_t$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} + (c-1+b_0) \left(y_{t-1} - \frac{-f}{c-1+b_0} - \frac{-(b+d+b_1)}{c-1+b_0} x_{t-1} \right) + u_t$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} + \theta(y_{t-1} - \alpha_0 - \alpha_1x_{t-1}) + u_t$$

This can also be derived by starting with

$$\Delta y_t = \gamma + b\Delta x_t + \theta(y_{t-1} - c_0 - c_1x_{t-1}) + u_t$$

And adding y_{t-2} and x_{t-2} :

$$\Delta y_t = \gamma + b\Delta x_t + \theta(y_{t-1} - c_0 - c_1x_{t-1}) + u_t + b_0y_{t-2} + b_1x_{t-2}$$

$$\Delta y_t = \gamma + b\Delta x_t + \theta(y_{t-1} - c_0 - c_1x_{t-1}) + u_t - \Delta b_0y_{t-1} + b_0y_{t-1} - \Delta b_1x_{t-1} + b_1x_{t-1}$$

$$\Delta y_t = \gamma + b\Delta x_t - \Delta b_0y_{t-1} - \Delta b_1x_{t-1} + \theta(y_{t-1} - c_0 - c_1x_{t-1}) + u_t + b_0y_{t-1} + b_1x_{t-1}$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} + \theta(y_{t-1} - \alpha_0 - \alpha_1x_{t-1}) + u_t$$

Extend further just one variable, so add x_{t-3} :

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} + \theta(y_{t-1} - \alpha_0 - \alpha_1x_{t-1}) + u_t + b_2x_{t-3}$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} + \theta(y_{t-1} - \alpha_0 - \alpha_1x_{t-1}) + u_t - \Delta b_2x_{t-2} + b_2x_{t-2}$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} - \Delta b_2x_{t-2} + \theta(y_{t-1} - \alpha_0 - \alpha_1x_{t-1}) + u_t + b_2x_{t-2}$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - b_1\Delta x_{t-1} - \Delta b_2x_{t-2} + \theta(y_{t-1} - \alpha_0 - \alpha_1x_{t-1}) + u_t - \Delta b_2x_{t-1} + b_2x_{t-1}$$

$$\Delta y_t = \gamma + b\Delta x_t - b_0\Delta y_{t-1} - (b_1 + b_2)\Delta x_{t-1} - \Delta b_2x_{t-2} + \theta(y_{t-1} - \alpha_0 - \Gamma x_{t-1}) + u_t$$

Adding lags is recursive. So the ECM form is:

$$\Delta y_t = a_0 + \sum_{i=1}^L a_{yi}\Delta y_{t-i} + \sum_{i=0}^M a_{xi}\Delta x_{t-i} - \gamma ECM_{t-1} + \epsilon_t$$

44.6.5 ECM Cointegration Orders

ECM requires the equilibrium equation regressors to have the same integration orders. If not, transform them to the same integration orders. Regressors of differing cointegration orders cannot be combined more generally (even beyond ECMs) as it is intuitively 'unbalanced'.

As ECM requires specific cointegration orders, need to test unit roots of each univariate time series to determine the integration orders - even if there is cointegration between them.

If the equilibrium regressors are $I(0)$, an ECM will overdifference and introduce artificial correlation. Generally, they are $I(1)$. If they are $I(2)$ or higher, the ECM equation will be more complicated. Alternatively, transform the variables, e.g. if y_t and x_t are cointegrated as $I(2)$, then Δy_t and Δx_t are cointegrated as $I(1)$.

44.6.6 Engle-Granger 2-Step Procedure

This only works with $I(1)$ cointegration - so transform variables if not.

(1) Test the equilibrium residual $\hat{v}_t$ for stationarity (use AIC to determine lags for ADF test) to establish cointegration.

(2) Insert $\hat{v}_t$ into the main equation and regress with OLS.

44.6.7 Cointegrated VAR and Vector ECM

In theory VECM is just a representation of cointegrated VAR. This representation is courtesy of Granger's representation theorem. So if you have cointegrated VAR it has VECM representation and vice versa.

In practice you need to determine the number of cointegrating relationships. When you fix that number you restrict certain coefficients of VAR model. So advantage of VECM over VAR (which you estimate ignoring VECM) is that the resulting VAR from VECM representation has more efficient coefficient estimates.

Johansen's test - look at ranks of the VAR matrix. Use trace and/or maximum eigenvalue tests to determine cointegration.

44.7 Vector Autoregressions

An extension of ARMA or ARDL.

The ARDL form is:

$$y_t = a + b_1 y_{t-1} + b_2 y_{t-2} + b_3 x_t + b_4 x_{t-1} + b_5 x_{t-2} + e_t$$

- (1) Select variables.
- (2) Test variables for stationarity.
- (3) Assume all variables are endogenous in the structural VAR equations:

$$y_t = a + b_1 y_{t-1} + b_2 y_{t-2} + b_3 x_t + b_4 x_{t-1} + b_5 x_{t-2} + e_t$$

$$x_t = c + d_1 y_{t-1} + d_2 y_{t-2} + d_3 y_t + d_4 x_{t-1} + d_5 x_{t-2} + \epsilon_t$$

- (4) Obtain the reduced VAR equations by substituting out the contemporaneous relationships - the more popular form, where all and only lags are the right hand side (similar to AR models):

$$y_t = e + f_1 y_{t-1} + f_2 x_{t-1} + f_3 y_{t-2} + f_4 x_{t-2} + u_t$$

$$x_t = g + h_1 y_{t-1} + h_2 x_{t-1} + h_3 y_{t-2} + h_4 x_{t-2} + v_t$$

Rewrite in matrix notation:

$$\begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix} + \begin{pmatrix} f_1 & f_2 \\ h_1 & h_2 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} + \begin{pmatrix} f_3 & f_4 \\ h_3 & h_4 \end{pmatrix} \begin{pmatrix} y_{t-2} \\ x_{t-2} \end{pmatrix} + \begin{pmatrix} u_t \\ v_t \end{pmatrix}$$

$$Z_t = A_0 + A_1 Z_{t-1} + A_2 Z_{t-2} + E_t$$

Optionally add exogenous variables like W:

$$Z_t = A_0 + A_1 Z_{t-1} + A_2 Z_{t-2} + A_3 W_t + E_t$$

- (5) Choose lag length by e.g. information criteria.
- (6) Estimate coefficients equation-by-equation with OLS.
- (7) As with AR, need to check A matrices for characteristic roots so that the equations don't diverge.
- (8) Test 'causal' relationships with joint hypotheses using implied restrictions and F-tests. This is the same as testing for Granger causality. Note there can be uni- and bi-directionality.
- (9) VAR coefficients are often difficult to interpret. So use impulse response functions and variance decompositions.

VAR advantages:

- Don't need to distinguish endogenous and exogenous variables.
- Simple estimation: equation-by-equation OLS.
- Generally better forecasts.

VAR disadvantages:

- No theory.
- Many parameters - curse of dimensionality.
- Hard to interpret coefficients (though IRFs help).
- Need to decide variables and lag lengths (but true for other models too).

45 Information theory

45.1 Bits and entropy

Bits are related to the number of states of the system:

$$\text{Bits} = \log N$$

All information is physical: computer atoms moved around as bits, photons in eye and brain, etc. Moreover, all physical systems contain information.

$$S = \text{Entropy} = k \log N$$

where N is the number of states and k is Boltzmann's constant. With heat engine, could extract energy from heat but the residual amount of molecular movement was entropy, which could not be reduced.

Coin tosses of heads and tails. m is the number of possibilities.

$$\binom{m}{m_h} = \binom{m}{m_t} = \frac{m!}{m_h!(m - m_h)!} = \frac{m!}{m_h!m_t!}$$

Using Stirling's approximation:

$$\log_2(n!) = \sum_j^n \log_2(j) \approx \int_1^n \log_2(x) dx$$

By integration by parts:

$$\int \log_a(x) dx = x \log_a(x) - \frac{x}{\ln a} + C$$

So,

$$\begin{aligned} \int \log_2(x) dx &= x \log_2(x) - \frac{x}{\ln 2} + C \\ &= x \log_2(x) - \frac{x \log_2(e)}{\log_2(2)} + C \\ &= x \log_2(x) - x \log_2(e) + C \end{aligned}$$

Stirling's approximation is then:

$$\log_2(n!) \approx n \log_2(n) - n \log_2(e)$$

Derive the fundamental entropy equation:

$$\begin{aligned} \log_2 N &= \log_2 \binom{m}{m_h} \\ &= \log_2(m!) - \log_2(m_h!) - \log_2((m - m_h)!) \\ &= \log_2(m!) - \log_2(m_h!) - \log_2(m_t!) \\ &\approx m \log_2(m) - m \log_2(e) - m_h \log_2(m_h) + m_h \log_2(e) - m_t \log_2(m_t) + m_t \log_2(e) \\ &= m \log_2(m) - m_h \log_2(m_h) - m_t \log_2(m_t) \\ &= -m_h (\log_2(m_h) - \log_2(m)) - m_t (\log_2(m_t) - \log_2(m)) \\ &= -m_h \log_2(m_h/m) - m_t \log_2(m_t/m) \\ &= m(-q_h \log_2 q_h - q_t \log_2 q_t) \\ &= mI \end{aligned}$$

Fundamental entropy equation:

$$I = -q_h \log_2 q_h - q_t \log_2 q_t = -\sum_i^k q(i) \log_2 q(i)$$

where q is probability. With fair coin, $H = 1$ (since it's one binary digit long).

$$\log_2 N = mI$$

$$2^{mI} = N$$

45.1.1 Intuition

Entropy measures variation. Variance already measures variability but only for cardinal values, not ordinal values.

Variation is equivalent to the minimum number of questions required to determine the distribution.

45.1.2 Why this equation form?

It is only the variability indicator that is continuous, linear, and monotonic.

45.2 Continuous distributions

The fundamental entropy equation is for discrete cases. It is not true that $-\sum p_i \log p_i \rightarrow -\int \rho \log \rho(x) dx$. If $H = \log N$ and $N \rightarrow \infty$ to cover the continuum of $\mathcal{R}$, then it's not true that $H = \log \infty = \infty$.

Suppose $\rho(x)$ is a continuous distribution, a PDF, and that $\Delta x = 1/N$. Then,

$$\bar{\rho}_i = \frac{p_i}{\Delta x} \rightarrow \Delta \bar{\rho} h o_i = p_i$$

Then,

$$\begin{aligned} -\sum p_i \log p_i &= -\sum p_i \log(\bar{\rho}_i \Delta x) \\ &= -\sum p_i \log\left(\bar{\rho} \frac{1}{N}\right) \\ &= -\sum p_i \log \bar{\rho}_i - \sum p_i \log \frac{1}{N} \\ &= -\sum \bar{\rho}_i \Delta x \log \bar{\rho}_i - \log \frac{1}{N} \end{aligned}$$

So,

$$\begin{aligned} \lim_{N \rightarrow \infty} -\sum p_i \log p_i &= \lim_{N \rightarrow \infty} -\sum \bar{\rho}_i \Delta x \log \bar{\rho}_i - \lim_{N \rightarrow \infty} \log \frac{1}{N} \\ &= \int \rho \log \rho dx + \underbrace{\lim_{N \rightarrow \infty} \log N}_{Diverges} \end{aligned}$$

To address divergence, use KL divergence,

$$H(\rho) - H(\bar{\rho}) = -\int \rho \log \rho dx + \int \bar{\rho} \log \bar{\rho} dx$$

Now assume $\bar{\rho}(x) = 1$, i.e. a uniform distribution. Then,

$$H(\rho) - H(1) = -\int \rho \log \rho dx + \int (1) \log (1) dx = -\int \rho \log \rho dx$$

So, $-\int \rho \log \rho dx$ is the variability of a distribution compared to a uniform distribution. Call this $H(\rho)$. As it's a comparison, it can be negative. To convert units from $x \rightarrow y(x)$, note

$$\rho(x) = \rho(y) \left| \frac{\partial y}{\partial x} \right| = \rho(y) J$$

where J is the Jacobian. Then,

$$\begin{aligned} H(\rho(x)) &= -\int \rho(x) \log \rho(x) dx \\ &= -\int \rho(y) \left| \frac{\partial y}{\partial x} \right| \log \left(\rho(y) \left| \frac{\partial y}{\partial x} \right| \right) dx \\ &= -\int \rho(y) \log \rho(y) dy - \int \rho(y) \log(J) dy \\ &= H(\rho(y)) - \int \rho(y) \log(J) dy \end{aligned}$$

So the conversion depends on the Jacobian.

45.3 Information processing and computation

Boole showed that AND, OR, NOT, COPY make up all digital logic. Digital computers just use these gates in combination.

45.4 Mutual information

$$I(X) = - \sum p(x_i) \log_2 p(x_i)$$

$$I(Y) = - \sum p(y_j) \log_2 p(y_j)$$

Joint information:

$$I(XY) = - \sum_i \sum_j p(x_i y_j) \log_2 p(x_i y_j)$$

Mutual information is about the information X has about Y and vice versa:

$$\begin{aligned} I(X : Y) &= I(X) + I(Y) - I(XY) \\ &= \sum_i \sum_j p(x_i y_j) \log \frac{p(x_i y_j)}{p(x_i) p(y_j)} \end{aligned}$$

Mutual information is the reduction in uncertainty knowing X value to predict Y and vice versa. If X and Y are mutually independent, then $I(X,Y)=0$, as there is no reduction in uncertainty.

45.5 Conditional information

$$p(Y|X) = \frac{p(XY)}{p(X)}$$

$$I(Y|x_i) = - \sum_j p(y_j|x_i) \log_2 p(y_j|x_i)$$

$$\begin{aligned} I(Y|X) &= - \sum_i p(x_i) I(Y|x_i) \\ &= - \sum_i p(x_i) \left(\sum_j p(y_j|x_i) \log_2 p(y_j|x_i) \right) \\ &= - \sum_i p(x_i) \left(\sum_j \frac{p(x_i y_j)}{p(x_i)} \log_2 \frac{p(x_i y_j)}{p(x_i)} \right) \\ &= - \sum_i \sum_j p(x_i y_j) \log_2 p(x_i y_j) - \log_2 p(x_i) \\ &= I(XY) - I(X) \end{aligned}$$

If X is the input information and Y is the output information, mutual information is information gain from the input-to-output process, or the channel

capacity:

$$\begin{aligned} I(X) - I(X|Y) &= I(X) - (I(XY) - I(Y)) \\ &= I(X) + I(Y) - I(XY) \\ &= I(X : Y) \end{aligned}$$

So mutual information is symmetric:

$$\begin{aligned} I(X : Y) &= I(X) - I(X|Y) \\ &= I(X) + I(Y) - I(XY) \\ &= I(Y) - I(Y|X) \end{aligned}$$

If information X is being transmitted and Y is received, the information lost during transmission is the conditional information:

$$I(X|Y) = I(XY) - I(Y)$$

where $I(XY)$ is all information transmitted+noise, and $I(Y)$ (noise) is calculated from marginal probabilities. $I(Y|X)$ is the noise introduced during transmission.

45.6 Shannon coding theorem

In a noisy channel, the maximum transmission is the mutual information. Trying to surpass mutual information, the error probability is 1.

A communication channel is defined by a conditional probability distribution of output signals given input signals, $p(y|x)$. The channel capacity is the mutual information across the channel, $I(X : Y)$, for a given distribution of sender signals, $p(x)$. Capacity is also sometimes used to refer to the maximum possible mutual information across the channel, given any distribution of sender signals: $C = \max p(x)I(X : Y)$. This determines the ultimate bound on the amount of information that can be transmitted across the channel. This indicates whether a channel is noisy or not and what its capacity is.

45.7 KL Divergence

KL Divergence of independent probabilities:

$$\begin{aligned} p(x) &= \prod_i p_i(x_i) \\ KL(p(x) \parallel q(x)) &= \sum_x p(x) \log \frac{p(x)}{q(x)} \\ &= \sum_x \prod_i p_i(x_i) \log \prod_i \frac{p_i(x_i)}{q_i(x_i)} = \sum_x \prod_i p_i(x_i) \sum_i \log \frac{p_i(x_i)}{q_i(x_i)} \end{aligned}$$

$$= \sum_i \sum_x \prod_i p_i(x_i) \log \frac{p_i(x_i)}{q_i(x_i)} = \sum_x \sum_i p_i(x_i) \log \frac{p_i(x_i)}{q_i(x_i)}$$

For Bernoulli, $p_i(x_i) = p_i(0 \text{ or } 1) = p_i$, so:

$$KL(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i} + (1 - p_i) \log \frac{1 - p_i}{1 - q_i}$$

KL divergence between Gaussians:

$$\begin{aligned} KL(q_i(t_i) \parallel p(t_i)) &= \int \log \frac{p(t)}{q(t)} q(t) dt \\ &= \int \left(-\frac{1}{2} \log(2\pi) - \log(\sigma_p) - \frac{1}{2} \left(\frac{t - \mu_p}{\sigma_p} \right)^2 + \frac{1}{2} \log(2\pi) + \log(\sigma_q) + \frac{1}{2} \left(\frac{t - \mu_q}{\sigma_q} \right)^2 \right) \\ &\quad \exp \left(-\frac{1}{2} \left(\frac{t - \mu_q}{\sigma_q} \right)^2 \right) dt \end{aligned}$$

$\sigma_p = 1$, $\mu_p = 0$ in the standard normal for p. Let $\sigma_q = \sigma$ and $\mu_p = \mu$:

$$\begin{aligned} &= \int \left(\log \sigma + \frac{1}{2} \left(\left(\frac{t - \mu}{\sigma} \right)^2 - t^2 \right) \right) \exp \left(-\frac{1}{2} \left(\frac{t - \mu}{\sigma} \right)^2 \right) dt \\ &= E_q \left(\log \sigma + \frac{1}{2} \left(\left(\frac{t - \mu}{\sigma} \right)^2 - t^2 \right) \right) = \log \sigma + \frac{1}{2\sigma^2} E_q(t - \mu)^2 - \frac{1}{2} \underbrace{E_q t^2}_{\sigma^2=1} \\ &= \log \sigma + \frac{1}{2\sigma^2} E_q(t - \mu)^2 - \frac{1}{2} = \log \sigma + \frac{1}{2\sigma^2} E_q(t^2 - 2\mu t + \mu^2) \\ &= \log \sigma + \frac{1}{2\sigma^2} ((E_q(t) = 1) - 2\mu(E_q(t) = 0) + \mu^2)) - \frac{1}{2} \\ &= \log \sigma + \frac{\mu^2 - \sigma^2 + 1}{2\sigma^2} \end{aligned}$$

So:

$$KL(q_i(t_i) \parallel p(t_i)) = \sum_j \left(\log \sigma + \frac{\mu^2 - \sigma^2 + 1}{2\sigma^2} \right)$$

45.8 Maximum entropy

Maximizes uncertainty.

The principle of maximum entropy addresses high-dimensional data well. High-dimensional data is $N \gg k$, where N is number of possible configurations and k is the amount of data.

Taxicab example. If times to get a cab are (6, 3, 4, 6, 2, 3...), then the probability $p(6) = x$ but this would be overfitting the data. Instead, invoke $p_{ME}(t)$ having a distribution that (1) satisfies certain constraints (e.g. $E(x) = \langle x \rangle = x_{average}$, $\sum_i p_i = 1$), and (2) maximizes entropy ($S = -\sum p_{ME}(x) \log p_{ME}(x)$).

So maximizing entropy is a constrained optimization problem using Lagrange multiplier (where the two gradients align):

$$\frac{\partial S}{\partial p_i} = \sum_i \lambda_i \frac{\partial g_i}{\partial p_i}$$

subject to

$$g_1(p) = \sum_i p_i i$$

$$g_2(p) = \sum_i p_i = 1$$

So,

$$\frac{\partial S}{\partial p_i} = -(\log p_i + 1)$$

$$\frac{\partial g_1}{\partial p_i} = i$$

$$\frac{\partial g_2}{\partial p_i} = 1$$

Combining,

$$-\log p_i - 1 = \lambda_1 i + \lambda_2$$

$$p_i = e^{-1-\lambda_1-\lambda_2} = \frac{e^{-\lambda_1 i}}{Z}$$

which is the analytic form, and $Z = e^{1+\lambda_2}$. Using the normalization constraint,

$$\sum_i p_i = 1 = \frac{1}{Z} \sum_i e^{-\lambda_1 i}$$

Solve for Z, which is the partition function:

$$Z = \sum_i e^{-\lambda_1 i}$$

$$= 1 + e^{-\lambda_1} + e^{-2\lambda_1} + \dots = \frac{1}{1 - e^{-\lambda_1}}$$

Further,

$$\frac{\partial Z}{\partial \lambda_1} = - \sum_i i e^{-\lambda_1 i}$$

Using the average constraint, solving, and substituting the above,

$$\frac{1}{Z} \sum_i i e^{-\lambda_1 i} = \langle x \rangle$$

$$\begin{aligned}
-\frac{1}{Z} \frac{\partial Z}{\partial \lambda_1} &= \langle x \rangle \\
\frac{\partial Z}{\partial \lambda_1} &= \frac{e^{-\lambda_1}}{(1 - e^{-\lambda_1})^2} \\
-\frac{1}{Z} \frac{\partial Z}{\partial \lambda_1} &= \frac{e^{-\lambda_1}}{1 - e^{-\lambda_1}} = \langle x \rangle
\end{aligned}$$

Solve $\lambda_1 \approx 0.22$. Then,

$$\begin{aligned}
p_i &= e^{-1-\lambda_1-\lambda_2} = \frac{e^{-\lambda_1 i}}{Z} \\
&\propto e^{-0.22x}
\end{aligned}$$

This is equivalent to a 'mechanistic' model built from first probability principles. The probability of catching a cab is $p(x) = (1-p)^x p$, and define $Z = 1/p$ (which is equivalent to the final p in $1-p)^x p$) and $\lambda_1 = -\log(1-p)$ (s.t. $e^{-\lambda_1} = (1-p)^x$).

Open system ecosystem. A few languages that are very popular and a few that are not. Can apply same form mathematically as taxicab example by having same two constraints and maximizing entropy. However, fitting the probability with best parameters is a poor fit to actual data. Let $p(n, \epsilon)$ with n being language popularity (number of projects) and ϵ be efficiency (amount of time per project). So $\langle n \rangle$ is average of projects and $\langle n\epsilon \rangle$ is the average amount of time per project. By analogy to the taxicab example,

$$\begin{aligned}
p(n, \epsilon) &= \frac{e^{-\lambda_1 n - \lambda_2 n\epsilon}}{Z} \\
\frac{\partial S}{\partial p_i} &= \sum_i \lambda_i \frac{\partial g_i}{\partial p_i} \\
g_1(p) &= \int \sum p(n,) n d\epsilon \\
g_2(p) &= \int \sum p(n, \epsilon) n \epsilon d\epsilon \\
g_3(p) &= \int \sum p(n, \epsilon) d\epsilon = 1
\end{aligned}$$

So,

$$\begin{aligned}
\frac{\partial g_1}{\partial p_i} &= n \\
\frac{\partial g_2}{\partial p_i} &= n\epsilon \\
\frac{\partial g_3}{\partial p_i} &= 1 \\
\frac{\partial S}{\partial p_i} &= \lambda_1 n + \lambda_2 n\epsilon + \lambda_3 = -\log p_i - 1
\end{aligned}$$

$$p_i \propto e^{-\lambda_1 n - \lambda_2 n \epsilon}$$

$$p(n, \epsilon) = \frac{e^{-\lambda_1 n - \lambda_2 n \epsilon}}{Z}$$

Integrate:

$$p(n) = \int p(n, \epsilon) d\epsilon = \int \frac{e^{-\lambda_1 n - \lambda_2 n \epsilon}}{Z} d\epsilon = \frac{e^{-\lambda' n}}{Z' n}$$

This distribution is the Fisher log series, which is a much better fit. It has integrated away a 'hidden' variable which is a further 'invisible' constraint. So low-popularity languages are associated with high ϵ or inefficiency. First constraint: the average number of projects is fixed (by whatever mechanism). Second constraint: programming time for a project is fixed (by efficiency of the language).

Mechanistic model for open source ecosystems. Can model popularity as linear growth or other ways. Another model is to randomly select a multiplicative factor from a given distribution each day: $p(1) = \alpha_1, p(2) = \alpha_2 \alpha_1, \dots p(t) = \prod_i \alpha_i$. Then,

$$\log(p) = \sum_i \log \alpha_i$$

Since the sum of random variables tends to normality by CLT, it is a log-normal distribution:

$$p(n) \propto \frac{1}{n} \exp\left(-\frac{(\log n - \mu)^2}{2\sigma^2}\right)$$

which fits slightly worse than the Fisher log series when statistically compared. In contrast to the taxicab example, the mechanistic model is mathematically different from the maximum entropy model.

Sears prices. Distribution of prices is log-normal. Let x be the log price:

$$\begin{aligned} p(x) &\propto e^{-(x-\mu)^2/(2\sigma^2)} \\ &= e^{-x^2/(2\sigma^2) + (2x\mu)/(2\sigma^2) - \mu^2/(2\sigma^2)} \\ &= e^{-\lambda_1 x^2 + \lambda_2 x + \lambda_3} \end{aligned}$$

So the log-normal distribution is the maximum entropy distribution which constrains $\langle x \rangle = \langle \log price \rangle$ and $\langle x^2 \rangle = \langle (x - \langle x \rangle)^2 \rangle = \langle (\log price)^2 \rangle$, which are the mean and variance. Inflation drives up the average, but historical data shows variance remains constant (because psychologically, people perceive prices multiplicatively, not additively). Each product may have ordered price distributions but they add incoherently and randomly to describe the price distribution only requiring two variables, mean and variance.

Conclusion. So from complexity arises simplicity because of dimensional reduction.

46 Algorithms

46.1 Matching

Ford-Fulkerson (FF) max-cut min-flow. Increase flow using augmenting paths (noting that reversing a flow means reducing it) using bottleneck values until it equals the minimum capacity cut (a sort of bottleneck cut). Many algorithms to implement, e.g. Edmonds-Karp, Dinic algorithm. FF reflects duality in linear programming as the maximum of the solution is the minimum of the obstacle.

Hall's theorem of perfect matchings is a corollary of Ford-Fulkerson with a source on one side and a sink on the other. It is a maximal matching. If there is a subset of k vertices on the left but less than k vertices on the right, then there is no perfect matching. Can therefore apply FF to tilings, where black-white tiles are expressed as a bipartite graph.

Gale-Shapley (addresses the 'Stable Marriage problem'). Finds one stable matching. Stable matchings aren't unique. Unstable matching: two vertices BOTH prefer each other to current arrangement. M proposes one by one down his preference list until married. W accepts if not married, but divorces and remarries if a better M arrives. A divorced M proposes again but doesn't propose to those who already rejected proposals (M has memory for efficiency). M doesn't leave partners. Notes:

- (a) Situation overall declines for Ms with time and improves for Ws.
- (b) Unfair as it's better for M - it lets M get their best choices - this can be proved by contradiction. Further, W get their worst preferences.
- (c) Time complexity of $O(n^2)$.

46.2 Shortest path

Dijkstra's algorithm finds the shortest path between two nodes. From the starting node, find distances to the connecting nodes, ordering by distance. Then find connecting nodes to the connecting nodes, again ordering by distance. If a shorter distance to a node is found, use that shorter distance and ignore any longer distances. Repeat until the end is reached. Time complexity is $\log$ of the number of vertices.

46.3 Algorithmic complexity

Time complexity for time, space complexity for memory usage. Add time and space complexities, e.g. if n for time and a further n due to space, then $O(n+n)=O(n)$.

46.3.1 Notation

$O(\cdot)$ notation. Algorithm is at most of O -scale. Only approximate worst-case scenarios asymptotically (when n is large). Drop lower order terms (e.g. scalars)

because time complexity or run time is about how data scales. Drop dominant terms, e.g. $O(n^2 + n)$ should be $O(n^2)$ as $O(n^2) \leq O(n^2 + n) \leq O(n^2 + n^2)$, though it can also be $O(n^3)$. Further properties:

- (1) If $f_1 \in O(n_1)$ and $f_2 \in O(n_2)$ then $f_1 + f_2 \in O(n_1 + n_2)$ and $f_1 f_2 \in O(n_1 n_2)$.
- (2) If $f \in O(n)$ then $n + f \in O(n)$. n will dominate. Example: $n^2 + 1000n \in O(n^2)$.
- (3) $n^a \in O(c^n)$ if $c > 1$. Exponentials dominate powers. Example: $n^{1000} \in O(1.01^n)$ eventually.
- (4) $(\ln c)^c \in O(n^a)$ if $a > 0$. Example: $(\log n)^{1000} \in O(n^{1/20})$. Powers dominate logs.

$\Omega(\cdot)$ notation is at least. $\Theta(\cdot)$ notation is the range: $\Theta(n) = O(n) \cup \Omega(n)$.

46.3.2 P and NP

P. Decision problems that can be solved in polynomial time. E.g. bipartite matching. Polynomial time problems can take a long time to solve but can be eventually solved with more powerful computers due to Moore's law.

NP. Non-deterministic. Decision problems verifiable in polynomial time. May or may not be solvable in polynomial time. E.g. clique problems. If solvable in polynomial time, it's simply P.

Co-NP where it's easy to exclude wrong answers. Unclear whether it's the same as NP.

NP-complete. Problems, not necessarily decision problems, that are at least as hard as any NP problem. E.g. protein folding, sudoku. A specific class of problems that are related such that if one is P, other NP-complete problems are also likely P and $P=NP$. Most computer scientists believe $P \neq NP$.

NP-hard. Problems that are at least as hard as NP-complete. So include NP-complete, EXP, and beyond. E.g. traveling salesperson problem.

EXP. Takes exponential time to just verify. E.g. checking a proposed solution to: what is the best first move in chess?

P-space (memory restriction), BPP (probabilistic), BQP (quantum), many others.

Proving is an NP problem (easy to verify but may or may not be solvable quickly). Proving $P=NP$ is therefore itself an NP problem.

47 Machine Learning

47.1 Computation

47.1.1 Curse of dimensionality

Inducing inputs. Train model on a subsample to reduce training time.

47.1.2 Computational Stability

There are often underflow and overflow problems. Example:

$$v_{i,k} = \frac{z_{i,k}}{\sum_l z_{i,l}}$$

where

$$\begin{aligned} z_{i,k} &= w_k g_k(x_i | \theta_k) \\ &= \frac{\exp(\log z_{i,k})}{\sum_l \exp(\log z_{i,l})} = \frac{\exp(-b) \exp(\log z_{i,k})}{\exp(-b) \sum_l \exp(\log z_{i,l})} = \frac{\exp(\log z_{i,k} - b)}{\sum_l \exp(\log z_{i,l} - b)} \end{aligned}$$

Where $b = \max_l \log z_{i,l}$.

In R: use e.g. `dnorm(..... log=T)` or log the PDF equation. Don't `log(dnorm(...))` which won't work since `dnorm(...)` itself is $0 \rightarrow \log(\text{dnorm})=\text{NaN}$.

Multimodality is a major problem. Different Q value convergences from EM algorithms using different points $\rightarrow$ want to use largest Q $\rightarrow$ use different initial points. Numerical instability if choose bad initial points (where e.g. $\log 0 = \text{NaN}$) or converges to such points. NOTE: this doesn't happen with MCMC where the priors are like a regularization term!

47.2 Regression

47.2.1 Parametric and non-parametric methods

Parametric methods rely on parameters for assumed distributions. This reduces complexity and allows simpler, faster inference - once the distribution has been specified by its parameters, can just compare new data with the specified distribution. However, if the dimensions or complexity of the distribution increases, need more parameters.

Non-parametric methods don't rely on parameters or assume distributions. So the same calculation is used regardless of complexity or dimensions. For any prediction however, need to re-process all data.

47.2.2 K-nearest neighbors

KNN is non-parametric. Regress, draw boundaries, and predict based on all observed data:

$$y = \frac{1}{N} \sum_i^N y_i$$

To smooth, use the Nadaraya-Watson method:

$$y = \frac{1}{N} \sum_i^N w_i y_i$$

$$w_i = \frac{K(x, x_i)}{\sum_i^N K(x, x_i)}$$

47.2.3 Gaussian process

Instead of overfitting the data x with no variance, assume Gaussian noise:

$$\hat{f}(x) = f(x) + \epsilon$$

$$\epsilon \sim N(0, s^2)$$

$$\hat{\Sigma} \text{ with } \hat{k}(x_i - x_j) = k(x_i - x_j) + s^2 \mathcal{I}_{x_i=x_j}$$

Train with a kernel like RBF:

$$\hat{k}(x_i - x_j) = \sigma^2 \exp\left(-\frac{(x_i - x_j)^2}{2l^2}\right) + s^2 \mathcal{I}_{x_i=x_j}$$

Optimize with gradient ascent:

$$\underset{\sigma^2, l, s^2}{\operatorname{argmax}} (f(x_1) \dots f(x_n) | \sigma^2, l, s^2) = \underset{\sigma^2, l, s^2}{\operatorname{argmax}} \mathcal{N}(f | 0, \Sigma)$$

47.3 Expectation-Maximization

Method of maximizing the likelihood.

An alternative to gradient descent that can be computed analytically. We can use stochastic gradient descent (SGD) from neural networks to solve for the parameters θ , but the covariance matrix Σ is generally only positive semi-definite so can cause singularities, and it is difficult to restrain it to positive definiteness. Σ might be restricted to a diagonal matrix, i.e. axial alignments only, but this is restrictive.

Apply to:

- Latent Variable Models and missing data

- Latent Dirichlet Allocation
- Baum-Welch algorithm for Hidden Markov models
- Mixture models and clustering

Assume $t \in [1, M]$ are latent variables that produce x observations. For any specific data point x_i :

$$\begin{aligned} \max_{\theta} p(x_i|\theta) &= \max_{\theta} \sum_{c=1}^M [p(x_i|t_i = c, \theta)p(t_i = c|\theta)] \\ &= \max_{\theta} \sum_{c=1}^M p(x_i, t_i = c|\theta) \end{aligned}$$

Where $p(t_i = c|\theta)$ is the probability of belonging to a specific latent variable. Assume conditional probabilities are independent, because we assume latent variable t independently results in x 's (see Latent Variable Model section). Assume the log is bounded at the lower end:

$$\log p(x|\theta) = \sum_i^N \log p(x_i|\theta) = \sum_i^N \log \sum_c^M p(x_i, t_i = c|\theta) \geq \mathcal{L}(\theta)$$

The lower bound uses Jensen's inequality (for concave functions):

$$\log \left(\sum_c \alpha_c v_c \right) \geq \sum_c \alpha_c \log(v_c)$$

Result:

$$\sum_i^N \log \sum_c^M \frac{q(t_i = c)}{q(t_i = c)} p(x_i, t_i = c|\theta) \geq \sum_i^N \sum_c^M q(t_i = c) \log \frac{p(x_i, t_i = c|\theta)}{q(t_i = c)}$$

Where $q(t_i = c)$ is α in Jensen's equality. So for any q (the 'variational parameter'):

$$\log p(x|\theta) \geq \mathcal{L}(\theta, q)$$

(1) Expectation step:

$$\underset{q}{\operatorname{argmin}} [\log p(x|\theta) - \mathcal{L}(\theta, q)]$$

So trying to minimize the gap between the lower bound and the function. Given θ , find the q that maximizes HEIGHT of the lower bound. q is the probability of the latent variable (actually the Bayesian posterior of the latent variable given x and θ).

$$\log p(x|\theta) - \mathcal{L}(\theta, q) = \sum_i^N \log p(x_i|\theta) - \sum_i^N \sum_c^M q(t_i = c) \log \frac{p(x_i, t_i = c|\theta)}{q(t_i = c)}$$

$$\begin{aligned}
&= \sum_i^N \left(\log p(x_i|\theta) \underbrace{\sum_c^M q(t_i = c)}_{=1} - \sum_c^M q(t_i = c) \log \frac{p(x_i, t_i = c|\theta)}{q(t_i = c)} \right) \\
&= \sum_i^N \sum_c^M q(t_i = c) \left(\log p(x_i|\theta) - \log \frac{p(x_i, t_i = c|\theta)}{q(t_i = c)} \right) \\
&= \sum_i^N \sum_c^M q(t_i = c) \log \frac{p(x_i|\theta)q(t_i = c)}{p(x_i, t_i = c|\theta)} = \sum_i^N \sum_c^M q(t_i = c) \log \frac{p(x_i|\theta)q(t_i = c)}{p(t_i = c|x_i, \theta)p(x_i|\theta)} \\
&= \sum_i^N \sum_c^M q(t_i = c) \log \frac{q(t_i = c)}{p(t_i = c|x_i, \theta)} = \sum_i^N KL(q(t_i) \parallel p(t_i = c|x_i, \theta))
\end{aligned}$$

So the gap is just the KL divergence, which is intuitive - it's the distance between distributions:

$$\log p(x|\theta) - \mathcal{L}(\theta, q) = \sum_i KL(q(t_i) \parallel p(t_i|x_i, \theta))$$

Where the KL divergence = 0 when the next q (q^{k+1}) becomes $p(t_i|x_i, \theta)$:

$$\underset{q}{\operatorname{argmax}} \mathcal{L}(\theta^k, q) = q^{k+1} = p(t_i|x_i, \theta)$$

(2) Maximization step:

$$\theta^{k+1} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta, q^{k+1})$$

So given q , maximize the likelihood and therefore optimize θ so move HORIZONTALLY along the lower bound.

$$\mathcal{L}(\theta, q) = \sum_i \sum_c q(t_i = c) \log \frac{p(x_i, t_i = c|\theta)}{q(t_i = c)}$$

The second term doesn't depend on θ :

$$\begin{aligned}
&= \sum_i \sum_c q(t_i = c) \log p(x_i, t_i = c|\theta) - \sum_i \sum_c q(t_i = c) \log q(t_i = c) \\
&= E_q[\log p(x, t|\theta)] + \text{constant} \\
&= E_q[\log(p(x|t, \theta)p(t|x, \theta))] + \text{constant} \\
&= E_q[\log(p(x|t, \theta)q(t))] + \text{constant}
\end{aligned}$$

So in the E step, optimize probability of latent variable. In the M step, maximize likelihood to optimize θ - it's an alternating algorithm like Gibbs:

$$\log p(X|\theta^k) = \mathcal{L}(\theta^k, q^{k+1}) \leq \mathcal{L}(\theta^{k+1}, q^{k+1}) \leq \log p(X|\theta^{k+1})$$

So the algorithm converges to a critical point. The EM algorithm is sensitive to initial conditions as there can be many local maxima which can trap the algorithm at suboptimal points, so try several different initializations.

So the EM algorithm produces results even with missing x variables. Further, use their latent variables to infer what the missing variable might be.

47.4 Clustering and Classification

47.4.1 Kernel Density Estimation

Easy in R: just use `density()` function - default is Gaussian kernel Given some

$$x_1, x_2, \dots, x_n \sim f$$

$$\hat{f}(x) = \frac{1}{n} \sum_i \frac{1}{h} g\left(\frac{|x - x_i|}{h}\right)$$

where g = kernel density and h = bandwidth The kernel density can be anything - e.g. Gaussian, box, triangle, etc. The bandwidth is like the standard deviation Example:

$$\hat{f}(x) = \frac{1}{n} \sum_i \frac{1}{h} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - x_i}{h}\right)^2\right)$$

Compare and contrast KDE with mixture models:

$$\hat{f}(x) = \sum_k^K \hat{w}_k g_k(x|\hat{\theta}_k)$$

Mixture model example:

$$\hat{f}(x) = \sum_k^K \hat{w}_k \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp\left(-\frac{1}{2\hat{\sigma}^2} (x - \hat{\mu}_k)^2\right)$$

Note: mixture models can provide different bandwidth/standard deviations for different clusters, and are more customizable. KDEs just use the same kernel everywhere en mass for all points.

47.5 Clustering

K-means Clustering. Randomly initialize μ vector. Then:

(1) Assignment step:

$$c_i = \underset{c}{\operatorname{argmin}} \|x_i - \mu_c\|^2$$

(2) Centroid-updating step:

$$\mu_c = \frac{\sum^n 1_{c_i=c} x_i}{\sum 1_{c_i=c}}$$

EM-GMM Clustering. GMM PDF:

$$p(x|t, \theta) = \sum_c \underbrace{\frac{1}{K}}_{v_c} \underbrace{\left(\frac{1}{\sqrt{2\pi}\sigma} \right)^p \exp \left(-\frac{1}{2\sigma^2} \|x_i - \mu_c\|^2 \right)}_{N(\mu_c, \sigma^2 I)}$$

Assume uniform weights v_k . Fix $\Sigma_c = I$, so $\sigma^2 = 1$. Then:

$$p(x|t, \theta) \propto \exp \left(-\frac{1}{2} \|x_i - \mu_c\|^2 \right)$$

(1) E-step:

$$p(t_i|x_i, \theta) \propto p(x_i|t_i, \theta)p(t_i|\theta)$$

$$p(t_i|x_i, \theta) \propto \exp \left(-\frac{1}{2} \|x_i - \mu_c\|^2 \right) \pi_c$$

Set $q(t_i = c) = p(t_i|x_i, \theta)$ as in GMMs, but since we force a choice into a $t_i = c$ instead of probabilities, conceptualize the choice with Dirac δ -functions. To maximize $p(t_i|x_i, \theta)$, minimize $\|x_i - \mu_c\|^2$.

$$c_i = \underset{c}{\operatorname{argmin}} \|x_i - \mu_c\|^2$$

(2) M-step - apply the GMM weighted average formula:

$$\mu_c = \frac{\sum^n 1_{c_i=c} x_i}{\sum 1_{c_i=c}}$$

This is the same as K-means, so K-means is just a specific case of GMMs with uniform weights v_k and fixed uniform unit variances.

K-means clustering algorithm is very generic, with few options so provides poor results. Particularly has trouble with heterogeneous clusters, such as those with non-circular clusters and different-sized clusters.

47.5.1 Classification

Can use logistic or multinomial models (with latent variables), Naive Bayes, semi-supervised methods, and others.

Naive Bayes classifiers. A special case of mixture models. Test and training sets:

- Training: $x_1, x_2, \dots, x_n$
- Test: $x_{n+1}, x_{n+2}, \dots, x_{n+m}$

$$p(x_i \in \text{class } k) = \frac{w_k g_k(x_i | \theta_k)}{\sum_l^K w_l g_l(x_i | \theta_l)}$$

and now use, e.g. Gaussian kernel

$$p(x \in \text{class } k) = \frac{w_k \left(\frac{1}{\sqrt{2\pi}} \right)^p |\Sigma_K|^{-1/2} \exp \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_K^{-1} (x_i - \mu_k) \right)}{\sum_l^K w_l \left(\frac{1}{\sqrt{2\pi}} \right)^p |\Sigma_l|^{-1/2} \exp \left(-\frac{1}{2} (x_i - \mu_k)^T \Sigma_l^{-1} (x_i - \mu_l) \right)}$$

With p features. $\hat{\mu}_k$, $\hat{\Sigma}_k$, $\hat{w}_k$ are estimated from the training set. Classify new observations as

$$c_i = \underset{k}{\operatorname{argmax}} p(x_i \in \text{class } k)$$

Above is same as EM algorithm by incorporating some assumptions EM algorithm Training data: E-step

$$v_{i,k} = \begin{cases} 1 & \text{if } c_i = k \\ 0 & \text{otherwise} \end{cases}$$

M-step

$$\mu_k = \frac{\sum_i^n v_{i,k} x_i}{\sum_i^n v_{i,k}}$$

$$\Sigma_K = \frac{1}{\sum_i^n v_{i,k}} \sum_i^n v_{i,k} (x_i - \mu_k)(x_i - \mu_k)^T$$

Postprocessing step

$$c_i = \underset{k}{\operatorname{argmax}} p(x_i \in k)$$

Testing data: combine with the parameters from the training to determine the test data parameters Why recast classification as an EM algorithm? Can do semi-supervised classification which might be more powerful: Combine training and test data For test $v_{i,k}$, treat as unknown and refine with

$$v_{i,k} = p(x_i \in k | w, \mu, \Sigma)$$

47.5.2 Discriminant Analysis

Linear discriminant analysis: boundaries between Gaussian distributions with the same Σ matrix. Quadratic discriminant analysis: different Σ s. If using mixture models, it is mixture discriminant analysis.

$$f(x) = \sum_k^2 w_k \left(\frac{1}{\sqrt{2\pi}} \right)^p |\Sigma|^{-1/2} \exp \left(-\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right)$$

$$p(x_i \in \text{class } 1)$$

$$= \frac{w_1 \left(\frac{1}{\sqrt{2\pi}}\right)^p |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)\right)}{w_1 \left(\frac{1}{\sqrt{2\pi}}\right)^p |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)\right) + w_2 \left(\frac{1}{\sqrt{2\pi}}\right)^p |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu_2)^T \Sigma^{-1}(x-\mu_2)\right)}$$

Postprocessing $ci=1$ iff $p(x_i \in \text{class 1}) > p(x_i \in \text{class 2})$ which is equivalent to

$$w_1 \left(\frac{1}{\sqrt{2\pi}}\right)^p |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)\right) > w_2 \left(\frac{1}{\sqrt{2\pi}}\right)^p |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu_2)^T \Sigma^{-1}(x-\mu_2)\right)$$

Since the vcov's are the same, they cancel out, and

$$\begin{aligned} \exp\left(-\frac{1}{2}((x-\mu_1)^T \Sigma^{-1}(x-\mu_1) - (x-\mu_2)^T \Sigma^{-1}(x-\mu_2))\right) &> \frac{w_2}{w_1} \\ -\frac{1}{2}((x-\mu_1)^T \Sigma^{-1}(x-\mu_1) - (x-\mu_2)^T \Sigma^{-1}(x-\mu_2)) &< -2\log \frac{w_2}{w_1} \\ x^T \Sigma^{-1} x - 2x^T \Sigma^{-1} \mu_1 + \mu_1^T \Sigma^{-1} \mu_1 - x^T \Sigma^{-1} x + 2x^T \Sigma^{-1} \mu_1 - \mu_2^T \Sigma^{-1} \mu_2 &< -2\log \frac{w_2}{w_1} \\ \underbrace{2x^T \Sigma^{-1}(\mu_2 - \mu_1)}_a &< -2\log \frac{w_2}{w_1} + \mu_2^T \Sigma^{-1} \mu_1 - \mu_2 \Sigma^{-1} \mu_1 = T \end{aligned}$$

This is linear discriminant analysis. Just feed in the x's. If Σ s are different, then some terms wouldn't cancel, leading to quadratic discriminant analysis.

47.6 Dimensionality Reduction

47.6.1 Principal Components Analysis

Probabilistic PCA. PCA proceeds represents higher dimensional points x in lower dimensions t . Probabilistic PCA is the reverse.

Let t be a latent variable. Then:

$$x_i = W t_i + b + \epsilon_i$$

Where $\epsilon \sim \mathcal{N}(0, \Sigma)$ and W is a matrix transform. Further, let $t_i \sim \mathcal{N}(0, I)$ so that $x_i | t_i, \theta \sim \mathcal{N}(W t_i + b, \Sigma)$. Then train with data by maximizing the likelihood with MLE or gradient descent:

$$p(x|\theta) = \prod_i^N p(x_i|\theta) = \prod_i^N \underbrace{\int p(x_i|t_i, \theta) p(t_i) dt_i}_{\text{Conjugacy: } \mathcal{N}(\mu_i, \Sigma_i)}$$

This leads to the same result as PCA. It is analytically tractable due to conjugacy and complete data points, but EM can deal if one or both of these

conditions not met.

Training the EM algorithm with the data (x's):

(1) E-step - conjugacy results in an analytical solution:

$$q(t_i) = p(t_i|x_i, \theta) = \frac{p(x_i|t_i, \theta)p(t_i)}{NC} = \mathcal{N}(\mu, \Sigma)$$

(2) M-step - maximize:

$$\begin{aligned} & E_q \sum_i \log p(x_i|t_i, \theta)p(t_i) \\ &= \sum_i E_q \log \left(\frac{1}{NC} \exp \left(-\frac{(x - Wt_i - b)^2}{2\sigma^2} \right) \exp \left(-\frac{t_i^2}{2} \right) \right) \\ &= \sum_i E_q \log \left(\frac{1}{NC} \right) + \sum_i E_q \log \left(\exp \left(\frac{-(x - Wt_i - b)^2 - \sigma^2 t_i^2}{2\sigma^2} \right) \right) \\ &= \sum_i E_q \log \left(\frac{1}{NC} \right) + \underbrace{\sum_i E_q \left(\frac{-(x - Wt_i - b)^2 - \sigma^2 t_i^2}{2\sigma^2} \right)}_{\text{Negative quadratic}} \end{aligned}$$

The negative quadratic can be maximized easily on a computer to obtain $\theta = (W, b, \Sigma)$

47.7 Hidden Markov Model

Popular for sequential data. Applications:

- Words (emission) $\rightarrow$ grammar (hidden)
- Handwriting (emission) $\rightarrow$ actual letters (hidden)
- Walls, etc. observed by robot (emission) $\rightarrow$ location (hidden)
- Sounds (emission) $\rightarrow$ words (hidden)
- Hidden Markov process (e.g. weather) and observed emission process (e.g. mood)

Hidden Markov Model:

$$p(x_1, x_2, \dots, x_n, z_1, z_2, \dots, z_n) = p(x_1|z_1)p(z_1)p(x_2|z_2)p(z_2|z_1)\dots p(x_n|z_n)p(z_n|z_{n-1})$$

Where $p(z_1)$ is the probability of the initial hidden state. The transition probabilities are $p(z_k|z_{k-1})$ - e.g. a Markov transition matrix of two weather states. The emission probabilities are $p(x_k|z_k)$ - e.g. mood depending on the hidden state.

Transition probabilities. Obtain conditional probabilities from known or assumed hidden mechanism(s) (e.g. weather changes). To generate unconditional probabilities, draw Markov graph, generate and solve transition equation (e.g. $S=0.8S+0.4R$ and $S+R=1$ where S =sunny and R =rainy).

Emission probabilities. Use Bayesian probabilities to guess hidden state (z_k) based on emission states (x_k): $p(weather|mood)$ based on data of both hidden and emission states.

Use maximum likelihood to determine weather history based on mood history. This is tedious, picking the highest likelihood graphical model.

However, as hidden states depend on their previous states, there is a curse of dimensionality. For a sequence of $n=2$ with 2 possible hidden states, there are 4 possibilities to calculate. For a sequence of $n=3$, with 2 hidden states, there are 8 possibilities, etc.

Use dynamic programming (recursion instead of naive computation which leads to a curse of dimensionality) to address this.

47.7.1 Forward-Backward Algorithm

Calculates by conditional probabilities the most likely hidden state for ONE PARTICULAR POINT IN TIME.

$$p(z_k|x) \propto p(z_k, x) = \underbrace{p(x_{k+1:n}|z_k)}_{Backward} \underbrace{p(z_k, x_{1:k})}_{Forward}$$

(1) Forward algorithm:

$$\begin{aligned} \underbrace{p(z_k, x_{1:k})}_{\alpha_k(z_k)} &= \sum_{z_{k-1}=1}^m p(z_k, z_{k-1}, x_{1:k}) \\ &= \sum_{z_{k-1}} \underbrace{p(x_k|z_k, z_{k-1}, x_{1:k-1})}_{=p(x_k|z_k)} \underbrace{p(z_k|z_{k-1}, x_{1:k-1})}_{=p(z_k|z_{k-1})} \underbrace{p(z_{k-1}, x_{1:k-1})}_{\alpha_{k-1}(z_{k-1})} \end{aligned}$$

The first term is a known emission probability, the second is a hidden Markov transition probability. It's recursive, so

$$\alpha_1(z_1) = p(x_1|z_1)p(z_1|z_0)\alpha_0(z_0)$$

And etc. Instead of an NP problem, it's now a P problem.

(2) Backward algorithm:

$$\underbrace{p(x_{k+1:n}|z_k)}_{\beta_k(z_k)} = \sum_{z_{k+1}=1}^m p(x_{k+1:n}, z_{k+1}|z_k)$$

$$= \sum_{z_{k+1}} \underbrace{p(x_{k+2:n}|z_{k+1}, z_k, x_{k+1})}_{\beta_{k+1}(z_{k+1})} \underbrace{p(x_{k+1}|z_{k+1}, z_k)}_{p(x_{k+1}|z_{k+1})} p(z_{k+1}|z_k)$$

The first term is the recursive term, the second is known, and the last is a transition probability. Same computational complexity as forward algorithm.

47.7.2 Viterbi Algorithm

Calculates by conditional probabilities the most likely hidden state for THE MOST LIKELY SEQUENCE THROUGHOUT TIME.

For each hidden state of each observation in the sequence, calculate the highest likelihood for that state, based on the immediately preceding likelihoods. Choose the highest-likelihood path in the end for the sequence.

Look for a recursion again:

$$\begin{aligned} \mu_k(z_k) &= \max_{z_{1:k-1}} p(z_{1:k}, x_{1:k}) \\ &= \max_{z_{1:k-1}} p(x_k|z_k) p(z_k|z_{k-1}) p(z_{1:k-1}, x_{1:k-1}) \\ &= \max_{z_{k-1}} [p(x_k|z_k) p(z_k|z_{k-1}) \max_{z_{1:k-2}} \underbrace{p(z_{1:k-1}, x_{1:k-1})}_{\mu_{k-1}(z_{k-1})}] \end{aligned}$$

47.7.3 Baum-Welch Algorithm

Application of EM algorithm and forward-backward algorithm.

(1) E-step:

(a) Forward algorithm

$$\alpha(X_k) = P[Y_{0:k}, X_k] = \sum_{X_{k-1}} \alpha(X_{k-1}) P(X_k|X_{k-1}) P(Y_k|X_k)$$

(b) Backward algorithm

$$\beta(X_k) = P[Y_{k+1:T}|X_k] = \sum_{X_{k+1}} \beta(X_{k+1}) P(X_{k+1}|X_k) P(Y_{k+1}|X_{k+1})$$

Combine to get the probability distribution of the state variable X at time k based on all observations:

$$\begin{aligned} P[X_k|Y_{0:T}] &= \frac{P[X_k, Y_{0:T}]}{P[Y_{0:T}]} = \frac{P[Y_{0:k}, Y_{k+1:T}, X_k]}{P[Y_{0:T}]} \\ &= \frac{P[Y_{0:k}, X_k] P[Y_{k+1:T}|X_k]}{P[Y_{0:T}]} = \frac{\alpha(X_k) \beta(X_k)}{P[Y_{0:T}]} \propto \alpha(X_k) \beta(X_k) \end{aligned}$$

(2) M-step:

$$\begin{aligned}\eta(X_k) &= P[X_k|Y_{0:T}] = \frac{\alpha(X_k)\beta(X_k)}{\sum_{X_k} \alpha(X_k)\beta(X_k)} \\ \xi(X_k, X_{k+1}) &= P[X_k, X_{k+1}|Y_{0:T}] \\ &= \frac{\alpha(X_k)\beta(X_{k+1})P[X_{k+1}|X_k]}{\sum_{X_k} \alpha(X_k)\beta(X_{k+1})P[X_{k+1}|X_k]}\end{aligned}$$

Now calculate the transition probabilities a^* and emission probabilities b^* (of seeing state v_k):

$$\begin{aligned}\pi_0^* &= \eta(X_0) \\ a_{ij}^* &= P[X_k = j|X_{k-1} = i] = \frac{\sum_k \xi(X_k = j, X_{k-1} = i)}{\sum_k \eta(X_{k-1} = i)} \\ b_i(v_k)^* &= P[Y_k = v_k|X_k = i] = \frac{\sum_k \eta(X_k = i)1_{Y_k=v_k}}{\sum_k \eta(X_k = i)}\end{aligned}$$

47.8 Latent Dirichlet Allocation

Used most often for topic modelling in documents.

Document d has topic probabilities θ_d (which represent all topics in the documents), where z_{dn} is the topic chosen from θ_d , and w_{dn} is the actually chosen word. z and θ are latent variables while w is observed data. D is the number of documents, T is the number of topics, and N is the number of words.

$$\begin{aligned}p(w, z, \theta) &= \prod_d p(\theta_d) \prod_n p(z_{dn}|\theta_d)p(w_{dn}|z_{dn}) \\ \theta_d &\sim Dir(\alpha)\end{aligned}$$

Where the Dirichlet PDF is:

$$p(x, \alpha) = \frac{1}{B(\alpha)} \prod_t \theta^{\alpha_t - 1}$$

Further:

$$\begin{aligned}p(z_{dn}|\theta_d) &= \theta_{dz_{dn}} \\ p(w_{dn}|z_{dn}) &= \phi_{z_{dn}w_{dn}}\end{aligned}$$

The last two are generally multinomial distributions.

$$\log p(w, z, \theta) = \sum_d (\alpha_t - 1) \log p(\theta_d) + \sum_d \sum_n \sum_t \underbrace{[z_{dn} = t]}_{Indicator} (\log \theta_{dt} + \log \phi_{tw_{dn}})$$

Now apply 'Variational EM': the EM algorithm but with mean field approximation in the E-step.

(1a) E-step for θ :

$$\begin{aligned}
& \underset{q(\theta)}{\operatorname{argmin}} KL(q(\theta)q(z) \parallel p(\theta, z|w)) \\
& \log q(\theta) = E_{q(z)} \log p(\theta, z, w) + \text{Constant} \\
& = E_{q(z)} \sum_d \left(\sum_t (\alpha_t - 1) \log \theta_{dt} + \sum_n \sum_t [z_{dn} = t] \log \theta_{dt} + \text{Constant} \right) + \text{Constant}
\end{aligned}$$

E is a linear operator:

$$\begin{aligned}
& = \sum_d \left(\sum_t (\alpha_t - 1) \log \theta_{dt} + \sum_n \sum_t \underbrace{E_{q(z)}[z_{dn} = t]}_{\gamma_{dn}} \log \theta_{dt} \right) + \text{Constant} \\
& = \sum_d \sum_t \left((\alpha_t - 1) + \sum_n \gamma_{dn} \right) \log \theta_{dt} + \text{Constant}
\end{aligned}$$

Therefore:

$$q(\theta) \propto \prod_d \underbrace{\left(\prod_t \theta_{dt}^{\alpha_t + \sum_n \gamma_{dn} - 1} \right)}_{\text{Dirichlet}}$$

This is a Dirichlet distribution:

$$\theta_d \sim \text{Dir}(\theta_d | \alpha + \sum_n \gamma_{dn})$$

So:

$$q(\theta) = \prod_d q(\theta_d)$$

(1b) E-step for z :

$$\begin{aligned}
& \underset{q(z)}{\operatorname{argmin}} KL(q(\theta)q(z) \parallel p(\theta, z|w)) \\
& \log q(z) = E_{q(\theta)} \log p(\theta, z, w) + \text{Constant}
\end{aligned}$$

Where:

$$\log p(w, z, \theta) = \sum_d \left(\text{Constant} + \sum_n \sum_t \underbrace{[z_{dn} = t]}_{\text{Indicator}} (\log \theta_{dt} + \log \phi_{tw_{dn}}) \right)$$

So with E as a linear operator:

$$\begin{aligned}
\log q(z) & = \sum_d \sum_n \left(\sum_t [z_{dn} = t] (E_{q(\theta)} \log \theta_{dt} + \log \phi_{tw_{dn}}) \right) + \text{Constant} \\
\log q(z) & = \prod_d \prod_n q(z_{dn})
\end{aligned}$$

Where:

$$q(z_{dn} = t) = \frac{\phi_t w_{dn} \exp(E_{q(\theta)} \log \theta_{dt})}{\sum_t \phi_t w_{dn} \exp(E_{q(\theta_{dt})} \log \theta_{dt})} = \gamma_{dn}$$

(2) M-step:

$$\begin{aligned} & \underset{\phi}{\operatorname{argmax}} E_{q(\theta)q(z)} \log p(\theta, z, w) \\ &= E_{q(\theta)q(z)} \sum_d \sum_n \sum_t [z_{dn} = t] \log \phi_{tw_{dn}} + \text{Constant} \end{aligned}$$

Maximize above with $\sum_w \phi_{tw} = 1$ by forming the Lagrangian L:

$$\begin{aligned} L &= E_{q(\theta)q(z)} \sum_d \sum_n \sum_t [z_{dn} = t] \log \phi_{tw_{dn}} + \sum_t \lambda_t (\sum_w \phi_{tw} - 1) \\ \frac{\partial L}{\partial \phi_{tw}} &= \sum_d \sum_n \sum_t \gamma_{dn} \frac{1}{\phi_{tw_{dn}}} [w_{dn} = w] + \lambda_t = 0 \\ \phi_{tw} &= \frac{\sum_d \sum_n \sum_t \gamma_{dn} [w_{dn} = w]}{-\lambda_t} \\ 1 &= \sum_w \phi_{tw} = \frac{\sum_w \sum_d \sum_n \sum_t \gamma_{dn} [w_{dn} = w]}{-\lambda_t} \end{aligned}$$

Then:

$$\lambda_t = \sum_w \sum_d \sum_n \sum_t \gamma_{dn} [w_{dn} = w]$$

So:

$$\phi_{tw} = \frac{\sum_d \sum_n \sum_t \gamma_{dn} [w_{dn} = w]}{\sum_w \sum_d \sum_n \sum_t \gamma_{dn} [w_{dn} = w]}$$

Can add priors for topic-word associations, correlated topics by having off-diagonal entries in Σ , e.g. in $\theta_d \sim N(\mu, \Sigma)$. The number of words can be chosen by adding another variable - generally, use a Gamma distribution.

Can use Gibbs sampling instead of Variational EM and mean-field approximation. In particular, use 'Collapsed Gibbs' sampling where one or more variables are marginalized out analytically to speed up MCMC.

47.9 Bayesian Optimization

Also called Sequential Model-Based Optimization. Estimates black-box functions (e.g. objective functions). More efficient than other techniques (e.g. manual, random, grid search) by reducing sampling requirements. Therefore, useful for hyperparameter-tuning, e.g. which activation functions and parameters, gradient descent parameters. Repeat two steps:

(1) Define a surrogate model (also called: response surface model) $\hat{f}(x)$ which

approximates the function to optimize but is easy to maximize. Maximize $\hat{f}(x)$ and then sample $f(x)$ at the maximum.

Generally assume Gaussian process and RBF covariance. Use Bayes' rule on Gaussian processes based on already-known points to build k , f , and Σ . Then calculate μ and σ for unknown x 's (so on $f(x)$ s). Can try estimating the maximum at this point based on this prediction function.

A random process where the $f(x)$ are jointly normal. Characterized by the mean and covariance matrix. Stationary if the mean is constant and the covariance only depends on the distance.

Pairwise covariances can be trained with many different kernels. Most common is the radial basis function (RBF):

$$Cov(x_1, x_2) = \sigma^2 \exp\left(-\frac{(x_1 - x_2)^2}{2l^2}\right)$$

If l is small, then oscillation frequency is high. If l is large, oscillation frequencies are low. Other common kernels:

$$\text{Rational quadratic: } Cov(x_1, x_2) = \sigma^2 \exp\left(1 + \frac{(x_1 - x_2)^2}{2\alpha l^2}\right)^{-\alpha}$$

$$\text{White noise: } Cov(x_1, x_2) = \sigma^2 \delta(x_1 - x_2)$$

Prediction

$$p(f(x)|f(x_1), f(x_2), \dots, f(x_n)) = \frac{p(f(x), f(x_1) \dots f(x_n))}{p(f(x_1) \dots f(x_n))}$$

$$\begin{aligned} p(f(x)|f(x_1), f(x_2), \dots, f(x_n)) &= \frac{N(f(x), f(x_1) \dots f(x_n) | \mu, \bar{\Sigma}_{(n+1)^2})}{\mathcal{N}(f(x_1) \dots f(x_n) | \mu, \Sigma_{n^2})} \\ &= \mathcal{N}(f(x) | \mu, \hat{\Sigma}) \end{aligned}$$

Where:

$$\begin{aligned} \Sigma &= \begin{pmatrix} k(0) & k(x_1 - x_2) & \dots & k(x_1 - x_n) \\ k(x_2 - x_1) & k(0) & \dots & k(x_2 - x_n) \\ \dots & \dots & \dots & \dots \\ k(x_n - x_1) & k(x_n - x_2) & \dots & k(0) \end{pmatrix} \\ \bar{\Sigma} &= \begin{pmatrix} k(0) & k(x - x_1) & \dots & \dots & k(x - x_n) \\ k(x_1 - x) & k(0) & \dots & \dots & k(x_1 - x_n) \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ k(x_n - x) & k(x_n - x_1) & \dots & \dots & k(0) \end{pmatrix} = \begin{pmatrix} k(0) & k' \\ k & \Sigma \end{pmatrix} \\ k(0) &= \sigma^2 \\ \mu &= k' \Sigma^{-1} f \end{aligned}$$

$$f = (f(x_1), f(x_2) \dots f(x_n))'$$

$$\hat{\Sigma} = \sigma^2 - k' \Sigma^{-1} k$$

Proof:

$$p(f(x)|f(x_1), f(x_2), \dots, f(x_n)) = \frac{p(f(x), f(x_1) \dots f(x_n))}{p(f(x_1) \dots f(x_n))} = \frac{p(f^*, f')}{p(f')}$$

$$= \frac{\mathcal{N}\left(\begin{pmatrix} f^* \\ f \end{pmatrix} \middle| 0, \begin{pmatrix} k(0) & k' \\ k & \Sigma \end{pmatrix}\right)}{\mathcal{N}(f|0, \Sigma)}$$

$$\propto \exp\left(-\frac{1}{2}(f^* \ f') \begin{pmatrix} k(0) & k' \\ k & \Sigma \end{pmatrix}^{-1} \begin{pmatrix} f^* \\ f \end{pmatrix} + f' \Sigma^{-1} f\right)$$

Temporarily rename the inverse matrix to be solved later:

$$= \exp\left(-\frac{1}{2}(f^* \ f') \begin{pmatrix} d & b' \\ b & A \end{pmatrix} \begin{pmatrix} f^* \\ f \end{pmatrix} + f' \Sigma^{-1} f\right)$$

$$= \exp\left(-\frac{1}{2}\left(f^* d + b f' f^* b' + f' A\right) \begin{pmatrix} f^* \\ f \end{pmatrix} + \text{constant}_0\right)$$

$$= \exp\left(-\frac{1}{2}((f^*)^2 d + b f' f^* + f f^* b' + f' f A + \text{constant}_1)\right)$$

$$= \exp\left(-\frac{1}{2}((f^*)^2 d + 2b f' f^* + \text{constant}_2)\right)$$

$$\propto \left(-\frac{d}{2}\left((f^*)^2 + \frac{2b' f f^*}{d} + \text{constant}_3\right)\right)$$

$$\propto \exp\left(-\frac{1}{2d^{-1}}\left(f^* + \frac{b' f}{d}\right)^2\right)$$

Since the above $= \mathcal{N}(f^*|\mu, \sigma^2)$,

$$\mu = -\frac{b' f}{d}$$

$$\sigma^2 = d^{-1}$$

Solve the inverse matrix:

$$\begin{pmatrix} k(0) & k' \\ k & \Sigma \end{pmatrix} \begin{pmatrix} d & b' \\ b & A \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & I \end{pmatrix}$$

- (1) $k(0)d + k'b = 1$
- (2) $k(0)b' + k'A = 0$
- (3) $kd + \Sigma b = 0$

$$(4) \quad kb' + \Sigma A = I$$

Take (3):

$$b = -\Sigma^{-1}kd$$

And insert into (1):

$$k(0)d + k'\Sigma^{-1}kd = 1$$

$$d = \frac{1}{k(0) + k'\Sigma^{-1}k}$$

$$b = \frac{-\Sigma^{-1}k}{k(0) + k'\Sigma^{-1}k}$$

Then:

$$\sigma^2 = k(0) + k'\Sigma^{-1}k$$

$$\mu = \frac{-b'f}{d} = \frac{k'\Sigma^{-1}f}{k(0) + k'\Sigma^{-1}k} (k(0) + k'\Sigma^{-1}k) = k'\Sigma^{-1}f$$

(2) Use an acquisition function (also called: selection function) $\mu(x)$ which estimates the gain or profit from optimization to determine where to sample function next. Balances exploration (search regions of high uncertainty) and exploitation (search where value already estimated high even if uncertainty is low). Three most common acquisition functions:

Maximum probability of improvement (MPI). Current best value is f^* . If Gaussian, the acquisition function using the surrogate model is:

$$\mu(x) = p(\hat{f} \geq f^* + \epsilon) = CDF \left(\frac{E(\hat{f}) - f^* - \epsilon}{\sigma(\hat{f})} \right)$$

Upper confidence bound (UCB). Tradeoff between points around the surrogate maximum and points with high variance. Based on Hoeffding inequalities - see below on multi-armed bandits.

$$E(\hat{f}(x)) + \eta\sigma(\hat{f}(x))$$

Expected improvement (EI).

$$EI_{y^*}(x) = \int \max(y^* - y, 0)p(y|x)dy$$

Where y^* is the minimum observed value so far, p is the surrogate function, y is the true objective and x is the hyperparameter.

47.9.1 Multi-armed bandit

Optimize gain through a choice of unknown and different rewards (can be varying distributions) by balancing exploration and exploitation. Often seen in reinforcement learning.

Naive strategies of only exploring or only exploiting are not optimal. Their regret estimates - maximum value minus value from that strategy - are high. Better strategies exist.

In a zero-regret strategy, $\frac{p}{T} \rightarrow 0$ as $T \rightarrow \infty$. Naive strategies are not zero-regret strategies.

ϵ -Greedy strategies involve uniform sampling at a low frequency (e.g. $\epsilon = 0.1$) periodically. ϵ -First strategies are a variation where the sampling occurs at the beginning. ϵ -Decreasing strategies involve more exploration at the beginning with ϵ decreasing through time.

In probability matching, the probability of choosing reflects the probability of that choice being optimal. This is Bayesian. Sample the posterior.

Upper Confidence Bound. Upper confidence bound (UCB). Using Hoeffding's inequality, sample at the rate of:

$$\hat{r} + \sqrt{\frac{2\ln(t)}{N_r}}$$

If the reward r has been high, the first term is high. If it has been less sampled then N_r is low so the second term is high.

With unlimited time to sample, the UCB will likely dominate over naive and ϵ -Greedy strategies. With very limited time to sample, the exploitation-only strategy may do better - particularly if the standard deviations are narrow so that limited sampling is more optimal than more thorough sampling.

Proof:

Markov's inequality:

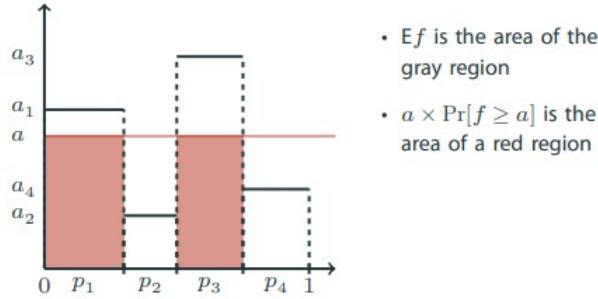
$$p(z \geq t) \leq \frac{E(z)}{t}$$

Intuition: a function $g \leq f$ where g is 1 if $f \geq a$ and g is 0 otherwise.

$$Ef \geq a \times \Pr[f \geq a]$$

Suppose f obtains values a_1, a_2, a_3, a_4 with probabilities

p_1, p_2, p_3, p_4



Derive Chebyshev's inequality from Markov's inequality:

$$p((z \geq E(z) + t) \cup (z \leq E(z) - t)) \leq \frac{Var(z)}{t^2}$$

If $\lambda > 0$ and

$$z \geq E(z) + t \rightarrow z - E(z) \geq t$$

Then:

$$e^{\lambda z} \geq e^{\lambda(E(z)+t)}$$

$$e^{\lambda(z-E(z))} \geq e^{\lambda t}$$

So:

$$p(z - E(z) \geq t) = p(e^{\lambda(z-E(z))} \geq e^{\lambda t})$$

By Markov's inequality,

$$p(e^{\lambda(z-E(z))} \geq e^{\lambda t}) \leq \frac{E(e^{\lambda(z-E(z))})}{e^{\lambda t}}$$

This proves the first of two Chernoff bounds:

$$p(z \geq E(z) + t) \leq \operatorname{argmin}_{\lambda \geq 0} \frac{E(e^{\lambda(z-E(z))})}{e^{\lambda t}} = \operatorname{argmin}_{\lambda \geq 0} \frac{MGF_{z-E(z)}}{e^{\lambda t}}$$

and the second of the bounds being proved analogously:

$$p(z \leq E(z) - t) \leq \operatorname{argmin}_{\lambda \geq 0} \frac{E(e^{\lambda(E(z)-z)})}{e^{\lambda t}} = \operatorname{argmin}_{\lambda \geq 0} \frac{MGF_{E(z)-z}}{e^{\lambda t}}$$

With z_i being independent:

$$E\left(\exp\left(\lambda \sum_i z_i\right)\right) = E\left(\prod_i e^{\lambda z_i}\right) = \prod_i E(e^{\lambda z_i})$$

$$p\left(\sum_i^n z_i \geq t\right) \leq \frac{\prod_i^n E(e^{\lambda z_i})}{e^{\lambda t}}$$

Let S be a Rademacher random variable, with $S = 1$ with half the time and $S = -1$ the other half. Then:

$$E(S^k) = \begin{cases} E(S^k) = 0 & \text{if } k \text{ is odd} \\ E(S^k) = 1 & \text{if } k \text{ is even} \end{cases}$$

Use the Taylor expansion:

$$e^x = \sum_0^\infty \frac{x^k}{k!}$$

To get:

$$E(e^{\lambda S}) = \sum_0^\infty \frac{\lambda^k E(S^k)}{k!} = \sum_{\text{even } k}^\infty \frac{\lambda^k}{k!} = \sum_0^\infty \frac{\lambda^{2k}}{2k!}$$

Use $2k! = 2^k k!$:

$$E(e^{\lambda S}) \leq \sum_0^\infty \frac{(\lambda^2)^k}{2^k k!} = \sum_0^\infty \left(\frac{\lambda^2}{2}\right)^k \frac{1}{k!} = \exp\left(\frac{\lambda^2}{2}\right)$$

Use symmetrization by creating an independent clone z' of z :

$$E_z(e^{\lambda(z-E_z(z))}) = E_z(e^{\lambda(z-E_{z'}(z))})$$

By Jensen's inequality ($f(E(z)) \leq E(f(z))$) on an exponential function:

$$E_z(e^{\lambda(z-E_{z'}(z))}) \leq E_z(E_{z'}(e^{\lambda(z-z')}))$$

$$E_z(e^{\lambda(z-E_{z'}(z))}) \leq E_z(e^{\lambda(z-z')})$$

$z - z'$ is symmetric about 0. With the Rademacher random variable S , $S(z - z')$ has same distribution as $z - z'$. So:

$$E_{z,z'}(e^{\lambda(z-z')}) = E_{z,z',S}(e^{\lambda S(z-z')}) = E_{z,z'}(E_S(e^{\lambda S(z-z')})|z, z')$$

Apply the result from above:

$$E(e^{\lambda S}) \leq \exp\left(\frac{\lambda^2}{2}\right)$$

To get:

$$E_S(e^{\lambda S(z-z')})|z, z') \leq \exp\left(\frac{\lambda^2(z-z')^2}{2}\right)$$

Since $|z - z'| \leq (b - a)$, $(z - z')^2 \leq (b - a)^2$. Then:

$$E_{z,z'}(e^{\lambda(z-z')})|z, z') \leq \exp\left(\frac{\lambda^2(b-a)^2}{2}\right)$$

which is a weaker version of Hoeffding's Lemma:

$$z \in [a, b]$$

$$E(e^{\lambda(z-E(z))}) \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right)$$

Use the Chernoff bound:

$$p\left(\frac{1}{n} \sum_i^n (z_i - E(z_i)) \geq t\right) = p\left(\sum_i^n (z_i - E(z_i)) \geq nt\right) \leq \frac{E(\exp(\lambda \sum_i^n (z_i - E(z_i))))}{e^{\lambda nt}}$$

Apply Hoeffding's Lemma:

$$= \frac{(\prod_i^n E(e^{\lambda(z_i - E(z_i))}))}{e^{\lambda nt}} \leq \frac{\prod_i^n \exp\left(\frac{\lambda^2(b-a)^2}{8}\right)}{e^{\lambda nt}}$$

Re-express to get first of two Hoeffding inequalities:

$$p\left(\frac{1}{n} \sum_i^n (z_i - E(z_i)) \geq t\right) \leq \underset{\lambda \geq 0}{\operatorname{argmin}} \exp\left(\frac{n\lambda^2(b-a)^2}{8} - \lambda nt\right) = \exp\left(-\frac{2nt^2}{(b-a)^2}\right)$$

The second Hoeffding inequality is derived analogously:

$$p\left(\frac{1}{n} \sum_i^n (z_i - E(z_i)) \leq -t\right) \leq \underset{\lambda \geq 0}{\operatorname{argmin}} \exp\left(\frac{n\lambda^2(b-a)^2}{8} - \lambda nt\right) = \exp\left(-\frac{2nt^2}{(b-a)^2}\right)$$

Take the second inequality and apply logs:

$$\ln p(E(\bar{z}) \geq \bar{z} + t) \leq -2nt^2$$

If let $p = t^{-4}$ then as $t \rightarrow \infty$, $t \rightarrow 0$:

$$\ln(t^{-4}) \leq -2nt^2$$

$$\frac{-4 \ln t}{n} \leq t^2$$

$$\sqrt{\frac{2 \ln t}{n}} \leq t$$

Where t is one-half the confidence interval, bounded by one-half the confidence interval.

48 Deep Learning

48.1 Neural Networks

48.1.1 Bayesian neural networks

Standard neural networks are trained using MLE for the weights. However, MLE yields point values without uncertainty and they're prone to overfitting.

Regularization can address this, and it is Gaussian priors if using L2 regularization (ridge regression) - but this just finds MAP which is not probabilistically valid. So use a Bayesian neural network, which uses posterior inference.

$$p(y|x_{train}, y_{train}) = \int \underbrace{p(y|x, w)}_{\text{NN output}} \underbrace{p(w|x_{train}, y_{train})}_{MCMC} dw$$

$$p(w|x_{train}, y_{train}) = \frac{p(y_{train}|x_{train}, w)p(w)}{NC}$$

Where $NC = \int p(y_{train}|x_{train}, w)p(w)dw$ and is analytically and computationally intractable. Test:

$$p(y_{test}|x_{test}, x_{train}, y_{train}) = \int p(y_{test}|x_{test}, w)p(w|x_{train}, y_{train})dw$$

Training is time-consuming. Two options: Langevin MC or Stochastic Variational Inference.

Langevin Monte Carlo. After initial w_0 :

$$w_{k+1} = \underbrace{w_k + \epsilon \nabla \log p(w_k|x_{train}, y_{train})}_{\text{Gradient descent}} + \eta_k$$

Where ϵ is the learning rate.

$$w_{k+1} = w_k + \epsilon \nabla \left(\underbrace{\log p(w_k)}_{\text{Weight decay } -C\|w_k\|^2} + \sum_i^N \underbrace{(1) \log p(y_i|x_i, w_k)}_{\text{Cross-entropy}} \right) + \eta_k$$

Where $\eta \sim \mathcal{N}(0, 2\epsilon I)$ which is noise proportional to ϵ . As $\epsilon \rightarrow 0$, get $p(w|x_{train}, y_{train})$.

Stochastic variational inference.

$$p(w|x, y) \approx q(w|\phi)$$

Optimize q :

$$q(w|\phi) = \underset{\phi}{\operatorname{argmin}} \operatorname{Dist}(q(w|\phi), p(w|x, y))$$

By minimizing KL Divergence:

$$\underset{\phi}{\operatorname{argmin}} KL(q(w|\phi) \parallel p(w|x, y))$$

So maximize the variational lower bound:

$$\underset{\phi}{\operatorname{argmax}} \mathcal{L}(\phi) = \int q(w|\phi) \log \frac{p(y|x, w)p(w)}{q(w|\phi)} dw$$

Can optionally use reparameterization trick here. Proceed without it and decompose:

$$\mathcal{L}(\phi) = \int q(w|\phi) \log p(y|x, w) dw - \int q(w|\phi) \log \frac{p(w)}{q(w|\phi)} dw$$

$$\mathcal{L}(\phi) = \int q(w|\phi) \log p(y|x, w) dw - KL(q(w|\phi) \parallel p(w))$$

Where the first term is a data term and the second is a regularization term.

48.1.2 Regularization

Several techniques:

- Penalize cost function using e.g. ridge, lasso, and elastic net.
- Dropout. Remove random neurons (=set activations to 0 for these neurons) so that there's no over-reliance on certain neurons. After training, rescale dropped neurons up to compensate for time absent. More generally, dropout is to add noise with the noise being the dropout rate. Drop-Connect, a variation of dropout, sets weights to 0 for selected neurons. Variational dropout is a generalization of dropout and uses a log-uniform prior to obtain same result as stochastic variational inference.
- Early stopping. Stop training after a fraction of iterations if cross-validation accuracy is declining.
- Stochastic/batch gradient descents. They won't perfectly fit training data.
- Data augmentation - which also increases amount of data

48.2 Autoencoders

48.2.1 General

This is a form of unsupervised learning. Compresses and subsequently expands data. Encode into very low dimension (latent space), embed in lower dimension, decode. MSE = reconstruction loss function that compares output to input.

Extracts features by reducing dimensionality, denoising, neural inpainting.

Much better than PCA at dimensionality reduction. If don't add hidden layers, reconstruction loss will be worse than PCA transform/inverse transform.

48.2.2 Evaluation

PCA is benchmark to compare autoencoder performance. Find PCA mean-squared error:

$$MSE = \frac{(true(test) - reconstructed)^2}{n}$$

Reconstructed is PCA transform of test set into reduced dimensions and then PCA inverse transform. Compare with MSE of autoencoder.

48.2.3 Variational autoencoders

Autoencoder latent space is full of gaps, and there is separation between clusters but can't interpolate well between these clusters. Autoencoder might maximize space between clusters for better classification, but VAEs can interpolate.

Encode, select μ and σ vectors from normal distribution with $z = \mu + \sigma x \epsilon$ where $\epsilon \sim N(0, 1)$ white noise. Then decode. If there are 2 dimensions, then 2 μ s and 2 σ s, etc.

VAEs use the reparameterization trick. If sample μ and/or σ from the distribution, backpropagation is blocked because there is a stochastic process. Instead, we use z which isolates the stochastic process as ϵ .

Results are regularized. Get 'degrees' of features, which make more sense, less volatile decoding (e.g. if a latent space isn't used in regular autoencoding, and it is encoded for test or new data, then it can be decoded as garbage). So can't just compare reconstructions loss of VAEs with AEs.

Loss function has two parts:

- Reconstruction loss (which forms clusters). Negative expected log likelihood.
- KL divergence (compresses clusters towards each other) of stochastic process from designated distribution.

Probability Distribution Outputs. If ML methods output $p(x)$, this is more useful than point estimates or groups of point estimates.

A simple algorithm $ML(x) = \log p(x)$ involves a normalization constant intractable at high dimensions:

$$p(x) = \frac{e^{ML(x)}}{NC}$$

RNNs apply the probability chain rule which calculates one pixel at a time so is too inefficient:

$$RNN(x_1, \dots, x_n) = p(x_1)p(x_2|x_1)\dots = p(x_1, \dots, x_n)$$

Independent pixels with the chain rule are too restrictive and lead to bad results because all the dependence structure is lost:

$$p(x_1, \dots, x_n) = p(x_1)\dots p(x_n)$$

Gaussian mixture modelling may work but requires a large number of Gaussians, so mixture modelling techniques are inefficient. GMM with infinite or large numbers of Gaussians like probabilistic PCA work:

$$p(x) = \int p(x|t)p(t)dt$$

With:

$$t \sim N(0, I)$$

$$x|t \sim N(\mu(t), \Sigma(t))$$

A simple set of functions like probabilistic PCA is:

$$\mu(t) = Wt + b$$

$$\Sigma(t) = \Sigma_0$$

Use CNNs for better, more detailed results:

$$p(x|w) = \int p(x|t, w)p(t)dt$$

$$t \sim N(0, I)$$

$$x|t \sim N(\mu(t, w), \Sigma(t, w))$$

But since this is parameter overload, simplify:

$$x|t \sim N(\mu(t, w), \text{diag } \sigma^2(t, w))$$

MLE to maximize with respect to w:

$$p(x|w) = \int p(x|t, w)p(t)dt$$

Though there is a latent variable t, EM is difficult. The E-step requires calculating the posterior distribution on latent variables with CNNs, which is intractable. MCMC is too slow - have to do it many times.

48.2.4 Variational autoencoder details

Approximate $p(t)$ from:

$$p(x|w) = \int p(x|t, w)p(t)dt$$

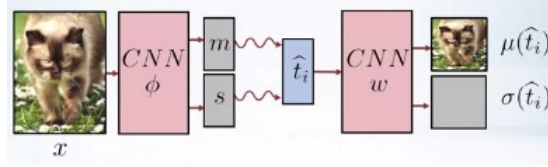
with $q(t)$ using the mean-field approximation from variational inference: $q(t) = q_i(t_i) \dots q_n(t_n)$ where $t_i \sim N(m_i, \text{diag } s_i^2)$. But this is intractable as it contains too many parameters.

If all q_i s are the same, it's too restrictive and get poor results.

So make all q_i s differ only by x_i :

$$t_i \sim N(m(x_i, \phi), \text{diag}(s^2(x_i, \phi)))$$

Then train:



If $s = 0$, then the VAE is just a normal AE.

Maximize the EM lower bound:

$$\begin{aligned} & \sum_i E_{q_i} \log \frac{p(x_i | t_i, w) p(t_i)}{q_i(t_i)} \\ &= \sum_i E_{q_i} \underbrace{\log p(x_i | t_i, w)}_{\text{Reconstruction loss}} + \underbrace{E_{q_i} \log \frac{p(t_i)}{q_i(t_i)}}_{-KL(q_i(t_i) \| p(t_i))} \end{aligned}$$

The first term is reconstruction loss and is the negative of the distance between x_i and $\mu(t_i)$ so maximized when the distance is minimized. Call it $f(w, \phi)$, as w is needed and ϕ is needed for t :

$$\sum_i E_{q_i} \log p(x_i | t_i, w) = f(w, \phi)$$

Where:

$$\begin{aligned} q_i(t_i) &= q(t_i | x_i, \phi) \\ t_i | x_i, \phi &\sim \mathcal{N}(m_i, \text{diag}(s_i^2)) \end{aligned}$$

Calculate gradient for gradient descent. Start with w :

$$\begin{aligned} \nabla_w f(w, \phi) &= \nabla_w \sum_i \int q(t_i | x_i, \phi) \log p(x_i | t_i, w) dt_i \\ &= \sum_i \nabla_w \int q(t_i | x_i, \phi) \log p(x_i | t_i, w) dt_i = \sum_i \int \nabla_w q(t_i | x_i, \phi) \log p(x_i | t_i, w) dt_i \\ &= \sum_i \int q(t_i | x_i, \phi) \nabla_w \log p(x_i | t_i, w) dt_i = \sum_i E_{q_i} \nabla_w \log p(x_i | t_i, w) \end{aligned}$$

Sample $\hat{t}_i \sim q(t_i|x_i, \phi)$. Then:

$$\approx \sum \nabla_w \log p(x_i|\hat{t}_i, w)$$

Which is a standard NN gradient. Use stochastic gradients:

$$\approx \frac{N}{n} \sum_s^n \nabla_w \log p(x_{is}|\hat{t}_{is}, w)$$

$$i_s \sim U[1...N]$$

Next calculate with respect to ϕ :

$$\begin{aligned} \nabla_\phi f(w, \phi) &= \nabla_\phi \sum_i \int q(t_i|x_i, \phi) \log p(x_i|t_i, w) dt_i \\ &= \sum_i \int \nabla_\phi q(t_i|x_i, \phi) \log p(x_i|t_i, w) dt_i \end{aligned}$$

But can't move ∇_ϕ further as it depends on q - q is conditional on ϕ . So use the log-derivative trick:

$$= \sum_i \int \frac{q(t_i|x_i, \phi)}{q(t_i|x_i, \phi)} \nabla_\phi q(t_i|x_i, \phi) \log p(x_i|t_i, w) dt_i$$

Since

$$\begin{aligned} \frac{d}{d\phi} \log g(\phi) &= \frac{\frac{dg(\phi)}{d\phi}}{g(\phi)} \\ \nabla \log g(\phi) &= \frac{\nabla g(\phi)}{g(\phi)} \end{aligned}$$

Then:

$$\begin{aligned} &= \sum_i \int q(t_i|x_i, \phi) \nabla_\phi \log q(t_i|x_i, \phi) \log p(x_i|t_i, w) dt_i \\ &= \sum_i E_{q_i} \nabla_\phi \log q(t_i|x_i, \phi) \log p(x_i|t_i, w) \end{aligned}$$

But $\log p(x_i|t_i, w)$ is really low in e.g. images, at least initially. So use the reparameterization trick: reparameterize t_i :

$$t_i = \epsilon_i \cdot s_i + m_i = g(\epsilon_i, x_i, \phi)$$

$$\epsilon_i \sim p(\epsilon_i) = \mathcal{N}(0, I)$$

Then:

$$\nabla_\phi f(w, \phi) = \sum_i \nabla_\phi E_{p(\epsilon_i)} \log p(x_i|g(\epsilon_i, x_i, \phi), w)$$

$$\begin{aligned}
&= \sum_i \int p(\epsilon_i) \nabla_{\phi} \log p(x_i | g(\epsilon_i, x_i, \phi), w) d\epsilon_i \\
&= \sum_i E_{p(\epsilon_i)} \nabla_{\phi} \log p(x_i | g(\epsilon_i, x_i, \phi), w)
\end{aligned}$$

The second term is a regularization term that is less negative when q and p are close (they're far for outliers and anomalous data). It is a difference between normal distributions so has an analytical solution:

$$\begin{aligned}
KL(q_i(t_i) \parallel p(t_i)) &= \int \log \frac{p(t)}{q(t)} q(t) dt \\
&= \int \left(-\frac{1}{2} \log(2\pi) - \log(\sigma_p) - \frac{1}{2} \left(\frac{t - \mu_p}{\sigma_p} \right)^2 + \frac{1}{2} \log(2\pi) + \log(\sigma_q) + \frac{1}{2} \left(\frac{t - \mu_q}{\sigma_q} \right)^2 \right) \\
&\quad \exp \left(-\frac{1}{2} \left(\frac{t - \mu_q}{\sigma_q} \right)^2 \right) dt
\end{aligned}$$

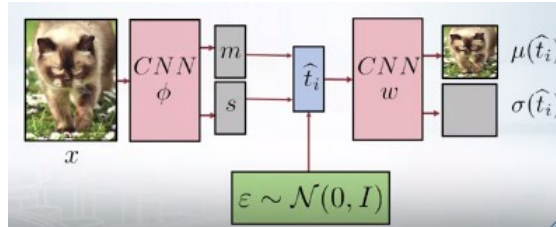
$\sigma_p = 1, \mu_p = 0$ in the standard normal for p. Let $\sigma_q = \sigma$ and $\mu_p = \mu$:

$$\begin{aligned}
&= \int \left(\log \sigma + \frac{1}{2} \left(\left(\frac{t - \mu}{\sigma} \right)^2 - t^2 \right) \right) \exp \left(-\frac{1}{2} \left(\frac{t - \mu}{\sigma} \right)^2 \right) dt \\
&= E_q \left(\log \sigma + \frac{1}{2} \left(\left(\frac{t - \mu}{\sigma} \right)^2 - t^2 \right) \right) = \log \sigma + \frac{1}{2\sigma^2} E_q(t - \mu)^2 - \frac{1}{2} \underbrace{E_q t^2}_{\sigma^2=1} \\
&= \log \sigma + \frac{1}{2\sigma^2} E_q(t - \mu)^2 - \frac{1}{2} = \log \sigma + \frac{1}{2\sigma^2} E_q(t^2 - 2\mu t + \mu^2) \\
&= \log \sigma + \frac{1}{2\sigma^2} ((E_q(t) = 1) - 2\mu(E_q(t) = 0) + \mu^2)) - \frac{1}{2} \\
&= \log \sigma + \frac{\mu^2 - \sigma^2 + 1}{2\sigma^2}
\end{aligned}$$

So:

$$KL(q_i(t_i) \parallel p(t_i)) = \sum_j \left(\log \sigma + \frac{\mu^2 - \sigma^2 + 1}{2\sigma^2} \right)$$

Which is computed without issue. So train:



49 Reinforcement Learning

49.1 Bellman equations

Bellman equations are about recursion, which is similar to agents recursively interacting with the environment.

49.1.1 Markov decision process

Discrete-time stochastic model.

Markovian property (only the present matters), together with state s , actions $A(s)$, model transition $T(s, a, s') \sim p(s'|s, a)$ (only depends on current state s), and rewards (a scalar value) $R(s, *) = R(s), R(s, a), R(s', a, s)$.

The MDP solution is the policy $\pi(s) = a$ and optimal policy π^* .

49.1.2 Rewards

Delayed rewards after a sequence of moves leads to the temporal credit assignment problem.

Sequence of rewards. Stationarity preferences. If

$$u(s_0, s_1, s_2, \dots) > u(s_0, s'_1, s'_2, \dots)$$

then

$$u(s_1, s_2, \dots) > u(s'_1, s'_2, \dots)$$

So,

$$u(s_0, s_1, s_2, \dots) = \sum_t^{\infty} R(s_t)$$

but with infinite horizons, this equals infinity. But

$$u(s_0, s_1, s_2, \dots) = \sum_t^{\infty} \gamma^t R(s_t)$$

can be finite where γ is between 0 and 1. The R can change a lot and is unknown, but can bound it:

$$\sum_t^{\infty} \gamma^t R(s_t) \leq \sum_t^{\infty} \gamma^t R_{max}(s_t) = \frac{R_{max}}{1 - \gamma}$$

49.1.3 Bellman equations

Optimal policy is

$$\pi^*(s) = \arg \max_a \sum_{s'} T(s, a, s') u(s')$$

Bellman equation

$$u(s) = R(s) + \gamma \max_a \sum_{s'} T(s, a, s') u(s')$$

Can rewrite:

$$V(s) = \max_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s') \right)$$

$$Q(s, a) = R(s, a) + \gamma \sum_{s'} T(s, a, s') \max_{a'} Q(s', a')$$

Moreover, define the continuation as

$$C(s, a) = \gamma \sum_{s'} T(s, a, s') \max_{a'} (R(s', a') + C(s', a'))$$

The value function V maximizes Q over all actions:

$$\begin{aligned} V(s) &= \max_a Q(s, a) \\ &= \max_a (R(s, a) + C(s, a)) \end{aligned}$$

where Q is decomposed into the reward and the continuation:

$$\begin{aligned} Q(s, a) &= R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s') \\ &= R(s, a) + C(s, a) \end{aligned}$$

49.2 Value and policy iteration

49.2.1 Value iteration

Iterating and separating out the truth, this keeps being added to the initial utility successively until the truth overwhelms (i.e. end at fixed point).

Initialize with $\hat{u}$ and propagate until convergence:

$$\hat{u}_{t+1}(s) = \underbrace{R(s)}_{\text{truth}} + \underbrace{\gamma \max_a \sum_{s'} T(s, a, s') \hat{u}_t(s')}_{\text{not truth}}$$

After convergence, extract the optimal policy from the actions taken in response to states.

49.2.2 Policy iteration

Initialize with $Q_0(s)$. Extract optimal policy at t:

$$\pi_t(s) = \arg \max_a Q_t(s, a)$$

Next, value iterate

$$Q_{t+1} = Q_t^\pi$$

which will drop out the reward. Repeat. Policy iteration will lead to

$$\lim_{t \rightarrow \infty} Q_t = Q^*$$

Proof of convergence. Begin with fixed policies π_1, π_2 associated respectively with Bellman operators B_1, B_2 :

$$B_1 V(s) = R(s, \pi_1(s)) + \gamma \sum_{s'} T(s, \pi_1(s), s') V(s')$$

$$B_2 V(s) = R(s, \pi_2(s)) + \gamma \sum_{s'} T(s, \pi_2(s), s') V(s')$$

The Bellman operators are monotonic. If $V_1 \geq V_2$ then $B_i V_1 \geq B_i V_2$ where $i = 1, 2$:

$$(B_i V_1 - B_i V_2)(s) = \gamma \underbrace{\sum_{s'} T(s, \pi_i(s), s')}_{\geq 0} \underbrace{(V_1(s) - V_2(s))}_{\geq 0} \geq 0$$

Let π_1 be an optimal policy of Q_1 and so $Q_1 = B_1 Q_1$. Let π_2 be a greedy policy of Q_1 and so $Q_1 \leq B_2 Q_1$. Policy iteration then follows:

1. $B_2 Q_1 \geq Q_1$
2. $B_2^{k+1} \geq B_2^k Q_1$
3. $B_2^k \geq Q_1$
4. $Q_2 \geq Q_1$

Policy function iteration is more direct because the policy is improved compared to value function iteration but more difficult (1) mathematically, and (2) the policy function contains 'less information' so is harder to maximize. So value function iteration is more common.

49.2.3 Linear programming

Value and policy iteration are almost primal-dual pairs in linear programming.

Linear programming solves Bellman equation in P-time after linearization.

Value function is non-linear because of max operator:

$$V(s) = \max_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s') \right)$$

The max operator is actually a supremum, the minimum of the acceptable values (which are max). So decompose the above into two equations which are collectively the 'primal' (in primal-dual pairs):

$$\begin{aligned} V_s &\geq R(s, a) + \gamma \sum_{s'} T V_{s'} \\ \min \sum_s V_s \end{aligned}$$

The 'dual' involves 'policy flow' q_{sa} which is similar to policy.

$$\max_{q_{sa}} \sum_s \sum_a q_{sa} R(s, a)$$

s.t.

$$1 + \gamma \sum_s \sum_a q_{sa} T = \sum_a q_{s'a}$$

49.3 Learning rates

Learning rate determines the 'size' of the gradient step in optimization. Should be large enough to not waste time but cannot be so large that it prevents convergence.

$$\begin{aligned} v_t(s) &= \frac{(t-1)v_{t-1}(s) + r_t(s)}{t} \\ &= \frac{t-1}{t} v_{t-1}(s) + \frac{1}{t} r_t(s) \\ &= v_{t-1}(s) - \frac{1}{t} (r_t(s) - v_{t-1}(s)) \\ &= v_{t-1}(s) - \alpha (r_t(s) - v_{t-1}(s)) \end{aligned}$$

where α is the learning rate.

$$\lim_{t \rightarrow \infty} v_t(s) \rightarrow \infty$$

assuming the learning rate is large enough to take on different values, i.e. $\sum \alpha_t$ diverges, but is not so big that it damps out everything else, i.e. $\sum \alpha_t^2$ converges.

49.4 Temporal difference learning

Sample different pathways through the MDP.

TD(0) (Monte Carlo) only takes one step in the MDP model, but it uses all samples (using expectations) in their observed probabilities to estimate the MDP model probabilities, including the value function and parameters like reward and learning rate. It therefore gives the maximum likelihood estimate.

TD(1) (Markov Decision Process) is based only on the estimates using only one trajectory within the MDP model, so a subset of data. So it estimates different values.

Taking a mix of TD($\lambda = 0$) and TD($\lambda = 1$) is better. This is $TD(\lambda)$, where $0 < \lambda < 1$. The first step E_1 below gets $1 - \lambda$ weight, the first two steps E_2 below get $\lambda(1 - \lambda)$, the third gets $\lambda^2(1 - \lambda)$, etc. where

$$\sum \lambda^{k-1}(1 - \lambda) = (1 - \lambda) \sum \lambda^{k-1} = (1 - \lambda) \frac{1}{1 - \lambda} = 1$$

The TD equations viewed as a weighted combination of k-step estimators:

$$\begin{aligned} E_1(TD(0)) : v(s_t) &= v(s_t) + \alpha_t(r_{t+1} + \gamma v(s_{t+1}) - v(s_t)) \\ E_2 : v(s_t) &= v(s_t) + \alpha_t(r_{t+1} + \gamma r_{t+2} + \gamma^2 v(s_{t+2}) - v(s_t)) \\ &\dots \\ E_k(TD(1)) : v(s_t) &= v(s_t) + \alpha_t(r_{t+1} + \dots + \gamma^{k-1} r_{t+k} + \dots - v(s_t)) \end{aligned}$$

TD(0) uses E_1 (over many parts of the MDP model), TD(1) only uses E_k (over one given trajectory in the MDP model).

49.5 Convergence

49.5.1 Q-learning

Maximize a' of the Q-function, using either the temporal difference form

$$Q_t(s_{t-1}, a_{t-1}) = Q_{t-1}(s_{t-1}, a_{t-1}) + \alpha(r_t + \gamma \max_{a'} Q_{t-1}(s_t, a') - Q_{t-1}(s_{t-1}, a_{t-1}))$$

or equivalently the Bellman equation form:

$$Q(s, a) = R(s, a) + \gamma \sum_{s'} T(s, a, s') \max_{a'} Q_{t-1}(s', a')$$

Value iteration is $Q_t = BQ_{t-1}$. The Bellman operator is $Q^* = BQ^*$.

49.5.2 Contraction mapping

Bellman operator is a contraction mapping so converges. (Note: $T = T(s, a, s')$):

$$\begin{aligned}
\|BQ_1 - BQ_2\| &= \max_{a,s} (BQ_1(s, a) - BQ_2(s, a)) \\
&= \max_{a,s} \left| R(s, a) + \gamma \sum_{s'} T \max_{a'} Q_1(s, a') - R(s, a) - \gamma \sum_{s'} T \max_{a'} Q_2(s, a') \right| \\
&= \max_{a,s} \left| \gamma \sum_{s'} T (\max_{a'} Q_1(s', a') - \max_{a'} Q_2(s', a')) \right| \\
&\leq \max_{a,s'} \left| \max_{a'} Q_1(s', a') - \max_{a'} Q_2(s', a') \right| \\
&\leq \max_{a',s'} |Q_1(s', a') - Q_2(s', a')|
\end{aligned}$$

where the last step uses:

$$\left| \max_a f(a) - \max_a g(a) \right| \leq \max_a |f(a) - g(a)|$$

Proof. WLOG

$$\max_a f(a) \geq \max_a g(a)$$

Then with $a_1 = \arg \max_a f(a)$, $a_2 = \arg \max_a g(a)$,

$$\begin{aligned}
\left| \max_a f(a) - \max_a g(a) \right| &\leq \max_a f(a) - \max_a g(a) = f(a_1) - g(a_2) \\
f(a_1) - g(a_2) &\leq f(a_1) - g(a_2)
\end{aligned}$$

49.6 Reward shaping

Can change Q-functions without changing optimal policy because the Bellman equation doesn't change on net. However, the rewards can be changed to tune behavior, make it more efficient, etc.

Scalar multiplication just scales everything and makes no net change

$$R'(s, a) = cR(s, a)$$

$$Q'(s, a) = cQ(s, a)$$

Scalar addition leads to no net change as terms cancel out

$$R'(s, a) = R(s, a) + c$$

$$Q'(s, a) = Q(s, a) + c + \gamma c + \gamma^2 c + \dots$$

$$Q(s, a) + \frac{c}{1 - \gamma} = R(s, a) + c + \gamma \sum_{s'} T \left(\max_{a'} Q'(s, a) + \frac{c}{1 - \gamma} \right)$$

$$\begin{aligned}
Q(s, a) + \frac{c}{1-\gamma} &= R(s, a) + c + \gamma \frac{c}{1-\gamma} + \gamma \sum_{s'} T \left(\max_{a'} Q'(s, a) \right) \\
Q(s, a) + \frac{c}{1-\gamma} &= R(s, a) + \frac{c}{1-\gamma} + \gamma \sum_{s'} T \left(\max_{a'} Q'(s, a) \right) \\
Q(s, a) &= R(s, a) + \gamma \sum_{s'} T \left(\max_{a'} Q'(s, a) \right)
\end{aligned}$$

Potential-based shaping

$$\begin{aligned}
R'(s, a) &= R(s, a) - \Psi(s) + \gamma \Psi(s') \\
R + \gamma R + \gamma^2 R + \dots
\end{aligned}$$

Telescoping sums cancel, so

$$Q'(s, a) = Q(s, a) - \Psi(s)$$

in which terms cancel out like scalar addition.

49.7 Exploration

49.7.1 Stochastic exploration: K-armed bandits

Metrics:

1. Optimal policy in limit
2. Max discounted rewards (e.g. Gittins index)
3. Max expected reward over finite horizon
4. Near-optimal arm (ϵ) with probability $1 - \gamma$ in time t
5. Near-max reward (ϵ) with probability $1 - \gamma$ in time t
6. Others

where some can be shown to be equivalent to others. Use Hoeffding and Union Bounds to make precise these notions.

49.7.2 Sequential exploration: R-max

Explore world to get known estimates and unknown states are bounded by R_{max} . Note: may not be efficient to explore everything or far if discount factor is low enough.

49.7.3 Simulation lemma

If parameters like rewards differ little, then value functions won't vary much.

49.7.4 Explore or exploit lemma

If all transitions are either accurately estimated or unknown, the optimal policy is either near-optimal or an unknown state which is reached quickly.

49.8 Generalization

Generalize dynamic programming as used in reinforcement learning.

$$Q(s, a) = F(w^a, f(s))$$

where w is a weight. Initialize weights. Then improve in a gradient step with

$$\Delta w_i^a = \underbrace{\alpha(r + \gamma \max_{a'} Q(s', a') - Q(s, a))}_{\text{Bellman TD-error}} \frac{\partial Q(s, a)}{\partial w_i^a}$$

Example with linear value function:

$$Q(s, a) = \sum_{i=1}^n f_i(s) w_i^a$$

An interpolator/extrapolator as a value function (an 'averager'):

$$V(s) = \sum_{s_b \in B} w(s, s_b) V(s_b)$$

where s_b is an anchor point in the set. But this is just an MDP:

$$\begin{aligned} V(s) &= \max_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s') \right) \\ &= \max_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') \sum_{s_b \in B} w(s', s_b) V(s_b) \right) \\ &= \max_a \left(R(s, a) + \gamma \sum_{s'} \underbrace{\sum_{s_b \in B} T(s, a, s') w(s', s_b)}_{T'(s, a, s_b)} V(s_b) \right) \\ &= \max_a \left(R(s, a) + \gamma \sum_{s'} T'(s, a, s') V(s_b) \right) \end{aligned}$$

which generalizes the linear function approximation to reinforcement learning, i.e. generalizes dynamic programming to RL.

49.9 Partially observable MDPs

Observable z represents the observable state. The probability of seeing z when the state is actually s is $O(s, z)$. $b(s)$ is the probability of observing state s . The probability of observing s' in the next time t is

$$\begin{aligned}
 b'(s') &= p(s'|b, a, z) \\
 &= \sum_s p(s|b, a, z) \underbrace{p(s'|b, a, z)}_{\text{Next state}} \\
 &= \sum_s b(s) p(s'|a, z, s) \\
 &= \sum_s b(s) \frac{p(z|s', a, s) p(s'|a, s)}{p(z|a, s)} \\
 &= \frac{\sum_s b(s) O(s', z) T(s, a, s')}{\sum_s \sum_{s'} b(s) O(s', z) T(s, a, s')}
 \end{aligned}$$

which is an intuitive probability.

49.9.1 Value iteration

Let

$$\begin{aligned}
 V_0(b) &= 0 \\
 V_t(b) &= \max_a (R(b, a) + \gamma \sum_z p(z|b, a) V_{t-1}(b'))
 \end{aligned}$$

where b' is the state estimation of b, a, z :

$$b' = SE(b, a, z)$$

and

$$R(b, a) = \sum_s p(s) R(s, a) = \sum_s b(s) R(s, a)$$

Claim that

$$V_t(b) = \max_{\alpha \in \Gamma_t} \alpha \cdot b = \max_{\alpha \in \Gamma_t} b(s) \cdot \alpha(s)$$

where Γ_t vectors are finite vectors which are piecewise linear and convex. It is a combination of highest linear pieces.

$$V_t(b) = \max_a V_t^a(b)$$

$$V_b^a(b) = \sum_z V_t^{a,z}(b) = Q$$

$$V_t^{a,z}(b) = \sum_s \left(R(s, a) \cdot \frac{b(s)}{|z|} \right) + \gamma p(z|b, a) V_{t-1}(SE(b, a, z))$$

where

$$V_{t-1}(SE(b, a, z)) = \max_{\alpha \in \Gamma_{t-1}} \frac{\sum_{s'} \alpha(s') O(s', z) \sum_s T(s, a, s') b(s)}{p(z|a, b)}$$

where $p(z|a, b)$ will cancel out, linearizing the problem, which is solved by linear programming.

49.9.2 Taxonomy

- Monte Carlo methods: uncontrolled and observable
- MDP: controlled and observable
- POMP: controlled and partially observable
- Hidden Markov model: uncontrolled and partially observable

49.9.3 Undecidability

The exact solution is undecidable. Since life is like a POMDP, life strategies may be undecidable.

49.9.4 Bayesian RL

Applies Bayesian probabilities to explore versus exploit tradeoff.

Reinforcement learning itself is planning in a continuous POMDP, with hidden states represented by MDP parameters.

49.9.5 Predictive state representation

Testing the world is equivalent to beliefs about the hidden states. Number of tests must bijectively map to the number of hidden state parameters. Theorem. Any n-state POMP can be represented by a PSR with no more than n states, each of which is no longer than n steps.

49.10 Generalizing to large state spaces

Address computational complexity when encountering large state spaces.

49.10.1 Temporal abstraction

Rescale the MDP with temporal abstraction:

$$V_{t+1}(s) = \max_o (R(s, o) + \sum_{s'} F(s, o, s') V_t(s'))$$

where o is options and

$$R = E(r_1 + \gamma r_2 + \gamma^2 r_3 + \dots)$$
$$F = E(\gamma^k)$$

where 'atomic' MDP steps are renormalized or grouped into coarse-grained MDP steps. This is semi-MDP, or SMDP. However, this is not Markovian as it depends on past rewards. Options are groups of actions. Options or coarse-graining can reduce exploration costs.

49.10.2 Goal abstraction

When there are simultaneous competing goals or objectives, need to balance by balancing, adding/average, voting, arbitration or other method. Gradient mass Q-learning: adding actions that lead to highest sum of Q functions.

49.10.3 State abstraction

Monte Carlo tree search. Map the MDP model and policy into a path in a tree search. Initialize with policy. Then at end of tree, look at options, choose one and see what the reward is by acting randomly beyond. Repeat for other branches/nodes. Then pick the best, and add it to current policy. Policy can then be optimized backward up the tree. 'Ends' are constraints. Useful for large state spaces. Planning time is independent of state space size.

49.11 Communication and coordination

Decentralized POMDPs: balance cost of communicating and coordinating with not.

Reinforcement learning: rewards + environment $\rightarrow$ behavior. Inverse RL: behavior + environment $\rightarrow$ infer reward. E.g. Max likelihood inverse RL, starting with reward based on observations and compute policy and then compute gradient on the reward function.

Policy shaping. Humans think in policies not rewards. Start with a human policy and gather data and improve policy over time.

50 Financial Math

50.1 Geometric Brownian motion

Model assuming asset prices will never be negative. A vanilla equity, such as a stock, always has this property. A standard Brownian motion cannot be used here, since there is a non-zero probability of the price becoming negative. A geometric Brownian motion is used instead, where the logarithm of the stock

price has stochastic behaviour.

A geometric Brownian motion (GBM) (also known as exponential Brownian motion) is a continuous-time stochastic process in which the logarithm of the randomly varying quantity follows a Brownian motion.

50.2 Risk neutral probability

The value of a call option is C_u if in the money and C_d if out of the money at an arbitrary strike price K . The theoretical current price is C . The call option is based on an underlying security is S in the present and it will be uS or dS with unknown probabilities. Can borrow money b at interest rate r . Then set up a system of equations,

$$aS_T + br = \begin{cases} auS + br = C_u & S_T = uS \\ adS + br = C_d & S_T = dS \end{cases}$$

Solving,

$$a = \frac{C_u - C_d}{(u - d)S}$$

$$b = \frac{uC_d - dC_u}{(u - d)r}$$

And the current price is

$$C = aS + b$$

Where the replicating portfolio with parameters a and b will replicate the results of C_u, C_d . The value of C can be calculated without knowing probabilities because it's based on the underlying so reflects its probabilities. Further,

$$C = aS + b = \frac{1}{r}(qC_u + (1 - q)C_d)$$

where

$$q = \frac{r - d}{u - d}$$

is the risk neutral probability. The r absorbs other risks, like political risks. If $r = u$ then must always get C_u ; if $r = d$ then must always get C_d . The q is an alternative probability measure, and it reflects risks that are difficult to calculate.

Value for risk-averse investors of any risky financial product:

$$Value = DiscountFactor \times E_P(Payoff) - X$$

where X is an adjustment for uncertainty. A product with very uncertain cash flows (understand very variable, scattered) will have a higher X .

This formula is not extremely powerful and certainly not fit for derivative pricing, as the X value is hard to compute and has to be calibrated for each market, and even for investors who have different levels of risk aversion. To get around this problem, financial research has come up with the concept of risk neutral measure or equivalent Martingale measure.

This concept consists in defining a probability measure that is completely theoretical and instead of reflecting only real world probabilities of payoffs, also reflects the risk aversion of investors for improbable extreme payoffs. With such a theoretical probability measure (let it be Q) we would have the following formula for the value of a financial product:

$$Value = DiscountFactor \times E_Q(Payoff)$$

where $E_Q(Payoff)$ designates the expected value of payoffs in the probability measure Q .

50.3 Fundamental theorems of asset pricing

First fundamental theorem. A market model does not have arbitrage opportunities if, and only if, there exists probability measure Q , equivalent to P , such that the discounted stock process $Z(t) = \frac{S(t)}{\beta(t)}$ is a Q -martingale.

Completeness. The market model is complete if, and only if, the martingale probability measure Q is unique.

Second fundamental theorem. The following are equivalent: (1) the market model is complete, (2) the martingale $Z(t)$ has the predictable representation property, and (3) the EMM Q that makes $Z(t) = S(t)/\beta(t)$ into a martingale is unique.

50.4 Put-call parity

50.5 Black-Scholes

50.5.1 PDE derivation: Delta hedging

Assume

$$dS(t) = \mu S(t)dt + \sigma S(t)dt$$

Let $C(t)$ be the value of the call option. Then use Ito's lemma

$$\begin{aligned}
dC(S(t), t) &= \frac{\partial C(S(t), t)}{\partial S(t)} dS(t) + \frac{\partial C(S(t), t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \sigma^2 S(t)^2 dt \\
&= \frac{\partial C(S(t), t)}{\partial S(t)} (\mu S(t) dt + \sigma S(t) dB(t)) + \frac{\partial C(S(t), t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \sigma^2 S(t)^2 dt \\
&= \left(\frac{\partial C(S(t), t)}{\partial t} + \mu S(t) \frac{\partial C(S(t), t)}{\partial S(t)} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \right) dt + \underbrace{\sigma S(t) \frac{\partial C(S(t), t)}{\partial S(t)} dB(t)}_{\text{Stochastic}}
\end{aligned}$$

where $dS(t) = rS(t)dt + \sigma S(t)dB(t)$. Let Π be a portfolio to be added to the call option

$$\Pi = \Delta S(t) + \alpha A$$

where Δ is the amount (positive or negative) of the underlying $S(t)$, and α is the amount borrowed from the bank A at interest rate r . Then differentiate

$$\begin{aligned}
d\Pi &= \Delta dS(t) + \alpha dA \\
&= \Delta dS(t) + \alpha r A dt \\
&= \Delta (\mu S(t) dt + \sigma S(t) dB(t)) + \alpha r A dt \\
&= (\Delta \mu S(t) + \alpha r A) dt + \underbrace{\Delta \sigma S(t) dB(t)}_{\text{Stochastic}}
\end{aligned}$$

Now add $dC(t) + d\Pi$ in a way that neutralizes the stochastic components:

$$\Delta \sigma S(t) + \sigma S(t) \frac{\partial C(t)}{\partial S(t)} = 0$$

$$\Delta = - \frac{\partial C(t)}{\partial S(t)}$$

Then,

$$d\Pi = \Delta dS(t) + \alpha dA = - \frac{\partial C(t)}{\partial S(t)} dS(t) + \alpha dA$$

which should be substituted into

$$\begin{aligned}
d(C(t) + \Pi) &= \left(\frac{\partial C(t)}{\partial t} + \mu S(t) \frac{\partial C(t)}{\partial S(t)} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2} - \frac{\partial C(t)}{\partial S(t)} \mu S(t) + \alpha r A \right) dt \\
&= \left(\frac{\partial C(t)}{\partial t} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2} + \alpha r A \right) dt
\end{aligned}$$

To avoid arbitrage, the option and portfolio differential must equal the value that this combination earns in the bank

$$d(C(t) + \Pi) = (C(t) + \Pi) r dt = \left(C(t) - \frac{\partial C(t)}{\partial S(t)} S(t) + \alpha A \right) r dt$$

Equating,

$$\left(C(t) - \frac{\partial C(t)}{\partial S(t)} S(t) + \alpha A \right) r dt = \left(\frac{\partial C(t)}{\partial t} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2} + \alpha r A \right) dt$$

So,

$$rC(t) - r \frac{\partial C(t)}{\partial S(t)} S(t) + r\alpha A = \frac{\partial C(t)}{\partial t} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2} + \alpha r A$$

$$rC(t) - r \frac{\partial C(t)}{\partial S(t)} S(t) = \frac{\partial C(t)}{\partial t} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2}$$

$$\frac{\partial C(t)}{\partial t} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2} + r \frac{\partial C(t)}{\partial S(t)} S(t) - rC(t) = 0$$

which is the Black-Scholes PDE. Intuitively, going a step back,

$$r \left(C(t) - \frac{\partial C(t)}{\partial S(t)} S(t) \right) = \frac{\partial C(t)}{\partial t} + \frac{1}{2} \sigma^2 S(t)^2 \frac{\partial^2 C(t)}{\partial S(t)^2}$$

where the left side is the interest growing on the option premium minus the amount borrowed in terms of the underlying.

50.5.2 PDE derivation: Replicating portfolio

To replicate a European call option, use this replicating portfolio:

$$V(t) = a(t)S(t) + b(t)\beta(t) = C(S(t), t)$$

Take the differential:

$$dV(t) = a(t)dS(t) + b(t)d\beta(t) = dC(S(t), t)$$

Then using Ito's lemma,

$$dC(S(t), t) = \frac{\partial C(S(t), t)}{\partial S(t)} dS(t) + \frac{\partial C(S(t), t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \sigma^2 S(t)^2 dt$$

where the last term uses $dS(t) = rS(t)dt + \sigma S(t)dB(t)$. Equate the above equations,

$$a(t)dS(t) + b(t)d\beta(t) = \frac{\partial C(S(t), t)}{\partial S(t)} dS(t) + \frac{\partial C(S(t), t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \sigma^2 S(t)^2 dt$$

$$\left(a(t) - \frac{\partial C(S(t), t)}{\partial S(t)} \right) dS(t) = \left(\frac{\partial C(S(t), t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \sigma^2 S(t)^2 dt \right) - b(t)d\beta(t)$$

The left side has a positive quadratic variation unless it is zero, and the right side has zero quadratic variation. If the two sides are equal, it must be that

$$a(t) = \frac{\partial C(S(t), t)}{\partial S(t)}$$

and

$$b(t)d\beta(t) = \left(\frac{\partial C(S(t), t)}{\partial t} dt + \frac{1}{2} \frac{\partial^2 C(S(t), t)}{\partial S(t)^2} \sigma^2 S(t)^2 \right) dt$$

Using $d\beta(t) = r\beta(t)dt$, obtain Black-Scholes call option results which will be derived later.

$$\begin{aligned} a(t) &= S(t)\Phi(h(t)) \\ b(t) &= K\Phi(h(t) - \sigma\sqrt{T-t}) \end{aligned}$$

50.5.3 PDE derivation: Backwards with Feynman-Kac

50.5.4 PDE solution: Derivation

Derive for European call option.

Compound interest ODE with constant interest rate is

$$\frac{dx(t)}{dt} = rx(t)$$

If the rate is subject to white noise,

$$\frac{dX(t)}{dt} = (r + \sigma\epsilon(t))X(t)$$

so,

$$dX(t) = rX(t)dt + \sigma X(t)\epsilon(t)dt$$

but since white noise is the derivative of Brownian motion

$$\epsilon(t) = \frac{dB(t)}{dt} = B'(t)$$

then

$$dX(t) = rX(t)dt + \sigma X(t)dB(t)$$

If $\sigma = 0$, the deterministic equation is recovered. The solution to the SDE will be geometric Brownian motion (GBM). Convert the above equation:

$$\frac{dX(t)}{X(t)} = rdt + \sigma dB(t)$$

But

$$\frac{dX(t)}{X(t)} = d(\ln(X(t)))$$

So,

$$\begin{aligned} d(\ln(X(t))) &= \frac{1}{X(t)}dX(t) + \frac{1}{2} \left(-\frac{1}{X^2(t)} \right) (dX(t))^2 \\ &= \frac{1}{X(t)}dX(t) + \frac{1}{2} \left(-\frac{1}{X^2(t)} \right) \sigma^2 X^2(t)dt \\ &= \frac{1}{X(t)}(\mu X(t)dt + \sigma X(t)dB(t)) - \frac{1}{2}\sigma^2 dt \\ &= \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dB(t) \end{aligned}$$

where $(dt)^2 = 0$, $dt dB = 0$ because t is of finite variation and

$$(dX)^2 = \sigma^2 X^2(t) dB^2(t) = \sigma^2 X^2(t) dt$$

So,

$$d(\ln(X(t))) = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB(t)$$

$$\ln(X(t)) = \ln(X(0)) + (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB(t)$$

$$X(t) = e^{\ln(X(0)) + (\mu - \frac{1}{2}\sigma^2)dt + \sigma dB(t)}$$

$$X(t) = X(0)e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma B(t)}$$

Rename $X(t) \rightarrow S(t)$, the underlying security price, and start with the Black-Scholes model where the mean is μ :

$$dS(t) = \mu S(t)dt + \sigma S(t)dB(t)$$

with $B(t)$ being P-Brownian motion. It is only a martingale if $\mu = r$. So convert this to Q-Brownian motion $W(t)$. Write

$$dS(t) = \sigma S(t) \left(\frac{\mu}{\sigma} dt + dB(t) \right)$$

and there is a unique Q s.t.

$$\frac{\mu}{\sigma} + dB(t) = \frac{r}{\sigma} dt + dW(t)$$

So,

$$\sigma B(t) = rt + \sigma W(t) - \mu t$$

Thus,

$$S(t) = S(0)e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma B(t)} = S(0)e^{(r - \frac{1}{2}\sigma^2)t + \sigma W(t)}$$

So the equivalent martingale measure (EMM) Q for $S(t)$ is

$$dS(t) = rS(t)dt + \sigma S(t)dW(t)$$

Then $S(T)$ has a log-normal distribution

$$S(T) \sim LN \left(\left(r - \frac{\sigma^2}{2} \right) T + \ln S(0), \sigma^2 T \right)$$

The price of a claim X at time T is

$$C(t) = e^{-r(T-t)} E_Q(X | \mathcal{F}_t)$$

Generalize so that the claim starts at time t instead of 0:

$$S(T) = S(t)e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma(W(T) - W(t))}$$

and so

$$\ln(S(T)) = \ln(S(t)) + (r - \frac{1}{2}\sigma^2)(T - t) + \sigma(W(T) - W(t))$$

whose expectation or mean is

$$E(\ln(S(T))) = E(\ln(S(t)) + (r - \frac{1}{2}\sigma^2)(T - t) + \sigma(W(T) - W(t))) = \ln(S(t)) + (r - \frac{1}{2}\sigma^2)(T - t)$$

and whose variance is

$$\text{Var}(\ln(S(T))) = \sigma^2(T - t)$$

Then

$$S(T) \sim LN\left(\left(r - \frac{\sigma^2}{2}\right)(T - t) + \ln S(t), \sigma^2(T - t)\right)$$

A call option pays $(S(T) - K)^+$ at time T . So,

$$C(t) = e^{-r(T-t)} E_Q((S(T) - K)^+ | \mathcal{F}_t)$$

$S(T) = e^Y$, $Y \sim \mathcal{N}(\mu, \sigma_1^2)$, with mean

$$\mu = (r - \frac{\sigma^2}{2})(T - t) + \ln S(t)$$

and variance

$$\sigma_1^2 = \sigma^2(T - t)$$

So, rewrite the above as

$$C(t) = e^{-r(T-t)} E_Q((e^Y - K)^+ | \mathcal{F}_t)$$

and with expectation under $N(\mu, \sigma_1^2)$,

$$E(e^Y - K)^+ = E((e^Y - K)I(e^Y > K)) = E(e^Y I(e^Y > K)) - KP(e^Y > K)$$

The first term can be calculated by a direct integration, or by changing the measure, with $dQ_1/dP = \Lambda = e^Y/Pe^Y$:

$$\begin{aligned} E(e^Y I(e^Y > K)) &= Ee^Y E((e^Y/Pe^Y)I(e^Y > K)) \\ &= Ee^Y E(\Lambda I(e^Y > K)) \\ &= Ee^Y E_{Q_1} I(e^Y > K) \\ &= Ee_{Q_1}^E(e^Y > K) \end{aligned}$$

Since the expectation of a log normal probability is $Ee^Y = e^{\mu - \sigma_1^2/2}$ (see the section on log normal distributions),

$$\frac{dQ_1}{dP} = \frac{e^Y}{Pe^Y} = e^{Y - \mu - \sigma_1^2/2}$$

which by

$$\begin{aligned}
\frac{dQ}{dP}(X) &= \Lambda(X) \\
&= \frac{\sigma_1}{\sigma_2} e^{\frac{(X-\mu_1)^2}{2\sigma_1^2} - \frac{(X-\mu_2)^2}{2\sigma_2^2}} \\
&= e^{\frac{(Y-(\mu+\sigma_1^2))^2}{2\sigma_1^2} - \frac{(Y-\mu)^2}{2\sigma_2^2}} \\
&= e^{Y-\mu-\sigma_1^2/2}
\end{aligned}$$

So $Q_1 \sim N(\mu + \sigma_1^2, \sigma_1^2)$. Therefore, the first term is

$$E(e^Y I(e^Y > K)) = e^{\mu+\sigma_1^2/2} (1 - \Phi((\ln K - \mu - \sigma_1^2)/\sigma_1))$$

The second term is

$$KP(e^Y > K) = K(1 - \Phi((\ln K - \mu)/\sigma_1))$$

Using the property $1 - \Phi(x) = \Phi(-x)$,

$$E(e^Y - K)^+ = e^{\mu+\sigma^2/2} \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right)$$

Now, the call option is

$$\begin{aligned}
C(t) &= e^{-r(T-t)} E(e^Y - K)^+ \\
&= e^{-r(T-t)} \left(e^{\mu+\sigma^2/2} \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right) \right) \\
&= e^{\mu+\sigma^2/2-r(T-t)} \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K e^{-r(T-t)} \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right) \\
&= e^{\mu-(r-\sigma^2/2)(T-t)} \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K e^{-r(T-t)} \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right) \\
&= e^{(r-\frac{\sigma^2}{2})(T-t)+\ln S(t)-(r-\sigma^2/2)(T-t)} \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K e^{-r(T-t)} \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right) \\
&= e^{\ln S(t)} \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K e^{-r(T-t)} \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right) \\
&= S(t) \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) - K e^{-r(T-t)} \Phi\left(\frac{\mu - \ln K}{\sigma_1}\right)
\end{aligned}$$

The first term is

$$\begin{aligned}
S(t) \Phi\left(\frac{\mu + \sigma_1^2 - \ln K}{\sigma_1}\right) &= S(t) \Phi\left(\frac{(r - \frac{\sigma^2}{2})(T-t) + \ln S(t) + \sigma^2(T-t) - \ln K}{\sigma\sqrt{T-t}}\right) \\
&= S(t) \Phi\left(\frac{\ln \frac{S(t)}{K} + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}\right) \\
&= S(t) \Phi(h(t))
\end{aligned}$$

The second term is

$$\begin{aligned}
Ke^{-r(T-t)}\Phi\left(\frac{\mu - \ln K}{\sigma_1}\right) &= Ke^{-r(T-t)}\Phi\left(\frac{(r - \frac{\sigma^2}{2})(T-t) + \ln S(t) - \ln K}{\sigma\sqrt{T-t}}\right) \\
&= Ke^{-r(T-t)}\Phi\left(\frac{(r + \frac{\sigma^2}{2})(T-t) + \ln \frac{S(t)}{\ln K} - \sigma^2(T-t)}{\sigma\sqrt{T-t}}\right) \\
&= Ke^{-r(T-t)}\Phi\left(h(t) - \frac{\sigma^2(T-t)}{\sigma\sqrt{T-t}}\right) \\
&= Ke^{-r(T-t)}\Phi\left(h(t) - \sigma\sqrt{T-t}\right)
\end{aligned}$$

So,

$$C(t) = S(t)\Phi(h(t)) - Ke^{-r(T-t)}\Phi(h(t) - \sigma\sqrt{T-t})$$

50.5.5 PDE solution: Binomial tree

50.5.6 PDE solution: Heat and diffusion equation

50.5.7 Greeks

50.6 Exotic options

Path-dependent options.

50.6.1 Asian options

Assume the underlying under Q-Brownian motion is

$$dS(t) = rS(t)dt + \sigma S(t)dB(t)$$

An Asian option involves either fixed strike price

$$C(T) = \left(\frac{1}{T} \int_0^T S(u)du - K\right)^+$$

or a floating strike price

$$C(T) = \left(\frac{1}{T} \int_0^T S(u)du - KS(T)\right)^+$$

Let the number of shares be

$$a(t) = \frac{1}{rT}(1 - e^{-r(T-t)})$$

and find a self-financing portfolio to replicate the average $\frac{1}{T} \int_0^T S(u)du$. So invest $a(t)$ into the risky asset to get $a(t)S(t)$ and the rest, i.e. $V(t) - a(t)S(t)$ into bonds at interest rate r):

$$\begin{aligned}
dV(t) &= a(t)dS(t) + r(V(t) - a(t)S(t))dt \\
&= rV(t)dt + a(t)(dS(t) - rS(t)dt)
\end{aligned}$$

Use e^{rt} as the numeraire, so find differential of $\frac{V(t)}{e^{rt}}$:

$$\begin{aligned} d(Ve^{-rt}) &= -re^{-rt}V(t)dt + e^{-rt}dV \\ &= -re^{-rt}V(t)dt + e^{-rt}(rV(t)dt + a(t)(dS(t) - rS(t)dt)) \\ &= e^{-rt}a(t)(dS(t) - rS(t)dt) \end{aligned}$$

Integrating both sides,

$$\begin{aligned} \int_0^T d(Ve^{-rt}) &= \int_0^T e^{-rt}a(t)(dS(t) - rS(t)dt) \\ V(T)e^{-rT} - V(0) &= \int_0^T e^{-rt}a(t)(dS(t) - rS(t)dt) \\ e^{-rT}V(T) &= V(0) + \int_0^T e^{-rt}a(t)(dS(t) - rS(t)dt) \\ V(T) &= V(0)e^{rT} + \int_0^T e^{-r(T-t)}a(t)(dS(t) - rS(t)dt) \end{aligned}$$

Since

$$d(e^{r(T-t)}S(t)a(t)) = e^{r(T-t)}a(t)dS(t) - re^{r(T-t)}S(t)a(t)dt + e^{r(T-t)}S(t)da(t)$$

then

$$\begin{aligned} \int_0^T d(e^{r(T-t)}S(t)a(t)) &= \int_0^T e^{r(T-t)}a(t)dS(t) - re^{r(T-t)}S(t)a(t)dt + \int_0^T e^{r(T-t)}S(t)da(t) \\ a(T)S(T) - e^{rT}a(0)S(0) &= \int_0^T e^{r(T-t)}a(t)(dS(t) - rS(t)dt) + \int_0^T e^{r(T-t)}S(t)da(t) \\ a(T)S(T) - e^{rT}a(0)S(0) - \int_0^T e^{r(T-t)}S(t)da(t) &= \int_0^T e^{r(T-t)}a(t)(dS(t) - rS(t)dt) \end{aligned}$$

Substituting back into the above,

$$V(T) = V(0)e^{rT} + a(T)S(T) - e^{rT}a(0)S(0) - \int_0^T e^{r(T-t)}S(t)da(t)$$

Now, $V(0) = a(0)S(0)$, so

$$V(T) = a(T)S(T) - \int_0^T e^{r(T-t)}S(t)da(t)$$

and $a(T) = \frac{1}{rT}(1 - e^{-r(T-T)}) = 0$, so

$$V(T) = - \int_0^T e^{r(T-t)}S(t)da(t)$$

Make a substitution of $da(t) \rightarrow dt$:

$$\begin{aligned}\frac{d}{dt}a(t) &= \frac{d}{dt} \left(\frac{1}{rT} (1 - e^{-r(T-t)}) \right) = -\frac{1}{T} e^{-r(T-t)} \\ da(t) &= -\frac{dt}{T} e^{-r(T-t)}\end{aligned}$$

Substituting,

$$\begin{aligned}V(T) &= \frac{1}{T} \int_0^T e^{r(T-t)} S(t) \frac{dt}{T} e^{-r(T-t)} \\ V(T) &= \frac{1}{T} \int_0^T S(t) dt\end{aligned}$$

This is not a martingale. Use a change of numeraire to move in the direction of a martingale. Let

$$Z(t) = \frac{V(t)}{S(t)}$$

and further,

$$Z(t) = \frac{V(t)}{S(t)} = \frac{e^{-rt}V(t)}{e^{-rt}S(t)} = (e^{-rt}V(t))(e^{-rt}S(t))^{-1}$$

This is actually two changes of numeraires: once with e^{rt} and second with $S(t)$. We want

$$dZ(t) = d((e^{-rt}V(t))(e^{-rt}S(t))^{-1})$$

Do this by taking the first part in the brackets, then the second. The first part is

$$\begin{aligned}d(e^{-rt}V(t)) &= e^{-rt}a(t)(rS(t)dt + \sigma S(t)dB(t) - rS(t)dt) \\ &= e^{-rt}a(t)(\sigma S(t)dB(t))\end{aligned}$$

The second part is

$$d(e^{-rt}S(t))^{-1} = \frac{-1}{(e^{-rt}S(t))^2} d(e^{-rt}S(t)) + \frac{1}{2} \frac{2}{(e^{-rt}S(t))^3} d(e^{-rt}S(t))^2$$

by Ito's lemma. The $d(e^{-rt}S(t))$ term is

$$d(e^{-rt}S(t)) = -re^{-rt}S(t)dt + e^{-rt}(rS(t)dt + \sigma S(t)dB(t)) = \sigma S(t)e^{-rt}dB(t)$$

Substitute this back into the second part above

$$\begin{aligned}d(e^{-rt}S(t))^{-1} &= \frac{-1}{(e^{-rt}S(t))^2} \sigma S(t)e^{-rt}dB(t) + \frac{1}{2} \frac{2}{(e^{-rt}S(t))^3} (\sigma S(t)e^{-rt}dB(t))^2 \\ &= \frac{-1}{(e^{-rt}S(t))^2} \sigma S(t)e^{-rt}dB(t) + \frac{1}{2} \frac{2}{(e^{-rt}S(t))^3} \sigma^2 (S(t)e^{-rt})^2 dt \\ &= \frac{-\sigma dB(t) + \sigma^2 dt}{e^{-rt}S(t)}\end{aligned}$$

Combine the first and second parts. Begin by applying Ito's lemma

$$\begin{aligned}
dZ(t) &= d((e^{-rt}V(t))(e^{-rt}S(t))^{-1}) \\
&= e^{-rt}d(e^{-rt}S(t))^{-1} + (e^{-rt}S(t))^{-1}d(e^{-rt}V(t)) + d(e^{-rt}V(t))d(e^{-rt}S(t))^{-1} \\
&= e^{-rt}V(t)\sigma\left(\frac{-\sigma dB(t) + \sigma^2 dt}{e^{-rt}S(t)}\right) + \frac{1}{e^{-rt}S(t)}e^{-rt}a(t)\sigma S(t)dB(t) + \sigma\left(\frac{-\sigma dB(t) + \sigma^2 dt}{e^{-rt}S(t)}\right)e^{-rt}a(t)\sigma S(t)dt \\
&= Z(t)\sigma(\sigma dt - dB(t)) + \sigma a(t)dB(t) + \sigma^2 a(t)dB(t)(\sigma dt - dB(t)) \\
&= \sigma^2 Z(t)dt - \sigma Z(t)dB(t) + \sigma a(t)dB(t) + \sigma^2 a(t)dt \\
&= Z(t)\sigma(\sigma dt - dB(t)) + \sigma a(t)dB(t) - \sigma^2 a(t)dt \\
&= Z(t)\sigma^2 dt - \sigma Z(t)dB(t) + \sigma a(t)dB(t) - \sigma^2 a(t)dt \\
&= \sigma^2(Z(t) - a(t))dt + \sigma(a(t) - Z(t))dB(t)
\end{aligned}$$

This is still not a martingale because the coefficient of $dt \neq 0$. Turn this into a martingale. Change to Q_1 -Brownian motion with

$$\begin{aligned}
B^1(t) &= B(t) - \sigma t \\
dB(t) &= dB^1(t) + \sigma
\end{aligned}$$

Substitute into the above,

$$\begin{aligned}
dZ(t) &= \sigma^2(Z(t) - a(t))dt + \sigma(a(t) - Z(t))dB(t) \\
&= \sigma^2(Z(t) - a(t))dt + \sigma(a(t) - Z(t))(dB^1(t) + \sigma) \\
&= -\sigma(Z(t) - a(t))dB^1(t)
\end{aligned}$$

which is a martingale.

An Asian call option at maturity (ie its payoff at time T) is

$$C(T) = (\bar{S} - K_1 S(T) - K_2)^+$$

Take $K_2 = 0$ because this is a floating Asian option ($K_1 = 0$ for a fixed).

$$C(T) = (V(T) - K_1 S(T))^+$$

At time t,

$$C(t) = e^{-r(T-t)}E_Q((V(T) - K_1 S(T))^+|\mathcal{F}_t)$$

Substitute $Z(t) = \frac{V(t)}{S(t)}$.

$$C(t) = e^{-r(T-t)}E_Q((Z(T)S(T) - K_1 S(T))^+|\mathcal{F}_t)$$

$$C(t) = e^{-r(T-t)}E_Q(S(T)(Z(T) - K_1)^+|\mathcal{F}_t)$$

but Q-Brownian motion is not a martingale as seen above. Instead, use $S(t)$ and e^{rt} as numeraires and convert to a Q_1 -Brownian motion which is a martingale.

$$C(t) = S(t)E_{Q_1}((Z(T) - K_1)^+|\mathcal{F}_t)$$

$$\begin{aligned}
C(t) &= S(t)u(Z(t), t) \\
d(u(Z(t), t)) &= \frac{\partial u(Z(t), t)}{\partial t} dt + \frac{u(Z(t), t)}{\partial Z(t)} dZ(t) + \frac{1}{2} \frac{\partial^2 u(Z(t), t)}{\partial Z(t)^2} (dZ(t))^2 \\
&= \frac{\partial u(Z(t), t)}{\partial t} dt + \frac{u(Z(t), t)}{\partial Z(t)} (-\sigma(Z(t) - a(t)) dB^1(t)) + \frac{1}{2} \frac{\partial^2 u(Z(t), t)}{\partial Z(t)^2} \sigma^2(Z(t) - a(t))^2 dt \\
&= \underbrace{\left(\frac{\partial u(Z(t), t)}{\partial t} + \frac{1}{2} \frac{\partial^2 u(Z(t), t)}{\partial Z(t)^2} \sigma^2(Z(t) - a(t))^2 \right)}_{=0} dt + \frac{u(Z(t), t)}{\partial Z(t)} (-\sigma(Z(t) - a(t)) dB^1(t))
\end{aligned}$$

where

$$\frac{\partial u(Z(t), t)}{\partial t} + \frac{1}{2} \frac{\partial^2 u(Z(t), t)}{\partial Z(t)^2} \sigma^2(Z(t) - a(t))^2 = 0$$

because the Asian call option is a martingale. This is a PDE, which is solved using boundary conditions

$$u(Z(T), T) = (Z(T) - K_1)^+$$

$$Z(0) = \frac{V(0)}{S(0)} = \frac{1}{rT} (1 - e^{-rT}) - e^{-rT} K_2 / S(0)$$

The PDE solution $u(Z(t), t)$ will then yield the price of the Asian call option

$$C(t) = S(t)u(Z(t), t)$$

50.6.2 Lookback options

50.6.3 Barrier options

50.7 Currency options

50.7.1 Foreign asset in domestic currency

There is a domestic market and foreign market. There is a foreign asset $U(t)$

$$dU(t) = r_f U(t) dt + \sigma_U U(t) dB_f(t)$$

Take $X(t)$ as the exchange rate, the price in the domestic currency of one unit of foreign currency.

$$dX(t) = \mu_X X(t) dt + \sigma_X X(t) dB_d(t)$$

The following should be a d-martingale

$$e^{-r_d t} X(t) e^{r_f t} = X(0) e^{(r_f - r_d + \mu_X - 1/2 \sigma_X^2) t + \sigma_X B_d(t)}$$

Since it should be a martingale,

$$\mu_X = r_d - r_f$$

So, the following should be a d-martingale (note $\tilde{U}(t)$ is the foreign asset in domestic currency):

$$\begin{aligned} e^{-r_d t} \tilde{U}(t) &= e^{-r_d t} X(t) U(t) \\ &= X(0) U(0) e^{-\frac{1}{2}(-r_d + r_f + \mu_X + \sigma_U^2 + \sigma_X^2)t + \sigma_U B_f(t) + \sigma_X B_d(t)} \\ &= X(0) U(0) e^{-\frac{1}{2}(\sigma_U^2 + \sigma_X^2)t + \sigma_U B_f(t) + \sigma_X B_d(t)} \end{aligned}$$

Assume

$$dB_f(t) + \rho \sigma_X dt = d\tilde{B}_d(t)$$

where $\tilde{B}_d(t)$ is a Brownian motion correlated to $B_d(t)$. Then,

$$\begin{aligned} dU(t) &= r_f U(t) dt + \sigma_U U(t) dB_f(t) \\ &= r_f U(t) dt + \sigma_U U(t) (d\tilde{B}_d(t) - \rho \sigma_X dt) \\ &= (r_f - \rho \sigma_U \sigma_X) U(t) dt + \sigma_U U(t) d\tilde{B}_d(t) \end{aligned}$$

Now, the foreign asset in domestic currency SDE is

$$\begin{aligned} d\tilde{U}(t) &= d(U(t)X(t)) \\ &= X(t)dU(t) + U(t)dX(t) + dX(t)dU(t) \\ &= \dots \\ &= r_d \tilde{U}(t) dt + (\sigma_U d\tilde{B}_d(t) + \sigma_X dB_d(t)) \tilde{U}(t) \end{aligned}$$

so the volatility is a combination of the foreign asset volatility and the exchange rate volatility.

50.7.2 Options on foreign currency

Call option on foreign currency with strike price in domestic currency K_d . Assume $X(t)$ is lognormal.

$$\begin{aligned} C(t) &= e^{-r_d(T-t)} E_{Q_d}((X(T) - K_d)^+ | \mathcal{F}_t) \\ &= X(t) e^{-r_f(T-t)} \Phi(h(t)) - K_d e^{-r_d(T-t)} \Phi(h(t) - \sigma_X \sqrt{T-t}) \\ h(t) &= \frac{\ln(X(t)/K_d) + (r_d - r_f + \sigma_S^2/2)(T-t)}{\sigma_X \sqrt{T-t}} \end{aligned}$$

50.7.3 Options on foreign assets in foreign currency

Call option on foreign asset denominated in domestic currency. Strike price in foreign currency.

$$\begin{aligned} C(t) &= e^{-r_d(T-t)} E_{Q_d}(X(T)(U(T) - K_f)^+ | \mathcal{F}_t) \\ &= U(t) (\Phi(h(t)) - K_f e^{-r_f(T-t)} \Phi(h(t) - \sigma_U \sqrt{T-t})) \\ h(t) &= \frac{\ln(U(t)/K_f) + (r_f + \sigma_U^2/2)(T-t)}{\sigma_U \sqrt{T-t}} \end{aligned}$$

50.7.4 Options on foreign assets in domestic currency

Call option on foreign asset denominated in domestic currency. Strike price in domestic currency.

$$\begin{aligned}
C(t) &= e^{-r_d(T-t)} E_{Q_d}((X(T)U(T) - K_d)^+ | \mathcal{F}_t) \\
&= e^{-r_d(T-t)} E_{Q_d}((\tilde{U}(t) - K_d)^+ | \mathcal{F}_t) \\
&= \tilde{U}(t)(\Phi(h(t)) - K_d e^{-r_f(T-t)} \Phi(h(t) - \bar{\sigma}\sqrt{T-t})) \\
h(t) &= \frac{\ln(\tilde{U}(t)/K_d) + (r_d + \bar{\sigma}^2/2)(T-t)}{\bar{\sigma}\sqrt{T-t}}
\end{aligned}$$

where $\bar{\sigma} = \sigma_U^2 + \sigma_X^2 + 2\rho\sigma_U\sigma_X$.

50.7.5 Quanto options

Foreign asset payoff converted to domestic currency at predetermined rate. Assume predetermined rate at $t=0$ here.

$$\begin{aligned}
C(t) &= e^{-r_d(T-t)} E_{Q_d}(X(0)(U(T) - K_f)^+ | \mathcal{F}_t) \\
&= X(0)e^{-r_d(T-t)} E_{Q_d}((U(T) - K_f)^+ | \mathcal{F}_t) \\
&= X(0)e^{-r_d(T-t)} (U(t)e^{\delta(T-t)} \Phi(h(t)) - K_f \Phi(h(t) - \sigma_U\sqrt{T-t})) \\
h(t) &= \frac{\ln(U(t)/K_f) + (\delta + \sigma_U^2/2)(T-t)}{\sigma_U\sqrt{T-t}}
\end{aligned}$$

where $\delta = r_f - \rho\sigma_U\sigma_X$.